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PART FIFTEEN - ENVIRONMENTAL AND OCCUPATIONAL HAZARDS

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HARRISON'S PRINCIPLES OF INTERNAL MEDICINE - 15TH EDITION

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DEDICATION

KURT J. ISSELBACHER

With this edition, the editors acknowledge the many contributions of our colleague Kurt J. Isselbacher, who served as an editor of *Harrison's* for nine editions, the sixth through the fourteenth, including Editor-in-Chief of the ninth and thirteenth editions. For more than three decades Dr. Isselbacher played a decisive role in ensuring that *Harrison's* epitomized the state of the art and science of internal medicine and the essence of accuracy and clarity. His indelible contributions to *Harrison's* are felt in the fifteenth edition and will endure into the future.

Dr. Isselbacher is a graduate of Harvard College and of the Harvard Medical School. His further training included a residency in medicine at the Massachusetts General Hospital and a research fellowship at the National Institutes of Health. Chosen to lead the Gastrointestinal Unit of the MGH at the remarkable age of 31, over the ensuing 30 years as Chief of that Unit, he was a leader in advancing both the clinical specialty of gastroenterology and the basic understanding of gastrointestinal disease. Under his leadership, the MGH Gastrointestinal Unit became renowned for its training program in academic gastroenterology as well as for being one of the world's leading centers for clinical and research activities in gastroenterology. In 1987, Dr. Isselbacher undertook new challenges as the first Director of the Cancer Center at the MGH. Bringing his characteristic insight and leadership to this new task, in a relatively short time, the MGH Cancer Center has emerged as a premier cancer research institute. Dr. Isselbacher holds the Mallinckrodt Distinguished Professorship of Medicine at Harvard Medical School, and he has been a powerful force for excellence in scholarship at this institution since his graduation. For almost 30 years he served as Chairman of the Executive Committee of Harvard's Departments of Medicine and played a pivotal role in the departments' growth and quest for excellence.

Dr. Isselbacher combines the attributes of an excellent scientist with those of a superb clinician and teacher. He has trained generations of physicians and investigators, including many who are now leaders in academic medicine. As the author of more than 400 scientific articles in leading journals, his research contributions include definition of enzymatic defects in absorptive disorders, delineation of biochemical mechanisms of absorption, malabsorption, protein synthesis, derangements of metabolism, and immunologic aspects of hepatic gastrointestinal disease. Kurt Isselbacher has been a recipient of many well-earned honors, including the Distinguished Achievement Award and the Friedenwald Medal of the American Gastroenterological Association, the John Phillips Memorial Award for distinguished contributions to clinical medicine from the American College of Physicians, as well as the Kober Medal of the Association of American Physicians. He is a member of the National Academy of Sciences, of its Institute of Medicine, and has served as President of the American Gastroenterological Association, the American Association for the Study of Liver Disease, and the Association of American Physicians.

Kurt Isselbacher exemplifies the highest values of medicine. A caring, empathic physician, he consistently combines compassion with incisive analysis in the care of patients. With contributions as a clinician, teacher, scientist, and editor, he has advanced the care of patients with gastrointestinal disorders and cancer while educating

generations of physicians.

JEAN DONALD WILSON

Jean Wilson served as editor of the ninth through the fourteenth editions of *Harrison's Principles of Internal Medicine* from 1978 to 1998; he was Editor-in-Chief of the twelfth edition. A native of Texas, Jean Wilson attended the University of Texas at Austin and the University of Texas Southwestern Medical School in Dallas. He trained as a resident in internal medicine and as a fellow in endocrinology and metabolism at Parkland Hospital. With the exception of 2 years of research in biochemistry in the intramural program of the National Institutes of Health, Dr. Wilson spent his entire career at the University of Texas Southwestern Medical School, where he now holds the Charles Cameron Sprague Distinguished Chair in Biomedical Science.

Dr. Wilson is one America's most distinguished biomedical scientists and is largely responsible for working out the mechanism of action and physiology of the male sex hormones from the embryo to the normal and diseased adult. Among his many important discoveries has been the 5-reductase reaction, whereby male target tissues convert testosterone to the more active androgen, dihydrotestosterone. He has been honored many times for his research, having received the Ernst Oppenheimer Memorial Award of the Endocrine Society, the Amory Prize of the American Academy of Arts and Sciences, the Lita Annenberg Hazen Award for Excellence in Clinical Research, the Henry Dale Medal of the Society for Endocrinology, the Gregory Pincus Award of the Worcester Foundation for Experimental Biology, the Fred Conrad Koch Award of the Endocrine Society, and the Kober Medal of the Association of American Physicians.

Amongst his memberships are the National Academy of Sciences, the Institute of Medicine of the National Academy of Sciences, and the American Academy of Arts and Sciences. He is a Fellow of the Royal College of Physicians. He has served as President of the American Society for Clinical Investigation, the Association of American Physicians, and the Endocrine Society. For two decades, Dr. Wilson directed the enormously successful MD/PhD program and for 8 years, the highly esteemed Endocrine-Metabolism Division at Southwestern.

Perhaps the finest thing that can be said of Jean Wilson is that he is a professor of internal medicine in the complete sense. He is, and always has been, a superb teacher and exemplary clinician while constantly maintaining a sterling career in research. Perhaps nothing describes him better than the enduring image of this renowned academic physician trimming callouses and ulcers in the Diabetic Foot Care Clinic at Parkland Memorial Hospital, his teaching hospital, where the patients are the medically indigent of Dallas.

Dr. Wilson is a man of diverse interests, a true intellectual. One of his great gifts and loves is scientific and medical editing. He served as Editor-in-Chief of the *Journal of Clinical Investigation* and of *William's Textbook of Endocrinology*. As an editor of *Harrison's* for two decades, Dr. Wilson made ample use of his conspicuous strengths as clinician, teacher, and scientist. His meticulous scholarship and high standards have

had an enormous impact, not only on the Endocrinology, Metabolism, and Genetics sections, for which he had primary responsibility, but on the entire book.

JOSEPH B. MARTIN

The editors wish to acknowledge the enormous contributions made by Joseph B. Martin, who edited the Neurology section of *Harrison's* from the tenth to the fourteenth editions. Dr. Martin followed Dr. Raymond D. Adams as editor of the Neurology section. In retrospect, the choice of Joseph Martin to replace Adams was prescient. It foresaw the transformation of neurology in the 1980s and 1990s from a largely descriptive discipline to one of the most dynamic and rapidly evolving areas of internal medicine. With his appointment as editor, the textbook had secured the foremost leader in the new field of molecular neurology to its ranks. Beginning with the tenth edition, Martin built upon the powerful didactic structure of the "syndromic approach" to neurology created by Adams and emphasized advances in molecular genetics and cell biology that reclassify neurologic diseases, clarify disease mechanisms, and offer new insights into clinical diagnosis and therapy. During his tenure, the neurology section of *Harrison's* became the best resource of its kind for the exposition of new discoveries in neurology and contributed substantially to the growing overall success of the textbook.

Born in Bassano, Alberta, Canada, Dr. Martin received his premedical and medical education at the University of Alberta, Edmonton, trained in neurology at Case Western University, and received the PhD from the University of Rochester. His career in academic medicine began in 1971 at McGill University in Montreal, where he established an independent laboratory focused on hypothalamic regulation of pituitary hormone secretion, and where he quickly rose to become Chair of the Department of Neurology and Neurosurgery. In 1978, he joined the faculty at Harvard Medical School as Bullard (later Julianne Dorn) Professor of Neurology and Chief of the Neurology Service at Massachusetts General Hospital. While at Harvard, he established the Huntington's Disease Center Without Walls, which in 1984 reported the spectacular finding of a genetic marker linked to Huntington's disease, thereby inaugurating the modern era of molecular neurogenetics. In 1989 Dr. Martin joined the University of California, San Francisco, initially serving as Dean of the School of Medicine and subsequently as Chancellor. Among his many achievements at UCSF was the conception of a major new research campus in San Francisco, which is fast becoming a reality. In July, 1997, he returned to Harvard as the Caroline Shields Walker Professor of Neurobiology and Clinical Neuroscience and Dean of the Faculty of Medicine. A wonderful teacher and physician, Joe Martin has inspired a generation of housestaff, students, and colleagues at Harvard and UCSF.

Dr. Martin has received many honors, including honorary degrees from five distinguished universities and the Abraham Flexner Award of the Association of American Medical Colleges. He serves or has served on the editorial boards of nineteen medical and neurology journals. He is a member of the Institute of Medicine of the National Academy of Sciences and has served as President of the American Neurological Association.

Dr. Martin's many contributions to this textbook were enhanced by his extraordinary organizational skills and by a clear and direct style of writing and editing that permitted him to distill complex concepts into easily readable prose accessible to a general medical readership. As an example, his chapter on neurogenetics has become an instant classic and a highlight of each new edition of the book. The editors greatly value their friendship with this remarkable man whose integrity and intellectual strengths have served *Harrison's* so well during the past two decades. Although he is no longer an editor, we are delighted that Dr. Martin will continue to contribute his expertise to *Harrison's* as an author.

NOTICE

Medicine is an ever-changing science. As new research and clinical experience broaden our knowledge, changes in treatment and drug therapy are required. The editors and the publisher of this work have checked with sources believed to be reliable in their efforts to provide information that is complete and generally in accord with the standards accepted at the time of publication. However, in view of the possibility of human error or changes in medical sciences, neither the editors nor the publisher nor any other party who has been involved in the preparation or publication of this work warrants that the information contained herein is in every respect accurate or complete, and they are not responsible for any errors or omissions or the results obtained from the use of such information. Readers are encouraged to confirm the information contained herein with other sources. For example and in particular, readers are advised to check the product information sheet included in the package of each drug they plan to administer to be certain that the information contained in this book is accurate and that changes have not been made in the recommended dose or in the contraindications for administration. This recommendation is particularly important in connection with new or infrequently used drugs.

PREFACE

The first edition of *Harrison's Principles of Internal Medicine* was published in the middle of the twentieth century, more than 50 years ago. In this fifteenth edition, the first of the new century, the text has undergone major revision to reflect further understanding of the biology and pathophysiology of disease and at the same time to retain those facts that, while not new, remain clinically useful and important. Virtually every chapter in this new edition has been completely or substantially rewritten, and a record 86 are new or have new authors. In this preface, we cannot describe all of these changes; however, we would like to call to the reader's attention those that are particularly noteworthy.

Part One, "Introduction to Clinical Medicine," contains new chapters dealing with decision making and cost awareness in clinical medicine. A growing number of patients are turning to alternative therapies, and these are discussed in a new chapter. New authors describe contemporary approaches to medical problems associated with pregnancy and the peripartum period. The chapters on medical ethics and on segments of the population that often present special problems^{3/4}adolescents, women, and the elderly^{3/4}have been revised and updated.

Part Two, "Cardinal Manifestations and Presentation of Disease," serves as a

comprehensive introduction to clinical medicine, examining current concepts of the pathophysiology and differential diagnosis to be considered in patients with these manifestations. Major symptoms are reviewed and correlated with specific disease states, and clinical approaches to patients presenting with these symptoms are summarized. New chapters have been prepared on chest discomfort, headache, hypothermia, shock, and disorders of smell, taste, and hearing. A new chapter succinctly outlines a rational approach to the febrile patient presenting to the emergency department. The sections on alterations in gastrointestinal and sexual function are almost entirely new.

Given the explosive advances in human genetics, including the completion of a working draft of the sequence of the entire human genome and its growing relevance to clinical practice, Part Three, "Genetics and Disease," has been expanded and completely rewritten with new chapters on human genetics, chromosomal genetics, genetic defects, mitochondrial dysfunction, genetic screening and counseling, as well as gene therapy.

Part Four, "Clinical Pharmacology," provides a sound theoretical basis for pharmacotherapy, so critical to every aspect of medical practice.

Part Five, "Nutrition," has been extensively revised, with five new authors contributing chapters. This section covers nutritional considerations related to clinical medicine, including nutritional requirements, assessment of nutritional status, protein-energy malnutrition, and enteral and parenteral nutrition. It contains a new chapter on obesity, which incorporates the results of rapidly developing basic research in this important field.

The core of *Harrison's* encompasses the disorders of the organ systems and is contained in Parts Six through Fifteen. These sections include succinct accounts of the pathophysiology of the diseases involving the major organ systems and emphasize clinical manifestations, diagnostic procedures, differential diagnosis, and treatment strategies. The treatment sections of virtually every chapter have been amplified and updated. They are supplemented by the liberal use of algorithms, and are clearly highlighted. Guidelines for disease management prepared by specialty societies are included for the first time.

Part Six, "Oncology and Hematology," includes twelve chapters with new authors, including a new chapter by Judah Folkman on angiogenesis. In addition, a new chapter has been added on the medical problems that can arise in patients cured of cancer, including disease-related and treatment-related sequelae. The chapters on myeloid and lymphoid neoplasms include the new World Health Organization classification schemes. A conscientious effort has been made to provide specific, up-to-date treatment recommendations. Where appropriate, diagnostic and management algorithms have been incorporated.

Changes in Part Seven, "Infectious Diseases," include the latest information on the pathology, genetics, and epidemiology of infectious diseases while focusing sharply on the needs of clinicians who must accurately diagnose and treat infections in their patients. Specific recommendations are offered for therapeutic regimens, including the drug of choice, dose, duration, and alternatives. Current figures and trends in

antimicrobial resistance are presented and considered in light of their impact on therapeutic choices. New authors cover the latest advances in the management of diseases such as infective endocarditis, meningococcal and gonococcal infections, and schistosomiasis. The overview of pathogenesis from earlier editions has been expanded to encompass viruses, fungi, and parasites as well as bacteria. The Atlas of Hematology includes a complete diagnostic set of spectacular color plates showing malaria-infected red blood cells.

In Part Eight, "Disorders of the Cardiovascular System," a new chapter on the prevention of atherosclerosis focuses not only on the importance of the traditional risk factors but also on the novel risk factors that influence plaque stability. Global risk assessment and management are described. Both primary and secondary prevention of atherosclerosis are discussed. Myocardial imaging by means of ultrasound or radionuclide techniques, at rest and during stress, plays an ever more critical role in assessment of patients with ischemic heart disease, and a new chapter focuses on the clinical use of these important technologies.

Despite major advances in its diagnosis and therapy, acute myocardial infarction remains the most common cause of death in industrialized nations. The chapter on acute myocardial infarction provides important new information on myocardial reperfusion therapy, thrombolysis, and primary coronary angioplasty and summarizes guidelines for acute coronary care and for risk stratification in the postinfarct patient. Unstable angina and congestive heart failure have emerged as two of the most common conditions leading to hospital admission in Western nations. Important advances in pathophysiology and therapy of these two very important conditions are included.

Enormous strides have been made in the use of lung transplantation for selected patients with end-stage, irreversible, pulmonary parenchymal and vascular disease, and Part Nine, "Disorders of the Respiratory System," provides a chapter that focuses on patient selection for this therapy. New chapters on interstitial and granulomatous lung diseases as well as on sleep apnea provide contemporary views of these conditions at the interface between basic science and clinical pulmonology.

In Part Ten, "Disorders of the Kidney and Urinary Tract," there has been considerable revision, with a new chapter on dialysis, incorporating the most recent advances.

In Part Eleven, "Disorders of the Gastrointestinal System," several new authors have contributed to the section on liver and biliary tract disease, and all chapters have been extensively revised. The section is pivoted by a new chapter on "Approach to the Patient with Liver Disease." Recent advances in the therapy of hepatitis B and C have been highlighted. New authors have contributed chapters on endoscopy, peptic ulcer disease, disorders of absorption, inflammatory bowel disease, and irritable bowel syndrome. Our new contributors include the leaders in gastroenterology and hepatology.

In Part Twelve, "Disorders of the Immune System, Connective Tissue, and Joints," the updating focuses on therapy. The chapter on "Introduction to the Immune System" has been completely rewritten and provides a comprehensive review of the human immune system, using the modern designations of innate versus adaptive immunity. The chapter on HIV disease and AIDS is comprehensive and up-to-date and includes coverage of

the natural history, epidemiology, and immunopathogenic mechanisms of HIV disease. In addition, the chapter contains both an organ system by organ system approach and a delineation of the major complications of HIV disease. The sections on therapy include a state-of-the-art discussion of the striking treatment advances of HIV infection with combinations of antiretroviral agents as well as the complications of such therapy.

Profound changes can be found in Part Thirteen, "Endocrinology and Metabolism." Many new authors have been recruited, and all chapters have been extensively revised under the direction of our new editor, Dr. J. Larry Jameson. Nine of these chapters are completely new, including those on the pituitary, thyroid, diabetes mellitus, and osteoporosis. These clinically demanding topics retain a traditional pathophysiologic approach that characterizes the field of endocrinology. In addition, new insights from genetics permeate this section, and the results of evidence-based medicine provide a firm foundation for medical decision making and treatment.

Part Fourteen, "Neurologic Disorders," has been thoroughly updated and expanded. The theme of genetics is emphasized throughout the section, and new chapters highlight the remarkable progress made during the "decade of the brain" in the 1990s that has elucidated the molecular basis of many neurologic and psychiatric diseases. One of the new chapters, written by 1997 Nobel Laureate Stanley B. Prusiner, summarizes the unique biology of prions and the clinical features of human prion disorders, including "mad cow disease."

The very latest information can be found on treatment of epilepsy, Parkinson's disease, and Alzheimer's disease. Coverage of immune-mediated disorders of the nervous system has been greatly expanded to include the many new insights into pathogenesis and treatment that have appeared since the fourteenth edition. The chapter on cerebrovascular diseases offers state-of-the-art information on prevention and treatment of stroke, the third leading killer in the developed world; this chapter is a mini-textbook of stroke and stroke therapy. Another feature of the fifteenth edition is a discussion of the acute neurologic disorders encountered in the setting of critical illness; this chapter should be of value to all physicians who care for hospitalized patients.

Throughout the book, there is an emphasis on the use of neuroimaging figures to illustrate the various disorders discussed. *Harrison's* exceptional collection of high-quality neuroimaging photographs sets a new standard for textbooks of medicine.

Finally, Part Fifteen, "Environmental and Occupational Hazards," has been expanded and reorganized.

In view of the requirements for continuing education for licensure and relicensure, as well as the emphasis on certification and recertification, a revision of the Pre-Test Self-Assessment and Review will again be published with this edition. It consists of several hundred questions based on *Harrison's*, along with answers and explanations for the answers. The *Companion Handbook* that was pioneered as a supplement to the eleventh edition of *Harrison's* has been updated and will appear shortly after the publication of this edition. A CD-ROM version of *Harrison's* has been available since the thirteenth edition. An expanded CD-ROM version of the fifteenth edition will be available and will be regularly updated. In 1998, *Harrison's* went online to provide a "living"

textbook of internal medicine. In addition to providing full search capabilities of the text, *Harrison's Online* offers daily updating, reports of clinical trials, practice guidelines, and concise reviews of timely topics, as well as new references with links to MEDLINE abstracts.

The fifteenth edition of *Harrison's* welcomes a new editor, Dr. J. Larry Jameson, who has taken on principal responsibility for the sections on Nutrition, Genetics, Endocrinology, and Metabolism and whose impact on this edition is already clear. Dr. Kurt J. Isselbacher, Dr. Jean D. Wilson, and Dr. Joseph B. Martin have left the editorial group. Their enormous contributions to *Harrison's* are cited elsewhere. Special thanks go to Dr. Robert F. Schrier who has prepared biographies of Nobel Prize Laureates in Physiology or Medicine. These brief essays remind us how deeply our current knowledge and practice of medicine depends on seminal contributions to biomedical science and informs about the lives of some of the most outstanding contributors.

We wish to express our appreciation to our many associates and colleagues, who, as experts in their fields, have helped us with constructive criticism and helpful suggestions. We acknowledge especially the contributions of:

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Finally, we continue to be indebted to two outstanding members of the McGraw-Hill organization: Mariapaz Ramos Englis, Senior Managing Editor, and Martin J. Wonsiewicz, Publisher. They are an effective team who have given the editors constant encouragement and sage advice and have been of enormous help in bringing this edition to fruition in a timely manner.

The Editors

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PART ONE -INTRODUCTION TO CLINICAL MEDICINE

1. THE PRACTICE OF MEDICINE - *The Editors*

WHAT IS EXPECTED OF THE PHYSICIAN

The practice of medicine combines both science and art. The role of *science in medicine* is clear. Science-based technology and deductive reasoning form the foundation for the solution to many clinical problems; the spectacular advances in genetics, biochemistry, and imaging techniques allow access to the innermost parts of the cell and the most remote recesses of the body. Highly advanced therapeutic maneuvers are increasingly a major part of medical practice. Yet skill in the most sophisticated application of laboratory technology and in the use of the latest therapeutic modality alone does not make a good physician. One must be able to identify the crucial elements in a complex history and physical examination and extract the key laboratory results from the crowded computer printouts of laboratory data in order to determine in a difficult case whether to "treat" or to "watch." Deciding when a clinical clue is worth pursuing, or when it should be dismissed as a "red herring," and estimating in any given patient whether a proposed treatment entails a greater risk than the disease are essential to the decision-making process that the skilled clinician must exercise many times each day. This combination of medical knowledge, intuition, and judgment defines the *art of medicine*, which is as necessary to the practice of medicine as is a sound scientific base.

The editors of the first edition of this book articulated what is expected of the physician in words that, although they reflect the gender bias of that era, still ring true as a universal principle:

No greater opportunity, responsibility, or obligation can fall to the lot of a human being than to become a physician. In the care of the suffering he needs technical skill, scientific knowledge, and human understanding. He who uses these with courage, with humility, and with wisdom will provide a unique service for his fellow man, and will build an enduring edifice of character within himself. The physician should ask of his destiny no more than this; he should be content with no less.

Tact, sympathy and understanding are expected of the physician, for the patient is no mere collection of symptoms, signs, disordered functions, damaged organs, and disturbed emotions. He is human, fearful, and hopeful, seeking relief, help and reassurance.

THE PATIENT-PHYSICIAN RELATIONSHIP

It may seem trite to emphasize that physicians need to approach patients not as "cases" or "diseases" but as individuals whose problems all too often transcend their physical complaints. Most patients are anxious and frightened. Physicians should instill confidence and reassurance, overtly and in their demeanor, but without an air of arrogance. A professional attitude, coupled with warmth and openness, can do much to alleviate the patients' anxiety and to encourage them to share parts of their history that may be embarrassing. Some patients "use" illness to gain attention or to serve as a

crutch to extricate themselves from a stressful situation; some even feign physical illness; others may be openly hostile. Whatever the patient's attitude, the physician needs to consider the setting in which an illness occurs -- in terms not only of the patients themselves but also of their families and social and cultural backgrounds. The ideal patient-physician relationship is based on thorough knowledge of the patient, on mutual trust, and on the ability to communicate with one another.

The direct, one-to-one patient-physician relationship, which has traditionally characterized the practice of medicine, is increasingly in jeopardy because of the increasing complexity of medicine and change in health care delivery systems. Often the management of the individual patient is a team effort involving a number of several different physicians and professional personnel. The patient can benefit greatly from such collaboration, but *it is the duty of the patient's principal physician to guide them through an illness*. To carry out this difficult task, this physician must be familiar with the techniques, skills, and objectives of specialist physicians and of colleagues in the fields allied to medicine. In giving the patient an opportunity to benefit from scientific advances, the primary physician must, in the last analysis, retain responsibility for the major decisions concerning diagnosis and treatment.

Patients are increasingly cared for by groups of physicians in clinics, hospitals, integrated health care delivery systems, and health maintenance organizations (HMOs). Whatever the potential advantages of such organized medical groups, there are also drawbacks, chiefly the loss of the clear identification of the physician who is primarily and continuously responsible for the patient. Even under these circumstances, it is essential for each patient to have a physician who has an overview of the problems and who is familiar with the patient's reaction to the illness, to the drugs given, and to the challenges that the patient faces.

The practice of medicine in a "managed care" setting puts additional stress on the classic paradigm of the patient-physician relationship. Many physicians must deal with a patient within a restricted time frame, with limited access to specialists, and under organizational guidelines that may compromise their ability to exercise their individual clinical judgment. As difficult as these restrictions may be, it is the ultimate responsibility of the physician to determine what is best for the patient. This responsibility cannot be relinquished in the name of compliance with organizational guidelines.

The physician must also bear in mind that the modern hospital constitutes an intimidating environment for most patients. Lying in a bed surrounded by air jets, buttons, and lights; invaded by tubes and wires; beset by the numerous members of the health care team -- nurses, nurses' aides, physicians' assistants, social workers, technologists, physical therapists, medical students, house officers, attending and consulting physicians, and many others; sharing rooms with other patients who have their own problems, visitors, and physicians; transported to special laboratories and imaging facilities replete with blinking lights, strange sounds, and unfamiliar personnel -- it is little wonder that patients may lose their sense of reality. In fact, the physician is often the only tenuous link between the patient and the real world, and a strong personal relationship with the physician helps to sustain the patient in such a stressful situation.

Many trends in contemporary society tend to make medical care impersonal. Some of these have been mentioned already and include (1) vigorous efforts to reduce the escalating costs of health care; (2) the growing number of managed care programs, which are intended to reduce costs but in which the patient may have little choice in selecting a physician; (3) increasing reliance on technologic advances and computerization for many aspects of diagnosis and treatment; (4) increased geographic mobility of both patients and physicians; (5) the need for numerous physicians to be involved in the care of most patients who are seriously ill; and (6) an increasing tendency on the part of patients to express their frustrations with the health care system by legal means (i.e., by malpractice litigation). Given these changes in the medical care system, it is a major challenge for physicians to maintain the *humane* aspects of medical care. The American Board of Internal Medicine has defined humanistic qualities as encompassing integrity, respect, and compassion. Availability, the expression of sincere concern, the willingness to take the time to explain all aspects of the illness, and a nonjudgmental attitude when dealing with patients whose cultures, lifestyles, attitudes, and values differ from those of the physician are just a few of the characteristics of the humane physician. Every physician will, at times, be challenged by patients who evoke strongly negative (or strongly positive) emotional responses. Physicians should be alert to their own reactions to such patients and situations and should consciously monitor and control their behavior so that the patients' best interests remain the principal motivation for their actions at all times.

An important aspect of patient care involves an appreciation of the "quality of life," a subjective assessment of what each patient values most. Such an assessment requires detailed, sometimes intimate knowledge of the patient, which can usually be obtained only through deliberate, unhurried, and often repeated conversations. It is in these situations that the time constraints of a managed care setting may prove problematic.

The famous statement of Dr. Francis Peabody is even more relevant today than when delivered more than three-quarters of a century ago:

*The significance of the intimate personal relationship between physician and patient cannot be too strongly emphasized, for in an extraordinarily large number of cases both the diagnosis and treatment are directly dependent on it. One of the essential qualities of the clinician is interest in humanity, **for the secret of the care of the patient is in caring for the patient.***

CLINICAL SKILLS

History Taking The written history of an illness should embody all the facts of medical significance in the life of the patient. Recent events should be given the most attention. The patient should, at some point, have the opportunity to tell his or her own story of the illness without frequent interruption and, when appropriate, receive expressions of interest, encouragement, and empathy from the physician. The physician must be alert to the possibility that any event related by the patient, however trivial or apparently remote, may be the key to the solution of the medical problem.

An informative history is more than an orderly listing of symptoms; something is always gained by listening to patients and noting the way in which they describe their

symptoms. Inflections of voice, facial expression, gestures, and attitude may reveal important clues to the meaning of the symptoms to the patient. Taking history often involves much data gathering. Patients vary in their medical sophistication and ability to recall facts. Medical history should therefore be corroborated whenever possible. The family and social history can also provide important insights into the types of diseases that should be considered. In listening to the history, the physician discovers not only something about the disease but also something about the patient. The process of history taking provides an opportunity to observe the patient's behavior and to watch for features to be pursued more thoroughly during the physical examination.

The very act of eliciting the history provides the physician with the opportunity to establish or enhance the unique bond that is the basis for the ideal patient-physician relationship. It is helpful to develop an appreciation of the patient's perception of the illness, the patient's expectations of the physician and the medical care system, and the financial and social implications of the illness to the patient. The confidentiality of the patient-physician relationship should be emphasized, and the patient should be given the opportunity to identify any aspects of the history that should not be disclosed.

Physical Examination Physical signs are objective indications of disease whose significance is enhanced when they confirm a functional or structural change already suggested by the patient's history. At times, however, the physical signs may be the only evidence of disease.

The physical examination should be performed methodically and thoroughly, with consideration for the patient's comfort and modesty. Although attention is often directed by the history to the diseased organ or part of the body, the examination of a new patient must extend from head to toe in an objective search for abnormalities. Unless the physical examination is systematic, important segments may be omitted. The results of the examination, like the details of the history, should be recorded at the time they are elicited, not hours later when they are subject to the distortions of memory. Skill in physical diagnosis is acquired with experience, but it is not merely technique that determines success in eliciting signs. The detection of a few scattered petechiae, a faint diastolic murmur, or a small mass in the abdomen is not a question of keener eyes and ears or more sensitive fingers but of a mind alert to these findings. Since physical findings are subject to changes, the physical examination should be repeated as frequently as the clinical situation warrants.

Laboratory Tests The availability of a wide array of laboratory tests has increased our reliance on these studies for the solution of clinical problems. The accumulation of laboratory data does not relieve the physician from the responsibility of careful observation, examination, and study of the patient. It is also essential to bear in mind the limitations of such tests. By virtue of their impersonal quality, complexity, and apparent precision, they often gain an aura of authority regardless of the fallibility of the tests themselves, the instruments used in the tests, and the individuals performing or interpreting them. Physicians must weigh the expense involved in the laboratory procedures they order relative to the value of the information they are likely to provide.

Single laboratory tests are rarely ordered. Rather, they are generally obtained as "batteries" of multiple tests, which are often useful. For example, abnormalities of

hepatic function may provide the clue to such nonspecific symptoms as generalized weakness and increased fatigability, suggesting the diagnosis of chronic liver disease. Sometimes a single abnormality, such as an elevated serum calcium level, points to particular diseases, such as hyperparathyroidism or underlying malignancy.

The thoughtful use of screening tests should not be confused with indiscriminate laboratory testing. The use of screening tests is based on the fact that a group of laboratory determinations can be carried out conveniently on a single specimen of blood at relatively low cost. Screening tests are most useful when they are directed towards common diseases or disorders in which the result directs other useful tests or interventions that would otherwise be costly to perform. Biochemical measurements, together with simple laboratory examinations such as blood count, urinalysis, and sedimentation rate, often provide the major clue to the presence of a pathologic process. At the same time, the physician must learn to evaluate occasional abnormalities among the screening tests that may not necessarily connote significant disease. An in-depth workup following a report of an isolated laboratory abnormality in a person who is otherwise well is almost invariably wasteful and unproductive. Among the more than 40 tests that are routinely performed on patients, one or two are often slightly abnormal. If there is no suspicion of an underlying illness, these tests are ordinarily repeated to ensure that the abnormality does not represent a laboratory error. If an abnormality is confirmed, it is important to consider its potential significance in the context of the patient's condition and other test results.

Imaging Techniques The availability of ultrasonography, a variety of scans that employ isotopes to visualize organs heretofore inaccessible, computed tomography, and magnetic resonance imaging has opened new diagnostic vistas and has benefited patients because these new techniques have largely supplanted more invasive ones. While the enthusiasm for noninvasive technology is understandable, the expense entailed in performing these tests is often substantial and should be considered when assessing the potential benefits of the information provided.

PRINCIPLES OF PATIENT CARE

Medical Decision-Making Both during and in particular after the physician has taken the history, performed the physical examination, and reviewed the laboratory and imaging data, the challenging process of the differential diagnosis and medical decision-making begins. Formulating a differential diagnosis requires not only a broad knowledge base but also the ability to assess the relative probabilities of various diseases and to understand the significance of missing diagnoses that may be less likely. Arriving at a diagnosis requires the application of the scientific method. Hypotheses are formed, data are collected, and objective conclusions are reached concerning whether to accept or reject a particular diagnosis. Analysis of the differential diagnosis is an iterative process. As new information or test results are acquired, the group of disease processes being considered can be contracted or expanded appropriately. Medical decision-making occurs throughout the diagnostic and treatment process. It involves the ordering of additional tests, requests for consults, and decisions regarding prognosis and treatment. This process requires an in-depth understanding of the natural history and pathophysiology of disease, explaining why these features are strongly emphasized in this textbook. As described below, medical decision-making

should be evidence-based, thereby ensuring that patients derive the full benefit of the scientific knowledge available to physicians.

Evidence-Based Medicine Sackett has defined evidence-based medicine as "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients." Rigorously obtained evidence is contrasted with anecdotal experience, which is often biased. Even the most experienced physicians can be influenced by recent experiences with selected patients, unless they are attuned to the importance of using larger, more objective studies for making decisions. The prospectively designed, double-blind, randomized clinical trial represents the "gold standard" for providing evidence regarding therapeutic decisions, but it is not the only source. Valuable evidence about the natural history of disease and prognosis can come from prospective cohort studies and analytic surveys. Persuasive evidence on the accuracy of diagnostic tests can be derived from cross-sectional studies of patients in whom a specific disorder is suspected. Evidence is strengthened immensely when it has been confirmed by multiple investigations, which can be compared with one another and presented in a meta-analysis or systemic overview.

In failing to apply the best and most current evidence, the physician places the patient at unnecessary risk. However, a knowledge of or rapid access to the best available evidence is not sufficient for optimal care. The physician must know whether the evidence is relevant to the patient in question and, when it is, the consequences of applying it in any particular situation. The skills and judgment required to apply sound evidence represent an increasing challenge. Indeed, one might redefine a "good doctor" as one who uses the ever-growing body of rigorously obtained evidence (the science of medicine) in a sensible, compassionate manner (the art of medicine).

While an understanding of biologic and physiologic mechanisms forms the basis of contemporary medicine, when a therapeutic modality is selected, the highest priority must often be placed on improving *clinical outcome* rather than interrupting what is believed to be the underlying process. For example, for decades patients who had suffered myocardial infarction were treated intuitively with drugs that suppress frequent ventricular extrasystoles, since these were believed to be harbingers of ventricular fibrillation and sudden death. Clinical trials, however, have provided firm evidence that the antiarrhythmic agents actually increase the risk of death in such patients. This finding suggests that the extrasystoles are *markers* of high risk rather than the *cause* of fatal events.

Practice Guidelines Physicians are faced with a large, increasing, and often bewildering body of evidence pointing to potentially useful diagnostic techniques and therapeutic choices. The intelligent and cost-effective practice of medicine consists of making selections most appropriate to a particular patient and clinical situation. Professional organizations and government agencies are developing formal clinical practice guidelines in an effort to aid physicians and other caregivers in this endeavor. When guidelines are current and properly applied, they can provide a useful framework for managing patients with particular diagnoses or symptoms. They can protect patients -- particularly those with inadequate health care benefits -- from receiving substandard care. Guidelines can also protect conscientious caregivers from inappropriate charges of malpractice and society from the excessive costs associated with the overuse of

medical resources. On the other hand, clinical guidelines tend to oversimplify the complexities of medicine. Different groups with differing perspectives may develop divergent recommendations regarding issues as basic as the need for periodic sigmoidoscopy in middle-aged persons. Furthermore, guidelines do not -- and cannot be expected to -- take into account the uniqueness of each individual and of his or her illness. The challenge for the physician is to integrate into clinical practice the useful recommendations offered by the experts who prepare clinical practice guidelines without accepting them blindly or being inappropriately constrained by them.

Assessing the Outcome of Treatment Clinicians generally use *objective* and readily measurable parameters to judge the outcome of a therapeutic intervention. For example, findings on physical or laboratory examination -- such as the level of blood pressure, the patency of a coronary artery on an angiogram, or the size of a mass on a radiologic examination -- can provide information of critical importance. However, patients usually seek medical attention for *subjective* reasons; they wish to obtain relief from pain, to preserve or regain function, and to enjoy life. The components of a patient's health status or quality of life can include bodily comfort, capacity for physical activity, personal and professional function, sexual function, cognitive function, and overall perception of health. Each of these important areas can be assessed by means of structured interviews or specially designed questionnaires. Such assessments also provide useful parameters by which the physician can judge the patient's subjective view of his or her disability and the response to treatment, particularly in chronic illness. The practice of medicine requires consideration and integration of both objective and subjective outcomes.

Care of the Elderly Over the next several decades, the practice of medicine will be greatly influenced by the health care needs of the growing elderly population. In the United States the population over age 65 will almost triple over the next 30 years. It is essential that we understand and appreciate the physiologic processes associated with aging; the different responses of the elderly to common diseases; and disorders that occur commonly with aging, such as depression, dementia, frailty, urinary incontinence, and fractures. The elderly have more adverse reactions to drugs, in large part due to altered pharmacokinetics and pharmacodynamics. Commonly used medications such as digoxin and aminoglycosides have prolonged half-lives in the elderly, and tissues such as the central nervous system are more sensitive to certain drugs, such as the benzodiazepines and narcotics. The large number of drugs used by the elderly increases the risk of unwanted interactions, especially when care is provided by several physicians in an uncoordinated manner.

Diseases in Women versus Men In the past, many epidemiologic studies and clinical trials focused on men. It is now appreciated that there are significant gender differences in diseases that afflict both men and women. Mortality rates are substantially higher in women than in men under the age of 50 suffering acute myocardial infarction. Hypertension is more prevalent in African-American women than in their male counterparts (and in African-American than in white males); osteoporosis is more common in women, reflecting the menopausal loss of estrogen; diseases involving the immune system, such as lupus erythematosus, multiple sclerosis, and primary biliary cirrhosis, occur more frequently in women; and the average life expectancy of women is greater than that of men. Recently, considerable attention has been paid to women's

health issues, a subject that regrettably did not receive sufficient attention in the past. Ongoing study should enhance our understanding of the mechanisms of gender differences in the course and outcome of certain diseases.

Iatrogenic Disorders In an *iatrogenic disorder*, the deleterious effects of a therapeutic or diagnostic maneuver cause pathology independent of the condition for which the intervention was performed. Adverse drug reactions occur in at least 5% of hospitalized patients, and the incidence increases with use of a large number of drugs. No matter what the clinical situation, it is the responsibility of the physician to use powerful therapeutic measures wisely, with due regard for their beneficial action, potential dangers, and cost. Every medical procedure, whether diagnostic or therapeutic, has the potential for harm, but it would be impossible to provide the benefits of modern scientific medicine if reasonable steps in diagnosis and therapy were withheld because of possible risks. *Reasonable* implies that the physician has weighed the pros and cons of a procedure and has concluded, on the basis of objective evidence whenever possible, that it is necessary for establishing a diagnosis, for the relief of discomfort, or for the cure of disease. However, the harm that a physician can do is not limited to the imprudent use of medication or procedures. Equally important are ill-considered or unjustified remarks. Many a patient has developed a cardiac neurosis because the physician ventured a grave prognosis on the basis of a misinterpreted finding of a heart murmur. Not only the diagnostic procedure or the treatment but the physician's words and behavior are capable of causing injury.

Informed Consent Patients often require diagnostic and therapeutic procedures that are painful and that pose some risk. For many such procedures, patients are required to sign a consent form. The patient must understand clearly the risks entailed in these procedures; this is the definition of *informed consent*. It is incumbent on the physician to explain the procedures in a clear and understandable manner and to ascertain that the patient comprehends both the nature of the procedure and the attendant risks. The dread of the unknown that is inherent in hospitalization can be mitigated by such explanations.

Incurability and Death No problem is more distressing than that presented by the patient with an incurable disease, particularly when premature death is inevitable. What should the patient and family be told, what measures should be taken to maintain life, what can be done to maintain the quality of life, and how is death to be defined?

The concept of incurable illness and terminal care often evokes examples of cancer. However, patients with many other end-stage diseases including chronic obstructive pulmonary disease, congestive heart failure, renal or hepatic failure, and overwhelming infection face similar issues. The same principles of terminal care should be applied in each of these cases. Doing seemingly small things, focused on the needs of the patient, can do much to restore comfort or dignity during a person's final weeks or days. In the same way that pain should be attentively managed with analgesia, every effort should be made to alleviate shortness of breath and to provide good skin care.

Although some would argue otherwise, there is no ironclad rule that the patient must immediately be told "everything," even if the patient is an adult with substantial family responsibilities. How much is told should depend on the individual's ability to deal with

the possibility of imminent death; often this capacity grows with time, and whenever possible, gradual rather than abrupt disclosure is the best strategy. A wise and insightful physician is often guided by an understanding of what a patient wants to know and when he or she wants to know it. The patient's religious beliefs may also be taken into consideration. The patient must be given an opportunity to talk with the physician and ask questions. Patients may find it easier to share their feelings about death with their physician, who is likely to be more objective and less emotional, than with family members. As William Osler wrote:

One thing is certain; it is not for you to don the black cap and, assuming the judicial function, take hope away from any patient...hope that comes to us all.

Even when the patient directly inquires, "Am I dying?" the physician must attempt to determine whether this is a request for information or a demand for reassurance. Only open communication between the patient and the physician can resolve this question and guide the physician in what to say and how to say it.

The physician should provide or arrange for emotional, physical, and spiritual support and must be compassionate, unhurried, and open. There is much to be gained by the laying on of hands. Pain should be adequately controlled, human dignity maintained, and isolation from the family avoided. These aspects of care tend to be overlooked in hospitals, where the intrusion of life-sustaining apparatus can so easily detract from attention to the whole person and encourage concentration instead on the life-threatening disease, against which the battle will ultimately be lost in any case. In the face of terminal illness, the goal of medicine must shift from *cure* to *care*, in the broadest sense of the term. In offering care to the dying patient, the physician must be prepared to provide information to family members and to deal with their guilt and grief. It is important for the doctor to assure the family that everything possible has been done.

"Do Not Resuscitate" Orders and Cessation of Therapy When carried out in a timely and expert manner, cardiopulmonary resuscitation is often useful in the prevention of sudden, unexpected death. However, unless there are reasons to the contrary, this procedure should not be used merely to prolong the life of a patient with terminal, incurable disease. The decision whether or not to resuscitate or even to treat an incurably and terminally ill patient must be reviewed frequently and must take into consideration any unexpected changes in the patient's condition. In this context, the administration of fluids or food is considered therapy that may be withdrawn or withheld. These decisions must also take into account both the underlying medical condition, especially its reversibility, and the wishes of the patient, especially if these have been expressed in a living will or advance directive. If the patient's wishes cannot be ascertained directly, a close relative or another surrogate who can be relied on to transmit the patient's wishes and to be guided by the patient's best interests should be consulted. The patient's autonomy -- whether the choice is to continue or discontinue treatment or to be resuscitated or not in the event of a cardiopulmonary arrest -- must be paramount. The courts have ruled that competent patients may refuse therapy and that an incompetent patient's previously stated wishes regarding life support should therefore be respected. The issues involving death and dying are among the most difficult in medicine. In approaching them rationally and consistently, the physician must

combine both the science and the art of medicine.

THE EXPANDING ROLE OF THE PHYSICIAN

Genetics and Medicine The genomic era is likely to lead to a revolution in the practice of medicine. Obtaining the DNA sequence of the entire human genome may help to elucidate the genetic components of common chronic diseases -- hypertension, diabetes, atherosclerosis, many cancers, dementias, and behavioral and autoimmune disorders. This information should make it possible to determine individual susceptibility to these conditions early in life and to implement individualized prevention programs. Subclassification of many diseases on a genetic basis may allow the selection of appropriate therapy for each patient. As the response to drugs becomes more predictable, pharmacotherapy should become more rational. In short, the completion of the Human Genome Project is likely to lead to a substantial increase in physicians' ability to influence their patient's health and well-being.

Patients will be best served if physicians play an active role in applying this powerful, sensitive new information rather than being passive bystanders who are intimidated by the new technology. This is a rapidly evolving field, and physicians and other health care professionals must remain updated to apply this new knowledge. Genetic testing requires wise counsel based on an understanding of the value and limitations of the tests as well as the implications of their results for specific individuals.

Medicine on the Internet The explosion in use of the Internet through personal computers is having an important impact on many practicing physicians. The Internet makes a wide range of information available to physicians almost instantaneously at any time of the day or night and from anywhere in the world. This medium holds enormous potential for delivering up-to-date information, practice guidelines, state-of-the-art conferences, journal contents, textbooks (including this text), and direct communications with other physicians and specialists, thereby expanding the depth and breadth of information available to the physician about the diagnosis and care of patients. Most medical journals are now accessible on-line, providing rapid and comprehensive sources of information. Patients, too, are turning to the Internet in increasing numbers to derive information about their illnesses and therapies and to join Internet-based support groups. Physicians are increasingly challenged by dealing with patients who are becoming more sophisticated in their understanding of illness. At this time, there is one critically important caveat. Virtually anything can be published on the Internet, thus circumventing the peer-review process that is an essential feature of quality publications. Physicians or patients who search the Internet for medical information must be aware of this danger. Notwithstanding this limitation, appropriate use of the Internet is revolutionizing information access for physicians and is a positive force in the practice of medicine.

Delivering Cost-Effective Medical Care As the cost of medical care has risen, it has become necessary to establish priorities in the expenditure of resources. In some instances, preventive measures offer the greatest return for the expenditure; outstanding examples include vaccination, improved sanitation, reduction in accidents and occupational hazards, and biochemical- and DNA-based screening of newborns. For example, the detection of phenylketonuria by newborn screening may result in a net

saving of many thousands of dollars.

As resources become increasingly constrained, the physician must weigh the possible benefits of performing costly procedures that provide only a limited life expectancy against the pressing need for more primary care for those persons who do not have adequate access to medical services. For the individual patient, it is important to reduce costly hospital admissions as much as possible if total health care is to be provided at a cost that most can afford. This policy, of course, implies and depends on close cooperation among patients, their physicians, employers, payers, and government. It is equally important for physicians to know the cost of the diagnostic procedures they order and the drugs and other therapies they prescribe and to monitor both costs and effectiveness. The medical profession should provide leadership and guidance to the public in matters of cost control, and physicians must take this responsibility seriously without being or seeming to be self-serving. However, the economic aspects of health care delivery must not interfere with the welfare of patients. The patient must be able to rely on the individual physician as his or her principal advocate in matters of health care.

Accountability Medicine is a satisfying but demanding profession. Physicians must understand the characteristics of the populations they serve, and they must appreciate their patients' social and cultural attitudes to health, disease, and death. As the public has become more educated and more sophisticated regarding health matters, their expectations of the health system in general and of their physicians in particular have risen. Physicians are expected to maintain mastery of their rapidly advancing fields (the *science* of medicine) while considering their patient's unique needs (the *art* of medicine). Thus, physicians are held accountable not only for the technical aspects of the care that they provide but also for their patient's satisfaction with the delivery and costs of care.

In the United States, there are increasing demands for physicians to account for the way in which they practice medicine by meeting certain standards prescribed by federal and state governments. The hospitalization of patients whose health care costs are reimbursed by the government and other third parties is subjected to utilization review. Thus the physician must defend the cause for and duration of a patient's hospitalization if it falls outside certain "average" standards. Authorization for reimbursement is increasingly based on documentation of the nature and complexity of an illness, as reflected by recorded elements of the history and physical examination. The purpose of these regulations is both to improve standards of health care and to contain spiraling health care costs. This type of review is being extended to all phases of medical practice and is profoundly altering the practice of medicine. Physicians are also expected to give evidence of their continuing competence through mandatory continuing education, patient-record audits, recertification by examination, or relicensing.

Continued Learning The conscientious physician must be a perpetual student because the body of medical knowledge is constantly expanding and being refined. The profession of medicine should be inherently linked to a career-long thirst for new knowledge that can be used for the good of the patient. It is the responsibility of a physician to pursue continually the acquisition of new knowledge by reading, attending conferences and courses, and consulting colleagues and the Internet. This is often a difficult task for a busy practitioner; however, such a commitment to continued learning is an integral part of being a physician and must be given the highest priority.

Research and Teaching The title *doctor* is derived from the Latin *docere*, "to teach," and physicians should share information and medical knowledge with colleagues, with students of medicine and related professions, and with their patients. The practice of medicine is dependent on the sum total of medical knowledge, which in turn is based on an unending chain of scientific discovery, clinical observation, analysis, and interpretation. Advances in medicine depend on the acquisition of new information, i.e., on research, which often involves patients; improved medical care requires the transmission of this information. As part of broader societal responsibilities, the physician should encourage patients to participate in ethical and properly approved clinical investigations if they do not impose undue hazard, discomfort, or inconvenience. To quote Osler once more:

To wrest from nature the secrets which have perplexed philosophers in all ages, to track to their sources the causes of disease, to correlate the vast stores of knowledge, that they may be quickly available for the prevention and cure of disease -- these are our ambitions.

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2. ETHICAL ISSUES IN CLINICAL MEDICINE - *Bernard Lo*

Physicians frequently confront ethical issues in clinical practice that are perplexing, time-consuming, and emotionally draining. Experience, common sense, and simply being a good person do not guarantee that physicians can identify or resolve ethical dilemmas. Knowledge about common ethical dilemmas is also essential.

FUNDAMENTAL ETHICAL GUIDELINES

Physicians should follow two fundamental but frequently conflicting ethical guidelines: respecting patient autonomy and acting in the patient's best interests.

RESPECTING PATIENT AUTONOMY

Competent, informed patients may refuse recommended interventions and choose among reasonable alternatives.

Informed Consent Informed consent requires physicians to discuss with patients the nature of the proposed care, the alternatives, the risks and benefits of each, the likely consequences, and to obtain the patient's agreement to care. Informed consent involves more than obtaining signatures on consent forms. Physicians need to educate patients, answer questions, make recommendations, and help them deliberate. Patients can be overwhelmed with medical jargon, needlessly complicated explanations, or too much information at once.

Nondisclosure of Information Physicians may consider withholding a serious diagnosis, misrepresenting it, or limiting discussions of prognosis or risks because they fear that a patient will develop severe anxiety or depression or refuse needed care. Patients should not be forced to receive information against their will. Most people, however, want to know their diagnosis and prognosis, even if they are terminally ill. Generally, physicians should provide relevant information, offer empathy and hope, and help patients cope with bad news.

Emergency Care Informed consent is not required when patients cannot give consent and when delay of treatment would place their life or health in peril. People are presumed to want such emergency care, unless they have previously indicated otherwise.

Futile Interventions Autonomy does not entitle patients to insist on whatever care they want. Physicians are not obligated to provide futile interventions that have no physiologic rationale or have already failed. For example, cardiopulmonary resuscitation would be futile in a patient with progressive hypotension despite maximal therapy. But physicians should be wary of using the term "futile" in looser senses to justify unilateral decisions to forego interventions when they believe that the probability of success is too low, no worthwhile goals can be achieved, the patient's quality of life is unacceptable, or the costs are too high. Such looser usages of the term are problematic because they may be inconsistent and mask value judgments.

ACTING IN THE BEST INTERESTS OF PATIENTS

The guideline of *beneficence* requires physicians to act for the patient's benefit. Laypeople do not possess medical expertise and may be vulnerable because of their illness. They justifiably rely on physicians to provide sound advice and to promote their well-being. Physicians encourage such trust. Hence, physicians have a fiduciary duty to act in the best interests of their patients. The interests of the patient should prevail over physicians' self-interest or the interests of third parties, such as hospitals or insurers. These fiduciary obligations of physicians contrast sharply with business relationships, which are characterized by "let the buyer beware," not by trust and reliance. The guideline of "*do no harm*" forbids physicians from providing ineffective interventions or acting without due care. This precept, while often cited, provides only limited guidance, because many beneficial interventions also have serious risks.

CONFLICTS BETWEEN BENEFICENCE AND AUTONOMY

Patients' refusals of care may thwart their own goals or cause them serious harm. For example, a young man with asthma may refuse mechanical ventilation for reversible respiratory failure. Simply to accept such refusals, in the name of respecting autonomy, seems morally constricted. Physicians can elicit patients' expectations and concerns, correct misunderstandings, and try to persuade them to accept beneficial therapies. If disagreements persist after discussions, the patient's informed choices and view of his or her best interests should prevail. While refusing recommended care does not render a patient incompetent, it may lead the physician to probe further to ensure that the patient is able to make informed decisions.

PATIENTS WHO LACK DECISION-MAKING CAPACITY

Patients may not be able to make informed decisions because of unconsciousness, dementia, delirium, or other conditions. Physicians should ask two questions regarding such patients: Who is the appropriate surrogate? What would the patient want done?

ASSESSING CAPACITY TO MAKE MEDICAL DECISIONS

All adults are considered legally competent unless declared incompetent by a court. In practice, physicians usually determine that patients lack the capacity to make health care decisions and arrange for surrogates to make them, without involving the courts. By definition, competent patients can express a choice and appreciate the medical situation, the nature of the proposed care, the alternatives, and the risks, benefits, and consequences of each. Their choices should be consistent with their values and should not result from delusions or hallucinations. Psychiatrists may help in difficult cases because they are skilled at interviewing mentally impaired patients and can identify treatable depression or psychosis. When impairments are fluctuating or reversible, decisions should be postponed if possible until the patient recovers decision-making capacity.

CHOICE OF SURROGATE

If a patient lacks decision-making capacity, physicians routinely ask family members to serve as surrogates. Most patients want their family members to be surrogates, and

family members generally know the patient's preferences and have the patient's best interests at heart. Patients may designate a particular individual to serve as proxy; such choices should be respected. Some states have established a prioritized list of which relative may serve as surrogate if the patient has not designated a proxy.

STANDARDS FOR SURROGATE DECISION MAKING

Advance Directives These are statements by competent patients to direct care if they lose decision-making capacity. They may indicate (1) what interventions they would refuse or accept or (2) who should serve as surrogate. Following the patient's advance directives, surrogate respects patients' autonomy.

Oral conversations are the most frequent form of advance directives. While such conversations are customarily followed in clinical practice, casual or vague comments may not be trustworthy.

Living wills direct physicians to forego or provide life-sustaining interventions if the patient develops a terminal condition or persistent vegetative state. Generally patients may refuse only interventions that "merely prolong the process of dying."

A *health care proxy* is someone appointed by the patient to make health care decisions if he or she loses decision-making capacity. It is more flexible and comprehensive than the living will, applying whenever the patient is unable to make decisions.

Physicians can encourage patients to provide advance directives, to indicate both what they would want and who should be surrogate, and to discuss their preferences with surrogates. In discussions with patients, physicians can ensure that advance directives are informed, up-to-date, and address likely clinical scenarios. Such discussions are best carried out in the ambulatory setting. The federal Patient Self-Determination Act requires hospitals and health maintenance organizations to inform patients of their right to make health care decisions and to provide advance directives.

Substituted Judgment In the absence of clear advance directives, surrogates and physicians should try to decide as the patient would under the circumstances, using all information that they know about the patient. While such substituted judgments try to respect the patient's values, they may be speculative or inaccurate. A surrogate may be mistaken about the patient's preferences, particularly when they have not been discussed explicitly.

Best Interests When the patient's preferences are unclear or unknown, decisions should be based on the patient's best interests. Patients generally take into account the quality of life as well as the duration of life when making decisions for themselves. It is understandable that surrogates would also consider quality of life of patients who lack decision-making capacity. Judgments about quality of life are appropriate if they reflect the patient's own values. Bias or discrimination may occur, however, if others project their values onto the patient or weigh the perceived social worth of the patient. Most patients with chronic illness rate their quality of life higher than their family members and physicians do.

Legal Issues Physicians need to know pertinent state laws regarding patients who lack decision-making capacity. A few state courts allow doctors to forego life-sustaining interventions only if patients have provided written advance directives or very specific oral ones.

Disagreements Disagreements may occur among potential surrogates or between the physician and surrogate. Physicians can remind everyone to base decisions on what the patient would want, not what they would want for themselves. Consultation with the hospital ethics committee or with another physician often helps resolve disputes. Such consultation is also helpful when patients have no surrogate and no advance directives. The courts should be used only as a last resort when disagreements cannot be resolved in the clinical setting.

DECISIONS ABOUT LIFE-SUSTAINING INTERVENTIONS

Although medical technology can save lives, it can also prolong the process of dying. Competent, informed patients may refuse life-sustaining interventions. Such interventions may also be withheld from patients who lack decision-making capacity on the basis of advance directives or decisions by appropriate surrogates. Courts have ruled that foregoing life-sustaining interventions is neither suicide nor murder.

MISLEADING DISTINCTIONS

People commonly draw distinctions that are intuitively plausible but prove untenable on closer analysis.

Extraordinary and Ordinary Care Some physicians are willing to forego "extraordinary" or "heroic" interventions, such as surgery, mechanical ventilation, or renal dialysis, but insist on providing "ordinary" ones, such as antibiotics, intravenous fluids, or feeding tubes. However, this distinction is not logical because all medical interventions have both risks and benefits. Any intervention may be withheld, if the burdens for the individual patient outweigh the benefits.

Withdrawing and Withholding Interventions Many health care providers find it more difficult to discontinue interventions than to withhold them in the first place. Although such emotions need to be acknowledged, there is no logical distinction between the two acts. Justifications for withholding interventions, such as refusal by patients or surrogates, are also justifications for withdrawing them. In addition, an intervention may prove unsuccessful or new information about the patient's preferences or condition may become available after the intervention is started. If interventions could not be discontinued, patients and surrogates might not even attempt treatments that might prove beneficial.

DO NOT RESUSCITATE (DNR) ORDERS

When a patient suffers a cardiopulmonary arrest, cardiopulmonary resuscitation (CPR) is initiated unless a DNR order has been made. Although CPR can restore people to vigorous health, it can also disrupt a peaceful death. After CPR is attempted on a general hospital service, only 14% of patients survive to discharge, and even fewer in

certain subgroups. DNR orders are appropriate if the patient or surrogate requests them or if CPR would be futile. To prevent misunderstandings, physicians should write DNR orders and the reasons for them in the medical record. "Slow" or "show" codes that merely appear to provide CPR are deceptive and therefore unacceptable. Although a DNR order signifies only that CPR will be withheld, the reasons that justify DNR orders may lead to a reconsideration of other plans for care.

ASSISTED SUICIDE AND ACTIVE EUTHANASIA

Proponents of these controversial acts believe that competent, terminally ill patients should have control over the end of life and that physicians should relieve refractory suffering. Opponents assert that such actions violate the sanctity of life, that suffering can generally be relieved, that abuses are inevitable, and that such actions are outside the physician's proper role. These actions are illegal throughout the United States, except that physician-assisted suicide is legal in Oregon under certain circumstances. Whatever their personal views, physicians should respond to patients' inquiries with compassion and concern. Physicians should elicit and address any underlying problems, such as physical symptoms, loss of control, or depression. Often, additional efforts to relieve distress are successful, and after this is done patients generally withdraw their requests for these acts.

CARE OF DYING PATIENTS

Patients often suffer unrelieved pain and other symptoms during their final days of life. Physicians may hesitate to order high doses of narcotics and sedatives, fearing they will hasten death. Relieving pain in terminal illness and alleviating dyspnea when patients forego mechanical ventilation enhances patient comfort and dignity. If lower doses of narcotics and sedatives have failed to relieve suffering, increasing the dose to levels that may suppress respiratory drive is ethically appropriate because the physician's intention is to relieve suffering, not hasten death. Physicians can also relieve suffering by spending time with dying patients, listening to them, and attending to their psychological distress.

CONFLICTS OF INTEREST

Acting in the patient's best interests may conflict with the physician's self-interest or the interests of third parties such as insurers or hospitals. The ethical ideal is to keep the patient's interests paramount. Even the appearance of a conflict of interest may undermine trust in the profession.

FINANCIAL INCENTIVES

In managed care systems, physicians may serve as gatekeepers or bear financial risk for expenditures. Although such incentives are intended to reduce inefficiency and waste, there is concern that physicians may withhold beneficial care in order to control costs. In contrast, physicians have incentives to provide more care than indicated when they receive fee-for-service reimbursement or when they refer patients to medical facilities in which they have invested. Regardless of financial incentives, physicians should recommend available care that is in the patient's best interests -- no more and

no less.

DENIALS OF COVERAGE

Utilization review programs designed to reduce unnecessary services may also deny coverage for care that the physician believes will benefit the patient. Physicians should inform patients when a plan is not covering standard care and act as patient advocates by appealing such denials of coverage. Patients may ask physicians to misrepresent their condition to help them obtain insurance coverage or disability. While physicians understandably want to help patients, such misrepresentation undermines physicians' credibility and violates their integrity.

GIFTS FROM PHARMACEUTICAL COMPANIES

Physicians may be offered gifts ranging from pens and notepads to lavish entertainment. Critics worry that any gift from drug companies may impair objectivity, increase the cost of health care, and give the appearance of conflict of interest. A helpful rule of thumb is to consider whether patients would approve if they knew physicians had accepted such gifts.

OCCUPATIONAL RISKS

Some health care workers, fearing fatal occupational infections, refuse to care for persons with HIV infection or multidrug-resistant tuberculosis. Such fears about personal safety need to be acknowledged, and institutions should reduce occupational risk by providing proper training, equipment, and supervision. Physicians should provide appropriate care within their clinical expertise, despite personal risk.

MISTAKES

Mistakes are inevitable in clinical medicine. They may cause serious harm to patients or result in substantial changes in management. Physicians and students may fear that disclosing such mistakes could damage their careers. Without disclosure, however, patients cannot understand their clinical situation or make informed choices about subsequent care. Similarly, unless attending physicians are informed of trainees' mistakes, they cannot provide optimal care and help trainees learn from mistakes.

LEARNING CLINICAL SKILLS

Learning clinical medicine, particularly learning to perform invasive procedures, may present inconvenience or risk to patients. To ensure patient cooperation, students may be introduced as physicians, or patients may not be told that trainees will be performing procedures. Such misrepresentation undermines trust, may lead to more elaborate deception, and makes it difficult for patients to make informed choices about their care. Patients should be told who is providing care, what benefits and burdens can be attributed to trainees, and how trainees are supervised. Most patients, when informed, allow trainees to play an active role in their care.

IMPAIRED PHYSICIANS

Physicians may hesitate to intervene when colleagues impaired by alcohol abuse, drug abuse, or psychiatric or medical illness place patients at risk. However, society relies on physicians to regulate themselves. If colleagues of an impaired physician do not take steps to protect patients, no one else may be in a position to do so.

CONFLICTS FOR TRAINEES

Medical students and residents may fear that they will receive poor grades or evaluations if they act on the patient's behalf by disclosing mistakes, avoiding misrepresentation of their role, and reporting impaired colleagues. Discussing such dilemmas with more senior physicians can help trainees check their interpretation of the situation and obtain advice and assistance.

ADDITIONAL ETHICAL ISSUES

MAINTAINING CONFIDENTIALITY

Maintaining the confidentiality of medical information respects patients' autonomy and privacy, encourages them to seek treatment and to discuss their problems candidly, and prevents discrimination. Physicians need to guard against inadvertent breaches of confidentiality, as when talking about patients in elevators. Maintaining confidentiality is not an absolute rule. The law may require physicians to override confidentiality in order to protect third parties, for example, reporting to government officials persons with specified infectious conditions, such as tuberculosis and syphilis; persons with gunshot wounds; and victims of elder abuse and domestic violence. Computerized medical records raise additional concerns because breaches of confidentiality may affect many patients.

ALLOCATING RESOURCES JUSTLY

Allocation of limited health care resources is problematic. Ideally, allocation decisions should be made as public policy, with physician input. At the bedside, physicians generally should act as patient advocates within constraints set by society, reasonable insurance coverage, and sound practice. *Ad hoc* rationing by the individual physician at the bedside may be inconsistent, discriminatory, and ineffective. In some cases, however, two patients may compete for the same limited resources, such as physician time or a bed in intensive care. When this occurs, physicians should ration their time and resources according to patients' medical needs and the probability of benefit.

ASSISTANCE WITH ETHICAL ISSUES

Discussing perplexing ethical issues with other members of the health care team, colleagues, or the hospital ethics committee often clarifies issues and suggests ways to improve communication and to deal with strong emotions. When struggling with difficult ethical issues, physicians may need to reevaluate their basic convictions, tolerate uncertainty, and maintain their integrity while respecting the opinions of others.

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3. DECISION-MAKING IN CLINICAL MEDICINE - *Daniel B. Mark*

To the medical student who requires 2 h to collect a patient's history and perform a physical examination, and several additional hours to organize them into a coherent presentation, the experienced clinician's ability to reach a diagnosis and decide on a management plan in a fraction of the time seems extraordinary. While medical knowledge and experience play a significant role in the senior clinician's ability to arrive at a differential diagnosis and plan quickly, much of the process involves skill in clinical decision-making. The first goal of this chapter is to provide an introduction to the study of clinical reasoning.

Equally bewildering to the student are the proper use of diagnostic tests and the integration of the results into the clinical assessment. The novice medical practitioner typically uses a "shotgun" approach to testing, hoping to hit a target without knowing exactly what that target is. The expert, on the other hand, usually has a specific target in mind and efficiently adjusts the testing strategy to it. The second goal of this chapter is to review briefly some of the crucial basic statistical concepts that govern the proper interpretation and use of diagnostic tests; quantitative tools available to assist in clinical decision-making will also be discussed.

CLINICAL DECISION-MAKING

CLINICAL REASONING

The most important clinical actions are not procedures or prescriptions but the judgments from which all other aspects of clinical medicine flow. In the modern era of large randomized trials, it is easy to overlook the importance of this elusive mental activity and focus instead on the algorithmic practice guidelines constructed to improve care. One reason for this apparent neglect is that much more research has been done on how doctors *should* make decisions (e.g., using a Bayesian model discussed below) than on how they actually *do*. Thus, much of what we know about clinical reasoning comes from empirical studies of nonmedical problem-solving behavior.

Despite the great technological advances of the twentieth century, uncertainty still plays a pivotal role in all aspects of medical decision-making. We may know that a patient does not have long to live, but we cannot be certain how long. We may prescribe a potent new receptor blocker to reverse the course of a patient's illness, but we cannot be certain that the therapy will do so without side effects. Uncertainty in medical outcomes creates the need for probabilities and other mathematical/statistical tools to help guide decision-making. (These tools are reviewed later in the chapter.)

Uncertainty is compounded by the information overload that characterizes modern medicine. Today's experienced clinician needs close to 2 million pieces of information to practice medicine. Doctors subscribe to an average of 7 journals, representing over 2500 new articles each year. Computers offer the obvious solution both for management of information and for better quantitation and management of the daily uncertainties of medical care. While the technology to computerize medical practice is available, many practical problems remain to be solved before patient information can be standardized and integrated with medical evidence on a single electronic platform.

The following three examples introduce the subject of clinical reasoning:

- A 46-year-old man presents to his internist with a chief complaint of hemoptysis. The physician knows that the differential diagnosis of hemoptysis includes over 100 different conditions, including cancer and tuberculosis ([Chap. 33](#)). The examination begins with some general background questions, and the patient is asked to describe his symptoms and their chronology. By the time the examination is completed, and even before any tests are run, the physician has formulated a working diagnostic hypothesis and planned a series of steps to test it. In an otherwise healthy and nonsmoking patient recovering from a viral bronchitis, the doctor's hypothesis would be that the acute bronchitis is responsible for the small amount of blood-streaked sputum the patient observed. In this case, a chest x-ray and purified protein derivative (PPD) skin test may be sufficient.
- A second 46-year-old patient with the same chief complaint who has a 100-pack-year smoking history, a productive morning cough, and episodes of blood-streaked sputum may generate the principal diagnostic hypothesis of carcinoma of the lung. Consequently, along with the chest x-ray and [PPD](#) skin test, the physician refers this patient for bronchoscopy.
- A third 46-year-old patient with hemoptysis who is from a developing country is evaluated with an echocardiogram as well, because the physician thinks she hears a soft diastolic rumble at the apex on cardiac auscultation, suggesting rheumatic mitral stenosis.

These three vignettes illustrate two aspects of expert clinical reasoning: (1) the use of cognitive shortcuts, or *heuristics*, as a way to organize the complex unstructured material that is collected in the clinical evaluation; and (2) the use of diagnostic hypotheses to consolidate the information and indicate appropriate management steps.

THE USE OF COGNITIVE SHORTCUTS

Heuristics reduce the complexity of a problem to a manageable level. Psychologists have found that people rely on three basic types of heuristics. For example, when assessing a patient, clinicians often weigh the probability that this patient's clinical features match those of the class of patients with the leading diagnostic hypotheses being considered. In other words, the clinician is searching for the diagnosis for which the patient appears to be a representative example; this cognitive shortcut is called the *representativeness heuristic*. It may take only a few characteristics from the history for an expert clinician using the representativeness heuristic to arrive at a sound diagnostic hypothesis. For example, an elderly patient with new-onset fever, cough productive of copious sputum, unilateral pleuritic chest pain, and dyspnea is readily identified as fitting the pattern for acute pneumonia, probably of bacterial origin. Evidence of focal pulmonary consolidation on the physical examination will increase the clinician's confidence in the diagnosis because it fits the expected pattern of acute bacterial pneumonia. Knowing this allows the experienced clinician to conduct an efficient, directed, and therapeutically productive patient evaluation although there may be little else in the history or physical examination of direct relevance. The inexperienced medical student or resident, who has not yet learned the patterns most prevalent in

clinical medicine, must work much harder to achieve the same result and is often at risk of missing the important clinical problem in a sea of compulsively collected but unhelpful data.

However, physicians using the representativeness heuristic can reach erroneous conclusions if they fail to consider the underlying prevalence of two competing diagnoses. Consider a patient with pleuritic chest pain, dyspnea, and a low-grade fever. A clinician might consider acute pneumonia and acute pulmonary embolism to be the two leading diagnostic alternatives. Clinicians using the representativeness heuristic might judge both diagnostic candidates to be equally likely, although to do so would be wrong if pneumonia was much more prevalent in the underlying population. Mistakes may also result from a failure to consider that a pattern based on a small number of prior observations will likely be less reliable than one based on larger samples.

A second commonly used cognitive shortcut, the *availability heuristic*, involves judgments made on the basis of how easily prior similar cases or outcomes can be brought to mind. For example, the experienced clinician may recall 20 elderly patients seen over the past few years who presented with painless dyspnea of acute onset and were found to have acute myocardial infarction. The novice clinician may spend valuable time seeking a pulmonary cause for the symptoms before considering and discovering the cardiac diagnosis. In this situation, the patient's clinical pattern does not fit the expected pattern of acute myocardial infarction, but experience with this atypical presentation, and the ability to recall it, can help direct the physician to the diagnosis.

Errors with the availability heuristic can come from several sources of recall bias. For example, rare catastrophes are likely to be remembered with a clarity and force out of proportion to their value, and recent experience is, of course, easier to recall and therefore more influential on clinical judgments.

The third commonly used cognitive shortcut, the *anchoring heuristic*, involves estimating a probability by starting from a familiar point (the anchor) and adjusting to the new case from there. For example, a clinician may judge the probability of colorectal cancer to be extremely high after an elevated screening carcinoembryonic antigen (CEA) result because the prediction of colorectal cancer is anchored to the test result. Yet, as discussed below, this prediction would be inaccurate if the clinical picture of the patient being tested indicates a low probability of disease (for example, a 30-year-old woman with no risk factors). Anchoring can be a powerful tool for diagnosis but is often used incorrectly (see "Measures of Disease Probability and Bayes' Theorem," below).

DIAGNOSTIC HYPOTHESIS GENERATION

Cognitive scientists studying the thought processes of expert clinicians have observed that clinicians group data into packets or "chunks," which are stored in their memories and manipulated to generate diagnostic hypotheses. Because short-term memory can typically hold only 7 to 10 items at a time, the number of packets that can be actively integrated into hypothesis-generating activities is similarly limited. The cognitive shortcuts discussed above play a key role in the generation of diagnostic hypotheses, many of which are discarded as rapidly as they are formed.

A diagnostic hypothesis sets a context for diagnostic steps to follow and provides testable predictions. For example, if the enlarged and quite tender liver felt on physical examination is due to acute hepatitis (the hypothesis), certain specific liver function tests should be markedly elevated (the prediction). If the tests come back normal, the hypothesis may need to be discarded or substantially modified.

One of the factors that makes teaching diagnostic reasoning so difficult is that expert clinicians do not follow a fixed pattern in patient examinations. From the outset, they are generating, refining, and discarding diagnostic hypotheses. The questions they ask in the history are driven by the hypotheses they are working with at the moment. Even the physical examination is driven by specific questions rather than a preordained checklist. While the student is palpating the abdomen of the alcoholic patient, waiting for a finding to strike him, the expert clinician is on a focused search mission. Is the spleen enlarged? How big is the liver? Is it tender? Are there any palpable masses or nodules? Each question focuses the attention of the examiner to the exclusion of all other inputs until answered, allowing the examiner to move on to the next specific question.

Negative findings are often as important as positive ones in establishing and refining diagnostic hypotheses. Chest discomfort that is not provoked or worsened by exertion in an active patient reduces the likelihood that chronic ischemic heart disease is the underlying cause. The absence of a resting tachycardia and thyroid gland enlargement reduces the likelihood of hyperthyroidism in a patient with paroxysmal atrial fibrillation.

While the representativeness and availability heuristics may play the major roles in shaping early diagnostic hypotheses, the acuity of a patient's illness can also be very influential. For example, clinicians are taught to consider aortic dissection routinely as a possible cause of acute severe chest discomfort along with myocardial infarction, even though the typical history of dissection is different from myocardial infarction and dissection is far less prevalent ([Chap. 247](#)). This recommendation is based on the recognition that a relatively rare but catastrophic diagnosis like aortic dissection is very difficult to make unless it is explicitly considered. If the clinician fails to elicit any of the characteristic features of dissection by history and finds equivalent blood pressures in both arms and no pulse deficits, he or she may feel comfortable in discarding the aortic dissection hypothesis. If, however, the chest x-ray shows a widened mediastinum, the hypothesis may be reinstated and a diagnostic test ordered [e.g., thoracic computed tomography (CT) scan, transesophageal echocardiogram] to evaluate it more fully. In noncritical situations, the prevalence of potential alternative diagnoses should play a much more prominent role in diagnostic hypothesis generation. The value of conducting a rapid systematic clinical survey of symptoms and organ systems to avoid missing important but inapparent clues cannot be overstated.

Because the generation and evaluation of appropriate diagnostic hypotheses is a skill that not all clinicians possess to an equal degree, errors in this process can occur, and in the patient with serious acute illness these may lead to tragic consequences. Consider the following hypothetical example. A 45-year-old male patient with a 3-week history of a "flu-like" upper respiratory infection (URI) presented to his physician with symptoms of dyspnea and a productive cough. Based on the presenting complaint, the clinician pulled out a "URI Assessment Form" to improve quality and efficiency of care. The physician quickly completed the examination components outlined on this

structured form, noting in particular the absence of fever and a clear chest examination. He then prescribed an antibiotic for presumed bronchitis, showed the patient how to breathe into a paper bag to relieve his "hyperventilation," and sent him home with the reassurance that his illness was not serious. After a sleepless night with significant dyspnea unrelieved by rebreathing into a bag, the patient developed nausea and vomiting and collapsed. He was brought into the Emergency Department in cardiac arrest and could not be resuscitated. Autopsy showed a posterior wall myocardial infarction and a fresh thrombus in an atherosclerotic right coronary artery. What went wrong? The clinician decided, even before starting the history, that the patient's complaints were not serious. He therefore felt confident that he could perform an abbreviated and focused examination using the URI assessment protocol rather than considering the full range of possibilities and performing appropriate tests to confirm or refute his initial hypotheses. In particular, by concentrating on the "URI," the clinician failed to elicit the full dyspnea history, which would have suggested a far more serious disorder, and did not even search for other symptoms that could have directed him to the correct diagnosis.

This example illustrates how patients can diverge from textbook symptoms and the potential consequences of being unable to adapt the diagnostic process to real-world challenges. The expert, while recognizing that common things occur commonly, approaches each evaluation on high alert for clues that the initial diagnosis may be wrong. Patients often provide information that "does not fit" with any of the leading diagnostic hypotheses being considered. Distinguishing real clues from false trails can only be achieved by practice and experience. A less experienced clinician who tries to be too efficient (as in the above example) can make serious judgment errors.

MAJOR INFLUENCES ON CLINICAL DECISION-MAKING

More than a decade of research on variations in clinician practice patterns has shed much light on forces that shape clinical decisions. The use of heuristic "shortcuts," as detailed above, provides a partial explanation, but several other key factors play an important role in shaping diagnostic hypotheses and management decisions. These factors can be grouped conceptually into three overlapping categories: (1) factors related to physician personal characteristics and practice style, (2) factors related to the practice setting, and (3) economic incentive factors.

Practice Style Factors One of the key roles of the physician in medical care is to serve as the patient's agent to ensure that necessary care is provided at a high level of quality. Factors that influence this role include the physician's knowledge, training, and experience. It is obvious that physicians cannot practice evidence-based medicine if they are unfamiliar with the evidence. As would be expected, specialists generally know the evidence in their field better than do generalists. Surgeons may be more enthusiastic about recommending surgery than medical doctors because their belief in the beneficial effects of surgery is stronger. For the same reason, invasive cardiologists are much more likely to refer chest pain patients for diagnostic catheterization than are noninvasive cardiologists or generalists. The physician beliefs that drive these different practice styles are based on personal experience, recollection, and interpretation of the available medical evidence. For example, heart failure specialists are much more likely than generalists to achieve target angiotensin-converting enzyme (ACE) inhibitor

therapy in their heart failure patients because they are more familiar with what the targets are (as defined by large clinical trials), have more familiarity with the specific drugs (including dosages and side effects), and are less likely to overreact to foreseeable problems in therapy such as a rise in creatinine levels or symptomatic hypotension. Other intriguing research has shown a wide distribution of acceptance times of antibiotic therapy for peptic ulcer disease following widespread dissemination of the "evidence" in the medical literature. Some gastroenterologists accepted this new therapy before the evidence was clear (reflecting, perhaps, an aggressive practice style), and some gastroenterologists lagged behind (a conservative practice style, associated in this case with older physicians). As a group, internists lagged several years behind gastroenterologists.

The opinion of influential leaders can also have an important effect on practice patterns. Such influence can occur at both the national level (e.g., expert physicians teaching at national meetings) and the local level (e.g., local educational programs, "curbside consultants"). Opinion leaders do not have to be physicians. When conducting rounds with clinical pharmacists, physicians are less likely to make medication errors and more likely to use target levels of evidence-based therapies.

The patient's welfare is not the only concern that drives clinical decisions. The physician's perception about the risk of a malpractice suit resulting from either an erroneous decision or a bad outcome creates a style of practice referred to as *defensive medicine*. This practice involves using tests and therapies with very small marginal returns to preclude future criticism in the event of an adverse outcome. For example, a 40-year-old woman who presents with a long-standing history of intermittent headache and a new severe headache along with a normal neurologic examination has a very low likelihood of structural intracranial pathology. Performance of a head [CT](#) or magnetic resonance imaging (MRI) scan in this situation would constitute defensive medicine. On the other hand, the results of the test could provide reassurance to an anxious patient.

Practice Setting Factors Factors in this category relate to the physical resources available to the physician's practice and the practice environment. *Physician-induced demand* is a term that refers to the repeated observation that physicians have a remarkable ability to accommodate to and employ the medical facilities available to them. A classic early study in this area showed that physicians in Boston had an almost 50% higher hospital admission rate than did physicians in New Haven, despite there being no obvious differences in the health of the cities' inhabitants. The physicians in New Haven were not aware of using fewer hospital beds for their patients, nor were the Boston physicians aware of using less stringent criteria to admit patients.

Other environmental factors that can influence decision-making include the local availability of specialists for consultations and procedures, "high tech" facilities such as angiography suites, a heart surgery program, and [MRI](#) machines.

Economic Incentives Economic incentives are closely related to the other two categories of practice-modifying factors. Financial issues can exert both stimulatory and inhibitory influences on clinical practice. In general, physicians are paid on a fee-for-service, capitation, or salary basis ([Chap. 4](#)). In fee-for-service, the more the physician does, the more the physician gets paid. The incentive in this case is to do

more. When fees are reduced (discounted fee-for-service), doctors tend to increase the number of services billed for. Capitation, in contrast, provides a fixed payment per patient per year, encouraging physicians to take on more patients but to provide each patient with fewer services. Expensive services are more likely to be affected by this type of incentive than inexpensive preventive services. Salary compensation plans pay physicians the same regardless of the amount of clinical work performed. The incentive here is to see fewer patients. Recognizing these powerful shapers of physician behavior, managed care plans have begun to explore combinations of the three reimbursement types with the goal of improving individual physician productivity while restraining their use of expensive tests and therapies.

In summary, expert clinical decision-making can be appreciated as a complex interplay between cognitive devices used to simplify large amounts of complex information interacting with physician biases reflecting education, training, and experience, all of which are shaped by powerful, sometimes perverse, external forces. In the next section, we will review a set of statistical tools and concepts that can assist in making clinical decisions under uncertainty.

QUANTITATIVE METHODS TO AID CLINICAL DECISION-MAKING

The process of medical decision-making can be divided into two parts: (1) defining the available courses of action and estimating the likely outcomes with each, and (2) assessing the desirability of the outcomes. The former task involves integrating key information about the patient along with relevant evidence from the medical literature to create the structure of a decision problem. The remainder of this chapter will present some quantitative tools to assist the clinician in these activities. These tools can be divided into those that assist the clinician in making better outcome predictions, which are then used to make decisions, and those that support the decision process directly. While these tools are not yet used routinely in daily clinical practice, the computerization of medicine is creating the required substrate for their future widespread dissemination.

QUANTITATIVE MEDICAL PREDICTIONS

Diagnostic Testing The purpose of performing a test on a patient is to reduce uncertainty about the patient's diagnosis or prognosis and to aid the clinician in making management decisions. Although diagnostic tests are commonly thought of as laboratory tests (e.g., measurement of serum amylase level) or procedures (e.g., colonoscopy or bronchoscopy), any technology that changes our understanding of the patient's problem qualifies as a diagnostic test. Thus, even the history and physical examination can be considered a form of diagnostic test. In clinical medicine, it is common to reduce the results of a test to a dichotomous outcome, such as positive or negative, normal or abnormal. In many cases, this simplification results in the waste of useful information. However, such simplification makes it easier to demonstrate some of the quantitative ways in which test data can be used.

To characterize the accuracy of diagnostic tests, four terms are routinely used ([Table 3-1](#)). The *true-positive rate*, i.e., the sensitivity, provides a measure of how well the test correctly identifies patients with disease. The *false-negative rate* is calculated as (1-sensitivity). The *true-negative rate*, i.e., the specificity, reflects how well the test

correctly identifies patients without disease. The *false-positive rate* is (1- specificity). A perfect test would have a sensitivity of 100% and a specificity of 100% and would completely separate patients with disease from those without it.

Calculating sensitivity and specificity require selection of a cutpoint value for the test to separate "normal" from "diseased" subjects. As the cutpoint is moved to improve sensitivity, specificity typically falls and vice versa. This dynamic tradeoff between more accurate identification of subjects with versus those without disease is often displayed graphically as a receiver operating characteristic (ROC) curve. An ROC curve plots sensitivity (y-axis) versus 1 -specificity (x-axis). Each point on the curve represents a potential cutpoint with an associated sensitivity and specificity value. The area under the ROC curve is often used as a quantitative measure of the information content of a test. Values range from 0.5 (no diagnostic information at all, test is equivalent to flipping a coin) to 1.0 (perfect test).

In the diagnostic testing literature, ROC areas are often used to compare alternative tests. The test with the highest area (i.e., closest to 1.0) is presumed to be the most accurate. However, ROC curves are not a panacea for evaluation of diagnostic test utility. Like Bayes' theorem, they are typically focused on only one possible test parameter (e.g., ST segment response in a treadmill exercise test) to the exclusion of other potentially relevant data. In addition, ROC area comparisons do not simulate the way test information is actually used in clinical practice. Finally, biases in the underlying population used to generate the ROC curves (e.g., related to an unrepresentative test sample) can bias the ROC area and the validity of a comparison among tests.

Measures of Disease Probability and Bayes' Theorem Unfortunately, there are no perfect tests; after every test is completed the true disease state of the patient remains uncertain. Quantitating this residual uncertainty can be done with Bayes' theorem. This theorem provides a simple mathematical way to calculate the posttest probability of disease from three parameters: the pretest probability of disease, the test sensitivity, and the test specificity ([Table 3-2](#)). The pretest probability is a quantitative expression of the confidence in a diagnosis before the test is performed. In the absence of more relevant information it is usually estimated from the prevalence of the disease in the underlying population. For some common conditions, such as coronary artery disease (CAD), nomograms and statistical models have been created to generate better estimates of pretest probability from elements of the history and physical examination. The posttest probability, then, is a revised statement of the confidence in the diagnosis, taking into account both what was known before and after the test.

To understand how Bayes' theorem creates this revised confidence statement, it is useful to examine a nomogram version of Bayes' theorem that uses the same three parameters to predict the posttest probability of disease ([Fig. 3-1](#)). In this nomogram, the accuracy of the diagnostic test in question is summarized by the likelihood ratio for a positive test, which is the ratio of the true-positive rate to the false-positive rate [or sensitivity/(1 - specificity)]. For example, a test with a sensitivity of 0.90 and a specificity of 0.90 has a likelihood ratio of $0.90/(1 - 0.90)$, or 9. Thus, for this hypothetical test, a "positive" result is 9 times more likely in a patient with the disease than in a patient without it. The more accurate the test, the higher the likelihood ratio. However, if sensitivity is excellent but specificity is less so, the likelihood ratio will be substantially

reduced (e.g., with a 90% sensitivity but a 60% specificity, the likelihood ratio is 2.25). Most tests in medicine have likelihood ratios for a positive result between 1.5 and 20.

Consider two tests commonly used in the diagnosis of [CAD](#), an exercise treadmill and an exercise thallium-201 single photon emission CT (SPECT) test ([Chap. 244](#)). Meta-analysis has shown the treadmill to have an average sensitivity of 66% and an average specificity of 84%, yielding a likelihood ratio of 4.1 $[0.66/(1 - 0.84)]$. If we use this test on a patient with a pretest probability of CAD of 10%, the posttest probability of disease following a positive result rises only to about 30%. If a patient with a pretest probability of CAD of 80% has a positive test result, the posttest probability of disease is about 95%.

The exercise thallium [SPECT](#) test is a more accurate test for the diagnosis of [CAD](#). For our purposes, assume that it has both a sensitivity and specificity of 90%, yielding a likelihood ratio of 9.0 $[0.90/(1 - 0.90)]$. If we again test our low pretest probability patient and he has a positive test, using [Fig. 3-1](#) we can demonstrate that the posttest probability of CAD rises from 10 to 50%. However, from a decision-making point of view, the more accurate test has not been able to improve diagnostic confidence enough to change management. In fact, the test has moved us from being fairly certain that the patient did not have CAD to being completely undecided (a 50:50 chance of disease). In a patient with a pretest probability of 80%, using the more accurate thallium SPECT test raises the posttest probability to 97% (compared with 95% for the exercise treadmill). Again, the more accurate test does not provide enough improvement in posttest confidence to alter management, and neither test has improved much upon what was known from clinical data alone.

If the pretest probability is low (e.g., $\leq 20\%$), even a positive result on a very accurate test will not move the posttest probability to a range high enough to rule in disease (e.g., $\geq 80\%$). Conversely, with a high pretest probability, a negative test will not adequately rule out disease. Thus, the largest gain in diagnostic confidence from a test occurs when the clinician is most uncertain before performing it (e.g., pretest probability between 30 and 70%). For example, if a patient has a pretest probability for [CAD](#) of 50%, a positive exercise treadmill test will move the posttest probability to 80% and a positive exercise thallium [SPECT](#) test will move it to 90% ([Fig. 3-1](#)).

Bayes' theorem, as presented above, employs a number of important simplifications that should be considered. First, few tests have only two useful outcomes, positive or negative, and many tests provide numerous pieces of data about the patient. Even if these can be integrated into a summary result, multiple levels of useful information may be present (e.g., strongly positive, positive, indeterminate, negative, strongly negative). While Bayes' theorem can be adapted to this more detailed test result format, it is computationally complex to do so. Second, Bayes' theorem assumes that the information from the test is completely unique and nonoverlapping with information used to estimate the pretest probability. This independence assumption, however, is often wrong. In many cases, test results are correlated with patient characteristics. For example, the findings of cardiomegaly and pulmonary edema on chest x-ray are correlated with the historic features of heart failure and with the physical findings of a displaced left ventricular apical impulse, an S_3 gallop, and rales. The unique predictive information contributed by the test in this case (the chest x-ray) is only a fraction of its

total information because much had already been learned about the probability of heart failure before the test was done.

Finally, it has long been thought that sensitivity and specificity are prevalence-independent parameters of test accuracy, and many texts still make this assertion. This statistically useful assumption, however, is clinically wrong. For example, a treadmill exercise test has a sensitivity in a population of patients with one-vessel CAD of around 30%, whereas the sensitivity in severe three-vessel CAD approaches 80%. Thus, the best estimate of sensitivity to use in a particular decision will often vary depending on the distribution of disease stages present in the tested population. A hospitalized population typically has a higher prevalence of disease and in particular a higher prevalence of more advanced disease stages than an outpatient population. As a consequence, test sensitivity will tend to be higher in hospitalized patients, whereas test specificity will be higher in outpatients.

Statistical Prediction Models Bayes' theorem, as presented above, deals with a clinical prediction problem that is unrealistically simple relative to most problems a clinician faces. Prediction models, based on multivariable statistical models, can handle much more complex problems and substantially enhance predictive accuracy for specific situations. Their particular advantage is the ability to take into account many overlapping pieces of information and assign a relative weight to each based on its unique contribution to the prediction in question. For example, a logistic regression model to predict the probability of CAD takes into account all of the relevant independent factors from the clinical examination and diagnostic testing instead of the small handful of data that clinicians can manage in their heads or with Bayes' theorem. However, despite this strength, the models are too complex computationally to use without a calculator or computer (although this limit may be overcome when medicine is practiced from a fully computerized platform.) To date, only a handful of prediction models have been developed and properly validated. The importance of independent validation in a population separate from the one used to develop the model cannot be overstated. Unfortunately, most published models have not been properly validated, making their utility in clinical practice uncertain at best.

When statistical models have been compared directly with expert clinicians, they have been found to be more consistent, as would be expected, but not significantly more accurate. Their biggest promise, then, would seem to be to make less-experienced clinicians more accurate predictors of outcome.

DECISION SUPPORT TOOLS

DECISION SUPPORT SYSTEMS

Over the past 30 years, many attempts have been made to develop computer systems to help clinicians make decisions and manage patients. Conceptually, computers offer a very attractive way to handle the vast information load that today's physicians face. The computer can help by making accurate predictions of outcome, simulating the whole decision process, or providing algorithmic guidance. Computer-based predictions using Bayesian or statistical regression models inform a clinical decision but do not actually reach a "conclusion" or "recommendation." Artificial intelligence systems attempt to

simulate or replace human reasoning with a computer-based analogue. To date, such approaches have achieved only limited success. Reminder or protocol-directed systems do not make predictions but use existing algorithms, such as practice guidelines, to guide clinical practice. In general, however, decision support systems have shown little impact on practice. Reminder systems, although not yet in widespread use, have shown the most promise, particularly in correcting drug dosing and in promoting guideline adherence. The full potential of these approaches will only be achieved when computers are fully integrated into medical practice.

DECISION ANALYSIS

Compared with the methods discussed above, decision analysis represents a completely different approach to decision support. Its principal application is in decision problems that are complex and involve a substantial risk, a high degree of uncertainty in some key area, or an idiosyncratic feature that does not "fit" the available evidence. Three general steps are involved. First, the decision problem must be clearly defined. Second, the elements of the decision must be made explicit. This involves specifying the alternatives being considered, their relevant outcomes, the probabilities attached to each outcome, and the relative desirability (called "utility") of each outcome. Cost can also be assigned to each branch of the decision tree, allowing calculation of cost-effectiveness ([Chap. 4](#)).

An example of a decision tree used to evaluate strategies for management of the risk of infective endocarditis after catheter-associated *Staphylococcus aureus* bacteremia is shown in [Fig. 3-2](#). Approximately 35,000 cases of *S. aureus* bacteremia occur each year in the United States. The development of complicating endocarditis, which occurs in about 6% of cases, is associated with high morbidity (31% mortality, 21% stroke rate) and medical costs. The three choices for management of the bacteremia are (1) transesophageal echocardiography (TEE), (2) a 4-week course of intravenous antibiotics (long-course), or (3) a 2-week course of intravenous antibiotics (short-course). In the TEE strategy, a 4-week course of antibiotics is given if endocarditis is evident and a 2-week course is given if it is not. With each strategy, there is a risk that the patient will develop endocarditis with or without major complications. In this analysis, the longest quality-adjusted survival (5.47 quality-adjusted life-years) was associated with the 4-week antibiotic course strategy, which also had the highest costs (\$14,136 per patient), whereas the lowest costs (\$9830 per patient) and worst outcomes (5.42 quality-adjusted life-years) were associated with the 2-week antibiotic course strategy. From a clinical point of view (ignoring costs), the 4-week antibiotic course was best. From a cost-effectiveness point of view, the TEE strategy (5.46 quality-adjusted life-years and \$10,051 per patient costs) provided the best balance of added benefits and costs. Thus, decision analysis can be extremely helpful in clarifying tradeoffs in outcomes and costs in difficult management areas such as the above where it is highly unlikely that an adequate randomized trial will ever be done.

The data needed to fill in a decision tree ([Fig. 3-2](#)) are typically cobbled together from a variety of sources, including the literature (randomized trials, meta-analyses, observational studies) and expert opinion. Once the decision tree is finished, the decision is "analyzed" by calculating the average value of each limb of the tree. The decision arm with the highest net value (or expected utility) is the preferred choice. The

value of this exercise, however, is not so much in developing a prescription for action as it is in exploring the key elements and pressure points of a complex or difficult decision. The process of building the decision tree forces the analyst to be explicit about the choices being considered and all their relevant outcomes. Areas of high uncertainty are readily identified. Sensitivity analyses are an integral part of decision analysis and involve systematically varying the value of each key parameter in the model alone (one-way sensitivity analysis,) in pairs (two-way), or in higher combinations (multivariable) to assess the impact on choice of preferred management strategy. In the above example, varying the incidence of endocarditis resulting from *S. aureus* bacteremia from 3% to over 50% had no impact on the choice of [TEE](#) as the preferred strategy.

User friendly personal computer-based software packages now make the creation and analysis of decision trees much more straightforward than in the past. However, the process is still too cumbersome and time-consuming to be used on a routine basis. When medicine is practiced from a fully computerized platform, a library of prestructured decision trees with user modifiable values can be made available to support practitioners working with individual patients.

CONCLUSIONS

In this era of evidence-based medicine, it is tempting to think that all the difficult decisions practitioners face have been or soon will be solved and digested into practice guidelines and computerized reminders. For the foreseeable future, however, such is not the case. Meta-analyses cannot generate evidence where there are no adequate randomized trials, and most of what clinicians face will never be thoroughly tested in a randomized trial. Excellent clinical reasoning skills and experience supplemented by well-designed quantitative tools and a keen appreciation for individual patient preferences will continue to be of paramount importance in the professional life of medical practitioners for years to come.

(Bibliography omitted in Palm version)

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4. ECONOMIC ISSUES IN CLINICAL MEDICINE - *Daniel B. Mark*

The United States has the distinction of having some of the best medical care of any technologically advanced country. We have many of the best hospitals and doctors in the world. The research pipeline is full of significant new therapeutic advances, with revolutionary genetic-based therapies perhaps only a decade away. Our citizens largely subscribe to the principle that excellent medical care should be available to all, regardless of ability to pay. Yet we also have over 43 million people (most of them employed and earning minimal wages) without any health insurance and many more who are inadequately insured. Since the collapse of the Clinton health care reform efforts in 1994, U.S. health policy has been directed by marketplace forces that have created powerful and sometimes perverse incentives in medicine: Health insurance companies that use every available means to avoid insuring sick people; "managed care" programs that really only manage costs; doctors who are provided incentives to provide less medical care; and pharmaceutical companies that develop powerful and expensive new drugs priced beyond the reach of many of the elderly and chronically ill who need them most.

Facing such powerful and chaotic forces, physicians tend to focus narrowly on what they are most comfortable with, taking care of individual patients and conducting academic investigations. Many doctors consider economics too arcane for them to grasp and therefore do not even try. Consequently, when presented with economic arguments and evidence they are often unable to discriminate the legitimate from the fallacious. More importantly, they are ill equipped to defend their patients' interests in the crucible of cost containment that characterizes the modern managed care era.

This chapter has two goals: first, to provide a brief introduction to some of the larger economic forces that shape modern medical practices, and second, to introduce the economic tools that are used for assessing the value of medical practices, including cost effectiveness analysis.

HEALTH CARE SPENDING AND FINANCING

HOW MUCH IS SPENT ON HEALTH CARE?

In 1997, the United States spent \$1.1 trillion on its health care system, representing 13.5% of the gross domestic product (GDP) (a crude measure of national income). Most of this (\$969 billion) was spent on personal health care: 34% went to hospitals, 20% to physicians, 7% to nursing homes, and 8% to outpatient pharmaceuticals. In comparison, Canada and Western European countries spend a substantially smaller portion (6 to 10%) of their national income on health care but their citizens appear to be equally healthy, at least by crude metrics such as life expectancy and infant mortality rates. Economists and politicians have for years used such data to argue that the United States spends too much on health care. The issue of how much to spend is an inherently political one, however, and the discipline of economics has little to say about it.

WHO PAYS FOR HEALTH CARE?

Two major factors are continually driving up the costs of medical care: introduction into medical practice of new medical technologies (drugs, devices, procedures) that have a high price tag, and the aging of the U.S. population (since older people require more medical care than younger ones). These costs are distributed unevenly across society. In 1997, the government paid about 47% of the total national health care bill (75% federal, 25% states), private insurance paid about 32%, and individuals paid 17%. The government, of course, gets its money from taxpayers and uses the health care segment of its budget to pay for the Medicare and Medicaid programs (discussed below). To respond to rising medical costs, the government can increase taxes or redistribute funds from other programs such as defense and education. Neither of these options are politically attractive. Alternatively, because of its size in the medical marketplace, the government can impose lower prices on providers to make the available funds go farther (see "Cost-Containment Strategies," below). Much of the private insurance bill is subsidized by employers through their employee benefits packages. As medical costs go up, health insurance costs also rise and businesses must either pass on higher premium and copayment costs to their employees, raise their prices (potentially impairing their competitive position in the marketplace), or reduce their profit margin (a very unpopular move with stockholders). Like the government, businesses may also negotiate lower prices with health care providers and/or health insurance plans.

PUBLIC FINANCING OF HEALTH CARE

The public sector (i.e., government as an agent for society) finances the Medicare and Medicaid programs as well as the Veteran's Administration Hospital system, the Department of Defense military care system, the Public Health Service, and the Indian Health Service. Of these, Medicare is by far the largest and most influential, with 39 million people receiving health insurance at a total cost in 1997 of \$214.6 billion (20% of total national health expenditures). The Medicare program was enacted in 1965 by Congress as an amendment to the Social Security Act of 1935 and was envisioned by President Lyndon B. Johnson as a first step toward universal health insurance in the United States, a key part of his "great society" plan. Its impact on the evolution of the U.S. health care system has been profound. The original congressional act provided health care insurance for the elderly (defined as those 65 and older) who were eligible for social security (i.e., retired workers who had paid into the system during their working years and their dependents). Amendments in 1972 extended coverage to the disabled of all ages (currently numbering around 5 million) and to patients with chronic renal failure (who currently number about 284,000).

Medicare consists of two related insurance programs. The Medicare Hospital Insurance Trust Fund (also known as Part A) covers hospital care and skilled nursing home care and is funded by compulsory federal payroll taxes on employers and employees. Medicare Part B, the Medical Supplementary Insurance Program, covers physician fees as well as laboratory and other diagnostic tests and is funded by general federal tax revenues and patient premiums. Both programs have substantial gaps in coverage, necessitating supplemental insurance (so-called Medigap policies) for those who can afford them. Because of its compulsory income redistribution feature, taking tax money from current workers to pay for health care for elderly citizens (many of whom are on fixed income close to the poverty level), Medicare is both a health insurance program

and a social welfare program designed to combat poverty in the disabled and elderly. In exchange for their tax money, the 150 million workers funding the program are promised the same type of social security when they become elderly (paid for by future generations of workers).

Medicaid is a social insurance program for the poor that is jointly run by the federal and state governments. The federal government gives each state a grant of money for the program based on that state's per capita income (in 1997, this amount totaled \$95 billion), and the states pay for the rest (\$65 billion in 1997). The program, like Medicare, was enacted by Congress in 1965 as a part of President Johnson's "great society" program. It is larger than Medicare in terms of eligible beneficiaries (41 million people) but smaller in terms of budget (\$160 billion, or 12% of the total national health expenditures). Because the requirements to qualify for Medicaid are stringent, many low-income individuals under age 65 (especially the working poor) do not qualify. Eligibility criteria are set by each state within general federal guidelines, and the income and asset tests individuals must meet to qualify vary widely among states. Many of the dollars in the Medicaid program actually pay for care for elderly and disabled Medicare beneficiaries who also qualify for Medicaid on the basis of poverty.

PRIVATE FINANCING OF HEALTH CARE

Approximately 70% of the non-elderly U.S. population is covered by some form of private medical insurance. The feasibility of group insurance for medical care was initially demonstrated in the 1930s by Blue Cross, a franchise of nonprofit groups providing hospitalization insurance in order to help prop up the financially strapped U.S. hospital industry. Blue Shield, a separate organization modeled after Blue Cross, started providing insurance for in-hospital physician services in 1939. During World War II, employee wages were frozen by the government and to entice workers, who were in short supply, some employers started offering health insurance as a fringe benefit. With the feasibility of employer-sponsored group health insurance demonstrated by the experience of the "Blues," commercial insurers began to enter the market. To win the support of doctors and hospitals, insurers agreed to pay "reasonable and customary charges" and to defer all medical management decisions to doctors. This "fee-for-service" reimbursement system, created in the post-World War II era, sowed the seeds of the tremendous inflation observed in the U.S. medical system during the 1970s and 1980s.

The original focus of indemnity insurance plans was to cover individuals against catastrophic financial losses from high medical care bills. Insurance is a contract for protection against specific hazards that are unpredictable for individuals but can be defined with confidence for large groups. "Major medical" health insurance was designed to provide coverage for catastrophic illness, a relatively rare event in most populations. Group coverage is less expensive than individual coverage because it allows the insurance company to diffuse the risk of a large payout among a big pool of individuals who will pay premiums but make no claims. When coverage is shifted from a focus on rare catastrophes to routine maintenance medical care (comprehensive insurance policies), health insurance becomes a means for payment of expected rather than unexpected care. The consequence is higher health insurance premiums. The early appeal of health maintenance organizations (HMOs) was that they appeared to

offer an economically efficient way to provide routine preventive care and to manage the occasional catastrophic illness.

MANAGED CARE

Managed care is a generic term that embraces a wide spectrum of systems for integrating the financing and delivery of health care. Managed care organizations (MCOs) contract with doctors and hospitals to provide comprehensive care to enrolled members for a fixed, prospectively set, premium. [HMOs](#) are a form of managed care originally organized between the 1940s and 1960s as an alternative to the prevailing fee-for-service-based private insurance. With the advent of serious medical inflation in the 1970s, the HMO model was promoted by the federal government as a way to control the growth in medical spending. Early enthusiasm for this initiative was limited; in 1984, only 5% of individuals with employer-based health insurance were in an HMO. However, by 1998 that figure had risen to 85%. The exponential growth of managed care started in the 1990s in part as an employer-driven response to the uncontrolled medical inflation of the previous two decades.

The massive increase in demand for managed care by employers and by the Medicare program produced a rapid, and sometimes bewildering, evolution in the managed care industry. One important trend has been the growth of for-profit (i.e., investor owned) managed care companies. Over half of [HMO](#) members now belong to a for-profit plan. Investment dollars from Wall Street have made it easier for these plans to respond quickly to increased employer demand for managed care options. However, compared with their not-for-profit counterparts, for-profit HMOs spend a smaller proportion of each premium dollar paying for health care for members (the paradoxically named "medical loss ratio"), since stockholders also have to be paid. As a result, for-profit HMOs are less successful than not-for-profit plans in providing preventive care (a presumed strength of managed care).

Another prevalent trend of the 1990s was the move from traditional [HMO](#) models to virtual HMOs, built from contractual relationships with community physicians and hospitals. The three HMO models are the staff model, the group model, and the Independent Practice Association (IPA). The staff model HMO is a vertically integrated organization. That is, it owns its own hospitals, employs all its physicians full time for a set salary, and is focused in a particular geographic area. The group model HMO, exemplified by Group Health Cooperative of Puget Sound, contracts with one or more large multispecialty group practices to care for its patients for a preset capitated reimbursement. These physicians do not care for non-HMO patients. In the IPA model, the HMO contracts with an association of self-employed physicians who maintain their own offices and see both HMO and non-HMO patients. The network model refers to a hybrid of the other three forms of HMO. IPA and network model HMOs now have the majority of HMO membership in the United States.

The other portion of the managed care industry is represented by point of service (POS) plans and preferred provider organizations (PPOs). POS plans incorporate key features of both [HMOs](#) and traditional fee-for-service plans. A patient may choose care from a provider network or go outside the network. Care within network requires only a minimal copayment, while care outside the network requires a deductible and a large (e.g., 30%)

copayment. The goal of the plan is to offer patients a choice but to provide major financial incentives to stay within the HMO portion of the plan. PPOs use a defined provider network (physicians, hospitals) that has agreed to accept discounted fee-for-service to care for enrolled members. PPOs may incorporate various managed care features, such as physician gatekeepers and utilization review.

THE UNINSURED AND UNDERINSURED

Data from the U.S. Census Bureau indicate that 43.4 million people had no health insurance for all of 1997 and 71.5 million people were without insurance for at least part of the year. The great majority of uninsured individuals either work for small employers who do not offer a health insurance benefit or, more commonly, cannot afford the premiums of the plan(s) that are offered. Underinsurance also has a significant impact on the working poor by requiring them to pay an excessive proportion of their family's income for health insurance premiums and out-of-pocket medical costs (deductibles, copayments, and uninsured care). Outpatient prescription medications are a major source of underinsurance. Prescription drug costs are now the fastest growing segment of the national medical budget and the least likely segment to be covered by insurance. The elderly are particularly affected, since Medicare does not currently cover outpatient prescriptions and even Medigap policies have limited coverage.

Some states have experimented with expanded coverage through their Medicaid programs to help the uninsured poor (such as the Oregon Medicaid program). For the foreseeable future, however, it does not appear that the federal government will address this problem comprehensively.

COST-CONTAINMENT STRATEGIES

Current projections from the federal government's Health Care Finance Administration (HCFA) are that health care expenditures will double (to \$2.2 trillion, or 16.2% of the [GDP](#)) by 2008. Over the past 30 years, the U.S. health care system has experimented with a vast array of cost-containment approaches. Conceptually, there are four major ways to control medical spending: (1) control prices, (2) control volume of care provided, (3) control the total budget available to pay for care, and (4) shift costs to another payer.

Two of the most important price control initiatives in medicine have been the Medicare Hospital Prospective Payment System and the Medicare Fee Schedule for physicians. In 1983, Medicare replaced its retrospective cost-based hospital reimbursement system with a prospective payment system. In this system, all hospitalizations are classified into one of approximately 500 Diagnosis Related Groups (DRGs) based on the principal discharge diagnosis for the hospitalization and a few selected additional factors such as age, the performance of surgery, and the presence of complications. Each DRG is assigned an average reimbursement (adjusted annually). If the hospital can provide care for less than this amount, they make a profit. If they spend more than this amount, they lose money. The DRG system was designed to promote efficiency and cost containment in hospital-based care. While it has helped to control Medicare costs, it has not reduced overall U.S. health care costs, probably because of substantial cost-shifting by hospitals to the private insurance sector.

Between 1975 and 1987, Medicare payments to physicians increased at an annual rate of 18%, well above the rate of inflation. While total spending for physician services accounts for less than 25% of the Medicare budget, physicians have control over aspects of care (use of procedures, length of stay, hospital admission) that extend their direct influence to over 75% of the Medicare budget. Recognizing the importance of physicians in cost containment, Congress directed the development of a new physician payment system based on the use of a resource-based relative value scale (RBRVS). The Medicare Fee Schedule, which was first used in 1992, has three components: (1) a measure of the total work (time and complexity) involved in each physician service and standardized across all specialties, (2) a practice expense to cover the cost of running an office, and (3) an amount to cover malpractice insurance costs. The Medicare Fee Schedule classifies all physician services using the American Medical Association's Current Procedural Terminology (CPT) codes. Each CPT code has an associated relative value units (RVUs) weight. The RVU weights are multiplied by a national conversion factor to generate the actual physician fee associated with the service in question.

Price controls are attractive for cost containment because they are less expensive administratively than volume controls and don't involve micromanagement of clinical care. Price controls alone, however, don't generally achieve control of costs because of compensatory responses of providers. For example, under Medicare prospective payment, hospitals have shifted much care to the outpatient setting, where [DRGs](#) are not used. Physicians have responded to lower fees by an increased volume and intensity of service.

Volume controls include various programs to limit the diffusion of expensive technologies (such as heart surgery) or extra hospital beds. Limits can be operationalized using either a regulatory approach [such as certificate of need (CON) programs] or a budgetary approach. Utilization review approaches attempt to discern which expensive care items are medically necessary and which are not.

Budgetary controls are simpler than either price or volume control approaches. In Canada, for example, hospitals have global annual budgets. How the money is spent is decided by each hospital. If the budget is exceeded, there are no guarantees that the shortfall will be covered.

Finally, payers can control their costs by cost-shifting to other willing payers. For example, as health insurance premiums rise, employers can choose to pass these costs on to employees. Hospitals and doctors who lose money caring for Medicare patients can try to make up their losses by charging more to private insurance patients. Insurance companies can choose to offer limited or no coverage for outpatient pharmaceuticals, shifting the full cost of expensive new medicines directly to patients.

MEDICAL ECONOMIC CONCEPTS AND TOOLS

MEDICAL COST CONCEPTS

Medical cost analysis is a field that borrows heavily from both economics and

accounting. Economics provides the theoretical structure that defines the key questions to be addressed, and accounting provides many of the measurement tools. Traditional economics has as one of its major axioms that societal resources are finite. For this reason, society must choose from among the many ways that resources can be used and not all of society's goals can be fulfilled. Economics has devised a theoretical framework and a set of tools (including cost-effectiveness analysis) to help define the major competing goals for societal resources and to assist in selecting from among the ones that most efficiently fulfill societal needs. "Cost" in economics refers not so much to money but rather to the lost opportunities that occur when the limited societal resources are expended in a particular way. For example, if our medical armamentarium is enhanced over the next decade by discovery of powerful but expensive therapies and these are incorporated into standard clinical practice, the ability of the country to invest in education, defense, or transportation may be compromised. This notion of cost as a lost opportunity to use resources in alternative ways is referred to as *opportunity cost*. While representing the purest economic notion of cost, there is no practical way to measure it.

Accountants, who are much more concerned with issues of measurement, have proposed a "gold standard" of cost measurement, *true accounting cost*, that involves enumerating all the individual resources consumed in the production of a particular medical good or service and assigning market prices for each of them. The total cost is then the sum of the dollar costs for all the component resources. Even this calculation, however, may be prohibitively difficult in "real world" applications, for several reasons. First, all medical care requires not only the easily identifiable components of personnel time and disposable supplies but also the infrastructure components such as the rent on the office building where the care is provided, the cost of utilities, and the expense of an office staff. Second, even if all the components can be identified, enumeration of exactly what is used may be prohibitively expensive. Finally, medicine does not have publicly available "market prices" that can be readily obtained for a medical cost analysis, the way one can obtain prices for automobiles or refrigerators. The reasons for this relate to the lack of a true competitive free market in medicine along with the severe price distortion created in medical charges by cost-shifting practices.

KEY COST TERMS

Several key sets of cost terms are used in medicine. As the volume of health care produced is increased or decreased, costs may exhibit either variable or fixed "behavior." *Variable costs* change with each unit shift in production volume (up or down). For example, each vaccination administered to a group of children increases costs (related to the dose of vaccine and the disposable syringe) in a predictable linear fashion. *Fixed costs* do not shift with short-term changes in the volume of care provided. For example, the rent on the clinical building and the cost of heating, lighting, and so forth do not change according to the number of individuals vaccinated per day. Some types of costs display hybrid features of both variable and fixed components. For example, clinic personnel costs (e.g., nurses, secretaries) may be fixed if these personnel are paid a salary regardless of clinic volume. If the clinic volume goes up so much that evening hours must be added, either new personnel must be hired or existing personnel must work overtime. Either of these changes would graft a variable component onto the fixed personnel costs.

Marginal cost is a concept often used by economists to refer to the cost of producing one more unit of a given health care good or service. For example, the costs of doing one more or one less diagnostic cardiac catheterization would be its marginal cost. For all practical purposes, this is the same as its variable costs (since fixed costs do not change with small changes in volume). While the concept of unit changes in volume is theoretically interesting, a more pragmatic issue is the cost effect of changing a group of patients from one strategy to another. Many experts use the term *incremental costs* to refer to this type of shift (although some use marginal and incremental synonymously). Incremental analysis is a key component of cost-effectiveness analysis (see below).

Another set of cost terms relates to the traceability of costs to the production of health care goods and services. *Direct costs*, such as nursing and physician personnel and disposable supplies, can be clearly linked to the health care provided and are under the control of the health care providers. *Indirect costs*, sometimes known as *overhead*, cannot. For example, the utility, laundry, maintenance, and administration costs of a hospital cannot be linked with the care of an individual patient and are generally not under the control of the physicians and nurses providing the medical care. The distinction of direct versus indirect is useful in cost-containment efforts, where the first step is to identify all major cost components and decide how they are to be controlled.

One common error in the evaluation of medical costs is to focus on the cost of a test or therapy in isolation. Virtually every major medical management decision creates downstream consequences. For example, if physicians order a screening diagnostic test and the result is abnormal, they will need to do a confirmatory or more definitive test. If they order a potent new antibiotic and a fraction of patients develop liver failure as an unexpected toxicity, the total cost of that course of antibiotic includes not only the cost of the drug itself but also the costs of treating the liver failure in the fraction of patients who develop it. Extra costs added as a consequence of some diagnostic or therapeutic decision are referred to as *induced costs*. Similarly, if a management decision produces downstream savings, these would be referred to as *induced savings*. For example, administration of HMG CoA reductase inhibitors to patients with hypercholesterolemia can prevent future myocardial infarctions and revascularization procedures, both of which entail expensive hospitalizations.

One final important cost concept relates to the societal costs of lost productivity (primarily lost time from work) due to illness. While economists often refer to these as indirect costs, confusion with the accounting concept of indirect costs (overhead) has led many to prefer the alternative term, *productivity costs*.

COST MEASUREMENT

Using varying degrees of simplification, medical costs can be measured using either bottom-up or top-down approaches. Bottom-up approaches build from component resources to calculate total cost for an episode or type of care. Microcosting is the gold standard approach. It involves careful enumeration of all resources consumed and detailed cost-accounting estimation of the costs for each component resource. A number of medical centers have now installed computer-based cost-accounting systems that perform a modified type of microcosting analysis. For difficult-to-obtain resource

use data (such as time required for a particular type of care by a given type of personnel), these systems use expert opinion in place of empirical data. The other extreme of the bottom-up category of approaches involves enumeration and costing for only the "big ticket" or expensive items, such as hospitalization episodes and costly tests and procedures.

The top-down methods of medical cost estimation calculate a cost estimate from aggregated data. One such approach uses hospital billing charge data and charge-to-cost conversion ratios (which each hospital produces annually in its Medicare Cost Report) to estimate hospital costs. Despite the approximations involved, this approach, which can be used for most nonfederal U.S. hospitals, has provided good agreement with bottom-up estimates in the few instances where formal comparisons have been made. The other top-down approach is the use of [DRG](#) assignments and reimbursement rates to provide standard cost weights for hospitalization episodes.

COST-EFFECTIVENESS ANALYSIS

Given a finite budget (for health care overall or for a particular health system), how can we use the available money to provide the most health benefits for our patients? For the clinician, who is less concerned with such policy issues, a prevalent question is whether a new treatment is economically attractive. The analysis method used to address this question is dependent on how the effectiveness and costs of the new therapy compare with those of "standard care" ([Fig. 4-1](#)). *Cost-effectiveness analysis* is used when effectiveness of the new treatment is greater and its costs are higher. This analysis calculates the ratio of added (or incremental) health benefits to added costs produced by a new therapy or strategy relative to some reference standard. The general formula is:

where C = costs and E = effectiveness.

The cost-effectiveness ratio provides a quantitative statement of the amount of money required to produce a single extra unit of benefit with the new therapy relative to usual care or some other relevant reference standard. The benefit can be calculated in any meaningful clinical unit, such as added survivors or extra patients with a correct diagnosis. However, the vast majority of cost-effectiveness analyses use the epidemiologic concept of life-years to express incremental benefit. Virtually all benchmarks for cost effectiveness relate to this endpoint. Because some therapies affect quality of life but not quantity, a more generally relevant effectiveness measure combines quality of life and life expectancy into a single composite metric, the quality-adjusted life year (QALY). Calculation of incremental dollars required to add an extra QALY is called *cost-utility analysis*. The QALY is a useful concept, but many details regarding measurement and interpretation remain controversial. The third form of economic efficiency analysis, *cost-benefit analysis*, requires conversion of health benefits into monetary equivalents. Because such conversions are controversial, this form of analysis is rarely used in medicine. In theory, the time horizon of a cost-effectiveness analysis should be long enough to capture all important cost and health consequences of the therapy or strategy being evaluated. Most often, analysts

use a lifetime time frame. Because very few empirical studies are long enough to observe lifetime outcomes (especially when chronic diseases are being studied), models are required to extrapolate from available data.

A cost-effectiveness analysis can be done from a variety of perspectives, but the most widely applicable perspective is societal. Other perspectives are often much narrower and may include unattractive qualities. For example, a managed care organization may be interested only in short-term costs and outcomes, knowing that patients tend to change their health insurance every few years.

The benchmarks for cost-effectiveness ratios are determined by comparison with other well-accepted therapies in widespread medical use. A useful benchmark is hemodialysis for chronic renal failure, since the federal government has paid for all renal failure patients to get dialysis since 1973 through the End Stage Renal Disease Program. Recent estimates are that it costs this Medicare program about \$50,000 to add 1 life-year to a chronic renal failure patient. Partly for this reason, many analysts use a cost-effectiveness ratio of <\$50,000 per added life-year to identify therapies that are economically attractive (i.e., have a favorable balance of extra costs to extra benefits), while therapies with ratios >\$100,000 per added life-year are deemed economically unattractive and therapies between \$50,000 and \$100,000 per added life-year are in the economic "gray zone."

Several caveats about cost-effectiveness analysis should be noted. First, cost-effectiveness analysis is descriptive, not prescriptive. It measures value that could be produced with available health care dollars but does not mandate how these dollars are to be used. If an expensive new therapy is introduced and is found to be very economically attractive by the above benchmarks, it will still not get used if there is no money in the budget to pay for it. Second, a cost-effectiveness ratio is only as good as the data that were used to calculate it. High-quality results can be obtained if economic analysis is prospectively incorporated into the design of large-scale multicenter randomized trials. Third, although cost-effectiveness ratios are often presented as deterministic (i.e., no variability), they often incorporate large amounts of uncertainty. This should be examined either with sensitivity analyses (varying each key parameter through a plausible range to see if the results are materially changed) or calculation of confidence limits.

MEDICAL ECONOMICS AND CLINICAL PRACTICE

In evaluating new therapies, three issues must be addressed: (1) is the new therapy significantly better than what is currently available? (2) how much does it cost and is it economically attractive? and (3) how many patients will need this therapy and is it affordable? The clinician should be primarily concerned with the answer to the first question. Although cost issues are now a reality of daily clinical life and cost-containment pressures are often substantial, decisions by clinicians that are based primarily on economic rather than clinical considerations put the physician in the role of the double agent (i.e., acting on behalf of both the patient and the payer) and compromise our fiduciary obligation to patients. The second question addresses cost effectiveness and, if favorable, can be used to support an argument by clinicians for adoption of the therapy. In the ideal world, at least, therapies that have a large database

of evidence demonstrating effectiveness and economic attractiveness should be given preference over therapies that do not have such supporting data. The final question is of primary concern to payers and health policy analysts. An effective therapy that is too expensive to use is of little more value than a therapy that has yet to be discovered.

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5. INFLUENCE OF ENVIRONMENTAL AND OCCUPATIONAL HAZARDS ON DISEASE - Howard Hu, Frank E. Speizer

Exposures to hazardous materials and processes in the home, the workplace, and the community can cause or exacerbate a multitude of diseases. Physicians commonly treat the sequelae of such diseases in the practice of medicine; however, unless the underlying connection with hazardous exposures is identified and mitigated, treatment of manifestations rather than the cause at best only ameliorates the condition. At worst, the neglect of hazardous exposures may lead to both failure of treatment and failure to recognize a public health problem with wide significance.

No existing surveillance or reporting system can estimate the total contribution of hazardous exposures to morbidity and mortality. However, careful histories have identified occupational factors as etiologic in more than 10% of all admissions to general internal medicine wards in hospitals, with even higher percentages when the primary illness is either respiratory or musculoskeletal. Estimates of the number of new cases of disease due to work in the United States range from 125,000 to 350,000 per year; these cases do not include 5.3 million work-related injuries.

Environmental exposures are increasingly associated with decrements in measures of health whose outcomes range from subclinical to clinically catastrophic. For example, exposure to lead at levels that are common in the general population has been associated with increased blood pressure and decreased creatinine clearance. Ambient air pollution with respect to levels of ozone and fine-particulate matter has been related to increased rates of hospital admission for respiratory and cardiovascular diseases and to increased mortality rates, respectively. Indoor exposure to radon and passive indoor exposure to environmental tobacco smoke have been linked with an increased risk of lung cancer. There is pressure on clinicians to be aware of and act on this type of information, which is suggestive but not necessarily conclusive with respect to causation.

Patients are becoming increasingly concerned about hazardous exposures. More than 15% of patients seen in one study conducted in a primary care clinic expressed the opinion that their health problems were work-related, and 75% of this subgroup of patients reported exposure to one or more recognized toxic agents. Patients often want answers to very specific questions, such as: Is the water in our town safe to drink? Could my breathing problem be related to the new roofing sealant used in my building at work? Physicians are consulted because they are the most trusted sources of information on health risks, including chemical risks. Unfortunately, few physicians have more than rudimentary training in environmental and occupational medicine. Therefore, it becomes important for primary care physicians to be able to recognize symptoms precipitated by exposure to environmental or occupational hazards and either to manage these cases or to make appropriate referrals.

Many manifestations of exposure-related illnesses are nonspecific (e.g., dizziness, headache) or are commonly encountered in general internal medicine (e.g., myocardial infarction, cancer). The establishment of a connection with an environmental or occupational hazard requires a high index of suspicion and the application of fundamental concepts of environmental/occupational medicine. Furthermore, early

recognition by physicians of unusual patterns of illness or of evidence of asymptomatic exposure to toxins with low-level effects (e.g., an elevated blood lead level) can alert health officials to the need for control measures. Case reports either sent to local authorities or published in the literature often prompt follow-up studies that can lead to the identification of new hazards. In many states and countries, the reporting by physicians of occupational/environmental diseases is mandatory. For instance, beginning in 1992, physicians in Massachusetts were required to report cases of pneumoconiosis, occupational asthma, carpal tunnel syndrome, and carbon monoxide poisoning, among other conditions. Identification of an environmental/occupational etiology of an illness may have important economic ramifications for the patient (e.g., the awarding of worker's compensation, which covers medical bills as well as lost wages). Finally, physicians are frequently asked to provide expert medical testimony during litigation on the causal relationship between toxic exposures and diseases. In this setting, the more knowledgeable the physician is about potential hazardous exposures, the better prepared he or she is to serve the patient.

THE ENVIRONMENTAL/OCCUPATIONAL HISTORY

For a physician, the most critical steps toward recognizing these disorders are remembering to consider them in the differential diagnosis and taking an appropriate environmental/occupational history as part of the medical workup. The level of detail that is called for depends on the clinical situation. *Information should always be obtained on current and major past occupations, and patients should be asked whether they think their health problem is related to their work or to any particular environment or exposure.* In the review of systems, patients should be asked if they have been exposed to dusts, fumes, chemicals, radiation, or loud noise. When patient and physician are confronted with an illness of uncertain etiology, these factors should be explored in more detail, with the environmental/occupational history as the point of departure. (A brief outline of a sample history is shown in [Table 5-1](#).)

The identification of specific chemical exposures can be difficult. Household products must list chemical ingredients on their labels, and this information may prove useful. For workplace exposures, the U.S. Occupational Safety and Health Administration (OSHA) requires chemical suppliers to provide material safety data sheets with their products and requires employers to retain these sheets and make them available to employees. The data sheets can be obtained by the physician or employee by a telephoned or written request; failure of an employer to provide them within 30 days of such a request is a violation of OSHA regulations and is punishable by fines. In addition to providing information on chemical ingredients and percent composition, the material safety data sheets provide basic information on toxicity. This information is seldom adequate from a clinical perspective but may indicate the general type of toxicity to be anticipated.

EVALUATION OF POSSIBLE CHEMICAL OR ENVIRONMENTAL HAZARDS

Given the wide variety of toxic exposures that may be uncovered during a workup, a clinician should routinely consult additional reference material to evaluate whether particular hazards may be associated with the illness at hand. Many sources of information exist. [OSHA](#) and some regional poison-control centers have extensive information on hazards and brief summary documents that can be transmitted over the

Internet or by telephone or facsimile. Depending on the area, other resources may include county and state health departments; regional offices of the National Institute for Occupational Safety and Health and the Environmental Protection Agency; the Consumer Products Safety Commission in Washington, DC; academic institutions; websites of these institutions; and individual toxicologists, occupational/environmental medicine specialists, or industrial hygienists. Sophisticated computerized databases are also available, including detailed listings on CD-ROM information systems. MEDLARS, the electronic database maintained by the National Library of Medicine, is accessible by modem or the Internet and is familiar to many physicians. Files other than MEDLINE, such as the Hazardous Substances Databank, provide specific toxicity information on chemicals and include toxicologic references not covered by MEDLINE. Many of these databases can also be accessed through the Internet.

As with any other illness, laboratory investigation may be crucial. For example, tests of carboxyhemoglobin level to document carbon monoxide exposure or of serum anticholinesterase level to document organophosphate pesticide absorption should be performed within hours of exposure. As in cases of acute drug overdose, it is useful to freeze samples of urine and serum from any patient suspected of having had an acute chemical exposure; such specimens can be analyzed at a later date by sensitive methods of detection. Use of other tests must rely on knowledge of the specific hazard or illness in question.

SUSPICIOUS SCENARIOS

Some medical problems or clinical scenarios demand a particularly high degree of suspicion of occupational or environmental factors as causative or contributing agents.

Respiratory Disease The contribution of occupational/environmental factors to respiratory disease is generally underrecognized, particularly among patients who smoke and among the elderly ([Chap. 254](#)). For instance, asthma related to chemical exposure may be treated without regard to cause or may be erroneously diagnosed as acute tracheobronchitis. A study of new-onset asthma among HMO members in Massachusetts found that 21% of these individuals met criteria for clinically significant asthma attributable to occupational exposures. The types of exposures and jobs in these cases varied widely; examples include exposure to smoke in a firefighter, to welding fumes in a technical school student, to cleaning compounds in a bartender, and to epoxy in an archery repairman. No single type of job or exposure predominated. Other examples of etiologic errors include shortness of breath from asbestosis that is attributed to chronic obstructive pulmonary disease and chemical pneumonitis that is misdiagnosed as a bacterial infection.

Cancer Many cancers are thought to be causally related to occupational and environmental factors in addition to tobacco. Some are particularly likely to have a chemical etiology or another environmental cause, including cancers of the skin (solar radiation, arsenic, coal tar, soot); lung (asbestos, arsenic, nickel, radon); pleura (almost exclusively asbestos); nasal cavity and sinuses (chromium, nickel, wood and leather dusts); liver (arsenic, vinyl chloride); bone marrow (benzene, ionizing radiation); and bladder (aromatic amines).

Coronary Disease and Hypertension Carbon monoxide exposure is common, particularly in homes with malfunctioning furnaces or in workplaces close to motor vehicle exhaust. By reducing oxygen transport by hemoglobin and inhibiting mitochondrial metabolism, carbon monoxide can aggravate coronary disease. Methylene chloride, a solvent used in paint stripping, is converted to carbon monoxide and thus poses the same risk. Exposure to carbon disulfide, a chemical used in the production of rayon, accelerates the rate of atherosclerotic plaque formation. Chronic lead exposure, even at modest levels, is a risk factor for the development of hypertension as well as abnormalities of cardiac conduction.

Hepatitis/Chronic Liver Disease In the absence of evidence that a viral infection, alcohol ingestion, or drug use is the main cause of hepatitis ([Chaps. 295, 296, and 297](#)), the involvement of a toxin must be considered. Toxin-induced hepatic injury may be cytotoxic, cholestatic, or both. The list of hepatotoxic agents is long, including organic synthetic compounds such as carbon tetrachloride (used in solvents and cleaning fluids) and methylene diamine (a resin hardener); pesticides such as chlordecone (Kepone); metals, particularly arsenic (used in pesticides and paints and found in well water); and natural toxins such as the pyrrolizidine alkaloids.

Kidney Disease Many chemical and environmental factors can cause renal injury ([Chap. 269](#)). The etiology of much chronic kidney disease, however, remains unknown. An increasing body of evidence now links chronic renal failure with hypertension to lead exposure. One study demonstrated that chelation therapy with EDTA slowed the progression of renal insufficiency in patients with a mildly elevated body lead burden. Some studies suggest that chronic exposure to hydrocarbons (e.g., gasoline, paints, solvents) may lead to various types of glomerulonephritis, including Goodpasture's syndrome. Environmental cadmium exposure has been found to promote calcium loss via urinary excretion, which results in skeletal demineralization and thus in an increased risk of fractures.

Peripheral Neuropathy Organic solvents such as *n*-hexane, heavy metals such as lead and arsenic, and some organophosphate compounds can damage the axons of peripheral nerves. Dimethylaminopropionitrile, an industrial catalyst, causes bladder neuropathy. Nerve entrapment syndromes of the upper extremity, such as carpal tunnel syndrome, may be caused by jobs that involve repetitive motion, especially those requiring the maintenance of awkward positions.

Central Nervous System Disorders Fatigue, memory loss, difficulty in concentration, and emotional lability have been linked to chronic exposure to solvents such as toluene and perchloroethylene. Painters, metal degreasers, plastics workers, and cleaners are commonly exposed to solvents and develop these symptoms at a high rate. Among the features that distinguish these patients are characteristic patterns on formal neurobehavioral testing and stabilization of symptoms with gradual improvement after discontinuation of the exposure. Other substances associated with neurobehavioral dysfunction include metals, particularly lead, mercury, arsenic, and manganese; pesticides, such as organophosphates and organochlorines; polychlorinated biphenyls (PCBs); and gases such as carbon monoxide.

Environmental factors are also suspected of contributing to other neurologic diseases,

such as degenerative disorders, motor neuron diseases, and extrapyramidal disorders. For example, a study in monozygotic and dizygotic twin pairs found a similarity in concordance indicating that environmental (as opposed to genetic) factors play a major etiologic role in cases of typical Parkinson's disease beginning after the age of 50 years.

Teratogenesis and Reproductive Problems Toxins can impair successful reproduction at a variety of levels. Examples include insecticides and herbicides, [PCBs](#) and polybrominated biphenyls (PBBs), ethylene oxide (a sterilizing gas used in hospitals), metals (lead, arsenic, cadmium, mercury), and solvents. Dibromochloropropane, a nematocide, suppresses spermatogenesis. Some toxins, such as PCBs, PBBs, and chlorinated pesticides, are concentrated in milk. Concern has arisen over the ability of specific organic pollutants, particularly pesticides, to persist in the environment and accumulate in human tissues. Some of these chemicals may disrupt endocrine function, and these effects may be related to phenomena such as the observed increases in the incidences of testicular cancer, breast cancer, and hypospadias.

Immunosuppression, Autoimmunity, and Hypersensitivity Evidence is increasing that exposures to some chemical agents can compromise the immune system, thereby leading to a generalized increase in the incidence of tumors (e.g., exposure to [PBBs](#)) or infections (e.g., respiratory infections after exposure to common air pollutants). Mercury, dieldrin, and methylcholanthrene are known to elicit autoimmune responses. Some chemicals are potent allergic sensitizers that cause dermal and respiratory problems ([Chaps. 60](#) and [254](#)).

BIOLOGICAL MARKERS

An increasing number of methods are available for measuring and interpreting toxic exposure, including (1) the internal dose of specific toxins and (2) markers of the biologic effects of toxins. Internal-dose markers are relevant for toxins that are sequestered in the human body, such as lead (in blood), arsenic (in hair), and other metals ([Chap. 395](#)), and for halogenated compounds (such as [PCBs](#)). Examples of markers of the biologic effects of toxins include depressed levels of acetylcholinesterase in serum after exposure to organophosphate pesticides, sister chromatid exchanges in peripheral lymphocytes after exposure to the carcinogen ethylene oxide, and DNA adducts after exposure to tobacco smoke carcinogens.

MANAGING A HAZARD-RELATED ILLNESS

Once a chemical or another environmental hazard has been identified as an important contributor to an illness, the next step is to prevent further exposure. Although for chronic diseases such as cancer this step may be irrelevant for the patient in question, prevention of further exposure may still be critical for other persons who have been similarly exposed. When prevention of further exposure is important, *the physician must be willing to become an active advocate for the patient*. This advocacy may involve writing a letter stating that the patient should no longer be exposed to a hazard or should remain out of work. Alternatively, it may involve contacting appropriate officials in government, industry, or labor or other advocates who can deal with a hazardous exposure. Treatment is dependent on the specific hazard.

In few areas of medicine does a physician deal with more scientific uncertainty. Comprehensive information on toxicants is available for only a small percentage of chemicals. In general, the physician should take a conservative approach (i.e., advise the patient to avoid a hazard likely to have contributed to illness) and should use common sense and up-to-date information to evaluate causal relationships.

LOW-LEVEL EXPOSURES AND THEIR EFFECTS

The subclinical effects of toxins that are widespread in our environment and our workplaces are of increasing concern. Given the absence of any demonstrable effect threshold, low-level exposure to carcinogens should be avoided; not only carcinogenic but also noncarcinogenic effects of chronic low-level exposure to these substances are important.

Perhaps lead provides the most important example of low-level noncarcinogenic effects that constitute a major public health problem. Multiple pathways of exposure, including the combustion of leaded gasoline, the use of lead-based paints and solder, and the presence of lead in cans containing food, have contributed to exposure of the entire population. Such low-level exposures can impair neurobehavioral development in infants and children and can raise blood pressure in adults. Furthermore, absorbed lead is stored in the skeleton and may reenter the circulation at times of heightened bone turnover (e.g., pregnancy, lactation, osteoporosis, hyperthyroidism). Subclinical toxic effects can be prevented if chronic low-level exposure is detected early and curtailed. In the case of lead, such exposure is detected by tests of blood lead level, which should be performed regularly in young children living in old housing and as a precautionary measure in adults with a history of lead exposure.

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6. WOMEN'S HEALTH - Anthony L. Komaroff, Celeste Robb-Nicholson, Andrea E. Dunaif

In recent years, the medical problems and health care of women have received increasing attention. There are poorly understood differences between men and women, both in morbidity and mortality and in the expression of diseases. Many research studies of disease prevention and pathophysiology have included only male subjects; most illnesses that can affect both sexes have not been as well studied in women. It also appears that women receive different care than men for certain common health problems. Finally, an increasing number of women are seeking health care in multidisciplinary women's health units that combine the expertise of gynecology, psychiatry, and internal medicine or family medicine.

MORBIDITY AND MORTALITY IN WOMEN

Morbidity Past studies have found that women experience more days of restricted activity than men at all ages, over and above the restricted activity caused by obstetric and gynecologic conditions. However, a study in 1998 concluded differently. Women make more visits to physicians, particularly for acute self-limited illnesses.

Mortality In the developed nations, women live longer than men. In the United States, as of 1996, the projected average life expectancy from birth is 79.1 years for females, and 73.1 years for males. Although there are more male fetuses conceived than female fetuses, females have a survival advantage when compared to males, in all age groups. The longer life expectancy of women versus men in developed countries is due in large part to the difference in mortality caused by ischemic heart disease (IHD).

As shown in [Table 6-1](#), the leading causes of death among young women in the United States are accidents, homicide, and suicide. During the middle years, breast cancer is a slightly more common cause of death than [IHD](#) and lung cancer. In women between ages 65 and 74, IHD, lung cancer, and cerebrovascular disease supercede breast cancer as the leading causes of death. Among women of all ages, IHD is the leading cause of death by a substantial margin, with a mortality rate five to sixfold higher than the rate for either lung or breast cancer. Nevertheless, polls find that U.S. women believe breast cancer poses the greatest threat to their lives.

Social Factors Influencing Morbidity and Mortality Gender differences in morbidity and mortality may be explained in part by psychosocial factors such as socially-defined gender roles, poverty, participation in the work force, health insurance, and lifestyle.

In the past 30 years in the United States, there has been a "feminization of poverty." One-third of families headed by women currently live in poverty, and the fraction is greater than one-half for African-American and Latino women. Almost a fifth of women over age 65 live below the poverty level. People of lower socioeconomic status experience poorer health and a higher mortality rate than those in higher income groups. The poor are more likely to smoke and less likely to have recommended preventive measures, including cancer screening. Lack of adequate health insurance is a major problem for many women; in general, they are more likely than men to have low-paying, part-time, non-union jobs that do not provide health insurance. Women who

are divorced or widowed may also lose health insurance that they had through their husbands.

PREVENTION (See also [Chap. 10](#))

Primary prevention and screening are crucial elements in improving the health of women. Based upon available literature and the consensus of experts, various authoritative organizations have published guidelines on preventive practices in women.

Most physicians believe that a baseline history and physical examination is useful to set the stage for preventive measures appropriate to each patient. In general, authorities recommend that blood pressure be measured every other year throughout life. Counseling on diet, smoking cessation, exercise, and use of seatbelts are of demonstrated value in the primary prevention of diseases and accidents. Counseling about safe sexual practices, alcohol abuse, and violence are also recommended.

Screening for glaucoma is recommended for African-American women over age 40 and for Caucasian women over age 50. Yearly examinations to test visual acuity are recommended for women over age 70.

Regular screening for breast, cervical, and colorectal cancer is recommended, but how often tests should be performed and which tools to use are still being debated. Most authorities recommend annual clinical breast examination in all women beginning at age 35 to 40. There is strong evidence to support the efficacy of annual mammography in women age 50 to 59. For women age 60 or older, the evidence for screening is less strong. The benefits of screening for women between the ages of 40 and 49 are still being debated.

Most authorities recommend Pap smear screening beginning at age 18 or when a woman becomes sexually active. After two or three consecutive normal Pap smears, most groups recommend Pap smear testing every three years. If Pap smears have been normal for 10 years, they can be discontinued in women after age 65.

Recommendations for colorectal cancer screening vary. For patients over 50, the American Cancer Society recommends yearly fecal occult blood testing and rectal examination combined with flexible sigmoidoscopy every 5 years, colonoscopy every 10 years, or double-contrast barium enema every 5 to 10 years.

Bone mineral testing has gained rapid acceptance as a screening tool for detecting osteoporosis, as well as for predicting the likelihood of the condition in the future. With the advent of multiple preventive and therapeutic strategies for osteoporosis, many authorities now recommend bone mineral testing to screen for the condition. A bone mineral density test is recommended for all women over age 65 as well as for all postmenopausal women who are at increased risk for developing osteoporosis ([Chap. 342](#)).

Cigarette smoking, a major risk factor for cardiovascular diseases and cancers in women, has been well studied ([Chap. 390](#)). Over the past 60 years there has been a sharp decline in smoking among men, but not among women; teenage women smoke at

higher rates than their male counterparts. "Low-yield" cigarettes are marketed heavily to women. The Nurses' Health Study showed that one-third of the excess risk of ischemic heart disease was eliminated two years after smoking cessation, and that all of the excess risk was eliminated by 10 to 14 years after smoking cessation.

The National Cholesterol Education Program recommends that total cholesterol and high-density lipoprotein (HDL) levels be measured once. If both are normal, a repeat test after 5 years is recommended. A meta-analysis of several small studies of women showed an increased risk of [IHD](#) in women with serum cholesterol greater than 265, a ratio of total cholesterol to HDL cholesterol greater than 4, or an elevated fasting triglyceride.

In various case-control and observational studies, postmenopausal estrogen therapy is associated with a 40 to 50% reduction in deaths due to [IHD](#), but its value in a prospective, randomized trial has not yet been documented.

Calcium and estrogen, as well as alendronate and the selective estrogen receptor modulators, tamoxifen and raloxifene, slow the development of osteoporosis and reduce the frequency of hip and vertebral fracture in postmenopausal women. In randomized clinical trials, both tamoxifen and raloxifene have been shown to reduce the risk of breast cancer in postmenopausal women.

Considerable research indicates that a relatively high dietary intake of various antioxidants (including vitamins E and C) is associated with lower rates of vascular disease and malignancies. Randomized trials of supplemental antioxidants are under way. Preliminary research indicates that regular aspirin use is associated with reduced rates of [IHD](#) and colorectal carcinoma.

GENDER DIFFERENCES IN DISEASE

Obviously, some diseases and conditions occur exclusively (or nearly exclusively) in women -- e.g., menopause and various breast and gynecological disorders. These are discussed elsewhere in this book ([Chaps. 52,89,336,337](#)). In this chapter, we seek to highlight some gender differences in diseases that occur in both women and men.

Ischemic Heart Disease (See also [Chap. 244](#)) Many persons think of [IHD](#) as a primary problem for men rather than women, perhaps because men have more than twice the total incidence of cardiovascular morbidity and mortality between the ages of 35 and 84. However, as stated earlier, in the United States IHD is among the leading causes of death among women as well as men ([Table 6-1](#)). The curve for the IHD mortality rate in women lags behind that for men by about a decade. Nevertheless, nearly 250,000 women die annually from IHD; after age 40, one in three women will die from heart disease. Although IHD mortality has been falling in men in the United States over the past 30 years, it has been increasing in women.

Why are [IHD](#) rates lower in women? They have a more favorable risk profile in some respects: higher [HDL](#) cholesterol levels, lower triglyceride levels, and less upper-body obesity than men. But women also have a less favorable risk profile in other respects: more obesity, higher blood pressure, higher plasma cholesterol levels, higher fibrinogen

levels, and more diabetes. The simplest explanation for the sex differential in IHD is the "cardioprotective" effect of estrogen, which can be due to improvement of the lipid profile, a direct vasodilatory effect, and perhaps other factors. HDL cholesterol levels appear to be a particularly important risk factor for IHD in women. HDL levels are higher in all age groups in women compared to men, and are higher in premenopausal and estrogen-treated postmenopausal women. Smoking is the most important risk factor for IHD in women.

[IHD](#) presents differently in men and women. In the Framingham study, angina was the most frequent initial symptom of IHD in females, occurring in 47% of women, whereas myocardial infarction was the most frequent initial symptom in males, occurring in 46% of men. The exercise electrocardiogram has a substantial false positive as well as false negative rate for women, compared to men.

Women, particularly African-American women, have a higher risk of morbidity and mortality than men following a myocardial infarction. Compared to men, women who obtain coronary artery bypass graft surgery have more advanced disease, a higher perioperative mortality rate, less relief of angina, and less graft patency; however, 5- and 10-year survival rates are similar. Women undergoing percutaneous transluminal coronary angioplasty have lower rates of clinical and angiographic success than men, but also a lower rate of restenosis and a better long-term outcome. Women may benefit less and have more frequent serious bleeding complications from thrombolytic therapy than do men. Factors such as older age, more comorbid conditions, and more severe [IHD](#) in women at the time of events or procedures appear to account for at least part of the gender differences observed. Women with IHD benefit at least as much as men, and perhaps more, from reductions in cholesterol level.

The incidence of [IHD](#) increases markedly at menopause, consistent with the hypothesis that estrogens are cardioprotective. A number of observational studies have supported this hypothesis by demonstrating significant decreases in IHD in women on hormone replacement therapy (HRT), both estrogen alone and estrogen-progestin combination therapy. However, the HERS, a recent clinical trial of HRT for the *secondary* prevention of IHD, showed no significant difference in cardiovascular events between therapy with combined continuous conjugated equine estrogen (0.625 mg qd) and that with medroxyprogesterone acetate (2.5 mg qd), compared to placebo over four years. Indeed, in the HRT group, there was about a 50% increase in cardiovascular events in the first year of the trial. The Women's Health Initiative is investigating directly the impact of various HRT modalities as a *primary* prevention of IHD risk. Until further data are available, caution should be exercised in prescribing HRT to women with a history of IHD, or for cardioprotection alone.

Hypertension (See also [Chap. 246](#)) Hypertension is more common in U.S. women than men, largely owing to the high prevalence of hypertension in older age groups and the longer survival rate for women. Both the effectiveness and the adverse effects of various antihypertensive drugs appear to be comparable in women and men. Benefits of treatment for severe hypertension have been dramatic in both women and men. However, in clinical trials of the treatment of mild to moderate hypertension, women have had a smaller decrease in morbidity and mortality than men, perhaps because women have a lower risk of myocardial infarction and stroke than men to begin with.

Older women benefit at least as much as men from treatment, as demonstrated by the Systolic Hypertension in Elderly study. The incidence of hypertension (above 140/90) appears to be low (less than 5%) with the current low-dose oral contraceptives. Postmenopausal estrogen therapy is not associated with increases in blood pressure.

Immunologically Mediated Diseases Several immunologically mediated diseases -- e.g., rheumatoid arthritis, systemic lupus erythematosus, multiple sclerosis, Graves' disease, and thyroiditis -- occur much more frequently in women than in men. In animal models of rheumatoid arthritis -- lupus and multiple sclerosis, for example -- it is the females of the species that are predominantly affected. On the other hand, animal studies indicate that females are less susceptible to infection.

In short, female animals appear to have more vigorous immune responses, with both beneficial and adverse consequences. Increasing evidence indicates that estrogens upregulate both cellular and humoral immunity. Also, some immunocytes contain estrogen, progesterin and androgen receptors, and the uterus produces a variety of cytokines, suggesting a complex interaction between the reproductive and immune systems.

Osteoporosis (See also [Chap. 342](#)) This condition is much more prevalent in postmenopausal women than in men of similar age. Osteoporotic hip fractures are a major cause of morbidity in elderly women. Men accumulate more bone mass and lose bone more slowly than women. Gender differences in bone mass are found as early as infancy. Calcium intake, vitamin D and estrogen all play important roles in osteoporosis; calcium intake is an important determinant of peak bone mass, particularly during adolescence. Vitamin D deficiency is surprisingly common in elderly women. Receptors for estrogens and androgens have been identified in bone. The aromatase enzyme system, which converts androgens to estrogens, is also present in bone.

Therapy with [HRT](#), or with calcium and vitamin D, has been shown to reduce the risk of osteoporotic fractures. Newer modalities, such as bisphosphonates (alendronate), calcitonin, and raloxifene, a selective estrogen receptor modulator, prevent bone loss and reduce the risk of osteoporotic fractures.

Alzheimer's Disease (See also [Chap. 362](#)) Alzheimer's disease (AD) affects approximately twice as many women as men, in part because women live longer. Several observational studies suggest that [HRT](#) may decrease the risk of AD and improve cognitive function in older women. These benefits are seen in both current as well as past HRT users. In a few experimental studies, estrogen replacement has been shown to be associated with improved memory compared to placebo treatment. Estrogens enhance neuronal growth and activity, providing a biologic basis for these putative cognitive effects of HRT. Prospective clinical trials, including the Women's Health Initiative, are underway to pursue these intriguing observations.

Diabetes Mellitus (See also [Chap. 333](#)) Estrogens enhance insulin sensitivity in women but not in men. Despite this, the prevalence of type 2 diabetes mellitus (DM) is higher in women, which is related in part to the higher prevalence of female obesity. Premenopausal women with DM lose the cardioprotective effect of female gender and have identical rates of [IHD](#) to those in males. This is partially explained by the presence

of several IHD risk factors in women with DM: obesity, hypertension and dyslipidemia. Recent evidence suggests that vascular responses differ in women with DM, as compared to normal women. Polycystic ovary syndrome and gestational diabetes mellitus -- common conditions in premenopausal women -- are associated with a significantly increased risk for type 2 DM.

Psychological Disorders (See also [Chap. 385](#)) Depression, anxiety panic disorder and eating disorders (bulimia and anorexia nervosa) occur more often in women than in men. Epidemiologic studies from both developed and developing nations consistently find major depression to be twice as common in women as in men, with the gender disparity becoming evident in early adolescence. Depression occurs in 10% of women during pregnancy and in 10 to 15% of women during the first several months of the postpartum period. The incidence of major depression diminishes after age 45, and does not increase with the onset of menopause. Depression in women also appears to have a worse prognosis than in men; episodes of depression last longer and there is a lower rate of spontaneous remission.

Social factors may account for the greater prevalence of some disorders in women; the traditionally subordinate role of women in society may generate feelings of helplessness and frustration which contribute to psychiatric illness. In addition, it is likely that biological factors, including hormonally influenced neurochemical changes, also play a role. The limbic system and hypothalamus -- areas of the brain thought to subserve appetite, satiety and emotion -- contain estradiol and testosterone receptors.

Alcohol and Drug Abuse (See also [Chap. 387](#)) One-third of Americans who suffer from alcoholism are women. Women alcoholics are less likely to be diagnosed than men; a greater proportion of men than women seek help for alcohol and drug abuse. Men are more likely to go to an alcohol or drug treatment facility, while women tend to approach a primary care physician or mental health professional for help under the guise of a psychosocial problem. Late-life alcoholism is more common in women than men. In 1997, an epidemiologic survey reported that, among women over age 59, an estimated 1.8 million were addicted to or abused alcohol, and over 2.8 million were addicted to or abused psychoactive or mood-altering prescription drugs.

On average, alcoholic women drink less than alcoholic men, but exhibit the same degree of impairment. Blood alcohol levels are higher in women than in men after drinking equivalent amounts of alcohol, adjusted for body weight. This greater bioavailability of alcohol in women is probably due to the higher proportion of body fat and lower total body water. Women also have a lower gastric "first-pass metabolism" of alcohol, associated with lower activity of gastric alcohol dehydrogenase. In addition, alcoholic women are more likely than alcoholic men to abuse tranquilizers, sedatives, and amphetamines. Women alcoholics have a higher mortality rate than do nonalcoholic women and alcoholic men. Compared to men, women also appear to develop alcoholic liver disease and other alcohol-related diseases with shorter drinking histories and lower levels of alcohol consumption. Alcohol abuse also poses special risks to women who are or wish to become pregnant, adversely affecting fertility and the health of the baby (fetal alcohol syndrome).

Finally, there is growing evidence that for several illicit drugs, women proceed more

rapidly to drug dependence than do men.

Human Immunodeficiency Virus Infection (See also [Chap. 309](#)) As of September 1998, the Centers for Disease Control and Prevention estimate that between 120,000 and 160,000 adolescent and adult women in the United States were living with HIV infection, including those with AIDS ([Table 6-1](#)). Between 1985 and 1998, the proportion of all U.S. AIDS cases reported among women more than tripled, from 7 to 23%. HIV infection was the fourth leading cause of death among U.S. women age 25 to 44 in 1997, and the second leading cause of death among African-American women in this age group. The CDC estimates that 30% of the approximately 40,000 new HIV infections in the United States each year are among women.

Between 1996 and 1997 the incidence of new AIDS cases in the United States decreased by 18% and that of AIDS-related deaths by 42%, largely because of advances in HIV therapies. The decline continued between 1997 and 1998, albeit at a slower rate. AIDS incidence and AIDS-related mortality fell by 11 and 20%, respectively. However, AIDS incidence and deaths are not decreasing as rapidly among women as among men. HIV and AIDS continue to affect women in racial/ethnic minorities and lower socioeconomic classes disproportionately. CDC estimates that 64% of new HIV infections in 1998 occurred among African-American women, 18% among Hispanic women, and 18% among white women. Of the new HIV infections among women in the United States in 1998, CDC estimates that 75% of women were infected through heterosexual sex and 25% of women through injection drug use.

Violence Against Women Violence against women in the United States is an enormous problem. Incidents of both rape and domestic violence are vastly underreported. Sexual assault is one of the most common crimes against women. One in five adult women in the United States reports having experienced sexual assault during her lifetime. Adult women are much more likely to be raped by a spouse, ex-spouse, or acquaintance than by a stranger.

Domestic violence is defined in the American Medical Association guidelines as "an ongoing, debilitating experience of physical, psychologic, and/or sexual abuse in the home, associated with increasing isolation from the outside world and limited personal freedom and accessibility to resources." It affects women of all ages, ethnic orientations, and socioeconomic groups. Based upon national crime statistics, every year an estimated 2 million women in the United States are severely injured and more than 1000 are killed by their current or former male partner. Domestic violence is the most common cause of physical injury in women, exceeding the combined incidence of all other types of injury (such as from rape, mugging, and auto accidents). Women who are young, single, pregnant, recently separated or divorced, or who have a history of substance abuse or mental illness, or a partner with substance abuse or mental illness, are at increased risk of domestic violence.

Domestic violence and sexual assault are associated with increased rates of physical and psychologic symptoms, medical office visits, and hospitalizations. Given this indirect presentation of the consequences of violence, and the high prevalence of unreported violence, clinicians should have a low threshold for pursuing the possibility of violence in female patients, particularly those with vague symptoms and psychological disorders.

The immediate treatment of rape and domestic violence focuses on assessing and treating physical injuries; providing emotional support; assessing and dealing with the risks of sexually transmitted infection and pregnancy; evaluating the safety of the patient and other family members; and documenting the patient's history and physical examination findings. In addition to dealing with the medical and psychological issues, appropriate care includes providing information about legal services, shelters and safe houses, hotlines, support groups, and counseling services.

RESEARCH IN WOMEN'S HEALTH

The growing recognition of the importance of women's health has spawned a number of research efforts, including large observational studies and clinical trials. The U.S. National Institutes of Health has introduced guidelines to mandate the inclusion of women in clinical studies, and the reporting of gender-specific data.

Studies of Prevention Large observational studies of men and women, such as the Rancho Bernardo Study and the Framingham Study, designed to analyze data specific to women have been on the increase. The Nurses' Health Study has been following more than 200,000 women, many for more than 20 years, prospectively collecting data to study the impact of smoking, diet, physical activity, medications, prevention and screening behaviors, and some psychosocial factors on the risk of various medical disorders, including breast cancer, [IHD](#), stroke, diabetes, and fracture, as well as causes of mortality.

These studies have set the stage for clinical trials such as the Postmenopausal Estrogens/Progestins Intervention (PEPI) Trial, the first multicenter, randomized, double-blind, placebo-control trial of the effects of three estrogen/progestin regimens on risk factors for cardiovascular disease, bone mineral density, and endometrial hyperplasia. The study found that estrogen, alone or in combination with progestin, increased serum levels of [HDL](#) and decreased low-density lipoprotein (LDL) and fibrinogen levels. While unopposed estrogen (without progestins) resulted in the most beneficial effects on lipids, it was also associated with an increased risk of endometrial hyperplasia.

In 1992, the NIH funded the Women's Health Initiative (WHI), a study of the health of postmenopausal women. The WHI, the largest research study ever funded by the NIH, involves over 160,000 postmenopausal women participating at 45 clinical centers across the United States through the year 2002. The WHI study includes both a prospective observational study and an interventional randomized trial involving over 63,000 women, which is designed to test the effects of a low-fat diet, hormone replacement therapy, and calcium and vitamin D supplementation on the risks for cardiovascular disease, breast cancer, and osteoporotic fractures.

Many other studies currently in progress promise new insights into the health of women within the next decade.

Pharmacologic Studies Historically, women have been underrepresented in drug trials, even though the majority of pharmaceuticals sold in the United States each year

are used by women. However, this has been rapidly changing. The FDA requires information on the safety and effectiveness of experimental drugs in women, on the effects of the menstrual cycle and menopause on a drug's pharmacokinetics, and on a drug's influence on the effectiveness of oral contraceptives. The increased emphasis on entering women into drug trials is likely to yield important information. Studies that have included women indicate that there are clinically significant differences in the way women respond to a number of frequently prescribed pharmaceuticals, including sedative-hypnotics, antidepressants, antipsychotics, anticonvulsants, and b-adrenergic blocking agents. The 1992 FDA Adverse Experience Report found that women have a higher frequency of adverse drug reactions than men. Other studies suggest that the efficacy of many drugs may be different in women compared to men. For example, women require lower doses of neuroleptics to control schizophrenia than men do. Women awaken from anesthesia faster than do men who are given the same doses of anesthetics, and they have a more powerful response to certain classes of analgesics than men. The reasons for these differences are not clear. However, these observations have spurred researchers to consider separating out the effects of gender in future clinical research in an effort to define "gender-based" biologic processes.

CONCLUSION

At the same time that the health of women is undergoing more rigorous study and women's clinics are becoming increasingly common and popular, a growing fraction of health professionals are women. The number of women physicians has increased by 300% between 1970 and 1990, and more than 40% of all U.S. medical students now are women. This infusion of women into the physician work force is likely to lead to a still greater recognition of the unique aspects of health and disease in women.

(Bibliography omitted in Palm version)

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7. MEDICAL DISORDERS DURING PREGNANCY - Robert L. Barbieri, John T. Repke

Approximately 4 million births occur in the United States each year. A significant proportion of these are complicated by one or more medical disorders. Two decades ago, many medical disorders were contraindications to pregnancy. Advances in obstetrics, neonatology, obstetric anesthesiology, and medicine have increased the expectation that pregnancy will result in an excellent outcome for both mother and fetus despite most of these conditions. Successful pregnancy requires important physiologic adaptations, such as a marked increase in cardiac output. Medical problems that interfere with the physiologic adaptations of pregnancy increase the risk for poor pregnancy outcome; conversely, in some instances pregnancy may adversely impact an underlying medical disorder.

HYPERTENSION (See also [Chap. 246](#))

In pregnancy, cardiac output increases by 40%, most of which is due to an increase in stroke volume. Heart rate increases by approximately 10 beats per minute during the third trimester. In the second trimester of pregnancy, systemic vascular resistance decreases and this is associated with a fall in blood pressure. During pregnancy, a blood pressure of 140/90 mmHg is considered to be abnormally elevated and is associated with a marked increase in perinatal morbidity and mortality. In all pregnant women, the measurement of blood pressure should be performed in the sitting position, because for many the lateral recumbent position is associated with a blood pressure lower than that recorded in the sitting position. The diagnosis of hypertension requires the measurement of two elevated blood pressures, at least 6 h apart. Hypertension during pregnancy is usually caused by preeclampsia, chronic hypertension, gestational hypertension, or renal disease.

PREECLAMPSIA

Approximately 5 to 7% of all pregnant women develop *preeclampsia*, the new onset of hypertension (blood pressure > 140/90 mmHg), proteinuria (>300 mg per 24 h), and pathologic edema. Although the precise placental factors that cause preeclampsia are unknown, the end result is vasospasm and endothelial injury in multiple organs. Preeclampsia is associated with abnormalities of cerebral circulatory autoregulation, which increase the risk of stroke at near-normal blood pressures. Risk factors for the development of preeclampsia include nulliparity, diabetes mellitus, a history of renal disease or chronic hypertension, a prior history of preeclampsia, extremes of maternal age (>35 years or <15 years), obesity, factor V Leiden mutation, angiotensinogen gene T235, antiphospholipid antibody syndrome, and multiple gestation.

There are no well-established strategies for the prevention of preeclampsia. Clinical trials have demonstrated that low-dose aspirin treatment does *not* prevent preeclampsia in either low- or high-risk women. Two meta-analyses reported that dietary calcium supplementation appeared to be effective in reducing the risk of developing preeclampsia. Subsequently, however, a large randomized clinical trial in low-risk women did not demonstrate a protective effect of calcium supplementation. Therefore, calcium supplementation may be considered in women at high risk for preeclampsia.

(see above). The observation that dietary intervention may reduce the risk of hypertension in men and nonpregnant women raises the possibility that dietary manipulations will be discovered that reduce the risk of preeclampsia.

Severe preeclampsia is the presence of new-onset hypertension and proteinuria accompanied by central nervous system dysfunction (headaches, blurred vision, seizures, coma), marked elevations of blood pressure ($>160/110$ mmHg), severe proteinuria (>5 g per 24 h), oliguria or renal failure, pulmonary edema, hepatocellular injury (ALT $>2\times$ the upper limits of normal), thrombocytopenia (platelet count $<100,000/\mu\text{L}$), or disseminated intravascular coagulation. Women with *mild preeclampsia* are those with the diagnosis of new-onset hypertension, proteinuria, and edema without evidence of severe preeclampsia. The *HELLP* (hemolysis, elevated liver enzymes, low platelets) syndrome is a special subgroup of severe preeclampsia and is a major cause of morbidity and mortality in this disease. The presence of platelet dysfunction and coagulation disorders further increases the risk of stroke.

TREATMENT

Preeclampsia resolves within a few weeks after delivery. For pregnant women with preeclampsia prior to 37 weeks' gestation, delivery reduces the mother's morbidity but exposes the fetus to the risk of premature delivery. The management of preeclampsia is challenging because it requires the clinician to balance the health of both mother and fetus simultaneously and to make management decisions that afford both the best opportunities for infant survival. In general, prior to term, women with *mild* preeclampsia can be managed conservatively with bed rest, close monitoring of blood pressure and renal function, and careful fetal surveillance. For women with *severe* preeclampsia, delivery is recommended after 32 weeks' gestation. This reduces maternal morbidity and slightly increases the risks associated with prematurity for the newborn. Prior to 32 weeks' gestation, the risks of prematurity for the fetus are great, and some authorities recommend conservative management to allow for continued fetal maturation. Expectant management of severe preeclampsia remote from term affords some benefits for the fetus with significant risks for the mother. Such management should be restricted to tertiary care centers where maternal-fetal medicine, neonatal medicine, and critical care medicine expertise are available.

The definitive treatment of preeclampsia is delivery of the fetus and placenta. For women with severe preeclampsia, aggressive management of blood pressures $>160/110$ mmHg reduces the risk of cerebrovascular accidents.

Intravenous labetalol or hydralazine are the drugs most commonly used to manage preeclampsia. Alternative agents such as calcium channel blockers may be used. Elevated arterial pressure should be reduced slowly to avoid hypotension and a decrease in blood flow to the fetus. *Angiotensin-converting enzyme (ACE) inhibitors as well as angiotensin-receptor blockers should be avoided in the second and third trimesters of pregnancy because of their adverse effects on fetal development.* Pregnant women treated with ACE inhibitors often develop oligohydramnios, which may be caused by decreased fetal renal function.

Magnesium sulfate is the treatment of choice for the prevention and treatment of

eclamptic seizures. Two large randomized clinical trials have demonstrated the superiority of magnesium sulfate over phenytoin and diazepam. Magnesium may prevent seizures by interacting with *N*-methyl-D-aspartate (NMDA) receptors in the central nervous system. Given the difficulty of predicting eclamptic seizures on the basis of disease severity, it is recommended that once the decision to proceed with delivery is made, all patients carrying a diagnosis of preeclampsia be treated with magnesium sulfate (see [Guideline](#)).

CHRONIC ESSENTIAL HYPERTENSION

Pregnancy complicated by chronic essential hypertension is associated with intrauterine growth restriction and increased perinatal mortality. Pregnant women with chronic hypertension are at increased risk for superimposed preeclampsia and abruptio placenta. Women with chronic hypertension should have a thorough prepregnancy evaluation, both to identify remediable causes of hypertension and to ensure that the prescribed antihypertensive agents are not associated with adverse pregnancy outcome (e.g., [ACE](#) inhibitors, angiotensin-receptor blockers). α -Methyldopa and labetalol are the most commonly used medications for the treatment of chronic hypertension in pregnancy. Baseline evaluation of renal function is necessary to help differentiate the effects of chronic hypertension versus superimposed preeclampsia should the hypertension worsen during pregnancy. There are no convincing data that demonstrate that treatment of mild chronic hypertension improves perinatal outcome.

GESTATIONAL HYPERTENSION

This is the development of elevated blood pressure during pregnancy or in the first 24 h post partum in the absence of preexisting chronic hypertension and other signs of preeclampsia. Uncomplicated gestational hypertension that does not progress to preeclampsia has not been associated with adverse pregnancy outcome or adverse long-term prognosis.

RENAL DISEASE (See also [Chap. 268](#))

Normal pregnancy is characterized by an increase in glomerular filtration rate and creatinine clearance. This occurs secondary to a rise in renal plasma flow and increase glomerular filtration pressures. Patients with underlying renal disease and hypertension may expect a worsening of hypertension during pregnancy. If superimposed preeclampsia develops, the additional endothelial injury results in a capillary leak syndrome that may make the management of these patients challenging. In general, patients with underlying renal disease and hypertension benefit from more aggressive management of blood pressure than do those with gestational hypertension. Preconception counseling is also essential for these patients so that accurate risk assessment can occur prior to the establishment of pregnancy and important medication changes and adjustments be made. In general, a prepregnancy serum creatinine level $<133 \mu\text{mol/L}$ ($<1.5 \text{ mg/dL}$) is associated with a favorable prognosis. When renal disease worsens during pregnancy, close collaboration between the nephrologist and the maternal-fetal medicine specialist is essential so that decisions regarding delivery can be weighed in the context of sequelae of prematurity for the neonate versus long-term sequelae for the mother with respect to future renal function.

Successful pregnancy after renal transplantation has been reported increasingly. Predictors for success include a normal-functioning transplanted kidney, absence of rejection for at least 2 years prior to the pregnancy, absence of hypertension, and preferably minimal doses of immunosuppressant medications. Pregnancies in women using cyclosporine are more likely to be complicated by renal insufficiency and/or the development of hypertension. Such patients require very careful maternal and fetal surveillance. Nearly half of these pregnancies deliver preterm, and 20% of neonates are small for their gestational age. Rejection occurs in approximately 10% of pregnancies, and approximately 15% of patients will have deterioration in their renal function that persists after delivery. While pregnancy is generally well tolerated in renal transplant recipients, controversy remains as to whether or not deterioration of graft function is accelerated by pregnancy. More aggressive management of blood pressure has been suggested in this group of patients in an effort to protect the grafted kidney.

Another subset of patients with chronic renal disease and hypertension are those patients whose pregnancies are complicated by systemic lupus erythematosus (SLE) ([Chap. 311](#)). In the past, SLE was considered to be a contraindication to pregnancy. With improved understanding of the effects of SLE on pregnancy, and vice versa, and with improved pharmacologic methods for managing SLE, successful pregnancy outcome is likely. Good prognostic factors for establishment of pregnancy in the presence of SLE are as follows:

1. Disease quiescence > 6 months
2. Normal blood pressure (with or without medication)
3. Normal renal function [creatinine < 133 $\mu\text{mol/L}$ (< 1.5 mg/dL)]
4. Absence of antiphospholipid antibodies
5. Minimal or no need for immunosuppressive drugs
6. Absence of prior adverse reproductive outcome

Previously a point of controversy, there is now increasing consensus that pregnancy and the postpartum period are times of increased lupus activity. In severe flares early in gestation, pregnancy termination is often recommended. If pregnancy termination is not an option, then medical therapy to manage the lupus flare should not be influenced by the pregnancy, provided informed consent for treatment is obtained from the patient. Pulsed glucocorticoid therapy, azathioprine, hydroxychloroquine, and cyclophosphamide have all been used successfully in pregnancy.

CARDIAC DISEASE

VALVULAR HEART DISEASE (See also [Chap. 236](#))

This is the most common cardiac problem complicating pregnancy.

Mitral Stenosis This is the valvular disease most likely to cause death during pregnancy. The pregnancy-induced increase in blood volume and cardiac output can cause pulmonary edema in women with mitral stenosis. Pregnancy associated with long-standing mitral stenosis may result in pulmonary hypertension. Sudden death has been reported when hypovolemia has been allowed to occur in this condition. Careful control of heart rate, especially during labor and delivery, minimizes the impact of tachycardia and reduced ventricular filling times on cardiac function. Pregnant women with mitral stenosis are at increased risk for the development of atrial fibrillation and other tachyarrhythmias. Medical management of severe mitral stenosis and atrial fibrillation with digoxin and beta blockers is recommended. Balloon valvulotomy can be carried out during pregnancy.

Mitral Regurgitation and Aortic Regurgitation These are both generally well tolerated during pregnancy. The pregnancy-induced decrease in systemic vascular resistance reduces the risk of cardiac failure with these conditions. As a rule, mitral valve prolapse does not present problems for the pregnant patient and aortic stenosis, unless very severe, is also well tolerated. In the most severe cases of aortic stenosis, limitation of activity or balloon valvuloplasty may be indicated.

For women with artificial valves contemplating pregnancy, it is important that warfarin be stopped and heparin initiated prior to conception. Warfarin therapy during the first trimester of pregnancy has been associated with fetal chondrodysplasia punctata. In the second and third trimester of pregnancy, warfarin may cause fetal optic atrophy and mental retardation.

CONGENITAL HEART DISEASE (See also [Chap. 234](#))

The presence of a congenital cardiac lesion in the mother increases the risk of congenital cardiac disease in the newborn. Prenatal screening of the fetus for congenital cardiac disease with ultrasound is recommended. Atrial or ventricular septal defect is usually well tolerated during pregnancy in the absence of pulmonary hypertension, provided that the woman's prepregnancy cardiac status is favorable. Use of air filters on intravenous sets during labor and delivery in patients with intracardiac shunts is generally recommended.

OTHER CARDIAC DISORDERS

Supraventricular tachycardia ([Chap. 230](#)) is a common cardiac complication of pregnancy. Treatment is the same as in the nonpregnant patient, and fetal tolerance of medications such as adenosine and calcium channel blockers is acceptable. When necessary, electrocardioversion may be performed and is generally well tolerated by mother and fetus.

Peripartum cardiomyopathy ([Chap. 238](#)) is a rare disorder of pregnancy associated with myocarditis, and its etiology remains unknown. Treatment is directed toward symptomatic relief and improvement of cardiac function. Many patients recover completely; others are left with a progressive dilated cardiomyopathy. Recurrence in a subsequent pregnancy has been reported, and women should be counseled to avoid pregnancy after a diagnosis of peripartum cardiomyopathy.

SPECIFIC HIGH RISK CARDIAC LESIONS

Marfan Syndrome (See also [Chap. 351](#)) This is an autosomal dominant disease, associated with a high risk of maternal morbidity. Approximately 15% of pregnant women with Marfan syndrome develop a major cardiovascular manifestation during pregnancy, with almost all women surviving. An aortic root diameter <40 mm is considered to be associated with a favorable outcome of pregnancy. Prophylactic therapy with beta blockers has been advocated, although large-scale clinical trials in pregnancy have not been performed.

Pulmonary Hypertension (See also [Chap. 260](#)) Maternal mortality in the setting of severe pulmonary hypertension is high, and primary pulmonary hypertension is a contraindication to pregnancy. Termination of pregnancy may be advisable in these circumstances to preserve the life of the mother. In the Eisenmenger syndrome, i.e., the combination of pulmonary hypertension with right-to-left shunting due to congenital abnormalities ([Chap. 234](#)), maternal and fetal death occur frequently. Systemic hypotension may occur after blood loss, prolonged Valsalva maneuver, or regional anesthesia; sudden death secondary to hypotension is a dreaded complication. Management of these patients is challenging, and invasive hemodynamic monitoring during labor and delivery is generally recommended.

In patients with pulmonary hypertension, vaginal delivery is less stressful hemodynamically than Cesarean section, which should be reserved for accepted obstetric indications.

DEEP VEIN THROMBOSIS AND PULMONARY EMBOLISM (See also [Chaps. 248 and 261](#))

A hypercoagulable state is characteristic of pregnancy, and deep vein thrombosis (DVT) is a common complication. Indeed, pulmonary embolism is the most common cause of maternal death in the United States. Activated protein C resistance caused by the factor V Leiden mutation increases the risk for DVT and pulmonary embolism during pregnancy. Approximately 25% of women with DVT during pregnancy carry the factor V Leiden allele. The presence of the factor V Leiden mutation also increases the risk for severe preeclampsia. If the fetus carries a factor V Leiden mutation, the risk of extensive placental infarction is very high. Additional genetic mutations associated with DVT during pregnancy include the prothrombin G20210A mutation (heterozygotes and homozygotes) and the methylenetetrahydrofolate reductase C677T mutation (homozygotes).

TREATMENT

Aggressive diagnosis and management of [DVT](#) and suspected pulmonary embolism optimize the outcome for mother and fetus. In general, all diagnostic and therapeutic modalities afforded the nonpregnant patient should be utilized in pregnancy. Anticoagulant therapy with heparin is indicated in pregnant women with DVT. Warfarin therapy is contraindicated in the first trimester due to its association with fetal chondrodysplasia punctata. In the second and third trimesters, warfarin may cause fetal

optic atrophy and mental retardation. In the initial treatment of DVT, heparin, which does not cross the placenta, may be administered as an intravenous bolus of approximately 100 IU per kilogram of body weight. Continuous heparin infusion is generally initiated at 1000 IU/h and then titrated to achieve a target activated partial thromboplastin time of 50 to 80 s. After initial intravenous anticoagulation, intermittent subcutaneous heparin therapy with 10,000 IU two or three times daily may be employed. When deep venous thromboembolism occurs in the postpartum period, heparin therapy for 7 to 10 days may be followed by warfarin therapy for 3 to 6 months. Warfarin is not contraindicated in breast-feeding women.

Low-molecular-weight heparins are of sufficient size and charge that they do not cross the placenta and may be substituted for unfractionated heparin in the pregnant patient. Recent concerns about low-molecular-weight heparin use and epidural hematoma suggest that caution be used in the anesthetic management of patients who had been receiving low-molecular-weight heparin near the onset of labor.

ENDOCRINE DISORDERS

DIABETES MELLITUS (See also [Chap. 333](#))

In pregnancy, the fetoplacental unit induces major metabolic changes, the purpose of which is to shunt glucose and amino acids to the fetus while the mother uses ketones and triglycerides to fuel her metabolic needs. These metabolic changes are accompanied by maternal insulin resistance, caused in part by placental production of steroids, a growth hormone variant, and placental lactogen. Although pregnancy has been referred to as a state of accelerated starvation, it is better characterized as accelerated ketosis. In pregnancy, after an overnight fast, plasma glucose is lower by 0.8 to 1.1 mmol/L (15 to 20 mg/dL) than in the nonpregnant state. This is due to the use of glucose by the fetus. In early pregnancy, fasting may result in circulating glucose concentrations in the range of 2.2 mmol/L (40 mg/dL) and may be associated with symptoms of hypoglycemia. In contrast to the decrease in maternal glucose concentration, plasma hydroxybutyrate and acetoacetate levels rise to two to four times normal after a fast.

TREATMENT

Pregnancy complicated by diabetes mellitus is associated with higher maternal and perinatal morbidity and mortality rates. Preconception counseling and treatment are important for the diabetic patient contemplating pregnancy. Optimizing preconception glucose control and attention to other dietary needs such as appropriate levels of folate can significantly reduce the risk of congenital fetal malformations. Folate supplementation reduces the incidence of fetal neural tube defects, which occur with greater frequency in fetuses of diabetic mothers. In addition, optimizing glucose control during key periods of organogenesis reduces other congenital anomalies including sacral agenesis, caudal dysplasia, renal agenesis, and ventricular septal defect.

Once pregnancy is established, glucose control should be managed more aggressively than in the nonpregnant state. In addition to dietary changes, this requires more frequent blood glucose monitoring and often involves additional injections of insulin or

conversion to an insulin pump. Fasting blood glucose levels should be maintained at <5.8 mmol/L (<105 mg/dL) with no values exceeding 7.8 mmol/L (140 mg/dL). Commencing in the third trimester, regular surveillance of maternal glucose control as well as assessment of fetal growth (obstetric sonography) and fetoplacental oxygenation (fetal heart rate monitoring or biophysical profile) optimize pregnancy outcome. Pregnant diabetic patients without vascular disease are at greater risk for delivering a macrosomic fetus, and attention to fetal growth via clinical and ultrasound examinations is important. Fetal macrosomia is associated with an increased risk of maternal and fetal birth trauma. Pregnant women with diabetes have an increased risk of developing preeclampsia, and those with vascular disease are at greater risk for developing intrauterine growth restriction, which is associated with an increased risk of fetal and neonatal death. Excellent pregnancy outcomes in patients with diabetic nephropathy and proliferative retinopathy have been reported with aggressive glucose control and intensive maternal and fetal surveillance.

Glycemic control may become more difficult to achieve as pregnancy progresses. Because of delayed pulmonary maturation of the fetuses of diabetic mothers, early delivery should be avoided unless there is biochemical evidence of fetal lung maturity. In general, efforts to control glucose and maintain the pregnancy until the estimated date of delivery result in the best overall outcome for both mother and newborn.

GESTATIONAL DIABETES

All pregnant women should be screened for gestational diabetes unless they are in a low-risk group. Women at low risk for gestational diabetes are those <25 years of age; those with a body mass index < 25 kg/m², no maternal history of macrosomia or gestational diabetes, and no diabetes in a first-degree relative; and those not members of a high-risk ethnic group (African American, Hispanic, Native American). A typical two-step strategy for establishing the diagnosis of gestational diabetes involves administration of a 50-g oral glucose challenge with a single serum glucose measurement at 60 min. If the serum glucose is < 7.8 mmol/L (<140 mg/dL), the test is considered normal. Serum glucose > 7.8 mmol/L (>140 mg/dL) warrants administration of a 100-g oral glucose challenge with serum glucose measurements obtained in the fasting state, and at 1, 2, and 3 h. Normal values are serum glucose concentrations <5.8 mmol/L (<105 mg/dL), 10.5 mmol/L (190 mg/dL), 9.1 mmol/L (165 mg/dL), and 8.0 mmol/L (145 mg/dL), respectively.

Pregnant women with gestational diabetes are at increased risk of preeclampsia, delivering infants who are large for their gestational age, and birth lacerations. Their fetuses are at risk of hypoglycemia and birth trauma (brachial plexus) injury.

TREATMENT

Gestational diabetes is first treated with dietary measures. Inability to maintain fasting glucose concentrations <5.8 mmol/L (<105 mg/dL) or 2-h postprandial glucose concentrations <6.7 mmol/L (<120 mg/dL) should prompt initiation of insulin therapy. Oral agents should not be used to treat diabetes in pregnancy. Patients with a diagnosis of gestational diabetes will benefit from postpartum follow-up as they are at increased risk for developing type 2 diabetes.

THYROID DISEASE (See also [Chap. 330](#))

In pregnancy, the estrogen-induced increase in thyroxine-binding globulin causes an increase in circulating levels of total T₃ and total T₄. The normal range of circulating levels of free T₄, free T₃, and thyroid stimulating hormone (TSH) remain unaltered by pregnancy.

The thyroid gland normally enlarges during pregnancy. Maternal hyperthyroidism occurs at a rate of approximately 2 per 1000 pregnancies and is generally well tolerated by pregnant women. Clinical signs and symptoms should alert the physician to the occurrence of this disease. Many of the physiologic adaptations to pregnancy may mimic subtle signs of hyperthyroidism. Although pregnant women are able to tolerate mild hyperthyroidism without adverse sequelae, more severe hyperthyroidism can cause spontaneous abortion or premature labor, and thyroid storm is associated with a significant risk of maternal mortality.

TREATMENT

Hyperthyroidism in pregnancy should be aggressively evaluated and treated. The treatment of choice is propylthiouracil. Because it crosses the placenta, the minimum effective dose should be used to maintain free T₄ in the upper normal range. Methimazole crosses the placenta to a greater degree than propylthiouracil and has been associated with fetal aplasia cutis. Radioiodine should not be used during pregnancy, either for scanning or treatment, because of effects on the fetal thyroid. In emergent circumstances, additional treatment with beta blockers and a saturated solution of potassium iodide may be necessary. Hyperthyroidism is most difficult to control in the first trimester of pregnancy and easiest to control in the third trimester.

The goal of therapy for *hypothyroidism* is to maintain the serum [TSH](#) in the normal range, and thyroxine is the drug of choice. Children born to women with an elevated serum TSH (and a normal total thyroxine) during pregnancy have impaired performance on neuropsychologic tests. During pregnancy, the dose of thyroxine required to keep the TSH in the normal range rises. In one study, the mean replacement dose of thyroxine required to maintain the TSH in the normal range was 0.1 mg daily before pregnancy, and it increased to 0.15 mg daily during pregnancy.

DISORDERS OF CALCIUM METABOLISM (See also [Chap. 340](#))

Serum *total* calcium concentration decreases throughout gestation due to a reduction in serum albumin concentration, while serum *ionized* calcium remains unchanged during pregnancy. Circulating parathyroid hormone concentration is slightly reduced throughout the course of pregnancy. Pregnancy has been described as a state of physiologic absorptive hypercalciuria. Estrogen and increased production of 1,25-dihydroxyvitamin D by both the kidney and the placenta mediate the increased absorption of calcium during pregnancy. Due to the fetal requirements for calcium, the National Institutes of Health has recommended that pregnant women receive 1500 mg/d of elemental calcium, slightly higher than the recommended daily intake of 1200 mg/d for nonpregnant adults.

HEMATOLOGIC DISORDERS

Pregnancy has been described as a state of physiologic anemia. Part of the reduction in hemoglobin concentration is dilutional, but iron and folate deficiencies are the major causes of correctable anemia during pregnancy. Folic acid food supplementation implemented in 1998 has reduced the risk of fetal neural tube defects.

In populations at high risk for hemoglobinopathies ([Chap. 106](#)), hemoglobin electrophoresis should be performed as part of the prenatal screen. Hemoglobinopathies can be associated with increased maternal and fetal morbidity and mortality. Management is tailored to the specific hemoglobinopathy and is generally the same for both pregnant and nonpregnant women. Prenatal diagnosis of hemoglobinopathies in the fetus is readily available and should be discussed with prospective parents either prior to or early in pregnancy.

Thrombocytopenia occurs commonly during pregnancy. The majority of cases are benign gestational thrombocytopenias, but the differential diagnosis should include immune thrombocytopenia ([Chap. 116](#)) and preeclampsia. Maternal thrombocytopenia may also be caused by catastrophic obstetric events such as retention of a dead fetus, sepsis, abruption placenta, and amniotic fluid embolism.

NEOPLASTIC DISEASES

Maternal neoplasms are rarely, if ever, transmitted to the fetus. The three most common cancers in pregnant women are cervical cancer (~1 case per 1000 pregnancies, depending on the country), breast cancer (~2 cases per 10,000 pregnancies), and lymphomas (Hodgkin's disease or non-Hodgkin's lymphomas). Cervical cancer may be missed when its early sign, vaginal bleeding, is attributed to the pregnancy. Pregnant women with vaginal bleeding should be examined, and suspicious cervical lesions biopsied. Conization is generally performed only after the first trimester because of the abortion risk.

Breast lumps may also be attributed to change associated with pregnancy. However, women with a dominant mass should undergo diagnostic evaluation (mammogram, ultrasound, biopsy). Resection of the primary lesion is safe, but radiation therapy is unsafe at any time during pregnancy. The fetus cannot be shielded from internal scattering of radiation; therapeutic doses are associated with spontaneous abortion, increased perinatal death, and defects in central nervous system and/or cognitive function. Tamoxifen is not safe for pregnant women.

Lymphoma is usually diagnosed on the basis of adenopathy or constitutional symptoms (fever, sweats, or weight loss). Staging evaluation is not undertaken during the first trimester; women in the first trimester should be counseled about termination of the pregnancy. Single-agent chemotherapy can be used in the second or third trimester as a temporizing measure. Vinblastine or doxorubicin have been used most commonly. Early induction of labor may permit the physician to maximize the survival chances of both the fetus and the mother. Survival rates for 28-week-old fetuses are about 75% and about 90% for 32-week-old fetuses.

Cancer survivors of reproductive age may desire children. Pregnancy may increase the risk of melanoma recurrence but does not influence breast cancer recurrence. Cancer treatment may deplete oocytes. Oocyte retrieval and storage of fertilized or nonfertilized eggs before cancer treatment may permit conception after the cancer has been treated successfully.

GASTROINTESTINAL AND LIVER DISEASE

Up to 90% of pregnant women experience nausea and vomiting during the first trimester of pregnancy. Occasionally, hyperemesis gravidarum requires hospitalization to prevent dehydration, and sometimes parenteral nutrition is required.

Crohn's disease may be associated with exacerbations in the second and third trimesters. Ulcerative colitis is associated with disease exacerbations in the first trimester and during the early postpartum period. Medical management of these diseases during pregnancy is identical to the management in the nonpregnant state ([Chap. 287](#)).

Exacerbation of gall bladder disease is commonly observed during pregnancy. In part this may be due to pregnancy-induced alteration in the metabolism of bile and fatty acids. Intrahepatic cholestasis of pregnancy is generally a third-trimester event. Profound pruritus may accompany this condition and may be associated with increased fetal mortality. It has been suggested that placental bile salt deposition may contribute to progressive uteroplacental insufficiency. Therefore, regular fetal surveillance should be undertaken once the diagnosis of intrahepatic cholestasis is made. Favorable results with ursodiol have been reported.

Acute fatty liver is a rare complication of pregnancy. Frequently confused with the [HELLP](#) syndrome (see "Preeclampsia," above) and severe preeclampsia, the diagnosis of acute fatty liver of pregnancy may be facilitated by imaging studies and laboratory evaluation. Acute fatty liver of pregnancy is generally characterized by markedly increased levels of bilirubin and ammonia and by hypoglycemia. Management of acute fatty liver of pregnancy is supportive; recurrence in subsequent pregnancies has been reported.

All pregnant women should be screened for hepatitis B. This information is important for pediatricians after delivery of the infant. All infants receive hepatitis B vaccine. Infants born to mothers who are carriers of hepatitis B surface antigen should also receive hepatitis B immune globulin as soon after birth as possible and preferably within the first 72 h.

INFECTIONS

BACTERIAL INFECTIONS

Other than bacterial vaginosis, the most common bacterial infections during pregnancy involve the urinary tract ([Chap. 280](#)). Many pregnant women have asymptomatic bacteriuria, most likely due to stasis caused by progestational effects on ureteral and

bladder smooth muscle and to compression effects of the enlarging uterus. In itself, this condition is not associated with an adverse outcome of pregnancy. However, if asymptomatic bacteriuria is left untreated, symptomatic pyelonephritis may occur. Indeed, approximately 75% of cases of pregnancy-associated pyelonephritis are the result of untreated asymptomatic bacteriuria. All pregnant women should be screened with a urine culture for asymptomatic bacteriuria at the first prenatal visit. Subsequent screening with nitrite/leukocyte esterase strips is indicated for high-risk women, such as those with sickle cell trait or a history of urinary tract infections. All women with positive screens should be treated.

Because of the association between bacterial vaginosis and preterm delivery, screening for bacterial vaginosis has been used in an effort to reduce risk. However, standard treatment for bacterial vaginosis does not reduce the risk of preterm delivery.

Abdominal pain and fever during pregnancy create a clinical dilemma. The diagnosis of greatest concern is intrauterine amniotic infection. While amniotic infection most commonly follows rupture of the membranes, this is not always the case. In general, antibiotic therapy is not recommended as a temporizing measure in these circumstances. If intrauterine infection is suspected, induced delivery with concomitant antibiotic therapy is generally indicated. Intrauterine amniotic infection is most often caused by pathogens such as *Escherichia coli* and group B streptococcus. In high-risk patients at term or in preterm patients, routine intrapartum prophylaxis of group B streptococcal disease is recommended. Penicillin G and ampicillin are the drugs of choice. In penicillin-allergic patients, clindamycin is recommended.

Postpartum infection is a significant cause of maternal morbidity and mortality. While rare after vaginal delivery, postpartum endomyometritis develops in 5% of patients having elective repeat cesarean section and in 25% of patients after emergency cesarean section following prolonged labor. Prophylactic antibiotics should be given to all patients undergoing cesarean section. As most cases of postpartum endomyometritis are polymicrobial, broad-spectrum antibiotic coverage with a penicillin, aminoglycoside, and metronidazole is recommended ([Chap. 167](#)). Most cases resolve within 72 h. Women who do not respond to antibiotic treatment for postpartum endomyometritis should be evaluated for septic pelvic thrombophlebitis. Imaging studies may be helpful in establishing the diagnosis, which is primarily a clinical diagnosis of exclusion. Patients with septic pelvic thrombophlebitis generally have tachycardia out of proportion to their fever and respond rapidly to intravenous administration of heparin.

All patients are screened prenatally for gonorrhea and chlamydial infections, and the detection of either should result in prompt treatment. Ceftriaxone and azithromycin are the agents of choice ([Chaps. 147](#) and [179](#)).

VIRAL INFECTIONS

Cytomegalovirus Infection Viral infection in pregnancy presents a significant challenge. The most common cause of congenital viral infection in the United States is cytomegalovirus (CMV) ([Chap. 185](#)). As many as 50 to 90% of women of childbearing age have antibodies to CMV, but only rarely does CMV reactivation result in neonatal infection. More commonly, primary CMV infection during pregnancy creates a risk of

congenital CMV. No currently accepted treatment of CMV during pregnancy has been demonstrated to protect the fetus effectively. Moreover, it is impossible to predict which fetus will sustain life-threatening CMV infection. Severe CMV disease in the newborn is characterized most often by petechiae, hepatosplenomegaly, and jaundice. Chorioretinitis, microcephaly, intracranial calcifications, hepatitis, hemolytic anemia, and purpura may also develop. Central nervous system involvement resulting in the development of psychomotor, ocular, auditory, and dental abnormalities over time have been described.

Rubella (See also [Chap. 195](#)) Rubella virus is a known teratogen; first-trimester rubella carries a high risk of fetal anomalies, though the risk decreases significantly later in pregnancy. Congenital rubella may be diagnosed by percutaneous umbilical blood sampling with the detection of IgM antibodies in fetal blood. All pregnant women should be screened for their immune status to rubella. Indeed, all women of childbearing age, regardless of pregnancy status, should have their immune status for rubella verified and be immunized if necessary. The incidence of congenital rubella in the United States is extremely low.

Herpesvirus (See also [Chap. 182](#)) The acquisition of genital herpes during pregnancy is associated with spontaneous abortion, prematurity, and congenital and neonatal herpes. A recent cohort study of pregnant women without evidence of previous herpes infection demonstrated that approximately 2% of the women acquired a new herpes infection during the pregnancy. Approximately 60% of the newly infected women had no clinical symptoms. Infection occurred equally in all three trimesters. If herpes seroconversion occurred early in pregnancy, the risk of transmission to the newborn was very low. In women who acquired genital herpes shortly before delivery, the risk of transmission was high. The risk of active genital herpes lesions at term can be reduced by prescribing acyclovir for the last 4 weeks of pregnancy to women who have had their first episode of genital herpes during the pregnancy. However, whether or not this strategy results in less viral shedding or enhanced fetal protection at delivery remains to be determined.

Herpesvirus infection in the newborn can be devastating. Disseminated neonatal herpes carries with it high mortality and morbidity rates from central nervous system involvement. It is recommended that pregnant women with active genital herpes lesions at the time of presentation in labor be delivered by cesarean section.

Parvovirus (See also [Chap. 187](#)) Parvovirus infection (human parvovirus B19) may occur during pregnancy. It rarely causes sequelae, but susceptible women infected during pregnancy may be at risk for fetal hydrops secondary to erythroid aplasia and profound anemia.

Toxoplasmosis (See also [Chap. 217](#)) In the United States, approximately 70% of women of childbearing age are susceptible to *Toxoplasma*. Most primary infections of toxoplasmosis in the United States come from eating undercooked meat. The diagnosis of congenital toxoplasmosis is possible through sampling of fetal umbilical blood. If there is no evidence of placental/fetal infection, single-drug treatment with spiramycin is recommended. Triple-drug therapy with spiramycin, pyrimethamine, and sulfa is recommended if there is evidence of fetal infection and the woman does not wish to

terminate the pregnancy or cannot terminate it because of advanced gestational age. Prenatal treatment has been shown to reduce the number of infants with severe infection.

Human Immunodeficiency Virus (See also [Chap. 309](#)) The predominant cause of HIV infection in children is transmission of the virus from the mother to the newborn during the perinatal period. Exposures, which increase the risk of mother-to-child transmission, include vaginal delivery, preterm delivery, trauma to the fetal skin, and maternal bleeding. Additionally, recent infection with high maternal viral load, low maternal CD4+T cell count, prolonged labor, prolonged length of membrane rupture, and the presence of other genital tract infections, such as syphilis or herpes, increase the risk of transmission. Breast feeding may also transmit HIV to the newborn and is therefore contraindicated in most developed countries for HIV-infected mothers. There is no clear evidence to suggest that the course of HIV disease is altered by pregnancy. There is also no clear evidence to suggest that uncomplicated HIV disease adversely impacts pregnancy other than by its inherent infection risk.

TREATMENT

The majority of cases of mother-to-child (vertical) transmission of HIV-1 occur during the intrapartum period. Mechanisms of vertical transmission include infection after rupture of the membranes and direct contact of the fetus with infected secretions or blood from the maternal genital tract. In women with HIV infection who are not receiving antiretroviral therapy, the rate of vertical transmission is approximately 25%. Cesarean section and treatment with zidovudine, administered both before and during delivery, decrease the rate of vertical transmission. In a meta-analysis, zidovudine treatment of both the mother during the prenatal and intrapartum periods and of the neonate at birth reduced the risk of vertical transmission to 7.3%. The combination of elective cesarean section plus zidovudine treatment reduced the risk of vertical transmission to 2%. The role of multiple drug therapy during pregnancy has not yet been established, pending safety data for the neonate.

SUMMARY

Maternal mortality has decreased steadily during the past 60 years. The maternal death rate has decreased from nearly 600/100,000 live births in 1935 to 8.5/100,000 live births in 1996. The most common causes of maternal death in the United States today are, in decreasing order of frequency, thromboembolic disease, hypertension, ectopic pregnancy, and hemorrhage. With improved diagnostic and therapeutic modalities as well as with advances in the treatment of infertility, more patients with medical complications will be seeking, and be in need of, complex obstetric care. Improving outcome of pregnancy in these women will be best obtained by assembling a team of internists and specialists in maternal-fetal medicine (high-risk obstetrics) to counsel these patients about the risks of pregnancy and to plan their treatment prior to conception. The importance of preconception counseling cannot be overstated. It is the responsibility of all physicians caring for women in the reproductive age group to assess their patient's reproductive plans as part of their overall health evaluation.

(Bibliography omitted in Palm version)

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8. ADOLESCENT HEALTH PROBLEMS - Mehul T. Dattani, Charles G. D. Brook

Adolescence marks the transition from childhood to adulthood. It is a time of dramatic physical and psychological change. Adolescents are particularly prone to risk-taking behaviors. In the United States, 73% of deaths among adolescents and young adults result from motor vehicle and other accidents, homicide, and suicide. As a result of sexual maturation, adolescents begin to experiment sexually and, consequently, are susceptible to sexually transmitted diseases (STDs) and unwanted pregnancies. For adolescents with underlying disease, the psychological consequences can be as important as the physical disabilities; denial or resentment of disease is common and can hamper treatment. Adolescence is also a time when many lifelong health-relevant behaviors are established, including dietary habits, exercise patterns, tobacco and alcohol use, and interactions with the health care system. The physician, working together with parents, can help to guide adolescents through this dynamic period of life.

PUBERTY

Puberty encompasses (1) the adolescent growth spurt, (2) development of secondary sexual characteristics, (3) attainment of fertility, and (4) establishment of individual sexual identity.

There is wide variation in the timing of puberty. Signs of puberty are first evident between 9 and 14 years of age (mean 11.5) in 95% of American boys ([Fig. 8-1A](#)). In girls, puberty begins earlier, with 95% of American girls entering puberty between 8 and 12 years of age (mean 10.5) ([Fig. 8-1B](#)). The age of menarche in girls from developed countries has decreased by approximately 2 to 3 months per decade over the past 100 to 150 years. This trend is likely the result of improvements in socioeconomic conditions, nutritional status, and general health and well-being. Currently in the United States, the average age of menarche is 12.8 years. Genetic factors also influence the course of puberty. Data from twin studies indicate that the average age of menarche is more similar in identical twin sisters than in nonidentical twins. Secondary sexual development occurs earlier in girls of Asian and African-Caribbean heritage than in girls of European heritage. Recognition of the progressively earlier onset of puberty, the ethnic variations, and wide age distribution in the timing of puberty is important for identifying precocious or delayed puberty and for counseling adolescents and parents about the natural course of physiologic changes.

HORMONAL CHANGES

Puberty is accompanied by dramatic changes in multiple hormonal systems including alterations in adrenal steroid production, maturation of the reproductive axis, and increased production and action of growth hormone (GH). Serum GH levels increase early in puberty as a consequence of the rise in gonadal steroids. GH in turn increases the level of insulin-like growth factor 1 (IGF-1), which enhances linear bone growth ([Chap. 328](#)). The prolonged pubertal exposure to gonadal steroids ultimately causes epiphyseal closure and limits further bone growth. Appetite increases in association with the growth spurt, and sleep patterns change with a tendency to stay up later and a desire to sleep later into the morning.

The development of secondary sexual characteristics is initiated by *adrenarche*, which usually occurs between 6 and 8 years of age and marks the time when the adrenal gland begins to produce greater amounts of androgens. Despite the search for hormonal mediators, the mechanism that controls adrenarche is unknown. It may well result from adrenal cell differentiation, characterized by the growth of the innermost zone of the adrenal cortex, the zona reticularis, which is the principal site of dehydroepiandrosterone (DHEA) production. The increase in adrenal androgen (DHEA, androstenedione) secretion precedes activation of the reproductive axis. Nonetheless, adrenarche and gonadarche are independent events: children with Addison's disease enter puberty at the normal time, and children with premature adrenarche achieve gonadarche at the expected age.

The sexual maturation process is greatly accelerated by the activation of the hypothalamic-pituitary axis, leading to gonadal stimulation and the production of sex steroids. The hypothalamic-pituitary-gonadal axis is controlled predominantly by gonadotropin-releasing hormone (GnRH), a decapeptide produced by the arcuate nucleus of the mediobasal hypothalamus. GnRH is released in a pulsatile fashion, leading in turn to the pulsatile secretion of the pituitary gonadotropins, luteinizing hormone (LH) and follicle-stimulating hormone (FSH). The so-called GnRH pulse-generator in the hypothalamus is active during fetal life and early infancy but is then quiescent during early childhood. In the early stages of puberty, the sensitivity to steroid inhibition is gradually lost, causing reactivation of GnRH secretion. Leptin, a hormone produced by adipose cells, may play a permissive role in this process, as leptin-deficient individuals fail to enter puberty. Early puberty is characterized by nocturnal surges of LH and FSH. As the reproductive axis matures, the characteristic patterns of feedback regulation are acquired. In males, testosterone inhibits hypothalamic GnRH and pituitary gonadotropin production ([Chap. 335](#)); in females, estrogen and progesterone feed back to generate the characteristic hormonal patterns of the menstrual cycle ([Chap. 336](#)).

FEMALE PUBERTY

The first sign of ovarian estradiol secretion is breast development, or *thelarche*. Pubic and axillary hair growth and the onset of apocrine sweat production result from adrenal androgen secretion, though they may be facilitated by estrogen. The progression of puberty is classified according to Tanner stages for breast and pubic hair development: stage 1 represents the preadolescent appearance, and stage 5 represents the adult appearance ([Fig. 8-1A](#)). Each of these aspects of puberty should be staged separately, as they are controlled by different underlying endocrine mechanisms. Concurrent with these outward signs of puberty are the changes in the size and shape of the uterus. The pubertal growth spurt is dependent on estradiol secretion, which leads to increased [GH](#) secretion. This in turn results in a doubling of the growth rate, and peak height velocity is usually coincident with breast stage 3. After menarche, a girl usually grows only an additional 5 cm.

MALE PUBERTY

In boys, growth of the testes is usually the first sign of puberty, reflecting the effects of pulsatile gonadotropin secretion on seminiferous tubule volume and, to some degree,

Leydig cell mass. Testosterone is converted to dihydrotestosterone by 5- α reductase. Both hormones act via the androgen receptor to induce growth of the external genitalia and pubic hair ([Fig. 8-1; Chap. 338](#)). The growth spurt in boys occurs at a testicular volume of about 10 to 12 mL (as measured by a Prader orchidometer). Testosterone also deepens the voice and increases muscle growth. Dihydrotestosterone stimulates prostate growth and beard growth and initiates recession of the temporal hairline. Although boys enter puberty approximately 6 to 12 months later than girls, they are potentially fertile at an earlier stage of puberty. Aromatization of testosterone to estradiol increases [GH](#) secretion, which acts synergistically with testosterone to induce a greater peak height velocity in boys than in girls.

DISORDERS OF PUBERTY

Precocious puberty is usually defined as an early onset of puberty in boys younger than 9 years or in girls younger than 8 years of age. Some authorities suggest that the lower limits of normal in girls be revised downward to age 7 for Caucasians and age 6 for African-American girls. *Premature thelarche* refers to breast development in the absence of other signs of puberty. It occurs most commonly in girls between infancy and 3 years of age and usually resolves spontaneously. Causes of precocious puberty are divided into central *gonadotropin-dependent* forms and peripheral *gonadotropin-independent* forms ([Table 8-1](#)). Central precocious puberty is much more common in girls than in boys, and the majority of these cases involve idiopathic activation of spontaneous [GnRH](#) pulses. It can also be caused by a variety of central nervous system tumors, structural lesions, and inflammatory conditions.

Delayed puberty is defined as the 3% of girls and boys who have not developed the first signs of puberty by 13.2 and 14.2 years, respectively. It is most commonly due to delayed activation of the hypothalamic-pituitary-gonadal axis ([Table 8-1](#)). Most individuals who meet this definition will progress through puberty normally, but at a later age. Short stature and delayed skeletal maturation are commonly seen in association with delayed puberty. Growth delay may have been evident earlier in childhood; the diagnosis of *constitutional delay of growth and puberty* can be suspected from a delayed bone age in a short child who is otherwise well. Individuals who experience delays in puberty may be emotionally as well as physically immature relative to their peers.

The main diagnostic challenge in delayed puberty is to distinguish those with constitutional delay, who will progress through puberty at a later age, from those with an underlying pathologic process. [LH](#) and [FSH](#) responses to [GnRH](#) do not differentiate constitutional delay from pathologic causes of *hypogonadotropic hypogonadism* ([Chap. 335](#)). Thus, constitutional delay is a diagnosis of exclusion and requires ongoing evaluation during development to assure that normal growth and development occur at a later time. Reassurance without hormonal treatment is appropriate for most individuals with presumed constitutional delay of puberty. Alternatively, an anabolic steroid (e.g., 50 to 100 mg per month testosterone enanthate, intramuscularly) in boys or estrogen (5 to 10 mg/d ethinyl estradiol, orally) in girls may be useful to induce growth and secondary sexual characteristics appropriate for age. Low-dose oral oxandrolone (2.5 mg/d), an anabolic steroid that is not aromatized to estrogen, is also used for boys because it does not accelerate skeletal maturation when used for short periods. After treatment for a year or more, hormonal treatments can be stopped and the function of the

reproductive axis can be reassessed.

PSYCHOLOGICAL CHANGES AND SOCIAL FACTORS

The adolescent years are characterized by a multitude of psychological changes, including (1) the development of abstract thinking, (2) greater independence from family, (3) the formation of a personal and sexual identity, (4) the establishment of a system of values, and (5) an increase in socialization. For most adolescents, these transitions occur relatively smoothly. For others, however, these years can be frustrating and tumultuous; parents and clinicians must be attuned to the needs of those who show signs of struggling with emotional, sexual, and social issues.

Young people tend to share their feelings openly, one of which is ambivalence. These contradictory feelings most often involve both a desire for greater autonomy and, at the same time, a need to cling to the emotional and physical security provided by the family. Adolescents are granted increasing responsibilities but still lack some of the social and legal privileges of adults. This feature of adolescence can lead to conflict and challenges to parental authority.

Adolescents have a strong desire to establish an identity that is increasingly independent of the family. This new identity is strongly influenced by peer groups, some of which are institutionalized (e.g., team sports). Role confusion is quite common in adolescence, and some young people move from one intense allegiance to another with alarming speed. These transitional arrangements are eventually replaced by more permanent attachments to individuals.

In an attempt to alleviate some of the transitions associated with adolescence, many cultures have traditionally used "rites of passage" to acknowledge and accelerate an adolescent's evolution to adulthood. Among Native American Great Plains cultures, for example, a boy was sent away from the village at the time of puberty to fast and receive a vision from a spirit; upon returning to the community, he took his place among the adult men. Similarly, it was traditional in many societies for girls to be secluded at the time of the first menstruation before returning a "full-grown woman." These rites provide a public recognition of the end of childhood, and the ritual leaves the young person with the conviction that he or she has undergone a personal transformation. A relative lack of these coming-of-age rituals in western cultures may contribute to the sense of alienation experienced by some adolescents in this part of the world.

During adolescence, gender identity must be renegotiated. Though prepubertal children have a relatively secure view of themselves as either a boy or a girl, experimentation with gender roles is a common feature of adolescence. For example, adolescents may explore, at least in fantasy, alternative gender roles (e.g., cross-dressing), homosexuality, or relationships with older men or women.

The hormonal changes of puberty influence behavior as well as causing physical changes. Rising levels of testosterone in boys and the increase in adrenal and ovarian androgens in girls increase libido. The mean age of sexual intercourse varies widely within and among cultures, but ranges between ages 15 and 18 for most groups. Boys generally report sexual intercourse about 1 year earlier than girls.

ADOLESCENT VIOLENCE

Adolescents and young adults are subject to much greater rates of violence, both as victims and perpetrators. Males are involved in violence much more commonly than females and account for >90% of homicides involving those 10 to 17 years of age. Ethnic and racial differences in rates of adolescent violence have been noted consistently. African Americans, Hispanics, and Native Americans are much more likely to be victims and perpetrators of lethal violence than are people of Asian or European ancestry. The origins of different rates of violence are complex. Higher rates of lethal aggression are associated with low socioeconomic status, high housing density, increased population turnover in neighborhoods, single-parent households, and socially disorganized communities. In many cases, these factors interact; increased violence leads to high population turnover and social disorganization.

Gangs represent a potentially volatile environment that is characterized by power struggles, initiation and detachment rituals, battles over territory, and escalating violence associated with retaliation. The increase in lethal violence has been attributed in part to easier access to firearms and a greater willingness to use firearms. A Centers for Disease Control and Prevention study in 1995 found that about one-fourth of students had carried a weapon to school during the preceding month and 8 to 10% had carried a gun. Many adolescents lack the abstract reasoning skills required to understand social mores and the consequence of gun use. Though firearms do not cause violence, handguns in particular provide a facile means to a lethal outcome; widespread reduction in access to handguns is essential to curb the current trend in adolescent homicide and serious injury.

Aggressive behavior can often be recognized in early childhood; bullying is a precursor to later antisocial behavior. Child abuse, antisocial parents, inadequate child-rearing practices, and dysfunctional interpersonal interactions between parents or among siblings are associated with aggressive behavior. The physician, along with teachers, clergy, and others in positions of authority, should be alert to a pattern of aggressive behavior or problems in the home. Though these issues are not easily remedied, appropriate interventions to improve family functioning and parenting may interrupt a pattern of violence, which is all too often perpetuated by the adolescent.

HEALTH PROBLEMS

Adolescence is generally a healthy period and is often accompanied by a feeling of immortality, which leads to risk-taking. When diseases of childhood or the consequences of their treatment extend into adolescence, or when disease strikes during adolescence, the sense of unfairness may be overwhelming. Anger and denial can lead to poor compliance with therapeutic regimens.

Relatively few diseases are unique to adolescents. Rather, diseases of childhood, including many inherited disorders and infectious diseases, extend into the adolescent period. Similarly, many of the disorders that affect teenagers are also seen in the adult population. The presentation and management of asthma, for example, is similar in adolescents and adults. Some of the diseases with relatively increased prevalence

during adolescence are summarized in [Table 8-2](#). These diseases should be borne in mind when considering the differential diagnosis. For example, when an adolescent presents with exertional chest pain, dyspnea, and syncope, hypertrophic cardiomyopathy or congenital heart disease should be considered as likely diagnoses, whereas coronary artery disease would be more likely in an adult.

SEXUALLY TRANSMITTED DISEASES

Sexually active adolescents are at greater risk of acquiring [STDs](#) than their adult counterparts ([Chap. 132](#)). Prevention of STDs in adolescence depends on adequate sexual education coupled with access to appropriate clinical services. Early age of first sexual intercourse is associated with (1) an increased number of lifetime sexual partners; (2) an increased risk of acquiring chronic STDs, such as herpes simplex, HIV, and hepatitis B; and (3) cervical cancer in women. In addition, pelvic inflammatory disease in adolescent females increases the likelihood of future ectopic pregnancy, tubal infertility, and chronic pelvic inflammation. A low rate of barrier contraceptive use, combined with ignorance about the acquisition and prevention of infectious diseases, also contributes to the increased risk of STDs among adolescents. Screening for STDs is recommended in sexually active teens ([Table 8-3](#)). Adolescents with sexually transmitted infections, particularly those who deny sexual activity, may be victims of sexual abuse.

CHILD SEXUAL ABUSE

Child sexual abuse is defined as the involvement of developmentally immature children and adolescents in sexual activities they do not comprehend, to which they are unable to give consent, or that violate social taboos or family roles. In a U.S. study in 1985, sexual abuse during childhood was reported by 27% of adult females and 16% of adult males. Females are more likely than males to have been sexually abused by a family member. Although there is a paucity of literature on male sexual abuse, it is probably more common than generally recognized. The psychological trauma appears to be similar for boys and girls. Sexual abuse during adolescence may merge with peer sexual assault, or "date rape." Sexual abuse in adolescent girls can be associated with a constant fear of pregnancy. Teenage pregnancy or [STD](#) may, in fact, be the first indication of ongoing abuse.

Psychological consequences of child sexual abuse often involve behavioral problems, psychiatric disturbances, or adjustment difficulties at the onset of adolescence, even though the actual abuse may have taken place at a younger age. Child sexual abuse may lead to low self-esteem and/or a degree of sexual disinhibition. The cognitive maturation that occurs with adolescence may bring about the realization and expression of these feelings. Young women who have been sexually abused have significantly higher rates of early-onset consensual sexual activity, teenage pregnancy, multiple sexual partners, unprotected intercourse, [STDs](#), and later sexual assault. Poor psychological outcome is related to the duration of abuse, the extent to which the abuse involves violence or coercion, and the perception that the child has cooperated with the abuser, with ensuing feelings of guilt. The impact of these sequelae can be reduced by supportive peer and family relationships. Disclosure of the abuse may help to ameliorate some of the psychological traumas associated with abuse.

SUBSTANCE ABUSE

Substance abuse and drug misuse among adolescents is a significant cause of morbidity and mortality ([Chaps. 386](#) to 389). The prevalence rates vary widely by region, ethnic group, age, and gender. The age of initiation into substance abuse has gradually declined. In 1997, rates among American teenagers for substance use or abuse, at some stage during their lifetimes, were: cigarettes smoking (70%), alcohol use (79%), marijuana use (47%), cocaine use (8%), anabolic steroids (4%), injected illegal drugs (2%), and other illegal drugs (17%), e.g., lysergic acid (LSD), phencyclidine (PCP), methylenedioxymethamphetamine (ecstasy), methamphetamine (ice), or heroin.

The forms of substance abuse change continuously. Anabolic steroids, for example, are now used by 3 to 5% of male high school seniors, with a 10% prevalence rate among male adolescent athletes. In addition to their use by athletes in an effort to increase muscle strength, nonathletes use anabolic steroids with a goal of achieving a more virile appearance. In contrast to popular views, anabolic steroids do not appear to enhance performance except at very high doses, which are associated with significant side effects ([Chap. 335](#)). Other performance-enhancing agents include human growth hormone and erythropoietin (EPO), but the high cost of these hormones limits their use.

In addition to the direct effect on health, substance abuse is associated with other risk-taking behaviors. The relationship of alcohol use and motor vehicle accidents, for example, is well documented. However, drug and alcohol use are also correlated with many other problems during adolescence including violence, suicide, depression, [STD](#), and unwanted pregnancies. Therefore, the presence of one form of risky behavior should prompt consideration of others.

SUICIDAL BEHAVIOR AND DEPRESSION

After motor vehicle accidents and homicide, suicide is the third leading cause of death in adolescents, and the rate has risen almost fourfold over the past 50 years. In 1988, the suicide rate among 15- to 19-year olds was 11.3 in 100,000. The causes for increased rates of suicide are not well understood, but one theory holds that modern society fosters increased social isolation and alienation. Nearly one-fourth of adolescents acknowledge seriously considering suicide, and 8% have actually attempted it. Attempted suicide is three times more common in females than males, with drug overdose or wrist-cutting being the most common means of suicide attempt. Completed suicide is three to five times more common in teenage boys than girls and usually involves firearms, hanging, or jumping from heights. Suicide is rare before puberty. Risk factors for suicide among adolescents include prior attempt of suicide, a history of depression or other major psychiatric disorder, history of substance abuse, medical illness, family history of suicidal behavior, and knowing someone who has committed suicide. Unfortunately, these and other risk factors are relatively common among nonsuicidal youth as well, making suicide difficult to predict in individual cases. Stressful events can precipitate depression and increase risk of suicide; these can include the death of a relative or friend, disciplinary crisis, rejection or humiliation, school difficulty, and anxiety about homosexuality. Apparently impulsive actions may be harbingers of more serious underlying mood disturbances, personality disorders, or substance abuse.

Major depression occurs in 4 to 6% of adolescents, and the *DSM-IV* criteria for diagnosis are the same as in adults ([Chap. 385](#)). Every depressed or suicidal adolescent should undergo psychiatric examination, whether hospitalized or not. Comprehensive evaluation requires exploration of the adolescent's history of mental health problems, symptoms of depression, level of functioning in school, interactions with friends and family, and evaluation for comorbid disorders. Indications for hospitalization include imminent risk of suicide as evidenced by an identified plan and access to lethal means, recurrent suicide attempts, the presence of severe depression or psychosis, substance abuse, and the need to remove the individual from an overwhelmingly stressful environment.

ADOLESCENT EATING DISORDERS

Many adolescents have voracious appetites in response to the increased energy and caloric requirements generated by the growth spurt. The unique physical, psychological, and social transitions of adolescence provide a context for the development and perpetuation of eating patterns. Adolescents with a body mass index (BMI), measured as weight (kg)/height (m²), greater than the 95th percentile for age and gender are overweight, and those between the 85th and 94th percentiles are at risk for becoming overweight. Based on the NHANES III survey for 1988 to 1994, there was evidence for a 6% increase in the prevalence of overweight adolescents compared to the previous decade. The increasing prevalence of obesity is multifactorial and involves patterns of eating behavior as well as alterations in activity level ([Chap. 77](#)). Physical activity among both girls and boys tends to decline steadily during adolescence. Regular involvement in enjoyable forms of exercise should be encouraged to help promote lifelong habits that involve physical activity.

Eating disorders such as anorexia nervosa or bulimia nervosa often have their onset during adolescence ([Chap. 78](#)). Control over dietary intake is perhaps one of the first mechanisms that adolescents use to establish autonomy and achieve independence from family. The majority of female adolescents and young adults in western cultures report feeling discontented with their body shape. Surveys of normal adolescent populations disclose a surprisingly high frequency of dieting and abnormal eating patterns. For instance, up to 79% binge, 70% consider themselves fat, 11% induce vomiting, 5% abuse laxatives, and about 3% meet diagnostic criteria for anorexia or bulimia nervosa. Eating disorders also occur in males, but much less frequently than in females.

PHYSICIAN-ADOLESCENT RELATIONSHIP

The transition from the pediatrician to an adult medical practice can be difficult for adolescents, their parents, and their physicians. The emergence of adolescent medicine as a specialty practice has helped to facilitate this transition and to focus on the special needs of this group. When adolescents transfer to an adult-based practice, the physician should first establish a relationship with the patient and his or her parents. Previous medical history should be reviewed and medical records obtained. The need for the teenager to be seen alone, and office policies concerning confidentiality, should be discussed and agreed to with the parent(s) and adolescent together.

Legal issues related to the medical care of minors arise frequently, and laws vary in different countries and from state to state ([Chap. 2](#)). As a general rule, anyone who has reached the age of majority (usually 18 years) may consent to treatment. Under this age, a parent or legal guardian must consent for medical intervention. However, there are several exceptions to this requirement. The delivery of medical care is generally accepted in an emergency, but it is important to document the nature of the emergency and any efforts to notify parents. Emancipated minors may also provide consent. This group includes those fulfilling adult roles (e.g., military service), married teens, and those who are financially independent and living separately from their parents. In addition, when the health of a minor is potentially endangered by disorders for which they may be reluctant to seek parental consent, such as substance abuse, pregnancy, or [STD](#), mature minors may generally provide consent. In these circumstances, the caregiver must assess the minor's maturity, ability to understand the risks and benefits of treatment, and capacity to provide informed consent. Mature or emancipated minors do not need to reveal consent or treatment to their parents.

Obtaining a medical history from an adolescent includes many elements that are distinct from an adult history. It should include, for example, schoolwork, home environment, and relationships with parents, siblings, and peers. Adolescents often lack knowledge about medical issues and may be reluctant to discuss sensitive topics with authority figures. Most will be nervous, even when these issues do not pertain. It can be useful, therefore, to provide printed forms or questionnaires. These not only serve to gather information in a relatively nonthreatening manner but also provide an indication of the kinds of issues that might be discussed with the physician. It is difficult to predict the topics that are paramount to the adolescent. Some may be preoccupied with concerns about the onset of acne, whereas others fear HIV or pregnancy. Adolescents may harbor guilt about sexual abuse or feel overwhelmed by peer pressure to engage in certain activities. The physician is well positioned to assist with many of these issues, if there is trust and an indication of interest and understanding. Because of these types of questions, it is important to interview the adolescent in private. Some parents will resist this approach, but it is necessary if the adolescent is to volunteer information that he or she is unwilling to discuss in the presence of parents. It is useful to reinforce the fact that conversations will be kept confidential. Adolescents are sometimes willing to bring concerns to the attention of nurses or other caregivers before raising these issues with a physician. It is helpful, therefore, to have another health care provider interact with the patient, if only briefly. In addition to direct questioning, general conversation about topical issues or inquiries about school or peers may provide insight into an adolescent's interests, activities, and potential risk factors. Because of the prevalence of substance abuse, risk-taking behavior, suicide, sexual orientation crises, [STDs](#), unwanted pregnancies, sexual abuse, depression, and eating disorders in the teenage years, these topics warrant specific inquiry as part of routine health assessment. The interview should also include adequate time for education, health care guidance, and counseling. It is also useful to provide written information about topics that are pertinent to the care of the adolescent.

The physical examination of the adolescent, while incorporating many elements of the adult examination, has several unique features. Foremost among these is the assessment of growth and sexual development. In addition to questionnaires that allow

the adolescent an opportunity to self-assess stages of pubertal development, the examination can be made less stressful by using it as opportunity to explain normal physiology. The issue of when to perform a pelvic examination as part of the routine health maintenance is controversial. Some advocate pelvic examinations in all sexually active young women as a means to detect [STDs](#) and for Pap smears. With the advent of urinary screening tests for chlamydia and gonorrhea, others suggest that pelvic examinations are not routinely necessary in the absence of specific indications. When a pelvic examination is performed, the patient should be asked whether she prefers her mother or a member of the health care team as an observer. The physical examination should also focus on diseases that tend to present during adolescence ([Table 8-2](#)).

Disorders such as hypertension, hyperlipidemia, and obesity are often first detected during adolescence. Strategies for disease prevention also include immunization, avoidance of cigarette smoking or excessive alcohol use, establishing good dietary habits, and engaging in regular exercise. General guidelines for adolescent preventive services (GAPS) are summarized in [Table 8-3](#).

SUMMARY

The term *adolescent* is derived from a Latin phrase meaning "to grow up." Adolescence is, in many ways, the culmination of development, with the achievement of identity and reproductive competence. Though these processes are triggered by internal physiologic events, they are intimately intertwined with the family and social environment. Physicians have an important role to facilitate these transitions by providing information and managing the diseases of adolescents. Moreover, it should be remembered that many adolescents view physicians as role models and will seek objective and informed advice about issues that reach beyond medicine.

(Bibliography omitted in Palm version)

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9. GERIATRIC MEDICINE - *Neil M. Resnick*

Of all the people who have ever lived to age 65, more than half are now alive. This statistic has important demographic and economic implications, and its impact on medical care is also substantial.

BIOLOGY OF AGING

Numerous molecular concomitants of aging have been described. For instance, there is an increase in chromosome structural abnormalities, DNA cross-linking, and frequency of single-strand breaks; a decline in DNA methylation; and loss of DNA telomeric sequences. The primary structure of proteins is unaltered, but posttranslational changes, such as deamidation, oxidation, cross-linking, and nonenzymatic glycation, increase. Mitochondrial structure also deteriorates, albeit not universally.

However, the biologic changes are clearer than the mechanisms that mediate them. In fact, although the senescent phenotype appears to be ubiquitous, biologists disagree about whether senescence even exists beyond zoos and civilized societies and whether it occurs at all in many species. There is little evolutionary rationale for a process that happens after reproduction is complete, particularly one associated with such a long and complex course. In nature, senescence is most notable for its absence; nearly all animals die of predation, disease, or environmental hazards rather than aging. The argument that different species have different maximum life spans can be explained without invoking a specific aging process: while growth and development are based on a genetic template, aging may reflect merely the accumulation of random damage rather than a specific mechanism.

If aging exists as a distinct process, there is consensus that the mechanisms are likely multifactorial, environmentally influenced, and species-specific, if not organ- and cell-specific, making the paucity of available human data particularly problematic. As a result, there are nearly as many theories of aging as investigators. Most theories overlap or are not mutually exclusive, and none is completely compatible with the dearth of data. As a group, the theories can be divided into two broad categories, based on whether they attribute aging to a genetic program or to progressive and random damage to homeostatic systems.

Enthusiasm for genetic theories of aging is fueled by several observations, including the dramatic species-specific differences in maximal life span, the strong correlation with survival among monozygotic compared to dizygotic twins, and the fact that single mutations can prolong life span by more than 50% in some nematodes and mice. However, all genetic theories must account for the fact that evolutionary selection pressure is minimal following completion of reproduction. Three genetic theories have recently been advanced, but few relevant data have yet been accrued. The first theory suggests that, since animals usually succumb to natural forces long before reaching their maximal life span, aging might reflect mutations that impair long-term survival. These mutations would accumulate in the genome because there is no selection pressure to delete them. A second theory, "pleiotropic antagonism," proposes that aging may be caused by the late and deleterious effects of genes that are conserved because of the survival advantages they confer prior to reproduction. The third theory applies to

ecological niches where extrinsic hazards are relatively low. In such an environment, evolution might select for mutations that retard the aging process since these might allow an animal to produce and protect many more litters. In support of this theory, the rate of aging in an isolated clan of Virginia opossums was calculated to be roughly half of that seen in their less fortunate cousins.

The "random damage" theories are based on the possibility that the balance between ongoing damage and repair is disrupted. The theories differ in the emphasis placed on increased damage (e.g., by free radicals, oxidation, or glycation) versus deficient repair, as well as in the mechanisms that might mediate each. However, all share the observation that cell and organ repair capacity declines with age. Some 40 years ago, Hayflick and Moorehead observed that the number of replications among cultured cells is finite. Subsequent research revealed that this replicative senescence was due to arrest of the cell cycle at the G₁/S phase, the point at which DNA synthesis begins. Recently, cell replication has also been linked to the length of telomeric DNA. Present at the termini of chromosomes, telomeric DNA prevents chromosomal instability, fragmentation, and rearrangement; anchors chromosomes to nuclear matrix; and provides a buffer between coding regions of DNA and the ends of the chromosomes. In addition, telomeric DNA is necessary for cell division. With each cell division, however, roughly 50 of the total 2000 base pairs of the telomere are lost. Telomeric shortening might thus result in loss of gene accessibility, which is necessary to repair ongoing cell damage caused by metabolism. Together with cytoplasmic factors mediating arrest of DNA synthesis, telomeric shortening could also limit the cell's ability to divide and thereby replace cells lost to apoptosis.

Many mechanisms previously postulated to mediate aging have not been borne out, including the somatic mutation theory (in which aging would result from cumulative spontaneous mutations), the error catastrophe theory (in which aging would result from errors in the synthesis of proteins critical to the synthesis of genetic material or protein-synthesizing machinery), and the intrinsic mutagenesis theory (in which aging is the result of ongoing intrinsic DNA rearrangements).

To date, the only intervention known to delay aging is caloric restriction. The salutary effect of restricting caloric intake by 30 to 40% has been documented in multiple species, from single-cell organisms to rodents. In rodents, it not only increases average life expectancy and maximum life span but also delays the onset of some typical age-associated diseases as well as deterioration of physiologic systems (e.g., immune responsiveness, glucose metabolism, muscle atrophy). Moreover, its impact is evident in both mitotic and postmitotic cells, in gene expression, and in protein turnover and cross-linking. Although the mechanism is still not determined, it is specific to caloric restriction rather than to reduction of any dietary component (e.g., fat intake) or supplements with vitamins or antioxidants. Unfortunately, adequate data from primates are not yet available, and the effect of caloric restriction in humans is still unknown.

PRINCIPLES OF GERIATRIC MEDICINE

Despite the biologic controversy, from a physiologic standpoint human aging is characterized by progressive constriction of the homeostatic reserve of every organ system. This decline, often referred to as *homeostenosis*, is evident by the third decade

and is gradual and progressive, although the rate and extent of decline vary. The decline of each organ system ([Table 9-1](#)) appears to occur independently of changes in other organ systems and is influenced by diet, environment, and personal habits as well as by genetic factors.

Several important principles follow from these facts: (1) Individuals become more dissimilar as they age, belying any stereotype of aging; (2) an *abrupt* decline in any system or function is always due to disease and not to "normal aging"; (3) "normal aging" can be attenuated by modification of risk factors (e.g., increased blood pressure, smoking, sedentary lifestyle); and (4) "healthy old age" is not an oxymoron. In fact, *in the absence of disease, the decline in homeostatic reserve causes no symptoms and imposes few restrictions on activities of daily living regardless of age.*

Appreciation of these facts may make it easier to understand the striking increases that have occurred in life expectancy. Average life expectancy is now 17 years at age 65, 11 years at age 75, 6 years at age 85, 4 years at age 90, and 2 years at age 100. Moreover, the bulk of these years is characterized by a lack of significant impairment ([Table 9-2](#)). Even beyond age 85, only 30% of people are impaired in any activity required for daily living and only 20% reside in a nursing home. Yet, as individuals age they are more likely to suffer from disease, disability, and the side effects of drugs, all of which, when combined with the decrease in physiologic reserve, make the older person more vulnerable to environmental, pathologic, and pharmacologic challenges.

The following concepts underlie the remainder of the chapter:

1. Disease presentation is often atypical in the elderly, especially in those more than 75 to 80 years old. Homeostatic strain caused by onset of a new disease often leads to symptoms associated with a different organ system, particularly one compromised by preexisting disease. For example, fewer than one-fourth of older patients with hyperthyroidism present with goiter, tremor, and exophthalmos; more likely are atrial fibrillation, confusion, depression, syncope, and weakness. Significantly, because the "weakest link" is so often the brain, the lower urinary tract, or the cardiovascular or musculoskeletal system, a limited number of presenting symptoms predominate -- acute confusion, depression, incontinence, falling, and syncope -- no matter what the underlying disease. Thus for the most common geriatric syndromes, regardless of the presenting symptom, the differential diagnosis is often largely similar. The corollary is equally important: The organ system usually associated with a particular symptom is less likely to be the source of that symptom in older individuals than in younger ones. Compared with middle-aged individuals, for example, acute confusion in older patients is less often due to a new brain lesion, depression to a psychiatric disorder, incontinence to bladder dysfunction, falling to a neuropathy, or syncope to heart disease.

2. Because of decreased physiologic reserve, older patients often develop symptoms at an earlier stage of their disease ([Fig. 9-1](#)). For example, heart failure may be precipitated by mild hyperthyroidism, cognitive dysfunction by mild hyperparathyroidism, urinary retention by mild prostatic enlargement, and nonketotic hyperosmolar coma by mild glucose intolerance. Paradoxically, therefore, treatment of the underlying disease may be easier because it is frequently less advanced at the time of presentation. A

corollary is that drug side effects can occur with drugs and drug doses unlikely to produce side effects in younger people ([Chap. 71](#)). For instance, an antihistamine (e.g., diphenhydramine) may cause confusion, loop diuretics may precipitate urinary incontinence, digoxin may induce depression even with normal serum levels, and over-the-counter sympathomimetics may precipitate urinary retention in men with mild prostatic obstruction.

Unfortunately, the predisposition to develop symptoms at an earlier stage of disease is often offset by the change in illness behavior that occurs with age. Raised at a time when symptoms and debility were accepted as normal consequences of aging, the elderly are less likely to seek attention until symptoms become disabling. Thus, any symptom, particularly those associated with a change in functional status, must be taken seriously and evaluated promptly.

3. Since many homeostatic mechanisms may be compromised concurrently, there are usually multiple abnormalities amenable to treatment, and small improvements in each may yield dramatic benefits overall. For instance, cognitive impairment in patients with Alzheimer's disease may respond much better to interventions that alleviate comorbidity than to prescription of donepezil ([Fig. 9-2](#)). Similar approaches apply to most other geriatric syndromes, including falls, incontinence, depression, delirium, syncope, and fracture. In each case, substantial functional improvement can result from treating the contributing factors even if -- as in Alzheimer's disease -- the disease itself is largely untreatable.

4. Many findings that are abnormal in younger patients are relatively common in older people -- e.g., bacteriuria, premature ventricular contractions, low bone mineral density, impaired glucose tolerance, and uninhibited bladder contractions. However, they may not be responsible for a particular symptom but only be incidental findings that result in missed diagnoses and misdirected therapy. For instance, the finding of bacteriuria should not end the search for a source of fever in an acutely ill older patient, nor should an elevated random blood sugar -- especially in an acutely ill patient -- be incriminated as the cause of neuropathy. On the other hand, certain other abnormalities must not be dismissed as due to old age -- e.g., there is no anemia, impotence, depression, or confusion of old age.

5. Because symptoms in older people are often due to multiple causes, the diagnostic "law of parsimony" often does not apply. For instance, fever, anemia, retinal embolus, and a heart murmur prompt almost a reflex diagnosis of infective endocarditis in a younger patient but may reflect aspirin-induced blood loss, a cholesterol embolus, insignificant aortic sclerosis, and a viral illness in an older patient. Moreover, even when the diagnosis is correct, treatment of a single disease in an older patient is unlikely to result in cure. For instance, in a younger patient, incontinence due to involuntary bladder contractions is treated effectively with a bladder relaxant medication. However, in an older patient with the same condition but who also has fecal impaction, takes medications that cloud the sensorium, and suffers from arthritis-associated impairments of mobility and manual dexterity, treatment of the bladder spasms alone is unlikely to restore continence. On the other hand, disimpaction, discontinuation of the offending medications, and treatment of the arthritis are likely to restore continence without the need for a bladder relaxant. Failure to recognize these principles often leads to

prescribing "ineffective" therapy and to unjustified therapeutic nihilism towards older patients.

6. Because the older patient is more likely to suffer the adverse consequences of disease, treatment -- and even prevention -- may be equally or even more effective. For instance, the survival benefits of exercise, as well as thrombolysis and beta-blocker therapy after a myocardial infarction, are as impressive in older patients as in younger ones; and treatment of hypertension and transient ischemic attacks, as well as immunization against influenza and pneumococcal pneumonia, are more effective in older patients. In addition, prevention in older patients must often be seen in a broader context. For instance, although interventions to increase bone density may be limited in older patients, fracture may still be prevented by efforts to improve balance, strengthen legs, reduce peripheral edema, treat other contributing medical conditions, replete nutritional deficits, eliminate environmental hazards, and remove adverse medications -- not so much those that affect bone metabolism, but rather those that induce orthostasis, confusion, and extrapyramidal stiffness.

In summary, optimal treatment of the older patient generally requires treating much more than the organ system usually associated with the disease or symptom, and often permits ignoring that system entirely.

EVALUATION

Evaluation of the older patient can be time-consuming, even when it is tailored to the problem. Yet, such initial investment can reduce subsequent morbidity and resource utilization and enhance patient and physician satisfaction. Additionally, the assessment can often be accomplished over several visits. Moreover, much can be gleaned from questionnaires filled out by the patient or caregiver in advance as well as from observation. For instance, greeting the patient in the waiting room allows the physician to note affective and cognitive response, the strength of the handshake, the ease of rising from a chair without using the arms, the length and steadiness of the stride, and the ability to follow directions to the examining room and to sit down safely in the examining room chair. Observing the patient dress or undress can also enhance detection of impaired cognition, fine motor skills, balance, and judgment. Such observations often provide more information than standard examinations and can shorten the clinical evaluation.

HISTORY TAKING IN ELDERLY PATIENTS

Most older patients are able to provide a reliable medical history; however, a multitude of complaints may make obtaining a history more difficult. If the patient is unable to comprehend or communicate, data should be sought from family, friends, and caregivers. The history should also include drug ingestion; dietary patterns; falling, incontinence, sexual dysfunction, depression and anxiety.

Advance Directives All older patients should be asked whether they have drafted advance health care directives, and, if they have, a copy should be placed in the record. Such directives may consist of a health care proxy or durable power of attorney for health care, in which patients designate a surrogate decision-maker who makes health

care decisions if the patient cannot, and/or a living will or medical directive, in which patients specify their desires for treatment in specific situations if they cannot communicate at the critical time.

Whether or not the patient has formally drafted these directives, it is useful to indicate in the record who should make health care decisions if the patient is no longer able to do so. Patients should then be encouraged to discuss their thoughts with the physician as well as the designated proxy. It is not feasible to cover all possible future complications in such discussions. Ascertaining patients' perspectives on specific interventions, such as resuscitation or intubation, is also difficult because preferences will likely differ depending on prognosis. For instance, a patient may not be interested in feeding tube placement following a massive stroke with little chance of recovery but would prefer the same intervention if it is short-term and helps ensure more rapid and complete recovery from an intercurrent illness such as pneumonia. More useful is a discussion that uses open-ended questions and empathic comments to elicit the patient's values and goals. Moreover, for any given condition, preferences may differ depending on baseline clinical status. For robust elderly individuals, recovery is a realistic goal, albeit the odds of complications are higher than for younger individuals. For the frail elderly patient with comorbidity that impairs functional status, reduction or alleviation of symptoms may be the goal. For patients with advanced dementia or terminal illness, palliation may be the most appropriate strategy. In each situation, however, early elicitation of a patient's preferences and values -- when the patient can still state them -- can often help both physicians and families in subsequent difficult decisions by giving surrogate decision-makers the sense that they are doing as the patient would have wanted.

PHYSICAL EXAMINATION

Certain features of the examination should receive special attention, depending in part on clues from the history. Weight and postural blood pressure should be measured at most visits. Vision and hearing should be checked; if hearing is impaired, excess cerumen should be removed from the external auditory canals prior to audiologic referral. Denture fit should be assessed, and the oral cavity should be inspected with the dentures removed. Although thyroid disease becomes more common with age, the sensitivity and specificity of related findings are substantially lower than in younger individuals; consequently, the physical examination can rarely corroborate or exclude thyroid dysfunction in older patients. The breasts should not be overlooked, since older women are more likely to have breast cancer and less likely to do breast self-examination. The systolic murmur of aortic sclerosis is common and may be difficult to differentiate from aortic stenosis, especially since the presence of a fourth heart sound in an elderly person does not imply significant cardiac disease, and the carotid upstroke normally increases owing to age-related arterial stiffening.

In inactive patients and those with fecal or urinary incontinence, one should check for fecal impaction. In patients with urinary incontinence -- especially men -- a distended bladder must be looked for, since it may be the only finding in urinary retention; perineal sensation and the bulbocavernosus reflex should also be tested. Patients who fall should be observed standing up from a chair, bending down, reaching up, walking 10 feet, turning, returning, and sitting again; abnormalities of gait and balance should be evaluated with the patient's eyes open and closed and in response to a sternal push. It

should be appreciated that "frontal release signs" (e.g., "snout," "glabellar," or palmomental reflexes) and absent ankle jerks and vibratory sense in the feet may be normal in the elderly.

MENTAL STATUS EXAMINATION

In addition to evaluating mood and affect, some form of cognitive testing is essential in all elderly patients, even if it involves only checking different components of the history for consistency. People with mild degrees of dementia usually retain their social graces and may mask intellectual impairment by a cheerful and cooperative manner. Thus, the examiner should always probe for content. For patients who follow the news, one can ask what stories they are particularly interested in and why; the same applies to reading, social events -- even the soap operas on television.

If there is any suspicion of a cognitive deficit after this kind of conversational probing, further questioning is indicated. An examination that tests only orientation as to person, place, and time is insufficient to detect mild or moderate intellectual impairment. As a quick screen, simply assessing orientation and asking the patient to draw a clock with the hands at a set time (e.g., 10 min before 2:00) can be very informative regarding cognitive status, visuospatial deficits, ability to comprehend and execute instructions in logical sequence, and presence or absence of perseveration. For slightly more detailed examinations, many practical mental status tests are available. The most widely used is the Mini-Mental Status Examination of Folstein ([Chap. 24](#)), which provides a numerical score that can be obtained in 5 to 10 min. Regardless of the test employed, the total score is less useful diagnostically than is knowledge of the specific domain of the deficit. As a general rule, disproportionate difficulty with immediate recall (e.g., of a list of three items) suggests depression, while predominant difficulty with recalling the items 5 min later suggests dementia. For patients with deficits of attention -- recognized by inability to spell simple words backwards, repeat five digits, or recite the months of the year backwards -- delirium is probably present, and the accuracy of the remainder of the test is dubious. However, the test can be interpreted accurately only in the context of a comprehensive evaluation.

EVALUATION OF FUNCTIONAL CAPACITY

Medical problem lists, a standard tool for assessing and following younger patients, often prove inadequate for older patients. Heart failure, stroke, and prostate cancer can describe a bedbound institutionalized person as well as a Supreme Court justice. Thus, it is essential to ascertain the patient's degree of functional incapacity owing to both medical and psychosocial problems. The functional assessment includes determination of the patient's ability to perform basic activities of daily life (ADL), which are those needed for personal self-care, as well as the ability to perform more complex tasks required for independent living, the instrumental activities of daily living (IADL). ADLs include bathing, dressing, toileting, feeding, getting in and out of chairs and bed, and walking. IADLs include shopping, cooking, money management, housework, using a telephone, and traveling outside the home. For frail patients, an assessment in the home by a trained observer may be required, but for most patients a questionnaire dealing with these activities can be completed by the family or patient. In either case, the physician must determine the cause of any impairment and whether it can be

treated. Assessment should conclude with determination of the socioeconomic circumstances and social support systems.

MANAGEMENT OF COMMON GERIATRIC CONDITIONS

Diseases more common in the elderly are covered elsewhere in the text. The medical problems discussed below do not usually present as clear-cut organ-specific diagnoses and are most common in the frail elderly, especially those over 80 years of age.

INTELLECTUAL IMPAIRMENT

The predominant causes of impaired mentation in older patients are delirium, dementia, and depression. Each condition is covered elsewhere in the text in detail ([Chaps. 24](#) and [362](#)), but their management in the elderly is discussed here.

Differentiating the causes of impaired mentation is important, but in older patients they frequently coexist. Thus, the most important first step is to search for and correct all factors that may contribute to cognitive impairment, even in patients with dementia ([Fig. 9-2](#)). Evidence of dangerous behavior should also be sought (e.g., leaving the stove on, wandering, and getting lost), and plans should be devised to deal with it. Although there is no specific pharmacologic treatment for Alzheimer's disease and agents such as donepezil are of limited efficacy, this does not mean that the physician has no further role in treating the patient and family. In addition to discontinuing all nonessential medications and treating new intercurrent illness, the physician should help the family and patient predict and deal with the disease; indeed, the family often needs the physician's support more than the patient does.

TREATMENT

Community services should be suggested as needed, including a visiting nurse, a home health aide to assist with personal hygiene, a homemaker to assist with housework, meal delivery, transportation services, day health centers, and respite care to ease the burden on family members. Support groups such as the Alzheimer's Association are often of value to the family and help them to anticipate problems. Signs of patient abuse by an overstressed caregiver should be watched for. Legal counsel should be recommended to help the patient and family devise plans for ongoing management and ultimate disposition of assets not already obtained; advance directives should be sought as soon as possible while the patient can still participate.

Finally, abrupt worsening of mentation or the onset of disruptive behavior should always prompt a search for new illness or medication. Exacerbation of cognitive dysfunction may occur with mild infections (e.g., subungual toe abscess, vaginitis, or pressure ulcer); with "therapeutic" levels of many drugs; with use of nonprescribed drugs or alcohol; with modest abnormalities of serum sodium, calcium, glucose, or thyroxine; with mild hypoxia; with borderline nutritional deficiencies; with subdural hematoma or "minor" stroke; and with the development of fecal impaction, urinary retention, pain, or change in environment, particularly in frail older patients. However, if a cause is not found and behavior does not respond to environmental manipulation (e.g., ignoring the behavior, distracting the patient, addressing situational "triggers," and providing a calm

environment), low doses of an antipsychotic medication may be helpful (e.g., haloperidol 0.25 to 2 mg/d orally; see below).

DEPRESSION

Depression of significant degree occurs in 5 to 10% of community-dwelling elderly but is often overlooked. At highest risk are individuals with recent medical illness (e.g., stroke or fracture), bereavement, lack of social supports, recent nursing home admission, or psychiatric history (including alcohol abuse). The diagnosis requires the presence of a depressed mood for at least two consecutive weeks plus at least four of the following eight symptoms: sleep disturbance, lack of interest, feelings of guilt, decreased energy, decreased concentration, decreased appetite, psychomotor agitation/retardation, and suicidal ideation. Also helpful diagnostically are a personal or family history of depression, anhedonia (loss of pleasure), and past response to an antidepressant. It is essential to bear in mind that depression in older patients is often caused or contributed to by drugs or a systemic illness. Although "subsyndromal" depression (fewer than four of the above symptoms) also causes substantial morbidity and health resource utilization, it appears to be less responsive than major depression to therapy.

TREATMENT

For the hospitalized patient in whom acute depression delays recovery or rehabilitation -- when correction of medical and pharmacologic contributing factors is ineffective and there is no prior history of mania or major depression -- methylphenidate, 5 to 10 mg at 8 A.M. and noon (to avoid insomnia) is often very effective, with benefits discernible within a few days. For patients with major depression, there is no ideal antidepressant drug. All are about equally effective, but the side effects differ (see below and [Chap. 385](#)). Consequently, one should become familiar with one or two agents for patients with psychomotor retardation (e.g., sertraline, desipramine) and for those with agitation (e.g., nortriptyline or nefazodone). Because of its potent anticholinergic and orthostatic side effects, amitriptyline should be avoided whenever possible in older patients. Initial low dosages should be increased slowly to avoid serious side effects; low doses of each medication (e.g., nortriptyline, 10 to 50 mg daily; desipramine, 25 to 75 mg daily; or sertraline 50 to 150 mg daily) are often effective in the elderly. Careful follow-up is required to anticipate and minimize anticholinergic side effects, orthostatic hypotension, sedating effects, confusion, bizarre mental symptoms, cardiovascular complications, and drug overdose with suicidal intent. Adverse drug reactions should not be assumed to be due to the aging process.

Cautious use of the monoamine oxidase inhibitors is sometimes of benefit when other antidepressants are ineffective. Neither monoamine oxidase inhibitors nor selective serotonin reuptake inhibitors should be used in combination with the cyclic compounds. Electroconvulsive therapy has been successful and is usually well tolerated by elderly patients who remain severely depressed despite drug treatment, particularly if they also have delusions.

URINARY INCONTINENCE

Transient Incontinence ([Table 9-3](#)) Because urinary continence requires adequate

mobility, mentation, motivation, and manual dexterity -- in addition to integrated control of the lower urinary tract -- problems outside the bladder can result in incontinence.

1. *Delirium*. A clouded sensorium impedes recognition of both the need to void and the location of the nearest toilet; once delirium clears, incontinence resolves.

2. *Infection*. Symptomatic urinary tract infection commonly causes or contributes to incontinence; asymptomatic infection does not.

3. *Atrophic urethritis/vaginitis*. Atrophic urethritis/vaginitis, characterized by the presence of vaginal telangiectasia, petechiae, erythema, or friability, commonly contributes to incontinence in women and responds to a several-month course of low-dose estrogen or vaginal estrogen creams.

4. *Pharmaceutical*. The drugs most commonly causing transient incontinence are listed in [Table 9-4](#).

5. *Psychologic*. Depression and psychosis are uncommon but treatable causes.

6. *Excess urine output*. Excess urine output may overwhelm the ability to reach a toilet in time. Causes include diuretics, alcohol, excess fluid intake, and metabolic abnormalities (e.g., hyperglycemia, hypercalcemia, diabetes insipidus); nocturnal incontinence may also result from mobilization of peripheral edema.

7. *Restricted mobility*. If mobility cannot be improved, access to a urinal or commode may restore continence. (See "Immobility," below.)

8. *Stool impaction*. This is a common cause of urinary incontinence, especially in hospitalized or immobile patients. Although the mechanism is unknown, a clue to its presence is the coexistence of both urinary and fecal incontinence. Disimpaction restores continence.

Established Incontinence ([Table 9-3](#)) The causes of established incontinence include irreversible functional deficits, such as *end-stage* Alzheimer's disease, and intrinsic lower urinary tract dysfunction. Lower urinary tract dysfunction should be sought after transient causes have been excluded.

Detrusor Overactivity This disorder (involuntary bladder contraction) accounts for two-thirds of geriatric incontinence in both sexes, regardless of whether patients are demented. Detrusor overactivity can be diagnosed presumptively in a woman when leakage occurs in the absence of stress maneuvers or urinary retention and is preceded by the abrupt onset of an intense urge to urinate that cannot be forestalled. In men, the symptoms are similar, but since detrusor overactivity often coexists with urethral obstruction, urodynamic testing should be done if prescription of a bladder relaxant is planned. Because detrusor overactivity may also be due to bladder stones or tumor, the abrupt onset of otherwise unexplained urge incontinence -- especially if accompanied by perineal/suprapubic discomfort or sterile hematuria -- should prompt cystoscopy and cytologic examination.

TREATMENT

The cornerstone of treatment is behavioral therapy with or without biofeedback. Patients without dementia are instructed to void every 1 to 2 h (while awake only) and to suppress urgency in between; once daytime continence is restored, the interval between voiding can be progressively increased. Demented patients are "prompted" to void at similar intervals. When drugs are necessary, they should be added to these regimens and monitored to avoid inducing urinary retention. Effective drugs include oxybutynin (2.5 to 5 mg three or four times daily, or sustained release, 5 to 20 mg once daily), dicyclomine (10 to 30 mg three times daily), tolterodine (1 to 2 mg twice daily), and imipramine or doxepin (25 to 100 mg at bedtime). If prescribed for older patients, DDAVP should be used cautiously -- especially in the setting of renal insufficiency or heart failure -- and it probably should not be given to patients with hyponatremia or urine output >2500 mL/d. Alternative treatments, such as neuromodulation, are under investigation.

Indwelling catheterization is rarely indicated for detrusor overactivity. If all measures fail, an external collection device or protective pad or undergarment may be required.

Stress Incontinence This disorder, the second most common cause of established incontinence in older women (it is rare in men), is characterized by symptoms and evidence of *instantaneous* leakage of urine in response to stress. Leakage is worse or occurs only during the day unless another abnormality (e.g., detrusor overactivity) is also present. On examination, with the bladder full and the perineum relaxed, instantaneous leakage upon coughing strongly suggests stress incontinence, especially if it reproduces symptoms and if urinary retention has been excluded by a postvoiding residual determination; a several-second delay suggests that leakage is instead caused by an involuntary bladder contraction induced by coughing.

TREATMENT

Surgery is the most effective treatment. For women who can comply indefinitely, pelvic muscle exercises are an option for mild to moderate stress incontinence, but they often require specialized training using vaginal cones or biofeedback. If not contraindicated, an α -adrenergic agonist (e.g., phenylpropanolamine) is also helpful in such cases, especially if combined with estrogen. Occasionally, a pessary or even a tampon (for women with vaginal stenosis) provides some relief.

Urethral Obstruction Rarely present in women, urethral obstruction (due to prostatic enlargement, urethral stricture, bladder neck contracture, or prostate cancer) is the second most common cause of established incontinence in older men. It can present as dribbling incontinence after voiding, urge incontinence due to detrusor overactivity (which coexists in two-thirds of cases), or overflow incontinence due to urinary retention. Renal ultrasound is recommended to exclude hydronephrosis in men whose postvoiding residual volume exceeds 100 to 200 mL; in older men for whom surgery is planned, urodynamic confirmation of obstruction is strongly advised.

TREATMENT

Surgical decompression is the most effective treatment for obstruction, especially if there is urinary retention. For a nonoperative candidate, intermittent or indwelling catheterization is used; a condom catheter is contraindicated when urinary retention is present. For a man with prostatic obstruction who is not in retention, treatment with an α -adrenergic antagonist (e.g., terazosin 5 to 10 mg daily) may lessen symptoms in a few weeks. The 5 α -reductase inhibitor finasteride may also ameliorate symptoms in a third or more of patients, but its impact is modest and not apparent for many months. Combined treatment with both agents has proved no better than treatment with an α blocker alone in most men.

Detrusor Underactivity Whether idiopathic or due to sacral lower motor nerve dysfunction, this is the least common cause of incontinence (<10% of cases). When it causes incontinence, detrusor underactivity is associated with urinary frequency, nocturia, and frequent leakage of small amounts. The elevated postvoiding residual volume (generally >450 mL) distinguishes it from detrusor overactivity and stress incontinence, but only urodynamic testing (rather than cystoscopy or intravenous urography) differentiates it from urethral obstruction in men; such testing is not usually required in women, in whom obstruction is rare.

TREATMENT

For the patient with a poorly contractile bladder, augmented voiding techniques (e.g., double voiding or applying suprapubic pressure) are often effective; pharmacologic agents (e.g., bethanechol) are rarely effective. If further emptying is needed or for the patient with an acontractile bladder, intermittent or indwelling catheterization is the only option. Antibiotics should be used for symptomatic upper tract infection, or as prophylaxis for recurrent symptomatic infections only in a patient using intermittent catheterization; they should not be used as prophylaxis with an indwelling catheter.

FALLS

Falls are a major problem for elderly people, especially women. Some 30% of community-dwelling elderly individuals fall each year, and the proportion increases with age. Nonetheless, falling must *not* be viewed as accidental, inevitable, or untreatable.

Causes of Falls Balance and ambulation require a complex interplay of cognitive, neuromuscular, and cardiovascular function and the ability to adapt rapidly to an environmental challenge. With age, balance becomes impaired and sway increases. The resulting vulnerability predisposes the older person to fall when challenged by an additional insult to *any* of these systems. Thus, a seemingly minor fall may be due to a serious problem, such as pneumonia or a myocardial infarction.

Much more commonly, however, falls are due to the complex interaction between a variably impaired patient and an environmental challenge. While a warped floorboard may pose little problem for a vigorous, unmedicated, alert person, it may be sufficient to precipitate a fall and hip fracture in the patient with impaired vision, strength, balance, or cognition. Thus, falls in older people are rarely due to a single cause, and effective prevention entails a comprehensive assessment of the patient's intrinsic deficits (usually diseases and medications), the routine activities, and the environmental obstacles.

Intrinsic deficits are those that impair sensory input, judgment, blood pressure regulation, reaction time, and balance and gait ([Table 9-5](#)). Medications and alcohol use are among the most common, significant, and reversible causes of falling. Other treatable contributors include postprandial hypotension (which peaks 30 to 60 min after a meal), insomnia, urinary urgency, foot problems, and peripheral edema [which can burden impaired leg strength and gait with an additional 2 to 5 kg (5 to 10 lb)].

Environmental obstacles are listed in [Table 9-6](#). Since most falls occur in or around the home, a visit by a visiting nurse, physical therapist, or physician often reaps substantial dividends.

Complications of Falls and Treatment One out of four people who fall suffers serious injury. About 5% of falls result in fractures, and an equal proportion cause serious soft tissue damage. Falls are the sixth leading cause of death for older people and a contributing factor in 40% of admissions to nursing homes. Resultant hip problems and fear of falls are major causes of loss of independence.

Subdural hematoma is a treatable but easily overlooked complication of falls that must be considered in any elderly patient presenting with new neurologic signs, including confusion alone, even in the absence of a headache. Dehydration, electrolyte imbalance, pressure sores, rhabdomyolysis, and hypothermia may also occur and endanger the patient's life following a fall.

The risk of falling is related to the number of contributory conditions. Because the relationship is multiplicative rather than additive, however, even minor improvement in a number of these factors will reduce the risk substantially. In addition, gait training by a physical therapist often alleviates fear of falling. Ensuring the availability of phones at floor level, a portable phone, or a lightweight radio call system is also important, as is detection and treatment of osteoporosis.

IMMOBILITY

The main causes of immobility are weakness, stiffness, pain, imbalance, and psychological problems. Weakness may result from disuse of muscles, malnutrition, electrolyte disturbances, anemia, neurologic disorders, or myopathies. The most common cause of stiffness in the elderly is osteoarthritis; however, Parkinson's disease, rheumatoid arthritis, gout, pseudogout, and antipsychotic drugs such as haloperidol may also contribute. Pain, whether from bone (e.g., osteoporosis, osteomalacia, Paget's disease, metastatic bone cancer, trauma), joints (e.g., osteoarthritis, rheumatoid arthritis, gout), bursa, muscle (e.g., polymyalgia rheumatica, intermittent claudication, or "pseudoclaudication"), or foot problems may immobilize the patient.

Imbalance and fear of falling are major causes of immobilization. Imbalance may result from general debility, neurologic causes (e.g., stroke; loss of postural reflexes; peripheral neuropathy due to diabetes mellitus, alcohol, or malnutrition; and vestibulocerebellar abnormalities), orthostatic or postprandial hypotension, or drugs (e.g., diuretics, antihypertensives, neuroleptics, and antidepressants) or may occur following prolonged bed rest. Psychological conditions such as severe anxiety or

depression may also contribute to immobilization.

Consequences In addition to thrombophlebitis and pulmonary embolus, there are multiple hazards of bed rest in the elderly. Deconditioning of the cardiovascular system occurs within days and involves fluid shifts, fluid loss, decreased cardiac output, decreased peak oxygen uptake, and increased resting heart rate. Striking changes also occur in skeletal muscle. At the cellular level, intracellular ATP and glycogen concentrations decrease, rates of protein degradation increase, and contractile velocity and strength decline, while at the whole-muscle level, atrophy, weakness, and shortening are seen. Pressure sores are another serious complication; mechanical pressure, moisture, friction, and shearing forces all predispose to their development. As a result, within days of being confined to bed, the risk of postural hypotension, falls, and skin breakdown rises. Moreover, these changes usually take weeks to months to reverse.

TREATMENT

The most important step is preventive -- to avoid bedrest whenever possible. When it cannot be avoided, several measures can be employed to minimize its consequences. Patients should be positioned as close to the upright position as possible several times daily. Range-of-motion exercises should begin immediately, and the skin over pressure points should be inspected frequently. Isometric and isotonic exercises should be performed while the patient is in bed, and whenever possible patients should assist their own positioning, transferring, and self-care. As mobility becomes feasible, graduated ambulation should begin. For individuals confined to a wheelchair, ring-shaped devices ("donuts") should not be used to prevent pressure ulcers since they cause venous congestion and edema and actually increase the risk.

If a pressure ulcer develops, therapy depends on its stage. Stage 1 ulcers are characterized by nonblanchable erythema of intact skin; stage 2 lesions involve an ulcer of the epidermis, dermis, or both; stage 3 ulcers extend to the subcutaneous tissue; and stage 4 lesions involve muscle, bone, and/or the supporting tissues. For stage 1 lesions, eliminating excess pressure and ensuring adequate nutrition and hygiene are sufficient. For the remaining types, the caregiver must also ensure that the wound stays clean and moist; thus, if saline dressings are used they should be changed when they are damp rather than dry. Synthetic dressings are more expensive than saline but are more effective because they require fewer changes (with less disruption of reepithelialization) and protect against contamination. Because bacterial colonization of pressure ulcers is universal, swab cultures should not be performed and topical treatment should be considered only for patients whose ulcers have not healed after 2 weeks of therapy. By contrast, associated cellulitis, osteomyelitis, or sepsis requires systemic therapy after cultures of blood and the wound border (by needle aspiration or biopsy) have been obtained. Surgical or enzymatic debridement is required for stage 3 and 4 lesions. In addition to a daily multivitamin, prescribing vitamin C (500 mg twice daily) is also useful. For debilitated patients, special mattresses are beneficial, including those that reduce pressure (e.g., static air mattress or foam) and those that relieve it (e.g., dynamic units that sequentially inflate and deflate).

In addition to treating all identified factors that contribute to immobility, consultation with

a physical therapist should be sought. Installing handrails, lowering the bed, and providing chairs of proper height with arms and rubber skid guards may allow the patient to be safely mobile in the home. A properly fitted cane or walker may be helpful.

IATROGENIC DRUG REACTIONS

For several reasons, older patients are two or three times more likely to have adverse drug reactions ([Chap. 71](#)). Drug clearance is often markedly reduced. This is due to a decrease in renal plasma flow and glomerular filtration rate and a reduced hepatic clearance. The last is due to a decrease in activity of the drug-metabolizing microsomal enzymes and an overall decline in blood flow to the liver with aging. The volume of distribution of drugs is also affected, since the elderly have a decrease in total-body water and a relative increase in body fat. Thus, water-soluble drugs become more concentrated, and fat-soluble drugs have longer half-lives. In addition, serum albumin levels decline, particularly in sick patients, so that there is a decrease in protein binding of some drugs (e.g., warfarin, phenytoin), leaving more free (active) drug available.

In addition to impaired drug clearance, which alters pharmacokinetics, older patients have altered responses to similar serum drug levels, a phenomenon known as *altered pharmacodynamics*. They are more sensitive to some drugs (e.g., opiates, anticoagulants) and less sensitive to others (e.g., β -adrenergic agents). Finally, the older patient with multiple chronic conditions is likely to be taking several drugs, including nonprescribed agents. Thus, adverse drug reactions and dosage errors are more likely to occur, especially if the patient has visual, hearing, or memory deficits.

Precautions to Avoid Drug Toxicity

Drug Selection and Administration Before initiating treatment, the physician should first ensure that the symptom requiring treatment is not itself due to another drug. For example, antipsychotic agents can cause symptoms that mimic depression (flat affect, restlessness, and pacing); such symptoms should prompt lowering of the dose rather than initiation of an antidepressant. In addition, drug therapy should be employed only after nonpharmacologic means have been considered or tried and only when the benefit clearly outweighs the risk.

Once pharmacotherapy has been decided upon, it should begin at less than the usual adult dosage and the dose should be increased slowly. However, given the marked variability in pharmacokinetics and pharmacodynamics in the elderly, dose escalation should continue until either a successful endpoint is reached or an intolerable side effect is encountered. The final dosage schedule should be kept as simple as possible, and the number of pills should be kept as low as possible. Serum drug levels are often useful in older patients, especially for monitoring drugs with narrow therapeutic indices such as phenytoin, theophylline, quinidine, aminoglycosides, lithium, and psychotropic agents such as nortriptyline. However, toxicity can occur even with "normal" therapeutic levels of some drugs (e.g., digoxin, phenytoin).

Over-the-Counter Agents Nearly three-quarters of the elderly regularly use nonprescribed drugs, many of which cause significant symptoms and/or interact with other medications. Frequent offenders include nonprescribed agents for insomnia (all of

which are anticholinergics), and nonsteroidal anti-inflammatory drugs (NSAIDs), which can hamper control of hypertension in addition to causing renal dysfunction and gastrointestinal bleeding. Gingko biloba, increasingly used as a "memory booster," may interfere with previously stable anticoagulation regimens. Because older patients often consider such agents "nostrums" rather than drugs, the physician must ask about them directly.

Sedative-Hypnotics If nonpharmacologic treatment of insomnia is unsuccessful, low-dose and short-term or intermittent use of an intermediate-acting agent whose metabolism is not affected by age (e.g., oxazepam, 10 to 30 mg/d) may be useful. Because of the increased risk of confusion and other adverse effects, benzodiazepines with either short (e.g., triazolam) or long duration of action (e.g., flurazepam and diazepam) should be avoided. Barbiturates should be avoided for the same reasons. An antidepressant should not be prescribed for insomnia unless the patient is depressed.

Antibiotics Serum creatinine is not a good index of renal function in old people; however, when it is elevated, special care must be taken with the administration of drugs normally excreted by the kidneys. Concentrations of relevant antibiotics should be measured directly.

Cardiac Drugs In older patients, digitalis, procainamide, and quinidine have prolonged half-lives and narrow therapeutic windows; toxicity is common at the usual dosages. For example, digoxin toxicity -- especially anorexia, confusion, or depression -- can occur even with therapeutic digoxin levels.

H₂Receptor Antagonists Most of these agents interfere with hepatic metabolism of other drugs, and all can produce confusion in the elderly. Because they are renally excreted, lower doses should be used to minimize the risk of toxicity in older individuals.

Antipsychotics and Tricyclic Antidepressants These drugs can produce anticholinergic side effects in old people (e.g., confusion, urinary retention, constipation, dry mouth). These can be minimized by switching to a nonanticholinergic agent (e.g., sertraline or nefazodone) or one with less anticholinergic effect (e.g., olanzapine, desipramine). In general, the least potent agents for psychosis (e.g., chlorpromazine) have the most sedating and anticholinergic effects and are the most likely to induce postural hypotension. By contrast, the most potent antipsychotic agents (e.g., haloperidol) have the least sedating, anticholinergic, and hypotensive side effects but cause extrapyramidal side effects, including dystonia, akathisia, rigidity, and tardive dyskinesia. The newer potent antipsychotics (e.g., risperidone, olanzapine, quetiapine, and clozapine) are relative exceptions to this rule. More specific for serotonin than dopamine D₂receptors, these medications may be safer for older demented patients, especially those with hallucinations associated with Lewy body dementia or in those receiving therapy for Parkinson's disease. Unfortunately, even these newer drugs lose their specificity at the higher doses that are commonly required in clinical practice. Thus all of these agents are potentially toxic. Moreover, since both depression and agitation often remit spontaneously, cautious discontinuation of these drugs should be considered periodically.

Glaucoma Medications Both topical beta blockers and carbonic anhydrase inhibitors can

cause systemic side effects. The latter can cause malaise and anorexia independent of the induced metabolic acidosis.

Anticoagulants Elderly patients benefit from anticoagulation as much as do younger individuals but are more vulnerable to serious bleeding and drug interactions. Hence, more careful monitoring and less aggressive anticoagulation are advisable.

Analgesics Both propoxyphene and meperidine are associated with a disproportionate risk of delirium, and propoxyphene also increases the risk of hip fracture. Of the [NSAIDs](#), indomethacin is most likely to induce confusion, fluid retention, and gastrointestinal bleeding. Each of these agents should be avoided in the elderly.

Avoidance of Overtreatment Drugs are frequently not indicated in some common clinical situations. For instance, antibiotics need not be given for asymptomatic bacteriuria unless obstructive uropathy, other anatomic abnormalities, or stones are also present. Ankle edema is often due to venous insufficiency, drugs such as [NSAIDs](#) or some calcium antagonists, or even inactivity or malnutrition in chairbound patients. Diuretics are usually not indicated unless edema is associated with heart failure. Fitted, pressure gradient stockings are often helpful. Regular exercise is much more useful for claudication than is pentoxifylline. Finally, since older patients generally tolerate aspirin and other NSAIDs less well than do younger patients, localized pain should be treated when possible with local measures such as injection, physical therapy, heat, ultrasound, or transcutaneous electrical stimulation ([Chap. 12](#)).

PREVENTION

Much can be done to prevent the progression and even the onset of disease in older people. Dietary inadequacies should be corrected. Daily calcium intake should approximate 1500 mg, and most elderly people should take 400 to 800 IU of vitamin D daily (contained in one to two multivitamin tablets). Tobacco and alcohol use should be minimized, since the benefits of discontinuing these accrue even to individuals over age 65. The importance of reviewing all of a patient's medications and discontinuing them whenever feasible cannot be overemphasized.

Hypertension, whether isolated systolic hypertension or combined systolic and diastolic hypertension, should be treated. Treatment reduces the risk of stroke and the risk of death due to cardiovascular causes substantially in this age group and may also reduce the risk of cognitive impairment. These benefits have been achieved using *low doses* of a thiazide-like diuretic (e.g., chlorthalidone, 12.5 to 25 mg/d) as the first step (alone effective in almost half of patients) and adding low-dose reserpine (0.05 to 0.1 mg/d) or atenolol (25 to 50 mg/d) only as needed. Benefits are dramatic, side effects are minimal, cost is trivial, and concerns about potential toxicity have not been borne out.

Because of the prevalence, functional impact, and ease of treatment, glaucoma should be screened for, and visual and auditory impairment should be corrected. Dentures should be assessed for their fit, and oral lesions beneath them should be detected.

Because thyroid dysfunction is more prevalent in the elderly, difficult to detect clinically, and treatable, serum levels of thyroid-stimulating hormone should be measured at least

once in asymptomatic older people and probably every 3 to 5 years thereafter. Serum cholesterol is worth measuring in patients with established coronary heart disease, but in those without apparent disease, screening for hypercholesterolemia is controversial. It seems reasonable to screen those who would be willing to comply with therapy, whose quality of life is good (from the patient's viewpoint), whose life expectancy exceeds several years (long enough to potentially benefit from therapy), and whose other risk factors -- for which benefit of treatment has been definitely established -- have already been addressed. A Papanicolaou test should be done in women who have not had one before, since the incidence of both preventable cervical carcinoma and associated death increases with age, especially in this group; it should be repeated triennially in all older women unless two previous tests have been normal. Screening for colon cancer is warranted until a minimum age of 80 to 85, at least in the community-dwelling elderly, although the optimal method is unclear. Immunizations for influenza, pneumococcal pneumonia, and tetanus should be current. Purified protein derivative (PPD) testing should be done on residents of chronic care facilities and on others at high risk of tuberculosis; those who have recently converted probably should be treated. Since responsiveness wanes with age, the test, if negative, should be repeated in a week to increase the chances of detecting all exposed patients. Because older women with breast cancer are more likely to die *of* it than *with* it, screening mammography is indicated every 1 to 2 years at least until age 75 and thereafter if a positive finding would result in therapeutic intervention. The relative risks and benefits of low-dose aspirin and (for women) estrogen replacement therapy have not yet been elucidated sufficiently in the elderly to warrant routine use, but they should be considered on an individual basis.

Exercise should be encouraged not only because of its beneficial effects on blood pressure, cardiovascular conditioning, glucose homeostasis, bone density, insomnia, functional status, and even longevity, but also because it may improve mood and social interaction, reduce constipation, and prevent falls. Resistance training should be encouraged as much as a walking program. Spinal flexion exercises should be avoided in patients with osteopenia; consultation with a physical therapist may be helpful.

Measures should be taken to prevent falling, as outlined in [Tables 9-5](#) and [9-6](#). Now that alendronate has proved effective in preventing vertebral and hip fractures in older women, bone density should be measured in women who are willing to take the drug and who do not already take estrogen. Counseling about driving is important, especially for patients with cognitive impairment.

Perhaps the most valuable preventive measure in old people is to take a careful history, focusing not only on the "chief complaint" but also on common and often hidden conditions such as falls, confusion, depression, alcohol abuse, sexual dysfunction, and incontinence. In addition, one should always identify the complications for which the specific patient is at risk and take steps to avert them. For instance, a patient with cognitive impairment who smokes is at risk not only for lung cancer but also for starting a fire, and a patient who requires narcotics is at risk for fecal impaction, delirium, urinary retention, and confusion. Community-dwelling patients who are at highest risk of rapid deterioration and institutionalization and who should be monitored more closely include those over age 80, those who live alone, those who are bereaved or depressed, and those who are intellectually impaired.

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10. PRINCIPLES OF DISEASE PREVENTION - Maureen T. Connelly, Thomas S. Inui

PERSPECTIVES ON PREVENTION

The primary goals of prevention in medicine are to prolong life, to decrease morbidity, and to improve quality of life -- all with the available resources. Working in partnership with patients, physicians play critical roles as educators, managers of access to screening and intervention services, and interpreters of divergent recommendations for promoting health. Despite evidence of the effectiveness of many preventive services in prolonging healthy life and decreasing medical costs, physicians frequently do not integrate appropriate preventive practices into their care. Obstacles to providing optimal preventive care include lack of appropriate training, doubt about the effectiveness of preventive interventions, skepticism about patients' commitment to change, limited reimbursement and time, and conflicting professional recommendations. Success achieved for populations may not be visible to individuals, and physicians may not appreciate the cumulative benefit of their efforts. Despite considerable success in some areas, such as the reduction of smoking by U.S. adults from 40 to 25% in the last 35 years, effective behavior change in other domains is often elusive, challenging and frustrating physicians and patients alike.

DEFINITIONS

This chapter will be devoted to a discussion of *primary* and *secondary* prevention. *Primary prevention*, including various forms of health promotion and vaccination, is care intended to minimize risk factors and the subsequent incidence of disease. *Secondary prevention* is screening for detection of early disease, for example the use of mammography to detect preclinical breast cancer. While the term secondary prevention is also sometimes used for the prevention of recurrent episodes of an existing illness, most would consider this activity to be *tertiary prevention*, care intended to ameliorate the course of established disease.

Deciding what types of primary and secondary preventive care clinicians should offer to their patients is not a trivial matter. The United States Preventive Services Task Force (USPSTF), The Canadian Task Force on the Periodic Health Examination, and the American College of Physicians, among other organizations, have critically reviewed the strength of available evidence for preventive practices and have made recommendations. Adopting an evidence-based approach to the development of preventive practices policy is an essential step to assuaging provider concerns about the validity of particular recommendations, to identifying the specific basis of controversies in prevention, and to reassuring patients that certain interventions will do more good than harm.

PRIMARY PREVENTION

RISK MODIFICATION

Of the more than 2 million deaths that occur in the United States each year, as many as half may be due to preventable causes ([Table 10-1](#)). Life-style and behavior play a

central role in the primary causes of morbidity and mortality for adults -- coronary heart disease, cancer, and injuries.

Tobacco The largest potentially modifiable risk to health is the abuse of tobacco products. Responsible for more than 400,000 deaths each year and an estimated annual cost to society as high as \$50 billion, tobacco abuse accounts for a substantial fraction of cardiovascular, cancer, and pulmonary morbidity and mortality. Recent evidence also suggests that passive exposure to tobacco smoke results in chronic pulmonary disease, cardiovascular disease, and lung cancer for some adults. Because of the addictive properties of nicotine, preventing the initiation of tobacco abuse is the tobacco control intervention of choice. Most adult smokers acquire their habit as teenagers, and primary efforts to discourage initial tobacco use must engage younger audiences. However, smoking cessation extends life, even in individuals who quit after age 65 or those with established disease.

Counseling regarding the health risks of tobacco and methods for quitting is advised by all prevention advisory panels. Because 70% of smokers come into contact with health professionals each year, the medical encounter provides an opportunity to address the health implications of tobacco abuse. Although 70% of smokers say that they want to stop smoking, many are not ready to make an immediate change. The role of the provider is to motivate smokers to attempt cessation, reduce barriers to smoking cessation, and advise effective methods for cessation, including an expanding array of pharmacotherapeutic agents. Ninety percent of successful quitters will stop smoking without the aid of programmatic interventions. Setting a date to quit, arranging follow-up visits or phone calls during the initial quitting period, providing literature, and considering the use of nicotine replacement systems and other effective medications, such as bupropion, are all interventions that may improve the quitting success rate. Compared to placebo, nicotine replacement systems and bupropion have approximately twice the success rate at 6 months.

Diet Mounting evidence suggests that modification of caloric intake, particularly the quality of calories, can result in decreased morbidity and mortality from cardiovascular disease, cancer, and diabetes. Excess weight is an independent risk factor for coronary disease, in addition to its contribution to the incidence of diabetes, hyperlipidemia, and hypertension. Between 20 and 30% of Americans are overweight, defined as 20% above the acceptable body-mass index (kg/m^2), and more than 40% of certain subpopulations, such as black, Native American, and Mexican-American women, are overweight. Despite concern about the risk of weight cycling, the health hazards of obesity appear to outweigh the potential harm of repeated weight loss and gain.

Americans derive excess calories from fats, particularly saturated fats, rather than from more beneficial sources such as complex carbohydrates, monounsaturated fats, and fiber. Since intake of saturated fat correlates with cholesterol level, and coronary heart disease is reduced by 2 to 3% for every 1% reduction in plasma cholesterol level, dietary modification will play a central role in decreasing the primary cause of mortality in America. Excess dietary fat intake has also been associated with breast, colon, prostate, and lung cancer in epidemiologic studies. The once widely accepted goals of reducing calories from all fats to 30% and from saturated fat to 10% have been challenged as the impact of types of fat (not simply fat itself) on morbidity and mortality

is further elucidated. Increasing the intake of dietary fiber, such as from plant, legume, and grain sources, may contribute specifically to a decrease in colon cancer incidence.

Dietary sodium restriction may benefit those who have salt-sensitive hypertension, although the need for such restriction in the general population is unclear. Calcium and vitamin D are protective against osteoporosis, particularly in young women prior to reaching menopause, and evidence suggests that females at all ages have an inadequate intake. Menstruating women are at risk for iron-deficiency anemia. To achieve the recommended daily intake of vitamins and minerals, a varied diet including fish, lean meats, dairy products, whole grains, and five to six servings of fruits and vegetables daily is recommended, rather than the use of vitamin supplements. However, certain nutrients, such as adequate folate to prevent neural tube defects in developing fetuses, are not readily obtained from the typical American diet and may be best found in supplements. While evidence supporting the use of antioxidants such as vitamins E and C is still incomplete, the recommended quantities of these micronutrients can be obtained from a balanced diet.

Alcohol and Drugs The use of alcohol and drugs accounts for more than 100,000 deaths annually. While the ability of health care providers to prevent the initiation of such behaviors has not been proven, screening for exposure and addiction could potentially direct medical effort to the prevention of alcohol and drug-associated problems such as injury, violence, and medical complications of drug abuse. Although instruments such as the CAGE questionnaire have proven to be valuable for detection of alcohol abuse, no comparable brief screening strategy is available for the routine identification of illicit drug abuse. Health care providers screen inadequately for both disorders, despite evidence for effective early treatment of addictions and their complications. Reviewing recent data that moderate alcohol consumption may lower the risk of heart disease may open a discussion with patients about appropriate use. Legal implications of identifying illicit drug use may hinder detection of this problem. When screening for these disorders is feasible, interventions that have proven effective include brief counseling, referral to ambulatory and in-patient treatment programs, use of 12-step and other community organizations, and appropriate use of medications such as methadone for heroin abuse.

Physical Activity Not only can increased physical activity decrease obesity, but avoiding a sedentary life-style can also decrease the incidence of cardiac disease, hypertension, diabetes, and osteoporotic fracture. It is estimated that only 22% of U.S. adults engage in at least light to moderate physical activity, such as walking for 30 min three to five times per week. A full quarter of the population pursues no vigorous physical activity at all. The magnitude of benefit derived from physical activity may be as great as a 35% reduction in coronary heart disease, and even light exercise is preferable to no exercise. At present, the intensity, frequency, duration, and type of physical activity required to achieve optimal cardiac benefit remain unclear. While earlier studies suggested that vigorous exercise was needed to achieve maximal risk reduction, recent studies suggest that regular moderate-intensity activity, such as walking for exercise on most days, is associated with a reduced risk of cardiac events. A sudden onset of vigorous activity in the unfit may increase the risk for myocardial infarction and sudden death. Patients should be informed that, despite previous physical inactivity, the incremental adoption of a regular fitness program can decrease their risk

of cardiovascular and other diseases to the level of those who have remained fit throughout their lives. Successful exercise programs are integrated into daily routines, self-directed, and injury-free.

Sexual Behavior Because of the substantial risks of infectious diseases and unwanted pregnancy from unprotected sexual activity, patients should be strongly advised to use barrier methods for all high-risk practices such as oral, anal, and vaginal intercourse as well as additional contraceptive methods when pregnancy would not be welcome.

Environment Physicians should adopt a broad construction of environmental risks to health, considering the physical, social, and occupational environments of their patients. Taking a complete exposure history, focusing on home, work, neighborhood, hobbies, and dietary habits, can help direct interventions and recommendations. While local circumstances will dictate specific risks to which patients should be alerted, such as regional infectious diseases or particular toxic exposures produced by local industry, certain general recommendations should be adopted universally for health promotion.

Since skin cancers, the vast majority of them secondary to sun exposure, constitute the most common form of malignancy, all patients should be counseled to avoid sun overexposure and to use sunscreens. Patients should be encouraged to consider potential toxin exposures, such as those due to air pollution, household smoking, or carbon monoxide and radon gases, and be informed of the medical symptoms and consequences of such exposures. Proper food preparation and storage decrease the incidence of food-borne infectious disease.

Unintended injury constitutes a significant preventable burden of morbidity and mortality and is the leading cause of death for the general population under 40. Automobile accidents are the leading cause of unintentional injuries. The risk of being involved in a disabling traffic accident may be as high as 30% in the course of an individual's lifetime, and 50% of deaths from automobile accidents could be prevented with regular seatbelt use. Physicians should recommend seatbelt use, as well as helmet use for motorcycle and bicycle riders, since evidence supports a higher likelihood of use among patients who receive such advice. Clinicians should also recommend against operating a motor vehicle after drinking, since alcohol (and illicit drugs) is a clear-cut risk cofactor.

Smoke detectors are underused, being found in only 80% of homes. Since most deaths due to fire occur in the residential setting, patients should be encouraged to install at least one on each floor of their home.

Attention to health hazards in the workplace can identify those at risk and prevent long-term consequences of exposure. Evaluation of the work environment should include questions about exposure to metals, dusts, fibers, chemicals, fumes, radiation, loud noises, extreme temperatures, and biologic agents.

Community and family violence, particularly through the misuse of firearms, is the second leading cause of death from unintentional injury. Firearms, especially handguns, are far more likely to injure a family member than an intruder and are associated with increased rates of suicide and harm to children. Patients should be encouraged to remove their weapons from the home and should be informed of the risks associated

with improper security and storage of firearms. At a minimum, trigger locks may prevent accidental injury from firearms. While community and family violence are epidemic in the United States, interventions to curtail violent behavior are not well established. Screening for exposure to relationship violence, developing plans for safe havens, and referrals to appropriate community and government agencies can prevent continued abuse.

IMMUNIZATION

As many as 70,000 deaths due to influenza, pneumococcal infections, and hepatitis B occur in the United States annually. Despite good availability and evidence for the cost-effectiveness of recommended vaccinations for adults, only 40% or fewer members of target populations are immunized. Factors explaining poor adherence to adult immunization guidelines include lack of confidence in vaccine efficacy among providers and patients, underestimation of the severity of the target diseases, incomplete reimbursement, lack of systems to identify and vaccinate high-risk populations, and the absence of an adult requirement for vaccination equivalent to our vaccination policies for school-age children. [Table 10-2](#) lists recommended adult immunizations.

CHEMOPROPHYLAXIS

There is significant supportive evidence for the use of certain medications in primary prevention. Therapy of this nature in the otherwise healthy person, however, is not risk-free. The use of aspirin for the prevention of cardiovascular disease or colorectal cancer, for example, is supported by evidence from cohort and, in the case of cardiovascular disease, randomized controlled trials. The potential for cerebral bleeds and gastrointestinal intolerance, however, must be balanced against a patient's individual risk for the target diseases. Although no randomized trials have measured the impact on mortality, postmenopausal hormone replacement therapy is another therapy given to healthy women for the prevention of future disease (coronary heart disease and osteoporosis), as well as to control menopausal symptoms. These benefits must be weighed against the risks of possible breast and endometrial carcinoma. Patient involvement in the decision-making process, perhaps even informed consent, is recommended to ensure compliance, proper use of medication, and sustained monitoring for side effects.

SECONDARY PREVENTION

SCREENING

Widespread screening for the presence of existing diseases should meet the following criteria:

1. The targeted disease must be sufficiently burdensome to the population that a screening program is warranted. Minor changes in relative risk should have a substantial impact on the absolute risk within the population.
2. The target disease must have a well-understood natural history with a long preclinical latent period.

3. The screening method must have acceptable technical performance parameters, detecting the disease at an earlier stage than would be possible without screening and minimizing false-positive and false-negative results.
4. Efficacious treatment for the target illness must be available.
5. Early detection must improve disease outcome.
6. Cost, feasibility, and acceptability of screening and early treatment should be established.

While physicians under-provide certain screening services that have met these criteria (for example, regular mammograms for women over age 50 years), it is also the case that some prevalent screening practices today are not solidly rooted in evidence. Screening tests such as mammography in women under 50 and measurement of prostate-specific antigen have been adopted for use by many clinicians despite lack of complete current evidence that these services will decrease the risk of morbidity or mortality or improve the quality of life. See [Table 10-2](#) recommendations of the [USPSTF](#) for screening of adults who are at average risk for target conditions. Recommendations for special-risk and vulnerable populations are available in the *USPSTF Guide*.

COMMUNITY HEALTH ADVOCACY

In addition to the direct clinical provision of preventive and health-promoting services, physicians can bring their knowledge, expertise, clinical experience, and influence to bear at the community level to promote health. Whether arguing for the denormalization of tobacco use or providing data about the health risks of local incinerators, physicians are important sources of information and support for improving health beyond the clinical office. Such activities are consistent with the overall objective of caring for patients and may have a substantial impact on decreasing the prevalence of the root causes of disease.

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11. ALTERNATIVE MEDICINE - Adriane Fugh-Berman

Alternatives to conventional medicine will always exist. Defined by its outsider status, alternative therapies of one era may be conventional therapies in another. Radiation treatment and the use of transcutaneous electrical nerve stimulation (TENS), now widely used in medicine, were once unconventional therapies. Even leeches have made a minor comeback; hirudin, a potent anticoagulant secreted by leeches, has been approved by the U.S. Food and Drug Administration (FDA) as an antithrombotic agent. Alternative medicine is any approach to a health problem that is different from those used by conventional medical practitioners; many alternative therapies complement rather than supplant conventional medicine.

Alternative medicine ranges from systems with distinct disease theories, diagnostic methods, and multiple treatment options (including traditional Chinese medicine and Ayurvedic medicine) to single-component panaceas, such as bee pollen ([Table 11-1](#)). While conventional western medicine is primarily based upon physiology and pathophysiology, alternative therapies may be based on alternative paradigms (e.g., eastern concepts of energy, called "qi" in Chinese medicine or "prana" in Ayurvedic medicine) or may be based on unproven biochemical hypotheses (e.g., high doses of vitamin C). Although many unconventional therapies are not supported by rigorous prospective clinical trials, "alternative" is not synonymous with "unproven." Data from well-designed and well-executed clinical studies support the use of certain alternative medicines in some settings (see below).

The use of unconventional therapies is widespread. In 1997, a telephone survey found that 42% of English-speaking adults in the United States used some sort of alternative therapy. Usage is especially high among people of color, reflecting the fact that, for many, alternative therapies are traditional medicine. Patients who use alternative medicine do so most often at opposite ends of the disease spectrum: either for symptom relief in mild or chronic illnesses or for life-threatening conditions (typically as adjuncts to conventional medicine).

PLACEBO OR NONSPECIFIC EFFECTS

To what extent does the *placebo effect* explain the popularity of alternative medicine? Now more commonly called *nonspecific effects*, this phenomenon encompasses numerous factors including the environment, the relationship between the patient and the practitioner, beliefs and expectations (of both the patient and practitioner), natural history of the disease, and individual variability. The nature of the placebo effect is itself under investigation.

It may be assumed that nonspecific effects enhance positive outcomes of both unconventional and conventional therapies. However, scientific evaluation must be the primary determinant of the physician's attitude toward alternative therapies.

HERBAL MEDICINE

Herbalism (also called *phytomedicine*, *phytotherapy*, or *botanical medicine*) is the medicinal use of plants or plant constituents. The use of plants as medicine predates

even human evolution; great apes have been noted to consume specific medicinal plants when they are ill. Many of the drugs in clinical practice are derived from plants, as are the majority of analgesics: lidocaine and novocaine from the coca plant (*Erythroxylum coca*); opioids from the poppy (*Papaver somniferum*); aspirin from meadowsweet (*Spirea ulmaria*), whence the "spir" part of its name derives. The progestin component of oral contraceptives comes from the Mexican yam (*Dioscorea villosa*); digoxin comes from foxglove (*Digitalis lanata*); cromolyn sodium is a khellin derivative from the Ayurvedic herb *Ammi visnaga*; and warfarin is a derivative of dicoumarin, from sweet clover (*Melilotus officinalis*). The ipecac that is kept in the medicine cabinet for poisonings comes from the root of a South American shrub (*Cephaelis ipecacuanha*); the benzoin that attaches bandages to skin is a gum resin from *Styrax benzoin*; and the witch hazel used to soothe hemorrhoids is an extract of *Hamamelis virginiana*. Fungi also have contributed to pharmaceuticals: penicillin was isolated from the fungus *Penicillium notatum*, cephalosporins were derived from a marine fungus (*Cephalosporium acremonium*), and lovastatin was derived from the fungus *Aspergillus terreus*.

Essentially every culture has a tradition of herbal medicine. In western herbalism, herbs are often used singly but may sometimes be combined. Chinese herbal medicine utilizes complex mixtures of many herbs, sometimes combined with animal materials. Herbs may be used to treat or prevent disease. As preventives, "tonics" support the function of specific organs, and "adaptogens" are nonspecific treatments that facilitate a return to homeostasis.

Some clinical trial data support the use of Saint John's wort (SJW) for depression, kava for anxiety, saw palmetto for benign prostatic hyperplasia (BPH), and ginkgo for increasing cerebral blood flow. Evidence also supports the use of garlic for lowering cholesterol, hawthorn to improve cardiac function, and echinacea for treating (but not preventing) upper respiratory infections.

Saint John's Wort (*Hypericum perforatum*) A meta-analysis of the use of [SJW](#) for mild to moderate depression examined 23 randomized trials (20 were double-blind) with a total of 1757 outpatients. In 15 placebo-controlled trials, SJW was found to be significantly more effective than placebo. In eight treatment-controlled trials, clinical improvement in those receiving SJW did not differ significantly from those receiving tricyclic antidepressants. The trials in this meta-analysis were heterogeneous and used varying diagnostic criteria and dosages of SJW.

Most clinical trials have been done with extracts of the flowering tops standardized to 0.3% hypericin (long thought to be the most active compound in [SJW](#)). However, hyperforin is most likely the prime active agent, and extracts standardized to 3% hyperforin have been tested in clinical trials and are available. The usual dose of both of these preparations is 300 mg tid.

In vitro, a high concentration of [SJW](#) inhibits the uptake of serotonin, norepinephrine, and dopamine. Among neurotransmitters, its strongest binding affinity is to gamma-aminobutyric acid (GABA) A and B receptors. Although SJW demonstrates monoamine oxidase (MAO) inhibition in vitro, this effect has not been demonstrated in vivo, nor have there been any reported cases of MAO inhibitor-associated hypertensive crises in humans.

using SJW (the herb may, however, potentiate serotonin reuptake inhibitors). Side effects of SJW include gastrointestinal symptoms, fatigue, and photosensitization.

Ginkgo (*Ginkgo biloba*) Ginkgo leaf extracts have been proposed for treating Alzheimer's or multi-infarct dementia. In several randomized controlled trials, a small but statistically significant effect after 3 to 6 months of treatment with 40 to 80 mg tid of standardized *G. biloba* extract (containing 22 to 27% flavonoid glycosides and 5 to 7% terpene lactones) was noted on objective measures of cognitive function in patients with Alzheimer's disease. Ginkgo appears to have vasoregulatory and antioxidant effects as well as inhibiting platelet-activating factor (PAF). Serious intracerebral bleeding associated with ginkgo use has been reported, including two subdural hematomas, one intracerebral hemorrhage, one subarachnoid hemorrhage, and one case of spontaneous hyphema. In most cases, these patients were receiving concurrent anticoagulant drugs.

Kava (*Piper methysticum*) Used in Polynesia as a ceremonial beverage, the roots and rhizomes of kava are used medicinally for anxiety and insomnia in Europe and in North America. Several placebo-controlled trials have shown significant anxiolytic activity of kava products (standardized to 70% kavalactones), usually in a dose of 70 mg tid. *Kavalactones* (also called *kavapyrones*) are muscle relaxants; they include kawain, dihydrokawain, methysticin, and dihydromethysticin (the latter two are potent inhibitors of norepinephrine uptake).

Therapeutic doses may result in mild gastrointestinal complaints or allergic skin reactions (incidence ~1.5%). Chronic use of high-dose kava may result in *kava dermatopathy*, a reversible ichthyosiform eruption that is often accompanied by eye irritation.

Ginseng (*Panax ginseng* and other *Panax* species) Ginseng is a popular herb in both western and eastern medicine. The place of ginseng root in the treatment of specific conditions remains to be shown in clinical trials. Ginseng contains ginsenosides, polyacetylenes, and sesquiterpenes. It appears to have some glucocorticoid-like actions and hypoglycemic activity and also affects neurotransmitter activity. Glucocorticoid administration blocks the effect of ginsenosides both in vitro and in vivo; ginsenosides increase adrenal cyclic AMP in intact but not hypophysectomized rats, so effects on adrenal secretion appear to be through the pituitary gland. Several cases of postmenopausal uterine bleeding have been reported from ginseng use, although ginseng does not contain known phytoestrogens.

Saw Palmetto (*Serenoa repens*) Saw palmetto fruits were used as food by Native Americans; their most common use today is to treat [BPH](#). A systemic review of randomized controlled trials of extracts of the fruit *S. repens* (alone or in combination with other herbs) identified 18 randomized trials (16 double-blind) that included 2939 men and lasted 4 to 48 weeks. Compared with placebo, *S. repens* improved urinary symptom scores, nocturia, and peak urine flow. In two studies that compared finasteride with *S. repens*, improvements in urinary symptoms scores were similar. Adverse effects due to *S. repens* were mild and infrequent.

Saw palmetto is usually administered in liposterolic extracts standardized to 70 to 95%

free fatty acids; the usual dose is 160 mg bid. Saw palmetto appears to have multiple mechanisms of action including inhibition of 5 α -reductase and inhibition of dihydrotestosterone binding to cytosolic androgen receptors.

Pollen/Bee Pollen Pollen may be collected directly from plants or their pollinators; bee pollen is flower pollen collected from bees. A double-blind placebo-controlled trial of a mixed-pollen extract in 60 patients with [BPH](#) found that subjective improvement was significantly better in the treated group; there was a significant decrease in residual urine and in diameter of the prostate on ultrasound. However, flow rate and volume were unchanged. Allergic reactions, hypereosinophilia, and eosinophilic gastroenteritis have been associated with bee pollen intake.

Echinacea Species Echinacea roots are used to treat or prevent infections. The three species used commercially are *Echinacea purpurea*, *E. angustifolia*, and *E. pallida*. A systemic review of 16 trials (8 prevention and 8 treatment trials on upper respiratory tract infections) with a total of 3396 participants found a wide variation in preparations and methodologic quality of trials. Although many available studies reported positive results of echinacea compared to placebo, reviewers concluded that the evidence is not strong enough to recommend a specific dose, product, or preparation.

Immunomodulatory effects are attributed to five classes of compounds in echinacea preparations: caffeic acid derivatives, alkylamides, polyacetylenes, glycoproteins, and polysaccharides. The alkylamides are regarded as the most active chemical constituent. Echinacea stimulates both humoral and cellular immunity; theoretically, it could worsen symptoms in atopic individuals or those with autoimmune disease.

Adverse Effects of Herbs Herbs have pharmacologic effects and can be associated with adverse effects or interactions. Many medicinal herbs (and pharmaceutical drugs) are therapeutic at one dose and toxic at another. The relative dearth of reports of adverse events and interactions attributed to herbal products reflects a combination of underreporting and the relatively nontoxic nature of most herbal usage.

The most dangerous plants used medicinally are aconite and any herbs containing unsaturated pyrrolizidine alkaloids (saturated pyrrolizidine alkaloids lack toxicity). Several herbs that do not contain pyrrolizidine alkaloids have also shown hepatotoxicity.

Aconite (*Aconitum* spp.), sometimes used in Chinese herb mixtures to treat pain or heart failure, contains aconitine and other C₁₉diterpenoid alkaloids. Proper curing of aconite reduces alkaloids by 90%, but even appropriately cured aconite can result in serious, sometimes fatal, cardiac arrhythmias. The first symptoms of aconite poisoning occur within 90 min of ingestion; the majority of patients present with neurologic symptoms (most commonly oral numbness or burning), progressing to peripheral paresthesia and generalized muscle weakness. Nausea and vomiting are also common.

Cardiovascular effects include bradycardia, hypotension, and arrhythmias (including ventricular or supraventricular tachycardia, bidirectional tachycardia, sinus bradycardia with first-degree heart block, bundle branch block with junctional escape rhythm, or torsade de pointes). Other symptoms may include chest pain, abdominal pain, diarrhea, hyperventilation, respiratory distress, dizziness, sweating, confusion, headache, and

excessive lacrimation. No specific antidote for aconite is known, and treatment is mainly supportive. Atropine may be given if symptoms of cholinergic excess are apparent. Antiarrhythmics are often helpful, but characteristically, electrical cardioversion is markedly unsuccessful in aconite poisoning.

Unsaturated pyrrolizidine alkaloids are hepatotoxic; children may be especially sensitive. Unsaturated pyrrolizidine alkaloids occur in comfrey (*Symphytum*), borage (*Borago officinalis*) leaf (seed oils are safe), coltsfoot (*Tussilago farfara*), and species of *Crotalaria* and *Senecio*. Liver toxicity has also been associated with chaparral (*Larrea divaricata*), germander (*Teucrium chamaedrys*), and a Chinese medicine called *jin bu huan* (which contains 36% levo-tetrahydropalmitine, a chemical present in *Stephania* and *Corydalis* genera).

Drug Interactions One of the most serious herb-drug interactions is increased risk of bleeding when warfarin is combined with anticoagulant herbs: cases of bleeding have been reported with ginkgo (*G. biloba*), garlic (*Allium sativum*), and the Chinese herbs danshen (*Salvia miltiorrhiza*) and dong quai (*Angelica sinensis*). The soluble fibers guar gum and psyllium can slow or reduce the absorption of many drugs, and anthranoid-containing laxatives, including senna (*Cassia senna* and *C. angustifolia*) and cascara sagrada (*Rhamnus purshiana*), can also reduce the absorption of many drugs. An Ayurvedic syrup, shankhapushpi, has been associated with reduced levels of phenytoin. Licorice (*Glycyrrhiza glabra*) can potentiate both oral and topical glucocorticoids.

Several herbs can interact with psychotropic drugs; the herb yohimbe (*Pausinystalia yohimbe*) (also available as the drug yohimbine; both forms are used to treat impotence) increases the risk of hypertension when combined with tricyclic antidepressants. Extrapyramidal effects have occurred in patients ingesting neuroleptics and betel nut (*Areca catechu*); mania has been induced in depressed patients who mix antidepressants and *P. ginseng*; and [SJW](#) (*H. perforatum*) combined with a serotonin reuptake inhibitor may produce a mild "serotonin syndrome" (with symptoms of nausea, vomiting, and confusion); the full-blown syndrome may include myoclonus, agitation, fever, abdominal cramping, and hypertension ([Chap. 385](#)).

Adulterants and Contaminants Herbal products may be contaminated, mislabeled, or contain misidentified plants. Medicinal plants from India and Sri Lanka can be contaminated with toxigenic fungi, including *Aspergillus* and *Fusarium*. Heavy metals have been detected in some Asian herbal products (metals are sometimes deliberately used in the preparation of Ayurvedic herbal medicines). Without any mention on the label, pharmaceutical drugs may also be incorporated into herbal products, a particular problem in Chinese herbal preparations imported from Hong Kong and Taiwan. Nonsteroidal anti-inflammatory drugs and benzodiazepines have been found in Chinese herbal products, including Miracle Herb, Tung Shueh, and Chuifong Toukuwan (since 1974 this notorious brand has incorporated at least 10 different drugs into the preparation). The absence of standard manufacturing practices creates risks that are difficult to quantify.

ACUPUNCTURE

Acupuncture was recognized in western medical texts a century ago; Sir William Osler's *Principles and Practice of Medicine*, first published in 1892, recommended acupuncture for both sciatica and lumbago; and the 1901 edition of *Gray's Anatomy* noted the use of acupuncture for sciatica.

Stimulation of acupuncture points may be done by needles, finger pressure, electrical stimulation, or heat (usually applied by a smoldering cone or rod of "moxa," made of the herb mugwort, *Artemisia vulgaris*). Acupuncture is effective in the treatment of nausea and vomiting. In 27 of 33 controlled trials, superiority of acupuncture point stimulation over placebo for nausea and vomiting of various etiologies was demonstrated. In substance abusers, the therapy may reduce withdrawal symptoms, but evidence is lacking about whether acupuncture has any long-term effect in preventing recidivism. An analysis of 16 randomized controlled trials of acupuncture for smoking cessation showed no beneficial effect of acupuncture over sham or no treatment. Limited preliminary data suggest a possible beneficial effect for acupuncture in stroke rehabilitation. Although acupuncture is known to stimulate endorphin release, and its use for pain is better accepted than for other conditions, controlled clinical trials of pain treatment have had mixed results.

Risks Inadequately sterilized acupuncture needles have been linked to infections, including HIV infection and an epidemic of hepatitis B. Two cases of fatal *Staphylococcus* sepsis have been reported. More than 100 cases of pneumothorax have been reported. Rare cases both of spinal trauma and cardiac tamponade (caused by penetration of a congenital sternal foramen) have been reported.

HOMEOPATHY

Originated in the early nineteenth century by Samuel Hahnemann, a German physician, homeopathy is based on the "doctrine of similars"; animal, vegetable, or mineral substances that cause symptoms in a well person are used to treat those same symptoms in a sick person. For example, poison ivy (*Rhus toxicodendron*) is used to treat varicella (chickenpox). Because conventional treatments aim to counter rather than reproduce symptoms, practitioners of homeopathy refer to conventional medicine as "allopathy."

Usually, remedies are used in highly dilute concentrations, and homeopaths believe that the most dilute remedies are the most potent. If analyzed chemically, many homeopathic remedies contain no detectable levels of the original substance. It is difficult to conceive of a scientifically testable hypothesis that could explain the putative effects of homeopathic medicine where a preparation containing few or no molecules of an active agent are said to have pharmacologic effects. A meta-analysis of 89 placebo-controlled trials of homeopathy found that the odds ratio was 2.45 (CI, 2.05 to 2.93) in favor of homeopathy over placebo. The quality of these studies was not uniform, and the studies with the best methodologic quality yielded significantly less positive results.

Risks A case of pancreatitis associated with intake of homeopathic medication has been reported. Potentially toxic levels of arsenic and cadmium have been found in "low potency" (less dilute) homeopathic preparations.

SPINAL MANIPULATION

Therapeutic manipulation of the body has ancient roots; Hippocrates, Aesculapius, and Galen all used some form of it. A physician, Andrew Taylor Still, originated osteopathy in 1892. Daniel David Palmer invented chiropractic in 1895. A meta-analysis of nine methodologically acceptable studies of spinal manipulation for low-back pain found a definite improvement at 3 weeks for patients with uncomplicated, acute back pain. For patients with chronic pain or sciatic nerve irritation, chiropractic was not helpful.

A meta-analysis of cervical manipulation for neck pain concluded that manipulation, in combination with other treatment, may produce short-term pain relief.

Risks Complications of spinal manipulation include vertebrobasilar accidents, disc herniations, vertebral fracture, spinal cord compression, and cauda equina syndrome. More than 80% of serious complications from chiropractic occur after cervical manipulation. It is impossible to determine the true rate of complications from chiropractic manipulations, but estimates vary from 1 in 400,000 to between 3 and 6 per 10 million. The incidence of cauda equina syndrome is thought to be less than 1 per 10 million manipulations.

MASSAGE

Several studies support the use of massage for reducing lymphedema; the technique matches the effectiveness of uniform-pressure pneumatic devices.

Numerous studies in hospital settings describe the use of infant massage to decrease hospital stays of premature babies. A meta-analysis of randomized trials found that massage interventions improved daily weight gain by 5 g, while gentle, still touch did not show a benefit. Methodologic concerns about the blinding undermine the conclusions.

Risks Massage has been used in an effort to prevent pressure sores, but it is not clearly effective and may increase tissue trauma when done over bony prominences.

MIND/BODY THERAPIES

Biofeedback Biofeedback uses instruments to translate information on physiologic function into audio or visual signals that patients use as cues to help them learn to affect functions not normally thought to be under voluntary control. Commonly used modalities include electromyographic (EMG) feedback of skeletal muscle contraction, thermal feedback of skin temperature (an indirect measure of peripheral blood flow), electroencephalographic (EEG) feedback, electrodermal response (EDR) (feedback of sweat gland activity on the fingers), and perineometry (feedback of contraction of pelvic floor muscles and anal sphincter).

Biofeedback treatment modalities may be combined. For example, for urinary incontinence, biofeedback may be used to measure pelvic muscle activity through urethral sphincter pressure and electromyography, circumvaginal muscle manometry and electromyography, and anorectal manometry and electromyography. Detrusor

pressure feedback may be measured by cystometry, and feedback on intraabdominal pressure may be used to help patients learn to simultaneously contract pelvic muscles while relaxing abdominal muscles (to avoid putting excess pressure on the bladder). Clinical trials support the use of biofeedback in the treatment of urinary incontinence (stress, urge, or mixed), fecal incontinence, migraine, tension headaches, and in stroke rehabilitation.

Hypnosis Traditional hypnosis utilizes the induction of a deep trance state to enhance suggestibility. Several clinical trials indicate that hypnosis is effective in chemotherapy-associated nausea and may be helpful in the treatment of irritable bowel syndrome and pain syndromes. Numerous uncontrolled trials suggest a beneficial effect of hypnosis on smoking cessation, but controlled trials are less impressive. A meta-analysis of hypnosis in nine randomized controlled trials of hypnotherapy found significant heterogeneity among the results of the individual studies, with conflicting results for the effectiveness of hypnotherapy compared to no treatment or to advice. Hypnotherapy was not more effective than rapid smoking (an aversive therapy in which cigarettes are smoked in quick succession) or psychological treatment.

DIETARY SUPPLEMENTS

The fundamental principles of good nutrition are to eat a balanced variety of foods and to maintain a balance of calories taken in with calories burned through activity. Dietary supplements have become a large and lucrative business that promotes the idea that our food is somehow lacking in specific nutrients and that additional intake of any number of food components will treat or prevent specific diseases, improve athletic and sexual performance, and make us live longer. These claims are largely either untested or unproven ([Table 11-2](#)). Unfortunately, the *Dietary Supplement Health and Education Act* (DSHEA) of 1994 prevents the [FDA](#) from monitoring the quality or safety of dietary supplements before marketing. Supplements do not even have to be dietary components to be sold as dietary supplements (the availability of over-the-counter hormones, including DHEA, progesterone topical creams, and organ extracts is particularly worrisome).

Even vitamin and mineral supplementation may have unexpected results. For example, high dietary consumption of carrots, sweet potatoes, greens, and other foods rich in β -carotene is associated with decreased risk of cardiovascular disease and cancer. However, two large prospective randomized controlled trials of β -carotene [the Alpha Tocopherol Beta Carotene (ATBC) Cancer Prevention Study and the Beta-Carotene and Retinal Efficacy Trial (CARET)] found that β -carotene increased rates of lung cancer in supplemented groups. The ATBC trial also failed to show a benefit in terms of cardiovascular disease, and β -carotene supplementation has been disappointing in other trials. β -carotene may well be a dietary marker for more beneficial carotenoids (including lycopene, lutein, α -carotene, and β -cryptoxanthin) that typically occur in mixtures in foods.

Similarly, while dietary intake of vitamin E is associated with a reduced risk of coronary heart disease and several observational studies (including the Health Professionals Follow-Up Study and the Nurses Health Study) found a protective effect of supplemental vitamin E, prospective placebo-controlled trials have had mixed results. In the

Cambridge Heart Antioxidant Study (CHAOS), a placebo-controlled trial of vitamin E supplementation in patients with coronary artery disease, vitamin E supplementation reduced nonfatal myocardial infarction but did not appear to decrease cardiovascular deaths or all-cause mortality. In the [ATBC](#) study, vitamin E failed to protect male smokers with a previous myocardial infarction from major coronary events and significantly increased risk of death from hemorrhagic stroke. Although vitamin E improves endothelium-dependent vasodilation significantly and attenuates the development of nitrate tolerance, prospective trials have found no benefit of vitamin E for angina.

Vitamin E is not a single compound; the term applies to eight related compounds in two groups, the tocopherols and the tocotrienols. Most North American dietary sources of vitamin E are composed of two-thirds γ -tocopherol and one-third α -tocopherol. γ -Tocopherol neutralizes both oxygen and nitrogen free radicals, while α -tocopherol is selective for oxygen free radicals. Most vitamin E supplements, however, contain only D- or D,L- α -tocopherol, and large doses of α -tocopherol displace γ -tocopherol in plasma and tissues.

While it is possible that tomorrow's trendy carotenoid or tocopherol will be more successful than yesterday's, it is more likely that carotenoids and tocopherols are most beneficial when consumed in the naturally occurring mixtures of nutrients found in foods.

Dietary supplements that are safe under one set of conditions may be harmful under other conditions. For example, nearly all physiologic reactions that generate free radicals are oxidation-reduction reactions, capable of either generating free radicals or reducing free radicals. While low doses of vitamin C and other substances have antioxidant effects, high levels may actually have a prooxidant effect.

There is clinical trial evidence supporting the supplemental use of some vitamins, minerals, and amino acids ([Table 11-2](#)); but at high doses these supplements should be treated with the same respect given pharmaceuticals.

SUMMARY

The incorporation of science into medicine began only in the middle of the nineteenth century; *evidence-based medicine* is a recent term and still not necessarily standard practice. There is no such thing as objective care of a patient. There are many ways in which we physicians communicate goals and our own beliefs in our therapies; it is likely that a substantial proportion of benefit from even rational interventions is due to nonspecific effects.

Many of our patients take alternative medicines; physicians need to adopt an open-minded, nonjudgmental attitude toward the practice. Inquire about all the medications and supplements a patient is taking. Increasingly, data are available to help guide decision-making about alternative medicines. A thorough knowledge of all therapies a patient is utilizing may help explain unexpected findings and is an important component of a holistic approach to patient care.

Conventional medicine often casts a wary eye on therapies outside its boundaries, but it

is incumbent upon physicians to evaluate evidence regarding alternative therapies with the same rigor with which we evaluate conventional therapies. Scientific evidence from controlled clinical trials supports specific applications of alternative medicine, and some therapies considered alternative today may well be incorporated into conventional medicine in the future.

(Bibliography omitted in Palm version)

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PART TWO -CARDINAL MANIFESTATIONS AND PRESENTATION OF DISEASE

SECTION 1 -PAIN

12. PAIN: PATHOPHYSIOLOGY AND MANAGEMENT - Howard L. Fields, Joseph B. Martin

The task of medicine is to preserve and restore health and to relieve suffering. Understanding pain is essential to both these goals. Because pain is universally understood as a signal of disease, it is the most common symptom that brings a patient to a physician's attention. The function of the pain sensory system is to detect, localize, and identify tissue-damaging processes. Since different diseases produce characteristic patterns of tissue damage, the quality, time course, and location of a patient's pain complaint and the location of tenderness provide important diagnostic clues and are used to evaluate the response to treatment.

THE PAIN SENSORY SYSTEM

Pain is an unpleasant sensation localized to a part of the body. It is often described in terms of a penetrating or tissue-destructive process (e.g., stabbing, burning, twisting, tearing, squeezing) and/or of a bodily or emotional reaction (e.g., terrifying, nauseating, sickening). Furthermore, any pain of moderate or higher intensity is accompanied by anxiety and the urge to escape or terminate the feeling. These properties illustrate the duality of pain: it is both sensation and emotion. When acute, pain is characteristically associated with behavioral arousal and a stress response consisting of increased blood pressure, heart rate, pupil diameter, and plasma cortisol levels. In addition, local muscle contraction (e.g., limb flexion, abdominal wall rigidity) is often present.

THE PRIMARY AFFERENT NOCICEPTOR

A peripheral nerve consists of the axons of three different types of neurons: primary sensory afferents, motor neurons, and sympathetic postganglionic neurons ([Fig. 12-1](#)). The cell bodies of primary afferents are located in the dorsal root ganglia in the vertebral foramina. The primary afferent axon bifurcates to send one process into the spinal cord and the other to innervate tissues. Primary afferents are classified by their diameter, degree of myelination, and conduction velocity. The largest-diameter fibers, A-beta (Ab), respond maximally to light touch and/or moving stimuli; they are present primarily in nerves that innervate the skin. In normal individuals, the activity of these fibers does not produce pain. There are two other classes of primary afferents: the small-diameter myelinated A-delta (Ad) and the unmyelinated (C fiber) axons ([Fig. 12-1](#)). These fibers are present in nerves to the skin and to deep somatic and visceral structures. Some tissues, such as the cornea, are innervated only by Ad and C afferents. Most Ad and C afferents respond maximally only to intense (painful) stimuli and produce the subjective experience of pain when they are electrically stimulated; this defines them as *primary afferent nociceptors (pain receptors)*. The ability to detect painful stimuli is completely abolished when Ad and C axons are blocked.

Individual primary afferent nociceptors can respond to several different types of noxious stimuli. For example, most nociceptors respond to heating, intense mechanical stimuli

such as a pinch, and application of irritating chemicals.

Sensitization When intense, repeated, or prolonged stimuli are applied in the presence of damaged tissue or inflammation, the threshold for activating primary afferent nociceptors is lowered and the frequency of firing is higher for all stimulus intensities. Inflammatory mediators such as bradykinin, some prostaglandins, and leukotrienes contribute to this process, which is called *sensitization*. In sensitized tissues normally innocuous stimuli can produce pain. Sensitization is a clinically important process that contributes to tenderness, soreness, and hyperalgesia. A striking example of sensitization is sunburned skin, in which severe pain can be produced by a gentle slap on the back or a warm shower.

Under normal conditions, viscera are relatively insensitive to noxious mechanical and thermal stimuli. Hollow viscera do generate significant discomfort when distended. Furthermore, when affected by a disease process with an inflammatory component, deep structures such as joints or hollow viscera characteristically become exquisitely sensitive to mechanical stimulation.

A large proportion of Ad and C afferents innervating viscera are completely insensitive in normal noninjured, noninflamed tissue. That is, they cannot be activated by known mechanical or thermal stimuli and are not spontaneously active. However, in the presence of inflammatory mediators, these afferents become sensitive to mechanical stimuli. Such afferents have been termed *silent nociceptors*, and their characteristic properties may explain how under pathologic conditions the relatively insensitive deep structures can become the source of severe and debilitating pain and tenderness.

Nociceptor-Induced Inflammation One important concept to emerge in recent years is that afferent nociceptors also have a neuroeffector function. Most nociceptors contain polypeptide mediators that are released from their peripheral terminals when they are activated ([Fig. 12-2](#)). An example is substance P, an 11-amino-acid peptide. Substance P is released from primary afferent nociceptors and has multiple biologic activities. It is a potent vasodilator, degranulates mast cells, is a chemoattractant for leukocytes, and increases the production and release of inflammatory mediators. Interestingly, depletion of substance P from joints reduces the severity of experimental arthritis. Primary afferent nociceptors are not simply passive messengers of threats to tissue injury but also play an active role in tissue protection through these neuroeffector functions.

CENTRAL PATHWAYS FOR PAIN

The Spinal Cord and Referred Pain The axons of primary afferent nociceptors enter the spinal cord via the dorsal root. They terminate in the dorsal horn of the spinal gray matter ([Fig. 12-3](#)). The terminals of primary afferent axons contact spinal neurons that transmit the pain signal to brain sites involved in pain perception. The axon of each primary afferent contacts many spinal neurons, and each spinal neuron receives convergent inputs from many primary afferents.

From a clinical standpoint, the convergence of many sensory inputs to a single spinal pain-transmission neuron is of great importance because it underlies the phenomenon of referred pain. All spinal neurons that receive input from the viscera and deep

musculoskeletal structures also receive input from the skin. The convergence patterns are determined by the spinal segment of the dorsal root ganglion that supplies the afferent innervation of a structure. For example, the afferents that supply the central diaphragm are derived from the third and fourth cervical dorsal root ganglia. Primary afferents with cell bodies in these same ganglia supply the skin of the shoulder and lower neck. Thus sensory inputs from both the shoulder skin and the central diaphragm converge on pain-transmission neurons in the third and fourth cervical spinal segments. *Because of this convergence and the fact that the spinal neurons are most often activated by inputs from the skin, activity evoked in spinal neurons by input from deep structures is mislocalized by the patient to a place that is roughly coextensive with the region of skin innervated by the same spinal segment.* Thus inflammation near the central diaphragm is usually reported as discomfort near the shoulder. This spatial displacement of pain sensation from the site of the injury that produces it is known as *referred pain*.

Ascending Pathways for Pain A majority of spinal neurons contacted by primary afferent nociceptors send their axons to the contralateral thalamus. These axons form the contralateral spinothalamic tract which lies in the anterolateral white matter of the spinal cord, the lateral edge of the medulla, and the lateral pons and midbrain. The spinothalamic pathway is crucial for pain sensation in humans. Interruption of this pathway produces permanent deficits in pain and temperature discrimination.

Spinothalamic tract axons connect to thalamic neurons that project to somatosensory cortex ([Fig. 12-4](#)). This pathway from spinal cord to thalamus to somatosensory cortex appears to be particularly important for the sensory aspects of pain, i.e., its location, intensity, and quality. Spinothalamic tract axons also connect to thalamic and cortical regions linked to emotional responses, such as the cingulate gyrus and frontal lobe. This pathway is thought to subserve the affective or unpleasant emotional dimension of pain.

PAIN MODULATION

The pain produced by similar injuries is remarkably variable in different situations and in different people. For example, athletes have been known to sustain serious fractures with only minor pain, and Beecher's classic World War II survey revealed that many men were unbothered by battle injuries that would have produced agonizing pain in civilian patients. Furthermore, even the suggestion of relief can have a significant analgesic effect (placebo). On the other hand, many patients find even minor injuries (such as venipuncture) unbearable, and the expectation of pain has been demonstrated to induce pain *without a noxious stimulus*.

The powerful effect of expectation and other psychological variables on the perceived intensity of pain implies the existence of brain circuits that can modulate the activity of the pain-transmission pathways. Although there are probably several circuits that can modulate pain, only one has been studied extensively. This circuit has links in the hypothalamus, midbrain, and medulla, and it selectively controls spinal pain-transmission neurons through a descending pathway ([Fig. 12-4](#)).

There is good evidence that this pain-modulating circuit contributes to the pain-relieving

effect of opioid analgesic medications. Each of the component structures of the pathway contains opioid receptors and is sensitive to the direct application of opioid drugs. Furthermore, lesions of the system reduce the analgesic effect of systemically administered opioids such as morphine. Along with the opioid receptor, the component nuclei of this pain-modulating circuit contain endogenous opioid peptides such as the enkephalins and δ -endorphin.

The most reliable way to activate this endogenous opioid-mediated modulating system is by prolonged pain and/or fear. There is evidence that pain-relieving endogenous opioids are released following operative procedures and in patients given a placebo for pain relief.

Pain modulation is bidirectional. Pain-modulating circuits not only produce analgesia but are also capable of increasing pain. Both pain-inhibiting and pain-facilitating neurons in the medulla project to and control spinal pain-transmission neurons. Since pain-transmission neurons can be activated by modulatory neurons, it is theoretically possible to generate a pain signal with no peripheral noxious stimulus. Some such mechanism could account for the finding that pain can be induced by suggestion alone and may provide a framework for understanding how psychological factors can contribute to chronic pain.

NEUROPATHIC PAIN

The normal nervous system transmits coded signals that result in pain. Thus lesions of the peripheral or central nervous system may result in a loss or impairment of pain sensation. Paradoxically, damage or dysfunction of the nervous system can produce pain. For example, damage to peripheral nerves, as occurs in diabetic neuropathy, or to primary afferents, as in herpes zoster, can result in pain that is referred to the body region innervated by the damaged nerves. Though rare, pain may also be produced by damage to the central nervous system, particularly the spinothalamic pathway or thalamus. Such neuropathic pains are often severe and are notoriously intractable to standard treatments for pain.

Neuropathic pains typically have an unusual burning, tingling, or electric shock-like quality and may be triggered by very light touch. These features are rare in other types of pain. On examination, a sensory deficit is characteristically present in the area of the patient's pain.

A variety of mechanisms contribute to neuropathic pain. As with sensitized primary afferent nociceptors, damaged primary afferents, including nociceptors, become highly sensitive to mechanical stimulation and begin to generate impulses in the absence of stimulation. There is evidence that this increased sensitivity and spontaneous activity is due to an increased concentration of sodium channels. Damaged primary afferents may also develop sensitivity to norepinephrine. Interestingly, spinal pain-transmission neurons cut off from their normal input may also become spontaneously active. Thus both central and peripheral nervous system changes may contribute to neuropathic pain.

Sympathetically Maintained Pain A certain percentage of patients with peripheral

nerve injury develop a severe burning pain (causalgia) in the region innervated by the nerve. The pain typically begins after a delay of hours to days or even weeks. The pain is accompanied by swelling of the extremity, periarticular osteoporosis, and arthritic changes in the distal joints. A similar syndrome called *reflex sympathetic dystrophy* can be produced without obvious nerve damage by a variety of injuries, including fractures of bone, soft tissue trauma, myocardial infarction, and stroke ([Chap. 366](#)). Although the pathophysiology of this condition is poorly understood, the pain can be relieved within minutes by blocking the sympathetic nervous system. This implies that sympathetic activity activates nociceptors even if they are not obviously damaged. These results also suggest that the sympathetic nervous system can, under some circumstances, play an active role in inflammation.

TREATMENT

The ideal treatment for any pain is to remove the cause. Sometimes this is possible, but more often after diagnosis and initiation of appropriate treatments for the cause, there is a lag period before the pain subsides. Furthermore, some conditions are so painful that rapid and effective analgesia is essential (e.g., the postoperative state, burns, trauma, cancer, sickle cell crisis). Analgesic medications are a first line of treatment in these cases, and their use should be familiar to all practitioners.

Aspirin, Acetaminophen, and Nonsteroidal Anti-Inflammatory Agents (NSAIDs)

These drugs are considered together because they are used for similar problems and may have a similar mechanism of action ([Table 12-1](#)). All these compounds inhibit cyclooxygenase (COX), and, except for acetaminophen, all have anti-inflammatory actions, especially at higher dosages. They are particularly effective for mild to moderate headache and for pain of musculoskeletal origin.

Since they are effective for these common types of pains and are available without prescription, [COX](#) inhibitors are by far the most commonly used analgesics. They are absorbed well from the gastrointestinal tract and, with occasional use, side effects are minimal. With chronic use, gastric irritation is a common side effect of aspirin and [NSAIDs](#) and is the problem that most frequently limits the dose that can be given. Gastric irritation is most severe with aspirin, which may cause erosion of the gastric mucosa, and because aspirin irreversibly acetylates platelets and interferes with coagulation of the blood, gastrointestinal bleeding is a risk. The NSAIDs are less problematic in this regard. Although toxic to the liver when taken in a high dose, acetaminophen rarely produces gastric irritation and does not interfere with platelet function. [Table 12-1](#) lists the dosages and durations of action of the commonly used drugs of this class.

The introduction of a parenteral form of [NSAID](#), ketorolac, extends the usefulness of this class of compounds in the management of acute severe pain. Ketorolac is sufficiently potent and rapid in onset to supplant opioids for many patients with acute severe headache and musculoskeletal pain.

There are two major classes of [COX](#): COX 1 is constitutively expressed, and COX 2 is induced in the inflammatory state. COX 2-selective drugs have recently been introduced for the treatment of arthritis and are associated with a significant reduction of gastric

irritation. Whether COX 2-selective drugs have analgesic actions equivalent to other [NSAIDs](#) remains to be demonstrated.

Opioid Analgesics Opioids are the most potent pain-relieving drugs currently available. Furthermore, of all analgesics, they have the broadest range of efficacy, providing the most reliable method for rapidly relieving pain. Although side effects are common, they are usually not serious except for respiratory depression and can be reversed rapidly with the narcotic antagonist naloxone. The physician should not hesitate to use opioid analgesics in patients with acute severe pain. [Table 12-1](#) lists the most commonly used opioid analgesics.

Opioids produce analgesia by actions in the central nervous system. They activate pain-inhibitory neurons and directly inhibit pain-transmission neurons. Most of the commercially available opioid analgesics act at the same opioid receptor (mu receptor), differing mainly in potency, speed of onset, duration of action, and optimal route of administration. Although the dose-related side effects (sedation, respiratory depression, pruritus, constipation) are similar among the different opioids, some side effects are due to accumulation of nonopioid metabolites that are unique to individual drugs. One striking example of this is normeperidine, a metabolite of meperidine. Normeperidine produces hyperexcitability and seizures that are not reversible with naloxone. Normeperidine accumulation is much greater in patients with renal failure.

The most rapid relief with opioids is obtained by intravenous administration; relief with oral administration is significantly slower. Common acute side effects include nausea, vomiting, and sedation. The most serious side effect is respiratory depression. Patients with any form of respiratory compromise must be kept under close observation following opioid administration; an oxygen saturation monitor may be useful. The opioid antagonist, naloxone, should be readily available. These effects are dose-related, and there is great variability among patients in the doses that relieve pain and produce side effects. Because of this, initiation of therapy requires titration to optimal dose and interval. The most important principle is to provide adequate pain relief. This requires asking the patient whether the drug has relieved the pain and, if so, when the relief wears off. *The most common error made by physicians in managing severe pain with opioids is to prescribe an inadequate dose. Since many patients are reluctant to complain, this practice leads to needless suffering.* In the absence of sedation at the expected time of peak effect, a physician should not hesitate to repeat the initial dose to achieve satisfactory pain relief.

An innovative approach to the problem of achieving adequate pain relief is the use of patient-controlled analgesia (PCA). PCA requires a device that delivers a baseline continuous dose of an opioid drug, and preprogrammed additional doses whenever the patient pushes a button. The device can be programmed to limit the total hourly dose so that overdosing is impossible. The patient can then titrate the dose to the optimal level. This approach is used most extensively for the management of postoperative pain, but there is no reason why it should not be used for any hospitalized patient with persistent severe pain. PCA is also used for short-term home care of patients with intractable pain, such as is caused by metastatic cancer.

Many physicians, nurses, and patients have a certain trepidation about using opioids

that is based on an exaggerated fear of patients becoming addicted. In fact, there is a vanishingly small chance of patients becoming addicted to narcotics as a result of their appropriate medical use.

The availability of new routes of administration has extended the usefulness of opioid analgesics. Most important is the availability of spinal administration. Opioids can be infused through a spinal catheter placed either intrathecally or epidurally. By applying opioids directly to the spinal cord, regional analgesia can be obtained using a relatively low total dose. In this way, such side effects as sedation, nausea, and respiratory depression can be minimized. This approach has been used extensively in obstetric procedures and for lower-body postoperative pain. Opioids can also be given intranasally (butorphanol), rectally, and transdermally (fentanyl), thus avoiding the discomfort of frequent injections in patients who cannot be given oral medication.

Opioid and Cyclooxygenase Inhibitor Combinations When used in combination, opioids and COX inhibitors have additive effects. Because a lower dose of each can be used to achieve the same degree of pain relief and their side effects are nonadditive, such combinations can be used to lower the severity of dose-related side effects. Fixed-ratio combinations of an opioid with acetaminophen carry a special risk. Dose escalation as a result of increased severity of pain or decreased opioid effect as a result of tolerance may lead to levels of acetaminophen that are toxic to the liver.

CHRONIC PAIN

PATIENT EVALUATION

Managing patients with chronic pain is intellectually and emotionally challenging. The patient's problem is often difficult to diagnose: such patients are demanding of the physician's time and often appear emotionally distraught. The traditional medical approach of seeking an obscure organic pathology is usually unhelpful. On the other hand, psychological evaluation and behaviorally based treatment paradigms are frequently helpful, particularly in the setting of a multidisciplinary pain-management center.

There are several factors that can cause, perpetuate, or exacerbate chronic pain. First, of course, the patient may simply have a disease that is characteristically painful for which there is presently no cure. Arthritis, cancer, migraine headaches, fibromyalgia, and diabetic neuropathy are examples of this. Second, there may be secondary perpetuating factors that are initiated by disease and persist after that disease has resolved. Examples include damaged sensory nerves, sympathetic efferent activity, and painful reflex muscle contraction. Finally, a variety of psychological conditions can exacerbate or even cause pain.

There are certain areas to which special attention should be paid in the medical history. Because depression is the most common emotional disturbance in patients with chronic pain, patients should be questioned about their mood, appetite, sleep patterns, and daily activity. A simple standardized questionnaire, such as the Beck Depression Inventory, can be a useful screening device. It is important to remember that major depression is a common, treatable, and potentially fatal illness.

Other clues that a significant emotional disturbance is contributing to a patient's chronic pain complaint include: pain that occurs in multiple unrelated sites; a pattern of recurrent, but separate, pain problems beginning in childhood or adolescence; pain beginning at a time of emotional trauma, such as the loss of a parent or spouse; a history of physical or sexual abuse; and past or present substance abuse.

On examination, special attention should be paid to whether the patient guards the painful area and whether certain movements or postures are avoided because of pain. Discovering a mechanical component to the pain can be useful both diagnostically and therapeutically. Painful areas should be examined for deep tenderness, noting whether this is localized to muscle, ligamentous structures, or joints. Chronic myofascial pain is very common, and in these patients deep palpation may reveal highly localized trigger points that are firm bands or knots in muscle. If injection of local anesthetic into these trigger points relieves the pain, it supports the diagnosis. A neuropathic component to the pain is indicated by evidence of nerve damage, such as sensory impairment, exquisitely sensitive skin, weakness and muscle atrophy, or loss of deep tendon reflexes. Evidence suggesting sympathetic nervous system involvement includes the presence of diffuse swelling, changes in skin color and temperature, and hypersensitive skin and joint tenderness compared with the normal side. Relief of the pain with a sympathetic block is diagnostic.

A guiding principle in evaluating patients with chronic pain is to assess both emotional and organic factors before initiating therapy. Addressing these issues together, rather than waiting to "rule out" organic causes of the pain, improves compliance in part because it assures patients that a psychological evaluation does not mean that the physician is questioning the validity of their complaint. Even when an organic cause for a patient's pain can be found, it is still wise to look for other factors. For example, cancer patients with painful bony metastases may also have pain due to nerve damage and significant depression. Optimal therapy requires that each of these factors be looked for and treated.

TREATMENT

Once the evaluation process has been completed and the likely causative and exacerbating factors identified, an explicit treatment plan should be developed. An important part of this process is to identify specific and realistic functional goals for therapy, such as getting a good night's sleep, being able to go shopping, or returning to work. A multidisciplinary approach that utilizes medications, counseling, physical therapy, nerve blocks, and even surgery may be required to improve the patient's quality of life. This may require referral to a pain clinic; however, this is not necessary for all chronic pain patients. For some, pharmacologic management alone can provide significant help.

Antidepressant Medications The tricyclic antidepressants ([Table 12-1](#)) are extremely useful for the management of patients with chronic pain. Although developed for the treatment of depression, the tricyclics have a spectrum of dose-related biologic activities that include the production of analgesia in a variety of clinical conditions. Although the mechanism is unknown, the analgesic effect of tricyclics has a more rapid onset and

occurs at a lower dose than is typically required for the treatment of depression. Furthermore, patients with chronic pain who are not depressed obtain pain relief with antidepressants. There is evidence that tricyclic drugs potentiate opioid analgesia, so they are useful adjuncts for the treatment of severe persistent pain such as occurs with malignant tumors. [Table 12-2](#) lists some of the painful conditions that respond to tricyclics. Tricyclics are of particular value in the management of neuropathic pain such as occurs in diabetic neuropathy and postherpetic neuralgia, for which there are few other therapeutic options.

The tricyclics that have been shown to relieve pain have significant side effects ([Table 12-1](#); [Chap 385](#)). Unfortunately, some of the serotonin-selective reuptake inhibitors such as fluoxetine (Prozac) that have fewer and less serious side effects have not been shown to provide pain relief. On the other hand, venlafaxine (Effexor), a nontricyclic antidepressant that blocks both serotonin and norepinephrine reuptake, appears to be useful in patients who cannot tolerate tricyclics.

Anticonvulsants and Antiarrhythmics ([Table 12-1](#)) These drugs are useful primarily for patients with neuropathic pain. Phenytoin (Dilantin) and carbamazepine (Tegretol) were first shown to relieve the pain of trigeminal neuralgia. This pain has a characteristic brief, shooting, electric shock-like quality. In fact, anticonvulsants seem to be helpful largely for pains that have such a lancinating quality. A new-generation anticonvulsant, gabapentin (Neurontin), which increases brain γ -aminobutyric acid levels, is effective for a broad range of neuropathic pains.

Antiarrhythmic drugs such as low-dose lidocaine and mexiletine (Mexilitil) are also effective for neuropathic pains. These drugs block the spontaneous activity of primary afferent nociceptors that appears when they are damaged.

Chronic Opioid Medication The long-term use of opioids is accepted for patients with pain due to malignant disease. Although its use for chronic pain of nonmalignant origin is controversial, it is clear that for many such patients opioid analgesics are the only option available for obtaining effective relief. This is understandable since opioids are the most potent and have the broadest range of efficacy of any analgesic medications. Although addiction is rare in patients who first use opioids for pain relief, some degree of tolerance and physical dependence are likely to occur with long-term use. Therefore, before embarking on opioid therapy, other options should be explored, and the limitations and risks of opioids should be explained to the patient. It is also important to point out that some opioid analgesic medications have mixed agonist-antagonist properties (e.g., pentazocine and butorphanol). From a practical standpoint, this means that they may worsen pain by inducing an abstinence syndrome in patients who are physically dependent on other opioid analgesics.

With long-term outpatient use of orally administered opioids it is desirable to use long-acting compounds such as levorphanol, methadone, or sustained-release morphine ([Table 12-1](#)). The pharmacokinetic profile of these drugs enables prolonged pain relief, minimizes side effects such as sedation that are associated with high peak plasma levels, and, perhaps, reduces the likelihood of rebound pain associated with a rapid fall in plasma opioid concentration. Constipation is a virtually universal side effect of opioid use and should be treated expectantly.

It is worth emphasizing, in conclusion, that many patients, especially those with chronic pain, seek medical attention primarily because they are suffering and because only physicians can provide the medications required for their relief. A primary responsibility of all physicians is to minimize the physical and emotional discomfort of their patients. Familiarity with pain mechanisms and analgesic medications is an important step toward accomplishing this aim.

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13. CHEST DISCOMFORT AND PALPITATIONS - Thomas H. Lee

CHEST DISCOMFORT

Chest discomfort is one of the most common challenges for clinicians in the office or emergency department. The differential diagnosis includes conditions affecting organs throughout the thorax and abdomen, with prognostic implications that vary from benign to life-threatening ([Table 13-1](#)). Failure to recognize potentially serious conditions such as acute ischemic heart disease, aortic dissection, or pulmonary embolism can lead to serious complications, including death. Conversely, overly conservative management of low-risk patients leads to unnecessary hospital admissions, tests, and procedures.

CAUSES OF CHEST DISCOMFORT

Myocardial Ischemia and Injury (See also [Chap. 244](#)) Myocardial ischemia occurs when the oxygen supply to the heart is not sufficient to meet metabolic needs. This mismatch can result from a decrease in oxygen supply, a rise in demand, or both. The most common underlying cause of myocardial ischemia is obstruction of coronary arteries by atherosclerosis; in the presence of such obstruction, transient ischemic episodes are usually precipitated by an increase in oxygen demand as a result of physical exertion. However, ischemia can also result from psychological stress, fever, or large meals or from compromised oxygen delivery due to anemia, hypoxia, or hypotension. Ventricular hypertrophy due to valvular heart disease, hypertrophic cardiomyopathy, or hypertension can predispose the myocardium to ischemia because of impaired penetration of blood flow from epicardial coronary arteries to the endocardium.

Angina Pectoris The chest discomfort of myocardial ischemia is a visceral discomfort that is usually described as a heaviness, pressure, or squeezing ([Table 13-2](#)). Other common adjectives for anginal pain are burning and aching. Some patients deny any "pain" but may admit to dyspnea or a vague sense of anxiety. The word "sharp" is sometimes used by patients to describe intensity rather than quality.

The location of angina pectoris is usually retrosternal; most patients do not localize the pain to any small area. The discomfort may radiate to the neck, jaw, teeth, arms, or shoulders, reflecting the common origin in the posterior horn of the spinal cord of sensory neurons supplying the heart and these areas. Some patients present with aching in sites of radiated pain as their only symptoms of ischemia. Occasional patients report epigastric distress with ischemic episodes. Less common is radiation to below the umbilicus or to the back.

Stable angina pectoris usually develops gradually with exertion, emotional excitement, or after heavy meals. Rest or treatment with sublingual nitroglycerin typically leads to relief within several minutes. In contrast, pain that is fleeting (lasting only a few seconds) is rarely ischemic in origin. Similarly, pain that lasts for several hours is unlikely to represent angina, particularly if the patient's electrocardiogram does not show evidence of ischemia.

Anginal episodes can be precipitated by any physiologic or psychological stress that

induces tachycardia. Most myocardial perfusion occurs during diastole, when there is minimal pressure opposing coronary artery flow from within the left ventricle. Since tachycardia decreases the percentage of the time in which the heart is in diastole, it decreases myocardial perfusion.

Unstable angina and myocardial infarction (See also [Chaps. 243 and 244](#)) Patients with these acute ischemic syndromes usually complain of symptoms similar in quality to angina pectoris, but more prolonged and severe. The onset of these syndromes may occur with the patient at rest, and sublingual nitroglycerin may lead to transient or no relief. Accompanying symptoms may include diaphoresis, dyspnea, nausea, and light-headedness.

The physical examination may be completely normal in patients with chest discomfort due to ischemic heart disease. Careful auscultation during ischemic episodes may reveal a third or fourth heart sound, reflecting myocardial systolic or diastolic dysfunction. A transient murmur of mitral regurgitation suggests ischemic papillary muscle dysfunction. Severe episodes of ischemia can lead to pulmonary congestion and even pulmonary edema.

Other Cardiac Causes Myocardial ischemia caused by hypertrophic cardiomyopathy, aortic stenosis, or other conditions leads to angina pectoris similar to that caused by coronary atherosclerosis. In such cases, a systolic murmur or other findings usually suggest abnormalities other than coronary atherosclerosis that may be contributing to the patient's symptoms.

Pericarditis (See also [Chap. 239](#)) The pain in pericarditis is believed to be due to inflammation of the adjacent parietal pleura, since most of the pericardium is believed to be insensitive to pain. Thus, infectious pericarditis, which usually involves adjoining pleura surfaces, tends to be associated with pain, while conditions that cause only local inflammation (e.g., myocardial infarction or uremia) and cardiac tamponade tend to result in mild or no chest pain.

The adjacent parietal pleura receives its sensory supply from several sources, so the pain of pericarditis can be experienced in areas ranging from the shoulder and neck to the abdomen and back. Most typically, the pain is retrosternal and is aggravated by coughing, deep breaths, or changes in position -- all of which lead to movements of pleural surfaces. The pain is often worse in the supine position and relieved by sitting upright and leaning forward. Less common is a steady aching discomfort that mimics acute myocardial infarction.

Diseases of the Aorta (See also [Chap. 247](#)) *Aortic dissection* is a potentially catastrophic condition that is due to spread within the wall of the aorta of a subintimal hematoma. The hematoma may begin with a tear in the intima of the aorta or with rupture of the vasa vasorum within the aortic media. This syndrome can occur with trauma to the aorta, including motor vehicle accidents or medical procedures in which catheters or intraaortic balloon pumps damage the intima of the aorta. Nontraumatic aortic dissections are rare in the absence of hypertension and/or conditions associated with deterioration of the elastic or muscular components of the media within the aorta's wall. Cystic medial degeneration is a feature of several inherited connective tissue

diseases, including Marfan and Ehlers-Danlos syndromes. About half of all aortic dissections in women under 40 years of age occur during pregnancy.

Almost all patients with acute dissections present with severe chest pain, although some patients with chronic dissections are identified without associated symptoms. Unlike the pain of ischemic heart disease, symptoms of aortic dissection tend to reach peak severity immediately, often causing the patient to collapse from its intensity. The adjectives used to describe the pain reflect the process occurring within the wall of the aorta -- "ripping" and "tearing" -- and the location usually correlates with the site and extent of the dissection. Thus, dissections that begin in the ascending aorta and extend to the descending aorta tend to cause pain in the front of the chest that extends into the back, between the shoulder blades.

Physical findings may also reflect extension of the aortic dissection that compromises flow into arteries branching off the aorta. Thus, loss of a pulse in one or both arms, cerebrovascular accident, or paraplegia can all be catastrophic consequences of aortic dissection. Hematomas that extend proximally and undermine the coronary arteries or aortic valve apparatus may lead to acute myocardial infarction or acute aortic insufficiency. Rupture of the hematoma into the pericardial space leads to pericardial tamponade.

Another abnormality of the aorta that can cause chest pain is a *thoracic aortic aneurysm*. Aortic aneurysms are frequently asymptomatic but can cause chest pain and other symptoms by compressing adjacent structures. This pain tends to be steady, deep, and sometimes severe.

Pulmonary Embolism (See also [Chap. 261](#)) Chest pain due to pulmonary embolism is believed to be due to distention of the pulmonary artery or infarction of a segment of the lung adjacent to the pleura. Massive pulmonary emboli may lead to substernal pain that is suggestive of acute myocardial infarction. More commonly, smaller emboli lead to focal pulmonary infarctions that cause pain that is lateral and pleuritic. Associated symptoms include dyspnea and, occasionally, hemoptysis. Tachycardia is usually present.

Pneumonia or Pleuritis Lung diseases that damage and cause inflammation of the pleura of the lung usually cause a sharp, knifelike pain that is aggravated by inspiration or coughing.

Gastrointestinal Conditions Esophageal pain from acid reflux from the stomach, spasm, obstruction, or injury can be difficult to discern from myocardial syndromes. Acid reflux typically causes a deep burning discomfort that may be exacerbated by alcohol, aspirin, or some foods; this discomfort is often relieved by antacid or other acid-reducing therapies. Acid reflux tends to be exacerbated by lying down and may be worse in early morning when the stomach is empty of food that might otherwise absorb gastric acid.

Esophageal spasm may occur in the presence or absence of acid reflux, and leads to a squeezing pain indistinguishable from angina. Prompt relief of esophageal spasm is often provided by antianginal therapies such as sublingual nifedipine, further promoting confusion between these syndromes. Chest pain can also result from injury to the

esophagus, such as a Mallory-Weiss tear caused by severe vomiting.

Chest pain can result from diseases of the gastrointestinal tract below the diaphragm, including *peptic ulcer disease*, *biliary disease*, and *pancreatitis*. These conditions usually cause abdominal pain as well as chest discomfort; symptoms are not likely to be associated with exertion. The pain of ulcer disease typically occurs 60 to 90 min after meals, when postprandial acid production is no longer neutralized by food in the stomach. Cholecystitis usually causes a pain that is described as aching, occurring an hour or more after meals.

Neuromusculoskeletal Conditions *Cervical disk disease* can cause chest pain by compression of nerve roots. Pain in a dermatomal distribution can also be caused by *intercostal muscle cramps* or by *herpes zoster*. Chest pain symptoms due to herpes zoster may occur before skin lesions are apparent.

Costochondral and chondrosternal syndromes are the most common causes of anterior chest musculoskeletal pain. Only occasionally are physical signs of costochondritis such as swelling, redness, and warmth (Tietze's syndrome) present. The pain of such syndromes is usually fleeting and sharp, but some patients experience a dull ache that lasts for hours. Direct pressure on the chondrosternal and costochondral junctions may reproduce the pain from these and other musculoskeletal syndromes. Arthritis of the shoulder and spine and bursitis may also cause chest pain. Some patients who have these conditions and myocardial ischemia blur and confuse symptoms of these syndromes.

Emotional and Psychiatric Conditions As many as 10% of patients who present to emergency departments with acute chest pain have panic disorder or other emotional conditions. The symptoms in these populations are highly variable, but frequently the discomfort is described as visceral tightness or aching that lasts more than 30 min. Some patients offer other atypical descriptions, such as pain that is fleeting, sharp, and/or localized to a small region. The electrocardiogram in patients with emotional conditions may be difficult to interpret if hyperventilation causes ST-T-wave abnormalities. A careful history may elicit clues of depression, prior panic attacks, somatization, agoraphobia, or other phobias.

Approach to the Patient

The evaluation of the patient with chest discomfort must accommodate two goals -- determining the diagnosis and assessing the safety of the immediate management plan. The latter issue is often dominant when the patient has acute chest discomfort, such as patients seen in the emergency department. In such settings, the clinician must focus on questions such as the safety of discharge to home, admission to a non-coronary care unit facility, or immediate exercise testing. [Table 13-3](#) displays a sequence of questions that can be used in the evaluation of the patient with chest discomfort, with the diagnostic entities that are most important for consideration at each stage of the evaluation.

Acute Chest Discomfort In patients with acute chest discomfort, the clinician must first assess the patient's respiratory and hemodynamic status. If either is compromised,

initial management should focus on stabilizing the patient before the diagnostic evaluation is pursued. If, however, the patient does not require emergent interventions, then a focused history, physical examination, and laboratory evaluation should be performed to assess the patient's risk of life-threatening conditions, including acute ischemic heart disease, aortic dissection, and pulmonary embolism.

The *history* should include questions about the quality and location of the chest discomfort ([Table 13-2](#)). The patient should also be asked about the nature of onset of the pain and its duration. Myocardial ischemia is usually associated with a gradual intensification of symptoms over a period of minutes. Pain that is fleeting or that lasts hours without being associated with electrocardiographic changes is not likely to be ischemic in origin.

The *physical examination* should include evaluation of blood pressure in both arms and of pulses in both legs. Poor perfusion of a limb may be due to an aortic dissection that has compromised flow to an artery branching from the aorta. Chest auscultation may reveal diminished breath sounds; a pleural rub; or evidence of pneumothorax, pulmonary embolism, pneumonia, or pleurisy. The cardiac examination should seek pericardial rubs, systolic and diastolic murmurs, and third or fourth heart sounds.

An *electrocardiogram* is an essential test for adults with chest discomfort that is not due to an obvious traumatic cause. The presence of electrocardiographic changes consistent with ischemia or infarction ([Chap. 226](#)) is associated with high risks of acute myocardial infarction or unstable angina ([Table 13-4](#)); such patients should be admitted to a unit with electrocardiographic monitoring and the capacity to respond to a cardiac arrest. The absence of such changes does not exclude acute ischemic heart disease, but the risk of life-threatening complications is low for patients with normal electrocardiograms or only nonspecific ST-T-wave changes. If these patients are not considered appropriate for immediate discharge, they are often candidates for early or immediate exercise testing.

Markers of myocardial injury are often obtained in the emergency department evaluation of acute chest discomfort. The most commonly used markers are creatine kinase (CK), CK-MB, and the cardiac troponins (I and T). Single values of these markers do not have high sensitivity for acute myocardial infarction or for prediction of complications. Hence, decisions to discharge patients home should not be made on the basis of single negative values of these tests.

Provocative tests for coronary artery disease are not appropriate for patients with ongoing chest pain. In such patients, rest myocardial perfusion scans can be considered; a normal scan reduces the likelihood of coronary artery disease. Clinicians frequently employ therapeutic trials with sublingual nitroglycerin or antacids, and a common error is to assume that a response to either of these interventions clarifies the diagnosis. While such information is often helpful, the patient's response may be due to the placebo effect. Hence, myocardial ischemia should never be considered excluded solely because of a response to antacid therapy. Similarly, failure of nitroglycerin to relieve pain does not exclude the diagnosis of coronary disease.

If the patient's history or examination is consistent with aortic dissection, imaging studies

to evaluate the aorta must be pursued promptly because of the high risk of catastrophic complications with this condition. A chest x-ray is not sufficient to exclude this diagnosis. Appropriate tests include a chest computed tomography scan with contrast or a magnetic resonance imaging scan in patients who are hemodynamically stable, or a transesophageal echocardiogram in patients who are less stable. Aortic angiography is no longer a first test at most institutions.

Acute pulmonary embolism should be considered in patients with respiratory symptoms, pleuritic chest pain, hemoptysis, or a history of venous thromboembolism or coagulation abnormalities. Initial tests usually include a lung scan and/or pulmonary arteriography.

If patients with acute chest discomfort show no evidence of life-threatening conditions, the clinician should then focus on serious chronic conditions with the potential to cause major complications, the most common of which is stable angina. Early use of treadmill exercise testing for such patients, whether in the office or the emergency department, is now an accepted management strategy for low-risk patients. Exercise testing is not appropriate, however, for patients who (1) report pain that is believed to be ischemic occurring at rest or (2) have electrocardiographic changes consistent with ischemia not known to be old.

Patients with sustained chest discomfort who do not have evidence for life-threatening conditions should be evaluated for evidence of conditions likely to benefit from acute treatment ([Table 13-3](#)). Pericarditis may be suggested by the history, physical examination, and electrocardiogram ([Table 13-2](#)). Clinicians should carefully assess blood pressure patterns and consider echocardiography in such patients to detect evidence of impending pericardial tamponade. Chest x-rays can be used to evaluate the possibility of pulmonary disease.

GUIDELINES AND CRITICAL PATHWAYS FOR ACUTE CHEST PAIN

Guidelines for the initial evaluation for patients with acute chest pain have been developed by the American College of Emergency Physicians (ACEP) and other organizations. The ACEP statement describes *rules* and *guidelines* about the data that should be recorded as part of the evaluation, and the actions that should follow from certain findings ([Table 13-5](#)). In the ACEP framework, *rules* are actions that are general principles of good practice, while *guidelines* are actions that should be considered but are not always followed. Hence, failure to follow a guideline is not necessarily improper care.

Other organizations, including the Agency for Health Care Policy and Research (AHCPR) and the National Heart Attack Alert Program, have also issued guidelines for management of patients with a high probability of acute ischemic heart disease. In these and other guidelines, patients with possible or probable acute myocardial infarction as suggested by the description of their pain or electrocardiographic findings are expected to be admitted to the hospital. The AHCPR guidelines for unstable angina note that not all patients with that syndrome require admission but recommend that patients with unstable angina be monitored electrocardiographically during their evaluation; that those with ongoing rest pain should be placed at bed rest during the initial phase of stabilization. The [ACEP](#) policy statement indicates that patients who are discharged

should be given a referral for follow-up care and instructions regarding treatment and circumstances that require a return to the emergency department.

Many medical centers have adopted critical pathways and other forms of guidelines to increase efficiency. These guidelines emphasize two strategies:

- Triage to non-coronary care unit monitored facilities such as intermediate care units or chest pain units of patients with a low risk for complications, such as patients without new ischemic changes on their electrocardiograms and without ongoing chest pain. Such patients can usually be safely observed in non-coronary care unit settings, undergo early exercise testing, or be discharged home. Risk stratification can be assisted through use of prospectively validated multivariate algorithms that have been published for acute ischemic heart disease and its complications.
- Shortening lengths of stay in the coronary care unit and hospital. Recommendations regarding the minimum length of stay in a monitored bed for a patient who has no further symptoms have decreased in recent years to 12 h or less if exercise testing or other risk stratification technologies are available.

NONACUTE CHEST DISCOMFORT

The management of patients who do not require admission to the hospital or who no longer require inpatient observation should seek to identify the cause of the symptoms and the likelihood of major complications. Cost-effectiveness analyses support use of noninvasive testing for coronary disease, such as exercise electrocardiography and stress echocardiography. These tests serve both to diagnose coronary disease and to identify patients with high-risk forms of coronary disease who may benefit from revascularization. Gastrointestinal causes of chest pain can be evaluated via endoscopy or radiology studies. Emotional and psychiatric conditions warrant appropriate evaluation and treatment; randomized trial data indicate that cognitive therapy and group interventions lead to decreases in symptoms for such patients.

PALPITATIONS

Palpitations are characterized by an awareness of the beating of the heart. Patients commonly describe "pounding" or "fluttering" heart beats or report a sensation that the heart is stopping or skipping beats. These symptoms may be caused by a change in the heart's rhythm or rate or by an increase in the force of its contractions. In many cases, this awareness reflects lack of competing sensory stimuli, such as when a person is lying in bed, unable to sleep.

Palpitations are often manifestations of psychiatric conditions, the most common of which are depression and panic disorder. For example, in one study of outpatients referred for ambulatory electrocardiographic monitoring to evaluate palpitations, 19% were found to have a psychiatric disorder. Patients with psychiatric disorders were more likely than other patients to report that their palpitations lasted longer than 15 min or were accompanied by ancillary symptoms. In this study, physicians usually recognized the emotional basis of the patients' symptoms but frequently did not refer the patient for specific therapy.

Palpitations can also be caused by virtually any cardiac arrhythmia as well as by other cardiac and noncardiac conditions. A markedly enlarged left ventricle can cause awareness of the heart beat by contact with the chest wall. Any condition associated with increased catecholamine levels can lead to palpitations both by increasing the forcefulness of cardiac contractions and by increasing the rate of premature beats.

Palpitations can be intermittent or sustained and regular or irregular. Patients with this complaint should be asked to describe their palpitations' onset, duration, associated symptoms and the circumstances in which they occur. Abrupt onset and termination after several minutes may reflect a sustained ventricular or supraventricular tachyarrhythmia. Gradual onset and termination of a pounding heart beat is more consistent with sinus tachycardia. Patients should try to replicate the rhythm of their palpitations by tapping on a table. This maneuver can help the physician determine the nature of any cardiac arrhythmia. Patients should also be taught to take their pulse so that they can more accurately report their approximate heart rate and whether the rhythm was regular.

DIFFERENTIAL DIAGNOSIS

Patients who report "skipped" beats or a "flopping" sensation often have atrial or ventricular extrasystoles ([Chap. 230](#)). These premature beats are followed by a compensatory pause, and the first heart beat after the pause may be unusually strong due to increased left ventricular volume and enhanced contractility (a phenomenon called *postextrasystolic potentiation*). Sustained bursts of rapid heart beats may be due to ventricular or supraventricular tachyarrhythmias. A sustained irregular rhythm suggests atrial fibrillation.

Conditions that cause marked left ventricular enlargement such as aortic regurgitation can cause an awareness of the heart beat that is sometimes positional. Presumably because of associated arrhythmias, hypertrophic cardiomyopathy, mitral valve prolapse, and other cardiac structural abnormalities are also associated with palpitations.

Palpitations can also be a prominent symptom in noncardiac conditions, including thyrotoxicosis, hypoglycemia, pheochromocytoma, and fever. The physiologic basis of palpitations with these conditions is either arrhythmia or increased catecholamine levels leading to greater myocardial contractility. Drugs that can precipitate arrhythmias and palpitations include tobacco, coffee, tea, alcohol, epinephrine, ephedrine, aminophylline, and atropine.

Approach to the Patient

The first goal in the evaluation of patients with palpitations is to exclude the possibility of life-threatening arrhythmias. The risk for such arrhythmias is highest in patients with coronary artery disease, congestive heart failure, or other structural cardiac abnormalities. The history, physical examination, and electrocardiogram should therefore be focused on stratifying patients according to the risk of such conditions. Palpitations are also more likely to reflect serious arrhythmias if they are associated with symptoms that suggest hemodynamic compromise, such as syncope, light-headedness,

dizziness, or shortness of breath.

The most common first test after the initial evaluation of palpitations is continuous electrocardiographic (Holter) monitoring. This test is especially useful if patients have palpitations on a daily basis. For patients with more sporadic palpitations, a variety of new technologies have become available to allow capture of electrocardiographic tracings at the time of their symptoms. These technologies include loop recorders, that can freeze the last several minutes of data when the patient presses a button, and telephonic monitors, which can be used to "call in" tracings when symptoms occur. If episodes are associated with physical stress, exercise electrocardiography can be used in an attempt to elicit an arrhythmia.

Most patients with palpitations do not have evidence of major arrhythmias or abnormal physiologic conditions associated with increased catecholamine levels. Patients with emotional or psychological causes of palpitations should be evaluated for possible cognitive and pharmaceutical therapy. Drugs and medications that may precipitate palpitations should be eliminated or reduced. A trial of beta blockers is often successful in reducing premature beats and symptoms. Regardless of the cause and treatment, the clinician should remain aware that palpitations are extremely bothersome symptoms for patients. Reassurance that a comprehensive evaluation has been performed and that the palpitations do not adversely affect the patient's prognosis is a critical part of the patient's care.

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14. ABDOMINAL PAIN - William Silen

The correct interpretation of acute abdominal pain is challenging. Since proper therapy may require urgent action, the unhurried approach suitable for the study of other conditions is sometimes denied. Few other clinical situations demand greater judgment, because the most catastrophic of events may be forecast by the subtlest of symptoms and signs. A meticulously executed, detailed history and physical examination is of great importance. The etiologic classification in [Table 14-1](#), although not complete, forms a useful basis for the evaluation of patients with abdominal pain.

The diagnosis of "acute or surgical abdomen" is not an acceptable one because of its often misleading and erroneous connotation. The most obvious of "acute abdomens" may not require operative intervention, and the mildest of abdominal pains may herald an urgently correctable lesion. Any patient with abdominal pain of recent onset requires early and thorough evaluation and accurate diagnosis.

SOME MECHANISMS OF PAIN ORIGINATING IN THE ABDOMEN

Inflammation of the Parietal Peritoneum The pain of parietal peritoneal inflammation is steady and aching in character and is located directly over the inflamed area, its exact reference being possible because it is transmitted by somatic nerves supplying the parietal peritoneum. The intensity of the pain is dependent on the type and amount of material to which the peritoneal surfaces are exposed in a given time period. For example, the sudden release into the peritoneal cavity of a small quantity of *sterile* acid gastric juice causes much more pain than the same amount of grossly contaminated neutral feces. Enzymatically active pancreatic juice incites more pain and inflammation than does the same amount of sterile bile containing no potent enzymes. Blood and urine are often so bland as to go undetected if exposure of the peritoneum has not been sudden and massive. In the case of bacterial contamination, such as in pelvic inflammatory disease, the pain is frequently of low intensity early in the illness until bacterial multiplication has caused the elaboration of irritating substances.

The rate at which the irritating material is applied to the peritoneum is important. Perforated peptic ulcer may be associated with entirely different clinical pictures dependent only on the rapidity with which the gastric juice enters the peritoneal cavity.

The pain of peritoneal inflammation is invariably accentuated by pressure or changes in tension of the peritoneum, whether produced by palpation or by movement, as in coughing or sneezing. The patient with peritonitis lies quietly in bed, preferring to avoid motion, in contrast to the patient with colic, who may writhe incessantly.

Another characteristic feature of peritoneal irritation is tonic reflex spasm of the abdominal musculature, localized to the involved body segment. The intensity of the tonic muscle spasm accompanying peritoneal inflammation is dependent on the location of the inflammatory process, the rate at which it develops, and the integrity of the nervous system. Spasm over a perforated retrocecal appendix or perforated ulcer into the lesser peritoneal sac may be minimal or absent because of the protective effect of overlying viscera. A slowly developing process often greatly attenuates the degree of muscle spasm. Catastrophic abdominal emergencies such as a perforated ulcer may be

associated with minimal or no detectable pain or muscle spasm in obtunded, seriously ill, debilitated elderly patients or in psychotic patients.

Obstruction of Hollow Viscera The pain of obstruction of hollow abdominal viscera is classically described as intermittent, or colicky. Yet the lack of a truly cramping character should not be misleading, because distention of a hollow viscus may produce steady pain with only very occasional exacerbations. It is not nearly as well localized as the pain of parietal peritoneal inflammation.

The colicky pain of obstruction of the small intestine is usually periumbilical or supraumbilical and is poorly localized. As the intestine becomes progressively dilated with loss of muscular tone, the colicky nature of the pain may diminish. With superimposed strangulating obstruction, pain may spread to the lower lumbar region if there is traction on the root of the mesentery. The colicky pain of colonic obstruction is of lesser intensity than that of the small intestine and is often located in the infraumbilical area. Lumbar radiation of pain is common in colonic obstruction.

Sudden distention of the biliary tree produces a steady rather than colicky type of pain; hence the term *biliary colic* is misleading. Acute distention of the gallbladder usually causes pain in the right upper quadrant with radiation to the right posterior region of the thorax or to the tip of the right scapula, and distention of the common bile duct is often associated with pain in the epigastrium radiating to the upper part of the lumbar region. Considerable variation is common, however, so that differentiation between these may be impossible. The typical subscapular pain or lumbar radiation is frequently absent. Gradual dilatation of the biliary tree, as in carcinoma of the head of the pancreas, may cause no pain or only a mild aching sensation in the epigastrium or right upper quadrant. The pain of distention of the pancreatic ducts is similar to that described for distention of the common bile duct but, in addition, is very frequently accentuated by recumbency and relieved by the upright position.

Obstruction of the urinary bladder results in dull suprapubic pain, usually low in intensity. Restlessness without specific complaint of pain may be the only sign of a distended bladder in an obtunded patient. In contrast, acute obstruction of the intravesicular portion of the ureter is characterized by severe suprapubic and flank pain that radiates to the penis, scrotum, or inner aspect of the upper thigh. Obstruction of the ureteropelvic junction is felt as pain in the costovertebral angle, whereas obstruction of the remainder of the ureter is associated with flank pain that often extends into the same side of the abdomen.

Vascular Disturbances A frequent misconception, despite abundant experience to the contrary, is that pain associated with intraabdominal vascular disturbances is sudden and catastrophic in nature. The pain of embolism or thrombosis of the superior mesenteric artery or that of impending rupture of an abdominal aortic aneurysm certainly may be severe and diffuse. Yet, just as frequently, the patient with occlusion of the superior mesenteric artery has only mild continuous diffuse pain for 2 or 3 days before vascular collapse or findings of peritoneal inflammation appear. The early, seemingly insignificant discomfort is caused by hyperperistalsis rather than peritoneal inflammation. Indeed, absence of tenderness and rigidity in the presence of continuous, diffuse pain in a patient likely to have vascular disease is quite characteristic of

occlusion of the superior mesenteric artery. Abdominal pain with radiation to the sacral region, flank, or genitalia should always signal the possible presence of a rupturing abdominal aortic aneurysm. This pain may persist over a period of several days before rupture and collapse occur.

Abdominal Wall Pain arising from the abdominal wall is usually constant and aching. Movement, prolonged standing, and pressure accentuate the discomfort and muscle spasm. In the case of hematoma of the rectus sheath, now most frequently encountered in association with anticoagulant therapy, a mass may be present in the lower quadrants of the abdomen. Simultaneous involvement of muscles in other parts of the body usually serves to differentiate myositis of the abdominal wall from an intraabdominal process that might cause pain in the same region.

REFERRED PAIN IN ABDOMINAL DISEASES

Pain referred to the abdomen from the thorax, spine, or genitalia may prove a vexing diagnostic problem, because diseases of the upper part of the abdominal cavity such as acute cholecystitis or perforated ulcer are frequently associated with intrathoracic complications. A most important, yet often forgotten, dictum is that the possibility of intrathoracic disease must be considered in every patient with abdominal pain, especially if the pain is in the upper part of the abdomen. Systematic questioning and examination directed toward detecting myocardial or pulmonary infarction, pneumonia, pericarditis, or esophageal disease (the intrathoracic diseases that most often masquerade as abdominal emergencies) will often provide sufficient clues to establish the proper diagnosis. Diaphragmatic pleuritis resulting from pneumonia or pulmonary infarction may cause pain in the right upper quadrant and pain in the supraclavicular area, the latter radiation to be distinguished from the referred subscapular pain caused by acute distention of the extrahepatic biliary tree. The ultimate decision as to the origin of abdominal pain may require deliberate and planned observation over a period of several hours, during which repeated questioning and examination will provide the diagnosis.

Referred pain of thoracic origin is often accompanied by splinting of the involved hemithorax with respiratory lag and decrease in excursion more marked than that seen in the presence of intraabdominal disease. In addition, apparent abdominal muscle spasm caused by referred pain will diminish during the inspiratory phase of respiration, whereas it is persistent throughout both respiratory phases if it is of abdominal origin. Palpation over the area of referred pain in the abdomen also does not usually accentuate the pain and in many instances actually seems to relieve it. Thoracic and abdominal disease frequently coexist and may be difficult or impossible to differentiate. For example, the patient with known biliary tract disease often has epigastric pain during myocardial infarction, or biliary colic may be referred to the precordium or left shoulder in a patient who has suffered previously from angina pectoris. **For an explanation of the radiation of pain to a previously diseased area, see Chap. 12.*

Referred pain from the spine, which usually involves compression or irritation of nerve roots, is characteristically intensified by certain motions such as cough, sneeze, or strain and is associated with hyperesthesia over the involved dermatomes. Pain referred to the abdomen from the testicles or seminal vesicles is generally accentuated by the

slightest pressure on either of these organs. The abdominal discomfort is of dull aching character and is poorly localized.

METABOLIC ABDOMINAL CRISES

Pain of metabolic origin may simulate almost any other type of intraabdominal disease. Several mechanisms may be at work. In certain instances, such as hyperlipidemia, the metabolic disease itself may be accompanied by an intraabdominal process such as pancreatitis, which can lead to unnecessary laparotomy unless recognized. C α I esterase deficiency associated with angioneurotic edema is often associated with episodes of severe abdominal pain. Whenever the cause of abdominal pain is obscure, a metabolic origin always must be considered. Abdominal pain is also the hallmark of familial Mediterranean fever ([Chap. 289](#)).

The problem of differential diagnosis is often not readily resolved. The pain of porphyria and of lead colic is usually difficult to distinguish from that of intestinal obstruction, because severe hyperperistalsis is a prominent feature of both. The pain of uremia or diabetes is nonspecific, and the pain and tenderness frequently shift in location and intensity. Diabetic acidosis may be precipitated by acute appendicitis or intestinal obstruction, so if prompt resolution of the abdominal pain does not result from correction of the metabolic abnormalities, an underlying organic problem should be suspected. Black widow spider bites produce intense pain and rigidity of the abdominal muscles and back, an area infrequently involved in intraabdominal disease.

NEUROGENIC CAUSES

Causalgic pain may occur in diseases that injure sensory nerves. It has a burning character and is usually limited to the distribution of a given peripheral nerve. Normal stimuli such as touch or change in temperature may be transformed into this type of pain, which is frequently present in a patient at rest. The demonstration of irregularly spaced cutaneous pain spots may be the only indication of an old nerve lesion underlying causalgic pain. Even though the pain may be precipitated by gentle palpation, rigidity of the abdominal muscles is absent, and the respirations are not disturbed. Distention of the abdomen is uncommon, and the pain has no relationship to the intake of food.

Pain arising from spinal nerves or roots comes and goes suddenly and is of a lancinating type ([Chap. 16](#)). It may be caused by herpes zoster, impingement by arthritis, tumors, herniated nucleus pulposus, diabetes, or syphilis. It is not associated with food intake, abdominal distention, or changes in respiration. Severe muscle spasm, as in the gastric crises of tabes dorsalis, is common but is either relieved or is not accentuated by abdominal palpation. The pain is made worse by movement of the spine and is usually confined to a few dermatomes. Hyperesthesia is very common.

Psychogenic pain conforms to none of the aforementioned patterns. Mechanism is hard to define. The most common problem is the hysterical adolescent or young person who develops abdominal pain and who frequently loses an appendix or other organs because of it. Ovulation or some other natural event that causes brief mild abdominal discomfort may be experienced as an abdominal catastrophe.

Psychogenic pain varies enormously in type and location but usually has no relation to meals. It is often markedly accentuated during the night. Nausea and vomiting are rarely observed. Spasm is seldom induced in the abdominal musculature and, if present, does not persist, especially if the attention of the patient can be distracted. Persistent localized tenderness is rare, and if found, the muscle spasm in the area is inconsistent or absent. Shallow respiration is the most common breathing abnormality; anxiety may produce a smothering or choking sensation. It occurs in the absence of thoracic splinting or change in the respiratory rate.

Approach to the Patient

Few abdominal conditions require such urgent operative intervention that an orderly approach need be abandoned, no matter how ill the patient. Only those patients with exsanguinating hemorrhage must be rushed to the operating room immediately, but in such instances, only a few minutes are required to assess the critical nature of the problem. Under these circumstances, all obstacles must be swept aside, adequate venous access for fluid replacement obtained, and the operation begun. Many patients of this type have died in the radiology department or the emergency room while awaiting such unnecessary examinations as electrocardiograms or abdominal films. *There are no contraindications to operation when massive hemorrhage is present.* This situation fortunately is relatively rare.

Nothing will supplant an orderly, painstakingly *detailed history*, which is far more valuable than any laboratory or radiographic examination. This kind of history is laborious and time-consuming, making it not especially popular, even though a reasonably accurate diagnosis can be made on the basis of the history alone in the majority of cases. Computer-aided diagnosis of abdominal pain provides no advantage over clinical assessment alone. In cases of *acute* abdominal pain, a diagnosis is readily established in most instances, whereas success is not so frequent in patients with *chronic* pain. Irritable bowel syndrome is one of the most common causes of abdominal pain and must always be kept in mind ([Chap. 288](#)). The *chronological sequence of events* in the patient's history is often more important than emphasis on the location of pain. If the examiner is sufficiently open-minded and unhurried, asks the proper questions, and listens, the patient will usually provide the diagnosis. Careful attention should be paid to the extraabdominal regions that may be responsible for abdominal pain. An accurate menstrual history in a female patient is essential. Narcotics or analgesics should *not* be withheld until a definitive diagnosis or a definitive plan has been formulated; obfuscation of the diagnosis by adequate analgesia is unlikely.

In the examination, simple critical inspection of the patient, e.g., of facies, position in bed, and respiratory activity, may provide valuable clues. The amount of information to be gleaned is directly proportional to the *gentleness* and thoroughness of the examiner. Once a patient with peritoneal inflammation has been examined brusquely, accurate assessment by the next examiner becomes almost impossible. Eliciting rebound tenderness by sudden release of a deeply palpating hand in a patient with suspected peritonitis is cruel and unnecessary. The same information can be obtained by gentle percussion of the abdomen (rebound tenderness on a miniature scale), a maneuver that can be far more precise and localizing. Asking the patient to cough will elicit true

rebound tenderness without the need for placing a hand on the abdomen. Furthermore, the forceful demonstration of rebound tenderness will startle and induce protective spasm in a nervous or worried patient in whom true rebound tenderness is not present. A palpable gallbladder will be missed if palpation is so brusque that voluntary muscle spasm becomes superimposed on involuntary muscular rigidity.

As in history taking, there is no substitute for sufficient time spent in the examination. Abdominal signs may be minimal but nevertheless, if accompanied by consistent symptoms, may be exceptionally meaningful. Abdominal signs may be virtually or totally absent in cases of pelvic peritonitis, so careful *pelvic and rectal examinations are mandatory in every patient with abdominal pain*. Tenderness on pelvic or rectal examination in the absence of other abdominal signs can be caused by operative indications such as perforated appendicitis, diverticulitis, twisted ovarian cyst, and many others.

Much attention has been paid to the presence or absence of peristaltic sounds, their quality, and their frequency. Auscultation of the abdomen is one of the least revealing aspects of the physical examination of a patient with abdominal pain. Catastrophes such as strangulating small intestinal obstruction or perforated appendicitis may occur in the presence of normal peristalsis. Conversely, when the proximal part of the intestine above an obstruction becomes markedly distended and edematous, peristaltic sounds may lose the characteristics of borborygmi and become weak or absent, even when peritonitis is not present. It is usually the severe chemical peritonitis of sudden onset that is associated with the truly silent abdomen. Assessment of the patient's state of hydration is important.

Laboratory examinations may be of great value in assessment of the patient with abdominal pain, yet with few exceptions they rarely establish a diagnosis. Leukocytosis should never be the single deciding factor as to whether or not operation is indicated. A white blood cell count greater than 20,000/uL may be observed with perforation of a viscus, but pancreatitis, acute cholecystitis, pelvic inflammatory disease, and intestinal infarction may be associated with marked leukocytosis. A normal white blood cell count is not rare in cases of perforation of abdominal viscera. The diagnosis of anemia may be more helpful than the white blood cell count, especially when combined with the history.

The urinalysis may reveal the state of hydration or rule out severe renal disease, diabetes, or urinary infection. Blood urea nitrogen, glucose, and serum bilirubin levels may be helpful. Serum amylase levels may be increased by many diseases other than pancreatitis, e.g., perforated ulcer, strangulating intestinal obstruction, and acute cholecystitis; thus, elevations of serum amylase do not rule out the need for an operation. The determination of the serum lipase may have greater accuracy than that of the serum amylase.

Plain and upright or lateral decubitus radiographs of the abdomen may be of value in cases of intestinal obstruction, perforated ulcer, and a variety of other conditions. They are usually unnecessary in patients with acute appendicitis or strangulated external hernias. In rare instances, barium or water-soluble contrast study of the upper part of the gastrointestinal tract may demonstrate partial intestinal obstruction that may elude diagnosis by other means. If there is any question of obstruction of the colon, oral

administration of barium sulfate should be avoided. On the other hand, in cases of suspected colonic obstruction (with perforation), contrast enema may be diagnostic.

In the absence of trauma, peritoneal lavage has been replaced as a diagnostic tool by ultrasound, computed tomography (CT), and laparoscopy. Ultrasonography has proved to be useful in detecting an enlarged gallbladder or pancreas, the presence of gallstones, an enlarged ovary, or a tubal pregnancy. Laparoscopy is especially helpful in diagnosing pelvic conditions, such as ovarian cysts, tubal pregnancies, salpingitis, and acute appendicitis. Radioisotopic scans (HIDA) may help differentiate acute cholecystitis from acute pancreatitis. A CT scan may demonstrate an enlarged pancreas, ruptured spleen, or thickened colonic or appendiceal wall and streaking of the mesocolon or mesoappendix characteristic of diverticulitis or appendicitis.

Sometimes, even under the best circumstances with all available aids and with the greatest of clinical skill, a definitive diagnosis cannot be established at the time of the initial examination. Nevertheless, despite lack of a clear anatomic diagnosis, it may be abundantly clear to an experienced and thoughtful physician and surgeon that on clinical grounds alone operation is indicated. Should that decision be questionable, watchful waiting with repeated questioning and examination will often elucidate the true nature of the illness and indicate the proper course of action.

(Bibliography omitted in Palm version)

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15. HEADACHE, INCLUDING MIGRAINE AND CLUSTER HEADACHE - Neil H. Raskin, Stephen J. Peroutka

Few of us are spared the experience of head pain. As many as 90% of individuals have at least one headache per year. Severe, disabling headache is reported to occur at least annually by 40% of individuals worldwide. A useful classification of the many causes of headache is shown in [Table 15-1](#). Headache is usually a benign symptom, but occasionally it is the manifestation of a serious illness such as brain tumor, subarachnoid hemorrhage, meningitis, or giant cell arteritis. In emergency settings, approximately 5% of patients with headache are found to have a serious underlying neurologic disorder. Therefore, it is imperative that the serious causes of headache be diagnosed rapidly and accurately.

PAIN-SENSITIVE STRUCTURES OF THE HEAD

Pain is most commonly due to tissue injury resulting in stimulation of peripheral nociceptors in an intact nervous system. Pain can also result from damage to or anomalous activation of pain-sensitive pathways of the peripheral or central nervous system. Headache may originate from either or both mechanisms. Relatively few cranial structures are pain-sensitive: the scalp, middle meningeal artery, dural sinuses, falx cerebri, and the proximal segments of the large pial arteries. The ventricular ependyma, choroid plexus, pial veins, and much of the brain parenchyma are pain-insensitive. Electrical stimulation of the midbrain in the region of the dorsal raphe has resulted in migraine-like headaches. Thus, whereas most of the brain is insensitive to electrode probing, a site in the midbrain represents a possible source of headache generation. Sensory stimuli from the head are conveyed to the central nervous system via the trigeminal nerves for structures above the tentorium in the anterior and middle fossae of the skull and via the first three cervical nerves for those in the posterior fossa and the inferior surface of the tentorium.

Headache can occur as the result of (1) distention, traction, or dilation of intracranial or extracranial arteries; (2) traction or displacement of large intracranial veins or their dural envelope; (3) compression, traction, or inflammation of cranial and spinal nerves; (4) spasm, inflammation, or trauma to cranial and cervical muscles; (5) meningeal irritation and raised intracranial pressure; or (6) other possible mechanisms such as activation of brainstem structures.

GENERAL CLINICAL CONSIDERATIONS

The quality, location, duration, and time course of the headache and the conditions that produce, exacerbate, or relieve it should be carefully reviewed. Ascertaining the *quality* of cephalic pain is occasionally helpful for diagnosis. Most tension-type headaches are described as tight "bandlike" pain or as dull, deeply located, and aching pain. Jabbing, brief, sharp cephalic pain, often occurring multifocally (ice pick-like pain), is the signature of a benign, nondescript disorder. A throbbing quality and tight muscles about the head, neck, and shoulder girdle are common nonspecific accompaniments of vascular headaches.

Pain *intensity* rarely has diagnostic value, although from the patient's perspective, it is

the single aspect of pain that is most important. Although meningitis, subarachnoid hemorrhage, and cluster headache produce intense cranial pain, most patients entering emergency departments with the most severe headache of their lives usually have migraine. Contrary to common belief, the headache produced by a brain tumor is not usually distinctive or severe.

Data regarding *location* of headache may be informative. If the source is an extracranial structure, as in giant cell arteritis, the correspondence with the site of pain is fairly precise. Inflammation of an extracranial artery causes pain and exquisite tenderness localized to the site of the vessel. Lesions of paranasal sinuses, teeth, eyes, and upper cervical vertebrae induce less sharply localized pain, but pain that is still referred in a regional distribution. Intracranial lesions in the posterior fossa cause pain that is usually occipitotemporal, and supratentorial lesions most often induce frontotemporal pain.

Duration and *time-intensity* curves of headaches are diagnostically useful. A ruptured aneurysm results in head pain that peaks in an instant, thunderclap-like; much less often, unruptured aneurysms may signal their presence in the same way. Cluster headache attacks reach their peak over 3 to 5 min, remain at maximal levels for about 45 min, and then taper off. Migraine attacks build up over hours, are maintained for several hours to days, and are characteristically relieved by sleep. Sleep disruption and early morning headaches that improve during the day are characteristics of headaches produced by brain tumors.

The analysis of facial pain requires a different approach. Trigeminal and, less commonly, glossopharyngeal neuralgia are frequent causes of facial pain ([Chap. 367](#)). "Neuralgias" are painful disorders characterized by paroxysmal, fleeting, often electric shock-like episodes that are frequently caused by demyelinating lesions of nerves (the trigeminal or glossopharyngeal nerves in cranial neuralgias). Certain maneuvers characteristically trigger paroxysms of pain. However, the most common cause of facial pain by far is dental; provocation by hot, cold, or sweet foods is typical. The application of a cold stimulus will repeatedly induce dental pain, whereas in neuralgic disorders, a refractory period usually occurs after the initial response so that pain cannot be repeatedly induced.

The effect of eating on facial pain may provide insight into its cause. Is it the chewing, swallowing, or taste of the food that elicits pain? Chewing points toward trigeminal neuralgia, temporomandibular joint dysfunction, or giant cell arteritis ("jaw claudication"), whereas swallowing *and* taste provocation point toward glossopharyngeal neuralgia. Pain upon swallowing is common among patients with carotidynia (see below) because the inflamed, tender carotid artery abuts the esophagus during deglutition.

Many patients with facial pain do not experience stereotypic neuralgias; the term *atypical facial pain* has been used in this setting. Vague, poorly localized, continuous facial pain is characteristic of nasopharyngeal carcinoma; a burning pain often develops as deafferentation occurs and evidence of cranial neuropathy appears. Burning facial pain may also occur with tumors of the fifth cranial nerve (meningioma or schwannoma) or with lesions of the pons that interrupt the dorsal root entry zone of the nerve (multiple sclerosis). In patients with facial pain, the finding of objective sensory loss is an important clue to a serious underlying disorder. Occasionally, the cause of a pain

problem cannot be resolved promptly, necessitating periodic follow- up until further signs appear.

CLINICAL EVALUATION OF ACUTE, NEW-ONSET HEADACHE

Patients who present with their first severe headache raise entirely different diagnostic possibilities than those with recurrent headaches over many years. In new-onset and severe headaches, the probability of finding a potentially serious cause is considerably greater than in recurrent headache. When a patient complains of an acute, new-onset headache, a number of causes should be considered including meningitis, subarachnoid hemorrhage, epidural or subdural hematoma, glaucoma, and purulent sinusitis. Clinical features of acute, new-onset headache caused by serious underlying conditions are summarized in [Table 15-2](#).

A complete neurologic examination is an essential first step in the evaluation. In most cases, an abnormal examination should be followed by a computed tomography (CT) or a magnetic resonance imaging (MRI) study. As a screening procedure for intracranial pathology in this setting, CT and MRI methods appear to be equally sensitive. A general evaluation of acute headache might include the investigation of cardiovascular and renal status by blood pressure monitoring and urine examination; eyes by fundoscopy, intraocular pressure measurement, and refraction; cranial arteries by palpation; and cervical spine by the effect of passive movement of the head and imaging.

The psychological state of the patient should also be evaluated since a relationship exists between head pain and depression. Many patients in chronic daily pain cycles become depressed; moreover, there is a greater-than-chance coincidence of migraine with both bipolar (manic depressive) and unipolar major depressive disorders. Drugs with antidepressant actions are also effective in the prophylactic treatment of both tension-type headache and migraine.

Underlying recurrent headache disorders may be activated by pain that follows otologic or endodontic surgical procedures. Treatment of the headache problem is largely ineffective until the cause of the primary problem is addressed. Thus, pain about the head as the result of diseased tissue or trauma may reawaken an otherwise quiescent migrainous syndrome.

Serious underlying conditions that are associated with headache are described below and in [Table 15-3](#).

MENINGITIS

In general, acute, severe headache with stiff neck and fever suggests meningitis. Lumbar puncture is mandatory. Often there is striking accentuation of pain with eye movement. Meningitis is particularly easy to mistake for migraine in that the cardinal symptoms of pounding headache, photophobia, nausea, and vomiting are present. **A detailed discussion of meningitis can be found in [Chaps. 372 to 374](#).*

INTRACRANIAL HEMORRHAGE

In general, acute, severe headache with stiff neck but without fever suggests subarachnoid hemorrhage. A ruptured aneurysm, arteriovenous malformation, or intraparenchymal hemorrhage may also present with only headache. Rarely, if the hemorrhage is small or below the foramen magnum, the head [CT](#) scan can be normal. Therefore, a lumbar puncture may be required to make the definitive diagnosis of a subarachnoid hemorrhage. **A detailed discussion of intracranial hemorrhage can be found in [Chap. 361](#).*

BRAIN TUMOR

Approximately 30% of patients with brain tumors consider headache to be their chief complaint. The head pain is usually nondescript -- an intermittent deep, dull aching of moderate intensity, which may worsen with exertion or change in position and may be associated with nausea and vomiting. This pattern of symptoms results from migraine far more often than from brain tumor. Headache of brain tumor disturbs sleep in about 10% of patients. Vomiting that precedes the appearance of headache by weeks is highly characteristic of posterior fossa brain tumors. A history of amenorrhea or galactorrhea should lead one to question whether a prolactin-secreting pituitary adenoma (or the polycystic ovary syndrome) is the source of headache. Headache arising de novo in a patient with known malignancy suggests either cerebral metastases and/or carcinomatous meningitis. Head pain appearing abruptly after bending, lifting, or coughing can be the clue to a posterior fossa mass (or a Chiari malformation). **A detailed discussion of brain tumors can be found in [Chap. 370](#).*

TEMPORAL ARTERITIS (See also [Chaps. 28](#) and [317](#))

Temporal (giant cell) arteritis is an inflammatory disorder of arteries that frequently involves the extracranial carotid circulation. This is a common disorder of the elderly; its annual incidence is 77:100,000 in individuals aged 50 and older. The average age of onset is 70 years, and women account for 65% of cases. About half of patients with untreated temporal arteritis develop blindness due to involvement of the ophthalmic artery and its branches; indeed, the ischemic optic neuropathy induced by giant cell arteritis is the major cause of rapidly developing bilateral blindness in patients over 60 years of age. Because treatment with glucocorticoids is effective in preventing this complication, prompt recognition of this disorder is important.

Typical presenting symptoms include headache, polymyalgia rheumatica ([Chap. 317](#)), jaw claudication, fever, and weight loss. Headache is the dominant symptom and often appears in association with malaise and muscle aches. Head pain may be unilateral or bilateral and is located temporally in 50% of patients but may involve any and all aspects of the cranium. Pain usually appears gradually over a few hours before peak intensity is reached; occasionally, it is explosive in onset. The quality of pain is only seldom throbbing; it is almost invariably described as dull and boring with superimposed episodic ice pick-like lancinating pains similar to the sharp pains that appear in migraine. Most patients can recognize that the origin of their head pain is superficial, external to the skull, rather than originating deep within the cranium (the pain site for migraineurs). Scalp tenderness is present, often to a marked degree; brushing the hair or resting the head on a pillow may be impossible because of pain. Headache is usually worse at night and is often aggravated by exposure to cold. Reddened, tender nodules or red

streaking of the skin overlying the temporal arteries may be found in patients with headache, as is tenderness of the temporal or, less commonly, the occipital arteries.

The erythrocyte sedimentation rate (ESR) is often, though not always, elevated; a normal ESR does not exclude giant cell arteritis. A temporal artery biopsy and the initiation of prednisone at 80 mg daily for the first 4 to 6 weeks should be instituted when clinical suspicion is high. The prevalence of migraine among the elderly is substantial, considerably higher than that of giant cell arteritis. Migraineurs often report amelioration of their headaches with prednisone, so that one must be cautious about interpreting the therapeutic response.

GLAUCOMA

Glaucoma may present with a prostrating headache associated with nausea and vomiting. The history will usually reveal that the headache started with severe eye pain. On physical examination, the eye is often red with a fixed, moderately dilated pupil. **A detailed discussion of glaucoma can be found in [Chap. 28](#).*

OTHER CAUSES OF HEADACHE

Systemic Illness There is hardly any illness that is never manifested by headache; however, some illnesses are frequently associated with headache. These include infectious mononucleosis, systemic lupus erythematosus, chronic pulmonary failure with hypercapnia (early morning headaches), Hashimoto's thyroiditis, inflammatory bowel disease, many of the illnesses associated with HIV, and the acute blood pressure elevations that occur in pheochromocytoma and in malignant hypertension. The last two examples are the exceptions to the generalization that hypertension per se is a very uncommon cause of headache; diastolic pressures of at least 120 mmHg are requisite for hypertension to cause headache. Persistent headache and fever are often the manifestations of an acute systemic viral infection; if the neck is supple in such a patient, lumbar puncture may be deferred. Some drugs and drug-withdrawal states, e.g., oral contraceptives, ovulation-promoting medications, and glucocorticoid withdrawal, are also associated with headache in some individuals.

Idiopathic Intracranial Hypertension (Pseudotumor Cerebri) Headache, clinically resembling that of brain tumor, is a common presenting symptom of pseudotumor cerebri, a disorder of raised intracranial pressure probably resulting from impaired cerebrospinal fluid CSF absorption by the arachnoid villi. Transient visual obscurations and papilledema with enlarged blind spots and loss of peripheral visual fields are additional manifestations. Most patients are young, female, and obese. They often have a history of exposure to provoking agents such as vitamin A and glucocorticoids. **Treatment of idiopathic intracranial hypertension is discussed in [Chap. 28](#).*

Cough A male-dominated (4:1) syndrome, cough headache is characterized by transient, severe head pain upon coughing, bending, lifting, sneezing, or stooping. Head pain persists for seconds to a few minutes. Many patients date the origins of the syndrome to a lower respiratory infection accompanied by severe coughing or to strenuous weight-lifting programs. Headache is usually diffuse but is lateralized in about

one-third of patients. The incidence of serious intracranial structural anomalies causing this condition is about 25%; the Chiari malformation ([Chap. 368](#)) is a common cause. Thus, [MRI](#) is indicated for most patients with cough headache. The benign disorder may persist for a few years; it responds dramatically to indomethacin at doses ranging from 50 to 200 mg daily. Approximately half of patients will also show a response to therapeutic lumbar puncture with removal of 40 mL of [CSF](#).

Many patients with migraine note that attacks of headache may be provoked by *sustained* physical exertion, such as during the third mile of a 5-mile run. Such headaches build up over hours, in contrast to cough headache. The term *effort migraine* has been used for this syndrome to avoid the ambiguous term *exertional headache*.

Lumbar Puncture Headache following lumbar puncture ([Chap. 356](#)) usually begins within 48 h but may be delayed for up to 12 days. Its incidence is between 10 and 30%. Head pain is dramatically positional; it begins when the patient sits or stands upright; there is relief upon reclining or with abdominal compression. The longer the patient is upright, the longer the latency before head pain subsides. It is worsened by head shaking and jugular vein compression. The pain is usually a dull ache but may be throbbing; its location is occipitofrontal. Nausea and stiff neck often accompany headache, and occasional patients report blurred vision, photophobia, tinnitus, and vertigo. The symptoms resolve over a few days but may on occasion persist for weeks to months.

Loss of [CSF](#) volume decreases the brain's supportive cushion, so that when a patient is upright there is probably dilation and tension placed on the brain's anchoring structures, the pain-sensitive dural sinuses, resulting in pain. Intracranial hypotension often occurs, but severe lumbar puncture headache may be present even in patients who have normal CSF pressure.

Treatment with intravenous caffeine sodium benzoate given over a few minutes as a 500-mg dose will promptly terminate headache in 75% of patients; a second dose given in 1 h brings the total success rate to 85%. An epidural blood patch accomplished by injection of 15 mL of autologous whole blood rarely fails for those who do not respond to caffeine. The mechanism for these treatment effects is not straightforward. The blood patch has an *immediate* effect, making it unlikely that sealing off a dural hole with blood clot is its mechanism of action.

Postconcussion Following seemingly trivial head injuries and particularly after rear-end motor vehicle collisions, many patients report varying combinations of headache, dizziness, vertigo, and impaired memory. Anxiety, irritability and difficulty with concentration are other hallmarks of this syndrome. Symptoms may remit after several weeks or persist for months and even years after the injury. Postconcussion headaches may occur whether or not a person was rendered unconscious by head trauma. Typically, the neurologic examination is normal with the exception of the behavioral abnormalities, and [CT](#) or [MRI](#) studies are unrevealing. Chronic subdural hematoma may on occasion mimic this disorder. Although the cause of postconcussive headache disorder is not known, it should not in general be viewed as a primary psychological disturbance. It often persists long after the settlement of pending lawsuits. The treatment is symptomatic support. Repeated encouragement that the syndrome

eventually remits is important.

Coital Headache This is another male-dominated (4:1) syndrome. Attacks occur periorgasmically, are very abrupt in onset, and subside in a few minutes if coitus is interrupted. These are nearly always benign events and usually occur sporadically; if they persist for hours or are accompanied by vomiting, subarachnoid hemorrhage must be excluded ([Chap. 361](#)).

PRINCIPAL CLINICAL VARIETIES OF RECURRENT HEADACHE

There is usually little difficulty in diagnosing the serious types of headaches listed above because of the clues provided by the associated symptoms and signs. It is when headache is chronic, recurrent, and unattended by other important signs of disease that the physician faces a challenging and unique medical problem. The following sections describe a variety of headache types, ranging from the most common (e.g., tension-type headache) to rare causes of recurrent headache.

TENSION-TYPE HEADACHE

The term *tension-type headache* is still commonly used to describe a chronic head pain syndrome characterized by bilateral tight, bandlike discomfort. Patients may report that the head feels as if it is in a vise or that the posterior neck muscles are tight. The pain typically builds slowly, fluctuates in severity, and may persist more or less continuously for many days. Exertion does not usually worsen the headache. The headache may be episodic or chronic (i.e., present more than 15 days per month). Tension-type headache is common in all age groups, and females tend to predominate. In some patients, anxiety or depression coexist with tension headache.

The pathophysiologic basis of tension-type headache remains unknown. Some investigators believe that periodic tension headache is biologically indistinguishable from migraine, whereas others believe that tension-type headache and migraine are two distinct clinical entities. Abnormalities of cervical and temporal muscle contraction are likely to exist, but the exact nature of the dysfunction has not yet been elucidated.

Relaxation almost always relieves tension-type headaches. Patients should be encouraged to find a means of relaxation, which, for a given individual, could include bed rest, massage, and/or formal biofeedback training. Pharmacologic treatment consists of either simple analgesics and/or muscle relaxants. Ibuprofen and naproxen sodium are useful treatments for most individuals. When simple over-the-counter analgesics such as acetaminophen, aspirin, ibuprofen, and/or other nonsteroidal anti-inflammatory drugs (NSAIDs) alone fail, the addition of butalbital and caffeine (in a combination compound such as Fiorinal, Fioricet) to these analgesics may be effective. A list of commonly used analgesics for tension-type headaches is presented in [Table 15-4](#). For chronic tension-type headache, prophylactic therapy is recommended. Low doses of amitriptyline (10 to 50 mg at bedtime) can provide effective prophylaxis.

MIGRAINE

Migraine, the most common cause of vascular headache, afflicts approximately 15% of

women and 6% of men. A useful definition of migraine is a benign and recurring syndrome of headache, nausea, vomiting, and/or other symptoms of neurologic dysfunction in varying admixtures ([Table 15-5](#)). Migraine can often be recognized by its activators (red wine, menses, hunger, lack of sleep, glare, estrogen, worry, perfumes, let-down periods) and its deactivators (sleep, pregnancy, exhilaration, sumatriptan). A classification of the many subtypes of migraine, as defined by the International Headache Society, is shown in [Table 15-1](#).

Severe headache attacks, regardless of cause, are more likely to be described as throbbing and associated with vomiting and scalp tenderness. Milder headaches tend to be nondescript -- tight, bandlike discomfort often involving the entire head -- the profile of tension-type headache.

Pathogenesis

Genetic Basis of Migraine Migraine has a definite genetic predisposition. Specific mutations leading to rare causes of vascular headache have been identified ([Table 15-6](#)). For example, the MELAS syndrome consists of a mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes and is caused by an A® G point mutation in the mitochondrial gene encoding for tRNA^{Leu(UUR)} at nucleotide position 3243. Episodic migraine-like headaches are another common clinical feature of this syndrome, especially early in the course of the disease. The genetic pattern of mitochondrial disorders is unique, since only mothers transmit mitochondrial DNA. Thus, all children of mothers with MELAS syndrome are affected with the disorder.

Familial hemiplegic migraine (FHM) is characterized by episodes of recurrent hemiparesis or hemiplegia during the aura phase of a migraine headache. Other associated symptoms may include hemianesthesia or paresthesia; hemianopic visual field disturbances; dysphasia; and variable degrees of drowsiness, confusion, and/or coma. In severe attacks, these symptoms can be quite prolonged and persist for days or weeks, but characteristically they last for only 30 to 60 min and are followed by a unilateral throbbing headache.

Approximately 50% of cases of [FHM](#) appear to be caused by mutations within the CACNL1A4 gene on chromosome 19, which encodes a P/Q type calcium channel subunit expressed only in the central nervous system. The gene is very large (>300 kb in length) and consists of 47 exons. Four distinct point mutations have been identified within the gene (in five different families) that cosegregate with the clinical diagnosis of FHM. Analysis of haplotypes in the two families with the same mutation suggest that each mutation arose independently rather than representing a founder effect. Thus, certain subtypes of FHM are caused by mutations in the CACNL1A4 gene. The function of the CACNL1A4 gene remains unknown, but it is likely to play a role in calcium-induced neurotransmitter release and/or contraction of smooth muscle. Different mutations within this gene are the cause of another neurogenetic disorder, episodic ataxia type 2 ([Chap. 364](#)).

In a genetic association study, a NcoI polymorphism in the gene encoding the D₂dopamine receptor (DRD2) was overrepresented in a population of patients with migraine with aura compared to a control group of nonmigraineurs, suggesting that

susceptibility to migraine with aura is modified by certain DRD2 alleles. In a Sardinian population, an association between different DRD2 alleles and migraine has also been demonstrated. Therefore, these initial studies suggest that variations in dopamine receptor regulation and/or function may alter susceptibility to migraine since molecular variations within the DRD2 gene have been associated with variations in dopaminergic function. However, since not all individuals with certain DRD2 genotypes suffer from migraine with aura, additional genes or factors must also be involved. Migraine is likely to be a complex disorder with polygenic inheritance and a strong environmental component.

The Vascular Theory of Migraine It was widely held for many years that the headache phase of migrainous attacks was caused by extracranial vasodilatation and that the neurologic symptoms were produced by intracranial vasoconstriction (i.e., the "vascular" hypothesis of migraine). Regional cerebral blood flow studies have shown that in patients with classic migraine there is, during attacks, a modest cortical hypoperfusion that begins in the visual cortex and spreads forward at a rate of 2 to 3 mm/min. The decrease in blood flow averages 25 to 30% (insufficient to explain symptoms on the basis of ischemia) and progresses anteriorly in a wavelike fashion independent of the topography of cerebral arteries. The wave of hypoperfusion persists for 4 to 6 h, appears to follow the convolutions of the cortex, and does not cross the central or lateral sulcus, progressing to the frontal lobe via the insula. Perfusion of subcortical structures is normal. Contralateral neurologic symptoms appear during temporoparietal hypoperfusion; at times, hypoperfusion persists in these regions after symptoms cease. More often, frontal spread continues as the headache phase begins. A few patients with classic migraine show no flow abnormalities; an occasional patient has developed focal ischemia sufficient to cause symptoms. However, focal ischemia does not appear to be *necessary* for focal symptoms to occur.

The ability of these changes to induce the symptoms of migraine has been questioned. Specifically, the decrease in blood flow that is observed does not appear to be significant enough to cause focal neurologic symptoms. Second, the increase in blood flow per se is not painful, and vasodilatation alone cannot account for the local edema and focal tenderness often observed in migraineurs. Moreover, in migraine without aura, no flow abnormalities are usually seen. Thus, it is unlikely that simple vasoconstriction and vasodilatation are the fundamental pathophysiologic abnormalities in migraine. However, it is clear that cerebral blood flow is altered during certain migraine attacks, and these changes may explain some, but clearly not all, of the clinical syndrome of migraine.

The Neuronal Theory of Migraine In 1941, the psychologist KS Lashley charted his own *fortification spectrum*, which is a migraine aura characterized by a slowly enlarging visual scotoma with luminous edges (see below). He was able to estimate that the evolution of his own scotoma proceeded across the occipital cortex at a rate of 3 mm/min. He speculated that a wavefront of intense excitation followed by a wave of complete inhibition of activity were propagated across the visual cortex. In 1944, the phenomenon that has come to be known as *spreading depression* was described by the Brazilian physiologist Leao in the cerebral cortex of laboratory animals. It is a slowly moving (2 to 3 mm/min), potassium-liberating depression of cortical activity, preceded by a wavefront of increased metabolic activity that can be produced by a variety of

experimental stimuli, including hypoxia, mechanical trauma, and the topical application of potassium. These observations suggest that neuronal abnormalities, most likely initiated in the brainstem, could be the cause of a migraine attack. More recently, both cortical and brainstem changes have been observed in positron emission tomography (PET) scan studies of migraine. Thus, the existence of a specific "brainstem generator" for migraine remains an intriguing possibility that might represent the pathophysiologic basis of migraine.

The Trigeminovascular System in Migraine Activation of cells in the trigeminal nucleus caudalis in the medulla (a pain-processing center for the head and face region) results in the release of vasoactive neuropeptides, including substance P and calcitonin gene-related peptide (CGRP), at vascular terminations of the trigeminal nerve. These peptide neurotransmitters have been proposed to induce a sterile inflammation that activates trigeminal nociceptive afferents originating on the vessel wall, further contributing to the production of pain. This mechanism also provides a potential mechanism for the soft tissue swelling and tenderness of blood vessels that attend migraine attacks. However, numerous pharmacologic agents that are effective in preventing or reducing inflammation in this animal model (e.g., selective 5-HT_{1D}agonists, NK-1 antagonists, endothelin antagonists) have failed to demonstrate any clinical efficacy in recent migraine trials.

5-Hydroxytryptamine in Migraine Pharmacologic and other data point to the involvement of the neurotransmitter 5-hydroxytryptamine (5-HT; also known as serotonin) in migraine. Approximately 40 years ago, methysergide was found to antagonize certain peripheral actions of 5-HT and was introduced as the first drug capable of preventing migraine attacks. Subsequently, it was found that platelet levels of 5-HT fall consistently at the onset of headache and that drugs that cause 5-HT to be released may trigger migrainous episodes. Such changes in circulating 5-HT levels proved to be pharmacologically trivial, however, and interest in the humoral role of 5-HT in migraine declined.

More recently, interest in the role of [5-HT](#) in migraine has been renewed due to the introduction of the triptan class of antimigraine drugs. The triptans are designed to stimulate selectively a particular subpopulation of 5-HT receptors. Molecular cloning studies have demonstrated that at least 14 specific 5-HT receptors exist in humans. The triptans (e.g., naratriptan, rizatriptan, sumatriptan, and zolmitriptan) are potent agonists of 5-HT_{1B}, 5-HT_{1D}, and 5-HT_{1F} receptors and are less potent at 5-HT_{1A} and 5-HT_{1E} receptors. A growing body of data indicates that the antimigraine efficacy of the triptans relates to their ability to stimulate 5-HT_{1B} receptors, which are located both on blood vessels and nerve terminals. Selective 5-HT_{1D} receptor agonists have, thus far, failed to demonstrate clinical efficacy in migraine. Triptans that are weak 5-HT_{1F} agonists are also effective in migraine; however, only 5-HT_{1B} efficacy is currently thought to be essential for antimigraine efficacy.

Physiologically, electrical stimulation near dorsal raphe neurons can result in migraine-like headaches. Blood flow in the pons and midbrain increases focally during migraine headache episodes; this alteration probably results from increased activity of cells in the dorsal raphe and locus caeruleus. There are projections from the dorsal raphe that terminate on cerebral arteries and alter cerebral blood flow. There are also

major projections from the dorsal raphe to important visual centers, including the lateral geniculate body, superior colliculus, retina, and visual cortex. These various serotonergic projections may represent the neural substrate for the circulatory and visual characteristics of migraine. The dorsal raphe cells stop firing during deep sleep, and sleep is known to ameliorate migraine; the antimigraine prophylactic drugs also inhibit activity of the dorsal raphe cells through a direct or indirect agonist effect.

Recent [PET](#) scan studies have demonstrated that midbrain structures near the dorsal raphe are differentially activated during a migraine attack. In one study of acute migraine, an injection of sumatriptan relieved the headache, but did not alter the brainstem changes noted on the PET scan. These data suggest that a "brainstem generator" may be the cause of migraine and that certain antimigraine medications may not interfere with the underlying pathologic process in migraine.

Dopamine in Migraine A growing body of biologic, pharmacologic, and genetic data support a role for dopamine in the pathophysiology of certain subtypes of migraine. Most migraine symptoms can be induced by dopaminergic stimulation. Moreover, there is dopamine receptor hypersensitivity in migraineurs, as demonstrated by the induction of yawning, nausea, vomiting, hypotension, and other symptoms of a migraine attack by dopaminergic agonists at doses that do not affect nonmigraineurs. Conversely, dopamine receptor antagonists are effective therapeutic agents in migraine, especially when given parenterally or concurrently with other antimigraine agents. As noted above, recent genetic data also suggest that molecular variations within dopamine receptor genes play a modifying role in the pathophysiology of migraine with aura. Therefore, modulation of dopaminergic neurotransmission should be considered in the therapeutic management of migraine.

The Sympathetic Nervous System in Migraine Biochemical changes occur within the sympathetic nervous system (SNS) of migraineurs before, during, and between migraine attacks. Factors that activate the SNS are all trigger factors for migraine. Specific examples include environmental changes (e.g., stress, sleep patterns, hormonal shifts, hypoglycemia) and agents that cause release and a secondary depletion of peripheral catecholamines [e.g., tyramine, phenylethylamine, fenfluramine, m-chlorophenylpiperazine (mCPP) and reserpine]. By contrast, effective therapeutic approaches to migraine share an ability to mimic and/or enhance the effects of norepinephrine in the peripheral SNS. For example, norepinephrine itself, sympathomimetics (e.g., isometheptene), monoamine oxidase inhibitors (MAOIs) and reuptake blockers alleviate migraine. Dopamine antagonists, prostaglandin synthesis inhibitors, and adenosine antagonists are pharmacologic agents effective in the acute treatment of migraine. These drugs block the negative feedback inhibition or norepinephrine release induced by endogenous dopamine, prostaglandins, and adenosine. Therefore, migraine susceptibility may relate to genetically based variations in the ability to maintain adequate concentrations of certain neurotransmitters within postganglionic sympathetic nerve terminals. This hypothesis has been called the *empty neuron theory* of migraine.

Clinical Features

Migraine without Aura (Common Migraine) In this syndrome no focal neurologic

disturbance precedes the recurrent headaches. Migraine without aura is by far the more frequent type of vascular headache. The International Headache Society criteria for migraine include moderate to severe head pain, pulsating quality, unilateral location, aggravation by walking stairs or similar routine activity, attendant nausea and/or vomiting, photophobia and phonophobia, and multiple attacks, each lasting 4 to 72 h.

Migraine with Aura (Classic Migraine) In this syndrome headache is associated with characteristic premonitory sensory, motor, or visual symptoms. Focal neurologic disturbances are more common during headache attacks than as prodromal symptoms. Focal neurologic disturbances without headache or vomiting have come to be known as *migraine equivalents* or *migraine accompaniments* and appear to occur more commonly in patients between the ages of 40 and 70 years. The term *complicated migraine* has generally been used to describe migraine with dramatic transient focal neurologic features or a migraine attack that leaves a persisting residual neurologic deficit.

The most common premonitory symptoms reported by migraineurs are visual, arising from dysfunction of occipital lobe neurons. Scotomas and/or hallucinations occur in about one-third of migraineurs and usually appear in the central portions of the visual fields. A highly characteristic syndrome occurs in about 10% of patients; it usually begins as a small paracentral scotoma, which slowly expands into a "C" shape. Luminous angles appear at the enlarging outer edge, becoming colored as the scintillating scotoma expands and moves toward the periphery of the involved half of the visual field, eventually disappearing over the horizon of peripheral vision. The entire process lasts 20 to 25 min. This phenomenon is pathognomonic for migraine, and has never been described in association with a cerebral structural anomaly. It is commonly referred to as a *fortification spectrum* because the serrated edges of the hallucinated "C" seemed to resemble a "fortified town with bastions all round it"; "spectrum" is used in the sense of an apparition or specter.

Basilar Migraine Symptoms referable to a disturbance in brainstem function, such as vertigo, dysarthria, or diplopia, occur as the only neurologic symptoms of the attack in about 25% of patients. A dramatic form of basilar migraine (Bickerstaff's migraine) occurs primarily in adolescent females. Episodes begin with total blindness accompanied or followed by admixtures of vertigo, ataxia, dysarthria, tinnitus, and distal and perioral paresthesia. In about one-quarter of patients, a confusional state supervenes. The neurologic symptoms usually persist for 20 to 30 min and are generally followed by a throbbing occipital headache. This basilar migraine syndrome is now known also to occur in children and in adults over age 50. An altered sensorium may persist for as long as 5 days and may take the form of confusional states superficially resembling psychotic reactions. Full recovery after the episode is the rule.

Carotidynia The carotidynia syndrome, sometimes called *lower-half headache* or *facial migraine*, is most common among older patients, with the incidence peaking in the fourth through sixth decades. Pain is usually located at the jaw or neck, although sometimes periorbital or maxillary pain occurs; it may be continuous, deep, dull, and aching, and it becomes pounding or throbbing episodically. There are often superimposed sharp, ice pick-like jabs. Attacks occur one to several times per week, each lasting several minutes to hours. Tenderness and prominent pulsations of the cervical carotid artery and soft tissue swelling overlying the carotid are usually present

ipsilateral to the pain; many patients also report throbbing ipsilateral headache concurrent with carotidynia attacks as well as between attacks. Dental trauma is a common precipitant of this syndrome. Carotid artery involvement also appears to be common in the more traditional forms of migraine; over 50% of patients with frequent migraine attacks are found to have carotid tenderness at several points on the side most often involved during hemicranial migraine attacks.

TREATMENT

Nonpharmacologic Approaches for All Migraineurs Migraine can often be managed to some degree by a variety of nonpharmacologic approaches ([Table 15-7](#)). The measures that apply to a given individual should be used routinely since they provide a simple, cost-effective approach to migraine management. Patients with migraine do not encounter more stress than headache-free individuals; overresponsiveness to stress appears to be the issue. Since the stresses of everyday living cannot be eliminated, lessening one's response to stress by various techniques is helpful for many patients. These include yoga, transcendental meditation, hypnosis, and conditioning techniques such as biofeedback. For most patients, this approach is, at best, an adjunct to pharmacotherapy. Avoidance of migraine trigger factors may also provide significant prophylactic benefits ([Table 15-7](#)). Unfortunately, these measures are unlikely to prevent all migraine attacks. When these measures fail to prevent an attack, then pharmacologic approaches are needed to abort an attack.

Pharmacologic Treatment of Acute Migraine The mainstay of pharmacologic therapy is the judicious use of one or more of the many drugs that are effective in migraine. The selection of the optimal regimen for a given patient depends on a number of factors, the most important of which is the severity of the attack ([Table 15-8](#)). Mild migraine attacks can usually be managed by oral agents; the average efficacy rate is 50-70%. Severe migraine attacks may require parenteral therapy. Most drugs effective in the treatment of migraine are members of one of three major pharmacologic classes: anti-inflammatory agents, 5-HT₁ agonists, and dopamine antagonists.

[Table 15-9](#) lists specific drugs effective in migraine. In general, an adequate dose of whichever agent is chosen should be used as soon as possible after the onset of an attack. If additional medication is required within 60 min because symptoms return or have not abated, the initial dose should be increased for subsequent attacks. Migraine therapy must be individualized for each patient; a standard approach for all patients is not possible. A therapeutic regimen may need to be constantly refined and personalized until one is identified that provides the patient with rapid, complete, and consistent relief with minimal side effects.

Nonsteroidal anti-inflammatory agents Both the severity and duration of a migraine attack can be reduced significantly by anti-inflammatory agents. Indeed, many undiagnosed migraineurs are self-treated with nonprescription anti-inflammatory agents ([Table 15-4](#)). A general consensus is that [NSAIDs](#) are most effective when taken early in the migraine attack. However, the effectiveness of anti-inflammatory agents in migraine is usually less than optimal in moderate or severe migraine attacks. The combination of acetaminophen, aspirin, and caffeine (Excedrin Migraine) has been approved for use by the U.S. Food and Drug Administration (FDA) for the treatment of mild to moderate

migraine. The combination of aspirin and metoclopramide has been shown to be equivalent to a single dose of sumatriptan. Major side effects of NSAIDs include dyspepsia and gastrointestinal irritation.

5-HT₁agonists

ORAL Stimulation of 5-HT₁ receptors can stop an acute migraine attack. Ergotamine and dihydroergotamine are nonselective receptor agonists, while the series of drugs known as triptans are selective 5-HT₁ receptor agonists. A variety of triptans (e.g., naratriptan, rizatriptan, sumatriptan, zolmitriptan) are now available for the treatment of migraine (Table 15-9).

Each of the triptan class of drugs has similar pharmacologic properties, but varies slightly in terms of clinical efficacy. Rizatriptan appears to be the fastest acting and most efficacious of the triptans currently available in the United States. Sumatriptan and zolmitriptan have similar rates of efficacy as well as time to onset, whereas naratriptan is the slowest acting and the least efficacious. Clinical efficacy appears to be related more to the t_{max} (time to peak plasma level) than to the potency, half-life, or bioavailability (Table 15-10). This observation is in keeping with a significant body of data indicating that faster-acting analgesics are more efficacious than slower-acting agents.

Unfortunately, monotherapy with a selective oral 5-HT₁ agonist does not result in rapid, consistent, and complete relief of migraine in all patients. Triptans are not effective in migraine with aura unless given after the aura is completed and the headache initiated. Side effects, although often mild and transient, occur in up to 89% of patients. Moreover, 5-HT₁ agonists are contraindicated in individuals with a history of cardiovascular disease. Recurrence of headache is a major limitation of triptan use, and occurs at least occasionally in 40 to 78% of patients.

Ergotamine preparations offer a nonselective means of stimulating 5-HT₁ receptors. A nonnauseating dose of ergotamine should be sought since a dose that provokes nausea is too high and may intensify head pain. Except for a sublingual formulation of ergotamine (Ergomar), oral formulations of ergotamine also contain 100 mg caffeine (theoretically to enhance ergotamine absorption and possibly to add additional vasoconstrictor activity). The average oral ergotamine dose for a migraine attack is 2 mg. Since the clinical studies demonstrating the efficacy of ergotamine in migraine predated the clinical trial methodologies used with the triptans, it is difficult to assess the clinical efficacy of ergotamine versus the triptans. In general, ergotamine appears to have a much higher incidence of nausea than triptans, but less headache recurrence.

NASAL The fastest acting nonparenteral antimigraine therapies that can be self-administered include nasal formulations of dihydroergotamine (Migranal) or sumatriptan (Imitrex Nasal). The nasal sprays result in substantial blood levels within 30 to 60 min. However, the nasal formulations suffer from inconsistent dosing, poor taste, and variable efficacy. Although in theory the nasal sprays might provide faster and more effective relief of a migraine attack than oral formulations, their reported efficacy is only approximately 50 to 60%.

PARENTERAL Parenteral administration of drugs such as dihydroergotamine (DHE-45

Injectable) and sumatriptan (Imitrex SC) is approved by the [FDA](#) for the rapid relief of a migraine attack. Peak plasma levels of dihydroergotamine are achieved 3 min after intravenous dosing, 30 min after intramuscular dosing, and 45 min after subcutaneous dosing. If an attack has not already peaked, subcutaneous or intramuscular administration of 1 mg dihydroergotamine suffices for about 80 to 90% of patients. Sumatriptan, 6 mg subcutaneously is effective in approximately 70 to 80% of patients.

Dopamine Antagonists

ORAL Oral dopamine antagonists should be considered as adjunctive therapy in migraine. Drug absorption is impaired during migrainous attacks because of reduced gastrointestinal motility. Delayed absorption occurs in the absence of nausea and is related to the severity of the attack and not its duration. Therefore, when oral [NSAIDs](#) and/or triptan agents fail, the addition of a dopamine antagonist such as metoclopramide, 10 mg, should be considered to enhance gastric absorption. In addition, dopamine antagonists decrease nausea/vomiting and restore normal gastric motility.

PARENTERAL Parenteral dopamine antagonists (e.g., chlorpromazine, prochlorperazine, metoclopramide) can also provide significant acute relief of migraine; they can be used in combination with parenteral [5-HT₁](#) agonists. A common intravenous protocol used for the treatment of severe migraine is the administration over 2 min of a mixture of 5 mg of prochlorperazine and 0.5 mg of dihydroergotamine.

Other Medications for Acute Migraine

ORAL The combination of acetaminophen, dichloralphenazone, and isometheptene (i.e., Midrin, Duradrin, generic), one to two capsules, has been classified by the [FDA](#) as "possibly" effective in the treatment of migraine. Since the clinical studies demonstrating the efficacy of this combination analgesic in migraine predated the clinical trial methodologies used with the triptans, it is difficult to assess the clinical efficacy of this sympathomimetic compound in comparison to other agents.

NASAL A nasal preparation of butorphanol is available for the treatment of acute pain. As with all narcotics, the use of nasal butorphanol should be limited to a select group of migraineurs, as described below.

PARENTERAL Narcotics are effective in the acute treatment of migraine. For example, intravenous meperidene (Demerol), 50 to 100 mg, is given frequently in the emergency room. This regimen "works" in the sense that the pain of migraine is eliminated. However, this regimen is clearly suboptimal in patients with recurrent headache for two major reasons. First, narcotics do not treat the underlying headache mechanism; rather, they act at the thalamic level to alter pain sensation. Second, the recurrent use of narcotics can lead to significant problems. In patients taking oral narcotics such as oxycodone (Percodan) or hydrocodone (Vicoden), narcotic addiction can greatly confuse the treatment of migraine. The headache that results from narcotic craving and/or withdrawal can be difficult to distinguish from chronic migraine. Therefore, it is recommended that narcotic use in migraine be limited to patients with severe, but infrequent, headaches that are unresponsive to other pharmacologic approaches.

Prophylactic Treatment of Migraine A substantial number of drugs are now available that have the capacity to stabilize migraine ([Table 15-11](#)). The decision of whether to use this approach depends on the frequency of attacks and on how well acute treatment is working. The occurrence of at least three attacks per month could be an indication for this approach. Drugs must be taken daily and there is usually a lag of at least 2 to 6 weeks before an effect is seen. The drugs that have been approved by the [FDA](#) for the prophylactic treatment of migraine include propranolol, timolol, sodium valproate, and methysergide. In addition, a number of other drugs appear to display prophylactic efficacy. This group of drugs includes amitriptyline, nortriptyline, verapamil, phenelzine, isocarbazid, and cyproheptadine. Phenelzine and methysergide are usually reserved for recalcitrant cases because of their serious potential side effects. Phenelzine is an [MAOI](#); therefore, tyramine-containing foods, decongestants, and meperidine are contraindicated. Methysergide may cause retroperitoneal or cardiac valvular fibrosis when it is used for more than 8 months, thus monitoring is required for patients using this drug; the risk of the fibrotic complication is about 1:1500 and is likely to reverse after the drug is stopped.

The probability of success with any one of the antimigraine drugs is 50 to 75%; thus, if one drug is assessed each month, there is a good chance that effective stabilization will be achieved within a few months. Many patients are managed adequately with low-dose amitriptyline, propranolol, or valproate. If these agents fail or lead to unacceptable side effects, then methysergide or phenelzine can be used. Once effective stabilization is achieved, the drug is continued for 5 to 6 months and then slowly tapered to assess the continued need. Many patients are able to discontinue medication and experience fewer and milder attacks for long periods, suggesting that these drugs may alter the natural history of migraine.

CLUSTER HEADACHE

A variety of names have been used for this condition, including *Raeder's syndrome*, *histamine cephalalgia*, and *sphenopalatine neuralgia*. *Cluster headache* is a distinctive and treatable vascular headache syndrome. The episodic type is most common and is characterized by one to three short-lived attacks of periorbital pain per day over a 4- to 8-week period, followed by a pain-free interval that averages 1 year. The chronic form, which may begin de novo or several years after an episodic pattern has become established, is characterized by the absence of sustained periods of remission. Each type may transform into the other. Men are affected seven to eight times more often than women; hereditary factors are usually absent. Although the onset is generally between ages 20 and 50, it may occur as early as the first decade of life. Propranolol and amitriptyline are largely ineffective. Lithium is beneficial for cluster headache and ineffective in migraine. The cluster syndrome is thus clinically, genetically, and therapeutically different from migraine. Nevertheless, mixed features of the two disorders are occasionally present, suggesting some common elements to their pathogenesis.

Clinical Features Periorbital or, less commonly, temporal pain begins without warning and reaches a crescendo within 5 min. It is often excruciating in intensity and is deep, nonfluctuating, and explosive in quality; only rarely is it pulsatile. Pain is strictly unilateral

and usually affects the same side in subsequent months. Attacks last from 30 min to 2 h; there are often associated symptoms of homolateral lacrimation, reddening of the eye, nasal stuffiness, lid ptosis, and nausea. Alcohol provokes attacks in about 70% of patients but ceases to be provocative when the bout remits; this on-off vulnerability to alcohol is pathognomonic of cluster headache. Only rarely do foods or emotional factors precipitate pain, in contrast to migraine.

There is a striking periodicity of attacks in at least 85% of patients. At least one of the daily attacks of pain recurs at about the same hour each day for the duration of a cluster bout. Onset is nocturnal in about 50% of the cases, and then the pain usually awakens the patient within 2 h of falling asleep.

Pathogenesis No consistent cerebral blood flow changes accompany attacks of pain. Perhaps the strongest evidence for a central mechanism is the periodicity of attacks; the existence of a central mechanism is also suggested by the observation that autonomic symptoms that accompany the pain are bilateral and are more severe on the painful side. The hypothalamus may be the site of activation in this disorder. The posterior hypothalamus contains cells that regulate autonomic functions, and the anterior hypothalamus contains cells (in the suprachiasmatic nuclei) that constitute the principal circadian pacemaker in mammals. Activation of both is necessary to explain the symptoms of cluster headache. The pacemaker is modulated via serotonergic dorsal raphe projections. It can be concluded tentatively that both migraine and cluster headache result from abnormal serotonergic neurotransmission, albeit at different loci.

TREATMENT

The most satisfactory treatment is the administration of drugs to prevent cluster attacks until the bout is over. Effective prophylactic drugs are prednisone, lithium, methysergide, ergotamine, sodium valproate, and verapamil. Lithium (600 to 900 mg daily) appears to be particularly useful for the chronic form of the disorder. A 10-day course of prednisone, beginning at 60 mg daily for 7 days followed by a rapid taper, may interrupt the pain bout for many patients. When ergotamine is used, it is most effective when given 1 to 2 h before an expected attack. Patients must be educated regarding the early symptoms of ergotism when ergotamine is used daily; a weekly limit of 14 mg should be adhered to.

For the attacks themselves, oxygen inhalation (9 L/min via a loose mask) is the most effective modality; 15 min of inhalation of 100% oxygen is often necessary. Sumatriptan, 6 mg subcutaneously, will usually shorten an attack to 10 to 15 min.

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16. BACK AND NECK PAIN - John W. Engstrom

The importance of back and neck pain in our society is underscored by the following: (1) the annual societal cost of back pain in the United States is estimated to be between \$20 and \$50 billion; (2) back symptoms are the most common cause of disability in patients under 45 years of age; (3) 50% of working adults, in one survey, admitted to having a back injury each year; and (4) approximately 1% of the U.S. population is chronically disabled because of back pain.

The enormous economic pressure to provide rational and efficient care of patients with back pain has resulted in clinical practice guidelines (CPGs) for these patients. CPGs are algorithms which guide evaluation or treatment at specific steps in patient care. CPGs for *acute low back pain* (ALBP) are based upon incomplete evidence (see algorithms, [Fig. 16-6](#)) but represent an attempt to standardize common medical practice. Major revisions in CPGs for back pain can be anticipated in the future. Management of patients with *chronic low back pain* (CLBP) is complex and not amenable to a simple algorithmic approach at this time.

ANATOMY OF THE SPINE

The anterior portion of the spine consists of cylindrical vertebral bodies separated by intervertebral disks and held together by the anterior and posterior longitudinal ligaments. The intervertebral disks are composed of a central gelatinous nucleus pulposus surrounded by a tough cartilagenous ring, the annulus fibrosis; disks are responsible for 25% of spinal column length ([Figs. 16-1](#) and [16-2](#)). The disks are largest in the cervical and lumbar regions where movements of the spine are greatest. The disks are elastic in youth and allow the bony vertebrae to move easily upon each other. Elasticity is lost with age. The function of the anterior spine is to absorb the shock of typical body movements such as walking and running.

The posterior portion of the spine consists of the vertebral arches and seven processes. Each arch consists of paired cylindrical pedicles anteriorly and paired laminae posteriorly ([Fig. 16-1](#)). The vertebral arch gives rise to two transverse processes laterally, one spinous process posteriorly, plus two superior and two inferior articular facets. The functions of the posterior spine are to protect the spinal cord and nerves within the spinal canal and to stabilize the spine by providing sites for the attachment of muscles and ligaments. The contraction of muscles attached to the spinous and transverse processes produces a system of pulleys and levers that results in flexion, extension, and lateral bending movements of the spine. Normal upright posture in humans places the center of gravity anterior to the spine. The graded contraction of well-developed paraspinal muscles attached to the laminae, transverse processes, and spinous processes is necessary to maintain normal upright posture.

The nerve roots exit at a level above their respective vertebral bodies in the cervical region (the C7 nerve root exits at the C6-C7 level) and below their respective vertebral bodies in the thoracic and lumbar regions (the T1 nerve root exits at the T1-T2 level). The spinal cord ends at the L1 or L2 level of the bony spine. Consequently, the lumbar nerve roots follow a long intraspinal course and can be injured anywhere from the upper lumbar spine to their exit at the intervertebral foramen. For example, it is common for

disk herniation at the L4-L5 level to produce compression of the S1 nerve root ([Fig. 16-3](#)). In contrast, cervical nerve roots follow a short intraspinal course and exit at the level of their respective spinal cord segments (upper cervical) or one segment below the corresponding levels (lower cervical cord). Cervical spine pathology can result in spinal cord compression, but lumbar spine pathology cannot.

Pain-sensitive structures in the spine include the vertebral body periosteum, dura, facet joints, annulus fibrosus of the intervertebral disk, epidural veins, and the posterior longitudinal ligament. Damage to these nonneural structures may cause pain. The nucleus pulposus of the intervertebral disk is not pain-sensitive under normal circumstances. Pain sensation is conveyed by the sinuvertebral nerve that arises from the spinal nerve at each spine segment and reenters the spinal canal through the intervertebral foramen at the same level. Disease of these diverse pain-sensitive spine structures may explain many cases of back pain without nerve root compression. The lumbar and cervical spine possess the greatest potential for movement and injury.

Approach to the Patient

Types of Back Pain An understanding of the nature of the pain as described by the patient is the essential first step in evaluation. Attention is also focused on identification of risk factors for serious underlying diseases that require specific evaluation.

Local pain is caused by stretching of pain-sensitive structures that compress or irritate sensory nerve endings. The site of the pain is near the affected part of the back.

Pain referred to the back may arise from abdominal or pelvic viscera. The pain is usually described as primarily abdominal or pelvic but is accompanied by back pain and usually unaffected by posture. The patient may occasionally complain of back pain only.

Pain of spine origin may be located in the back or referred to the buttocks or legs. Diseases affecting the upper lumbar spine tend to refer pain to the lumbar region, groin, or anterior thighs. Diseases affecting the lower lumbar spine tend to produce pain referred to the buttocks, posterior thighs, or rarely the calves or feet. Provocative injections into pain-sensitive structures of the spine (diskography) may produce leg pain that does not follow a dermatomal distribution. The exact pathogenesis of this "sclerotomal" pain is unclear, but it may explain many instances in which combined back and leg pain is unaccompanied by evidence of nerve root compression.

Radicular back pain is typically sharp and radiates from the spine to the leg within the territory of a nerve root (see "Lumbar Disk Disease," below). Coughing, sneezing, or voluntary contraction of abdominal muscles (lifting heavy objects or straining at stool) may elicit the radiating pain. The pain may increase in postures that stretch the nerves and nerve roots. Sitting stretches the sciatic nerve (L5 and S1 roots) because the nerve passes posterior to the hip. The femoral nerve (L2, L3, and L4 roots) passes anterior to the hip and is not stretched by sitting. The description of the pain alone often fails to distinguish clearly between sclerotomal pain and radiculopathy.

Pain associated with muscle spasm, although of obscure origin, is commonly associated with many spine disorders. The spasms are accompanied by abnormal posture, taut

paraspinal muscles, and dull pain.

Back pain at rest or unassociated with specific postures should raise the index of suspicion for an underlying serious cause (e.g., spine tumor, fracture, infection, or referred pain from visceral structures). Knowledge of the circumstances associated with the onset of back pain is important when weighing possible serious underlying causes for the pain. Some patients involved in accidents or work-related injuries may exaggerate their pain for the purpose of compensation or for psychological reasons.

Examination of the Back A physical examination that includes the abdomen and rectum is advisable. Back pain referred from visceral organs may be reproduced during palpation of the abdomen (pancreatitis, abdominal aortic aneurysm) or percussion over the costovertebral angles (pyelonephritis, adrenal disease, L1-L2 transverse process fracture).

The normal spine ([Fig. 16-2](#)) displays a thoracic kyphosis, lumbar lordosis, and cervical lordosis. Exaggeration of these normal alignments may result in hyperkyphosis (lameback) of the thoracic spine or hyperlordosis (swayback) of the lumbar spine. Spasm of lumbar paraspinal muscles results in flattening of the usual lumbar lordosis. Inspection may reveal lateral curvature of the spine (scoliosis) or an asymmetry in the appearance of the paraspinal muscles, suggesting muscle spasm. Taut paraspinal muscles limit the motion of the lumbar spine. Back pain of bony spine origin is often reproduced by palpation or percussion over the spinous process of the affected vertebrae.

Forward bending is frequently limited by paraspinal muscle spasm. Flexion of the hips is normal in patients with lumbar spine disease, but flexion of the lumbar spine is limited and sometimes painful. Lateral bending to the side opposite the injured spinal element may stretch the damaged tissues, worsen pain, and limit motion. Hyperextension of the spine (with the patient prone or standing) is limited when nerve root compression or bony spine disease is present.

Pain from hip disease may mimic the pain of lumbar spine disease. The first movement is typically internal rotation of the hip. Manual internal and external rotation at the hip with the knee and hip in flexion (Patrick sign) may reproduce the pain, as may percussion of the heel (of an outstretched leg) with the palm of the examiner's hand.

In the supine position passive flexion of the thigh on the abdomen while the knee is extended produces stretching of the L5 and S1 nerve roots and the sciatic nerve because the nerve passes posterior to the hip. Passive dorsiflexion of the foot during the maneuver adds to the stretch. While flexion to at least 80° is normally possible without causing pain, tight hamstrings commonly limit motion, may result in pain, and are readily identified by the patient. This *straight leg-raising (SLR) sign* is positive if the maneuver reproduces the patient's usual back or limb pain. Eliciting the SLR sign in the sitting position may help determine if the finding is reproducible. The patient may describe pain in the low back, buttocks, posterior thigh, or lower leg, but the key feature is reproduction of the patient's usual pain. The *crossed SLR* sign is positive when performance of the maneuver on one leg reproduces the patient's pain symptoms in the opposite leg or buttocks. The nerve or nerve root lesion is always on the side of the

pain. The *reverse* SLR sign is elicited by standing the patient next to the examination table and passively extending each leg while the patient continues to stand. This maneuver stretches the L2-L4 nerve roots and the femoral nerve because the nerves pass anterior to the hip. The reverse SLR test is positive if the maneuver reproduces the patient's usual back or limb pain.

The neurologic examination includes a search for weakness, muscle atrophy, focal reflex changes, diminished sensation in the legs, and signs of spinal cord injury. Findings with specific nerve root lesions are shown in [Table 16-1](#) and are discussed below.

Laboratory Studies Routine laboratory studies such as a complete blood count, erythrocyte sedimentation rate, chemistry panel, and urinalysis are rarely needed for the initial evaluation of acute (<3 months), nonspecific, low back pain. If risk factors for a serious underlying disease are present, then laboratory studies (guided by the history and examination) are indicated ([Fig. 16-6B](#)).

Plain films of the lumbar or cervical spine are helpful when risk factors for vertebral fracture (trauma, chronic steroid use) are present. *In the absence of risk factors, routine x-rays of the lumbar spine in the setting of acute, nonspecific, low back pain are expensive and rarely helpful.* Magnetic resonance imaging (MRI) and computed tomography (CT)-myelography have emerged as the radiologic tests of choice for evaluation of most serious diseases involving the spine. In general, the definition of soft tissue structures by MRI is superior, whereas CT-myelography provides optimal imaging of bony lesions in the region of the lateral recess and intervertebral foramen and is tolerated by claustrophobic patients. With rare exceptions, conventional myelography and bone scan are inferior to MRI and CT-myelography.

Electromyography (EMG) can be used to assess the functional integrity of the peripheral nervous system ([Chap. 357](#)) in the setting of back pain. Sensory nerve conduction studies are normal when focal sensory loss is due to nerve root damage because the nerve roots are proximal to the nerve cell bodies in the dorsal root ganglia. The diagnostic yield of needle EMG is higher than that of nerve conduction studies for radiculopathy. Denervation changes in a myotomal (segmental) distribution are detected by sampling multiple muscles supplied by different nerve roots and nerves; the pattern of muscle involvement indicates the nerve root(s) responsible for the injury. Needle EMG provides objective information about motor nerve fiber injury when the clinical evaluation of weakness is limited by pain or poor effort. EMG and nerve conduction studies will be normal when only limb pain or sensory nerve root injury or irritation is present. Mixed nerve somatosensory evoked potentials and F-wave studies are of uncertain value in the evaluation of radiculopathy.

CAUSES OF BACK PAIN

CONGENITAL ANOMALIES OF THE LUMBAR SPINE

Spondylolysis is a bony defect in the pars interarticularis (a segment near the junction of the pedicle with the lamina) of the vertebra; the etiology of the defect may be a stress fracture in a congenitally abnormal segment. The defect (usually bilateral) is best

visualized on oblique projections in plain x-rays or by [CT](#) scan and occurs in the setting of a single injury, repeated minor injuries, or growth.

Spondylolisthesis is the anterior slippage of the vertebral body, pedicles, and superior articular facets, leaving the posterior elements behind. Spondylolisthesis is associated with spondylolysis and degenerative spine disease and occurs more frequently in women. The slippage may be asymptomatic but may also cause low back pain, nerve root injury (the L5 root most frequently), or symptomatic spinal stenosis. Tenderness may be elicited near the segment that has "slipped" forward (most often L4 on L5 or occasionally L5 on S1). A "step" may be present on deep palpation of the posterior elements of the segment above the spondylolisthetic joint. The trunk may be shortened and the abdomen protuberant as a result of extreme forward displacement of L4 on L5 in severe degrees of spondylolisthesis. In these cases, cauda equina syndrome may occur ([Chap. 368](#)).

TRAUMA

Trauma is an important cause of acute low back pain. A patient complaining of back pain and inability to move the legs may have a spinal fracture or dislocation, and, with fractures above L1, spinal cord compression. In such cases care must be taken to avoid further damage to the spinal cord or nerve roots. The back should be immobilized pending results of plain x-rays.

Sprains and Strains The terms *low back sprain*, *strain*, or *mechanically induced muscle spasm* are used for minor, self-limited injuries associated with lifting a heavy object, a fall, or a sudden deceleration such as occurs in an automobile accident. These terms are used loosely and do not clearly describe a specific anatomic lesion. The pain is usually confined to the lower back, and there is no radiation to the buttocks or legs. Patients with low back pain and paraspinal muscle spasm often assume unusual postures.

Vertebral Fractures Most traumatic fractures of the lumbar vertebral bodies result from compression or flexion injuries producing anterior wedging or compression. With more severe trauma, the patient may sustain a fracture-dislocation or a "burst" fracture involving not only the vertebral body but posterior elements as well. Traumatic vertebral fractures are caused by falls from a height (a pars interarticularis fracture of the L5 vertebra is common), sudden deceleration in an automobile accident, or direct injury. Neurologic impairment is commonly associated with these injuries, and early surgical treatment is indicated ([Chap. 369](#)).

When fractures are atraumatic, the bone is presumed to be weakened by a pathologic process. The cause is usually postmenopausal (type 1) or senile (type 2) osteoporosis ([Chap. 342](#)). Underlying systemic disorders such as osteomalacia, hyperparathyroidism, hyperthyroidism, multiple myeloma, metastatic carcinoma, or glucocorticoid use may also weaken the vertebral body. The clinical context, neurologic signs, and x-ray appearance of the spine establish the diagnosis. Antiresorptive drugs including biphosphonates, alendronate, transdermal estrogen, and tamoxifen have been shown to reduce the risk of osteoporotic fractures.

LUMBAR DISK DISEASE

This disorder is a common cause of chronic or recurrent low back and leg pain. Disk disease is most likely to occur at the L4-L5 and L5-S1 levels, but upper lumbar levels are involved occasionally. The cause of the disk injury is often unknown; the risk is increased in overweight individuals. Degeneration of the nucleus pulposus and the annulus fibrosus increases with age and may be asymptomatic or painful. A sneeze, cough, or trivial movement may cause the nucleus pulposus to prolapse, pushing the frayed and weakened annulus posteriorly. In severe disk disease, the nucleus may protrude through the annulus (herniation) or become extruded to lie as a free fragment in the spinal canal.

The mechanism by which intervertebral disk injury causes back pain is controversial. The inner annulus fibrosus and nucleus pulposus are normally devoid of innervation. Inflammation and production of proinflammatory cytokines within the protruding or ruptured disk may trigger or perpetuate back pain. Ingrowth of nociceptive (pain) nerve fibers into inner portions of diseased intervertebral disk may be responsible for chronic "diskogenic" pain. Nerve root injury (*radiculopathy*) from disk herniation may be due to compression, inflammation, or both; pathologically, varying degrees of demyelination and axonal loss are usually present.

The symptoms of a ruptured intervertebral disk include back pain, abnormal posture, limitation of spine motion (particularly flexion), or radicular pain. A dermatomal pattern of sensory loss or a reduction in or loss of a deep tendon reflex is more suggestive of a specific root lesion than the pattern of pain. Motor findings (focal weakness, muscle atrophy, or fasciculations) occur less frequently than sensory or reflex changes, but a myotomal pattern of involvement can suggest specific nerve root injury. Lumbar disk disease is usually unilateral ([Fig. 16-4](#)), but bilateral involvement does occur with large central disk herniations that compress several nerve roots at the same level. Clinical manifestations of specific lumbosacral nerve root lesions are summarized in [Table 16-1](#). There is evidence to suggest that lumbar disk herniation with a nonprogressive nerve root deficit can be managed conservatively (i.e., nonsurgically) with a successful outcome. The size of the disk protrusion may naturally decrease over time.

Degeneration of the intervertebral disk without frank extrusion of disk tissue may give rise to low back pain only. There may be referred pain in the leg, buttock, or hip with little or no discomfort in the back and no signs of nerve root involvement. Lumbar disk syndromes are usually unilateral, but large central disk herniations can cause bilateral symptoms and signs and may produce a cauda equina syndrome.

Breakaway weakness describes a variable power of muscle contraction by a patient who is asked to provide maximal effort. The weakness may be due to pain or a combination of pain and underlying true weakness. Breakaway weakness without pain is due to lack of effort; patients who exhibit breakaway weakness should be asked if testing a specific muscle is painful. In uncertain cases, [EMG](#) can determine whether or not true weakness is present.

The differential diagnosis of lumbar disk disease includes a variety of serious and treatable conditions, including epidural abscess, hematoma, or tumor. Fever, constant

pain uninfluenced by position, sphincter abnormalities, or signs of spinal cord disease suggest an etiology other than lumbar disk disease. Bilateral absence of ankle reflexes can be a normal finding in old age or a sign of bilateral S1 radiculopathy. An absent deep tendon reflex or focal sensory loss may reflect injury to a nerve root, but other sites of injury along the nerve must also be considered. For example, an absent knee reflex may be due to a femoral neuropathy rather than an L4 nerve root injury. A focal decrease in sensation over the foot and distal lateral calf may result from a peroneal or lateral sciatic neuropathy rather than an L5 nerve root injury. Focal muscle atrophy may reflect loss of motor axons from a nerve root or peripheral nerve injury, an anterior horn cell disease, or disuse.

An [MRI](#) scan or [CT](#)-myelogram is necessary to establish the location and type of pathology. Simple MRI yields exquisite views of intraspinal and adjacent soft tissue anatomy and is more likely to establish a specific anatomic diagnosis than plain films or myelography. Bony lesions of the lateral recess or intervertebral foramen may be seen with optimal clarity on CT-myelographic studies.

The correlation of neuroradiologic findings to symptoms, particularly pain, is often problematic. As examples, contrast-enhancing tears in the annulus fibrosus or disk protrusions are widely accepted as common sources of back pain. However, one recent study found that over half of asymptomatic adults have annular tears on lumbar spine MR imaging, nearly all of which demonstrate contrast enhancement. Furthermore, asymptomatic disk protrusions are common in adults, and many of these abnormalities enhance with contrast. These observations strongly suggest that MRI findings of disk protrusion, tears in the annulus fibrosus, or contrast enhancement are common incidental findings that by themselves should not dictate management decisions for patients with back pain. The presence or absence of persistent disk herniation 10 years after surgical or conservative treatment has no bearing on a successful clinical outcome.

There are four indications for intervertebral disk surgery: (1) progressive motor weakness from nerve root injury demonstrated on clinical examination or [EMG](#), (2) bowel or bladder disturbance or other signs of spinal cord disease, (3) incapacitating nerve root pain despite conservative treatment for at least 4 weeks, and (4) recurrent incapacitating pain despite conservative treatment. The latter two criteria are more subjective and less well established than the others. Surgical treatment should also be considered if the pain and/or neurologic findings do not substantially improve over 4 to 12 weeks.

Surgery is preceded by MRI scan or CT-myelogram to define the location and type of pathology. The usual surgical procedure is a partial hemilaminectomy with excision of the involved and prolapsed intervertebral disk. Arthrodesis of the involved lumbar segments is considered only in the presence of significant spinal instability (i.e., degenerative spondylolisthesis or isthmic spondylolysis).

OTHER CAUSES OF LOW BACK PAIN

Spinal stenosis is an anatomic diagnosis reflecting a narrowed lumbar or cervical spinal canal. Classic *neurogenic claudication* occurs in the setting of moderate to severe spinal stenosis and typically consists of back and buttock or leg pain induced by walking or

standing. The pain is relieved by sitting. Symptoms in the legs are usually bilateral. Focal weakness, sensory loss, or reflex changes may occur when associated with radiculopathy. Unlike vascular claudication, the symptoms are often provoked by standing without walking. Unlike lumbar disk disease, the symptoms are usually relieved by sitting. Severe neurologic deficits, including paralysis and urinary incontinence, occur rarely. Spinal stenosis usually results from acquired (75%), congenital, or mixed acquired/congenital factors. Congenital forms (achondroplasia, idiopathic) are characterized by short, thick pedicles that produce both spinal canal and lateral recess stenosis. Acquired factors that may contribute to spinal stenosis include degenerative diseases (spondylosis, spondylolisthesis, scoliosis), trauma, spine surgery (postlaminectomy, fusion), metabolic or endocrine disorders (epidural lipomatosis, osteoporosis, acromegaly, renal osteodystrophy, hypoparathyroidism), and Paget's disease. [MRI](#) or [CT](#)-myelography provide the best definition of the abnormal anatomy ([Fig. 16-5](#)).

Conservative treatment includes nonsteroidal anti-inflammatory drugs (NSAIDs), exercise programs, and symptomatic treatment of acute pain exacerbations. Surgical therapy is considered when medical therapy does not relieve pain sufficiently to allow for activities of daily living or when significant focal neurologic signs are present. Between 65 and 80% of properly selected patients treated surgically experience >75% relief of back and leg pain. Up to 25% develop recurrent stenosis at the same spinal level or an adjacent level 5 years after the initial surgery; recurrent symptoms usually respond to a second surgical decompression.

Facet joint hypertrophy can produce unilateral radicular symptoms, due to bony compression, that are indistinguishable from disk-related radiculopathy. Patients may exhibit stretch signs, focal motor weakness, hyporeflexia, or sensory loss. Hypertrophic superior or inferior facets can often be visualized radiologically. Foraminotomy results in long-term relief of leg and back pain in 80 to 90% of patients.

Lumbar adhesive arachnoiditis with radiculopathy is the result of a fibrotic process following an inflammatory response to local tissue injury within the subarachnoid space. The fibrosis results in nerve root adhesions, producing back and leg pain associated with motor, sensory, and reflex changes. Myelography-induced arachnoiditis has become rare with the abandonment of oil-based contrast. Other causes of arachnoiditis include multiple lumbar operations, chronic spinal infections, spinal cord injury, intrathecal hemorrhage, intrathecal injection of steroids and anesthetics, and foreign bodies. The spine [MRI](#) appearance of arachnoiditis includes nerve roots clumping together centrally and adherent to the dura peripherally, or loculations of cerebrospinal fluid (CSF) within the thecal sac that obscure nerve root visualization. Treatment is often unsatisfactory. Microsurgical lysis of adhesions, dorsal rhizotomy, and dorsal root ganglionectomy have resulted in poor outcomes. Dorsal column stimulation for pain relief has produced varying results. Epidural steroid injections have been of limited value.

ARTHRITIS

Arthritis is a major cause of spine pain.

Spondylosis Osteoarthritic spine disease typically occurs in later life and primarily involves the cervical and lumbosacral spine. Patients often complain of back pain that is increased by motion and associated with stiffness or limitation of motion. The relationship between clinical symptoms and radiologic findings is usually not straightforward. Pain may be prominent when x-ray findings are minimal; alternatively, large osteophytes can be seen in asymptomatic patients in middle and later life. Hypertrophied facets and osteophytes may compress nerve roots in the lateral recess or intervertebral foramen. Osteophytes arising from the vertebral body may cause or contribute to central spinal canal stenosis. Loss of intervertebral disk height reduces the vertical dimensions of the intervertebral foramen; the descending pedicle may compress the nerve root exiting at that level. Osteoarthritic changes in the lumbar spine may rarely compress the cauda equina.

Ankylosing Spondylitis (See also [Chap. 315](#)) This distinctive arthritic spine disease typically presents with the insidious onset of low back and buttock pain. Patients are often males below age 40. Associated features include morning back stiffness, nocturnal pain, pain unrelieved by rest, an elevated sedimentation rate, and the histocompatibility antigen HLA-B27. The differential diagnosis includes tumor and infection. Onset at a young age and back pain characteristically improving with exercise suggest ankylosing spondylitis. Loss of the normal lumbar lordosis and exaggeration of thoracic kyphosis are seen as the disease progresses. Inflammation and erosion of the outer fibers of the annulus fibrosus at the point of contact with the vertebral body are followed by ossification and bone growth. Bony growth (syndesmophyte) bridges adjacent vertebral bodies and results in reduced spine mobility in all planes. The radiologic hallmarks of the disease are periarticular destructive changes, sclerosis of the sacroiliac joints, and bridging of vertebral bodies by bone to produce the fused "bamboo spine." Similar restricted movement may accompany Reiter's syndrome, psoriatic arthritis, and chronic inflammatory bowel disease. Stress fractures through the spontaneously ankylosed posterior bony elements of the rigid, osteoporotic spine may result in focal spine pain, spinal cord compression or cauda equina syndrome. Occasional atlantoaxial subluxation with spinal cord compression occurs. Bilateral ankylosis of the ribs to the spine and a decrease in the height of axial thoracic structures may cause marked impairment of respiratory function.

OTHER DESTRUCTIVE DISEASES

Neoplasm (See also [Chap. 370](#)) Back pain is the most common neurologic symptom among patients with systemic cancer. One-third of patients with undiagnosed back or neck pain and known systemic cancer have epidural extension or metastasis of tumor, and one-third have pain associated with vertebral metastases alone. About 11% have back pain unrelated to metastatic disease. Metastatic carcinoma (breast, lung, prostate, thyroid, kidney, gastrointestinal tract), multiple myeloma, and non-Hodgkin's and Hodgkin's lymphomas frequently involve the spine. Back pain may be the presenting symptom because the primary tumor site may be overlooked or asymptomatic. The pain tends to be constant, dull, unrelieved by rest, and worse at night. In contrast, mechanical low back pain is usually improved with rest. Plain x-rays usually, though not always, show destructive lesions in one or several vertebral bodies without disk space involvement. [MRI](#) or [CT](#)-myelography are the studies of choice in the setting of suspected spinal metastasis, but the trend of evidence favors the use of MRI. The procedure of

choice is the study most rapidly available because the patient may worsen during a diagnostic delay.

Infection *Vertebral osteomyelitis* is usually caused by staphylococci, but other bacteria or the tubercle bacillus (Pott's disease) may be the responsible organism. A primary source of infection, most often from the urinary tract, skin, or lungs, can be identified in 40% of patients. Intravenous drug use is a well-recognized risk factor. Back pain exacerbated by motion and unrelieved by rest, spine tenderness over the involved spine segment, and an elevated erythrocyte sedimentation rate are the most common findings. Fever or elevated white blood cell count are found in a minority of patients. Plain radiographs may show a narrowed disk space with erosion of adjacent vertebrae; these diagnostic changes may take weeks or months to appear. [MRI](#) and [CT](#) are sensitive and specific for osteomyelitis; MRI definition of soft tissue detail is exquisite. CT scan may be more readily available and better tolerated by some patients with severe back pain.

Spinal epidural abscess ([Chap. 368](#)) presents with back pain (aggravated by palpation or movement) and fever. The patient may exhibit nerve root injury or spinal cord compression accompanied by a sensory level, incontinence, or paraplegia. The abscess may track over multiple spinal levels and is best delineated by spine [MRI](#).

Osteoporosis and Osteosclerosis Considerable loss of bone may occur with or without symptoms in association with medical disorders, including hyperparathyroidism, chronic glucocorticoid use, or immobilization. Compression fractures occur in up to half of patients with severe osteoporosis. The risk of osteoporotic vertebral fracture is 4.5 times greater over 3 years among patients with a baseline fracture compared with osteoporotic controls. The sole manifestation of a compression fracture may be focal lumbar or thoracic aching (often after a trivial injury) that is exacerbated by movement. Other patients experience thoracic or upper lumbar radicular pain. Focal spine tenderness is common. When compression fractures are found, treatable risk factors should be sought. Compression fractures above the midthoracic region suggest malignancy.

Osteosclerosis is readily identifiable on routine x-ray studies (e.g., Paget's disease) and may or may not produce back pain. Spinal cord or nerve root compression may result from bony encroachment on the spinal canal or intervertebral foramina. Single dual-beam photon absorptiometry or quantitative [CT](#) can be used to detect small changes in bone mineral density. **For further discussion of these bone disorders, see [Chaps. 341 to 343](#).*

REFERRED PAIN FROM VISCERAL DISEASE

Diseases of the pelvis, abdomen, or thorax may produce referred pain to the posterior portion of the spinal segment that innervates the diseased organ. Occasionally, back pain may be the first and only sign. In general, pelvic diseases refer pain to the sacral region, lower abdominal diseases to the lumbar region (around the second to fourth lumbar vertebrae), and upper abdominal diseases to the lower thoracic or upper lumbar region (eighth thoracic to the first and second lumbar vertebrae). Local signs (pain with spine palpation, paraspinal muscle spasm) are absent, and minimal or no pain

accompanies normal spine movements.

Low Thoracic and Upper Lumbar Pain in Abdominal Disease Peptic ulcer or tumor of the posterior stomach or duodenum typically produces epigastric pain ([Chaps. 285 and 90](#)), but midline back or paraspinal pain may occur if retroperitoneal extension is present. Back pain due to peptic ulcer may be precipitated by ingestion of an orange, alcohol, or coffee and relieved by food or antacids. Fatty foods are more likely to induce back pain associated with biliary disease. Diseases of the pancreas may produce back pain to the right of the spine (head of the pancreas involved) or to the left (body or tail involved). Pathology in retroperitoneal structures (hemorrhage, tumors, pyelonephritis) may produce paraspinal pain with radiation to the lower abdomen, groin, or anterior thighs. A mass in the iliopsoas region often produces unilateral lumbar pain with radiation toward the groin, labia, or testicle. The sudden appearance of lumbar pain in a patient receiving anticoagulants suggests retroperitoneal hemorrhage.

Isolated low back pain occurs in 15 to 20% of patients with a contained rupture of an abdominal aortic aneurysm (AAA). The classic clinical triad of abdominal pain, shock, and back pain in an elderly man occurs in fewer than 20% of patients. Two of these three features are present in two-thirds of patients, and hypotension is present in half. Ruptured AAA has a high mortality rate; the typical patient is an elderly male smoker with back pain. The diagnosis is initially missed in at least one-third of patients because the symptoms and signs can be nonspecific. Common misdiagnoses include nonspecific back pain, diverticulitis, renal colic, sepsis, and myocardial infarction. A careful abdominal examination revealing a pulsatile mass (present in 50 to 75% of patients) is an important physical finding.

Lumbar Pain with Lower Abdominal Diseases Inflammatory bowel disorders (colitis, diverticulitis) or colonic neoplasms may produce lower abdominal pain, midlumbar back pain, or both. The pain may have a beltlike distribution around the body. A lesion in the transverse or initial descending colon may refer pain to the middle or left back at the L2-L3 level. Sigmoid colon disease may refer pain to the upper sacral or midline suprapubic regions or left lower quadrant of the abdomen.

Sacral Pain in Gynecologic and Urologic Disease Pelvic organs rarely cause low back pain, except for gynecologic disorders involving the uterosacral ligaments. The pain is referred to the sacral region. Endometriosis or uterine carcinoma may invade the uterosacral ligaments; malposition of the uterus may cause uterosacral ligament traction. The pain associated with endometriosis begins during the premenstrual phase and often continues until it merges with menstrual pain. Malposition of the uterus (retroversion, descensus, and prolapse) may lead to sacral pain after standing for several hours.

Menstrual pain may be felt in the sacral region. The poorly localized, cramping pain can radiate down the legs. Other pelvic sources of low back pain include neoplastic invasion of pelvic nerves, radiation necrosis, and pregnancy. Pain due to neoplastic infiltration of nerves is typically continuous, progressive in severity, and unrelieved by rest at night. Radiation therapy of pelvic tumors may produce sacral pain from late radiation necrosis of tissue or nerves. Low back pain with radiation into one or both thighs is common in the last weeks of pregnancy.

Urologic sources of lumbosacral back pain include chronic prostatitis, prostate carcinoma with spinal metastasis, and diseases of the kidney and ureter. Lesions of the bladder and testes do not usually produce back pain. The diagnosis of metastatic prostate carcinoma is established by rectal examination, spine imaging studies ([MRI](#) or [CT](#)), and measurement of prostate-specific antigen (PSA) ([Chap. 95](#)). Infectious, inflammatory, or neoplastic renal diseases may result in ipsilateral lumbosacral pain, as can renal artery or vein thrombosis. Ureteral obstruction due to renal stones may produce paraspinal lumbar pain.

Postural Back Pain There is a group of patients with chronic, nonspecific low back pain in whom no anatomic or pathologic lesion can be found despite exhaustive investigation. These individuals complain of vague, diffuse back pain with prolonged sitting or standing that is relieved by rest. The physical examination is unrevealing except for "poor posture." Imaging studies and laboratory evaluations are normal. Exercises to strengthen the paraspinal and abdominal muscles are sometimes therapeutic.

Psychiatric Disease Chronic low back pain ([CLBP](#)) may be encountered in patients with compensation hysteria, malingering, substance abuse, chronic anxiety states, or depression. Many patients with CLBP have a history of psychiatric illness (depression, anxiety, substance abuse) or childhood trauma (physical or sexual abuse) that antedates the onset of back pain. Preoperative psychological assessment has been used to exclude patients with marked psychological impairment who are at high risk for a poor surgical outcome. It is important to be certain that the back pain in these patients does not represent serious spine or visceral pathology in addition to the impaired psychological state.

Unidentified The cause of low back pain occasionally remains unclear. Some patients have had multiple operations for disk disease but have persistent pain and disability. The original indications for surgery may have been questionable with back pain only, no definite neurologic signs, or a minor disk bulge noted on [CT](#) or [MRI](#). Scoring systems based upon neurologic signs, psychological factors, physiologic studies, and imaging studies have been devised to minimize the likelihood of unsuccessful surgical explorations and to avoid selection of patients with psychological profiles that predict poor functional outcomes.

TREATMENT

Acute Low Back Pain A practical approach to the management of low back pain is to consider acute and chronic presentations separately. [ALBP](#) is defined as pain of less than 3 months' duration. Full recovery can be expected in 85% of adults with ALBP unaccompanied by leg pain. Most of these patients exhibit "mechanical" symptoms -- pain that is aggravated by motion and relieved by rest.

Observational, population-based studies have been used to justify a minimalist approach to individual patient care. These studies share a number of limitations: (1) a true placebo control group is often lacking; (2) patients who consult different provider groups (generalists, orthopedists, neurologists) are assumed to have similar etiologies

for their back pain; (3) no information is provided about the details of treatment within each provider group or between provider groups; and (4) no attempt to tabulate serious causes of [ALBP](#) is made. The appropriateness of specific diagnostic procedures or therapeutic interventions for low back pain cannot be assessed from these studies.

The proposed algorithms ([Fig. 16-6](#)) for management of [ALBP](#) in adults draw considerably from published guidelines. However, it must be emphasized that current [CPGs](#) for the treatment of low back pain are based on incomplete evidence -- for example, there is a paucity of well-designed studies documenting the natural history of disk lesions associated with a focal neurologic deficit. Guidelines should not substitute for sound clinical judgment.

The initial assessment excludes serious causes of spine pathology that require urgent intervention, including infection, cancer, and trauma. Risk factors for a possible serious underlying cause of back pain include: age > 50 years, prior diagnosis of cancer or other serious medical illness, bed rest without relief, duration of pain >1 month, urinary incontinence or recent nocturia, focal leg weakness or numbness, pain radiating into the leg(s) from the back, intravenous drug use, chronic infection (pulmonary or urinary), pain increasing with standing and relieved by sitting, history of spine trauma, and glucocorticoid use. Clinical signs associated with a possible serious etiology include unexplained fever, well-documented and unexplained weight loss, positive [SLR](#) sign or reverse SLR sign, crossed SLR sign, percussion tenderness over the spine or costovertebral angle, an abdominal mass (pulsatile or nonpulsatile), a rectal mass, focal sensory loss (saddle anesthesia or focal limb sensory loss), true leg weakness, spasticity, and asymmetric leg reflexes. Laboratory studies are unnecessary unless a serious underlying cause ([Fig. 16-6](#), Algorithms A and B) is suspected. Plain spine films are rarely indicated in the first month of symptoms unless a spine fracture is suspected.

The roles of bed rest, early exercise, and traction in the treatment of acute uncomplicated low back pain have been the subject of recent prospective studies. Clinical trials fail to demonstrate any benefit of prolonged (>2 days) bed rest for [ALBP](#). There is evidence that bed rest is also ineffective for patients with sciatica or for acute back pain with findings of nerve root injury. Theoretical advantages of early ambulation for ALBP include maintenance of cardiovascular conditioning, improved disk and cartilage nutrition, improved bone and muscle strength, and increased endorphin levels. A recent trial did not show benefit from an early vigorous exercise program, but the benefits of less vigorous exercise or other exercise programs remain unknown. The early resumption of normal physical activity (without heavy manual labor) is likely to be beneficial. Well-designed clinical studies of traction that include a sham traction group have failed to show a benefit of traction for ALBP. Despite this knowledge, one survey of physicians' perceptions of effective treatment identified strict bed rest for >3 days, trigger point injections (see below), and physical therapy (PT) as beneficial for more than 50% of patients with ALBP. In many instances, the behavior of treating physicians does not reflect the current medical literature.

Proof is lacking to support the treatment of acute back and neck pain with acupuncture, transcutaneous electrical nerve stimulation, massage, ultrasound, diathermy, or electrical stimulation. Cervical collars can be modestly helpful by limiting spontaneous and reflex neck movements that exacerbate pain. Evidence regarding the efficacy of ice

or heat is lacking, but these interventions are optional given the lack of negative evidence, low cost, and low risk. Biofeedback has not been studied rigorously. Facet joint, trigger point, and ligament injections are not recommended in the treatment of [ALBP](#).

A beneficial role for specific exercises or modification of posture has not been validated by rigorous clinical studies. As a practical matter, temporary suspension of activity known to increase mechanical stress on the spine (heavy lifting, prolonged sitting, bending or twisting, straining at stool) may be helpful.

Patient education is an important part of treatment. Studies reveal that patient satisfaction and the likelihood of follow-up increase when patients are educated about prognosis, treatment methods, activity modifications, and strategies to prevent future exacerbations. In one study, patients who felt they did not receive an adequate explanation for their symptoms wanted more diagnostic tests. Evidence for the efficacy of structured education programs ("back school") is inconclusive; in one controlled study, patients attending back school had a shorter duration of sick leave during the initial episode but not during subsequent episodes. Recent large, controlled, randomized studies of back school for primary prevention of low back injury and pain have failed to demonstrate a benefit.

Medications used in the treatment of [ALBP](#) include [NSAIDs](#), acetaminophen, muscle relaxants, and opioids. NSAIDs are superior to placebo for back pain relief. Acetaminophen is superior to placebo in the treatment of other types of pain but has not been compared against placebo for low back pain. Muscle relaxants provide short-term (4 to 7 days) benefit compared with placebo, but drowsiness often limits their daytime use. The efficacy of muscle relaxants compared to NSAIDs or in combination with NSAIDs is unclear. Opioid analgesics have not been shown to be more effective than NSAIDs or acetaminophen for relief of ALBP or likelihood of return to work. Short-term use of opioids in selected patients unresponsive to or intolerant of acetaminophen or NSAIDs may be helpful. There is no evidence to support the use of oral glucocorticoids or tricyclic antidepressants in treatment of ALBP.

The role of diagnostic and therapeutic nerve root blocks for patients with acute back or neck pain remains controversial. Equivocal data suggests that epidural steroids may occasionally produce short-term pain relief in patients with [ALBP](#) and radiculopathy, but proof is lacking for pain relief beyond 1 month. Epidural anesthetics, steroids, or opioids are not indicated as initial treatment for ALBP without radiculopathy. Diagnostic selective nerve root blocks have been advocated to determine if pain originates from a nerve root. However, these studies may be falsely positive due to a placebo effect, in patients with a painful lesion located distally along the peripheral nerve, or from anesthesia of the sinuvertebral nerve. Therapeutic selective nerve root blocks are an option after brief conservative measures fail, particularly when temporary relief of pain may be important for patient function. Needle position is confirmed under fluoroscopic guidance with nonionic contrast before injection of glucocorticoid and local anesthetic.

A short course of spinal manipulation or [PT](#) for symptomatic relief of uncomplicated [ALBP](#) is an option. A prospective, randomized study comparing PT, chiropractic manipulation, and education interventions for patients with ALBP found

modest trends toward benefit with both PT and chiropractic manipulation at 1 year. Costs per year were equivalent in the PT/chiropractic group and ~\$280 less for the group treated with the education booklet alone. The extent to which this modest improvement in symptoms and outcome is worth the cost must be determined for each patient. Extended duration of treatment or treatment of patients with radiculopathy is of unknown value and carries potential risk. The appropriate frequency or duration of spinal manipulation has not been addressed adequately.

Chronic Low Back Pain [CLBP](#) is defined as pain lasting longer than 12 weeks. Patients with CLBP account for 50% of back pain costs. Overweight individuals appear to be at particular risk. Other risk factors include: female gender, older age, prior history of back pain, restricted spinal mobility, pain radiating into a leg, high levels of psychological distress, poor self-rated health, minimal physical activity, smoking, job dissatisfaction, and widespread pain. Combinations of these premorbid factors have been used to predict which individuals with [ALBP](#) are likely to develop CLBP. The initial approach to these patients is similar to that for ALBP, and the differential diagnosis of CLBP includes most of the conditions described in this chapter. Treatment of this heterogeneous group of patients is directed toward the underlying cause when possible; the ultimate goal is to restore function to the greatest extent possible.

Many conditions that produce [CLBP](#) can be identified by the combination of neuroimaging and electrophysiologic studies. Spine [MRI](#) or [CT](#)-myelography are the techniques of choice but are generally not indicated within the first month after initial evaluation in the absence of risk factors for a serious underlying cause. Imaging studies should be performed only in circumstances where the results are likely to influence surgical or medical treatment.

Diskography is of questionable value in the evaluation of back pain. No additional anatomic information is provided beyond what is available by [MRI](#). Reproduction of the patient's typical pain with the injection is often used as evidence that a specific disk is the pain generator, but it is not known whether this information has any value in selecting candidates for surgery. There is no proven role for thermography in the assessment of radiculopathy.

The diagnosis of nerve root injury is most secure when the history, examination, results of imaging studies, and the [EMG](#) are concordant. The correlation between [CT](#) and EMG for localization of nerve root injury is between 65 and 73%. Up to one-third of asymptomatic adults have a disk protrusion detected by CT or [MRI](#) scans. Thus, surgical intervention based solely upon radiologic findings and pain increases the likelihood of an unsuccessful outcome.

[CLBP](#) can be treated with a variety of conservative measures. Acute and subacute exacerbations are managed with [NSAIDs](#) and comfort measures. There is no good evidence to suggest that one NSAID is more effective than another. Bed rest should not exceed 2 days. Activity tolerance is the primary goal, while pain relief is secondary. Exercise programs can reverse type II muscle fiber atrophy in paraspinal muscles and strengthen trunk extension. Supervised, intensive physical exercise or "work hardening" regimens (under the guidance of a physical therapist) have been effective in returning some patients to work, improving walking distances, and diminishing pain. The benefit

can be sustained with home exercise regimens; compliance with the exercise regimen strongly influences outcome. The role of manipulation, back school, or epidural steroid injections in the treatment of CLBP is unclear. Up to 30% of "blind" epidural steroid injections miss the epidural space even when performed by an experienced anesthesiologist. There is no strong evidence to support the use of acupuncture or traction in this setting. A reduction in sick leave days, long-term health care utilization, and pension expenditures may offset the initial expense of multidisciplinary treatment programs. In one study comparing 3 weeks of hydrotherapy versus routine ambulatory care, hydrotherapy resulted in diminished duration and intensity of back pain, reduced analgesic drug consumption, improved spine mobility, and improved functional score. Functional score returned to baseline at the 9-month follow-up, but all other beneficial effects were sustained. Percutaneous electrical nerve stimulation (PENS) has been shown to provide significant short-term relief of CLBP, but additional studies regarding long-term efficacy and cost are necessary.

PAIN IN THE NECK AND SHOULDER

Approach to the Patient

In one recent epidemiologic survey, the 6-month prevalence of disabling neck pain was 4.6% among adults. Neck pain commonly arises from diseases of the cervical spine and soft tissues of the neck. Neck pain arising from the cervical spine is typically precipitated by neck movements and may be accompanied by focal spine tenderness and limitation of motion. Pain arising from the brachial plexus, shoulder, or peripheral nerves can be confused with cervical spine disease, but the history and examination usually identify a more distal origin for the pain. Cervical spine trauma, disk disease, or spondylosis may be asymptomatic or painful and can produce a myelopathy, radiculopathy, or both. The nerve roots most commonly affected are C7 and C6.

TRAUMA TO THE CERVICAL SPINE

Unlike injury to the low back, trauma to the cervical spine (fractures, subluxation) places the spinal cord at risk for compression. Motor vehicle accidents, violent crimes, or falls account for 87% of spinal cord injuries, which can have devastating consequences ([Chap. 369](#)). Emergency immobilization of the neck prior to complete assessment is mandatory to minimize further spinal cord injury from movement of unstable cervical spine segments.

Whiplash injury is due to trauma (usually automobile accidents) causing cervical musculoligamentous sprain or strain due to hyperflexion or hyperextension. This diagnosis should not be applied to patients with fractures, disk herniation, head injury, or altered consciousness. One prospective study found that 18% of patients with whiplash injury had persistent injury-related symptoms 2 years after the car accident. Such patients were older, had a higher incidence of inclined or rotated head position at impact, greater intensity of initial neck and head pain, greater number of initial symptoms, and more osteoarthritic changes on cervical spine x-rays at baseline compared to patients who ultimately recovered. Objective data on the pathology of neck soft tissue injuries is lacking. Patients with severe initial injury are at increased risk for poor long-term outcome.

CERVICAL DISK DISEASE

Herniation of a lower cervical disk is a common cause of neck, shoulder, arm, or hand pain. Neck pain (worse with movement), stiffness, and limited range of neck motion are common. With nerve root compression, pain may radiate into a shoulder or arm. Extension and lateral rotation of the neck narrows the intervertebral foramen and may reproduce radicular symptoms (Spurling's sign). In young individuals, acute cervical nerve root compression from a ruptured disk is often due to trauma. Subacute radiculopathy is less likely to be related to a specific traumatic incident and may involve both disk disease and spondylosis. Cervical disk herniations are usually posterolateral near the lateral recess and intervertebral foramen. The usual patterns of reflex, sensory, and motor changes that accompany specific cervical nerve root lesions are listed in [Table 16-2](#). When evaluating patients with suspected cervical radiculopathy it is important to consider the following: (1) overlap in function between adjacent nerve roots is common, (2) the anatomic pattern of pain is the most variable of the clinical features, and (3) the distribution of symptoms and signs may be evident in only part of the injured nerve root territory.

Surgical management of cervical herniated disks usually consists of an anterior approach with discectomy followed by anterior interbody fusion. A simple posterior partial laminectomy with discectomy is an alternative approach. The risk of subsequent radiculopathy or myelopathy at cervical segments adjacent to the fusion is 3% per year and 26% at 10 years. Although the risk is sometimes portrayed as a late complication of cervical surgery, it may also reflect the natural history of degenerative cervical spine disease in this subpopulation of patients.

CERVICAL SPONDYLOSIS

Osteoarthritis of the cervical spine may produce neck pain that radiates into the back of the head, shoulders, or arms. Arthritic or other pathologic conditions of the upper cervical spine may be the source of headaches in the posterior occipital region (supplied by the C2-C4 nerve roots). Cervical spondylosis with osteophyte formation in the lateral recess or hypertrophic facet joints may produce a monoradiculopathy ([Fig. 16-7](#)). Narrowing of the spinal canal by osteophytes, ossification of the posterior longitudinal ligament, or a large central disk may compress the cervical spinal cord. In some patients, a combination of radiculopathy and myelopathy occur. An electrical sensation elicited by neck flexion and radiating down the spine from the neck (Lhermitte's symptom) usually indicates cervical or upper thoracic (T1-T2) spinal cord involvement. When little or no neck pain accompanies the cord compression, the diagnosis may be confused with amyotrophic lateral sclerosis ([Chap. 365](#)), multiple sclerosis ([Chap. 371](#)), spinal cord tumors ([Chap. 368](#)), or syringomyelia ([Chap. 368](#)). The possibility of this treatable cervical spinal cord disease must be considered even when the patient presents with leg complaints only. Furthermore, lumbar radiculopathy or polyneuropathy may mask an associated cervical myelopathy. [MRI](#) or [CT](#)-myelography can define the anatomic abnormalities, and [EMG](#) and nerve conduction studies can quantify the severity and localize the levels of motor nerve root injury.

OTHER CAUSES OF NECK PAIN

Rheumatoid arthritis (RA) ([Chap. 312](#)) of the cervical apophyseal joints results in neck pain, stiffness, and limitation of motion. In typical cases with symmetric inflammatory polyarthritis, the diagnosis of RA is straightforward. In advanced RA, synovitis of the atlantoaxial joint (C1-C2; [Fig. 16-2](#)) may damage the transverse ligament of the atlas, producing forward displacement of the atlas on the axis (atlantoaxial subluxation). Radiologic evidence of atlantoaxial subluxation occurs in 30% of patients with RA. Not surprisingly, the degree of subluxation correlates with the severity of erosive disease. When subluxation is present, careful neurologic assessment is important to identify early signs of myelopathy. Occasional patients develop high spinal cord compression leading to quadriplegia, respiratory insufficiency, and death. Although low back pain is common among RA patients, the frequency of facet disease, fracture, and spondylolisthesis is no greater than among age- and sex-matched controls with mechanical low back pain.

Ankylosing spondylitis can cause neck pain and on occasion atlantoaxial subluxation; when spinal cord compression is present or threatened, surgical intervention is indicated. Herpes zoster produces neck and posterior occipital pain in a C2-C3 distribution prior to the outbreak of vesicles. Neoplasms metastatic to the cervical spine, infections (osteomyelitis and epidural abscess), and metabolic bone diseases may also be the cause of neck pain. Neck pain may also be referred from the heart in the setting of coronary artery ischemia (cervical angina syndrome).

THORACIC OUTLET

The thoracic outlet is an anatomic region containing the first rib, the subclavian artery and vein, the brachial plexus, the clavicle, and the lung apex. Injury to these structures may result in posture or task-related pain around the shoulder and supraclavicular region. There are at least three subtypes of thoracic outlet syndrome (TOS). *True neurogenic TOS* results from compression of the lower trunk of the brachial plexus by an anomalous band of tissue connecting an elongate transverse process at C7 with the first rib. Neurologic deficits include weakness of intrinsic muscles of the hand and diminished sensation on the palmar aspect of the fourth and fifth digits. [EMG](#) and nerve conduction studies confirm the diagnosis. Definitive treatment consists of surgical division of the anomalous band compressing either the lower trunk of the brachial plexus or ventral rami of the C8 or T1 nerve roots. The weakness and wasting of intrinsic hand muscles typically does not improve, but surgery halts the insidious progression of weakness. The *arterial TOS* results from compression of the subclavian artery by a cervical rib; the compression results in poststenotic dilatation of the artery and thrombus formation. Blood pressure is reduced in the affected limb, and signs of emboli may be present in the hand; neurologic signs are absent. Noninvasive ultrasound techniques confirm the diagnosis. Treatment is with thrombolysis or anticoagulation (with or without embolectomy) and surgical excision of the cervical rib compressing the subclavian artery or vein. The *disputed TOS* includes a large number of patients with chronic arm and shoulder pain of unclear cause. The lack of sensitive and specific findings on physical examination or laboratory markers for this condition frequently results in diagnostic uncertainty. The role of surgery in disputed TOS is controversial; conservative approaches often include multidisciplinary pain management. Treatment is often unsuccessful.

BRACHIAL PLEXUS AND NERVES

Pain from injury to the brachial plexus or arm peripheral nerves can occasionally be confused with pain of cervical spine origin. Neoplastic infiltration of the lower trunk of the brachial plexus may produce shoulder pain radiating down the arm, numbness of the fourth and fifth fingers, and weakness of intrinsic hand muscles innervated by the ulnar and median nerves. Postradiation fibrosis (breast carcinoma is the most common setting) or a Pancoast tumor of the lung ([Chap. 88](#)) may produce similar findings. A Horner's syndrome is present in two-thirds of patients with a Pancoast tumor. Suprascapular neuropathy may produce severe shoulder pain, weakness, and wasting of the supraspinatus and infraspinatus muscles. *Acute brachial neuritis* is often confused with radiculopathy. It consists of the acute onset of severe shoulder or scapular pain followed over days to weeks by weakness of the proximal arm and shoulder girdle muscles innervated by the upper or middle trunks or cords of the brachial plexus. The onset is often preceded by an infection or immunization. Separation of this syndrome from cervical radiculopathy is important because slow, complete recovery of brachial neuritis occurs in 75% of patients after 2 years and in 89% after 3 years. Occasional cases of carpal tunnel syndrome produce pain and paresthesia extending into the forearm, arm, and shoulder resembling a C5 or C6 root lesion. Lesions of the radial or ulnar nerve can mimic a radiculopathy at C7 or C8, respectively. [EMG](#) and nerve conduction studies can accurately localize lesions to the nerve roots, brachial plexus, or nerves. **For further discussion of peripheral nerve disorders, see [Chap. 377](#).*

SHOULDER

Pain in the shoulder region can be difficult to separate clearly from neck pain. If the symptoms and signs of radiculopathy are absent, then the differential diagnosis includes mechanical shoulder pain (tendonitis, bursitis, rotator cuff tear, dislocation, adhesive capsulitis, and cuff impingement under the acromion) and referred pain (subdiaphragmatic irritation, angina, Pancoast tumor). Mechanical pain is often worse at night, associated with local shoulder tenderness, and aggravated by abduction, internal rotation, or extension of the arm. The pain of shoulder disease may at times radiate into the arm or hand, but the sensory, motor, and reflex changes that indicate disease of the nerve roots, plexus, or peripheral nerves are absent.

TREATMENT

A paucity of well-designed clinical trials exists for the treatment of neck pain. Symptomatic treatment of neck pain can include the use of analgesic medications and/or a soft cervical collar. Current indications for cervical disk surgery are similar to those for lumbar disk surgery; because of the risk of spinal cord injury with cervical spine disease, an aggressive approach is generally indicated whenever spinal cord injury is threatened. Surgical management of cervical herniated disks usually consists of an anterior approach with discectomy followed by anterior interbody fusion. A simple posterior partial laminectomy with discectomy is an acceptable alternative approach. The cumulative risk of subsequent radiculopathy or myelopathy at cervical segments adjacent to the fusion is approximately 3% per year and 26% per decade. Although this

risk is sometimes portrayed as a late complication of surgery, it may also reflect the natural history of degenerative cervical spine disease. Nonprogressive cervical radiculopathy (associated with a focal neurologic deficit) due to a herniated cervical disk may be treated conservatively with a high rate of success. Cervical spondylosis with bony, compressive cervical radiculopathy is generally treated with surgical decompression to interrupt the progression of neurologic signs. Cervical spondylotic myelopathy is typically managed with either anterior decompression and fusion or laminectomy. Outcomes in both surgical groups vary, but late functional deterioration occurs in 20 to 30% of patients; a prospective, controlled study comparing different surgical interventions is sorely needed.

(Bibliography omitted in Palm version)

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SECTION 2 -ALTERATIONS IN BODY TEMPERATURE

17. FEVER AND HYPERTHERMIA - Charles A. Dinarello, Jeffrey A. Gelfand

Body temperature is controlled by the hypothalamus. Neurons in both the preoptic anterior hypothalamus and the posterior hypothalamus receive two kinds of signals: one from peripheral nerves that reflect warmth/cold receptors and the other from the temperature of the blood bathing the region. These two types of signals are integrated by the thermoregulatory center of the hypothalamus to maintain normal temperature. In a neutral environment, the metabolic rate of humans consistently produces more heat than is necessary to maintain the core body temperature at 37°C. Therefore, the hypothalamus controls temperature by mechanisms of heat loss.

A normal body temperature is ordinarily maintained, despite environmental variations, because the hypothalamic thermoregulatory center balances the excess heat production derived from metabolic activity in muscle and the liver with heat dissipation from the skin and lungs. According to recent studies of healthy individuals 18 to 40 years of age, the mean oral temperature is $36.8^{\circ}\pm 0.4^{\circ}\text{C}$ ($98.2^{\circ}\pm 0.7^{\circ}\text{F}$), with low levels at 6 A.M. and higher levels at 4 to 6 P.M. The maximum normal oral temperature is 37.2°C (98.9°F) at 6 A.M. and 37.7°C (99.9°F) at 4 P.M.; these values define the 99th percentile for healthy individuals. In light of these studies, *an A.M. temperature of $>37.2^{\circ}\text{C}$ (98.9°F) or a P.M. temperature of $>37.7^{\circ}\text{C}$ (99.9°F) would define a fever.* The normal daily temperature variation is typically 0.5°C (0.9°F). However, in some individuals recovering from a febrile illness, this daily variation can be as great as 1.0°C . During a febrile illness, diurnal variations are usually maintained but at higher levels. Daily temperature swings do not occur in patients with hyperthermia (see below). Rectal temperatures are generally 0.4°C (0.7°F) higher than oral readings. The lower oral readings are probably attributable to mouth breathing, which is a particularly important factor in patients with respiratory infections and rapid breathing. Lower esophageal temperatures closely reflect core temperature. Tympanic membrane (TM) thermometers measure radiant heat energy from the tympanic membrane and nearby ear canal and display that absolute value (unadjusted mode) or a value automatically calculated from the absolute reading on the basis of nomograms relating the radiant temperature measured to actual core temperatures obtained in clinical studies (adjusted mode). These measurements, although convenient, may be more variable than directly determined oral or rectal values. Studies in adults show that readings are lower with unadjusted-mode than with adjusted-mode TM thermometers and that unadjusted-mode TM values are 0.8°C (1.6°F) lower than rectal temperatures.

In women who menstruate, the A.M. temperature is generally lower in the 2 weeks before ovulation; it then rises by about 0.6°C (1°F) with ovulation and remains at that level until menses occur. Seasonal variation in body temperature has been described but may reflect a metabolic change and is not common. Body temperature is elevated in the postprandial state, but this elevation does not represent fever. Pregnancy and endocrinologic dysfunction also affect body temperature. The daily temperature variation appears to be fixed in early childhood; in contrast, elderly individuals can exhibit a reduced ability to develop fever, with only a modest fever even in severe infections.

FEVER VERSUS HYPERTHERMIA

FEVER

Fever is an elevation of body temperature that exceeds the normal daily variation and occurs *in conjunction with an increase in the hypothalamic set point* -- for example, from 37°C to 39°C. This shift of the set point from "normothermic" to febrile levels very much resembles the resetting of the home thermostat to a higher level in order to raise the ambient temperature in a room. Once the hypothalamic set point is raised, neurons in the vasomotor center are activated and vasoconstriction commences. The individual first notices vasoconstriction in the hands and feet. Shunting of blood away from the periphery to the internal organs essentially decreases heat loss from the skin, and the person feels cold. For most fevers, body temperature increases by 1 to 2°C. Shivering, which increases heat production from the muscles, may begin at this time; however, shivering is not required if heat conservation mechanisms raise blood temperature sufficiently. Heat production from the liver also increases. In humans, behavioral instincts (e.g., putting on more clothing or bedding) lead to a reduction of exposed surfaces, which helps raise body temperature.

The processes of heat conservation (vasoconstriction) and heat production (shivering and increased metabolic activity) continue until the temperature of the blood bathing the hypothalamic neurons matches the new thermostat setting. Once that point is reached, the hypothalamus maintains the temperature at the febrile level by the same mechanisms of heat balance that are operative in the afebrile state. When the hypothalamic set point is again reset downward (due to either a reduction in the concentration of pyrogens or the use of antipyretics), the processes of heat loss through vasodilation and sweating are initiated. Behavioral changes triggered at this time include the removal of insulating clothing or bedding. Loss of heat by sweating and vasodilation continues until the blood temperature at the hypothalamic level matches the lower setting.

A fever of $>41.5^{\circ}\text{C}$ (106.7°F) is called *hyperpyrexia*. This extraordinarily high fever can develop in patients with severe infections but most commonly occurs in patients with central nervous system hemorrhages. In the preantibiotic era, fever due to a variety of infectious diseases rarely exceeded 106°F , and there has been speculation that this natural "thermal ceiling" is mediated by neuropeptides functioning as central antipyretics.

In some rare cases, the hypothalamic set point is elevated as a result of local trauma, hemorrhage, tumor, or intrinsic hypothalamic malfunction. The term *hypothalamic fever* is sometimes used to describe elevated temperature caused by abnormal hypothalamic function. However, most patients with hypothalamic damage have *subnormal*, not *supranormal*, body temperatures. These patients do not respond properly to mild environmental temperature changes. For example, when exposed to only mildly cold conditions, their core temperature falls quickly rather than over the normal period of a few hours. In the very few patients in whom elevated core temperature is suspected to be due to hypothalamic damage, diagnosis depends on the demonstration of other abnormalities in hypothalamic function, such as the production of hypothalamic releasing factors, abnormal response to cold, and absence of circadian temperature and

hormonal rhythms.

HYPERTHERMIA

Hyperthermia is characterized by *an unchanged (normothermic) setting of the thermoregulatory center* in conjunction with an uncontrolled increase in body temperature that exceeds the body's ability to lose heat. Exogenous heat exposure and endogenous heat production are two mechanisms by which hyperthermia can result in dangerously high internal temperatures. Excessive heat production can easily cause hyperthermia despite physiologic and behavioral control of body temperature. For example, over-insulating clothing can result in an elevated core temperature, and work or exercise in hot environments can produce heat faster than peripheral mechanisms can lose it.

Although most patients with elevated body temperature have fever, there are a few circumstances in which elevated temperature represents not fever but hyperthermia ([Table 17-1](#)). *Heat stroke*, caused by thermoregulatory failure in association with a warm environment, may be categorized as exertional or nonexertional. *Exertional heat stroke* typically occurs in younger individuals exercising at ambient temperatures and/or humidities that are higher than normal. Even in normal individuals, dehydration or the use of common medications (e.g., over-the-counter antihistamines with anticholinergic side effects) may help to precipitate exertional heat stroke. *Nonexertional* or *classic heat stroke* typically occurs in elderly individuals, particularly during heat waves. For example, in Chicago in July 1995, 465 deaths were certified as heat related. The elderly, the bedridden, persons taking anticholinergic or antiparkinsonian drugs or diuretics, and individuals confined to poorly ventilated and non-air-conditioned environments are most susceptible.

Drug-induced hyperthermia has become increasingly common as a result of the increased use of prescription psychotropic drugs and illicit drugs. Drug-induced hyperthermia may be caused by monoamine oxidase inhibitors, tricyclic antidepressants, and amphetamines and by the illicit use of phencyclidine, lysergic acid diethylamide (LSD), or cocaine.

Malignant hyperthermia occurs in individuals with an inherited abnormality of skeletal-muscle sarcoplasmic reticulum that causes a rapid increase in intracellular calcium levels in response to halothane and other inhalational anesthetics or to succinylcholine. Elevated temperature, increased muscle metabolism, rigidity, rhabdomyolysis, acidosis, and cardiovascular instability develop rapidly. This condition is often fatal. The *neuroleptic malignant syndrome* can occur with phenothiazines and other drugs such as haloperidol and is characterized by muscle rigidity, autonomic dysregulation, and hyperthermia. This disorder appears to be caused by the inhibition of central dopamine receptors in the hypothalamus, which results in increased heat generation and decreased heat dissipation. Thyrotoxicosis and pheochromocytoma can also cause increased thermogenesis.

It is important to distinguish between fever and hyperthermia since hyperthermia can be rapidly fatal and characteristically does not respond to antipyretics. However, there is no rapid way to make this distinction. Hyperthermia is often diagnosed on the basis of the

events immediately preceding the elevation of core temperature -- e.g., heat exposure or treatment with drugs that interfere with thermoregulation. However, in addition to the clinical history of the patient, the physical aspects of some forms of hyperthermia may alert the clinician. For example, in patients with heat stroke syndromes and in those taking drugs that block sweating, the skin is hot but dry. Moreover, antipyretics do not reduce the elevated temperature in hyperthermia, whereas in fever -- and even in hyperpyrexia -- adequate doses of either aspirin or acetaminophen usually result in some decrease in body temperature.

PYROGENS

The term *pyrogen* is used to describe any substance that causes fever. *Exogenous* pyrogens are derived from outside the patient; most are microbial products, microbial toxins, or whole microorganisms. The classic example of an exogenous pyrogen is the lipopolysaccharide endotoxin produced by all gram-negative bacteria. Endotoxins are potent not only as pyrogens but also as inducers of various pathologic changes in gram-negative infections. Another group of potent bacterial pyrogens is produced by gram-positive organisms and includes the enterotoxins of *Staphylococcus aureus* and the group A and B streptococcal toxins, also called *superantigens*. One staphylococcal toxin of clinical importance is the toxic shock syndrome toxin associated with isolates of *S. aureus* from patients with toxic shock syndrome. Like the endotoxins of gram-negative bacteria, the toxins produced by staphylococci and streptococci cause fever in experimental animals when injected intravenously at concentrations of <1 ug/kg of body weight. Endotoxin is a highly pyrogenic molecule in humans: a dose of 2 to 3 ng/kg produces fever and generalized symptoms of malaise in volunteers.

PYROGENIC CYTOKINES

Cytokines are small proteins (molecular mass, 10,000 to 20,000 Da) that regulate immune, inflammatory, and hematopoietic processes. For example, stimulation of lymphocyte proliferation during an immune response to vaccination is the result of the cytokines interleukin (IL) 2, IL-4, and IL-6. Another cytokine, granulocyte colony-stimulating factor, stimulates granulocytopoiesis in the bone marrow. Some cytokines cause fever and hence are called *pyrogenic cytokines*. From a historic point of view, the field of cytokine biology began in the 1940s with laboratory investigations into fever induction by products of activated leukocytes. These fever-producing molecules were called *endogenous pyrogens*. When endogenous pyrogens were purified from activated leukocytes, they were shown to possess various biologic activities, which are now recognized as the properties of the various cytokines.

The known pyrogenic cytokines include IL-1, IL-6, tumor necrosis factor (TNF), ciliary neurotropic factor (CNTF), and interferon (IFN) α . Others probably exist. Each cytokine is encoded by a separate gene, and each pyrogenic cytokine has been shown to cause fever in laboratory animals and in humans. When injected into humans, IL-1, IL-6, and TNF produce fever at low doses (10 to 100 ng/kg).

The synthesis and release of endogenous pyrogenic cytokines are induced by a wide spectrum of exogenous pyrogens, most of which have recognizable bacterial or fungal sources. Viruses also induce pyrogenic cytokines by infecting cells. However, in the

absence of microbial infection, inflammation, trauma, tissue necrosis, or antigen-antibody complexes can induce the production of [IL-1](#), [TNF](#), and/or IL-6, which -- individually or in combination -- trigger the hypothalamus to raise the set point to febrile levels. The cellular sources of pyrogenic cytokines are primarily monocytes, neutrophils, and lymphocytes, although many other types of cells can synthesize these molecules when stimulated.

ELEVATION OF THE HYPOTHALAMIC SET POINT BY CYTOKINES

During fever, levels of prostaglandin E₂ (PGE₂) are elevated in hypothalamic tissue and the third cerebral ventricle. The concentrations of PGE₂ are highest near the circumventricular vascular organs (organum vasculosum of lamina terminalis) -- networks of enlarged capillaries surrounding the hypothalamic regulatory centers. Destruction of these organs reduces the ability of pyrogens to produce fever. Most studies in animals have failed to show, however, that pyrogenic cytokines pass from the circulation into the brain itself. Thus, it appears that both exogenous and endogenous pyrogens interact with the endothelium of these capillaries and that this interaction is the first step in initiating fever -- i.e., in raising the set point to febrile levels.

The key events in the production of fever are illustrated in [Fig. 17-1](#). As has been mentioned, several cell types can produce pyrogenic cytokines. Pyrogenic cytokines such as [IL-1](#), IL-6, and [TNF](#) are released from the cells and enter the systemic circulation. Although the systemic effects of these circulating cytokines lead to fever by inducing the synthesis of [PGE₂](#), they also induce PGE₂ in peripheral tissues. The increase in PGE₂ in the periphery accounts for the nonspecific myalgias and arthralgias that often accompany fever. However, it is the induction of PGE₂ in the brain that starts the process of raising the hypothalamic set point for core temperature.

There are four receptors for [PGE₂](#), and each signals the cell in different ways. Of the four receptors, the third (EP-3) is essential for fever: when the gene for this receptor is deleted in mice, no fever follows the injection of [IL-1](#) or endotoxin. Deletion of the other PGE₂ receptor genes leaves the fever mechanism intact. Although PGE₂ is essential for fever, it is not a neurotransmitter. Rather, the release of PGE₂ from the brain side of the hypothalamic endothelium triggers the PGE₂ receptor on glial cells, and this stimulation results in the rapid release of cyclic adenosine 5'-monophosphate (cyclic AMP), which is a neurotransmitter. As shown in [Fig. 17-1](#), the release of cyclic AMP from the glial cells activates neuronal endings from the thermoregulatory center that extend into the area. The elevation of cyclic AMP is thought to account for changes in the hypothalamic set point either directly or indirectly by inducing the release of monoamine neurotransmitters. Since receptors for endotoxin are in many ways similar to IL-1 receptors, the activation of endotoxin receptors on the hypothalamic endothelium also results in PGE₂ production and fever.

PRODUCTION OF CYTOKINES IN THE CENTRAL NERVOUS SYSTEM

Several viral diseases produce active infection in the brain. Glial and possibly neuronal cells synthesize [IL-1](#), [TNF](#), and IL-6. [CNTF](#) is also synthesized by neural as well as neuronal cells. What role in the production of fever is played by these cytokines produced in the brain itself? In experimental animals, the concentrations of cytokine

required to cause fever are several orders of magnitude lower with direct injection into the brain than with intravenous injection. Therefore, central nervous system production of these cytokines apparently can raise the hypothalamic set point, bypassing the circumventricular organs involved in fever caused by circulating cytokines. Central nervous system cytokines may account for the hyperpyrexia of central nervous system hemorrhage, trauma, or infection.

Approach to the Patient

History It is in the diagnosis of a febrile illness that the science and art of medicine come together. In no other clinical situation is a meticulous history more important. Painstaking attention must be paid to the chronology of symptoms in relation to the use of prescription drugs (including drugs or herbs taken without a physician's supervision) or treatments such as surgical or dental procedures. The exact nature of any prosthetic materials and/or implanted devices should be ascertained. A careful occupational history should include exposures to animals; toxic fumes; potential infectious agents; possible antigens; or other febrile or infected individuals in the home, workplace, or school. A history of the geographic areas in which the patient has lived and a travel history should include locations during military service. Information on unusual hobbies, dietary proclivities (such as raw or poorly cooked meat, raw fish, and unpasteurized milk or cheeses), and household pets should be elicited, as should that on sexual orientation and practices, including precautions taken or omitted. Attention should be directed to the use of tobacco, marijuana, intravenous drugs, or alcohol; trauma; animal bites; tick or other insect bites; and prior transfusions, immunizations, drug allergies, or hypersensitivities. A careful family history should include information on family members with tuberculosis, other febrile or infectious diseases, arthritis or collagen vascular disease, or unusual familial symptomatology such as deafness, urticaria, fevers and polyserositis, bone pain, or anemia. Ethnic origin may be critical. For example, blacks are more likely than persons in other groups to have hemoglobinopathies. Turks, Arabs, Armenians, and Sephardic Jews are especially likely to have familial Mediterranean fever.

Physical Examination A meticulous physical examination should be repeated on a regular basis. All the vital signs are relevant. The temperature may be taken orally or rectally, but the site used should be consistent. Axillary temperatures are notoriously unreliable. Particular attention should be paid to daily (or sometimes more frequent) physical examination, which should continue until the diagnosis is certain and the anticipated response has been achieved. Special attention should be paid to the skin, lymph nodes, eyes, nail beds, cardiovascular system, chest, abdomen, musculoskeletal system, and nervous system. Rectal examination is imperative. The penis, prostate, scrotum, and testes should be examined carefully and the foreskin, if present, retracted. Pelvic examination must be part of every complete physical examination of a woman, with a search for such causes of fever as pelvic inflammatory disease and tubo-ovarian abscess.

Laboratory Tests Few signs and symptoms in medicine have as many diagnostic possibilities as fever. If the history, epidemiologic situation, or physical examination suggests more than a simple viral illness or streptococcal pharyngitis, then laboratory testing is indicated. The tempo and complexity of the workup will depend on the pace of

the illness, diagnostic considerations, and the immune status of the host. If findings are focal or if the history, epidemiologic setting, or physical examination suggests certain diagnoses, the laboratory examination can be focused. If fever is undifferentiated, the diagnostic nets must be cast farther, and certain guidelines are indicated, as follows.

CLINICAL PATHOLOGY The workup should include a complete blood count; a differential count should be performed manually or with an instrument sensitive to the identification of eosinophils, juvenile or band forms, toxic granulations, and Dohle bodies, the last three of which are suggestive of bacterial infection. Neutropenia may be present with some viral infections, particularly parvovirus B19 infection; drug reactions; systemic lupus erythematosus; typhoid; brucellosis; and infiltrative diseases of the bone marrow, including lymphoma, leukemia, tuberculosis, and histoplasmosis. Lymphocytosis may occur with typhoid, brucellosis, tuberculosis, and viral disease. Atypical lymphocytes are documented in many viral diseases, including infection with Epstein-Barr virus, cytomegalovirus, or HIV; dengue; rubella; varicella; measles; and viral hepatitis. This abnormality also occurs in serum sickness and toxoplasmosis. Monocytosis is a feature of typhoid, tuberculosis, brucellosis, and lymphoma. Eosinophilia may be associated with hypersensitivity drug reactions, Hodgkin's disease, adrenal insufficiency, and certain metazoan infections. If the febrile illness appears to be severe or is prolonged, the smear should be examined carefully for malarial or babesial pathogens (where appropriate) as well as for classic morphologic features, and the erythrocyte sedimentation rate should be determined. Urinalysis, with examination of urinary sediment, is indicated. It is axiomatic that any abnormal fluid accumulation (pleural, peritoneal, joint), even if previously sampled, merits reexamination in the presence of undiagnosed fever. Joint fluids should be examined for bacteria as well as crystals. Bone marrow biopsy (not simple aspiration) for histopathologic studies (as well as culture) is indicated when marrow infiltration by pathogens or tumor cells is possible. Stool should be inspected for occult blood; an inspection for fecal leukocytes, ova, or parasites also may be indicated.

CHEMISTRY Electrolyte, glucose, blood urea nitrogen, and creatinine levels should be measured. Liver function tests are usually indicated if efforts to identify the cause of fever do not point to the involvement of another organ. Additional assessments (e.g., measurement of creatine phosphokinase or amylase) can be added as the workup progresses.

MICROBIOLOGY Smears and cultures of specimens from the throat, urethra, anus, cervix, and vagina should be assessed when there are no localizing findings or when findings suggest the involvement of the pelvis or the gastrointestinal tract. If respiratory tract infection is suspected, sputum evaluation (Gram's staining, staining for acid-fast bacilli, culture) is indicated. Cultures of blood, abnormal fluid collections, and urine are indicated when fever is thought to reflect more than uncomplicated viral illness. Cerebrospinal fluid should be examined and cultured if meningismus, severe headache, or a change in mental status is noted.

RADIOLOGY A chest x-ray is usually part of the evaluation for any significant febrile illness.

Outcome of Diagnostic Efforts In most cases of fever, either the patient recovers

spontaneously or the history, physical examination, and initial screening laboratory studies lead to a diagnosis. When fever continues for 2 to 3 weeks, during which time repeat physical examinations and laboratory tests are unrevealing, the patient is provisionally diagnosed as having fever of unknown origin ([Chap. 125](#)).

TREATMENT

The Decision to Treat Fever Most fevers are associated with self-limited infections, most commonly of viral origin. In these cases, the general cause of the fever is easily identified. The routine use of antipyretics given automatically as "standing," "routine," or "prn" orders to treat low-grade fevers in adult patients on hospital wards is entirely unacceptable. This practice masks not only fever but also other important clinical indicators of a patient's course. The assumption underlying any decision to reduce fever with antipyretics is that there is no diagnostic benefit to be gained by allowing the fever to persist. However, there may be such a diagnostic benefit. For example, the daily highs and lows of normal temperature are exaggerated in most fevers, but the usual times of peak and trough temperatures may be reversed in typhoid fever and disseminated tuberculosis. Temperature-pulse dissociation (relative bradycardia) occurs in typhoid fever, brucellosis, leptospirosis, some drug-induced fevers, and factitious fever. In newborns, the elderly, patients with chronic renal failure, and patients taking glucocorticoids, fever may not be present despite infection, or core temperature may be hypothermic. Hypothermia is observed in patients with septic shock.

Some febrile diseases have characteristic patterns. With *relapsing* fevers, febrile episodes are separated by intervals of normal temperature; when paroxysms occur on the first and third days, the fever is called *tertian*. *Plasmodium vivax* causes tertian fevers. *Quartan* fevers are associated with paroxysms on the first and fourth days and are seen with *P. malariae*. Other relapsing fevers are related to *Borrelia* infections and rat-bite fever, which are both associated with days of fever followed by a several-day afebrile period and then a relapse of days of fever. Pel-Ebstein fever, with fevers lasting 3 to 10 days followed by afebrile periods of 3 to 10 days, is classic for Hodgkin's disease and other lymphomas. Another characteristic fever is that of cyclic neutropenia, in which fevers occur every 21 days and accompany the neutropenia. There is no periodicity of fever in patients with familial Mediterranean fever.

Mechanisms of Antipyretic Agents The synthesis of [PGE₂](#) depends on the constitutively expressed enzyme cyclooxygenase. The substrate for cyclooxygenase is arachidonic acid released from the cell membrane, and this release is the rate-limiting step in the synthesis of PGE₂. Inhibitors of cyclooxygenase are potent antipyretics. The antipyretic potency of various drugs is directly correlated with the inhibition of brain cyclooxygenase. Acetaminophen is a poor cyclooxygenase inhibitor in peripheral tissue and is without noteworthy anti-inflammatory activity; in the brain, however, acetaminophen is oxidized by the p450 cytochrome system, and the oxidized form inhibits cyclooxygenase activity.

Oral aspirin and acetaminophen are equally effective in reducing fever in humans. Nonsteroidal anti-inflammatory agents (NSAIDs) such as indomethacin and ibuprofen are also excellent antipyretics. Chronic high-dose therapy with antipyretics such as aspirin or the NSAIDs used in arthritis does not reduce normal core body temperature.

Thus, PGE₂ appears to play no role in normal thermoregulation.

As effective antipyretics, glucocorticoids act at two levels. First, similar to the cyclooxygenase inhibitors, glucocorticoids reduce PGE₂ synthesis by inhibiting the activity of phospholipase A₂, which is needed to release arachidonic acid from the cell membrane. Second, glucocorticoids block the transcription of the mRNA for the pyrogenic cytokines.

Drugs that interfere with vasoconstriction (phenothiazines, for example) can act as antipyretics, as can drugs that block muscle contractions. However, these agents are not true antipyretics since they can also reduce core temperature independently of hypothalamic control.

Indications and Regimens for the Treatment of Fever The objectives in treating fever are first to reduce the elevated hypothalamic set point and second to facilitate heat loss. There is no evidence that fever itself facilitates the recovery from infection or acts as an adjuvant to the immune system. In fact, peripheral PGE₂ production is a potent immunosuppressant. Hence, treating fever and its symptoms does no harm and does not slow the resolution of common viral and bacterial infections. Reducing fever with antipyretics also reduces systemic symptoms of headache, myalgias, and arthralgias.

Oral aspirin and NSAIDs effectively reduce fever but can adversely affect platelets and the gastrointestinal tract. Therefore, acetaminophen is preferred to all of these agents as an antipyretic. In children, acetaminophen must be used because aspirin increases the risk of Reye's syndrome. If the patient cannot take oral antipyretics, parenteral preparations of NSAIDs and rectal suppository preparations of various antipyretics can be used.

Treatment of fever in some patient groups is recommended. Fever increases the demand for oxygen (i.e., for every increase of 1°C over 37°C, there is a 13% increase in oxygen consumption) and can aggravate preexisting cardiac, cerebrovascular, or pulmonary insufficiency. Elevated temperature can induce mental changes in patients with organic brain disease. Children with a history of febrile or nonfebrile seizure should be aggressively treated to reduce fever, although it is unclear what triggers the febrile seizure and there is no correlation between absolute temperature elevation and onset of a febrile seizure in susceptible children.

In hyperpyrexia, the use of cooling blankets facilitates the reduction of temperature; however, cooling blankets should not be used without oral antipyretics. In hyperpyretic patients with central nervous system disease or trauma, reducing core temperature mitigates the ill effects of high temperature on the brain.

Treating Hyperthermia A high core temperature in a patient with an appropriate history (e.g., environmental heat exposure or treatment with anticholinergic or neuroleptic drugs, tricyclic antidepressants, succinylcholine, or halothane) along with appropriate clinical findings (dry skin, hallucinations, delirium, pupil dilation, muscle rigidity, and/or elevated levels of creatine phosphokinase) suggests hyperthermia. The attempt to lower the already normal hypothalamic set point is of little use. Physical cooling with sponging, fans, cooling blankets, and even ice baths should be initiated immediately in conjunction

with the administration of intravenous fluids and appropriate pharmacologic agents (see below). If insufficient cooling is achieved by external means, internal cooling can be achieved by gastric or peritoneal lavage with iced saline. In extreme circumstances, hemodialysis or even cardiopulmonary bypass with cooling of blood may be performed.

Malignant hyperthermia should be treated immediately with cessation of anesthesia and intravenous administration of dantrolene sodium. The recommended dose of dantrolene is 1 to 2.5 mg/kg of body weight given intravenously every 6 h for at least 24 to 48 h -- until oral dantrolene can be administered, if needed. Procainamide should also be administered to patients with malignant hyperthermia because of the likelihood of ventricular fibrillation in this syndrome. Dantrolene at similar doses is indicated in the neuroleptic malignant syndrome and in drug-induced hyperthermia and may even be useful in the hyperthermia of thyrotoxicosis. The neuroleptic malignant syndrome may also be treated with bromocriptine, levodopa, amantadine, or nifedipine or by induction of muscle paralysis with curare and pancuronium. Tricyclic antidepressant overdose may be treated with physostigmine.

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18. FEVER AND RASH - Elaine T. Kaye, Kenneth M. Kaye

The acutely ill patient with fever and rash ([Fig. 18-CD1](#)) often presents a diagnostic challenge for physicians. The distinctive appearance of an eruption in concert with a clinical syndrome may facilitate a prompt diagnosis and the institution of life-saving therapy or critical infection-control interventions.

Approach to the Patient

A thorough history of patients with fever and rash includes the following relevant information: immune status, medications taken within the previous month, specific travel history, immunization status, exposure to domestic pets and other animals, history of animal or arthropod bites, existence of cardiac abnormalities, presence of prosthetic material, recent exposure to ill individuals, and exposure to sexually transmitted diseases. The history should also include the site of onset of the rash and its direction and rate of spread.

A thorough physical examination entails close attention to the rash, with an assessment and precise definition of its salient features. First, it is critical to determine the *type* of lesions that make up the eruption. *Macules* are flat lesions defined by an area of changed color (i.e., a blanchable erythema). *Papules* are raised, solid lesions <5 mm in diameter; *plaques* are lesions >5 mm in diameter with a flat, plateau-like surface; and *nodules* are lesions >5 mm in diameter with a more rounded configuration. *Wheals* (urticaria, hives) are papules or plaques that are pale pink and may appear annular (ringlike) as they enlarge; classic (nonvasculitic) wheals are transient, lasting only 24 to 48 h in any defined area. *Vesicles* (<5 mm) and *bullae* (>5 mm) are circumscribed, elevated lesions containing fluid. *Pustules* are raised lesions containing purulent exudate; vesicular processes such as varicella or herpes simplex may evolve to pustules. *Nonpalpable purpura* is a flat lesion that is due to bleeding into the skin; if <3 mm in diameter, the purpuric lesions are termed *petechiae*; if >3 mm, they are termed *ecchymoses*. *Palpable purpura* is a raised lesion that is due to inflammation of the vessel wall (vasculitis) with subsequent hemorrhage. An *ulcer* is a defect in the skin extending at least into the upper layer of the dermis, and an *eschar* (tache noire) is a necrotic lesion covered with a black crust.

Other pertinent features of rashes include their *configuration* (i.e., annular or target), the *arrangement* of their lesions, and their *distribution* (i.e., central or peripheral). **For further discussion, see [Chaps. 55 and 57](#).*

CLASSIFICATION OF RASH

This chapter reviews rashes that reflect systemic disease but does not include localized skin eruptions (i.e., cellulitis, impetigo) that may also be associated with fever ([Chap. 128](#)). Rashes are classified herein on the basis of the morphology and distribution of lesions. For practical purposes, this classification system is based on the most typical disease presentations. However, morphology may vary as rashes evolve, and the presentation of diseases with rashes is subject to many variations ([Chap. 57](#)). For instance, the classic petechial rash of Rocky Mountain spotted fever (RMSF) may initially consist of blanchable erythematous macules distributed peripherally; at times,

the rash associated with RMSF may not be predominantly acral, or a rash may not develop at all.

Diseases with fever and rash may be classified by type of eruption: centrally distributed maculopapular, peripheral, confluent desquamative erythematous, vesiculobullous, urticarial, nodular, purpuric, ulcerated, or eschars ([Table 18-1](#)). For a more detailed discussion of each disease associated with a rash, the reader is referred to the chapter dealing with that specific disease. (Reference chapters and color plates are cited in the text and listed in [Table 18-1](#).)

Centrally Distributed Maculopapular Eruptions Centrally distributed rashes, in which lesions are primarily truncal, are the most common type of eruption. The rash of *measles* (rubeola) starts at the hairline 2 to 3 days into the illness and moves down the body, sparing the palms and soles ([Chap. 194](#)). It begins as discrete erythematous lesions, which become confluent as the rash spreads. Koplik's spots (1- to 2-mm white or bluish lesions with an erythematous halo on the buccal mucosa) are pathognomonic for measles and are generally seen during the first 2 days of symptoms. They should not be confused with Fordyce's spots (ectopic sebaceous glands), which have no erythematous halos and are found in the mouth of healthy individuals. Koplik's spots may briefly overlap with the measles exanthem.

German measles (rubella) also spreads from the hairline downward; unlike that of measles, however, the rash of rubella tends to clear from originally affected areas as it migrates and may be pruritic ([Chap. 195](#)). Forchheimer spots (palatal petechiae) may develop but are nonspecific since they also develop in mononucleosis ([Chap. 184](#)) and scarlet fever ([Chap. 140](#)). Postauricular and suboccipital adenopathy and arthritis are common among adults with German measles. Exposure of pregnant women to ill individuals should be avoided, as rubella causes severe congenital abnormalities. Numerous strains of enteroviruses ([Chap. 193](#)), primarily echoviruses and coxsackieviruses, cause nonspecific syndromes of fever and eruptions that may mimic rubella or measles. Patients with infectious mononucleosis caused by Epstein-Barr virus or with primary infection caused by HIV ([Chap. 309](#)) may exhibit pharyngitis, lymphadenopathy, and a nonspecific maculopapular exanthem.

The rash of *erythema infectiosum* (fifth disease), which is caused by human parvovirus B19, primarily affects children 3 to 12 years old; it develops after fever has resolved as a bright blanchable erythema on the cheeks ("slapped cheeks") with perioral pallor ([Chap. 187](#)). A more diffuse rash (often pruritic) appears the next day on the trunk and extremities and then rapidly develops into a lacy reticular eruption that may wax and wane (especially with temperature change) over 3 weeks. Adults with fifth disease often have arthritis, and fetal hydrops can develop in association with this condition in pregnant women.

Exanthem subitum (roseola, [Fig. 18-CD2](#)) is most common among children under 3 years of age ([Chap. 185](#)). As in erythema infectiosum, the rash usually appears after fever has subsided. It consists of 2- to 3-mm rose-pink macules and papules that rarely coalesce, occur initially on the trunk and sometimes on the extremities (sparing the face), and fade within 2 days.

Though drug reactions have many manifestations, including urticaria, exanthematous *drug-induced eruptions* ([Chap. 59](#)) are most common and are often difficult to distinguish from viral exanthems. Eruptions elicited by drugs are usually more intensely erythematous and pruritic than viral exanthems, but this distinction is not reliable. A history of new medications and an absence of prostration may help to distinguish a drug-related rash from an eruption of another etiology. Rashes may persist for up to 2 weeks after administration of the offending agent is discontinued. Certain populations are more prone than others to drug rashes. Of HIV-infected patients, 50 to 60% develop a rash in response to sulfa drugs; 50 to 100% of patients with mononucleosis due to Epstein-Barr virus develop a rash when given ampicillin.

Rickettsial illnesses ([Chap. 177](#)) should be considered in the evaluation of individuals with centrally distributed maculopapular eruptions. The usual setting for *epidemic typhus* is a site of war or natural disaster in which people are exposed to body lice. A diagnosis of recrudescent typhus should be considered in European immigrants to the United States. However, an indigenous form of typhus, presumably transmitted by flying squirrels, has been reported in the southeastern United States. *Endemic typhus* or *leptospirosis* (the latter caused by a spirochete; [Chap. 174](#)) may be seen in urban environments where rodents proliferate. Outside the United States, other rickettsial diseases cause a spotted-fever syndrome and should be considered in residents of or travelers to endemic areas. Similarly, *typhoid fever*, a nonrickettsial disease caused by *Salmonella typhi* ([Chap. 156](#)), is usually acquired during travel outside the United States.

Some centrally distributed maculopapular eruptions have distinctive features. Erythema chronicum migrans (ECM), the rash of Lyme disease ([Chap. 176](#)), typically manifests as singular or multiple annular plaques. Untreated ECM lesions usually fade within a month but may persist for more than a year. *Erythema marginatum*, the rash of acute rheumatic fever ([Chap. 235](#)), has a distinctive pattern of enlarging and shifting transient annular lesions.

Collagen vascular diseases may cause fever and rash. Patients with *systemic lupus erythematosus* ([Chap. 311](#)) typically develop a sharply defined, erythematous eruption in a butterfly distribution on the cheeks (malar rash) as well as many other skin manifestations. *Still's disease* ([Chap. 326](#)) manifests as an evanescent salmon-colored rash on the trunk and proximal extremities that coincides with fever spikes.

Peripheral Eruptions These rashes are alike in that they are most prominent peripherally or begin in peripheral (acral) areas before spreading centripetally. Early diagnosis and therapy are critical in *RMSF* ([Chap. 177](#)) because of its grave prognosis if untreated. Lesions evolve from macular to petechial, start on the wrists and ankles, spread centripetally, and appear on the palms and soles only later in the disease. The rash of *secondary syphilis* ([Chap. 172](#)), which may be diffuse but is prominent on the palms and soles, should be considered in the differential diagnosis of pityriasis rosea, especially in sexually active patients. *Atypical measles* ([Chap. 194](#)) is seen in individuals contracting measles who received the killed measles vaccine between 1963 and 1967 in the United States and who were not subsequently protected with the live vaccine. *Hand-foot-and-mouth disease* ([Chap. 193](#)) is distinguished by tender vesicles distributed peripherally and in the mouth; outbreaks commonly occur within families. The classic

target lesions of *erythema multiforme* appear symmetrically on the elbows, knees, palms, and soles. In relatively severe cases, these lesions may spread diffusely and involve mucosal surfaces. Lesions may develop on the hands and feet in *endocarditis* ([Chap. 126](#)).

Confluent Desquamative Erythemas These eruptions consist of diffuse erythema frequently followed by desquamation. The eruptions caused by group A *Streptococcus* or *Staphylococcus aureus* are toxin mediated. Certain disease features may provide diagnostic clues. *Scarlet fever* ([Chap. 140](#)) usually follows pharyngitis; patients have a facial flush, a "strawberry" tongue, and accentuated petechiae in body folds (Pastia's lines). *Kawasaki disease* ([Chaps. 57 and 317](#)) presents in the pediatric population as fissuring of the lips, a strawberry tongue, conjunctivitis, adenopathy, and sometimes cardiac abnormalities. *Streptococcal toxic shock syndrome* ([Chap. 140](#), [Fig. 18-CD3](#)) manifests with hypotension, multiorgan failure, and often a severe group A streptococcal infection (e.g., necrotizing fasciitis, [Fig. 18-CD4](#)). *Staphylococcal toxic shock syndrome* ([Chap. 139](#)) also presents with hypotension and multiorgan failure, but usually only *S. aureus* colonization -- not a severe *S. aureus* infection -- is documented. *Staphylococcal scalded-skin syndrome* ([Chap. 139](#)) is seen primarily in children and in immunocompromised adults. Generalized erythema is often evident during the prodrome of fever and malaise; profound tenderness of the skin is distinctive. In the exfoliative stage, the skin can be induced to form bullae with light lateral pressure (Nikolsky's sign). In a mild form, a scarlatiniform eruption mimics scarlet fever, but the patient does not exhibit a strawberry tongue or circumoral pallor. In contrast to the staphylococcal scalded-skin syndrome, in which the cleavage plane is superficial in the epidermis, *toxic epidermal necrolysis* ([Chap. 59](#)) ([Fig. 18-CD5](#)) involves sloughing of the entire epidermis, resulting in severe disease. *Exfoliative erythroderma syndrome* ([Chaps. 56 and 59](#)) is a serious reaction associated with systemic toxicity that is often due to eczema, psoriasis, mycosis fungoides, or a severe drug reaction.

Vesiculobullous Eruptions *Varicella* ([Chap. 183](#)) is highly contagious, often occurring in winter or spring. At a given time within a given region of the body, varicella lesions are in different stages of development. In immunocompromised hosts, varicella vesicles may lack the characteristic erythematous base or may appear hemorrhagic. *Rickettsialpox* ([Chap. 177](#)) is often documented in urban settings and is characterized by vesicles. It can be distinguished from varicella by an eschar at the site of the mouse-mite bite and the papule/plaque base of each vesicle. Disseminated *Vibrio vulnificus* infection ([Chap. 159](#)) or *ecthyma gangrenosum* due to *Pseudomonas aeruginosa* ([Chap. 155](#)) should be considered in immunosuppressed individuals with sepsis and hemorrhagic bullae.

Urticarial Eruptions Individuals with classic urticaria ("hives") usually have a hypersensitivity reaction without associated fever. In the presence of fever, urticarial eruptions are usually due to *urticarial vasculitis* ([Chap. 317](#)). Unlike individual lesions of classic urticaria, which last up to 48 h, these lesions may last up to 5 days. Etiologies include serum sickness (often induced by drugs such as penicillins, sulfas, salicylates, or barbiturates), connective-tissue disease (e.g., systemic lupus erythematosus or Sjogren's syndrome), and infection (e.g., with hepatitis B virus, coxsackievirus A9, or parasites). Malignancy may be associated with fever and chronic urticaria ([Chap. 57](#)).

Nodular Eruptions In immunocompromised hosts, nodular lesions often represent disseminated infection. Patients with disseminated *candidiasis* (often due to *Candida tropicalis*) may have a triad of fever, myalgias, and eruptive nodules ([Chap. 205](#)). Disseminated *cryptococcosis* lesions ([Chap. 204](#)) may resemble molluscum contagiosum. Necrosis of nodules should raise the suspicion of *aspergillosis* ([Chap. 206](#)) or *mucormycosis* ([Chap. 207](#)). *Erythema nodosum* presents with exquisitely tender nodules on the lower extremities. *Sweet's syndrome* ([Chap. 57](#)) should be considered in individuals with multiple nodules and plaques, often so edematous that they give the appearance of vesicles or bullae. Sweet's syndrome may affect either healthy individuals or persons with lymphoproliferative disease.

Purpuric Eruptions *Acute meningococcemia* ([Chap. 146](#)) classically presents in children as a petechial eruption, but initial lesions may appear as blanchable macules or urticaria. *RMSF* should be considered in the differential diagnosis of acute meningococcemia. *Echovirus 9 infection* ([Chap. 193](#)) may mimic acute meningococcemia; patients should be treated as if they have bacterial sepsis since prompt differentiation of these conditions may be impossible. Large ecchymotic areas of *purpura fulminans* ([Chaps. 124 and 146](#)) reflect severe underlying disseminated intravascular coagulation, which may be due to infectious or noninfectious causes. The lesions of *chronic meningococcemia* ([Chap. 146](#)) may have a variety of morphologies, including petechial. Purpuric nodules may develop on the legs and resemble erythema nodosum but lack its exquisite tenderness. Lesions of *disseminated gonococcemia* ([Chap. 147](#)) are distinctive, sparse, countable hemorrhagic pustules, usually located near joints. The lesions of chronic meningococcemia and those of gonococcemia may be indistinguishable in terms of appearance and distribution. *Viral hemorrhagic fever* ([Chaps. 198 and 199](#)) should be considered in patients with an appropriate travel history and a petechial rash. *Thrombotic thrombocytopenic purpura* ([Chaps. 57, 108, and 116](#)) is a noninfectious cause of fever and petechiae. *Cutaneous small-vessel vasculitis (leukocytoclastic vasculitis)* typically manifests as palpable purpura and has a wide variety of causes ([Chap. 57](#)).

Eruptions with Ulcers or Eschars The presence of an ulcer or eschar in the setting of a more widespread eruption can provide an important diagnostic clue. For example, the presence of an eschar may suggest the diagnosis of scrub typhus or rickettsialpox in the appropriate setting. In other illnesses (e.g., anthrax), an ulcer or eschar may be the only skin manifestation.

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19. APPROACH TO THE ACUTELY ILL INFECTED FEBRILE PATIENT - Tamar F. Barlam, Dennis L. Kasper

The physician treating the acutely ill febrile patient must be able to recognize infections that require emergent attention. If such infections are not adequately evaluated and treated at initial presentation, the opportunity to alter an adverse outcome may be lost. In this chapter, the clinical presentations of and approach to patients with relatively common infectious disease emergencies are discussed. These infectious processes are discussed in detail in other chapters. **Noninfectious causes of fever are not covered in this chapter; information on the approach to fever of unknown origin, including that eventually shown to be of noninfectious etiology, is presented in [Chap. 125](#).*

GENERAL CONSIDERATIONS

APPEARANCE

A physician must have a consistent approach to acutely ill patients. Even before the history is elicited and a physical examination performed, an immediate assessment of the patient's general appearance yields valuable information. The perceptive physician's subjective sense that a patient is septic or toxic often proves accurate. Visible agitation or anxiety in a febrile patient can be a harbinger of critical illness.

HISTORY

Presenting symptoms are frequently nonspecific. In addition to a general description of symptoms, it is important to obtain a sense of disease progression. Detailed questions should be asked about the onset and duration of symptoms and about changes in severity or rate of progression over time. Host factors and comorbid conditions may enhance the risk of infection with certain organisms or of a more fulminant course than is usually seen. Lack of splenic function, alcoholism with significant liver disease, intravenous drug use, HIV infection, diabetes, malignancy, and chemotherapy all predispose to specific infections and frequently to increased severity. The patient should be questioned about factors that might help identify a nidus for invasive infection, such as recent upper respiratory tract infections, influenza, or varicella; prior trauma; disruption of cutaneous barriers due to lacerations, burns, surgery, or decubiti; and the presence of foreign bodies, such as nasal packing after rhinoplasty, barrier contraceptives, tampons, arteriovenous fistulas, or prosthetic joints. Travel, contact with pets or other animals, or activities that might result in tick exposure can lead to diagnoses that would not otherwise be considered. Recent dietary intake, medication use, social contact with ill individuals, vaccination history, and menstrual history may be relevant. A review of systems should focus on any neurologic signs or sensorium alterations, rashes or skin lesions, and focal pain or tenderness and should also include a general review of respiratory, gastrointestinal, or genitourinary symptoms. It is especially important to determine the duration and progression of these symptoms in order to gain an appreciation of the pace and urgency of the process.

PHYSICAL EXAMINATION

A complete physical examination should be performed, with special attention to some

areas that are sometimes given short shrift in routine examinations. Assessment of the patient's general appearance and vital signs, skin and soft tissue examination, and the neurologic evaluation are of particular importance.

The patient may appear either anxious and agitated or lethargic and apathetic. Fever is usually present, although the elderly and compromised hosts, such as those who are uremic or cirrhotic and patients who are taking glucocorticoids or nonsteroidal anti-inflammatory agents, may be afebrile despite serious underlying infection. Measurement of blood pressure, heart rate, and respiratory rate helps determine the degree of hemodynamic and metabolic compromise. The patient's airway must be evaluated to rule out the risk of obstruction from an invasive oropharyngeal infection.

The etiologic diagnosis may become evident in the context of a thorough skin examination. Petechial rashes are typically seen with meningococcemia or Rocky Mountain spotted fever (RMSF); erythroderma is usual with toxic shock syndrome (TSS) and drug fever. The soft tissue and muscle examination is critical. Areas of erythema or duskiness, edema, and tenderness may indicate underlying necrotizing fasciitis, myositis, or myonecrosis. The neurologic examination must include a careful assessment of mental status for signs of early encephalopathy. Evidence of nuchal rigidity or focal neurologic findings should be sought. Focal findings, depressed mental status, or papilledema should be evaluated by brain imaging prior to lumbar puncture, which, in this setting, could initiate herniation.

SPECIFIC PRESENTATIONS

For most infections, there is time for careful evaluation, diagnostic testing, and consultation with other physicians. However, the infections considered below according to common clinical presentation can have rapidly catastrophic outcomes, and their immediate recognition can be life-saving. Recommended therapeutic regimens are presented in [Table 19-1](#).

SEPSIS WITHOUT AN OBVIOUS FOCUS OF PRIMARY INFECTION

These patients initially have a brief prodrome of nonspecific symptoms and signs that progresses quickly to hemodynamic instability with hypotension, tachycardia, tachypnea, or respiratory distress. A patient may display altered mental status. Disseminated intravascular coagulation (DIC) with clinical evidence of a hemorrhagic diathesis is a poor prognostic sign.

Septic Shock (See also [Chap. 124](#)) Patients with bacteremia leading to septic shock may have a primary site of infection (e.g., pneumonia, pyelonephritis, or cholangitis) that is not evident initially. Elderly patients with comorbid conditions, hosts compromised by malignancy and neutropenia, or patients who have recently undergone a surgical procedure or hospitalization are at increased risk for an adverse outcome. Gram-negative bacteremia with organisms such as *Pseudomonas aeruginosa*, *Aeromonas hydrophila*, or *Escherichia coli* and gram-positive infection with organisms such as *Staphylococcus aureus* or group A streptococci can present as intractable hypotension and multiorgan failure. Treatment can usually be initiated empirically on the basis of the presentation ([Table 124-3](#)).

Overwhelming Infection in Asplenic Patients (See also [Chap. 124](#)) Patients without splenic function are at risk for overwhelming bacterial sepsis. Asplenic patients succumb to sepsis at 600 times the rate of the general population; 50 to 70% of cases occur within the first 2 years after splenectomy, with a mortality rate of up to 80%. However, in the asplenic individual, an increased risk of overwhelming sepsis continues throughout life. In asplenia, encapsulated bacteria cause the majority of infections, and adults are at lower risk than children because they are more likely to have antibody to these organisms. *Streptococcus pneumoniae* infection is most common, but the risk of infection with *Haemophilus influenzae* or *Neisseria meningitidis* is also high. Severe clinical manifestations of infections due to *E. coli*, *S. aureus*, group B streptococci, *P. aeruginosa*, *Capnocytophaga*, *Babesia*, and *Plasmodium* have been described.

Babesiosis (See also [Chap. 214](#)) A history of recent travel to endemic areas should raise the possibility of infection with *Babesia*. Between 1 and 4 weeks after a tick bite, the patient experiences chills, fatigue, anorexia, myalgia, arthralgia, nausea, and headache; ecchymosis and/or petechiae are occasionally seen. The tick that most commonly transmits *Babesia*, *Ixodes scapularis*, also transmits *Borrelia burgdorferi* (the agent of Lyme disease) and *Ehrlichia*, and co-infection can occur, resulting in more severe disease. Infection with the European species *Babesia divergens* is more frequently fulminant than that due to the U.S. species *B. microti*, causing a febrile syndrome with hemolysis, jaundice, hemoglobinemia, and renal failure and a mortality rate of >50%. Severe babesiosis is especially common in asplenic hosts but does occur in hosts with normal splenic function.

Other Sepsis Syndromes Tularemia ([Chap. 161](#)) is seen throughout the United States, but primarily in Arkansas, Oklahoma, and Missouri, in association with wild rabbit, tick, and tabanid fly contact. The uncommon typhoidal form can be associated with gram-negative septic shock and a mortality rate of >30%. In the United States, plague ([Chap. 162](#)) is found primarily in New Mexico, Arizona, and Colorado after contact with ground squirrels, prairie dogs, or chipmunks. The septic form is particularly rare and is associated with shock, multiorgan failure, and a 30% mortality rate. These rare infections should be considered in the appropriate epidemiologic setting.

SEPSIS WITH SKIN MANIFESTATIONS (See also [Chap. 18](#))

Maculopapular rashes may reflect early meningococcal or rickettsial disease but are usually associated with nonemergent infections. Exanthems are usually viral.

Petechiae Petechial rashes caused by viruses are seldom associated with hypotension or a toxic appearance, although severe measles can be an exception. In other settings, petechial rashes require more urgent attention.

Meningococcemia (See also [Chap. 146](#)) Almost three-quarters of patients with bacteremic *N. meningitidis* infection have a rash. Meningococcemia most often affects young children (i.e., those 6 months to 5 years old, often in daycare). However, sporadic cases and outbreaks occur in schools (grade school through college) and army barracks. Between 10 and 20% of all cases have a fulminant course, with shock, [DIC](#), and multiorgan failure. Of these patients, 50 to 60% die, and survivors often require

extensive debridement or amputation of gangrenous extremities. Patients may exhibit fever, headache, nausea, vomiting, myalgias, change in mental status, and meningismus. However, the rapidly progressive form of disease is not usually associated with meningitis. The rash is initially pink, blanching, and maculopapular, appearing on the trunk and extremities, but then becomes hemorrhagic, forming petechiae. Petechiae are first seen at the ankles, wrists, axillae, mucosal surfaces, and palpebral and bulbar conjunctiva, with subsequent spread to the lower extremities and trunk. A cluster of petechiae may be seen at pressure points, e.g., where a blood pressure cuff has been inflated. In rapidly progressive meningococemia, the petechial rash quickly becomes purpuric ([Plate IID-44](#)) and patients develop DIC. Hypotension with petechiae for <12 h is associated with significant mortality. The mortality rate can exceed 90% in patients without meningitis who have rash, hypotension, and a normal or low white blood cell count and erythrocyte sedimentation rate. A better prognosis has been reported in cases where antibiotics are given before admission by the primary care provider. This observation suggests that early initiation of treatment may be life-saving.

Rocky Mountain spotted fever (See also [Chap. 177](#)) **RMSF** occurs throughout the United States. A history of tick bite is common; however, if such a history is lacking, a history of travel or outdoor activity (e.g., camping in tick-infested areas) can be ascertained. RMSF is caused by *Rickettsia rickettsii*. For the first 3 days, headache, fever, malaise, myalgias, nausea, vomiting, and anorexia are present. By day 3, half of patients have skin findings. Blanching macules develop initially on the wrists and ankles and then spread over the legs and trunk. The lesions become hemorrhagic and are frequently petechial. The rash spreads to palms and soles later in the course ([Plate IID-45](#)). The centripetal spread is a classic feature of RMSF. However, 10 to 15% of patients with RMSF never develop a rash. The patient can be hypotensive and develop noncardiogenic pulmonary edema, confusion, lethargy, and encephalitis progressing to coma. The cerebrospinal fluid (CSF) contains 10 to 100 cells/uL, usually with a predominance of mononuclear cells. The CSF glucose level is often normal; the protein concentration may be slightly elevated. Renal and hepatic injury and bleeding secondary to vascular damage are noted. Untreated infection has a mortality rate of 30%.

Purpura Fulminans (See also [Chaps. 124 and 146](#)) This is the cutaneous manifestation of **DIC** and presents as large ecchymotic areas and hemorrhagic bullae. Progression of petechiae to purpura and ecchymoses is associated with congestive heart failure, septic shock, acute renal failure, acidosis, hypoxia, hypotension, and death. Purpura fulminans has primarily been associated with *N. meningitidis* but, in the splenectomized patient, has been described in association with *S. pneumoniae* and *H. influenzae*.

Ecthyma Gangrenosum Septic shock caused by *P. aeruginosa* and *A. hydrophila* can be associated with ecthyma gangrenosum ([Plate IID-57C](#), [Fig. 19-CD1](#)): hemorrhagic vesicles surrounded by a rim of erythema with central necrosis and ulceration. These gram-negative bacteremias are most common among patients with neutropenia, extensive burns, and hypogammaglobulinemia.

Other Emergent Infections Associated with Rash *Vibrio vulnificus* and other noncholera *Vibrio* bacteremic infections ([Chap. 159](#)) can cause focal skin lesions and overwhelming sepsis in the host with liver disease. After ingestion of contaminated

shellfish, there is a sudden onset of malaise, chills, fever, and hypotension. The patient develops bullous or hemorrhagic skin lesions, usually on the lower extremities, and 75% of patients have leg pain. The mortality rate can be as high as 50%. *Capnocytophaga canimorsus* ([Chap. 127](#)) can cause septic shock in asplenic patients. Infection with this fastidious gram-negative rod typically presents after a dog bite as fever, chills, myalgia, vomiting, diarrhea, dyspnea, confusion, and headache. Findings can include an exanthem or erythema multiforme ([Plate IIE-67](#)), cyanotic mottling or peripheral cyanosis, petechiae, and ecchymosis. About 30% of patients with this fulminant form die of overwhelming sepsis and [DIC](#), and survivors may require amputation to treat gangrene.

Erythroderma TSS([Chaps. 139](#) and [140](#)) is usually associated with erythroderma ([Fig. 18-CD3](#)). The patient presents with fever, malaise, myalgias, nausea, vomiting, diarrhea, and confusion. There is a sunburn-type rash that may be subtle and patchy but is usually diffuse and is found on the face, trunk, and extremities. Erythroderma, which desquamates after 1 to 2 weeks, is more common in *Staphylococcus*-associated than in *Streptococcus*-associated TSS. Hypotension develops rapidly after onset of symptoms, often within hours. Multiorgan failure is seen. Often there is no indication of a primary focal infection. Colonization rather than overt infection of the vagina or a postoperative wound, for example, is typical with staphylococcal TSS, and the mucosal areas appear hyperemic but not infected. Early renal failure may distinguish this syndrome from other septic shock syndromes. Clinical evaluation constitutes the diagnosis because TSS is defined by the clinical criteria of fever, rash, hypotension, and multiorgan involvement. The mortality rate is 5% for menstruation-associated TSS, 10 to 15% for nonmenstrual TSS, and 30 to 70% for streptococcal TSS.

SEPSIS WITH A SOFT TISSUE/MUSCLE PRIMARY FOCUS (See also [Chap. 128](#))

Necrotizing Fasciitis This infection may arise at a site of minimal trauma or postoperative incision and may also be associated with recent varicella, childbirth, or muscle strain. The most common causes of necrotizing fasciitis are group A streptococci alone ([Chap. 140](#)) and a mixed facultative and anaerobic flora ([Chap. 128](#)). Diabetes mellitus, peripheral vascular disease, and intravenous drug use are associated risk factors. Use of nonsteroidal anti-inflammatory agents adversely affects granulocyte chemotaxis, phagocytosis, and bacterial killing, allowing progression of skin or soft tissue infections. The patient may have bacteremia and hypotension without other organ-system failure. Physical findings are minimal compared to the severity of pain and the degree of fever. The examination is often unremarkable except for soft tissue edema and erythema. The infected area is red, hot, shiny, swollen, and exquisitely tender. In untreated infection, the overlying skin develops blue-gray patches after 36 h, and cutaneous bullae and necrosis develop after 3 to 5 days. Necrotizing fasciitis due to a mixed flora, but not that due to group A streptococci, can be associated with gas production. Without treatment, pain decreases because of thrombosis of the small blood vessels and destruction of the peripheral nerves -- an ominous sign. The mortality rate is >30% overall, >70% in association with [TSS](#), and nearly 100% without surgical intervention. Life-threatening necrotizing fasciitis may also be due to *Clostridium perfringens* ([Chap. 145](#)); in this condition, the patient is extremely toxic and the mortality rate is high. Within 48 h, rapid tissue invasion and systemic toxicity associated with hemolysis and death ensue. The distinction between this entity and clostridial

myonecrosis is made by muscle biopsy.

Clostridial Myonecrosis (See also [Chap. 145](#)) Myonecrosis is often associated with trauma or surgery but can be spontaneous. The incubation period is usually 12 to 24 h long, and massive necrotizing gangrene develops within hours of onset. Systemic toxicity, shock, and death can occur within 12 h. The patient's pain and toxic appearance are out of proportion to physical findings. On examination, the patient is febrile, apathetic, tachycardic, and tachypneic and may express a feeling of impending doom. Hypotension and renal failure develop later, and hyperalertness is evident preterminally. The skin over the affected area is bronze-brown, mottled, and edematous. Bullous lesions with serosanguineous drainage and a mousy or sweet odor can be present. Crepitus can occur secondary to gas production in muscle tissue. The mortality rate is >65% with spontaneous myonecrosis, which is often associated with *C. septicum* and underlying malignancy. The mortality rates associated with trunk and limb infection are 63% and 12%, respectively, and any delay in surgical treatment increases the risk of death.

NEUROLOGIC INFECTIONS WITH OR WITHOUT SEPTIC SHOCK

Bacterial Meningitis (See also [Chap. 372](#)) Bacterial meningitis is one of the most common infectious emergencies involving the central nervous system. Although hosts with cell-mediated immune deficiency, including transplant recipients, diabetic patients, the elderly, and cancer patients treated with certain chemotherapeutic agents, are at particular risk for *Listeria monocytogenes* meningitis, most cases in adults are due to *S. pneumoniae* (30 to 50%) and *N. meningitidis* (10 to 35%). An early presentation of headache, meningismus, and fever is classic but is seen in only half of patients. The elderly can present without fever or meningeal signs despite lethargy and confusion. Cerebral dysfunction is evidenced by confusion, delirium, and lethargy that can progress to coma. The presentation is fulminant, with sepsis and brain edema, in some cases; papilledema at presentation is unusual and suggests another diagnosis (e.g., an intracranial lesion). Focal signs, including cranial nerve palsies (IV, VI, VII), can be seen in 10 to 20% of cases; 50 to 60% of patients have bacteremia. A poor neurologic outcome is associated with coma at any time during the course or with a CSF glucose level of <0.6 mmol/L (<10 mg/dL). Mortality is associated with coma, respiratory distress, shock, a CSF protein level of >2.5 g/L, a peripheral white blood cell count of <5000/uL, and a serum sodium level of <135 mmol/L.

Suppurative Intracranial Infections (See also [Chap. 372](#)) Other rare intracranial lesions that present with sepsis and hemodynamic instability are subdural empyema, septic cavernous sinus thrombosis, and septic superior sagittal sinus thrombosis. Rapid recognition of the toxic patient with central neurologic signs is crucial to improvement of the dismal prognosis of these entities.

Subdural Empyema This infection arises from the paranasal sinus in 60 to 70% of cases. Microaerophilic streptococci and staphylococci are the predominant etiologic organisms. The patient is toxic, with fever, headache, and nuchal rigidity. Of all patients, 75% have focal signs and 6 to 20% die.

Septic Cavernous Sinus Thrombosis This condition follows a facial or sphenoid sinus

infection; 70% of cases are due to staphylococci and the remainder to aerobic or anaerobic streptococci. A unilateral or retroorbital headache progresses to a toxic appearance and fever within days. Three-quarters of patients have unilateral periorbital edema that becomes bilateral and then progresses to ptosis, proptosis, ophthalmoplegia, and papilledema. The mortality rate is as high as 30%.

Septic Thrombosis of the Superior Sagittal Sinus This infection spreads from the ethmoid or maxillary sinuses. Its bacterial causes include *S. pneumoniae*, other streptococci, and staphylococci. The fulminant course is characterized by headache, nausea, vomiting, rapid progression to confusion and coma, nuchal rigidity, and brainstem signs. If the sinus is totally thrombosed, the mortality rate exceeds 80%.

Brain Abscess (See also [Chap. 372](#)) Brain abscess often occurs without systemic signs. Almost half of patients are afebrile, and presentations are more consistent with a space-occupying lesion in the brain; 70% have headache, 50% have focal neurologic signs, and 25% have papilledema. Abscesses can present as single or multiple lesions resulting from contiguous foci or hematogenous infection, such as unrecognized endocarditis. The infection progresses over several days from cerebritis to an abscess with a mature capsule. Abscesses arising hematogenously are especially apt to rupture into the ventricular space, causing a sudden and severe deterioration in clinical status and high mortality. Otherwise, mortality is low but morbidity is high (30 to 55%). Patients presenting with stroke and a parameningeal infectious focus, such as sinusitis or otitis, may have a brain abscess, and physicians must maintain a high level of suspicion. Prognosis worsens in patients with a fulminant course, delayed diagnosis, abscess rupture into the ventricles, multiple abscesses, or abnormal neurologic status at presentation.

Cerebral Malaria (See also [Chap. 214](#)) This entity should be urgently considered if patients who have recently traveled to areas endemic for malaria present with a febrile illness and lethargy or other neurologic signs. Fulminant malaria is caused by *Plasmodium falciparum* and is associated with temperatures of $>40^{\circ}\text{C}$ ($>104^{\circ}\text{F}$), hypotension, jaundice, adult respiratory distress syndrome, and bleeding. By definition, any patient with a change in mental status or repeated seizure in the setting of fulminant malaria has cerebral malaria. In adults this nonspecific febrile illness progresses to coma over several days; occasionally, coma occurs within hours and death within 24 h. Nuchal rigidity and photophobia are rare. On physical examination, symmetric encephalopathy is typical, and upper motor neuron dysfunction with decorticate and decerebrate posturing can be seen with advanced disease. Unrecognized infection results in a 30% mortality rate.

Spinal Epidural Abscesses (See also [Chap. 368](#)) Patients with spinal epidural abscesses often present with back pain and develop neurologic deficits late in their course. At-risk patients include those with diabetes mellitus; intravenous drug use; recent spinal trauma, surgery, or epidural anesthesia; and other comorbid conditions, such as HIV infection. The thoracic or lumbar spine is the most common location, and staphylococci are the most common etiologic agents; in HIV-infected intravenous drug users, therapy must cover gram-negative rods and methicillin-resistant *S. aureus*. If a patient gives a history of antecedent back pain and has new neurologic symptoms, this diagnosis must immediately be considered. Almost 60% of patients have fever and

almost 90% have back pain. Paresthesia, bowel and bladder dysfunction, radicular pain, and weakness are frequent neurologic complaints, and examination of the patient may reveal abnormal reflexes and motor and sensory deficits. Rapid recognition and treatment, including immediate drainage, can prevent or minimize permanent neurologic sequelae.

FOCAL SYNDROMES WITH A FULMINANT COURSE

Infection at virtually any primary focus (e.g., osteomyelitis, pneumonia, pyelonephritis, or cholangitis) can result in bacteremia and sepsis. [TSS](#) has been associated with focal infections such as septic arthritis, peritonitis, sinusitis, and wound infection. Death occurs secondary to septic shock or toxin production with hemodynamic instability and multiorgan failure. Rapid clinical deterioration and death can be associated with destruction of the primary site of infection, as is seen in endocarditis and in necrotizing infections of the oropharynx (in which edema suddenly compromises the airway).

Rhinocerebral Mucormycosis (See also [Chap. 207](#)) Patients with diabetes or malignancy are at risk for invasive rhinocerebral mucormycosis. Patients present with low-grade fever, dull sinus pain, diplopia, decreased mental status, decreased ocular motion, chemosis, proptosis, dusky or necrotic nasal turbinates, and necrotic hard-palate lesions that respect the midline. Without rapid recognition and intervention, the process continues an inexorable invasive course with high mortality.

Acute Bacterial Endocarditis (See also [Chap. 126](#)) This entity presents with a much more aggressive course than subacute endocarditis. Bacteria such as *S. aureus*, *S. pneumoniae*, *L. monocytogenes*, *Haemophilus* spp., and streptococci of groups A, B, and G attack native valves. Mortality rates range from 10 to 40%. The host may have comorbid conditions such as underlying malignancy, diabetes mellitus, intravenous drug use, or alcoholism. The patient presents with fever, fatigue, and malaise <2 weeks after onset of infection. On physical examination, a changing murmur and congestive heart failure may be noted. Hemorrhagic macules on palms or soles (*Janeway lesions*, [Fig. 19-CD2](#)) sometimes develop. Petechiae, Roth's spots ([Fig. 19-CD3](#)), splinter hemorrhages ([Fig. 19-CD4](#)), and splenomegaly are unusual. Rapid valvular destruction, particularly of the aortic valve, results in pulmonary edema and hypotension. Myocardial abscesses can form, eroding through the septum or into the conduction system and causing life-threatening arrhythmias or high-degree conduction block. Large friable vegetations can result in major arterial emboli, metastatic infection, or tissue infarction. Emboli can lead to stroke, change in mental status, visual disturbances, aphasia, ataxia, headache, meningismus, brain abscess, cerebritis, spinal cord infarct with paraplegia, arthralgia, osteomyelitis, splenic abscess, septic arthritis, and hematuria. Rapid intervention is crucial for a successful outcome.

DIAGNOSTIC WORKUP OF THE ACUTELY ILL PATIENT

After a quick clinical assessment, diagnostic material should be obtained rapidly and antibiotic and supportive treatment begun. In the sepsis syndromes, blood (for cultures; baseline complete blood count with differential; measurement of serum electrolytes, blood urea nitrogen, serum creatinine, and serum glucose; and liver function tests) can be obtained at the time an intravenous line is placed and before antibiotics are

administered. For patients with possible acute endocarditis, three sets of blood cultures should be performed. Asplenic patients should have a blood smear examined to confirm the presence of Howell-Jolly bodies (indicating the absence of splenic function) and a buffy coat examined for bacteria; these patients can have $>10^6$ organisms per milliliter of blood (compared to 10^4 /mL in patients with an intact spleen). Blood smears from patients with possible cerebral malaria or babesiosis must be examined for the diagnosis and quantitation of parasitemia. Blood smears may also be diagnostic in ehrlichiosis.

Patients with meningitis should have [CSF](#) obtained before the initiation of antibiotic therapy. *If focal neurologic signs, abnormal mental status, or papilledema mandates brain imaging before a lumbar puncture, antibiotics should be administered prior to imaging but after blood for cultures has been drawn.* If CSF cultures are negative, laboratory examination of CSF by latex agglutination or immunoprecipitation can be attempted to make an etiologic diagnosis. However, blood cultures will provide the diagnosis in 50 to 70% of cases.

Focal abscesses necessitate immediate computed tomography or magnetic resonance imaging as part of an evaluation for surgical intervention. Other diagnostic procedures, such as cultures of wounds or scraping of skin lesions, should not delay the initiation of treatment for more than minutes. Once emergent evaluation, diagnostic procedures, and (if appropriate) surgical consultation (see below) have been completed, other laboratory tests can be conducted. Appropriate radiography, computed axial tomography, magnetic resonance imaging, urinalysis, erythrocyte sedimentation rate determination, and transthoracic or transesophageal echocardiography may all prove important.

TREATMENT

[Table 19-1](#) lists first-line treatments for the infections considered in this chapter. (For a more detailed discussion of treatment, see specific chapters.) In addition to the initiation of parenteral antibiotic therapy, several of these infections require urgent surgical attention. General surgery for possible necrotizing fasciitis or myonecrosis, neurosurgical evaluation for subdural empyema or spinal epidural abscess, otolaryngologic surgery for possible mucormycosis, and cardiothoracic surgery for critically ill patients with acute endocarditis are as important as the rapid commencement of antibiotic therapy. For infections such as necrotizing fasciitis and clostridial myonecrosis, rapid surgical intervention supercedes other diagnostic or therapeutic maneuvers.

Acutely ill febrile patients require close observation, aggressive supportive measures, and -- in most cases -- admission to intensive care units. Adjunctive treatments, such as intravenous immunoglobulin administration for [TSS](#), can be considered after initial stabilization. The most important task of the physician is to recognize the acute infectious emergency and proceed with appropriate urgency.

(Bibliography omitted in Palm version)

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20. HYPOTHERMIA AND FROSTBITE - Daniel F. Danzl

HYPOTHERMIA

Accidental hypothermia occurs when there is an unintentional drop in the body's core temperature below 35°C (95°F). At this temperature, many of the compensatory physiologic mechanisms to conserve heat begin to fail. *Primary accidental hypothermia* is a result of the direct exposure of a previously healthy individual to the cold. The mortality rate is much higher for those patients who develop *secondary hypothermia* as a complication of a serious systemic disorder.

CAUSES

Primary accidental hypothermia is geographically and seasonally pervasive. Although most cases occur in the winter months and in colder climates, it is surprisingly common in warmer regions as well. In the United States, hypothermia accounts for more than 700 deaths each year, half of which occur in people age 65 or older.

Multiple variables make individuals at the extremes of age, the elderly and neonates, particularly vulnerable to hypothermia ([Table 20-1](#)). The elderly have diminished thermal proprioception and are more susceptible to immobility, malnutrition, and systemic illnesses that interfere with heat generation or conservation. Dementia, psychiatric illness, and socioeconomic factors often compound these problems by impeding adequate measures to prevent hypothermia. Neonates have high rates of heat loss because of their increased surface-to-mass ratio and their lack of effective shivering and adaptive behavioral responses. In addition, malnutrition can contribute to heat loss because of diminished subcutaneous fat and because of its association with depleted energy stores used for thermogenesis.

Individuals whose occupations or hobbies entail extensive exposure to cold weather are clearly at increased risk for hypothermia. Military history is replete with hypothermic tragedies. Hunters, sailors, skiers, and climbers also are at great risk of exposure, whether it involves injury, changes in weather, or lack of preparedness.

Ethanol causes vasodilatation (which increases heat loss), reduces thermogenesis and gluconeogenesis, and may impair judgment or lead to obtundation. Hypothermia is not an uncommon feature in Wernicke's encephalopathy and may mask its other manifestations. A number of medications are associated with altered thermal regulation. Phenothiazines, barbiturates, benzodiazepines, cyclic antidepressants, and many other medications reduce centrally-mediated vasoconstriction. Up to one-quarter of patients admitted to an intensive care unit because of drug overdose are hypothermic.

Anesthetics can block the shivering responses; their effects may be compounded when patients are not covered adequately in the operating or recovery rooms.

Several types of endocrine dysfunction can lead to hypothermia. Hypothyroidism -- particularly when extreme, as in myxedema coma -- reduces the metabolic rate and impairs thermogenesis and behavioral responses. Myxedema is more common in women than in men and may be occult. Adrenal insufficiency and hypopituitarism can also increase susceptibility to hypothermia. Hypoglycemia, most commonly caused by

insulin or oral hypoglycemic drugs, is associated with hypothermia, in part the result of neuroglycopenic effects on hypothalamic function. Increased osmolality and metabolic derangements associated with uremia, diabetic ketoacidosis, and lactic acidosis can lead to altered hypothalamic thermoregulation.

Neurologic injury from trauma, cerebrovascular accident, subarachnoid hemorrhage, or hypothalamic lesions increases susceptibility to hypothermia. Agenesis of the corpus callosum, or Shapiro syndrome, is one cause of episodic hypothermia, characterized by profuse perspiration followed by a rapid fall in temperature. Acute spinal cord injury disrupts the autonomic pathways that lead to shivering and prevents cold-induced reflex vasoconstrictive responses.

Hypothermia associated with sepsis is a poor prognostic sign. Hepatic failure causes decreased glycogen stores and gluconeogenesis, as well as a diminished shivering response. In acute myocardial infarction associated with low cardiac output, hypothermia may be reversed after adequate resuscitation. With extensive burns, psoriasis, erythrodermas, and other skin diseases, increased peripheral blood flow leads to excessive heat loss.

THERMOREGULATION

Heat loss occurs through five mechanisms: radiation (55 to 65% of heat loss), conduction (10 to 15% of heat loss, but much greater in cold water), convection (increase in the wind), respiration, and evaporation (which are affected by the ambient temperature and the relative humidity).

The preoptic anterior hypothalamus normally orchestrates thermoregulation ([Chap. 17](#)). The immediate defense of thermoneutrality is via the autonomic nervous system ([Chap. 72](#)), whereas delayed control is mediated by the endocrine system. Autonomic nervous system responses include the release of norepinephrine, increased muscle tone, and shivering, leading to thermogenesis and an increase in the basal metabolic rate. Cutaneous cold thermoreception causes direct reflex vasoconstriction to conserve heat. Prolonged exposure to cold also stimulates hypothalamic release of thyrotropin releasing hormone; this leads to increased levels of thyroid stimulating hormone (TSH), which stimulates the thyroid gland to produce thyroxine, a hormone that increases metabolic rate.

CLINICAL PRESENTATION

In most cases of hypothermia, the history of exposure to environmental factors, such as prolonged exposure to the outdoors without adequate clothing, makes the diagnosis straightforward. In urban settings, however, the presentation is often more subtle and the clinician may focus on other disease processes, toxin exposures, or psychiatric diagnoses.

After initial stimulation by hypothermia, there is progressive depression of all organ systems. The timing of the appearance of these clinical manifestations varies widely ([Table 20-2](#)). Without knowing the core temperature, it can be difficult to interpret other vital signs. For example, a tachycardia disproportionate to the core temperature

suggests secondary hypothermia resulting from hypoglycemia, hypovolemia, or a toxin overdose. Because carbon dioxide production declines progressively, the respiratory rate should be low; persistent hyperventilation suggests a central nervous system (CNS) lesion or one of the organic acidoses. A markedly depressed level of consciousness in a patient with mild hypothermia should raise suspicion of an overdose or CNS dysfunction due to infection or trauma.

Physical examination findings can also be altered by hypothermia. For instance, the assumption that areflexia is solely attributable to hypothermia can obscure and delay the diagnosis of a spinal cord injury. Patients with hypothermia may be confused or combative; these symptoms abate more rapidly with rewarming than with the use of restraints. A classic example of maladaptive behavior in patients with hypothermia is paradoxical undressing, which involves the inappropriate removal of clothing in response to a cold stress. The cold-induced ileus and abdominal rectus spasm can mimic, or mask, the presentation of an acute abdomen ([Chap. 14](#)).

When a patient in hypothermic cardiac arrest is first discovered, cardiopulmonary resuscitation is indicated, unless (1) a do-not-resuscitate status is verified, (2) obviously lethal injuries are identified, or (3) the depression of a frozen chest wall is not possible. As the resuscitation proceeds, the prognosis is grave if there is evidence of widespread cell lysis, as reflected by potassium levels exceeding 10 mEq/L. Other findings that may preclude continuing resuscitation include a core temperature $<12^{\circ}\text{C}$, a pH <6.5 , or evidence of intravascular thrombosis with a fibrinogen value <50 mg/dL. The decision to terminate resuscitation before rewarming the patient to 35°C is extremely difficult. There are no validated prognostic indicators for recovery from hypothermia. A history of asphyxia with secondary cooling is the most important negative predictor of survival.

DIAGNOSIS AND STABILIZATION

Hypothermia is confirmed by measuring the core temperature, preferably at two sites. Rectal probes should be placed to a depth of 15 cm and not adjacent to cold feces. A simultaneous esophageal measurement will be falsely high during heated inhalation therapy. The probe should be placed 24 cm below the larynx. The greatest discordance between the readings is usually during the transition phase before effective rewarming. Relying solely on infrared tympanic thermography is not advisable.

After a diagnosis of hypothermia is established, cardiac monitoring should be instituted, along with attempts to limit further heat loss. If the patient is in ventricular fibrillation, one sequence of 3 defibrillation attempts (2 J/kg) should be administered. If unsuccessful, active rewarming should be continued past 30° to 32°C . Supplemental oxygenation is always warranted, since tissue oxygenation is adversely affected by the leftward shift of the oxyhemoglobin dissociation curve. Pulse oximetry may be unreliable in patients with vasoconstriction. If protective airway reflexes are absent, gentle endotracheal intubation should be performed. Adequate pre-oxygenation will prevent ventricular arrhythmias.

Insertion of a gastric tube prevents dilatation secondary to decreased bowel motility. Indwelling bladder catheters facilitate monitoring of cold-induced diuresis. Dehydration is commonly encountered with chronic hypothermia, and most patients benefit from a bolus of crystalloid. Normal saline containing 5% dextrose is preferable to lactated

Ringer's solution, as the liver in hypothermic patients inefficiently metabolizes lactate. The placement of a pulmonary artery catheter, although of potential value, risks perforation of the less compliant pulmonary artery. The use of a central venous catheter should be avoided because of right atrial irritability.

Arterial blood gases should not be corrected for temperature ([Chap. 50](#)). This is termed the ectothermic or alpha-stat approach, which maximizes enzymatic function and maintains the normal distribution of charged metabolic intermediates. An uncorrected pH of 7.42 and a P_{CO_2} of 40 mmHg reflects appropriate alveolar ventilation and acid-base balance at any core temperature. Acid-base imbalances should be corrected gradually, since the bicarbonate buffering system is inefficient. When the P_{CO_2} increases 10 mmHg at 28°C, it doubles the pH decline of 0.08 that is normally induced at 37°C.

The severity of anemia may be underestimated because the hematocrit increases 2% for each 1°C drop in temperature. White blood cell sequestration and bone marrow suppression are common, potentially masking an infection. Although hypokalemia is more common in chronic hypothermia, hyperkalemia also occurs; the expected electrocardiographic changes can be obscured by hypothermia. Patients with renal insufficiency, metabolic acidoses, or rhabdomyolysis are most at risk for electrolyte disturbances.

Coagulopathies are common because cold inhibits the enzymatic reactions required for activation of the intrinsic cascade. In addition, the production of thromboxane B_2 by platelets is temperature-dependent, and platelet function is impaired. The administration of platelets and fresh frozen plasma is, therefore, not effective. The prothrombin or partial thromboplastin times reported by the laboratory appear deceptively normal and contrast with the observed coagulopathy. This contradiction appears because all coagulation tests are routinely performed at 37°C, and the enzymes are thus rewarmed.

REWARMING STRATEGIES

The key initial decision is whether to rewarm the patient passively or actively. *Passive external rewarming* simply involves covering and insulating the patient in a warm environment. With the head covered, the rate of rewarming is usually 0.5° to 2.0°C per hour. This technique is ideal for previously healthy patients who develop acute, mild primary accidental hypothermia. The patient must have sufficient fuel and glycogen to support endogenous thermogenesis.

There are reservations about the application of heat directly to the extremities of patients with chronic severe hypothermia. Extinguishing peripheral vasoconstriction in the dehydrated patient may precipitate core temperature "afterdrop" -- the continual decline in the core temperature after removal of the patient from the cold. This phenomenon results from conductive temperature equilibration and a circulatory convective mechanism. Rewarming frostbitten extremities before stabilization of the core temperature causes a significant core temperature afterdrop. In contrast, truncal heat application may minimize the risk of afterdrop.

Active rewarming is necessary under the following circumstances: core temperature < 32°C (poikilothermia), cardiovascular instability, age extremes, CNS

dysfunction, endocrine insufficiency, or any suspicion of secondary hypothermia. *Active external rewarming* is best accomplished with forced-air heating blankets. Other options include radiant heat sources and hot packs. Monitoring a patient with hypothermia in a heated tub is extremely difficult. Electric blankets should be avoided because vasoconstricted skin is easily burned. Widely available *active core rewarming* options include heated inhalation, heated infusion, and lavage (gastric, colonic, mediastinal, thoracic, pleural). The therapeutic options also include hemodialysis, venovenous, and continuous arteriovenous rewarming, in addition to formal cardiopulmonary bypass.

Arteriovenous anastomoses (AVA) rewarming provides exogenous heat by immersion of the hands, forearms, feet, and calves in 44° to 45°C water. Airway rewarming with heated humidified oxygen (40° to 45°C) is a convenient option via mask or endotracheal tube. Although airway rewarming provides less heat than some other forms of active core rewarming, it eliminates respiratory heat loss and adds 1° to 2°C to the overall rewarming rate. Crystalloids should be heated to 40° to 42°C. The quantity of heat provided is significant only during massive volume resuscitations. The most efficient method for heating and delivering fluid or blood is with a countercurrent in-line heat exchanger. Heated irrigation of the gastrointestinal tract or bladder transfers minimal heat because of the limited available surface area. These methods should be reserved for patients in cardiac arrest and then used in combination with all available active rewarming techniques. Closed thoracic lavage is far more efficient in severely hypothermic patients with cardiac arrest. The hemithoraces are irrigated through two large-bore thoracostomy tubes that are inserted into the left or both of the hemithoraces. Thoracostomy tubes should not be placed in the left chest of a spontaneously perfusing patient for purposes of rewarming. Peritoneal lavage with the dialysate at 40° to 45°C efficiently transfers heat when delivered through two catheters with outflow suction. Like peritoneal dialysis, standard hemodialysis is especially useful for patients with electrolyte abnormalities, rhabdomyolysis, or toxin ingestions.

With extracorporeal venovenous rewarming, the blood is removed from a central venous catheter, heated to 40°C, and returned through a second central or peripheral venous catheter. Continuous arteriovenous rewarming involves the use of percutaneously inserted femoral arterial and contralateral femoral venous 8.5 Fr catheters. The blood pressure must be at least 60 mmHg. Heparin-bonded tubing obviates the need for systemic anticoagulation. Full circulatory support with an oxygenator can only be provided through formal cardiopulmonary bypass (CPB). Femoral flow rates of 2 to 3 L/min elevate the core temperature 1° to 2°C every 3 to 5 min. CPB should be considered in nonperfusing patients without documented contraindications to resuscitation. Circulatory support may also be the only effective option in patients with completely frozen extremities, or those with significant tissue destruction coupled with rhabdomyolysis.

There is no evidence that extremely rapid rewarming improves survival in perfusing patients. The best strategy is usually a combination of passive, truncal active, and active core rewarming techniques.

DRUG THERAPY

When a patient is hypothermic, target organs and the cardiovascular system respond

minimally to most medications. Moreover, cumulative doses can cause toxicity during rewarming because of increased binding of drugs to proteins, and impaired metabolism and excretion. As an example, the administration of repeated doses of digoxin or insulin would be ineffective while the patient is hypothermic, and the residual drugs are potentially toxic during rewarming.

Any pharmacologic manipulation of the depressed and vasoconstricted cardiovascular system should generally be avoided. If the hypotension does not respond to crystalloid infusion and rewarming, low-dose dopamine (2 to 5 ug/kg per min) support should be considered. Atrial arrhythmias should initially be monitored without intervention, as the ventricular response will be slow, and most will convert spontaneously during rewarming. When indicated, bretylium tosylate is the class III ventricular antiarrhythmic of choice. During ventricular fibrillation, it should initially be administered at a dose of 10 mg/kg. Bretylium uniquely increases the ventricular arrhythmia threshold at low temperatures, although the wisdom of prophylaxis is unresolved.

Initiating empirical therapy for adrenal insufficiency is usually not warranted unless there is a history suggesting steroid dependence, hypoadrenalism, or a failure to rewarm with standard therapy. However, the administration of parenteral levothyroxine to euthyroid patients with hypothermia is potentially hazardous. Because laboratory results can be delayed and confounded by the presence of the sick euthyroid syndrome ([Chap. 330](#)), historical clues or physical findings suggestive of hypothyroidism should be sought. When myxedema is the cause of hypothermia, the relaxation phase of the Achilles reflex is prolonged more than the contraction phase.

Hypothermia obscures most of the symptoms and signs of infection, notably fever and leukocytosis. Shaking rigors from infection may be mistaken for shivering. Except in mild cases, extensive cultures and repeated physical examinations are essential. Unless an infectious source is identified, empirical antibiotic prophylaxis is most warranted in the elderly, neonates, and immunocompromised patients.

Preventive measures should be discussed with high-risk individuals, such as the elderly or people whose work frequently exposes them to extreme cold. The importance of layered clothing and headgear, adequate shelter, increased caloric intake, and the avoidance of ethanol should be emphasized, along with access to rescue services.

FROSTBITE

Peripheral cold injuries include both freezing and nonfreezing injuries to tissue. Frostbite occurs when the tissue temperature drops below 0°C. Ice crystal formation subsequently distorts and destroys the cellular architecture. Once the vascular endothelium is damaged, stasis progresses rapidly to microvascular thrombosis. Tissue freezes quickly when in contact with thermal conductors such as metal or volatile solutions. Other predisposing factors include constrictive clothing or boots, immobility, or vasoconstrictive medications.

Clinically, it is most practical to classify frostbite as superficial or deep. Superficial does not entail tissue loss. Classically, frostbite is retrospectively graded like a burn once the resultant pathology is demarcated over time. First-degree frostbite causes only

anesthesia and erythema. The appearance of superficial vesiculation surrounded by edema and erythema is considered second degree ([Plates IIA-18,IIA-19](#)). Hemorrhagic vesicles reflect a serious injury to the microvasculature, and indicate third-degree frostbite. Fourth-degree injuries damage subcuticular, muscular, and osseous tissues.

PATHOPHYSIOLOGY

Peripheral cold injury involves a cascade of events. Endothelial cells are very susceptible to cold injury. In the prefreeze phase, plasma leaks and there is the development of microvascular vasoconstriction. The radiation of heat from underlying tissues initially prevents crystallization. The freeze phase usually begins with extracellular fluid crystallization. Water exits the cell and causes intracellular dehydration, hyperosmolality, and ultimately cellular shrinkage and demise. Damaged tissue releases thromboxane A₂ and prostaglandin F_{2a}, which produce platelet aggregation, leukocyte immobilization, and vasoconstriction.

After the tissue thaws, the second phase of the cascade causes progressive dermal ischemia. The microvasculature begins to collapse, arteriovenous shunting increases tissue pressures, and there is progressive formation of edema. Finally, thrombosis, ischemia, and superficial necrosis appear. The development of mummification and demarcation may take weeks to months.

CLINICAL PRESENTATION

The initial presentation of frostbite can be deceptively benign. The symptoms always include a sensory deficiency affecting light touch, pain, and temperature perception. The acral areas and distal extremities are the most common insensate areas. Some patients complain of a clumsy or "chunk of wood" sensation in the extremity.

Deep frostbitten tissue can appear waxy, mottled, yellow, or violaceous-white. Favorable presenting signs include some warmth or sensation with normal color. The injury is often superficial if the subcutaneous tissue is pliable or if the dermis can be rolled over bony prominences.

The two most common nonfreezing peripheral cold injuries are *chilblain (pernio)* and *immersion (trench) foot*. Chilblain results from neuronal and endothelial damage induced by repetitive exposure to dry cold. Young females, particularly those with a history of Raynaud's phenomenon, are most at risk. Persistent vasospasticity and vasculitis can cause erythema, mild edema, and pruritus. Eventually plaques, blue nodules, and ulcerations develop. These lesions typically involve the dorsa of the hands and feet. In contrast, immersion (trench) foot results from repetitive exposure to wet cold above the freezing point. The feet initially appear cyanotic, cold, and edematous. The subsequent development of bullae is often indistinguishable from frostbite. This vesiculation rapidly progresses to ulceration and liquefaction gangrene. Patients with milder cases complain of hyperhidrosis, cold sensitivity, and painful ambulation for many years.

Various ancillary tests have been used in an attempt to diagnose the severity of peripheral cold injuries. None consistently predicts the extent of injury at presentation. For example, angiography and magnetic resonance imaging can demonstrate the

patency of large vessels but not the microvasculature. Ultrasonography and digital plethysmography are also insensitive. Thermography and technetium scintigraphy help evaluate perfusion several days after rewarming.

TREATMENT

Frozen tissue should be rapidly and completely thawed by immersion in circulating water at 37° to 40°C. Rapid rewarming often produces an initial hyperemia. The early formation of clear distal large blebs is more favorable than smaller proximal dark hemorrhagic blebs. A common error is the premature termination of thawing, since the reestablishment of perfusion is intensely painful. Parenteral narcotics will be necessary with deep frostbite. If cyanosis persists after rewarming, the tissue compartment pressures should be monitored carefully.

Numerous experimental antithrombotic and vasodilatory treatment regimens have been evaluated. There is no conclusive evidence that dextran, heparin, steroids, calcium channel blockers, or hyperbaric oxygen salvage tissue. A treatment protocol for frostbite is summarized in [Table 20-3](#).

Unless infection develops, any decision regarding debridement or amputation should be deferred until there is clear evidence of demarcation, mummification, and sloughing. The most common symptomatic sequelae reflect neuronal injury and the persistently abnormal sympathetic tone, including paresthesias, thermal misperception, and hyperhidrosis. Delayed findings include nail deformities, cutaneous carcinomas, and epiphyseal damage in children.

(Bibliography omitted in Palm version)

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SECTION 3 -NERVOUS SYSTEM DYSFUNCTION

21. FAINTNESS, SYNCOPE, DIZZINESS, AND VERTIGO - *Robert B. Daroff, Mark D. Carlson*

Syncope is defined as transient loss of consciousness due to reduced cerebral blood flow. Syncope is associated with postural collapse and spontaneous recovery. It may occur suddenly, without warning, or may be preceded by symptoms of varying duration. These include faintness or lightheadedness, "dizziness" without true vertigo, a feeling of warmth, diaphoresis, nausea, and visual blurring occasionally proceeding to blindness. These presyncopal symptoms may increase in severity until loss of consciousness occurs or may resolve prior to loss of consciousness if the cerebral ischemia is corrected. The differentiation of syncope from seizure is an important, sometimes difficult, diagnostic problem.

Syncope may be benign when it occurs as a result of normal cardiovascular reflex effects on heart rate and vascular tone or malignant when due to a life-threatening arrhythmia. Syncope may occur as a single event or may be recurrent. Recurrent, unexplained syncope, particularly in an individual with structural heart disease, is associated with a high risk for death (40% mortality within 2 years).

SYNCOPE

At the beginning of a syncopal attack, the patient is nearly always in the upright position, either sitting or standing. A cardiac etiology, such as an arrhythmia, is exceptional in this respect. The patient is warned of the impending faint by a sense of "feeling bad," of giddiness, and of movement or swaying of the floor or surrounding objects. The patient becomes confused and may yawn, visual spots and dimming may occur, and the ears may ring. Nausea and vomiting sometimes accompany these symptoms. There is a striking pallor or ashen gray color of the face, and generalized perspiration ensues. In some patients, a gradual onset with presyncopal symptoms may allow time for protection against injury; in others the syncope is sudden and without warning. The onset varies from instantaneous to 10 to 30 s, rarely longer.

The depth and duration of unconsciousness vary. Sometimes the patient remains partly aware of the surroundings, or there may be profound coma. The patient may remain in this state for seconds or minutes. Usually the patient lies motionless with skeletal muscles relaxed, but a few clonic jerks of the limbs and face may occur shortly after consciousness is lost. Sphincter control is usually maintained, in contrast to a seizure. The pulse is feeble or apparently absent, the blood pressure may be low or undetectable, and breathing may be almost imperceptible. Once the patient is in a horizontal position, gravity no longer hinders the flow of blood to the brain. The strength of the pulse may then improve, color begins to return to the face, breathing becomes quicker and deeper, and consciousness is regained. There is usually an immediate recovery of consciousness. Some patients, however, may be keenly aware of physical weakness, and rising too soon may precipitate another faint. In other patients, particularly those with tachyarrhythmias, there may be no residual symptoms following the initial syncope. Headache and drowsiness, which with mental confusion are the usual sequelae of a seizure, do not follow a syncopal attack.

PATHOPHYSIOLOGY

The more common types of faint are reducible to a few simple mechanisms. Syncope results from a sudden impairment of brain metabolism, usually brought about by hypotension with reduction of cerebral blood flow. Several mechanisms subserve circulatory adjustments to the upright posture. Approximately three-fourths of the systemic blood volume is contained in the venous bed, and any interference in venous return may lead to a reduction in cardiac output. Cerebral blood flow may still be maintained, as long as systemic arterial vasoconstriction occurs, but when this adjustment fails, serious hypotension with resultant cerebral underperfusion to less than half of normal results in syncope. Normally, the pooling of blood in the lower parts of the body is prevented by: (1) pressor reflexes that induce constriction of peripheral arterioles and venules, (2) reflex acceleration of the heart by means of aortic and carotid reflexes, and (3) improvement of venous return to the heart by activity of the muscles of the limbs. Placing a normal person on a tilt table to relax the muscles and tilting upright slightly diminishes cardiac output and allows the blood to accumulate in the legs to a slight degree; this may then be followed by a slight transitory fall in systolic arterial pressure and, in patients with defective vasomotor reflexes, may produce faints.

CAUSES OF SYNCOPE

Transiently decreased cerebral blood flow is usually due to one of three general mechanisms: disorders of vascular tone or blood volume, cardiovascular disorders including cardiac arrhythmias, or cerebrovascular disease ([Table 21-1](#)). Not infrequently, however, the cause of syncope is multifactorial.

Disorders of Vascular Tone or Blood Volume Disorders of autonomic control of the heart and circulation account for at least half of syncopal episodes. These disorders share common pathophysiologic mechanisms: a cardioinhibitory component (e.g., bradycardia due to increased efferent vagal activity), a vasodepressor component (e.g., inappropriate vasodilatation due to sympathetic withdrawal), or both.

Vasovagal (Vasodepressor, Neurocardiogenic) Syncope This form of syncope is the common faint that may be experienced by normal persons and accounts for approximately half of all episodes of syncope. It is frequently recurrent and commonly precipitated by a hot or crowded environment, alcohol, extreme fatigue, severe pain, hunger, prolonged standing, and emotional or stressful situations. Episodes are often preceded by a presyncopal prodrome lasting seconds to minutes. Vasovagal syncope rarely occurs in the supine position. The individual is usually sitting or standing and experiences weakness, nausea, diaphoresis, lightheadedness, blurred vision, and often a forceful heart beat with tachycardia followed by cardiac slowing prior to loss of consciousness. The individual appears pallid and has decreasing blood pressure prior to syncope. The duration of unconsciousness is rarely longer than a few minutes if the conditions that provoke the episode are reversed. Consciousness is usually regained shortly after assuming a recumbent posture, but unconsciousness may be prolonged if an individual remains upright. Although commonly benign, vasovagal syncope can be associated with prolonged asystole and hypotension, resulting in injury.

Vasovagal syncope occurs in the setting of increased sympathetic activity and venous pooling. Under these conditions, vigorous myocardial contraction of a relatively empty left ventricle activates ventricular mechanoreceptors and vagal afferent nerve fibers, inhibiting sympathetic efferent activity and increasing parasympathetic efferent activity. The resultant vasodilatation and bradycardia induce hypotension and syncope.

The central nervous system (CNS) mechanisms responsible for vasovagal syncope are not clear. Animal studies, not confirmed in humans, suggest that endogenous opiates (endorphins) may play a role. Serotonin (5-hydroxytryptamine) participates in blood pressure regulation and may also be involved in inhibition of sympathetic efferent activity (and arterial vasodilatation) associated with vasovagal syncope.

Although the reflex involving myocardial mechanoreceptors is the mechanism usually accepted as responsible for vasovagal syncope, other reflexes may also be operative. Patients with transplanted (denervated) hearts have experienced cardiovascular responses identical to those present during vasovagal syncope. This should not be possible if the response depends solely on the reflex mechanisms described above unless the transplanted heart has become reinnervated. Moreover, vasovagal syncope often occurs in response to stimuli (fear, emotional stress, or pain) that may not be associated with venous pooling in the lower extremities, which suggests a cognitive or cortical component to the reflex. Thus, a variety of afferent and efferent responses may cause vasovagal syncope.

Postural (Orthostatic) Hypotension This occurs in patients who have a chronic defect in, or variable instability of, vasomotor reflexes. Systemic arterial blood pressure falls on assumption of upright posture due to loss of vasoconstriction reflexes in resistance and capacitance vessels of the lower extremities. Although the syncopal attack differs little from vasodepressor syncope, the effect of posture is critical. Sudden rising from a recumbent position or standing quietly are precipitating circumstances. *Orthostatic hypotension may be the cause of syncope in up to 30% of the elderly; polypharmacy with antihypertensive or antidepressant drugs is often a contributor in these patients.*

Postural syncope may occur in otherwise normal persons with defective postural reflexes. Patients with *idiopathic postural hypotension* may be identified by a characteristic response to upright tilt on a table. Initially, the blood pressure diminishes slightly before stabilizing at a lower level. Shortly thereafter, the compensatory reflexes fail and the systemic arterial pressure falls precipitously. The condition is often familial.

Orthostatic hypotension, often accompanied by disturbances in sweating, impotence, and sphincter difficulties, is also a primary feature of the autonomic nervous system disorders discussed in [Chap. 366](#) and listed in [Table 366-1](#). The most common causes of neurogenic orthostatic hypotension are chronic diseases of the peripheral nervous system that involve postganglionic unmyelinated fibers (e.g., diabetic, nutritional, and amyloid polyneuropathy). Much less common are the multisystem atrophies, which are [CNS](#) disorders in which orthostatic hypotension is associated with (1) parkinsonism (Shy-Drager syndrome), (2) progressive cerebellar degeneration, or (3) a more variable parkinsonian and cerebellar syndrome (striatonigral degeneration). Very rarely, an acute postganglionic dysautonomia has been reported that appears to represent a variant of Guillain-Barre syndrome ([Chap. 378](#)).

There are several additional causes of postural syncope: (1) After physical deconditioning (such as after prolonged illness with recumbency, especially in elderly individuals with reduced muscle tone) or after prolonged weightlessness, as in space flight; (2) after sympathectomy that has abolished vasopressor reflexes; and (3) in patients receiving antihypertensive or vasodilator drugs and those who are hypovolemic because of diuretics, excessive sweating, diarrhea, vomiting, hemorrhage, or adrenal insufficiency.

Carotid Sinus Hypersensitivity Syncope due to carotid sinus hypersensitivity is precipitated by pressure on the carotid sinus baroreceptors, which are located just cephalad to the bifurcation of the common carotid artery. This typically occurs in the setting of shaving, a tight collar, or turning the head to one side. Carotid sinus hypersensitivity occurs predominantly in men, most of whom are 50 years of age or older. Activation of carotid sinus baroreceptors gives rise to impulses carried via the nerve of Hering, a branch of the glossopharyngeal nerve, to the medulla oblongata. These afferent impulses activate efferent sympathetic nerve fibers to the heart and blood vessels, cardiac vagal efferent nerve fibers, or both. In patients with carotid sinus hypersensitivity, these responses may cause sinus arrest or atrioventricular (AV) block (a cardioinhibitory response), vasodilatation (a vasodepressor response), or both (a mixed response). Although originally described in 1933, the mechanisms responsible for the syndrome are not clear, and validated diagnostic criteria do not exist; some authorities have questioned its very existence.

Situational Syncope A variety of activities, including cough, deglutition, micturition, and defecation, are associated with syncope in susceptible individuals. These syndromes are caused, at least in part, by abnormal autonomic control and may involve a cardioinhibitory response, a vasodepressor response, or both. Cough, micturition, and defecation are associated with maneuvers (such as Valsalva, straining, and coughing) that may contribute to hypotension and syncope by decreasing venous return. Increased intracranial pressure secondary to the increased intrathoracic pressure may also contribute by decreasing cerebral blood flow.

Cough syncope typically occurs in men with chronic bronchitis or chronic obstructive lung disease during or immediately after prolonged coughing fits. Micturition syncope occurs predominantly in middle-aged and older men, particularly those with prostatic hypertrophy and obstruction of the bladder neck; loss of consciousness usually occurs at night during or immediately after voiding. Deglutition and defecation syncope occur in men and women. Deglutition syncope may be associated with esophageal disorders, particularly esophageal spasm. In some individuals, particular foods and carbonated or cold beverages initiate episodes by activating esophageal sensory receptors that trigger reflex sinus bradycardia or AV block. Defecation syncope is probably secondary to a Valsalva maneuver in older individuals with constipation.

Glossopharyngeal Neuralgia Syncope due to glossopharyngeal neuralgia is preceded by pain in the oropharynx, tonsillar fossa, or tongue. Loss of consciousness is usually associated with asystole rather than vasodilatation. The mechanism is thought to involve activation of afferent impulses in the glossopharyngeal nerve which terminate in the nucleus solitarius of the medulla and, via collaterals, activate the dorsal motor

nucleus of the vagus nerve.

Cardiovascular Disorders Cardiac syncope results from a sudden reduction in cardiac output, caused most commonly by a cardiac arrhythmia. In normal individuals, heart rates between 30 and 180 beats per minutes (bpm) do not reduce cerebral blood flow, especially if the person is in the supine position. As the heart rate decreases, ventricular filling time and stroke volume increase to maintain normal cardiac output. At rates below 30 bpm, stroke volume can no longer increase to compensate adequately for the decreased heart rate. At rates greater than approximately 180 bpm, ventricular filling time is inadequate to maintain adequate stroke volume. In either case, cerebral hypoperfusion and syncope may occur. Upright posture, cerebrovascular disease, anemia, and coronary, myocardial, or valvular disease all reduce the tolerance to alterations in rate.

Bradyarrhythmias ([Chap. 229](#)) may occur as a result of an abnormality of impulse generation (e.g., sinoatrial arrest) or impulse conduction (e.g., [AV](#) block). Either may cause syncope if the escape pacemaker rate is insufficient to maintain cardiac output. Syncope due to bradyarrhythmias may occur abruptly, without presyncopal symptoms, and recur several times daily. Patients with *sick sinus syndrome* may have sinus pauses (>3 s), and those with syncope due to high-degree AV block (*Stokes-Adams-Morgagni syndrome*) may have evidence of conduction system disease (e.g., prolonged PR interval, bundle branch block). However, the arrhythmia is often transitory, and the surface electrocardiogram or continuous electrocardiographic monitor (Holter monitor) taken later may not reveal the abnormality. The *bradycardia-tachycardia syndrome* is a common form of sinus node dysfunction in which syncope generally occurs as a result of marked sinus pauses following termination of paroxysmal supraventricular tachycardia. Drugs are a common cause for bradyarrhythmias, particularly in patients with underlying structural heart disease. Digoxin, β -adrenergic receptor antagonists, calcium channel blockers, and many antiarrhythmic drugs may suppress sinoatrial node impulse generation or slow AV nodal conduction.

Syncope due to a *tachyarrhythmia* ([Chap. 230](#)) is usually preceded by palpitation or lightheadedness but may occur abruptly with no warning symptoms. *Supraventricular tachyarrhythmias* are unlikely to cause syncope in individuals with structurally normal hearts but may do so if they occur in patients with: (1) heart disease that also compromises cardiac output, (2) cerebrovascular disease, (3) a disorder of vascular tone or blood volume, or (4) a rapid ventricular rate. These tachycardias result most commonly from paroxysmal atrial flutter, atrial fibrillation, or reentry involving the [AV](#) node or accessory pathways that bypass part or all of the AV conduction system. Patients with the *Wolff-Parkinson-White syndrome* may experience syncope when a very rapid ventricular rate occurs due to reentry across an accessory AV connection.

In patients with structural heart disease, ventricular tachycardia, sometimes associated with ventricular fibrillation, is a common cause of syncope, particularly in patients with a prior myocardial infarction. Patients with aortic valvular stenosis and hypertrophic obstructive cardiomyopathy are also at risk for ventricular tachycardia. Individuals with abnormalities of ventricular repolarization (prolongation of the QT interval) are at risk to develop polymorphic ventricular tachycardia (*torsade de pointes*). Those with the inherited form of this syndrome often have a family history of sudden death in young

individuals. Genetic markers can identify some patients with familial long-QT syndrome but the clinical utility of these markers remains unproven. Drugs (i.e., certain antiarrhythmics and erythromycin) and electrolyte disorders (i.e., hypokalemia, hypocalcemia, hypomagnesemia) can prolong the QT interval and predispose to torsade de pointes. Antiarrhythmic medications may precipitate ventricular tachycardia, particularly in patients with structural heart disease.

In addition to arrhythmias, syncope may also occur with a variety of structural cardiovascular disorders. Episodes are usually precipitated when the cardiac output cannot increase to compensate adequately for peripheral vasodilatation. Peripheral vasodilatation may be appropriate, such as following exercise, or may occur due to inappropriate activation of left ventricular mechanoreceptor reflexes, as occurs in aortic outflow tract obstruction (aortic valvular stenosis or hypertrophic obstructive cardiomyopathy). Obstruction to forward flow is the most common reason that cardiac output cannot increase. Pericardial tamponade is a rare cause of syncope. Syncope occurs in up to 10% of patients with massive pulmonary embolism and may occur with exertion in patients with severe primary pulmonary hypertension. The cause is an inability of the right ventricle to provide appropriate cardiac output in the presence of obstruction or increased pulmonary vascular resistance. Loss of consciousness is usually accompanied by other symptoms such as chest pain and dyspnea. Atrial myxoma, a prosthetic valve thrombus, and, rarely, mitral stenosis may impair left ventricular filling, decrease cardiac output, and cause syncope.

Cerebrovascular Disease Cerebrovascular disease alone rarely causes syncope but may lower the threshold for syncope in patients with other causes. The vertebrobasilar arteries, which supply brainstem centers responsible for maintaining consciousness, are usually involved when cerebrovascular disease causes or contributes to syncope. An exception is the rare patient with tight bilateral carotid stenosis and recurrent syncope, often precipitated by standing or walking. Most patients who experience lightheadedness or syncope due to cerebrovascular disease also have symptoms of focal neurologic ischemia, such as arm or leg weakness, diplopia, ataxia, dysarthria, or sensory disturbances. Basilar artery migraine is a rare disorder that causes syncope in adolescents.

DIFFERENTIAL DIAGNOSIS

Anxiety Attacks and the Hyperventilation Syndrome Anxiety, such as occurs in panic attacks, is frequently interpreted as a feeling of faintness or dizziness resembling presyncope. The symptoms are not accompanied by facial pallor and are not relieved by recumbency. The diagnosis is made on the basis of the associated symptoms such as a feeling of impending doom, air hunger, palpitations, and tingling of the fingers and perioral region. Attacks can often be reproduced by hyperventilation, resulting in hypocapnia, alkalosis, increased cerebrovascular resistance, and decreased cerebral blood flow. The release of epinephrine in anxiety states also contributes to the symptoms.

Seizures A seizure may be heralded by an aura, which is caused by a focal seizure discharge and hence has localizing significance. The aura is usually followed by a rapid return to normal or by a loss of consciousness. Injury from falling is frequent in a seizure

and rare in syncope, because only in seizures are protective reflexes abolished instantaneously. Tonic-convulsive movements are characteristic of seizures and usually do not occur with syncope, although, as stated above, brief tonic-clonic seizure-like activity can accompany fainting episodes. The period of unconsciousness tends to be longer in seizures than in syncope. Urinary incontinence is frequent in seizures and rare in syncope. The return of consciousness is prompt in syncope, slow after a seizure. Mental confusion, headache, and drowsiness are common sequelae of seizures; physical weakness with a clear sensorium characterizes the postsyncopal state. Repeated spells of unconsciousness in a young person at a rate of several per day or month are more suggestive of epilepsy than syncope.

Hypoglycemia Severe hypoglycemia is usually due to a serious disease such as a tumor of the islets of Langerhans; advanced adrenal, pituitary, or hepatic disease; or to excessive administration of insulin.

Acute Hemorrhage Hemorrhage, usually within the gastrointestinal tract, is an occasional cause of syncope. In the absence of pain and hematemesis, the cause of the weakness, faintness, or even unconsciousness may remain obscure until the passage of a black stool.

Hysterical Fainting The attack is usually unattended by an outward display of anxiety. Lack of change in pulse and blood pressure or color of the skin and mucous membranes distinguish it from the vasodepressor faint.

Approach to the Patient

The diagnosis of syncope is often challenging. The cause may only be apparent at the time of the event, leaving few, if any, clues when the patient is seen later by the physician. In dealing with patients who have fainted, the physician should think first of those causes of fainting that constitute a therapeutic emergency. Among them are massive internal hemorrhage or myocardial infarction, which may be painless, and cardiac arrhythmias. In elderly persons, a sudden faint, without obvious cause, should arouse the suspicion of complete heart block or a tachyarrhythmia, even though all findings are negative when the patient is seen.

An algorithmic approach to syncope is presented in [Fig. 21-1](#). A careful history is the most important diagnostic tool, both to suggest the correct cause and to exclude other important potential causes ([Table 21-1](#)). Although no single element of the history is specific for a particular etiology of syncope, the nature of the events and their time course immediately prior to, during, and after an episode often provide valuable etiologic clues. Loss of consciousness in particular situations, such as during venipuncture, micturition, or in association with volume depletion, suggests an abnormality of vascular tone. The position of the patient at the time of the syncopal episode is very important; syncope in the supine position is unlikely to be vasovagal and suggests an arrhythmia or a seizure. Syncope due to carotid sinus syndrome may occur when the individual is wearing a shirt with a tight collar, turning the head (turning to look while driving in reverse), or manipulating the neck (as in shaving). The patient's medications must be noted, including nonprescription drugs or health store supplements, with particular attention to recent changes.

The physical examination should include evaluation of heart rate and blood pressure in the supine, sitting, and standing positions. In patients with unexplained recurrent syncope, an attempt to reproduce an attack may assist in diagnosis. Anxiety attacks induced by hyperventilation can be reproduced readily by having the patient breathe rapidly and deeply for 2 to 3 min. Cough syncope may be reproduced by inducing the Valsalva maneuver. Carotid sinus massage should generally be avoided, even in patients with suspected carotid sinus hypersensitivity; it is a risky procedure that can cause a transient ischemic attack (TIA) or stroke in susceptible individuals.

Diagnostic Tests The choice of diagnostic tests should be guided by the history and the physical examination. Measurements of serum electrolytes, glucose, and the hematocrit may help to establish the cause of syncope. Cardiac enzymes should be evaluated if myocardial ischemia is suspected. Blood and urine toxicology screens may reveal the presence of alcohol or other drugs. In patients with possible adrenocortical insufficiency, plasma aldosterone and mineralocorticoid levels should be obtained.

Although the surface electrocardiogram is unlikely to provide a definitive diagnosis, it may provide clues to the cause of syncope *and should be performed in almost all patients*. The presence of conduction abnormalities (PR prolongation and bundle branch block) suggests a bradyarrhythmia, whereas pathologic Q waves or prolongation of the QT interval suggests a ventricular tachyarrhythmia. Inpatients should undergo continuous electrocardiographic monitoring; outpatients should wear a Holter monitor for 24 to 48 h. Whenever possible, symptoms should be correlated with the occurrence of arrhythmias. Continuous electrocardiographic monitoring may establish the cause of syncope in as many as 15% of patients. Cardiac event monitors may be useful in patients with infrequent symptoms, particularly in patients with presyncope. The presence of a late potential on a signal-averaged electrocardiogram is associated with increased risk for ventricular tachyarrhythmias in patients with a prior myocardial infarction. Low-voltage (visually inapparent) T wave alternans is also associated with development of sustained ventricular arrhythmias.

Invasive cardiac electrophysiologic testing provides diagnostic and prognostic information regarding sinus node function, [AV](#) conduction, and supraventricular and ventricular arrhythmias. Abnormal findings include prolongation of the sinus node recovery time, prolongation of the histioventricle (HV) interval, induction of a supraventricular arrhythmia associated with hypotension, or induction of a supraventricular arrhythmia. Prolongation of the sinus node recovery time (>1500 ms) is a specific finding (85 to 100%) for diagnosis of sinus node dysfunction but has a low sensitivity; continuous electrocardiographic monitoring is usually more effective for diagnosing this abnormality. Prolongation of the HV interval and conduction block below the His bundle indicate that His-Purkinje disease may be responsible for syncope. Although an HV interval >100 ms is abnormal, this finding is not common in patients with syncope, and some patients with shorter intervals are also at risk for AV block. Programmed stimulation for ventricular arrhythmias is most useful in patients who have experienced a myocardial infarction; the sensitivity and specificity of this technique is lower in patients with normal hearts or those with heart disease other than coronary artery disease.

Upright tilt table testing is indicated for recurrent syncope, a single syncopal episode that caused injury, or a single syncopal event in a "high-risk" setting (pilot, commercial vehicle driver, etc.), whether or not there is a history of preexisting heart disease or prior vasovagal episodes. In susceptible patients, upright tilt at an angle between 60 and 80° for 30 to 60 min induces a vasovagal episode. The protocol can be shortened if upright tilt is combined with intravenous administration of drugs that cause venous pooling or increase adrenergic stimulation (isoproterenol, nitroglycerin, edrophonium, or adenosine). The sensitivity and specificity of tilt table testing is difficult to ascertain because of the lack of validated criteria. Moreover, the reflexes responsible for vasovagal syncope can be elicited in most, if not all, individuals given the appropriate stimulus. The reported accuracy of the test ranges from 30 to 80%, depending on the population studied and the techniques used. Whereas the reproducibility of a negative test is 85 to 100%, the reproducibility of a positive tilt table test is only between 62 and 88%.

A variety of other tests may be useful to determine the presence of structural heart disease that may cause syncope. The echocardiogram with Doppler examination detects valvular, myocardial, and pericardial abnormalities. The echocardiogram is the "gold standard" for the diagnosis of hypertrophic cardiomyopathy and atrial myxoma. Cardiac cine magnetic resonance (MR) imaging provides an alternative noninvasive modality that may be useful for patients in whom diagnostic-quality echocardiographic images cannot be obtained. This test is also indicated for patients suspected of having arrhythmogenic right ventricular dysplasia or right ventricular outflow tract ventricular tachycardia. Both are associated with right ventricular structural abnormalities that are better visualized on MR imaging than by echocardiogram. Exercise testing may detect ischemia or exercise-induced arrhythmias. In some patients, cardiac catheterization may be necessary to diagnose the presence or severity of coronary artery disease or valvular abnormalities. Ultrafast computed tomographic scan, ventilation-perfusion scan, or pulmonary angiography are indicated in patients in whom syncope may be due to pulmonary embolus.

In possible cases of cerebrovascular syncope, a variety of neuroimaging tests may be indicated, including Doppler ultrasound studies of the carotid and vertebral basilar systems, MR imaging, MR angiography, and x-ray angiography of the cerebral vasculature ([Chaps. 358 and 361](#)). Electroencephalography is indicated if seizures are suspected.

TREATMENT

The treatment of syncope is directed toward the underlying cause. This discussion will focus on the treatment of disorders of autonomic control. **Arrhythmias are discussed in [Chaps. 229 and 230](#), valvular heart diseases in [Chap. 236](#), and cerebrovascular disorders in [Chap. 361](#).*

Certain precautions should be taken regardless of the cause of syncope. At the first sign of symptoms, patients should make every effort to avoid injury should they lose consciousness. Patients with frequent episodes, or those who have experienced syncope without warning symptoms should avoid situations in which sudden loss of consciousness might result in injury (e.g., climbing ladders, swimming alone, operating

heavy machinery, driving). Patients should lower their head to the extent possible, and preferably should lie down. Lowering the head by bending at the waist should be avoided because it may further compromise venous return to the heart. When appropriate, family members or other close contacts should be educated as to the problem. This will ensure appropriate therapy and may prevent delivery of inappropriate therapy (chest compressions associated with cardiopulmonary resuscitation) that may inflict trauma.

Patients who have lost consciousness should be placed in a position that maximizes cerebral blood flow, offers protection from trauma, and secures the airway. Whenever possible, the patient should be placed supine with the head turned to the side to prevent aspiration and the tongue from blocking the airway. Assessment of the pulse and direct cardiac auscultation may assist in determining if the episode is associated with a bradyarrhythmia or tachyarrhythmia. Clothing that fits tightly around the neck or waist should be loosened. Peripheral stimulation, such as by sprinkling cold water on the face, may be helpful. Patients should not be given anything by mouth or be permitted to rise until the sense of physical weakness has passed.

Patients with vasovagal syncope should be instructed to avoid situations or stimuli that have caused them to lose consciousness. Episodes associated with intravascular volume depletion may be prevented by salt and fluid loading prior to provocative events. β -Adrenoceptor antagonists, the most widely used agents, mitigate the increase in myocardial contractility that stimulates left ventricular mechanoreceptors and also block central serotonin receptors. Disopyramide, a vagolytic with negative inotropic properties, and another vagolytic, transdermal scopolamine, are used to treat vasovagal syncope. Paroxetine, a serotonin reuptake inhibitor used for depression, appears to be an effective treatment, as are theophylline and ephedrine. Midodrine, an α -agonist, has been a first-line agent for some patients. Permanent cardiac pacing is effective for patients with frequent episodes of vasovagal syncope and is indicated for those with prolonged asystole associated with vasovagal episodes.

Patients with orthostatic hypotension should be instructed to rise slowly and systematically (supine to seated, seated to standing) from the bed or a chair. Movement of the legs prior to rising facilitates venous return from the lower extremities. Whenever possible, medications that aggravate the problem (vasodilators, diuretics, etc.) should be discontinued. Elevation of the head of the bed [20 to 30 cm (8 to 12 in.)] and use of elastic stockings may help.

Therapeutic modalities include devices that prevent lower limb blood pooling, such as an antigravity or g suit or elastic stockings; salt loading; and a variety of pharmacologic agents including sympathomimetic amines, monamine oxidase inhibitors, beta blockers, and levodopa. **The treatment of orthostatic hypotension secondary to central or peripheral disorders of the autonomic nervous system is discussed in Chap. 366.*

Glossopharyngeal neuralgia is treated with carbamazepine, which is effective for the syncope as well as for the pain. Patients with carotid sinus syndrome should be instructed to avoid clothing and situations that stimulate carotid sinus baroreceptors. Patients should turn their entire body, rather than just their head, to look to one side. Those with intractable syncope due to the cardioinhibitory response to carotid sinus

stimulation should undergo permanent pacemaker implantation.

Patients with syncope should be hospitalized when the episode may have resulted from a life-threatening abnormality or if recurrence with significant injury seems likely. These individuals should be admitted to a bed with continuous electrocardiographic monitoring. Patients who are known to have a normal heart and for whom the history strongly suggests vasovagal or situational syncope may be treated as outpatients if the episodes are neither frequent nor severe.

DIZZINESS AND VERTIGO

Dizziness is a common and often vexing symptom. Patients use the term to encompass a variety of sensations, including those that seem semantically appropriate (e.g., lightheadedness, faintness, spinning, giddiness, etc.) and those that are misleadingly inappropriate, such as mental confusion, blurred vision, headache, or tingling. Moreover, some individuals with gait disorders complain of dizziness despite the absence of vertigo or other abnormal cephalic sensations. The causes include peripheral neuropathy, myelopathy, spasticity, parkinsonian rigidity, and cerebellar ataxia. In this context, the term *dizziness* is being used to describe disturbed mobility. There may be mild associated lightheadedness, particularly with impaired sensation from the feet or poor vision; this is known as *multiple-sensory-defect dizziness* and occurs in elderly individuals who complain of dizziness only during ambulation. Decreased position sense (secondary to neuropathy or myelopathy) and poor vision (from cataracts or retinal degeneration) create an overreliance on the aging vestibular apparatus. A less precise but sometimes comforting designation to patients is *benign dysequilibrium of aging*. Thus, a careful history is necessary to determine exactly what a patient who states, "Doctor, I'm dizzy," is experiencing. After eliminating the misleading symptoms or gait disturbance, "dizziness" usually means either *faintness* (presyncope) or *vertigo* (an illusory or hallucinatory sense of movement of the body or environment, most often a feeling of spinning). Operationally, dizziness is classified into three categories: (1) faintness, (2) vertigo, and (3) miscellaneous head sensations.

FAINTNESS

Prior to an actual faint (syncope), there are often prodromal presyncopal symptoms (faintness) reflecting ischemia to a degree insufficient to impair consciousness (see above).

VERTIGO

Vertigo is usually due to a disturbance in the vestibular system. The end organs of this system, situated in the bony labyrinths of the inner ears, consist of the three semicircular canals and the otolithic apparatus (utricle and saccule) on each side. The canals transduce angular acceleration, while the otoliths transduce linear acceleration and static gravitational forces, the latter providing a sense of head position in space. The neural output of the end organs is conveyed to the vestibular nuclei in the brainstem via the eighth cranial nerve. The principal projections from the vestibular nuclei are to the nuclei of cranial nerves III, IV, and VI, the spinal cord, the cerebral cortex, and the cerebellum. The vestibuloocular reflex (VOR) serves to maintain visual

stability during head movement and depends on direct projections from the vestibular nuclei to the sixth cranial nerve (abducens) nuclei in the pons and, via the medial longitudinal fasciculus, to the third (oculomotor) and fourth (trochlear) cranial nerve nuclei in the midbrain. These connections account for the nystagmus (to-and-fro oscillation of the eyes) that is an almost invariable accompaniment of vestibular dysfunction. The vestibular nerves and nuclei project to areas of the cerebellum (primarily the flocculus and nodulus) that modulate the VOR. The vestibulospinal pathways assist in the maintenance of postural stability. Projections to the cerebral cortex, via the thalamus, provide conscious awareness of head position and movement.

The vestibular system is one of three sensory systems subserving spatial orientation and posture; the other two are the visual system (retina to occipital cortex) and the somatosensory system that conveys peripheral information from skin, joint, and muscle receptors. The three stabilizing systems overlap sufficiently to compensate (partially or completely) for each other's deficiencies. Vertigo may represent either physiologic stimulation or pathologic dysfunction in any of the three systems.

Physiologic Vertigo This occurs when (1) the brain is confronted with a mismatch among the three stabilizing sensory systems; (2) the vestibular system is subjected to unfamiliar head movements to which it has never adapted, such as in seasickness; or (3) unusual head/neck positions, such as the extreme extension when painting a ceiling. Intersensory mismatch explains carsickness, height vertigo, and the visual vertigo most commonly experienced during motion picture chase scenes; in the latter, the visual sensation of environmental movement is unaccompanied by concomitant vestibular and somatosensory movement cues. *Space sickness*, a frequent transient effect of active head movement in the weightless zero-gravity environment, is another example of physiologic vertigo.

Pathologic Vertigo This results from lesions of the visual, somatosensory, or vestibular systems. Visual vertigo is caused by new or incorrect spectacles or by the sudden onset of an extraocular muscle paresis with diplopia; in either instance, [CNS](#) compensation rapidly counteracts the vertigo. Somatosensory vertigo, rare in isolation, is usually due to a peripheral neuropathy that reduces the sensory input necessary for central compensation when there is dysfunction of the vestibular or visual systems.

The most common cause of pathologic vertigo is vestibular dysfunction. The vertigo is frequently accompanied by nausea, jerk nystagmus, postural unsteadiness, and gait ataxia. Since vertigo increases with rapid head movements, patients tend to hold their heads still.

Labyrinthine Dysfunction This causes severe rotational or linear vertigo. When rotational, the hallucination of movement, whether of environment or self, is directed away from the side of the lesion. The fast phases of nystagmus beat away from the lesion side, and the tendency to fall is toward the side of the lesion.

When the head is straight and immobile, the vestibular end organs generate a tonic resting firing frequency that is equal from the two sides. With any rotational acceleration, the anatomic positions of the semicircular canals on each side necessitate an increased firing rate from one and a commensurate decrease from the other. This change in

neural activity is ultimately projected to the cerebral cortex, where it is summed with inputs from the visual and somatosensory systems to produce the appropriate conscious sense of rotational movement. After cessation of movement, the firing frequencies of the two end organs reverse; the side with the initially increased rate decreases, and the other side increases. A sense of rotation in the opposite direction is experienced; since there is no actual head movement, this hallucinatory sensation is *physiologic postrotational vertigo*.

Any disease state that changes the firing frequency of an end organ, producing unequal neural input to the brainstem and ultimately the cerebral cortex, causes vertigo. The symptom can be conceptualized as the cortex inappropriately interpreting the abnormal neural input from the brainstem as indicating actual head rotation. Transient abnormalities produce short-lived symptoms. With a fixed unilateral deficit, central compensatory mechanisms ultimately diminish the vertigo. Since compensation depends on the plasticity of connections between the vestibular nuclei and the cerebellum, patients with brainstem or cerebellar disease have diminished adaptive capacity, and symptoms may persist indefinitely. Compensation is always inadequate for severe fixed bilateral lesions despite normal cerebellar connections: these patients are permanently symptomatic.

Acute unilateral labyrinthine dysfunction is caused by infection, trauma, and ischemia. Often, no specific etiology is uncovered, and the nonspecific terms *acute labyrinthitis*, *acute peripheral vestibulopathy*, or *vestibular neuritis* are used to describe the event. The attacks are brief and leave the patient for some days with a mild positional vertigo. Infection with herpes simplex virus type 1 has been implicated. It is impossible to predict whether a patient recovering from the first bout of vertigo will have recurrent episodes.

Acute bilateral labyrinthine dysfunction is usually the result of toxins such as drugs or alcohol. The most common offending drugs are the aminoglycoside antibiotics which damage the fine hair cells of the vestibular end organs and may cause a permanent disorder of equilibrium.

Recurrent unilateral labyrinthine dysfunction, in association with signs and symptoms of cochlear disease (progressive hearing loss and tinnitus), is usually due to Meniere's disease ([Chap. 29](#)). When auditory manifestations are absent, the term *vestibular neuronitis* denotes recurrent monosymptomatic vertigo. [TIAs](#) of the posterior cerebral circulation (vertebrobasilar insufficiency) very infrequently cause recurrent vertigo without concomitant motor, sensory, visual, cranial nerve, or cerebellar signs.

Positional vertigo is precipitated by a recumbent head position, either to the right or to the left. Benign paroxysmal positional (or positioning) vertigo (BPPV) of the posterior semicircular canal is particularly common. Although the condition may be due to head trauma, usually no precipitating factors are identified. It generally abates spontaneously after weeks or months. The vertigo and accompanying nystagmus have a distinct pattern of latency, fatigability, and habituation that differs from the less common central positional vertigo ([Table 21-2](#)) due to lesions in and around the fourth ventricle. Moreover, the pattern of nystagmus in posterior canal BPPV is distinctive. The lower eye displays a large-amplitude torsional nystagmus, and the upper eye has a lesser degree of torsion combined with upbeating nystagmus. If the eyes are directed to the

upper ear, the vertical nystagmus in the upper eye increases in amplitude.

Vertigo of vestibular nerve origin may occur with diseases that involve the nerve in the petrous bone or the cerebellopontine angle. Except that it is less severe and less frequently paroxysmal, it has many of the characteristics of labyrinthine vertigo. The adjacent auditory division of the eighth cranial nerve also may be affected, which explains the frequent association of vertigo with tinnitus and deafness. The function of the eighth cranial nerve may be disturbed by tumors of the lateral recess (especially schwannomas), less frequently by meningeal inflammation in this region and, rarely, by an abnormal vessel that compresses the nerve.

Schwannomas involving the eighth cranial nerve (*acoustic neuroma*) grow slowly and produce such a gradual reduction of labyrinthine output that central compensatory mechanisms can prevent or minimize the vertigo; auditory symptoms of hearing loss and tinnitus are the most common manifestations. While lesions of the brainstem or cerebellum can cause acute vertigo, associated signs and symptoms usually permit distinction from a labyrinthine etiology ([Table 21-3](#)). However, labyrinthine ischemia, presumably due to occlusion of the labyrinthine branch of the internal auditory artery, may be the sole manifestation of vertebrobasilar insufficiency; patients with this syndrome present with the abrupt onset of severe vertigo, nausea and vomiting without tinnitus or hearing loss. Occasionally, an acute lesion of the vestibulocerebellum may present with monosymptomatic vertigo indistinguishable from a labyrinthopathy.

Vestibular epilepsy, vertigo secondary to temporal lobe epileptic activity, is rare and almost always intermixed with other epileptic manifestations.

Psychogenic vertigo, usually a concomitant of panic attacks or agoraphobia (fear of large open spaces, crowds, or leaving the safety of home), should be suspected in patients so "incapacitated" by their symptoms that they adopt a prolonged housebound status. Most patients with organic vertigo attempt to function despite their discomfort. Organic vertigo is accompanied by nystagmus; a psychogenic etiology is almost certain when nystagmus is absent during a vertiginous episode.

Miscellaneous Head Sensations This designation is used, primarily for purposes of initial classification, to describe dizziness that is neither faintness nor vertigo. Cephalic ischemia or vestibular dysfunction may be of such low intensity that the usual symptomatology is not clearly identified. For example, a small decrease in blood pressure or a slight vestibular imbalance may cause sensations different from distinct faintness or vertigo but that may be identified properly during provocative testing techniques. Other causes of dizziness in this category are hyperventilation syndrome, hypoglycemia, and the somatic symptoms of a clinical depression; these patients should have normal neurologic examinations and vestibular function tests.

Approach to the Patient

The most important diagnostic tool is a careful history focused on the meaning of "dizziness" to the patient. Is it faintness? Is there a sensation of spinning? If either of these is affirmed and the neurologic examination is normal, appropriate investigations for the multiple etiologies of cephalic ischemia or vestibular dysfunction are undertaken.

When the meaning of "dizziness" is uncertain, provocative tests may be helpful. These office procedures simulate either cephalic ischemia or vestibular dysfunction. Cephalic ischemia is obvious if the dizziness is duplicated during maneuvers that produce orthostatic hypotension. Further provocation involves the Valsalva maneuver, which decreases cerebral blood flow and should reproduce ischemic symptoms.

The simplest provocative test for vestibular dysfunction is rapid rotation and abrupt cessation of movement in a swivel chair. This always induces vertigo that the patients can compare with their symptomatic dizziness. The intense induced vertigo may be unlike the spontaneous symptoms, but shortly thereafter, when the vertigo has all but subsided, a lightheadedness supervenes that may be identified as "my dizziness." When this occurs, the dizzy patient, originally classified as suffering from "miscellaneous head sensations," is now properly diagnosed as having mild vertigo secondary to a vestibulopathy.

Patients with symptoms of positional vertigo should be appropriately tested ([Table 21-2](#)); positional testing is more sensitive with special spectacles that preclude visual fixation (Frenzel lenses).

A final provocative test, requiring the use of Frenzel lenses, is vigorous head shaking in the horizontal plane for about 10 s. If nystagmus develops after the shaking stops, even in the absence of vertigo, vestibular dysfunction is demonstrated. The maneuver can then be repeated in the vertical plane. If the provocative tests establish the dizziness as a vestibular symptom, an evaluation of vestibular vertigo is undertaken.

Evaluation of Patients with Pathologic Vestibular Vertigo The evaluation depends on whether a central etiology is suspected ([Table 21-3](#)). If so, [MR](#) imaging of the head is mandatory. Such an examination is rarely helpful in cases of recurrent monosymptomatic vertigo with a normal neurologic examination. Typical [BPPV](#) requires no investigation after the diagnosis is made ([Table 21-2](#)).

Vestibular function tests serve to (1) demonstrate an abnormality when the distinction between organic and psychogenic is uncertain, (2) establish the side of the abnormality, and (3) distinguish between peripheral and central etiologies. The standard test is electronystagmography (calorics), where warm and cold water (or air) are applied, in a prescribed fashion, to the tympanic membranes, and the slow-phase velocities of the resultant nystagmus from the right and left ears are compared. A velocity decrease from one side indicates hypofunction ("canal paresis"). An inability to induce nystagmus with ice water denotes a "dead labyrinth." Some institutions have the capability of quantitatively determining various aspects of the vestibuloocular reflex using computer-driven rotational chairs and precise oculographic recording of the eye movements.

Hyperventilation is the cause of dizziness in many anxious individuals; tingling of the hands and face may be absent. Forced hyperventilation for 1 min is indicated for patients with enigmatic dizziness and normal neurologic examinations. Similarly, depressive symptoms (which patients usually insist are "secondary" to the dizziness) must alert the examiner to a clinical depression as the *cause*, rather than the effect, of

the dizziness.

[CNS](#) disease can produce dizzy sensations of all types. Consequently, a neurologic examination is always required even if the history or provocative tests suggest a cardiac, peripheral vestibular, or psychogenic etiology. Any abnormality on the neurologic examination should prompt appropriate neurodiagnostic studies.

TREATMENT

Treatment of acute vertigo consists of bed rest and vestibular suppressant drugs such as antihistaminics (meclizine, dimenhydrinate, promethazine), or a tranquilizer with GABA-ergic effects (diazepam). If the vertigo persists beyond a few days, most authorities advise ambulation in an attempt to induce central compensatory mechanisms, despite the short-term discomfort to the patient. Chronic vertigo of labyrinthine origin may be treated with a systematized vestibular rehabilitation program to facilitate central compensation (see also [Table 21-4](#)).

[BPPV](#) is often self-limited but, when persistent, responds dramatically to specific repositioning exercise programs designed to empty particulate debris from the posterior semicircular canal. One of these exercises, the Epley procedure, is graphically demonstrated, in four languages, on a website for use in both physician's offices and self-treatment (<http://www.charite.de/ch/neuro/vertigo.html>).

Prophylactic measures to prevent recurrent vertigo are variably effective. Antihistamines are commonly utilized. Meniere's disease may respond to a diuretic or, more effectively, to a very low salt diet (1 g/day).

There are a variety of inner ear surgical procedures for refractory Meniere's disease, but these are only rarely necessary.

(Bibliography omitted in Palm version)

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22. WEAKNESS, MYALGIAS, DISORDERS OF MOVEMENT, AND IMBALANCE - *Richard K. Olney, Michael J. Aminoff*

Normal motor function requires integrated muscle activity with appropriate modulation by neuronal activity in the cerebral cortex, basal ganglia, cerebellum, and spinal cord. Symptoms and signs of motor system dysfunction may include weakness, fatigue, myalgias, spasms, cramps, dyskinetic movement, ataxia, imbalance, or disorders in the initiation or planning of movement.

WEAKNESS

Weakness is a reduction in normal power of one or more muscles. Patients may use the term differently; thus one or more specific examples of weakness should be elicited during the history. Increased fatigability or limitation in function due to pain is often confused with weakness by patients. *Increased fatigability* is the inability to sustain the performance of an activity that should be normal for a person of the same age, gender, and size.

Weakness is described commonly by severity and distribution. Paralysis and the suffix "-plegia" indicate weakness that is so severe that it is complete or nearly complete. "Paresis" refers to weakness that is mild or moderate. The prefix "hemi-" refers to one half of the body, "para-" to both legs, and "quadri-" to all four limbs.

Tone is the resistance of a muscle to passive stretch. Central nervous system (CNS) abnormalities that cause weakness generally produce *spasticity*, an increase in tone due to upper motor neuron disease. Spasticity is velocity-dependent, has a sudden release after reaching a maximum (the "clasp-knife" phenomenon), and predominantly affects antigravity muscles (i.e., upper limb flexors and lower limb extensors). Spasticity is distinct from rigidity and paratonia, two other types of increased tone. *Rigidity* is increased tone that is present throughout the range of motion (a "lead pipe" or "plastic" stiffness) and affects flexors and extensors equally. In some patients, rigidity has a cogwheel quality that is enhanced by voluntary movement of the contralateral limb (reinforcement). Rigidity occurs with certain extrapyramidal disorders. *Paratonia*, also referred to as *gegenhalten*, is increased tone that varies irregularly in a manner that may seem related to the degree of relaxation, is present throughout the range of motion, and affects flexors and extensors equally. Paratonia usually results from disease of the frontal lobes. Weakness with decreased tone (flaccidity) or normal tone occurs with disorders of the *motor unit*, that is, a single lower motor neuron and all of the muscle fibers it innervates.

Three basic patterns of weakness can usually be recognized based on the signs summarized in [Table 22-1](#). One results from upper motor neuron pathology, and the other two from disorders of the motor unit (lower motor neuron and myopathic weakness). Fasciculations and early atrophy help to distinguish lower motor neuron (neurogenic) weakness from myopathic weakness. A *fasciculation* is a visible or palpable twitch within a single muscle due to the spontaneous discharge of one motor unit. Neurogenic weakness also produces more prominent hypotonia and greater depression of tendon reflexes than myopathic weakness.

PATHOGENESIS

Upper Motor Neuron Weakness This pattern of weakness results from disorders that affect the upper motor neurons or their axons in the cerebral cortex, subcortical white matter, internal capsule, brainstem, or spinal cord ([Fig. 22-1](#)). Both the pyramidal and bulbospinal pathways contribute to normal strength, tone, coordination, and gait. Upper motor neuron lesions produce weakness through decreased activation of the lower motor neurons. In general, distal muscle groups are affected more severely than proximal ones, and axial movements are spared unless the lesion is severe and bilateral. With corticobulbar involvement, weakness is usually observed only in the lower face and tongue; extraocular, upper facial, pharyngeal, and jaw muscles are almost always spared. With bilateral corticobulbar lesions, *pseudobulbar palsy* often develops, in which dysarthria, dysphagia, dysphonia, and emotional lability accompany bilateral facial weakness. Spasticity accompanies upper motor neuron weakness but may not be present in the acute phase.

Upper motor neuron lesions also affect the ability to perform rapid repetitive movements. Such movements are slow and coarse, but normal rhythmicity is maintained. Finger-nose-finger and heel-knee-shin are performed slowly but adequately.

Lower Motor Neuron Weakness This pattern results from disorders of cell bodies of lower motor neurons in the brainstem motor nuclei and the anterior horn of the spinal cord, or from dysfunction of the axons of these neurons as they pass to skeletal muscle ([Fig. 22-2](#)).

Lower motor weakness is produced by a decrease in the number of motor units that can be activated, through a loss of the α motor neurons or disruption of their connections to muscle. With a decreased number of motor units, fewer muscle fibers are activated with full effort and maximum power is reduced. Loss of γ motor neurons does not cause weakness but decreases tension on the muscle spindles. Muscle tone and tendon reflexes depend on γ motor neurons, muscle spindles, spindle afferent fibers, and the α motor neurons. A tap on a tendon stretches muscle spindles and activates the primary spindle afferent fibers. These monosynaptically stimulate the α motor neurons in the spinal cord, producing a brief muscle contraction, which is the familiar tendon reflex.

When a motor unit becomes diseased, especially in anterior horn cell diseases, it may spontaneously discharge, producing a fasciculation. These isolated small twitches may be seen or felt clinically or recorded by electromyography (EMG) ([Chap. 357](#)). When α motor neurons or their axons degenerate, the denervated muscle fibers spontaneously discharge in a manner that cannot be seen or felt but can be recorded with EMG. These small single muscle fiber discharges are called *fibrillation potentials*. If significant lower motor neuron weakness is present, recruitment of motor units is delayed or reduced, with fewer than normal activated at a given discharge frequency. This contrasts with upper motor neuron weakness, in which a normal number of motor units are activated at a given frequency but in which the maximum discharge frequency is decreased.

Myopathic Weakness This pattern of weakness is produced by disorders within the motor unit that affect the muscle fibers or the neuromuscular junctions.

Two types of muscle fibers exist. Type I muscle fibers are rich in mitochondria and oxidative enzymes, produce relatively low force, but have low energy demands that can be supplied by ongoing aerobic metabolism. They produce sustained postural and nonforceful movements. Type II muscle fibers are rich in glycolytic enzymes, can produce relatively high force, but have high energy demands that cannot be supplied for long by ongoing aerobic metabolism. Thus, these units can be activated maximally for only brief periods of time to produce high-force movements.

For graded voluntary movements, type I muscle fibers are activated earlier in recruitment. For each muscle fiber, if the nerve terminal releases a normal number of acetylcholine molecules presynaptically and a sufficient number of postsynaptic acetylcholine receptors are opened, the end plate reaches threshold and thereby generates an action potential that spreads across the muscle fiber membrane and into the transverse tubular system. This electrical excitation activates intracellular events that produce an energy-dependent contraction of the muscle fiber (excitation-contraction coupling).

Myopathic weakness is produced by a decrease in the number or contractile force of muscle fibers activated within the motor unit. With muscular dystrophies, inflammatory myopathies, or myopathies with muscle fiber necrosis, decreased numbers of muscle fibers survive within many motor units. As demonstrated with [EMG](#), the size of each motor unit action potential is decreased so that motor units must be recruited more rapidly than normal to produce the power necessary for a certain movement. Neuromuscular junction diseases, such as myasthenia gravis, produce weakness in a similar manner, although the loss of muscle fibers within the motor unit is functional rather than actual. Furthermore, the number of muscle fibers activated can vary over time, depending on the state of rest of the neuromuscular junctions. Thus, fatigable weakness is suggestive of myasthenia gravis or another neuromuscular junction disease. Some myopathies produce weakness through loss of contractile force of muscle fibers or through relatively selective involvement of the type II muscle fibers. These may not affect the size of individual motor unit action potentials observed with EMG and are detected by a discrepancy between the electrical activity and force of a muscle.

Integrated Movements Most purposeful movements require the integrated coordination of many muscle groups. Consider a simple movement, such as grasping a ball. The primary movement is a flexion of the thumb and fingers of one hand, with opposition of the thumb and little finger. This requires the contraction of several muscles, including flexor digitorum superficialis, flexor digitorum profundus, flexor pollicis longus, flexor pollicis brevis, opponens pollicis, and opponens digiti minimi. These prime movers for this action are called *agonists*. In order for the grasping to be smooth and forceful, the thumb and finger extensors need to relax at the same rate as the flexors contract. The muscles that act in a directly opposing manner to the agonists are *antagonists*. A secondary action of the thumb and finger flexors is to flex the wrist; because wrist flexion tends to weaken finger flexion if both occur, activation of wrist extensors assists the grasping movement. Muscles that produce such complementary movements are *synergists*. Finally, the arm needs to be held in a stable position as the grasp occurs, so that the ball is not knocked away before it is secured. Muscles that stabilize the arm

position are *fixators*.

The coordination of activity by agonists, antagonists, synergists, and fixators is regulated by a three-level hierarchy of motor control. The lowest level of control is mediated through segmental reflexes in the spinal cord. These reflexes facilitate agonists and reciprocally inhibit the antagonists. Spinal segments also control rhythmic patterns of movement that involve more than a single pair of agonists and antagonists. For example, the lumbosacral spinal cord contains the basic programming for cyclical stepping movements that involve the synergistic activation of different muscle groups over time. The intermediate level of control is mediated through the descending bulbospinal pathways, which integrate visual, proprioceptive, and vestibular feedback into the execution of an action. For example, the locomotor center in the midbrain is required to modify the cyclical stepping movements in order that balance be maintained and forward movement occur. The highest level of control is mediated by the cerebral cortex. Superimposition of this highest level of control is necessary for activities such as walking to be goal-directed. Precise movements that are learned and improved through practice are also initiated and controlled by the motor cortex. Although only the agonists are directly activated, during the course of a complex sequence of actions such as playing the piano, the sequential activation of different groups of agonists for each note or chord is a part of the learned motor program. Further, the execution of these actions also involves input from the basal ganglia and cerebellar hemispheres to facilitate agonists, synergists, and fixators and to inhibit undesired antagonists.

Apraxia is a disorder of planning and initiating a skilled or learned movement ([Chap. 25](#)). Unilateral apraxia of the right hand may be due to a lesion of the left frontal lobe (especially anterior or inferior), the left temporoparietal region (especially the supramarginal gyrus), or their connections. Left body apraxia is produced by lesions of these regions in the right hemisphere or by lesions in the corpus callosum that disconnect the right temporoparietal or frontal regions from those on the left. Bilateral apraxia is often due to bilateral frontal lobe lesions or diffuse bilateral hemispheric disease.

Approach to the Patient

The mode of onset, distribution, and associated features of weakness should be carefully defined. When there is a discrepancy between the history and physical findings, it is usually because the patient complains of weakness, whereas symptoms are actually due to other causes, such as incoordination or pain limiting effort. Power may be examined in a variety of ways. The patient is asked to push or pull in a specified direction against resistance, and the strength in each muscle group is graded from 0 to 5 by the scale developed by the Medical Research Council ([Table 22-2](#)). A second method is indirect testing through observation of task performance such as holding the arms outstretched. This is especially useful in detecting mild, asymmetric upper motor neuron weakness through the observation of a downward drift with pronation of the forearm on one side. A third method is functional testing, which involves quantitation of activities. Common tests include counting the number of times a person can perform a deep-knee bend or step on a stool or chair, or timing the length of time the arms can be held abducted to 90 degrees. When performed serially, functional tests provide useful estimates of changes in the patient's status over time.

Other elements of the motor examination include appraisal of muscular bulk, inspection for fasciculations, and assessment of tone. Fasciculations are most easily determined by observing relaxed limbs that are illuminated from behind, but they can also be palpated as irregular low-amplitude twitches within the muscle. Tone is assessed by passive movement of each limb at its various joints and at several different speeds. In the clinical context of weakness, tone may be spastic or flaccid. The presence of cogwheel rigidity, lead-pipe rigidity, or paratonia suggests a disorder of integrated movements, rather than true weakness.

Hemiparesis Hemiparesis results from an upper motor neuron lesion above the midcervical spinal cord; most lesions that produce hemiparesis are located above the foramen magnum. The presence of language disorders, cortical sensory disturbances, cognitive abnormalities, disorders of visual-spatial integration, apraxia, or seizures indicates a cortical lesion. Homonymous visual field defects reflect either a cortical or a subcortical hemispheric lesion. A "pure motor" hemiparesis of the face, arm, and/or leg is due to a small, discrete lesion in the posterior limb of the internal capsule, cerebral peduncle, or upper pons. Some brainstem lesions produce the classic findings of ipsilateral cranial nerve signs and contralateral hemiparesis. These "crossed paralyses" are discussed further in [Chap. 361](#). The absence of cranial nerve signs or facial weakness suggests that a hemiparesis is due to a lesion in the high cervical spinal cord, especially if associated with ipsilateral loss of proprioception and contralateral loss of pain and temperature sense (the Brown-Sequard syndrome). However, most spinal cord lesions produce quadriparesis or paraparesis.

Acute or episodic hemiparesis usually has a vascular pathogenesis, either ischemia or a primary hemorrhage ([Chap. 361](#)). Less commonly, hemorrhage may occur into brain tumors ([Chap. 370](#)) or from rupture of normal vessels due to trauma ([Chap. 369](#)); the trauma may be trivial in patients who are anticoagulated or elderly. Less likely possibilities include a focal inflammatory lesion from multiple sclerosis ([Chap. 371](#)), abscess, or sarcoidosis ([Chap. 318](#)). Evaluation begins immediately with a computed tomography (CT) scan of the brain ([Fig. 22-3](#)). If CT is normal and an ischemic stroke is unlikely, magnetic resonance imaging (MRI) of the brain or cervical spine may be indicated.

Subacute hemiparesis that evolves over days or weeks has a long differential diagnosis. A common cause is subdural hematoma; this readily treatable condition must always be considered, especially in elderly or anticoagulated patients, even in the absence of a history of trauma ([Chap. 369](#)). Infectious possibilities include cerebral bacterial abscess ([Chap. 372](#)), fungal granuloma or meningitis ([Chap. 374](#)), and parasitic infection. Weakness from malignant primary and metastatic neoplasms may evolve over days to weeks ([Chap. 370](#)). AIDS ([Chap. 309](#)) may present with subacute hemiparesis due to toxoplasmosis or primary [CNS](#) lymphoma. Noninfectious inflammatory processes, such as multiple sclerosis ([Chap. 371](#)) or, less commonly, sarcoidosis, are further considerations. If the brain [MRI](#) is normal and if cortical and hemispheric signs are not present, MRI of the cervical spine may be required.

Chronic hemiparesis that evolves over months is usually due to a neoplasm ([Chap. 370](#)), an unruptured arteriovenous malformation ([Chap. 361](#)), a chronic subdural

hematoma ([Chap. 369](#)), or a degenerative disease ([Chaps. 363](#) to 366). The initial diagnostic test is often an [MRI](#) of the brain, especially if the clinical findings suggest brainstem pathology. If MRI of the brain is normal, the possibility of a foramen magnum or high cervical spinal cord lesion should be considered.

Paraparesis An intraspinal lesion at or below the upper thoracic spinal cord level is most commonly responsible. A sensory level over the trunk identifies the approximate level of the cord lesion. Paraparesis can also result from lesions at other locations that disturb upper motor neurons (especially parasagittal lesions and hydrocephalus) and lower motor neurons (anterior horn cell disorders, cauda equina syndromes, and occasionally peripheral neuropathies).

Acute or episodic paraparesis due to spinal cord disease may be difficult to distinguish from disorders affecting lower motor neurons or cerebral hemispheres. Recurrent episodes of paraparesis are often due to multiple sclerosis or to vascular malformations of the spinal cord. With acute spinal cord disease, the upper motor neuron deficit is usually associated with incontinence and a sensory disturbance of the lower limbs that extends rostrally to a level on the trunk; tone is typically flaccid, and tendon reflexes absent. In such cases, the diagnostic approach starts with an imaging study of the spinal cord ([Fig. 22-3](#)). Compressive lesions (particularly epidural tumor, abscess, or hematoma), spinal cord infarction (proprioception is usually spared), an arteriovenous fistula or other vascular anomaly, and transverse myelitis, among other causes may be responsible ([Chap. 368](#)). Diseases of the cerebral hemispheres that produce acute paraparesis include anterior cerebral artery ischemia (shoulder shrug also affected), superior sagittal sinus or cortical venous thrombosis, and acute hydrocephalus. If upper motor neuron signs are associated with drowsiness, confusion, seizures, or other hemispheric signs but not a sensory level over the trunk, the diagnostic approach starts with an [MRI](#) of the brain. Paraparesis is part of the cauda equina syndrome, which may result from trauma to the low back, a midline disk herniation, or intraspinal tumor; although sphincters are affected, hip flexion is often spared, as is sensation over the anterolateral thighs. Rarely, paraparesis is caused by a rapidly evolving peripheral neuropathy such as Guillain-Barre syndrome or by a myopathy. In such cases, electrophysiologic studies are diagnostically helpful and refocus the subsequent evaluation ([Chaps. 378](#) and [381](#)).

Subacute or chronic paraparesis with spasticity is caused by upper motor neuron disease. When paraparesis evolves over weeks or months with lower limb sensory loss and sphincter involvement, possible spinal cord disorders include multiple sclerosis, intraparenchymal tumor, chronic spinal cord compression from degenerative disease of the spine, subacute combined degeneration due to vitamin B₁₂ deficiency, viral infections (especially human T cell leukemia/lymphoma virus I), and hereditary or other degenerative diseases. Primary progressive multiple sclerosis usually presents in the fourth or fifth decade as progressive paraparesis ([Chap. 371](#)). Gliomas of the spinal cord typically produce a progressive myelopathy that is painful ([Chap. 370](#)). The clinical approach begins with an [MRI](#) of the spinal cord. If the imaging study is normal and spasticity is present, MRI of the brain may be indicated. If hemispheric signs are present, parasagittal meningioma or chronic hydrocephalus is likely and MRI of the brain is the initial test. Progression over months to years is typical of degenerative disorders such as primary lateral sclerosis ([Chap. 365](#)) and hereditary disorders such as

familial spastic paraparesis and adrenomyeloneuropathy ([Chap. 368](#)). In the rare situations when a chronic paraparesis is due to a lower motor neuron or myopathic etiology, the localization is usually suspected on clinical grounds by the absence of spasticity and confirmed by [EMG](#) and nerve conduction tests.

Quadriparesis or Generalized Weakness Generalized weakness may be due to disorders of the central nervous system or of the motor unit. Although the terms *quadriparesis* and *generalized weakness* are often used interchangeably, quadriparesis is more often chosen when an upper motor neuron cause is suspected and generalized weakness when a disease of the motor unit is likely. Weakness from [CNS](#) disorders is usually associated with changes in consciousness or cognition, with increased muscle tone and muscle stretch reflexes, and with alterations of sensation. Most neuromuscular causes of intermittent weakness are associated with normal mental function, diminished muscle tone, and hypoactive muscle stretch reflexes. Exceptions are some causes of acute quadriparesis due to upper motor neuron disorders in which transient hypotonia is present. The major causes of intermittent weakness are listed in [Table 22-3](#). A patient with generalized fatigability without objective weakness may have the *chronic fatigue syndrome* ([Chap. 384](#)).

Acute Quadriparesis Acute quadriparesis with onset over minutes may result from disorders of upper motor neurons (e.g., anoxia, hypotension, brainstem or cervical cord ischemia, trauma, and systemic metabolic abnormalities) or muscle (electrolyte disturbances, certain inborn errors of muscle energy metabolism, toxins, or periodic paralyses). Onset over hours to weeks may, in addition to the above, be due to lower motor neuron disorders. Guillain-Barre syndrome ([Chap. 378](#)) is the most common lower motor neuron weakness that progresses over days to several weeks; the finding of an elevated protein level in the cerebrospinal fluid is helpful but may be absent early in the course. If stupor or coma is present, the evaluation begins with a [CT](#) scan of the brain. If upper motor neuron signs are present but the patient is alert, the initial test is usually an [MRI](#) of the cervical cord. If weakness is lower motor neuron, myopathic, or uncertain in origin, the clinical approach starts with blood studies for muscle enzymes and electrolytes and an [EMG](#) and nerve conduction study.

Subacute or Chronic Quadriparesis When quadriparesis due to upper motor neuron disease develops over weeks, months, or years, the distinction between disorders of the cerebral hemispheres, brainstem, and cervical spinal cord is usually possible by clinical criteria alone. The diagnostic approach begins with an [MRI](#) of the clinically suspected site of pathology. Lower motor neuron disease usually presents with weakness that is most profound distally, whereas myopathic weakness is typically proximal; the evaluation then begins with [EMG](#) and nerve conduction studies.

Monoparesis This is usually due to lower motor neuron disease, with or without associated sensory involvement. Upper motor neuron weakness occasionally presents with a monoparesis of distal and nonantigravity muscles. Myopathic weakness is rarely limited to one limb.

Acute Monoparesis Distinguishing between upper and lower motor neuron disorders may be difficult clinically because tone and reflexes are frequently decreased in both at presentation. If the weakness is predominantly in distal and nonantigravity muscles and

not associated with sensory impairment or pain, focal cortical ischemia is likely ([Chap. 361](#)); in this setting, diagnostic possibilities are similar to those for acute hemiparesis. Sensory loss and pain usually accompany acute lower motor neuron weakness. The distribution of weakness is commonly localized to a single nerve root or peripheral nerve within one limb but occasionally reflects involvement of the brachial or lumbosacral plexus. If lower motor neuron weakness is suspected, or if the pattern of weakness is uncertain, the clinical approach begins with an [EMG](#) and nerve conduction study.

Subacute or Chronic Monoparesis Weakness with atrophy of one limb that develops over weeks or months is almost always lower motor neuron in origin. If the weakness is associated with numbness, a peripheral nerve or spinal root origin is likely; uncommonly, the brachial or lumbosacral plexus is affected. If numbness is absent, anterior horn cell disease is likely. In either case, an electrodiagnostic study is indicated. If upper rather than lower motor neuron signs are present, a tumor, vascular malformation, or other cortical lesion affecting the precentral gyrus may be responsible. Alternatively, if the leg is affected, a small thoracic cord lesion, often a tumor or multiple sclerosis, may be present. In these situations, the approach begins with an imaging study of the suspicious area.

Distal Weakness Involvement of two or four limbs distally suggests lower motor neuron or peripheral nerve disease. Acute distal lower limb weakness occurs occasionally from an acute toxic polyneuropathy or cauda equina syndrome. Distal symmetric weakness usually develops over weeks, months, or years and is due to metabolic, toxic, hereditary, degenerative, or inflammatory diseases of peripheral nerves ([Chap. 377](#)). With peripheral nerve disease, weakness is usually less severe than numbness. Anterior horn cell disease may begin distally but is typically asymmetric and is not associated with numbness ([Chap. 365](#)). Rarely, myopathies also present with distal weakness ([Chap. 381](#)). The first step in evaluation is an electrodiagnostic study ([Fig. 22-3](#)).

Proximal Weakness Proximal weakness of two or four limbs suggests a disorder of muscle or, less commonly, neuromuscular junction or anterior horn cell. Myopathy often produces symmetric weakness of the pelvic or shoulder girdle muscles ([Chap. 381](#)). Diseases of the neuromuscular junction (such as myasthenia gravis) may present with symmetric proximal weakness ([Chap. 380](#)), often associated with ptosis, diplopia, or bulbar weakness and fluctuating in severity during the day. Extreme fatigability present in some cases of myasthenia gravis may even suggest episodic weakness, but strength rarely returns fully to normal. The proximal weakness of anterior horn cell disease is most often asymmetric, but may be symmetric if familial ([Chap. 365](#)). Numbness does not occur with any of these diseases. The evaluation usually begins with determination of the serum creatine kinase level and electrophysiologic studies.

Weakness in a Restricted Distribution In some patients, weakness does not fit any of the above patterns. Examples include weakness limited to the extraocular, hemifacial, bulbar, or respiratory muscles. If unilateral, restricted weakness is usually due to lower motor neuron or peripheral nerve disease, such as in a facial palsy ([Chap. 367](#)) or an isolated superior oblique muscle paresis ([Chap. 28](#)). Relatively symmetric weakness of extraocular or bulbar muscles is usually due to a myopathy ([Chap. 381](#)) or neuromuscular junction disorder ([Chap. 380](#)). Bilateral facial palsy with areflexia

suggests Guillain-Barre syndrome ([Chap. 378](#)). Worsening of relatively symmetric weakness with fatigue is characteristic of neuromuscular junction disorders ([Chap. 380](#)). Asymmetric bulbar weakness is usually due to motor neuron disease. Weakness limited to respiratory muscles is uncommon and is usually due to motor neuron disease, myasthenia gravis, or polymyositis/dermatomyositis ([Chap. 382](#)).

MYALGIAS, SPASMS, AND CRAMPS

Spontaneous or exercise-related discomfort from muscles is usually benign and is rarely caused by a definable neuromuscular disease. However, a number of disorders of the motor system are characteristically painful. Some terms for muscular discomfort or involuntary contractions, such as myalgias, spasms, and cramps, are often used interchangeably by patients but have a more specific meaning to physicians. Other terms, such as aching, heaviness, and stiffness, are less specific. *Myalgias* are pains that are felt in muscle; the term does not imply an involuntary contraction. *Spasms* and *cramps* refer to episodes of involuntary contraction of one or more muscles. Cramps are usually painful, whereas spasms are not necessarily uncomfortable.

MYALGIAS

Proximal or generalized weakness associated with myalgias is usually due to an inflammatory, metabolic, endocrine, or toxic myopathy ([Chap. 381](#)). Spontaneous myalgias not accompanied by objective weakness are often without a clear cause unless associated with a well-defined systemic illness. Myalgias are a common manifestation of fever or infection, especially influenza. Muscle pains and stiffness with elevated serum creatine kinase concentration is common in hypothyroidism, even in patients without objective weakness. *Polymyalgia rheumatica* ([Chap. 317](#)) is characterized by diffuse myalgias and joint stiffness that predominantly affect the pelvic and shoulder girdles in a patient over 50 years of age who has anorexia, mild weight loss, and low-grade fever. Limitation of activity from the myalgias and joint stiffness also leads to disuse atrophy and may give the impression of weakness. However, [EMG](#), serum creatine kinase levels, and muscle biopsy are normal. The erythrocyte sedimentation rate is elevated in most patients, and features of giant-cell arteritis are present in 25%. Diffuse myalgias are common in many rheumatologic diseases, in which the diagnosis and treatment are based on other symptoms and signs. Myalgias are occasionally present in dermatomyositis/polymyositis, but most patients have weakness without significant pain. *Fibromyalgia* (fibrositis, fibromyositis) is associated with pain and tenderness of muscle and adjacent connective tissue ([Chap. 325](#)). Fatigue, insomnia, and depression are often present, but objective weakness, elevation of serum creatine kinase level, or elevation of the erythrocyte sedimentation rate does not occur. The diagnosis is dependent upon identifying characteristic focal "trigger points."

Focal Myalgias Focal muscle pain is often traumatic. Rupture of muscle tendons such as the biceps or gastrocnemius muscle may produce visible muscle shortening. Many such tears resolve without surgery but leave an abnormal appearance to the muscle belly. Nontraumatic focal muscle pain is often related to adjacent nonmuscular disorders (e.g., unilateral gastrocnemius pain due to deep venous thrombosis). Rarely, focal muscle pain may be caused by ischemic infarction or bacterial myositis, if acute, or by

neoplasm, parasitic infection, sarcoidosis, or other inflammation or infection, if subacute or chronic.

Exertional Myalgias Myalgias following unaccustomed, strenuous physical activity occur in normal individuals are often associated with laboratory evidence for muscle damage, such as an elevation of serum creatine kinase, edema of muscles on [MRI](#), necrosis of muscle fibers on biopsy, and rarely myoglobinuria. Similar symptoms and laboratory abnormalities characterize certain metabolic disorders of muscle, such as carnitine palmitoyl transferase and glycolytic pathway enzyme deficiencies. The association of objective weakness during an episode of myalgias suggests a metabolic muscle disease. The development of an acute contracture (the inability to relax a muscle due to energy depletion) with the myalgias suggests a metabolic muscle disease with a glycolytic enzyme deficiency ([Chap. 383](#)). Exertional myalgias with muscle fiber necrosis also occur in muscular dystrophy with partial deficiency of dystrophin, and certain mitochondrial cytopathies ([Chap. 383](#)). Exertional myalgias with elevated creatine kinase concentration but without weakness also occur in hypothyroidism, and when confined to the legs may be due to vascular or neurogenic intermittent claudication. Most patients with exertional myalgias and no weakness do not have a definable abnormality.

SPASMS AND CRAMPS

Involuntary contraction of muscle may occur with disorders of the [CNS](#), lower motor neuron, or muscle. Contractions that originate within the CNS and are associated with upper motor neuron signs are usually referred to as spasms and generally affect the flexors or extensors of one or more limbs. Those that originate within the CNS and are not associated with upper motor neuron signs include movement disorders discussed below, as well as the rare stiff-person syndrome and tetanus. Muscle rigidity from active muscle contraction can occur in the malignant hyperthermia syndrome, usually associated with general anesthesia. In the neuroleptic malignant syndrome, muscle rigidity arises from CNS overactivity and is present in muscle. Involuntary contractions that originate in the lower motor neurons are usually cramps, occasionally tetany, or rarely neuromyotonia. Spasms that originate in muscle or muscle membrane are usually a delayed relaxation after voluntary contraction, either myotonia or rarely a contracture. These conditions may be difficult to distinguish clinically but are often well characterized by [EMG](#) studies.

Stiff-Person Syndrome This rare syndrome is characterized by slowly progressive muscle stiffness and superimposed spasms. The stiffness commonly begins in the low back and spreads over months up the spine and into the limbs but not into the jaw. The gait becomes stiff, and there is hyperlordosis of the lumbar spine. Spasms are often produced by startle. Emotional stress tends to worsen the stiffness as well as the frequency and severity of spasms. The spontaneous motor activity disappears during sleep. The syndrome is often associated with diabetes mellitus and can be paraneoplastic, accompanying Hodgkin's lymphoma, small cell cancer of the lung, and breast cancer. Most patients have a serum antibody against glutamic acid decarboxylase, an enzyme responsible for synthesis of the inhibitory neurotransmitter g-aminobutyric acid (GABA). Stiffness results from loss of descending brainstem or segmental spinal inhibitory influences on the lower motor neurons. [EMG](#) studies reveal

continuous motor unit activity that is similar to voluntary effort with preservation of the silent period to muscle stretch. Stiffness and spasms typically respond partially to treatment with baclofen or benzodiazepines.

Tetanus This rare hyperexcitable state results from exposure to tetanus toxin in patients infected with *Clostridium tetani* ([Chap. 143](#)). Painful spasms typically begin with jaw closure (trismus) and soon become generalized. [EMG](#) studies reveal continuous motor unit activity that is similar to voluntary effort except for loss of the silent period to muscle stretch.

Cramps These are the most common type of involuntary muscle contraction. Cramps are a painful contraction of a single muscle that produces a palpable knot within the muscle for seconds to minutes and is relieved by passive stretch of the muscle or spontaneously. [EMG](#) studies reveal motor unit activity that has too high a discharge frequency to be voluntary. If cramps are associated with weakness, the weakness is almost always lower motor neuron in origin. When strength is normal, no definable condition is usually found, although dehydration, hypothyroidism ([Video 330-1](#)), or uremia is occasionally present. If prominent, membrane stabilizing drugs, such as carbamazepine, may provide symptomatic benefit.

Tetany Tetany is characterized by contraction of distal muscles of the hands (carpal spasm with extension of interphalangeal joints and adduction and flexion of the metacarpophalangeal joints) and feet (pedal spasm) and is associated with tingling around the mouth and distally in the limbs. Tetany with carpopedal spasms is a common manifestation of hypocalcemia or respiratory alkalosis (even from hyperventilation). [EMG](#) studies reveal single or more often grouped motor unit discharges at low discharge frequency.

Neuromyotonia (Isaac's Syndrome) Neuromyotonia is characterized by muscle stiffness at rest that persists during sleep and by delayed relaxation after voluntary effort. Distal limb muscles are usually affected most severely, but all skeletal muscle may be involved. Gait may be stiff, and close inspection of the muscle reveals undulation of the overlying skin due to continuous muscle fiber contractions (myokymia). The continuous muscle fiber activity generates heat, and excessive sweating is common. [EMG](#) studies commonly reveal myokymic discharges, especially in familial cases. Rarely, EMGs record high-frequency neuromyotonic discharges. Autoantibodies against voltage-gated potassium channels have been demonstrated in some cases, and plasma exchange may be effective.

Myotonia This is a nonpainful delay in the relaxation of muscle after voluntary activity. Delay in opening the hand after a forceful grip (grip myotonia) is common. These disorders are usually familial and worsen in cold weather. [EMG](#) demonstrates a waxing and waning discharge of individual muscle fibers.

Contracture A painful inability to relax a muscle after voluntary activity due to energy depletion characterizes certain metabolic disorders with failure of energy production, such as myophosphorylase deficiency (McArdle's disease). [EMG](#) studies reveal electrical silence.

MOVEMENT DISORDERS

Movement disorders are neurologic syndromes in which abnormal movements (or *dyskinesias*) occur due to a disturbance of fluency and speed of voluntary movement or the presence of unintended extra movements. Because they are so distinct from the pyramidal disorders that cause upper motor neuron weakness, movement disorders are often referred to as *extrapyramidal diseases*. *Hyperkinetic movement disorders* are those in which an excessive amount of spontaneous motor activity is seen or in which abnormal involuntary movements occur. *Hypokinetic movement disorders* are characterized by *akinesia* or *bradykinesia*, in which purposeful motor activity is absent or reduced. This is often described as "poverty of movement."

PATHOGENESIS

Movement disorders result from disease of the basal ganglia, paired subcortical gray matter structures consisting of the caudate and the putamen (which together are called the striatum), the internal and external segments of the globus pallidus, the subthalamic nucleus, and the substantia nigra. The major interconnections and neurotransmitters involved in basal ganglia circuits are illustrated in [Fig. 22-4A](#). An understanding of this circuitry can explain, in part, the perturbation that occurs in both the hypo- and hyperkinetic disorders.

Parkinson's disease ([Video 361-1](#)) ([Chap. 363](#)), the prototypic hypokinetic movement disorder, results from a loss of dopaminergic neurons in the substantia nigra pars compacta. This leads to less excitation of striatal neurons that express the D₁ type of dopamine receptors and less inhibition of D₂ striatal neurons, both contributing to reduced facilitation of cortically initiated movement ([Fig. 22-4B](#)). The resting tremor of Parkinson's disease is less readily explained by this model but may result from effects on cholinergic interneurons in the striatum. *Huntington's disease* ([Chap. 362](#)), a hyperkinetic movement disorder, may be explained by selective loss of D₂ striatal neurons, resulting in disinhibition of cortically initiated movements without normal feedback control. The pathogenesis of hemiballismus is similar -- a direct lesion of the glutamatergic neurons in the subthalamic nucleus (usually from a stroke) leads to disinhibition of thalamocortical projections.

Approach to the Patient

An algorithm for the interpretation of abnormal movements is illustrated in [Fig. 22-5](#). The initial step is to determine if the movement disorder is due to an excess or a poverty of movement (i.e., a hyperkinetic or a hypokinetic movement disorder).

Hyperkinetic Movement Disorders Abnormal involuntary movements are divided into those that are rhythmical and those that are irregular. Those that are rhythmical are termed *tremors*, with the uncommon exception of *palatal and segmental myoclonus*. Tremors are divided into three types: rest, postural, and intention tremor. A *rest tremor* is maximal at rest and becomes less prominent with activity. It is characteristic of parkinsonism, a hypokinetic movement disorder, and is therefore commonly associated with bradykinesia and cogwheel rigidity. A rest tremor that develops acutely is usually due to toxins [such as exposure to 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine

(MPTP)] or dopamine blocking drugs (such as phenothiazines). If insidious in onset, the diagnostic approach is the same as for Parkinson's disease ([Chap. 363](#)). A *postural tremor* is maximal while limb posture is actively maintained against gravity; it is lessened by rest and is not markedly enhanced during voluntary movement toward a target. A postural tremor that develops acutely is usually due to toxic or metabolic factors (for example, hyperthyroidism) or stress. The insidious onset of a postural tremor suggests a benign or familial essential tremor ([Chap. 363](#)). An *intention tremor* is most prominent during voluntary movement toward a target and is not present during postural maintenance or at rest. It is a sign of cerebellar disease ([Chap. 364](#)). *Asterixis*, which may superficially resemble a tremor, is an intermittent inhibition of muscle contraction that occurs with metabolic encephalopathy ([Chap. 376](#)). This leads, for example, to a momentary and repetitive partial flexion of the wrists during attempted sustained wrist extension.

Involuntary movements that are irregular are characterized further by their speed and site of occurrence and by whether they can be suppressed voluntarily. The slowest are athetosis and dystonia. *Athetosis* is a slow, writhing, sinuous movement that occurs nearly continuously in distal muscles. *Dystonia* is a slowly varying but nearly continuous deviation of posture about one or more joints; it may occur in a proximal or distal limb or in axial structures. Dystonia is a more sustained deviation of posture than athetosis, although these two phenomena overlap considerably. The further evaluation of athetosis and dystonia are discussed in [Chap. 363](#).

Among the rapid irregular movements, *tics* are controlled with voluntary effort, while the others are not. Tics often occur repetitively in a single location but are sometimes multifocal ([Chap. 363](#)).

Chorea, hemiballismus, and myoclonus are rapid, irregular jerks that cannot be consciously suppressed. *Hemiballismus* is the most distinctive among them. It is manifest as a sudden and often violent flinging movement of a proximal limb, usually an arm ([Chap. 363](#)). Hemiballismus usually develops acutely due to infarction of the contralateral subthalamic nucleus but occasionally develops subacutely or chronically due to other lesions of this nucleus.

Chorea is a rapid, jerky, irregular movement that tends to occur in the distal limbs or face but may also occur in proximal limb and axial structures. Acute or subacute onset is usually toxic due to excess levodopa or dopamine-agonist therapy or, less often, neuroleptics, birth control pills, pregnancy (chorea gravidarum), hyperthyroidism, or the antiphospholipid syndrome. In children, it may be associated with rheumatic fever and, in such cases, is referred to as *Sydenham's chorea*. The gradual onset of chorea is typical of degenerative neurologic diseases, such as Huntington's chorea ([Chap. 362](#)).

Myoclonus is a rapid, brief, irregular movement that is usually multifocal. Myoclonus can occur spontaneously at rest, in response to sensory stimuli, or with voluntary movements. It is a symptom that occurs in a wide variety of metabolic and neurologic disorders. Posthypoxic intention myoclonus is a special myoclonic syndrome that occurs as a sequel to transient cerebral anoxia. Myoclonus may result from lipid storage disease, encephalitis, Creutzfeldt-Jakob disease, or metabolic encephalopathies due to respiratory failure, chronic renal failure, hepatic failure, or electrolyte imbalance.

Myoclonus is also a feature of certain types of epilepsy, as discussed in [Chap. 360](#). *Palatal and segmental myoclonus* are uncommon rhythmic forms of myoclonus that may resemble tremor; they are caused by structural disease of the brainstem or spinal cord at the level of the abnormal movement.

Hypokinetic Movement Disorders These syndromes are manifest as bradykinesia, with a masked, expressionless facial appearance, loss of associated limb movements during walking, and rigid en bloc turning. If bradykinesia is associated only with a rest tremor, cogwheel rigidity, or impairment of postural reflexes (especially with a tendency to fall backwards), Parkinson's disease is likely ([Chap. 363](#)). If cognitive, language, upper motor neuron, sensory, or autonomic signs are also present, a *multisystem degenerative neurologic disease* is present. **These disorders are discussed in [Chaps. 363, 364, and 366](#).*

IMBALANCE AND DISORDERS OF GAIT

Imbalance is the impaired ability to maintain the intended orientation of the body in space. It is generally manifest as difficulty in maintaining an upright posture while standing or walking; a severe imbalance may also affect the ability to maintain posture while seated. Patients with imbalance commonly complain of a feeling of unsteadiness or dysequilibrium. Whereas imbalance and unsteadiness are synonymous, *dysequilibrium* implies the additional component of impaired spatial orientation even while lying down. Patients with dysequilibrium commonly also experience *vertigo*, defined as an hallucination of rotatory movement.

PATHOGENESIS

Imbalance and Limb Ataxia Imbalance results from disorders of the spinal cord (spinocerebellar) or vestibular sensory input, the integration of these inputs in the brainstem or midline cerebellum, or the motor output to the spinal neurons that control axial and proximal muscles. Limb ataxia results from disorders of the spinocerebellar and corticopontocerebellar inputs, the integration of these inputs in the intermediate and lateral cerebellum, or the output to the spinal neurons (via the red nucleus and rubrospinal tract) or to the cortex. These pathways ensure adequate speed, fluency, and integration of limb movements. The lateral cerebellar hemispheres coordinate a complex feedback circuit that modulates cortically initiated limb movement.

Sensory ataxia is caused by lesions that affect the peripheral sensory fibers, dorsal root ganglia cells, posterior columns of the spinal cord, lemniscal system in the brainstem, thalamus, or parietal cortex; relevant anatomy is discussed in [Chap. 23](#). Impairment of the proprioceptive sensory feedback to the cerebellum, basal ganglia, and cortex produces sensory ataxia. Sensory ataxia results in imbalance and disturbs the fluency and integration of movements that can be partially alleviated by visual feedback.

Disorders of Gait Walking is one of the most complicated motor activities. Essentially all structures discussed in this chapter participate in normal walking. Cyclical stepping movements produced by the lumbosacral spinal cord centers are modified by cortical, basal ganglionic, brainstem, and cerebellar influences based on proprioceptive, vestibular, and visual feedback.

Approach to the Patient

Examination of coordination, balance, and gait is typically performed at the same time. The finger-nose-finger and the heel-knee-shin maneuvers are observed for signs of incoordination in general and dysmetria in particular. *Dysmetria* consists of irregular errors in the amplitude and force of limb movements. This is accentuated near the target or point of intention and hence termed *intention tremor*. The patient is also asked to maintain the arms outstretched against a resistance that is suddenly removed; excessive *rebound* indicates cerebellar dysfunction. The ability of the patient to rapidly and repetitively tap the hands and feet is assessed for speed and rhythmicity. Errors in rhythm (irregular rate, velocity, or force) indicate *dysdiadochokinesia*. Slow, coarse, but rhythmical movements indicate upper motor neuron disorders. The patient is asked to demonstrate how to comb the hair or brush the teeth to assess the ability to initiate and execute a simple sequence of activity. Balance is examined by having the patient stand stationary with the feet together. If this position can be maintained, the eyes are closed for 5 to 10 s. Accentuation of sway or actual loss of balance is assessed. If balance is momentarily lost, several trials may be necessary to determine if the loss is consistently in the same direction. Walking along an uncrowded space, such as a hallway, is observed. Symmetry of arm swing and various phases of the gait cycle are observed. Walking is then performed for several steps on the heels, on the toes, and in tandem.

Imbalance An algorithm for interpretation of imbalance is presented in [Fig. 22-6](#).

Cerebellar ataxia results from disorders of the cerebellum or of its afferent inputs or efferent projections. Abnormalities of the midline cerebellar vermis or the flocculonodular lobe produce truncal ataxia which is usually revealed during the process of rising from a chair, assuming the upright stance with the feet together, or performing some other activity while standing. Once a desired position is reached, imbalance may be surprisingly mild. As walking begins, the imbalance recurs. Patients usually learn to lessen the imbalance by walking with the legs widely separated. The imbalance is usually not lateralized and may be accompanied by symmetric nystagmus.

Abnormalities of the intermediate and lateral portions of the cerebellum typically produce impaired limb movements rather than truncal ataxia. If involvement is asymmetric, lateralized imbalance is common and usually associated with asymmetric nystagmus. Clinical signs of cerebellar limb ataxia include dysmetria, intention tremor, dysdiadochokinesia, and abnormal rebound. Muscle tone is often modestly reduced; this contributes to the abnormal rebound due to decreased activation of segmental spinal cord reflexes and also to pendular reflexes, i.e., a tendency for a tendon reflex to produce multiple swings to and fro after a single tap. **For further discussion of cerebellar diseases, see [Chap. 364](#).*

Imbalance with vestibular dysfunction is characterized by a consistent tendency to fall to one side. The patient commonly complains of vertigo rather than imbalance, especially if the onset is acute. Acute vertigo associated with lateralized imbalance but no other neurologic signs is often due to disorders of the semicircular canal ([Chap. 21](#)); the presence of other neurologic signs suggests brainstem ischemia ([Chap. 361](#)) or multiple sclerosis ([Chap. 371](#)). When the vestibular dysfunction is peripheral, positional

nystagmus and vertigo tend to resolve if a provocative position is maintained (extinction) or repeated (habituation). Lateralized imbalance of gradual onset or persisting for more than 2 weeks, accompanied by nystagmus, may result from lesions of the semicircular canal or vestibular nerve, brainstem, or cerebellum.

Imbalance with sensory ataxia is characterized by marked worsening when visual feedback is removed. The patient can often assume the upright stance with feet together cautiously with eyes open. With eye closure, balance is rapidly lost (positive Romberg sign) in various directions at random. Sensory examination reveals impairment of proprioception at the toes and ankles, usually associated with an even more prominent abnormality of vibratory perception. Prompt evaluation for vitamin B₁₂ deficiency is important, as this disorder is reversible if recognized early ([Chap. 368](#)). Depression or absence of reflexes points to peripheral nerve disorders ([Chap. 377](#)). Spasticity with extensor plantar responses suggests posterior column and spinal cord disorders ([Chap. 368](#)). Rarely, sensory ataxia produces lateralized imbalance. In these cases, the disorder is usually in the parietal lobe or thalamus ([Chap. 23](#)), but may also be due to an asymmetric sensory neuropathy ([Chap. 377](#)) or posterior column disease ([Chap. 368](#)).

Sensory limb ataxia is similar to cerebellar limb ataxia but is markedly worse when the eyes are closed. Examination also reveals abnormal proprioception and vibratory perception. The approach focuses on localizing the proprioceptive impairment to the peripheral nerves ([Chap. 377](#)), the posterior columns of the spinal cord ([Chap. 368](#)), or rarely the parietal lobe.

Other forms of imbalance occur, but the fundamental problem is usually a primary disorder of strength, extrapyramidal function, or cortical initiation of movement.

Abnormal Gait Each of the disorders discussed in this chapter produces a characteristic gait disturbance. If the neurologic examination is normal except for an abnormal gait, diagnosis may be difficult even for the experienced clinician.

Hemiparetic gait characterizes spastic hemiparesis. In its most severe form, an abnormal posture of the limbs is produced by spasticity. The arm is adducted and internally rotated, with flexion of the elbow, wrist, and fingers and with extension of the hip, knee, and ankle. Forward swing of the spastic leg during walking requires abduction and circumduction at the hip, often with contralateral tilt of the trunk to prevent the toes catching on the floor as the leg is advanced. In its mildest form, the affected arm is held in a normal position, but swings less than the normal arm. The affected leg is flexed less than the normal leg during its forward swing and is more externally rotated. A hemiparetic gait is a common residual sign of a stroke ([Chap. 361](#)).

Paraparetic gait ([Video 361-3](#)) is a walking pattern in which both legs are moved in a slow, stiff manner with circumduction, similar to the leg movement in a hemiparetic gait. In many patients, the legs tend to cross with each forward swing ("scissoring"). A paraparetic gait is a common sign of spinal cord disease ([Chap. 368](#)) and also occurs in cerebral palsy.

Steppage gait is produced by weakness of ankle dorsiflexion. Because of the partial or

complete foot drop, the leg must be lifted higher than usual to avoid catching the toe on the floor during the forward swing of the leg. If unilateral, steppage gait is usually due to L5 radiculopathy, sciatic neuropathy, or peroneal neuropathy ([Chap. 377](#)). If bilateral, it is the common result of a distal polyneuropathy or lumbosacral polyradiculopathy ([Chap. 377](#)).

Waddling gait results from proximal lower limb weakness, most often from myopathy ([Chap. 381](#)) but occasionally from neuromuscular junction disease ([Chap. 380](#)) or a proximal symmetric spinal muscular atrophy ([Chap. 365](#)). With weakness of hip flexion, the trunk is tilted away from the leg that is being moved to lift the hip and provide extra distance between the foot and the floor, and the pelvis is rotated forward to assist with forward motion of the leg. Because pelvic girdle weakness is customarily bilateral, the pelvic lift and rotation alternates from side to side, giving the waddling appearance to the gait.

Parkinsonian gait ([Video 361-1](#)) is characterized by a forward stoop, with modest flexion at the hips and knees. The arms are flexed at the elbows and adducted at the shoulders, often with a 4- to 6-Hz resting pronation-supination tremor but little other movement, even during walking. Walking is initiated slowly by leaning forward and maintained with short rapid steps, during which the feet shuffle along the floor. The pace tends to accelerate (festination) as the upper body gradually leans further ahead of the feet, whether movement is forward (propulsion) or backward (retropulsion). The postural instability leads to falls ([Chap. 363](#)).

Apraxic gait ([Video 361-4](#)) results from bilateral frontal lobe disease with impaired ability to plan and execute sequential movements. This gait superficially resembles that of parkinsonism, in that the posture is stooped and any steps taken are short and shuffling. However, initiation and maintenance of walking are impaired in a different manner. Each movement that is required for walking can usually be performed, if tested in isolation while sitting or lying. However, when asked to step forward while standing, a long pause often occurs before any attempt is made to flex at the hip and advance, as if the patient is "glued to the ground." Once walking is initiated, it is not maintained, even in an abnormal festinating manner. Rather, after one or several steps are taken, walking is stopped for several seconds or longer. The process is then repeated. Dementia and incontinence may coexist.

Choreoathetotic gait is characterized by an intermittent, irregular movement that disrupts the smooth flow of a normal gait. Flexion or extension movements at the hip are common and unpredictable but readily observed as a pelvic lurch ([Chap. 363](#)).

Cerebellar ataxic gait ([Video 361-3](#)) is a broad-based gait disorder in which the speed and length of stride varies irregularly from step to step. With midline cerebellar disease, as in alcoholics, posture is erect but the feet are separated; lower limb ataxia is commonly present as well. Assumption of a particular stance or a change in position may cause instability, yet balance can usually be maintained well with the eyes open or closed. Walking may be rapid, but cadence is irregular. Although patients commonly lack confidence in the stability of their walking, only minimal support is often required for reassurance. With disease of the cerebellar hemispheres, limb ataxia and nystagmus are commonly present as well ([Chap. 364](#)).

Sensory ataxic gait may resemble a cerebellar gait, with its broad-based stance and difficulty with change in position. However, although balance may be maintained with the eyes open, loss of visual input through eye closure results in rapid loss of balance with a fall (positive Romberg sign), unless the physician assists the patient.

Vestibular gait is one in which the patient consistently tends to fall to one side, whether walking or standing. Cranial nerve examination demonstrates an obviously asymmetric nystagmus. The possibilities of unilateral sensory ataxia and hemiparesis are excluded by the findings of normal proprioception and strength ([Chap. 21](#)).

Astasia-abasia is a typical hysterical gait disorder. Although the patient usually has normal coordination of leg movements in bed or while sitting, the patient is unable to stand or walk without assistance. If distracted, stationary balance is sometimes maintained and several steps are taken normally, followed by a dramatic demonstration of imbalance with a lunge toward the examiner's arms or a nearby bed.

(Bibliography omitted in Palm version)

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23. NUMBNESS, TINGLING, AND SENSORY LOSS - Arthur K. Asbury

NORMAL SENSATION

Normal somatic sensation reflects a continuous day and night monitoring process that occupies considerable moment-to-moment nervous system capacity. Little of this activity reaches consciousness under ordinary conditions. In contrast, disordered sensation, particularly if experienced as painful, is alarming and dominates the sufferer's attention. Abnormalities of sensation, especially if painful, tend to make those suffering seek medical help. The physician must be able to recognize abnormal sensations by how they are described, know their type and likely site of origin, and understand their implications. **For a consideration of pain, see [Chap. 12](#).*

Positive and Negative Phenomena Abnormal sensory phenomena may be divided into two categories, positive and negative. The prototypical positive phenomenon is tingling (pins-and-needles), and the principal negative phenomenon is numbness. In addition to tingling, positive sensory phenomena include other altered sensations that are often described as pricking, bandlike, lightning-like shooting feelings (lancinations), aching, knifelike, twisting, drawing, pulling, tightening, burning, searing, electrical, or raw feelings. These descriptors are frequently the actual words used by patients. Such sensations may or may not be experienced as painful.

Positive phenomena usually result from trains of impulses generated at a site or sites of lowered threshold or heightened excitability along a sensory pathway, either peripheral or central. The nature and severity of an abnormal sensation depend on the number, rate, timing, and distribution of ectopic impulses and the type and function of nervous tissue in which they arise. Because positive phenomena represent excessive activity in sensory pathways, they are not necessarily associated with any sensory deficit (loss) upon examination.

Negative phenomena represent loss of sensory function and are characterized by diminished or absent feeling, often experienced as numbness. In contrast to positive phenomena, negative phenomena are accompanied by abnormal findings on sensory examination. In disorders affecting peripheral sensation, it is estimated that at least half the afferent axons innervating a given site are lost or functionless before sensory deficit can be demonstrated by clinical examination. This estimate probably varies according to how rapidly sensory nerve fibers have lost function. If the rate of loss is slow and chronic, lack of cutaneous feeling may be unnoticed by the patient and difficult to demonstrate on examination, even though few sensory fibers are functioning. Rapidly evolving sensory abnormality usually evokes both positive and negative phenomena and is readily recognized by patients. Subclinical degrees of sensory dysfunction not demonstrable on clinical sensory examination may be revealed by sensory nerve conduction studies or somatosensory cerebral evoked potentials ([Chap. 357](#)). Sensory symptoms may be either positive or negative, but sensory signs on examination are always a measure of negative phenomena.

Terminology Words used to characterize sensory disturbance are descriptive and have been arrived at mainly by convention. Paresthesia and dysesthesia are general terms used to denote sensory symptoms (positive phenomena) and are usually stated in the

plural form. *Paresthesias* usually refer to tingling or pins-and-needles sensations but may also include a wide variety of other abnormal sensations, excepting pain. Sometimes "paresthesias" carry the implication that the abnormal sensations are perceived without an apparent stimulus. *Dysesthesia* is a more general term used to subsume all types of abnormal sensations, even painful ones, whether a stimulus is evident or not.

While dysesthesias and paresthesias refer to sensations described by patients, another set of terms refers to sensory abnormalities found on examination. These include *hypesthesia* or *hypoesthesia* (reduction of cutaneous sensation to a specific type of testing such as pressure, light touch, and warm or cold stimuli); *anesthesia* (complete absence of skin sensation to the same stimuli plus pinprick); and *hypalgesia* (referring to reduced pain perception, i.e., nociception, such as the pricking quality elicited by a pin). *Hyperesthesia* means pain in response to touch. Similarly, *allodynia* describes the situation in which a nonpainful stimulus, once perceived, is experienced as painful, even excruciating. An example is elicitation of a painful sensation by application of a vibrating tuning fork. *Hyperalgesia* denotes severe pain in response to a mildly noxious stimulus, and *hyperpathia*, a broad term, encompasses all the phenomena described by hyperesthesia, allodynia, and hyperalgesia. With hyperpathia, the threshold for a sensory stimulus is increased and the perception is delayed but once felt, is unduly painful.

Disorders of deep sensation, arising from muscle spindles, tendons, and joints, affect proprioception (position sense). Manifestations include imbalance (particularly with eyes closed or in the dark), clumsiness of precision movements, and unsteadiness of gait, which are referred to collectively as *sensory ataxia* ([Chap. 22](#)). Other findings on examination usually, but not invariably, include reduced or absent joint position and vibratory sensibility and absent deep tendon reflexes in the affected limbs. Romberg's sign is positive, which means that the patient sways or topples when asked to stand with feet close together and eyes closed. In severe states of deafferentation involving deep sensation, the patient cannot walk or stand unaided or even sit unsupported. Continuous, sometimes wormlike involuntary movements, called *pseudoathetosis*, of the outstretched hands and fingers occur, particularly with eyes closed. Such patients are severely disabled.

Anatomy of Sensation Cutaneous afferent innervation is conveyed by a rich variety of receptors, both naked nerve endings (nociceptors and thermoreceptors) and encapsulated terminals (mechanoreceptors). Each type of receptor has its own set of sensitivities to specific stimuli, size and distinctness of receptive fields, and adaptational qualities. Much of the knowledge about these receptors has come from the development of techniques to study single intact nerve fibers intraneurally in awake unanesthetized human subjects. It is possible not only to record from single nerve fibers, large or small, but also to stimulate single fibers in isolation. A single impulse, whether elicited by a natural stimulus or evoked by electrical microstimulation, in a large myelinated afferent fiber may be both perceived and localized.

Afferent fibers of all sizes in peripheral nerve trunks traverse the dorsal roots and enter the dorsal horn of the spinal cord ([Fig. 23-1](#)). From there the smaller fibers take a different route to the parietal cortex than the larger fibers. The polysynaptic projections

of the smaller fibers (unmyelinated and small myelinated), which subserve mainly nociception, temperature sensibility, and touch, cross and ascend in the opposite anterior and lateral columns of the spinal cord, through the brainstem, to the ventral posterolateral (VPL) nucleus of the thalamus, and ultimately project to the postcentral gyrus of the parietal cortex ([Chap. 12](#)). This is referred to as the *spinothalamic pathway*, or *anterolateral system*. The larger fibers, which subserve tactile and position sense and kinesthesia, project rostrally in the posterior column on the same side of the spinal cord and make their first synapse in the gracile or cuneate nuclei of the lower medulla. The second-order neuron decussates and ascends in the medial lemniscus located medially in the medulla and in the tegmentum of the pons and midbrain and synapses in the VPL. The third-order neuron projects to parietal cortex; this large fiber system is referred to as the *posterior column-medial lemniscal pathway* (lemniscal, for short). Note that although the lemniscal and the anterolateral pathways both project up the spinal cord to the thalamus, it is the (crossed) anterolateral pathway that is referred to as the *spinothalamic tract*, by convention.

Although the fiber types and functions that make up the spinothalamic and lemniscal systems are relatively well known, it has been found that many other fibers, particularly those associated with touch, pressure, and position sense, ascend in a diffusely distributed pattern both ipsilaterally and contralaterally in the anterolateral quadrants of the spinal cord. This explains why an individual with a complete lesion of the posterior columns of the spinal cord may have little sensory deficit on examination.

EXAMINATION OF SENSATION

The main tasks of the sensory examination are tests of primary sensation. By convention these include the sense of pain, touch, vibration, joint position, and thermal sensation, both hot and cold ([Table 23-1](#)). Detailed descriptions of how to perform the various tests of the sensory examination can be found in standard texts (see "Bibliography").

Some general principles pertain. First, the examiner must depend on subjective patient response, particularly when using cutaneous stimuli (pin, touch, vibration, warm or cold). This factor may complicate the interpretation of the sensory examination. Second, with complaints of numbness, patients should be asked to outline on themselves the borders of numb areas. Third, some patients are only partially examinable. In a stuporous patient, sensory examination is reduced to observing the briskness of withdrawal in response to a pinch or other noxious stimulus. Comparison of response on one side of the body to the other is essential. In the alert but uncooperative patient, cutaneous sensation may be unexaminable. However, it is usually possible to get some idea of proprioceptive function by noting the patient's best performance of movements requiring balance and precision. Fourth, sensory examination of a patient who has no neurologic complaints should be abbreviated and may consist of pin, touch, and vibration testing in the hands and feet plus evaluation of stance and gait, including the Romberg maneuver. Evaluation of stance and gait also tests the integrity of motor and cerebellar systems.

Primary Sensation (See [Table 23-1](#)) The sense of pain is usually tested with a pin, asking the patient to focus on the pricking or unpleasant quality of the stimulus and not just the pressure or touch sensation elicited. Areas of hypalgesia should be mapped by

proceeding radially from the most hypalgesic site ([Figs. 23-2](#) and [23-3](#)).

Temperature sensation, to both hot and cold, is probably best tested with water flasks filled with water of the desired temperature, using a thermometer to verify the temperature. This is impractical in most settings. An alternative way to test cold sensation is to touch a metal object, such as a tuning fork at room temperature, to the skin. For testing warm temperatures, the tuning fork or other metal object may be held under warm water of the desired temperature and then used. Both cold and warm should be tested because different receptors respond to each.

Touch is usually tested with a wisp of cotton or a fine camelhair brush. In general, it is better to avoid testing touch on hairy skin because of the profusion of sensory endings that surround each hair follicle.

Joint position testing is a measure of proprioception, one of the most important functions of the sensory system. With the patient keeping eyes closed, joint position is tested in the great toe and in the fingers. If errors are made in recognizing the direction of passive movements of the toe or the finger, more proximal joints should be tested. A test of proximal joint position sense, primarily at the shoulder, is performed by asking the patient to bring the two index fingers together with the arms extended and the eyes closed. Normal individuals should be able to do this quite accurately, with errors of a centimeter or less.

The sense of vibration is tested with a tuning fork, preferably a large one that vibrates at 128 Hz. Vibration is usually tested at bony prominences, beginning distally at the malleoli of the ankles, and at the knuckles. If abnormalities are found, more proximal sites can be examined. Vibratory thresholds at the same site in the patient and the examiner can be compared for control purposes.

Quantitative Sensory Testing Effective sensory testing devices have been developed over the past two decades. Quantitative sensory testing is particularly useful for serial evaluation of cutaneous sensation in clinical trials. Threshold testing for touch and vibratory and thermal sensation is the most widely used application.

Cortical Sensation Cortical sensory testing includes two-point discrimination, touch localization, and bilateral simultaneous stimulation and tests for graphesthesia and stereognosis, to name the most commonly used methods. Abnormalities of these sensory tests, in the presence of normal primary sensation in an alert cooperative patient, signify a lesion of the parietal cortex or thalamocortical projections to the parietal lobe. If primary sensation is altered, these cortical discriminative functions will usually be abnormal, too. Comparisons should always be made between analogous sites on the two sides of the body because the deficit with a specific parietal lesion is likely to be hemilateral. Side-to-side comparisons hold true for all cortical sensory testing.

Two-point discrimination is tested by special calipers, the points of which may be set from 2 mm to several centimeters apart and then applied simultaneously to the site to be tested. The pulp of the fingertips is a common site to test; a normal individual can distinguish about 3-mm separation of points there.

Touch localization is usually carried out by light pressure with the examiner's fingertip, asking the patient, whose eyes are closed, to identify the site of touch. It is usual to ask the patient to touch the same site with a fingertip.

Bilateral simultaneous stimulation at analogous sites (e.g., the dorsa of both hands) can be carried out to determine whether the perception of touch is extinguished consistently on one side or the other. The phenomenon is referred to as *extinction* on bilateral simultaneous stimulation.

Graphesthesia means the capacity to recognize with eyes closed letters or numbers drawn by the examiner's fingertip on the palm of the hand. Once again, the comparison of one side with the other is of prime importance. Inability to recognize numbers or letters is termed *agraphesthesia*.

Stereognosis refers to the ability to identify common objects by palpation, recognizing their shape, texture, and size. Common standard objects are the best test objects, such as a marble, a paper clip, or coins. Patients with normal stereognosis should be able to distinguish a dime from a penny and a nickel from a quarter without looking. Patients should only be allowed to feel the object with one hand at a time. If they are unable to identify it in one hand, it should be placed in the other for comparison. Individuals unable to identify common objects and coins in one hand who can do so in the other are said to have *astereognosis* of the abnormal hand.

LOCALIZATION OF SENSORY ABNORMALITIES

Sensory symptoms and signs can result from lesions at almost any level of the nervous system, including parietal cortex, deep white matter, thalamus, brainstem, spinal cord, spinal root, peripheral nerve, and sensory receptor. Noting the distribution and nature of sensory symptoms and signs is the most important way to localize their source. The extent, configuration, symmetry, quality, and severity are the key observations.

Dysesthesias without sensory findings by examination can be difficult to interpret. To illustrate, tingling dysesthesias in an acral distribution (hands and feet) can have more than one interpretation. Distal dysesthesias can be systemic in origin, e.g., secondary to hyperventilation, or can be induced by a medication, such as the diuretic acetazolamide. Distal dysesthesias can also be an early event in an evolving polyneuropathy or can herald a myelopathy, such as with vitamin B₁₂ deficiency. Sometimes distal dysesthesias have no definable basis. In contrast, dysesthesias that correspond to a particular peripheral nerve territory denote a lesion of that nerve trunk. For instance, dysesthesias restricted to the fifth digit and the adjacent one-half of the fourth finger on one hand reliably point to disorder of the ulnar nerve, most commonly at the elbow.

Nerve and Root In focal nerve trunk lesions severe enough to cause a deficit, sensory abnormalities are readily mapped and generally have discrete boundaries ([Figs. 23-2](#) and [23-3](#)). Root lesions, referred to as radicular, are frequently accompanied by deep, aching pain along the course of the related nerve trunk. With compression of a fifth lumbar (L5) or first sacral (S1) root, as may occur with a ruptured intervertebral disc, sciatica is a frequent manifestation. With a lesion affecting a single root, sensory deficit

in the distribution of that root is often minimal or not demonstrable at all. This is because adjacent root territories overlap extensively.

Polyneuropathies are generally graded, distal, and symmetric in distribution of deficit ([Chap. 377](#)). Dyesthesias begin in the toes and ascend symmetrically, followed by numbness. When dyesthesias reach the knees, they have usually also appeared in the fingertips. The process appears to be nerve length-dependent, and the deficit is often described as "stocking-glove" in type. Although most polyneuropathies are pansenory and affect all modalities of sensation, selective sensory dysfunction according to nerve fiber size may occur. In polyneuropathies that affect small nerve fibers selectively, the hallmark is burning, painful dysesthesias with reduced pinprick and thermal sensation but with sparing of proprioception, motor function, and even deep tendon jerks. Touch is variably involved, but when spared, the sensory pattern is referred to as *sensory dissociation*. Sensory dissociation patterns can be seen with spinal cord lesions (see below) as well as with small fiber neuropathies. In contrast to small fiber polyneuropathies, large fiber polyneuropathies are characterized by position sense deficit, imbalance, absent tendon jerks, and variable motor dysfunction but preservation of most cutaneous sensation. Dyesthesias, if present at all, tend to be tingling or bandlike.

Spinal Cord (See [Chap. 368](#)) If the spinal cord is transected, all sensation is lost below the level of transection. Bladder and bowel function are also lost, as is motor function. Hemisection of the spinal cord produces the Brown-Sequard syndrome, which involves absent pain and temperature sensation on the opposite side below the lesion, and loss of proprioceptive sensation and loss of motor power on the same side below the lesion (see [Figs. 23-1](#) and [368-1](#)). Dissociated sensory deficit patterns (see above) are also a sign of spinothalamic tract involvement in the spinal cord, especially if the deficit is unilateral and has an upper level on the torso. Bilateral spinothalamic tract involvement occurs with lesions affecting the center of the spinal cord, such as happens with expansion of the central canal in syringomyelia. Sensory dissociation is characteristic of syringomyelia.

Brainstem Harlequin patterns of sensory disturbance, in which one side of the face and the opposite side of the body are affected, localize to the lateral medulla. Here a small lesion may damage both the ipsilateral descending trigeminal tract and ascending spinothalamic fibers subserving the opposite arm, leg, and hemitorso (see "Lateral medullary syndrome" in [Fig. 361-7](#)). In the tegmentum of the pons and midbrain, where the lemniscal and spinothalamic tracts merge, a lesion here causes pansenory loss on the contralateral body.

Thalamus Hemisensory disturbance with tingling numbness from head to foot is often thalamic in origin but can also be anterior parietal. If abrupt in onset, the lesion is likely to be due to a small stroke (lacunar infarction), particularly if localized to the thalamus. Occasionally, with lesions affecting the [VPL](#) or adjacent white matter, a syndrome of thalamic pain, also called *Dejerine-Roussy syndrome*, may ensue. This persistent unrelenting hemipainful state is often described in dramatic terms such as "like the flesh is being torn from my limbs" or "as though that side is bathed in acid" ([Chap. 12](#)).

Cortex With lesions of the parietal lobe, either of the cortex or of subjacent white matter,

the most prominent symptoms are contralateral hemineglect, hemi-inattention, and a tendency not to use the affected hand and arm. Tests of primary sensation may be normal or altered. Anterior parietal infarction may present as a pseudothalamic syndrome with crossed hemilateral loss of primary sensation. Dysesthesias or a sense of numbness may also occur, and rarely a painful state.

Focal Sensory Seizures These are generally due to lesions in or near the postcentral gyrus. Symptoms of focal sensory seizures are usually combinations of numbness and tingling, but frequently additional more complex sensations are present, such as a rushing feeling, a sense of warmth, a sense of movement without visible motion, or other unpleasant dysesthesias. Duration of seizures is variable; they may be transient, lasting only seconds, or they may persist for hours. Focal motor features (clonic jerking) may supervene, and seizures can become generalized with loss of consciousness. Likely sites of symptoms are unilaterally in the lips, face, digits, or foot, and symptoms may spread as in a Jacksonian march. On occasion, symptoms may occur in a symmetric bilateral fashion, for instance, in both hands; this results from involvement of the second sensory area (unilaterally) located in the rolandic area at and just above the Sylvian fissure.

(Bibliography omitted in Palm version)

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24. ACUTE CONFUSIONAL STATES AND COMA - Allan H. Ropper

Confusional states and coma are among the most common problems in general medicine. They account for a substantial portion of admissions to emergency wards and are a frequent cause of distress on all hospital services. Because clouding of consciousness and a diminished level of consciousness frequently coexist and result from many of the same diseases, they are presented together here, but from a medical perspective they have different clinical characteristics and physiologic explanations.

The basis of consciousness has long been a topic of great interest to psychologists and philosophers and is the subject of a vast literature. Physicians have been mainly concerned with impairments in the level of consciousness (coma, stupor, drowsiness) and with alterations of consciousness, meaning an inability to think coherently, i.e., with accustomed clarity and speed. The latter, broadly termed *confusion* relates to lessened awareness, perception, apperception (the interpretation of perceptions), thinking, expression in language and action, and all forms of intellection that are dependent on the continuous integration of mental processes. Normal awareness provides a background to our inner mental life, which flows from infancy to death like a "stream of thought," to use William James's metaphor. Self-awareness requires that a person experience these thoughts and be able to reflect and operate upon them.

Almost all instances of diminished alertness can be traced to widespread abnormalities of the cerebral hemispheres or to reduced activity of a special thalamocortical alerting system termed the *reticular activating system* (RAS). The proper functioning of this system, its ascending projections to the cortex, the cortex itself, and corticothalamic connections are required to maintain alertness and coherence of thought.

THE CONFUSIONAL STATE

Confusion is a mental and behavioral state of reduced comprehension, coherence, and capacity to reason. Inattention, as defined by the inability to sustain uninterrupted thought and actions, and disorientation are its earliest outward signs. As the state of confusion worsens, there are more global mental failings, including impairments of memory, perception, comprehension, problem solving, language, praxis, visuospatial function, and various aspects of emotional behavior that are each attributable to particular regions of the brain. In other instances an apparent confusional state may arise from an isolated deficit in mental function such as an impairment of language (*aphasia*), loss of memory (*amnesia*), or lack of appreciation of spatial relations of self and the external environment (*agnosia*), but the attributes of the problem are then quite different ([Chap. 25](#)). Confusion is also a feature of dementia, in which case the chronicity of the process, as in the instance of Alzheimer's disease, distinguishes it from an acute encephalopathy ([Chap. 26](#)).

The confused patient is usually subdued, not inclined to speak, and is physically inactive. A state of confusion that is accompanied by agitation, hallucinations, tremor, and illusions (misperceptions of environmental sight, sound, or touch) is termed *delirium*, as typified by delirium tremens from alcohol or drug withdrawal. In psychiatric circles, delirium often refers, albeit imprecisely, to all acute states of confusion with clouding of consciousness and incoherence of thought.

Approach to the Patient

Confusion and delirium always signify a disorder of the nervous system. They may be the major manifestation of a head injury; a seizure; drug toxicity (or drug withdrawal); a metabolic disorder resulting from hepatic, renal, pulmonary or cardiac failure; a systemic infection; meningitis or encephalitis; or a chronic dementing disease.

The search for these manifold causes begins with a careful history emphasizing the patient's condition before the onset of confusion. The clinical examination should focus on signs of diminished attentiveness, disorientation, and drowsiness and on the presence of localizing neurologic signs. From the clinical data the clinician is directed to the appropriate laboratory tests discussed further on. Often, even after all diagnostic tests are completed, one may still not know the cause of a confusional state. The proper approach is to observe the patient in the hospital for a number of days under stable conditions. New clues may appear or an obscure confusion perhaps related to a medication, may clear up, while other causes such as renal or hepatic failure may worsen and lead to coma.

Orientation and memory are tested by asking the patient in a forthright manner the date, inclusive of month, day, year, and day of week; the precise place; and some items of generally acknowledged and universally known information (the names of the President and Vice President, a recent national catastrophe, the state capital). Further probing may be necessary to reveal a defect -- why is the patient in the hospital; what is his or her address, zip code, telephone number, social security number? Problems of increasing complexity may be pursued, but they usually provide little additional information. Attention and coherence of thought can be gauged by the clarity of and speed of responses while the history is being given but are examined more explicitly by having the patient repeat strings of numbers (most adults easily retain seven digits forward and four backward), spell a word such as "world" backwards, and perform serial calculations -- tests of serial subtraction of 3 from 30 or 7 from 100 are useful. It is the inability to sustain coherent mental activity in performing tasks such as these that exposes the most subtle confusional states.

Other salient neurologic findings are the level of alertness, which fluctuates if there is drowsiness; indications of focal damage of the cerebrum such as hemiparesis, hemianopia, and aphasia; or adventitious movements of myoclonus or partial convulsions. The language of the confused patient may be disorganized and rambling, even to the extent of incorporating paraphasic words. These features, along with impaired comprehension that is due mainly to inattention, may be mistaken for aphasia.

One of the most specific signs of a metabolic encephalopathy is asterixis, which is an arrhythmic flapping tremor that is typically elicited by asking the patient to hold the arms outstretched with the wrists and hands fully extended. After a few seconds, there is a large jerking lapse in the posture of the hand and then a rapid return to the original position. The same movements can be appreciated in any tonically held posture, even of the tongue, and in extreme form the movements may intrude on voluntary limb motion. Bilateral asterixis always signifies a metabolic encephalopathy, e.g., from hepatic failure, hypercapnia, or from drug ingestion, especially with anticonvulsant

medications. Myoclonic jerking and tremor in an awake patient are typical of uremic encephalopathy or the use of antipsychotic drugs such as lithium, phenothiazines, or butyrophenones; myoclonus with coma may also signify anoxic cerebral damage.

Confusion in the postoperative period is common but at times so subtle as to escape attention. Cardiac and orthopedic procedures are particularly likely to produce disorientation or delirium in susceptible patients. Often a careful history will reveal that a mild but compensated dementia existed prior to the operation. Medications, particularly those with anticholinergic activity (including meperidine), inadvertent withdrawal from sleeping pills or alcohol, fever, and any of the endogenous metabolic derangements listed above may be responsible, or a stroke may have occurred.

Frequently confusion cannot be attributed to any single factor and it clears in several days. In many cases, particularly in the elderly, transient confusion and drowsiness arise with a febrile infection of the urinary tract, lungs, blood, or peritoneum. The term *septic encephalopathy* is currently used to describe this association, but the mechanism by which infection or inflammation leads to cerebral dysfunction is unknown. Fever can also alter brain function in a way that makes preexisting focal signs worse.

Distinguishing dementia from an acute confusional state is a great problem, especially in the elderly, since the two may coexist if a fever, other acute medical problem, or a poorly tolerated medication supervenes in a mildly demented patient, producing a so-called beclouded dementia. The memory loss of dementia brings about a confusional state that varies little in severity from hour to hour and day to day. Poor mental performance is derived mainly from incomplete recollection, inadequate access to names and ideas, and the inability to retain new information, thus affecting orientation and factual knowledge. In contrast to the acute confusional states, attention, alertness, and coherence are preserved until the most advanced stages. Eventually dementia produces a chronic confusion with breakdown of all types of mental performance, and the distinction from an acute encephalopathy depends mainly on the longstanding nature of the condition.

Treatment of the confusional state requires that all unnecessary medication be stopped, metabolic alterations be rectified, and infection be treated. Skilled nursing and a quiet room with a window are important. Careful explanations should be given at regular intervals to the family. In the elderly, regular reorientation and active measures to avoid risk factors (sleep deprivation, immobility, and vision and hearing impairments) reduce the number and severity of episodes of delirium in hospitalized patients.

COMA AND RELATED DISORDERS OF CONSCIOUSNESS

The unnatural situation of reduced alertness and responsiveness represents a continuum that in severest form is called *coma*, a deep sleeplike state from which the patient cannot be aroused. *Stupor* defines lesser degrees of unarousability in which the patient can be awakened only by vigorous stimuli, accompanied by motor behavior that leads to avoidance of uncomfortable or aggravating stimuli. *Drowsiness*, which is familiar to all persons, simulates light sleep and is characterized by easy arousal and the persistence of alertness for brief periods. Drowsiness and stupor are usually attended by some degree of confusion. In clinical practice these terms should be

supplemented by a narrative description of the level of arousal and of the type of responses evoked by various stimuli precisely as observed at the bedside. Such an account is preferable to ambiguous terms such as semicoma or obtundation, the definitions of which differ between physicians.

Several other neurologic conditions render patients apparently unresponsive and simulate coma, and certain other subsyndromes of coma must be considered separately because of their special significance. Among the latter, the *vegetative state* signifies an awake but unresponsive state. Most of these patients were earlier comatose and after a period of days or weeks emerge to an unresponsive state in which their eyelids are open, giving the appearance of wakefulness. Yawning, grunting, swallowing, as well as limb and head movements persist, but there are few, if any, meaningful responses to the external and internal environment -- in essence, an "awake coma." Although respiratory and autonomic functions are retained, the term "vegetative" is nonetheless unfortunate as it is subject to misinterpretation by lay persons. Always there are accompanying signs that indicate extensive damage in both cerebral hemispheres, e.g., decerebrate or decorticate limb posturing and absent responses to visual stimuli (see below). Cardiac arrest and head injuries are the most common causes of the vegetative state ([Chaps. 369](#) and [376](#)). The prognosis for regaining mental faculties once the vegetative state has supervened for several months is almost nil hence the term *persistent vegetative state*. Most instances of dramatic recovery, when investigated carefully, are found to yield to the usual rules for prognosis, but it must be acknowledged that rare instances of awakening to a condition of dementia and paralysis have been documented.

Certain other clinical states are prone to be misinterpreted as stupor or coma. *Akinetic mutism* refers to a partially or fully awake patient who is able to form impressions and think but remains immobile and mute, particularly when unstimulated. The condition may result from damage in the regions of the medial thalamic nuclei, the frontal lobes (particularly situated deeply or on the orbitofrontal surfaces), or from hydrocephalus. The term *abulia* is used to describe a mental and physical slowness and lack of impulse to activity that is in essence a mild form of akinetic mutism, with the same anatomic origins. *Catatonia* is a curious hypomobile and mute syndrome associated with a major psychosis. In the typical form patients appear awake with eyes open but make no voluntary or responsive movements, although they blink spontaneously, swallow, and may not appear distressed. As often, the eyes are half-open as if the patient is in a fog or light sleep. There are signs that indicate voluntary attempts to appear less than fully responsive, though it may take some ingenuity on the part of the examiner to demonstrate these. Eyelid elevation is actively resisted, blinking occurs in response to a visual threat, and the eyes move concomitantly with head rotation, all signs belying a brain lesion. It is characteristic but not invariable for the limbs to retain the posture, no matter how bizarre, in which they have been placed by the examiner ("waxy flexibility," or catalepsy.) Upon recovery, such patients have some memory of events that occurred during their catatonic stupor. The appearance is superficially similar to akinetic mutism, but clinical evidence of brain damage is lacking.

The *locked-in state* describes a pseudocoma in which an awake patient has no means of producing speech or volitional limb, face, and pharyngeal movements in order to indicate that he or she is awake, but vertical eye movements and lid elevation remain

unimpaired, thus allowing the patient to signal. Such individuals have written entire treatises using Morse code. Infarction or hemorrhage of the ventral pons, which transects all descending corticospinal and corticobulbar pathways, is the usual cause. A similar awake but deafferented state occurs as a result of total paralysis of the musculature in severe cases of Guillain-Barre syndrome ([Chap. 378](#)), critical illness neuropathy ([Chap. 376](#)), and pharmacologic neuromuscular blockade.

THE ANATOMY AND PHYSIOLOGY OF UNCONSCIOUSNESS

To the extent that all complex waking behaviors require the widespread participation of the cerebral cortex, consciousness cannot exist without the activity of these structures. A loosely grouped aggregation of neurons located in the upper brainstem and medial thalamus, the [RAS](#), maintains the cerebral cortex in a state of wakeful consciousness. It follows that the principal causes of coma are (1) lesions that damage a substantial portion of the RAS; (2) destruction of large portions of both cerebral hemispheres; and (3) suppression of thalamocerebral function by drugs, toxins, or by internal metabolic derangements such as hypoglycemia, anoxia, azotemia, or hepatic failure.

The classic animal experiments of Moruzzi and Magoun, published in 1949, and subsequent human clinicopathologic observations have established that the regions of the reticular formation that are critical to the maintenance of wakefulness extend from the caudal midbrain to the lower thalamus. A most important practical consideration derives from the anatomic proximity of the [RAS](#) to structures that are concerned with pupillary function and eye movements. Pupillary enlargement and loss of vertical and adduction movements of the globes suggest that upper brainstem damage may be the source of coma. Although circumscribed lesions confined to one or both cerebral hemispheres do not affect the brainstem RAS, a large mass on one side of the brain may cause coma by secondarily compressing the upper brainstem and consequently producing abnormalities of the pupils and eye movements (see discussion of transtentorial herniation below). This type of indirect effect is most typical of cerebral hemorrhages and of rapidly expanding tumors within a cerebral hemisphere. In all cases the degree of diminished alertness also relates to the rapidity of evolution and the extent of compression of the RAS.

The neurons of the [RAS](#) are thought to project rostrally to the cortex primarily via thalamic relay nuclei that in turn exert a tonic influence on the activity of the entire cerebral cortex. The behavioral arousal effected by somesthetic, auditory, and visual stimuli depends upon the rich reciprocal innervation that the RAS receives from these sensory systems. The relays between the RAS and the thalamic and cortical areas utilize a variety of neurotransmitters. Of these, the effect of arousal on acetylcholine and on the biogenic amines has been studied more extensively. Cholinergic fibers connect the midbrain to other areas of the upper brainstem, thalamus, and cortex. Serotonin and norepinephrine also subserve important functions in regulation of the sleep-wake cycle ([Chap. 27](#)). Their roles in arousal and coma have not been clearly established, although the alerting effects of amphetamines are likely to be mediated by catecholamine release.

Coma Due to Cerebral Mass Lesions and Herniations The cranial cavity is separated into compartments by infoldings of the dura -- the two cerebral hemispheres are

separated by the falx, and the anterior and posterior fossae by the tentorium. *Herniation* refers to displacement of brain tissue away from a mass and into a compartment that it normally does not occupy. Many of the signs associated with coma, and indeed coma itself, can be attributed to these tissue shifts. Herniation can be *transfalcial* (displacement of the cingulate gyrus under the falx and across the midline), *transtentorial* (displacement of the medial temporal lobe into the tentorial opening), and *foraminal* (downward forcing of the cerebellar tonsils into the foramen magnum; [Fig. 24-1](#)).

Uncal transtentorial herniation refers to impaction of the anterior medial temporal gyrus (the uncus) into the anterior portion of the tentorial opening. The displaced tissue compresses the third nerve as it traverses the subarachnoid space and results in enlargement of the ipsilateral pupil (putatively because the fibers subserving parasympathetic pupillary function are located peripherally in the nerve). The coma that follows may be due to lateral compression of the midbrain against the opposite tentorial edge by the displaced parahippocampal gyrus ([Fig. 24-2](#)). In some cases the lateral displacement causes compression of the opposite cerebral peduncle, producing a Babinski response and hemiparesis contralateral to the original hemiparesis (the Kernohan-Woltman sign). In addition to compressing the upper brainstem, tissue shifts, including herniations, may compress major blood vessels, particularly the anterior and posterior cerebral arteries as they pass over the tentorial reflections, thus producing brain infarctions. The distortions may also entrap portions of the ventricular system, resulting in regional hydrocephalus.

Central transtentorial herniation denotes a symmetric downward movement of the upper thalamic region through the tentorial opening. Miotic pupils and drowsiness are the heralding signs. Both temporal and central herniations are thought to cause progressive compression of the brainstem from above: first the midbrain, then the pons, and finally the medulla. The result is a sequential appearance of neurologic signs that corresponds to the affected level.

A direct relationship between the various configurations of transtentorial herniations and coma is, at best, tenuous. The orderly progression of signs from midbrain to medulla is often bypassed in catastrophic lesions where all brainstem functions are lost almost simultaneously. It is also clear that displacement of deep brain structures by a mass in any direction, with or without herniation, compresses the region of the [RAS](#) and results in coma. Furthermore, drowsiness and stupor typically occur with moderate lateral shifts at the level of the diencephalon (thalamus) well before transtentorial or other herniations are evident. Lateral shift is easily quantified on axial images of computed tomography (CT) and magnetic resonance imaging (MRI) scans ([Fig. 24-2](#)). In cases of *acutely appearing masses*, a fairly consistent and simple relationship exists between the degree of horizontal displacement of midline structures and the level of consciousness. Specifically, horizontal displacement of the pineal calcification of 3 to 5 mm is generally associated with drowsiness, 6 to 8 mm with stupor, and >9 mm with coma. At the same time, intrusion of the medial temporal lobe into the tentorial opening may be apparent as an obliteration of the cisterns that surround the upper brainstem.

Coma and Confusional States Due to Metabolic Disorders A large variety of systemic metabolic abnormalities cause coma by interrupting the delivery of energy

substrates (hypoxia, ischemia, hypoglycemia) or by altering neuronal excitability (drug and alcohol intoxication, anesthesia, and epilepsy). The same metabolic abnormalities that produce coma may in milder form induce widespread cortical dysfunction and an acute confusional state. Thus, in metabolic encephalopathies, clouded consciousness and coma are a continuum. Neuropathologic changes in the various metabolic failures are variable -- very evident in hypoxia-ischemia, manifest as astrocytic changes in hepatic coma, and negligible in renal and other metabolic encephalopathies.

Cerebral neurons are fully dependent on cerebral blood flow (CBF) and the related delivery of oxygen and glucose. CBF approximates 75 mL per 100 g/min in gray matter and 30 mL per 100 g/min in white matter (mean = 55 mL per 100 g/min); oxygen consumption is 3.5 mL per 100 g/min, and glucose utilization is 5 mg per 100 g/min. Brain stores of glucose provide energy for approximately 2 min after blood flow is interrupted, and oxygen stores last 8 to 10 s after the cessation of blood flow. Simultaneous hypoxia and ischemia exhaust glucose more rapidly. The electroencephalogram (EEG) rhythm in these circumstances becomes diffusely slowed, typical of metabolic encephalopathies, and as conditions of substrate delivery worsen, eventually all recordable brain electrical activity ceases. In almost all instances of metabolic encephalopathy, the global metabolic activity of the brain is reduced in proportion to the degree of unconsciousness.

Conditions such as hyponatremia, hyperosmolality, hypercapnia, hypercalcemia, and hepatic and renal failure are associated with a variety of alterations in neurons and astrocytes. It should be stated at the outset that the reversible effects of these conditions on the brain are not understood, but they may in different circumstances impair energy supplies, change ion fluxes across neuronal membranes, and cause neurotransmitter abnormalities. For example, the high brain ammonia concentration that is associated with hepatic coma interferes with cerebral energy metabolism and with the Na⁺, K⁺-ATPase pump, increases the number and size of astrocytes, alters nerve cell function, and causes increased concentrations of potentially toxic products of ammonia metabolism; it may also result in abnormalities of neurotransmitters, including possible "false" neurotransmitters that may be active at receptor sites. Apart from hyperammonemia, which of these mechanisms is of critical importance is not clear. The mechanism of the encephalopathy of renal failure is also not known. Unlike ammonia, urea itself does not produce central nervous system (CNS) toxicity. A multifactorial causation has been proposed, including increased permeability of the blood-brain barrier to toxic substances such as organic acids and an increase in brain calcium or cerebrospinal fluid (CSF) phosphate content. Likewise, the basis of confusion and drowsiness that commonly accompanies the septic state has not been clarified.

Coma and seizures are a common accompaniment of any large shifts in sodium and water balance. These changes in osmolality may be the result of a number of systemic medical disorders including diabetic ketoacidosis, the nonketotic hyperosmolar state, and hyponatremia from any cause (e.g., water intoxication, excessive secretion of antidiuretic hormone or atrial natriuretic peptides). The volume of brain water correlates with the level of consciousness in these states, but other factors also play a role. Sodium levels below 125 mmol/L induce confusion, and below 115 mmol/L are associated with coma and convulsions. In hyperosmolar coma the serum osmolality generally exceeds 350 mosmol/L. *As in most other metabolic encephalopathies, the*

severity of neurologic change depends to a large degree on the rapidity with which the serum changes occur. Hypercapnia depresses the level of consciousness in proportion to the rise in CO₂ tension in the blood and depends very much on the rapidity of change. The pathophysiology of other metabolic encephalopathies such as hypercalcemia, hypothyroidism, vitamin B₁₂ deficiency, and hypothermia are incompletely understood but must also reflect derangements of [CNS](#) biochemistry and membrane function.

Epileptic Coma Although all metabolic derangements in some way alter neuronal electrophysiologic function, epilepsy is the only primary excitatory disturbance of brain electrical activity that is encountered in clinical practice. Continuous, generalized electrical discharges of the cortex (*seizures*) are associated with coma even in the absence of epileptic motor activity (*convulsions*). The self-limited coma that follows seizures, termed the *postictal state*, may be due to exhaustion of energy reserves or effects of locally toxic molecules that are the byproduct of seizures. The postictal state produces a pattern of continuous, generalized slowing of the background [EEG](#) activity similar to that of other metabolic encephalopathies.

Pharmacologic Coma This class of encephalopathy is in large measure reversible and leaves no residual damage providing hypoxia does not supervene. Many drugs and toxins are capable of depressing nervous system function. Some produce coma by affecting both the brainstem nuclei, including the [RAS](#), and the cerebral cortex. The combination of cortical and brainstem signs, which occurs in certain drug overdoses, may lead to an incorrect diagnosis of structural brainstem disease.

Approach to the Patient

The diagnosis and management of coma depend on knowledge of its main causes (see "Differential Diagnosis," below) and on interpretation of salient clinical signs, notably brainstem reflexes and motor function. Acute respiratory and cardiovascular problems should be attended to prior to neurologic assessment. A complete medical evaluation, except for the vital signs, funduscopy, and examination for nuchal rigidity, may be deferred until the neurologic evaluation has established the severity and nature of coma.

History In many cases, the cause of coma is immediately evident (e.g., trauma, cardiac arrest, or known drug ingestion). In the remainder, historic information about the onset of coma is often sparse, but certain historic points are especially useful: (1) the circumstances and rapidity with which neurologic symptoms developed; (2) the details of any immediately preceding medical and neurologic symptoms (confusion, weakness, headache, fever, seizures, dizziness, double vision, or vomiting); (3) the use of medications, illicit drugs, or alcohol; and (4) chronic liver, kidney, lung, heart, or other medical disease. Direct interrogation or telephone calls to family and observers on the scene are an important part of the initial evaluation. Ambulance technicians often provide the most useful information in an enigmatic case.

General Physical Examination The temperature, pulse, respiratory rate and pattern, and blood pressure should be measured quickly as the evaluation is getting under way. Fever suggests a systemic infection, bacterial meningitis, or encephalitis; only rarely is it attributable to a brain lesion that has disturbed temperature-regulating centers. A slight elevation in temperature may follow vigorous convulsions. High body temperature, 42 to

44°C, associated with dry skin should arouse the suspicion of heat stroke or anticholinergic drug intoxication. Hypothermia is observed with bodily exposure to lowered environmental temperature; alcoholic, barbiturate, sedative, or phenothiazine intoxication; hypoglycemia; peripheral circulatory failure; or hypothyroidism. Hypothermia itself causes coma only when the temperature is <31°C. Tachypnea may indicate acidosis or pneumonia. Aberrant respiratory patterns that may reflect brainstem disorders are discussed below. Marked hypertension, a sign of hypertensive encephalopathy or a rapid rise in intracranial pressure, may occur acutely after head injury. Hypotension is characteristic of coma from alcohol or barbiturate intoxication, internal hemorrhage, myocardial infarction, sepsis, profound hypothyroidism, or Addisonian crisis. The funduscopic examination is invaluable in detecting subarachnoid hemorrhage (subhyaloid hemorrhages), hypertensive encephalopathy (exudates, hemorrhages, vessel-crossing changes, papilledema), and increased intracranial pressure (papilledema). Generalized cutaneous petechiae suggest thrombotic thrombocytopenic purpura, meningococcemia, or a bleeding diathesis from which an intracerebral hemorrhage arises.

Neurologic Assessment The patient should be observed first without examiner intervention. Patients who toss about, reach up toward the face, cross their legs, yawn, swallow, cough, or moan are close to being awake. Lack of restless movements on one side or an outturned leg at rest suggests a hemiplegia. Intermittent twitching movements of a foot, finger, or facial muscle may be the only sign of seizures. Multifocal myoclonus almost always indicates a metabolic disorder, particularly azotemia, anoxia, or drug ingestion (lithium and haloperidol are particularly prone to cause this sign), or the rarer conditions of spongiform encephalopathy and Hashimoto disease. In a drowsy and confused patient bilateral asterixis is a certain sign of metabolic encephalopathy or drug ingestion.

The terms *decorticate rigidity* and *decerebrate rigidity*, or "posturing," describe stereotyped arm and leg movements occurring spontaneously or elicited by sensory stimulation. Flexion of the elbows and wrists and supination of the arm (decortication) suggests severe bilateral damage rostral to the midbrain, whereas extension of the elbows and wrists with pronation (decerebration) indicates damage to motor tracts in the midbrain or caudal diencephalon. The less frequent combination of arm extension with leg flexion or flaccid legs is associated with lesions in the pons. These concepts have been adapted from animal work and cannot be applied with the same precision to coma in humans. In fact, acute and widespread cerebral disorders of any type, regardless of location, frequently cause limb extension, and almost all such extensor posturing becomes predominantly flexor as time passes. Thus, posturing alone cannot be utilized for precise anatomic localization. Posturing may also be unilateral and may coexist with purposeful limb movements, usually reflecting incomplete damage to the motor system.

Level of Arousal and Elicited Movements If the patient is not aroused by a conversational volume of voice, a sequence of increasingly intense stimuli is used to determine the patient's threshold of arousal and the optimal motor response of each limb. It should be recognized that the results of this testing may vary from minute to minute and that serial examinations are most useful. Tickling the nostrils with a cotton wisp is a moderate stimulus to arousal -- all but deeply stuporous and comatose patients will move the head away and rouse to some degree. Using the hand to remove

an offending stimulus such as this one represents an even lesser degree of unresponsiveness.

Responses to noxious stimuli should be appraised critically. Stereotyped posturing indicates severe dysfunction of the corticospinal system. Abduction-avoidance movement of a limb is usually purposeful and denotes an intact corticospinal system extending from the contralateral cortex to the ipsilateral spinal cord. Pressure on the knuckles or bony prominences and pinprick are humane forms of noxious stimulus; pinching the skin causes unsightly ecchymoses and is generally not necessary but may be useful in eliciting abduction withdrawal movements of the limbs. Conversely, consistent (obligatory) adduction and flexion of stimulated limbs may be reflexive in origin and implies damage to the corticospinal system. Brief clonus or twitching may occur at the end of extensor posturing movements and should not be mistaken for convulsions.

Brainstem Reflexes Assessment of brainstem damage is essential to the localization of the lesion in coma ([Fig. 24-3](#)). The brainstem reflexes that are conveniently assessed are pupillary responses to light, spontaneous and elicited eye movements, corneal responses, and the respiratory pattern. As a rule, when these brainstem activities are preserved, particularly the pupil reactions and eye movements, coma must necessarily be ascribed to bilateral hemispherical disease. The converse, however, is not always true as a mass in the hemispheres may be the proximate cause of coma but nonetheless produce brainstem signs.

PUPILS Pupillary reactions are examined with a bright, diffuse light (not an ophthalmoscope); if the response is absent, this should be confirmed by observation through a magnifying lens. Reaction to light is often difficult to appreciate in pupils <2 mm in diameter, and bright room lighting mutes pupillary reactivity. Normally reactive and round pupils of midsize (2.5 to 5 mm) essentially exclude midbrain damage, either primary or secondary to compression. One unreactive and enlarged pupil (>6 mm) or one that is poorly reactive signifies a compression or stretching of the third nerve from the effects of a mass above. Enlargement of the pupil contralateral to a mass may occur but is infrequent. It may be found in cases of subdural hematoma or brain hemorrhage, possibly as a result of compression of the midbrain or third nerve against the opposite tentorial margin. An oval and slightly eccentric pupil is a transitional sign that accompanies early midbrain-third nerve compression. The most extreme pupillary sign, bilaterally dilated and unreactive pupils, indicates severe midbrain damage, usually from compression by a mass or from ingestion of drugs with anticholinergic activity. The use of mydriatic eye drops, by a previous examiner or self-administered by the patient, and direct ocular trauma are among the causes of misleading pupillary enlargement.

Unilateral miosis in coma has been attributed to dysfunction of sympathetic efferents originating in the posterior hypothalamus and descending in the tegmentum of the brainstem to the cervical cord. Reactive and bilaterally small (1 to 2.5 mm) but not pinpoint pupils are seen in metabolic encephalopathies or in deep bilateral hemispherical lesions such as hydrocephalus or thalamic hemorrhage. Very small but reactive pupils (<1 mm) characterize narcotic or barbiturate overdoses but also occur with extensive pontine hemorrhage. The response to naloxone and the presence of reflex eye movements (see below) distinguish these. The unilaterally small pupil of the Horner

syndrome is detected by failure of the pupil to enlarge in the dark. It is an occasional finding with a large cerebral hemorrhage that affects the thalamus.

OCULAR MOVEMENTS Eye movements are the second sign of importance in determining if the brainstem has been damaged. Abnormalities, implicate both midbrain and pontine functions, thus permitting the analysis of a large portion of the brainstem. The eyes are first observed by elevating the lids and noting the resting position and spontaneous movements of the globes. Lid tone, tested by lifting the eyelids and noting their resistance to opening and the speed of closure, is reduced progressively as coma deepens. Horizontal divergence of the eyes at rest is normal in drowsiness. As coma deepens, the ocular axes may become parallel again. An abducted eye indicates a medial rectus paresis due to third nerve dysfunction and has the same significance as pupillary enlargement. An adducted eye indicates lateral rectus paresis due to a sixth nerve lesion and, when bilateral, is often a sign of increased intracranial pressure. With few exceptions, vertical separation of the ocular axes (one eye lower than the other, i.e. skew deviation) results from pontine or cerebellar lesions but may also be a manifestation of a partial third nerve palsy.

Spontaneous eye movements in coma often take the form of conjugate horizontal roving. This finding alone exonerates the midbrain and pons and has the same meaning as normal reflex eye movements (see below). Cyclic vertical downward movements are seen in some circumstances. "Ocular bobbing" describes a brisk downward and slow upward movement of the eyes associated with loss of horizontal eye movements and is diagnostic of bilateral pontine damage, characteristically from thrombosis of the basilar artery. "Ocular dipping" is a slower, arrhythmic downward movement followed by a faster upward movement in patients with normal reflex horizontal gaze; it usually indicates diffuse cortical anoxic damage. The eyes may turn down and inward as a result of thalamic and upper midbrain lesions, typically with thalamic hemorrhage or dilatation of the third ventricle from hydrocephalus. Conjugate horizontal ocular deviation to one extreme at rest indicates damage to the pons on the side of the gaze paresis or a lesion in the frontal lobe on the opposite side. This phenomenon may be summarized by the following maxim: *The eyes look toward a hemispherical lesion and away from a brainstem lesion*. On rare occasions, the eyes may turn paradoxically away from the side of a deep hemispherical lesion ("wrong-way eyes"). Many other complex and interesting eye movements are known but do not have the same salience in coma as the ones already mentioned.

Oculocephalic reflexes are automatic movements of the eyes elicited by moving the head from side to side or vertically. As the activity of the hemispheres is subdued from whatever cause, eye movements are evoked in the direction opposite to the head movement ([Fig. 24-3](#)). These movements, called somewhat inappropriately "doll's eyes" (which more accurately refers to the reflex elevation of the eyelids with flexion of the neck) are suppressed by visual fixation, which requires the patient to be awake. Induced adduction of the globes tends to be less complete than abduction, hence subtle abnormalities in the doll's-eye maneuver should be interpreted with caution. Oculocephalic reflexes are generated by brainstem mechanisms originating in the labyrinths and in cervical proprioceptors and require the undiminished activity of the third nerve nucleus in the midbrain, the contralateral sixth nerve nucleus in the pons, and the medial longitudinal fasciculus (MLF) that runs virtually the length of the

brainstem and links the two. Preservation of reflex eye movements (particularly adduction) therefore informs the examiner that coma is probably not due to an upper brainstem lesion and by implication that the origin of unconsciousness lies in the cerebral diencephalic structures. However, the opposite -- the absence of eye movements -- may signify either damage within the brainstem or profound metabolic depression of all neuronal function including the brainstem nuclei. Metabolic causes of depressed neuronal function include overdoses of phenytoin, tricyclic antidepressants, barbiturates, alcohol, phenothiazines, diazepam, and neuromuscular blocking agents. The presence of normal pupillary size and light reaction will distinguish most drug-induced comas from structural brainstem damage.

Thermal, or "caloric," stimulation of the vestibular apparatus (oculovestibular response) provides a more intense stimulus that may be used to confirm the absence of the oculocephalic reflex but gives fundamentally the same information. The test is performed by irrigating the external auditory canal with cool water in order to induce convection currents in the labyrinths. After a brief latency, the result is tonic deviation of both eyes (lasting 30 to 120 s) to the side of cool-water irrigation. The integrity of the third and sixth nerve complexes and brainstem pathways from the labyrinths to the midbrain are thereby confirmed, thus excluding a brainstem lesion as the cause of coma. If the cerebral hemispheres are functioning, as in catatonic or hysterical pseudocoma, an obligate rapid corrective nystagmus is generated away from the side of tonic deviation. (The acronym "COWS" has been used to remind generations of medical students of the direction of compensatory nystagmus -- "cold water opposite, warm water same"). The absence of this nystagmus despite conjugate deviation of the globes signifies that the cerebral hemispheres are damaged or profoundly suppressed.

By touching the cornea with a wisp of cotton, a response consisting of brief bilateral lid closure is normally observed. Although the corneal reflexes are rarely useful alone, they may corroborate eye-movement abnormalities because they also depend on the integrity of pontine pathways. The response is lost if the reflex connections between the fifth (afferent) and both seventh (efferent) cranial nerves within the pons are damaged. [CNS](#)depressant drugs diminish or eliminate the corneal responses soon after reflex eye movements are paralyzed but before the pupils become unreactive to light. The corneal (and pharyngeal) response may be lost for a time on the side of an acute hemiplegia.

RESPIRATION Respiratory patterns have received much attention in coma diagnosis but are of less localizing value in comparison to other brainstem signs. Shallow, slow, but regular breathing suggests metabolic or drug depression. Cheyne-Stokes respiration in its classic cyclic form, ending with a brief apneic period, signifies bihemispherical damage or metabolic suppression and commonly accompanies light coma. Rapid, deep (Kussmaul) breathing usually implies metabolic acidosis but may also occur with pontomesencephalic lesions and, of course, severe pneumonia. Agonal gasps reflect bilateral lower brainstem damage and are well known as the terminal respiratory pattern of severe brain damage. A number of other cyclic breathing variations are of lesser significance for localization.

LABORATORY STUDIES AND IMAGING

The following studies are most useful in the diagnosis of confusional states and coma: chemical-toxicologic analysis of blood and urine, cranial [CT](#) or [MRI](#), [EEG](#), and [CSF](#) examination. Arterial blood-gas analysis is helpful in patients with lung disease and acid-base disorders. Chemical blood determinations are obtained routinely to disclose metabolic, toxic, or drug-induced encephalopathies. The metabolic aberrations commonly encountered in clinical practice require measurements of electrolytes, glucose, calcium, osmolality, and renal (blood urea nitrogen) and hepatic (NH_3) function. Toxicologic analysis is necessary in any case of coma where the diagnosis is not immediately clear. However, the presence of exogenous drugs or toxins, especially alcohol, does not exclude the possibility that other factors, particularly head trauma, are also contributing to the clinical state. An ethanol level of 43 mmol/L (200 mg/dL) in nonhabituated patients generally causes confusion and impaired mental activity and of >65 mmol/L (300 mg/dL) is associated with stupor. The development of tolerance may allow the chronic alcoholic to remain awake at levels >87 mmol/L (400 mg/dL).

The increased availability of [CT](#) and [MRI](#) has focused attention on causes of coma that are radiologically detectable (e.g., hemorrhages, tumors, or hydrocephalus). Resorting primarily to this approach, although at times expedient, is imprudent because most cases of coma (and confusion) are metabolic or toxic in origin. The notion that a normal CT scan excludes anatomic lesions as the cause of coma is also erroneous. Bilateral hemisphere infarction, small brainstem lesions, encephalitis, meningitis, mechanical shearing of axons as a result of closed head trauma, absent cerebral perfusion associated with brain death, sagittal sinus thrombosis, and subdural hematomas that are isodense to adjacent brain are some of the lesions that may not be visible. Nevertheless, if the source of coma remains unknown, a scan should be obtained.

The EEG is useful in metabolic or drug-induced confusional states but is rarely diagnostic, with the important exceptions of coma due to clinically unrecognized seizures, to herpesvirus encephalitis and Creutzfeldt-Jakob disease. The amount of background slowing of the EEG is a useful reflection of the severity of any diffuse encephalopathy. Predominant high-voltage slowing (d or triphasic waves) in the frontal regions is typical of metabolic coma, as from hepatic failure, and widespread fast (b) activity implicates sedative drugs (diazepines, barbiturates). A pattern of "a coma," defined by widespread, variable 8- to 12-Hz activity, superficially resembles the normal a rhythm of waking but is unresponsive to environmental stimuli. It results from pontine or diffuse cortical damage and has a poor prognosis. Most importantly, EEG recordings reveal coma that is due to persistent epileptic discharges that are not clinically manifested as convulsions. Normal a activity on the EEG may also alert the clinician to the locked-in syndrome or to hysteria or catatonia.

Lumbar puncture is used more judiciously than in prior decades in cases of coma or confusion because neuroimaging scans effectively exclude intracerebral hemorrhage and most cases of subarachnoid hemorrhages. However, examination of the [CSF](#) is indispensable in the diagnosis of meningitis and encephalitis and in instances of suspected subarachnoid hemorrhage in which the scan is normal. Lumbar puncture should therefore not be deferred if meningitis is a possibility. Xanthochromia, indicating preexisting blood in the CSF, is documented by spinning the CSF in a large tube and comparing the supernatant to water. Measurement of the opening pressure within the subarachnoid space is of further help in interpreting abnormalities of the cell count and

protein content of the CSF.

DIFFERENTIAL DIAGNOSIS OF COMA ([Table 24-1](#))

In most instances confusion and coma are part of an obvious medical problem such as overt drug ingestion, hypoxia, stroke, trauma, or liver or kidney failure. Attention is then appropriately focused on the primary illness. Some general rules are helpful. Illnesses that cause sudden onset of coma are due to drug ingestion or to cerebral hemorrhage, trauma, cardiac arrest, epilepsy, or basilar artery embolism. Coma that appears subacutely is usually related to a preceding medical or neurologic problem, including the secondary brain swelling that surrounds a preexisting lesion such as a tumor or cerebral infarction.

The structural causes of coma can also be conceptualized in three broad categories: those without focal or lateralizing neurologic signs (e.g., metabolic encephalopathies); meningitis syndromes, characterized by stiff neck and an excess of cells in the spinal fluid (e.g., bacterial meningitis, subarachnoid hemorrhage); and those with prominent focal signs (e.g., stroke, cerebral hemorrhage). These are elaborated in [Table 24-1](#).

Cerebrovascular diseases cause the greatest difficulty in coma diagnosis. These are described in more detail in [Chap. 361](#) but may be summarized as follows: (1) basal ganglia and thalamic hemorrhage (acute but not instantaneous onset, vomiting, headache, hemiplegia, and characteristic eye signs); (2) pontine hemorrhage (sudden onset, pinpoint pupils, loss of reflex eye movements and corneal responses, ocular bobbing, posturing, hyperventilation, and excessive sweating); (3) cerebellar hemorrhage (occipital headache, vomiting, gaze paresis, and inability to stand); (4) basilar artery thrombosis (neurologic prodrome or warning spells, diplopia, dysarthria, vomiting, eye movement and corneal response abnormalities, and asymmetric limb paresis); and (5) subarachnoid hemorrhage (precipitous coma after headache and vomiting). The most common stroke, infarction in the territory of the middle cerebral artery, does not cause coma acutely but the surrounding edema may expand and act as a mass in a limited number of patients with large infarcts. The syndrome of acute hydrocephalus may accompany many intracranial diseases, particularly subarachnoid hemorrhage. Acute symmetric enlargement of both lateral ventricles causes headache and sometimes vomiting that may progress quickly to coma, with extensor posturing of the limbs, bilateral Babinski signs, small nonreactive pupils, and impaired vertical oculocephalic movements in the vertical direction.

If the history and examination do not suggest a large cerebral lesion or meningitic syndrome or a metabolic or drug cause, then information obtained from [CT](#) or [MRI](#) may be needed as outlined in [Table 24-1](#). As mentioned earlier, the majority of medical causes of coma can be established without a neuroimaging study.

BRAIN DEATH

This is a state in which there has been cessation of cerebral blood flow; as a result, global ischemia of the brain occurs while respiration is maintained by artificial means and the heart continues to function. It is the only type of brain damage that is unequivocally recognized as death. Many roughly equivalent criteria have been

advanced for the diagnosis of brain death, and it is essential to adhere to those endorsed as standards by the local medical community. Ideal criteria are simple, can be conducted at the bedside, and allow no chance of diagnostic error. They contain three essential elements: (1) widespread cortical destruction shown by deep coma -- unresponsiveness to all forms of stimulation; (2) global brainstem damage demonstrated by absent pupillary light reaction and the loss of oculovestibular and corneal reflexes; and (3) lower brainstem destruction indicated by complete apnea. The pulse rate is also invariant and unresponsive to atropine. Most patients have diabetes insipidus, but in some it develops only hours or days after the clinical signs of brain death. The pupils are often enlarged and may be mid-sized but should not be constricted. The absence of deep tendon reflexes is not required because the spinal cord may remain functional.

The proof that apnea is due to irreversible medullary damage requires that the P_{CO_2} be high enough to stimulate respiration during a test of spontaneous breathing (apnea test). This can be done safely in most patients by the use, prior to removing the ventilator, of diffusion oxygenation. This is accomplished by preoxygenation with 100% oxygen and then sustained during the test by a tracheal cannula connected to an oxygen supply. CO_2 tension increases approximately 0.3 to 0.4 kPa/min (2 to 3 mmHg/min) during apnea. At the end of the period of observation, typically several minutes in duration, arterial P_{CO_2} should be at least >6.6 to 8.0 kPa (50 to 60 mmHg) for the test to be valid.

The possibility of profound drug-induced or hypothermic depression of the nervous system should be excluded, and some period of observation, usually 6 to 24 h, is desirable during which this state is shown to be sustained. It is particularly advisable to delay clinical testing for up to 24 h if a cardiac arrest has caused brain death or if the inciting disease is not known. An isoelectric [EEG](#) may be used as a confirmatory test for total cerebral damage but is not absolutely necessary. Radionuclide brain scanning, cerebral angiography, or transcranial Doppler measurements may also be used to demonstrate the absence of cerebral blood flow, but with the exception of the latter, they are cumbersome and have not been correlated extensively with pathology.

There is no compelling reason to demonstrate brain death except when organ transplantation is involved. Although it is largely accepted in western society that the respirator can be disconnected from a brain-dead patient, problems frequently arise because of inadequate explanation and preparation of the family by the physician. Moreover, there is no proscription in reasonable medical practice to removing such support from patients who are not brain dead but whose condition is nonetheless hopeless and are likely to live for only a brief time.

TREATMENT

The immediate goal in acute coma is the prevention of further nervous system damage. Hypotension, hypoglycemia, hypercalcemia, hypoxia, hypercapnia, and hyperthermia should be corrected rapidly and assiduously. An oropharyngeal airway is adequate to keep the pharynx open in drowsy patients who are breathing normally. Tracheal intubation is indicated if there is apnea, upper airway obstruction, hypoventilation, or emesis, or if the patient is liable to aspirate because of coma. Mechanical ventilation is

required if there is hypoventilation or if there is an intracranial mass and a need to induce hypocapnia in order to lower intracranial pressure (ICP) as described below. Intravenous access is established and naloxone and dextrose are administered if narcotic overdose or hypoglycemia are even remote possibilities, and thiamine is given with glucose in order to avoid eliciting Wernicke disease in malnourished patients. In cases of suspected basilar thrombosis with brainstem ischemia, intravenous heparin or a thrombolytic agent is often utilized, keeping in mind that cerebellar and pontine hemorrhages resemble basilar artery occlusion. Physostigmine, when used by experienced physicians and with careful monitoring, may awaken patients with anticholinergic-type drug overdose, but many physicians believe that this is justified only to treat cardiac arrhythmias resulting from these overdoses. The use of benzodiazepine antagonists offers some prospect of improvement after overdoses of soporific drugs and has transient benefit in hepatic encephalopathy. Intravenous administration of hypotonic solutions should be monitored carefully in any serious acute brain illness because of the potential for exacerbating brain swelling. Cervical spine injuries must not be overlooked, particularly prior to attempting intubation or the evaluation of oculoccephalic responses. Headache accompanied by fever and meningismus indicates an urgent need for examination of the [CSF](#) to diagnose meningitis, and it is worth reemphasizing that lumbar puncture should not be delayed while awaiting a [aCT](#) scan. If the lumbar puncture in a case of suspected meningitis is delayed for any reason, an antibiotic such as a third-generation cephalosporin should be administered as soon as possible, preferably after obtaining blood cultures.

Enlargement of one pupil usually indicates secondary midbrain or third nerve compression by a hemispherical mass and requires that [ICP](#) be reduced ([Chap. 376](#)). Surgical evacuation of the mass may be appropriate in some cases (e.g., subdural and epidural hematoma.). Medical management to reduce ICP begins with the infusion of normal saline (safe because it is slightly hyperosmolar to serum). Therapeutic hyperventilation may be used to reduce ICP by inducing an arterial P_{CO_2} of 3.7 to 4.2 kPa (28 to 32 mmHg), but its effects are brief. Hyperosmolar therapy with mannitol or an equivalent agent is the mainstay of ICP reduction. It is used simultaneously with hyperventilation in critical cases. A ventricular puncture is necessary to decompress hydrocephalus if medical measures fail to improve alertness. The routine use of high-dose barbiturates and other neuronal-sparing agents soon after cardiac arrest or head trauma has not been shown in clinical studies to be beneficial, and glucocorticoids, although often still used, have no proven value except in cases of brain tumor with edema.

PROGNOSIS

The prediction of the outcome of coma must be considered in reference to long-term care and medical resources. One hopes to avoid the emotionally painful, hopeless outcomes associated with patients who are left severely disabled or vegetative. Several general rules pertain. The uniformly pessimistic outcome of the persistent vegetative state has already been mentioned. Children and young adults may have ominous early clinical findings such as abnormal brainstem reflexes and yet recover, so that temporization in offering a prognosis in this group of patients is wise. Metabolic comas have a far better prognosis than traumatic comas. All schemes for prognosis in adults should be taken as approximate indicators, and medical judgments must be tempered

by factors such as age, underlying systemic disease, and general medical condition. In an attempt to collect prognostic information from large numbers of patients with head injury, the Glasgow Coma Scale was devised; empirically it has predictive value in cases of brain trauma ([Chap. 369](#)). For anoxic and metabolic coma, clinical signs such as the pupillary and motor responses after 1 day, 3 days, and 1 week have been shown to have predictive value ([Chap. 376](#)). The absence of the cortical waves of the somatosensory evoked potentials has also proved a strong indicator of poor outcome in coma from any cause.

(Bibliography omitted in Palm version)

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25. APHASIAS AND OTHER FOCAL CEREBRAL DISORDERS - M.-Marsel Mesulam

The cerebral cortex of the human brain contains approximately 20 billion neurons spread over an area of 2 m². The *primary sensory* areas provide an obligatory portal for the entry of sensory information into cortical circuitry, whereas the *primary motor* areas provide a final common pathway for coordinating complex motor acts. The primary sensory and motor areas constitute <10% of the cerebral cortex. The rest is subsumed by unimodal, heteromodal, paralimbic, and limbic areas, collectively known as the *association cortex* (Fig. 25-1). The association cortex mediates the integrative processes that subserve cognition, emotion, and comportment. A systematic testing of these mental functions is necessary for the effective clinical assessment of the association cortex and its diseases.

According to current thinking, there are no centers for "hearing words," "perceiving space," or "storing memories." Cognitive and behavioral functions (domains) are coordinated by intersecting *large-scale neural networks* that contain interconnected cortical and subcortical components. The network approach to higher cerebral function has at least four implications of clinical relevance: (1) a single domain such as language or memory can be disrupted by damage to any one of several areas, as long as these areas belong to the same network; (2) damage confined to a single area can give rise to multiple deficits, involving the functions of all networks that intersect in that region; (3) damage to a network component may give rise to minimal or transient deficits if other parts of the network undergo compensatory reorganization; and (4) individual anatomic sites within a network display a relative (but not absolute) specialization for different behavioral aspects of the relevant function. Five anatomically defined large-scale networks are most relevant to clinical practice: a perisylvian network for language; a parietofrontal network for spatial cognition; an occipitotemporal network for face and object recognition; a limbic network for retentive memory; and a prefrontal network for attention and comportment.

THE LEFT PERISYLVIAN NETWORK FOR LANGUAGE: APHASIAS AND RELATED CONDITIONS

Language allows the communication and reshaping of thoughts and experiences by linking them to arbitrary symbols known as words. The neural substrate of language is composed of a distributed network centered in the perisylvian region of the *left* hemisphere. The posterior pole of this network is known as *Wernicke's area* and includes the posterior third of the superior temporal gyrus and a surrounding rim of the inferior parietal lobule. An essential function of Wernicke's area is to transform sensory inputs into their neural word representations so that these can establish the distributed associations that give the word its meaning. The anterior pole of the language network, known as *Broca's area*, includes the posterior part of the inferior frontal gyrus and a surrounding rim of prefrontal heteromodal cortex. An essential function of this area is to transform neural word representations into their articulatory sequences so that the words can be uttered in the form of spoken language. The sequencing function of Broca's area also appears to involve the ordering of words into sentences that contain a meaning-appropriate *syntax* (grammar). Wernicke's and Broca's areas are interconnected with each other and with additional perisylvian, temporal, prefrontal, and posterior parietal regions, making up a neural network subserving the various aspects of

language function. Damage to any one of these components or to their interconnections can give rise to language disturbances (*aphasia*). Aphasia should be diagnosed only when there are deficits in the formal aspects of language such as naming, word choice, comprehension, spelling, and syntax. Dysarthria and mutism do not, by themselves, lead to a diagnosis of aphasia. The language network shows a left hemisphere dominance pattern in the vast majority of the population. In approximately 90% of right handers and 60% of left handers, aphasia occurs only after lesions of the left hemisphere. In some individuals no hemispheric dominance for language can be discerned, and in some others (including a small minority of right handers) there is a right hemisphere dominance for language. A language disturbance occurring after a right hemisphere lesion in a right hander is called *crossed aphasia*.

Clinical Examination The clinical examination of language should include the assessment of naming, spontaneous speech, comprehension, repetition, reading, and writing. A deficit of naming (*anomia*) is the single most common finding in aphasic patients. When asked to name common objects (pencil or wristwatch) or their parts (eraser, lead, stem, band), the patient may fail to come up with the appropriate word, may provide a circumlocutious description of the object ("the thing for writing"), or may come up with the wrong word (*paraphasia*). If the patient offers an incorrect but legitimate word ("pen" for "pencil"), the naming error is known as a *semantic paraphasia*; if the word approximates the correct answer but is phonetically inaccurate ("plentil" for "pencil"), it is known as a *phonemic paraphasia*. Asking the patient to name body parts, geometric shapes, and component parts of objects (lapel of coat, cap of pen) can elicit mild forms of anomia in patients who can otherwise name common objects. In most anomias, the patient cannot retrieve the appropriate name when shown an object but can point to the appropriate object when the name is provided by the examiner. This is known as a one-way (or retrieval-based) naming deficit. A two-way naming deficit exists if the patient can neither provide nor recognize the correct name, indicating the presence of a language comprehension impairment. *Spontaneous speech* is described as "fluent" if it maintains appropriate output volume, phrase length, and melody or as "nonfluent" if it is sparse, halting, and average phrase length is below four words. The examiner should also note if the speech is paraphasic or circumlocutious; if it shows a relative paucity of substantive nouns and action verbs versus function words (prepositions, conjunctions); and if word order, tenses, suffixes, prefixes, plurals, and possessives are appropriate. *Comprehension* can be tested by assessing the patient's ability to follow conversation, by asking yes-no questions ("Can a dog fly?", "Does it snow in summer?") or asking the patient to point to appropriate objects ("Where is the source of illumination in this room?"). Statements with embedded clauses or passive voice construction ("If a tiger is eaten by a lion, which animal stays alive?") help to assess the ability to comprehend complex syntactic structure. Commands to close or open the eyes, stand up, sit down, or roll over should not be used to assess overall comprehension since appropriate responses aimed at such axial movements can be preserved in patients who otherwise have profound comprehension deficits.

Repetition is assessed by asking the patient to repeat single words, short sentences, or strings of words such as "No ifs, ands, or buts." The testing of repetition with tongue-twisters such as "hippopotamus" or "Irish constabulary" provides a better assessment of dysarthria and pallilalia than aphasia. Aphasic patients may have little difficulty with tongue-twisters but have a particularly hard time repeating a string of

function words. It is important to make sure that the number of words does not exceed the patient's attention span. Otherwise, the failure of repetition becomes a reflection of the narrowed attention span rather than an indication of an aphasic deficit. *Reading* should be assessed for deficits in reading aloud as well as comprehension. *Writing* is assessed for spelling errors, word order, and grammar. *Alexia* describes an inability to either read aloud or comprehend single words and simple sentences; *agraphia* (or dysgraphia) is used to describe an acquired deficit in the spelling or grammar of written language.

The correspondence between individual deficits of language function and lesion location does not display a rigid one-to-one relationship and should be conceptualized within the context of the distributed network model. Nonetheless, the classification of aphasic patients into specific clinical syndromes helps to determine the most likely anatomic distribution of the underlying neurologic disease and has implications for etiology and prognosis ([Table 25-1](#)). Aphasic syndromes can be divided into "central" syndromes, which result from damage to the two epicenters of the language network (Broca's and Wernicke's areas), and "disconnection" syndromes, which arise from lesions that interrupt the functional connectivity of these centers with each other and with the other components of the language network. The syndromes outlined below are idealizations; pure syndromes occur rarely.

Wernicke's Aphasia Comprehension is impaired for spoken and written language. Language output is fluent but is highly paraphasic and circumlocutious. The tendency for paraphasic errors may be so pronounced that it leads to strings of neologisms, which form the basis of what is known as "jargon aphasia." Speech contains large numbers of function words (e.g., prepositions, conjunctions) but few substantive nouns or verbs that refer to specific actions. The output is therefore voluminous but uninformative. For example, a patient attempts to describe how his wife accidentally threw away something important, perhaps his dentures: "We don't need it anymore, she says. And with it when that was downstairs was my teethtick...a...den...dentith...my dentist. And they happened to be in that bag...see? How could this have happened? How could a thing like this happen...So she says we won't need it anymore...I didn't think we'd use it. And now if I have any problems anybody coming a month from now, four months from now, or six months from now, I have a new dentist. Where my two...two little pieces of dentist that I use...that I...all gone. If she throws the whole thing away...visit some friends of hers and she can't throw them away."

Gestures and pantomime do not improve communication. The patient does not seem to realize that his or her language is incomprehensible and may appear angry and impatient when the examiner fails to decipher the meaning of a severely paraphasic statement. In some patients this type of aphasia can be associated with severe agitation and paranoid behaviors. One area of comprehension that may be preserved is the ability to follow commands aimed at axial musculature. The dissociation between the failure to understand simple questions ("What is your name") in a patient who rapidly closes his or her eyes, sits up, or rolls over when asked to do so is characteristic of Wernicke's aphasia and helps to differentiate it from deafness, psychiatric disease, or malingering. Patients with Wernicke's aphasia cannot express their thoughts in meaning-appropriate words and cannot decode the meaning of words in any modality of input. This aphasia therefore has expressive as well as receptive components.

Repetition, naming, reading, and writing are also impaired.

The lesion site most commonly associated with Wernicke's aphasia is the posterior portion of the language network and tends to involve at least parts of Wernicke's area. An embolus to the inferior division of the middle cerebral artery, and to the posterior temporal or angular branches in particular, is the most common etiology ([Chap. 361](#)). Intracerebral hemorrhage, severe head trauma, or neoplasm are other causes. A coexisting right hemi- or superior quadrantanopia is common, and mild right nasolabial flattening may be found, but otherwise the examination is often unrevealing. The paraphasic, neologistic speech in an agitated patient with an otherwise unremarkable neurologic examination may lead to the suspicion of a primary psychiatric disorder such as schizophrenia or mania, but the other components characteristic of acquired aphasia and the absence of prior psychiatric disease usually settle the issue. Some patients with Wernicke's aphasia due to intracerebral hemorrhage or head trauma may improve as the hemorrhage or the injury heals. In most other patients, prognosis for recovery is guarded.

Broca's Aphasia Speech is nonfluent, labored, interrupted by many word-finding pauses, and usually dysarthric. It is impoverished in function words but enriched in meaning-appropriate nouns and verbs. Abnormal word order and the inappropriate deployment of *bound morphemes* (word endings used to denote tenses, possessives, or plurals) lead to a characteristic agrammatism. Speech is telegraphic and pithy but quite informative. In the following passage, a patient with Broca's aphasia describes his medical history: "I see...the dotor, dotor sent me...Bosson. Go to hospital. Dotor...kept me beside. Two, tee days, doctor send me home."

Output may be reduced to a grunt or single word ("yes" or "no"), which is emitted with different intonations in an attempt to express approval or disapproval. In addition to fluency, naming and repetition are also impaired. Comprehension of spoken language is intact, except for syntactically difficult sentences with passive voice structure or embedded clauses. Reading comprehension is also preserved, with the occasional exception of a specific inability to read small grammatical words such as conjunctions and pronouns. The last two features indicate that Broca's aphasia is not just an "expressive" or "motor" disorder and that it may also involve a comprehension deficit for function words and syntax. Patients with Broca's aphasia can be tearful, easily frustrated, and profoundly depressed. Insight into their condition is preserved, in contrast to Wernicke's aphasia. Even when spontaneous speech is severely dysarthric, the patient may be able to display a relatively normal articulation of words when singing. This dissociation has been used to develop specific therapeutic approaches (melodic intonation therapy) for Broca's aphasia. Additional neurologic deficits usually include right facial weakness, hemiparesis or hemiplegia, and a buccofacial apraxia characterized by an inability to carry out motor commands involving oropharyngeal and facial musculature (e.g., patients are unable to demonstrate how to blow out a match or suck through a straw). Visual fields are intact. The cause is most often infarction of Broca's area (the inferior frontal convolution; [Fig. 25-1](#)) and surrounding anterior perisylvian and insular cortex, due to occlusion of the superior division of the middle cerebral artery ([Chap. 361](#)). Mass lesions including tumor, intracerebral hemorrhage, or abscess may also be responsible. Small lesions confined to the posterior part of Broca's area may lead to a nonaphasic and often reversible deficit of speech articulation, usually

accompanied by mild right facial weakness. When the cause of Broca's aphasia is stroke, recovery of language function generally peaks within 2 to 6 months, after which time further progress is limited.

Global Aphasia Speech output is nonfluent, and comprehension of spoken language is severely impaired. Naming, repetition, reading, and writing are also impaired. This syndrome represents the combined dysfunction of Broca's and Wernicke's areas and usually results from strokes that involve the entire middle cerebral artery distribution in the left hemisphere. Most patients are initially mute or say a few words, such as "hi" or "yes." Related signs include right hemiplegia, hemisensory loss, and homonymous hemianopia. Occasionally, a patient with a lesion in Wernicke's area will present with a global aphasia that soon resolves into Wernicke's aphasia.

Conduction Aphasia Speech output is fluent but paraphasic, comprehension of spoken language is intact, and repetition is severely impaired. Naming and writing are also impaired. Reading aloud is impaired, but reading comprehension is preserved. The lesion sites spare Broca's and Wernicke's areas but may induce a functional disconnection between the two so that neural word representations formed in Wernicke's area and adjacent regions cannot be conveyed to Broca's area for assembly into corresponding articulatory patterns. Occasionally, a Wernicke's area lesion gives rise to a transient Wernicke's aphasia that rapidly resolves into a conduction aphasia. The paraphasic output in conduction aphasia interferes with the ability to express meaning, but this deficit is not nearly as severe as the one displayed by patients with Wernicke's aphasia. Associated neurologic signs in conduction aphasia vary according to the primary lesion site.

Nonfluent Transcortical Aphasia (Transcortical Motor Aphasia) The features are similar to Broca's aphasia, but repetition is intact and agrammatism may be less pronounced. The neurologic examination may be otherwise intact, but a right hemiparesis can also exist. The lesion site disconnects the intact language network from prefrontal areas of the brain and usually involves the anterior watershed zone between anterior and middle cerebral artery territories or the supplementary motor cortex in the territory of the anterior cerebral artery.

Fluent Transcortical Aphasia (Transcortical Sensory Aphasia) Clinical features are similar to those of Wernicke's aphasia, but repetition is intact. The lesion site disconnects the intact core of the language network from other temporoparietal association areas. Associated neurologic findings may include hemianopia. Cerebrovascular lesions (e.g., infarctions in the posterior watershed zone) or neoplasms that involve the temporoparietal cortex posterior to Wernicke's area are the most common causes.

Isolation Aphasia This rare syndrome represents a combination of the two transcortical aphasias. Comprehension is severely impaired, and there is no purposeful speech output. The patient may parrot fragments of heard conversations (*echolalia*), indicating that the neural mechanisms for repetition are at least partially intact. This condition represents the pathologic function of the language network when it is isolated from other regions of the brain. Broca's and Wernicke's areas tend to be spared, but there is damage in surrounding frontal, parietal, and temporal cortex. Lesions are patchy and

can be associated with anoxia, carbon monoxide poisoning, or complete watershed zone infarctions.

Anomic Aphasia This form of aphasia may be considered the "minimal dysfunction" syndrome of the language network. Articulation, comprehension, and repetition are intact, but confrontation naming, word finding, and spelling are impaired. Speech is enriched in function words but impoverished in substantive nouns and verbs denoting specific actions. Language output is fluent but paraphasic, circumlocutious, and uninformative. The lesion sites can be anywhere within the left hemisphere language network, including the middle and inferior temporal gyri. *Anomic aphasia is the single most common language disturbance seen in head trauma, metabolic encephalopathy, and Alzheimer's disease.* The language impairment of Alzheimer's disease almost always leads to fluent aphasias (e.g., anomic, Wernicke's, conduction, or fluent transcortical aphasia). The insidious onset and relentless progression of nonfluent language disturbances (Broca's or nonfluent transcortical aphasia) can be seen in *primary progressive aphasia*, a degenerative syndrome most commonly associated with focal nonspecific neuronal loss or Pick's disease.

Pure Word Deafness This is not a true aphasic syndrome because the language deficit is modality-specific. The most common lesions are either bilateral or left-sided in the superior temporal gyrus. The net effect of the underlying lesion is to interrupt the flow of information from the unimodal auditory association cortex to Wernicke's area. Patients have no difficulty understanding written language and can express themselves well in spoken or written language. They have no difficulty interpreting and reacting to environmental sounds since primary auditory cortex and subcortical auditory relays are intact. Since auditory information cannot be conveyed to the language network, however, it cannot be decoded into neural word representations and the patient reacts to speech as if it were in an alien tongue that cannot be deciphered. Patients cannot repeat spoken language but have no difficulty naming objects. In time, patients with pure word deafness teach themselves lip reading and may appear to have improved. There may be no additional neurologic findings, but agitated paranoid reactions are frequent in the acute stages. Cerebrovascular lesions are the most frequent cause.

Pure Alexia Without Agraphia This is the visual equivalent of pure word deafness. The lesions (usually a combination of damage to the left occipital cortex and to a posterior sector of the corpus callosum -- the splenium) interrupt the flow of visual input into the language network. There is usually a right hemianopia, but the core language network remains unaffected. The patient can understand and produce spoken language, name objects in the left visual hemifield, repeat, and write. However, the patient acts as if illiterate when asked to read even the simplest sentence because the visual information from the written words (presented to the intact left visual hemifield) cannot reach the language network. Objects in the left hemifield may be named accurately because they activate nonvisual associations in the right hemisphere, which, in turn, can access the language network through transcallosal pathways anterior to the splenium. Patients with this syndrome may also lose the ability to name colors, although they can match colors. This is known as a *color anomia*. The most common etiology of pure alexia is a vascular lesion in the territory of the posterior cerebral artery or an infiltrating neoplasm in the left occipital cortex that involves the optic radiations as well as the crossing fibers of the splenium. Since the posterior cerebral artery also supplies medial temporal components

of the limbic system, the patient with pure alexia may also experience an amnesia, but this is usually transient because the limbic lesion is unilateral.

Aphemia There is an acute onset of severely impaired fluency (often mutism), which cannot be accounted by corticobulbar, cerebellar, or extrapyramidal dysfunction. Recovery is the rule and involves an intermediate stage of hoarse whispering. Writing, reading, and comprehension are intact, so this is not a true aphasic syndrome. Partial lesions of Broca's area or subcortical lesions that undercut its connections with other parts of the brain may be present. Occasionally, the lesion site is on the medial aspects of the frontal lobes and may involve the supplementary motor cortex of the left hemisphere.

Apraxia This generic term designates a complex motor deficit that cannot be attributed to pyramidal, extrapyramidal, cerebellar, or sensory dysfunction and that does not arise from the patient's failure to understand the nature of the task. The form that is most frequently encountered in clinical practice is known as *ideomotor apraxia*. Commands to perform a specific motor act ("cough," "blow out a match") or to pantomime the use of a common tool (a comb, hammer, straw, or toothbrush) in the absence of the real object cannot be followed. The patient's ability to comprehend the command is ascertained by demonstrating multiple movements and establishing that the correct one can be recognized. Some patients with this type of apraxia can imitate the appropriate movement (when it is demonstrated by the examiner) and show no impairment when handed the real object, indicating that the sensorimotor mechanisms necessary for the movement are intact. Some forms of ideomotor apraxia represent a disconnection of the language network from pyramidal motor systems: commands to execute complex movements are understood but cannot be conveyed to the appropriate motor areas, even though the relevant motor mechanisms are intact. *Buccofacial apraxia* involves apraxic deficits in movements of the face and mouth. *Limb apraxia* encompasses apraxic deficits in movements of the arms and legs. Ideomotor apraxia is almost always caused by lesions in the left hemisphere and is commonly associated with aphasic syndromes, especially Broca's aphasia and conduction aphasia. Its presence cannot be ascertained in patients with language comprehension deficits. The ability to follow commands aimed at axial musculature ("close the eyes," "stand up") is subserved by different pathways and may be intact in otherwise severely aphasic and apraxic patients. Patients with lesions of the anterior corpus callosum can display a special type of ideomotor apraxia confined to the left side of the body. Since the handling of real objects is not impaired, ideomotor apraxia, by itself, causes no limitation of daily living activities.

Ideational apraxia refers to a deficit in the execution of a goal-directed sequence of movements in patients who have no difficulty executing the individual components of the sequence. For example, when asked to pick up a pen and write, the sequence of uncapping the pen, placing the cap at the opposite end, turning the point towards the writing surface, and writing may be disrupted, and the patient may be seen trying to write with the wrong end of the pen or even with the removed cap. These motor sequencing problems are usually seen in the context of confusional states and dementias rather than focal lesions associated with aphasic conditions. *Limb-kinetic apraxia* involves a clumsiness in the actual use of tools that cannot be attributed to sensory, pyramidal, extrapyramidal, or cerebellar dysfunction. This condition can

emerge in the context of focal premotor cortex lesions or *corticobasal ganglionic degeneration*.

Gerstmann's Syndrome The combination of *acalculia* (impairment of simple arithmetic), *dysgraphia* (impaired writing), *finger anomia* (an inability to name individual fingers such as the index or thumb), and *right-left confusion* (an inability to tell whether a hand, foot, or arm of the patient or examiner is on the right or left side of the body) is known as Gerstmann's syndrome. In making this diagnosis it is important to establish that the finger and left-right naming deficits are not part of a more generalized anomia and that the patient is not otherwise aphasic. When Gerstmann's syndrome is seen in isolation, it is commonly associated with damage to the inferior parietal lobule (especially the angular gyrus) in the left hemisphere.

Aprosodia Variations of melodic stress and intonation influence the meaning and impact of spoken language. For example, the two statements "He *is* clever." and "He is clever?" contain an identical word choice and syntax but convey vastly different messages because of differences in the intonation and stress with which the statements are uttered. This aspect of language is known as *prosody*. Damage to perisylvian areas in the right hemisphere can interfere with speech prosody and can lead to syndromes of aprosodia. Ross has pointed out that damage to right hemisphere regions corresponding to Wernicke's area yields a greater impairment in the decoding of speech prosody, whereas damage to right hemisphere regions corresponding to Broca's area yields a greater impairment in the ability to introduce meaning-appropriate prosody into spoken language. The latter deficit is the most common type of aprosodia identified in clinical practice -- the patient produces grammatically correct language with accurate word choice but the statements are uttered in a monotone that interferes with the ability to convey the intended stress and affect. Patients with this type of aprosodia give the mistaken impression of being depressed or indifferent.

Subcortical Aphasias Damage to subcortical components of the language network (e.g., the striatum and thalamus of the left hemisphere) can also lead to aphasia. The resulting syndromes contain combinations of deficits in the various aspects of language but rarely fit the specific patterns described in [Table 25-1](#). An anomic aphasia accompanied by dysarthria or a fluent aphasia with hemiparesis should raise the suspicion of a subcortical lesion site.

THE PARIETOFRONTAL NETWORK FOR SPATIAL ORIENTATION: NEGLECT AND RELATED CONDITIONS

Hemispatial Neglect Adaptive orientation to significant events within the extrapersonal space is subserved by a large-scale network containing three major cortical components. The *cingulate cortex* provides access to a limbic-motivational mapping of the extrapersonal space, the *posterior parietal cortex* to a sensorimotor representation of salient extrapersonal events, and the *frontal eye fields* to motor strategies for attentional behaviors ([Fig. 25-2](#)). Subcortical components of this network include the striatum and the thalamus. Contralateral hemispatial neglect represents one outcome of damage to any of the cortical or subcortical components of this network. *The traditional view that hemispatial neglect always denotes a parietal lobe lesion is inaccurate.* In keeping with this anatomic organization, the clinical manifestations of

neglect display three behavioral components: sensory events (or their mental representations) within the neglected hemispace have a lesser impact on overall awareness; there is a paucity of exploratory and orienting acts directed toward the neglected hemispace; and the patient behaves as if the neglected hemispace was motivationally devalued.

According to one model of spatial cognition, the right hemisphere directs attention within the *entire* extrapersonal space, whereas the left hemisphere directs attention mostly within the contralateral right hemisphere. Consequently, unilateral left hemisphere lesions do not give rise to much contralesional neglect since the ipsilateral attentional mechanisms of the right hemisphere can compensate for the loss of the *contralaterally* directed attentional functions of the left hemisphere. Unilateral right hemisphere lesions, however, give rise to severe contralesional left hemispatial neglect because the unaffected left hemisphere does not contain ipsilateral attentional mechanisms. This model is consistent with clinical experience, which shows that contralesional neglect is more common, severe, and lasting after damage to the right hemisphere than after damage to the left hemisphere. Severe neglect for the right hemispace is rare, even in left handers with left hemisphere lesions.

Patients with severe neglect may fail to dress, shave, or groom the left side of the body; may fail to eat food placed on the left side of the tray; and may fail to read the left half of sentences. When the examiner draws a large circle [12 to 16 cm (5 to 6 in.) in diameter] and asks the patient to place the numbers 1 to 12 as if the circle represented the face of a clock, there is a tendency to crowd the numbers on the right side and leave the left side empty. When asked to copy a simple line drawing, the patient fails to copy detail on the left; and when asked to write, there is a tendency to leave an unusually wide margin on the left.

Two bedside tests that are useful in assessing neglect are *simultaneous bilateral stimulation* and *visual target cancellation*. In the former, the examiner provides either unilateral or simultaneous bilateral stimulation in the visual, auditory, and tactile modalities. Following right hemisphere injury, patients who have no difficulty detecting unilateral stimuli on either side experience the bilaterally presented stimulus as coming only from the right. This phenomenon is known as *extinction* and is a manifestation of the sensory-representational aspect of hemispatial neglect. In the target detection task, targets (e.g., A's) are interspersed with foils (e.g., other letters of the alphabet) on a 21.5 × 28.0 cm (8.5 × 11 in.) sheet of paper and the patient is asked to circle all the targets. A failure to detect targets on the left is a manifestation of the exploratory deficit in hemispatial neglect. Hemianopia, by itself, does not interfere with performance in this task since the patient is free to turn the head and eyes to the left. The normal tendency in target detection tasks is to start from the left upper quadrant and move systematically in horizontal or vertical sweeps. Some patients show a tendency to start the process from the right and proceed in a haphazard fashion. This represents a subtle manifestation of left neglect, even if the patient eventually manages to detect all the appropriate targets. Some patients with neglect may also deny the existence of hemiparesis and may even deny ownership of the paralyzed limb, a condition known as *anosognosia*.

Cerebrovascular lesions and neoplasms in the right hemisphere are the most common

causes of hemispatial neglect. Depending on the site of the lesion, the patient with neglect may also have hemiparesis, hemihypesthesia, and hemianopia on the left, but these are not invariant findings. The majority of patients display considerable improvement of hemispatial neglect, usually within the first several weeks.

Balint's Syndrome, Simultanagnosia, Dressing Apraxia, and Construction Apraxia

Bilateral involvement of the network for spatial attention, especially its parietal components, leads to a state of severe spatial disorientation known as *Balint's syndrome*. Balint's syndrome involves deficits in the orderly visuomotor scanning of the environment (*oculomotor apraxia*) and in accurate manual reaching toward visual targets (*optic ataxia*). The third and most dramatic component of Balint's syndrome is known as *simultanagnosia* and reflects an inability to integrate visual information in the center of gaze with more peripheral information. The patient gets stuck on the detail that falls in the center of gaze without attempting to scan the visual environment for additional information. The patient with simultanagnosia "misses the forest for the trees." Complex visual scenes cannot be grasped in their entirety, leading to severe limitations in the visual identification of objects and scenes. For example, a patient who is shown a table lamp and asked to name the object may look at its circular base and call it an ash tray. Some patients with simultanagnosia report that objects they look at may suddenly vanish, probably indicating an inability to look back at the original point of gaze after brief saccadic displacements. Movement and distracting stimuli greatly exacerbate the difficulties of visual perception. Simultanagnosia can sometimes occur without the other two components of Balint's syndrome.

A modification of the letter cancellation task described above can be used for the bedside diagnosis of simultanagnosia. In this modification, some of the targets (e.g., A's) are made to be much larger than the others [7.5 to 10 cm vs. 2.5 cm (3 to 4 in. vs. 1 in.) in height], and all targets are embedded among foils. Patients with simultanagnosia display a counterintuitive but characteristic tendency to miss the larger targets ([Fig. 25-3](#)). This occurs because the information needed for the identification of the larger targets cannot be confined to the immediate line of gaze and requires the integration of visual information across a more extensive field of view. The greater difficulty in the detection of the larger targets also indicates that poor acuity is not responsible for the impairment of visual function and that the problem is central rather than peripheral. Balint's syndrome results from bilateral dorsal parietal lesions; common settings include watershed infarction between the middle and posterior cerebral artery territories, hypoglycemia, sagittal sinus thrombosis, or atypical forms of Alzheimer's disease. In patients with Balint's syndrome due to stroke, bilateral visual field defects (usually inferior quadrantanopias) are common.

Another manifestation of bilateral (or right sided) dorsal parietal lobe lesions is *dressing apraxia*. The patient with this condition is unable to align the body axis with the axis of the garment and can be seen struggling as he or she holds a coat from its bottom or extends his or her arm into a fold of the garment rather than into its sleeve. Lesions that involve the posterior parietal cortex also lead to severe difficulties in copying simple line drawings. This is known as a *construction apraxia* and is much more severe if the lesion is in the right hemisphere. In some patients with right hemisphere lesions, the drawing difficulties are confined to the left side of the figure and represent a manifestation of hemispatial neglect; in others, there is a more universal deficit in reproducing contours

and three-dimensional perspective. Dressing apraxia and construction apraxia represent special instances of a more general disturbance in spatial orientation.

THE OCCIPITOTEMPORAL NETWORK FOR FACE AND OBJECT RECOGNITION: PROSOPAGNOSIA AND OBJECT AGNOSIA

Perceptual information about faces and objects is initially encoded in primary (striate) visual cortex and adjacent (upstream) peristriate visual association areas. This information is subsequently relayed first to the downstream visual association areas of occipitotemporal cortex and then to other heteromodal and paralimbic areas of the cerebral cortex. Bilateral lesions in the fusiform and lingual gyri of occipitotemporal cortex disrupt this process and interfere with the ability of otherwise-intact perceptual information to activate the distributed multimodal associations that lead to the recognition of faces and objects. The resultant face and object recognition deficits are known as *prosopagnosia* and *visual object agnosia*.

The patient with prosopagnosia cannot recognize familiar faces, including, sometimes, the reflection of his or her own face in the mirror. This is not a perceptual deficit since prosopagnosic patients can easily tell if two faces are identical or not. Furthermore, a prosopagnosic patient who cannot recognize a familiar face by visual inspection alone can use auditory cues to reach appropriate recognition if allowed to listen to the person's voice. The deficit in prosopagnosia is therefore modality-specific and reflects the existence of a lesion that prevents the activation of otherwise intact multimodal templates by relevant visual input. Damasio has pointed out that the deficit in prosopagnosia is not limited to the recognition of faces but that it can also extend to the recognition of individual members of larger generic object groups. For example, prosopagnosic patients characteristically have no difficulty with the generic identification of a face as a face or of a car as a car, but they cannot recognize the identity of an individual face or the make of an individual car. This reflects a visual recognition deficit for proprietary features that characterize individual members of an object class. When recognition problems become more generalized and extend to the generic identification of common objects, the condition is known as visual object agnosia. In contrast to prosopagnosic patients, those with object agnosia cannot recognize a face as a face or a car as a car. It is important to distinguish visual object agnosia from anomia. The patient with anomia cannot name the object but can describe its use. In contrast, the patient with visual agnosia is unable either to name a visually presented object or to describe its use. The characteristic lesions in prosopagnosia and visual object agnosia consist of bilateral infarctions in the territory of the posterior cerebral arteries. Associated deficits can include visual field defects (especially superior quadrantanopias) or a centrally based color blindness known as achromatopsia. Rarely, the responsible lesion is unilateral. In such cases, prosopagnosia is associated with lesions in the right hemisphere and object agnosia with lesions in the left.

THE LIMBIC NETWORK FOR MEMORY: AMNESIAS

Limbic and paralimbic areas (such as the hippocampus, amygdala, and entorhinal cortex), the anterior and medial nuclei of the thalamus, the medial and basal parts of the striatum, and the hypothalamus collectively constitute a distributed network known as the *limbic system*. The behavioral affiliations of this network include the coordination of

emotion, motivation, autonomic tone, and endocrine function. An additional area of specialization for the limbic network, and the one which is of most relevance to clinical practice, is that of declarative (conscious) memory for recent episodes and experiences. A disturbance in this function is known as *amnesic state*. In the absence of deficits in motivation, attention, language, or visuospatial function, the clinical diagnosis of a persistent global amnesic state is always associated with bilateral damage to the limbic network, usually within the hippocampo-entorhinal complex or the thalamus.

The memory disturbance in the amnesic state is multimodal and includes retrograde and anterograde components. The *retrograde amnesia* involves an inability to recall experiences that occurred before the onset of the amnesic state. Relatively recent events are more vulnerable to retrograde amnesia than more remote events. A patient who comes to the emergency room complaining that he cannot remember his identity but who can remember the events of the previous day is almost certainly not suffering from a neurologic cause of memory disturbance. The second and most important component of the amnesic state is the *anterograde amnesia*, which indicates an inability to store, retain, and recall new knowledge. Patients with amnesic states cannot remember what they ate a few minutes ago or the details of an important event they may have experienced a few hours ago. In the acute stages, there may also be a tendency to fill in memory gaps with inaccurate, fabricated, and often implausible information. This is known as *confabulation*. Patients with the amnesic syndrome forget that they forget and tend to deny the existence of a memory problem when questioned.

The patient with an amnesic state is almost always disoriented, especially to time. Accurate temporal orientation and accurate knowledge of current news rule out a major amnesic state. Memory can be tested with a list of four to five words read aloud by the examiner up to five times or until the patient can immediately repeat the entire list without intervening delay. In the next phase of testing, the patient is allowed to concentrate on the words and to rehearse them internally for 1 min before being asked to recall them. Accurate performance in this phase indicates that the patient is motivated and sufficiently attentive to hold the words on-line for at least 1 min. The final phase of the testing involves a retention period of 5 to 10 min, during which the patient is engaged in other tasks. Adequate recall at the end of this interval requires off-line storage, retention, and retrieval. Amnesic patients fail this phase of the task and may even forget that they were given a list of words to remember. Accurate recognition of the words by multiple choice in a patient who cannot recall them indicates a less severe memory disturbance that affects mostly the retrieval stage of memory.

Many neurologic diseases can give rise to an amnesic state. These include tumors (of the sphenoid wing, posterior corpus callosum, thalamus, or medial temporal lobe), infarctions (in the territories of the anterior or posterior cerebral arteries), head trauma, herpes simplex encephalitis, Wernicke-Korsakoff encephalopathy, paraneoplastic limbic encephalitis, and degenerative dementias such as Alzheimer's or Pick's disease. The one common denominator of all these diseases is that they lead to the bilateral lesions within one or more components in the limbic network, most commonly the hippocampus, entorhinal cortex, the mammillary bodies of the hypothalamus, and the limbic thalamus. Occasionally, unilateral left-sided lesions can give rise to an amnesic state, but the memory disorder tends to be transient. Depending on the nature and distribution of the underlying neurologic disease, the patient may also have visual field deficits, eye

movement limitations, or cerebellar findings. In many patients, such as those with *transient global amnesia* ([Chap. 26](#)), there are no associated neurologic findings; this sometimes leads incorrectly to the diagnosis of a psychiatric disorder.

Although the limbic network is the site of damage for amnestic states, it is almost certainly not the storage site for memories. Memories are stored in widely distributed form throughout the association cortex. The role attributed to the limbic network is to bind these distributed fragments into coherent events and experiences that can sustain conscious recall. Damage to the limbic network does not necessarily destroy memories but interferes with their conscious (declarative) recall in coherent form. The individual fragments of information remain preserved despite the limbic lesions and can sustain what is known as *implicit memory*. For example, patients with amnestic states can acquire new motor or perceptual skills, even though they may have no conscious knowledge of the experiences that led to the acquisition of these skills.

THE PREFRONTAL NETWORK FOR ATTENTION AND COMPORTMENT

Approximately one-third of all the cerebral cortex in the human brain is located in the frontal lobes. The frontal lobes can be subdivided into motor-premotor, dorsolateral prefrontal, medial prefrontal, and orbitofrontal components. The terms *frontal lobe syndrome* and *prefrontal cortex* refer only to the last three of these four components. These are the parts of the cerebral cortex that show the greatest phylogenetic expansion in primates and especially in humans. The dorsolateral prefrontal, medial prefrontal, and orbitofrontal areas, and the subcortical structures with which they are interconnected (i.e., the head of the caudate and the dorsomedial nucleus of the thalamus), collectively make up a large-scale network that coordinates exceedingly complex aspects of human cognition and comportment.

The prefrontal network plays an important role in behaviors that require an integration of thought with emotion and motivation. There is no simple formula for summarizing the diverse functional affiliations of the prefrontal network. Its integrity appears important for the simultaneous awareness of context, options, consequences, relevance, and emotional impact so as to allow the formulation of adaptive inferences, decisions, and actions. Damage to this part of the brain impairs mental flexibility, reasoning, hypothesis formation, abstract thinking, foresight, judgment, the on-line (attentive) holding of information, and the ability to inhibit inappropriate responses. Behaviors impaired by prefrontal cortex lesions, especially those related to the manipulation of mental content, are often referred to as "executive functions."

Even very large bilateral prefrontal lesions may leave all sensory, motor, and basic cognitive functions intact while leading to isolated but dramatic alterations of personality and comportment. The most common clinical manifestations of damage to the prefrontal network take the form of two relatively distinct syndromes. In the *frontal abulic syndrome*, the patient shows a loss of initiative, creativity, and curiosity and displays a pervasive emotional blandness and apathy. In the *frontal disinhibition syndrome*, the patient becomes socially disinhibited and shows severe impairments of judgment, insight, and foresight. The dissociation between intact intellectual function and a total lack of even rudimentary common sense is striking. Despite the preservation of all essential memory functions, the patient cannot learn from experience and continues to

display inappropriate behaviors without appearing to feel emotional pain, guilt, or regret when such behaviors repeatedly lead to disastrous consequences. The impairments may emerge only in real-life situations when behavior is under minimal external control and may not be apparent within the structured environment of the medical office. Testing judgment by asking patients what they would do if they detected a fire in a theater or found a stamped and addressed envelope on the road is not very informative since patients who answer these questions wisely in the office may still act very foolishly in the more complex real-life setting. The physician must therefore be prepared to make a diagnosis of frontal lobe disease on the basis of historic information alone even when the office examination of mental state may be quite intact.

The abulic syndrome tends to be associated with damage to the dorsolateral prefrontal cortex, and the disinhibition syndrome with the medial prefrontal or orbitofrontal cortex. These syndromes tend to arise almost exclusively after bilateral lesions, most frequently in the setting of head trauma, stroke, ruptured aneurysms, hydrocephalus, tumors (including metastases, glioblastoma, and falx or olfactory groove meningiomas), or focal degenerative diseases. Unilateral lesions confined to the prefrontal cortex may remain silent until the pathology spreads to the other side. The emergence of developmentally primitive reflexes such as grasping, rooting, and sucking are seen primarily in patients with large structural lesions that extend into the premotor components of the frontal lobes or in the context of metabolic encephalopathies. The vast majority of patients with prefrontal lesions and frontal lobe behavioral syndromes do not display these reflexes.

Damage to the frontal lobe disrupts a variety of attention-related functions including working memory (the transient on-line holding of information), concentration span, verbal fluency, the scanning and retrieval of stored information, the inhibition of immediate but inappropriate responses, and mental flexibility. The capacity for focusing on a trend of thought and the ability to voluntarily shift the focus of attention from one thought or stimulus to another can become impaired. Digit span (which should be seven forward and five reverse) is decreased; the recitation of the months of the year in reverse order (which should take less than 15 s) is slowed; and the number of words starting with a, f, or s that can be generated in 1 min (normally 12 or more per letter) is diminished even in nonaphasic patients. Characteristically, there is a progressive slowing of performance as the task proceeds; e.g., the patient asked to count backwards by 3s may say "100, 97, 94...91,...88," etc., and may not complete the task. In go-no go tasks (where the instruction is to raise the finger upon hearing one tap but to keep it still upon hearing two taps), the patient shows a characteristic inability to keep still in response to the "no go" stimulus; mental flexibility (tested by the ability to shift from one criterion to another in sorting or matching tasks) is impoverished; distractibility by irrelevant stimuli is increased; and there is a pronounced tendency for impersistence and perseveration.

These attentional deficits disrupt the orderly registration and retrieval of new information and lead to *secondary* memory deficits. Such memory deficits can be differentiated from the *primary* memory impairments of the amnesic state by showing that they improve when the attentional load of the task is decreased. Working memory (also known as immediate memory) is an attentional function based on the temporary on-line holding of information. It is closely associated with the integrity of the prefrontal network and the ascending reticular activating system. Retentive memory, on the other hand, depends

on the stable (off-line) storage of information and is associated with the integrity of the limbic network. The distinction of the underlying neural mechanisms is illustrated by the observation that severely amnesic patients who cannot remember events that occurred a few minutes ago may have intact if not superior working memory capacity as shown in tests of digit span.

Lesions in the caudate nucleus or in the dorsomedial nucleus of the thalamus (subcortical components of the prefrontal network) can also produce a frontal lobe syndrome. This is one reason why the mental state changes associated with degenerative basal ganglia diseases, such as Parkinson's or Huntington's disease, may take the form of a frontal lobe syndrome. Because of its widespread connections with other regions of association cortex, one essential computational role of the prefrontal network is to function as an integrator, or "orchestrator," for other networks. Bilateral multifocal lesions of the cerebral hemispheres, none of which are individually large enough to cause specific cognitive deficits such as aphasia or neglect, can collectively interfere with the connectivity and integrating function of prefrontal cortex. A frontal lobe syndrome is the single most common behavioral profile associated with a variety of bilateral multifocal brain diseases including metabolic encephalopathy, multiple sclerosis, vitamin B₁₂ deficiency, and others. In fact, the vast majority of patients with the clinical diagnosis of a frontal lobe syndrome tend to have lesions that do not involve prefrontal cortex but involve either the subcortical components of the prefrontal network or its connections with other parts of the brain. In order to avoid making a diagnosis of "frontal lobe syndrome" in a patient with no evidence of frontal cortex disease, it is advisable to use the diagnostic term *frontal network syndrome*, with the understanding that the responsible lesions can lie anywhere within this distributed network.

The patient with frontal lobe disease raises potential dilemmas in differential diagnosis: the abulia and blandness may be misinterpreted as depression, and the disinhibition as mania or acting-out. Appropriate intervention may be delayed while a treatable tumor keeps expanding. An informed approach to frontal lobe disease and its comportmental manifestations may help to avoid such errors.

CARING FOR THE PATIENT WITH DEFICITS OF HIGHER CEREBRAL FUNCTION

Some of the deficits described in this chapter are so complex that they may bewilder not only the patient and family but also the physician. It is imperative to carry out a systematic clinical evaluation in order to characterize the nature of the deficits and explain them in lay terms to the patient and family. Such an explanation can allay at least some of the anxieties, address the mistaken impression that the deficit (e.g., social disinhibition or inability to recognize family members) is psychologically motivated, and lead to practical suggestions for daily living activities. The consultation of a skilled neuropsychologist may aid in the formulation of diagnosis and management. Patients with simultanagnosia, for example, may benefit from the counterintuitive instruction to stand back when they cannot find an item so that a greater search area falls within the immediate field of gaze. In some patients, the history may be more important than the bedside examination. For example, patients with frontal lobe disease can be extremely irritable and abusive to spouses and yet display all the appropriate social graces during the visit to the medical office.

Reactive depression is common in patients with higher cerebral dysfunction and should be treated. These patients may be sensitive to the usual doses of antidepressants or anxiolytics and deserve a careful titration of dosage. Brain damage may cause a dissociation between feeling states and their expression, so that a patient who may superficially appear jocular could still be suffering from an underlying depression that deserves to be treated. In many cases, agitation may be controlled with reassurance. In other cases, treatment with benzodiazepines or sedating antidepressants may become necessary. The use of neuroleptics for the control of agitation should be reserved for refractory cases since extrapyramidal side effects are frequent in patients with coexisting brain damage.

Spontaneous improvement of cognitive deficits due to acute neurologic lesions is common. It is most rapid in the first few weeks but may continue for up to 2 years, especially in young individuals with single brain lesions. The mechanisms for this recovery are incompletely understood. Some of the initial deficits appear to arise from remote dysfunction (diaschisis) in parts of the brain that are interconnected with the site of initial injury. Improvement in these patients may reflect, at least in part, a normalization of the remote dysfunction. Other mechanisms may involve functional reorganization in surviving neurons adjacent to the injury or the compensatory use of homologous structures, e.g., the right superior temporal gyrus with recovery from Wernicke's aphasia. In some patients with large lesions involving Broca's and Wernicke's areas, only Wernicke's area may show contralateral compensatory reorganization (or bilateral functionality), giving rise to a situation where a lesion that should have caused a global aphasia becomes associated with a residual Broca aphasia. Prognosis for recovery from aphasia is best when Wernicke's area is spared. Cognitive rehabilitation procedures have been used in the treatment of higher cortical deficits. There are few controlled studies, but some do show a benefit of rehabilitation in the recovery from hemispatial neglect and aphasia. Some types of deficits may be more prone to recovery than others. For example, patients with nonfluent aphasias are more likely to benefit from speech therapy than patients with fluent aphasias and comprehension deficits. In general, lesions that lead to a denial of illness (e.g., anosognosia) are associated with cognitive deficits that are more resistant to rehabilitation. The recovery of higher cortical dysfunction is rarely complete. Periodic neuropsychological assessment is necessary for quantifying the pace of the improvement and for generating specific recommendations for cognitive rehabilitation, modifications in the home environment, and the timetable for returning to school or work.

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26. MEMORY LOSS AND DEMENTIA - *Thomas D. Bird*

DEFINITION

Dementia is a serious and common problem that affects more than 4 million Americans and costs society more than \$50 billion annually. Ten percent of persons over age 70 and 20 to 40% of individuals over age 85 have clinically identifiable memory loss. Dementia is a syndrome with many causes. A simple definition of dementia is a deterioration in cognitive abilities that impairs the previously successful performance of activities of daily living. Memory is the most common and most important cognitive ability that is lost. Other mental faculties may also be affected such as attention, judgment, comprehension, orientation, learning, calculation, problem solving, mood, and behavior. Agitation or withdrawal, hallucinations, delusions, insomnia, and loss of inhibitions are also common. Individuals with mental retardation and psychosis may become demented if a decline in intellectual function occurs. Many common forms of dementia are progressive, but some dementing illnesses are static and unchanging. Dementia is a chronic condition, whereas delirium is an acute confusional state associated with a change in level of consciousness (ranging from lethargy to agitation).

Memory is a complex function of the brain that has fascinated philosophers and scientists for centuries. Memory is currently viewed as a mental process that uses several storage buffers of differing capacity and duration ([Table 26-1](#)). Sensory memory lasts for about 250 ms in the visual mode (iconic memory) and 1 to 2 s in the auditory mode (echoic memory). Immediate (short-term or primary) memory has a duration of about half a minute and a limited capacity of approximately 5 to 10 items. Immediate memory is highly vulnerable to distraction, requiring attention and vigilance to maintain the content. It is often tested at the bedside by asking the patient to recall several digits forward and backward. Recent, or secondary, memory has been called both "short-term" and "long-term." It has a duration of minutes to weeks and exhibits a larger storage capacity than immediate memory. On entering this buffer, information undergoes a process of consolidation of variable duration. Recent memory is commonly tested in the clinical setting by asking a patient to recall three words after 3 to 5 min. Remote, or long-term, memory stores information lasting weeks to a lifetime and contains most of our personal experiences and knowledge. Some information appears to be stored accurately for an indefinite time, whereas other items fade or become distorted. Memory function includes registration (encoding or acquisition), retention (storage or consolidation), stabilization, and retrieval (decoding or recall). Registration and retrieval are conscious processes. Animal experiments have shown that long-term memory requires new protein synthesis, and the stabilization process probably involves physical changes at neuronal synapses.

Several additional classifications of memory are sometimes used by psychologists, particularly in reference to the content or use of the memory stores. Reference memory refers to a filing system that contains recent and remote information gained from previous experience. Working memory refers to an active process that is being updated continually by current experience. Episodic memory contains information about events occurring in a specific place and time. Semantic memory contains unchanging facts, principles, associations, and rules (for example, state capitals and the number of days in a week). Declarative (explicit) memory refers to facts about the world and past personal

events that must be consciously retrieved to be remembered. Procedural (implicit) memory, in contrast, is involved in learning and retaining a skill or procedure such as how to ride a bicycle, get dressed, or drive a car. Abilities stored in procedural memory become automatic and do not require conscious implementation.

Finally, the term *executive function* refers to mental activity involved in planning, initiating, and regulating behavior. It is considered the central organizing function of the brain that results in systematic, goal-directed activity. Executive functions are active in nonroutine situations where reflex or automatic behavior is not adequate. The anatomic and physiologic substrates of executive function are presumed to involve the frontal lobes ([Chap. 25](#)). Deficits in executive function occur frequently in patients with dementia.

FUNCTIONAL ANATOMY AND PATHOGENESIS

Dementia results from disorders of cerebral neuronal circuits and is a result of the total quantity of neuronal loss combined with the specific location of such loss ([Chap. 25](#)). The anatomic basis of memory was initially clarified from study of the alcohol/thiamine deficiency syndrome of Korsakoff and the consequences of temporal lobe surgery performed for the treatment of epilepsy. In Korsakoff's syndrome lesions in the hypothalamus, mammillary bodies, and dorsomedial nuclei of the thalamus showed that these areas were important for learning, recall, and recognition. Unilateral temporal lobe surgery for epilepsy produced mild to moderate amnesia for either verbal or nonverbal material. Bilateral medial temporal lobe excision involving the hippocampal formation, the parahippocampal gyrus, and part of the amygdala produced a severe anterograde learning disorder, i.e., an inability to store new memories, often with retained ability to recall old ones. The components of the medial temporal lobe memory system include the hippocampus and adjacent cortex, including the entorhinal, perirhinal, and parahippocampal regions ([Fig. 26-1](#)). This includes a circular pathway of neurons from the entorhinal cortex to the dentate gyrus, CA3 and CA1 neurons of the hippocampus to the subiculum, and back to the entorhinal cortex; this pathway is heavily damaged in Alzheimer's disease (AD). This system is fast, has limited capacity, and performs a crucial function at the time of learning and establishing declarative memory. Its role continues after learning during a lengthy period of reorganization and consolidation whereby memory stored in neocortex eventually becomes independent of the medial temporal lobe memory system. This process, by which the burden of long-term (permanent) memory storage is gradually assumed by neocortex, assures that the medial temporal lobe system is always available for the acquisition of new information. Recent functional brain imaging studies indicate that learning and memory involve many of the same regions of the cortex that process sensory information and control motor output. The forms of perceptual and motor learning that can occur without conscious recollections are mediated in part by contractions and expansions of representations in the sensory and motor cortex. One study, for example, has shown that the cortical representation of the fingers of the left hand of musical string players is larger than that in control individuals, suggesting that the representation of different parts of the body in the primary somatosensory cortex of humans depends on use and changes to conform to the current needs and experiences of the individual. Discrete cortical regions exist in which object knowledge (such as words related to color, animals, tools, or action) is organized as a distributed system in which the attributes of an object are stored close to

the regions of the cortex that mediate perception of those attributes ([Chap. 25](#)). That is, brain regions active during object identification are partly dependent on the intrinsic properties of the object. Procedural (implicit) memory appears to involve centers outside the hippocampus such as amygdala, cerebellum, and sensory cortex. Different frontal regions are activated for different kinds of memory storage. Functional magnetic resonance imaging (MRI) studies show that the magnitude of focal activation in left prefrontal-temporal regions or right prefrontal-bilateral parahippocampal regions predicts how well verbal or visual stimuli, respectively, will be remembered.

Biochemically, the cholinergic system plays an important role in memory. Anticholinergic agents such as atropine and scopolamine interfere with memory. Choline acetyl transferase (the enzyme catalyzing the formation of acetylcholine) and nicotinic cholinergic receptors are known to be deficient in the cortex of patients with [AD](#). The brains of patients with AD show severe neuronal loss in the nucleus basalis of Meynert, a major source of cholinergic input to the cerebral cortex. These findings form the basis for the use of cholinesterase inhibitors in the treatment of AD, with benefit presumably arising from increased available levels of acetylcholine. Behavior and mood are modulated by noradrenergic, serotonergic, and dopaminergic pathways; and norepinephrine has been shown to be reduced in the brainstem locus coeruleus in patients with AD. Neurotrophins are also postulated to play a role in memory in part by preserving cholinergic neurons.

Long-term potentiation (LTP), which refers to a long-lasting enhancement of synaptic transmission resulting from repetitive stimulation of excitatory synapses, is presumed to be involved in memory acquisition and storage. LTP occurs in the hippocampus and is mediated by *N*-methyl-D-aspartate (NMDA) receptors as well as cyclic AMP-responsive element binding protein (CREB). Gene knockout mouse models have been useful in the definition of secondary messenger systems that play a role in hippocampal LTP. For example, disruption of either calcium/calmodulin-dependent protein kinase or cytoplasmic tyrosine kinase (*fyn*) results in deficient hippocampal LTP and impaired spatial learning. In contrast, mice in which a neuronal glycoprotein *thy-1* has been inactivated show regionally selective impairment of LTP but intact spatial learning, suggesting that LTP in the entorhinal projection to the dentate gyrus of the hippocampus ([Fig. 26-1](#)) may not be necessary for some forms of spatial learning. Disruption of hippocampal levels of CREB impairs long-term memory in rats.

Most diseases causing dementia do not have highly restricted regions of pathology. Disorders such as [AD](#) appear to eventually represent relatively diffuse neuronal deterioration throughout the cerebral cortex, whereas multi-infarct dementia associated with recurrent strokes causes more focal damage in a random patchwork of cortical regions. Diffuse white matter damage may disrupt intracerebral connections and cause dementia syndromes such as those associated with leukodystrophies, multiple sclerosis, and Binswanger's disease. Subcortical structures such as the caudate, putamen, thalamus, and substantia nigra also modulate cognition and behavior in ways that are not yet well understood. Some investigators distinguish between cortical and subcortical types of dementia. A cortical dementia such as AD primarily presents as memory loss and is often associated with aphasia or other disturbance of language. Patients with subcortical dementia such as Huntington's disease (HD) are less likely to have memory and language problems and more likely to have difficulties with attention,

judgment, awareness, and behavior. Both the clinical and anatomic characteristics of the cortical and subcortical dementias show considerable overlap, and the conditions are often not distinct.

Lesions of some relatively specific cortical-subcortical pathways may have significant effects on behavior ([Chap. 25](#)). The dorsolateral prefrontal cortex has connections with the dorsolateral caudate, globus pallidus, and thalamus. Lesions of these pathways result in poor organization and planning, perseveration, and decreased cognitive flexibility with impaired judgment. The lateral orbital frontal cortex connects with the ventromedial caudate, globus pallidus, and thalamus. Lesions of these connections cause irritability, impulsiveness, and distractibility. The anterior cingulate cortex connects with the nucleus accumbens, globus pallidus, and thalamus. Interruption of these connections produces apathy and poverty of speech or even akinetic mutism.

The single strongest risk factor for dementia is increasing age. The prevalence of disabling memory loss increases with each decade over age 50 and is associated most often with the microscopic changes of [AD](#) at autopsy. Slow accumulation of mutations in neuronal mitochondria is also hypothesized to contribute to the increasing prevalence of dementia with age. Yet many centenarians have intact memory function and no evidence of clinically significant dementia. Whether dementia is an inevitable consequence of normal human aging remains controversial.

DIFFERENTIAL DIAGNOSIS

The many causes of dementia are listed in [Table 26-2](#). The frequency of each condition depends on the age group under study, the country of origin, and perhaps racial or ethnic variations. [AD](#) is the most common cause of dementia in western countries, affecting more than half of demented patients. Vascular disease is the second most common cause of dementia in the United States, affecting 10 to 20%; but it is more common than AD in some Asian countries. Dementia associated with chronic alcoholism and Parkinson's disease (PD) represent the next two most common categories. Chronic intoxications including those resulting from prescription drugs are an important, potentially treatable cause of dementia. Other disorders listed in the table are uncommon but important because many are reversible. The classification of dementing illnesses into two broad groups of reversible and irreversible disorders is a useful approach to the differential diagnosis of dementia.

Subtle cumulative memory loss is a natural part of aging. This frustrating experience, often the source of jokes and humor, is referred to as *benign forgetfulness of the elderly*. Benign means that it is not so progressive or serious that it impairs reasonably successful and productive daily functioning, although the distinction between benign and more significant memory loss can be difficult to make. A proportion of persons with benign memory loss progress to frank dementia, usually caused by [AD](#). It remains unclear why some individuals show progression and others do not. It was once assumed that a cumulative loss of hippocampal neurons with normal aging might underlie this forgetfulness, but recent quantitative neuronal counts indicate that this "natural" neuronal loss may not occur.

Alzheimer's disease is a slowly progressive dementing illness associated with diffuse

cortical atrophy and specific neuropathologic hallmarks of amyloid plaques and neurofibrillary tangles. Although quite common in the elderly, it remains a diagnosis of exclusion to be confirmed definitively only at autopsy. The clinical diagnosis of [AD](#) established by experienced neurologists proves to be correct at autopsy approximately 85 to 90% of the time. **This condition is described in greater detail in [Chap. 362](#).*

Two major types of vascular dementia can be identified ([Chap. 362](#)). The first, often called multi-infarct dementia, results from an accumulation of discrete cerebral strokes that produce disabling deficits of memory, behavior, and other cognitive abilities. Such patients usually give a history of sudden, separate stroke episodes with stepwise deterioration. On examination, focal neurologic deficits such as hemiparesis, unilateral Babinski reflex, aphasia, or visual field defect are common. Brain imaging shows multiple areas of stroke, which may have been ischemic or hemorrhagic. A second, more subtle and insidious type of vascular dementia, Binswanger's disease, is a dementing illness associated with diffuse, subcortical white matter damage often occurring in patients with chronic hypertension and/or severe atherosclerosis. The white matter changes are dramatically visualized by [MRI](#) and have also been called leukoencephalopathy. The pathogenesis of Binswanger's disease is unknown. Because [AD](#) and vascular dementia are common, occasional patients may have both conditions.

An inherited form of vascular dementia is CADASIL (cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy). It is caused by a mutation in the notch 3 gene and produces characteristic dense bodies in the media of arterioles in brain and skin. Affected persons have diffuse white matter deficits on brain [MRI](#) associated with migraine and recurrent stroke without hypertension.

Frontotemporal dementia (FTD) may represent 10 to 20% of persons with presenile dementia (onset before age 65). Initial symptoms are behavioral, such as disinhibition, apathy, or agitation with relatively intact memory. Brain imaging studies show focal lobar atrophy of the frontal and/or temporal lobes. Pick's disease is a form of FTD. Some cases are familial and associated with neuronal neurofibrillary tangle formation and mutations in the tau gene.

Dementia commonly accompanies chronic alcoholism ([Chap. 387](#)). This situation may be a result of associated malnutrition, especially of B vitamins and particularly thiamine. However, other as yet poorly defined aspects of chronic alcohol ingestion may also produce cerebral damage and atrophy. A rare idiopathic syndrome of dementia and seizures with degeneration of the corpus callosum has been reported primarily in male Italian drinkers of red wine (Marchiafava-Bignami disease).

Thiamine (vitamin B₁) deficiency causes Wernicke's encephalopathy. The clinical presentation is a malnourished individual (frequently but not necessarily alcoholic) with confusion, ataxia, and diplopia from ophthalmoplegia (Charcot's triad). Thiamine deficiency damages the thalamus, mammillary bodies, midline cerebellum, periaqueductal gray matter of the midbrain, and peripheral nerves. Damage to medial thalamic regions correlates most closely with memory loss. Prompt administration of parenteral thiamine (100 mg intravenously for 3 days followed by daily oral dosage) may reverse the disease if given in the first few days of symptom onset. However, prolonged

untreated thiamine deficiency can result in an irreversible dementia/amnestic syndrome (Korsakoff's psychosis) or even death.

In Korsakoff's syndrome, the patient is unable to recall new information despite normal immediate memory, attention span, and level of consciousness. Memory for new events is seriously impaired, whereas memory of knowledge prior to the illness is relatively intact. Patients are easily confused, disoriented, and incapable of recalling new information for more than a brief interval. Superficially, they may be conversant, entertaining, able to perform simple tasks, and follow immediate commands. Confabulation is common, although not always present, and may result in obviously erroneous statements and elaborations. There is no specific treatment because the previous thiamine deficiency has produced irreversible damage to the medial thalamic nuclei and mammillary bodies. Mammillary body atrophy may be visible on high-resolution [MRI](#).

Vitamin B₁₂ deficiency, as can occur in pernicious anemia, causes a macrocytic anemia and may also damage the nervous system ([Chaps. 107](#) and [368](#)). Neurologically it most commonly produces a spinal cord syndrome (myelopathy) affecting the posterior columns (loss of position and vibratory sense) and the lateral corticospinal tracts (hyperactive tendon reflexes and Babinski responses); it also damages peripheral nerves, resulting in sensory loss with depressed tendon reflexes. Damage to cerebral myelinated fibers may also cause dementia. The mechanism of neurologic damage is unclear but may be related to a deficiency of S-adenosylmethionine (required for methylation of myelin phospholipids) due to reduced methionine synthase activity or accumulation of methylmalonate and propionate, providing abnormal substrates for fatty acid synthesis in myelin. The neurologic signs of vitamin B₁₂ deficiency are usually associated with macrocytic anemia, but on occasion may occur in its absence. Treatment with parenteral vitamin B₁₂ (1000 ug intramuscularly daily for a week, weekly for a month, and monthly for life for pernicious anemia) stops progression of the disease if instituted promptly, but reversal of advanced nervous system damage will not occur.

Deficiency of nicotinic acid (pellagra) is associated with sun-exposed skin rash, glossitis, and angular stomatitis ([Chap. 75](#)). Severe dietary deficiency of nicotinic acid along with other B vitamins such as pyridoxine may result in spastic paraparesis, peripheral neuropathy, fatigue, irritability, and dementia. This syndrome has been seen in prisoner-of-war and concentration camps. Low serum folate levels appear to be a rough index of malnutrition, but isolated folate deficiency has not been proven to be a specific cause of dementia.

Approximately 20% of patients with [PD](#) ([Chap. 363](#)) eventually develop dementia. Treatment with L-dopa neither accelerates nor prevents this process. Some PD patients with dementia have cytoplasmic neuronal inclusions (Lewy bodies) or [AD](#) changes in the cerebral cortex, but others have no specific identifiable cortical pathology. Progressive supranuclear palsy is a dementing illness associated with parkinsonian features of rigidity, bradykinesia, and postural instability. Resting tremor is often absent, there is a vertical gaze palsy, and patients are resistant to treatment with L-dopa.

Infections of the central nervous system (CNS) usually cause delirium and other acute neurologic syndromes ([Chap. 24](#)). However, some chronic CNS infections such as

tuberculosis or cryptococcosis may produce a dementing illness ([Chap. 374](#)). Between 20 and 30% of patients in the advanced stages of infection with HIV become demented ([Chap. 309](#)). Cardinal features include psychomotor retardation, apathy, and impaired memory. This condition may result from secondary opportunistic infections but can also be caused by direct involvement of CNS neurons with HIV, where there is a multinucleated giant cell encephalitis and diffuse pallor of white matter. The neuronal toxicity may be mediated by cytokines or the direct neurotoxic effect of the gp 120 envelope glycoprotein. In the absence of CNS opportunistic infection, elevated $\delta 2$ -microglobulin in cerebrospinal fluid (CSF) is a useful marker for HIV dementia. Herpes simplex encephalitis ([Chap. 373](#)) has a predilection for the inferior temporal lobes and may present as subacute confusion and disorientation but more often as an acute syndrome rather than a chronic dementia. Computed tomography (CT), [MRI](#), and electroencephalogram (EEG) may all demonstrate the temporal lobe location of the lesions. CSF usually shows increased protein and a lymphocytic pleocytosis. CNS syphilis ([Chap. 172](#)) was a common cause of dementia in the preantibiotic era; it is uncommon now but can still be encountered in individuals with multiple sex partners. Characteristic CSF changes consist of pleocytosis, increased protein, and a positive Venereal Disease Research Laboratory (VDRL) test.

Prion disorders such as Creutzfeldt-Jakob disease (CJD) ([Chap. 375](#)) are rare conditions (approximately 1 per million population) that commonly produce dementia. CJD is typically a rapidly progressive disease associated with dementia, rigidity, and myoclonus, causing death in less than 1 to 2 years. These clinical characteristics may also rarely be seen in [AD](#), and the differential diagnosis usually depends on the slower progression of AD and the markedly abnormal periodic [EEG](#) discharges seen in CJD. Ataxia or cortical blindness may also accompany CJD. The transmissible agent, or prion, consists principally of an abnormal isoform of a host-encoded protein, the prion protein, which has undergone a physical conformational change and accumulates in affected brains. Bovine spongiform encephalopathy in the United Kingdom is thought to have resulted from cattle feed containing sheep tissues contaminated with infectious prions. Immunoassay for a 14-3-3 brain protein in [CSF](#) may be a useful marker for transmissible spongiform encephalopathies in patients with dementia.

Primary and metastatic neoplasms of the [CNS](#) ([Chap. 370](#)) usually produce focal neurologic findings and seizures rather than dementia. However, if tumor growth begins in the frontal or temporal lobes, the initial manifestations may be memory loss or behavioral changes. A rare paraneoplastic syndrome of dementia associated with occult carcinoma (usually small cell lung cancer) has been termed *limbic encephalitis* ([Chap. 101](#)). In this syndrome, confusion, agitation, seizures, poor memory, and frank dementia may occur in association with sensory neuropathy. The [CSF](#) often shows an increase in cells and protein. There is neuronal loss and perivascular lymphocytic infiltration in the hippocampus, amygdala, and cingulate and frontal cortex. Circulating antineuronal nuclear antibodies may be present. There is no specific treatment.

The syndrome of normal-pressure hydrocephalus ([Chap. 362](#)) is frequently discussed but difficult to diagnose. Clinically, a triad of memory loss, gait disturbance, and bladder incontinence is typical. The gait abnormality is often the initial symptom, and the dementia is usually mild. On imaging studies, the lateral ventricles are enlarged but there is minimal or no cortical atrophy. Lumbar puncture shows a normal or slightly

elevated opening pressure with normal [CSF](#). The condition may be idiopathic or the result of previous meningitis or subarachnoid blood from a ruptured aneurysm or head trauma. The pathogenetic mechanism is presumably a block of normal CSF flow over the convexity and delayed absorption into the venous system, with resulting stretch and distortion of white matter tracts within the corona radiata. Some individuals improve with ventricular shunting but many do not. The condition is difficult to distinguish from [AD](#) ([Chap. 362](#)).

A nonconvulsive seizure disorder may underlie a syndrome of confusion, clouding of consciousness, and garbled speech. Psychiatric disease is often suspected, but an [EEG](#) demonstrates the seizure discharges. If recurrent or persistent, the condition may be termed *complex partial status epilepticus*. The cognitive disturbance often responds to anticonvulsant therapy. The etiology may be previous small strokes or head trauma; some cases are idiopathic.

It is important to recognize systemic diseases that indirectly affect the brain and produce chronic confusion or dementia. Such conditions include dysthyroid states (especially hypothyroidism), vasculitis, and hepatic, renal, or pulmonary disease. Hepatic encephalopathy may begin with irritability and confusion and slowly progress to agitation, lethargy, and coma ([Chap. 376](#)).

Isolated angiitis of the [CNS](#) (CNS granulomatous angiitis) ([Chaps. 317](#) and [361](#)) occasionally causes a chronic encephalopathy associated with confusion, disorientation, and clouding of consciousness. Headache is common, and strokes and cranial neuropathies may occur. Brain imaging studies may be normal or nonspecifically abnormal. Studies of [CSF](#) reveal a mild pleocytosis or elevation in the protein level in half of the cases. Cerebral angiography often shows multifocal stenosis and narrowing of vessels. A few patients have only small-vessel disease that is not revealed on angiography. The angiographic appearance is not specific and may be mimicked by atherosclerosis, infection, or other causes of vascular disease. Brain or meningeal biopsy demonstrates abnormal arteries with endothelial cell proliferation and infiltrates of mononuclear cells. Autoantibodies and immune complexes are not present, and a cell-mediated process appears most likely. The prognosis is poor, but some patients respond to glucocorticoids or chemotherapy.

Chronic metal intoxications may also produce a dementing syndrome. The key to diagnosis is the elicitation of a history of exposure at work, home, or even as a consequence of a medical procedure such as dialysis. Lead poisoning has highly variable neurologic manifestations. Fatigue, depression, and confusion may be associated with episodic abdominal pain and peripheral neuropathy. Gray lead lines may appear in the gums. There is usually an associated anemia with basophilic stippling of red cells. The clinical presentation can resemble that of acute intermittent porphyria, including elevated levels of urine porphyrins as a result of the inhibition of δ -aminolevulinic acid dehydratase. Chronic lead poisoning from inadequately fired glazed pottery has been reported. The treatment is chelation therapy with agents such as ethylene diaminetetraacetic acid (EDTA). Chronic mercury poisoning may produce dementia, peripheral neuropathy, ataxia, and a fine tremulousness that may progress to a cerebellar intention tremor or choreoathetosis. The confusion and memory loss of chronic arsenic intoxication is also associated with nausea, weight loss, peripheral

neuropathy, pigmentation and scaling of the skin, and transverse white lines of the fingernails (Mee's lines). Treatment is chelation therapy with dimercaprol (BAL). Aluminum poisoning has been best documented with the dialysis dementia syndrome in which water used during renal dialysis was contaminated with excessive amounts of aluminum. This resulted in a progressive encephalopathy associated with confusion, memory loss, agitation, and, later, lethargy and stupor. Speech arrest and myoclonic jerking was common and associated with severe and generalized [EEG](#) changes. The condition was often fatal. There were no specific pathologic findings, but elevated brain aluminum content was documented. The condition has been eliminated by use of deionized water for dialysis. Although aluminum injected into experimental animals may produce neurofibrillary tangles, patients with dialysis dementia had neither tangles nor amyloid plaques, and there has been no direct association of aluminum poisoning with [AD](#).

Recurrent head trauma in professional boxers may lead to dementia, sometimes called the "punch drunk" syndrome or *dementia pugilistica*. The symptoms can be progressive and may begin late in a boxer's career or even long after retirement. The severity of the syndrome correlates with the length of the boxing career and the total number of bouts. Early in the condition there occurs a personality change associated with social instability and sometimes paranoia and delusions. Later, memory loss progresses to full dementia, often associated with parkinsonian signs and ataxia or intention tremor. At autopsy, the cerebral cortex may show changes similar to [AD](#), although neurofibrillary tangles are usually more predominant than amyloid plaques (which are usually diffuse rather than neuritic). There may also be loss of neurons in the substantia nigra. Chronic subdural hematoma is also occasionally associated with dementia, often in the context of underlying cortical atrophy from conditions such as [AD](#) or [HD](#). In these latter cases, evacuation of the subdural hematoma does not alter the underlying degenerative process.

Head injury ([Chap. 369](#)) may also be associated with temporary amnesia. The memory disturbance may include events that occurred both before the injury (retrograde amnesia) and during the postinjury period (posttraumatic or anterograde amnesia). Retrograde amnesia after severe head injury may extend back for hours or weeks before the injury; remote memory is usually intact. As patients recover, the extent of retrograde amnesia shrinks and may disappear. Often, retrograde amnesia causes permanent inability to recall the few minutes before the head injury, implying disruption of the immediate memory system and failure to register long-term memory. The length of posttraumatic amnesia generally corresponds to the length of the postconcussive confusional state, but posttraumatic amnesia may persist even in the presence of normal immediate memory and digit span. The duration of posttraumatic amnesia indicates the severity of head injury; the ability to learn new material is often the last cognitive deficit to recover. There are reports of recovery from retrograde amnesia occurring months or years after the initial brain insult; the recovery is sometimes stimulated by hypnosis, amobarbital interview, or electrical stimulation. One theory of such recovery envisions a resetting of distorted patterns of neuronal matrices subserving memory.

Transient global amnesia (TGA) is characterized by sudden onset of complete anterograde loss of memory and learning abilities, usually occurring in persons over age

50. Onset of memory loss may occur in the context of an emotional stimulus or physical exertion. During the attack the individual is alert and communicative, general cognition seems intact, and there are no other neurologic signs or symptoms. The patient may seem confused and repeatedly ask about present events. The ability to form new memories returns after a period of hours, and the individual returns to normal but has no recall for the period of the attack. Frequently no cause can be determined, but cerebrovascular disease, epilepsy (7% in one study), migraine, or cardiac arrhythmia sometimes may be implicated. A Mayo Clinic review of 277 patients with TGA found a past history of migraine in 14% and cerebrovascular disease in 11%, but these conditions were not temporally related to the TGA episodes. About one-fourth of the patients had recurrent attacks, but they were not at increased risk for subsequent stroke. Rare instances of permanent memory loss after sudden onset have been reported.

Psychogenic amnesia for personally important memories is common, although whether this amnesia results from deliberate avoidance of unpleasant memories or from unconscious repression may be impossible to establish. The event-specific amnesia is particularly common after violent crimes such as homicide of a close relative or friend or sexual abuse. It also may occur with severe drug or alcohol intoxication and sometimes with schizophrenia. More prolonged psychogenic amnesia occurs in fugue states that also commonly follow severe emotional stress. The patient with a fugue state suffers from a sudden loss of personal identity and may be found wandering far from home. In contrast to organic amnesia, fugue states are associated with amnesia for personal identity and events closely associated with the personal past. At the same time, memory for other recent events and the ability to learn and use new information are preserved. The episodes usually last hours or days and occasionally weeks or months while the patient takes on a new identity. On recovery, there is a residual amnesic gap for the period of the fugue.

Psychiatric diseases may mimic dementia. Severely depressed individuals may appear demented, a phenomenon called *pseudodementia*. Unlike cortical dementias, memory and language are usually intact when carefully tested in depressed persons. The patients may feel confused and are unable to accomplish routine tasks. Vegetative symptoms are common, such as insomnia, lack of energy, poor appetite, and concern with bowel function. The psychosocial milieu may suggest prominent reasons for depression. The patients respond to antidepressant treatment. Schizophrenia is usually not difficult to distinguish from dementia, but occasionally the distinction can be problematic. (Kraepelin's original term for schizophrenia was *dementia praecox*.) Schizophrenia usually has a much earlier age of onset (second and third decades) than most dementing illnesses. It is associated with intact memory, and the delusions and hallucinations of schizophrenia are usually more complex and bizarre than those of dementia. Some individuals with chronic schizophrenia develop an unexplained progressive dementia late in life that is not related to [AD](#). Memory loss may also be part of a conversion reaction. In this situation, patients commonly complain bitterly of memory loss, but careful cognitive testing either does not confirm the deficits or demonstrates inconsistent or unusual patterns of cognitive problems. The patients' behavior and "wrong" answers to questions often indicate that they both understand the question and know the answer.

Clouding of cognition by chronic drug or medication use, often prescribed by physicians, is an important cause of dementia. Sedatives, tranquilizers, and analgesics used to treat insomnia, pain, anxiety, or agitation may cause confusion, memory loss, and lethargy, especially in the elderly. Discontinuation of such medication often improves mentation.

Approach to the Patient

The approach to the patient with dementia should always keep two major questions in the forefront: What is the most accurate diagnosis, and is there a treatable or reversible condition? A broad overview of this approach is shown in [Table 26-3](#).

History The history should concentrate on the onset, duration, and tempo of the memory loss. Acute or subacute confusion may represent delirium and suggests intoxication, infection, or metabolic derangement. An elderly person with slowly progressive memory loss over several years is likely to have [AD](#). Initial symptoms often are difficulty with managing money, driving, shopping, following instructions, or finding one's way around town. A change in personality with disinhibition and intact memory may suggest [FTD](#). A history of sudden stroke with an irregular stepwise progression suggests multi-infarct dementia. Stroke is also commonly associated with a history of hypertension, atrial fibrillation, peripheral vascular disease, and diabetes. Rapid progression with rigidity and myoclonus suggests [CJD](#). Seizures may indicate stroke or neoplasm. Trouble in walking may suggest [PD](#) or normal-pressure hydrocephalus, especially the latter when associated with bladder incontinence. A history of multiple sex partners or intravenous drug use may indicate [CNS](#) infection, especially with HIV. A history of recurrent head trauma could indicate chronic subdural hematoma, dementia pugilistica, or normal-pressure hydrocephalus. Alcoholism may suggest malnutrition and thiamine deficiency. A remote history of gastric surgery resulting in loss of intrinsic factor might indicate vitamin B₁₂ deficiency. Certain occupations such as working in a battery or chemical factory might indicate heavy metal intoxication. Careful review of medication intake, especially of sedatives and tranquilizers, may raise the issue of chronic drug intoxication. A positive family history of dementia would be elicited in [HD](#), familial AD, and inherited FTD. The recent death of a loved one, insomnia, or poor appetite suggest depression.

Physical Examination A careful examination is essential to document the dementia, look for other signs of nervous system involvement, and search for clues suggesting other systemic disease. Cognitive function should be assessed in terms of orientation, recent and remote memory, and calculation. Many of the simple, commonly used bedside tests of cognitive function (such as serial 7s, digits forward and backward) are most useful when they are performed normally; this makes the diagnosis of dementia unlikely. Mistakes on these simple tests are more difficult to interpret and are of less diagnostic importance. Drawing a clock and the trail-making test are frequently used tests of immediate memory and visual-spatial abilities. The mini-mental status exam (MMSE) is an easily administered 30-points test of cognitive function ([Table 26-4](#)). It is used to quickly indicate a dementing process, provide a rough assessment of its severity, and follow progression of the illness. The MMSE is influenced by culture and education and is less useful in the early and late stages of dementia. Language function should be tested by the ability to read, write, comprehend, and name objects. Resting tremor, cogwheel rigidity, bradykinesia, and festinating gait indicate a parkinsonian

syndrome. Gait ataxia or apraxia (inability to initiate and coordinate steps in a sequential fashion) suggests normal-pressure hydrocephalus. Confusion, sixth cranial nerve paresis, and ataxia suggests thiamine deficiency. Myoclonic jerks are present in [CJD](#) but also occur in [AD](#). Hemiparesis or other focal neurologic deficits may occur in multi-infarct dementia or brain tumor. Bilateral hyperactive tendon reflexes, Babinski responses, and loss of vibration and position sensation suggest a myelopathy, such as occurs in vitamin B₁₂ deficiency. Stocking-glove sensory loss and diminished tendon reflexes suggest a peripheral neuropathy, which could indicate underlying diabetes, vitamin deficiency, or heavy metal intoxication. Dry cool skin, hair loss, and bradycardia suggest hypothyroidism. Confusion associated with repetitive stereotyped movements may indicate ongoing seizure activity. Hearing impairment or visual loss may produce confusion and disorientation misinterpreted as dementia. Such sensory deficits are common in the elderly.

Laboratory Tests The use of multiple laboratory tests in the evaluation of dementia is controversial. The physician does not want to miss a treatable cause, yet no single treatable cause stands out as common; thus a screen must employ multiple different tests, each of which has a low yield. Therefore, cost/benefit ratios are difficult to assess, and many laboratory screening algorithms for dementia discourage multiple tests. Nevertheless, even a test with only a 1 to 2% positive rate is probably worth undertaking if the alternative is missing a reversible or treatable cause of dementia. [Table 26-3](#) lists most screening tests for dementia. Neuroimaging studies ([CT](#) and [MRI](#)) are especially controversial because of their cost. However, they are clearly of value to identify primary and secondary neoplasms, locate areas of infarction, or suggest normal-pressure hydrocephalus or diffuse white matter disease. They also lend support to the diagnosis of [AD](#), especially if there is hippocampal atrophy in addition to diffuse cortical atrophy, and focal lobar atrophy may suggest [FTD](#). However, attempts to relate cognition to neuroimaging measures of atrophy and white matter changes have shown only modest correlations. A diagnosis of AD is reached primarily by exclusion of other causes of dementia. (The indications for apolipoprotein E testing for AD are discussed in [Chap. 362](#).) Serum levels of vitamin B₁₂ and TSH, complete blood count, electrolyte measurements, and a [VDRL](#) test are reasonable routine screening measures because they detect treatable conditions. Lumbar puncture need not be done routinely in the evaluation of dementia but is indicated if [CNS](#) infection is a serious consideration, for example, in patients with delirium, fever, or nuchal rigidity. [CSF](#) levels of tau protein are increased and those of Abamyloid are decreased in some patients with AD; however, the clinical usefulness of these changes is not yet clear. Formal psychometric testing is not necessary in every patient with dementia but can be used to document the severity of dementia, suggest psychogenic causes, and provide a semiquantitative method for following the disease course. [EEG](#) is rarely helpful except to suggest [CJD](#) (repetitive bursts of diffuse high voltage sharp waves) or an underlying nonconvulsive seizure disorder (epileptiform discharges). Brain biopsy (including meninges) is not commonly advised except to diagnose vasculitis, potentially treatable neoplasms, unusual infections (such as sarcoid), or in young persons where the diagnosis is in doubt. Angiography is not likely to be of use except when multiple strokes or cerebral vasculitis is a possible cause of the dementia.

TREATMENT

The two major goals of management are, first, to treat any correctable cause of the dementia and, second, to provide comfort and support to the patient and caregivers. Treatment of underlying causes might include thyroid replacement for hypothyroidism; vitamin therapy for thiamine and B₁₂ deficiency; antibiotics for opportunistic infections; ventricular shunting for normal-pressure hydrocephalus; and appropriate surgical, radiation, and/or chemotherapy for CNS neoplasms. Removal of sedating or cognition-impairing drugs and medications is often beneficial. If the patient is depressed rather than demented (pseudodementia), the depression should be vigorously treated. Patients with degenerative diseases such as AD and HD may also be depressed, and that portion of their condition may respond to antidepressant therapy. Antidepressants should be used with caution in demented patients because they may produce delirium. Antidepressants that have a low incidence of cognitive side effects, such as selective serotonin reuptake inhibitors, and tricyclic antidepressants with low anticholinergic activity such as desipramine and nortriptyline, are advisable. Anticonvulsants are used to control seizures. Agitation, hallucinations, delusions, and confusion are difficult to treat. These behavioral problems represent major causes for nursing home placement and institutionalization. Drugs such as phenothiazines, risperidone, haloperidol, and benzodiazepines may ameliorate the behavior problems but have untoward side effects such as sedation, rigidity, and dyskinesias. Medications that may calm agitation and insomnia without worsening dementia include low-dose haloperidol (0.5 to 2 mg), trazodone, buspirone, and propranolol. Olanzapine is increasingly used for patients with hallucinations. When patients do not respond, it is usually a mistake to advance to higher doses or to use anticholinergics or sedatives (such as barbiturates or benzodiazepines).

Cholinesterase inhibitors are being used to treat AD, and other drugs, such as estrogen, anti-inflammatory agents, and vitamin E are being investigated for the treatment or prevention of AD. These approaches are reviewed in [Chap. 362](#).

A proactive approach has been shown to reduce the occurrence of delirium in hospitalized patients. This scheme includes frequent orientation, cognitive activities, sleep enhancement measures, vision and hearing aids, and correction of dehydration.

Nondrug behavior therapy has an important place in the management of dementia. The primary goal is to make the life of the patient with dementia comfortable, uncomplicated, and safe. Preparing lists, schedules, calendars, and labels can be helpful. It is also useful to stress familiar routines, short-term tasks, brief walks, and simple physical exercises. For many patients with dementia, the memory for facts is worse than that for routine activities, and they still may be able to take part in remembered physical activities such as walking, bowling, dancing, and golf. Patients with dementia usually object to losing control over familiar tasks such as driving, cooking, and handling finances. Attempts to help or take over may be greeted with complaints, depression, or anger. Hostile responses on the part of the caretaker are useless and sometimes harmful. Explanation, reassurance, distraction, and calm statements are more productive responses in this setting. Eventually, tasks such as finances and driving must be assumed by others, and the patient will conform and adjust. Safety is an important issue that includes not only driving but the environment of the kitchen, bathroom, and sleeping area. These areas need to be monitored, supervised, and made as safe as possible. A move to a retirement home, assisted-living center, or nursing home can

initially increase confusion and agitation. Repeated reassurance, reorientation, and careful introduction to the new personnel will help to smooth the process. Provision of activities that are known to be enjoyable to the patient can be of considerable benefit. Attention should also be paid to frustration and depression in family members and caregivers. Caregiver guilt and burn-out are common. Family members often feel overwhelmed and helpless and may vent their frustrations on the patient, each other, and healthcare providers. Caregivers should be encouraged to take advantage of day-care facilities and respite breaks. Education and counseling about dementia are important. Local and national support groups can be of considerable help, such as the Alzheimer's Disease and Related Disorders Association.

(Bibliography omitted in Palm version)

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27. SLEEP DISORDERS - Charles A. Czeisler, John W. Winkelman, Gary S. Richardson

Disturbed sleep is among the most frequent health complaints physicians encounter. More than one-half of adults in the United States experience at least intermittent sleep disturbances. For most, it is an occasional night of poor sleep and/or daytime sleepiness. However, at least 15 to 20% of adults report chronic sleep disturbance or misalignment of circadian timing, which can lead to serious impairment of daytime functioning. In addition, such problems may contribute to or exacerbate medical or psychiatric conditions. Thirty years ago, many such complaints were treated with hypnotic medications without further diagnostic evaluation. Since then, a distinct class of sleep and arousal disorders has been identified, and the field of sleep disorders medicine is now an established clinical discipline. However, most physicians still only receive, on average, 1 h of education in sleep disorders in their medical school curriculum.

PHYSIOLOGY OF SLEEP AND WAKEFULNESS

Most adults sleep 7 to 8 h per night, although the timing, duration, and internal structure of sleep vary among healthy individuals and as a function of age. At the extremes, infants and the elderly have frequent interruptions of sleep. In the United States, adults of intermediate age tend to have one consolidated sleep episode per day, although in some cultures sleep may be divided into a midafternoon nap and a shortened night sleep. Two principal neurobiologic systems govern the sleep-wake cycle: one that actively generates sleep and sleep-related processes and another that times sleep within the 24-h day. Either intrinsic abnormalities in these systems or extrinsic disturbances (environmental, drug- or illness-related) can lead to sleep or circadian rhythm disorders.

STATES AND STAGES OF SLEEP

States and stages of human sleep are defined on the basis of characteristic patterns in the electroencephalogram (EEG), the electrooculogram (EOG -- a measure of eye-movement activity), and the surface electromyogram (EMG) measured on the chin and neck. The continuous recording of this array of electrophysiologic parameters to define sleep and wakefulness is termed *polysomnography*.

Polysomnographic profiles define two states of sleep: (1) rapid-eye-movement (REM) sleep, and (2) non-rapid-eye-movement (NREM) sleep. NREM sleep is in turn subdivided into four stages, characterized by increasing arousal threshold and slowing of the cortical [EEG](#). REM sleep is characterized by a low-amplitude, mixed-frequency EEG similar to that of NREM stage 1 sleep. The [EOG](#) shows bursts of REM similar to those seen during eyes-open wakefulness. Chin [EMG](#) activity is absent, reflecting the brainstem-mediated muscle atonia that is characteristic of that state.

ORGANIZATION OF HUMAN SLEEP

Normal nocturnal sleep in adults displays a consistent organization from night to night ([Fig. 27-1](#)). After sleep onset, sleep usually progresses through [NREM](#) stages 1 to 4

within 45 to 60 min. Slow-wave sleep predominates in the first third of the night and comprises 15 to 25% of total nocturnal sleep time in young adults. The percentage of slow-wave sleep is influenced by several factors, most notably age (see below). Prior sleep deprivation increases the rapidity of sleep onset and both the intensity and amount of slow-wave sleep.

The first [REM](#) sleep episode usually occurs in the second hour of sleep. More rapid onset of REM sleep in a young adult (particularly if less than 30 min) may suggest pathology such as endogenous depression, narcolepsy, circadian rhythm disorders, or drug withdrawal. [NREM](#) and REM alternate through the night with an average period of 90 to 110 min (the "ultradian" sleep cycle). Overall, REM sleep constitutes 20 to 25% of total sleep, and NREM stages 1 and 2 are 50 to 60% (increasing in elderly subjects).

Age has a profound impact on sleep state organization ([Fig. 27-1](#)). Slow-wave sleep is most intense and prominent during childhood, decreasing sharply at puberty and across the second and third decades of life. After age 30, there is a progressive, almost linear decline in the amount of slow-wave sleep, and the amplitude of delta [EEG](#) activity comprising slow-wave sleep is reduced. In the otherwise healthy older person, slow-wave sleep may be completely absent, particularly in males.

A different age profile exists for [REM](#) sleep. In infancy, REM sleep may comprise 50% of total sleep time, and the percentage is inversely proportional to developmental age. The amount of REM sleep falls off sharply over the first postnatal year as a mature REM-[NREM](#) cycle develops. During the rest of life into extreme old age, REM sleep occupies a relatively constant percentage of total sleep time.

NEUROANATOMY OF SLEEP

Lesion studies in animals and neurologic diseases in humans have suggested distinct neuroanatomic sites in the generation of normal sleep and wakefulness. Experimental studies in animals have variously implicated the medullary reticular formation, the thalamus, and the basal forebrain in the generation of sleep, while the brainstem reticular formation, the midbrain, the subthalamus, the thalamus, and the basal forebrain have all been suggested to play a role in the generation of wakefulness or [EEG](#) arousal ([Chap. 24](#)).

Current hypotheses suggest that the capacity for sleep and wakefulness generation is distributed along an axial "core" of neurons extending from the brainstem rostrally to the basal forebrain. Complex commingling of neuronal groups occurs at many points along this brainstem-forebrain axis. It was recently discovered that a cluster of g-aminobutyric acid (GABA) and galaninergic ventrolateral preoptic (VLPO) neurons, which innervate monoaminergic cell groups in the tuberomammillary nucleus that contribute to the ascending arousal system, are activated during sleep. This has led to the hypothesis that these hypothalamic VLPO neurons may play a key role in sleep regulation.

Moreover, the neuroanatomic correlates of [REM](#) sleep appear to be discretely localized. Specific regions in the pons are associated with the neurophysiologic correlates of REM sleep. Small lesions in the dorsal pons result in the loss of the descending muscle inhibition normally associated with REM sleep; microinjections of the cholinergic agonist

carbachol into the pontine reticular formation appear to produce a state with all of the features of REM sleep. These experimental manipulations are mimicked by pathologic conditions in humans and animals. In narcolepsy, for example, abrupt, complete, or partial paralysis (cataplexy) occurs in response to a variety of stimuli. In dogs with this condition, physostigmine, a central cholinesterase inhibitor, increases the frequency of cataplectic attacks, while atropine decreases their frequency. Conversely, in REM sleep behavior disorder (see below), patients suffer from incomplete motor inhibition during REM sleep, resulting in involuntary, occasionally violent movement during REM sleep.

NEUROCHEMISTRY OF SLEEP

Early experimental studies that focused on the raphe nuclei of the brainstem appeared to implicate serotonin as the primary sleep-promoting neurotransmitter, while catecholamines were considered to be responsible for wakefulness. Subsequent work has demonstrated that the raphe-serotonin system may facilitate sleep but is not necessary for its expression. Extensive pharmacologic studies of sleep and wakefulness suggest roles for other neurotransmitters as well. Cholinergic neurotransmission is known to play a role in [REM](#) sleep generation. The alerting influence of caffeine implicates adenosine, whereas the hypnotic effect of benzodiazepines and barbiturates suggests a role for endogenous ligands of the [GABA](#) receptor complex.

A variety of sleep-promoting substances have been identified, although it is not known whether or not they are involved in the endogenous sleep-wake regulatory process. These include prostaglandin D₂, delta sleep-inducing peptide, muramyl dipeptide, interleukin 1, fatty acid primary amides, and melatonin. The hypnotic effect of these substances is commonly limited to [NREM](#) or slow-wave sleep, although peptides that increase [REM](#) sleep have also been reported. Many putative "sleep factors," including interleukin 1 and prostaglandin D₂, are immunologically active as well, suggesting a link between immune function and sleep-wake states.

PHYSIOLOGY OF CIRCADIAN RHYTHMICITY

The sleep-wake cycle is the most evident of the many 24-h rhythms in humans. Prominent daily variations also occur in endocrine, thermoregulatory, cardiac, pulmonary, renal, gastrointestinal, and neurobehavioral functions. However, in evaluating a daily variation, it is important to distinguish between those rhythmic components passively evoked by periodic environmental or behavioral changes (e.g., the increase in blood pressure and heart rate upon assumption of the upright posture) and those actively driven by an endogenous oscillatory process (e.g., the circadian variation in plasma cortisol that persists under a variety of environmental and behavioral conditions).

The suprachiasmatic nuclei (SCN) of the hypothalamus act as the central neural pacemaker driving endogenous circadian rhythms in mammals. Bilateral destruction of these nuclei results in a loss of endogenous circadian rhythmicity that can only be restored by transplantation of the same structure from a donor animal. The genetically determined period of this endogenous neural oscillator, which averages ~24.2 h in humans, is normally synchronized to the 24-h period of the environmental light-dark cycle. Entrainment of mammalian circadian rhythms by the light-dark cycle is mediated

via the retinohypothalamic tract, a monosynaptic pathway that links the retina to the SCN. Humans are exquisitely sensitive to the resetting effects of light, even at low intensity.

The timing and internal architecture of sleep are directly coupled to the output of the endogenous pacemaker. Paradoxically, the endogenous circadian rhythms of sleep tendency, sleepiness, and [REM](#) sleep propensity all peak near the habitual wake time, just after the nadir of the endogenous circadian temperature cycle, whereas the circadian wake propensity rhythm peaks 1 to 3 h before the habitual bedtime. These rhythms are thus timed to oppose the homeostatic decline of sleep tendency during the habitual sleep episode and the rise of sleep tendency throughout the usual waking day, respectively. Misalignment of the output of the endogenous circadian pacemaker with the desired sleep-wake cycle can, therefore, induce insomnia, as well as decrements of alertness and neurobehavioral performance in night-shift workers and after jet lag.

BEHAVIORAL CORRELATES OF SLEEP STATES AND STAGES

Polysomnographic staging of sleep correlates with behavioral changes during specific states and stages. During the transitional state between wakefulness and sleep (stage 1 sleep), subjects may respond to faint auditory or visual signals without "awakening." Furthermore, memory incorporation is inhibited at the onset of [NREM](#) stage 1 sleep, and individuals aroused from that transitional sleep stage frequently deny having been asleep. Such transitions occur spontaneously after chronic partial sleep deprivation (e.g., 4 to 6 h of sleep per night) and acute total sleep deprivation (e.g., 24 h of wakefulness), notwithstanding attempts to remain continuously awake (see "Shift-Work Sleep Disorder," below).

Awakenings from [REM](#) sleep are associated with recall of vivid dream imagery more than 80% of the time. The reliability of dream recall increases with REM sleep episodes occurring later in the night. Imagery may also be reported after [NREM](#) sleep interruptions, though these typically lack the detail and vividness of REM sleep dreams. The incidence of NREM sleep dream recall can be increased by selective REM sleep deprivation, suggesting that REM sleep and dreaming per se are not inexorably linked.

PHYSIOLOGIC CORRELATES OF SLEEP STATES AND STAGES

All major physiologic systems are influenced by sleep. Changes in cardiovascular function include a decrease in blood pressure and heart rate during [NREM](#), and particularly during slow-wave sleep. During [REM](#) sleep, phasic activity (bursts of eye movements) is associated with variability in both blood pressure and heart rate mediated principally by the vagus. Cardiac dysrhythmias may occur selectively during REM sleep. Respiratory function also changes ([Chap. 263](#)). In comparison to relaxed wakefulness, respiratory rate becomes more regular during NREM sleep (especially slow-wave sleep) and tonic REM sleep and becomes very irregular during phasic REM sleep. Minute ventilation decreases in NREM sleep out of proportion to the decrease in metabolic rate at sleep onset, resulting in a higher P_{CO_2} .

Endocrine function also varies with sleep. The most prominent changes are apparent in neuroendocrine parameters. Slow-wave sleep is associated with secretion of growth

hormone, while sleep in general is associated with augmented secretion of prolactin. Sleep has a complex effect on the secretion of luteinizing hormone (LH): during puberty, sleep is associated with increased LH secretion, whereas sleep in the mature woman inhibits LH secretion in the early follicular phase of the menstrual cycle. Sleep onset (and probably slow-wave sleep) is associated with inhibition of thyroid-stimulating hormone and of the adrenocorticotrophic hormone-cortisol axis, an effect that is superimposed on the circadian rhythms in the two systems.

The pineal hormone melatonin is secreted predominantly at night in both day- and night-active species, reflecting the direct modulation of pineal activity by the circadian pacemaker through a circuitous neural pathway from the [SCN](#) to the pineal gland. Melatonin secretion is not dependent upon the occurrence of sleep, persisting in individuals kept awake at night. In addition, exogenous melatonin increases sleepiness and may potentiate sleep when administered to good sleepers attempting to sleep during daylight hours at a time when endogenous melatonin levels are low. However, there is little evidence to support the use of melatonin as a hypnotic in such individuals during nighttime hours, when endogenous melatonin levels are high and sleep is already consolidated. A large-scale, double-blind clinical trial is needed to evaluate the efficacy of melatonin as a sleep-promoting therapeutic for patients with insomnia.

Sleep is also associated with alterations of thermoregulatory function. [NREM](#) sleep is associated with an attenuation of thermoregulatory responses to either heat or cold stress, and animal studies of thermosensitive neurons in the hypothalamus document an NREM-sleep-dependent reduction of the thermoregulatory set-point. [REM](#) sleep is associated with complete absence of thermoregulatory responsiveness, effectively resulting in functional poikilothermy. However, the potential adverse impact of this failure of thermoregulation is blunted by inhibition of REM sleep by extreme ambient temperatures.

DISORDERS OF SLEEP AND WAKEFULNESS

Approach to the Patient

Patients may seek help from a physician because of one of several symptoms: (1) an acute or chronic inability to sleep adequately at night (insomnia); (2) chronic fatigue, sleepiness, or tiredness during the day; or (3) a behavioral manifestation associated with sleep itself. Complaints of insomnia or excessive daytime sleepiness should be viewed as symptoms (much like fever or pain) of underlying disorders. Knowledge of the differential diagnosis of these presenting complaints is essential to identify the underlying medical disorder. Only then can appropriate treatment, rather than nonspecific approaches (e.g., over-the-counter sleeping aids) be applied. Diagnoses of exclusion, such as primary insomnia, should be made only after other diagnoses have been ruled out. [Table 27-1](#) outlines the diagnostic and therapeutic approach to the patient with a complaint of excessive daytime sleepiness.

A careful history is essential in the evaluation of the patient with a sleep complaint. In particular, the duration, severity, and consistency of the complaint are important, along with the patient's estimate -- in the case of an insomnia complaint -- of the consequences of reported sleep loss on subsequent waking function. Information from a

friend or family member can be an invaluable aid in assessing the symptoms and the severity of the complaint for daytime functioning, as some patients may be unaware of, or will underreport, such potentially embarrassing symptoms as heavy snoring or falling asleep while driving.

Completion by the patient of a day-by-day sleep-work-drug log for at least 2 weeks can help the physician better understand the nature of the complaint. Work times and sleep times (including daytime naps and nocturnal awakenings) as well as drug and alcohol use, including caffeine and hypnotics, should be noted each day. The sleep times should be plotted to facilitate recognition of circadian rhythm sleep disorders such as delayed sleep phase syndrome (see below).

Polysomnography is necessary for the diagnosis of specific disorders such as narcolepsy and sleep apnea and may be of utility in other settings as well. In addition to the three electrophysiologic variables used to define sleep states and stages, the standard clinical polysomnogram includes measures of respiration (respiratory effort, air flow, and oxygen saturation), anterior tibialis [EMG](#), and electrocardiogram. Evaluation of penile tumescence during nocturnal sleep can also help determine whether the cause of erectile dysfunction in a patient is psychogenic or organic ([Chap. 51](#)).

INSOMNIA

Insomnia is the complaint of inadequate sleep; it can be classified according to the nature of sleep disruption and the duration of the complaint. The nature of the sleep disruption provides important information about the possible etiology of the insomnia and is also central to the selection of specific and appropriate treatment. Insomnia is subdivided into difficulty falling asleep (*sleep onset insomnia*), frequent or sustained awakenings (*sleep maintenance insomnia*), early morning awakenings (*sleep offset insomnia*), or persistent sleepiness despite sleep of adequate duration (*nonrestorative sleep*). Similarly, the duration of the symptom is an important determinant of the nature of appropriate treatment. An insomnia complaint lasting one to several nights (within a single episode) is termed *transient insomnia*. Transient insomnia is typically the result of situational stress or a change in sleep schedule or environment (e.g., jet lag). *Short-term insomnia* lasts from a few days to 3 weeks. Disruption of this duration is usually associated with more protracted stress, such as recovery from surgery or short-term illness. *Long-term insomnia*, or *chronic insomnia*, lasts for months or years and, in contrast with short-term insomnia, requires a thorough evaluation of underlying causes (see below). Chronic insomnia is often a waxing and waning disorder, with spontaneous or stressor-induced exacerbations.

While an occasional night of poor sleep, typically in the setting of stress or excitement about external events, is both common and without lasting consequences, persistent insomnia can have important adverse consequences in the form of impaired daytime function and increased risk of injury due to accidents. There is also clear evidence of increased risk of the development of major depression with insomnia of at least 1 year's duration. In addition, there is emerging evidence that individuals with chronic insomnia have increased utilization of health care resources, even after controlling for comorbid medical and psychiatric disorders.

Extrinsic Insomnia A number of sleep disorders are the result of extrinsic factors that interfere with sleep. *Transient situational insomnia* can occur after a change in the sleeping environment (e.g., in an unfamiliar hotel or hospital bed) or before or after a significant life event, such as a change of occupation, loss of a loved one, illness, or anxiety over a deadline or examination. Increased sleep latency, frequent awakenings from sleep, and early morning awakening can all occur. Recovery generally occurs rapidly, usually within a few weeks. Treatment is usually symptomatic, with intermittent use of hypnotics and resolution of the underlying stress. *Inadequate sleep hygiene* is characterized by a behavior pattern prior to sleep and/or a bedroom environment that is not conducive to sleep. Noise and/or light in the bedroom can interfere with sleep, as can a bed partner with periodic limb movements during sleep or one who snores loudly. Clocks can heighten the anxiety about the time it has taken to fall asleep. Drugs that act on the central nervous system, large meals, vigorous exercise, or hot showers just before sleep may interfere with sleep onset. Many individuals participate in stressful work-related activities in the evening, producing a state incompatible with sleep onset. In preference to hypnotic medications, patients should be counseled to avoid stressful activities before bed, develop a soporific bedtime ritual, and to prepare and reserve the bedroom environment for sleeping. Consistent, regular rising times should be maintained daily, including weekends.

Psychophysiologic Insomnia Persistent *psychophysiologic insomnia* is a behavioral disorder in which patients are preoccupied with a perceived inability to sleep adequately at night. The sleep disturbance is often triggered by an emotionally stressful event; however, the poor sleep habits and beliefs about sleep acquired during the stressful period persist long after the initial incident. Such patients become hyperaroused by their own persistent efforts to sleep and/or the sleep environment, and the insomnia is a conditioned or learned response. They may be able to fall asleep more easily at unscheduled times (when not trying) or outside the home environment. Polysomnographic recording in patients with psychophysiologic insomnia reveals an objective sleep disturbance, often with an abnormally long sleep latency; frequent nocturnal awakenings; and an increased amount of stage 1 transitional sleep. Rigorous attention should be paid to sleep hygiene and correction of counterproductive, arousing behaviors before bedtime. Behavioral therapies are the treatment modality of choice for psychophysiologic insomnia, with only intermittent use of medications. When patients are awake longer than 20 min, they should read or perform other relaxing activities to distract themselves from insomnia-related anxiety. In addition, bedtime and waketime should be scheduled to restrict time in bed to be equal to their perceived total sleep time. This will generally produce sleep deprivation, greater sleep drive, and, eventually, better sleep. Time in bed can then be gradually expanded.

Medication-, Drug-, or Alcohol-Dependent Insomnia Disturbed sleep can result from ingestion of a wide variety of agents. Caffeine is perhaps the most common pharmacologic cause of insomnia. It produces increased latency to sleep onset, more frequent arousals during sleep, and a reduction in total sleep time for up to 8 to 14 h after ingestion. As few as three to five cups of coffee can significantly disturb sleep in some patients; therefore, a 1- to 2-month trial without caffeine should be attempted in patients with these symptoms. Similarly, alcohol and nicotine can interfere with sleep, despite the fact that many patients use them to relax and promote sleep. Although alcohol can increase drowsiness and shorten sleep latency, even moderate amounts of

alcohol increase awakenings in the second half of the night. In addition, alcohol ingestion prior to sleep is contraindicated in patients with sleep apnea because of the inhibitory effects of alcohol on upper airway muscle tone. Acutely, amphetamines and cocaine suppress both [REM](#) sleep and total sleep time, which return to normal with chronic use. Withdrawal leads to a REM sleep rebound.

A number of prescribed medications can produce insomnia. Antidepressants, sympathomimetics, and glucocorticoids are common causes. In addition, severe rebound insomnia can result from the acute withdrawal of hypnotics, especially following the use of high doses of benzodiazepines with a short half-life. For this reason, hypnotic doses should be low to moderate, the total duration of hypnotic therapy should usually be limited to 2 to 3 weeks, and prolonged drug tapering is encouraged.

Altitude Insomnia Sleep disturbance is a common consequence of exposure to high altitude. Periodic breathing of the Cheyne-Stokes type occurs during [NREM](#) sleep about half the time at high altitude, with restoration of a regular breathing pattern during [REM](#) sleep. Both hypoxia and hypocapnia are thought to be involved in the development of periodic breathing. Frequent awakenings and poor quality sleep characterize altitude insomnia, which is generally worst on the first few nights at high altitude but may persist. Treatment with acetazolamide can decrease time spent in periodic breathing and substantially reduce hypoxia during sleep.

Restless Legs Syndrome (RLS) Patients with this sensory-motor disorder report a creeping or crawling dysesthesia deep within the calves or feet, or sometimes even in the upper extremities, that is associated with an irresistible urge to move the affected limbs. For most patients with RLS, the dysesthesias and restlessness are much worse in the evening or night compared to the daytime and frequently interfere with the ability to fall asleep. The disorder is exacerbated by inactivity and temporarily relieved by movement. In contrast, paresthesia secondary to peripheral neuropathy persists with activity. The severity of this chronic disorder may wax and wane with time and can be exacerbated by sleep deprivation, caffeine, and pregnancy. The prevalence is thought to be 5% of adults. Roughly one-third of patients will have multiple affected family members, possibly with an autosomal dominant pattern. Iron deficiency and renal failure may actually cause RLS, which is then considered secondary RLS. The symptoms of RLS are exquisitely sensitive to dopaminergic drugs (e.g., L-dopa or dopamine agonists). Narcotics, benzodiazepines, and certain anticonvulsants may also be of therapeutic value. Most patients with restless legs also experience periodic limb movement disorder during sleep, although the reverse is not the case.

Periodic Limb Movement Disorder *Periodic limb movement disorder*, previously known as *nocturnal myoclonus*, is the principal objective polysomnographic finding in 17% of patients with insomnia and 11% of those with excessive daytime somnolence ([Fig. 27-2](#)). It is often unclear whether it is an incidental finding or the cause of disturbed sleep. Stereotyped, 0.5- to 5.0-s extensions of the great toe and dorsiflexion of the foot recur every 20 to 40 s during [NREM](#) sleep, in episodes lasting from minutes to hours. Most such episodes occur during the first half of the night. The disorder occurs in a wide variety of sleep disorders (including narcolepsy, sleep apnea, [REM](#) sleep behavior disorder, and various forms of insomnia) and may be associated with frequent arousals and an increased number of sleep-stage transitions. The incidence increases with age:

44% of people over age 65 without a sleep complaint have >five periodic leg movements per hour of sleep. The pathophysiology is not well understood, though individuals with high spinal transections can exhibit periodic leg movements during sleep, suggesting the existence of a spinal generator. Polysomnography with bilateral surface [EMG](#) recording of the anterior tibialis is used to establish the diagnosis. Treatment options include dopaminergic medications or benzodiazepines.

Insomnia Associated with Mental Disorders Approximately 80% of patients with psychiatric disorders describe sleep complaints. There is considerable heterogeneity, however, in the nature of the sleep disturbance both between conditions and among patients with the same condition.

Depression can be associated with sleep onset insomnia, sleep maintenance insomnia, and/or early morning wakefulness. However, hypersomnia occurs in some depressed patients, especially adolescents and those with either bipolar or seasonal (fall/winter) depression ([Chap. 385](#)). Indeed, sleep disturbance is an important vegetative sign of depression and may commence before any mood changes are perceived by the patient. Consistent polysomnographic findings in depression include decreased [REM](#) sleep latency, lengthened first REM sleep episode, and shortened first [NREM](#) sleep episode; however, these findings are not specific for depression, and the extent of these changes varies with age and symptomatology. Depressed patients also show decreased slow-wave sleep and reduced sleep continuity.

In *mania* and *hypomania*, sleep latency is increased and total sleep time can be reduced. Patients with *anxiety disorders* tend not to show the changes in [REM](#) sleep and slow-wave sleep seen in endogenously depressed patients. Finally, *chronic alcoholics* lack slow-wave sleep, have decreased amounts of REM sleep (as an acute response to alcohol), and have frequent arousals throughout the night. This is associated with impaired daytime alertness. The sleep of chronic alcoholics may remain disturbed for years after discontinuance of alcohol usage. Sleep architecture and physiology are disturbed in *schizophrenia* (with a decreased amount of stage 4 sleep and a lack of augmentation of REM sleep following REM sleep deprivation); chronic schizophrenics often show day-night reversal, sleep fragmentation, and insomnia.

Insomnia Associated with Neurologic Disorders A variety of neurologic diseases result in sleep disruption through both indirect, nonspecific mechanisms (e.g., pain in cervical spondylosis or low back pain) or by impairment of central neural structures involved in the generation and control of sleep itself.

For example, *dementia* from any cause has long been associated with disturbances in the timing of the sleep-wake cycle, often characterized by nocturnal wandering and an exacerbation of symptomatology at night (so-called sundowning).

Epilepsy may rarely present as a sleep complaint ([Chap. 360](#)). Often the history is of abnormal behavior, at times with convulsive movements, during sleep, and the differential diagnosis includes [REM](#) sleep behavior disorder, sleep apnea syndrome, and periodic movements of sleep (see above). Diagnosis requires nocturnal [EEG](#) recording. Other neurologic diseases associated with abnormal movements, such as *Parkinson's disease*, *hemiballismus*, *Huntington's chorea*, and *Gilles de la Tourette syndrome*, are

also associated with disrupted sleep, presumably through secondary mechanisms. However, the abnormal movements themselves are greatly reduced during sleep. Headache syndromes may show sleep-associated exacerbations (*migraine* or *cluster headache*) ([Chap. 15](#)) by unknown mechanisms.

Fatal familial insomnia is a rare hereditary disorder caused by bilateral degeneration of anterior and dorsomedial nuclei of the thalamus. Insomnia is a prominent early symptom. Progressively, the syndrome produces autonomic dysfunction, dysarthria, myoclonus, coma, and death. The pathogenesis is a mutation in the prion protein ([Chap. 375](#)).

Insomnia Associated with Other Medical Disorders A number of medical conditions are associated with disruptions of sleep. The association is frequently nonspecific, e.g., that between sleep disruption and chronic pain from rheumatologic disorders. Attention to this association is important in that sleep-associated symptoms are the presenting complaint of many such patients. Treatment of the underlying medical disorder or symptom is the most useful approach to such patients. As noted above, sleep disruption can also result from the appropriate use of drugs such as glucocorticoids.

Among the most prominent associations is that between sleep disruption and *asthma*. In many asthmatics there is a prominent daily variation in airway resistance that results in marked increases in asthmatic symptoms at night, especially during sleep. In addition, treatment of asthma with theophylline-based compounds, adrenergic agonists, or glucocorticoids can independently disrupt sleep. When sleep disruption is a prominent side effect of asthma treatment, inhaled steroids (e.g., beclomethasone) that do not disrupt sleep may provide a useful alternative.

Cardiac ischemia may also be associated with sleep disruption. The ischemia itself may result from increases in sympathetic tone as a result of sleep apnea. Patients may present with complaints of nightmares or vivid, disturbing dreams, with or without awareness of the more classic symptoms of angina or of the sleep-disordered breathing. Treatment of the sleep apnea may substantially improve the angina and the nocturnal sleep quality. *Paroxysmal nocturnal dyspnea* can also occur as a consequence of sleep-associated cardiac ischemia that causes pulmonary congestion exacerbated by the recumbent posture.

Chronic obstructive pulmonary disease is also associated with sleep disruption, as is *cystic fibrosis*, *menopause*, *hyperthyroidism*, *gastroesophageal reflux*, *chronic renal failure*, and *liver failure*.

EVALUATION OF DAYTIME SLEEPINESS

Daytime impairment due to sleep loss may be difficult to quantify in the clinical setting for several reasons. First, sleepiness is not necessarily proportional to subjectively assessed sleep deprivation. In obstructive sleep apnea, for example, the repeated brief interruptions of sleep associated with resumption of respiration at the end of apneic episodes result in significant waking impairment, despite the fact that the patient may be unaware of the sleep fragmentation. Second, subjective descriptions of waking impairment vary from patient to patient. Patients may describe themselves as "sleepy,"

"fatigued," or "tired" and may have a clear sense of the meaning of those terms, while others may use the same terms to describe a completely different condition. Third, sleepiness, particularly when profound, may affect judgment in a manner analogous to ethanol, such that subjective awareness of the condition and the consequent cognitive and motor impairment is reduced. Finally, patients may be reluctant to admit that sleepiness is a problem, both because they are generally unaware of what constitutes normal alertness and because sleepiness is generally viewed pejoratively, ascribed more often to a deficit in motivation than to an inadequately addressed physiologic sleep need.

In assessing sleepiness in the clinical setting, specific questioning about the occurrence of sleep episodes during normal waking hours, both intentional and unintentional, can overcome the inconsistencies among subjective characterizations and help to interpret the adverse impact of sleepiness on daytime function. Specific areas to be addressed include the occurrence of inadvertent sleep episodes while driving or in other safety-related settings, sleepiness while at work or school (and the relationship of sleepiness to work and school performance), and the effect of sleepiness on social and family life. Evidence for significant daytime impairment [in association either with the diagnosis of a primary sleep disorder, such as narcolepsy or sleep apnea, or with imposed or self-selected sleep-wake schedules (see "Shift-Work Sleep Disorder," below)] raises the question of the physician's responsibility to notify motor vehicle licensing authorities of the increased risk of sleepiness-related vehicle accidents. As with epilepsy, legal requirements vary from state to state, and existing legal precedents do not provide a consistent interpretation of the balance between the physician's responsibility and the patient's right to privacy. At a minimum, physicians should document discussions with the patient regarding the increased risk of operating a vehicle, as well as a recommendation that driving be suspended until successful treatment or schedule modification can be instituted.

The distinction between fatigue and sleepiness can be useful in the differentiation of patients with complaints of fatigue or tiredness in the setting of disorders such as fibromyalgia, chronic fatigue syndrome ([Chap. 384](#)), or endocrine deficiencies such as hypothyroidism or Addison's disease. While patients with these disorders can typically distinguish their daytime symptoms from the sleepiness that occurs with sleep deprivation, substantial overlap can occur. This is particularly true when the primary disorder also results in chronic sleep disruption (e.g., sleep apnea in hypothyroidism) or in abnormal sleep (e.g., fibromyalgia).

While clinical evaluation of the complaint of excessive sleepiness is usually adequate, objective quantification is sometimes necessary for diagnostic purposes or for the evaluation of treatment response. Assessment of daytime functioning as an index of the adequacy of sleep can be made with the multiple sleep latency test (MSLT), which involves repeated measurement of sleep latency (time to onset of sleep) under standardized conditions during a day following quantified nocturnal sleep. The average latency across four to six tests (administered every 2 h across the waking day) is taken as an objective measure of daytime sleep tendency. Disorders of sleep that result in pathologic daytime somnolence can be reliably distinguished with the MSLT. In addition, the multiple measurements of sleep onset may identify direct transitions from wakefulness to [REM](#) sleep that are suggestive of specific pathologic conditions (e.g.,

narcolepsy).

NARCOLEPSY

Narcolepsy is both a disorder of the ability to sustain wakefulness voluntarily and a disorder of [REM](#) sleep regulation ([Table 27-2](#)). The classic "narcolepsy tetrad" consists of excessive daytime somnolence plus three specific symptoms related to an intrusion of REM sleep characteristics (e.g., muscle atonia, vivid dream imagery) into the transition between wakefulness and sleep: (1) sudden weakness or loss of muscle tone without loss of consciousness, often elicited by emotion (cataplexy); (2) hallucinations at sleep onset (hypnagogic hallucinations) or upon awakening (hypnopompic hallucinations); and (3) muscular paralysis upon awakening (sleep paralysis). The severity of cataplexy varies, as patients may have two to three attacks per day or per decade. The extent and duration of an attack may also vary, from a transient sagging of the jaw lasting a few seconds to rare cases of flaccid paralysis of the entire voluntary musculature for up to 20 to 30 min. Symptoms of narcolepsy typically begin in the second decade, although the onset ranges from ages 5 to 50. Once established, the disease is chronic without remissions. Secondary forms of narcolepsy have been described (e.g., after head trauma).

Narcolepsy affects about 1 in 4000 people in the United States and appears to have a genetic basis. Recently, two independent discoveries have revealed that hypothalamic neurons containing the neuropeptide orexin (hypocretin) may play an important role in the regulation of sleep/wakefulness: (1) a mutation in the orexin (hypocretin) receptor 2 gene has been associated with canine narcolepsy; and (2) orexin "knockout" mice that are genetically unable to produce this neuropeptide exhibit a phenotype, as assessed by behavioral and electrophysiologic criteria, that is similar to human narcolepsy. In addition, modafinil, a drug recently approved by the U.S. Food and Drug Administration (FDA) for the treatment of narcolepsy, activates orexin-containing neurons. However, the inheritance pattern of narcolepsy in humans is more complex than that of the canine model. A high rate of discordance in identical twins indicates that one or more nonheritable factors contributes to its development. First-degree relatives of narcoleptic patients nonetheless have about a 1% incidence of narcolepsy, much higher than the general population but much lower than is seen in the animal models. Of note, nearly all narcoleptics with cataplexy are positive for the human leukocyte antigen DQB*0106 (ordinarily found in 20 to 30% of the general population) ([Chap. 306](#)).

Diagnosis Definition of the essential and distinctive features of narcolepsy has continued to evolve, and the diagnostic criteria continue to be a matter of debate. Certainly, objective verification of excessive daytime somnolence, typically with [MSLT](#) mean sleep latencies <8 min, is an essential if nonspecific diagnostic feature. Other conditions that cause excessive sleepiness, such as sleep apnea or chronic sleep restriction, must be rigorously excluded. The other objective diagnostic feature of narcolepsy is the presence of [REM](#) sleep in at least two of the naps during the MSLT. This excessive REM "pressure" is also manifested by the appearance of REM sleep immediately or within minutes after sleep onset in 50% of narcoleptic patients, a rarity in unaffected individuals maintaining a conventional sleep-wake schedule. The REM-related symptoms of the classic narcolepsy tetrad are variably present. There is increasing evidence that narcoleptics with cataplexy (one-half to two-thirds of patients)

may represent a more homogeneous group than those without this symptom. However, a history of cataplexy can be difficult to establish reliably. Hypnagogic and hypnopompic hallucinations and sleep paralysis are often found in nonnarcoleptic individuals and may be present in only one-half of narcoleptics. Nocturnal sleep disruption is commonly observed in narcolepsy but is also a nonspecific symptom. Similarly, history of "automatic behavior" during wakefulness (a trance-like state during which simple motor behaviors persist) is not specific for narcolepsy and serves principally to corroborate the presence of daytime somnolence.

TREATMENT

The treatment of narcolepsy is symptomatic. Somnolence is treated with stimulants. Methylphenidate has long been considered the drug of choice by most; the usual initial dose is 10 mg bid, increasing as needed to a maximum of 20 mg qid. Pemoline, frequently used as an alternative due to its longer half-life, may be less effective and has recently been associated with fatal hepatic failure in several children. Dextroamphetamine, 10 mg bid, and methamphetamine are also frequently used alternatives. Recently, modafinil, a novel wake-promoting agent, has been approved by the [FDA](#) for treatment of the excessive daytime somnolence in narcolepsy; the dose is 200-400 mg/d given as a single dose. It is a long-acting agent that may cause fewer side effects than other medications.

Treatment of the [REM](#)-related phenomena cataplexy, hypnagogic hallucinations, and sleep paralysis requires the potent REM sleep suppression produced by antidepressant medications. The tricyclic antidepressants [e.g., protriptyline (10-40 mg/d) and clomipramine (25-50 mg/d)] and the selective serotonin reuptake inhibitors (SSRIs) [e.g., fluoxetine (10-20 mg/d)] are commonly used for this purpose in the United States. Efficacy of the antidepressants is limited largely by anticholinergic side effects (tricyclics) and by sleep disturbance and sexual dysfunction (SSRIs). Adequate nocturnal sleep time and planned daytime naps (when possible) are important preventative measures in narcolepsy.

SLEEP APNEA SYNDROMES

Respiratory dysfunction during sleep is a common, serious cause of excessive daytime somnolence as well as of disturbed nocturnal sleep. An estimated 2 to 5 million people in the United States have a reduction or cessation of breathing for 10 to 150 s, from thirty to several hundred times every night during sleep. These episodes may be due to either an occlusion of the airway (*obstructive sleep apnea*), absence of respiratory effort (*central sleep apnea*), or a combination of these factors (*mixed sleep apnea*) ([Fig. 27-2](#)). Failure to recognize and treat these conditions appropriately may lead to: significant, and often disabling, impairment of daytime alertness; increased risk of sleep-related motor vehicle accidents; hypertension and other serious cardiovascular complications; and increased mortality. Sleep apnea is particularly prevalent in overweight men and in the elderly, yet it is estimated to remain undiagnosed in 80 to 90% of affected individuals. This is unfortunate since effective treatments are available. **Readers are referred to Chap. 263 for a comprehensive review of the diagnosis and treatment of patients with these conditions.*

PARASOMNIAS

The term *parasomnia* refers to abnormal behaviors that arise from, or occur during, sleep. A continuum of parasomnias arise from [NREM](#) sleep, from brief confusional arousals to sleepwalking and night terrors. The presenting complaint is usually related to the behavior itself, but the parasomnias can disturb sleep continuity or lead to mild impairments in daytime alertness. Only one parasomnia is known to occur in [REM](#) sleep, i.e., REM sleep behavior disorder (RBD; see below).

Sleepwalking (Somnambulism) Patients affected by this disorder carry out automatic motor activities that range from simple to complex. Individuals may leave the bed, walk, urinate inappropriately, eat, or exit from the house while remaining only partially aware. Full arousal may be difficult, and some patients may respond to attempted awakening with agitation or even violence. Sleepwalking arises from stage 3 or 4 [NREM](#) sleep and is most common in children and adolescents, when these sleep stages are most robust. Episodes are usually isolated but may be recurrent in 1 to 6% of patients. The cause is unknown, though it has a familial basis in roughly one-third of cases.

Sleep Terrors This disorder, also called *pavor nocturnus*, occurs primarily in young children during the first several hours after sleep onset, in stages 3 and 4 of [NREM](#) sleep. The child suddenly screams, exhibiting autonomic arousal with sweating, tachycardia, and hyperventilation. The individual may be difficult to arouse and rarely recalls the episode on awakening in the morning. Recurrent attacks are rare, and treatment is usually by way of reassurance of parents. Both sleep terrors and sleepwalking represent abnormalities of arousal. In contrast, *nightmares* (dream anxiety attacks) occur during [REM](#) sleep and cause full arousal, with intact memory for the unpleasant episode.

REM Sleep Behavior Disorder [RBD](#) is a rare condition that is distinct from other parasomnias in that it occurs during [REM](#) sleep. It primarily afflicts men of middle age or older, many of whom have a history of prior neurologic disease. In fact, over one-third of patients will go on to develop Parkinson's disease within 10 to 20 years. Presenting symptoms are of agitated or violent behavior during sleep, reported by a bed partner. In contrast to typical somnambulism, injury to patient or bed partner is not uncommon, and, upon awakening, the patient reports vivid, often unpleasant, dream imagery. The principal differential diagnosis is that of nocturnal seizures, which can be excluded with polysomnography. In RBD, seizure activity is absent on the [EEG](#), and disinhibition of the usual motor atonia is observed in the [EMG](#) during REM sleep, at times associated with complex motor behaviors. The pathogenesis is unclear, but damage to brainstem areas mediating descending motor inhibition during REM sleep may be responsible. In support of this hypothesis are the remarkable similarities between RBD and the sleep of animals with bilateral lesions of the pontine tegmentum in areas controlling REM sleep motor inhibition. Treatment with clonazepam provides sustained improvement in almost all reported cases.

Sleep Bruxism Bruxism is an involuntary, forceful grinding of teeth during sleep that affects 10 to 20% of the population. The patient is usually unaware of the problem. The typical age of onset is 17 to 20 years, and spontaneous remission usually occurs by age 40. Sex distribution appears to be equal. Treatment is dictated by the risk of dental injury. In many cases, the diagnosis is made during dental examination, damage is

minor, and no treatment is indicated. In more severe cases, treatment with a rubber tooth guard is necessary to prevent disfiguring tooth injury. Stress management or, in some cases, biofeedback can be useful when bruxism is a manifestation of psychological stress. There are anecdotal reports of benefit using benzodiazepines.

Sleep Enuresis Bedwetting, like sleepwalking and night terrors, is another parasomnia that occurs during slow-wave sleep in the young. Before age 5 or 6, nocturnal enuresis should probably be considered a normal feature of development. The condition usually improves spontaneously at puberty, has a prevalence in late adolescence of 1 to 3%, and is rare in adulthood. The age threshold for initiation of treatment depends on parental and patient concern about the problem. Persistence of enuresis into adolescence or adulthood may reflect a variety of underlying conditions. In older patients with enuresis a distinction must be made between primary and secondary enuresis, the latter being defined as bedwetting in patients who have been fully continent for 6 to 12 months. Treatment of primary enuresis is reserved for patients of appropriate age (older than 5 or 6 years) and consists of bladder training exercises and behavioral therapy. Urologic abnormalities are more common in primary enuresis and must be assessed by urologic examination. Important causes of secondary enuresis include emotional disturbances, urinary tract infections or malformations, cauda equina lesions, epilepsy, sleep apnea, and certain medications. Symptomatic pharmacotherapy is usually accomplished with intranasal desmopressin, or oral oxybutynin chloride or imipramine.

Miscellaneous Parasomnias Other clinical entities fulfill the definition of a parasomnia in that they occur selectively during sleep and are associated with some degree of sleep disruption. Examples include *jactatio capitis nocturna* (nocturnal headbanging), sleep talking, nocturnal paroxysmal dystonia, and nocturnal leg cramps.

CIRCADIAN RHYTHM SLEEP DISORDERS

A subset of patients presenting with either insomnia or hypersomnia may have a disorder of sleep *timing* rather than sleep *generation*. Disorders of sleep timing can either be organic (i.e., due to an intrinsic defect in the circadian pacemaker or its input from entraining stimuli) or environmental (i.e., due to a disruption of exposure to entraining stimuli from the environment). Regardless of etiology, the symptoms reflect the influence of the underlying circadian pacemaker on sleep-wake function. Thus, effective therapeutic approaches should aim to entrain the oscillator at an appropriate phase.

RAPID TIME-ZONE CHANGE (JET LAG) SYNDROME

More than 60 million people experience transmeridian air travel annually, which is often associated with excessive daytime sleepiness, sleep onset insomnia, and frequent arousals from sleep, particularly in the latter half of the night. Gastrointestinal discomfort is common. The syndrome is transient, typically lasting 2 to 14 d depending on the number of time zones crossed, the direction of travel, and the traveler's age and phase-shifting capacity. Travelers who spend more time outdoors reportedly adapt more quickly than those who remain in hotel rooms, presumably due to bright (outdoor) light exposure.

SHIFT-WORK SLEEP DISORDER

More than 7 million workers in the United States regularly work at night, either on a permanent or rotating schedule. In addition, each week millions of Americans elect to remain awake at night to meet deadlines, drive long distances, or participate in recreational activities, leading to both sleep loss and misalignment of their circadian rhythms with respect to their sleep-wake cycle. Chronic shift workers have higher rates of cardiac, gastrointestinal, and reproductive disorders. Studies of regular night-shift workers indicate that the circadian timing system usually fails to adapt successfully to such inverted schedules. This leads to a misalignment between the desired work-rest schedule and the output of the pacemaker and in disturbed daytime sleep. Consequent sleep deprivation, increased length of time awake prior to work, and misalignment of circadian phase produce decreased alertness and performance, increased reaction time, and increased risk of performance lapses, thereby resulting in greater safety hazards among night workers and other sleep-deprived individuals.

Sleep onset is associated with marked attenuation in perception of both auditory and visual stimuli and lapses of consciousness. The sleepy individual may thus attempt to perform routine and familiar motor tasks during the transition state between wakefulness and sleep (stage 1 sleep) in the absence of adequate sensory input from the environment. Motor vehicle operators are especially vulnerable to sleep-related accidents since the sleep-deprived driver or operator often fails to heed the warning signs of fatigue. Such attempts to override the powerful biologic drive for sleep by the sheer force of will can yield a catastrophic outcome when sleep processes intrude involuntarily upon the waking brain. Such intrusions typically last only seconds but are known on occasion to persist for longer durations. These frequent brief intrusions of stage 1 sleep into behavioral wakefulness are a major component of the impaired psychomotor performance seen with sleepiness. Such intrusions and their associated performance lapses, which are preceded by a markedly increased subjective sense of sleepiness, will inevitably occur if the need for sleep is not satiated. There is a marked increase in the risk of sleep-related, fatal-to-the-driver highway crashes in the early morning and late afternoon hours, coincident with peaks in the daily rhythm of sleep tendency.

Safety programs should promote education about sleep and increase awareness of the hazards associated with night work and should be aimed at minimizing both circadian disruption and sleep deprivation. The work schedule should minimize: (1) exposure to night work, (2) the frequency of shift rotation so that shifts do not rotate more than once every 2 to 3 weeks, (3) the number of consecutive night shifts, and (4) the duration of night shifts. In fact, shift durations of greater than 18 h should be universally recognized as increasing the risk of sleep-related errors and performance lapses. Caffeine is undoubtedly the most widely used wake-promoting drug, but it cannot forestall sleep indefinitely and does not protect users from sleep-related performance lapses. Postural changes, exercise, and strategic placement of nap opportunities can sometimes temporarily reduce the risk of fatigue-related performance lapses. Properly timed exposure to bright light can facilitate rapid adaptation to night-shift work, where feasible. An adequate number of safe highway rest areas, shoulder rumble strips, and strict enforcement and compliance monitoring of hours-of-service policies are needed to

reduce the risk of sleep-related transportation crashes. Such steps can lead to improvements in performance and to reduced accident rates both at work and on the roadways.

DELAYED SLEEP PHASE SYNDROME

Delayed sleep phase syndrome is characterized by: (1) reported sleep onset and wake times intractably later than desired, (2) actual sleep times at nearly the same clock hours daily, and (3) essentially normal all-night polysomnography except for delayed sleep onset. Patients exhibit an abnormally delayed endogenous circadian phase, with the temperature minimum during the constant routine occurring later than normal. This delayed phase could be due to: (1) an abnormally long intrinsic period of the endogenous circadian pacemaker; (2) an abnormally reduced phase-advancing capacity of the pacemaker; or (3) an irregular prior sleep-wake schedule, characterized by frequent nights when the patient chooses to remain awake well past midnight (for social, school, or work reasons). In most cases, it is difficult to distinguish among these factors, since patients with an abnormally long intrinsic period are more likely to "choose" such late-night activities because they are unable to sleep at that time. Patients tend to be young adults. This self-perpetuating condition can persist for years and does not usually respond to attempts to reestablish normal bedtime hours.

Treatment methods involving bright-light phototherapy during the morning hours or melatonin administration in the evening hours show promise in these patients, although the relapse rate among such patients is very high.

ADVANCED SLEEP PHASE SYNDROME

Advanced sleep phase syndrome is the converse of the delayed sleep phase syndrome and tends to occur in the elderly. Patients with this condition report excessive daytime sleepiness during the evening hours, when they have great difficulty remaining awake, even in social settings. The patients awaken from 3 to 5 A.M. each day, often several hours before their desired wake times. Although such patients have not been studied extensively, familial inheritance of this condition has been reported. Some of these patients may benefit from bright-light phototherapy during the evening hours, designed to reset the circadian pacemaker to a later hour.

NON-24-H SLEEP-WAKE DISORDER

This condition can occur when the maximal phase-advancing capacity of the circadian pacemaker is not adequate to accommodate the difference between the 24-h geophysical day and the intrinsic period of the pacemaker in the patient. Alternatively, patients' self-selected exposure to artificial light may drive the circadian pacemaker to a longer than 24-h schedule. Affected patients are not able to maintain a stable phase relationship between the output of the pacemaker and the 24-h day. Such patients typically present with an incremental pattern of successive delays in sleep onsets and wake times, progressing in and out of phase with local time. When the patient's endogenous rhythms are out of phase with the local environment, insomnia coexists with excessive daytime sleepiness. Conversely, when the endogenous rhythms are in phase with the local environment, symptoms remit. The intervals between symptomatic

periods may last several weeks to several months. Blind individuals unable to perceive light are particularly susceptible to this disorder. Melatonin administration has been reported to improve sleep, and in some cases even to induce synchronization of the circadian pacemaker.

MEDICAL IMPLICATIONS OF CIRCADIAN RHYTHMICITY

Understanding the role of circadian rhythmicity in the pathophysiology of illness may lead to improvements in diagnosis and treatment. For example, prominent circadian variations have been reported in the incidence of *acute myocardial infarction*, *sudden cardiac death*, and *stroke*, the leading causes of death in the United States. Platelet aggregability is increased after arising in the early morning hours, coincident with the peak incidence of these cardiovascular events. A better understanding of the possible role of circadian rhythmicity in the acute destabilization of a chronic condition such as atherosclerotic disease could improve the understanding of the pathophysiology.

Diagnostic and therapeutic procedures may also be affected by the time of day at which data are collected. Examples include blood pressure, body temperature, the dexamethasone suppression test, and plasma cortisol levels. The timing of chemotherapy administration has been reported to have an effect on the outcome of treatment. Few physicians realize the extent to which routine measures are affected by the time (or sleep/wake state) when the measurement is made.

In addition, both the toxicity and effectiveness of drugs can vary during the day. For example, more than a fivefold difference has been observed in mortality rates following administration of toxic agents to experimental animals at different times of day. Anesthetic agents are particularly sensitive to time-of-day effects. Finally, the physician must be increasingly aware of the public health risks associated with the ever-increasing demands made by the duty-rest-recreation schedules in our round-the-clock society.

(Bibliography omitted in Palm version)

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SECTION 4 -DISORDERS OF EYES, EARS, NOSE, AND THROAT

28. DISORDERS OF THE EYE - Jonathan C. Horton

THE HUMAN VISUAL SYSTEM

The visual system provides a supremely efficient means for the rapid assimilation of information from the environment to aid in the guidance of behavior. The act of seeing begins with the capture of images focused by the cornea and lens upon a light-sensitive membrane in the back of the eye, called the *retina*. The retina is actually part of the brain, banished to the periphery to serve as a transducer for the conversion of patterns of light energy into neuronal signals. Light is absorbed by photopigment in two types of receptors: rods and cones. In the human retina there are 100 million rods and 5 million cones. The rods operate in dim (scotopic) illumination. The cones function under daylight (photopic) conditions. The cone system is specialized for color perception and high spatial resolution. The majority of cones are located within the macula, the portion of the retina serving the central 10° of vision. In the middle of the macula a small pit termed the *fovea*, packed exclusively with cones, provides best visual acuity.

Photoreceptors hyperpolarize in response to light, activating bipolar, amacrine, and horizontal cells in the inner nuclear layer. After processing of photoreceptor responses by this complex retinal circuit, the flow of sensory information ultimately converges upon a final common pathway: the ganglion cells. These cells translate the visual image impinging upon the retina into a continuously varying barrage of action potentials that propagates along the primary optic pathway to visual centers within the brain. There are a million ganglion cells in each retina, and hence a million fibers in each optic nerve.

Ganglion cell axons sweep along the inner surface of the retina in the nerve fiber layer, exit the eye at the optic disc, and travel through the optic nerve, optic chiasm, and optic tract to reach targets in the brain. The majority of fibers synapse upon cells in the lateral geniculate body, a thalamic relay station. Cells in the lateral geniculate body project in turn to the primary visual cortex. This massive afferent retinogeniculocortical sensory pathway provides the neural substrate for visual perception. Although the lateral geniculate body is the main target of the retina, separate classes of ganglion cells project to other subcortical visual nuclei involved in different functions. The pupillary reflex is mediated by input to the pretectal olivary nuclei in the midbrain. These pretectal nuclei send their output to the ipsilateral and contralateral Edinger-Westphal nuclei of the oculomotor nuclear complex. Cells in the Edinger-Westphal nuclei provide parasympathetic innervation to the iris sphincter via an interneuron in the ciliary ganglion. Circadian rhythms are timed by a retinal projection to the suprachiasmatic nucleus. Visual orientation and eye movements are served by retinal input to the superior colliculus. Gaze stabilization and optokinetic reflexes are governed by a group of small retinal targets known collectively as the *brainstem accessory optic system*. Finally, there is a sizeable retinal projection to the pulvinar, a large thalamic visual nucleus of obscure function.

The eyes must be rotated constantly within their orbits to place and maintain targets of visual interest upon the fovea. This activity, called *foveation*, or looking, is governed by an elaborate efferent motor system. Each eye is moved by six extraocular muscles,

supplied by cranial nerves from the oculomotor (III), trochlear (IV), and abducens (VI) nuclei. Activity in these ocular motor nuclei is coordinated by pontine and midbrain mechanisms for smooth pursuit, saccades, and gaze stabilization during head and body movements. Large regions of the frontal and parietooccipital cortex control these brainstem eye movement centers by providing descending supranuclear input.

Visual function can be disturbed in myriad ways. The eyes are mounted in a prominent position on the head, where they are vulnerable to trauma, exposure, and infection. Vision can be damaged by diseases intrinsic to the eye, such as glaucoma, cataract, or retinal detachment. Many neurologic diseases produce ocular symptoms, because extensive areas of the cortex, thalamus, cerebellum, and brainstem are devoted to visual perception or to the execution of eye movements. In genetic disorders, eye manifestations are common and often help the clinician to recognize a rare syndrome. Finally, the eyes are affected frequently by acquired systemic diseases.

The eye is a specialized organ, requiring unique optical instruments for proper examination. The slit lamp and ophthalmoscope proffer a beautiful, magnified view of the transparent anatomy of the eye and afford the only opportunity for direct inspection of blood vessels in a living subject. Some physicians do not acquire sufficient facility with these instruments to care for patients with eye problems. This is regrettable, for although it may be determined that a patient requires referral to an ophthalmologist, the initial evaluation of ocular symptoms lies within the purview of all physicians, and the assessment of visual acuity, pupils, eye movements, visual fields, and the fundi remain part of any general physical examination.

CLINICAL ASSESSMENT OF VISUAL FUNCTION

REFRACTIVE STATE

In approaching the patient with reduced vision, the first step is to decide whether refractive error is responsible. In *emmetropia*, parallel rays from infinity are focused perfectly upon the retina. Sadly, this condition is enjoyed by only a minority of the population. In *myopia*, the globe is too long, and light rays come to a focal point in front of the retina. Near objects can be seen clearly, but distant objects require a diverging lens in front of the eye. In *hyperopia*, the globe is too short, and hence a converging lens is used to supplement the refractive power of the eye. In *astigmatism*, the corneal surface is not perfectly spherical, necessitating a cylindrical corrective lens. In recent years it has become possible to correct refractive error with the excimer laser by performing either LASIK (laser in situ keratomileusis) or PRK (photorefractive keratectomy) to alter the curvature of the cornea.

With the onset of middle age, *presbyopia* develops as the lens within the eye becomes unable to increase its refractive power to accommodate upon near objects. To compensate for presbyopia, the emmetropic patient must use reading glasses. The patient already wearing glasses for distance correction usually switches to bifocals. The only exception is the myopic patient, who may achieve clear vision at near simply by removing glasses containing the distance prescription.

Refractive errors usually develop slowly and remain stable after adolescence, except in

unusual circumstances. For example, the acute onset of diabetes mellitus can produce sudden myopia because of fluid imbibition and swelling of the lens induced by hyperglycemia. Testing vision through a pinhole aperture is a useful way to screen quickly for refractive error. If the visual acuity is better through a pinhole than with the unaided eye, the patient needs a refraction to obtain best corrected visual acuity.

VISUAL ACUITY

The Snellen chart is used to test acuity at a distance of 6 m (20 ft). For convenience, a scale version of the Snellen chart, called the Rosenbaum card, is held at 36 cm (14 in) from the patient ([Fig. 28-1](#)). All subjects should be able to read the 6/6 m (20/20 ft) line with each eye using their refractive correction, if any. Patients who need reading glasses because of presbyopia must wear them for accurate testing with the Rosenbaum card. If 6/6 (20/20) acuity is not present in each eye, the deficiency in vision must be explained. For acuity worse than 6/240 (20/800), the ability to count fingers, see hand motions, or perceive a bright light should be recorded. Legal blindness is defined by the Internal Revenue Service as a best corrected acuity of 6/60 (20/200) or less in the better eye, or a binocular visual field subtending 20° or less. For driving the laws vary by state, but most require a corrected acuity of 6/12 (20/40) in at least one eye. Patients with a homonymous hemianopia should not drive.

PUPILS

The pupils should be tested individually in dim light with the patient fixating upon a distant target. If they respond briskly to light, there is no need to check the near response, because isolated loss of constriction (miosis) to accommodation does not occur. For this reason, the ubiquitous abbreviation PERRLA (pupils equal, round, and reactive to light and accommodation) implies a wasted effort with the last step. However, it is important to test the near response if the light response is poor or absent. Light-near dissociation occurs with neurosyphilis (Argyll Robertson pupil), lesions of the dorsal midbrain (obstructive hydrocephalus, pineal region tumors), and after aberrant regeneration (oculomotor nerve palsy, Adie's tonic pupil).

An eye with no light perception has no pupillary response to direct light stimulation. If the retina or optic nerve is only partially injured, the direct pupillary response will be weaker than the consensual pupillary response evoked by shining a light into the other eye. This *relative afferent pupillary defect* (Marcus Gunn pupil) can be elicited with the swinging flashlight test ([Fig. 28-2](#)). It is an extremely useful sign in retrobulbar optic neuritis and other optic nerve diseases, where it may be the sole objective evidence for disease.

Subtle inequality in pupil size, up to 0.5 mm, is a fairly common finding in normal persons. The diagnosis of essential or physiologic anisocoria is secure as long as the relative pupil asymmetry remains constant as ambient lighting varies. Anisocoria that increases in dim light indicates a sympathetic paresis of the iris dilator muscle. The triad of miosis with ipsilateral ptosis and anhidrosis constitutes Horner's syndrome, although anhidrosis is an inconstant feature. Brainstem stroke, carotid dissection, or neoplasm impinging upon the sympathetic chain are occasionally identified as the cause of Horner's syndrome, but most of cases are idiopathic.

Anisocoria that increases in bright light suggests a parasympathetic palsy. The first concern is an oculomotor nerve paresis. This possibility is excluded if the eye movements are full and the patient has no ptosis or diplopia. Acute pupillary dilation (mydriasis) can occur from damage to the ciliary ganglion in the orbit. Common mechanisms are infection (herpes zoster, influenza), trauma (blunt, penetrating, surgical), or ischemia (diabetes, temporal arteritis). After denervation of the iris sphincter the pupil does not respond well to light, but the response to near is often relatively intact. When the near stimulus is removed, the pupil redilates very slowly compared with the normal pupil, hence the term *tonic pupil*. In Adie's syndrome, a tonic pupil occurs in conjunction with weak or absent tendon reflexes in the lower extremities. This benign disorder, which occurs predominantly in healthy young women, is assumed to represent a mild dysautonomia. Tonic pupils are also associated with Shy-Drager syndrome, segmental hypohidrosis, diabetes, and amyloidosis. Occasionally, a tonic pupil is discovered incidentally in an otherwise completely normal, asymptomatic individual. The diagnosis is confirmed by placing a drop of dilute (0.125%) pilocarpine into each eye. Denervation hypersensitivity produces pupillary constriction in a tonic pupil, whereas the normal pupil shows no response. Pharmacologic dilation from accidental or deliberate instillation of anticholinergic agents (atropine, scopolamine drops) into the eye can also produce pupillary mydriasis. In this situation, normal strength (1%) pilocarpine causes no constriction.

Both pupils are affected equally by systemic medications. They are small with narcotic use (morphine, heroin) and large with anticholinergics (scopolamine). Parasympathetic agents (pilocarpine, demecarium bromide) used to treat glaucoma produce miosis. In any patient with an unexplained pupillary abnormality, a slit-lamp examination is helpful to exclude surgical trauma to the iris, an occult foreign body, perforating injury, intraocular inflammation, adhesions (synechia), angle-closure glaucoma, and iris sphincter rupture from blunt trauma.

EYE MOVEMENTS AND ALIGNMENT

Eye movements are tested by asking the patient with both eyes open to pursue a small target such as a penlight into the cardinal fields of gaze. Normal ocular versions are smooth, symmetric, full, and maintained in all directions without nystagmus. Saccades, or quick refixation eye movements, are assessed by having the patient look back and forth between two stationary targets. The eyes should move rapidly and accurately in a single jump to their target. Ocular alignment can be judged by holding a penlight directly in front of the patient at about 1 m. If the eyes are straight, the corneal light reflex will be centered in the middle of each pupil. To test eye alignment more precisely, the cover test is useful. The patient is instructed to gaze upon a small fixation target in the distance. One eye is covered suddenly while observing the second eye. If the second eye shifts to fixate upon the target, it was misaligned. If it does not move, the first eye is uncovered and the test is repeated on the second eye. If neither eye moves, the eyes are aligned orthotropically. If the eyes are orthotropic in primary gaze but the patient complains of diplopia, the cover test should be performed with the head tilted or turned in whatever direction elicits the patient's diplopia. With practice the examiner can detect an ocular deviation (heterotropia) as small as 1 to 2° with the cover test. Deviations can be measured by placing prisms in front of the misaligned eye to determine the power

required to neutralize the fixation shift evoked by covering the other eye.

STEREOPSIS

Stereoacuity is determined by presenting targets with retinal disparity separately to each eye using polarized images. The most popular office tests measure a range of thresholds from 800 to 40 seconds of arc. Normal stereoacuity is 40 seconds of arc. If a patient achieves this level of stereoacuity, one is assured that the eyes are aligned orthotropically and that vision is intact in each eye. Random dot stereograms have no monocular depth cues and provide an excellent screening test for strabismus and amblyopia in children.

COLOR VISION

The retina contains three classes of cones, with visual pigments of differing peak spectral sensitivity: red (560 nm), green (530 nm), and blue (430 nm). The red and green cone pigments are encoded on the X chromosome; the blue cone pigment on chromosome 7. Mutations of the blue cone pigment are exceedingly rare. Mutations of the red and green pigments cause congenital X-linked color blindness in 8% of males. Affected individuals are not truly color blind; rather, they differ from normal subjects in how they perceive color and how they combine primary monochromatic lights to match a given color. Anomalous trichromats have three cone types, but a mutation in one cone pigment (usually red or green) causes a shift in peak spectral sensitivity, altering the proportion of primary colors required to achieve a color match. Dichromats have only two cone types and will therefore accept a color match based upon only two primary colors. Anomalous trichromats and dichromats have 6/6 (20/20) visual acuity, but their hue discrimination is impaired. Ishihara color plates can be used to detect red-green color blindness. The test plates contain a hidden number, visible only to subjects with color confusion from red-green color blindness. Because color blindness is almost exclusively X-linked, it is worth screening only male children.

The Ishihara plates are often used to detect acquired defects in color vision, although they are intended as a screening test for congenital color blindness. Acquired defects in color vision frequently result from disease of the macula or optic nerve. For example, patients with a history of optic neuritis often complain of color desaturation long after their visual acuity has returned to normal. Color blindness can also occur from bilateral strokes involving the ventral portion of the occipital lobe (cerebral achromatopsia). Such patients can perceive only shades of gray and may also have difficulty recognizing faces (prosopagnosia). Infarcts of the dominant occipital lobe sometimes give rise to color anomia. Affected patients can discriminate colors, but they cannot name them.

VISUAL FIELDS

Vision can be impaired by damage to the visual system anywhere from the eyes to the occipital lobes. One can localize the site of the lesion with considerable accuracy by mapping the visual field deficit by finger confrontation and then correlating it with the topographic anatomy of the visual pathway ([Fig. 28-3](#)). More quantitative data can be obtained by formal perimetric examination of the visual fields. In kinetic perimetry, the patient faces a tangent screen or a hemispheric bowl (Goldmann perimeter) while the

examiner moves a small light target from the periphery towards the center. Such manual techniques have largely been supplanted by computer-driven perimeters (Humphrey, Octopus) that present a target of variable intensity at fixed positions in the visual field ([Fig. 28-3A](#)). By generating an automated printout of light thresholds, these static perimeters provide a sensitive means of detecting scotomas in the visual field. They are also useful for serial assessment of visual function in chronic diseases such as glaucoma or pseudotumor cerebri.

The crux of visual field analysis is to decide whether a lesion is before, at, or behind the optic chiasm. If a scotoma is confined to one eye, it must be due to a lesion anterior to the chiasm, involving either the optic nerve or retina. Retinal lesions produce scotomas that correspond optically to their location in the fundus. For example, a superior-nasal retinal detachment results in an inferior-temporal field cut. Damage to the macula causes a central scotoma ([Fig. 28-3B](#)).

Optic nerve disease produces characteristic patterns of visual field loss. Glaucoma selectively destroys axons that enter the superotemporal or inferotemporal poles of the optic disc, resulting in arcuate scotomas shaped like a Turkish scimitar, which emanate from the blind spot and curve around fixation to end flat against the horizontal meridian ([Fig. 28-3C](#)). This type of field defect mirrors the arrangement of the nerve fiber layer in the temporal retina. The superb acuity of humans is achieved by thrusting aside all retinal elements at the fovea except photoreceptors, to minimize absorption and scattering of light. To avoid passing over the fovea, axons from cells in the temporal retina must follow an indirect course arching around the fovea to reach the optic disc. Arcuate or nerve fiber layer scotomas also occur from optic neuritis, ischemic optic neuropathy, optic disc drusen, and branch retinal artery or vein occlusion.

Damage to the entire upper or lower pole of the optic disc causes an altitudinal field cut that follows the horizontal meridian ([Fig. 28-3D](#)). This pattern of visual field loss is typical of ischemic optic neuropathy but also occurs from retinal vascular occlusion, advanced glaucoma, and optic neuritis.

About half the fibers in the optic nerve originate from ganglion cells serving the macula. Damage to papillomacular fibers causes a cecocentral scotoma encompassing the blind spot and macula ([Fig. 28-3E](#)). If the damage is irreversible, pallor eventually appears in the temporal portion of the optic disc. Temporal pallor from a cecocentral scotoma may develop in optic neuritis, nutritional optic neuropathy, toxic optic neuropathy, Leber's hereditary optic neuropathy, and compressive optic neuropathy. It is worth mentioning that the temporal side of the optic disc is slightly more pale than the nasal side in most normal individuals. Therefore, it can sometimes be difficult to decide whether the temporal pallor visible on fundus examination represents a pathologic change. Pallor of the nasal rim of the optic disc is a less equivocal sign of optic atrophy.

At the optic chiasm, fibers from nasal ganglion cells decussate into the contralateral optic tract. Crossed fibers are damaged more by compression than uncrossed fibers. As a result, mass lesions of the sellar region cause a temporal hemianopia in each eye. Tumors anterior to the optic chiasm, such as meningiomas of the tuberculum sellae, produce a junctional scotoma characterized by an optic neuropathy in one eye and a superior temporal field cut in the other eye ([Fig. 28-3G](#)). More symmetric compression

of the optic chiasm by a pituitary adenoma ([Fig. 28-4](#)), meningioma, craniopharyngioma, glioma, or aneurysm results in a bitemporal hemianopia ([Fig. 28-3H](#)). The insidious development of a bitemporal hemianopia often goes unnoticed by the patient and will escape detection by the physician unless each eye is tested separately.

It is difficult to localize a postchiasmal lesion accurately, because injury anywhere in the optic tract, lateral geniculate body, optic radiations, or visual cortex can produce a homonymous hemianopia, i.e., a temporal hemifield defect in the contralateral eye and a matching nasal hemifield defect in the ipsilateral eye ([Fig. 28-3I](#)). A unilateral postchiasmal lesion leaves the visual acuity in each eye unaffected, although the patient may read the letters on only the left or right half of the eye chart. Lesions of the optic radiations tend to cause poorly matched or incongruous field defects in each eye. Damage to the optic radiations in the temporal lobe (Meyer's loop) produces a superior quadrantic homonymous hemianopia ([Fig. 28-3J](#)), whereas injury to the optic radiations in the parietal lobe results in an inferior quadrantic homonymous hemianopia ([Fig. 28-3K](#)). Lesions of the primary visual cortex give rise to dense, congruous hemianopic field defects. Occlusion of the posterior cerebral artery supplying the occipital lobe is a frequent cause of total homonymous hemianopia. Some patients with hemianopia after occipital stroke have macular sparing, because the macular representation at the tip of the occipital lobe is supplied by collaterals from the middle cerebral artery ([Fig. 28-3L](#)). Destruction of both occipital lobes produces cortical blindness. This condition can be distinguished from bilateral prechiasmal visual loss by noting that the pupil responses and optic fundi remain normal.

RED OR PAINFUL EYE

Corneal Abrasions These are seen best by placing a drop of fluorescein in the eye and looking with the slit lamp using a cobalt-blue light. A penlight with a blue filter will suffice if no slit lamp is available. Damage to the corneal epithelium is revealed by yellow fluorescence of the exposed basement membrane underlying the epithelium. It is important to check for foreign bodies. To search the conjunctival fornices, the lower lid should be pulled down and the upper lid everted. A foreign body can be removed with a moistened cotton-tipped applicator after placing a drop of topical anesthetic, such as proparacaine, in the eye. Alternatively, it may be possible to flush the foreign body from the eye by irrigating copiously with saline or artificial tears. If the corneal epithelium has been abraded, antibiotic ointment and a patch should be applied to the eye. A drop of an intermediate-acting cycloplegic, such as cyclopentolate hydrochloride 1%, helps to reduce pain by relaxing the ciliary body. The eye should be reexamined the next day. Minor abrasions may not require patching and cycloplegia.

Subconjunctival Hemorrhage This results from rupture of small vessels bridging the potential space between the episclera and conjunctiva. Blood dissecting into this space can produce a spectacular red eye, but vision is not affected and the hemorrhage resolves without treatment. Subconjunctival hemorrhage is usually spontaneous but can occur from blunt trauma, eye rubbing, or vigorous coughing. Occasionally it is a clue to an underlying bleeding disorder.

Pinguecula This is a small, raised conjunctival nodule at the temporal or nasal limbus. In adults such lesions are extremely common and have little significance, unless they

become inflamed (pingueculitis). A *pterygium* resembles a pinguecula but has crossed the limbus to encroach upon the corneal surface. Removal is justified when symptoms of irritation or blurring develop, but recurrence is a common problem.

Blepharitis This refers to inflammation of the eyelids. The most common form occurs in association with acne rosacea or seborrheic dermatitis. The eyelid margins are usually colonized heavily by staphylococcus. Upon close inspection, they appear greasy, ulcerated, and crusted with scaling debris that clings to the lashes. Treatment consists of warm compresses, strict eyelid hygiene, and topical antibiotics such as erythromycin. An external *hordeolum* (sty) is caused by staphylococcal infection of the superficial accessory glands of Zeis or Moll located in the eyelid margins. An internal hordeolum occurs after suppurative infection of the oil-secreting meibomian glands within the tarsal plate of the eyelid. Systemic antibiotics, usually tetracyclines, are sometimes necessary for treatment of meibomian gland inflammation (meibomitis) or chronic, severe blepharitis. A *chalazion* is a painless, granulomatous inflammation of a meibomian gland that produces a pealike nodule within the eyelid. It can be incised and drained, or injected with glucocorticoids. Basal cell, squamous cell, or meibomian gland carcinoma should be suspected for any nonhealing, ulcerative lesion of the eyelids.

Dacrocystitis An inflammation of the lacrimal drainage system, this can produce epiphora (tearing) and ocular injection. Gentle pressure over the lacrimal sac evokes pain and reflux of mucus or pus from the tear puncta. Dacrocystitis usually occurs after obstruction of the lacrimal system. It is treated with topical and systemic antibiotics, followed by probing or surgery to reestablish patency. *Entropion* (inversion of the eyelid) or *ectropion* (sagging or eversion of the eyelid) can also lead to epiphora and ocular irritation.

Conjunctivitis This is the most common cause of a red, irritated eye. Pain is minimal, and the visual acuity is reduced only slightly. The most common viral etiology is adenovirus infection. It causes a watery discharge, mild foreign-body sensation, and photophobia. Bacterial infection tends to produce a more mucopurulent exudate. Mild cases of infectious conjunctivitis are usually treated empirically with broad-spectrum topical ocular antibiotics, such as sulfacetamide 10%, polymixin-bacitracin-neomycin, or trimethoprim-polymixin combination. Smears and cultures are usually reserved for severe, resistant, or recurrent cases of conjunctivitis. To prevent contagion, patients should be admonished to wash their hands frequently, not to touch their eyes, and to avoid direct contact with others.

Allergic Conjunctivitis This condition is extremely common and often mistaken for infectious conjunctivitis. Three forms of allergic conjunctivitis are recognized, with closely overlapping manifestations. *Hay fever conjunctivitis* has a seasonal incidence, related to the release of airborne antigens into the air by plants. IgE-mediated activation of mast cells in the conjunctiva causes itching, redness, and edema. *Vernal conjunctivitis* is also seasonal, becoming worse during warm months. It affects exclusively children or adolescents and is more common in boys. The cause is unknown, but airborne antigens are thought to trigger symptoms. Itching, photophobia, epiphora, and mucous discharge are typical. The palpebral conjunctiva may become hypertrophic with giant excrescences called cobblestone papillae. Irritation from contact lenses or any chronic foreign body can also induce formation of cobblestone papillae.

Atopic conjunctivitis occurs in subjects with atopic dermatitis or asthma. Symptoms caused by allergic conjunctivitis can be alleviated with cold compresses, topical vasoconstrictors, antihistamines, and mast-cell stabilizers such as cromolyn sodium. Topical glucocorticoid solutions provide dramatic relief of immune-mediated forms of conjunctivitis, but their long-term use is ill-advised because of the complications of glaucoma, cataract, and secondary infection. Topical nonsteroidal anti-inflammatory agents (NSAIDs) such as ketorolac tromethamine are a better alternative.

Keratoconjunctivitis Sicca Also known as dry eye, it produces a burning, foreign-body sensation, injection, and photophobia. In mild cases the eye appears surprisingly normal, but tear production measured by wetting of a filter paper (Schirmer strip) is deficient. A variety of systemic drugs, including antihistaminic, anticholinergic, and psychotropic medications, result in dry eye by reducing lacrimal secretion. Disorders that involve the lacrimal gland directly, such as sarcoidosis or Sjogren's syndrome, also cause dry eye. Patients may develop dry eye after radiation therapy if the treatment field includes the orbits. Problems with ocular drying are also common after lesions affecting cranial nerves V or VII. Corneal anesthesia is particularly dangerous, because the absence of a normal blink reflex exposes the cornea to injury without pain to warn the patient. Dry eye is managed by frequent and liberal application of artificial tears and ocular lubricants. In severe cases the tear puncta can be plugged or cauterized to reduce lacrimal outflow.

Keratitis This is a threat to vision because of the risk of corneal clouding, scarring, and perforation. Worldwide, the two leading causes of blindness from keratitis are trachoma from chlamydial infection and vitamin A deficiency related to malnutrition. In the United States, contact lenses play a major role in corneal infection and ulceration. They should not be worn by anyone with an active eye infection. In evaluating the cornea, it is important to differentiate between a superficial infection (*keratoconjunctivitis*) and a deeper, more serious ulcerative process. The latter is accompanied by greater visual loss, pain, photophobia, redness, and discharge. Slit-lamp examination shows disruption of the corneal epithelium, a cloudy infiltrate or abscess in the stroma, and an inflammatory cellular reaction in the anterior chamber. In severe cases, pus settles at the bottom of the anterior chamber, giving rise to a hypopyon. Immediate empirical antibiotic therapy should be initiated after corneal scrapings are obtained for Gram's stain, Giemsa stain, and cultures. Fortified topical antibiotics are most effective, supplemented with subconjunctival antibiotics as required. The most frequent bacterial pathogens are *Staphylococcus*, *Streptococcus* (particularly *S. pneumoniae*), *Pseudomonas*, Enterobacteriaceae, *Haemophilus*, and *Neisseria*. For *Neisseria*, systemic antibiotics should be given in addition to topical antibiotics to eliminate systemic infection. A fungal etiology should always be considered in the patient with keratitis. Fungal infection is common in warm humid climates, especially after penetration of the cornea by plant or vegetable material.

Herpes Simplex The *herpes viruses* are a major cause of blindness from keratitis. Most adults in the United States have serum antibodies to herpes simplex, indicating prior viral infection ([Chap. 182](#)). Primary ocular infection is generally caused by herpes simplex type 1, rather than type 2. It manifests as a unilateral follicular blepharoconjunctivitis, easily confused with adenoviral conjunctivitis unless telltale vesicles appear on the periorcular skin or conjunctiva. A dendritic pattern of corneal

epithelial ulceration revealed by fluorescein staining is pathognomonic for herpes infection but is seen in only a minority of primary infections. Recurrent ocular infection arises from reactivation of the latent herpes virus. Viral eruption in the corneal epithelium may result in the characteristic herpes dendrite. Involvement of the corneal stroma produces edema, vascularization, and iridocyclitis. Herpes keratitis is treated with topical antiviral agents, cycloplegics, and oral acyclovir. Topical glucocorticoids are effective in mitigating corneal scarring but must be used with extreme caution because of the danger of corneal melting and perforation. Topical glucocorticoids also carry the risk of prolonging infection and inducing glaucoma.

Herpes Zoster Herpes zoster from reactivation of latent varicella (chickenpox) virus causes a dermatomal pattern of painful vesicular dermatitis. Ocular symptoms can occur after zoster eruption in any branch of the trigeminal nerve but are particularly common when vesicles form on the nose, reflecting nasociliary (V1) nerve involvement (Hutchinson's sign). Herpes zoster ophthalmicus produces corneal dendrites, which can be difficult to distinguish from those seen in herpes simplex. Stromal keratitis, anterior uveitis, raised intraocular pressure, ocular motor nerve palsies, acute retinal necrosis, and postherpetic scarring and neuralgia are other common sequelae. Herpes zoster ophthalmicus is treated with antiviral agents and cycloplegics. In severe cases, glucocorticoids may be added to prevent permanent visual loss from corneal scarring.

Episcleritis This is an inflammation of the episclera, a thin layer of connective tissue between the conjunctiva and sclera. Episcleritis resembles conjunctivitis but is a more localized process and discharge is absent. Most cases of episcleritis are idiopathic, but some occur in the setting of an autoimmune disease. *Scleritis* refers to a deeper, more severe inflammatory process, frequently associated with a connective tissue disease such as rheumatoid arthritis, lupus erythematosus, polyarteritis nodosa, Wegener's granulomatosis, or relapsing polychondritis. The inflammation and thickening of the sclera can be diffuse or nodular. In anterior forms of scleritis, the globe assumes a violet hue and the patient complains of severe ocular tenderness and pain. With posterior scleritis the pain and redness may be less marked, but there is often proptosis, choroidal effusion, reduced motility, and visual loss. Episcleritis and scleritis should be treated with [NSAIDs](#). If these agents fail, topical or even systemic glucocorticoid therapy may be necessary, especially if an underlying autoimmune process is active.

Uveitis Involving the anterior structures of the eye, this is called *iritis* or *iridocyclitis*. The diagnosis requires slit-lamp examination to identify inflammatory cells floating in the aqueous humor or deposited upon the corneal endothelium (keratic precipitates). Anterior uveitis develops in sarcoidosis, ankylosing spondylitis, juvenile rheumatoid arthritis, inflammatory bowel disease, psoriasis, Reiter's syndrome, and Behcet's disease. It is also associated with herpes infections, syphilis, Lyme disease, onchocerciasis, tuberculosis, and leprosy. Although anterior uveitis can occur in conjunction with many diseases, no cause is found to explain the majority of cases. For this reason, laboratory evaluation is usually reserved for patients with recurrent or severe anterior uveitis. Treatment is aimed at reducing inflammation and scarring by judicious use of topical glucocorticoids. Dilation of the pupil reduces pain and prevents the formation of synechiae.

Posterior Uveitis This is diagnosed by observing inflammation of the vitreous, retina, or

choroid on fundus examination. It is more likely than anterior uveitis to be associated with an identifiable systemic disease. Some patients have panuveitis, or inflammation of both the anterior and posterior segments of the eye. Posterior uveitis is a manifestation of autoimmune diseases such as sarcoidosis, Behcet's disease, Vogt-Koyanagi-Harada syndrome, and inflammatory bowel disease (see [Plate IV-1](#)). It also accompanies diseases such as toxoplasmosis, onchocerciasis, cysticercosis, coccidioidomycosis, toxocariasis, and histoplasmosis; infections caused by organisms such as *Candida*, *Pneumocystis carinii*, *Cryptococcus*, *Aspergillus*, herpes, and cytomegalovirus (see [Plate IV-2](#)); and other diseases such as syphilis, Lyme disease, tuberculosis, cat-scratch disease, Whipple's disease, and brucellosis. In multiple sclerosis, chronic inflammatory changes can develop in the extreme periphery of the retina (pars planitis or intermediate uveitis).

Acute Angle-Closure Glaucoma This is a rare and frequently misdiagnosed cause of a red, painful eye. Susceptible eyes have a shallow anterior chamber, either because the eye has a short axial length (hyperopia) or a lens enlarged by the gradual development of cataract. When the pupil becomes mid-dilated, the peripheral iris blocks aqueous outflow via the anterior chamber angle and the intraocular pressure rises abruptly, producing pain, injection, corneal edema, obscurations, and blurred vision. In some patients, ocular symptoms are overshadowed by nausea, vomiting, or headache, prompting a fruitless workup for abdominal or neurologic disease. The diagnosis is made by measuring the intraocular pressure during an acute attack or by performing gonioscopy to reveal the narrowed chamber angle by means of a specially mirrored contact lens. Acute angle closure is treated with oral or intravenous acetazolamide, topical beta blockers, apraclonidine, and pilocarpine to induce miosis. If these measures fail, a laser can be used to create a hole in the peripheral iris to relieve pupillary block. Many physicians are reluctant to dilate patients routinely for fundus examination because they fear precipitating an angle-closure glaucoma. The risk is actually remote and more than outweighed by the potential benefit to patients of discovering a hidden fundus lesion visible only through a fully dilated pupil. Moreover, a single attack of angle closure after pharmacologic dilation rarely causes any permanent damage to the eye and serves as an inadvertent provocative test to identify patients with narrow angles who would benefit from prophylactic laser iridectomy.

Endophthalmitis This occurs from bacterial, viral, fungal, or parasitic infection of the internal structures of the eye. It is usually acquired by hematogenous seeding from a remote site. Chronically ill, diabetic, or immunosuppressed patients, especially those with a history of indwelling intravenous catheters or positive blood cultures, are at greatest risk for endogenous endophthalmitis. Although most patients have ocular pain and injection, visual loss is sometimes the only symptom. Septic emboli, from a diseased heart valve or a dental abscess, that lodge in the retinal circulation can give rise to endophthalmitis. White-centered retinal hemorrhages (Roth's spots) are considered pathognomonic for subacute bacterial endocarditis, but they also appear in leukemia, diabetes, and many other conditions. Endophthalmitis also occurs as a complication of ocular surgery, occasionally months or even years after the operation. An occult penetrating foreign body or unrecognized trauma to the globe should be considered in any patient with unexplained intraocular infection or inflammation.

TRANSIENT OR SUDDEN VISUAL LOSS

Amaurosis Fugax This term refers to a transient ischemic attack of the retina. Because neural tissue has a high rate of metabolism, interruption of blood flow to the retina for more than a few seconds results in *transient monocular blindness*, a term used interchangeably with amaurosis fugax. Patients describe a rapid fading of vision like a curtain descending, sometimes affecting only a portion of the visual field. Amaurosis fugax usually occurs from an embolus that becomes stuck within a retinal arteriole (see [Plate IV-3](#)). If the embolus breaks up or passes, flow is restored and vision returns quickly to normal without permanent damage. With prolonged interruption of blood flow, the inner retina suffers infarction. Ophthalmoscopy reveals zones of whitened, edematous retina following the distribution of branch retinal arterioles. Complete occlusion of the central retinal artery produces arrest of blood flow and a milky retina with a cherry-red fovea (see [Plate IV-4](#)). Emboli are composed of either cholesterol (Hollenhorst plaque), calcium, or platelet-fibrin debris. The most common source is an atherosclerotic plaque in the carotid artery or aorta, although emboli can also arise from the heart, especially in patients with diseased valves, atrial fibrillation, or wall motion abnormalities.

In rare instances, amaurosis fugax occurs from low central retinal artery perfusion pressure in a patient with a critical stenosis of the ipsilateral carotid artery and poor collateral flow via the circle of Willis. In this situation, amaurosis fugax develops when there is a dip in systemic blood pressure or a slight worsening of the carotid stenosis. Sometimes there is contralateral motor or sensory loss, indicating concomitant hemispheric cerebral ischemia.

Retinal arterial occlusion also occurs rarely in association with retinal migraine, lupus erythematosus, anticardiolipin antibodies (see [Plate IV-4](#)), anticoagulant deficiency states (protein S, protein C, and antithrombin III deficiency), pregnancy, intravenous drug abuse, blood dyscrasias, dysproteinemias, and temporal arteritis.

Amaurosis fugax warns of a patient at high risk for stroke. The carotid arteries should be studied by ultrasound. Endarterectomy for a stenosis of $\geq 60\%$, even in asymptomatic patients, has been shown to reduce the subsequent rate of ipsilateral stroke ([Chap. 361](#)). Therapy with aspirin, warfarin, or other anticoagulants is appropriate in selected patients. If no carotid lesion is found, cardiac ultrasound should be performed. Ambulatory electrocardiographic monitoring may reveal that intermittent atrial fibrillation is giving rise to emboli.

Marked *systemic hypertension* causes sclerosis of retinal arterioles, splinter hemorrhages, focal infarcts of the nerve fiber layer (cotton-wool spots), and leakage of lipid and fluid (hard exudate) into the macula (see [Plate IV-5](#)). In hypertensive crisis, sudden visual loss can result from vasospasm of retinal arterioles and consequent retinal ischemia. In addition, acute hypertension may produce visual loss from ischemic swelling of the optic disc. Patients with acute hypertensive retinopathy should be treated by lowering the blood pressure. However, the blood pressure should not be reduced precipitously, because there is a danger of optic disc infarction from sudden hypoperfusion.

Impending *branch or central retinal vein occlusion* can produce prolonged visual

obscurations that resemble those described by patients with amaurosis fugax. The veins appear engorged and phlebitic, with numerous retinal hemorrhages (see [Plate IV-6](#)). In some patients, venous blood flow recovers spontaneously, while others evolve a frank obstruction with extensive retinal bleeding ("blood and thunder" appearance), infarction, and visual loss. Venous occlusion of the retina is often idiopathic, but hypertension, diabetes, and glaucoma are prominent risk factors. The benefit of treatment with anticoagulants is unproven and carries the risk of hemorrhage into the vitreous. Polycythemia, thrombocythemia, or other factors leading to an underlying hypercoagulable state should be corrected.

Anterior Ischemic Optic Neuropathy (AION) This is caused by insufficient blood flow through the posterior ciliary arteries supplying the optic disc. It produces sudden, painless, monocular visual loss, although patients occasionally report premonitory obscurations. The optic disc appears swollen and surrounded by nerve fiber layer splinter hemorrhages (see [Plate IV-7](#)). AION is divided into two forms: arteritic and nonarteritic. The nonarteritic form of AION is most common. No specific cause can be identified, although diabetes and hypertension are frequent risk factors. No treatment is available. About 5% of patients, especially those over age 60, develop the arteritic form of AION in conjunction with giant cell (temporal) arteritis ([Chap. 317](#)). It is urgent to recognize arteritic AION so that high doses of glucocorticoids can be instituted immediately to prevent blindness in the second eye. Symptoms of polymyalgia rheumatica may be present, and the sedimentation rate is usually elevated. In a patient with visual loss from suspected arteritic AION, temporal artery biopsy is helpful to confirm the diagnosis, but glucocorticoids should be started without waiting for the biopsy to be completed. The diagnosis of arteritic AION is difficult to sustain in the face of a normal sedimentation rate and a negative temporal artery biopsy, but such cases do occur rarely.

Posterior Ischemic Optic Neuropathy This is an infrequent cause of acute visual loss. It is induced by the combination of severe anemia and hypotension, causing infarction of the retrobulbar optic nerve. Cases have been reported after major blood loss during surgery, exsanguinating trauma, gastrointestinal bleeding, and renal dialysis. The fundus usually appears normal, although optic disc swelling develops if the process extends far enough anteriorly. Vision can be salvaged in some patients by prompt blood transfusion and reversal of hypotension.

Optic Neuritis This is a common inflammatory disease of the optic nerve. In the Optic Neuritis Treatment Trial (ONTT), the mean age of patients was 32 years, 77% were female, 92% had ocular pain (especially with eye movements), and 35% had optic disc swelling. In most patients, the demyelinating event was retrobulbar and the ocular fundus appeared normal on initial examination (see [Plate IV-8](#)), although optic disc pallor slowly developed over subsequent months.

Virtually all patients experience a gradual recovery of vision after a single episode of optic neuritis, even without treatment. This rule is so reliable that failure of vision to improve considerably after a first attack of optic neuritis casts doubt upon the original diagnosis. Treatment of optic neuritis is controversial because the favorable prognosis for visual recovery has made it difficult to demonstrate any benefit from glucocorticoids. The [ONTT](#) showed that patients treated with a conventional dose of oral glucocorticoids

(prednisone, 1 mg/kg per day for 14 days) did no better than patients treated with a placebo. A recent Danish trial of oral high-dose methylprednisolone (500 mg daily for 5 days, followed by a 10-day taper) reported a slight response at 1 and 3 weeks but none at 8 weeks. From these studies, it is apparent that oral glucocorticoids have little to offer in the treatment of optic neuritis. According to the ONTT, even high-dose intravenous methylprednisolone (250 mg every 6 h for 3 days) followed by oral prednisone (1 mg/kg per day for 11 days) makes no difference in final acuity (measured 6 months after the attack), although the recovery of visual function occurs more rapidly.

For some patients, optic neuritis remains an isolated event. However, the [ONTT](#) showed that the 5-year cumulative probability of developing clinically definite multiple sclerosis following optic neuritis is 30%. Remarkably, intravenous glucocorticoids were associated with a reduced rate of development of multiple sclerosis over a 2-year follow-up period, especially in the subgroup of patients with multiple foci of demyelination on their magnetic resonance (MR) scan. However, by the end of a 3-year follow-up period, patients treated with intravenous glucocorticoids versus placebo showed no difference in the rate of multiple sclerosis. Moreover, intravenous glucocorticoids did not reduce the likelihood of subsequent attacks of optic neuritis. To summarize, the organizers of the ONTT recommend an MR scan in patients with optic neuritis. If two or more foci of demyelination are found or visual loss is severe, they suggest treatment with intravenous glucocorticoids. The potential benefits of intravenous glucocorticoids are: (1) a slightly faster recovery of visual function, and (2) a potential reduction in the risk of subsequent neurologic events that would signify multiple sclerosis. Critics of the ONTT have questioned these recommendations, pointing out that: (1) visual outcome is the same in the long run, (2) evidence indicating a reduced risk of eventual multiple sclerosis with intravenous glucocorticoid treatment is based upon follow-up data in a rather small number of patients, and (3) the protection against multiple sclerosis is transient, and no longer apparent beyond 2 years of follow-up. In cases of unilateral optic neuritis, the decision whether to obtain an MR scan or to treat with intravenous glucocorticoids should be based upon clinical judgment and careful discussion with the patient. In cases of bilateral, simultaneous optic neuritis, the rationale for intravenous glucocorticoids is stronger.

Leber's Hereditary Optic Neuropathy This is a disease of young men, characterized by onset over a few weeks of painless, severe, central visual loss in one eye, followed weeks or months later by the same process in the other eye. Acutely, the optic disc appears mildly plethoric with surface capillary telangiectases, but no vascular leakage on fluorescein angiography. Eventually optic atrophy ensues. There is no treatment. Leber's optic neuropathy is caused by a point mutation at codon 11778 in the mitochondrial gene encoding nicotinamide adenine dinucleotide dehydrogenase (NADH) subunit 4. Subsequently, additional mutations responsible for the disease have been identified, most in mitochondrial genes encoding proteins involved in electron transport. Mitochondrial mutations causing Leber's neuropathy are inherited from the mother by all her children, but usually only sons develop symptoms. This curious male predilection is a mystery.

Toxic Optic Neuropathy This can result in acute visual loss with bilateral optic disc swelling and central or cecocentral scotomas. Such cases have been reported to result from exposure to ethambutol, methyl alcohol (moonshine), ethylene glycol (antifreeze),

or carbon monoxide. In toxic optic neuropathy, visual loss can also develop gradually and produce optic atrophy without a phase of acute optic disc edema (see [Plate IV-9](#)). Many agents have been implicated as a cause of toxic optic neuropathy, but the evidence supporting the association for many is weak. The following is a partial list of potential offending drugs or toxins: disulfiram, ethchlorvynol, chloramphenicol, amiodarone, monoclonal anti-CD3 antibody, ciprofloxacin, digitalis, streptomycin, lead, arsenic, thallium, D-penicillamine, isoniazid, emetine, and sulfonamides. Deficiency states, induced either by starvation, malabsorption, or alcoholism, can lead to insidious visual loss. Thiamine, vitamin B₁₂, and folate levels should be checked in any patient with unexplained, bilateral central scotomas and optic pallor.

Papilledema This connotes bilateral optic disc swelling from raised intracranial pressure (see [Plate IV-10](#)). Headache is a frequent, but not invariable, accompaniment. All other forms of optic disc swelling, e.g., from optic neuritis or ischemic optic neuropathy, should be called "optic disc edema." This convention is arbitrary but serves to avoid confusion. Often it is difficult to differentiate papilledema from other forms of optic disc edema by fundus examination alone. Transient visual obscurations are a classic symptom of papilledema. They can occur in only one eye or simultaneously in both eyes. They usually last seconds but can persist for minutes if the papilledema is fulminant. Obscurations follow abrupt shifts in posture or happen spontaneously. When obscurations are prolonged or spontaneous, the papilledema is more threatening. Visual acuity is not affected by papilledema unless the papilledema is severe, long-standing, or accompanied by macular edema and hemorrhage. Visual field testing shows enlarged blind spots and peripheral constriction ([Fig. 28-3F](#)). With unremitting papilledema, peripheral visual field loss progresses in an insidious fashion while the optic nerve develops atrophy. In this setting, reduction of optic disc swelling is an ominous sign of a dying nerve rather than an encouraging indication of resolving papilledema.

Evaluation of papilledema requires computed tomography (CT) or [MR](#) imaging to exclude an intracranial lesion. MR angiography is appropriate in selected cases to search for a dural venous sinus occlusion or an arteriovenous shunt. If neuroradiologic studies are negative, the subarachnoid opening pressure should be measured by lumbar puncture. An elevated pressure, with normal cerebrospinal fluid, points by exclusion to the diagnosis of *pseudotumor cerebri* (idiopathic intracranial hypertension). The majority of patients are young, female, and obese. Treatment with a carbonic anhydrase inhibitor such as acetazolamide lowers intracranial pressure by reducing the production of cerebrospinal fluid. Weight reduction is vital but often unsuccessful. If acetazolamide and weight loss fail, and visual field loss is progressive, lumboperitoneal shunting or optic nerve sheath fenestration should be undertaken without delay to prevent blindness. Occasionally, emergency surgery is required for sudden blindness caused by fulminant papilledema.

Optic Disc Drusen These are refractile deposits within the substance of the optic nerve head (see [Plate IV-11](#)). They are unrelated to drusen of the retina, which occur in age-related macular degeneration. Optic disc drusen are most common in people of northern European descent, with an incidence of 0.3 to 0.4%. Their diagnosis is obvious when they are visible as glittering particles upon the surface of the optic disc. However, in many patients they are hidden beneath the surface, producing an elevated optic disc with blurred margins that is easily mistaken for papilledema. It is important to recognize

pseudo-papilledema due to optic disc drusen to avoid an unnecessary evaluation for papilledema. Ultrasound or [CT](#) scanning are sensitive for detection of buried optic disc drusen because they contain calcium. In most patients, optic disc drusen are an incidental, innocuous finding, but they can produce visual obscurations. On perimetry they give rise to enlarged blind spots and arcuate scotomas from damage to the optic disc. With increasing age, drusen tend to become more exposed on the disc surface as optic atrophy develops. Hemorrhage, choroidal neovascular membrane, and [AION](#) are more likely to occur in patients with optic disc drusen. No treatment for drusen is available.

Vitreous Degeneration This occurs in all individuals with advancing age, leading to chronic and acute visual symptoms. Opacities develop in the vitreous, casting annoying shadows upon the retina. As the eye moves, these distracting "floaters" move synchronously, with a slight lag caused by inertia of the vitreous gel. Vitreous traction upon the retina causes mechanical stimulation, resulting in perception of flashing lights. This photopsia is brief and confined to one eye, in contrast to the bilateral, prolonged scintillations of cortical migraine. Contraction of the vitreous can result in sudden separation from the retina, heralded by an alarming shower of floaters and photopsia. This process, known as *vitreous detachment*, is a frequent involutional event in the elderly. It is not harmful unless it damages the retina. A careful examination of the dilated fundus is mandatory in any patient complaining of floaters or photopsia to search for peripheral tears or holes. If such a lesion is found, laser application or cryotherapy can forestall a retinal detachment. Occasionally a tear ruptures a retinal blood vessel, causing vitreous hemorrhage and sudden loss of vision. On attempted ophthalmoscopy the fundus is hidden by a dark red haze of blood. Ultrasound is required to examine the interior of the eye for a retinal tear or detachment. If the hemorrhage does not resolve spontaneously, the vitreous can be removed surgically. Vitreous hemorrhage also occurs from the fragile neovascular vessels that proliferate on the surface of the retina in diabetes, sickle cell anemia, and other ischemic ocular diseases.

Retinal Detachment This produces symptoms of floaters, flashing lights, and a scotoma in the peripheral visual field corresponding to the detachment (see [Plate IV-12](#)). If the detachment includes the fovea, there is an afferent pupil defect and the visual acuity is reduced. In most eyes, retinal detachment starts with a hole, flap, or tear in the peripheral retina (rhegmatogenous retinal detachment). Patients with peripheral retinal thinning (lattice degeneration) are particularly vulnerable to this process. Once a break has developed in the retina, liquified vitreous is free to enter the subretinal space, separating the retina from the pigment epithelium. The combination of vitreous traction upon the retinal surface and passage of fluid behind the retina leads inexorably to detachment. Patients with a history of myopia, trauma, or prior cataract extraction are at greatest risk for retinal detachment. The diagnosis is confirmed by ophthalmoscopic examination of the dilated eye.

Classic Migraine (See also [Chap. 15](#)) This usually occurs with a visual aura lasting about 20 min. In a typical attack, a small central disturbance in the field of vision marches toward the periphery, leaving a transient scotoma in its wake. The expanding border of migraine scotoma has a scintillating, dancing, or zig-zag edge, resembling the bastions of a fortified city, hence the term "fortification spectra." Patients' descriptions of fortification spectra vary widely and can be confused with amaurosis fugax. Migraine

patterns usually last longer and are perceived in both eyes, whereas amaurosis fugax is briefer and occurs in only one eye. Migraine phenomena also remain visible in the dark or with the eyes closed. Generally they are confined to either the right or left visual hemifield, but sometimes both fields are involved simultaneously. Patients often have a long history of stereotypic attacks. After the visual symptoms recede, headache develops in most patients.

Transient ischemic attacks from *vertebrobasilar insufficiency* result in acute homonymous visual symptoms. Many patients mistakenly describe symptoms in their left or right eye, when in fact they are occurring in the left or right hemifield of both eyes. Interruption of blood supply to the visual cortex causes a sudden fogging or graying of vision, occasionally with flashing lights or other positive phenomena that mimic migraine. Cortical ischemic attacks are briefer in duration than migraine, occur in older patients, and are not followed by headache. There may be associated signs of brainstem ischemia, such as diplopia, vertigo, numbness, weakness, or dysarthria.

Stroke This occurs when interruption of blood supply from the posterior cerebral artery to the visual cortex is prolonged. The only finding on examination is a homonymous visual field defect that stops abruptly at the vertical meridian. Occipital lobe stroke is usually due to thrombotic occlusion of the vertebrobasilar system, embolus, or dissection. Lobar hemorrhage, tumor, abscess, and arteriovenous malformation are other common causes of hemianopic cortical visual loss.

Factitious (Functional, Nonorganic) Visual Loss This is claimed by hysterics or malingers. The latter comprise the vast majority, seeking sympathy, special treatment, or financial gain by feigning loss of sight. The diagnosis is suspected when the history is atypical, physical findings are lacking or contradictory, inconsistencies emerge on testing, and a secondary motive can be identified. In our litigious society, the fraudulent pursuit of recompense has spawned an epidemic of factitious visual loss.

CHRONIC VISUAL LOSS

Cataract This is a clouding of the lens sufficient to reduce vision. Most cataracts develop slowly as a result of aging, leading to gradual impairment of vision. The formation of cataract occurs more rapidly in patients with a history of ocular trauma, uveitis, or diabetes mellitus. Cataracts are acquired in a variety of genetic diseases, such as myotonic dystrophy, neurofibromatosis type 2, and galactosemia. Radiation therapy and glucocorticoid treatment can induce cataract as a side effect. The cataracts associated with radiation or glucocorticoids have a typical posterior subcapsular location. Cataract can be detected by noting an impaired red reflex when viewing light reflected from the fundus with an ophthalmoscope or by examining the dilated eye using the slit lamp.

The only treatment for cataract is surgical extraction of the opacified lens. Over a million cataract operations are performed each year in the United States. The operation is generally done under local anesthesia on an outpatient basis. Remarkable technical innovations have made it possible to aspirate the cataract while leaving the lens capsule intact (extracapsular cataract extraction), rather than removing the entire lens with its capsule (intracapsular cataract extraction). A plastic or silicone intraocular lens is then

placed within the empty lens capsule in the posterior chamber, substituting for the natural lens, and leading to rapid recovery of sight. More than 95% of patients who undergo cataract extraction can expect an improvement in vision. In many patients, the lens capsule remaining in the eye after cataract extraction eventually turns cloudy, causing a secondary loss of vision. A small opening is made in the lens capsule with a laser to restore clarity.

Glaucoma This is a slowly progressive, insidious optic neuropathy, usually associated with chronic elevation of intraocular pressure. In Americans of African descent it is the leading cause of blindness. The mechanism whereby raised intraocular pressure injures the optic nerve is not understood. Axons entering the inferotemporal and superotemporal aspects of the optic disc are damaged first, producing typical nerve fiber bundle or arcuate scotomas on perimetric testing. As fibers are destroyed, the neural rim of the optic disc shrinks and the physiologic cup within the optic disc enlarges (see [Plate IV-13](#)). This process is referred to colloquially as pathologic "cupping." The cup-to-disc diameter is expressed as a ratio, e.g., 0.2/1. The cup-to-disc ratio ranges widely in normal individuals, making it difficult to diagnose glaucoma reliably simply by observing an unusually large or deep optic cup. Careful documentation of serial prospective examinations is helpful. In the patient with physiologic cupping, the large cup remains stable, whereas in the patient with glaucoma it expands relentlessly over the years. Detection of visual field loss on formal perimetry also contributes to the diagnosis of glaucoma. Finally, most patients with glaucoma have raised intraocular pressure. However, a surprising number of patients with typical glaucomatous cupping and visual field loss have intraocular pressures that apparently never exceed the normal limit of 20 mmHg (so-called low-tension glaucoma).

In acute angle-closure glaucoma, the eye is red and painful due to abrupt, severe elevation of intraocular pressure. Such cases account for only a handful of patients with glaucoma. Most patients with glaucoma have open, nonoccludable anterior chamber angles. The cause of raised intraocular pressure in these patients is uncertain. Recent studies have implicated mutations in a gene encoding a glycoprotein expressed in the trabecular meshwork. This structure serves as a filter to drain aqueous from the eye. Because the elevation of intraocular pressure develops gradually and is less marked than in angle-closure glaucoma, there is no pain or ocular injection. The central visual field and foveal acuity are spared until end-stage disease is reached. For these reasons, severe and irreversible damage can occur before either the patient or physician recognizes the diagnosis. Screening of patients for glaucoma by noting the cup-to-disc ratio on ophthalmoscopy and by measuring intraocular pressure (using a Schiotz, Tonopen, air-puff, or Goldmann tonometer) is vital. Glaucoma is treated with topical adrenergic agonists (epinephrine, dipivefrin, apraclonidine, brimonidine), cholinergic agonists (pilocarpine), beta blockers (betaxolol, carteolol, levobunolol, metipranolol, and timolol), and prostaglandin analogues (latanaprost). Occasionally, systemic absorption of beta blocker from eye drops can be sufficient to cause side effects of bradycardia, hypotension, heart block, bronchospasm, impotence, or depression. Topical or oral carbonic anhydrase inhibitors are used to lower intraocular pressure by reducing aqueous production. Laser treatment of the trabecular meshwork in the anterior chamber angle improves aqueous outflow from the eye. If medical or laser treatments fail to halt optic nerve damage from glaucoma, a filter must be constructed surgically (trabeculectomy) to release aqueous from the eye in a controlled fashion.

Macular Degeneration This is a major cause of gradual, painless, bilateral central visual loss in the elderly. The old term, "senile macular degeneration," misinterpreted by many patients as an unflattering reference, has been replaced with "age-related macular degeneration." It occurs in a nonexudative (dry) form and an exudative (wet) form. The nonexudative process begins with the accumulation of extracellular deposits, called drusen, underneath the retinal pigment epithelium. On ophthalmoscopy, they are pleomorphic but generally appear as small discrete yellow lesions clustered in the macula (see [Plate IV-14](#)). With time they become larger, more numerous, and confluent. The retinal pigment epithelium becomes focally detached and atrophic, causing visual loss by interfering with photoreceptor function. There is currently no way to prevent the development of age-related macular degeneration. Concoctions of various vitamins (A, C, and E) and minerals (zinc, copper, and selenium) have been marketed, without good evidence that they retard the process of macular degeneration.

Exudative macular degeneration, which develops in only a minority of patients, occurs when neovascular vessels from the choroid grow through defects in Bruch's membrane into the potential space beneath the retinal pigment epithelium. Leakage from these vessels produces elevation of the retina and pigment epithelium, with distortion (metamorphopsia) and blurring of vision. Although onset of these symptoms is usually gradual, bleeding from subretinal choroidal neovascular membranes sometimes causes acute visual loss. The neovascular membranes can be difficult to see on fundus examination because they are beneath the retina. Fluorescein or indocyanine green angiography is extremely useful for their detection. In some patients, prompt laser ablation of choroidal neovascular membranes seen on fluorescein angiography can halt the exudative process. However, the neovascular membranes frequently recur, requiring constant vigilance and repeated photocoagulation.

Major or repeated hemorrhage under the retina from neovascular membranes results in fibrosis, development of a round (disciform) macular scar, and permanent loss of central vision. Surgical attempts to remove subretinal membranes in age-related macular degeneration have not improved vision in most patients. However, outcomes have been more encouraging for patients with choroidal neovascular membranes from ocular histoplasmosis syndrome.

Central Serous Chorioretinopathy This primarily affects males between the ages of 20 and 50. Leakage of serous fluid from the choroid causes small, localized detachment of the retinal pigment epithelium and the neurosensory retina. These detachments produce acute or chronic symptoms of metamorphopsia and blurred vision when the macula is involved. They are difficult to visualize with a direct ophthalmoscope because the detached retina is transparent and only slightly elevated. Diagnosis of central serous chorioretinopathy is made easily by fluorescein angiography, which shows dye streaming into the subretinal space. The cause of central serous chorioretinopathy is unknown. Symptoms may resolve spontaneously if the retina reattaches, but recurrent detachment is common. Laser photocoagulation has benefited some patients with this condition.

Diabetic Retinopathy A rare disease until 1921, when the discovery of insulin resulted in a dramatic improvement in life expectancy for patients with diabetes mellitus, it is now

a leading cause of blindness in the United States. The retinopathy of diabetes takes years to develop but eventually appears in nearly all cases. Regular surveillance of the dilated fundus is crucial for any patient with diabetes. In advanced diabetic retinopathy, the proliferation of neovascular vessels leads to blindness from vitreous hemorrhage, retinal detachment, and glaucoma (see [Plate IV-15](#)). These complications can be avoided in most patients by administration of panretinal laser photocoagulation at the appropriate point in the evolution of the disease. **For further discussion of the manifestations and management of diabetic retinopathy, see Chap. 333.*

Retinitis Pigmentosa This is a general term for a disparate group of rod and cone dystrophies characterized by progressive night blindness (nyctalopia), visual field constriction with a ring scotoma, loss of acuity, and an abnormal electroretinogram (ERG). It occurs sporadically or in an autosomal recessive, dominant, or X-linked pattern. Irregular black deposits of clumped pigment in the peripheral retina, called bone spicules because of their vague resemblance to the spicules of cancellous bone, give the disease its name (see [Plate IV-16](#)). The name is actually a misnomer because retinitis pigmentosa is not an inflammatory process. Most cases are due to a mutation in the gene for rhodopsin, the rod photopigment, or in the gene for peripherin, a glycoprotein located in photoreceptor outer segments. There is no effective treatment for retinitis pigmentosa. Vitamin A (15,000 IU/day) slightly retards the deterioration of the ERG but has no beneficial effect upon visual acuity or visual fields. Some forms of retinitis pigmentosa occur in association with rare, hereditary systemic diseases (olivopontocerebellar degeneration, Bassen-Kornzweig disease, Kearns-Sayre syndrome, Refsum's disease). Chronic treatment with chloroquine, hydroxychloroquine, and phenothiazines (especially thioridazine) can produce visual loss from a toxic retinopathy that resembles retinitis pigmentosa.

Epiretinal Membrane This is a fibrocellular tissue that grows across the inner surface of the retina, causing metamorphopsia and reduced visual acuity from distortion of the macula. With the ophthalmoscope one can see a crinkled, cellophane-like membrane on the retina. Epiretinal membrane is most common in patients over 50 years of age and is usually unilateral. Most cases are idiopathic, but some occur as a result of hypertensive retinopathy, diabetes, retinal detachment, or trauma. When visual acuity is reduced to the level of about 6/24 (20/80), vitrectomy and surgical peeling of the membrane to relieve macular puckering are recommended. Contraction of an epiretinal membrane sometimes gives rise to a *macular hole*. Most macular holes, however, are caused by local vitreous traction within the fovea. Vision is usually depressed to the level of 6/30 (20/100) or worse. Vitrectomy may improve visual acuity in some patients with macular hole. Fortunately, fewer than 10% of patients with a macular hole develop a hole in their other eye.

Melanoma and Other Tumors Melanoma is the most common primary tumor of the eye (see [Plate IV-17](#)). It causes photopsia, an enlarging scotoma, and loss of vision. A small melanoma is often difficult to differentiate from a benign choroidal nevus. Careful serial examinations are required to document a malignant pattern of growth. Treatment of melanoma is controversial. Options include enucleation, local resection, and irradiation. *Metastatic tumors* to the eye outnumber primary tumors of uveal origin. Breast and lung carcinoma have a special propensity to spread to the choroid or iris. Leukemia and lymphoma also commonly invade ocular tissues. Sometimes their only

sign on eye examination is cellular debris in the vitreous, which can masquerade as a chronic posterior uveitis. *Retrobulbar tumor* of the optic nerve (meningioma, glioma) or *chiasmal tumor* (pituitary adenoma, meningioma) produces gradual visual loss with few objective findings, except for optic disc pallor. Rarely, sudden expansion of a pituitary adenoma from infarction and bleeding (*pituitary apoplexy*) causes acute retrobulbar visual loss, with headache, nausea, and ocular motor nerve palsies. In any patient with visual field loss or optic atrophy, [CT](#) or [MR](#) scanning should be considered if the cause remains unknown after careful review of the history and thorough examination of the eye ([Fig. 28-4](#)).

PROPTOSIS

When the globes appear asymmetric, the clinician must first decide which eye is abnormal. Is one eye recessed within the orbit (*enophthalmos*) or is the other eye protuberant (*exophthalmos*, or *proptosis*)? A small globe or a Horner's syndrome can give the appearance of enophthalmos. True enophthalmos occurs commonly after trauma, from atrophy of retrobulbar fat, or fracture of the orbital floor. The position of the eyes within the orbits is measured using a Hertel exophthalmometer, a hand-held instrument that records the position of the anterior corneal surface relative to the lateral orbital rim. If this instrument is not available, relative eye position can be judged by bending the patient's head forward and looking down upon the orbits. A proptosis of only 2 mm in one eye is detectable from this perspective. The development of proptosis implies a space-occupying lesion in the orbit. [ACT](#) or [MR](#) scan should be obtained in any patient with proptosis, unless the diagnosis of Graves' ophthalmopathy is certain.

Graves' Ophthalmopathy This is the leading cause of proptosis in adults ([Chap. 330](#)). The proptosis is often asymmetric and can even appear to be unilateral. Orbital inflammation and engorgement of the extraocular muscles, particularly the medial rectus and the inferior rectus, account for the protrusion of the globe. Corneal exposure, lid retraction, conjunctival injection, restriction of gaze, diplopia, and visual loss from optic nerve compression are cardinal symptoms. Acute Graves' ophthalmopathy should be treated with oral prednisone (60 mg/day) for 1 month, followed by a taper over several months. Chronic manifestations can be managed by topical lubricants, eyelid surgery, eye muscle surgery, or radiation treatment. Optic nerve compression should be relieved promptly with glucocorticoids and orbital decompression to prevent permanent visual loss.

Orbital Pseudotumor This is an idiopathic, inflammatory orbital syndrome, frequently confused with Graves' ophthalmopathy. Symptoms are pain, limited eye movements, proptosis, and congestion. Evaluation for sarcoidosis, Wegener's granulomatosis, and other types of orbital vasculitis or collagen-vascular disease is negative. Imaging often shows swollen eye muscles (orbital myositis) with enlarged tendons. By contrast, in Graves' ophthalmopathy the tendons of the eye muscles are usually spared. The Tolosa-Hunt syndrome may be regarded as an extension of orbital pseudotumor through the superior orbital fissure into the cavernous sinus. The diagnosis of orbital pseudotumor is difficult. Biopsy of the orbit frequently yields nonspecific evidence of fat infiltration by lymphocytes, plasma cells, and eosinophils. A dramatic response to a therapeutic trial of systemic glucocorticoids indirectly provides the best confirmation of the diagnosis.

Orbital Cellulitis This causes pain, lid erythema, proptosis, conjunctival chemosis, restricted motility, decreased acuity, afferent pupillary defect, fever, and leukocytosis. It often arises from a paranasal sinus, especially by contiguous spread of infection from the ethmoid sinus through the thin lamina papyracea of the medial orbit. A history of recent upper respiratory tract infection, chronic sinusitis, thick mucous secretions, or dental disease is significant in any patient with suspected orbital cellulitis. Blood cultures should be obtained, but they are usually negative. Most patients respond to empiric therapy with broad-spectrum intravenous antibiotics. Occasionally, orbital cellulitis follows an overwhelming course, with massive proptosis, blindness, septic cavernous sinus thrombosis, and meningitis. To avert this disaster, orbital cellulitis should be managed aggressively in the early stages, with immediate antibiotic therapy and imaging of the orbits. Prompt surgical drainage of an orbital abscess or paranasal sinusitis is indicated if optic nerve function deteriorates despite antibiotics.

Tumors Tumors of the orbit cause painless, progressive proptosis. The most common primary tumors are hemangioma, lymphangioma, neurofibroma, dermoid cyst, adenoid cystic carcinoma, optic nerve glioma, optic nerve meningioma, and benign mixed tumor of the lacrimal gland. Metastatic tumor to the orbit occurs frequently in breast carcinoma, lung carcinoma, and lymphoma. Diagnosis by fine-needle aspiration followed by urgent radiation therapy can sometimes preserve vision.

Carotid Cavernous Fistulas With anterior drainage through the orbit these produce proptosis, diplopia, glaucoma, and tortuous, red conjunctival vessels. Direct fistulas usually result from trauma. They are easily diagnosed because of the dramatic signs produced by high-flow, high-pressure shunting. Indirect fistulas, or dural arteriovenous malformations, are more likely to occur spontaneously, especially in older women. The signs are more subtle and the diagnosis is frequently missed. The combination of slight proptosis, diplopia, enlarged muscles, and an injected eye is often mistaken for thyroid ophthalmopathy. A bruit heard upon auscultation of the head, or reported by the patient, is a valuable diagnostic clue. Imaging shows an enlarged superior ophthalmic vein in the orbits. Carotid cavernous shunts can be eliminated by intravascular embolization.

PTOSIS

Blepharoptosis This is an abnormal drooping of the eyelid. Unilateral or bilateral ptosis can be congenital, from dysgenesis of the levator palpebrae superioris, or from abnormal insertion of its aponeurosis into the eyelid. Acquired ptosis can develop so gradually that the patient is unaware of the problem. Inspection of old photographs is helpful in dating the onset. A history of prior trauma, eye surgery, contact lens use, diplopia, systemic symptoms (e.g., dysphagia or peripheral muscle weakness), or a family history of ptosis should be sought. Fluctuating ptosis that worsens late in the day is typical of myasthenia gravis. Examination should focus upon evidence for proptosis, eyelid masses or deformities, inflammation, pupil inequality, or limitation of motility. The width of the palpebral fissures is measured in primary gaze to quantitate the degree of ptosis. The ptosis will be underestimated if the patient is compensating by lifting the brow with the frontalis muscle.

Mechanical Ptosis This occurs in many elderly patients from stretching and

redundancy of eyelid skin and subcutaneous fat (dermatochalasis). The extra weight of these sagging tissues causes the lid to droop. Enlargement or deformation of the eyelid from infection, tumor, trauma, or inflammation also results in ptosis on a purely mechanical basis.

Aponeurotic Ptosis This is an acquired dehiscence or stretching of the aponeurotic tendon, which connects the levator muscle to the tarsal plate of the eyelid. It occurs commonly in older patients, presumably from loss of connective tissue elasticity. Aponeurotic ptosis is also a frequent sequela of eyelid swelling from infection or blunt trauma to the orbit, cataract surgery, or hard contact lens usage.

Myogenic Ptosis The causes of *myogenic ptosis* include myasthenia gravis ([Chap. 380](#)) and a number of rare myopathies that manifest with ptosis. The term *chronic progressive external ophthalmoplegia* refers to a spectrum of systemic diseases caused by mutations of mitochondrial DNA. As the name implies, the most prominent findings are symmetric, slowly progressive ptosis and limitation of eye movements. In general, diplopia is a late symptom because all eye movements are reduced equally. In the *Kearns-Sayre* variant, retinal pigmentary changes and abnormalities of cardiac conduction develop. Peripheral muscle biopsy shows characteristic "ragged-red fibers." *Oculopharyngeal dystrophy* is a distinct autosomal dominant disease with onset in middle age, characterized by ptosis, limited eye movements, and trouble swallowing. *Myotonic dystrophy*, another autosomal dominant disorder, causes ptosis, ophthalmoparesis, cataract, and pigmentary retinopathy. Patients have muscle wasting, myotonia, frontal balding, and cardiac abnormalities.

Neurogenic Ptosis This results from a lesion affecting the innervation to either of the two muscles that open the eyelid: Muller's muscle or the levator palpebrae superioris. Examination of the pupil helps to distinguish between these two possibilities. In Horner's syndrome, the eye with ptosis has a smaller pupil and the eye movements are full. In an oculomotor nerve palsy, the eye with the ptosis has a larger, or a normal, pupil. If the pupil is normal but there is limitation of adduction, elevation, and depression, a pupil-sparing oculomotor nerve palsy is likely (see next section). Rarely, a lesion affecting the small, central subnucleus of the oculomotor complex will cause bilateral ptosis with normal eye movements and pupils.

DOUBLE VISION

The first point to clarify is whether diplopia persists in either eye after covering the fellow eye. If it does, the diagnosis is monocular diplopia. The cause is usually intrinsic to the eye and therefore has no dire implications for the patient. Corneal aberrations (e.g., keratoconus, pterygium), uncorrected refractive error, cataract, or foveal traction may give rise to monocular diplopia. Occasionally it is a symptom of malingering or psychiatric disease. Diplopia alleviated by covering one eye is binocular diplopia and is caused by disruption of ocular alignment. Inquiry should be made into the nature of the double vision (purely side-by-side versus partial vertical displacement of images), mode of onset, duration, intermittency, diurnal variation, and associated neurologic or systemic symptoms. If the patient has diplopia while being examined, motility testing should reveal a deficiency corresponding to the patient's symptoms. However, subtle limitation of ocular excursions is often difficult to detect. For example, a patient with a

slight left abducens nerve paresis may appear to have full eye movements, despite a complaint of horizontal diplopia upon looking to the left. In this situation, the cover test provides a more sensitive method for demonstrating the ocular malalignment. It should be conducted in primary gaze, and then with the head turned and tilted in each direction. In the above example, a cover test with the head turned to the right will maximize the fixation shift evoked by the cover test.

Occasionally, a cover test performed in an asymptomatic patient during a routine examination will reveal an ocular deviation. If the eye movements are full and the ocular misalignment is equal in all directions of gaze (concomitant deviation), the diagnosis is strabismus. In this condition, which affects about 1% of the population, fusion is disrupted in infancy or early childhood. To avoid diplopia, vision is suppressed from the nonfixating eye. In some children, this leads to impaired vision (amblyopia, or "lazy" eye) in the deviated eye.

Binocular diplopia occurs from a wide range of processes: infectious, neoplastic, metabolic, degenerative, inflammatory, and vascular. One must decide if the diplopia is neurogenic in origin or due to restriction of globe rotation by local disease in the orbit. Orbital pseudotumor, myositis, infection, tumor, thyroid disease, and muscle entrapment (e.g., from a blowout fracture) cause restrictive diplopia. The diagnosis is confirmed by performing a forced duction test in the office. After applying topical anesthesia, the physician grasps the eye with forceps and pulls it toward the direction of deficient motion. If rotation of the globe is prevented by tethering, a restrictive process is at work. The utility of this test is limited by its unpopularity with patients; in practice, the diagnosis of restriction is made by recognizing other associated signs and symptoms of local orbital disease.

Myasthenia Gravis (See also [Chap. 380](#)) This is a major cause of diplopia. The diplopia is often intermittent, variable, and not confined to any single ocular motor nerve distribution. The pupils are always normal. Fluctuating ptosis may be present. Many patients have a purely ocular form of the disease, with no evidence of systemic muscular weakness. The diagnosis can be confirmed by an intravenous edrophonium injection or by an assay for antiacetylcholine receptor antibodies. Negative results from these tests do not exclude the diagnosis. *Botulism* from food or wound poisoning can mimic ocular myasthenia.

After restrictive orbital disease and myasthenia gravis are excluded, a lesion of a cranial nerve supplying innervation to the extraocular muscles is the most likely cause of binocular diplopia.

Oculomotor Nerve ([Video 28-1](#)) The third cranial nerve innervates the medial, inferior, and superior recti; inferior oblique; levator palpebrae superioris; and the iris sphincter. Total palsy of the oculomotor nerve causes ptosis, a dilated pupil, and leaves the eye "down and out" because of the unopposed action of the lateral rectus and superior oblique. This combination of findings is obvious. More challenging is the diagnosis of an early or partial oculomotor nerve palsy. In this setting, any combination of ptosis, pupil dilation, and weakness of the eye muscles supplied by the oculomotor nerve may be encountered. Frequent serial examinations during the evolving phase of the palsy and a high index of suspicion help ensure that the diagnosis is not missed. The advent of an

oculomotor nerve palsy with any degree of pupil involvement in an otherwise healthy patient, especially when accompanied by pain, raises the specter of a circle of Willis aneurysm. If an [MR](#) scan shows no compressive lesion, an arteriogram must be performed to rule out an aneurysm of either the posterior communicating artery or the basilar artery. If the pupil is entirely normal, with all other components of an oculomotor palsy present, aneurysm is so rare that an angiogram is seldom indicated.

A lesion of the oculomotor nucleus in the rostral midbrain produces signs that differ from those caused by a lesion of the nerve itself. There is bilateral ptosis because the levator muscle is innervated by a single central subnucleus. There is also weakness of the contralateral superior rectus, because it is supplied by the oculomotor nucleus on the other side. Occasionally both superior recti are weak. Isolated nuclear oculomotor palsy is quite rare. Usually neurologic examination reveals additional signs to suggest brainstem damage from infarction, hemorrhage, tumor, or infection.

Injury to structures surrounding fascicles of the oculomotor nerve descending through the midbrain has given rise to a number of classic eponymic designations. In *Nothnagel's syndrome*, injury to the superior cerebellar peduncle causes ipsilateral oculomotor palsy and contralateral cerebellar ataxia. In *Benedikt's syndrome*, injury to the red nucleus results in ipsilateral oculomotor palsy and contralateral tremor, chorea, and athetosis. *Claude's syndrome* incorporates features of both the aforementioned syndromes, by injury to both the red nucleus and the superior cerebellar peduncle. Finally, in *Weber's syndrome*, injury to the cerebral peduncle causes ipsilateral oculomotor palsy with contralateral hemiparesis.

In the subarachnoid space the oculomotor nerve is vulnerable to aneurysm, meningitis, tumor, infarction, and compression. In cerebral herniation the nerve becomes trapped between the edge of the tentorium and the uncus of the temporal lobe. Oculomotor palsy can also occur from midbrain torsion and hemorrhages during herniation. In the cavernous sinus, oculomotor palsy arises from carotid aneurysm, carotid cavernous fistula, cavernous sinus thrombosis, tumor (pituitary adenoma, meningioma, metastasis), herpes zoster infection, and the Tolosa-Hunt syndrome.

The etiology of an isolated, pupil-sparing oculomotor palsy often remains obscure, even after neuroimaging and extensive laboratory testing. Most cases are thought to result from microvascular infarction of the nerve, somewhere along its course from the brainstem to the orbit. Usually the patient complains of pain. Diabetes, hypertension, and vascular disease are major risk factors. Spontaneous recovery over a period of months is the rule. If this fails to occur, or if new findings develop, the diagnosis of microvascular oculomotor nerve palsy should be reconsidered. Aberrant regeneration is common when the oculomotor nerve is injured by trauma or compression (tumor, aneurysm). Miswiring of sprouting fibers to the levator muscle and the rectus muscles results in elevation of the eyelid upon downgaze or adduction. The pupil also constricts upon attempted adduction, elevation, or depression of the globe. Aberrant regeneration is not seen after oculomotor palsy from microvascular infarct and hence vitiates that diagnosis.

Trochlear Nerve The fourth cranial nerve originates in the midbrain, just caudal to the oculomotor nerve complex. Fibers exit the brainstem dorsally and cross to innervate the

contralateral superior oblique. The principal actions of this muscle are to depress and to intort the globe. A palsy therefore results in hypertropia and excyclotorsion. The cyclotorsion is seldom noticed by patients. Instead, they complain of vertical diplopia, especially upon reading or looking down. The vertical diplopia is also exacerbated by tilting the head toward the side with the muscle palsy, and alleviated by tilting it away. This "head tilt test" is a cardinal diagnostic feature.

Isolated trochlear nerve palsy occurs from all the causes listed above for the oculomotor nerve, except aneurysm. The trochlear nerve is particularly apt to suffer injury after closed head trauma. The mechanism is unknown, but the free edge of the tentorium may impinge upon the nerve during a concussive blow. Most isolated trochlear nerve palsies are idiopathic and hence diagnosed by exclusion as "microvascular." Spontaneous improvement occurs over a period of months in most patients. A base-down prism (conveniently applied to the patient's glasses as a stick-on Fresnel lens) may serve as a temporary measure to alleviate diplopia. If the palsy does not resolve, the eyes can be realigned by surgically adjusting other eye muscles.

Abducens Nerve (Video 28-2) The sixth cranial nerve innervates the lateral rectus muscle. A palsy produces horizontal diplopia, worse on gaze to the side of the lesion. A nuclear lesion has different consequences, because the abducens nucleus contains interneurons that project via the medial longitudinal fasciculus to the medial rectus subnucleus of the contralateral oculomotor complex. Therefore, an abducens nuclear lesion produces a complete lateral gaze palsy, from weakness of both the ipsilateral lateral rectus and the contralateral medial rectus. *Foville's syndrome* following dorsal pontine injury includes lateral gaze palsy, ipsilateral facial palsy, and contralateral hemiparesis incurred by damage to descending corticospinal fibers. *Millard-Gubler syndrome* from ventral pontine injury is similar, except for the eye findings. There is lateral rectus weakness only, instead of gaze palsy, because the abducens fascicle is injured rather than the nucleus. Infarct, tumor, hemorrhage, vascular malformation, and multiple sclerosis are the most common etiologies of brainstem abducens palsy.

After leaving the ventral pons, the abducens nerve runs forward along the clivus to pierce the dura at the petrous apex, where it enters the cavernous sinus. Along its subarachnoid course it is susceptible to meningitis, tumor (meningioma, chordoma, carcinomatous meningitis), subarachnoid hemorrhage, trauma, and compression by aneurysm or dolichoectatic vessels. At the petrous apex, mastoiditis can produce deafness, pain, and ipsilateral abducens palsy (*Gradenigo's syndrome*). In the cavernous sinus, the nerve can be affected by carotid aneurysm, carotid cavernous fistula, tumor (pituitary adenoma, meningioma, nasopharyngeal carcinoma), herpes infection, and Tolosa-Hunt syndrome.

Unilateral or bilateral abducens palsy is a classic sign of raised intracranial pressure. The diagnosis can be confirmed if papilledema is observed on fundus examination. The mechanism is still debated but is probably related to rostral-caudal displacement of the brainstem. The same phenomenon accounts for abducens palsy from low intracranial pressure (e.g., after lumbar puncture, spinal anesthesia, or spontaneous dural cerebrospinal fluid leak).

Treatment of abducens palsy is aimed at prompt correction of the underlying cause.

However, the cause remains obscure in many instances, despite diligent evaluation. As mentioned above for isolated trochlear or oculomotor palsy, most cases are assumed to represent microvascular infarcts because they often occur in the setting of diabetes or other vascular risk factors. Some cases may develop as a postinfectious mononeuritis (e.g., following a viral flu). Patching one eye or applying a temporary prism will provide relief of diplopia until the palsy resolves. If recovery is incomplete, eye muscle surgery can nearly always realign the eyes, at least in primary position. A patient with an abducens palsy that fails to improve should be reevaluated for an occult etiology (e.g., chordoma, carcinomatous meningitis, carotid cavernous fistula, myasthenia gravis).

Multiple Ocular Motor Nerve Palsies These should not be attributed to spontaneous microvascular events affecting more than one cranial nerve at a time. This remarkable coincidence does occur, especially in diabetic patients, but the diagnosis is made only in retrospect after exhausting all other diagnostic alternatives. Neuroimaging should focus on the cavernous sinus, superior orbital fissure, and orbital apex, where all three ocular motor nerves are in close proximity. In the diabetic or compromised host, fungal infection (*Aspergillus*, Mucorales, *Cryptococcus*) is a frequent cause of multiple nerve palsies. In the patient with systemic malignancy, carcinomatous meningitis is a likely diagnosis. Cytologic examination may be negative despite repeated sampling of the cerebrospinal fluid. The cancer-associated Lambert-Eaton myasthenic syndrome can also produce ophthalmoplegia. Giant cell (temporal) arteritis occasionally manifests as diplopia from ischemic palsies of extraocular muscles ([Figs. 28-CD1](#) and [28-CD2](#)). Fisher syndrome, an ocular variant of Guillain-Barre, can produce ophthalmoplegia with areflexia and ataxia. Often the ataxia is mild, and the areflexia is overlooked because the physician's attention is focused upon the eyes.

Supranuclear Disorders of Gaze These are often mistaken for multiple ocular motor nerve palsies. For example, Wernicke's encephalopathy can produce nystagmus and a partial deficit of horizontal and vertical gaze that mimics a combined abducens and oculomotor nerve palsy. The disorder occurs in malnourished or alcoholic patients and can be reversed by giving thiamine. Infarct, hemorrhage, tumor, multiple sclerosis, encephalitis, vasculitis, and Whipple's disease are other important causes of supranuclear gaze palsy.

The *frontal eye field* of the cerebral cortex is involved in generation of saccades to the contralateral side. After hemispheric stroke, the eyes usually deviate towards the lesioned side because of the unopposed action of the frontal eye field in the normal hemisphere. With time, this deficit resolves. Seizures generally have the opposite effect: the eyes deviate conjugately away from the irritative focus. *Parietal lesions* disrupt smooth pursuit of targets moving toward the side of the lesion. Bilateral parietal lesions produce *Balint's syndrome*, characterized by impaired eye-hand coordination (optic ataxia), difficulty initiating voluntary eye movements (ocular apraxia), and visuospatial disorientation (simultanagnosia).

Horizontal Gaze Descending cortical inputs mediating horizontal gaze ultimately converge at the level of the pons. Neurons in the paramedian pontine reticular formation are responsible for controlling conjugate gaze toward the same side. They project directly to the ipsilateral abducens nucleus. A lesion of either the paramedian pontine reticular formation or the abducens nucleus causes an ipsilateral conjugate gaze palsy.

Lesions at either locus produce nearly identical clinical syndromes, with the following exception: vestibular stimulation (oculocephalic maneuver or caloric) will succeed in driving the eyes conjugately to the side in a patient with a lesion of the paramedian pontine reticular formation, but not in a patient with a lesion of the abducens nucleus.

Internuclear Ophthalmoplegia This results from damage to the medial longitudinal fasciculus ascending from the abducens nucleus in the pons to the oculomotor nucleus in the midbrain (hence, "internuclear"). Damage to fibers carrying the conjugate signal from abducens interneurons to the contralateral medial rectus motoneurons results in a failure of adduction on attempted lateral gaze. For example, a patient with a left internuclear ophthalmoplegia will have slowed or absent adducting movements of the left eye. A patient with bilateral injury to the medial longitudinal fasciculus will have bilateral internuclear ophthalmoplegia. Multiple sclerosis is the most common cause, although tumor, stroke, trauma, or any brainstem process may be responsible. ***One-and-a half syndrome*** is due to a combined lesion of the medial longitudinal fasciculus and the abducens nucleus on the same side. The patient's only horizontal eye movement is abduction of the eye on the other side.

Vertical Gaze This is controlled at the level of the midbrain. The neuronal circuits affected in disorders of vertical gaze are not well elucidated, but lesions of the rostral interstitial nucleus of the medial longitudinal fasciculus and the interstitial nucleus of Cajal cause supranuclear paresis of upgaze, downgaze, or all vertical eye movements. Distal basilar artery ischemia is the most common etiology. ***Skew deviation*** refers to a vertical misalignment of the eyes, usually constant in all positions of gaze. The finding has poor localizing value because skew deviation has been reported after lesions in widespread regions of the brainstem and cerebellum.

Parinaud's Syndrome Also known as dorsal midbrain syndrome, this is a distinct supranuclear vertical gaze disorder from damage to the posterior commissure. It is a classic sign of hydrocephalus from aqueductal stenosis. Pineal region tumors (germinoma, pineoblastoma), cysticercosis, and stroke also cause Parinaud's syndrome. Features include loss of upgaze (and sometimes downgaze), convergence-retraction nystagmus on attempted upgaze, downwards ocular deviation ("setting sun" sign), lid retraction (Collier's sign), skew deviation, pseudoabducens palsy, and light-near dissociation of the pupils. Disorders of vertical gaze, especially downwards saccades, are an early feature of progressive supranuclear palsy. Smooth pursuit is affected later in the course of the disease. Parkinson's disease, Huntington's chorea, and olivopontocerebellar degeneration can also affect vertical gaze.

Nystagmus This is a rhythmical oscillation of the eyes, occurring physiologically from vestibular and optokinetic stimulation or pathologically in a wide variety of diseases. Abnormalities of the eyes or optic nerves, present at birth or acquired in childhood, can produce a complex, searching nystagmus with irregular pendular (sinusoidal) and jerk features. This nystagmus is commonly referred to as ***congenital sensory nystagmus***. It is a poor term, because even in children with congenital lesions, the nystagmus does not appear until several months of age. ***Congenital motor nystagmus***, which looks similar to congenital sensory nystagmus, develops in the absence of any abnormality of the sensory visual system. Visual acuity is also reduced in congenital motor nystagmus, probably by the nystagmus itself, but seldom below a level of 20/200.

Jerk Nystagmus This is characterized by a slow drift off the target, followed by a fast corrective saccade. By convention, the nystagmus is named after the quick phase. Jerk nystagmus can be downbeat, upbeat, horizontal (left or right), and torsional. The pattern of nystagmus may vary with gaze position. Some patients will be oblivious to their nystagmus. Others will complain of blurred vision, or a subjective, to-and-fro movement of the environment (oscillopsia) corresponding to their nystagmus. Fine nystagmus may be difficult to see upon gross examination of the eyes. Observation of nystagmoid movements of the optic disc on ophthalmoscopy is a sensitive way to detect subtle nystagmus. The slit lamp is also useful.

Gaze-Evoked Nystagmus This is the most common form of jerk nystagmus. When the eyes are held eccentrically in the orbits, they have a natural tendency to drift back to primary position. The subject compensates by making a corrective saccade to maintain the deviated eye position. Many normal patients have mild gaze-evoked nystagmus. Exaggerated gaze-evoked nystagmus can be induced by drugs (sedatives, anticonvulsants, alcohol); muscle paresis; myasthenia gravis; demyelinating disease; and cerebellopontine angle, brainstem, and cerebellar lesions.

Vestibular Nystagmus *Vestibular nystagmus* results from dysfunction of the labyrinth (Meniere's disease), vestibular nerve, or vestibular nucleus in the brainstem. Peripheral vestibular nystagmus often occurs in discrete attacks, with symptoms of nausea and vertigo. There may be associated tinnitus and hearing loss. Sudden shifts in head position may provoke or exacerbate symptoms.

Downbeat Nystagmus *Downbeat nystagmus* occurs from lesions near the craniocervical junction (Chiari malformation, basilar invagination). It has also been reported in brainstem or cerebellar stroke, lithium or anticonvulsant intoxication, alcoholism, and multiple sclerosis. *Upbeat nystagmus* is associated with damage to the pontine tegmentum, from stroke, demyelination, or tumor.

Opsoclonus This rare, dramatic disorder of eye movements consists of bursts of consecutive saccades (saccadomania). When the saccades are confined to the horizontal plane, the term *ocular flutter* is preferred. It can occur from viral encephalitis, trauma, or a paraneoplastic effect of neuroblastoma, breast carcinoma, and other malignancies. It has also been reported as a benign, transient phenomenon in otherwise healthy patients.

(Bibliography omitted in Palm version)

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29. DISORDERS OF SMELL, TASTE, AND HEARING - Anil K. Lalwani, James B. Snow, Jr.

SMELL

The sense of smell determines the flavor and palatability of food and drink. It serves, along with the trigeminal system, as a monitor of inhaled chemicals, including dangerous substances such as natural gas, smoke, and air pollutants. Loss of or decreased ability to smell affects approximately 1% of people under age 60 and more than half of the population beyond this age.

DEFINITIONS

Smell is the perception of odor by the nose. *Taste* is the perception of salty, sweet, sour, or bitter by the tongue. Related sensations during eating such as somatic sensations of coolness, warmth, and irritation are mediated through the trigeminal, glossopharyngeal, and vagal afferents in the nose, oral cavity, tongue, pharynx, and larynx. *Flavor* is the complex interaction of taste, smell, and somatic sensation.

Terms relating to disorders of smell include *anosmia*, an absence of the ability to smell; *hyposmia*, a decreased ability to smell; *hyperosmia* (an increased sensitivity to an odorant); *dysosmia* (distortion in the perception of an odor); *phantosmia*, perception of an odorant where none is present; and *agnosia*, inability to classify, contrast, or identify odor sensations verbally, even though the ability to distinguish between odorants or to recognize them may be normal. An odor stimulus is referred to as an *odorant*. Each category of smell dysfunction can be further subclassified as total (applying to all odorants) or partial (dysfunction of only select odorants).

PHYSIOLOGY OF SMELL

The *olfactory neuroepithelium* is located in the superior part of the nasal cavities. It contains an orderly arrangement of bipolar olfactory receptor cells, microvillar cells, sustentacular cells, and basal cells. The dendritic process of the bipolar cell has a bulb-shaped knob, or vesicle, that projects into the mucous layer and bears six to eight cilia. It is the cilia that contain the odorant receptors. The arrangement of cilia increases overall exposure to the environment and translates into each bipolar cell containing 56 cm² (9 in.²) of surface area to receive stimulus.

The microvillar cells are located adjacent to the receptor cells on the surface of the neuroepithelium. The sustentacular cells, unlike their counterparts in the respiratory epithelium, are not specialized to secrete mucus. Although they form a tight barrier separating the neurons from the outside environment, their complete function is unknown. The basal cells are progenitors of other cell types in the olfactory neuroepithelium, including the bipolar receptor cells. There is a regular turnover of the bipolar receptor cells, which function as the primary sensory neurons. In addition, with injury to the cell body or its axon, the receptor cell is replaced by a differentiated basal cell which reestablishes a central neural connection. Hence these primary sensory neurons are unique among sensory systems in that they are regularly replaced and regenerate after injury.

The unmyelinated axons of the receptor cells form the fila of the olfactory nerve, pass through the cribriform plate, and terminate within spherical masses of neuropil, termed *glomeruli*, in the olfactory bulb. The glomeruli are the focus of a high degree of convergence of information, since many more fibers enter than leave them. The main second-order neurons are the mitral cells. The primary dendrite of each mitral cell extends into a single glomerulus. Axons of the mitral cells project along with the axons of adjacent tufted cells to the limbic system, including the anterior olfactory nucleus, the prepiriform cortex, the periamygdaloid cortex, the olfactory tubercle, the nucleus of the lateral olfactory tract, and the corticomедial nucleus of the amygdala. Cognitive awareness of smell requires stimulation of the prepiriform cortex or the amygdaloid nuclei.

A secondary potential site of olfactory chemosensation is located in the epithelium of the vomeronasal organ, a tubular structure that opens on the ventral aspect of the nasal septum. Sensory neurons located in the vomeronasal organ detect pheromones, nonvolatile chemical signals that in lower mammals trigger innate and stereotyped reproductive and social behaviors, as well as neuroendocrine changes. The neurons from the organ project to the accessory olfactory bulbs and not the main olfactory bulb, as in the olfactory neuroepithelium. Whether humans use the vomeronasal organ to detect and respond to chemical signals from others remains controversial. Recent work in delineating the molecular pathology of Kallman syndrome suggests that development of the olfactory and vomeronasal system is required for normal sexual maturation ([Chap. 335](#)).

The sensation of smell begins with introduction of an odorant to the cilia of the bipolar neuron. Most odorants are hydrophobic; as they move from the air phase of the nasal cavity to the aqueous phase of the olfactory mucous, they are transported toward the cilia by small water-soluble proteins called *odorant-binding proteins* and reversibly bind to receptors on the cilia surface. Binding causes conformational changes in the receptor protein, which induces a chain of biochemical events and results in generation of action potentials in the primary neurons. Transduction depends on the activation of G protein-coupled second messengers. Intensity appears to be coded by the amount of firing in the afferent neurons.

Basic elements of the genetic coding involved in smell are now becoming understood. Olfactory receptor proteins belong to the large family of G protein-coupled receptors that also includes rhodopsins; α - and β -adrenergic receptors; muscarinic acetylcholine receptors; and neurotransmitter receptors for dopamine, serotonin, and substance P. Members of the G protein-coupled receptors are characterized by the presence of 7 putative α -helical transmembrane domains composed of 20 to 28 hydrophobic amino acid residues. In mammals, there are probably 300 to 1000 olfactory receptor genes belonging to 20 different families located on various chromosomes in clusters. The receptor genes are present at more than 25 different human chromosomal locations. The gene clusters have likely risen as a result of repeated duplication of individual genes or clusters of genes. Each olfactory neuron seems to express only one or, at most, a few receptor genes thus providing the molecular basis of odor discrimination. While the receptors are expressed in several tissues, including the olfactory neuroepithelium and mammalian germ cells, their primary role appears to be odorant

recognition and discrimination. Bipolar cells that express similar receptors appear to be scattered across discrete spatial zones. These similar cells converge on a select few number of glomeruli in the olfactory bulb. The result is a potential spatial map of how we receive odor stimulus, much like the tonotopic organization of how we perceive sound.

DISORDERS OF THE SENSE OF SMELL

Disorders of the sense of smell are caused by conditions that interfere with the access of the odorant to the olfactory neuroepithelium (transport loss), injure the receptor region (sensory loss), or damage central olfactory pathways (neural loss). At the present time, there are no clinical tests to differentiate these different types of olfactory losses. Fortunately, the history of the disease provides important clues to the cause. The leading causes of olfactory disorders are summarized in [Table 29-1](#); the most common etiologies are head trauma and viral infections. Head trauma is a frequent cause of anosmia in children and young adults, whereas viral etiologies predominate in older adults.

Cranial trauma is followed by unilateral or bilateral impairment of smell in up to 15% of cases; anosmia is more common than hyposmia. Olfactory dysfunction is more common when trauma is associated with loss of consciousness, moderately severe head injury (grades II to V), and skull fracture. Frontal injuries and fractures disrupt the cribriform plate and olfactory axons that perforate it. Sometimes there is an associated cerebrospinal fluid (CSF) rhinorrhea resulting from a tearing of the dura overlying the cribriform plate and paranasal sinuses. Anosmia may also follow blows to the occiput. Once traumatic anosmia develops, it is usually permanent; only 10% of patients ever improve or recover. Perversion of the sense of smell may occur as a transient phase in the recovery process.

Viral infections destroy the olfactory neuroepithelium, which is replaced by respiratory epithelium. Parainfluenza virus type 3 appears to be especially detrimental to human olfaction. HIV infection is associated with subjective distortion of taste and smell, which may become more severe as the disease progresses. The loss of taste and smell may play an important role in the development and progression of HIV-associated wasting. Congenital anosmias are rare but important. Kallmann syndrome is an X-linked disorder characterized by congenital anosmia and hypogonadotropic hypogonadism resulting from a failure of migration from the olfactory placode of olfactory receptor neurons and neurons synthesizing gonadotropin-releasing hormone ([Chap. 328](#)). The responsible gene (*KAL*) has been cloned. Anosmia can also occur in albinos. The receptor cells are present but are hypoplastic, lack cilia, and do not project above the surrounding supporting cells.

Meningioma of the inferior frontal region is the most frequent neoplastic cause of anosmia; loss of smell may be the only neurologic abnormality at presentation. Rarely, anosmia can occur with glioma of the frontal lobe. Occasionally, pituitary adenomas, craniopharyngiomas, suprasellar meningiomas, and aneurysms of the anterior part of the circle of Willis extend forward and damage olfactory structures. These tumors and hamartomas may also induce seizures with olfactory hallucinations, indicating involvement of the uncus of the temporal lobe.

Dysosmia, subjective distortions of olfactory perception, may occur with intranasal disease that partially impairs smell or may represent a phase in the recovery from a neurogenic anosmia. Most dysosmic disorders consist of disagreeable or foul odors, and they may be accompanied by distortions of taste. Dysosmia is associated with depression.

Approach to the Patient

The history of the onset and course of the disorder is often paramount in establishing an etiology. Unilateral anosmia is rarely a complaint and is only recognized by separate testing of smell in each nasal cavity. Bilateral anosmia, on the other hand, brings patients to medical attention. Anosmic patients usually complain of a loss of the sense of taste even though their taste thresholds may be within normal limits. In actuality, they are complaining of a loss of flavor detection, which is mainly an olfactory function. The physical examination should include a complete examination of the ears, upper respiratory tract, and head and neck. A neurologic examination emphasizing the cranial nerves and cerebellar and sensorimotor function is essential. The patient's general mood should be assessed, and any signs of depression should be noted.

The sensory evaluation of olfactory function is necessary to corroborate the patient's complaint, evaluate the efficacy of treatment, and assess the degree of permanent impairment. The degree to which qualitative sensations are present can be assessed by any of several methods. The Odor Stix test uses a commercially available odor-producing magic marker-like pen held approximately 8 to 15 cm (3 to 6 in.) from the patient's nose to check for gross perception of the odorant. Another gross perception of odorant test, the 30-cm alcohol test, uses a freshly opened isopropyl alcohol packet held approximately 30 cm (12 in.) from the patient's nose. There is a commercially available scratch-and-sniff card containing three odors available for testing olfaction grossly. A superior test is the University of Pennsylvania Smell Identification Test (UPSIT). This consists of a 40-item, forced choice, microencapsulated odor, scratch-and-sniff paradigm. For example, one of the items reads, "This odor smells most like (a) chocolate, (b) banana, (c) onion, or (d) fruit punch," and the patient is instructed to answer one of the alternatives. The test is highly reliable, is sensitive to age and sex differences, and provides an accurate quantitative determination of the olfactory deficit. Persons with a total loss of smell function score in the range of 7 to 19 out of 40. The average score for total anosmics is slightly higher than that expected on the basis of chance because of the inclusion of some odorants that act by trigeminal stimulation.

The second step is to establish a detection threshold for the odorant phenyl ethyl alcohol, using a graduated stimulus. Sensitivity for each side of the nose is determined with a detection threshold for phenyl ethyl methyl ethyl carbinol. Nasal resistance can also be measured with anterior rhinomanometry for each side of the nose.

Computed tomography (CT) or magnetic resonance imaging (MRI) of the head is required to rule out paranasal sinusitis, neoplasms of the anterior cranial fossa, nasal cavity, or paranasal sinuses and unsuspected fractures of the anterior cranial fossa. Bone abnormalities are best seen with CT. MRI is useful in evaluating olfactory bulbs, ventricles, and other soft tissue of the brain. Coronal CT is optimal for assessing

cribriform plate, anterior cranial fossa, and sinus anatomy.

Techniques have been developed to biopsy the olfactory neuroepithelium, but in view of the widespread degeneration of the olfactory neuroepithelium and intercalation of respiratory epithelium in the olfactory area of adults with no apparent olfactory dysfunction, biopsy material must be interpreted cautiously.

TREATMENT

Therapy for patients with transport olfactory losses due to allergic rhinitis, bacterial rhinitis and sinusitis, polyps, neoplasms, and structural abnormalities of the nasal cavities can be undertaken rationally and with a high likelihood for improvement. Allergy management, antibiotic therapy, topical and systemic glucocorticoid therapy, and surgery for nasal polyps, deviation of the nasal septum, and chronic hyperplastic sinusitis are frequently effective in restoring the sense of smell.

There is no treatment with demonstrated efficacy for sensorineural olfactory losses. Fortunately, spontaneous recovery often occurs. Zinc and vitamin therapy (especially with vitamin A) are advocated by some. Profound zinc deficiency can produce loss and distortion of the sense of smell but is not a clinically important problem except in very limited geographic areas ([Chap. 75](#)). The epithelial degeneration associated with vitamin A deficiency can cause anosmia, but in western societies the prevalence of vitamin A deficiency is low. Exposure to cigarette smoke and other airborne toxic chemicals can cause metaplasia of the olfactory epithelium. Spontaneous recovery can occur if the insult is discontinued. Patient counseling is therefore helpful in these cases.

As mentioned above, more than half of people over age 60 suffer from olfactory dysfunction. No effective treatment exists for presbyosmia, but patients are often reassured to learn that this problem is common in their age group. In addition, early recognition and counseling can help patients to compensate for the loss of smell. The incidence of natural gas-related accidents is disproportionately high in the elderly, perhaps due in part to the gradual loss of smell. Mercaptan, the pungent odor in natural gas, is an olfactory stimulant and does not activate taste receptors. Many elderly with olfactory dysfunction experience a decrease in flavor sensation and find it necessary to hyperflavor food, usually by increasing the amount of salt in their diet. The physician can assist patients in developing healthy strategies to deal with the decreased sense of smell.

TASTE

Compared with disorders of smell, gustatory disorders are uncommon and their pathogenesis poorly understood. Many patients with a loss of olfactory sensitivity also complain of a loss of the sense of taste. On testing, most of these patients have normal detection thresholds for taste.

DEFINITIONS

Disturbances of the sense of taste may be categorized as *total ageusia* -- total absence of gustatory function or inability to detect the qualities of sweet, salt, bitter, or sour;

partial ageusia -- ability to detect some of but not all the qualitative gustatory sensations; *specific ageusia* -- inability to detect the taste quality of certain substances; *total hypogeusia* -- decreased sensitivity to all tastants; *partial hypogeusia* -- decreased sensitivity to some tastants; and *dysgeusia* or *phantogeusia* -- distortion in the perception of a tastant, i.e., the perception of the wrong quality when a tastant is presented or the perception of a taste when there has been no tastant ingested. Confusions of sour and bitter are common and, at times, may be semantic misunderstandings. Frequently, however, they have physiologic or pathophysiologic bases. Other taste quality confusions occur between sour and salty and bitter. It may be possible to differentiate between the loss of flavor recognition in patients with olfactory losses who complain of a loss of taste as well as smell by asking if they are able to taste sweetness in sodas, saltiness in potato chips, etc.

PHYSIOLOGY OF TASTE

The taste receptor cells are located in the taste buds, spherical groups of cells arranged in a pattern resembling the segments of a citrus fruit. At the surface, the taste bud has a pore into which microvilli of the receptor cells project. Unlike the olfactory system, the receptor cell is not the primary neuron. Instead, gustatory afferent nerve fibers contact individual taste receptor cells. Transduction depends on activation of G protein-coupled second messengers but differs in details for each taste quality.

The sense of taste is mediated through the facial, glossopharyngeal, and vagal nerves. The gustatory system consists of at least five receptor populations. Taste buds are located in the papillae along the lateral margin and dorsum of the tongue at the junction of the dorsum and the base of the tongue, and in the palate, epiglottis, larynx, and esophagus. The chorda tympani branch of the facial nerve subserves taste from the anterior two-thirds of the tongue. The posterior third of the tongue is supplied by the lingual branch of the glossopharyngeal nerve. Afferents from the palate travel with the greater superficial petrosal nerve to the geniculate ganglion and then via the facial nerve to the brainstem. The internal branch of the superior laryngeal nerve of the vagus nerve contains the taste afferents from the larynx, including the epiglottis and esophagus.

The central connections of the nerves terminate in the brainstem in the nucleus of the tractus solitarius. The central pathway from the nucleus of the tractus solitarius projects to the ipsilateral parabrachial nuclei of the pons. Two divergent pathways project from the parabrachial nuclei. One ascends to the gustatory relay in the dorsal thalamus, synapses, and continues to the cortex of the insula. There is also evidence for a direct pathway from the parabrachial nuclei to the cortex. (Olfaction and gustation appear to be unique among sensory systems in that at least some fibers bypass the thalamus.) The other pathway from the parabrachial nuclei goes to the ventral forebrain, including the lateral hypothalamus, substantia innominata, central nucleus of the amygdala, and the stria terminalis.

Tastants gain access to the receptor cells through the taste pore. Four classes of taste are recognized: sweet, salt, sour, and bitter. Individual gustatory afferent fibers almost always respond to a number of different chemicals. Response patterns of gustatory afferent axons can be grouped into classes based on the stimulus chemical that produces the largest response. For example, for sucrose-best response neurons, the

second-best stimulus is almost always sodium chloride. The fact that individual gustatory afferent fibers respond to a large number of different chemicals led to the *across-fiber-pattern* theory of gustatory coding, while the best-stimulus analysis led to the concept of *labeled* afferents. It appears that labeled fibers are important for establishing gross quality, but the across-fiber pattern within a best-stimulus category, and perhaps among categories, is needed for discriminating chemicals within qualities. For example, sweetness may be carried by sucrose-best neurons, but the differentiation of sucrose and fructose may require a comparison of the relative activity among sucrose-best, salt-best, and quinine-best neurons. As with olfaction and other sensory systems, intensity appears to be encoded by the quantity of neural activity.

DISORDERS OF THE SENSE OF TASTE

Disorders of the sense of taste are caused by conditions that interfere with the access of the tastant to the receptor cells in the taste bud (transport loss), injure receptor cells (sensory loss), or damage gustatory afferent nerves and central gustatory pathways (neural loss) ([Table 29-2](#)). *Transport gustatory losses* result from xerostomia due to many causes, including Sjogren's syndrome, radiation therapy, heavy-metal intoxication, and bacterial colonization of the taste pore. *Sensory gustatory losses* are caused by inflammatory and degenerative diseases in the oral cavity; a vast number of drugs, particularly those that interfere with cell turnover such as antithyroid and antineoplastic agents; radiation therapy to the oral cavity and pharynx; viral infections; endocrine disorders; neoplasms; and aging. *Neural gustatory losses* occur with neoplasms, trauma, and surgical procedures in which the gustatory afferents are injured. Taste buds degenerate when their gustatory afferents are transected but remain when their somatosensory afferents are severed. Patients with renal disease have increased thresholds for sweet and sour tastes, which resolves with dialysis.

A side effect of medication is the single most common cause of taste dysfunction in clinical practice. The mechanism may be a change in the composition of saliva, an effect on receptor function or signal transduction, or disruption of the central processing of gustatory input. Unfortunately, the responsible mechanism is not well understood for most medications. Xerostomia, regardless of the etiology, can be associated with taste dysfunction. It is associated with poor oral clearance, poor dental hygiene, and can adversely affect the oral mucosa, all leading to dysgeusia. However, severe salivary gland failure does not necessarily lead to taste complaints. Xerostomia, along with the use of antibiotics or glucocorticoids, and compromised immune function can lead to overgrowth of *Candida*; overgrowth alone, without thrush or overt signs of infection can be associated with bad taste or hypogeusia. When taste dysfunction occurs in a patient at risk for fungal overgrowth, a trial of nystatin or other anti-fungal medication is warranted.

Upper respiratory infections and head trauma can lead to both smell and taste dysfunction; taste is more likely to improve than smell. The mechanism of taste disturbance in these situations is not well understood. Trauma to the chorda tympani branch of the facial nerve during middle ear surgery or third molar extractions is relatively common and can cause dysgeusia. Bilateral chorda tympani injuries are usually associated with hypogeusia, whereas unilateral lesions produce only limited symptoms, perhaps because responses from taste receptors are disinhibited by the

glossopharyngeal nerve.

Finally, aging itself may be associated with reduced taste sensitivity. The taste dysfunction may be limited to a single compound and may be mild. While many older patients may acknowledge loss of taste when asked, they are unlikely to seek medical attention for taste disturbance alone.

Approach to the Patient

Patients who complain of loss of taste should be evaluated for both gustatory and olfactory function. Clinical assessment of taste is not as well developed or standardized as that of smell. The first step is to perform suprathreshold whole-mouth taste testing for quality, intensity, and pleasantness perception of four taste qualities: sweet, salty, sour, and bitter. Most commonly used reagents for taste testing are sucrose, citric acid or hydrochloric acid, caffeine or quinine (sulfate or hydrochloride), and sodium chloride. The taste stimuli should be freshly prepared. For quantification, detection thresholds are obtained by applying graduated dilutions to the tongue quadrants or by whole-mouth sips. Electric taste testing (*electrogustometry*) is used clinically to identify taste deficits in specific quadrants of the tongue, following precise applications of stimuli. Regional gustatory testing may also be performed to assess for the possibility of loss localized to one or more receptor fields as a result of a peripheral or central lesion.

Once there is objective evidence of a disorder of taste, it is important to establish an anatomic diagnosis before proceeding to an etiologic diagnosis. The history of the disease often provides important clues to the cause. For example, absence of taste on the anterior two-thirds of the tongue associated with a facial paralysis indicates that the lesion is proximal to the juncture of the chorda tympani branch with the facial nerve in the mastoid.

TREATMENT

Therapy for gustatory loss is limited. Nonetheless, some etiologies of taste dysfunction are amenable to intervention. Taste disturbance related to drugs can often be resolved by changing the prescribed medication. Xerostomia can be treated with artificial saliva, providing some benefits to patients with a disturbed salivary milieu. Oral pilocarpine may be beneficial for a variety of forms of xerostomia. Appropriate treatment of bacterial and fungal infections of the oral cavity can be of great help in improving taste function. Taste dysfunction following trauma may resolve spontaneously without intervention and is more likely to do so than posttraumatic smell dysfunction. Altered taste due to surgical stretch injury of chorda tympani nerve usually improves within 3 to 4 months, while dysfunction is usually permanent with transection of the nerve. In most patients with idiopathic cases of altered taste sensitivity, the problem either remains stable or worsens. Zinc and vitamin therapy for gustatory losses is advocated by some but lacks demonstrated efficacy. No effective therapeutic strategies exist for the sensorineural disorders of taste.

HEARING

Hearing loss is one of the most common sensory disorders in humans. Nearly 10% of

the adult population has some hearing loss. For many, this impairment presents early in life. However, hearing loss can present at any age. Between 30 and 35% of individuals over the age of 65 have a hearing loss of sufficient magnitude to require a hearing aid.

PHYSIOLOGY OF HEARING ([Fig. 29-1](#))

Hearing occurs by air conduction and bone conduction. In air conduction, sound waves reach the ear by propagation in air, enter the external auditory canal, and set the tympanic membrane in motion, which in turn moves the malleus, incus, and stapes of the middle ear. Movement of the footplate of the stapes causes pressure changes in the fluid-filled inner ear eliciting a traveling wave in the basilar membrane of the cochlea. The tympanic membrane and the ossicular chain in the middle ear serve as an impedance-matching mechanism, improving the efficiency of energy transfer from air to the fluid-filled inner ear. Hearing by bone conduction occurs when the sounding source, in contact with the head, results in vibration of the bones of the skull, including the temporal bone, producing a traveling wave in the basilar membrane.

Stereocilia of the hair cells of the organ of Corti, which rests on the basilar membrane, are in contact with the tectorial membrane and are deformed by the traveling wave. A point of maximal displacement of the basilar membrane is determined by the frequency of the stimulating tone. High-frequency tones cause maximal displacement of the basilar membrane near the base of the cochlea. As the frequency of the stimulating tone decreases, the point of maximal displacement moves toward the apex of the cochlea.

The inner and outer hair cells of the organ of Corti have different innervation patterns, but both are mechanoreceptors. The afferent innervation relates principally to the inner hair cells, and the efferent innervation relates principally to outer hair cells. The motility of the outer hair cells alters the micromechanics of the inner hair cells creating a cochlear amplifier, which explains the exquisite sensitivity and frequency selectivity of the cochlea.

The current concept of cochlear transduction is that displacement of the tips of the stereocilia allows potassium to flow into the cell, resulting in its depolarization. The potassium influx opens calcium channels near the base of the cell, stimulating transmitter release. The neurotransmitter at the hair cell and cochlear nerve dendrite interface is thought to be glutamate. The action potential in the eighth nerve occurs 0.5 ms after the onset of the cochlear microphonic potential. Each of the cochlear nerve neurons can be activated at a frequency and intensity specific for that cell. This specificity is maintained at each point of the central auditory pathway: dorsal and ventral cochlear nuclei, trapezoid body, superior olivary complex, lateral lemniscus, inferior colliculus, medial geniculate body, and auditory cortex. At low frequencies, individual auditory nerve fibers can respond more or less synchronously with the stimulating tone. At higher frequencies, phase-locking occurs so that neurons alternate in response to particular phases of the cycle of the sound wave. Intensity is encoded by the amount of neural activity in individual neurons, the number of neurons that are active, and the specific neurons that are activated.

GENETIC CAUSES OF HEARING LOSS

More than half of childhood hearing impairment is thought to be hereditary; hereditary hearing impairment (HHI) can also manifest later in life. HHI may be classified as either nonsyndromic, when hearing loss is the only clinical abnormality, or syndromic, when hearing loss is associated with anomalies in other organ systems. Nearly two-thirds of HHIs are nonsyndromic and the remaining one-third are syndromic. Between 70 and 80% of nonsyndromic HHI is inherited in an autosomal recessive manner; another 15 to 20% is autosomal dominant. Less than 5% is X-linked or maternally inherited via the mitochondria.

Over 60 loci harboring genes for nonsyndromic [HHI](#) have been mapped, with equal numbers of dominant and recessive modes of inheritance; 14 different genes have been cloned ([Table 29-3](#)). The hearing genes fall into the categories of structural proteins (MYO7A, MYO15, TECTA, DIAPH1), transcription factors (POU3F4, POU4F3), ion channels (KCNQ4, PDS), and gap junction proteins (Cx26, Cx30, Cx31). Several of these genes, including connexin 26 (Cx26), TECTA, and MYO7A, cause both autosomal dominant and recessive forms of nonsyndromic HHI. In general, the hearing loss associated with dominant genes has its onset in adolescence or adulthood and varies in severity, whereas the hearing loss associated with recessive inheritance is congenital and profound. Connexin 26 is particularly important because it is associated with nearly 20% of cases of childhood deafness; in heterozygotes the onset of hearing loss may be in adolescence or adulthood. Two frame-shift mutations, 30delG and 167delT, account for >50% of the cases, making population screening feasible. The 167delT mutation is highly prevalent in Ashkenazi Jews; it is predicted that 1 in 1765 individuals in this population will be homozygous and affected. The hearing loss can also vary among the members of the same family, suggesting that other genes or factors likely influence the auditory phenotype.

The contribution of genetics to presbycusis (see below) is also becoming better understood. In addition to connexin 26, several other nonsyndromic genes are associated with hearing loss that progresses with age. It is likely that presbycusis has both environmental and genetic components.

Over 200 syndromes are associated with hearing loss. Common syndromic forms of hearing loss include Usher syndrome (retinitis pigmentosa and hearing loss), Waardenburg syndrome (pigmentary abnormality and hearing loss), Pendred syndrome (thyroid organification defect and hearing loss), Alport syndrome (renal disease and hearing loss), Jervell and Lange-Nielsen syndrome (prolonged QT interval and hearing loss), neurofibromatosis type 2 (bilateral acoustic schwannoma), and mitochondrial disorders [mitochondrial encephalopathy, lactic acidosis, and stroke-like episodes (MELAS); myoclonic epilepsy and ragged red fibers (MERRF); progressive external ophthalmoplegia (PEO)] ([Table 29-4](#)).

Rapid progress in understanding the basis of these and related disorders has revealed a fascinating complexity, including evidence for genetic heterogeneity (different genes resulting in a similar clinical phenotype), allelic disorders (distinct phenotypes associated with different mutations in the same gene), and polygenic modifiers ([Chap. 65](#)).

DISORDERS OF THE SENSE OF HEARING

Hearing loss can result from disorders of the auricle, external auditory canal, middle ear, inner ear, or central auditory pathways ([Fig. 29-2](#)). *In general, lesions in the auricle, external auditory canal, or middle ear cause conductive hearing losses, whereas lesions in the inner ear or eighth nerve cause sensorineural hearing losses.*

Conductive Hearing Loss This results from obstruction of the external auditory canal by cerumen, debris, and foreign bodies; swelling of the lining of the canal; atresia of the ear canal; neoplasms of the canal; perforations of the tympanic membrane; disruption of the ossicular chain, as occurs with necrosis of the long process of the incus in trauma or infection; otosclerosis; or fluid, scarring, or neoplasms in the middle ear.

Cholesteatoma, i.e., the presence of stratified squamous epithelium in the middle ear or mastoid, occurs frequently in adults. A cholesteatoma is a benign, slowly growing lesion that destroys bone and normal ear tissue. Major theories of pathogenesis of acquired cholesteatoma include traumatic implantation and invasion, immigration and invasion through a perforation, and metaplasia following chronic infection and irritation. On examination, there is often a perforation filled with cheesy white squamous debris. A chronically draining ear that fails to respond to appropriate antibiotic therapy should raise the suspicion of a cholesteatoma. Conductive hearing loss secondary to ossicular erosion is common. Surgery is required to remove this insidiously growing and destructive disease process.

Conductive hearing loss in the presence of a normal ear canal and intact tympanic membrane is suggestive of ossicular pathology. Fixation of the stapes from *otosclerosis* is a common cause of low-frequency conductive hearing loss. It occurs with equal frequency in men and women and has a simple autosomal dominant inheritance with incomplete penetrance. Hearing impairment usually presents between the late teens to the forties. In women, the hearing loss is often first noticeable during pregnancy, as the otosclerotic process is accelerated during pregnancy. A hearing aid or a short outpatient surgical procedure (stapedectomy) can provide adequate auditory rehabilitation. Extension of otosclerosis beyond the stapes footplate to involve the cochlea (cochlear otosclerosis) can lead to mixed or sensorineural hearing loss. Fluoride therapy to prevent hearing loss associated with cochlear otosclerosis remains controversial.

Eustachian tube dysfunction is extremely common in adults and may predispose to acute otitis media (AOM) or serous otitis media (SOM). Trauma, AOM, or chronic otitis media are the usual factors responsible for tympanic membrane perforation. While small perforations often heal spontaneously, larger defects usually require surgical intervention. Tympanoplasty is highly effective (>90%) in the repair of tympanic membrane perforations. Otoscopy is usually sufficient to diagnose AOM, SOM, chronic otitis media, cerumen impaction, tympanic membrane perforation, and eustachian tube dysfunction.

Sensorineural Hearing Loss Damage to the hair cells of the organ of Corti may be caused by intense noise, viral infections, ototoxic drugs (e.g., salicylates, quinine and its synthetic analogues, aminoglycoside antibiotics, loop diuretics such as furosemide and ethacrynic acid, and cancer chemotherapeutic agents such as cisplatin), fractures of the temporal bone, meningitis, cochlear otosclerosis (see above), Meniere's disease, and

aging. Congenital malformations of the inner ear may be the cause of hearing loss in some adults. Genetic predisposition alone or in concert with environmental influences may also be responsible.

Presbycusis (age-associated hearing loss) is the most common cause of sensorineural hearing loss in adults. In the early stages, it is characterized by symmetric, gentle to sharply sloping high-frequency hearing loss. With progression, the hearing loss involves all frequencies. More importantly, the hearing impairment is associated with significant loss in clarity. There is a loss of discrimination for phonemes, recruitment (abnormal growth of loudness), and particular difficulty in understanding speech in noisy environments. Hearing aids may provide limited rehabilitation once the word recognition score deteriorates below 50%. Significant advancements and improvements in cochlear implants have made them the treatment of choice when hearing aids prove inadequate (<30% word recognition score with optimal amplification).

Meniere's disease is characterized by episodic vertigo, fluctuating sensorineural hearing loss, tinnitus, and aural fullness. Tinnitus and/or deafness may be absent during the initial attacks of vertigo, but they invariably appear as the disease progresses and are increased in severity during an acute attack. The annual incidence of Meniere's disease is 0.5 to 7.5 per 1000; onset is most frequently in the fifth decade of life but may also occur in young adults or the elderly. Histologically, there is distention of the endolymphatic system (endolymphatic hydrops) leading to degeneration of vestibular and cochlear hair cells. This may result from endolymphatic sac dysfunction secondary to infection, trauma, autoimmune disease, inflammatory causes, or tumor; an idiopathic etiology constitutes the largest category and is most accurately referred to as Meniere's disease. Although any pattern of hearing loss can be observed, typically, low-frequency, unilateral sensorineural hearing impairment is present. [MRI](#) should be obtained to exclude retrocochlear pathology such as cerebellopontine angle tumors or demyelinating disorders. Therapy is directed towards the control of vertigo. A low-salt diet is the mainstay of treatment of the control of rotatory vertigo. Diuretics, a short course of glucocorticoids, and intratympanic gentamicin may also be useful adjuncts in recalcitrant cases. Surgical therapy of vertigo is reserved for unresponsive cases and includes endolymphatic sac decompression, labyrinthectomy, and vestibular nerve section. Both labyrinthectomy and vestibular nerve section abolish rotatory vertigo in >90% of the cases. Unfortunately, there is no effective therapy for hearing loss, tinnitus, or aural fullness associated with Meniere's disease.

Sensorineural hearing loss may also result from any neoplastic, vascular, demyelinating, infectious, or degenerative disease or trauma affecting the central auditory pathways. Human immunodeficiency virus leads to both peripheral and central auditory system pathology and is associated with sensorineural hearing impairment.

An individual can have both conductive and sensory hearing loss termed *mixed hearing loss*. Mixed hearing losses are due to pathology that can affect the middle and inner ear simultaneously such as otosclerosis involving the ossicles and the cochlea, head trauma, chronic otitis media, cholesteatoma, middle ear tumors, and some inner ear malformations.

Trauma resulting in temporal bone fractures may be associated with conductive,

sensorineural, and mixed hearing loss. If the fracture spares the inner ear, there may simply be conductive hearing loss due to rupture of the tympanic membrane or disruption of the ossicular chain. These abnormalities are amenable to surgical correction. Profound hearing loss and severe vertigo are associated with temporal bone fractures involving the inner ear. A perilymphatic fistula associated with leakage of inner-ear fluid into the middle ear can occur and may require surgical repair. An associated facial nerve injury is not uncommon. [CT](#) is best suited to assess fracture of the traumatized temporal bone, evaluate the ear canal, and determine the integrity of the ossicular chain and the involvement of the inner ear. [CSF](#) leaks that accompany temporal bone fractures are usually self-limited; the use of prophylactic antibiotics is controversial.

Tinnitus is defined as the perception of a sound when there is no sound in the environment. It may have a buzzing, roaring, or ringing quality and may be pulsatile (synchronous with the heartbeat). Tinnitus is often associated with either a conductive or sensorineural hearing loss. The pathophysiology of tinnitus is not well understood. The cause of the tinnitus can usually be determined by finding the cause of the associated hearing loss. Tinnitus may be the first symptom of a serious condition such as a vestibular schwannoma. Pulsatile tinnitus requires evaluation of the vascular system of the head to exclude vascular tumors such as glomus jugulare tumors, aneurysms, and stenotic arterial lesions; it may also occur with [SOM](#).

Approach to the Patient

The goals in the evaluation of a patient with auditory complaints are to determine: (1) the nature of the hearing impairment (conductive vs. sensorineural), (2) the severity of the impairment (mild, moderate, severe, profound), (3) the anatomy of the impairment (external ear, middle ear, inner ear, or central auditory pathway pathology, and (4) the etiology. Initially, the history and the physical examination are critical in the identification of the underlying pathology leading to the auditory deficit. The history should elicit characteristics of the hearing loss, including the duration of deafness, nature of onset (sudden vs. insidious), rate of progression (rapid vs. slow), and involvement of the ear (unilateral vs. bilateral). The presence or absence of tinnitus, vertigo, imbalance, aural fullness, otorrhea, headache, facial nerve dysfunction, and head and neck paresthesias should be ascertained. Information regarding head trauma, exposure to ototoxins, occupational or recreational noise exposure, and family history of hearing impairment may also be important. A sudden onset of unilateral hearing loss, with or without tinnitus, may represent a viral infection of the inner ear or a vascular accident. Patients with unilateral hearing loss (sensory or conductive) usually complain of reduced hearing, poor sound localization, and difficulty hearing clearly with background noise. Gradual progression of a hearing deficit is common with otosclerosis, noise-induced hearing loss, vestibular schwannoma, or Meniere's disease. Small vestibular schwannomas typically present with asymmetric hearing impairment, tinnitus, and imbalance (rarely vertigo); cranial neuropathy, in particular of the trigeminal or facial nerve, may accompany larger tumors. In addition to hearing loss, Meniere's disease may be associated with episodic vertigo, tinnitus, and aural fullness. Hearing loss with otorrhea is most likely due to chronic otitis media or cholesteatoma.

Family history may be crucial in delineating a genetic basis of hearing impairment. The

history may also help identify environmental risk factors that lead to hearing impairment in a family. Sensitivity to aminoglycoside ototoxicity, maternally transmitted through a mitochondrial mutation, can be ascertained through a careful family history ([Chap. 67](#)). Susceptibility to noise-induced hearing loss or age-related hearing loss (presbycusis) may also be genetically determined.

The physical examination should evaluate the auricle, external ear canal, and tympanic membrane. The external ear canal of the elderly is often dry and fragile; it is preferable to clean cerumen with wall-mounted suction and cerumen loops and to avoid irrigation. In examining the eardrum, the topography of the tympanic membrane is more critical than the presence or absence of the highly touted light reflex. In addition to the pars tensa (the lower two-thirds of the eardrum), the pars flaccida above the short process of the malleus should also be examined for retraction pockets that may be evidence of chronic eustachian tube dysfunction or cholesteatoma. Insufflation of the ear canal is necessary to assess tympanic membrane mobility and compliance. Careful inspection of the nose, nasopharynx, and upper respiratory tract is indicated. Unilateral serous effusion in the adult should prompt a fiberoptic examination of the nasopharynx to exclude neoplasms. Cranial nerves should be carefully evaluated with special attention to facial and trigeminal nerves, which are commonly disturbed with tumors involving the cerebellopontine angle.

The Weber and Rinne tuning fork tests are used to differentiate conductive from sensorineural hearing losses and to confirm the findings of audiologic evaluation. Rinne's test compares the ability to hear by air conduction with the ability to hear by bone conduction. The tines of a vibrating tuning fork are held near the opening of the external auditory canal, and then the stem is placed on the mastoid process; for direct contact, it may be placed on teeth or dentures. The patient is asked to indicate whether the tone is louder by air conduction or bone conduction. Normally, and in the presence of sensorineural hearing loss, a tone is heard louder by air conduction than by bone conduction; however, with conductive hearing loss of ≥ 30 dB (see "Audiologic Assessment," below), the bone-conduction stimulus is perceived as louder than the air-conduction stimulus. The Rinne test is most sensitive in detecting mild conductive hearing losses if a 256-Hz tuning fork is used. The Weber test may be performed with a 256- or 512-Hz fork. The stem of a vibrating tuning fork is placed on the head in the midline and the patient asked whether the tone is heard in both ears or better in one ear than in the other. With a unilateral conductive hearing loss, the tone is perceived in the affected ear. With a unilateral sensorineural hearing loss, the tone is perceived in the unaffected ear. As a general rule, a 5-dB difference in hearing between the two ears is required for lateralization. The combined information from the Weber and Rinne tests permits a tentative conclusion as to whether a conductive or sensorineural hearing loss is present; however, these tests are associated with significant false-positive and -negative responses and therefore should be utilized only as screening tools.

LABORATORY ASSESSMENT OF HEARING

Audiologic Assessment The minimum audiologic assessment for hearing loss should include the measurement of pure tone air-conduction and bone-conduction thresholds, speech reception threshold, discrimination score, tympanometry, acoustic reflexes, and acoustic-reflex decay. This test battery provides a comprehensive screening evaluation

of the whole auditory system and allows one to determine whether further differentiation of a sensory (cochlear) from a neural (retrocochlear) hearing loss is indicated.

Pure tone audiometry assesses hearing acuity for pure tones. The test is administered by an audiologist and is performed in a sound-attenuated chamber. The pure tone stimulus is delivered with an audiometer, an electronic device that allows the presentation of specific frequencies (generally between 250 and 8000 Hz) at specific intensities. Air and bone conduction thresholds are established for each ear. Air conduction thresholds are established by presenting the stimulus in air with the use of headphones. Bone conduction thresholds are accomplished by placing the stem of a vibrating tuning fork or an oscillator of an audiometer in contact with the head. In the presence of a hearing loss, broad-spectrum noise is presented to the nontest ear for *masking* purposes so that responses are based on perception from the ear under test.

The responses are measured in decibels. An *audiogram* is a plot of intensity in decibels required to achieve threshold versus frequency. A decibel (dB) is equal to 20 times the logarithm of the ratio of the sound pressure required to achieve threshold in the patient to the sound pressure required to achieve threshold in a normal hearing person. Therefore, a change of 6 dB represents doubling of sound pressure, and a change of 20 dB represents a ten-fold change in sound pressure. Loudness, which depends on the frequency, intensity, and duration of a sound, doubles with approximately each 10-dB increase in sound pressure level. Pitch, on the other hand, does not directly correlate with frequency. The perception of pitch changes slowly in the low and high frequencies. In the middle tones, which are important for human speech, pitch varies more rapidly with changes in frequency.

Pure tone audiometry establishes the presence and severity of hearing impairment, unilateral vs. bilateral involvement, and the type of hearing loss. Conductive hearing losses with a large mass component, as is often seen in middle-ear effusions, produce elevation of thresholds that predominate in the higher frequencies. Conductive hearing losses with a large stiffness component, as in fixation of the footplate of the stapes in early otosclerosis, produce threshold elevations in the lower frequencies. Often, the conductive hearing loss involves all frequencies, suggesting involvement of both stiffness and mass. In general, sensorineural hearing losses such as presbycusis affect higher frequencies more than lower frequencies. An exception is Meniere's disease, which is characteristically associated with low-frequency sensorineural hearing loss. Noise-induced hearing loss has an unusual pattern of hearing impairment in which the loss at 4000 Hz is greater than at higher frequencies. Vestibular schwannomas characteristically affect the higher frequencies, but any pattern of hearing loss can be observed.

Speech recognition requires greater synchronous neural firing than is necessary for appreciation of pure tones. *Speech audiometry* tests the clarity with which one hears. The *speech reception threshold* (SRT) is defined as the intensity at which speech is recognized as a meaningful symbol and is obtained by presenting two-syllable words with an equal accent on each syllable. The intensity at which the patient can repeat 50% of the words correctly is the SRT. Once the SRT is determined, discrimination or word recognition ability is tested by presenting one-syllable words at 25 to 40 dB above the speech reception threshold. The words are phonetically balanced in that the phonemes

(speech sounds) occur in the list of words at the same frequency that they occur in ordinary conversational English. An individual with normal hearing or conductive hearing loss can repeat 88 to 100% of the phonetically balanced words correctly. Patients with a sensorineural hearing loss have variable loss of discrimination depending on the severity of hearing loss and the site of lesion. Further, as a general rule, neural lesions are associated with more deterioration in discrimination ability than are lesions in the inner ear. For example, in a patient with mild asymmetric sensorineural hearing loss, a clue to the diagnosis of vestibular schwannoma is the presence of greater than expected deterioration in discrimination ability. Deterioration in discrimination ability at higher intensities above the SRT also suggests a lesion in the eighth nerve or central auditory pathways.

Tympanometry measures the impedance of the middle ear to sound and is particularly useful in the identification and diagnosis of middle-ear effusions. A sounding source and microphone are introduced into the ear canal with an airtight seal. The amount of sound that is absorbed through the middle ear or reflected from the middle ear is measured at the microphone. In conductive hearing losses, more sound is reflected than in the normal middle ear. The pressure in the ear canal can be increased or decreased from atmospheric pressure. A *tympanogram* is the graphic representation of change in impedance or compliance as the pressure in the ear canal is changed. It provides information about the status of the tympanic membrane and the ossicular chain. Normally, the middle ear is most compliant at atmospheric pressure, and the compliance decreases as the pressure is increased or decreased; this pattern is seen with normal hearing or in the presence of sensorineural hearing loss. Compliance that does not change with change in pressure suggests middle-ear effusion. With a negative pressure in the middle ear, as with eustachian tube obstruction, the point of maximal compliance occurs with negative pressure in the ear canal. A tympanogram in which no point of maximal compliance can be obtained is most commonly seen with discontinuity of the ossicular chain. A reduction in the maximal compliance peak can be seen in otosclerosis.

During tympanometry, an intense tone (80 dB above the hearing threshold) elicits contraction of the stapedius muscle. The change in compliance of the middle ear with contraction of the stapedius muscle can be detected. The presence or absence of this *acoustic reflex* is important in the anatomic localization of facial nerve paralysis as well as hearing loss. Normal or elevated acoustic reflex thresholds in an individual with significant sensorineural hearing impairment suggests a cochlear hearing loss. Assessment of *acoustic reflex decay* helps differentiate sensory from neural hearing losses. In neural hearing loss, the reflex adapts or decays with time.

Otoacoustic emissions (OAE) can be measured with sensitive microphones inserted into the external auditory canal. The emissions may be spontaneous or evoked with sound stimulation. The presence of OAEs indicates that the outer hair cells of the organ of Corti are intact and can be used to assess auditory thresholds and to distinguish sensory from neural hearing losses.

Evoked Responses *Electrocochleography* measures the earliest evoked potentials generated in the cochlea and the auditory nerve. Receptor potentials recorded include the cochlear microphonic, generated by the outer hair cells of the organ of Corti, and the

summating potential, generated by the inner hair cells in response to sound. The whole nerve action potential representing the composite firing of the first-order neurons can also be recorded during electrocochleography. Clinically, the test is useful in the diagnosis of Meniere's disease where an elevation of the ratio of summating potential to action potential is seen.

Brainstem auditory evoked responses (BAERs) are useful in differentiating the site of sensorineural hearing loss ([Chap. 356](#)). In response to sound, five distinct electrical potentials arising from different stations along the peripheral and central auditory pathway can be recorded with computer averaging from scalp surface electrodes. BAERs are valuable in situations in which patients cannot or will not give reliable voluntary thresholds. They are also used to assess the integrity of the auditory nerve and brainstem in various clinical situations, including intraoperative monitoring and in determination of brain death.

Imaging Studies The choice of radiologic tests is largely determined by whether the goal is to evaluate the bony anatomy of the external, middle, and inner ear or to image the auditory nerve and brain. Axial and coronal [CT](#) of the temporal bone with fine 1-mm cuts is ideal for determining the caliber of the external auditory canal, integrity of the ossicular chain, and presence of middle-ear or mastoid disease; it can also detect inner-ear malformations. CT is also ideal for the detection of bone erosion often seen in the presence of chronic otitis media and cholesteatoma. [MRI](#) is superior to CT for imaging of retrocochlear pathology such as vestibular schwannoma, meningioma, other lesions of the cerebellopontine angle, demyelinating lesions of the brainstem, and brain tumors. Recent experience suggests that both CT and MRI are equally capable of identifying inner-ear malformations and assessing cochlear patency for preoperative evaluation of patients for cochlear implantation.

TREATMENT

In general, conductive hearing losses are amenable to surgical intervention and correction, while sensorineural hearing losses are permanent. The diagnosis of conductive hearing loss is usually straightforward, and the etiology of the conductive deficit is often apparent on physical examination. Atresia of the ear canal can be surgically repaired, often with significant improvement in hearing. Tympanic membrane perforations due to chronic otitis media or trauma can be repaired with an outpatient tympanoplasty. Likewise, conductive hearing loss associated with otosclerosis can be treated by stapedectomy, which is successful in 90 to 95% of cases. Tympanostomy tubes allow the prompt return of normal hearing in individuals with middle-ear effusions. Hearing aids are effective and well-tolerated in patients with conductive hearing losses.

Patients with mild, moderate, and severe sensorineural hearing losses are regularly rehabilitated with hearing aids of varying configuration and strength. Hearing aids have been improved to provide greater fidelity and have been miniaturized. The current generation of hearing aids can be placed entirely within the ear canal, thus reducing the stigma associated with their use. In general, the more severe the hearing impairment, the larger the hearing aid required for auditory rehabilitation. Digital hearing aids lend themselves to individual programming, and multiple and directional microphones at the ear level may be helpful in noisy surroundings. Since all hearing aids amplify noise as

well as speech, the only absolute solution to the problem found thus far is to place the microphone closer to the speaker than the noise source. This arrangement is not possible with a self-contained, cosmetically acceptable device. It is cumbersome and requires a user-friendly environment.

In many situations, including lectures and the theater, hearing-impaired persons benefit from assistive devices that are based on the principle of having the speaker closer to the microphone than any source of noise. Assistive devices include infrared and FM transmission as well as an electromagnetic loop around the room for transmission to the individual's hearing aid. Hearing aids with telecoils can also be used with properly equipped telephones in the same way.

In the event that the hearing aid provides inadequate rehabilitation, cochlear implants are appropriate. Criteria for implantation include severe to profound hearing loss with word recognition score $\leq 30\%$ under best aided conditions. Children with congenital and acquired profound hearing impairment are also appropriate candidates for cochlear implantation. Worldwide, more than 20,000 deaf individuals (including 4000 children) have received cochlear implants. Cochlear implants are neural prostheses that convert sound energy to electrical energy and can be used to stimulate the auditory division of the eighth nerve directly. In most cases of profound hearing impairment, the auditory hair cells are lost but the ganglionic cells of the auditory division of the eighth nerve are preserved. Cochlear implants consist of electrodes that are inserted into the cochlea through the round window, speech processors that extract acoustical elements of speech for conversion to electrical currents, and a means of transmitting the electrical energy through the skin. Patients with implants experience sound that helps with speech reading, allows open-set word recognition, and helps in modulating the person's own voice. Usually, within 3 months after implantation, adult patients can understand speech without visual cues. With the current generation of multichannel cochlear implants, nearly 75% of patients are able to converse on the telephone. It is anticipated that improvements in the electrode design and speech processors will permit further enhancement in understanding speech, especially in the presence of background noise.

For individuals who have had both eighth nerves destroyed by trauma or bilateral vestibular schwannomas (e.g., neurofibromatosis type 2), brainstem auditory implants placed near the cochlear nucleus may provide auditory rehabilitation. It is hoped that additional advances may provide benefits similar to those with the cochlear implant.

Tinnitus can often accompany hearing loss. The treatment of tinnitus is particularly problematic. Therapy is usually directed towards minimizing the appreciation of tinnitus. Relief of the tinnitus may be obtained by masking it with background music. Hearing aids are also helpful in tinnitus suppression, as are tinnitus maskers, devices that present a sound to the affected ear that is more pleasant to listen to than the tinnitus. The use of a tinnitus masker is often followed by several hours of inhibition of the tinnitus. Antidepressants have also shown beneficial effect in helping patients deal with tinnitus.

Tinnitus and background noise can significantly affect understanding of speech in individuals with hearing impairment. Hard-of-hearing individuals often benefit from a reduction in unnecessary noise (e.g., radio or television) to enhance the signal-to-noise

ratio. Speech comprehension is aided by lip reading; therefore, the impaired listener should be seated so that the face of the speaker is well illuminated and can be seen at all times. Speaking directly into the ear is occasionally helpful, but usually more is lost in communication than gained when the speaker's face cannot be seen. Speech should be slow enough to make each word distinct, but overly slow speech is distracting and loses contextual and speech-reading benefits. Although speech should be in a loud, clear voice, one should be aware that in sensorineural hearing losses in general and in elderly hard-of-hearing persons in particular, recruitment (the ability to hear loud sounds normally loud) may be troublesome. Above all, optimal communication cannot take place without both parties giving it their full and undivided attention.

PREVENTION

Conductive hearing losses may be prevented by prompt and appropriate antibiotic therapy of adequate duration for [AOM](#) and by ventilation of the middle ear with tympanostomy tubes in middle-ear effusions lasting 12 weeks or longer. Loss of vestibular function and deafness due to aminoglycoside antibiotics can largely be prevented by careful monitoring of serum peak and trough levels.

Some 10 million Americans have noise-induced hearing loss, and 20 million are exposed to hazardous noise in their employment. Noise-induced hearing loss can be prevented by avoidance of exposure to loud noise or by regular use of ear plugs or fluid-filled ear muffs to attenuate intense sound. Noise-induced hearing loss results from recreational as well as occupational activities and begins in adolescence. High-risk activities for noise-induced hearing loss include wood and metal working with electrical equipment and target practice and hunting with small firearms. All internal-combustion and electric engines, including snow and leaf blowers, snowmobiles, outboard motors, and chain saws, require protection of the user with hearing protectors. Virtually all noise-induced hearing loss is preventable through education, which should begin before the teenage years. Programs of industrial conservation of hearing are required when the exposure over an 8-h period averages 85 dB on the A scale. Workers in such noisy environments can be protected with preemployment audiologic assessment, the mandatory use of hearing protectors, and annual audiologic assessments.

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(Bibliography omitted in Palm version)

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30. INFECTIONS OF THE UPPER RESPIRATORY TRACT - *Marlene Durand, Michael Joseph*

Infections of the upper respiratory tract include some of the most common infectious diseases encountered by internists and other primary-care physicians. Pharyngitis, laryngitis, rhinitis, sinusitis, otitis externa, and otitis media account for millions of visits to physicians annually. Although these infections are usually mild enough to be treated on an outpatient basis, the primary-care physician must be able to recognize their serious complications, such as peritonsillar abscess from pharyngitis, subperiosteal abscess from frontal sinusitis, and temporal-bone osteomyelitis from invasive otitis externa. The physician must also identify potentially life-threatening infections of the head and neck, such as epiglottitis, Ludwig's angina, and rhinocerebral mucormycosis.

INFECTIONS OF THE NOSE AND FACE

Skin infections that commonly affect the nose and face include folliculitis, furunculosis, impetigo, and erysipelas. These are discussed in detail elsewhere ([Chap. 128](#), in particular [Table 128-1](#)).

Infection of the mucosal surface of the nose is most commonly due to respiratory viruses (e.g., rhinovirus) and presents as acute rhinitis. There are several rare, chronic intranasal infections. *Ozena*, or atrophic rhinitis, is characterized by atrophied mucosa overlaid by foul-smelling dry crusts (Greek *ozein*, "stench"). *Klebsiella ozaenae* is often isolated from nasal cultures, but whether it is a cause of illness or merely a colonizer is unclear. Intranasal irrigation with aminoglycosides (e.g., tobramycin ophthalmic solution) or oral administration of ciprofloxacin has resulted in clinical improvement in some cases. *Klebsiella rhinoscleromatis* causes *rhinoscleroma*, a chronic granulomatous disease of the upper respiratory tract mucosa that is seen in inhabitants of parts of Africa, Asia, and Latin America; it has been described in two patients positive for HIV. Mikulicz cells (foamy histiocytes) are seen in the submucosa of biopsy specimens. Rhinoscleroma can be treated with streptomycin, trimethoprim-sulfamethoxazole, a quinolone, or tetracycline for 2 months. *Pseudomonas mallei* causes *glanders*, a respiratory disease of horses. Infection is rare in humans; nasal inoculation may produce a purulent nasal discharge followed by granulomatous intranasal lesions that ulcerate. Treatment is with sulfadiazine.

Neonatal congenital syphilis may present as rhinitis (snuffles), and the generalized osteochondritis that follows may result in a "saddle-nose" deformity. In *leprosy*, *Mycobacterium leprae* infiltrates the nasal mucosa and may cause chronic nasal congestion and nosebleeds. Involvement of the nasal cartilage may also result in a saddle nose.

Rhinosporidium seeberi is a fungus-like organism, not yet cultured, that causes *rhinosporidiosis*. Pedunculated nasal masses that grow over months or years cause obstruction and a foul odor and must be surgically excised. *Blastomyces dermatitidis*, a fungus prevalent in the Mississippi and Ohio River valleys, usually causes pulmonary disease but may cause chronic ulcerative lesions of the skin and nasal mucosa. *Mucormycosis*, a life-threatening fungal illness that occurs primarily in diabetic patients, may present as black eschars in the nasal cavity (see "Fungal Sinusitis").

THE COMMON COLD

The common cold is a mild, self-limited viral infection of the upper respiratory tract. Adults average two to four colds per year and children six to eight. The most common causes are rhinovirus (40% of cases) and coronavirus (at least 10%), but parainfluenza virus, respiratory syncytial virus, influenza virus, and adenoviruses also account for some cases. Rhinovirus alone has more than 100 different immunotypes, and this diversity has hampered efforts to identify an effective therapy or vaccine. There is no specific treatment for the common cold, although antihistamines, decongestants, and ipratropium bromide nasal spray provide some relief of symptoms. One study found that zinc gluconate lozenges, taken every 2 h, may reduce the duration of symptoms but are associated with nausea in 20% of patients. The value of vitamin C in preventing colds has not been proved.

SINUSITIS

The paranasal sinuses are aerated cavities in the bones of the face that develop as outpouches of the nasal cavity and communicate with this cavity throughout life. The maxillary and ethmoid sinuses are present at birth; the frontal and sphenoid sinuses develop after ages 2 and 7, respectively. Like the nose, the sinuses are lined with respiratory epithelium that includes mucus-producing goblet cells and ciliated cells. The mucous blanket is carried toward the sinus openings (ostia) at a speed of up to 1 cm/min by the beating of the cilia. The ostia are small; the ethmoid sinus ostia, for example, are only 1 to 2 mm in diameter. Delay in the mucociliary transport time or -- more important -- obstruction of the ostia may lead to retained secretions and sinusitis.

Sinusitis is a common problem. In the United States, this infection accounts for millions of office visits annually. The most common type, maxillary sinusitis, is followed in frequency by ethmoid, frontal, and sphenoid sinusitis. A viral infection of the upper respiratory tract is the most common precursor of sinusitis, although only about 0.5% of such infections are complicated by clinically evident acute bacterial sinusitis. Sinusitis develops primarily through ostial obstruction due to mucosal edema. Viral upper respiratory infections also increase the amount of mucus produced and may damage ciliated cells, thereby delaying mucus transport time. Allergic rhinitis is another common cause of ostial obstruction, either by mucosal edema or by polyps. Nasotracheal or nasogastric intubation can result in obstruction of the ostia and is a major risk factor for nosocomial sinusitis in intensive care units. Dental infections may cause 5 to 10% of all cases of maxillary sinusitis; the roots of the upper back teeth (second bicuspid, first and second molars) abut the floor of the maxillary sinus. Other causes of sinusitis include barotrauma from deep-sea diving or airplane travel, mucus abnormalities (e.g., cystic fibrosis), and chemical irritants. Foreign bodies, tumors (e.g., midline granuloma, intranasal lymphoma, or squamous cell carcinoma), and granulomatous diseases (e.g., Wegener's granulomatosis or rhinoscleroma) may all cause sinusitis secondary to obstruction.

ACUTE BACTERIAL SINUSITIS

Manifestations Symptoms of acute sinusitis include purulent nasal or postnasal

drainage, nasal congestion, and sinus pain or pressure whose location depends on the sinus involved. Maxillary sinus pain is often perceived as being located in the cheek or upper teeth; ethmoid sinus pain, between the eyes or retroorbital; frontal sinus pain, above the eyebrow; and sphenoid sinus pain, in the upper half of the face or retroorbital with radiation to the occiput. Sinus pain is frequently worse when the patient bends over or is supine. Fever develops in about half of patients with acute maxillary sinusitis.

Diagnosis The diagnosis of bacterial sinusitis may be difficult, as symptoms may resemble those of the inciting viral upper respiratory infection. The persistence of cold symptoms for 7 to 10 days (or longer than usual for a particular patient) is the most consistent clinical feature of bacterial sinusitis, according to some authors. Four-view sinus x-rays are helpful in the diagnosis of acute sinusitis: radiologic opacity, an air-fluid level, or ≥ 4 mm of sinus mucosal thickening correlates well with active bacterial infection. Computed tomography (CT) of the sinus is much more sensitive than routine radiography, particularly for ethmoid and sphenoid disease. Its use should be reserved for complicated cases and for cases in hospitalized patients, however. In light of the finding that sinus CT often shows reversible acute changes in patients with common colds, it is apparent that routine early use of CT would lead to overdiagnosis of bacterial sinusitis.

Etiology The bacteriology of acute community-acquired maxillary sinusitis has been well defined by studies using direct sinus puncture and aspiration. In children and adults, *Streptococcus pneumoniae* and *Haemophilus influenzae* (not type b), the most common pathogens, cause about one-third and one-fourth of cases, respectively. In children, *Moraxella catarrhalis* is also important, accounting for 20% of cases. Rhinoviruses, influenza viruses, and parainfluenza viruses are found alone or with bacteria in one-fifth of adult cases.

TREATMENT

Empirical therapy for acute bacterial sinusitis should be directed against the common bacterial pathogens; sinus puncture is not indicated in routine cases, and cultures of nasal drainage are not very reliable. Amoxicillin (500 mg orally, three times daily for 10 to 14 days) or trimethoprim-sulfamethoxazole may be effective in the treatment of first-time cases.

¹The dosages and durations of antimicrobial therapy given in this chapter are appropriate for adults with normal renal function and should be adjusted for children, for patients with impaired renal function, and in light of the response to treatment. Other effective but more expensive antibiotics include amoxicillin/clavulanate, cefuroxime axetil, and clarithromycin. Treatment should be given for 1 to 2 weeks. Intravenous administration of antibiotics may be necessary for the treatment of patients with severe disease who appear toxic. In nosocomial sinusitis, *Staphylococcus aureus* and gram-negative bacilli are most common, and sinus cultures are indicated as an aid in tailoring therapy. Initial broad-spectrum intravenous therapy (e.g., with nafcillin and ceftriaxone) should be adjusted on the basis of culture results. Surgery to widen the ostia and drain thick secretions may be essential in severe acute sinusitis, particularly when ethmoid, frontal, or sphenoid disease fails to respond to initial intravenous therapy.

CHRONIC BACTERIAL SINUSITIS

Manifestations Chronic sinusitis is characterized by symptoms of sinus inflammation lasting ≥ 3 months. Most experts believe that this condition is caused by dysfunction of the mucociliary blanket, usually as a result of repeated past infections, rather than by the persistence of bacterial infection. Patients report constant sinus pressure, nasal congestion, and postnasal drainage, especially in the morning. A temperature of $\geq 38^{\circ}\text{C}$ ($\geq 100.5^{\circ}\text{F}$) is rare and may signify a superimposed acute bacterial infection. Many patients also note a change in nasal discharge (to thick and green) with acute exacerbations.

Diagnosis Sinus [CT](#) should be used in all cases of chronic sinusitis to define the extent of disease and to help exclude other diagnoses, such as an obstructing tumor. Patients should be evaluated for allergies and immunodeficiencies (e.g., hypogammaglobulinemia). Evaluation by an otolaryngologist is essential, as this specialist will be able to obtain additional information by an office nasal endoscopic examination. Surgery, now usually done endoscopically, may be necessary to correct blockage of the sinus ostia. This blockage occurs most often in the osteomeatal complex that drains the maxillary, frontal, and anterior ethmoid sinuses. Samples of sinus secretions obtained intraoperatively should be cultured for anaerobes, aerobes, and fungi. Fungal sinusitis may mimic chronic bacterial infection (see "Fungal Sinusitis").

Etiology The bacteriology of chronic sinusitis is not well defined. Nearly all patients with chronic disease, especially those who have had prior sinus surgery, have sinus cultures positive for bacteria. Such cultures may represent colonization rather than infection, however. Patients who have received multiple courses of antibiotics may be colonized by *S. aureus* or by *Pseudomonas* and other gram-negative bacilli. Anaerobes have been isolated from 100% of sinus specimens in some studies but from as few as 2% in others.

TREATMENT

The need for antibiotic therapy must be assessed on an individual basis, with antibiotics chosen in light of recent culture results.

COMPLICATIONS OF BACTERIAL SINUSITIS

Orbital complications of sinusitis, such as orbital cellulitis and orbital abscess, usually arise from ethmoid sinusitis, since the ethmoid is separated from the orbit by only a very thin bone (the lamina papyracea). Patients present with fever, unilateral periorbital edema and erythema, conjunctival injection and chemosis, and proptosis. Eye movement may be decreased; with orbital abscess, the eye is often fixed in the "down and out" position. [CT](#) or magnetic resonance imaging (MRI) should be used to rule out an orbital abscess. Treatment of orbital infections should include immediate drainage of any abscess, intravenous administration of broad-spectrum antibiotics -- e.g., nafcillin (1.5 to 2.0 g every 4 h) plus ceftriaxone (2 g/d) -- for at least 7 days, and a consideration of sinus drainage surgery.

Another extracranial complication of sinusitis is frontal subperiosteal abscess (Pott's puffy tumor) from frontal sinusitis. Patients present with a tender doughy swelling over the forehead. Treatment consists of surgical drainage of the abscess and the frontal sinus and 6 weeks of intravenous antibiotic therapy directed at the isolated organisms.

Intracranial complications such as epidural abscess, subdural empyema, meningitis, cerebral abscess, and dural-vein thrombophlebitis may result from sinusitis, particularly from frontal or sphenoid infections. Because the sphenoid sinus sits between the two cavernous sinuses, sphenoid sinusitis is a major cause of cavernous sinus thrombophlebitis.

FUNGAL SINUSITIS

Fungal sinusitis is categorized as noninvasive or invasive. *Noninvasive* disease is chronic and occurs in immunocompetent hosts. It has two forms that are analogous to the noninvasive pulmonary diseases of aspergilloma and allergic bronchopulmonary aspergillosis. A fungus ball (aspergilloma) inside a sinus may cause symptoms of obstruction without invading the mucosa. Typically, only one sinus (often maxillary) is affected, and patients have unilateral symptoms and opacification of only that sinus on [CT](#). Treatment is surgical only, unless special fungal stains show tissue invasion on histopathology. Allergic fungal sinusitis was first described in 1983 and is seen mainly in patients with a history of nasal polyposis and asthma. It is characterized by extremely thick sinus mucus ("allergic mucin") that, on histopathologic examination, is found to contain numerous Charcot-Leyden crystals, eosinophils, and rare fungal hyphae. There is no evidence of tissue invasion. Surgical removal of the inspissated mucus is often curative. Antifungal therapy is not indicated.

Invasive fungal sinusitis presents differently in immunocompetent and immunocompromised hosts. In immunocompromised individuals, fungal disease has an acute presentation. Rhinocerebral mucormycosis is a life-threatening infection due to fungi of the order Mucorales (*Rhizopus*, *Rhizomucor*, *Mucor*, *Absidia*, *Cunninghamella*). Mucormycosis usually involves diabetic patients (70% of cases), half of whom are in ketoacidosis at presentation. Other patients at risk include those with hematologic malignancies, transplant recipients, and patients receiving chronic glucocorticoid or iron chelation (deferoxamine) therapy. Mucormycosis in patients taking deferoxamine is generally due to *Cunninghamella* and is almost always fatal. Symptoms and signs of mucormycosis may be explained by the fungal predilection for blood vessels and nerves and for invasion into the orbital apex and cavernous sinus, with consequent compromise of cranial nerves II through VI. Patients most frequently present with unilateral ocular findings of 1 to 5 days' duration that may mimic bacterial orbital cellulitis: lid swelling and erythema (sometimes bluish in appearance), ptosis, proptosis, decreased extraocular movement, and impaired vision. Retroorbital or periorbital pain is a prominent complaint. There may be either increased or decreased sensation in the first division of the fifth cranial nerve on the involved side; facial palsy with involvement of cranial nerve VII has also been described. Patients may be afebrile and appear nontoxic. Individuals in whom mucormycosis is a consideration should undergo an immediate examination by an otolaryngologist, who will look for intranasal black eschars or necrotic turbinates. If found, these sites should be biopsied and frozen tissue sections examined by a pathologist. In rare cases, the nasal passage appears normal, but biopsy of the middle

turbinate reveals invasive fungi. The finding of tissue invasion by broad-based, nonseptate hyphae necessitates extensive surgical debridement and intravenous therapy with amphotericin B or liposomal amphotericin ([Chaps. 200](#) and [207](#)). *Aspergillus* and other filamentous fungi may also cause invasive sinus disease.

Immunocompetent hosts with invasive fungal sinusitis, in contrast, have slowly progressive disease. Fungi in ethmoid and sphenoid sinuses may invade the orbital apex, causing proptosis, ptosis, limitation of eye movement, and decreased vision. Patients may mistakenly be treated with glucocorticoids for presumed optic neuritis or orbital pseudotumor until sinus disease is recognized and biopsies are undertaken. Treatment consists of surgical debridement of the involved sinuses and prolonged intravenous therapy with amphotericin B. In all cases of invasive fungal sinusitis, follow-up [CT](#) and [MRI](#) should be conducted frequently to evaluate the progression of disease.

Mortality from invasive fungal sinusitis is high, even among immunocompetent hosts.

EAR AND MASTOID INFECTIONS

AURICULAR CELLULITIS AND PERICHONDritis

Auricular cellulitis usually presents as a swollen, erythematous, hot, tender ear. The lobule is especially swollen and red. There may be a history of minor trauma to the ear (e.g., involving earrings, cotton swabs, or scratching). Treatment consists of warm compresses and intravenous administration of antibiotics active against *S. aureus* and streptococci -- e.g., cefazolin (1 g every 8 h) or nafcillin.

Perichondritis, an infection of the perichondrium of the ear, is often accompanied by infection of the underlying cartilage of the pinna (chondritis). Associated interruption of the blood supply to the cartilage may lead to ear deformity. Patients present with a swollen, hot, red, and exquisitely tender pinna, usually with sparing of the lobule. The most common antecedents of the infection are burns, significant trauma to the ear (e.g., as a result of boxing), or ear piercing through the pinna. *Pseudomonas aeruginosa* and *S. aureus* are the most common pathogens. Perichondritis should be treated with antibiotics, such as intravenous ticarcillin/clavulanic acid (3.1 g every 4 h) or intravenous nafcillin plus oral ciprofloxacin, for several weeks. Incision and drainage may be helpful for culture and for resolution of infection, which is often slow. This infection must be distinguished from relapsing polychondritis, a rheumatologic condition ([Chap. 325](#)).

OTITIS EXTERNA

The external auditory canal is about 2.5 cm long and is lined by skin. Beneath this skin is cartilage in the lateral half of the canal, temporal bone in the medial half. The skin in the bony portion lacks a subcutaneous layer and is attached directly to the periosteum, an important feature in the pathogenesis of invasive otitis externa (see below). Cerumen, secreted by glands, acidifies the canal and suppresses bacterial growth. However, desquamated skin and retained moisture make the canal especially susceptible to the hydrophilic organism *P. aeruginosa*.

Acute otitis externa, or swimmer's ear, occurs mostly in the summer and may be due to a decrease in canal acidity and resulting bacterial overgrowth. The ear is pruritic and painful, and the canal appears swollen and red. The most common pathogens are *P. aeruginosa*, *S. aureus*, and streptococci. Treatment consists of cleansing of the ear with alcohol-acetic acid mixtures and the administration of topical antibiotic ear drops, such as polymyxin-neomycin (4 drops four times daily for 5 days). Herpes zoster in the external canal causes severe otalgia and is often accompanied by ipsilateral facial paralysis due to the involvement of the geniculate ganglion of cranial nerve VII (Ramsay Hunt syndrome). Reports suggest that treatment with intravenous acyclovir decreases the incidence of permanent facial-nerve palsy, but the results of relevant controlled trials have not yet been published.

Chronic otitis externa causes pruritus rather than ear pain and is often due to irritation from either repeated minor trauma to the canal (e.g., scratching or use of cotton swabs) or drainage of a chronic middle-ear infection. In the latter situation, treatment of chronic otitis media with oral antibiotics will also cover this condition.

Invasive ("malignant") otitis externa is a potentially life-threatening infection, almost always due to *P. aeruginosa*, that slowly invades from the external canal into adjacent soft tissues, mastoid, and temporal bone and eventually spreads across the base of the skull. It occurs primarily in diabetic patients whose diabetes, unlike that of patients with mucormycosis, is usually under control. There is a history of weeks to months of ear pain and drainage, often misdiagnosed as chronic otitis media (an entity that is rarely painful). Examination reveals an edematous canal, with granulation tissue in the posterior wall about halfway down the canal (the region of the cartilage-bone junction). Trismus or partial facial paralysis (cranial nerve VII) is evident in some instances. Cranial nerves IX, X, and XI are occasionally affected as well. Fever is rare in invasive otitis externa and, when it does develop, is usually low-grade.

Laboratory studies generally reveal a normal white blood cell count but a high erythrocyte sedimentation rate. [CT](#) and [MRI](#) studies are essential for defining the extent of bone and soft-tissue involvement. CT shows bony destruction of the skull base in advanced cases. For culture, biopsies of granulation tissue in the canal or of deeper tissues are preferable to swab specimens of ear drainage, which may be unreliable. In nearly all cases, antibiotics should be withheld until a deep-tissue specimen is obtained for culture and pathologic examination. Once this specimen has been collected, empirical therapy with intravenous antibiotics active against *Pseudomonas* -- e.g., ticarcillin (3 g every 4 h), piperacillin, or ceftazidime, plus an aminoglycoside) -- may be started intraoperatively. To avoid ototoxicity, ciprofloxacin should be substituted for the aminoglycoside if cultures grow a *Pseudomonas* strain that is sensitive to this drug. In more than 95% of cases, *P. aeruginosa* is the pathogen involved; in the remaining cases, the pathogens include *Staphylococcus epidermidis*, *Aspergillus*, *Fusobacterium*, and *Actinomyces*. Systemic antibiotic treatment should be continued for 6 to 8 weeks. In early cases due to sensitive *Pseudomonas* strains, oral ciprofloxacin alone (750 mg twice daily for 6 weeks) may follow the initial 2 weeks of combination intravenous therapy.

ACUTE OTITIS MEDIA

The middle ear is connected to the nasopharynx via the eustachian tube. When this tube is blocked, fluid collects in the middle-ear and mastoid cavities, providing a culture medium for any bacteria present. Acute otitis media (AOM), or middle-ear infection, may result. Viral upper respiratory infections, which can cause edema of the eustachian tube mucosa, often precede or accompany episodes of AOM. Otitis media, like upper respiratory tract infections, is most common in fall, winter, and spring. The incidence of AOM declines with age. More than two-thirds of children under age 3 have had at least one episode of AOM; the prevalence among adults is only 0.25%.

Symptoms include ear pain, fever, and decreased hearing acuity. On examination, the tympanic membrane moves poorly with insufflation and is usually red, opaque, bulging, or retracted. Spontaneous perforations of the tympanic membrane and otorrhea are occasionally documented.

The bacteriology of [AOM](#) has been delineated for pediatric disease: *S. pneumoniae* (35%), *H. influenzae* (25%), and *M. catarrhalis* (15%) are the most common organisms. Viruses, either alone or with bacteria, are found in one-quarter of pediatric cases. Small studies of AOM in adults have also found *S. pneumoniae* (21%) and *H. influenzae* (26%) to be the most common pathogens. More than 90% of *H. influenzae* infections are due to nontypable strains: those due to type b may be accompanied by bacteremia or meningitis.

TREATMENT

Treatment of otitis media is empirical, as diagnostic tympanocentesis is indicated only for patients who appear toxic, who are immunocompromised, or whose infection is refractory to initial therapy. Although about one-third of *H. influenzae* strains and at least three-quarters of *M. catarrhalis* strains are β -lactamase producers, most authorities still find amoxicillin therapy to be successful in routine cases. Other drugs effective against most β -lactamase-positive strains include amoxicillin/clavulanate (875 mg by mouth twice daily for 7 to 10 days), trimethoprim-sulfamethoxazole, erythromycin/sulfisoxazole, clarithromycin, and second-generation oral cephalosporins (e.g., loracarbef, cefpodoxime proxetil, and cefuroxime axetil). Penicillin resistance in pneumococci, now a major problem, is not mediated by β -lactamase ([Chap. 138](#)). Strains exhibiting intermediate resistance may respond to therapy with high-dose amoxicillin or to clindamycin, erythromycin, or trimethoprim-sulfamethoxazole. Quinolones such as levofloxacin, although not approved for use in children, may be effective in adults. Serious infections or those due to highly resistant strains require treatment with parenteral ceftriaxone or vancomycin. Adjunctive treatment of [AOM](#) with antihistamines is of no proven benefit.

Recurrent episodes of [AOM](#) in children are due to the same pathogens that cause primary AOM (*S. pneumoniae*, *H. influenzae*, and *M. catarrhalis*). Most early recurrences (75%), however, are not relapses but are due to different organisms or to different strains of the organism that caused the initial episode. The pattern of recurrent AOM in adults is presumably similar but has not been well studied. Treatment for recurrent AOM should include drugs with activity against resistant strains. Patients with frequent recurrences (e.g., three episodes within 6 months) may benefit from antibiotic prophylaxis with once-daily amoxicillin or sulfisoxazole during the winter months,

although this benefit must be weighed against the risk of selecting more antibiotic-resistant strains of bacteria.

SEROUS OTITIS MEDIA

Otitis media with effusion, or serous otitis media, is characterized by the persistence of middle-ear fluid for several months without other signs of infection. This condition is associated with a 25-dB hearing loss in the affected ear. Cultures of middle-ear fluid are usually negative. Although some clinical trials have found that effusions resolve sooner in antibiotic-treated children than in controls, antibiotics are generally not recommended because the risk of increasing antibiotic resistance in the population is thought to outweigh the small benefit observed. Adenoidectomy, myringotomy, or tympanostomy tubes have been shown to decrease the duration of effusion in children.

CHRONIC SUPPURATIVE OTITIS MEDIA

In chronic suppurative otitis media, patients have painless hearing loss and intermittent purulent ear drainage. On examination, there is a central perforation in the tympanic membrane and purulent drainage from the middle ear. If a cholesteatoma is present, the perforation is peripheral. Culture of draining fluid reveals *P. aeruginosa* (40%), *S. aureus* (20%), *Klebsiella* (20%), and other enteric gram-negative bacilli. Anaerobes are found in 50% of cases, usually in mixed culture with aerobes. [CT](#) should be used to help evaluate a surgically treatable nidus of infection, such as a cholesteatoma or mastoid sequestrum. For therapeutic purposes, patients are divided into two groups: those with and without cholesteatoma. Those in the former group are cured with surgical excision of the cholesteatoma. Those without cholesteatoma require repeated courses of topical antibiotic drops for relapse of "active" (draining) disease, and true cures are rare. The role of systemic antibiotics is unclear. In one study, a course of intravenous antibiotics produced long-term success in 78% of children without cholesteatoma who had persistent otorrhea despite topical and oral antibiotic therapy.

Tuberculous otitis media is rare and is frequently misdiagnosed. It mimics nontuberculous chronic suppurative otitis media, but ear drainage fails to respond to routine antibiotics. On examination, the tympanic membrane often has multiple perforations, and "pearly" or "flabby" tissue is seen in the middle ear. Only 30% of patients have evidence of active tuberculosis on chest x-ray. Treatment is the same as for other types of extrapulmonary tuberculosis.

MASTOIDITIS

The mastoid is the portion of the temporal bone posterior to the ear that contains a honeycomb of air cells lined with respiratory epithelium. These air cells connect with the middle ear. Fluid in the middle ear, a prelude to otitis media, is almost always accompanied by fluid in the mastoid. True mastoiditis, however, has become rare in the antibiotic era, probably because of prompt treatment of otitis media.

Mastoiditis is characterized by erosion of the bony partitions between the mastoid air cells. Patients with acute mastoiditis present with pain, tenderness, and swelling over the mastoid. When there is an overlying subperiosteal abscess or cellulitis, the pinna is

pushed out and forward. [CT](#) may show bony destruction or a drainable mastoid abscess.

The reported bacteriology of mastoiditis has varied. Some cases involve organisms similar to those implicated in [AOM](#) (*S. pneumoniae*, *H. influenzae*); others are attributable to *S. aureus* and gram-negative bacilli, including *Pseudomonas*. Ideally, therapy should be guided by the results of cultures of middle-ear fluid obtained by tympanocentesis. Initial broad-spectrum therapy, such as that with intravenous ticarcillin/clavulanate plus gentamicin or ciprofloxacin, can later be narrowed.

COMPLICATIONS OF OTITIS MEDIA AND MASTOIDITIS

Extracranial complications include hearing loss, labyrinthitis and resulting vertigo, and facial-nerve palsy. Additional complications from mastoiditis develop when infection tracks under the periosteum of the temporal bone to cause a subperiosteal abscess or breaks through the mastoid tip to cause a neck abscess deep to the sternocleidomastoid muscle (Bezold's abscess). Intracranial complications include epidural abscess, dural venous thrombophlebitis (usually sigmoid sinus), meningitis, and brain abscess.

INFECTIONS OF THE ORAL CAVITY AND PHARYNX

ORAL CAVITY INFECTIONS

The oral cavity extends from the lips to the circumvallate papillae of the tongue and is heavily colonized with viridans streptococci and anaerobes. These organisms can cause several infections in this area. *Gingivitis* is an infection of the gums, the earliest form of periodontal disease. Anaerobes residing in the mouth, especially anaerobic gram-negative rods such as *Prevotella intermedia*, are the most common pathogens. Patients with *Vincent's angina*, also called *acute necrotizing ulcerative gingivitis* or *trench mouth*, have halitosis and ulcerations of the interdental papillae. Oral anaerobes are the cause, and therapy with oral penicillin plus metronidazole or with clindamycin alone is effective in both this condition and gingivitis.

Ludwig's angina is a rapidly spreading, life-threatening cellulitis of the sublingual and submandibular spaces that usually starts in an infected lower molar. Patients are febrile and may drool the secretions they cannot swallow. A brawny, boardlike edema in the sublingual area pushes the tongue up and back. Airway obstruction may result as the infection spreads to the supraglottic tissues. Treatment with intravenous antibiotics active against streptococci and oral anaerobes -- e.g., ampicillin/sulbactam (3 g every 6 h) or high-dose penicillin plus metronidazole -- should be followed by oral antibiotic therapy, with a total treatment duration of 14 days. Airway monitoring is also essential. Intubation or tracheostomy may be necessary.

Noma, or *cancrum oris*, is a fulminant gangrenous infection of the oral and facial tissues that occurs in severely malnourished and debilitated patients and is especially common among children. Beginning as a necrotic ulcer in the gingiva of the mandible, noma is caused by oral anaerobes, especially fusospirochetal organisms (e.g., *Fusobacterium nucleatum*). It is treated with high-dose penicillin, debridement, and correction of the underlying malnutrition.

Herpes simplex commonly causes cold sores of the lips but may also cause painful vesicles on the tongue and buccal mucosa. Primary infection may require intravenous hydration and should be treated with acyclovir. *Thrush*, or oropharyngeal candidiasis, is an infection caused by *Candida* spp. such as *C. albicans*. It occurs in neonates, patients who have received prolonged antibiotic therapy, and immunocompromised patients. More than 90% of patients with AIDS develop thrush. Patients with thrush report a "burning" tongue or "raw" throat and, on examination, have white plaques on the tongue and oral mucosa. Treatment consists of topical antifungal agents (clotrimazole, nystatin) or oral fluconazole. Therapy for fluconazole-resistant thrush in patients with AIDS may be difficult; itraconazole oral solution or amphotericin B oral suspension may be effective.

PHARYNGITIS

Most cases of pharyngitis are thought to be viral. Many occur as part of common colds caused by rhinovirus, coronavirus, or parainfluenza virus. Patients have a scratchy or sore throat as well as coryza and cough. The pharynx is inflamed and edematous, but no exudate is evident. Influenza virus and adenovirus may cause a particularly severe sore throat, along with fever and myalgias. In infection with either of the latter viruses, there is pharyngeal erythema and edema; however, adenovirus infection also commonly causes an exudate, thus mimicking streptococcal pharyngitis. *Infectious mononucleosis* due to Epstein-Barr virus often causes a severe sore throat. Exudative pharyngitis or tonsillitis is documented in half of mononucleosis cases and may also mimic streptococcal infection. *Herpangina*, caused by coxsackievirus, is characterized by fever, sore throat, myalgias, and a vesicular enanthem on the soft palate between the uvula and the tonsils. There are usually only two to six lesions, which begin as small papules that vesiculate and then ulcerate. Fever and nonexudative pharyngitis are common symptoms of the acute retroviral syndrome that develops several weeks after infection with [HIV](#).

The most important bacterial cause of pharyngitis is group A *Streptococcus* (*S. pyogenes*). This organism is responsible for about 15% of all cases of pharyngitis and can cause important complications, both suppurative (peritonsillar and retropharyngeal abscess) and nonsuppurative (scarlet fever, streptococcal toxic shock syndrome, rheumatic fever, acute poststreptococcal glomerulonephritis). Fever, severe sore throat, cervical adenopathy, and inflammation of the tonsils and pharynx (which are covered with exudate) are classic findings. However, many cases of streptococcal pharyngitis are mild, with minimal erythema and no exudate, and mimic the pharyngitis of the common cold. Although some patients may in fact have viral pharyngitis and may simply be colonized with group A streptococci, these individuals must nevertheless be treated for presumed streptococcal pharyngitis. Diagnosis is made by culture. Rapid antigen tests are now available. These tests are less sensitive than they are specific: a positive test may be considered equivalent to a positive culture, but a negative test requires culture confirmation. Either a single dose of intramuscular benzathine penicillin (1.2 million units) or a 10-day course of oral penicillin (250 mg four times daily) or erythromycin is necessary to eradicate the organism. Sensitivity to erythromycin should be verified if this agent is used, as an increase in erythromycin resistance has been noted, especially in Europe. Other antibiotics active against streptococci may be used

(e.g., amoxicillin, cefuroxime), and one trial showed that 4 days of cefuroxime therapy was as effective as 10 days of penicillin treatment in eradicating the organism. However, studies of the prevention of rheumatic fever are available only for penicillin ([Chap. 235](#)).

Other bacterial causes of pharyngitis include groups C and G *Streptococcus*, *Neisseria gonorrhoeae* ([Chap. 147](#)), *Arcanobacterium haemolyticum*, *Yersinia enterocolitica*, and -- very rarely -- *Corynebacterium diphtheriae* ([Chap. 141](#)). In addition, *Mycoplasma pneumoniae* ([Chap. 178](#)) and *Chlamydia pneumoniae* ([Chap. 179](#)) can cause pharyngitis.

A peritonsillar abscess (*quinsy*) may follow untreated streptococcal pharyngitis. Oral anaerobes also play a role in quinsy. Patients have a severe sore throat and speak with a "hot-potato" voice. Examination reveals pronounced unilateral peritonsillar swelling and erythema causing deviation of the uvula. Immediate aspiration by an otolaryngologist is required in conjunction with antibiotic therapy -- e.g., ampicillin/sulbactam (3 g intravenously every 6 h), penicillin plus metronidazole, or clindamycin.

LARYNGITIS, CROUP, AND EPIGLOTTITIS

LARYNGITIS

Laryngitis is characterized by hoarseness. Most cases of acute laryngitis are caused by viruses (rhinovirus, influenza virus, parainfluenza virus, coxsackievirus, adenovirus, or respiratory syncytial virus). Acute laryngitis may also be associated with group A *Streptococcus* and *M. catarrhalis*. Laryngitis must be differentiated from epiglottitis (see below). The goal of treatment is merely the relief of symptoms except when throat cultures are positive for group A *Streptococcus* (in which case penicillin should be used).

Chronic laryngitis due to infection is rare and must be distinguished from hoarseness of neoplastic etiology. *Tuberculous laryngitis* may be mistaken for laryngeal cancer when assessed by direct laryngoscopy. Laryngeal and supraglottic lesions include mucosal hyperemia and thickening, nodules, and ulcerations. In one study, a history of fever and night sweats was rare, and the most common chest radiographic finding was apical thickening and fibrosis. Biopsy reveals granulomas with acid-fast bacilli. Cultures should be performed to confirm the diagnosis and evaluate the sensitivities of the pathogen. Laryngeal tuberculosis is highly contagious and should be managed with the same precautions and therapy used for active pulmonary disease ([Chap. 169](#)). Fungal infections causing laryngitis include histoplasmosis ([Chap. 201](#)), blastomycosis ([Chap. 203](#)), and candidiasis ([Chap. 205](#)). *Histoplasma* and *Blastomyces* may cause nodules on the larynx, with or without ulcerations. *Candida* may cause laryngitis, along with thrush, in immunosuppressed patients or in patients with chronic mucocutaneous candidiasis.

CROUP

Croup, or acute laryngotracheobronchitis, is an infection of the upper and lower respiratory tract that causes marked subglottic edema. It mainly affects 2- and

3-year-old children and usually follows the onset of upper respiratory tract infection by 1 to 2 days. Symptoms include fever, hoarseness, a "seal's bark" cough, and inspiratory stridor. The most common etiology is parainfluenza virus, although croup may also be caused by other respiratory viruses (e.g., influenza or respiratory syncytial virus).

Croup must be differentiated from epiglottitis (see below). Epiglottitis usually progresses more rapidly and produces a more toxic appearance. Neck x-rays may be helpful but do not reliably exclude epiglottitis. In croup, the anterior-posterior neck x-ray shows subglottic edema (the "hourglass sign"); in epiglottitis, the lateral neck view shows a thick epiglottis.

Patients with severe croup should be hospitalized, monitored for hypoxemia through pulse oximetry, and watched for airway obstruction requiring intubation. Humidification is commonly prescribed, but few controlled trials have assessed its benefit. Nebulized racemic epinephrine provides temporary (2-h) improvement in patients with marked stridor, but such patients must be observed for rebound edema. Glucocorticoid therapy, either nebulized or parenteral, is clearly beneficial, and its effects are often evident within 1 h. One trial found that treatment with a single dose of either intramuscular dexamethasone or nebulized budesonide reduced the need for hospitalization of children with moderately severe croup by more than 50%.

EPIGLOTTITIS

Acute epiglottitis (supraglottitis) is a life-threatening, rapidly progressive cellulitis of the epiglottis that may cause complete airway obstruction. It begins as a cellulitis between the tongue base and the epiglottis that pushes the epiglottis posteriorly. The epiglottis itself then becomes swollen, threatening the airway. Before the introduction of *H. influenzae* type b (Hib) vaccine, epiglottitis was most common among children 2 to 4 years old. The disease is now rare in children, since the vaccine has reduced the incidence of invasive disease due to Hib by more than 95%. The incidence in adults has not changed.

The typical young child with epiglottitis has a several-hour history of fever, irritability, dysphonia, and dysphagia and presents sitting forward and drooling. Adolescents and adults usually have a less fulminant presentation, with symptoms (especially sore throat) of 1 or 2 days' duration. Adults may present with dyspnea (25%), drooling (15%), and stridor (10%). Epiglottitis constitutes a medical emergency, as airway occlusion may occur suddenly. Lateral neck films showing an enlarged epiglottis (the "thumb sign") are helpful if positive but may be falsely negative. The value of obtaining these films has also been questioned because doing so may cause a critical delay in securing the airway. Direct viewing of the pharynx by use of a tongue blade should not be attempted, as immediate laryngospasm and airway obstruction may result. Instead, a child with suspected epiglottitis should be transported -- while sitting up -- to the operating room for visualization of the epiglottis with a fiberoptic laryngoscope, with preparations made for immediate airway control. If the epiglottis is cherry-red, an uncuffed endotracheal tube should be placed. Diagnosis in adults is also made by direct viewing of the epiglottis with a flexible fiberoptic laryngoscope, again only after preparations are made to secure the airway.

All patients must be closely monitored in an intensive care unit and should be given antibiotics active against *H. influenzae*. Before [Hib](#) vaccine became available, this organism was responsible for nearly all pediatric cases and was isolated from the blood of almost 100% of the affected children. In adults, blood cultures are positive in about 25% of cases, all of which are due to *H. influenzae*. Other pathogens isolated from the pharynx of adults with epiglottitis include *H. parainfluenzae*, *S. pneumoniae*, group A *Streptococcus*, and (rarely) *S. aureus*; the correlation between throat and epiglottitis cultures is unclear, however. Children may be treated with intravenous cefuroxime, ceftriaxone, ampicillin/sulbactam, or trimethoprim-sulfamethoxazole. Adults may be treated for at least 7 days with cefuroxime, ampicillin/sulbactam (3 g intravenously every 6 h), or nafcillin plus ceftriaxone; those highly allergic to penicillin may be given clindamycin plus either trimethoprim-sulfamethoxazole or ciprofloxacin. If the patient with *H. influenzae* epiglottitis has household contacts that include an unvaccinated child under age 4, all members of the household and the patient should receive prophylactic rifampin to eradicate the carriage of *H. influenzae*.

DEEP NECK INFECTIONS

Deep neck infections may be life-threatening because of airway compromise, involvement of the carotid sheath, or spread into the mediastinum.

SUBMANDIBULAR SPACE INFECTIONS

See *Ludwig's angina* above (under "Oral Cavity Infections").

LATERAL PHARYNGEAL SPACE INFECTIONS

The lateral pharyngeal space, also called the parapharyngeal or pharyngomaxillary space, is in the superior lateral portion of the neck and extends from the hyoid bone to the base of the skull. It lies deep to the lateral wall of the pharynx and is lateral to the tonsil and carotid sheath and medial to the parotid gland. Infection in this space may follow tonsillitis, pharyngitis with adenoid involvement, parotitis, mastoiditis, or periodontal infection.

On presentation, most patients appear toxic and have fever, sore throat, pain on swallowing, and leukocytosis. Infection confined to the posterior (retrostyloid) portion of the lateral pharyngeal space causes swelling of the lateral pharyngeal wall, which may be missed because it is behind the palatopharyngeal arch. Involvement of the anterior portion of this space causes medial displacement of the tonsil, swelling over the parotid gland, and trismus. Rigidity of the neck or torticollis toward the opposite side may develop. Diagnosis is confirmed by [CT](#) with contrast.

Treatment includes securing of the airway, surgical drainage in the operating room, and administration for at least 10 days of intravenous antibiotics active against streptococci and oral anaerobes (e.g., ampicillin/sulbactam, 3 g every 6 h). Major complications result from involvement of the carotid sheath and the vessels it contains. These complications are frequently fatal and include jugular vein thrombophlebitis, erosion into the carotid artery, and mediastinitis. Jugular vein thrombophlebitis is characterized by high fevers, chills, and neck tenderness at the angle of the mandible. When it is caused

by *Fusobacterium necrophorum*, it may be accompanied by sepsis and septic pulmonary emboli (*Lemierre's syndrome*). Erosion into the carotid artery is usually heralded by repeated small bleeds into the mouth. The involvement of adjacent cranial nerves may result in ipsilateral Horner's syndrome, hoarseness, or unilateral tongue paresis. Extension of infection along the carotid sheath into the posterior mediastinum results in mediastinitis and a mortality of 50%. [MRI](#) is useful in delineating carotid and jugular involvement.

RETROPHARYNGEAL SPACE INFECTIONS

The retropharyngeal space lies between the pharynx and the prevertebral fascia and extends from the base of the skull into the mediastinum. Infection in this space may result from the spread of lateral pharyngeal space infection or from the lymphatic spread of infection in more cephalad sites (posterior sinuses, adenoids, nasopharynx) to the retropharyngeal lymph nodes. Retropharyngeal abscess is most common among infants and young children, probably because the retropharyngeal nodes later involute. Retropharyngeal abscess may also follow trauma to the posterior pharynx (e.g., endoscopy in adults, lollipop-stick perforation in children) or may result from anterior extension of infection from cervical osteomyelitis.

Symptoms include fever, marked difficulty and pain with swallowing, and a "hot-potato" voice. Physical examination may document drooling, nuchal rigidity, and bulging of the posterior pharyngeal wall. Advanced cases include dyspnea and stridor. Diagnosis may be confirmed by a lateral neck soft-tissue x-ray or [CT](#) scan. Treatment requires securing of the airway and emergency surgical drainage. Intravenous antibiotics should be given; the agents chosen should be active against streptococci, oral anaerobes, *S. aureus*, and *H. influenzae* (e.g., ampicillin/sulbactam alone or clindamycin plus ceftriaxone). Potential complications include airway obstruction, intraoral rupture of the abscess causing aspiration pneumonia, and mediastinitis.

(Bibliography omitted in Palm version)

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31. ORAL MANIFESTATIONS OF DISEASE - John S. Greenspan

A thorough oral examination, to include the oral and pharyngeal soft tissues as well as the teeth, is an important part of the physical examination. The common oral diseases are due to infection by bacteria, fungi, or viruses. The complex development of the orofacial structures leads to close interposition of diverse tissues, which are prone to developmental anomalies, growth disturbances, and neoplasia.

DISEASES OF THE TEETH

DENTAL CARIES, PULPAL AND PERIAPICAL DISEASE, AND COMPLICATIONS

Dental caries is a destructive disease of the hard tissues of the teeth due to infection with *Streptococcus mutans* and other bacteria. In the United States, fewer than half of those 17 years and younger now have carious lesions, although in many segments of the population and in developing countries the disease is more common. Artificial fluoridation of water to a level of 1 part per million, fluoride-containing toothpastes, and topical fluoride administration have reduced the incidence. Conversely, retention of teeth and the aging of the population have led to an increase in root caries. Increasing numbers of individuals surviving cancer therapy and other special populations (diabetic patients and those with xerostomia due to Sjogren's syndrome or to medications) may experience severe caries unless appropriate topical fluoride prophylaxis is used. Treatment of caries involves removal of the softened and infected hard tissues, sealing of the exposed dentine, and restoration of the lost tooth structure with silver amalgam, composite plastic, gold, or porcelain.

If the carious lesion progresses, infection of the dental pulp may occur, causing *acute pulpitis*. The tooth may become sensitive to hot or cold. When severe continuous throbbing pain ensues, pulp damage is irreversible, and root canal therapy becomes necessary. The contents of the pulp chamber and root canals are removed, followed by thorough cleaning, antisepsis, and filling with an inert material. Alternatively, extraction of the tooth may be indicated.

If the pulpitis is not treated successfully, infection may spread beyond the tooth apex into the periodontal ligament. Acute inflammation causes pain on chewing or on percussion, and a *periapical abscess* may form. Chronic inflammation may be painless or produce only slight pain, and a *periapical granuloma* may form within the alveolar bone. Proliferation of epithelial cell rests may convert the granuloma into a *periapical cyst*. Periapical radiolucencies may occur with the granuloma or the cyst but not with the abscess, unless it forms as a complication of one or two lesions. The pus in the periapical abscess may track through the alveolar bone into soft tissues, causing cellulitis and bacteremia, or may discharge into the oral cavity (*parulis* or *gumboil*), into the maxillary sinus, or through the skin of the face or submandibular area. A severe form of cellulitis, *Ludwig's angina*, originates from an infected mandibular molar, involves the submandibular space, and extends throughout the floor of the mouth, with elevation of the tongue, dysphagia, and difficulty breathing. Glottal edema may occur, necessitating tracheotomy.

EFFECT OF SYSTEMIC FACTORS ON TEETH

Enamel hypoplasia of the primary and/or permanent teeth, manifested by alterations ranging from white spots to gross defects in the surface structure of the crowns, may be caused by disturbances of calcium and phosphate metabolism such as are found in vitamin D-resistant rickets, hypoparathyroidism, gastroenteritis, and celiac disease. Premature birth or high fevers may also give rise to enamel hypoplasia. Tetracycline, when given during the second half of pregnancy, in infancy, and in childhood up to 8 years of age, causes both a permanent discoloration of the teeth and enamel hypoplasia. Daily ingestion of more than 1.5 mg fluoride can result in enamel discoloration (*mottling*). Larger teeth are associated with maternal diabetes, maternal hypothyroidism, and large birth size. Tooth size is reduced in Down's syndrome. Systemic disease may give rise to pain that simulates pulpal disease. Maxillary sinusitis is frequently manifested as pain in the maxillary teeth, including sensitivity to thermal changes and percussion. Angina pectoris may result in pain referred to the lower jaw, probably through the vagus nerve.

PERIODONTAL DISEASES

In adults, chronic destructive periodontal disease (*pyorrhea*) is responsible for more loss of teeth than caries, particularly in the aged. However, the prevalence and incidence of periodontal disease also appears to be declining in the United States. The most common form of periodontal disease starts as inflammation of the marginal gingiva (*gingivitis*), which is painless, although the gingiva may bleed on brushing. The disease spreads to involve the periodontal ligament, alveolar bone is slowly resorbed, and periodontal ligament attachment between tooth and bone is lost. The soft tissue separates from the tooth surface, causing "pocket" formation with bleeding on probing and during chewing. Acute inflammation may become superimposed on this chronic process, with the production of pus and the formation of a *periodontal abscess*. Ultimately, extreme bone loss, tooth mobility, and recurrent abscess formation lead to tooth exfoliation or may mandate tooth extraction.

Gingivitis and periodontitis are infections associated with the accumulation of *bacterial plaque*, which may become mineralized (*calculus*) and can be prevented by appropriate *oral hygiene* measures, including tooth brushing, flossing, antibacterial mouth rinses, and the removal of impacted food debris. Poorly fabricated or deteriorated restorations may contribute through overextended or inadequate margins. Therapy consists of removal of plaque and calculus, debridement of the pocket lining and superficial infected cementum, and elimination of other contributing factors.

Periodontal disease appears to be a group of conditions, including *adult periodontitis*, associated with *Porphyromonas gingivalis*, *Prevotella intermedia*, and other gram-negative organisms. *Localized juvenile periodontitis* (LJP) causes rapid, severe pocketing and bone loss and is associated with *Actinobacillus actinomycetemcomitans*, *Capnocytophaga*, *Eikenella corrodens*, and other anaerobes. *Acute necrotizing ulcerative gingivitis* (ANUG) involves sudden inflammation of the gingivae with necrosis, tissue loss, pain, bleeding, and halitosis and is associated with *P. intermedia* and spirochetes. ANUG and an aggressive and rapid form of periodontitis (*necrotizing ulcerative periodontitis*) are seen in association with HIV infection. Some of these cases progress to a destructive gangrene-like lesion of oral soft tissues and bone (*necrotizing*

stomatitis) resembling the *noma* seen in severely malnourished populations. Therapy involves local antibacterial measures, debridement, and, in severe cases, systemic antibiotics effective against anaerobes.

Host factors may be involved in the pathogenesis of periodontal disease in other populations as well. Severe periodontal disease may occur in persons with *Down's syndrome* and *diabetes mellitus*. During pregnancy there may be severe gingivitis and the formation of localized *pyogenic granulomas*. Certain drugs, notably the anticonvulsant *phenytoin* and the calcium channel blocker *nifedipine*, cause *fibrous hyperplasia* of the gingiva, which may cover the teeth, interfere with eating, and be unsightly. *Idiopathic familial gingival fibromatosis* may appear similar. Surgery may correct both conditions; change in medication may reverse the drug-induced form. The oral cavity is a significant reservoir for *Helicobacter pylori*. Uncontrolled diabetes mellitus leads to an exacerbation of oral infection, notably periodontal disease. In individuals genetically predisposed to diabetes, periodontal disease may also precipitate or exacerbate the diabetes. Oral infection has been proposed to contribute to coronary atherosclerosis as well as pregnancy outcomes such as premature labor and low birthweight.

Periapical and periodontal bacterial infections can cause transient bacteremia after tooth extraction and even routine dental prophylaxis. Antibiotic coverage is appropriate in patients with heart valves susceptible to infection or those with prosthetic joints.

DISEASES OF THE ORAL MUCOSA

INFECTIONS

Most oral mucosal diseases involve microorganisms ([Table 31-1](#)).

PIGMENTED LESIONS See [Table 31-2](#)

DERMATOLOGIC DISEASES See [Tables 31-1, 31-2](#), and [31-3](#) and [Chaps. 55, 56, 57, 58, 59](#), and [60](#)

DISEASES OF THE TONGUE See [Table 31-4](#)

HALITOSIS See [Table 31-5](#)

HIV DISEASE AND AIDS (See [Table 31-6](#) and also [Chaps. 191](#) and [309](#))

Immunosuppression induced by HIV infection predisposes to numerous oral infections, neoplasms, and autoimmune and idiopathic lesions. *Oral candidiasis* ([Plate IID-43](#)) and *hairy leukoplakia* ([Plate IID-42](#)) [a benign epithelial hyperplasia associated with Epstein-Barr virus (EBV)] are common features of HIV disease and often precede or accompany full-blown AIDS. Oral Kaposi's sarcoma and lymphoma are diagnostic of AIDS. Oral candidiasis is easily treated with topical or systemic antifungals: nystatin oral pastilles, clotrimazole oral troches, nystatin vaginal tablets used orally, fluconazole, and ketoconazole. While most oral lesions of HIV disease are also found in the general population. Necrotizing ulcerative periodontal disease and hairy leukoplakia are strongly

associated with HIV infection and are otherwise very rare.

HEMATOLOGIC AND NUTRITIONAL DISEASE

Gingival bleeding, necrotic ulcers, and enlargement due to malignant infiltrates are seen in all forms of leukemia, particularly *monocytic leukemia*. In *agranulocytosis* severe oral mucosal ulcers are seen, while in *thrombocytopenia* oral petechiae, ecchymoses, and gingival bleeding occur. In *Plummer-Vinson syndrome* ([Chap. 105](#)), atrophy of oral mucosa, particularly the tongue papillae, causes redness and soreness as well as dysphagia and is associated with increased susceptibility to oral cancer. A smooth tongue can also be seen in *pernicious anemia* ([Chap. 107](#)). Severe oral mucositis with ulcers, candidiasis, bacterial infections, and xerostomia complicate radiation therapy for head and neck cancers. Chemotherapy may also cause mucositis. Although now rarely seen in the United States, oral features of vitamin deficiency include oral mucositis and ulcers, glossitis, and burning sensations in the tongue (*B group vitamin deficiency*) and petechiae, gingival swelling, bleeding, and ulceration as well as loosening of teeth (*scurvy* of vitamin C deficiency).

DISEASES OF THE SALIVARY GLANDS

The major and minor salivary glands can be involved in mumps, sarcoidosis, tuberculosis, lymphoma, and Sjogren's syndrome ([Chap. 314](#)). The latter may cause dry eyes and dry mouth (*xerostomia*) and be associated with features of connective tissue diseases, including rheumatoid arthritis or systemic lupus erythematosus. Xerostomia may also be due to medications such as diuretics, antihistamines, or tricyclic antidepressants as well as radiation therapy for head and neck cancer. Without lysozyme-rich saliva, *cervical or incisal caries* and oral candidiasis may develop. Management includes fluoride mouth rinses and topical applications, saliva substitutes, salivary stimulation with sugarless candies, and the avoidance of sugar-containing drinks or food. Candidiasis is treated with nystatin or other antifungals. Salivary stones (*sialolithiasis*), usually in the duct of a major salivary gland, cause *sialoadenitis* with pain and swelling, often on eating, especially tart foods such as lemons.

The most common neoplasm of the salivary glands is the *pleomorphic adenoma*, which is benign but will recur unless fully resected; malignant tumors include *mucoepidermoid carcinoma*, *adenoid cystic carcinoma*, and *adenocarcinoma*. The pleomorphic adenoma causes a firm, slowly growing mass in the parotid, palate, or cheek, whereas malignant tumors grow faster and can cause ulceration and invade nerves, producing numbness or facial paralysis.

NEUROLOGIC DISTURBANCES AND OROFACIAL PAIN

The mouth and face may be the site of pain from a number of vascular, neurologic, muscle/connective tissue, or joint conditions. Interdisciplinary diagnosis and management programs involving neurologists, restorative dentists, oral surgeons, otorhinolaryngologists, and other specialists, together with new imaging techniques to diagnose or exclude organic lesions, have begun to clarify this complex field. *Temporal arteritis* causes pain in the face, jaws, and tongue and may mimic temporomandibular joint disease. Glucocorticoids may provide relief. *Myofascial pain* is a dull, constant ache

with local tenderness in the muscles of the jaws and difficulty in opening the mouth. Teeth clenching and grinding (*bruxism*) may play a role. *Arthralgia* of the temporomandibular joint causes local pain, which may extend to the face and head. Both myofascial pain and arthralgia can be relieved with heat, rest, and anti-inflammatory agents. Displacement of the meniscus or condyle may cause pain, clenching, or locking of the mandible in the open position. The joint may become involved in *osteoarthritis* with minimal symptoms, whereas *rheumatoid arthritis* causes pain and swelling in the joint and limitation of movement. *Ankylosis* may occur, necessitating condylectomy ([Chap. 312](#)).

Trigeminal neuralgia (tic douloureux) causes sudden, severe, unilateral lancinating pain initiated by touching a "trigger zone" or occurring spontaneously. Confusion with pulpal or periapical pain is common, leading to inappropriate endodontic or surgical therapy. Many cases respond to carbamazepine and phenytoin, but for a few, surgical intervention to decompress the trigeminal nerve is indicated. Similar symptoms in the distribution of the ninth cranial nerve (tongue, pharynx, soft palate) are due to *glossopharyngeal neuralgia*, which may be triggered by swallowing and may produce referred pain in the temporomandibular joint. *Postherpetic neuralgia* may follow trigeminal herpes zoster ([Chap. 367](#)) and cause burning, aching, and long-lasting pain. *Facial palsy* is usually unilateral and may be due to trauma, surgical intervention, tumor, or infection of the seventh cranial nerve. *Bell's palsy* is a form with acute onset and unknown cause, possibly viral infection such as herpes zoster. The corner of the mouth droops, and there may be difficulty in speech, eating, and in closing the eye. The symptoms usually disappear spontaneously, but residual facial immobility and lip drooping may persist. Abnormal or reduced *taste sensation* may be due to xerostomia, disturbances of the facial and glossopharyngeal nerves or their central connections, aging, or the wearing of dentures. Disease involving the hypoglossal nerve may cause atrophy of the tongue muscles with protrusion, if bilateral, or deviation toward the affected side, if unilateral. Numb chin (mental neuropathy) may be a sign of primary neural disease, but in the cancer patient it is often a harbinger of tumor relapse or progression.

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SECTION 5 -ALTERATIONS IN CIRCULATORY AND RESPIRATORY FUNCTIONS

32. DYSPNEA AND PULMONARY EDEMA - Roland H. Ingram, Jr., Eugene Braunwald

DYSPNEA

Breathing is controlled by central and peripheral mechanisms that adjust ventilation appropriate to increased metabolic demands during physical activity and increase ventilation in excess of metabolic demands in conditions such as anxiety and fear. A normal resting person is unaware of the act of breathing, and while he or she may become conscious of breathing during mild to moderate exertion, no discomfort is experienced. However, during and following exhausting exertion, an individual may become unpleasantly aware of breathing yet feel reasonably assured that the sensation will be transitory and is appropriate to the level of exercise. Therefore, as a cardinal symptom of diseases affecting the cardiorespiratory system, *dyspnea* is defined as an *abnormally uncomfortable awareness of breathing*.

Although dyspnea is not painful in the usual sense of the word, it is, like pain, involved with both the perception of a sensation and the reaction to that perception. Patients experience a number of uncomfortable sensations related to breathing and use an even larger number of verbal expressions to describe these sensations, such as "cannot get enough air," "air does not go all the way down," "smothering feeling or tightness or tiredness in the chest," and a "choking sensation." It may be necessary, therefore, to review the patient's history meticulously in order to ascertain whether the more abstruse descriptions do, in fact, represent dyspnea. Once it is established that a patient does have dyspnea, it is of paramount importance to define the circumstances in which it occurs and to assess associated symptoms. There are situations in which breathing appears labored but in which dyspnea does not occur. For example, the hyperventilation associated with metabolic acidemia is rarely accompanied by dyspnea. On the other hand, patients with apparently normal breathing patterns may complain of shortness of breath.

QUANTITATION OF DYSPNEA

The gradation of dyspnea is based on the amount of physical exertion required to produce the sensation. In assessing the severity of dyspnea, it is important to obtain a clear understanding of the patient's general physical condition, work history, and recreational habits. For example, the development of dyspnea in a trained runner upon running 2 mi may signify a much more serious disturbance than a similar degree of breathlessness in a sedentary person upon running a fraction of this distance. Interindividual variation in perception must also be considered. Some patients with severe disease may complain of only mild dyspnea; others with mild disease may experience more severe shortness of breath. Some patients with lung or heart disease may have such reduced capabilities due to other disease (e.g., peripheral vascular insufficiency or severe osteoarthritis of the hips or knees) that exertional dyspnea is precluded despite serious impairment of pulmonary or cardiac function.

Some patterns of dyspnea are not directly related to physical exertion. Sudden and

unexpected dyspneic episodes at rest can be associated with pulmonary emboli, spontaneous pneumothorax, hypercapnea secondary to breath holding, or anxiety. Nocturnal episodes of severe paroxysmal dyspnea are characteristic of left ventricular failure. Dyspnea upon assuming the supine posture, *orthopnea* (see below and [Chap. 232](#)), thought to be mainly characteristic of congestive heart failure, may also occur in some patients with asthma and chronic obstruction of the airways and is a regular finding in the rare occurrence of bilateral diaphragmatic paralysis. *Trepopnea* is used to describe the unusual circumstance in which dyspnea occurs only in a lateral decubitus position, most often in patients with heart disease, while *platypnea* is dyspnea that occurs only in the upright position. Positional alterations in ventilation-perfusion relationships ([Chap. 250](#)) have been invoked to explain these patterns.

MECHANISMS OF DYSPNEA (See [Fig. 32-1](#))

Dyspnea occurs whenever the work of breathing is excessive. Increased force generation is required of the respiratory muscles to produce a given volume change if the chest wall or lungs are less compliant or if resistance to airflow is increased. Increased work of breathing also occurs when the ventilation is excessive for the level of activity. Although an individual is more apt to become dyspneic when the work of breathing is increased, the work theory does not account for the perceptual difference between a deep breath with a normal mechanical load and a normal-sized breath with an increased mechanical load. The work might be the same with both breaths, but the normal one with the increased load will be associated with discomfort. In fact, with respiratory loading, such as adding a resistance at the mouth, there is an increase in respiratory center output that is disproportionate to the increase in the work of breathing. It has been postulated that whenever the force that muscles actually generate during breathing approaches some fraction of their maximal force-generating ability, which may vary among individuals, dyspnea ensues due to transduction of mechanical to neural stimuli.

In all likelihood, several different mechanisms operate to different degrees in the various clinical situations in which dyspnea occurs. In some circumstances, dyspnea is evoked by stimulation of receptors in the upper respiratory tract; in others it may originate from receptors in the lungs, airways, respiratory muscles, chest wall, or some combination of these structures. In any event, dyspnea is characterized by an excessive or abnormal activation of the respiratory centers in the brainstem. This activation comes about from stimuli transmitted from or through a variety of structures and pathways, including (1) intrathoracic receptors via the vagi; (2) afferent somatic nerves, particularly from the respiratory muscles and chest wall, but also from other skeletal muscles and joints; (3) chemoreceptors in the brain, aortic and carotid bodies, and elsewhere in the circulation; (4) higher (cortical) centers; and perhaps (5) afferent fibers in the phrenic nerves. In general, despite the interindividual variations described above, there is a reasonable correlation between the severity of dyspnea and the magnitude of disturbances of pulmonary or cardiac function that are responsible.

The mechanisms responsible for dyspnea may vary in different conditions ([Table 32-1](#)).

DIFFERENTIAL DIAGNOSIS

Obstructive Disease of Airways (See also [Chaps. 252](#) and [258](#)) Obstruction to airflow can be present anywhere from the extrathoracic airways out to the small airways in the periphery of the lung. Large extrathoracic airway obstruction can occur acutely, as with aspiration of food or a foreign body or with angioedema of the glottis. An allergic history together with a few scattered hives should raise the possibility of glottic edema. Acute upper airway obstruction is a medical emergency. More chronic forms can occur with tumors or with fibrotic stenosis following tracheostomy or prolonged endotracheal intubation. Whether acute or chronic, the cardinal symptom is dyspnea, and the characteristic signs are stridor and retraction of the supraclavicular fossae with *inspiration*.

Obstruction of intrathoracic airways can occur acutely and intermittently or can be present chronically with worsening during respiratory infections. Acute intermittent obstruction with wheezing is typical of *asthma* ([Chap. 252](#)). Chronic cough with expectoration is typical of *chronic bronchitis* ([Chap. 258](#)) and *bronchiectasis* ([Chap. 256](#)). Most often there are prolongation of expiration and coarse rhonchi that are generalized in chronic bronchitis and may be localized in the case of bronchiectasis. Intercurrent infection results in worsening of the cough, increased expectoration of purulent sputum, and more severe dyspnea. During such episodes, the patient may complain of nocturnal paroxysms of dyspnea with wheezing relieved by cough and expectoration of sputum. Despite the fact that severe limitation of expiratory flow and hyperinflation of the lung are characteristic of these diseases, the sensory experience is often that of an inability to take in a sufficiently deep breath rather than difficulty in exhaling.

The patient with predominant *emphysema* is characterized by many years of exertional dyspnea progressing to dyspnea at rest ([Chap. 258](#)). Although a parenchymal disease by definition, emphysema is invariably accompanied by obstruction of airways.

Diffuse Parenchymal Lung Diseases (See also [Chap. 259](#)) This category includes a large number of diseases ranging from acute pneumonia to chronic disorders such as sarcoidosis and the various forms of *pneumoconiosis* ([Chap. 254](#)). History, physical findings, and radiographic abnormalities often provide clues to the diagnosis. The patients are often tachypneic with arterial P_{CO_2} and P_{O_2} values below normal. Exertion often further reduces the arterial P_{O_2} . Lung volumes are decreased, and the lungs are stiffer, i.e., less compliant than normal.

Pulmonary Vascular Occlusive Diseases (See also [Chap. 261](#)) Repeated episodes of dyspnea at rest often occur with recurrent pulmonary emboli. Evidence of a source for emboli, such as phlebitis of a lower extremity or the pelvis, is quite helpful in leading the physician to suspect the diagnosis. Arterial blood gases are most often abnormal, but lung volumes are frequently normal or only minimally abnormal.

Diseases of the Chest Wall or Respiratory Muscles (See also [Chap. 263](#)) The physical examination establishes the presence of a chest wall disease such as severe kyphoscoliosis, pectus excavatum, or ankylosing spondylitis. Although all three of these deformities may be associated with dyspnea, only severe kyphoscoliosis regularly interferes with ventilation sufficiently to produce chronic cor pulmonale and respiratory failure.

Both weakness and paralysis of respiratory muscles can lead to respiratory failure and dyspnea ([Chap. 263](#)), but most often the signs and symptoms of the neurologic or muscular disorder are more prominently manifested in other systems.

Heart Disease In patients with cardiac disease, exertional dyspnea occurs most commonly as a consequence of an elevated pulmonary capillary pressure, which in turn may be due to left ventricular dysfunction ([Chaps. 231](#) and [232](#)), reduced left ventricular compliance, and mitral stenosis. The elevation of hydrostatic pressure in the pulmonary vascular bed tends to upset the Starling equilibrium (see "Pulmonary Edema," below) with resulting transudation of liquid into the interstitial space, reducing the compliance of the lungs and stimulating J (juxtacapillary) receptors in the alveolar interstitial space. When it is prolonged, pulmonary venous hypertension results in thickening of the walls of small pulmonary vessels and an increase in perivascular cells and fibrous tissue, causing a further reduction in compliance. The competition for space among vessels, airways, and increased liquid within the interstitial space compromises the lumina of small airways, increasing the airways' resistance. Diminution in compliance and an increase in the airways' resistance increase the work of breathing. In advanced congestive heart failure, usually involving elevation of both pulmonary and systemic venous pressures, hydrothorax may develop, interfering further with pulmonary function and intensifying dyspnea.

Orthopnea, i.e., dyspnea in the supine position, is the result of the alteration of gravitational forces when this position is assumed, which elevates pulmonary venous and capillary pressures. These, in turn, increase the pulmonary closing volume ([Chap. 250](#)) and reduce the vital capacity.

Paroxysmal (Nocturnal) Dyspnea Also known as *cardiac asthma*, this condition is characterized by attacks of severe shortness of breath that generally occur at night and usually awaken the patient from sleep. The attack is precipitated by stimuli that aggravate previously existing pulmonary congestion; frequently, the total blood volume is augmented at night because of the reabsorption of edema from dependent portions of the body during recumbency. A sleeping patient can tolerate relatively severe pulmonary engorgement and may awaken only when actual pulmonary edema and bronchospasm have developed, with the feeling of suffocation and with wheezing respirations.

Two other forms of nocturnal dyspnea must be distinguished from that due to heart failure. Chronic bronchitis is characterized by mucus hypersecretion and, after a few hours sleep, secretions can accumulate and produce dyspnea and wheezing, both of which are relieved by cough and expectoration of sputum. Asthma patients have circadian variations in their degree of airway obstruction. The obstruction becomes most severe between 2 A.M. and 4 A.M. and can be sufficiently severe that the patient awakens with a sense of suffocation, extreme dyspnea, and wheezing. Although there is a prominent inflammatory component to nocturnal asthma, inhaled bronchodilators usually improve symptoms quickly.

*Cheyne-Stokes respiration**[See Chap. 232](#)

Diagnosis The diagnosis of cardiac dyspnea depends on the recognition of heart disease on the basis of the clinical examination supplemented by noninvasive testing. There may be a history of antecedent myocardial infarction; third and fourth heart sounds may be audible; and/or there may be evidence of left ventricular enlargement, jugular neck vein distention, and/or peripheral edema. Often there are radiographic signs of heart failure, with evidence of interstitial edema, pulmonary vascular redistribution, and accumulation of liquid in the septal planes and pleural cavity. Transthoracic echocardiography is particularly useful in establishing the diagnosis of structural heart disease, which can be responsible for dyspnea. Specifically, left atrial and/or left ventricular dilatation, left ventricular hypertrophy, a reduced left ventricular ejection fraction, and disorders of left ventricular wall motion may be clues to the presence of a cardiac etiology of otherwise unexplained dyspnea.

DIFFERENTIATION BETWEEN CARDIAC AND PULMONARY DYSPNEA

In most patients with dyspnea there is obvious clinical evidence of disease of the heart and/or lungs. Like patients with cardiac dyspnea, patients with chronic obstructive lung disease may also waken at night with dyspnea, but, as pointed out above, this is usually associated with sputum production; the dyspnea is relieved after these patients rid themselves of secretions. The difficulty in the distinction between cardiac and pulmonary dyspnea may be compounded by the coexistence of diseases involving both organ systems.

In patients in whom the etiology of dyspnea is not clear, it is desirable to carry out pulmonary function testing, for these tests may be helpful in determining whether dyspnea is produced by heart disease, lung disease, abnormalities of the chest wall, or anxiety ([Chap. 250](#)). In addition to the usual means of assessing patients for heart disease, determination of the ejection fraction at rest and during exercise by echocardiography or radionuclide ventriculography is helpful in the differential diagnosis of dyspnea. The left ventricular ejection fraction is depressed in left ventricular failure, while the right ventricular ejection fraction may be low at rest or may decline during exercise in patients with severe lung disease. Both left and right ventricular ejection fractions are normal at rest and during exercise in dyspnea due to anxiety or malingering. Careful observation during the performance of an exercise treadmill test will often help in the identification of the patient who is malingering or whose dyspnea is secondary to anxiety. Under these circumstances, the patient usually complains of severe shortness of breath but appears to be breathing either effortlessly or totally irregularly. Cardiopulmonary testing, in which the patient's maximal functional exercise capacity is assessed while measurements of the electrocardiogram, blood pressure, oxygen consumption, arterial saturation (oximetry), and ventilation are carried out, is useful in the differentiation between cardiac and pulmonary dyspnea ([Table 32-2](#)).

ANXIETY NEUROSIS

Dyspnea experienced by a patient with an anxiety neurosis is difficult to evaluate. The signs and symptoms of acute and chronic hyperventilation do not serve to distinguish between anxiety neurosis and other processes, such as recurrent pulmonary emboli. Another potentially confusing situation is seen when chest pain and electrocardiographic changes accompany the hyperventilation syndrome. When present and attributable to

this condition, often referred to as *neurocirculatory asthenia* ([Chap. 13](#)), the chest pain is often sharp, fleeting, and in various loci, and the electrocardiographic changes are most often seen during repolarization. Frequent sighing respirations and an irregular breathing pattern point to a psychogenic origin of the dyspnea. Anxiety and depression in association with heart or lung disease can serve to intensify dyspnea symptoms beyond what would be expected for a given degree of dysfunction.

PULMONARY EDEMA (See [Table 32-3](#))

CARDIOGENIC PULMONARY EDEMA (See [Table 32-3](#), IA)

An increase in pulmonary venous pressure, which results initially in engorgement of the pulmonary vasculature, is common in most instances of dyspnea in association with congestive heart failure. The lungs become less compliant, the resistance of small airways increases, and there is an increase in lymphatic flow that apparently serves to maintain a constant pulmonary extravascular liquid volume. Mild tachypnea is present. If the increase in intravascular pressure is sufficient both in magnitude and duration, there is a net gain of liquid in the extravascular space, i.e., *interstitial* edema. At this point symptoms worsen, tachypnea increases, gas exchange deteriorates further, and radiographic changes, such as Kerley B lines and loss of distinct vascular margins, are seen. At this stage, the capillary endothelial intercellular junctions widen and allow passage of macromolecules into the interstices.

Further elevations in intravascular pressure disrupt the tight junctions between alveolar lining cells, and *alveolar* edema ensues, with outpouring of liquid that contains both red blood cells and macromolecules. With yet more severe disruption of the alveolar-capillary membrane, edematous liquid floods the alveoli and airways. At this point, full-blown clinical pulmonary edema with bilateral wet rales and rhonchi occurs, and the chest radiograph may show diffuse haziness of the lung fields with greater density in the more proximal hilar regions. Typically, the patient is anxious and perspires freely, and the sputum is frothy and blood-tinged. Gas exchange is more severely compromised with worsening hypoxia. Without effective treatment (described in [Chap. 232](#)), progressive acidemia, hypercapnia, and respiratory arrest ensue.

The earlier sequence of liquid accumulation described above follows the Starling law of capillary-interstitial liquid exchange:

The pressures tending to move liquid out of the vessel are P_c and p_{IF} , which are normally more than offset by pressures tending to move liquid back into the vasculature, i.e., the algebraic sum of P_{IF} and p_{pl} . Implicit in the preceding equation is that lymphatic flow can increase in the case of imbalance of forces and result in no net accumulation of interstitial liquid. Further elevations in P_c not only increase the outward movement of liquid in each capillary region but also recruit more of the capillary bed, which increases K . These two effects lead to liquid filtration that exceeds clearance capability by the lymphatics, and liquid accumulates in the loose interstitial spaces of the lung. Even greater increases in P_c open first the loose endothelial intercellular junctions and later the tight alveolar intercellular junctions with an increase in permeability to

macromolecules. This secondary disruption of both the function and structure of the alveolar-capillary membrane leads to alveolar flooding.

NONCARDIOGENIC PULMONARY EDEMA (See [Table 32-3](#), IB IC, II, III, and IV)

Several clinical conditions are associated with pulmonary edema based on an imbalance of Starling forces other than through primary elevations of pulmonary capillary pressure. Although diminished plasma oncotic pressure in hypoalbuminemic states (e.g., severe liver disease, nephrotic syndrome, protein-losing enteropathy) might be expected to lead to pulmonary edema, the balance of forces normally so strongly favors resorption that even in these conditions some elevation of capillary pressure is usually necessary before interstitial edema develops. Increased negativity of interstitial pressure has been implicated in the genesis of unilateral pulmonary edema following rapid evacuation of a large pneumothorax. In this situation, the findings may be apparent only by radiography, but occasionally the patient experiences dyspnea with physical findings localized to the edematous lung. It has been proposed that large negative intrapleural pressures during acute severe asthma may be associated with the development of interstitial edema. Lymphatic blockade secondary to fibrotic and inflammatory diseases or lymphangitic carcinomatosis may lead to interstitial edema. In such instances, both clinical and radiographic manifestations are dominated by the underlying disease process.

Other conditions characterized by increases in the interstitial liquid content of the lungs appear to be associated primarily with disruption of the alveolar-capillary membranes. Any number of spontaneously occurring or environmental toxic insults, including diffuse pulmonary infections, aspiration, and shock (particularly due to sepsis and hemorrhagic pancreatitis and following cardiopulmonary bypass), are associated with diffuse pulmonary edema that clearly does not have a hemodynamic origin. **These conditions, which may lead to the acute respiratory distress syndrome, are discussed in [Chap. 265](#).*

Other Forms of Pulmonary Edema There are three forms of pulmonary edema whose precise mechanism remains unexplained. *Narcotic overdose* is a well-recognized antecedent to pulmonary edema. Although illicit use of parenteral heroin is the most frequent cause, parenteral and oral overdoses of legitimate preparations of morphine, methadone, and dextropropoxyphene have also been associated with pulmonary edema. The earlier idea that injected impurities lead to the disorder is untenable. Available evidence suggests that there are alterations in the permeability of alveolar and capillary membranes rather than an elevation of pulmonary capillary pressure.

Exposure to high altitude in association with severe physical exertion is a well-recognized setting for pulmonary edema in unacclimatized yet otherwise healthy persons. Acclimatized high-altitude natives also develop this syndrome upon return to high altitude after a relatively brief sojourn at low altitudes. The syndrome is far more common in persons under the age of 25 years. The mechanism for high-altitude pulmonary edema (HAPE) remains obscure, and studies have been conflicting, some suggesting pulmonary venous constriction and others indicating pulmonary arteriolar constriction as the prime mechanisms. A role for hypoxia at high altitude is suggested by the fact that patients respond to the administration of oxygen and/or return to lower altitudes. Hypoxia per se does not alter permeability of the alveolar-capillary membrane.

Hence increased cardiac output and pulmonary arterial pressures with exercise combined with hypoxic pulmonary arteriolar constriction, which is more prominent in young persons, may combine to make this an example of prearteriolar, high-pressure pulmonary edema.

Neurogenic pulmonary edema has been described in patients with central nervous system disorders and without apparent preexisting left ventricular dysfunction. Although most experimental equivalents have implicated sympathetic nervous system activity, the mechanism whereby sympathetic efferent activity leads to pulmonary edema is a matter of speculation. It is known that a massive adrenergic nervous discharge leads to peripheral vasoconstriction with elevation of blood pressure and shifts of blood to the central circulation. In addition, it is probable that a reduction in left ventricular compliance also occurs, and both factors serve to increase left atrial pressures sufficiently to induce pulmonary edema on a hemodynamic basis. Some experimental evidence suggests that stimulation of adrenergic receptors increases capillary permeability directly, but this effect is relatively minor as compared with the imbalance of Starling forces.

TREATMENT OF PULMONARY EDEMA See [Chap. 232](#)

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33. COUGH AND HEMOPTYSIS - Steven E. Weinberger, Eugene Braunwald

COUGH

Cough is an explosive expiration that provides a normal protective mechanism for clearing the tracheobronchial tree of secretions and foreign material. When excessive or bothersome, it is also one of the most common symptoms for which medical attention is sought. Reasons for the latter include discomfort from the cough itself, interference with normal lifestyle, and concern for the cause of the cough, especially fear of cancer or AIDS.

MECHANISM

Coughing may be initiated either voluntarily or reflexively. As a defensive reflex it has both afferent and efferent pathways. The *afferent limb* includes receptors within the sensory distribution of the trigeminal, glossopharyngeal, superior laryngeal, and vagus nerves. The *efferent limb* includes the recurrent laryngeal nerve and the spinal nerves. The cough starts with a deep inspiration followed by glottic closure, relaxation of the diaphragm, and muscle contraction against a closed glottis. The resulting markedly positive intrathoracic pressure causes narrowing of the trachea. Once the glottis opens, the large pressure differential between the airways and the atmosphere coupled with tracheal narrowing produces rapid flow rates through the trachea. The shearing forces that develop aid in the elimination of mucus and foreign materials.

ETIOLOGY

Cough can be initiated by a variety of airway irritants, which enter the tracheobronchial tree by inhalation (smoke, dust, fumes) or by aspiration (upper airway secretions, gastric contents, foreign bodies). When cough is due to irritation by upper airway secretions (as with postnasal drip) or gastric contents (as with gastroesophageal reflux), the initiating factor may go unrecognized and the cough can be persistent. Additionally, prolonged exposure to such irritants may initiate airway inflammation, which can itself trigger cough and sensitize the airway to other irritants. Cough associated with gastroesophageal reflux is due only in part to aspiration of gastric contents, whereas vagally mediated reflex mechanisms appear to be responsible in many patients.

Any disorder resulting in inflammation, constriction, infiltration, or compression of airways can be associated with cough. Inflammation commonly results from airway infections, ranging from viral or bacterial bronchitis to bronchiectasis. In viral bronchitis, airway inflammation sometimes persists long after resolution of the typical acute symptoms, thereby producing a prolonged cough, lasting for weeks. Pertussis infection is also a possible cause of persistent cough in adults; however, diagnosis is generally made on clinical grounds ([Chap. 152](#)). Asthma is a common cause of cough. Although the clinical setting commonly suggests when a cough is secondary to asthma, some patients present with cough in the absence of wheezing or dyspnea, thus making the diagnosis more subtle ("cough variant asthma"). A neoplasm infiltrating the airway wall, such as bronchogenic carcinoma or a carcinoid tumor, is commonly associated with cough. Airway infiltration with granulomas may also trigger a cough, as seen with endobronchial sarcoidosis or tuberculosis. Compression of airways results from extrinsic

masses, including lymph nodes, mediastinal tumors, and aortic aneurysms.

Examples of parenchymal lung disease potentially producing cough include interstitial lung disease, pneumonia, and lung abscess. Congestive heart failure may be associated with cough, probably as a consequence of interstitial as well as peribronchial edema. A nonproductive cough complicates the use of angiotensin-converting enzyme (ACE) inhibitors in 5 to 20% of patients taking these agents. Onset is usually within 1 week of starting the drug but can be delayed up to 6 months. Although the mechanism is not known with certainty, it may relate to accumulation of bradykinin or substance P, both of which are degraded by ACE.

The most common causes of cough can be categorized according to the duration of the cough. Acute cough (<3 weeks) is most often due to upper respiratory infection (especially the common cold, acute bacterial sinusitis, and pertussis), but more serious disorders, such as pneumonia, pulmonary embolus, and congestive heart failure, can also present in this fashion. Chronic cough (>3 weeks) in a smoker raises the possibilities of chronic obstructive lung disease or bronchogenic carcinoma. In a nonsmoker who has a normal chest radiograph and is not taking an [ACE](#) inhibitor, the most common causes of chronic cough are postnasal drip, asthma, and gastroesophageal reflux.

Approach to the Patient

A detailed *history* frequently provides the most valuable clues for etiology of the cough. Particularly important questions include:

1. Is the cough acute or chronic?
2. At its onset, were there associated symptoms suggestive of a respiratory infection?
3. Is it seasonal or associated with wheezing?
4. Is it associated with symptoms suggestive of postnasal drip (nasal discharge, frequent throat clearing, a "tickle in the throat") or gastroesophageal reflux (heartburn or sensation of regurgitation)? (The absence of such suggestive symptoms does not exclude either of these diagnoses, particularly in the case of gastroesophageal reflux.)
5. Is it associated with fever or sputum? If sputum is present, what is its character?
6. Does the patient have any associated diseases or risk factors for disease (e.g., cigarette smoking, risk factors for infection with HIV, environmental exposures)?
7. Is the patient taking an [ACE](#) inhibitor?

The general *physical examination* may point to a nonpulmonary cause of cough, such as heart failure, primary nonpulmonary neoplasm, or AIDS. Examination of the oropharynx may provide suggestive evidence for postnasal drip, including oropharyngeal mucus or erythema, or a "cobblestone" appearance to the mucosa. Auscultation of the chest may demonstrate inspiratory stridor (indicative of upper airway

disease), rhonchi or expiratory wheezing (indicative of lower airway disease), or inspiratory crackles (suggestive of a process involving the pulmonary parenchyma, such as interstitial lung disease, pneumonia, or pulmonary edema).

Chest radiography may be particularly helpful in suggesting or confirming the cause of the cough. Important potential findings include the presence of an intrathoracic mass lesion, a localized pulmonary parenchymal infiltrate, or diffuse interstitial or alveolar disease. An area of honeycombing or cyst formation may suggest bronchiectasis, while symmetric bilateral hilar adenopathy may suggest sarcoidosis.

Pulmonary function testing ([Chap. 250](#)) is useful for assessing the functional abnormalities that accompany certain disorders producing cough. Measurement of forced expiratory flow rates can demonstrate reversible airflow obstruction characteristic of asthma. When asthma is considered but flow rates are normal, bronchoprovocation testing with methacholine or cold-air inhalation can demonstrate hyperreactivity of the airways to a bronchoconstrictive stimulus. Measurement of lung volumes and diffusing capacity is useful primarily for demonstration of a restrictive pattern, often seen with any of the diffuse interstitial lung diseases.

If *sputum* is produced, gross and microscopic examination may provide useful information. Purulent sputum suggests chronic bronchitis, bronchiectasis, pneumonia, or lung abscess. Blood in the sputum may be seen in the same disorders, but its presence also raises the question of an endobronchial tumor. Gram and acid-fast stains and cultures may demonstrate a particular infectious pathogen, while sputum cytology may provide a diagnosis of a pulmonary malignancy.

More specialized studies are helpful in specific circumstances. *Fiberoptic bronchoscopy* is the procedure of choice for visualizing an endobronchial tumor and collecting cytologic and histologic specimens. Inspection of the tracheobronchial mucosa can demonstrate endobronchial granulomas often seen in sarcoidosis, and endobronchial biopsy of such lesions or transbronchial biopsy of the lung interstitium can confirm the diagnosis. Inspection of the airway mucosa by bronchoscopy can also demonstrate the characteristic appearance of endobronchial Kaposi's sarcoma in patients with AIDS. *High-resolution computed tomography* (HRCT) can confirm the presence of interstitial disease and frequently suggests a diagnosis based on the pattern of disease. It is the procedure of choice for demonstrating dilated airways and confirming the diagnosis of bronchiectasis.

A diagnostic algorithm for evaluation of chronic cough is presented in [Fig. 33-1](#).

COMPLICATIONS

Common complications of coughing include chest and abdominal wall soreness, urinary incontinence, and exhaustion. On occasion, paroxysms of coughing may precipitate syncope (cough syncope; [Chap. 21](#)), consequent to markedly positive intrathoracic and alveolar pressures, diminished venous return, and decreased cardiac output. Although cough fractures of the ribs may occur in otherwise normal patients, their occurrence should at least raise the possibility of pathologic fractures, which are seen with multiple myeloma, osteoporosis, and osteolytic metastases.

TREATMENT

Definitive treatment of cough depends on determining the underlying cause and then initiating specific therapy. Elimination of an exogenous inciting agent (cigarette smoke, [ACE](#) inhibitors) or an endogenous trigger (postnasal drip, gastroesophageal reflux) is usually effective when such a precipitant can be identified. Other important management considerations are treatment of specific respiratory tract infections, bronchodilators for potentially reversible airflow obstruction, chest physiotherapy to enhance clearance of secretions in patients with bronchiectasis, and treatment of endobronchial tumors or interstitial lung disease when such therapy is available and appropriate.

Symptomatic or nonspecific therapy of cough should be considered when: (1) the cause of the cough is not known or specific treatment is not possible, and (2) the cough performs no useful function or causes marked discomfort. An irritative, nonproductive cough may be suppressed by an antitussive agent, which increases the latency or threshold of the cough center. Such agents include codeine (15 mg qid) or nonnarcotics such as dextromethorphan (15 mg qid). These drugs provide symptomatic relief by interrupting prolonged, self-perpetuating paroxysms. However, a cough productive of significant quantities of sputum should usually not be suppressed, since retention of sputum in the tracheobronchial tree may interfere with the distribution of ventilation, alveolar aeration, and the ability of the lung to resist infection.

Other agents working by a variety of mechanisms have also been used to control cough, but objective information assessing their benefit is meager. The inhaled anticholinergic agent, ipratropium bromide (2 to 4 puffs qid), has been used with the rationale of inhibiting the efferent limb of the cough reflex. Inhaled glucocorticoids, ideally administered with a spacer and dosed according to the particular agent, have been used for patients in whom airway inflammation is thought to be playing a role in the cough.

HEMOPTYSIS

Hemoptysis is defined as the expectoration of blood from the respiratory tract, a spectrum that varies from blood-streaking of sputum to coughing up large amounts of pure blood. *Massive hemoptysis* is variably defined as the expectoration of >100 to >600 mL over a 24-h period, although the patient's estimation of the amount of blood is notoriously unreliable. Expectoration of even relatively small amounts of blood is a frightening symptom and can be a marker for potentially serious disease, such as bronchogenic carcinoma. Massive hemoptysis, on the other hand, can represent an acutely life-threatening problem. Large amounts of blood can fill the airways and the alveolar spaces, not only seriously disturbing gas exchange but potentially causing the patient to suffocate.

ETIOLOGY

Because blood originating from the nasopharynx or the gastrointestinal tract can mimic blood coming from the lower respiratory tract, it is important to determine initially that the

blood is not coming from one of these alternative sites. Clues that the blood is originating from the gastrointestinal tract include a dark red appearance and an acidic pH, in contrast to the typical bright red appearance and alkaline pH of true hemoptysis.

The bronchial arteries, which are part of the high-pressure systemic circulation, originate either from the aorta or from intercostal arteries and are the source of bleeding in bronchitis or bronchiectasis or with endobronchial tumors.

An etiologic classification of hemoptysis can be based on the site of origin within the lungs ([Table 33-1](#)). The most common site of bleeding is the airways, i.e., the tracheobronchial tree, which can be affected by inflammation (acute or chronic bronchitis, bronchiectasis) or by neoplasm (bronchogenic carcinoma, endobronchial metastatic carcinoma, or bronchial carcinoid tumor). Blood originating from the pulmonary parenchyma can be either from a localized source, such as an infection (pneumonia, lung abscess, tuberculosis), or from a process diffusely affecting the parenchyma (as with a coagulopathy or with an autoimmune process such as Goodpasture's syndrome). Disorders primarily affecting the pulmonary vasculature include pulmonary embolic disease and those conditions associated with elevated pulmonary venous and capillary pressures, such as mitral stenosis or left ventricular failure.

Although the relative frequency of the different etiologies of hemoptysis varies from series to series, most recent studies indicate that bronchitis and bronchogenic carcinoma are the two most common causes. Despite the lower frequency of tuberculosis and bronchiectasis seen in recent compared to older series, these two disorders still represent the most common causes of massive hemoptysis in several series. Even after extensive evaluation, a sizable proportion of patients (up to 30% in some series) have no identifiable etiology for their hemoptysis. These patients are classified as having idiopathic or cryptogenic hemoptysis, and subtle airway or parenchymal disease is presumably responsible for the bleeding.

Approach to the Patient

The *history* is extremely valuable. Hemoptysis that is described as blood-streaking of mucopurulent or purulent sputum often suggests bronchitis. Chronic production of sputum with a recent change in quantity or appearance favors an acute exacerbation of chronic bronchitis. Fever or chills accompanying blood-streaked purulent sputum suggests pneumonia, whereas a putrid smell to the sputum raises the possibility of lung abscess. When sputum production has been chronic and copious, the diagnosis of bronchiectasis should be considered. Hemoptysis following the acute onset of pleuritic chest pain and dyspnea is suggestive of pulmonary embolism.

A history of previous or coexisting disorders should be sought, such as renal disease (seen with Goodpasture's syndrome or Wegener's granulomatosis), lupus erythematosus (with associated pulmonary hemorrhage from lupus pneumonitis), or a previous malignancy (either recurrent lung cancer or endobronchial metastasis from a nonpulmonary primary tumor). In a patient with AIDS, endobronchial or pulmonary parenchymal Kaposi's sarcoma should be considered. Risk factors for bronchogenic carcinoma, particularly smoking and asbestos exposure, should be sought. Patients

should be questioned about previous bleeding disorders, treatment with anticoagulants, or use of drugs that can be associated with thrombocytopenia.

The *physical examination* may also provide helpful clues to the diagnosis. For example, examination of the lungs may demonstrate a pleural friction rub (pulmonary embolism), localized or diffuse crackles (parenchymal bleeding or an underlying parenchymal process associated with bleeding), evidence of airflow obstruction (chronic bronchitis), or prominent rhonchi, with or without wheezing or crackles (bronchiectasis). Cardiac examination may demonstrate findings of pulmonary arterial hypertension, mitral stenosis, or heart failure. Skin examination may reveal Kaposi's sarcoma, arteriovenous malformations of Osler-Rendu-Weber disease, or lesions suggestive of systemic lupus erythematosus.

Diagnostic evaluation of hemoptysis starts with a chest radiograph to look for a mass lesion, findings suggestive of bronchiectasis ([Chap. 256](#)), or focal or diffuse parenchymal disease (representing either focal or diffuse bleeding or a focal area of pneumonitis). Additional initial screening evaluation often includes a complete blood count, a coagulation profile, and assessment for renal disease with a urinalysis and measurement of blood urea nitrogen and creatinine levels. When sputum is present, examination by Gram and acid-fast stains (along with the corresponding cultures) is indicated.

Fiberoptic bronchoscopy is particularly useful for localizing the site of bleeding and for visualization of endobronchial lesions. When bleeding is massive, rigid bronchoscopy is often preferable to fiberoptic bronchoscopy because of better airway control and greater suction capability. In patients with suspected bronchiectasis, [HRCT](#) is now the diagnostic procedure of choice, having replaced bronchography.

A diagnostic algorithm for evaluation of nonmassive hemoptysis is presented in [Fig. 33-2](#).

TREATMENT

The rapidity of bleeding and its effect on gas exchange determine the urgency of management. When the bleeding is confined to either blood-streaking of sputum or production of small amounts of pure blood, gas exchange is usually preserved; establishing a diagnosis is the first priority. When hemoptysis is massive, maintaining adequate gas exchange, preventing blood from spilling into unaffected areas of lung, and avoiding asphyxiation are the highest priorities. Keeping the patient at rest and partially suppressing cough may help the bleeding to subside. If the origin of the blood is known and is limited to one lung, the bleeding lung should be placed in the dependent position, so that blood is not aspirated into the unaffected lung.

With massive bleeding, the need to control the airway and maintain adequate gas exchange may necessitate endotracheal intubation and mechanical ventilation. In patients in danger of flooding the lung contralateral to the side of hemorrhage despite proper positioning, isolation of the right and left mainstem bronchi from each other can be achieved by selectively intubating the nonbleeding lung (often with bronchoscopic guidance) or by using specially designed double-lumen endotracheal tubes. Another

option involves inserting a balloon catheter through a bronchoscope by direct visualization and inflating the balloon to occlude the bronchus leading to the bleeding site. This technique not only prevents aspiration of blood into unaffected areas but also may promote tamponade of the bleeding site and cessation of bleeding.

Other available techniques for control of significant bleeding include laser phototherapy, electrocautery, embolotherapy, and surgical resection of the involved area of lung. With bleeding from an endobronchial tumor, the neodymium:yttrium-aluminum-garnet (Nd:YAG) laser can often achieve at least temporary hemostasis by coagulating the bleeding site. Electrocautery, which uses an electric current for thermal destruction of tissue, can be used similarly for management of bleeding from an endobronchial tumor. Embolotherapy involves an arteriographic procedure in which a vessel proximal to the bleeding site is cannulated, and a material such as Gelfoam is injected to occlude the bleeding vessel. Surgical resection is a therapeutic option either for the emergent therapy of life-threatening hemoptysis that fails to respond to other measures or for the elective but definitive management of localized disease subject to recurrent bleeding.

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34. APPROACH TO THE PATIENT WITH A HEART MURMUR - Patrick T. O'Gara, Eugene Braunwald

Auscultation of the heart constitutes the final step in the cardiovascular examination and, for many patients with established or suspected cardiac disease, represents a defining moment in the doctor-patient relationship. The examiner must bring to this exercise an integrated approach that incorporates pertinent information from several sources. The auscultatory findings must be interpreted in the context of the history and general physical examination and with the observations made regarding the venous wave forms and major arterial pulses. In this way, abnormalities of heart sounds, adventitious sounds, and murmurs can be placed in their proper perspective.

In many patients, a heart murmur is the only or the most conspicuous finding on physical examination. The recognition of a heart murmur usually leads to additional testing, such as electrocardiography, chest radiography, and echocardiography, and may result in referral to a cardiologist. The differential diagnosis of a heart murmur should begin with an unbiased and systematic evaluation of its major attributes: timing, duration, intensity, quality, frequency, configuration, location, radiation, and response to maneuvers (see [Table 225-1](#)). Laboratory testing can be pursued thereafter to clarify any remaining ambiguity and to provide additional anatomic and physiologic information that will impact on patient management.

Heart murmurs are defined in terms of their timing within the cardiac cycle. *Systolic murmurs* begin with or after the first heart sound (S_1) and terminate at or before the component (A_2 or P_2) of the second heart sound (S_2) that corresponds to their side of origin (left or right). *Diastolic murmurs* begin with or after the associated component of S_2 and end at or before the subsequent S_1 . *Continuous murmurs* are not confined to either phase of the cardiac cycle but rather begin in systole and proceed through S_2 into all or part of diastole.

The appropriate timing of heart murmurs is the first critical step in their identification. The distinction between S_1 and S_2 , and, therefore, systole and diastole, is usually a straightforward process but can be difficult in the setting of a tachyarrhythmia, in which case the heart sounds can be distinguished by simultaneous palpation of the carotid arterial pulse. The upstroke should closely follow S_1 . The principal causes of heart murmurs are shown in [Table 34-1](#), and the critical importance of the timing of heart murmurs in the differential diagnosis is shown in [Fig. 34-1](#).

SYSTOLIC HEART MURMURS

Systolic heart murmurs derive from the increased turbulence associated with (1) enhanced or accelerated flow across a normal semilunar valve, through a normal ventricular outflow tract, or into a dilated great vessel, (2) normal flow across a structurally abnormal semilunar valve or through a narrowed ventricular outflow tract, (3) flow across an incompetent atrioventricular valve, and (4) flow across the interventricular septum. One approach to their differential diagnosis further subdivides these murmurs according to their time of onset and duration within the systolic phase of the cardiac cycle.

EARLY SYSTOLIC MURMURS

Early systolic murmurs begin with S₁ and extend for a variable period of time, ending well before S₂. Their causes are relatively few in number. *Acute severe mitral regurgitation* into a normal-sized, relatively noncompliant left atrium results in an early and attenuated systolic murmur that is decrescendo in configuration and usually best heard at or just medial to the apical impulse ([Chap. 236](#)). These characteristics reflect the rapid rise in left atrial pressure caused by the sudden volume load into a nondilated chamber and contrast sharply with the auscultatory features of chronic mitral regurgitation. Clinical settings in which this occurs include: (1) papillary muscle rupture complicating acute myocardial infarction, (2) infective endocarditis, (3) rupture of chordae tendineae, and (4) blunt chest wall trauma.

Acute mitral regurgitation from papillary muscle rupture usually accompanies an inferior, posterior, or lateral infarction. The murmur is associated with a precordial thrill in approximately one-half of cases and is to be distinguished from that associated with postinfarction ventricular septal rupture. The latter is more commonly (90%) accompanied by a thrill at the left sternal edge, is holosystolic, and complicates anterior infarctions as often as inferior-posterior damage. The recognition of either of these mechanical defects mandates aggressive medical stabilization and emergent surgical intervention ([Chap. 243](#)).

The other potential causes of acute severe mitral regurgitation may be distinguished on the basis of associated findings. Spontaneous chordal rupture usually occurs on a substrate of myxomatous replacement, such as that underlying most forms of mitral valve prolapse ([Chap. 236](#)). This lesion may be part of a more generalized process, as can occur with the Marfan or Ehlers-Danlos syndromes, or it may be an isolated phenomenon. Infective endocarditis is associated with fever, peripheral embolic lesions, and positive blood cultures and most commonly occurs on a previously abnormal valvular apparatus ([Chap. 126](#)). Trauma is usually self-evident but may be disarmingly trivial ([Chap. 240](#)). It can result in papillary muscle contusion and rupture, chordal interruption, or leaflet avulsion or perforation.

Echocardiography should be performed in all cases of suspected acute severe mitral regurgitation to define the responsible mechanism, estimate the severity, and provide a preliminary assessment as to the feasibility of surgical repair (versus replacement).

Other causes of early systolic murmurs include congenital, small muscular ventricular septal defects. The duration of the murmur is attenuated by the closure of the defect during systolic contraction. The murmur is localized to the left sternal edge and is commonly of grade IV/VI or V/VI intensity. Signs of pulmonary hypertension or left ventricular volume overload are absent. Patients with anatomically large, uncorrected ventricular septal defects accompanied by pulmonary hypertension may also have murmurs confined to early systole. The elevated pulmonary vascular resistance attenuates the degree of shunting as pressures within the right and left ventricles equalize during the latter half of systole.

Tricuspid regurgitation with normal pulmonary artery pressures, such as that caused by infective endocarditis in injection drug users, may produce an early systolic murmur.

The murmur is soft, best heard at the lower left sternal edge, and may accentuate with inspiration (Carvallo's sign). Regurgitant c-v waves may be visible in the jugular venous pulse.

MIDSYSTOLIC MURMURS

Midsystolic murmurs begin at a short interval following S₁, end before S₂, and are usually crescendo-decrescendo in configuration ([Fig. 34-1C](#)). Semilunar valve stenosis is the classic prototype. With aortic valve stenosis, the murmur is usually loudest in the second right intercostal space (aortic area) and radiates along the carotid arteries ([Chap. 236](#)). The intensity of the murmur varies directly with the cardiac output; aortic valve stenosis with severe heart failure may produce a misleadingly soft systolic murmur. With a normal cardiac output, a systolic thrill is usually indicative of severe stenosis with a peak gradient in excess of 50 to 60 mmHg. An accompanying early systolic ejection click may be audible in younger patients with a bicuspid valve; its presence localizes the obstruction to the valvular (as opposed to the sub- or supra- valvular) level. The midsystolic murmur of aortic stenosis may be well transmitted to the apex, especially in older patients, where it becomes less harsh and slightly higher pitched (Gallavardin effect). The murmur of aortic stenosis should increase following a postpremature beat, whereas a mitral regurgitant murmur would not be expected to change in intensity.

Sclerosis of the aortic valve produces a murmur of similar location, radiation, and configuration, albeit without the usual signs of hemodynamic significance. The carotid upstroke is well preserved, the murmur peaks in midsystole and is not accompanied by a thrill, and only a modest gradient is estimated by Doppler echocardiography. Noncritical sclerodegenerative thickening of the aortic valve leaflets is perhaps the most common cause of a midystolic murmur in older adults. The similar midsystolic murmur of pulmonic valve stenosis, usually introduced by an ejection click, is best appreciated in the second and third left intercostal spaces (pulmonic area). The murmur lengthens and the intensity of P₂ diminishes with increasing degrees of stenosis. A midsystolic murmur in the aortic position can also be detected in hyperdynamic states (fever, thyrotoxicosis, pregnancy, anemia) and in the presence of isolated aortic regurgitation with the augmented flow into a dilated proximal aorta.

Crescendo-decrescendo midsystolic murmurs usually of grade II/VI intensity heard in the pulmonic area may be innocent if unaccompanied by any other signs of cardiac disease in children or young adults. They may also reflect enhanced flow into a normal pulmonary artery in hyperkinetic states or augmented flow into a dilated pulmonary artery. The latter may occur with an atrial septal defect, in which case splitting of S₂ is usually abnormal (fixed). Still's murmur is a vibratory, medium frequency, mid-systolic murmur heard best between the lower left sternal edge and the apex in normal children and young adults. It is generated by vibrations of the pulmonic valve leaflets at their attachments or by vibrations of a left ventricular false tendon.

The midsystolic murmur of hypertrophic cardiomyopathy ([Chap. 238](#)) is usually loudest between the left sternal edge and apex, of grade II/VI to III/VI intensity, and crescendo-decrescendo in configuration. In contrast to aortic valve stenosis, the murmur does *not* radiate into the neck and the carotid upstrokes are brisk and full and may even

be bifid. The intensity of the murmur associated with hypertrophic cardiomyopathy increases following maneuvers that decrease left ventricular volume (strain phase of the Valsalva maneuver, standing, amyl nitrite) or increase myocardial contractility (inotropic therapy). Conversely, the intensity of the systolic murmur decreases with maneuvers that increase ventricular volume (squatting, passive leg raising), impair contractility (beta-adrenoreceptor blockade), or raise preload and systemic afterload (squatting). Among these several maneuvers, auscultation in the standing and squatting positions, if possible, is perhaps the most sensitive technique to elicit a dynamic change in the intensity of the murmur associated with hypertrophic obstructive cardiomyopathy.

LATE SYSTOLIC MURMURS

A late systolic murmur begins well after the onset of ejection and is usually best heard at the left ventricular apex or between the apex and the left sternal edge. When introduced by a nonejection click, it is usually indicative of systolic prolapse of the mitral valve leaflet(s) into the left atrium. The click and murmur move closer to S₁ following maneuvers that decrease left ventricular volume (standing, Valsalva) and move oppositely upon increases in volume (leg raising, squatting). The intensity of the murmur augments with increases in systemic afterload (squatting, pressor agents) and decrease with vasodilation (amyl nitrite). Isometric exercise, which also delays the onset of the murmur, accentuates the intensity.

HOLOSYSTOLIC MURMURS

These murmurs, also termed *pansystolic murmurs*, begin with S₁ and continue through systole to S₂ ([Fig. 34-1B](#)). They are, with rare exception, indicative of atrioventricular valve regurgitation or of a ventricular septal defect; the differential diagnosis is shown in [Fig. 34-2](#). The murmur of mitral regurgitation is loudest at the left ventricular apex. Its radiation reflects the direction of the regurgitant jet. With a flail posterior mitral leaflet due to ruptured chordae tendineae, for example, the jet is directed anterosuperiorly, and the murmur radiates prominently to the base of the heart, where it might be confused with aortic valve stenosis unless the carotid upstrokes are carefully examined. Conversely, a flail anterior leaflet is associated with a posteriorly directed jet, which radiates into the axilla and the back. It may even strike the spine and be transmitted to the base of the neck. Severe mitral regurgitation is usually associated with a systolic thrill, a soft S₃, and a short diastolic rumbling murmur best appreciated in the left lateral decubitus position.

The holosystolic murmur of tricuspid regurgitation is generally softer (grades I to III/VI) than that of mitral regurgitation, is loudest at the left lower sternal edge, and increases in intensity upon inspiration. Associated signs include prominent "c-v" waves in the jugular venous pulse, systolic hepatic pulsations, and peripheral edema. Among the several causes of tricuspid regurgitation, annular dilatation from right ventricular enlargement in the setting of pulmonary artery hypertension is the most common.

Ventricular septal defect ([Chap. 234](#)) also produces a holosystolic murmur, the intensity of which varies inversely with the anatomic size of the defect. It is usually accompanied by a palpable thrill along the mid-left sternal border. The murmur of a ventricular septal defect is louder than that due to tricuspid regurgitation and does not share the latter's

inspiratory increase in intensity or associated peripheral signs.

DIASTOLIC HEART MURMURS

Like systolic murmurs, diastolic murmurs also can be subcategorized according to their time of onset.

EARLY DIASTOLIC MURMURS ([Fig. 34-1E](#))

Early diastolic murmurs result from semilunar valve incompetence and begin at the valve closure sound (A_2 or P_2), which reflects their site of origin. They are generally high pitched and decrescendo in configuration, especially in states of chronic regurgitation, in which their duration is a crude index of the severity of the lesion. The murmur of aortic regurgitation is generally, but not always, best heard in the second intercostal space at the left sternal edge. There is a tendency for the murmur associated with primary valvular pathology (e.g., rheumatic deformity, congenital bicuspid valve, endocarditis) to radiate more prominently along the *left* sternal border and to be well transmitted to the apex, while the murmur associated with primary aortic root pathology (e.g., annuloaortic ectasia, aortic dissection) radiates more often along the right sternal edge. It is occasionally necessary to examine the patient sitting forward in full expiration to appreciate the murmur, a maneuver that brings the aortic root closer to the anterior chest wall. Severe aortic regurgitation may be accompanied by a lower-pitched mid- to late-diastolic murmur at the apex (Austin Flint murmur), which is generally thought to reflect turbulence at the mitral inflow area from the mixing of the regurgitant (aortic) and forward (mitral) streams, and should be distinguished from mitral stenosis (see above). In the absence of significant heart failure, chronic severe aortic regurgitation is accompanied by several peripheral signs of significant diastolic runoff, including a wide systemic pulse pressure and water-hammer carotid upstrokes (Corrigan's pulse).

The murmur associated with *acute* aortic regurgitation is notably shorter in duration, lower pitched, and can be difficult to appreciate in the presence of tachycardia. Peripheral signs of significant diastolic runoff may be absent. These attributes reflect the abrupt rise in diastolic pressure within the noncompliant left ventricle, with a correspondingly rapid decline in the aortic diastolic-left ventricular pressure gradient.

The murmur of pulmonic valve regurgitation (Graham Steell murmur) begins with a loud (palpable) pulmonic closure sound (P_2) and is best heard in the pulmonic area with radiation along the left sternal border. Typically, it is high pitched, with a decrescendo quality, and is indicative of significant pulmonary artery hypertension with a diastolic pulmonary artery-right ventricular pressure gradient. Its increase in intensity upon inspiration is one means by which to distinguish it from aortic regurgitation. Signs of right ventricular pressure and volume overload are also usually present. With significant mitral stenosis, an early decrescendo diastolic murmur along the left sternal border is not uncommon and is almost always due to aortic rather than pulmonic regurgitation, despite the coexistence of pulmonary artery hypertension.

Pulmonic valve regurgitation in the absence of pulmonary artery hypertension can occur on a congenital basis and rarely with infective endocarditis. In these instances, the early diastolic murmur is softer and lower pitched than the classic Graham Steell murmur. It

begins at or even after P₂, which should be easily separable from A₂ and thus produce appreciation of an early diastolic pause.

MIDDIASTOLIC MURMURS

Middiastolic murmurs usually result from obstruction and/or augmented flow across the atrioventricular valves. The classic example is that of mitral stenosis due to rheumatic deformity ([Fig. 34-1F](#)). In the absence of extensive calcification, the first heart sound (S₁) is loud and the murmur begins after the opening snap; the time interval between S₂ and the opening snap is inversely related to the left atrial-left ventricular pressure gradient. The murmur is low pitched and best heard with the bell of the stethoscope over the apex, particularly in the left lateral decubitus position. While its intensity does not reflect the severity of the obstruction accurately, the duration of the murmur does provide some indication as to the magnitude of the obstruction. A longer murmur denotes persistence of a left atrioventricular pressure gradient over a greater proportion of the diastolic time interval. Presystolic accentuation of the murmur ([Fig. 34-1A](#)) is frequently appreciated in the presence of sinus rhythm and reflects a further increase in transmitral flow consequent to mechanical atrial systole.

The murmur associated with tricuspid stenosis shares many of these features, but it is best heard at the lower left sternal border and, like most right-sided events, increases in intensity upon inspiration. The observant examiner may discern a prolonged y descent in the jugular venous pulse. Signs of right heart failure may predominate.

There are several other causes of mid-diastolic murmurs that are important to distinguish from mitral stenosis. *Left atrial myxomas* ([Chap. 240](#)) may masquerade as mitral stenosis, but the diastolic murmur is not accompanied by an opening snap or pre-systolic accentuation. Augmented flow across the mitral valve in diastole, such as occurs with severe mitral regurgitation or with large left to right intra-cardiac (ventricular septal defect) or great vessel (patent ductus arteriosus) shunts may produce a short, low pitched mid-diastolic apical murmur. The murmur usually follows a soft S₃ that is lower pitched and later in timing than the opening snap ([Fig. 34-1G](#)). Severe tricuspid regurgitation can also result in enhanced diastolic tricuspid flow and produce a right-sided filling complex similar to that which accompanies severe mitral regurgitation. The Austin Flint murmur of severe aortic regurgitation has been previously described and occurs in the presence of chronic severe aortic regurgitation.

CONTINUOUS MURMURS

Continuous murmurs begin in systole, peak near S₂, and continue into all or part of diastole ([Fig. 34-1H](#)). Accordingly, they reflect the persistence of flow between two chambers during both phases of the cardiac cycle. The differential diagnosis of continuous murmurs is shown in [Table 34-1](#). Two innocent variants are the cervical venous hum and the mammary souffle. The former is audible in healthy children and young adults in the right supraclavicular fossa and can be abolished by compression over the internal jugular vein. Its diastolic component may be louder than its systolic counterpart. A mammary souffle represents augmented arterial flow through engorged breasts and becomes audible during the late third trimester of pregnancy or in the early postpartum period. Firm pressure with the diaphragm of the stethoscope can eliminate

the diastolic portion of the murmur. The murmur dissipates with time after delivery.

The classic continuous murmur is that due to a patent ductus arteriosus. It is best heard at or just above and to the left of the pulmonic area and may be audible in the back. Over time, a large uncorrected shunt may lead to elevation of the pulmonary vascular resistance, with resultant pulmonary artery hypertension and diminution or elimination of the diastolic component. A continuous murmur can also signify a ruptured congenital sinus of Valsalva aneurysm, which occurs either spontaneously or as a complication of infective endocarditis. Here, a high-pressure fistula is created between the aorta and a cardiac chamber, usually the right atrium or ventricle. The murmur is loudest along the right or left sternal border and is frequently accompanied by a thrill. Notably, the diastolic component is louder than the systolic component. It can be difficult to distinguish continuous murmurs from the temporally separate systolic and diastolic murmurs of mixed aortic valve disease or isolated severe aortic regurgitation. The emphasis is on the envelopment of S₂ by continuous murmurs and a gap between the to-and-fro murmurs of aortic valve disease.

A variety of other lesions can result in continuous murmurs. A coronary arteriovenous fistula sometimes produces a faint, continuous murmur with a louder diastolic component at the left sternal border or left ventricular apex. Severe atherosclerotic disease of a major systemic artery may produce a continuous bruit, the presence of which signifies very high-grade obstruction. Patients with peripheral pulmonary (branch) stenosis or with pulmonary atresia with extensive bronchial collaterals may also have continuous murmurs best heard in the back or along the lateral thoracic cage. Similar findings are present in patients with severe aortic coarctation, a lesion that should be identifiable on the basis of weak and delayed lower extremity pulses and upper extremity hypertension. The continuous murmurs emanate from the enlarged collateral (intercostal) arteries.

Approach to the Patient

It is widely recognized that, despite the importance placed on them by medical schools and training program directors, the auscultatory skills of medical students and residents have declined considerably since the advent of Doppler echocardiography. Recent surveys indicate that trainees fail to correctly identify up to 80% of adventitious sounds and murmurs. Diagnostic errors are as frequent among third-year medical residents as they are for first-year residents. Few training programs provide a dedicated educational curriculum for cardiac auscultation. In the aggregate, these deficiencies lead to an over-reliance on the use of echocardiography and increase the costs of evaluating patients with heart murmurs.

In many patients the cause of a heart murmur can be readily elucidated from careful assessment of the murmur itself, as described in this chapter, when considered in the light of the history, general physical examination, and other features of the cardiac examination, as described in [Chap. 225](#). When the diagnosis is in doubt, or when additional pathoanatomic and physiologic data are necessary in assessing the patient and planning treatment, transthoracic Doppler echocardiography is of great value in identifying not only the etiology of the murmur but also the severity of the responsible lesion ([Fig. 34-3](#)).

The majority of heart murmurs are midsystolic and soft (Grades I to II/VI). When such a murmur occurs in an asymptomatic child or young adult *without* other evidence of heart disease on clinical examination, it is usually benign and echocardiography is not generally required. On the other hand, echocardiographic examination is indicated in patients with loud systolic murmurs (³III/VI), especially those that are holosystolic or late systolic, in most patients with diastolic or continuous murmurs, and in patients with additional unexplained abnormal physical findings on cardiac examination.

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35. APPROACH TO THE PATIENT WITH HYPERTENSION - Gordon H. Williams

DEFINITION

Since there is no dividing line between normal and high blood pressure, arbitrary levels have been established to define persons who have an increased risk of developing a morbid cardiovascular event and/or will benefit from medical therapy. These definitions should take into account not only the level of diastolic pressure but also systolic pressure, age, sex, race, and concomitant diseases. For example, patients with a diastolic pressure >90 mmHg have a significant reduction in morbidity and mortality rate if they receive adequate therapy. These, then, are patients who have hypertension and who should be considered for treatment.

The level of *systolic* pressure is also important in assessing the influence of arterial pressure on cardiovascular morbidity. Some data suggest that it may be more important than diastolic pressure. For example, males with normal diastolic pressures (<82 mmHg) but elevated systolic pressures (>158 mmHg) have a cardiovascular mortality rate 2.5 times higher than individuals who have similar diastolic pressures but whose systolic pressures clearly are normal (<130 mmHg). A reduction in mortality and morbidity with treatment, specifically in the elderly, has been documented in these patients. This beneficial effect results mainly from a reduction in strokes and occurs in women as well. Other significant demographic factors that modify the influence of blood pressure on the frequency of morbid cardiovascular events are age, race, and sex, with young black males being most adversely affected by hypertension.

When hypertension is suspected, blood pressure should be measured at least twice during two separate examinations after the initial screening. In adults, a *diastolic* pressure below 85 mmHg is considered to be normal; one between 85 and 89 mmHg is high normal; one of 90 to 99 mmHg represents stage 1 or mild hypertension; one of 100 to 109 mmHg represents stage 2 or moderate hypertension; and one of ≥ 110 mmHg represents stage 3 or severe hypertension. A *systolic* pressure below 130 mmHg indicates normal blood pressure; one between 130 and 139 mmHg indicates high normal; one between 140 and 159 mmHg indicates stage 1 or mild hypertension; one between 160 and 179 mmHg indicates stage 2 or moderate hypertension; and one ≥ 180 mmHg indicates stage 3 or severe hypertension. Increasing use of 12- or 24-h blood pressure monitoring may provide additional useful information in patients who are difficult to classify. However, normal values for this procedure and its usefulness in relation to therapeutic outcomes are not currently known. A useful classification of hypertension derived from the Joint Committee on Detection, Evaluation, and Treatment of High Blood Pressure is shown in [Table 35-1](#).

Arterial pressure fluctuates in most persons, whether they are normotensive or hypertensive. Patients who are classified as having *labile hypertension* are those who sometimes, but not always, have arterial pressures in the hypertensive range. These patients are often considered to have borderline hypertension.

Sustained hypertension can become accelerated or enter a malignant phase, although that is unusual in treated patients. Though a patient with *malignant hypertension* often has a blood pressure above 200/140, the condition is defined by the presence of

papilledema, usually accompanied by retinal hemorrhages and exudates, rather than by the absolute pressure level. *Accelerated hypertension* is defined as a significant recent increase over previous hypertensive levels associated with evidence of vascular damage on fundusoscopic examination but without papilledema.

PATIENT EVALUATION

In evaluating patients with hypertension, the initial history, physical examination, and laboratory tests should be directed at (1) uncovering correctable secondary forms of hypertension ([Chap. 246](#)), (2) establishing a pretreatment baseline, (3) assessing factors that may influence the type of therapy or be changed adversely by therapy, (4) determining if target organ damage is present, and (5) determining whether other risk factors for the development of arteriosclerotic cardiovascular disease are present ([Chap. 241](#)). Ideally, this evaluation would also determine the underlying mechanism(s) in essential hypertension, particularly if such information leads to a more specific therapeutic program. Unfortunately, at present this aspect of the evaluation is limited by lack of knowledge of some of the underlying mechanisms, by uncertainty as to the correct treatment for a distinct subset even if the underlying mechanisms are known, or by the prohibitive cost of defining a subset of hypertensive patients even if specific therapy were available. However, with the accumulation of additional information, this sixth component of the evaluation of patients with hypertension may become increasingly important.

Symptoms and Signs Most patients with hypertension have no specific symptoms referable to their blood pressure elevation and are identified only in the course of a physical examination. When symptoms do bring the patient to the physician, they fall into three categories. They are related to (1) the elevated pressure itself, (2) the hypertensive vascular disease, and (3) the underlying disease, in the case of secondary hypertension. Though popularly considered a symptom of elevated arterial pressure, headache is characteristic only of severe hypertension; most commonly such headaches are localized to the occipital region and are present when the patient awakens in the morning but subside spontaneously after several hours. Other complaints that may be related to elevated blood pressure include dizziness, palpitations, easy fatigability, and impotence. Complaints referable to vascular disease include epistaxis, hematuria, blurring of vision owing to retinal changes, episodes of weakness or dizziness due to transient cerebral ischemia, angina pectoris, and dyspnea due to cardiac failure. Pain due to dissection of the aorta or to a leaking aneurysm is an occasional presenting symptom.

Examples of symptoms related to the underlying disease in secondary hypertension are polyuria, polydipsia, and muscle weakness secondary to hypokalemia in patients with primary aldosteronism or weight gain, and emotional lability in patients with Cushing's syndrome. The patient with a pheochromocytoma may present with episodic headaches, palpitations, diaphoresis, and postural dizziness.

History A strong family history of hypertension, along with the reported finding of intermittent pressure elevation in the past, favors the diagnosis of essential hypertension. Secondary hypertension often develops before the age of 35 or after 55. A history of use of adrenal steroids or estrogens is of obvious significance. A history of

repeated urinary tract infections suggests chronic pyelonephritis, although this condition may occur in the absence of symptoms; nocturia and polydipsia suggest renal or endocrine disease, while trauma to either flank or an episode of acute flank pain may be a clue to the presence of renal injury. A history of weight gain is compatible with Cushing's syndrome, and one of weight loss is compatible with pheochromocytoma. A number of aspects of the history aid in determining whether vascular disease has progressed to a dangerous stage. These include angina pectoris and symptoms of cerebrovascular insufficiency, congestive heart failure, and/or peripheral vascular insufficiency. Other risk factors that should be asked about include cigarette smoking, diabetes mellitus, lipid disorders, and a family history of early deaths due to cardiovascular disease. Finally, aspects of the patient's lifestyle that could contribute to the hypertension or affect its treatment should be assessed, including diet, physical activity, family status, work, and educational level.

Physical Examination The physical examination starts with the patient's general appearance. For instance, are the round face and truncal obesity of Cushing's syndrome present? Is muscular development in the upper extremities out of proportion to that in the lower extremities, suggesting coarctation of the aorta? The next step is to compare the blood pressures and pulses in the two upper extremities and in the supine and standing positions (for at least 2 min). A rise in diastolic pressure when the patient goes from the supine to the standing position is most compatible with essential hypertension; a fall, in the absence of antihypertensive medications, suggests secondary forms of hypertension. The patient's height and weight should be recorded. Detailed examination of the ocular fundi is mandatory, as funduscopic findings provide one of the best indications of the duration of hypertension and of prognosis. A useful guide is the Keith-Wagener-Barker classification of funduscopic changes ([Table 35-2](#)); the specific changes in each fundus should be recorded and a grade assigned. Palpation and auscultation of the carotid arteries for evidence of stenosis or occlusion are important; narrowing of a carotid artery may be a manifestation of hypertensive vascular disease, and it may also be a clue to the presence of a renal arterial lesion, since these two lesions may occur together. In examination of the heart and lungs, evidence of left ventricular hypertrophy and cardiac decompensation should be sought. Is there a left ventricular lift? Are third and fourth heart sounds present? Are there pulmonary rales? A third heart sound and pulmonary rales are unusual in uncomplicated hypertension. Their presence suggests ventricular dysfunction. Chest examination also includes a search for extracardiac murmurs and palpable collateral vessels that may result from coarctation of the aorta.

The most important part of the abdominal examination is auscultation for bruits originating in stenotic renal arteries. Bruits due to renal arterial narrowing nearly always have a diastolic component or may be continuous and are best heard just to the right or left of the midline above the umbilicus or in the flanks; they are present in many patients with renal artery stenosis due to fibrous dysplasia and in 40 to 50% of those with functionally significant stenosis due to arteriosclerosis. The abdomen should also be palpated for an abdominal aneurysm and for the enlarged kidneys of polycystic renal disease. The femoral pulses must be carefully felt, and, if they are decreased and/or delayed in comparison with the radial pulse, the blood pressure in the lower extremities must be measured. Even if the femoral pulse is normal to palpation, arterial pressure in the lower extremities should be recorded at least once in patients in whom hypertension

is discovered before the age of 30 years. Finally, examination of the extremities for edema and a search for evidence of a previous cerebrovascular accident and/or other intracranial pathology should be performed.

Laboratory Investigation There is controversy as to what laboratory studies should be performed in patients presenting with hypertension. In general, the disagreement centers on how extensively the patient should be evaluated for secondary forms of hypertension or subsets of essential hypertension. The *basic* laboratory studies that should be performed in all patients with sustained hypertension are described in ([Table 35-3](#)). **The secondary studies that should be added if (1) the initial evaluation indicates a form of secondary hypertension and/or (2) arterial pressure is not controlled after initial therapy are discussed in Chap. 246.*

Renal status is evaluated by assessing the presence of protein, blood, and glucose in the urine and measuring serum creatinine and/or blood urea nitrogen. Microscopic examination of the urine is also helpful. The serum potassium level should be measured both as a screen for mineralocorticoid-induced hypertension and to provide a baseline before diuretic therapy is begun. A blood glucose determination is helpful both because diabetes mellitus may be associated with accelerated arteriosclerosis, renal vascular disease, and diabetic nephropathy in patients with hypertension and because primary aldosteronism, Cushing's syndrome, and pheochromocytoma all may be associated with hyperglycemia. Furthermore, since antihypertensive therapy with diuretics, for example, can raise the blood glucose level, it is important to establish a baseline. The possibility of hypercalcemia may also be investigated. Serum cholesterol, high-density lipoprotein cholesterol, and triglyceride levels identify other factors that predispose to the development of arteriosclerosis.

An electrocardiogram should be obtained in all cases to permit assessment of cardiac status, particularly if left ventricular hypertrophy is present, and to provide a baseline. The echocardiogram is more sensitive than either the electrocardiogram or physical examination in determining whether cardiac hypertrophy is present. However, a complete, detailed echocardiographic study is expensive. Thus, in some circumstances, a cheaper, limited echocardiogram may be a useful addition to the *baseline* evaluation of a hypertensive patient, particularly as left ventricular hypertrophy is an independent cardiovascular risk factor and its presence suggests the need for vigorous antihypertensive therapy. Furthermore, while a substantial increase in arterial pressure usually correlates with the presence of left ventricular hypertrophy, a mild increase may not. Thus, one cannot use the blood pressure as a surrogate marker for the presence or absence of left ventricular hypertrophy. On the other hand, because of the cost of an echocardiogram and the uncertainty as to whether the resultant information would modify therapy, it is unclear that routine *follow-up* echocardiograms during therapy are justified. Furthermore, there are no data to suggest that reversal of left ventricular hypertrophy produces benefits beyond that conferred by blood pressure reduction. The chest roentgenogram may also be helpful by providing the opportunity to identify aortic dilation or elongation and the rib notching that occurs in coarctation of the aorta.

Certain clues from the history, physical examination, and basic laboratory studies may suggest an unusual cause for the hypertension and dictate the need for special studies as outlined in [Chap 246](#).

TREATMENT

See [Chap. 246](#).

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36. HYPOXIA AND CYANOSIS - Eugene Braunwald

HYPOXIA

The fundamental purpose of the cardiorespiratory system is to deliver O_2 (and substrates) to the cells and to remove CO_2 (and other metabolic products) from them. Proper maintenance of this function depends on intact cardiovascular and respiratory systems and a supply of inspired gas containing adequate O_2 . When hypoxia occurs consequent to respiratory failure, P_{aCO_2} usually rises ([Chap. 250](#)), and the hemoglobin-oxygen ($Hb-O_2$) dissociation curve (see [Fig. 106-2](#)) is displaced to the right. Under these conditions, the P_{aO_2} declines. Arterial hypoxemia, i.e., a reduction of O_2 saturation of arterial blood (S_{aO_2}), and consequent cyanosis are likely to be more marked when such depression of P_{aO_2} results from pulmonary disease than when the depression occurs as the result of a decline in the fraction of oxygen in inspired air (F_{IO_2}). In this situation P_{aCO_2} falls secondary to anoxia-induced hyperventilation and the $Hb-O_2$ dissociation curve is displaced to the left, limiting the decline in S_{aO_2} .

CAUSES OF HYPOXIA

Anemic Hypoxia Any reduction in the hemoglobin concentration of the blood is attended by a corresponding decline in the O_2 -carrying capacity of the blood. In anemic hypoxia, the P_{aO_2} is normal; but as a consequence of the reduction of the hemoglobin concentration, the absolute quantity of O_2 transported per unit volume of blood is diminished. As the anemic blood passes through the capillaries and the usual quantity of O_2 is removed from it, the P_{O_2} in the venous blood declines to a greater degree than would normally be the case.

Carbon Monoxide Intoxication (See also [Chap. 396](#)) Hemoglobin that is combined with carbon monoxide (carboxyhemoglobin, COHb) is unavailable for O_2 transport. In addition, the presence of COHb shifts the $Hb-O_2$ dissociation curve to the left (see [Fig. 106-2](#)) so that O_2 is unloaded only at lower tensions. By such formation of COHb, a given degree of reduction in O_2 -carrying power produces a far greater degree of tissue hypoxia than the equivalent reduction in hemoglobin due to simple anemia.

Respiratory Hypoxia Arterial unsaturation is a common finding in advanced pulmonary disease. The most common cause of respiratory hypoxia is ventilation-perfusion mismatch, which results from perfusion of poorly ventilated alveoli. As discussed in [Chap. 250](#), it may also be caused by hypoventilation, and it is then associated with an elevation of P_{aCO_2} . These two forms of respiratory hypoxia may be recognized because they are usually correctable by inspiring 100% O_2 for several minutes. A third cause is shunting of blood across the lung from right to left by perfusion of nonventilated portions of the lung, as in pulmonary atelectasis or through arteriovenous connections in the lung. The low P_{aO_2} in this situation is correctable only in part by an F_{IO_2} of 100%.

Hypoxia Secondary to High Altitude As one ascends rapidly to 3000 m (approximately 10,000 ft), the alveolar P_{O_2} declines to about 60 mmHg, and impaired memory and other cerebral symptoms of hypoxia may develop. At higher altitudes, arterial saturation declines rapidly and symptoms become more serious; and at 5000 m (approximately 15,000 ft) unacclimatized individuals usually cease to be able to function

normally.

Hypoxia Secondary to Right-to-Left Extrapulmonary Shunting From a physiologic viewpoint, this cause of hypoxia resembles intrapulmonary right-to-left shunting but is caused by congenital cardiac malformations such as tetralogy of Fallot, transposition of the great arteries, and Eisenmenger's syndrome ([Chap. 234](#)). As in pulmonary right-to-left shunting, the P_{aO_2} cannot be restored to normal with inspiration of 100% O_2 .

Circulatory Hypoxia As in anemic hypoxia, the P_{aO_2} is usually normal, but venous and tissue P_{O_2} values are reduced as a consequence of reduced tissue perfusion. Generalized circulatory hypoxia occurs in heart failure ([Chap. 232](#)) and in most forms of shock ([Chap. 38](#)).

Specific Organ Hypoxia Decreased perfusion of any organ resulting in localized circulatory hypoxia may occur secondary to organic arterial obstruction or as a consequence of vasoconstriction ([Chap. 248](#)). The latter is seen in the upper extremities in Raynaud's phenomenon. Ischemic hypoxia with accompanying pallor occurs in organic arterial obliterative disease. Localized hypoxia also may result from venous obstruction and the resultant congestion and reduced arterial inflow. Edema, which increases the distance through which O_2 diffuses before it reaches cells, also can cause localized hypoxia. In an attempt to maintain adequate perfusion to more vital organs, constriction may reduce perfusion in all limbs in patients with heart failure or hypovolemic shock.

Increased O_2 Requirements If the O_2 consumption of the tissues is elevated without a corresponding increase in perfusion, tissue hypoxia ensues and the P_{O_2} in venous blood becomes reduced. Ordinarily, the clinical picture of patients with hypoxia due to an elevated metabolic rate is quite different from that in other types of hypoxia; the skin is warm and flushed, owing to increased cutaneous blood flow that dissipates the excessive heat produced, and cyanosis is usually absent.

Exercise is a classic example of increased tissue O_2 requirements. These increased demands are normally met by several mechanisms operating simultaneously: (1) increasing the cardiac output and ventilation and thus O_2 delivery to the tissues; (2) preferentially directing the blood to the exercising muscles by changing vascular resistances in various circulatory beds, directly and/or reflexly; (3) increasing O_2 extraction from the delivered blood and widening the arteriovenous O_2 difference; and (4) reducing the pH of the tissues and capillary blood, thereby unloading more O_2 from hemoglobin. If the capacity of these mechanisms is exceeded, then hypoxia, especially of the exercising muscles, will result.

Improper Oxygen Utilization Cyanide ([Chap. 396](#)) and several other similarly acting poisons cause cellular hypoxia. The tissues are unable to utilize O_2 , and as a consequence, the venous blood tends to have a high O_2 tension. This condition has been termed *histotoxic hypoxia*.

EFFECTS OF HYPOXIA

Changes in the central nervous system, particularly the higher centers, are especially

important consequences of hypoxia. Acute hypoxia causes impaired judgment, motor incoordination, and a clinical picture closely resembling that of acute alcoholism. When hypoxia is long-standing, fatigue, drowsiness, apathy, inattentiveness, delayed reaction time, and reduced work capacity occur. As hypoxia becomes more severe, the centers of the brainstem are affected, and death usually results from respiratory failure. With the reduction of P_{aO_2} , cerebrovascular resistance decreases and cerebral blood flow increases, increasing O_2 delivery to the brain as a compensatory mechanism. However, when the reduction of P_{aO_2} is accompanied by hyperventilation and a reduction of P_{aCO_2} , cerebrovascular resistance rises, cerebral blood flow falls, and hypoxia is intensified. Hypoxia also causes pulmonary arterial constriction, which shunts blood away from poorly ventilated areas toward better-ventilated portions of the lung. However, it also increases pulmonary vascular resistance and right ventricular afterload.

Glucose is normally broken down to pyruvic acid. However, the further breakdown of pyruvate and the generation of adenosine triphosphate (ATP) consequent to it require O_2 ; and in the presence of hypoxia increasing proportions of pyruvate are reduced to lactic acid, which cannot be broken down further, causing metabolic acidosis. Under these circumstances, the total energy obtained from the breakdown of carbohydrate is greatly reduced, and the quantity of energy available for the production of ATP becomes inadequate.

An important component of the respiratory response to hypoxia originates in special chemosensitive cells in the carotid and aortic bodies, and in the respiratory center in the brainstem. The stimulation of these cells by hypoxia increases ventilation, with a loss of CO_2 , and leads to respiratory alkalosis. When combined with the metabolic acidosis resulting from the production of lactic acid, the serum bicarbonate level declines ([Chap. 50](#)).

Diminished P_{O_2} in any tissue results in local vasodilatation, and the diffuse vasodilatation that occurs in generalized hypoxia raises the cardiac output. In patients with underlying heart disease, the requirements of the peripheral tissues for an increase of cardiac output with hypoxia may precipitate congestive heart failure. In patients with ischemic heart disease, a reduced P_{aO_2} may intensify myocardial ischemia and further impair left ventricular function.

One of the important mechanisms of compensation for chronic hypoxia is an increase in the hemoglobin concentration and in the number of red blood cells in the circulating blood, i.e., the development of polycythemia secondary to erythropoietin production ([Chap. 110](#)).

CYANOSIS

Cyanosis refers to a bluish color of the skin and mucous membranes resulting from an increased quantity of reduced hemoglobin, or of hemoglobin derivatives, in the small blood vessels of those areas. It is usually most marked in the lips, nail beds, ears, and malar eminences. Cyanosis, especially if developed recently, is more commonly detected by a family member than the patient. The florid skin characteristic of polycythemia vera ([Chap. 110](#)) must be distinguished from the true cyanosis discussed here. A cherry-colored flush, rather than cyanosis, is caused by COHb ([Chap. 396](#)). The

degree of cyanosis is modified by the color of the cutaneous pigment and the thickness of the skin, as well as by the state of the cutaneous capillaries. The accurate clinical detection of the presence and degree of cyanosis is difficult, as proved by oximetric studies. In some instances, central cyanosis can be detected reliably when the SaO_2 has fallen to 85%; in others, particularly in dark-skinned persons, it may not be detected until it has declined to 75%. In the latter case, examination of the mucous membranes in the oral cavity and the conjunctivae rather than examination of the skin is more helpful in the detection of cyanosis.

The increase in the quantity of reduced hemoglobin in the mucocutaneous vessels that produces cyanosis may be brought about either by an increase in the quantity of venous blood as the result of dilatation of the venules and venous ends of the capillaries or by a reduction in the SaO_2 in the capillary blood. In general, cyanosis becomes apparent when the mean capillary concentration of reduced hemoglobin exceeds 40 g/L (4 g/dL). It is the *absolute* rather than the *relative* quantity of reduced hemoglobin that is important in producing cyanosis. Thus, in a patient with severe anemia, the relative amount of reduced hemoglobin in the venous blood may be very large when considered in relation to the total amount of hemoglobin in the blood. However, since the concentration of the latter is markedly reduced, the *absolute* quantity of reduced hemoglobin may still be small, and therefore patients with severe anemia and even *marked* arterial desaturation do not display cyanosis. Conversely, the higher the total hemoglobin content, the greater is the tendency toward cyanosis; thus, patients with marked polycythemia tend to be cyanotic at higher levels of SaO_2 than patients with normal hematocrit values. Likewise, local passive congestion, which causes an increase in the total amount of reduced hemoglobin in the vessels in a given area, may cause cyanosis. Cyanosis also is observed when nonfunctional hemoglobin such as methemoglobin or sulfhemoglobin ([Chap. 106](#)) is present in blood.

Cyanosis may be subdivided into central and peripheral types. In the *central* type, the SaO_2 is reduced or an abnormal hemoglobin derivative is present, and the mucous membranes and skin are both affected. *Peripheral* cyanosis is due to a slowing of blood flow and abnormally great extraction of O_2 from normally saturated arterial blood. It results from vasoconstriction and diminished peripheral blood flow, such as occurs in cold exposure, shock, congestive failure, and peripheral vascular disease. Often in these conditions the mucous membranes of the oral cavity or those beneath the tongue may be spared. Clinical differentiation between central and peripheral cyanosis may not always be simple, and in conditions such as cardiogenic shock with pulmonary edema there may be a mixture of both types.

DIFFERENTIAL DIAGNOSIS

Central Cyanosis ([Table 36-1](#)) Decreased SaO_2 results from a marked reduction in the PaO_2 . This reduction may be brought about by a decline in the FI_{O_2} without sufficient compensatory alveolar hyperventilation to maintain alveolar P_{O_2} . Cyanosis does not occur to a significant degree in an ascent to an altitude of 2500 m (8000 ft) but is marked in a further ascent to 5000 m (16,000 ft). The reason for this difference becomes clear on studying the S shape of the Hb- O_2 dissociation curve (see [Fig. 106-1](#)). At 2500 m (8000 ft) the FI_{O_2} is about 120 mmHg, the alveolar P_{O_2} is approximately 80 mmHg, and the SaO_2 is nearly normal. However, at 5000 m (16,000 ft) the FI_{O_2} and

alveolar P_{O_2} are about 85 and 50 mmHg, respectively, and the Sa_{O_2} is only about 75%. This leaves 25% of the hemoglobin in the arterial blood in the reduced form, an amount likely to be associated with cyanosis in the absence of anemia. Similarly, a mutant hemoglobin with a low affinity for O_2 (e.g., Hb Kansas) causes lowered Sa_{O_2} saturation and resultant central cyanosis ([Chap. 106](#)).

Seriously *impaired pulmonary function*, through perfusion of unventilated or poorly ventilated areas of the lung or alveolar hypoventilation, is a common cause of central cyanosis ([Chap. 250](#)). This condition may occur acutely, as in extensive pneumonia or pulmonary edema, or chronically with chronic pulmonary diseases (e.g., emphysema). In the last situation, secondary polycythemia is generally present, and clubbing of the fingers may occur. However, in many types of chronic pulmonary disease with fibrosis and obliteration of the capillary vascular bed, cyanosis does not occur because there is relatively little perfusion of underventilated areas.

Another cause of reduced Sa_{O_2} is *shunting of systemic venous blood into the arterial circuit*. Certain forms of congenital heart disease are associated with cyanosis ([Chap. 234](#)). Since blood flows from a higher-pressure to a lower-pressure region, for a cardiac defect to result in a right-to-left shunt, it must ordinarily be combined with an obstructive lesion distal to the defect or with elevated pulmonary vascular resistance. The most common congenital cardiac lesion associated with cyanosis in the adult is the combination of ventricular septal defect and pulmonary outflow tract obstruction (*tetralogy of Fallot*). The more severe the obstruction, the greater the degree of right-to-left shunting and resultant cyanosis. In patients with patent ductus arteriosus, pulmonary hypertension, and right-to-left shunt, *differential cyanosis* results; that is, cyanosis occurs in the lower but not in the upper extremities. *The mechanisms for the elevated pulmonary vascular resistance that may produce cyanosis in the presence of intra- and extracardiac communications without pulmonic stenosis (Eisenmenger syndrome) are discussed in Chap. 234.*

Pulmonary arteriovenous fistulae ([Chap. 57](#)) may be congenital or acquired, solitary or multiple, microscopic or massive. The severity of cyanosis produced by these fistulae depends on their size and number. They occur with some frequency in hereditary hemorrhagic telangiectasia. Sa_{O_2} reduction and cyanosis may also occur in some patients with cirrhosis, presumably as a consequence of pulmonary arteriovenous fistulas or portal vein-pulmonary vein anastomoses.

In patients with cardiac or pulmonary right-to-left shunts, the presence and severity of cyanosis depend on the size of the shunt relative to the systemic flow as well as on the Hb- O_2 saturation of the venous blood. With increased extraction of O_2 from the blood by the exercising muscles, the venous blood returning to the right side of the heart is more unsaturated than at rest, and shunting of this blood or its passage through lungs incapable of normal oxygenation intensifies the cyanosis. Also, since the systemic vascular resistance falls with exercise, the right-to-left shunt is augmented by exercise in patients with congenital heart disease and communications between the two sides of the heart. Secondary polycythemia occurs frequently in patients with arterial O_2 unsaturation and contributes to the cyanosis.

Cyanosis can be caused by small amounts of circulating methemoglobin and by even

smaller amounts of sulfhemoglobin ([Chap. 106](#)). Although they are uncommon causes of cyanosis, these abnormal hemoglobin pigments should be sought by spectroscopy when cyanosis is not readily explained by malfunction of the circulatory or respiratory systems. Generally, digital clubbing does not occur with them. The diagnosis of methemoglobinemia can be suspected if the patient's blood remains brown after being mixed in a test tube and exposed to air.

Peripheral Cyanosis Probably the most common cause of peripheral cyanosis is the normal vasoconstriction resulting from exposure to cold air or water. When cardiac output is low, as in severe congestive heart failure or shock, cutaneous vasoconstriction occurs as a compensatory mechanism so that blood is diverted from the skin to more vital areas such as the central nervous system and heart ([Chap. 232](#)), and intense cyanosis associated with cool extremities may result. Even though the arterial blood is normally saturated, the reduced volume flow through the skin and the reduced P_{O_2} at the venous end of the capillary result in cyanosis.

Arterial obstruction to an extremity, as with an embolus, or arteriolar constriction, as in cold-induced vasospasm (Raynaud's phenomenon, [Chap. 248](#)), generally results in pallor and coldness, but there may be associated cyanosis. Venous obstruction, as in thrombophlebitis, dilates the subpapillary venous plexuses and thereby intensifies cyanosis.

Approach to the Patient

Certain features are important in arriving at the cause of cyanosis:

1. The history, particularly the onset (cyanosis present since birth is usually due to congenital heart disease), and possible exposure to drugs or chemicals that may produce abnormal types of hemoglobin.
2. Clinical differentiation of central as opposed to peripheral cyanosis. Objective evidence by physical or radiographic examination of disorders of the respiratory or cardiovascular systems. Massage or gentle warming of a cyanotic extremity will increase peripheral blood flow and abolish peripheral but not central cyanosis.
3. The presence or absence of clubbing of the digits (see below). Clubbing without cyanosis is frequent in patients with infective endocarditis and ulcerative colitis; it may occasionally occur in healthy persons, and in some instances it may be occupational, e.g., in jackhammer operators. The combination of cyanosis and clubbing is frequent in patients with congenital heart disease and right-to-left shunting and is seen occasionally in persons with pulmonary disease such as lung abscess or pulmonary arteriovenous fistulae. In contrast, peripheral cyanosis or acutely developing central cyanosis is *not* associated with clubbed digits.
4. Determination of P_{aO_2} tension and S_{aO_2} and spectroscopic and other examinations of the blood for abnormal types of hemoglobin (critical in the differential diagnosis of cyanosis).

CLUBBING

The selective bullous enlargement of the distal segments of the fingers and toes due to proliferation of connective tissue, particularly on the dorsal surface, is termed *clubbing*; there is increased sponginess of the soft tissue at the base of the nail. Clubbing may be hereditary, idiopathic, or acquired and associated with a variety of disorders, including cyanotic congenital heart disease, infective endocarditis, and a variety of pulmonary conditions (among them primary and metastatic lung cancer, bronchiectasis, lung abscess, cystic fibrosis, and mesothelioma), as well as with some gastrointestinal diseases (including regional enteritis, chronic ulcerative colitis, and hepatic cirrhosis).

Clubbing in patients with primary and metastatic lung cancer, mesothelioma, bronchiectasis, and hepatic cirrhosis may be associated with *hypertrophic osteoarthropathy*. In this condition, the subperiosteal formation of new bone in the distal diaphyses of the long bones of the extremities causes pain and symmetric arthritis-like changes in the shoulders, knees, ankles, wrists, and elbows. The diagnosis of hypertrophic osteoarthropathy may be confirmed by bone radiographs. Although the mechanism of clubbing is unclear, it appears to be secondary to a humoral substance that causes dilation of the vessels of the fingertip.

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37. EDEMA - Eugene Braunwald

Edema is defined as a clinically apparent increase in the interstitial fluid volume, which may expand by several liters before the abnormality is evident. Therefore, a weight gain of several kilograms usually precedes overt manifestations of edema, and a similar weight loss from diuresis can be induced in a slightly edematous patient before "dry weight" is achieved. *Ascites* ([Chap. 46](#)) and *hydrothorax* refer to accumulation of excess fluid in the peritoneal and pleural cavities, respectively, and are considered to be special forms of edema. *Anasarca* refers to gross, generalized edema.

Depending on its cause and mechanism, edema may be localized or have a generalized distribution; it is recognized in its generalized form by puffiness of the face, which is most readily apparent in the periorbital areas, and by the persistence of an indentation of the skin following pressure; this is known as "pitting" edema. In its more subtle form, it may be detected by noting that after the stethoscope is removed from the chest wall, the rim of the bell leaves an indentation on the skin of the chest for a few minutes. When the ring on a finger fits more snugly than in the past or when a patient complains of difficulty in putting on shoes, particularly in the evening, edema may be present.

PATHOGENESIS

About one-third of the total-body water is confined to the extracellular space. Approximately 25% of the latter, in turn, is composed of the plasma volume, and the remainder is interstitial fluid.

Starling Forces The forces that regulate the disposition of fluid between these two components of the extracellular compartment are frequently referred to as the *Starling forces* (see p. 202). The hydrostatic pressure within the vascular system and the colloid oncotic pressure in the interstitial fluid tend to promote movement of fluid from the vascular to the extravascular space. In contrast, the colloid oncotic pressure contributed by the plasma proteins and the hydrostatic pressure within the interstitial fluid, referred to as the *tissue tension*, promote the movement of fluid into the vascular compartment. Consequently there is a movement of water and diffusible solutes from the vascular space at the arteriolar end of the capillaries.

Fluid is returned from the interstitial space into the vascular system at the venous end of the capillary and by way of the lymphatics, and unless these channels are obstructed, lymph flow tends to increase with increases in net movement of fluid from the vascular compartment to the interstitium. These flows are usually balanced so that a steady state exists in the sizes of the intravascular and interstitial compartments, and yet a large exchange between them occurs. However, should any one of the hydrostatic or oncotic pressure gradients be altered significantly, a further net movement of fluid between the two components of the extracellular space will take place. The development of edema then depends on one or more alterations in the Starling forces so that there is increased flow of fluid from the vascular system into the interstitium or into a body cavity.

Edema due to increase in capillary pressure may result from an elevation of venous pressure due to obstruction in venous drainage. This increase in capillary pressure may be generalized, as occurs in congestive heart failure. The Starling forces may be

imbalanced when the colloid oncotic pressure of the plasma is reduced, owing to any factor that may induce hypoalbuminemia, such as saline expansion, malnutrition, liver disease, loss of protein into the urine or into the gastrointestinal tract, or a severe catabolic state.

Capillary Damage Edema may also result from damage to the capillary endothelium, which increases its permeability and permits the transfer of protein into the interstitial compartment. Injury to the capillary wall can result from drugs, viral or bacterial agents, and thermal or mechanical trauma. Increased capillary permeability may also be a consequence of a hypersensitivity reaction and is characteristic of immune injury. Damage to the capillary endothelium is presumably responsible for inflammatory edema, which is usually nonpitting, localized, and accompanied by other signs of inflammation -- redness, heat, and tenderness.

To formulate a hypothesis about the pathophysiology of an edematous state, it is important to discriminate between the *primary* events, such as localized or generalized venous or lymphatic obstruction, reduction of cardiac output, hypoalbuminemia, trapping of fluid in spaces such as the pleural or peritoneal cavities, or an increase in capillary permeability, and the predictable *secondary* consequences, which include the renal retention of salt and water in an attempt to restore the plasma volume when the latter has been reduced, as in venous obstruction (see below). Both the primary event and the secondary consequences contribute to the formation of edema. In some cases the primary event is the renal retention of salt and water. Examples are renal failure, nephrotic syndrome, glomerulonephritis, and early hepatic failure.

Reduction of Effective Arterial Volume In many forms of edema the *effective arterial blood volume*, an as yet poorly defined parameter of the filling of the arterial tree, is reduced, and as a consequence a series of physiologic responses designed to restore it to normal are set into motion. A key element of these responses is the retention of salt and therefore of water, principally by the renal proximal tubule ([Fig. 37-1](#)), and in many instances this repairs the deficit of the effective arterial blood volume; often this deficit is repaired without the development of overt edema. If, however, the retention of salt and water is insufficient to restore and maintain the effective arterial blood volume, the stimuli are not dissipated, the retention of salt and water continues, and edema may ultimately develop. This sequence of events is operative in dehydration and hemorrhage. Although in these conditions there is a reduction of effective arterial blood volume and activation of the entire sequence shown in the center of [Fig. 37-2](#), including the diminished excretion of salt and water, because the net sodium and water balance is negative rather than positive, edema does *not* occur. In most conditions that lead to edema, the mechanisms responsible for maintaining a normal effective osmolality in the body fluids operate efficiently so that sodium retention promotes thirst and secretion of the antidiuretic hormone. In edematous states, isotonic expansion of the extracellular fluid space may be massive, while the intracellular fluid volume is unchanged.

Reduced Cardiac Output A reduction of cardiac output, whatever the cause, is associated with a lowering of the effective arterial blood volume as well as of renal blood flow, constriction of the efferent renal arterioles, and an elevation of the filtration fraction, i.e., the ratio of glomerular filtration rate to renal plasma flow. In severe heart failure there is a reduction in the glomerular filtration rate. Activation of the sympathetic

nervous system and of the renin-angiotensin systems are responsible for renal vasoconstriction. The finding that α -adrenergic blocking agents and/or angiotensin-converting enzyme (ACE) inhibitors augment renal blood flow and induce diuresis supports the role of these two systems in elevating renal vascular resistance and salt and water retention.

Renal Factors Reduced cardiac output lowers effective arterial blood volume. There is increased tubular reabsorption of glomerular filtrate in both the proximal and distal tubules ([Fig. 37-1](#)). Alterations in intrarenal hemodynamics appear to play a significant role. Heart failure and other conditions, such as nephrotic syndrome and cirrhosis that reduce effective arterial blood volume, cause renal efferent arteriolar constriction. This, in turn, reduces the hydrostatic pressure while the increased filtration fraction raises the colloid osmotic pressure in the peritubular capillaries, thus enhancing salt and water reabsorption in the proximal tubule as well as in the ascending limb of the loop of Henle.

In addition, the diminished renal blood flow characteristic of states in which the effective arterial blood volume is reduced is translated by the renal juxtaglomerular cells into a signal for increased renin release ([Chap. 331](#)). The mechanisms responsible for this release include a baroreceptor response: reduced renal perfusion results in incomplete filling of the renal arterioles and diminished stretch of the juxtaglomerular cells, a signal that provides for the elaboration or release, or both, of renin. A second mechanism for renin release involves the macula densa; as a result of reduced glomerular filtration, the sodium chloride load reaching the distal renal tubules is reduced. This is sensed by the macula densa, which signals the neighboring juxtaglomerular cells to secrete renin. A third mechanism involves the sympathetic nervous system and circulating catecholamines. Activation of the α -adrenergic receptors in the juxtaglomerular cells stimulates renin release. These three mechanisms generally act in concert.

The Renin-Angiotensin-Aldosterone (RAA) System (See [Chap. 331](#)) Renin, an enzyme with a molecular weight of about 40,000, acts on its substrate, angiotensinogen, an α_2 globulin synthesized by the liver, to release angiotensin I, a decapeptide, which is broken down to angiotensin II (AII), an octapeptide. This has generalized vasoconstrictor properties; it is especially active on the efferent arterioles and independently increases Na^+ reabsorption in the proximal tubule. The RAA system has long been recognized as a hormone system. However, it also operates locally. Both circulating and intrarenally produced AII contribute to renal vasoconstriction and to salt and water retention. These renal effects of AII are mediated by activation of AII type 1 receptors, which can be blocked by specific antagonists such as losartan. AII also enters the circulation and stimulates the production of aldosterone by the zona glomerulosa of the adrenal cortex. In patients with heart failure, not only is aldosterone secretion elevated but the biologic half-life of aldosterone is prolonged, which further increases the plasma level of the hormone. A depression of hepatic blood flow, particularly during exercise, secondary to a reduction in cardiac output, is responsible for the reduced hepatic catabolism of aldosterone. Aldosterone, in turn, enhances Na^+ reabsorption (and K^+ excretion) by the collecting tubule. The activation of the RAA system is most striking in the early phase of acute, severe heart failure and is less intense in patients with chronic, stable, compensated heart failure.

Although increased quantities of aldosterone are secreted in heart failure and in other

edematous states and although blockade of the action of aldosterone by spironolactone (an aldosterone antagonist) or amiloride (a blocker of epithelial Na⁺ channels) often induces a moderate diuresis in edematous states, persistent augmented levels of aldosterone (or other mineralocorticoids) alone do not always promote accumulation of edema, as witnessed by the lack of striking fluid retention in most instances of primary aldosteronism ([Chap. 331](#)). Furthermore, although normal individuals retain some salt and water with the administration of potent mineralocorticoids, such as deoxycorticosterone acetate or fludrocortisone, this accumulation is self-limiting, despite continued exposure to the steroid, a phenomenon known as *mineralocorticoid escape*. The failure of normal individuals who receive large doses of mineralocorticoids to accumulate large quantities of extracellular fluid and to develop edema is probably a consequence of an increase in glomerular filtration rate (pressure natriuresis) and through the action of natriuretic substance(s) (see below). The continued secretion of aldosterone may be more important in the accumulation of fluid in edematous states because patients with edema secondary to heart failure, nephrotic syndrome, and cirrhosis are generally unable to repair the deficit in effective arterial blood volume. As a consequence they do not develop pressure natriuresis.

Blockade of the [RAA](#) system, by blocking [Ang](#) receptors or inhibiting ACE, reduces efferent arteriolar resistance and increases renal blood flow. This action (combined in patients with heart failure with a rise in cardiac output secondary to afterload reduction) as well as reduction in the secretion of aldosterone cause diuresis. However, in patients with moderate or severe impairment of renal function or with renal artery stenosis, interference with the RAA system can cause paradoxical sodium retention due to intensification of renal failure.

Arginine Vasopressin (AVP) and Endothelin (See also [Chap. 329](#)) The secretion of AVP occurs in response to increased intracellular osmolar concentration and by stimulating V₂ receptors increases the reabsorption of free water in the renal distal tubule and collecting duct, thereby increasing total-body water. Circulating AVP is elevated in many patients with heart failure secondary to a nonosmotic stimulus associated with decreased effective arterial volume. Such patients fail to show the normal reduction of AVP with a reduction of osmolality, contributing to hyponatremia and edema formation.

Endothelin This is a potent peptide vasoconstrictor released by endothelial cells; its concentration is elevated in heart failure and contributes to renal vasoconstriction, Na⁺ retention, and edema in heart failure.

Natriuretic Peptides Atrial distention and/or a sodium load cause release into the circulation of atrial natriuretic peptide (ANP), a polypeptide; a high-molecular-weight precursor of ANP is stored in secretory granules within atrial myocytes. Release of ANP causes (1) excretion of sodium and water by augmenting glomerular filtration rate, inhibiting sodium reabsorption in the proximal tubule, and inhibiting release of renin and aldosterone; and (2) arteriolar and venous dilatation by antagonizing the vasoconstrictor actions of [Ang](#), [AVP](#), and sympathetic stimulation. Thus, ANP has the capacity to oppose sodium retention and arterial pressure elevation in hypervolemic states.

The closely related brain natriuretic peptide (BNP) is stored primarily in cardiac ventricular myocardium and is released when ventricular diastolic pressure rises. Its

actions are similar to those of [ANP](#). Circulating levels of ANP and BNP are elevated in congestive heart failure but obviously not sufficient to prevent edema formation. In addition, in edematous states (particularly heart failure), there is abnormal resistance to the actions of natriuretic peptides.

CLINICAL CAUSES OF EDEMA

Obstruction of Venous (and Lymphatic) Drainage of a Limb In this condition the hydrostatic pressure in the capillary bed upstream to the obstruction increases so that an abnormal quantity of fluid is transferred from the vascular to the interstitial space. Since the alternative route (i.e., the lymphatic channels) may also be obstructed, an increased volume of interstitial fluid in the limb develops, i.e., there is a trapping of fluid in the extremity, causing local edema at the expense of the blood volume in the remainder of the body, thereby reducing effective arterial blood volume and leading to the consequences shown in [Fig. 37-2](#).

When venous and lymphatic drainage are obstructed in a limb, fluid accumulates in the interstitium at the expense of plasma volume. The latter stimulates the retention of salt and water until the deficit in plasma volume has been corrected. Tissue tension rises in the affected limb until it counterbalances the primary alterations in the Starling forces, at which time no further fluid accumulates. The net effect is a local increase in the volume of interstitial fluid. This same sequence occurs in ascites and hydrothorax, in which fluid is trapped or accumulates in the cavitary space, depleting the intravascular volume and leading to secondary salt and fluid retention, as already described.

Congestive Heart Failure (See also [Chap. 232](#)) In this disorder the defective systolic emptying of the chambers of the heart and/or the impairment of ventricular relaxation promotes an accumulation of blood in the heart and venous circulation at the expense of the effective arterial volume, and the aforementioned sequence of events ([Fig. 37-2](#)) is initiated. In mild heart failure, a small increment of total blood volume may repair the deficit of arterial volume and establish a new steady state. Through the operation of Starling's law of the heart, an increase in the volume of blood within the chambers of the heart promotes a more forceful contraction and may thereby increase the cardiac output (See [Fig. 232-1](#)). However, if the cardiac disorder is more severe, retention of fluid cannot repair the deficit in effective arterial blood volume. The increment in blood volume accumulates in the venous circulation, and the increase in capillary and lymphatic hydrostatic pressures promotes the formation of edema. In heart failure, a reduction occurs in baroreflex-mediated inhibition of the vasomotor center, which causes activation of renal vasoconstrictor nerves and the [RAA](#) system, causing sodium and water retention.

Incomplete ventricular emptying (systolic heart failure) and/or inadequate ventricular relaxation (diastolic heart failure) both lead to an elevation of ventricular diastolic pressure. If the impairment of cardiac function involves the right ventricle, pressures in the systemic veins and capillaries may rise, thereby augmenting the transudation of fluid into the interstitial space and enhancing the likelihood of peripheral edema in the presence of the accumulation of sodium and water, as described above. The elevated systemic venous pressure is transmitted to the thoracic duct with consequent reduction of lymph drainage, further increasing the accumulation of edema.

If the impairment of cardiac function (incomplete ventricular emptying and/or inadequate relaxation) involves the left ventricle primarily, then pulmonary venous and capillary pressures rise [leading in some instances to pulmonary edema ([Chap. 32](#))], as does pulmonary artery pressure; this in turn interferes with the emptying of the right ventricle, leading to an elevation of right ventricular diastolic and of central and systemic venous pressures, enhancing the likelihood of formation of peripheral edema. Pulmonary edema impairs gas exchange and may induce hypoxia, which embarrasses cardiac function still further, sometimes causing a vicious cycle.

Nephrotic Syndrome and Other Hypoalbuminemic States (See also [Chap. 274](#)) The primary alteration in this disorder is a diminished colloid oncotic pressure due to massive losses of protein into the urine. This promotes a net movement of fluid into the interstitium, causes hypovolemia, and initiates the edema-forming sequence of events described above, including activation of the [RAA](#) system. With severe hypoalbuminemia and the consequent reduced colloid osmotic pressure, the salt and water that are retained cannot be restrained within the vascular compartment, total and effective arterial blood volumes decline, and hence the stimuli to retain salt and water are not abated. A similar sequence of events occurs in other conditions that lead to severe hypoalbuminemia, including severe nutritional deficiency states, protein-losing enteropathy, congenital hypoalbuminemia, and severe, chronic liver disease. However, in the nephrotic syndrome, impaired renal Na^+ excretion contributes to edema, even in the absence of severe hypoalbuminemia.

Cirrhosis (See also [Chaps. 46](#) and [299](#)) This condition is characterized by hepatic venous outflow blockade, which in turn causes expansion of the splanchnic blood volume and increased hepatic lymph formation. Intrahepatic hypertension acts as a potent stimulus for renal Na^+ retention and perhaps systemic vasodilation and a reduction of effective arterial blood volume as well. These alterations are frequently complicated by hypoalbuminemia secondary to reduced hepatic synthesis and reduce the effective arterial blood volume even further, leading to activation of the [RAA](#) system, of renal sympathetic nerves, and other salt- and water-retaining mechanisms. The concentration of circulating aldosterone is elevated by the liver's failure to metabolize this hormone. Initially, the excess interstitial fluid is localized preferentially upstream to the congested portal venous system and obstructed hepatic lymphatics, i.e., in the peritoneal cavity. In later stages, particularly when there is severe hypoalbuminemia, peripheral edema may develop. The excess production of prostaglandins (PGE_2 and PGI_2) in cirrhosis attenuates renal Na^+ retention. When the synthesis of these substances is inhibited by nonsteroidal anti-inflammatory agents, renal function deteriorates and Na^+ retention increases.

Drug-Induced Edema A large number of widely used drugs can cause edema ([Table 37-1](#)). Mechanisms include renal vasoconstriction (nonsteroidal anti-inflammatory agents and cyclosporine), arteriolar dilatation (vasodilators), augmented renal sodium reabsorption (steroid hormones) and capillary damage (interleukin 2).

Idiopathic Edema This syndrome, which occurs almost exclusively in women, is characterized by periodic episodes of edema (unrelated to the menstrual cycle), frequently accompanied by abdominal distention. Diurnal alterations in weight occur with

orthostatic retention of sodium and water, so that the patient may weigh several pounds more after having been in the upright posture for several hours. Such large diurnal weight changes suggest an increase in capillary permeability that appears to fluctuate in severity and to be aggravated by hot weather. There is some evidence that a reduction in plasma volume occurs in this condition with secondary activation of the [RAA](#) system and impaired suppression of [AVP](#) release. Idiopathic edema should be distinguished from cyclical or premenstrual edema, in which the sodium and water retention may be secondary to excessive estrogen stimulation. There are also some cases in which the edema appears to be "diuretic-induced." It has been postulated that in these patients, chronic diuretic administration leads to mild blood volume depletion, which causes chronic hyperreninemia and juxtaglomerular hyperplasia. Salt-retaining mechanisms appear to overcompensate for the direct effects of the diuretics. *Acute* withdrawal of diuretics can then leave the sodium-retaining forces unopposed, leading to fluid retention and edema. Decreased dopaminergic activity and reduced urinary kallikrein and kinin excretion have been reported in this condition and may also be of pathogenetic importance.

TREATMENT

The treatment of idiopathic cyclic edema includes a reduction in salt intake, rest in the supine position for several hours each day, the wearing of elastic stockings (which are put on before arising in the morning), and an attempt to understand any underlying emotional problems. A variety of pharmacologic agents including [ACE](#) inhibitors, progesterone, the dopamine receptor agonist bromocriptine, and the sympathomimetic amine dextroamphetamine have all been reported to be useful when administered to patients who do not respond to simpler measures. Diuretics may be helpful initially but may lose their effectiveness with continuous administration; accordingly, they should be employed sparingly, if at all. Discontinuation of diuretics paradoxically leads to diuresis in "diuretic-induced" edema, described above.

DIFFERENTIAL DIAGNOSIS

The differences between the three major causes of generalized edema are shown in [Table 37-2](#).

Localized edema can usually be readily differentiated from generalized edema. The great majority of patients with generalized edema suffer from advanced cardiac, renal, hepatic, or nutritional disorders. Consequently, the differential diagnosis of generalized edema should be directed toward identifying or excluding these several conditions.

LOCALIZED EDEMA (See also [Chap. 248](#))

Edema originating from inflammation or hypersensitivity is usually readily identified. Localized edema due to venous or lymphatic obstruction may be caused by thrombophlebitis, chronic lymphangitis, resection of regional lymph nodes, filariasis, etc. Lymphedema is particularly intractable because restriction of lymphatic flow results in increased protein concentration in the interstitial fluid, a circumstance that aggravates retention of fluid.

EDEMA OF HEART FAILURE (See also [Chap. 232](#))

The presence of heart disease, as manifested by cardiac enlargement and gallop rhythm, together with evidence of cardiac failure, such as dyspnea, basilar rales, venous distention, and hepatomegaly, usually provides an indication on clinical examination that edema results from heart failure. Noninvasive tests such as echocardiography and radionuclide angiography may be helpful in establishing the diagnosis of heart failure.

EDEMA OF THE NEPHROTIC SYNDROME (See also [Chap. 274](#))

Marked proteinuria (>3.5 g/d), hypoalbuminemia (<35 g/L), and in some instances hypercholesterolemia are present. This syndrome may occur during the course of a variety of kidney diseases, which include glomerulonephritis, diabetic glomerulosclerosis, and hypersensitivity reactions. A history of previous renal disease may or may not be elicited.

EDEMA OF ACUTE GLOMERULONEPHRITIS AND OTHER FORMS OF RENAL FAILURE

The edema occurring during the acute phases of glomerulonephritis is characteristically associated with hematuria, proteinuria, and hypertension. Although some evidence supports the view that the fluid retention is due to increased capillary permeability, in most instances the edema in this disease results from primary retention of sodium and water by the kidneys owing to renal insufficiency. This state differs from congestive heart failure in that it is characterized by a normal (or sometimes even increased) cardiac output and a normal arterial-mixed venous oxygen difference. Patients with edema due to renal failure commonly have evidence of pulmonary congestion on chest roentgenograms before cardiac enlargement is significant, but they usually do not develop orthopnea. Patients with chronic impairment of renal function may also develop edema due to primary renal retention of sodium and water.

EDEMA OF CIRRHOSIS (See also [Chap. 299](#))

Ascites and biochemical and clinical evidence of hepatic disease (collateral venous channels, jaundice, and spider angiomas) characterize edema of hepatic origin. The ascites is frequently refractory to treatment because it collects as a result of a combination of obstruction of hepatic lymphatic drainage, portal hypertension, and hypoalbuminemia. Edema may also occur in other parts of the body in these patients as a result of hypoalbuminemia. Furthermore, the sizable accumulation of ascitic fluid may increase intraabdominal pressure and impede venous return from the lower extremities; hence, it tends to promote accumulation of edema in this region as well.

EDEMA OF NUTRITIONAL ORIGIN

A diet grossly deficient in protein over a prolonged period may produce hypoproteinemia and edema. The latter may be intensified by the development of beriberi heart disease, also of nutritional origin, in which multiple peripheral arteriovenous fistulas result in reduced effective systemic perfusion and effective arterial blood volume, thereby enhancing edema formation ([Chap. 75](#)). Edema may actually become intensified when

these famished subjects are first provided with an adequate diet. The ingestion of more food may increase the quantity of salt ingested, which is then retained along with water. So-called "refeeding edema" may also be linked to increased release of insulin, which directly increases tubular sodium reabsorption. In addition to hypoalbuminemia, hypokalemia and caloric deficits may be involved in the edema of starvation.

OTHER CAUSES OF EDEMA

These include hypothyroidism, in which the edema (myxedema) may be located typically in the pretibial region and which may also be associated with periorbital puffiness. Exogenous hyperadrenocortism, pregnancy, and administration of estrogens and vasodilators, particularly the calcium antagonist nifedipine, may also all cause edema.

DISTRIBUTION OF EDEMA

The distribution of edema is an important guide to the cause. Thus, edema limited to one leg or to one or both arms is usually the result of venous and/or lymphatic obstruction. Edema resulting from hypoproteinemia characteristically is generalized, but it is especially evident in the very soft tissues of the eyelids and face and tends to be most pronounced in the morning because of the recumbent posture assumed during the night. Less common causes of facial edema include trichinosis, allergic reactions, and myxedema. Edema associated with heart failure, on the other hand, tends to be more extensive in the legs and to be accentuated in the evening, a feature also determined largely by posture. When patients with heart failure have been confined to bed, edema may be most prominent in the presacral region. Unilateral edema occasionally results from lesions in the central nervous system affecting the vasomotor fibers on one side of the body; paralysis also reduces lymphatic and venous drainage on the affected side.

ADDITIONAL FACTORS IN DIAGNOSIS

The color, thickness, and sensitivity of the skin are significant. Local tenderness and increase in temperature suggest inflammation. Local cyanosis may signify a venous obstruction. In individuals who have had repeated episodes of prolonged edema, the skin over the involved areas may be thickened, indurated, and often red.

Measurement or estimation of the venous pressure is of importance in evaluating edema. Elevation in an isolated part of the body usually reflects localized venous obstruction. Generalized elevation of systemic venous pressure usually indicates the presence of congestive heart failure. Ordinarily, a significant generalized increase in venous pressure can be recognized by the level at which cervical veins collapse ([Chap. 225](#)). In patients with obstruction of the superior vena cava, edema is confined to the face, neck, and upper extremities, where the venous pressure is elevated compared with that in the lower extremities. Measurement of venous pressure in the upper extremities is also useful in patients with massive edema of the lower extremities and ascites; it is elevated in the upper extremities when the edema is on a cardiac basis (e.g., constrictive pericarditis or tricuspid stenosis) but is normal when it is secondary to cirrhosis. Severe heart failure may cause ascites that may be distinguished from the ascites caused by hepatic cirrhosis by the jugular venous pressure, which usually is

elevated in heart failure and normal in cirrhosis.

Determination of the concentration of serum albumin aids importantly in identifying those patients in whom edema is due, at least in part, to diminished intravascular colloid oncotic pressure. The presence of proteinuria also affords useful clues. The absence of proteinuria excludes nephrotic syndrome but cannot exclude nonproteinuric causes of renal failure. Slight to moderate proteinuria is the rule in patients with heart failure.

Approach to the Patient

An important first question is whether the edema is localized or generalized. If it is localized, those phenomena that may be responsible should be concentrated upon. Hydrothorax and ascites are forms of localized edema. Either may be a consequence of local venous or lymphatic obstruction, as in inflammatory or neoplastic disease.

If the edema is generalized, it should be determined, first, if there is serious hypoalbuminemia, e.g., serum albumin < 25 g/L. If so, the history, physical examination, urinalysis, and other laboratory data will help evaluate the question of cirrhosis, severe malnutrition, protein-losing gastroenteropathy, or the nephrotic syndrome as the underlying disorder. If hypoalbuminemia is not present, it should be determined if there is evidence of congestive heart failure of a severity to promote generalized edema. Finally, it should be determined whether the patient has an adequate urine output, or if there is significant oliguria or even anuria. **These abnormalities are discussed in [Chaps. 47, 269, and 270](#).*

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38. SHOCK - Ronald V. Maier

Shock is the clinical syndrome that results from inadequate tissue perfusion. Irrespective of cause, the hypoperfusion-induced imbalance between the delivery of and requirements for oxygen and substrate leads to cellular dysfunction. The cellular injury created by the inadequate delivery of oxygen and substrates also induces the production and release of inflammatory mediators that further compromise perfusion through functional and structural changes within the microvasculature. This leads to a vicious cycle in which impaired perfusion is responsible for cellular injury which causes maldistribution of blood flow, further compromising cellular perfusion; the latter causes multiple organ failure and, if the process is not interrupted, leads to the death of the patient. The clinical manifestations of shock are the result, in part, of sympathetic neuroendocrine responses to hypoperfusion as well as the breakdown in organ function induced by severe cellular dysfunction.

When very severe and/or persistent, inadequate oxygen delivery leads to irreversible cell injury, and only rapid restoration of oxygen delivery can reverse the progression of the shock state. The fundamental approach to management, therefore, is to recognize overt and impending shock in a timely fashion and to intervene emergently to restore perfusion. Except in cases of cardiogenic shock, this requires the expansion or reexpansion of blood volume. Control of any inciting pathologic process, e.g., continued hemorrhage, impairment of cardiac function, or infection, must occur simultaneously.

Clinical shock is usually accompanied by hypotension, i.e., a mean arterial pressure <60 mmHg in previously normotensive persons. Multiple classification schemes have been developed in an attempt to synthesize the seemingly dissimilar processes leading to shock. Strict adherence to a classification scheme may be difficult from a clinical standpoint because of the frequent combination of two or more causes of shock in any individual patient, but the classification shown in [Table 38-1](#) provides a useful reference point from which to discuss and further delineate the underlying processes. The individual classes are discussed below.

PATHOGENESIS AND ORGAN RESPONSE

MICROCIRCULATION

Normally when cardiac output falls, systemic vascular resistance rises to maintain a level of systemic pressure that is adequate to allow perfusion of the heart and brain at the expense of other tissues, especially muscle, skin, and the gastrointestinal tract. Systemic vascular resistance is determined primarily by the luminal diameter of arterioles. The metabolic rates of the heart and brain are high, and their stores of energy substrate are low. These organs are critically dependent on a continuous supply of oxygen and nutrients, and neither tolerates severe ischemia for more than brief periods. Autoregulation, i.e., the maintenance of blood flow over a wide range of perfusion pressures, is critical in sustaining cerebral and coronary perfusion despite significant hypotension. However, when mean arterial pressure drops to 60 mmHg, flow to these organs falls and their function deteriorates.

Arteriolar vascular smooth muscle has both α - and β -adrenergic receptors ([Chap. 72](#)).

The α_1 receptors mediate vasoconstriction, while the β_2 receptors mediate vasodilation. Efferent sympathetic fibers release norepinephrine, which acts primarily on α_1 receptors in one of the most fundamental compensatory responses to reduced perfusion pressure. Other constrictor substances that are increased in most forms of shock include angiotensin II, vasopressin, endothelin-1, and thromboxane A_2 . Both norepinephrine and epinephrine are released by the adrenal medulla, and the concentrations of these catecholamines in the blood stream rise. Circulating vasodilators in shock include prostacyclin (PGI_2), nitric oxide (NO), and, importantly, products of local metabolism such as adenosine that match flow to the metabolic needs of the tissue. The balance between these various vasoconstrictor and vasodilator influences acting upon the microcirculation determines local perfusion.

Transport to cells depends on microcirculatory flow; capillary permeability; the diffusion of oxygen, carbon dioxide, nutrients, and products of metabolism through the interstitium; and the exchange of these products across cell membranes. Impairment of the microcirculation, which is central to the pathophysiologic responses in the late stages of all forms of shock, results in the derangement of cellular metabolism, which is ultimately responsible for organ failure.

The normal response to mild or moderate hypovolemia is an attempt at restitution of intravascular volume through alterations in hydrostatic pressure and osmolarity. Constriction of arterioles leads to reductions in both the capillary hydrostatic pressure and the number of capillary beds perfused, thereby limiting the capillary surface area across which filtration occurs. When filtration is reduced while intravascular oncotic pressure remains constant or rises, there is net reabsorption of fluid into the vascular bed, in accord with Starling's law of capillary-interstitial liquid exchange ([Chap. 32](#)). Metabolic changes (including hyperglycemia and elevations in the products of glycolysis, lipolysis, and proteolysis) raise extracellular osmolarity, leading to an osmotic gradient between cells and interstitium that increases interstitial and intravascular volume at the expense of intracellular volume.

CELLULAR RESPONSES

Interstitial transport of nutrients is impaired, leading to a decline of intracellular high-energy phosphate stores. Mitochondrial dysfunction and uncoupling of oxidative phosphorylation are the most likely causes for decreased amounts of ATP. As a consequence, there is an accumulation of anaerobic metabolites including hydrogen ions, lactate, and other products of anaerobic metabolism. As shock progresses, these vasodilator metabolites override vasomotor tone, causing further hypotension and hypoperfusion. Dysfunction of cell membranes is thought to represent a common end-stage pathophysiologic pathway in the various forms of shock. Normal cellular transmembrane potential falls, and there is an associated increase in intracellular sodium and water, leading to cell swelling, which interferes further with microvascular perfusion.

NEUROENDOCRINE RESPONSE

Hypovolemia, hypotension, and hypoxia are sensed by baroreceptors and chemoreceptors, which contribute further to an autonomic response that attempts to

restore blood volume, maintain central perfusion, and mobilize metabolic substrates. Hypotension disinhibits the vasomotor center, resulting in increased adrenergic output and reduced vagal activity. Release of norepinephrine induces peripheral and splanchnic vasoconstriction, a major contributor to the maintenance of central organ perfusion, while reduced vagal activity increases the heart rate and cardiac output. The effects of circulating epinephrine released by the adrenal medulla in shock are largely metabolic, causing increased glycogenolysis and gluconeogenesis and reduced pancreatic insulin release.

Severe pain and other severe stress cause the hypothalamic release of adrenocorticotrophic hormone (ACTH). This stimulates cortisol secretion, which contributes to decreased peripheral uptake of glucose and amino acids, enhances lipolysis, and increases gluconeogenesis. Increased pancreatic secretion of glucagon during stress accelerates hepatic gluconeogenesis and further elevates blood glucose concentration. These hormonal actions act synergistically in the maintenance of blood volume. The importance of the cortisol response to stress is illustrated by the profound circulatory collapse that occurs in hypoadrenal patients (see below).

Renin release is increased in response to adrenergic discharge and reduced perfusion of the juxtaglomerular apparatus in the kidney. Renin induces the formation of angiotensin I, which is then converted to angiotensin II, an extremely potent vasoconstrictor and stimulator of aldosterone release by the adrenal cortex and of vasopressin by the posterior pituitary. Aldosterone contributes to the maintenance of intravascular volume by enhancing renal tubular reabsorption of sodium, resulting in the excretion of a low-volume, concentrated, sodium-free urine. Vasopressin has a direct action on vascular smooth muscle, contributing to vasoconstriction, and acts on the distal renal tubules to enhance water reabsorption.

CARDIOVASCULAR RESPONSE

Three variables -- ventricular filling (preload), the resistance to ventricular ejection (afterload), and myocardial contractility -- are paramount in controlling stroke volume ([Chap. 231](#)). Cardiac output, the major determinant of tissue perfusion, is the product of stroke volume and heart rate. Hypovolemia leads to decreased ventricular preload, which in turn reduces the stroke volume. An increase in heart rate is a useful but limited compensatory mechanism to maintain cardiac output. A shock-induced reduction in myocardial compliance is frequent, reducing ventricular end-diastolic volume and hence stroke volume at any given ventricular filling pressure. Restoration of intravascular volume then returns stroke volume to normal but only at elevated filling pressures. In addition, sepsis, ischemia, myocardial infarction, severe tissue trauma, hypothermia, general anesthesia, prolonged hypotension, and acidemia may all impair myocardial contractility and also reduce the stroke volume at any given ventricular end-diastolic volume. The resistance to ventricular ejection is influenced importantly by the systemic vascular resistance, which is elevated in most forms of shock. However, resistance is depressed in the early hyperdynamic stage of septic shock (see below), thereby allowing the cardiac output to be maintained.

The venous system contains nearly two-thirds of the total circulating blood volume, most in the small veins, and serves as a dynamic reservoir for autoinfusion of blood. Active

venoconstriction as a consequence of α -adrenergic activity is an important compensatory mechanism for the maintenance of venous return and therefore of ventricular filling during shock. On the other hand, venous dilatation, as occurs in neurogenic shock, reduces ventricular filling and hence stroke volume and cardiac output (see below).

PULMONARY RESPONSE

The response of the pulmonary vascular bed to shock parallels that of the systemic vascular bed, and the relative increase in pulmonary vascular resistance, particularly in septic shock, may exceed that of the systemic vascular resistance. Shock-induced tachypnea reduces tidal volume and increases both dead space and minute ventilation. Relative hypoxia and the subsequent tachypnea induce a respiratory alkalosis. Recumbency and involuntary restriction of ventilation secondary to pain reduce functional residual capacity and may lead to atelectasis. Shock is recognized as a major cause of acute lung injury and subsequent acute respiratory distress syndrome (ARDS; [Chap. 265](#)). These disorders are characterized by noncardiogenic pulmonary edema secondary to diffuse pulmonary capillary endothelial and alveolar epithelial injury, hypoxemia, and bilateral diffuse pulmonary infiltrates. Hypoxemia results from perfusion of underventilated and nonventilated alveoli. Loss of surfactant and lung volume in combination with increased interstitial and alveolar edema reduce lung compliance. The work of breathing and the oxygen requirements of respiratory muscles increase.

RENAL RESPONSE

Acute renal failure ([Chap. 269](#)), a serious complication of shock and hypoperfusion, occurs less frequently than heretofore because of early aggressive volume repletion. Acute tubular necrosis is now more frequently seen as a result of the interactions of shock, sepsis, the administration of nephrotoxic agents (such as aminoglycosides and angiographic contrast media), and rhabdomyolysis; the latter may be particularly severe in skeletal muscle trauma. The physiologic response of the kidney to hypoperfusion is to conserve salt and water. In addition to decreased renal blood flow, increased afferent arteriolar resistance accounts for diminished glomerular filtration rate, which together with increased ADH and aldosterone is responsible for reduced urine formation. Toxic injury causes necrosis of tubular epithelium and tubular obstruction by cellular debris with back-leak of filtrate. The depletion of renal ATP stores that occurs with prolonged renal hypoperfusion is related to subsequent impairment of renal function.

METABOLIC DERANGEMENTS

During shock, there is disruption of the normal cycles of carbohydrate, lipid, and protein metabolism. Through the citric acid cycle, alanine in conjunction with lactate (which is converted from pyruvate in the periphery in the presence of oxygen deprivation) enhances the hepatic production of glucose. With reduced availability of oxygen, the breakdown of glucose to pyruvate and ultimately lactate represents an inefficient cycling of substrate with minimal net energy production. An elevated plasma lactate/pyruvate ratio is consistent with anaerobic metabolism and reflects inadequate tissue perfusion. Decreased clearance of exogenous triglycerides coupled with increased hepatic

lipogenesis causes a significant rise in serum triglyceride concentrations. There is increased protein catabolism, a negative nitrogen balance, and, if the process is prolonged, severe muscle wasting.

INFLAMMATORY RESPONSES

Activation of an extensive network of proinflammatory mediator systems plays a significant role in the progression of shock and contributes importantly to the development of organ injury and failure.

Multiple humoral mediators are activated during shock and tissue injury. The complement cascade, activated through both the classic and alternate pathways, generates the anaphylatoxins C3a and C5a. Direct complement fixation to injured tissues can progress to the C5-C9 attack complex, causing further cell damage. Activation of the coagulation cascade causes microvascular thrombosis, with subsequent lysis leading to repeated episodes of ischemia and reperfusion. Components of the coagulation system, such as thrombin, are potent proinflammatory mediators that cause expression of adhesion molecules on endothelial cells and activation of neutrophils, leading to microvascular injury. Coagulation also activates the kallikrein-kininogen cascade, contributing to hypotension.

Eicosanoids are vasoactive and immunomodulatory products of arachidonic acid metabolism that include cyclooxygenase-derived prostaglandins and thromboxane A₂ as well as lipoxygenase-derived leukotrienes and lipoxins. Thromboxane A₂ is a potent vasoconstrictor that contributes to the pulmonary hypertension and acute tubular necrosis of shock. [PGI₂](#) and prostaglandin E₂ are potent vasodilators that enhance capillary permeability and edema formation. The cysteinyl leukotrienes LTC₄ and LTD₄ are pivotal mediators of the vascular sequelae of anaphylaxis, as well as of shock states resulting from sepsis or tissue injury. LTB₄ is a potent neutrophil chemoattractant and secretagogue that stimulates the formation of reactive oxygen species. Lipoxins are endogenous autocooids that inhibit leukotriene-mediated responses. Platelet-activating factor, an ether-linked, arachidonyl-containing phospholipid mediator, also carries potent bioactivities that include pulmonary vasoconstriction, bronchoconstriction, systemic vasodilation, increased capillary permeability, and the priming of macrophages and neutrophils to produce enhanced levels of inflammatory mediators.

Tumor necrosis factor (TNF) α , produced by activated macrophages, reproduces many components of the shock state including hypotension, lactic acidosis, and respiratory failure. Interleukin (IL) 1 is also produced by tissue-fixed macrophages and is critical to the inflammatory response occurring in hypoperfusion and septic states. Chemokines also participate in the systemic inflammatory response. For example, IL-8 is a potent neutrophil chemoattractant and activator that upregulates adhesion molecules on the neutrophil to enhance aggregation and adherence to the vascular endothelium. The endothelium normally produces nitric oxide (NO), a potent vasodilator. The inflammatory response stimulates the inducible isoform of NO synthase (iNOS), which is thought to overexpress toxic NO and oxygen-derived free radicals and contributes to the hyperdynamic cardiovascular response that occurs in sepsis.

Multiple inflammatory cells, including neutrophils, macrophages, and platelets, are a

major contributor to inflammation-induced injury. Margination of activated neutrophils in the microcirculation is a common pathologic finding in shock, causing secondary injury due to the release of potentially toxic oxygen radicals and proteases. Adhesion molecules are expressed on the surface of the endothelium and on cytokine-stimulated neutrophils. Tissue-fixed macrophages produce virtually all major components of the inflammatory response and orchestrate the progression and duration of the response.

Approach to the Patient

The underlying problem in all forms of shock is inadequate tissue perfusion and an imbalance between delivery and cellular needs of oxygen and metabolic substrate. It is important to recognize the onset of hypoperfusion at the earliest possible time in order to institute aggressive resuscitation and correction of the underlying etiology.

Monitoring Patients in shock require care in an intensive care unit. Careful and continuous assessment of the physiologic status is necessary. Arterial pressure through an indwelling line, pulse, and respiratory rate should be monitored continuously; a Foley catheter should be inserted to follow urine flow; and mental status assessed frequently.

Although there is ongoing debate as to the indications for using the flow-directed pulmonary artery catheter (PAC, Swan-Ganz catheter) in the management of patients in shock, most intensivists believe that the ability to predict the hemodynamic profiles of patients in shock accurately without a PAC is poor. The PAC is placed percutaneously via the subclavian or jugular vein through the central venous circulation and right heart into the pulmonary artery. There are ports both proximal in the right atrium and distal in the pulmonary artery to provide access for infusions and for cardiac output measurements. Right atrial and pulmonary artery pressures are measured, and the pulmonary capillary wedge pressure (PCWP) serves as an approximation of the left atrial pressure. Normal hemodynamic parameters are shown in [Table 228-3](#) and [Table 38-2](#).

Cardiac output is determined by the thermodilution technique, and high-resolution thermistors can also be used to determine right ventricular end-diastolic volume to monitor further the response of the right heart to fluid resuscitation. [APAC](#) with an oximeter port offers the additional advantage of on-line monitoring of the mixed venous oxygen saturation, an important index of tissue perfusion. Systemic and pulmonary vascular resistances are calculated as the ratio of the pressure drop across these vascular beds to the cardiac output ([Chap. 228](#)). Determinations of oxygen content in arterial and venous blood, together with cardiac output and hemoglobin concentration allow calculation of oxygen delivery, oxygen consumption, and oxygen-extraction ratio ([Table 38-3](#)). The hemodynamic patterns associated with the various form of shock are shown in [Table 38-4](#).

In resuscitation from shock, it is critical to restore tissue perfusion and optimize oxygen delivery, hemodynamics, and cardiac function rapidly. A goal of therapy is to achieve normal mixed venous oxygen saturation and arteriovenous oxygen-extraction ratio. To enhance oxygen delivery, red cell mass, arterial oxygen saturation, and cardiac output may be augmented singly or simultaneously. An increase in oxygen delivery not accompanied by an increase in oxygen consumption implies that oxygen availability is

adequate and that oxygen consumption is not flow-dependent. Conversely, an elevation of oxygen consumption with increased cardiac output implies that the oxygen supply is inadequate. A reduction in systemic vascular resistance accompanying an increase in cardiac output indicates that compensatory vasoconstriction is reversing due to improved tissue perfusion. The determination of stepwise expansion of blood volume on cardiac performance allows identification of the optimum preload.

SPECIFIC FORMS OF SHOCK

HYPOVOLEMIC SHOCK

This most common form of shock results either from the loss of red blood cell mass and plasma from hemorrhage or from the loss of plasma volume alone arising from extravascular fluid sequestration or gastrointestinal, urinary, and insensible losses. The signs and symptoms of nonhemorrhagic hypovolemic shock are the same as those of hemorrhagic shock, although they may have a more insidious onset. The normal physiologic response to hypovolemia is to maintain perfusion of the brain and heart while restoring an effective circulating blood volume. There is an increase in sympathetic activity, hyperventilation, collapse of venous capacitance vessels, release of stress hormones, and expansion of intravascular volume through the recruitment of interstitial and intracellular fluid and reduction of urine output.

Mild hypovolemia ($\leq 20\%$ of the blood volume) generates mild tachycardia but relatively few external signs, especially in a supine resting young patient ([Table 38-5](#)). With moderate hypovolemia (~ 20 to 40% of the blood volume) the patient becomes increasingly anxious and tachycardic; although normal blood pressure may be maintained in the supine position, there may be significant postural hypotension and tachycardia. If hypovolemia is severe ($\geq 40\%$ of the blood volume), the classic signs of shock appear; the blood pressure declines and becomes unstable even in the supine position, and the patient develops marked tachycardia, oliguria, and agitation or confusion. Perfusion of the central nervous system is well maintained until shock becomes severe. Hence, mental obtundation is an ominous clinical sign. The transition from mild to severe hypovolemic shock can be insidious or extremely rapid. If severe shock is not reversed rapidly, especially in elderly patients and those with comorbid illnesses, death is imminent. A very narrow time frame separates the derangements found in severe shock that can be reversed with aggressive resuscitation from those of progressive decompensation and irreversible cell injury.

Diagnosis Hypovolemic shock is readily diagnosed when there are signs of hemodynamic instability and the source of volume loss is obvious. The diagnosis is more difficult when the source of blood loss is occult, as into the gastrointestinal tract, or when plasma volume alone is depleted. After acute hemorrhage, hemoglobin and hematocrit values do not change until compensatory fluid shifts have occurred or exogenous fluid is administered. Thus, an initial normal hematocrit does not disprove the presence of significant blood loss. Plasma losses cause hemoconcentration, and free water loss leads to hypernatremia. These findings should suggest the presence of hypovolemia.

It is essential to distinguish between hypovolemic and cardiogenic shock (see below)

because definitive therapy differs significantly. Both forms are associated with a reduced cardiac output and a compensatory sympathetic mediated response characterized by tachycardia and elevated systemic vascular resistance. However, the findings in cardiogenic shock of jugular venous distention, rales, and an S₃ gallop distinguish it from hypovolemic shock and signify that volume expansion is undesirable.

TREATMENT

Initial resuscitation requires rapid reexpansion of the circulating blood volume along with interventions to control ongoing losses. In accordance with Starling's law ([Chap. 231](#)), stroke volume and cardiac output rise with the increase in preload. After resuscitation, the compliance of the ventricles may remain reduced due to increased interstitial fluid in the myocardium. Therefore, elevated filling pressures are required to maintain adequate ventricular performance.

Volume resuscitation is initiated with the rapid infusion of isotonic saline or a balanced salt solution such as Ringer's lactate through large-bore intravenous lines. No distinct benefit from the use of colloid has been demonstrated. The infusion of 2 to 3 L over 10 to 30 min should restore normal hemodynamic parameters. Continued hemodynamic instability implies that shock has not been reversed and/or that there are significant ongoing blood or volume losses. Continuing blood loss, with hemoglobin concentrations declining to 100 g/L (10 g/dL), should initiate blood transfusion, preferably as fully cross-matched blood. In extreme emergencies, type-specific or O-negative packed red cells may be transfused. In the presence of severe and/or prolonged hypovolemia, inotropic support with dopamine or dobutamine ([Chap. 72](#)) may be required to maintain adequate ventricular performance, after blood volume has been restored. Infusion of norepinephrine to increase arterial pressure by raising peripheral resistance is inappropriate, other than as a temporizing measure in severe shock while blood volume is reexpanded.

Successful resuscitation also requires support of respiratory function. Supplemental oxygen should be provided, and endotracheal intubation may be necessary to maintain arterial oxygenation. Following resuscitation from isolated hemorrhagic shock, end-organ damage is frequently less than following septic or traumatic shock. This may be due to the absence of the massive activation of inflammatory mediator response systems and the consequent nonspecific organ injury seen in the latter conditions.

TRAUMATIC SHOCK

Shock following trauma is, in large measure, due to hypovolemia. However, even when hemorrhage has been controlled, patients can continue to suffer loss of plasma volume into the interstitium of injured tissues. These fluid losses are compounded by injury-induced inflammatory responses, which contribute to the secondary microcirculatory injury. This causes secondary tissue injury and maldistribution of blood flow, intensifying tissue ischemia and leading to multiple organ system failure. Trauma to the heart, chest, or head can also contribute to the shock. For example, pericardial tamponade or tension pneumothorax impairs ventricular filling, while myocardial contusion depresses myocardial contractility.

TREATMENT

Inability to maintain a systolic blood pressure ≥ 90 mmHg after trauma-induced hypovolemia is associated with a mortality rate of ~50%. To prevent decompensation of homeostatic mechanisms, therapy must be promptly administered.

The initial management of the seriously injured patient requires attention to the "ABCs" of resuscitation: assurance of an airway (A), adequate ventilation (breathing, B), and establishment of an adequate blood volume to support the circulation (C). Control of hemorrhage requires immediate attention. Early stabilization of fractures, debridement of devitalized or contaminated tissues, and evacuation of hematomata all reduce the subsequent inflammatory response to the initial insult and minimize subsequent organ injury.

INTRINSIC CARDIOGENIC SHOCK

This form of shock is caused by failure, often sudden, of the heart as an effective pump. It occurs most commonly as a complication of acute myocardial infarction (AMI; [Chap. 243](#)), but it may also be seen in patients with severe brady- or tachyarrhythmias, valvular heart disease, or in the terminal stage of chronic heart failure of any cause, including ischemic heart disease and dilated cardiomyopathy. Cardiogenic shock is characterized by a low cardiac output, diminished peripheral perfusion, pulmonary congestion, and elevation of systemic vascular resistance and pulmonary vascular pressures. Acute right heart failure can arise as the result of right ventricular infarction or may complicate the acute respiratory distress syndrome and severe pulmonary hypertension of any etiology. As a consequence of right ventricular failure, left ventricular preload falls, and this, in turn, reduces systemic perfusion. In contrast to other forms of shock, absolute or relative hypovolemia is usually not present in cardiogenic shock.

The ineffective contractile activity of either the right or left side of the heart leads to the accumulation of blood in the venous circulation upstream to the failing ventricle. Cardiogenic shock with left-sided heart failure increases fluid in the lungs that can overwhelm the capacity of the pulmonary lymphatics and causes interstitial and sometimes alveolar edema. Interstitial lung edema usually occurs at pulmonary capillary pressures >18 mmHg, and overt pulmonary alveolar edema develops at pressures >24 mmHg ([Chap. 32](#)). Pulmonary edema impacts cardiac function further by impairing diffusion of oxygen, setting up a vicious cycle. The increase in interstitial and intraalveolar fluid causes a progressive reduction in lung compliance, thereby increasing the work of ventilation while increasing perfusion of poorly ventilated alveoli.

In establishing the diagnosis of cardiogenic shock, a history of cardiac disease or of [AMI](#) is of value. Associated physical findings include those of hemodynamic instability, peripheral vasoconstriction, and pulmonary and/or systemic venous congestion, as well as findings specific to the underlying cardiac abnormalities. An electrocardiogram may provide evidence of AMI or preexisting cardiac disease. The chest x-ray may show pulmonary edema and cardiomegaly. Transthoracic or transesophageal echocardiograms assist in the diagnosis of structural abnormalities and/or functional impairment of contractility. Serum cardiac markers will support the diagnosis of acute

cardiac injury. Hemodynamic monitoring is usually necessary. Placement of a [PAC](#) is helpful and will show a reduced cardiac output and an elevated [PCWP](#), and direct measurement of right atrial pressure allows calculation of systemic vascular resistance which is elevated.

TREATMENT

For all forms of cardiogenic shock, preload, afterload, and contractility should be modified using the information provided by the [PAC](#). [APCWP](#) of 15 to 20 mmHg should be the initial goal. If the PCWP is excessively elevated, inotropic agents may provide significant reduction. The goal is to increase contractility without significant increases in heart rate. Dopamine and norepinephrine exert both inotropic and vasoconstrictor actions ([Chap. 72](#)) that are useful in the presence of persistent hypotension. Dobutamine, a positive inotropic agent with vasodilator properties, may be substituted when arterial pressure has been restored. Pulmonary congestion may be responsive to intravenous furosemide. Patients with an inadequate response to these measures can be supported by using intraaortic balloon counterpulsation to permit recovery of myocardial function. Additional measures to consider in cases of refractory cardiogenic shock include urgent myocardial revascularization in patients with [AMI](#) ([Chap. 243](#)), correction of anatomic cardiac defects such as rupture of the papillary muscles of the interventricular septum, the placement of ventricular assist devices, and even urgent cardiac transplantation.

COMPRESSIVE CARDIOGENIC SHOCK

With compression, the heart and surrounding structures are less compliant and, thus, normal filling pressures generate inadequate diastolic filling. Blood or fluid within the poorly distensible pericardial sac may cause tamponade ([Chap. 239](#)). Any cause of increased intrathoracic pressure, such as tension pneumothorax, herniation of abdominal viscera through a diaphragmatic hernia, or excessive positive pressure ventilation to support pulmonary function, can also cause compressive cardiogenic shock. Acute right heart failure with a sudden decline in cardiac output can be caused by pulmonary embolism obstructing right ventricular outflow and impairing left ventricular filling. Although initially responsive to increased filling pressures produced by volume expansion, as compression increases, cardiogenic shock occurs.

The diagnosis of compressive cardiogenic shock is most frequently based on clinical findings, the chest radiograph, and an echocardiogram. The diagnosis of compressive cardiac shock may be more difficult to establish in the setting of trauma when hypovolemia and cardiac compression are present simultaneously. The classic findings of pericardial tamponade include the triad of hypotension, neck vein distention, and muffled heart sounds ([Chap. 239](#)). Pulsus paradoxus, i.e., an inspiratory reduction in systolic pressure >10 mmHg, may also be noted. The diagnosis is confirmed by echocardiography, and treatment consists of immediate pericardiocentesis. A tension pneumothorax produces ipsilateral decreased breath sounds, tracheal deviation away from the affected thorax, and jugular venous distention. Radiographic findings include increased intrathoracic volume, depression of the diaphragm of the affected hemithorax, and shifting of the mediastinum to the contralateral side. Chest decompression must be carried out immediately. Release of air and restoration of normal cardiovascular

dynamics is both diagnostic and therapeutic.

SEPTIC SHOCK (See also [Chap. 124](#))

This form of shock is caused by the systemic response to a severe infection. It occurs most frequently in elderly or immunocompromised patients and in those who have undergone an invasive procedure in which bacterial contamination has occurred. Infections of the lung, abdomen, or urinary tract are most common, and approximately half of the patients have bacteremia. Gram-positive and -negative bacteria, viruses, fungi, rickettsiae, and protozoa have all been reported to produce the clinical picture of septic shock, and the overall response is generally independent of the specific type of invading organism. The clinical findings in septic shock are a consequence of the combination of metabolic and circulatory derangements driven by the systemic infection and the release of toxic components of the infectious organisms, e.g., the endotoxin of gram-negative bacteria or the exotoxins and enterotoxins of gram-positive bacteria. Organism toxins lead to the release of cytokines, including [IL-1](#) and [TNF- \$\alpha\$](#) , from tissue macrophages. Tissue factor expression and fibrin deposition are increased, and disseminated intravascular coagulation may develop. The inducible form of NO synthase is stimulated, and NO, a powerful vasodilator, is released. Hemodynamic changes in septic shock occur in two characteristic patterns: early, or hyperdynamic, and late, or hypodynamic, septic shock.

Hyperdynamic Response In hyperdynamic septic shock, tachycardia is present, the cardiac output is normal, and the systemic vascular resistance is reduced while the pulmonary vascular resistance is elevated. The extremities are usually warm. However, splanchnic vasoconstriction with decreased visceral flow is present. The venous capacitance is increased, which decreases venous return. With volume expansion cardiac output becomes supranormal. Myocardial contractility is depressed in septic shock by mediators including NO, [IL-1](#), and/or [TNF- \$\alpha\$](#) . Inflammatory mediator-induced processes include increased capillary permeability and continued loss of intravascular volume.

In septic shock, in contrast to other types of shock, total oxygen delivery may be increased while oxygen extraction is reduced due to maldistribution of microcirculatory perfusion and impaired utilization. In this setting the presence of a normal mixed venous oxygen saturation is not indicative of adequate peripheral perfusion, and even though the cardiac output may be elevated, it is still inadequate to meet the total metabolic needs. The toxicity of the infectious agents and their byproducts and the subsequent metabolic dysfunction drive the progressive deterioration of cellular and organ function. Acute respiratory distress syndrome, thrombocytopenia, and neutropenia are common complications.

Hypodynamic Response As sepsis progresses, vasoconstriction occurs and the cardiac output declines. The patient usually becomes markedly tachypneic, febrile, diaphoretic, and obtunded, with cool, mottled, and often cyanotic extremities. Oliguria, renal failure, and hypothermia develop; there may be striking increases in serum lactate.

TREATMENT

Aggressive volume expansion with a crystalloid solution to a [PCWP](#) of approximately 15 mmHg and the restoration of arterial oxygenation with inspired oxygen and frequently with mechanical ventilation are the highest priorities. In the presence of hypodynamic septic shock, augmentation of cardiac output may require inotropic support with dopamine or norepinephrine in the presence of hypotension or with dobutamine if arterial pressure is normal. Antibiotics should be administered, either appropriate for the results of cultures or empirical therapy based on the likely source of infection. Surgical debridement or drainage may also be necessary to control the infection.

NEUROGENIC SHOCK

Interruption of sympathetic vasomotor input after a high cervical spinal cord injury, inadvertent cephalad migration of spinal anesthesia, or severe head injury may result in neurogenic shock. In addition to arteriolar dilatation, venodilation causes pooling in the venous system, which decreases venous return and cardiac output. The extremities are often warm, in contrast to the usual vasoconstriction-induced coolness in hypovolemic or cardiogenic shock. Treatment involves a simultaneous approach to the relative hypovolemia and to the loss of vasomotor tone. Large volumes of fluid may be required to restore normal hemodynamics. Once hemorrhage has been ruled out, norepinephrine may be necessary to augment vascular resistance.

HYPOADRENAL SHOCK (See also [Chap. 331](#))

The normal host response to the stress of illness, operation, or trauma requires that the adrenal glands hypersecrete cortisol in excess of that normally required. Hypoadrenal shock occurs in settings in which unrecognized adrenal insufficiency complicates the host response to the stress induced by acute illness or major surgery. Adrenocortical insufficiency may occur as a consequence of the chronic administration of high doses of exogenous glucocorticoids. Recent studies have shown that prolonged stays in a critical state in an intensive care setting may also induce a relative hypoadrenal state. Other, less common causes include adrenal insufficiency secondary to idiopathic atrophy, tuberculosis, metastatic disease, bilateral hemorrhage, and amyloidosis. The shock produced by adrenal insufficiency is characterized by reductions in systemic vascular resistance, hypovolemia, and reduced cardiac output. The diagnosis of adrenal insufficiency may be established by means of an [ACTH](#) stimulation test ([Chap. 331](#)).

TREATMENT

In the hemodynamically unstable patient, dexamethasone sodium phosphate, 4 mg, should be given intravenously. This agent is preferred because unlike hydrocortisone it does not interfere with the [ACTH](#) stimulation test. If the diagnosis of adrenal insufficiency has been established, hydrocortisone, 100 mg every 6 to 8 h, can be given and tapered to a maintenance level as the patient achieves hemodynamic stability. Simultaneous volume resuscitation and pressor support is required.

ADJUNCTIVE THERAPIES

As described above, the sympathomimetic amines dobutamine, dopamine, and norepinephrine are widely used in the treatment of all forms of shock. The clinical

pharmacology of these agents is described in [Chap. 72](#).

POSITIONING

Positioning of the patient may be a valuable adjunct in the initial treatment of hypovolemic shock. Elevating the foot of the bed (i.e., placing it on "shock blocks") and assumption of the Trendelenburg position without flexion at the knees are effective but may increase work of breathing and risk for aspiration. Simply elevating both legs may be the optimal approach.

PNEUMATIC ANTISHOCK GARMENT (PASG)

The PASG and the military antishock trousers (MAST) are inflatable external compression devices that can be wrapped around the legs and abdomen and have been widely used in the prehospital setting as a means of providing temporary support of central hemodynamics in shock. They cause an increase in systemic vascular resistance and blood pressure by arterial compression, without causing a significant change in cardiac output. While the use of PASG has been recommended in noncardiogenic forms of shock, the most appropriate use appears to be as a means to tamponade bleeding and augment hemostasis. Inflation of the suit provides splinting of fractures of the pelvis and lower extremities and arrests hemorrhage from fractures.

REWARMING

Hypothermia is a potential adverse consequence of massive volume resuscitation. The infusion of large volumes of refrigerated blood products and room-temperature crystalloid solutions can rapidly drop core temperatures if fluid is not run through warming devices. Hypothermia may depress cardiac contractility and thereby further impair cardiac output and oxygen delivery. Hypothermia, particularly temperatures $<35^{\circ}\text{C}$, directly impairs the coagulation pathway, sometimes causing a significant coagulopathy. Rapid rewarming significantly decreases the requirement for blood products and an improvement in cardiac function. The most effective method for rewarming is extracorporeal countercurrent warmers through femoral artery and vein cannulation. This process does not require a pump and can rewarm from 30° to 36°C in <30 min.

(Bibliography omitted in Palm version)

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39. CARDIOVASCULAR COLLAPSE, CARDIAC ARREST, AND SUDDEN CARDIAC DEATH - Robert J. Myerburg, Agustin Castellanos

OVERVIEW AND DEFINITIONS

The vast majority of naturally occurring sudden deaths are caused by cardiac disorders. The magnitude of sudden *cardiac* death (SCD) as a public health problem is highlighted by estimates that more than 300,000 deaths occur each year in the United States by this mechanism, accounting for 50% of all cardiac deaths. SCD is a direct consequence of cardiac arrest, which is often reversible if responded to promptly. Since resuscitation techniques and emergency rescue systems are available to save patients who have out-of-hospital cardiac arrest, which was uniformly fatal in the past, understanding the SCD problem has practical importance.

[SCD](#) must be defined carefully. In the context of time, "sudden" is defined, for most clinical and epidemiologic purposes, as 1 h or less between the onset of the terminal clinical event, or an abrupt change in clinical status, and death. An exception is unwitnessed deaths in which pathologists may expand the definition of time to 24 h after the victim was last seen to be alive and stable.

Because of community-based interventions, victims may remain biologically alive for days or even weeks after a cardiac arrest that has resulted in irreversible central nervous system damage. Confusion in terms can be avoided by adhering strictly to definitions of death, cardiac arrest, and cardiovascular collapse ([Table 39-1](#)). Death is biologically, legally, and literally an absolute and irreversible event. Death may be delayed in a survivor of cardiac arrest, but "survival after sudden death" is an irrational term. Currently, the accepted definition of SCD is *natural death due to cardiac causes*, heralded by abrupt loss of consciousness within 1 h of the onset of acute symptoms, in an individual who may have known *preexisting* heart disease but in whom the *time* and *mode* of death are *unexpected*. When biologic death of the cardiac arrest victim is delayed because of interventions, the relevant pathophysiologic event remains the sudden and unexpected cardiac arrest that leads ultimately to death, even though delayed by artificial methods. The language used should reflect the fact that the index event was a cardiac arrest and that death was due to its delayed consequences.

ETIOLOGY, INITIATING EVENTS, AND CLINICAL EPIDEMIOLOGY

Clinical and epidemiologic studies have identified populations at high risk for [SCD](#). In addition, a large body of pathologic data provides information on the underlying *structural abnormalities* in victims of SCD, and studies of clinical physiology have begun to identify a group of *transient functional factors* that may convert a long-standing underlying structural abnormality from a stable to an unstable state ([Table 39-2](#)). This information is developing into an understanding of the causes and mechanisms of SCD.

Cardiac disorders constitute the most common causes of sudden *natural* death. After an initial peak incidence of sudden death between birth and 6 months of age (the sudden infant death syndrome), the incidence of sudden death declines sharply and remains low through childhood and adolescence. Among adolescents and young adults, the incidence of [SCD](#) is approximately 1 per 100,000 population per year. The incidence

begins to increase in adults over the age of 30 years, reaching a second peak in the age range of 45 to 75 years, when the incidence approximates 1 to 2 per 1000 per year among the unselected adult population. Increasing age within this range is a powerful risk factor for sudden *cardiac* death, and the proportion of cardiac causes among all sudden natural deaths increases dramatically with advancing years. From 1 to 13 years of age, only one of five sudden *natural* deaths is due to cardiac causes. Between 14 and 21 years of age, the proportion increases to 30%, and then to 88% in the middle-aged and elderly.

Young and middle-aged men and women have very different susceptibilities to [SCD](#), but the gender differences decrease with advancing age. In the 45- to 64-year-old age group, the male SCD excess is nearly 7:1. It falls to 2:1 or less in the 65- to 74-year-old age group. The difference in risk for SCD parallels the risks for other manifestations of coronary heart disease in men and women. As the gender gap for manifestations of coronary heart disease closes in the seventh and eighth decades of life, the excess risk of SCD in males also narrows. Despite the lower incidence among younger women, coronary risk factors such as cigarette smoking, diabetes, hyperlipidemia, and hypertension are highly influential, and SCD remains an important clinical and epidemiologic problem.

Hereditary factors contribute to the risk of [SCD](#), but largely in a nonspecific manner; they represent expressions of the hereditary predisposition to coronary heart disease. A few specific syndromes, such as congenital long QT interval syndromes ([Chap. 230](#)), right ventricular dysplasia, and the syndrome of right bundle branch block and non-ischemic ST-segment elevations (Brugada syndrome), are characterized by specific hereditary risk of SCD. There are also recent data suggesting a familial predisposition to SCD as a specific pattern of coronary heart disease expression.

The major categories of structural causes of, and functional factors contributing to, the [SCD](#) syndrome are listed in [Table 39-2](#). Worldwide, and especially in western cultures, coronary atherosclerotic heart disease is the most common structural abnormality associated with SCD. Up to 80% of all SCDs in the United States are due to the consequences of coronary atherosclerosis. The cardiomyopathies (dilated and hypertrophic, collectively; [Chap. 239](#)) account for another 10 to 15% of SCDs, and all the remaining diverse etiologies cause only 5 to 10% of these events. Transient ischemia in the previously scarred or hypertrophied heart, hemodynamic and fluid and electrolyte disturbances, fluctuations in autonomic nervous system activity, and transient electrophysiologic changes caused by drugs or other chemicals (e.g., proarrhythmia) have all been implicated as mechanisms responsible for transition from electrophysiologic stability to instability. In addition, reperfusion of ischemic myocardium may cause transient electrophysiologic instability and arrhythmias.

PATHOLOGY

Data from postmortem examinations of [SCD](#) victims parallel the clinical observations on the prevalence of coronary heart disease as the major structural etiologic factor. More than 80% of SCD victims have pathologic findings of coronary heart disease. The pathologic description often includes a combination of long-standing, extensive atherosclerosis of the epicardial coronary arteries and acute active coronary lesions,

which include a combination of fissured or ruptured plaques, platelet aggregates, hemorrhage, and thrombosis. In one study, chronic coronary atherosclerosis involving two or more major vessels with ³75% stenosis was observed in 75% of the victims. In another study, atherosclerotic plaque fissuring, platelet aggregates, and/or acute thrombosis were observed in 95 of 100 individuals who had pathologic studies after SCD.

As many as 70 to 75% of males who die suddenly have prior myocardial infarctions (MIs), but only 20 to 30% have recent acute MIs. A high incidence of left ventricular (LV) hypertrophy coexists with prior MIs.

CLINICAL DEFINITION OF FORMS OF CARDIOVASCULAR COLLAPSE ([Table 39-1](#))

Cardiovascular collapse is a general term connoting loss of effective blood flow due to acute dysfunction of the heart and/or peripheral vasculature. Cardiovascular collapse may be caused by vasodepressor syncope (vasovagal syncope, postural hypotension with syncope, neurocardiogenic syncope -- [Chap. 21](#)), a transient severe bradycardia, or cardiac arrest. The latter is distinguished from the transient forms of cardiovascular collapse in that it usually requires an intervention to achieve resuscitation. In contrast, vasodepressor syncope and many primary bradyarrhythmic syncopal events are transient and non-life-threatening, with spontaneous return of consciousness.

The most common electrical mechanism for true cardiac arrest is ventricular fibrillation (VF), which is responsible for 65 to 80% of cardiac arrests. Severe persistent bradyarrhythmias, asystole, and pulseless electrical activity (an organized electrical activity without mechanical response, formerly called electromechanical dissociation) cause another 20 to 30%. Sustained ventricular tachycardia (VT) with hypotension is a less common cause. Acute low cardiac output states, having precipitous onset, also may present clinically as a cardiac arrest. The causes include massive acute pulmonary emboli, internal blood loss from ruptured aortic aneurysm, intense anaphylaxis, cardiac rupture after myocardial infarction, and unexpected fatal arrhythmia due to electrolyte disturbances.

CLINICAL CHARACTERISTICS OF CARDIAC ARREST

PRODROME, ONSET, ARREST, DEATH

[SCD](#) may be presaged by days, weeks, or months of increasing angina, dyspnea, palpitations, easy fatigability, and other nonspecific complaints. However, these *prodromal complaints* are generally predictive of any major cardiac event; they are not specific for predicting SCD.

The *onset of the terminal event*, leading to cardiac arrest, is defined as an acute change in cardiovascular status preceding cardiac arrest by up to 1 h. When the onset is instantaneous or abrupt, the probability that the arrest is cardiac in origin is >95%. Continuous ECG recordings, fortuitously obtained at the onset of a cardiac arrest, commonly demonstrate changes in cardiac electrical activity in the minutes or hours before the event. There is a tendency for the heart rate to increase and for advanced

grades of premature ventricular contractions (PVCs) to evolve. Most cardiac arrests that are caused by [VF](#) begin with a run of sustained or nonsustained [VT](#), which then degenerates into VF.

Sudden unexpected loss of effective circulation may be separated into "arrhythmic events" and "circulatory failure." Arrhythmic events are characterized by a high likelihood of patients being awake and active immediately prior to the event, are dominated by [VF](#) as the electrical mechanism, and have a short duration of terminal illness (<1 h). In contrast, circulatory failure deaths occur in patients who are inactive or comatose, have a higher incidence of asystole than VF, have a tendency to a longer duration of terminal illness, and are dominated by noncardiac events preceding the terminal illness.

The onset of cardiac arrest may be characterized by typical symptoms of an acute cardiac event, such as prolonged angina or the pain of onset of [MI](#), acute dyspnea or orthopnea, or the sudden onset of palpitations, sustained tachycardia, or light-headedness. However, in many patients, the onset is precipitous, with minimal forewarning.

Cardiac arrest is, by definition, abrupt. Mentation may be impaired in patients with sustained [VT](#) during the onset of the terminal event. However, complete loss of consciousness is a *sine qua non* in cardiac arrest. Although rare spontaneous reversions occur, it is usual that cardiac arrest progresses to death within minutes (i.e., [SCD](#) has occurred) if active interventions are not undertaken promptly.

The probability of achieving successful resuscitation from cardiac arrest is related to the interval from onset to institution of resuscitative efforts, the setting in which the event occurs, the mechanism ([VF](#), [VT](#), pulseless electrical activity, asystole), and the clinical status of the patient prior to the cardiac arrest. Those settings in which it is possible to institute prompt cardiopulmonary resuscitation (CPR) provide a better chance of a successful outcome. However, the outcome in intensive care units and other in-hospital environments is heavily influenced by the patient's preceding clinical status. The immediate outcome is good for cardiac arrest occurring in the intensive care unit in the presence of an acute cardiac event or transient metabolic disturbance, but the outcome for patients with far-advanced chronic cardiac disease or advanced noncardiac diseases (e.g., renal failure, pneumonia, sepsis, diabetes, cancer) is not much more successful in hospital than in the out-of-hospital setting.

The success rate for initial resuscitation and survival to hospital discharge after an out-of-hospital cardiac arrest depends in part on the mechanism of the event. When the mechanism is [VT](#), the outcome is best; [VF](#) is the next most successful; and asystole and pulseless electrical activity generate dismal outcome statistics ([Fig. 39-1](#)). Advanced age also influences adversely the chances of successful resuscitation.

Progression to biologic death is a function of the mechanism of cardiac arrest and the length of the delay before interventions. [VF](#) or asystole without [CPR](#) within the first 4 to 6 min has a poor outcome, and there are few survivors among patients who had no life support activities for the first 8 min after onset. Outcome statistics are improved by lay bystander intervention (basic life support -- see below) prior to definitive interventions

(advanced life support -- defibrillation) and even more by early defibrillation. In regard to the latter, the notion that deployment of automatic external defibrillators in communities (e.g., police vehicles, large buildings, stadiums, etc.) will result in improved survival is currently being evaluated.

Death during the hospitalization after a successfully resuscitated cardiac arrest relates closely to the severity of central nervous system injury. Anoxic encephalopathy and infections subsequent to prolonged respirator dependence account for 60% of the deaths. Another 30% occur as a consequence of low cardiac output states that fail to respond to interventions. Recurrent arrhythmias are the least common cause of death, accounting for only 10% of in-hospital deaths.

In the setting of acute [MI](#), it is important to distinguish between primary and secondary cardiac arrests. *Primary* cardiac arrests refer to those that occur in the absence of hemodynamic instability, and *secondary* cardiac arrests are those that occur in patients in whom abnormal hemodynamics dominate the clinical picture before cardiac arrest. The success rate for immediate resuscitation in primary cardiac arrest during acute MI in a monitored setting should approach 100%. In contrast, as many as 70% of patients with secondary cardiac arrest succumb immediately or during the same hospitalization.

IDENTIFICATION OF PATIENTS AT RISK FOR SUDDEN CARDIAC DEATH

Primary prevention of cardiac arrest depends on the ability to identify individual patients at high risk. One must view the problem in the context of the total number of events and the population pools from which they are derived. The annual incidence of [SCD](#) among an unselected adult population is 1 to 2 per 1000 population ([Fig. 39-2A](#)), largely reflecting the prevalence of those coronary heart disease patients among whom SCD is the first clinically recognized manifestation (20 to 25% of first coronary events are SCD). The incidence (percent per year) increases progressively with the addition of identified coronary risk factors to populations free of prior coronary events. The most powerful factors are age, elevated blood pressure, [LV](#) hypertrophy, cigarette smoking, elevated serum cholesterol level, obesity, and nonspecific electrocardiographic abnormalities. These coronary risk factors are not specific for SCD but rather represent increasing risk for all coronary deaths. The proportion of coronary deaths that are sudden remains at approximately 50% in all risk categories. Despite the marked *relative* increased risk of SCD with addition of multiple risk factors (from 1 to 2 per 1000 population per year in an unselected population to as much as 50 to 60 per 1000 in subgroups having multiple risk factors for coronary artery disease), the *absolute* incidence remains relatively low when viewed as the relationship between the number of individuals who have a preventive intervention and the number of events that can be prevented. Specifically, a 50% reduction in annual SCD risk would be a huge *relative* decrease but would require an intervention in up to 200 unselected individuals to prevent one sudden death. These figures highlight the importance of primary prevention of coronary heart disease. Control of coronary risk factors may be the only practical method to prevent SCD in major segments of the population, because of the paradox that the majority of events occur in the large unselected subgroups rather than in the specific high-risk subgroups (compare "Events/Year" with "Percent/Year" in [Fig. 39-2A](#)). Under most conditions of higher level of risk, particularly those indexed to a recent major cardiovascular event (e.g., [MI](#), recent onset of heart failure, survival after out-of-hospital cardiac arrest), the highest risk of

sudden death occurs within the initial 6 to 18 months and then decreases toward baseline risk of the underlying disease ([Fig. 39-2B](#)). Accordingly, preventive interventions are most likely to be effective when initiated early.

For patients with acute or prior clinical manifestations of coronary heart disease, high-risk subgroups having a much higher ratio of [SCD](#) risk to population base can be identified. The acute, convalescent, and chronic phases of [MI](#) provide large population subsets with more highly focused risk ([Chap. 243](#)). The potential risk of cardiac arrest from the onset through the first 72 h after acute MI (the acute phase) may be as high as 15 to 20%. The highest risk of SCD in relation to MI is found in the subgroup that has experienced sustained [VT](#) or [VF](#) during the convalescent phase (3 days to 8 weeks) after MI. A greater than 50% mortality in 6 to 12 months has been observed among these patients, when managed with conservative medical therapy, and at least 50% of the deaths are sudden. Aggressive intervention techniques may reduce this incidence.

After the acute phase of [MI](#), long-term risk for total mortality and [SCD](#) are predicted by a number of factors. The most important for both SCD and non-SCD is the extent of myocardial damage sustained during the acute event. This is measured by the degree of reduction in the ejection fraction (EF), functional capacity, and/or the occurrence of heart failure. Increasing *frequency* of postinfarction [PVCs](#), with a plateau above the range of 10 to 30 PVCs per hour on 24-h ambulatory monitor recordings, also indicates increased risk, but advanced *forms* (salvos, nonsustained [VT](#)) may be more powerful predictors. PVCs interact strongly with decreased left ventricular EF. The combination of frequent PVCs, salvos or nonsustained VT, and an EF $\leq$ 35% identifies patients who have an annual risk of greater than 20%. The risk falls off sharply with decreasing PVC frequency and the absence of advanced forms, as well as with higher EF. Despite the risk implications of postinfarction PVCs, improved outcome as a result of PVC suppression has not been demonstrated ([Chap. 230](#)).

The extent of underlying disease due to any cause and/or prior clinical expression of risk of [SCD](#) (i.e., survival after out-of-hospital cardiac arrest not associated with acute [MI](#)) identify patients at very high risk for subsequent (recurrent) cardiac arrest. Survival after out-of-hospital cardiac arrest predicts up to a 30% 1-year recurrent cardiac arrest rate in the absence of specific interventions (see below).

A general rule is that the risk of [SCD](#) is approximately one-half the total cardiovascular mortality rate. As shown in [Fig. 39-2A](#), the very high risk subgroups provide more focused population fractions ("Percent/Year") for predicting cardiac arrest or SCD; but the impact on the overall population, indicated by the absolute number of preventable events ("Events/Year"), is considerably smaller. The requirements for achieving a major population impact are effective prevention of the underlying diseases and/or new epidemiologic probes that will allow better resolution of subgroups within large general populations.

TREATMENT

The individual who collapses suddenly is managed in four stages: (1) the initial response and basic life support; (2) advanced life support; (3) postresuscitation care; and (4) long-term management. The initial response and basic life support can be

carried out by physicians, nurses, paramedical personnel, and trained lay persons. There is a requirement for increasingly specialized skills as the patient moves through the stages of advanced life support, postresuscitation care, and long-term management.

Initial Response and Basic Life Support The initial response will confirm whether a sudden collapse is indeed due to a cardiac arrest. Observations for respiratory movements, skin color, and the presence or absence of pulses in the carotid or femoral arteries will promptly determine whether a life-threatening cardiac arrest has occurred. As soon as a cardiac arrest is suspected or confirmed, contacting an emergency rescue system (e.g., 911) should be the immediate priority.

Agonal respiratory movements may persist for a short time after the onset of cardiac arrest, but it is important to observe for severe stridor with a persistent pulse as a clue to aspiration of a foreign body or food. If this is suspected, a Heimlich maneuver (see below) may dislodge the obstructing body. A precordial blow, or "thump," delivered firmly by the clenched fist to the junction of the middle and lower third of the sternum may occasionally revert VT or VF, but there is concern about converting VT to VF. Therefore, it has been recommended to use precordial thumps as an advanced life support technique when monitoring and defibrillation are available. This conservative application of the technique remains controversial.

The third action during the initial response is to clear the airway. The head is tilted back and chin lifted so that the oropharynx can be explored to clear the airway. Dentures or foreign bodies are removed, and the Heimlich maneuver is performed if there is reason to suspect that a foreign body is lodged in the oropharynx. If respiratory arrest precipitating cardiac arrest is suspected, a second precordial thump is delivered after the airway is cleared.

Basic life support, more popularly known as CPR, is intended to maintain organ perfusion until definitive interventions can be instituted. The elements of CPR are the maintenance of ventilation of the lungs and compression of the chest. Mouth-to-mouth respiration may be used if no specific rescue equipment is immediately available (e.g., plastic oropharyngeal airways, esophageal obturators, masked Ambu bag). Conventional ventilation techniques during CPR require the lungs to be inflated 10 to 12 times per minute, i.e., once every fifth chest compression when two persons are performing the resuscitation and twice in succession every 15 chest compressions when one person is carrying out both ventilation and chest wall compression.

Chest compression is based on the assumption that cardiac compression allows the heart to maintain a pump function by sequential filling and emptying of its chambers, with competent valves maintaining forward direction of flow. The palm of one hand is placed over the lower sternum, with the heel of the other resting on the dorsum of the lower hand. The sternum is depressed, with the arms remaining straight, at a rate of approximately 80 to 100 per minute. Sufficient force is applied to depress the sternum 3 to 5 cm, and relaxation is abrupt.

Advanced Life Support Advanced life support is intended to achieve adequate ventilation, control cardiac arrhythmias, stabilize blood pressure and cardiac output, and restore organ perfusion. The activities carried out to achieve these goals include (1)

intubation with an endotracheal tube, (2) defibrillation/cardioversion and/or pacing, and (3) insertion of an intravenous line. Ventilation with O₂ (room air if O₂ is not immediately available) may promptly reverse hypoxemia and acidosis. The speed with which defibrillation/cardioversion is carried out is an important element for successful resuscitation. When possible, immediate defibrillation should precede intubation and insertion of an intravenous line; [CPR](#) should be carried out while the defibrillator is being charged. As soon as a diagnosis of [VT](#) or [VF](#) is obtained, a 200-J shock should be delivered. Additional shocks at higher energies, up to a maximum of 360 J, are tried if the initial shock does not successfully abolish VT or VF. Epinephrine, 1 mg intravenously, is given after failed defibrillation, and attempts to defibrillate are repeated. The dose of epinephrine may be repeated after intervals of 3 to 5 min (see [Fig. 39-3A](#)).

If the patient is less than fully conscious upon reversion, or if two or three attempts fail, prompt intubation, ventilation, and arterial blood gas analysis should be carried out. Intravenous NaHCO₃, which was formerly used in large quantities, is no longer considered routinely necessary and may be dangerous in larger quantities. However, the patient who is persistently acidotic after successful defibrillation and intubation should be given 1 meq/kg NaHCO₃ initially and an additional 50% of the dose repeated every 10 to 15 min.

After initial unsuccessful defibrillation attempts, or with persistent electrical instability, a bolus of 1 mg/kg lidocaine is given intravenously ([Chap. 243](#)), and the dose is repeated in 2 min in those patients who have persistent ventricular arrhythmias or remain in [VF](#). This is followed by a continuous infusion at a rate of 1 to 4 mg/min. If lidocaine fails to provide control, other antiarrhythmic therapies should be tried. For persistent, hemodynamically unstable ventricular arrhythmias, intravenous amiodarone has emerged as the treatment of choice (150 mg over 10 min, followed by 1 mg/min for up to 6 h, and 0.5 mg/min thereafter) ([Fig. 39-3A](#)). Intravenous procainamide (loading infusion of 100 mg/5 min to a total dose of 500 to 800 mg, followed by continuous infusion at 2 to 5 mg/min) may be tried for persisting, hemodynamically stable arrhythmias; or bretylium tosylate (loading dose 5 to 10 mg/kg in 5 min; maintenance dose 0.5 to 2 mg/min) may be tried as an alternative for unstable arrhythmias. Intravenous calcium gluconate is no longer considered safe or necessary for routine administration. It is used only in patients in whom acute hyperkalemia is known to be the triggering event for resistant VF, in the presence of known hypocalcemia, or in patients who have received toxic doses of calcium channel antagonists.

Cardiac arrest secondary to bradyarrhythmias or asystole is managed differently ([Fig. 39-3B](#)). The patient is promptly intubated, [CPR](#) is continued, and an attempt is made to control hypoxemia and acidosis. Epinephrine and/or atropine are given intravenously or by an intracardiac route. External pacing devices are now available to attempt to establish a regular rhythm, but the prognosis is generally very poor in this form of cardiac arrest, even with successful electrical pacing. Pulseless electrical activity (PEA) is treated similarly to bradyarrhythmias, but its outcome is also dismal. The one exception is bradyarrhythmic/asystolic cardiac arrest secondary to airway obstruction. This form of cardiac arrest may respond promptly to removal of foreign bodies by the Heimlich maneuver or, in hospitalized patients, by intubation and suctioning of obstructing secretions in the airway.

Postresuscitation Care This phase of management is determined by the clinical setting of the cardiac arrest. *Primary* VF in acute MI (Chap. 243) is generally very responsive to life-support techniques and easily controlled after the initial event. Patients are maintained on a lidocaine infusion at the rate of 2 to 4 mg/min for 24 to 72 h after the event. In the in-hospital setting, respirator support is usually not necessary or is needed for only a short time, and hemodynamics stabilize promptly after defibrillation or cardioversion. In *secondary* VF in acute MI (those events in which hemodynamic abnormalities predispose to the potentially fatal arrhythmia), resuscitative efforts are less often successful, and in those patients who are successfully resuscitated, the recurrence rate is high. The clinical picture and outcome are dominated by hemodynamic instability and the ability to control hemodynamic dysfunction. Bradyarrhythmias, asystole, and pulseless electrical activity are commonly secondary events in hemodynamically unstable patients.

The outcome after in-hospital cardiac arrest associated with *non-cardiac* diseases is poor, and in the few successfully resuscitated patients, the postresuscitation course is dominated by the nature of the underlying disease. Patients with cancer, renal failure, acute central nervous system disease, and uncontrolled infections, as a group, have a survival rate of less than 10% after in-hospital cardiac arrest. Some major exceptions are patients with transient airway obstruction, electrolyte disturbances, proarrhythmic effects of drugs, and severe metabolic abnormalities, most of whom may have an excellent chance of survival if they can be resuscitated promptly and maintained while the transient abnormalities are being corrected.

Long-Term Management after Survival of Out-of-Hospital Cardiac Arrest Patients who do not suffer irreversible injury of the central nervous system and who achieve hemodynamic stability should have extensive diagnostic and therapeutic testing to guide long-term management. This aggressive approach is driven by the fact that statistics from the 1970s indicated survival after out-of-hospital cardiac arrest was followed by a 30% recurrent cardiac arrest rate at 1 year, 45% at 2 years, and a total mortality rate of almost 60% at 2 years. Historical comparisons suggest that these dismal statistics may be significantly improved by newer interventions, but the magnitude of the improvement is unknown because of the lack of concurrently controlled intervention studies.

Among those patients in whom an acute transmural MI is the cause of out-of-hospital cardiac arrest, the management is the same as in any other patient who suffers cardiac arrest during the acute phase of a documented MI (Chap. 243). For almost all other categories of patients, however, extensive diagnostic studies are carried out to determine etiology, functional impairment, and electrophysiologic instability as guides to future management. In general, patients who have out-of-hospital cardiac arrest due to chronic ischemic heart disease, without an acute MI, are evaluated to determine whether transient ischemia or chronic electrophysiologic instability was the more likely cause of the event. If there is reason to suspect an ischemic mechanism, coronary revascularization by angioplasty or bypass surgery, plus drugs (most commonly beta blockers), are used to reduce ischemic burden.

Electrophysiologic instability has been identified by the use of programmed electrical stimulation to determine whether sustained VT or VF can be induced (Chap. 230). If so, this information can be used as a baseline against which to evaluate drug efficacy for

prevention of inducibility. The rationale for this approach is the assumption that suppression of inducibility predicts long-term benefit by the drug that achieves such suppression. For patients for whom successful drug therapy could not be identified by this technique, insertion of an implantable cardioverter-defibrillator (ICD), antiarrhythmic surgery (e.g., coronary bypass surgery, aneurysmectomy, cryoablation), or empiric amiodarone therapy have been recommended ([Chap. 230](#)). Primary surgical success, defined as surviving the procedure and reverting to a noninducible status without drug therapy, is better than 90% when patients are selected for ability to be mapped in the operating room. However, only a small fraction of patients meet the criteria. In addition, VT/VF *cannot* be induced in a number of survivors of cardiac arrest (30 to 50%), and inducible arrhythmias can be suppressed by drugs in no more than 20 to 30% of those whose arrhythmias can be induced. Because of these limitations of drug therapy and surgical approaches, ICD therapy has evolved into the most commonly used strategy for cardiac arrest survivors. ICDs have long been recognized to have very good success rates for sensing and reverting life-threatening arrhythmias, but improvement in long-term total survival outcomes remained lacking until a number of studies solidified the benefit of ICD therapy for specific subgroups. After empiric amiodarone therapy had been suggested to be as good as, or better than, conventional antiarrhythmic drug therapy for survivors of cardiac arrest, ICDs were demonstrated to be superior to amiodarone. Moreover, ICDs were also found to be superior for high risk patients with VT after myocardial infarction.

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SECTION 6 -ALTERATIONS IN GASTROINTESTINAL FUNCTION

40. DYSPHAGIA - Raj K. Goyal

Dysphagia is defined as a sensation of "sticking" or obstruction of the passage of food through the mouth, pharynx, or esophagus. It should be distinguished from other symptoms related to swallowing. *Aphagia* signifies complete esophageal obstruction, which is usually due to bolus impaction and represents a medical emergency. *Difficulty in initiating a swallow* occurs in disorders of the voluntary phase of swallowing. However, once initiated, swallowing is completed normally. *Odynophagia* means painful swallowing. Frequently, odynophagia and dysphagia occur together. *Globus pharyngeus* is the sensation of a lump lodged in the throat. However, no difficulty is encountered when swallowing is performed. *Misdirection of food*, resulting in nasal regurgitation and laryngeal and pulmonary aspiration of food during swallowing, is characteristic of oropharyngeal dysphagia. *Phagophobia*, meaning fear of swallowing, and *refusal to swallow* may occur in hysteria, rabies, tetanus, and pharyngeal paralysis due to fear of aspiration. Painful inflammatory lesions that cause odynophagia may also cause refusal to swallow. Some patients may feel the food as it goes down the esophagus. This esophageal sensitivity is not associated with either food sticking or obstruction, however. Similarly, the *feeling of fullness in the epigastrium* that occurs after a meal or after swallowing air should not be confused with dysphagia.

PHYSIOLOGY OF SWALLOWING

The process of swallowing begins with a voluntary (oral) phase during which a bolus of food is pushed into the pharynx by the contraction of the tongue. The bolus then activates oropharyngeal sensory receptors that initiate the involuntary (pharyngeal and esophageal) phase, or deglutition reflex. The deglutition reflex is a complex series of events and serves both to propel food through the pharynx and the esophagus and to prevent its entry into the airway. When the bolus is propelled backward by the tongue, the larynx moves forward and the upper esophageal sphincter opens. As the bolus moves into the pharynx, contraction of the superior pharyngeal constrictor against the contracted soft palate initiates a peristaltic contraction that proceeds rapidly downward to move the bolus through the pharynx and the esophagus. The lower esophageal sphincter opens as the food enters the esophagus and remains open until the peristaltic contraction has swept the bolus into the stomach. Peristaltic contraction in response to a swallow is called *primary peristalsis*. It involves inhibition followed by sequential contraction of muscles along the entire swallowing passage. The inhibition that precedes the peristaltic contraction is called *deglutitive inhibition*. Local distention of the esophagus from food activates intramural reflexes in the smooth muscle and results in *secondary peristalsis*, which is limited to the thoracic esophagus. *Tertiary contractions* are nonperistaltic because they occur simultaneously over a long segment of the esophagus. Tertiary contractions may occur in response to a swallow or esophageal distention, or they may occur spontaneously.

PATHOPHYSIOLOGY OF DYSPHAGIA

The normal transport of an ingested bolus through the swallowing passage depends on the size of the ingested bolus; the luminal diameter of the swallowing passage; the force

of peristaltic contraction; and deglutitive inhibition, including normal relaxation of upper and lower esophageal sphincters during swallowing. Dysphagia caused by a large bolus or luminal narrowing is called *mechanical dysphagia*, whereas dysphagia due to weakness of peristaltic contractions or to impaired deglutitive inhibition causing nonperistaltic contractions and impaired sphincter relaxation is called *motor dysphagia*.

Mechanical Dysphagia Mechanical dysphagia can be caused by a very large food bolus, intrinsic narrowing, or extrinsic compression of the lumen. In an adult, the esophageal lumen can distend up to 4 cm in diameter. When the esophagus cannot dilate beyond 2.5 cm in diameter, dysphagia to normal solid food can occur. Dysphagia is always present when the esophagus cannot distend beyond 1.3 cm. Circumferential lesions produce dysphagia more consistently than do lesions that involve only a portion of circumferences of the esophageal wall, as uninvolved segments retain their distensibility. The causes of mechanical dysphagia are listed in [Table 40-1](#). Common causes include carcinoma, peptic and other benign strictures, and lower esophageal ring.

Motor Dysphagia Motor dysphagia may result from difficulty in initiating a swallow or from abnormalities in peristalsis and deglutitive inhibition due to diseases of the esophageal striated or smooth muscle.

Diseases of the striated muscle involve the pharynx, upper esophageal sphincter, and cervical esophagus. The striated muscle is innervated by a somatic component of the vagus with cell bodies of the lower motor neurons located in the nucleus ambiguus. These neurons are cholinergic and excitatory and are the sole determinant of the muscle activity. Peristalsis in the striated muscle segment is due to sequential central activation of neurons innervating muscles at different levels along the esophagus. Motor dysphagia of the pharynx results from neuromuscular disorders causing muscle paralysis, simultaneous nonperistaltic contraction, or loss of opening of the upper esophageal sphincter. Loss of opening of the upper sphincter is caused by paralysis of geniohyoid and other suprahyoid muscles or loss of deglutitive inhibition of the cricopharyngeus muscle. Because each side of the pharynx is innervated by ipsilateral nerves, a unilateral lesion of motor neurons leads to unilateral pharyngeal paralysis. Although lesions of striated muscle also involve the cervical part of the esophagus, the clinical manifestations of pharyngeal dysfunction usually overshadow those due to esophageal involvement.

Diseases of the smooth-muscle segment involve the thoracic part of the esophagus and the lower esophageal sphincter. The smooth muscle is innervated by the parasympathetic component of the vagal preganglionic fibers and postganglionic neurons in the myenteric ganglia. The vagal pathway consists of parallel excitatory and inhibitory pathways that use acetylcholine and nitric oxide as neurotransmitters, respectively. The activation of inhibitory nerves causes inhibition that is followed by rebound contraction. These pathways are involved in the resting tone of the lower esophageal sphincter as well as swallow-induced lower esophageal sphincter opening and inhibition followed by peristaltic contractions in the esophageal body. Dysphagia results when the peristaltic contractions are weak or nonperistaltic or when the lower sphincter fails to relax normally. Loss of contractile power occurs due to muscle weakness, as in scleroderma. The nonperistaltic contractions and impaired relaxation of

the lower esophageal sphincter result from a defect in inhibitory vagal innervation and account for dysphagia in achalasia.

The causes of motor dysphagia are also listed in [Table 40-1](#). Important causes are pharyngeal paralysis, cricopharyngeal achalasia, scleroderma of the esophagus, achalasia, and diffuse esophageal spasm and related motor disorders.

Approach to the Patient

History The history can provide a presumptive diagnosis in over 80% of patients. The type of food causing dysphagia provides useful information. Difficulty only with solids implies mechanical dysphagia with a lumen that is not severely narrowed. In advanced obstruction, dysphagia occurs with liquids as well as solids. In contrast, motor dysphagia due to achalasia and diffuse esophageal spasm is equally affected by solids and liquids from the very onset. Patients with scleroderma have dysphagia to solids that is unrelated to posture and to liquids while recumbent but not upright. When peptic stricture develops in patients with scleroderma, dysphagia becomes more persistent.

The duration and course of dysphagia are helpful in diagnosis. Transient dysphagia may be due to an inflammatory process. Progressive dysphagia lasting a few weeks to a few months is suggestive of carcinoma of the esophagus. Episodic dysphagia to solids lasting several years indicates a benign disease characteristic of a lower esophageal ring.

The site of dysphagia described by the patient helps to determine the site of esophageal obstruction; the lesion is at or below the perceived location of dysphagia.

Associated symptoms provide important diagnostic clues. Nasal regurgitation and tracheobronchial aspiration with swallowing are hallmarks of pharyngeal paralysis or a tracheoesophageal fistula. Tracheobronchial aspiration unrelated to swallowing may be secondary to achalasia, Zenker's diverticulum, or gastroesophageal reflux.

Severe weight loss that is out of proportion to the degree of dysphagia is highly suggestive of carcinoma. When hoarseness precedes dysphagia, the primary lesion is usually in the larynx. Hoarseness following dysphagia may suggest involvement of the recurrent laryngeal nerve by extension of esophageal carcinoma. Sometimes hoarseness may be due to laryngitis secondary to gastroesophageal reflux. Association of laryngeal symptoms and dysphagia also occurs in various neuromuscular disorders. Hiccups may rarely occur with a lesion in the distal portion of the esophagus. Unilateral wheezing with dysphagia indicates a mediastinal mass involving the esophagus and a large bronchus.

Chest pain with dysphagia occurs in diffuse esophageal spasm and related motor disorders. Chest pain resembling diffuse esophageal spasms may occur in esophageal obstruction due to a large bolus. A prolonged history of heartburn and reflux preceding dysphagia indicates peptic stricture. A history of prolonged nasogastric intubation, ingestion of caustic agents, ingestion of pills without water, previous radiation therapy, or associated mucocutaneous diseases may provide the cause of esophageal stricture. If odynophagia is present, candidal or herpes esophagitis or pill-induced esophagitis

should be suspected.

In patients with AIDS or other immunodeficiency states, esophagitis due to opportunistic infections such as *Candida*, herpes simplex virus, or cytomegalovirus and tumors such as Kaposi's sarcoma and lymphoma should be suspected.

Physical Examination Physical examination is important in motor dysphagia due to skeletal muscle, neurologic, and oropharyngeal diseases. Signs of bulbar or pseudobulbar palsy, including dysarthria, dysphonia, ptosis, tongue atrophy, and hyperactive jaw jerk, in addition to evidence of generalized neuromuscular disease, should be sought. The neck should be examined for thyromegaly or a spinal abnormality. A careful inspection of the mouth and pharynx should disclose lesions that may interfere with passage of food because of pain or obstruction. Changes in the skin and extremities may suggest a diagnosis of scleroderma and other collagen-vascular diseases or mucocutaneous diseases such as pemphigoid or epidermolysis bullosa, which may involve the esophagus. Cancer spread to lymph nodes and liver may be evident. Pulmonary complications of acute aspiration pneumonia or chronic aspiration may be present.

Diagnostic Procedures Dysphagia is nearly always a symptom of organic disease rather than a functional complaint. If oropharyngeal dysphagia is suspected, videofluoroscopy of oropharyngeal swallowing should be obtained. If mechanical dysphagia is suspected on clinical history, barium swallow, esophagogastrosocopy and endoscopic biopsies are the diagnostic procedures of choice. Barium swallow and esophageal motility studies are diagnostic tests for motor dysphagia. Esophagogastrosocopy may be needed in patients with motor dysphagia to exclude an associated structural abnormality ([Chap. 284](#)).

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41. NAUSEA, VOMITING, AND INDIGESTION - William L. Hasler

Nausea is the subjective feeling of a need to vomit. Vomiting (emesis) is the oral expulsion of upper gastrointestinal contents resulting from contractions of gut and thoracoabdominal wall musculature. *Vomiting* is contrasted with regurgitation, the effortless passage of gastric contents into the mouth. *Rumination* is the repeated regurgitation of stomach contents, which are often rechewed and then reswallowed. In contrast to vomiting, these phenomena often exhibit some volitional control. *Indigestion* is a nonspecific term that encompasses a variety of upper abdominal complaints including nausea, vomiting, heartburn, regurgitation, and dyspepsia (upper abdominal discomfort or pain). Individuals with ulcer-like dyspepsia report epigastric burning or gnawing discomfort. Dysmotility-like dyspepsia is characterized by postprandial fullness, bloating, eructation (belching), anorexia (loss of appetite), and early satiety (an inability to complete a meal due to premature fullness).

NAUSEA AND VOMITING

MECHANISMS

Vomiting is coordinated by the brain stem and is effected by neuromuscular responses in the gut, pharynx, and thoracoabdominal wall. The mechanisms underlying nausea are poorly understood. Because nausea requires conscious perception, the sensation is probably mediated by the cerebral cortex. Electroencephalographic studies show activation of temporofrontal cortical regions with induction of nausea.

Coordination of Emesis Animal studies suggested that vomiting was coordinated by a single locus in the medullary reticular formation. However, further work has shown that no one "vomiting center" exists and that several brain stem nuclei initiate emesis, including the nucleus tractus solitarius; the dorsal vagal and phrenic nuclei; medullary nuclei that regulate respiration; and nuclei that control pharyngeal, facial, and tongue movements. The neurotransmitters involved in coordinating emesis are uncertain; however, neurokinin NK₁, serotonin, and vasopressin pathways are postulated.

Somatic and visceral muscles exhibit stereotypic responses during emesis. Inspiratory thoracic and abdominal wall muscles contract, producing high intrathoracic and intraabdominal pressures that facilitate expulsion of gastric contents. The gastric cardia herniates across the diaphragm, and the larynx moves upward to promote oral propulsion of the vomitus. Under normal conditions, distally migrating upper gut contractions are regulated by an electrical phenomenon, the slow wave, which cycles at 3 cycles per minute in the stomach and 11 cycles per minute in the duodenum. With emesis, slow waves are replaced by orally propagating spike activity, which induces retrograde contractions that assist in the oral expulsion of small intestinal contents.

Activators of Emesis Emetic stimuli act at several anatomic sites. Emesis provoked by noxious thoughts or smells originates in the cerebral cortex, whereas cranial nerves mediate vomiting after gag reflex activation. Motion sickness and inner ear disorders act on the labyrinthine apparatus, while gastric irritants and emetogenic anticancer agents such as cisplatin stimulate gastroduodenal vagal afferent nerves. Nongastric visceral afferents are activated by small intestinal and colonic obstruction and mesenteric

ischemia. The area postrema, a medullary nucleus, responds to bloodborne emetic stimuli and is termed the *chemoreceptor trigger zone*. Many emetic drugs act on the area postrema as do bacterial toxins and metabolic disorders such as uremia, hypoxia, and ketoacidosis.

Neurotransmitters that mediate induction of vomiting are selective for these anatomic sites. Labyrinthine disorders stimulate vestibular cholinergic muscarinic M₁ and histaminergic H₁ receptors, whereas gastroduodenal vagal afferent stimuli activate serotonin 5-HT₃ receptors. The area postrema is richly served by nerve fibers acting on diverse receptor subtypes including 5-HT₃, M₁, H₁, and dopamine D₂. Optimal pharmacologic management of the patient with vomiting requires an understanding of these pathways.

DIFFERENTIAL DIAGNOSIS

Nausea and vomiting are caused by conditions within and outside the gut as well as by drugs and circulating toxins ([Table 41-1](#)).

Intraperitoneal Disorders Visceral obstruction and inflammation of hollow and solid viscera may produce vomiting as the main symptom. Gastric obstruction results from ulcer disease and malignancy, whereas small bowel and colonic obstructions occur as a consequence of adhesions, benign or malignant tumors, volvulus, intussusception, or inflammatory diseases such as Crohn's disease. The superior mesenteric artery syndrome, occurring after weight loss or prolonged bed rest, results when the duodenum is compressed by the overlying superior mesenteric artery. Abdominal irradiation evokes emesis by impairing intestinal contractile function and by inducing strictures. Biliary colic causes nausea likely by action on visceral afferent nerves. Vomiting with pancreatitis, cholecystitis, and appendicitis is due to localized visceral irritation and induction of ileus. Enteric infections with viruses or bacteria such as *Staphylococcus aureus* and *Bacillus cereus* are among the most common causes of acute vomiting, especially in children. Opportunistic infections such as cytomegalovirus or herpes simplex induce emesis in immunocompromised individuals.

Disorders of gastrointestinal motor function also commonly cause nausea and vomiting. Gastroparesis is defined as a delay in emptying of food from the stomach and occurs after vagotomy for peptic ulcer, with pancreatic adenocarcinoma, or in systemic diseases such as diabetes, scleroderma, and amyloidosis. Idiopathic gastroparesis develops in the absence of systemic illness and may follow a viral prodrome suggesting an infectious etiology. Intestinal pseudoobstruction is characterized by disruption of intestinal and colonic motor activity and leads to intestinal retention of food residue and secretions, bacterial overgrowth, nutrient malabsorption, and development of nausea, vomiting, bloating, pain, and alteration of bowel pattern. Intestinal pseudoobstruction may be idiopathic, may be inherited as a familial visceral myopathy or neuropathy, or may result from systemic disease or be a paraneoplastic consequence of malignancy (especially small cell lung carcinoma).

Extraperitoneal Disorders Myocardial infarction and congestive heart failure are cardiac causes of nausea and vomiting. Nausea and vomiting occur after 25% of surgical operations, both within and outside the peritoneum. Postoperative emesis is

more common after laparotomy and orthopedic surgery than after laparoscopy and is more prevalent in women. Increased intracranial pressure from tumors, bleeding, abscess, or obstruction to cerebrospinal fluid outflow produces prominent vomiting with or without concurrent nausea. Motion sickness, labyrinthitis, and Meniere's disease evoke symptoms via labyrinthine pathways. Cyclic vomiting syndrome is a rare disorder of unknown etiology that produces episodes of intractable nausea and vomiting, usually in children. The syndrome shows a strong association with migraine headaches, suggesting that some cases may be migraine variants. Patients with psychiatric illnesses, including anorexia nervosa, bulimia nervosa, and depression, may report significant nausea. Psychogenic vomiting occurs most commonly in women with other emotional problems.

Medications and Metabolic Disorders Drugs are frequent causes of vomiting and may act on the stomach (analgesics, erythromycin) or area postrema (digoxin, opiates, anti-Parkinsonian drugs). Agents that cause emesis include antibiotics, antiarrhythmics, antihypertensives, oral hypoglycemics, and contraceptives. Cancer chemotherapy causes vomiting that is acute (within hours of administration), delayed (after 1 or more days), or anticipatory. Acute emesis resulting from highly emetogenic agents such as cisplatin is mediated by 5-HT₃ pathways, whereas delayed emesis is independent of 5-HT₃. Anticipatory nausea often responds better to anxiolytic therapy than to antiemetics.

Metabolic disorders are in the differential diagnosis in certain settings. Pregnancy is the most prevalent endocrinologic cause of nausea, occurring in 70% of women in the first trimester. Hyperemesis gravidarum is a severe form of nausea of pregnancy that can produce significant fluid loss and electrolyte disturbances. Uremia, ketoacidosis, adrenal insufficiency, as well as parathyroid and thyroid disease are other metabolic causes of emesis.

Circulating toxins evoke symptoms through effects on the area postrema. Endogenous toxins are generated in fulminant liver failure, whereas exogenous enterotoxins may be produced by enteric bacterial infection. Ethanol intoxication is a common toxic cause of nausea and vomiting.

Approach to the Patient

History and Physical Examination The history helps determine the etiology of unexplained nausea and vomiting. Drugs and toxins often cause acute symptoms, while established illnesses evoke chronic complaints. Vomiting within 1 h of eating characterizes pyloric obstruction, whereas emesis in the late postprandial period is reported with intestinal obstruction. Gastroparesis can produce nausea within minutes of food consumption but, in severe cases, leads to vomiting of meal residue ingested hours or days previously. Blood in the vomitus raises suspicion of an ulcer or malignancy; feculent emesis is noted with distal intestinal or colonic obstruction. Bilious vomiting excludes gastric obstruction, whereas emesis of undigested food is consistent with a pharyngoesophageal process such as Zenker's diverticulum or achalasia. Relief of abdominal pain by emesis characterizes small bowel obstruction, but vomiting has no effect on pancreatitis or cholecystitis pain. Pronounced weight loss raises concern about malignancy or obstruction. Fevers suggest inflammation, while an intracranial source is

considered if there are headaches or visual field changes. Vertigo or tinnitus indicate labyrinthine disease.

The physical examination complements the history. Abdominal auscultation may reveal absent bowel sounds with ileus. High-pitched rushes suggest bowel obstruction, while a succession splash on abrupt lateral movement of the patient is found with gastroparesis or pyloric obstruction. Tenderness or involuntary guarding raises suspicion of inflammation, whereas fecal blood suggests mucosal injury from ulcer, ischemia, or tumor. Neurologic etiologies present with papilledema, visual field loss, or focal neural abnormalities. Neoplasm is suggested by palpable masses or adenopathy.

The history and examination can characterize complications of emesis. Reports of lightheadedness with demonstration of orthostatic hypotension and reduced skin turgor indicate intravascular fluid loss. Hematemesis, especially with repeated vomiting, suggests a Mallory-Weiss tear of the gastroesophageal junction, while pulmonary abnormalities raise concern for aspiration of vomitus.

Diagnostic Testing With intractable symptoms or an elusive diagnosis, selected diagnostic tests can direct clinical management. Electrolyte replenishment is indicated if hypokalemia or metabolic alkalosis is found. Detection of iron-deficiency anemia mandates a search for mucosal injury. Pancreaticobiliary disease is suggested by abnormal pancreatic enzymes or liver biochemistries, whereas endocrinologic or rheumatologic etiologies are diagnosed by specific hormone or serologic testing. If luminal obstruction is considered, supine and upright abdominal radiographs may show intestinal air-fluid levels with reduced colonic air. Ileus is characterized by diffusely dilated air-filled bowel loops.

If initial testing is unrevealing, additional anatomic studies may be indicated. Upper endoscopy detects ulcer disease or gastroesophageal malignancy, and small bowel barium radiography diagnoses partial small bowel obstruction. Colonoscopy or barium enema can detect colonic obstruction. Ultrasound or computed tomography of the abdomen defines intraperitoneal inflammatory processes, while computed tomography or magnetic resonance imaging of the head can delineate intracranial sources of nausea and vomiting.

Gastrointestinal motility testing may detect a functional gastrointestinal disorder responsible for symptoms when investigation of anatomic abnormalities is negative. Gastroparesis is most commonly diagnosed with gastric scintigraphy, by which emptying of a radiolabeled meal is measured. A noninvasive means of quantitating gastric slow wave activity with cutaneous electrodes placed over the stomach, electrogastrography, has been proposed as an alternate means of diagnosing abnormal gastric emptying. With intestinal pseudoobstruction, small bowel barium radiography often suggests the diagnosis. Manometry of the small intestine may provide confirmation of the diagnosis as well as complementary information by characterizing the motor abnormality as neuropathic or myopathic based on contractile patterns. Such investigation can obviate the need for open biopsy of the intestine to evaluate for smooth muscle or neuronal degeneration.

TREATMENT

General Principles Therapy of vomiting is tailored to the underlying disease, with the medical or surgical correction of abnormalities if possible. Hospitalization is considered for severe dehydration, especially if oral fluid replenishment cannot be sustained. Once oral intake is tolerated, nutrients are restarted as liquids that are low in fat, as lipids delay gastric emptying. Foods high in indigestible residues are avoided because these also prolong gastric retention.

Antiemetic Medications Drugs that act on the central nervous system serve as antiemetic agents ([Table 41-2](#)). Antihistamines such as meclizine and dimenhydrinate and anticholinergic drugs such as scopolamine act on labyrinthine-activated pathways and are useful in the treatment of motion sickness and inner ear disorders. Phenothiazine and butyrophenone dopamine D₂antagonists are used to treat emesis evoked by area postrema stimuli and are effective for many medication, toxic, and metabolic etiologies. Dopamine antagonists freely cross the blood-brain barrier and may cause anxiety, dystonic reactions, hyperprolactinemic effects (galactorrhea and sexual dysfunction), and irreversible tardive dyskinesia.

Other drug classes have antiemetic properties. Serotonin 5-HT₃antagonists such as ondansetron and granisetron are useful in the treatment of postoperative vomiting and after radiation therapy but are mainly used to prevent cancer chemotherapy-induced emesis. The usefulness of 5-HT₃antagonists to control other causes of refractory emesis is less well established. Antidepressant drugs are established therapeutic options for patients with functional bowel disorders such as irritable bowel syndrome. Low-dose tricyclic antidepressants provide moderate symptomatic benefit in patients with unexplained nausea of a functional nature.

Gastrointestinal Motor Stimulants Drugs that stimulate gastric emptying are indicated for gastroparesis ([Table 41-2](#)). Cisapride, a serotonin 5-HT₄agonist that stimulates cholinergic nerves in the stomach, has become the preferred drug for outpatient management of gastroparesis. The drug is well tolerated but exhibits very rare drug interactions with selected antibiotics, antifungals, and other agents that predispose to fatal cardiac arrhythmias. Metoclopramide, a combined 5-HT₄agonist and D₂antagonist, is efficacious in the treatment of gastroparesis, but anti-dopaminergic side effects limit its use in 20% of patients. Erythromycin, a macrolide antibiotic, potently increases gastroduodenal motility by action on receptors for motilin, an endogenous stimulant of fasting motor activity. Erythromycin may be most useful when given intravenously to inpatients with refractory gastroparesis; however, oral forms of the drug also have some effect. Domperidone, a D₂antagonist not available in the United States, has prokinetic and antiemetic effects but does not cross into most other brain regions; thus, anxiety and dystonic reactions are rare. The main side effects of domperidone are induction of hyperprolactinemia through effects on pituitary regions served by a porous blood-brain barrier.

Patients with refractory upper gut motility disorders pose significant therapeutic challenges. Liquid suspensions of prokinetic drugs may be beneficial inasmuch as liquids empty from the stomach more rapidly than pills. Metoclopramide can be administered subcutaneously in patients who do not respond to oral drugs. Intestinal pseudoobstruction may respond to the somatostatin analogue octreotide, which induces

propagative small intestinal motor complexes. Placement of a feeding jejunostomy reduces hospitalizations and improves overall health in some patients with gastroparesis who do not respond to drug therapy. Surgical options are limited for refractory cases, but postvagotomy gastroparesis may improve with near-total resection of the stomach. Electrical pacing of the stomach may also be useful.

Selected Clinical Settings Cancer chemotherapeutic agents such as cisplatin are intensely emetogenic. Given prophylactically, 5-HT₃ antagonists prevent chemotherapy-induced acute vomiting in most cases ([Table 41-2](#)). Optimal antiemetic effects often are obtained with a 5-HT₃ antagonist in combination with a glucocorticoid. In high doses, metoclopramide is effective in controlling chemotherapy-evoked emesis, whereas benzodiazepines such as lorazepam are most useful in reducing anticipatory nausea and vomiting. In contrast, delayed emesis 1 to 5 days after chemotherapy is more refractory to treatment. Agents that act as neurokinin NK₁ antagonists in the brain stem may be potent antiemetic and antinausea drugs during both the acute and the delayed periods after chemotherapy. Cannabinoids such as tetrahydrocannabinol have been advocated for cancer-associated emesis, but these drugs produce significant side effects and are no more effective than antidopaminergic agents.

The clinician should exercise caution in the management of the patient with nausea of pregnancy. Studies of the teratogenic effects of available antiemetic agents have provided conflicting results. Few controlled trials have been performed in the nausea of pregnancy, although antihistamines such as meclizine and antidopaminergics such as prochlorperazine are more efficacious than placebo. As a consequence, alternative therapies such as pyridoxine or ginger have been recommended.

INDIGESTION

MECHANISMS

Most patients with indigestion have symptoms of a functional nature that result from gastroesophageal acid reflux or from gastric abnormalities including dysfunctional motor activity and afferent hypersensitivity; these symptoms comprise the syndrome functional dyspepsia. Some cases are a consequence of a more serious organic illness.

Gastroesophageal Acid Reflux Acid reflux results from selected physiologic defects. In scleroderma and pregnancy, lower esophageal sphincter (LES) tone is low, but most patients with acid reflux have normal LES pressures. Many individuals show frequent transient LES relaxations during which acid bathes the esophagus. The role of hiatal hernias is controversial -- although most reflux patients exhibit hiatal hernias, most individuals with hiatal hernias do not have excess heartburn.

Gastric Motor Dysfunction Disturbed gastric motility is purported to cause acid reflux in some patients with indigestion. Delayed gastric emptying also is found in 25 to 50% of individuals with functional dyspepsia. The relation of these defects to symptom induction is uncertain as many studies show poor correlation between symptom severity and the degree of motor dysfunction. Abnormal gastric fundic relaxation may cause dyspeptic symptoms such as bloating, fullness, nausea, and early satiety.

Visceral Afferent Hypersensitivity Disturbed gastric sensory function also may cause functional dyspepsia. Visceral afferent hypersensitivity was first demonstrated in patients with irritable bowel syndrome who had heightened perception of rectal balloon inflation without changes in rectal compliance. Patients with dyspepsia may experience discomfort with fundic distention to lower pressures than healthy control subjects.

Other Factors *Helicobacter pylori* has a clear etiologic role in peptic ulcer disease, but ulcers cause only a minority of cases of dyspepsia. The importance of *H. pylori* in the genesis of functional dyspepsia is controversial, but most investigators believe it is of minor importance. Analgesics cause dyspepsia; nitrates, calcium channel blockers, theophylline, and progesterone promote acid reflux. Other exogenous factors that induce acid reflux include ethanol, tobacco, and caffeine via [LES](#) relaxation. Finally, functional dyspepsia is exacerbated by stress, suggesting a pathogenic role for psychological factors.

DIFFERENTIAL DIAGNOSIS

Functional Causes Gastroesophageal reflux disease (GERD) is prevalent in Western society. Heartburn is reported once monthly by 40% of Americans and daily by 7%. Functional dyspepsia, defined as ≥ 3 months of dyspepsia without an organic cause, also is common. Nearly 25% of the populace has abdominal discomfort at least six times yearly, consistent with functional dyspepsia, but only 10 to 20% consult physicians. The clinician must distinguish these illnesses, which have a benign course, from conditions that have deleterious consequences.

Ulcer Disease In most cases of [GERD](#), the esophagus is not damaged. However, 5% of patients develop esophageal ulcers, and some form esophageal strictures. Functional dyspepsia is the cause of symptoms in 60% of individuals with dyspepsia. However, 15 to 25% of cases stem from ulcers of the stomach or duodenum. The most common causes of ulcer disease are gastric infection with *H. pylori* and use of nonsteroidal anti-inflammatory drugs. Other rare causes of gastroduodenal ulcer include Crohn's disease and Zollinger-Ellison syndrome, a condition resulting from gastrin overproduction by an endocrine tumor ([Chap. 285](#)).

Malignancy Patients with dyspepsia often seek care because of fear of cancer. However, <2% of cases result from gastroesophageal malignancy. Esophageal squamous cell carcinoma occurs most often in those patients with histories of tobacco or ethanol intake. Other risk factors include prior caustic ingestion, achalasia, and the hereditary disorder tylosis. Esophageal adenocarcinoma usually complicates long-standing acid reflux. Eight to 20% of patients with [GERD](#) exhibit glandular mucosal metaplasia of the squamous epithelium in the lower esophagus, termed *Barrett's metaplasia*. This condition predisposes to esophageal adenocarcinoma. Gastric malignancies include adenocarcinoma, which is more prevalent in certain Asian societies, and lymphoma (see [Chap. 90](#)).

Other Causes Alkaline reflux esophagitis produces [GERD](#)-like symptoms in patients who have had surgery for peptic ulcer disease. Opportunistic fungal or viral esophageal infections may produce heartburn or chest discomfort but more often cause painful swallowing. Although biliary colic is in the differential diagnosis of dyspepsia, most

patients with true biliary colic report discrete episodes of right upper quadrant or epigastric pain rather than chronic burning discomfort, nausea, and bloating. Lactose intolerance resulting from intestinal lactase deficiency produces gas, bloating, discomfort, and diarrhea. Lactase deficiency occurs in 15% of Caucasians of northern European descent but is more common in African Americans and Asians. Pancreatic disease (chronic pancreatitis and malignancy), hepatocellular carcinoma, celiac sprue, Menetrier's disease, infiltrative diseases (sarcoidosis and eosinophilic gastroenteritis), mesenteric ischemia, thyroid and parathyroid disease, and abdominal wall strain cause dyspepsia. Extraperitoneal etiologies of indigestion include congestive heart failure and tuberculosis.

Approach to the Patient

History and Physical Examination [GERD](#) classically produces heartburn, a substernal warmth beginning in the epigastrium that moves toward the neck. Heartburn often is exacerbated by meals and may awaken the patient. Associated symptoms include regurgitation of acid and water brash, the reflex release of salty salivary secretions into the mouth. Atypical symptoms include pharyngitis, asthma, cough, bronchitis, hoarseness, and chest pain that mimics angina. Some patients with acid reflux on esophageal pH testing do not report heartburn and instead note abdominal pain or other symptoms.

Individuals with ulcer-like dyspepsia have epigastric gnawing or burning that is relieved by meals or acid suppression. Dysmotility-like dyspepsia is a fullness or pain that is aggravated by eating and associated with nausea, bloating, eructation, and early satiety. There is overlap among the different dyspepsia subclasses and with other functional disorders such as irritable bowel syndrome.

The physical examination of individuals with functional causes of indigestion is usually normal. In atypical [GERD](#), pharyngeal erythema and wheezing over the lung fields may be present. Poor dentition may occur with prolonged acid regurgitation. Patients with functional dyspepsia may have epigastric tenderness or abdominal distension.

Discrimination between functional and organic causes of indigestion mandates exclusion of selected historical and examination features. Odynophagia suggests esophageal infection, while dysphagia promotes concern about a benign or malignant esophageal blockage. Other features that raise alarm include unexplained weight loss, recurrent vomiting with evidence of dehydration, occult or gross gastrointestinal bleeding, and a palpable mass or adenopathy.

Diagnostic Testing Because indigestion is prevalent in the community and because most cases result from functional illness, a general principle of diagnostic testing is to perform only limited and directed testing of selected individuals.

Once alarm factors are excluded, patients with typical [GERD](#) do not need further evaluation and are treated empirically. Upper endoscopy is indicated to exclude mucosal injury in patients with atypical symptoms, symptoms unresponsive to acid-suppressing drugs, or alarm factors. In patients with >5 years of heartburn, endoscopy is performed to screen for Barrett's metaplasia. Upper gastrointestinal

barium radiography has a slightly higher sensitivity for detecting strictures and rings than endoscopy; however, benign esophageal obstructions may be dilated with an endoscopic approach. Ambulatory esophageal pH testing is considered for atypical symptoms such as unexplained chest pain and for the symptoms that are unresponsive to appropriate medications. Esophageal manometry is most commonly ordered when surgical treatment of GERD is considered. A low [LES](#) pressure may predict failure with drug therapy and identify patients who may require surgery. The demonstration of disordered esophageal body peristalsis may affect the decision to operate or modify the type of operation chosen. Manometry with provocative testing may clarify the diagnosis in patients with atypical symptoms. Blinded perfusion of saline and then acid into the esophagus, known as the Bernstein test, can delineate whether unexplained chest discomfort results from acid reflux.

The approach to unexplained dyspepsia is dependent on the patient's age, symptom profile, and findings on examination. In individuals <45 years of age without alarm factors, blood serology for *H. pylori* is obtained to exclude the organism as a cause of ulcer disease. Upper endoscopy in this patient subset is reserved for those who fail to respond to treatment of *H. pylori*-positive or -negative dyspepsia. Upper endoscopy is performed as the initial diagnostic test in any individual with alarm factors or in patients >45 years of age because of the elevated risk of gastroesophageal malignancy with advancing age.

Further testing is indicated only if other factors are present. If there is blood loss, a blood count is obtained to exclude anemia. Thyroid chemistries or calcium levels screen for metabolic disease. With suspected pancreaticobiliary causes, blood is obtained for amylase, lipase, and liver chemistry determination. If biochemical abnormalities are found, abdominal ultrasound or computed tomography may give important information. Patients with dysmotility-like dyspepsia may selectively exhibit delayed gastric emptying; thus, gastric scintigraphy can be considered when drug treatment fails. Hydrogen breath testing after lactose ingestion may be performed for suspected lactase deficiency.

TREATMENT

General Principles In mild dyspepsia, reassurance that a careful evaluation revealed no serious organic disease may be the only intervention required. Drugs that cause acid reflux or dyspepsia should be stopped if possible. Patients with [GERD](#) should limit ethanol, caffeine, chocolate, and tobacco use because of their effects on the [LES](#). Other measures with efficacy in GERD include ingestion of a low-fat diet, avoidance of snacks before bedtime, and elevation of the head of the bed.

Specific therapy for organic diseases should be offered when possible. In disorders such as biliary colic, surgery is appropriate; whereas lactase deficiency and celiac sprue respond to special diets. Some illnesses such as peptic ulcer disease require specific medical regimens to effect cure. However, as most patients present with functional causes of indigestion, medications that reduce gastric acid, stimulate upper gut motility, or blunt gastric sensitivity are indicated.

Acid Suppressing or Neutralizing Medications Drugs that reduce or neutralize

gastric acid are the most prescribed agents for [GERD](#). Histamine H₂receptor antagonists such as cimetidine, ranitidine, famotidine, and nizatidine are useful in the treatment of mild to moderate GERD. For uncomplicated heartburn, H₂receptor antagonists are given for 4 weeks before endoscopy is considered. For severe symptoms or for many cases of erosive or ulcerative esophagitis, proton pump inhibitors such as omeprazole and lansoprazole are needed. These drugs, which inhibit gastric H⁺, K⁺-ATPase, are more potent than H₂receptor antagonists. Liquid antacids are useful for short-term control of mild GERD but are less effective for severe disease unless given at high doses that produce side effects (diarrhea with magnesium-containing agents and constipation with aluminum-containing agents). Sucralfate is a salt of aluminum hydroxide and sucrose octasulfate and buffers acid and binds pepsin and bile salts. Its efficacy in GERD and functional dyspepsia is unproven.

Acid suppressing drugs are advocated for first-line therapy of *H. pylori* negative dyspepsia, especially with ulcer-like symptoms. Ranitidine is of benefit in the treatment of functional dyspepsia versus placebo. In young patients without alarm symptoms, a 4-week trial of an H₂receptor antagonist or proton pump inhibitor is given. Endoscopy is performed only if symptoms do not improve.

***Helicobacter pylori* Eradication** Regimens to eradicate *H. pylori* are recommended for young patients with dyspepsia without alarm symptoms in whom the bacterium has been detected by serology. Several drug combinations show efficacy, but most include 10 to 14 days of a proton pump inhibitor or bismuth subsalicylate in concert with two antibiotics. If symptoms resolve, no further intervention is required. Most patients who respond to this "treatment-first" approach have underlying ulcer disease. The usefulness of *Helicobacter* eradication in patients with functional dyspepsia is unproven, but evidence suggests that <15% of cases relate to *H. pylori*. No evidence demonstrates that *H. pylori* eradication is useful in the treatment of [GERD](#).

Gastrointestinal Motor Stimulants Cisapride is superior to placebo in treating [GERD](#) and can be prescribed as sole therapy or as an adjunct to an acid-suppressing drug. Other motor stimulants such as metoclopramide, erythromycin, and domperidone are of limited use in the treatment of GERD.

Prokinetic agents are frequently used for treatment of functional dyspepsia. Cisapride and domperidone relieve symptoms more effectively than placebo. In general, these drugs are more potent than acid-reducing agents in the treatment of functional dyspepsia and may be given instead of acid suppressants as the initial empirical treatment of young patients with dyspepsia without alarm symptoms who are not infected with *H. pylori*. Patients with dysmotility-like dyspepsia may respond preferentially to motor-stimulating drugs.

Other Options In patients with [GERD](#) who do not respond to drug therapy, antireflux surgery may be offered. Operations include the Nissen fundoplication, in which the proximal stomach is wrapped completely around the [LES](#) to increase LES pressure, and the Belsey procedure, in which the wrap encircles 270° of the circumference of the LES. The latter is selected if esophageal peristalsis is suboptimal when a 360° wrap might cause dysphagia. Funduplications can be performed laparoscopically, thereby reducing the morbidity and the postoperative recuperation period.

Some patients with functional dyspepsia do not respond to acid suppressants or prokinetic drugs but may respond to low-dose tricyclic antidepressant therapy. The mechanism of action of these agents in functional dyspepsia is unknown but may involve blunting of visceral pain processing in the brain. Gas and bloating may be the most troubling symptoms in some patients with indigestion and can be difficult to treat. Successes with dietary exclusion of gas-producing foods such as legumes and use of the surface-active compound simethicone or the gas-absorptive agent activated charcoal have been reported. Psychological treatments have been proposed for functional dyspepsia; however, convincing data on their efficacy are lacking.

(Bibliography omitted in Palm version)

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42. DIARRHEA AND CONSTIPATION - David A. Ahlquist, Michael Camilleri

Diarrhea and constipation are exceedingly common and together exact an enormous toll in terms of morbidity, loss of work productivity, and consumption of medical resources. Worldwide, more than one billion people suffer one or more episodes of acute diarrhea each year. Among the 100 million persons affected annually by acute diarrhea in the United States, nearly half must restrict activities, 10% consult physicians, 250,000 require hospitalization, and roughly 3000 die (primarily the elderly). The economic burden to society is estimated at more than \$20 billion. Because of poor sanitation and more limited access to health care, acute infectious diarrhea remains one of the most common causes of mortality in developing countries, particularly among children, accounting for 5 to 8 million deaths per year. Population statistics on chronic diarrhea and constipation are more uncertain, perhaps due to variable definitions and reporting, but the frequency of these conditions is also high. Based on United States population surveys, prevalence rates for chronic diarrhea range from 2 to 7% and for chronic constipation from 3 to 17%. Diarrhea and constipation are among the most common patient complaints faced by internists and primary care physicians, and they account for nearly 50% of referrals to gastroenterologists.

Although diarrhea and constipation may present as mere nuisance symptoms at one extreme, they can be severe or life-threatening at the other. Even mild symptoms may signal a serious underlying gastrointestinal lesion, like colorectal cancer, or systemic disorder, like thyroid disease. Given the heterogeneous causes and potential severity of these common complaints, it is imperative for clinicians to appreciate the pathophysiology, etiologic classification, diagnostic strategies, and therapeutic principles of diarrhea and constipation so that rational and cost-effective care can be delivered.

NORMAL PHYSIOLOGY

The human small intestine and colon perform important functions including the secretion and absorption of water and electrolytes, the storage and subsequent transport of intraluminal contents aborally, and the salvage of some nutrients after bacterial metabolism of carbohydrate that are not absorbed in the small intestine. The main motor functions are summarized in [Table 42-1](#). Alterations in fluid and electrolyte handling contribute significantly to diarrhea. Alterations in motor and sensory functions of the human colon result in highly prevalent syndromes such as irritable bowel syndrome, chronic diarrhea, and chronic constipation.

NEURAL CONTROL

The small intestine and colon have intrinsic and extrinsic innervation. The *intrinsic innervation* also called the enteric nervous system, comprises myenteric, submucosal, and mucosal neuronal layers. The function of these layers is modulated by interneurons through the actions of neurotransmitter amines or peptides, including acetylcholine, opioids, norepinephrine, serotonin, ATP, and nitric oxide. The myenteric plexus regulates smooth muscle function, and the submucosal plexus affects secretion and absorption.

The *extrinsic innervations* of the small intestine and colon are part of the autonomic

nervous system and also modulate both motor and secretory functions. The parasympathetic nerve supply conveys both visceral sensory as well as excitatory pathways to the motor components of the colon. Parasympathetic fibers via the vagus nerve reach the small intestine and proximal colon along the branches of the superior mesenteric artery. The distal colon is supplied by sacral parasympathetic nerves (S₂₋₄) via the pelvic plexus; these fibers course through the wall of the colon as ascending intracolonic fibers as far as, and in some instances including, the proximal colon. The chief excitatory neurotransmitters controlling motor function are acetylcholine and the tachykinins, such as substance P. The sympathetic nerve supply modulates motor functions and reaches the small intestine and colon alongside the arterial arcades of the superior and inferior mesenteric vessels. Sympathetic input to the gut is generally excitatory to sphincters and inhibitory to nonsphincteric muscle. Visceral afferents convey sensation from the gut to the central nervous system; initially, they course along sympathetic fibers, but as they approach the spinal cord they separate, have cell bodies in the dorsal root ganglion, and enter the dorsal horn of the spinal cord. Afferent signals are conveyed to the brain along the lateral spinothalamic tract and the nociceptive dorsal column pathway and are then perceived. Other afferent fibers synapse in the prevertebral ganglia and reflexly modulate intestinal motility.

INTESTINAL FLUID ABSORPTION AND SECRETION

On an average day, 9 L of fluid enters the gastrointestinal tract; approximately 1 L of residual fluid reaches the colon; the stool excretion of fluid constitutes about 0.2 L/d. The colon has a large capacitance and functional reserve and may recover up to four times its usual volume of 0.8 L/d, provided the rate of flow permits reabsorption to occur. Thus, the colon can partially compensate for intestinal absorptive or secretory disorders.

In the colon, sodium absorption is predominantly electrogenic, and uptake takes place at the apical membrane; it is also compensated by the pumping out functions of the basolateral sodium pump. A variety of neural and non-neural mediators regulate colonic fluid and electrolyte balance, including cholinergic, adrenergic and serotonergic mediators. Angiotensin and aldosterone also influence colonic absorption, reflecting the common embryologic development of the distal colonic epithelium and the renal tubules.

ILEOCOLONIC STORAGE AND SALVAGE

The distal ileum acts as a reservoir, emptying intermittently by bolus movements. This action allows time for salvage of fluids, electrolytes, and nutrients. Segmentation by haustra compartmentalizes the colon and facilitates mixing, retention of residue, and formation of solid stools. In health, the ascending and transverse regions of colon function as reservoirs (average transit, 15 h), and the descending colon acts as a conduit (average transit, 3 h). The colon is efficient at conserving sodium and water, a function that is particularly important in sodium-depleted patients in whom the small intestine alone is unable to maintain sodium balance. Diarrhea or constipation may result from alteration in the reservoir function of the proximal colon, or the propulsive function of the left colon. Constipation may also result from disturbances of the rectal or sigmoid reservoir, typically as a result of dysfunction of the pelvic floor or the coordination of defecation.

SMALL INTESTINAL MOTILITY

During fasting, the motility of the small intestine is characterized by a cyclical event called the migrating motor complex (MMC), which serves to clear nondigestible residue from the small intestine. This organized, propagated series of contractions lasts on average 4 min, occurs every 60 to 90 min, and usually involves the entire small intestine. After food ingestion, the small intestine produces irregular, mixing contractions of relatively low amplitude, except in the distal ileum where more powerful contractions occur intermittently and empty the ileum by bolus transfers.

COLONIC MOTILITY AND TONE

The small intestinal [MMC](#) only rarely continues into the colon. However, short duration or phasic contractions mix colonic contents, and high amplitude propagated contractions (HAPCs) are sometimes associated with mass movements through the colon and occur approximately five times per day, usually on awakening in the morning and postprandially. Increased frequency of HAPCs may result in diarrhea. The predominant phasic contractions are irregular and nonpropagated and serve as a "mixing" function.

Colonic tone refers to the background contractility upon which phasic contractile activity (typically contractions lasting less than 15 s) is superimposed. It is an important cofactor in the colon's capacitance (volume accommodation) and sensation.

COLONIC MOTILITY AFTER MEAL INGESTION

After meal ingestion, colonic phasic and tonic contractility increase for a period of approximately 2 h. The initial phase (about 10 min) is mediated by the vagus nerve in response to mechanical distention of the stomach. The subsequent response of the colon requires caloric stimulation and is at least in part mediated by hormones, e.g., gastrin and serotonin.

DEFECATION

Tonic contraction of the puborectalis muscle, which forms a sling around the rectoanal junction, is important to maintain continence; during defecation, sacral parasympathetic nerves relax this muscle, facilitating the straightening of the rectoanal angle ([Fig. 42-1](#)). Distention of the rectum results in transient relaxation of the internal anal sphincter via intrinsic and reflex sympathetic innervation. As sigmoid and rectal contractions increase the pressure within the rectum, the rectosigmoid angle opens by more than 15°. Voluntary relaxation of the external anal sphincter (striated muscle innervated by the pudendal nerve) permits the evacuation of feces; this evacuation process can be augmented by an increase in intraabdominal pressure created by the Valsalva maneuver.

DIARRHEA

DEFINITION

Diarrhea is loosely defined as passage of abnormally liquid or unformed stools at an

increased frequency. For adults on a typical Western diet, stool weight exceeding 200 g/d can generally be considered diarrheal. Because of the fundamental importance of duration to diagnostic considerations, diarrhea may be further defined as *acute* if <2 weeks, *persistent* if 2 to 4 weeks, and *chronic* if >4 weeks in duration.

Two common conditions, usually associated with the passage of stool totaling <200 g/d, must be distinguished from diarrhea, as diagnostic and therapeutic algorithms differ. *Pseudodiarrhea*, or the frequent passage of small volumes of stool, often is associated with rectal urgency and accompanies the irritable bowel syndrome or anorectal disorders like proctitis. *Fecal incontinence* is the involuntary discharge of rectal contents and is most often caused by neuromuscular disorders or structural anorectal problems. Diarrhea and urgency, especially if severe, may aggravate or cause incontinence. Pseudodiarrhea and fecal incontinence occur at prevalence rates comparable to or higher than that of chronic diarrhea and should always be considered in patients complaining of "diarrhea." A careful history and physical examination generally allow these conditions to be discriminated from true diarrhea.

ACUTE DIARRHEA

More than 90% of cases of acute diarrhea are caused by infectious agents; these cases are often accompanied by vomiting, fever, and abdominal pain. The remaining 10% or so are caused by medications, toxic ingestions, ischemia, and other conditions.

Infectious Agents Most infectious diarrheas are acquired by fecal-oral transmission via direct personal contact or, more commonly, via ingestion of food or water contaminated with pathogens from human or animal feces. In the immunologically competent person, the resident fecal microflora, containing more than 500 taxonomically distinct species, are rarely the source of diarrhea and may actually play a role in suppressing the growth of ingested pathogens. Acute infection or injury occurs when the ingested agent overwhelms the host's mucosal immune and nonimmune (gastric acid, digestive enzymes, mucus secretion, peristalsis, and suppressive resident flora) defenses. Established clinical associations with specific enteropathogens may offer diagnostic clues.

In the United States, high risk groups are recognized:

1. *Travelers*. Nearly 40% of tourists to endemic regions of Latin America, Africa, and Asia develop so-called traveler's diarrhea, most commonly due to enterotoxigenic *Escherichia coli* as well as to *Campylobacter*, *Shigella*, and *Salmonella*. Visitors to Russia (especially St. Petersburg) may have increased risk of *Giardia*-associated diarrhea; visitors to Nepal may acquire *Cyclospora*. Campers, backpackers, and swimmers in wilderness areas may become infected with *Giardia*.
2. *Consumers of certain foods*. Diarrhea closely following food consumption at a picnic, banquet, or restaurant may suggest infection with *Salmonella*, *Campylobacter*, or *Shigella* from chicken; enterohemorrhagic *E. coli* (O157:H7) from undercooked hamburger; *Bacillus aureus* from fried rice; *Staphylococcus aureus* or *Salmonella* from mayonnaise or creams; *Salmonella* from eggs; and *Vibrio* species, *Salmonella*, or acute hepatitis A or B from seafood, especially if raw.

3. *Immunodeficient persons*. Individuals at risk for diarrhea include those with either primary immunodeficiency (e.g., IgA deficiency, common variable hypogammaglobulinemia, chronic granulomatous disease) or the much more common secondary immunodeficiency states (e.g., AIDS, senescence, pharmacologic suppression). Common enteropathogens often cause a more severe and protracted diarrheal illness; and, particularly in persons with AIDS, opportunistic infections, such as by *Mycobacterium* species, certain viruses (cytomegalovirus, adenovirus, and herpes simplex), and protozoa (*Cryptosporidium*, *Isospora belli*, Microsporidia, and *Blastocystis hominis*) may also play a role ([Chap. 309](#)). In patients with AIDS, agents transmitted venereally per rectum (e.g., *Neisseria gonorrhoeae*, *Treponema pallidum*, *Chlamydia*) may contribute to proctocolitis.

4. *Daycare participants and their family members*. Infections with *Shigella*, *Giardia*, *Cryptosporidium*, rotavirus, and other agents are very common and should be considered.

5. *Institutionalized persons*. Infectious diarrhea is one of the most frequent categories of nosocomial infections in many hospitals and long-term care facilities; the causes are a variety of microorganisms but most commonly *Clostridium difficile*.

The pathophysiology underlying acute diarrhea by infectious agents produces specific clinical features that may also be helpful in diagnosis ([Table 42-2](#)). Profuse watery diarrhea secondary to small bowel hypersecretion occurs with ingestion of preformed bacterial toxins, enterotoxin-producing bacteria, and enteroadherent pathogens. Diarrhea associated with marked vomiting and minimal or no fever may occur abruptly within a few hours after ingestion of the former two types; vomiting is usually less, and abdominal cramping or bloating is greater; fever is higher with the latter. Cytotoxin-producing and invasive microorganisms all cause high fever and abdominal pain. Invasive bacteria and *Entamoeba histolytica* often cause bloody diarrhea (referred to as *dysentery*). *Yersinia* invades the terminal ileal and proximal colon mucosa and may cause especially severe abdominal pain with tenderness mimicking acute appendicitis.

Finally, infectious diarrhea may be associated with systemic manifestations. Reiter's syndrome (arthritis, urethritis, and conjunctivitis) may accompany or follow infections by *Salmonella*, *Campylobacter*, *Shigella*, and *Yersinia*. Yersiniosis may also lead to an autoimmune-type thyroiditis, pericarditis, and glomerulonephritis. Both enterohemorrhagic *E. coli* (O157:H7) and *Shigella* can lead to the *hemolytic-uremic syndrome* with an attendant high mortality rate. Acute diarrhea can also be a major symptom of several systemic infections including *viral hepatitis*, *listeriosis*, *legionellosis*, and *toxic shock syndrome*.

Other Causes Side effects from medications are probably the most common noninfectious cause of acute diarrhea, and etiology may be suggested by a temporal association between use and symptom onset. Although innumerable medications may produce diarrhea, some of the more frequently incriminated include antibiotics, cardiac antidysrhythmics, antihypertensives, nonsteroidal anti-inflammatory drugs, certain antidepressants, chemotherapeutic agents, bronchodilators, antacids, and laxatives.

Occlusive or nonocclusive *ischemic colitis* typically occurs in persons older than 50 years of age, often presents as acute lower abdominal pain preceding watery, then bloody diarrhea, and generally results in acute inflammatory changes in the sigmoid or left colon while sparing the rectum. Acute diarrhea may accompany colonic *diverticulitis* and *graft-versus-host disease*. Acute diarrhea, often associated with systemic compromise, can follow ingestion of toxins including organophosphate insecticides, amanita and other mushrooms, arsenic, and preformed environmental toxins in seafoods, like ciguatera and scombroid. The conditions causing chronic diarrhea can also be confused with acute diarrhea early in their course. This confusion may occur with inflammatory bowel disease and some of the other inflammatory chronic diarrheas that may have an abrupt rather than insidious onset and exhibit features that mimic infection.

Approach to the Patient

The decision to evaluate acute diarrhea depends on its severity and duration and on various host factors ([Fig. 42-2](#)). Most episodes of acute diarrhea are mild and self-limited, and they do not justify the cost and potential morbidity of diagnostic or pharmacologic interventions. Indications for evaluation include profuse diarrhea with dehydration, grossly bloody stools, fever $\geq 38.5^{\circ}\text{C}$, duration $>48\text{ h}$ without improvement, new community outbreaks, associated severe abdominal pain in patients older than 50 years of age, and elderly (≥ 70 years) or immunocompromised patients. In some patients with moderately severe febrile diarrhea with fecal leukocytes (or increased fecal levels of the leukocyte proteins lactoferrin or calprotectin) present or with dysentery, a diagnostic evaluation might be eschewed in favor of an empiric antibiotic trial (see below).

The cornerstone of diagnosis in those suspected of severe acute infectious diarrhea is microbiologic analysis of the stool. Workup includes cultures for bacterial and viral pathogens, direct inspection for ova and parasites, and immunoassays for certain bacterial toxins (*C. difficile*), viral antigens (rotavirus), and protozoal antigens (*Giardia*, *E. histolytica*). The aforementioned clinical and epidemiologic associations may assist in focusing the evaluation. If a particular pathogen or set of possible pathogens is so implicated, then either the whole panel of routine studies may not be necessary or, in some instances, special cultures may be appropriate as for enterohemorrhagic and other types of *E. coli*, *Vibrio* species, and *Yersinia*. Molecular diagnosis of pathogens in stool can be made by identification of unique DNA sequences; and evolving microarray technologies could lead to a more rapid, sensitive, specific, and cost-effective diagnostic approach in the future.

Persistent diarrhea is commonly due to *Giardia*, but additional causative organisms that should be considered include *C. difficile* (especially if antibiotics had been administered), *E. histolytica*, *Cryptosporidium*, *Campylobacter*, and others. If stool studies are unrevealing, then flexible sigmoidoscopy with biopsies and upper endoscopy with duodenal aspirates and biopsies may be indicated.

Structural examination by sigmoidoscopy, colonoscopy, or abdominal CT scanning (or other imaging approaches) may be appropriate in patients with uncharacterized persistent diarrhea to exclude inflammatory bowel disease, or as an initial approach in

patients with suspected noninfectious acute diarrhea such as might be caused by ischemic colitis, diverticulitis, or partial bowel obstruction.

TREATMENT

Fluid and electrolyte replacement are of central importance to all forms of acute diarrhea. Fluid replacement alone may suffice for mild cases. Oral sugar-electrolyte solutions (sport drinks or designed formulations) should be instituted promptly with severe diarrhea to limit dehydration, which is the major cause of death. Profoundly dehydrated patients, especially infants and the elderly, require intravenous rehydration.

In moderately severe nonfebrile and nonbloody diarrhea, antiperistaltic antisecretory agents like loperamide can be useful adjuncts to control symptoms. Such agents should be avoided with febrile dysentery, which may be exacerbated or prolonged by them. Bismuth subsalicylate may reduce symptoms of vomiting and diarrhea but should not be used to treat immunocompromised patients because of the risk of bismuth encephalopathy.

Judicious use of antibiotics is appropriate in selected instances of acute diarrhea and may reduce its severity and duration ([Fig. 42-2](#)). Many physicians treat moderately to severely ill patients with febrile dysentery empirically without diagnostic evaluation using a quinolone, such as ciprofloxacin (500 mg bid for 3 to 5 d). Empiric treatment can also be considered for suspected giardiasis with metronidazole (250 mg qid for 7d). Selection of antibiotics and dosage regimens is otherwise dictated by specific pathogens and conditions found ([Chaps. 131,153,156-162](#)). Antibiotic coverage is indicated whether or not a causative organism is discovered in patients that are immunocompromised, have mechanical heart valves or recent vascular grafts, or are elderly. Antibiotic prophylaxis is indicated for certain patients traveling to high-risk countries in whom the likelihood or seriousness of acquired diarrhea would be especially high, including those with immunocompromise, inflammatory bowel disease, or gastric achlorhydria. Use of trimethoprim/sulfamethoxazole or ciprofloxacin may reduce bacterial diarrhea in such travelers by 90%.

CHRONIC DIARRHEA

Diarrhea lasting more than 4 weeks warrants evaluation to exclude serious underlying pathology. In contrast to acute diarrhea, most of the many causes of chronic diarrhea are noninfectious. The classification of chronic diarrhea by pathophysiologic mechanism facilitates a rational approach to management ([Table 42-3](#)).

Secretory Causes Secretory diarrheas are due to derangements in fluid and electrolyte transport across the enterocolic mucosa. They are characterized clinically by watery, large-volume fecal outputs that are typically painless and persist with fasting. Because there is no malabsorbed solute, stool osmolality is accounted for by normal endogenous electrolytes with no fecal osmotic gap.

Medications Side effects from regular ingestion of drugs and toxins are the most common secretory causes of chronic diarrhea. Hundreds of prescription and over-the-counter medications (see "Other Causes of Acute Diarrhea," above) may

produce unwanted diarrhea. Surreptitious or habitual use of stimulant laxatives [e.g., senna, cascara, bisacodyl, ricinoleic acid (castor oil)] must also be considered. Chronic ethanol consumption may cause a secretory-type diarrhea due to enterocyte injury with impaired sodium and water absorption as well as to rapid transit and other alterations. Inadvertent ingestion of certain environmental toxins (e.g., arsenic) may lead to chronic rather than acute forms of diarrhea. Certain bacterial infections may occasionally persist and be associated with a secretory-type diarrhea.

Bowel resection, mucosal disease, or enterocolic fistula These conditions may result in a secretory-type diarrhea because of inadequate surface for resorption of secreted fluids and electrolytes. Unlike other secretory diarrheas, this subset of conditions tends to worsen with eating. With disease (e.g., Crohn's ileitis) or resection of <100 cm of terminal ileum, dihydroxy bile acids may escape absorption and stimulate colonic secretion (cholorrheic diarrhea). This mechanism may contribute to so-called *idiopathic secretory diarrhea*, in which bile acids are functionally malabsorbed from a normal-appearing terminal ileum. Partial bowel obstruction, ostomy stricture, or fecal impaction may paradoxically lead to increased fecal output due to hypersecretion.

Hormones Although uncommon, the classic examples of secretory diarrhea are those mediated by hormones. *Metastatic gastrointestinal carcinoid tumors* or, rarely, *primary bronchial carcinoids* may produce watery diarrhea alone or as part of the carcinoid syndrome that comprises episodic flushing, wheezing, dyspnea, and right-sided valvular heart disease. Diarrhea is due to the release into the circulation of potent intestinal secretagogues including serotonin, histamine, prostaglandins, and various kinins. Pellagra-like skin lesions may rarely occur as the result of serotonin overproduction with niacin depletion. *Gastrinoma*, one of the most common neuroendocrine tumors, most typically presents with refractory peptic ulcers, but diarrhea occurs in up to one-third of cases and may be the only clinical manifestation in 10%. While various secretagogues released with gastrin may play a role, the diarrhea most often results from fat maldigestion owing to pancreatic enzyme inactivation by low intraduodenal pH. The watery diarrhea hypokalemia achlorhydria (WDHA) syndrome, also called *pancreatic cholera*, is due to a non-b cell pancreatic adenoma, referred to as a VIPoma, that secretes vasoactive intestinal peptide (VIP) and a host of other peptide hormones including pancreatic polypeptide, secretin, gastrin, gastrin-inhibitory polypeptide, neurotensin, calcitonin, and prostaglandins. The secretory diarrhea is often massive with stool volumes >3 L/d; daily volumes as high as 20 L have been reported. Life-threatening dehydration, neuromuscular dysfunction from associated hypokalemia, hypomagnesemia, or hypercalcemia, flushing, and hyperglycemia may accompany vipoma. *Medullary carcinoma of the thyroid* may present with watery diarrhea caused by calcitonin, other secretory peptides, or prostaglandins. This tumor occurs sporadically or, in 25 to 50% of cases, as a feature of multiple endocrine neoplasia type IIa with pheochromocytomas and hyperparathyroidism. Prominent diarrhea is often associated with metastatic disease and poor prognosis. *Systemic mastocytosis*, which may be associated with the skin lesion urticaria pigmentosa, may cause diarrhea that is either secretory and mediated by histamine, or inflammatory and due to intestinal filtration by mast cells. Large *colorectal villous adenomas* may rarely be associated with a secretory diarrhea that may cause hypokalemia, can be inhibited by NSAIDs, and is apparently mediated by prostaglandins.

Congenital defects in ion absorption Rarely, these defects cause watery diarrhea from birth and include defective $\text{Cl}^-/\text{HCO}_3^-$ -exchange (*congenital chloridorrhea*) with alkalosis and defective Na^+/H^+ -exchange with acidosis. Some hormone deficiencies may be associated with watery diarrhea, such as occurs with adrenocortical insufficiency (Addison's disease) that may be accompanied by hyperpigmentation.

Osmotic Causes Osmotic diarrhea occurs when ingested, poorly absorbable, osmotically active solutes draw enough fluid lumenward to exceed the resorptive capacity of the colon. Fecal water output increases in proportion to such a solute load. Osmotic diarrhea characteristically ceases with fasting or with discontinued oral intake of the offending agent.

Osmotic laxatives Ingestion of magnesium-containing antacids, health supplements, or laxatives may induce osmotic diarrhea typified by a stool osmotic gap: $2([\text{Na}] + [\text{K}]) \ll 290 \text{ mosm/kg}$. Anionic laxatives containing sulfates or phosphates produce osmotic diarrhea without an osmotic gap, as sodium accompanies the anionic solutes; direct measurement of stool sulfates and phosphates may be necessary to confirm the cause of diarrhea.

Carbohydrate malabsorption Carbohydrate malabsorption due to acquired or congenital defects in brush-border disaccharidases and other enzymes leads to osmotic diarrhea with a low pH. One of the most common causes of chronic diarrhea in adults is *lactase deficiency*, which affects three-fourths of non-Caucasians worldwide and 5 to 30% of persons in the United States; most learn to avoid milk products without an intervention. Some sugars, such as sorbitol, are universally malabsorbed, and diarrhea ensues with ingestion of ample medications, gum, or candies sweetened with these nonabsorbable sugars. Lactulose, used to acidify stools in patients with hepatic failure, also causes diarrhea on this basis.

Steatorrheal Causes Fat malabsorption may lead to greasy, foul-smelling, difficult-to-flush diarrhea often associated with weight loss and nutritional deficiencies due to concomitant malabsorption of amino acids and vitamins. Increased fecal output is caused by the osmotic effects of fatty acids, especially after bacterial hydroxylation, and, to a lesser extent, by the burden of neutral fat. Quantitatively, steatorrhea is defined as stool fat exceeding the normal 7 g/d; daily fecal fat averages 15 to 25 g with small intestinal diseases and often exceeds 40 g with pancreatic exocrine insufficiency. Intraluminal maldigestion, mucosal malabsorption, or lymphatic obstruction may produce steatorrhea.

Intraluminal maldigestion This condition most commonly results from pancreatic exocrine insufficiency, which occurs when >90% of pancreatic secretory function is lost. *Chronic pancreatitis*, usually a sequela of ethanol abuse, most frequently causes pancreatic insufficiency. Other causes include *cystic fibrosis*, *pancreatic duct obstruction*, and rarely, *somatostatinoma*. Bacterial overgrowth in the small intestine may deconjugate bile acids and alter micelle formation that impair fat digestion; it occurs with stasis from a blind-loop, small bowel diverticulum, or dysmotility and is especially likely in the elderly. Finally, cirrhosis or biliary obstruction may lead to mild steatorrhea due to deficient intraluminal bile acid concentration.

Mucosal Malabsorption Mucosal malabsorption occurs from a variety of enteropathies, but most prototypically and perhaps most commonly from *celiac sprue*. This gluten-sensitive enteropathy characterized by villous atrophy and crypt hyperplasia in the proximal small bowel often presents with fatty diarrhea associated with multiple nutritional deficiencies of varying severity and affects all ages. *Tropical sprue* may produce a similar histologic and clinical syndrome, but it occurs in residents of or travelers to tropical climates; its often abrupt onset and response to antibiotics suggest an infectious etiology. *Whipple's disease*, due to the actinomycete *Treponema whipplei* and histiocytic infiltration of the small bowel mucosa, is a less common cause of steatorrhea that most typically occurs in young or middle-aged men; it is frequently associated with arthralgias, fever, lymphadenopathy, and extreme fatigue and may affect the central nervous system and endocardium. A similar clinical and histologic picture results from *Mycobacterium avium intracellulare* infection in patients with AIDS. *Abetalipoproteinemia* is a rare defect of chylomicron formation and fat malabsorption in children associated with acanthocytic erythrocytes, ataxia, and retinitis pigmentosa. Several other conditions may cause mucosal malabsorption including infections, especially with protozoa like *Giardia*, numerous medications (e.g., colchicine, cholestyramine, neomycin), and chronic ischemia.

Postmucosal lymphatic obstruction The pathophysiology of this condition, which is due to the rare *congenital intestinal lymphangiectasia* or to *acquired lymphatic obstruction* secondary to trauma, tumor, or infection, leads to the unique constellation of fat malabsorption with enteric losses of protein (often causing edema) and lymphocytes (with resultant lymphocytopenia) that enter the portal circulation directly. Carbohydrate and amino acid absorption are preserved.

Inflammatory Causes Inflammatory diarrheas are generally accompanied by pain, fever, bleeding, or other manifestations of inflammation. The mechanism of diarrhea may not only be exudation but, depending on lesion site, may include fat malabsorption, disrupted fluid/electrolyte absorption, and hypersecretion or hypermotility from release of cytokines and other inflammatory mediators. The unifying feature on stool analysis is the presence of leukocytes or leukocyte-derived proteins such as calprotectin. With severe inflammation, exudative protein loss can lead to anasarca (generalized edema). Any middle-aged or older person with chronic inflammatory-type diarrhea, especially with blood, should be carefully evaluated to exclude a colorectal or large enteric tumor.

Idiopathic inflammatory bowel disease The illnesses in this category, which include *Crohn's disease* and *chronic ulcerative colitis*, are among the most common organic causes of chronic diarrhea in adults and range in severity from mild to fulminant and life threatening. They may be associated with uveitis, polyarthralgias, cholestatic liver disease (primary sclerosing cholangitis), and various skin lesions (erythema nodosum, pyoderma gangrenosum). *Microscopic colitis*, including *collagenous colitis*, is an increasingly recognized cause of chronic watery diarrhea; biopsy of a normal appearing colorectum is required for histologic diagnosis.

Primary or secondary forms of immunodeficiency Immunodeficiency may lead to prolonged infectious diarrhea. With common, variable *hypogammaglobulinemia*, diarrhea is particularly prevalent and often the result of giardiasis.

Eosinophilic gastroenteritis Eosinophil infiltration of the mucosa, muscularis, or serosa at any level of the gastrointestinal tract may cause diarrhea, pain, vomiting, or ascites. Affected patients often have an atopic history, Charcot-Leyden crystals due to extruded eosinophil contents may be seen on microscopic inspection of stool, and peripheral eosinophilia is present in 50 to 75% of patients. While hypersensitivity to certain foods occurs in adults, true food allergy causing chronic diarrhea is rare.

Other Causes Chronic inflammatory diarrhea may be caused by *radiation enterocolitis*, *chronic graft-versus-host disease*, *Behcet's syndrome*, and *Cronkite-Canada syndrome*, among others.

Dysmotile Causes Rapid transit may accompany many diarrheas as a secondary or contributing phenomenon, but primary dysmotility is an unusual etiology of true diarrhea. Stool features often suggest a secretory diarrhea, but mild steatorrhea up to 14 g of fat per day can be produced by maldigestion from rapid transit alone. *Hyperthyroidism*, *carcinoid syndrome*, and certain drugs (e.g., prostaglandins, prokinetic agents) may produce hypermotility with resultant diarrhea. Primary visceral neuromyopathies or idiopathic acquired intestinal pseudo-obstruction may lead to stasis with secondary bacterial overgrowth causing diarrhea. *Diabetic diarrhea*, often accompanied by peripheral and generalized autonomic neuropathies, may occur in part because of intestinal dysmotility.

The exceedingly common *irritable bowel syndrome* (10% point prevalence, 1 to 2% per year incidence) is characterized by disturbed intestinal and colonic motor and sensory responses to various stimuli. Symptoms of stool frequency typically cease at night, alternate with periods of constipation, are accompanied by abdominal pain relieved with defecation, and rarely result in weight loss or true diarrhea.

Factitial Causes Factitial diarrhea accounts for up to 15% of unexplained diarrheas referred to tertiary care centers. Either as a form of *Munchausen syndrome* (deception or self-injury for secondary gain) or *bulimia*, some patients covertly self-administer laxatives alone or in combination with other medications (e.g., diuretics) or surreptitiously add water or urine to stool sent for analysis. Such patients are typically women, often with histories of psychiatric illness and disproportionately from careers in health care. Hypotension and hypokalemia are common co-presenting features. Such patients often deny this possibility when confronted, but they do benefit from psychiatric counseling when they acknowledge their behavior.

Approach to the Patient

The laboratory tools available to evaluate the very common problem of chronic diarrhea are extensive, and many are costly and invasive. As such, the diagnostic evaluation must be rationally directed by a careful history and physical examination, and simple triage tests are often warranted before complex investigations are launched ([Fig. 42-3](#)). The history, physical examination, and routine blood studies should attempt to characterize the mechanism of diarrhea, identify diagnostically helpful associations, and assess the patient's fluid/electrolyte and nutritional status. Patients should be questioned about the onset, duration, pattern, aggravants (especially diet), relieving factors, and stool characteristics of their diarrhea. The presence or absence of fecal

incontinence, fever, weight loss, pain, certain exposures (travel, medications, contacts with diarrhea), and common extraintestinal manifestations (skin changes, arthralgias, oral aphtha) should be noted. Physical findings may offer clues such as a thyroid mass, wheezing, heart murmurs, edema, hepatomegaly, abdominal masses, lymphadenopathy, mucocutaneous abnormalities, perianal fistulae, or anal sphincter laxity. Peripheral blood counts may reveal leukocytosis that suggests inflammation; anemia that reflects blood loss or nutritional deficiencies; or eosinophilia that may occur with parasitoses, neoplasia, collagen-vascular disease, allergy, or eosinophilic gastroenteritis. Blood chemistries may demonstrate electrolyte, hepatic, or other metabolic disturbances.

A therapeutic trial is often appropriate, definitive, and highly cost-effective when a specific diagnosis is suggested on the initial physician encounter. For example, chronic watery diarrhea, which ceases with fasting in an otherwise healthy young adult, may justify a trial of a lactose-restricted diet; bloating and diarrhea persisting since a mountain backpacking trip may warrant a trial of metronidazole for likely giardiasis; and postprandial diarrhea persisting since an ileal resection might be treated with cholestyramine before further evaluation. Persistent symptoms require additional investigation.

Certain diagnoses may be suggested on the initial encounter, e.g., idiopathic inflammatory bowel disease; however, additional focused evaluations may be necessary to confirm the diagnosis and characterize the severity or extent of disease so that treatment can be best guided. Patients suspected of having irritable bowel syndrome should be initially evaluated with proctosigmoidoscopy and mucosal biopsies; those with normal findings might be reassured and, as indicated, treated empirically with antispasmodics, antidiarrheals, bulk agents, anxiolytics, or antidepressants. Any patient who presents with chronic diarrhea and hematochezia should be evaluated with stool microbiologic studies and colonoscopy.

In an estimated two-thirds of cases, the cause for chronic diarrhea remains unclear after the initial encounter, and further testing is required. Quantitative stool collection and analyses can yield important objective data that may establish a diagnosis or characterize the type of diarrhea as a triage for focused additional studies ([Fig. 42-3](#)). If stool weight exceeds 200 g/d, additional stool analyses should be performed that might include electrolyte concentration, pH, occult blood testing, leukocyte inspection (or leukocyte protein assay), fat quantitation, and laxative screens.

For secretory diarrheas (watery, normal osmotic gap), possible medication-related side effects or surreptitious laxative use should be reconsidered. Microbiologic studies should be done including fecal bacterial cultures (including media for *Aeromonas* and *Pleisiomonas*), inspection for ova and parasites, and *Giardia* antigen assay (the most sensitive test for giardiasis). Small bowel bacterial overgrowth can be excluded by intestinal aspirates with quantitative cultures or with glucose or xylose breath tests involving measurement of breath hydrogen or other metabolite (e.g., $^{14}\text{CO}_2$). However, interpretation of these breath tests may be confounded by disturbances of intestinal transit. When suggested by history or other findings, screens for peptide hormones should be pursued (e.g., serum gastrin, [VIP](#), calcitonin, and thyroid hormone/thyroid stimulating hormone, or urinary 5-hydroxyindolacetic acid and histamine). Upper

endoscopy and colonoscopy with biopsies and small bowel barium x-rays are helpful to rule out structural or occult inflammatory disease.

Further evaluation of osmotic diarrhea should include tests for lactose intolerance and magnesium ingestion, the two most common causes. Low fecal pH suggests carbohydrate malabsorption; lactose malabsorption can be confirmed by lactose breath testing or by a therapeutic trial with lactose exclusion and observation of the effect of lactose challenge (e.g., a quart of milk). Lactase determination on small bowel biopsy is generally not available. If fecal Mg^{2+} or laxative levels are elevated, then inadvertent or surreptitious ingestion should be considered and psychiatric help should be sought.

For those with proven fatty diarrhea, endoscopy with small bowel biopsy (including aspiration for *Giardia* and quantitative cultures) should be performed; if this procedure is unrevealing, a small bowel radiograph is often an appropriate next step. If small bowel studies are negative or if pancreatic disease is suspected, pancreatic exocrine insufficiency should be excluded with direct tests, such as the secretin-cholecystokinin stimulation test, or by indirect tests, such as assay of fecal chymotrypsin activity or a bentiromide test.

Chronic inflammatory-type diarrheas should be suspected by the presence of blood or leukocytes in the stool. Such findings warrant stool cultures, inspection for ova and parasites, *C. difficile* toxin assay, colonoscopy with biopsies, and if indicated, small bowel oral contrast studies.

TREATMENT

Treatment of chronic diarrhea depends on the specific etiology and may be curative, suppressive, or empiric. If the cause can be eradicated, treatment is curative as with resection of a colorectal cancer, antibiotic administration for Whipple's disease, or discontinuation of an offending drug. For many chronic conditions, diarrhea can be controlled by suppression of the underlying mechanism. Examples include elimination of dietary lactose for lactase deficiency or gluten for celiac sprue, use of glucocorticoids or other anti-inflammatory agents for idiopathic inflammatory bowel diseases, adsorptive agents such as cholestyramine for ileal bile acid malabsorption, proton pump inhibitors such as omeprazole for the gastric hypersecretion of gastrinomas, somatostatin analogues such as octreotide for malignant carcinoid, prostaglandin inhibitors such as indomethacin for medullary carcinoma of the thyroid, and pancreatic enzyme replacement for pancreatic insufficiency. When the specific cause or mechanism of chronic diarrhea evades diagnosis, empiric therapy may be beneficial. Mild opiates such as diphenoxylate or loperamide are often helpful in mild or moderate watery diarrhea. For those with more severe diarrhea, codeine or tincture of opium may be beneficial. Such antimotility agents should be avoided with inflammatory bowel disease, as toxic megacolon may be precipitated. Clonidine, an α_2 -adrenergic agonist, may allow control of diabetic diarrhea. For all patients with chronic diarrhea, fluid and electrolyte repletion is an important component of management (see "Acute Diarrhea," above). Replacement of fat-soluble vitamins may also be necessary in patients with chronic steatorrhea.

CONSTIPATION

DEFINITION

Constipation is a common complaint in clinical practice and usually refers to persistent, difficult, infrequent, or seemingly incomplete defecation. Because of the wide range of normal bowel habits, constipation is difficult to define precisely. Most persons have at least three bowel movements per week; however, stool frequency alone is not a sufficient criterion for the diagnosis of constipation because many constipated patients describe a normal frequency of defecation but subjective complaints of excessive straining, hard stools, lower abdominal fullness, and a sense of incomplete evacuation. The individual patient's symptoms must be analyzed in detail to ascertain what is meant by "constipation" or "difficulty" with defecation.

Stool form and consistency are well correlated with the time elapsed from the preceding defecation. Hard, pelleted stools occur with slow transit, while loose watery stools are associated with rapid transit. Small, pelleted stools are more difficult to expel than large ones.

The perception of hard stools or excessive straining is more difficult to assess objectively, and the need for enemas or digital disimpaction is a clinically useful way to corroborate the patient's perceptions of difficult defecation.

Psychosocial factors may also be important. A person whose parents attached great importance to daily defecation will become greatly concerned when he or she misses a daily bowel movement; some children withhold stool to gain attention; and some adults are simply too busy or too embarrassed to interrupt their work when the call to have a bowel movement is sensed.

CAUSES

Pathophysiologically, chronic constipation generally results from inadequate fiber intake or from disordered colonic transit or anorectal function as a result of a neurogastroenterologic disturbance, certain drugs, or in association with a large number of systemic diseases that affect the gastrointestinal tract ([Table 42-4](#)). Constipation of recent onset may be a symptom of significant organic disease such as tumor or stricture. In *idiopathic constipation*, a subset of patients exhibit delayed emptying of the ascending and transverse colon with prolongation of transit (often in the proximal colon) and a reduced frequency of propulsive colonic contractions ([HAPCs](#)). *Outlet obstruction to defecation* (also called *evacuation disorders*) may cause delayed colonic transit, which is usually corrected by biofeedback retraining of the disordered defecation. Constipation of any cause may be exacerbated by chronic illnesses that lead to physical or mental impairment and result in inactivity or physical immobility.

Approach to the Patient

A careful history should explore the patient's symptoms and confirm whether he or she is indeed constipated based on frequency (e.g., <3 bowel movements per week), consistency (lumpy/hard), excessive straining, prolonged defecation time, or need to support the perineum or digitate the anorectum. In the vast majority of cases (probably >90%), there is no underlying cause (e.g., cancer, depression, or hypothyroidism), and

constipation responds to ample hydration, exercise, and supplementation of dietary fiber (15 to 25 g/d). A good diet and medication history and attention to psychosocial issues are key. Physical examination and, particularly, a rectal examination should exclude most of the important diseases that present with constipation and possibly indicate features suggesting an evacuation disorder (e.g., high anal sphincter tone).

There is broad consensus on the selection of patients for further investigation. The presence of weight loss, rectal bleeding, or anemia with constipation mandates either sigmoidoscopy plus barium enema or colonoscopy alone, particularly in patients over 40 years of age, to exclude structural diseases such as cancer or strictures. Colonoscopy alone is most cost effective in this setting since it provides an opportunity to biopsy mucosal lesions, perform polypectomy, or dilate strictures. Barium enema has advantages over colonoscopy in the patient with isolated constipation, since it is less costly and identifies colonic dilatation and all significant mucosal lesions or strictures that are likely to present with constipation. Melanosis coli, or pigmentation of the colon mucosa, indicates the use of anthraquinone laxatives such as cascara or senna; however, this is usually apparent from a careful history. An unexpected disorder such as megacolon or cathartic colon may also be detected by colonic radiographs. Measurement of serum calcium and thyroid stimulating hormone levels will identify rare patients with metabolic disorders.

Patients with more troublesome constipation may not respond to fiber alone and may be helped by a bowel training regimen: taking an osmotic laxative and evacuating with enema or glycerine suppository as needed. After breakfast, a distraction-free 15 to 20 min on the toilet without straining is encouraged. Excessive straining may lead to development of hemorrhoids, and, if there is weakness of the pelvic floor or injury to the pudendal nerve, may result in obstructed defecation from descending perineum syndrome several years later. Those few who do not benefit from the simple measures delineated above or require long-term treatment with stimulant laxatives with the attendant risk of developing laxative abuse syndrome are assumed to have severe or intractable constipation and should have further investigation ([Fig. 42-4](#)).

INVESTIGATION OF SEVERE CONSTIPATION

A small minority (probably <5%) of all patients with constipation have cases that are considered severe or "intractable"; these are the patients most likely to be seen by gastroenterologists or in referral centers. Further observation of the patient may occasionally reveal a previously unrecognized cause, such as an evacuation disorder, laxative abuse, malingering, or psychiatric disorder. In these patients, recent studies suggest that evaluations of the physiologic function of the colon and pelvic floor and of psychological status aid in the rational choice of treatment. Even among these highly selected patients with severe constipation, a cause can be identified in only about 30% (see below).

Measurement of Colonic Transit Radiopaque marker transit tests are easy, repeatable, generally safe, inexpensive, reliable, and highly applicable in evaluating constipated patients in clinical practice. There are several validated methods that are very simple. For example, radiopaque markers are ingested, and an abdominal flat film taken 5 d later should indicate passage of 80% of the markers out of the colon. This test

does not provide useful information about the transit profile of the stomach and small bowel, and avoidance of laxatives or enemas during the testing period is essential.

Radioscintigraphy with a delayed-release capsule containing radiolabeled particles has been used to noninvasively characterize normal, accelerated, or delayed colonic function over 24 to 48 h with low radiation exposure. This approach simultaneously assesses gastric, small bowel, and colonic transit. The disadvantages are the greater cost and the need for specific materials prepared in a nuclear medicine laboratory.

Anorectal and Pelvic Floor Tests Pelvic floor dysfunction is suggested by the inability to evacuate the rectum, a feeling of persistent rectal fullness, rectal pain, the need to extract stool from the rectum digitally, application of pressure on the posterior wall of the vagina, support of the perineum during straining, and excessive straining. These significant symptoms should be contrasted with the sense of incomplete rectal evacuation, which is common in irritable bowel syndrome.

Patients with clinically suspected obstruction of defecation should also be evaluated by a psychologist to identify eating disorders or a "need to control," to provide stress management or relaxation training, and to identify depression.

A simple clinical test in the office to document a nonrelaxing puborectalis muscle is to have the patient strain to expel the index finger during a digital rectal exam. Motion of the puborectalis posteriorly during straining indicates proper coordination of the pelvic floor muscles.

Measurement of perineal descent is relatively easy to gauge clinically by placing the patient in the left decubitus position and watching the perineum to assess either paucity or lack of descent (<1.5 cm, a sign of pelvic floor dysfunction) or perineal ballooning during straining relative to bony landmarks (>4 cm, suggesting excessive perineal descent).

A useful overall test of evacuation is the balloon expulsion test. A urinary catheter is placed in the rectum, the balloon is inflated to 50 ml with water, and a determination is made about whether the patient can expel it while seated on a toilet or in the left lateral decubitus position. In the lateral position, the weight needed to facilitate expulsion of the balloon (normal, 0 to 200 g) is determined.

Anorectal manometry is not often contributory in the evaluation of patients presenting with severe constipation, except when an excessively high resting or squeeze anal sphincter tone suggests anismus (anal sphincter spasm). This test also identifies rare syndromes, such as adult Hirschsprung's disease, by the absence of the rectoanal inhibitory reflex or the presence of occult incontinence.

Defecography (a dynamic barium enema including lateral views obtained during barium expulsion) reveals "soft abnormalities" in many patients; the most relevant findings are the measured changes in rectoanal angle, anatomic defects of the rectum, and enteroceles or rectoceles. In a very small proportion of patients, significant anatomic defects associated with intractable constipation respond best to surgical treatment. These defects include severe intussusception with complete outlet obstruction due to

funnel-shaped plugging at the anal canal or an extremely large rectocele that is preferentially filled during attempts at defecation instead of expulsion of the barium through the anus. In summary, defecography requires an interested and experienced radiologist, and abnormalities are not pathognomonic for pelvic floor dysfunction. More commonly, outlet obstruction results from a nonrelaxing puborectalis muscle, which impedes rectal emptying, rather than from defects identified by defecography.

Dynamic imaging studies such as proctography during defecation or scintigraphic expulsion of artificial stool help measure perineal descent and the rectoanal angle during rest, squeezing and straining, and scintigraphic expulsion quantitates the amount of "artificial stool" emptied. Failure of the rectoanal angle to increase significantly ($\sim 15^\circ$) during straining confirms pelvic floor dysfunction.

Neurologic testing (EMG) is more helpful in the evaluation of patients with incontinence than of those with symptoms suggesting obstructed defecation. The absence of neurologic signs in the lower extremities suggests that any documented denervation of the puborectalis results from pelvic (e.g., obstetric) injury or from stretching of the pudendal nerve by chronic, long-standing straining.

Ultrasonography identifies sphincter or rectal wall defects and may help select patients for surgical correction. Spinal-evoked responses during electrical rectal stimulation or stimulation of external anal sphincter contraction by applying magnetic stimulation over the lumbosacral cord identify patients with limited sacral neuropathies with sufficient residual nerve conduction to attempt biofeedback training.

In summary, a balloon expulsion test is an important screening test for anorectal dysfunction. If positive, an anatomic evaluation of the rectum or anal sphincters and an assessment of pelvic floor relaxation are the tools for evaluating patients in whom obstructed defecation is suspected.

TREATMENT

After the cause of constipation is characterized, a treatment decision can be made. Slow transit constipation requires aggressive medical or surgical treatment; anismus or pelvic floor dysfunction usually responds to biofeedback management ([Fig. 42-4](#)). However, only about 30% of patients with severe constipation are found to have such a physiologic disorder.

Patients with slow transit constipation are treated with bulk, osmotic, and stimulant laxatives, including fiber, psyllium, milk of magnesia, lactulose, polyethylene glycol (colonic lavage solution), and bisacodyl. If a 2- to 3-month trial of medical therapy fails and patients continue to have documented slow transit constipation unassociated with obstructed defecation, colectomy with ileorectostomy is indicated. The decision to resort to surgery is facilitated in the presence of megacolon and megarectum. The complications after surgery include small bowel obstruction (11%) and fecal soiling, particularly at night during the first postoperative year.

Patients who have a combined disorder should pursue pelvic floor retraining (biofeedback and muscle relaxation), psychological counseling, and dietetic advice first,

followed by colectomy and ileorectostomy if colonic transit studies do not normalize with biofeedback alone. In patients with pelvic floor dysfunction alone, biofeedback training has a 70 to 80% success rate, measured by the acquisition of comfortable stool habits. Attempts to manage pelvic floor dysfunction with operations (internal anal sphincter or puborectalis muscle division) have achieved only mediocre success and have been largely abandoned.

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43. WEIGHT LOSS - Carol M. Reife

Significant unintentional weight loss in a previously healthy individual is often a harbinger of underlying systemic disease. During the routine medical history, therefore, inquiry should always be made about changes in weight; loss of 5% of body weight over 6 to 12 months should prompt further evaluation.

PHYSIOLOGY OF WEIGHT REGULATION

The normal individual maintains weight at a remarkably stable "set point," given the wide variation in daily caloric intake and level of activity. Because of the physiologic importance of maintaining energy stores, voluntary weight loss is difficult to achieve and sustain.

Appetite and metabolism are regulated by an intricate network of neural and hormonal factors. The hypothalamic feeding and satiety centers play a central role in these processes ([Chap. 77](#)). Neuropeptides, like corticotropin-releasing hormone (CRH), α -melanocyte stimulating hormone (α -MSH), and cocaine and amphetamine-related transcript (CART) induce anorexia by acting centrally on satiety centers. Epinephrine and norepinephrine cause a decrease in food intake and an increase in metabolic rate ([Chap. 72](#)). Amphetamines and related drugs used to suppress appetite act by releasing norepinephrine in the central nervous system. The gastrointestinal peptides glucagon, somatostatin, and particularly cholecystokinin induce a decrease in food intake by acting through a vagal mechanism to signal satiety. Hypoglycemia decreases levels of insulin which reduces glucose utilization and inhibits activity of the satiety center.

Leptin plays a central role in the long-term maintenance of weight homeostasis ([Chap. 77](#)). Leptin is produced by adipose tissue and acts on the hypothalamus to decrease food intake and increase energy expenditure. It suppresses expression of hypothalamic neuropeptide Y, a potent appetite stimulatory peptide. In parallel, leptin increases the expression of α -MSH, which decreases appetite by acting on the MC4R melanocortin receptor. Thus, leptin activates a series of downstream neural pathways that alter food-seeking behavior and metabolism. However, leptin deficiency, which occurs in conjunction with the loss of adipose tissue, stimulates appetite and induces other adaptive responses including inhibition of hypothalamic thyrotropin releasing hormone (TRH) and gonadotropin releasing hormone (GnRH).

A variety of cytokines, including tumor necrosis factor α (TNF- α), interleukin (IL) (IL-6), IL-1, interferon γ (IFN- γ), ciliary neurotrophic factor (CNTF), and leukemia inhibitory factor (LIF), can contribute to cachexia ([Chap. 17](#)). In addition to causing anorexia, these factors may induce fever, depress myocardial function, modulate immune and inflammatory responses, and induce a variety of specific metabolic alterations. TNF- α , for example, preferentially mobilizes fat but spares skeletal muscle. Levels of one or more of these cytokines may be increased in patients with cancer, sepsis, chronic inflammatory conditions, AIDS, and congestive heart failure.

Weight loss occurs when energy expenditure exceeds calories available for energy utilization. In most individuals, approximately half of food energy is utilized for basal

processes such as maintenance of body temperature. In a 70-kg person, basal activity consumes about 1800 kcal/d. About 40% of caloric intake is used for physical activity, although athletes may use more than 50% during vigorous exercise. About 10% of caloric intake is used for dietary thermogenesis, the energy expended for digestion, absorption, and metabolism of food.

Mechanisms of weight loss include decreased food intake, malabsorption, loss of calories, and increased energy requirements ([Fig. 43-1](#)). Changes in weight may reflect alterations in either tissue mass or body fluid content. A deficit of 3500 kcal generally correlates with the loss of 1 lb (0.45 kg) of body fat, but one must also consider water weight (2.2 lb/L) gained or lost. Weight loss that persists over weeks to months is almost invariably due to loss of tissue mass.

Food intake may be influenced by a wide variety of visual, olfactory, and gustatory stimuli as well by genetic, psychological, and social factors. Absorption may be impaired because of pancreatic insufficiency, cholestasis, celiac sprue, intestinal tumors, radiation injury, inflammatory bowel disease, infection, or medication effect. Manifestations of these disease processes may be suggested by changes in stool frequency and consistency. Calories also may be lost due to vomiting or diarrhea, glucosuria in diabetes mellitus, or fistulous drainage. Resting energy expenditure decreases with age and can be affected by thyroid status. Beginning at about age 60, body weight declines by an average of 0.5% per year. Body composition is also affected by aging; adipose tissue increases and lean muscle mass decreases with age.

SIGNIFICANCE OF WEIGHT LOSS

Unintentional weight loss, especially in the elderly, is not uncommon and is associated with increased morbidity and mortality rates, even after comorbid conditions have been taken into account. Prospective studies indicate that significant involuntary weight loss is associated with a mortality rate of 25% over the next 18 months. Retrospective studies of significant weight loss in the elderly document mortality rates of 9 to 38% over a 2- to 3-year period.

Cancer patients with weight loss have decreased performance status, response to chemotherapy, and median survival ([Chap. 79](#)). Marked degrees of weight loss also predispose to infection. Patients undergoing elective surgery, who have lost more than 10 lb (4.5 kg) in 6 months, have higher surgical mortality rates. Vitamin and nutrient deficiencies also can accompany significant weight loss ([Chap. 74](#)).

CAUSES OF WEIGHT LOSS

The list of possible causes of weight loss is extensive ([Table 43-1](#)). In the elderly, the most common causes of weight loss are depression, cancer, and benign gastrointestinal disease. Lung and gastrointestinal cancer are the most common malignancies in patients presenting with weight loss. In younger individuals, diabetes mellitus, hyperthyroidism, psychiatric disturbances including eating disorders, and infection, especially with HIV, should be considered.

The cause of involuntary weight loss is rarely occult. Careful history and physical

examination, in association with directed diagnostic testing, will identify the cause of weight loss in 75% of patients. The etiology of weight loss will not be found in the remaining patients, despite extensive testing. Patients with negative evaluations tend to have lower mortality rates than those found to have organic disease.

Patients with medical causes of weight loss usually have signs or symptoms that suggest involvement of a particular organ system. Gastrointestinal tumors, including those of the pancreas and liver, may affect food intake early in the course of illness, causing weight loss before other symptoms are apparent. Lung cancer may present with post-obstructive pneumonia, dyspnea, or cough and hemoptysis; however, it may be silent and should be considered even in those without a history of cigarette smoking. Depression and isolation can cause profound weight loss, especially in the elderly. Chronic pulmonary disease and congestive heart failure can produce anorexia and may also increase resting energy expenditure. Weight loss may be the presenting sign of infectious diseases such as HIV infection, tuberculosis, endocarditis, and fungal and parasitic infections. Hyperthyroidism or pheochromocytoma increase metabolism; elderly patients with apathetic hyperthyroidism may present with weight loss alone. New onset diabetes mellitus is often accompanied by weight loss, reflecting glucosuria and loss of the anabolic actions of insulin. Adrenal insufficiency may be suggested by increased pigmentation, hyponatremia, and hyperkalemia.

Approach to the Patient

Before extensive evaluation is undertaken, it is important to confirm that weight loss has occurred. Almost half of patients who claim significant weight loss have no actual change in weight when it is measured objectively. If weight loss is present, efforts should be made to determine the time interval over which it has occurred. In the absence of documentation, changes in belt notch size or the fit of clothing may help confirm loss of weight. Not infrequently, patients who have actually sustained significant weight loss are unaware that it has occurred. Routine documentation of weight during office visits is therefore important.

The review of systems should focus on signs or symptoms that are associated with disorders that commonly cause weight loss. These include fever, pain, shortness of breath or cough, palpitations, changes in pattern of urination, and evidence of neurologic disease. Gastrointestinal disturbances, including difficulty eating, dysphagia, anorexia, nausea, and change in bowel habits, should be sought. Use of cigarettes, alcohol, and all medications should be reviewed, and patients should be questioned about previous illness or surgery as well as diseases in family members. Risk factors for HIV infection should be assessed. Signs of depression, evidence of dementia, and social factors, including financial issues that might affect food intake, should be considered.

Physical examination should begin with weight determination and documentation of vital signs. The skin should be examined for pallor, jaundice, turgor, scars from prior surgery, and stigmata of systemic disease. The search for oral thrush or dental disease, thyroid gland enlargement, adenopathy, and respiratory or cardiac abnormalities and a detailed examination of the abdomen often lead to clues for further evaluation. Rectal examination, including prostate exam and testing of stool for occult blood, should be

performed in men; and all women should have a pelvic examination, even if they have had a hysterectomy. Neurologic examination should include mental status assessment and screening for depression.

Laboratory testing should confirm or exclude possible diagnoses elicited from the history and physical examination ([Table 43-2](#)). An initial phase of testing should include a complete blood count with differential, serum chemistry tests including glucose, electrolytes, renal and liver tests, calcium, thyroid stimulating hormone (TSH), urinalysis, and chest x-ray. Patients at risk for HIV infection should have HIV antibody testing. In all cases, recommended cancer screening tests appropriate for the gender and age group, such as mammograms and Pap smears, should be updated ([Chap. 80](#)). If gastrointestinal signs or symptoms are present, upper and/or lower endoscopy and abdominal imaging with either computed tomography (CT) or magnetic resonance imaging (MRI) have a relatively high yield, consistent with the high prevalence of gastrointestinal disorders in patients with weight loss. If an etiology of weight loss is not found, careful clinical follow-up, rather than persistent undirected testing, is reasonable.

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44. GASTROINTESTINAL BLEEDING - Loren Laine

Bleeding from the gastrointestinal (GI) tract may present in 5 ways. *Hematemesis* is vomitus of red blood or "coffee-grounds" material. *Melena* is black, tarry, foul-smelling stool. *Hematochezia* is the passage of bright red or maroon blood from the rectum. *Occult GI bleeding (GIB)* may be identified in the absence of overt bleeding by special examination of the stool (e.g., guaiac testing). Finally, patients may present only with *symptoms of blood loss or anemia* such as lightheadedness, syncope, angina, or dyspnea.

SOURCES OF GASTROINTESTINAL BLEEDING

UPPER GASTROINTESTINAL SOURCES OF BLEEDING ([Table 44-1](#))

The annual incidence of hospital admissions for upper GIB (UGIB) in the United States and Europe is approximately 0.1%, with a mortality rate of ~10%. Patients rarely die from exsanguination; rather, they die due to decompensation from other underlying illnesses. The mortality rate for patients under 60 years of age in the absence of malignancy or organ failure is <1%.

Peptic ulcers are the most common cause of [UGIB](#), accounting for about 50% of cases. Mallory-Weiss tears account for 5 to 15% of cases. The proportion of patients bleeding from varices varies widely from ~5 to 30%, depending on the population. Hemorrhagic or erosive gastropathy [e.g., due to nonsteroidal anti-inflammatory drugs (NSAIDs) or alcohol] and erosive esophagitis often cause mild UGIB, but major bleeding is rare.

Peptic Ulcers Clinical features that predict poorer outcome include hemodynamic instability, the number of units of blood transfused, red blood in the emesis and the stool, increasing age, and the presence of concurrent illness. Characteristics of the ulcer at endoscopy also provide important prognostic information. One-third of patients with active bleeding or a non-bleeding visible vessel have further bleeding that requires urgent surgery if they are treated conservatively. These patients clearly benefit from endoscopic therapy with bipolar electrocoagulation, heater probe, or injection therapy (e.g., absolute alcohol, 1:10,000 epinephrine), with reductions in bleeding, hospital stay, mortality rate, and costs. In contrast, patients with clean-based ulcers have rates of recurrent bleeding approaching zero. If there is no other reason for hospitalization, such patients may be discharged on the first hospital day, following stabilization. Patients without clean-based ulcers should usually remain in the hospital for 3 days, since most episodes of recurrent bleeding occur within 3 days.

Various pharmacologic agents have been assessed in the past for the treatment of ulcer bleeding without clearcut benefit. However, in recent controlled trials in Europe and Asia, high-dose intravenous omeprazole used to raise intragastric pH to 6 to 7 and enhance clot stability decreased further bleeding (but not mortality), even after the use of appropriate endoscopic therapy.

Approximately one-third of patients with a bleeding ulcer will rebleed within the next 1 to 2 years. Prevention of recurrent bleeding focuses on the three main factors in ulcer pathogenesis, *Helicobacter pylori*, [NSAIDs](#), and acid. Eradication of *H. pylori* in patients

with bleeding ulcers dramatically decreases rates of rebleeding to < 5%. If a bleeding ulcer develops in a patient taking NSAIDs, the NSAIDs should be discontinued if possible. If NSAIDs must be continued, initial treatment should be with a proton pump inhibitor, and subsequent prophylactic therapy with a proton pump inhibitor or misoprostol should be continued as long as the patient is taking NSAIDs. Changing from a standard NSAID to a COX-2-specific inhibitor should markedly lower the risk of recurrent [UGIB](#). Patients with bleeding ulcers unrelated to *H. pylori* or NSAIDs should remain on full-dose antisecretory therapy indefinitely. **Peptic ulcers are discussed in [Chap. 285](#).*

Mallory-Weiss Tears The classic history is vomiting, retching, or coughing preceding hematemesis, especially in an alcoholic patient. Bleeding from these tears, which are usually on the gastric side of the gastroesophageal junction, stops spontaneously in 80 to 90% of patients and recurs in only 0 to 5%. Endoscopic therapy is effective for actively bleeding Mallory-Weiss tears. Angiographic therapy with intra-arterial infusion of vasopressin or embolization also may be useful. Rarely, operative therapy with oversewing of the tear may be required. **Mallory-Weiss tears are discussed in [Chap. 284](#).*

Esophageal Varices Patients with [UGIB](#) and clinical evidence suggesting the possibility of liver disease should undergo early endoscopy to determine if varices are the sources of bleeding, because patients with variceal hemorrhage have poorer outcomes than patients with other sources of UGIB. Endoscopic therapy at this time decreases further bleeding, and repeated sessions with endoscopic therapy to eradicate esophageal varices significantly reduces rebleeding and mortality. Endoscopic ligation therapy is the endoscopic therapy of choice for esophageal varices because it has less rebleeding, a lower mortality rate, fewer local complications, and requires fewer treatment sessions to achieve variceal eradication as compared to sclerotherapy.

Acute treatment with octreotide (50 ug bolus and 50 ug/h intravenous infusion for 2 to 5 days) or somatostatin may help in the control of acute bleeding, and these agents have replaced vasopressin as the medical therapy of choice for acute variceal bleeding. Over the long term, treatment with nonselective beta blockers (e.g., propranolol) has also been shown to decrease recurrent bleeding from esophageal varices. These agents commonly are given along with chronic endoscopic therapy.

In patients who have persistent or recurrent bleeding despite endoscopic and medical therapy, more invasive therapy is warranted. Transjugular intrahepatic portosystemic shunt (TIPS) decreases rebleeding more effectively than endoscopic therapy, although hepatic encephalopathy is more common and the mortality rates are comparable. Most patients with TIPS have shunt stenosis within 1 to 2 years and require re-instrumentation. Therefore, TIPS is most appropriate for patients with more severe liver disease and those in whom transplant is anticipated. Patients with milder, well-compensated cirrhosis probably should undergo decompressive surgery (e.g., distal splenorenal shunt).

Portal hypertension is also responsible for bleeding from gastric varices, ectopic varices in the small and large intestine, and portal hypertensive gastropathy and enterocolopathy.

Hemorrhagic and Erosive Gastropathy ("Gastritis") Hemorrhagic and erosive gastropathy or gastritis refers to endoscopically visualized subepithelial hemorrhages and erosions. These are mucosal lesions and thus do not cause major bleeding. They develop in various clinical settings, the most important of which are ingestion of [NSAIDs](#), alcohol, and stress. Half of patients who chronically ingest NSAIDs have erosions (15 to 30% have ulcers), while up to 20% of actively drinking alcoholic patients with symptoms of [UGIB](#) have evidence of subepithelial hemorrhages or erosions.

Stress-related gastric mucosal injury occurs only in extremely sick patients: those who have experienced serious trauma, major surgery, burns covering more than one-third of the body surface area, major intracranial disease, and severe medical illness (ventilator dependency, coagulopathy). Significant bleeding probably does not develop unless ulceration occurs. The mortality rate in these patients is quite high because of their serious underlying illnesses.

The incidence of bleeding from stress-related gastric mucosal injury or ulceration has decreased dramatically in recent years, most likely due to better care of critically ill patients. Pharmacologic prophylaxis for bleeding may be considered in the high-risk patients mentioned above. The best clinical data suggest that intravenous H₂-receptor antagonist therapy is the treatment of choice, although sucralfate also is effective. Prophylactic therapy decreases bleeding, but it does not lower the mortality rate.

Other Causes Other, less frequent causes of [UGIB](#) include erosive duodenitis, neoplasms, aortoenteric fistulas, vascular lesions [including hereditary hemorrhagic telangiectasias (Osler-Weber-Rendu) and gastric antral vascular ectasia ("watermelon stomach")], Dieulafoy's lesion (in which an aberrant vessel in the mucosa bleeds from a pinpoint mucosal defect), prolapse gastropathy (prolapse of proximal stomach into esophagus with retching, especially in alcoholics), and hemobilia and hemosuccus pancreaticus (bleeding from the bile duct or pancreatic duct).

SMALL INTESTINAL SOURCES OF BLEEDING

Small intestinal sources of bleeding (bleeding from sites beyond the reach of the standard upper endoscope) are difficult to diagnose and are responsible for the majority of cases of obscure [GIB](#). Fortunately, small intestinal bleeding is uncommon. The most common causes are vascular ectasias and tumors (e.g., adenocarcinoma, leiomyoma, lymphoma, benign polyps, carcinoid, metastases, and lipoma). Other less common causes include Crohn's disease, infection, ischemia, vasculitis, small bowel varices, diverticula, Meckel's diverticula, duplication cysts, and intussusception. [NSAIDs](#) induce small intestinal erosions and ulcers and may be a relatively common cause of chronic, obscure GIB.

Meckel's diverticulum is the most common cause of significant lower GIB (LGIB) in children, decreasing in frequency as a cause of bleeding with age. In adults younger than 40 to 50 years, small bowel tumors often account for obscure [GIB](#), while in patients older than 50 to 60 years, vascular ectasias are usually responsible.

Vascular ectasias should be treated with endoscopic therapy if possible. Surgical

therapy can be used for vascular ectasias isolated to a segment of the small intestine when endoscopic therapy is unsuccessful; estrogen/progesterone compounds may also be tried. Isolated lesions, such as tumors, diverticula, or duplications, generally are treated with surgical resection.

COLONIC SOURCES OF BLEEDING

The incidence of hospitalizations for [LGIB](#) is about one-fifth that for [UGIB](#). Hemorrhoids are probably the most common cause of LGIB; anal fissures also cause minor bleeding and pain. If these local anal processes, which rarely require hospitalization, are excluded, the most common causes of LGIB in adults are diverticula, vascular ectasias (especially in the proximal colon of patients > 70 years), neoplasms (adenomatous polyps and adenocarcinoma), and colitis -- most commonly infectious or idiopathic inflammatory bowel disease, but occasionally ischemic or radiation-induced. Uncommon causes include post-polypectomy bleeding, solitary rectal ulcer syndrome, [NSAID](#)-induced ulcers or colitis, other neoplasms, trauma, ectopic varices (most commonly rectal), lymphoid nodular hyperplasia, vasculitis, and aorto-colic fistulas. In children and adolescents, the most common colonic causes of significant [GIB](#) are inflammatory bowel disease and juvenile polyps.

Diverticular bleeding is abrupt in onset, usually painless, sometimes massive, and often from the right colon; minor and occult bleeding is not characteristic. Clinical reports suggest that bleeding colonic diverticula stop bleeding spontaneously in approximately 80% of patients, and rebleed in 20 to 25% of patients. Intraarterial vasopressin may halt the bleeding, at least temporarily. If bleeding persists or recurs, segmental surgical resection is indicated.

Bleeding from right colonic vascular ectasias in the elderly may be overt or occult; it tends to be chronic and only occasionally is hemodynamically significant. Endoscopic hemostatic therapy may be useful in the treatment of vascular ectasias, as well as discrete bleeding ulcers and post-polypectomy bleeding, while endoscopic polypectomy, if possible, is used for bleeding colonic polyps. Surgical therapy is generally required for major, persistent, or recurrent bleeding from the wide variety of colonic sources of GIB that cannot be treated medically or endoscopically.

Approach to the Patient

Measurement of the heart rate and blood pressure is the best way to assess a patient with [GIB](#). Clinically significant bleeding leads to postural changes in heart rate or blood pressure, tachycardia, and, finally, recumbent hypotension. Patients also may have a vasovagal reaction with bradycardia during bleeding episodes.

In contrast, the hemoglobin does not fall immediately with acute [GIB](#), due to proportionate reductions in plasma and red cell volumes (i.e., "people bleed whole blood"). Thus, hemoglobin may be normal or only minimally decreased at the initial presentation of a severe bleeding episode. As extravascular fluid enters the vascular space to restore volume, the hemoglobin falls, but this process may take up to 72 h. Patients with slow, chronic GIB may have very low hemoglobin values despite normal blood pressure and heart rate. With the development of iron deficiency anemia, the

mean corpuscular volume will be low and red blood cell distribution width will be increased.

Differentiation of Upper from Lower GIB Hematemesis indicates an upper [GI](#) source of bleeding (above the ligament of Treitz). Melena indicates that blood has been present in the GI tract for at least 14 h. Thus, the more proximal the bleeding site, the more likely melena will occur. Hematochezia usually represents a lower GI source of bleeding, although an upper GI lesion may bleed so rapidly that blood does not remain in the bowel long enough for melena to develop. When hematochezia is the presenting symptom of [UGIB](#), it is associated with hemodynamic instability and dropping hemoglobin. Bleeding lesions of the small bowel may present as melena or hematochezia.

A non-bloody nasogastric aspirate may be seen in up to 16% of patients with [UGIB](#) -- usually from a duodenal source. Even a bile-stained appearance does not exclude a bleeding post-pyloric lesion since reports of bile in the aspirate are incorrect in about 50% of cases. Testing of aspirates that are not grossly bloody for occult blood is of no clinical value. Other clues to UGIB include hyperactive bowel sounds and an elevated BUN (due to volume depletion and absorbed blood proteins).

Diagnostic Evaluation of the Patient with [GIB](#)

UPPER GIB (Fig. 44-1) The history and physical exam seldom are diagnostic of the source of GIB. Upper endoscopy is the test of choice in patients with [UGIB](#), and should be performed urgently in patients with hemodynamic instability (hypotension, tachycardia, or postural changes in heart rate or blood pressure). Early routine endoscopy is also beneficial in cases of milder bleeding for management decisions. Patients with major bleeding and high risk endoscopic findings (varices, ulcers with active bleeding or a visible vessel) benefit from endoscopic hemostatic therapy, while patients with low-risk lesions (e.g., clean based ulcers, non-bleeding Mallory-Weiss tears, erosive or hemorrhagic gastropathy) who have stable vital signs and hemoglobin, and no other medical problems, can be discharged home.

LOWER [GIB](#) (Fig. 44-2) Patients with presumed [LGIB](#) may undergo early sigmoidoscopy for the detection of obvious, low-lying lesions. However, the procedure is difficult with brisk bleeding, and it often is impossible to identify the area of bleeding. Sigmoidoscopy is useful primarily in patients < 40 years with relatively minor bleeding. Patients with hematochezia and hemodynamic instability should have upper endoscopy to rule out an upper [GI](#) source before evaluation of the lower GI tract.

Colonoscopy after an oral lavage solution is the procedure of choice in patients with [LGIB](#) unless bleeding is too massive or unless sigmoidoscopy has disclosed an obvious actively bleeding lesion.⁹⁹ ^{99m}Tc-labeled red cell scan allows repeated imaging for up to 24 h and may identify the general location of bleeding. However, radionuclide scans should be interpreted with caution because results are highly variable. In active LGIB, angiography can detect the site of bleeding (extravasation of contrast into the gut) and permits treatment with intraarterial infusion of vasopressin or embolization. Even after bleeding has stopped, angiography may identify lesions with abnormal vasculature such as vascular ectasias or tumors.

GIB OF OBSCURE ORIGIN Obscure GIB is defined as recurrent acute or chronic bleeding for which no source has been identified by routine endoscopic and contrast studies. Push enteroscopy, with a specially designed enteroscope or a pediatric colonoscope to inspect the entire duodenum and part of the jejunum, is generally the next step. Push enteroscopy may identify probable bleeding sites in 20 to 40% of patients with obscure GIB. If enteroscopy is negative or unavailable, a specialized radiographic examination of the small bowel (e.g., enteroclysis) should be performed.

Patients with recurrent bleeding who require transfusions or repeated hospitalizations warrant further investigations.⁹⁹M Tc-labeled red blood cell scintigraphy should be employed. Angiography is useful even if bleeding has subsided, since it may disclose vascular anomalies or tumor vessels.⁹⁹M Tc-pertechnetate scintigraphy for diagnosis of Meckel's diverticulum should be done, especially in the evaluation of young patients with **LGIB**. When all tests are unrevealing, intraoperative endoscopy is indicated in patients with severe recurrent or persistent bleeding requiring repeated transfusions.

OCCULTGIB Occult GIB is manifested by either a positive test for fecal occult blood or iron deficiency anemia. Unless a patient has upper **GI** symptoms, evaluation of occult bleeding generally should begin with colonoscopy, particularly in patients older than 40 years. If evaluation of the colon is negative, some perform upper endoscopy only if iron deficiency anemia or upper GI symptoms are present, while others recommend upper endoscopy in all patients since up to 25 to 40% of these patients have some abnormality noted on upper endoscopy. If standard endoscopic tests are unrevealing, enteroscopy and/or enteroclysis may be considered in patients with iron-deficiency anemia.

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45. JAUNDICE - Daniel S. Pratt, Marshall M. Kaplan

Jaundice, or icterus, is a yellowish discoloration of tissue resulting from the deposition of bilirubin. Tissue deposition of bilirubin occurs only in the presence of serum hyperbilirubinemia and is a sign of either liver disease or, less often, a hemolytic disorder. The degree of serum bilirubin elevation can be estimated by physical examination. Slight increases in serum bilirubin are best detected by examining the sclerae which have a particular affinity for bilirubin due to their high elastin content. The presence of scleral icterus indicates a serum bilirubin of at least 3.0 mg/dL. The ability to detect scleral icterus is made more difficult if the examining room has fluorescent lighting. If the examiner suspects scleral icterus, a second place to examine is underneath the tongue. As serum bilirubin levels rise, the skin will eventually become yellow in light-skinned patients and even green if the process is longstanding; the green color is produced by oxidation of bilirubin to biliverdin.

The differential diagnosis for yellowing of the skin is limited. In addition to jaundice, it includes carotenoderma, the use of the drug quinacrine, and excessive exposure to phenols. Carotenoderma is the yellow color imparted to the skin by the presence of carotene; it occurs in healthy individuals who ingest excessive amounts of vegetables and fruits that contain carotene, such as carrots, leafy vegetables, squash, peaches, and oranges. Unlike jaundice, where the yellow coloration of the skin is uniformly distributed over the body, in carotenoderma the pigment is concentrated on the palms, soles, forehead, and nasolabial folds. Carotenoderma can be distinguished from jaundice by the sparing of the sclerae. Quinacrine causes a yellow discoloration of the skin in 4 to 37% of patients treated with it. Unlike carotene, quinacrine can cause discoloration of the sclerae.

Another sensitive indicator of increased serum bilirubin is darkening of the urine, which is due to the renal excretion of conjugated bilirubin. Patients often describe their urine as tea or cola colored. Bilirubinuria indicates an elevation of the direct serum bilirubin fraction and therefore the presence of liver disease.

Increased serum bilirubin levels occur when an imbalance exists between bilirubin production and clearance. A logical evaluation of the patient who is jaundiced requires an understanding of bilirubin production and metabolism.

PRODUCTION AND METABOLISM OF BILIRUBIN (See also [Chap. 294](#))

Bilirubin, a tetrapyrrole pigment, is a breakdown product of heme (ferroprotoporphyrin IX). About 70 to 80% of the 250 to 300 mg of bilirubin produced each day is derived from the breakdown of hemoglobin in senescent red blood cells. The remainder comes from prematurely destroyed erythroid cells in bone marrow and from the turnover of hemoproteins such as myoglobin and cytochromes found in tissues throughout the body.

The formation of bilirubin occurs in reticuloendothelial cells, primarily in the spleen and liver. The first reaction, catalyzed by the enzyme heme oxygenase, oxidatively cleaves the bridge of the porphyrin group and opens the heme ring. The end products of this reaction are biliverdin, carbon monoxide, and iron. The second reaction, catalyzed by

the cytosolic enzyme biliverdin reductase, reduces the central methylene bridge of biliverdin and converts it to bilirubin. Bilirubin formed in the reticuloendothelial cells is virtually insoluble in water. To be transported in blood, it must be solubilized. This is accomplished by its reversible, noncovalent binding to albumin. Unconjugated bilirubin bound to albumin is transported to the liver, where it, but not the albumin, is taken up by hepatocytes via a process that at least partly involves carrier-mediated membrane transport.

In the cytosol of the hepatocyte, unconjugated bilirubin is coupled predominantly to the protein ligandin (formerly called the Y protein). Ligandin was initially thought to be a transport protein facilitating the movement of bilirubin from the sinusoidal membrane to the endoplasmic reticulum. It is now thought to slow the cytosolic diffusion of bilirubin and to reduce its efflux back into serum. In the endoplasmic reticulum, bilirubin is solubilized by conjugation to glucuronic acid, forming bilirubin monoglucuronide and diglucuronide. The conjugation of glucuronic acid to bilirubin is catalyzed by bilirubin uridine-diphosphate (UDP) glucuronosyltransferase.

The now hydrophilic bilirubin conjugates diffuse from the endoplasmic reticulum to the canalicular membrane, where bilirubin monoglucuronide and diglucuronide are actively transported into canalicular bile by an energy-dependent mechanism involving the multiple organic ion transport protein/multiple drug resistance protein. The conjugated bilirubin excreted into bile drains into the duodenum and passes unchanged through the proximal small bowel. Conjugated bilirubin is not taken up by the intestinal mucosa. When the conjugated bilirubin reaches the distal ileum and colon, it is hydrolyzed to unconjugated bilirubin by bacterial β -glucuronidases. The unconjugated bilirubin is reduced by normal gut bacteria to form a group of colorless tetrapyrroles called urobilinogens. About 80 to 90% of these products are excreted in feces, either unchanged or oxidized to orange derivatives called urobilins. The remaining 10 to 20% of the urobilinogens are passively absorbed, enter the portal venous blood, and are reexcreted by the liver. A small fraction (usually less than 3 mg/dL) escapes hepatic uptake, filters across the renal glomerulus, and is excreted in urine.

MEASUREMENT OF SERUM BILIRUBIN

The terms direct- and indirect-reacting bilirubin are based on the original van den Bergh reaction. This assay, or a variation of it, is still used in most clinical chemistry laboratories to determine the serum bilirubin level. In this assay, bilirubin is exposed to diazotized sulfanilic acid, splitting into two relatively stable dipyrromethene azopigments that absorb maximally at 540 nm, allowing for photometric analysis. The direct fraction is that which reacts with diazotized sulfanilic acid in the absence of an accelerator substance such as alcohol. The direct fraction provides an approximate determination of the conjugated bilirubin in serum. The total serum bilirubin is the amount that reacts after the addition of alcohol. The indirect fraction is the difference between the total and the direct bilirubin and provides an estimate of the unconjugated bilirubin in serum.

With the van den Bergh method, the normal serum bilirubin concentration usually is <1 mg/dL (17 μ mol/L). Up to 30%, or 0.3 mg/dL (5.1 μ mol/L), of the total may be direct-reacting (conjugated) bilirubin. Total serum bilirubin concentrations are between 0.2 and 0.9 mg/dL in 95% of a normal population.

Several new techniques, although less convenient to perform, have added considerably to our understanding of bilirubin metabolism. First, they demonstrate that in normal people or those with Gilbert's syndrome, almost 100% of the serum bilirubin is unconjugated; less than 3% is monoconjugated bilirubin. Second, in jaundiced patients with hepatobiliary disease, the total serum bilirubin concentration measured by these new, more accurate methods is lower than the values found with diazo methods. This suggests that there are diazo-positive compounds distinct from bilirubin in the serum of patients with hepatobiliary disease. Third, these studies indicate that in jaundiced patients with hepatobiliary disease, monoglucuronides of bilirubin predominate over the diglucuronides. Fourth, part of the direct-reacting bilirubin fraction includes conjugated bilirubin that is covalently linked to albumin. This albumin-linked bilirubin fraction (*delta fraction* or *biliprotein*) represents an important fraction of total serum bilirubin in patients with cholestasis and hepatobiliary disorders. Albumin-bound conjugated bilirubin is formed in serum when hepatic excretion of bilirubin glucuronides is impaired and the glucuronides are present in serum in increasing amounts. By virtue of its tight binding to albumin, the clearance rate of albumin-bound bilirubin from serum approximates the half-life of albumin, 12 to 14 days, rather than the short half-life of bilirubin, about 4 h.

The prolonged half-life of albumin-bound conjugated bilirubin explains two previously unexplained enigmas in jaundiced patients with liver disease: (1) that some patients with conjugated hyperbilirubinemia do not exhibit bilirubinuria during the recovery phase of their disease because the bilirubin is bound to albumin and therefore not filtered by the renal glomeruli and (2) that the elevated serum bilirubin level declines more slowly than expected in some patients who otherwise appear to be recovering satisfactorily. Late in the recovery phase of hepatobiliary disorders, all the conjugated bilirubin may be in the albumin-linked form. Its value in serum falls slowly because of the long half-life of albumin.

MEASUREMENT OF URINE BILIRUBIN

Unconjugated bilirubin is always bound to albumin in the serum, is not filtered by the kidney, and is not found in the urine. Conjugated bilirubin is filtered at the glomerulus and the majority is reabsorbed by the proximal tubules; a small fraction is excreted in the urine. Any bilirubin found in the urine is conjugated bilirubin. The presence of bilirubinuria implies the presence of liver disease. A urine dipstick test (Ictotest) gives the same information as fractionation of the serum bilirubin. This test is very accurate. A false-negative test is possible in patients with prolonged cholestasis due to the predominance of conjugated bilirubin covalently bound to albumin.

THE EVALUATION OF JAUNDICE

The bilirubin present in serum represents a balance between input from production of bilirubin and hepatic/biliary removal of the pigment. Hyperbilirubinemia may result from (1) overproduction of bilirubin; (2) impaired uptake, conjugation, or excretion of bilirubin; or (3) regurgitation of unconjugated or conjugated bilirubin from damaged hepatocytes or bile ducts. An increase in unconjugated bilirubin in serum results from either overproduction, impairment of uptake, or conjugation of bilirubin. An increase in conjugated bilirubin is due to decreased excretion into the bile ductules or backward

leakage of the pigment. The initial steps in evaluating the patient with jaundice are to determine (1) whether the hyperbilirubinemia is predominantly conjugated or unconjugated in nature, and (2) whether other biochemical liver tests are abnormal. The thoughtful interpretation of limited data will allow for a rational evaluation of the patient ([Fig. 45-1](#)). This discussion will focus solely on the evaluation of the adult patient with jaundice.

ISOLATED ELEVATION OF SERUM BILIRUBIN

Unconjugated Hyperbilirubinemia The differential diagnosis of an isolated unconjugated hyperbilirubinemia is limited ([Table 45-1](#)). The critical determination is whether the patient is suffering from a hemolytic process resulting in an overproduction of bilirubin (hemolytic disorders and ineffective erythropoiesis) or from impaired hepatic uptake/conjugation of bilirubin (drug effect or genetic disorders).

Hemolytic disorders that cause excessive heme production may be either inherited or acquired. Inherited disorders include spherocytosis, sickle cell anemia, and deficiency of red cell enzymes such as pyruvate kinase and glucose-6-phosphate dehydrogenase. In these conditions, the serum bilirubin rarely exceeds 5 mg/dL. Higher levels may occur when there is coexistent renal or hepatocellular dysfunction, or in acute hemolysis such as a sickle cell crisis. In evaluating jaundice in patients with chronic hemolysis, it is important to remember the high incidence of pigmented (calcium bilirubinate) gallstones found in these patients, which increases the likelihood of choledocholithiasis as an alternative explanation for hyperbilirubinemia.

Acquired hemolytic disorders include microangiopathic hemolytic anemia (e.g., hemolytic-uremic syndrome), paroxysmal nocturnal hemoglobinuria, and immune hemolysis. Ineffective erythropoiesis occurs in cobalamin, folate, and iron deficiencies.

In the absence of hemolysis, the physician should consider a problem with the hepatic uptake or conjugation of bilirubin. Certain drugs, including rifampicin and probenecid, may cause unconjugated hyperbilirubinemia by diminishing hepatic uptake of bilirubin. Impaired bilirubin conjugation occurs in three genetic conditions: *Crigler-Najjar syndrome, types I and II*, and *Gilbert's syndrome*. *Crigler-Najjar type I* is an exceptionally rare condition found in neonates and characterized by severe jaundice (bilirubin > 20 mg/dL) and neurologic impairment due to kernicterus, frequently leading to death in infancy or childhood. These patients have a complete absence of bilirubin [UDP](#)glucuronosyltransferase activity, usually due to mutations in the critical 3' domain of the UDP glucuronosyltransferase gene, and are totally unable to conjugate, hence cannot excrete bilirubin. The only effective treatment is orthotopic liver transplantation. Use of gene therapy and allogeneic hepatocyte infusion are experimental approaches of future promise for this devastating disease.

Crigler-Najjar type II is somewhat more common than type I. Patients live into adulthood with serum bilirubin levels that range from 6 to 25 mg/dL. In these patients, mutations in the bilirubin [UDP](#)glucuronosyltransferase gene cause reduced but not completely absent activity of the enzyme. Bilirubin UDP glucuronosyltransferase activity can be induced by the administration of phenobarbital, which can reduce serum bilirubin levels in these patients. Despite marked jaundice, these patients usually survive into adulthood,

although they may be susceptible to kernicterus under the stress of intercurrent illness or surgery.

Gilbert's syndrome is also marked by the impaired conjugation of bilirubin due to reduced bilirubin [UDP](#)glucuronosyltransferase activity. Molecular analyses show that Gilbert's syndrome is due to reduced expression of UDP glucuronosyltransferase activity caused by lengthening of the TATAA box from A(TA)₆TAA to A(TA)₇TAA in the promoter element of the gene. This results in mild unconjugated hyperbilirubinemia with serum levels almost always less than 6 mg/dL. The serum levels may fluctuate and jaundice is often identified only during periods of fasting. Unlike both Crigler-Najjar syndromes, Gilbert's syndrome is very common. The reported incidence is 3 to 7% of the population with males predominating over females by a ratio of 2-7:1.

Conjugated Hyperbilirubinemia Elevated conjugated hyperbilirubinemia is found in two rare inherited conditions: *Dubin-Johnson syndrome* and *Rotor's syndrome* ([Table 45-1](#)). Patients with both conditions present with asymptomatic jaundice, typically in the second generation of life. The defect in Dubin-Johnson syndrome is a point mutation in the gene for the canalicular multispecific organic anion transporter. These patients have altered excretion of bilirubin into the bile ducts. Rotor's syndrome seems to be a problem with the hepatic storage of bilirubin. Differentiating between these syndromes is possible, but clinically unnecessary, due to their benign nature.

ELEVATION OF SERUM BILIRUBIN WITH OTHER LIVER TEST ABNORMALITIES

The remainder of this chapter will focus on the evaluation of the patient with a conjugated hyperbilirubinemia in the setting of other liver test abnormalities. This group of patients can be divided into those with a primary hepatocellular process and those with intra- or extrahepatic cholestasis. Being able to make this differentiation will guide the physician's evaluation ([Fig. 45-1](#)). This differentiation is made on the basis of the history and physical examination as well as the pattern of liver test abnormalities.

History A complete medical history is perhaps the single most important part of the evaluation of the patient with unexplained jaundice. Important considerations include the use of or exposure to any chemical or medication, either physician-prescribed or over-the-counter, such as herbal and vitamin preparations and other drugs such as anabolic steroids. The patient should be carefully questioned about possible parenteral exposures, including transfusions, intravenous and intranasal drug use, tattoos, and sexual activity. Other important questions include recent travel history, exposure to people with jaundice, exposure to possibly contaminated foods, occupational exposure to hepatotoxins, alcohol consumption, the duration of jaundice, and the presence of any accompanying symptoms such as arthralgias, myalgias, rash, anorexia, weight loss, abdominal pain, fever, pruritis, and changes in the urine and stool. While none of these latter symptoms are specific for any one condition, they can suggest a particular diagnosis. A history of arthralgias and myalgias predating jaundice suggests hepatitis, either viral or drug-related. Jaundice associated with the sudden onset of severe right upper quadrant pain and shaking chills suggests choledocholithiasis and ascending cholangitis.

Physical Examination The general assessment should include assessment of the

patient's nutritional status. Temporal and proximal muscle wasting suggests longstanding diseases such as pancreatic cancer or cirrhosis. Stigmata of chronic liver disease, including spider nevi, palmar erythema, gynecomastia, caput medusae, Dupuytren's contractures, parotid gland enlargement, and testicular atrophy are commonly seen in advanced alcoholic (Laennec's) cirrhosis and occasionally in other types of cirrhosis. An enlarged left supraclavicular node (Virchow's node) or periumbilical nodule (Sister Mary Joseph's nodule) suggest an abdominal malignancy. Jugular venous distention, a sign of right-sided heart failure, suggests hepatic congestion. Right pleural effusion, in the absence of clinically apparent ascites, may be seen in advanced cirrhosis.

The abdominal examination should focus on the size and consistency of the liver, whether the spleen is palpable and hence enlarged, and whether there is ascites present. Patients with cirrhosis may have an enlarged left lobe of the liver which is felt below the xiphoid and an enlarged spleen. A grossly enlarged nodular liver or an obvious abdominal mass suggests malignancy. An enlarged tender liver could be viral or alcoholic hepatitis or, less often, an acutely congested liver secondary to right-sided heart failure. Severe right upper quadrant tenderness with respiratory arrest on inspiration (Murphy's sign) suggests cholecystitis or, occasionally, ascending cholangitis. Ascites in the presence of jaundice suggests either cirrhosis or malignancy with peritoneal spread.

Laboratory Tests When the physician encounters a patient with unexplained jaundice, there are a battery of tests that are helpful in the initial evaluation. These include total and direct serum bilirubin with fractionation, aminotransferases, alkaline phosphatase, albumin, and prothrombin time tests. Enzyme tests [alanine aminotransferase (ALT), aspartate aminotransferase (AST), and alkaline phosphatase] are helpful in differentiating between a hepatocellular process and a cholestatic process (see [Table 293-1](#) and [Fig. 45-1](#)), a critical step in determining what additional workup is indicated. Patients with a hepatocellular process generally have a disproportionate rise in the aminotransferases compared to the alkaline phosphatase. Patients with a cholestatic process have a disproportionate rise in the alkaline phosphatase compared to the aminotransferases. The bilirubin can be prominently elevated in both hepatocellular and cholestatic conditions and therefore is not necessarily helpful in differentiating between the two.

In addition to the enzyme tests, all jaundiced patients should have additional blood tests, specifically an albumin and a prothrombin time, to assess liver function. A low albumin suggests a chronic process such as cirrhosis or cancer. A normal albumin is suggestive of a more acute process such as viral hepatitis or choledocholithiasis. An elevated prothrombin time indicates either vitamin K deficiency due to prolonged jaundice and malabsorption of vitamin K or significant hepatocellular dysfunction. The failure of the prothrombin time to correct with parenteral administration of vitamin K indicates severe hepatocellular injury.

The results of the bilirubin, enzyme, albumin, and prothrombin time tests will usually indicate whether a jaundiced patient has a hepatocellular or a cholestatic disease. The causes and evaluation of each of these is quite different.

Hepatocellular Conditions Hepatocellular diseases that can cause jaundice include viral hepatitis, drug or environmental toxicity, alcohol, and end-stage cirrhosis from any cause ([Table 45-2](#)). Wilson's disease should be considered in young adults. Autoimmune hepatitis is typically seen in young to middle-aged women, but may affect men and women of any age. Alcoholic hepatitis can be differentiated from viral and toxin-related hepatitis by the pattern of the aminotransferases. Patients with alcoholic hepatitis typically have an [AST:ALT](#) ratio of at least 2:1. The AST rarely exceeds 300 U/L. Patients with acute viral hepatitis and toxin-related injury severe enough to produce jaundice typically have aminotransferases greater than 500 U/L, with the ALT greater than or equal to the AST. The degree of aminotransferase elevation can occasionally help in differentiating between hepatocellular and cholestatic processes. While ALT and AST values less than 8 times normal may be seen in either hepatocellular or cholestatic liver disease, values 25 times normal or higher are seen primarily in acute hepatocellular diseases. Patients with jaundice from cirrhosis can have normal or only slight elevations of the aminotransferases.

When the physician determines that the patient has a hepatocellular disease, appropriate testing for acute viral hepatitis includes a hepatitis A IgM antibody, a hepatitis B surface antigen and core IgM antibody, and a hepatitis C viral RNA test. It can take many weeks for the hepatitis C antibody to become detectable, making it an unreliable test if acute hepatitis C is suspected. Depending on circumstances, studies for hepatitis D, E, Epstein-Barr virus (EBV), and cytomegalovirus (CMV) may be indicated. Ceruloplasmin is the initial screening test for Wilson's disease. Testing for autoimmune hepatitis usually includes an antinuclear antibody and measurement of specific immunoglobulins.

Drug-induced hepatocellular injury can be classified either as predictable or unpredictable. Predictable drug reactions are dose-dependent and affect all patients who ingest a toxic dose of the drug in question. The classic example is acetaminophen hepatotoxicity. Unpredictable or idiosyncratic drug reactions are not dose-dependent and occur in a minority of patients. A great number of drugs can cause idiosyncratic hepatic injury. Environmental toxins are also an important cause of hepatocellular injury. Examples include industrial chemicals such as vinyl chloride, herbal preparations containing pyrrolizidine alkaloids (Jamaica bush tea), and the mushrooms *Amanita phalloides* or *verna* containing highly hepatotoxic amatoxins.

Cholestatic Conditions When the pattern of the liver tests suggests a cholestatic disorder, the next step is to determine whether it is intra- or extrahepatic cholestasis ([Fig. 45-1](#)). Distinguishing intrahepatic from extrahepatic cholestasis may be difficult. History, physical examination, and laboratory tests are often not helpful. The next appropriate test is an ultrasound. The ultrasound is inexpensive, does not expose the patient to ionizing radiation, and can detect dilation of the intra- and extrahepatic biliary tree with a high degree of sensitivity and specificity. The absence of biliary dilatation suggests intrahepatic cholestasis, while the presence of biliary dilatation indicates extrahepatic cholestasis. False-negative results occur in patients with partial obstruction of the common bile duct or in patients with cirrhosis or primary sclerosing cholangitis (PSC) where scarring prevents the intrahepatic ducts from dilating.

Although ultrasonography may indicate extrahepatic cholestasis, it rarely identifies the

site or cause of obstruction. The distal common bile duct is a particularly difficult area to visualize by ultrasound because of overlying bowel gas. Appropriate next tests include computed tomography (CT) and endoscopic retrograde cholangiopancreatography (ERCP). CT scanning is better than ultrasonography for assessing the head of the pancreas and for identifying choledocholithiasis in the distal common bile duct, particularly when the ducts are not dilated. ERCP is the gold standard for identifying choledocholithiasis. It is performed by introducing a side-viewing endoscope perorally into the duodenum. The ampulla of Vater is visualized and a catheter is advanced through the ampulla. Injection of dye allows for the visualization of the common bile duct and the pancreatic duct. The success rate for cannulation of the common bile duct ranges from 80 to 95%, depending on the operator's experience. Beyond its diagnostic capabilities, ERCP allows for therapeutic interventions, including the removal of common bile duct stones and the placement of stents. In patients in whom ERCP is unsuccessful, transhepatic cholangiography can provide the same information. Magnetic resonance cholangiopancreatography (MRCP) is a rapidly developing, noninvasive technique for imaging the bile and pancreatic ducts; this may replace ERCP as the initial diagnostic test in cases where the need for intervention is felt to be small.

In patients with apparent *intrahepatic cholestasis*, the diagnosis is often made by serologic testing in combination with percutaneous liver biopsy. The list of possible causes of intrahepatic cholestasis is long and varied ([Table 45-3](#)). A number of conditions that typically cause a hepatocellular pattern of injury can also present as a cholestatic variant. Both hepatitis B and C can cause a cholestatic hepatitis (fibrosing cholestatic hepatitis) that has histologic features that mimic large duct obstruction. This disease variant has been reported in patients who have undergone solid organ transplantation. Hepatitis A, alcoholic hepatitis, [EBV](#), and [CMV](#) may also present as cholestatic liver disease.

Drugs may cause intrahepatic cholestasis, a variant of drug-induced hepatitis. Drug-induced cholestasis is usually reversible after eliminating the offending drug, although it may take many months for cholestasis to resolve. Drugs most commonly associated with cholestasis are the anabolic and contraceptive steroids. Cholestatic hepatitis has been reported with chlorpromazine, imipramine, tolbutamide, sulindac, cimetidine, and erythromycin estolate. It also occurs in patients taking trimethoprim, sulfamethoxazole, and penicillin-based antibiotics such as ampicillin, dicloxacillin, and clavulanic acid. Rarely, cholestasis may be chronic and associated with progressive fibrosis despite early discontinuation of the drug. Chronic cholestasis has been associated with chlorpromazine and prochlorperazine.

Primary biliary cirrhosis is a disease predominantly of middle-aged women in which there is a progressive destruction of interlobular bile ducts. The diagnosis is made by the presence of the antimitochondrial antibody that is found in 95% of patients. [Primary sclerosing cholangitis \(PSC\)](#) is characterized by the destruction and fibrosis of larger bile ducts. The disease may involve only the intrahepatic ducts and present as intrahepatic cholestasis. However, in 65% of patients with PSC, both intra- and extrahepatic ducts are involved. The diagnosis of PSC is made by [ERCP](#). The pathognomonic findings are multiple strictures of bile ducts with dilatations proximal to the strictures. Approximately 75% of patients with PSC have inflammatory bowel disease.

The *vanishing bile duct syndrome* and *adult bile ductopenia* are rare conditions in which there are a decreased number of bile ducts seen in liver biopsy specimens. The histologic picture is similar to that found in primary biliary cirrhosis. This picture is seen in patients who develop chronic rejection after liver transplantation and in those who develop graft-versus-host disease after bone marrow transplantation. Vanishing bile duct syndrome also occurs in rare cases of sarcoidosis, in patients taking certain drugs including chlorpromazine, and idiopathically. There are also familial forms of intrahepatic cholestasis, including the *familial intrahepatic cholestatic syndromes, I-III*. Benign recurrent cholestasis is an autosomal recessive disease that appears to be due to mutations in a P type ATPase, which probably acts as a bile acid transporter. The disease is marked by recurrent episodes of jaundice and pruritis; the episodes are self-limited but can be debilitating. *Cholestasis of pregnancy* occurs in the second and third trimesters and resolves after delivery. Its cause is unknown, but the condition is probably inherited and cholestasis can be triggered by estrogen administration.

Other causes of intrahepatic cholestasis include total parenteral nutrition (TPN), nonhepatobiliary sepsis, benign postoperative cholestasis, and a paraneoplastic syndrome associated with a number of different malignancies, including Hodgkin's disease, medullary thyroid cancer, hypernephroma, renal sarcoma, T cell lymphoma, prostate cancer, and several GI malignancies. In patients developing cholestasis in the intensive care unit, the major considerations should be sepsis, shock liver, and TPN jaundice. Jaundice occurring after bone marrow transplantation is most likely due to venoocclusive disease or graft-versus-host disease.

Causes of *extrahepatic cholestasis* can be split into malignant and benign ([Table 45-3](#)). Malignant causes include pancreatic, gallbladder, ampullary, and cholangiocarcinoma. The latter is most commonly associated with [PSC](#) and is exceptionally difficult to diagnose because its appearance is often identical to PSC. Pancreatic and gallbladder tumors, as well as cholangiocarcinoma, are rarely resectable and have poor prognoses. Ampullary carcinoma has the highest surgical cure rate of all the tumors that present as painless jaundice. Hilar lymphadenopathy due to metastases from other cancers may cause obstruction of the extrahepatic biliary tree.

Choledocholithiasis is the most common cause of extrahepatic cholestasis. The clinical presentation can range from mild right upper quadrant discomfort with only minimal elevations of the enzyme tests to ascending cholangitis with jaundice, sepsis, and circulatory collapse. [PSC](#) may occur with clinically important strictures limited to the extrahepatic biliary tree. In cases where there is a dominant stricture, patients can be effectively managed with serial endoscopic dilatations. Chronic pancreatitis rarely causes strictures of the distal common bile duct, where it passes through the head of the pancreas. AIDS cholangiopathy is a condition, usually due to infection of the bile duct epithelium with [CMV](#) or cryptosporidium, which has a cholangiographic appearance similar to PSC. These patients usually present with greatly elevated serum alkaline phosphatase levels, mean of 800 IU/L, but the bilirubin is often near normal. These patients do not typically present with jaundice.

SUMMARY

The goal of this chapter is not to provide an encyclopedic review of all of the conditions that can cause jaundice. Rather, it is intended to provide a framework that helps a physician to evaluate the patient with jaundice in a logical way ([Fig. 45-1](#)).

Simply stated, the initial step is to obtain appropriate blood tests to determine if the patient has an isolated elevation of serum bilirubin. If so, is the bilirubin elevation due to an increased unconjugated or conjugated fraction? If the hyperbilirubinemia is accompanied by other liver test abnormalities, is the disorder hepatocellular or cholestatic? If cholestatic, is it intra- or extrahepatic? All of these questions can be answered with a thoughtful history, physical examination, and interpretation of laboratory and radiologic tests and procedures.

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46. ABDOMINAL SWELLING AND ASCITES - Robert M. Glickman

ABDOMINAL SWELLING

Abdominal swelling or distention is a common problem in clinical medicine and may be the initial manifestation of a systemic disease or of otherwise unsuspected abdominal disease. *Subjective* abdominal enlargement, often described as a sensation of fullness or bloating, is usually transient and is often related to a functional gastrointestinal disorder when it is not accompanied by objective physical findings of increased abdominal girth or local swelling. *Obesity* and lumbar lordosis, which may be associated with prominence of the abdomen, may usually be distinguished from true increases in the volume of the peritoneal cavity by history and careful physical examination.

Clinical History Abdominal swelling may first be noticed by the patient because of a progressive increase in belt or clothing size, the appearance of abdominal or inguinal hernias, or the development of a localized swelling. Often, considerable abdominal enlargement has gone unnoticed for weeks or months, either because of coexistent obesity or because the ascites formation has been insidious, without pain or localizing symptoms. Progressive abdominal distention may be associated with a sensation of "pulling" or "stretching" of the flanks or groins and vague low back pain. Localized pain usually results from involvement of an abdominal organ (e.g., a passively congested liver, large spleen, or colonic tumor). Pain is uncommon in cirrhosis with ascites, and when it is present, pancreatitis, hepatocellular carcinoma, or peritonitis should be considered. Tense ascites or abdominal tumors may produce increased intraabdominal pressure, resulting in indigestion and heartburn due to gastroesophageal reflux or dyspnea, orthopnea, and tachypnea from elevation of the diaphragm. A coexistent pleural effusion, more commonly on the right, presumably due to leakage of ascitic fluid through lymphatic channels in the diaphragm, also may contribute to respiratory embarrassment. The patient with diffuse abdominal swelling should be questioned about increased alcohol intake, a prior episode of jaundice or hematuria, or a change in bowel habits. Such historic information may provide the clues that will lead one to suspect an occult cirrhosis, a colonic tumor with peritoneal seeding, congestive heart failure, or nephrosis.

Physical Examination A carefully executed general physical examination can yield valuable clues concerning the etiology of abdominal swelling. Thus palmar erythema and spider angiomas suggest an underlying cirrhosis, while supraclavicular adenopathy (Virchow's node) should raise the question of an underlying gastrointestinal malignancy.

Inspection of the abdomen is important. By noting the abdominal contour, one may be able to distinguish localized from generalized swelling. The tensely distended abdomen with tightly stretched skin, bulging flanks, and everted umbilicus is characteristic of ascites. A prominent abdominal venous pattern with the direction of flow away from the umbilicus often is a reflection of portal hypertension; venous collaterals with flow from the lower part of the abdomen toward the umbilicus suggest obstruction of the inferior vena cava; flow downward toward the umbilicus suggests superior vena cava obstruction. "Doming" of the abdomen with visible ridges from underlying intestinal loops is usually due to intestinal obstruction or distention. An epigastric mass, with evident peristalsis proceeding from left to right, usually indicates underlying pyloric obstruction.

A liver with metastatic deposits may be visible as a nodular right upper quadrant mass moving with respiration.

Auscultation may reveal the high-pitched, rushing sounds of early intestinal obstruction or a succussion sound due to increased fluid and gas in a dilated hollow viscus. Careful auscultation over an enlarged liver occasionally reveals the harsh bruit of a vascular tumor, especially a hepatocellular carcinoma, or the leathery friction rub of a surface nodule. A venous hum at the umbilicus may signify portal hypertension and an increased collateral blood flow around the liver. A fluid wave and flank dullness that shifts with change in position of the patient are important signs that indicate the presence of peritoneal fluid. In obese patients, small amounts of fluid may be difficult to demonstrate; on occasion, the fluid may be detected by abdominal percussion with patients on their hands and knees. Small amounts of ascites often can only be detected by ultrasound examination of the abdomen, which can detect as little as 100 mL of fluid. Careful percussion should serve to distinguish generalized abdominal enlargement from localized swelling due to an enlarged uterus, ovarian cyst, or distended bladder. Percussion also can outline an abnormally small or large liver. Loss of normal liver dullness may result from massive hepatic necrosis; it also may be a clue to free gas in the peritoneal cavity, as from perforation of a hollow viscus.

Palpation is often difficult with massive ascites, and ballottement of overlying fluid may be the only method of palpating the liver or spleen. A slightly enlarged spleen in association with ascites may be the only evidence of an occult cirrhosis. When there is evidence of portal hypertension, a soft liver suggests that obstruction to portal flow is extrahepatic; a firm liver suggests cirrhosis as the likely cause of the portal hypertension. A very hard or nodular liver is a clue that the liver is infiltrated with tumor, and when accompanied by ascites, it suggests that the latter is due to peritoneal seeding. The presence of a hard periumbilical nodule (Sister Mary Joseph's nodule) suggests metastatic disease from a pelvic or gastrointestinal primary tumor. A pulsatile liver and ascites may be found in tricuspid insufficiency.

An attempt should be made to determine whether a mass is solid or cystic, smooth or irregular, and whether it moves with respiration. The liver, spleen, and gallbladder should descend with respiration unless they are fixed by adhesions or extension of tumor beyond the organ. A fixed mass not descending with respiration may indicate that it is retroperitoneal. Tenderness, especially if localized, may indicate an inflammatory process such as an abscess; it also may be due to stretching of the visceral peritoneum or tumor necrosis. Rectal and pelvic examinations are mandatory; they may reveal otherwise undetected masses due to tumor or infection.

Radiographic and laboratory examinations are essential for confirming or extending the impressions gained on physical examination. Upright and recumbent films of the abdomen may demonstrate the dilated loops of intestine with fluid levels characteristic of intestinal obstruction or the diffuse abdominal haziness and loss of psoas margins suggestive of ascites. Ultrasonography is often of value in detecting ascites, determining the presence of a mass, or evaluating the size of the liver and spleen. Computed tomography (CT) scanning provides similar information. CT scanning is often necessary to visualize the retroperitoneum, pancreas, and lymph nodes. A plain film of the abdomen may reveal the distended colon of otherwise unsuspected ulcerative colitis

and give valuable information as to the size of the liver and spleen. An irregular and elevated right side of the diaphragm may be a clue to a liver abscess or hepatocellular carcinoma. Studies of the gastrointestinal tract with barium or other contrast media are usually necessary in the search for a primary tumor.

ASCITES

The evaluation of a patient with ascites requires that the cause of the ascites be established. In most cases ascites appears as part of a well-recognized illness, that is, cirrhosis, congestive heart failure, nephrosis, or disseminated carcinomatosis. In these situations, the physician should determine that the development of ascites is indeed a consequence of the basic underlying disease and not due to the presence of a separate or related disease process. This distinction is necessary even when the cause of ascites seems obvious. For example, when the patient with compensated cirrhosis and minimal ascites develops progressive ascites that is increasingly difficult to control with sodium restriction or diuretics, the temptation is to attribute the worsening of the clinical picture to progressive liver disease. However, an occult hepatocellular carcinoma, portal vein thrombosis, spontaneous bacterial peritonitis, or even tuberculosis may be responsible for the decompensation. The disappointingly low success in diagnosing tuberculous peritonitis or hepatocellular carcinoma in the patient with cirrhosis and ascites reflects the too-low index of suspicion for the development of such superimposed conditions. Similarly, the patient with congestive heart failure may develop ascites from a disseminated carcinoma with peritoneal seeding.

Diagnostic paracentesis (50 to 100 mL) should be part of the routine evaluation of the patient with ascites. The fluid should be examined for its gross appearance; protein content, cell count, and differential cell count should be determined; and Gram's and acid-fast stains and culture should be performed. Cytologic and cell-block examination may disclose an otherwise unsuspected carcinoma. [Table 46-1](#) presents some of the features of ascitic fluid typically found in various disease states. In some disorders, such as cirrhosis, the fluid has the characteristics of a transudate (<25 g protein per liter and a specific gravity of <1.016); in others, such as peritonitis, the features are those of an exudate. Rather than the total protein content of ascites, many authors prefer the use of a *serum-ascites albumin gradient (SAG)* to characterize ascites. The gradient correlates directly with portal pressure. A gradient >1.1 g/dL (high gradient) is characteristic of uncomplicated cirrhotic ascites and differentiates ascites due to portal hypertension from ascites not due to portal hypertension >95% of the time. A gradient <1.1 g/dL (low gradient) suggests that the ascites is not due to portal hypertension with >95% accuracy and mandates a search for other causes ([Table 46-1](#)). Although there is variability of the ascitic fluid in any given disease state, some features are sufficiently characteristic to suggest certain diagnostic possibilities. For example, blood-stained fluid with >25 g protein per liter is unusual in uncomplicated cirrhosis but is consistent with tuberculous peritonitis or neoplasm. Cloudy fluid with a predominance of polymorphonuclear cells and a positive Gram's stain are characteristic of bacterial peritonitis; if most cells are lymphocytes, tuberculosis should be suspected. The complete examination of each fluid is most important, for occasionally only one finding may be abnormal. For example, if the fluid is a typical transudate but contains >250 white blood cells per microliter, the finding should be recognized as atypical for cirrhosis and should warrant a search for tumor or infection. This is especially true in the evaluation of cirrhotic ascites where

occult peritoneal infection may be present with only minor elevations in the white blood cell count of the peritoneal fluid (300 to 500 cells per microliter). Since Gram's stain of the fluid may be negative in a high proportion of such cases, careful culture of the peritoneal fluid is mandatory. Bedside inoculation of blood culture flasks with ascitic fluid results in a dramatically increased incidence of positive cultures when bacterial infection is present (90 versus 40% positivity with conventional cultures done by the laboratory). Direct visualization of the peritoneum (laparoscopy) may disclose peritoneal deposits of tumor, tuberculosis, or metastatic disease of the liver. Biopsies are taken under direct vision, often adding to the diagnostic accuracy of the procedure.

Chylous ascites refers to a turbid, milky, or creamy peritoneal fluid due to the presence of thoracic or intestinal lymph. Such a fluid shows Sudan-staining fat globules microscopically and an increased triglyceride content by chemical examination. Opaque milky fluid usually has a triglyceride concentration of >1000 mg/dL. A turbid fluid due to leukocytes or tumor cells may be confused with chylous fluid (pseudochylous), and it is often helpful to carry out alkalization and ether extraction of the specimen. Alkali tend to dissolve cellular proteins and thereby reduce turbidity; ether extraction leads to clearing if the turbidity of the fluid is due to lipid. Chylous ascites is most often the result of lymphatic obstruction from trauma, tumor, tuberculosis, filariasis ([Chap. 221](#)), or congenital abnormalities. It also may be seen in the nephrotic syndrome.

Rarely, ascitic fluid may be *mucinous* in character, suggesting either pseudomyxoma peritonei ([Chap. 289](#)) or rarely a colloid carcinoma of the stomach or colon with peritoneal implants.

On occasion, ascites may develop as a seemingly isolated finding in the absence of a clinically evident underlying disease. Then, a careful analysis of ascitic fluid may indicate the direction the evaluation should take. A useful framework for the workup starts with an analysis of whether the fluid is classified as a high (transudate) or low (exudate) gradient fluid. *High gradient (transudative) ascites* of unclear etiology is most often due to occult cirrhosis, right-sided venous hypertension raising hepatic sinusoidal pressure, or hypoalbuminemic states such as nephrosis or protein-losing enteropathy. Cirrhosis with well-preserved liver function (normal albumin) resulting in ascites invariably is associated with significant portal hypertension ([Chap. 298](#)). Evaluation should include liver function tests, liver-spleen scan, or other hepatic imaging procedure (i.e., CT or ultrasound) to detect nodular changes in the liver or a colloid shift of isotope to suggest portal hypertension. On occasion, a wedged hepatic venous pressure can be useful to document portal hypertension. Finally, if clinically indicated, a liver biopsy will confirm the diagnosis of cirrhosis and perhaps suggest its etiology. Other etiologies may result in hepatic venous congestion and resultant ascites. Right-sided cardiac valvular disease and particularly constrictive pericarditis should raise a high index of suspicion and may require cardiac imaging and cardiac catheterization for definitive diagnosis. Hepatic vein thrombosis is evaluated by visualizing the hepatic veins with imaging techniques (Doppler ultrasound, angiography, CT scans, magnetic resonance imaging) to demonstrate obliteration, thrombosis, or obstruction by tumor. Uncommonly, transudative ascites may be associated with benign tumors of the ovary, particularly fibroma (Meigs' syndrome) with ascites and hydrothorax.

Low gradient (exudative) ascites should initiate an evaluation for primary peritoneal

processes, most importantly infection and tumor. Routine bacteriologic culture of ascitic fluid often yields a specific organism causing infectious peritonitis. Tuberculous peritonitis ([Table 46-1](#)) is best diagnosed by peritoneal biopsy, either percutaneously or via laparoscopy. Histologic examination invariably shows granulomata that may contain acid-fast bacilli. Since cultures of peritoneal fluid and biopsies for tuberculosis may require 6 weeks, characteristic histology with appropriate stains allows antituberculosis therapy to be started promptly. Similarly, the diagnosis of peritoneal seeding by tumor can usually be made by cytologic analysis of peritoneal fluid or by peritoneal biopsy if cytology is negative. Appropriate diagnostic studies can then be undertaken to determine the nature and site of the primary tumor. Pancreatic ascites ([Table 46-1](#)) is invariably associated with an extravasation of pancreatic fluid from the pancreatic ductal system, most commonly from a leaking pseudocyst. Ultrasound or CT examination of the pancreas followed by visualization of the pancreatic duct by direct cannulation [viz., endoscopic retrograde cholangiopancreatography (ERCP)] usually discloses the site of leakage and permits resective surgery to be carried out.

An analysis of the physiologic and metabolic factors involved in the production of ascites (detailed in [Chap. 298](#)), coupled with a complete evaluation of the nature of the ascitic fluid, invariably discloses the etiology of the ascites and permits appropriate therapy to be instituted.

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SECTION 7 -ALTERATIONS IN RENAL AND URINARY TRACT FUNCTION

47. AZOTEMIA AND URINARY ABNORMALITIES - *Bradley M. Denker, Barry M. Brenner*

Body homeostasis is maintained predominantly through the cellular processes that together comprise normal kidney function. Disturbances to any of these functions can lead to a constellation of abnormalities that may be detrimental to survival. The clinical manifestations of these diseases will depend upon the pathophysiology of the renal injury and will often be initially identified as a complex of symptoms, abnormal physical findings, and laboratory changes that will allow the identification of specific syndromes. These renal syndromes (summarized in [Table 47-1](#)) may arise as the consequence of a systemic illness or can occur as a primary renal disease. Nephrologic syndromes usually consist of several elements that reflect the underlying pathologic processes and the duration of the disease and typically include one or more of the following features: (1) disturbances in urine volume (oliguria, anuria, polyuria); (2) abnormalities of urine sediment [red blood cells (RBC); white blood cells, casts, and crystals]; (3) abnormal excretion of serum proteins (proteinuria); (4) reduction in glomerular filtration rate (GFR) (azotemia); (5) presence of hypertension and/or expanded total body volume (edema); (6) electrolyte abnormalities, or (7) in some syndromes, fever/pain. The combination of these findings should permit identification of one of the major nephrologic syndromes ([Table 47-1](#)) and will allow the differential diagnoses to be narrowed and the appropriate diagnostic evaluation and therapeutic course to be determined. Each of these syndromes and their associated diseases are discussed in more detail in subsequent chapters. This chapter will focus on several aspects of renal abnormalities that are critically important to distinguishing these processes: (1) reduction in GFR leading to azotemia, (2) alterations of the urinary sediment and/or protein excretion, and (3) abnormalities of urinary volume.

AZOTEMIA

ASSESSMENT OF GLOMERULAR FILTRATION RATE

Monitoring the [GFR](#) is important in both the hospital and outpatient settings, and several different methodologies are available (discussed below). In most acute clinical circumstances a measured GFR is not available, and it is necessary to estimate the GFR from the serum creatinine level in order to provide appropriate doses of drugs that are excreted into the urine. Serum creatinine is the most widely used marker for GFR and is related directly to the urine creatinine excretion and inversely to the serum creatinine (U_{Cr}/P_{Cr}). Based upon this relationship and some important caveats (discussed below), the GFR will fall proportionately with the increase in P_{Cr} . Failure to account for GFR reductions in drug dosing can lead to significant morbidity and mortality from drug toxicities (e.g., digoxin, aminoglycosides). In the outpatient setting, serial determinations of GFR are helpful for following the progression of chronic renal insufficiency, but again, the serum creatinine is often used as a surrogate for GFR (although much less accurate; see below). In patients with chronic progressive renal insufficiency there is an approximately linear relationship between $1/P_{Cr}$ and time. The slope of this line will remain constant for an individual patient, and when values are obtained that do not fall on this line, an investigation for a superimposed acute process

(e.g., volume depletion, drug reaction) should be initiated. It should be emphasized that the signs and symptoms of uremia will develop at significantly different levels of serum creatinine depending upon the patient (size, age, and sex), the underlying renal disease, existence of concurrent diseases, and true GFR. In general, patients do not develop symptomatic uremia until renal insufficiency is usually quite severe ($\text{GFR} < 15 \text{ mL/min}$) and in some patients it does not occur until the $\text{GFR} < 5 \text{ mL/min}$.

A reduced [GFR](#) leads to retention of nitrogenous waste products (azotemia) such as serum urea nitrogen and creatinine. Azotemia may result from reduced renal perfusion, intrinsic renal disease, or postrenal processes (ureteral obstruction; see below and [Fig. 47-1](#)). Precise determination of GFR is problematic as both commonly used markers (urea and creatinine) have characteristics that affect their accuracy as markers of clearance. Urea clearance is generally an underestimate of GFR because of tubule urea reabsorption and may be as low as one-half of GFR measured by other techniques.

Creatinine is a small, freely filtered solute that varies little from day to day (since it is derived from muscle metabolism of creatine). However, serum creatinine can increase acutely from dietary ingestion of cooked meat. Creatinine can be secreted by the proximal tubule through an organic cation pathway. There are many clinical settings where a creatinine clearance is not available, and decisions concerning drug dosing must be made based on the serum creatinine. A formula that allows an estimate of creatinine clearance in men that accounts for age-related decreases in [GFR](#), body weight, and sex has been derived by Cockcroft-Gault:

This value should be multiplied 0.85 for women, since a lower fraction of the body weight is composed of muscle. The gradual loss of muscle from chronic illness, chronic use of glucocorticoids, or malnutrition can mask significant changes in [GFR](#) with small or imperceptible changes in serum creatinine. More accurate determinations of GFR are available using inulin clearance or radionuclide-labeled markers such as ^{125}I -iothalamate or EDTA. These methods are highly accurate due to precise quantitation and the absence of any renal reabsorption/secretion and should be used to follow GFR in patients in whom creatinine is not likely to be a reliable indicator (patients with decreased muscle mass secondary to age, malnutrition, concurrent illnesses).

Approach to the Patient

Once it has been established that [GFR](#) is reduced, the physician must decide if this represents acute or chronic renal failure. The clinical situation, history, and laboratory data often make this an easy distinction. However, the laboratory abnormalities characteristic of chronic renal failure, including anemia, hypocalcemia, and hyperphosphatemia, are often also present in patients presenting with acute renal failure. Radiographic evidence of renal osteodystrophy ([Chap. 270](#)) would be seen only in chronic renal failure but is a very late finding, and these patients are usually on dialysis. The urinalysis and renal ultrasound can occasionally facilitate distinguishing acute from chronic renal failure. An approach to the evaluation of azotemic patients is shown in [Fig. 47-1](#). Patients with advanced chronic renal insufficiency often have some proteinuria, nonconcentrated urine (isosthenuria), and small kidneys on ultrasound

characterized by increased echogenicity and cortical thinning. Treatment should be directed toward slowing the progression of renal disease and providing symptomatic relief for edema, acidosis, anemia, and hyperphosphatemia, as discussed in [Chap. 270](#). Acute renal failure ([Chap. 269](#)) can result from processes affecting renal blood flow (prerenal azotemia), intrinsic renal diseases (affecting vessels, glomeruli, or tubules), or postrenal processes (obstruction to urine flow in ureters, bladder, or urethra) ([Chap. 281](#)).

Prerenal Failure Decreased renal perfusion accounts for 40 to 80% of acute renal failure and, if appropriately treated, is readily reversible. The etiologies of prerenal azotemia include any cause of decreased circulating blood volume including volume loss (gastrointestinal hemorrhage, burns, diarrhea, diuretics), volume sequestration (pancreatitis, peritonitis, rhabdomyolysis), or decreased effective circulating volume (cardiogenic shock, sepsis). Renal perfusion can also be affected by reductions in cardiac output from peripheral vasodilatation (sepsis, drugs) or profound renal vasoconstriction [severe heart failure, hepatorenal syndrome, drugs (such as nonsteroidal anti-inflammatory drugs (NSAIDs))]. True, or "effective," hypovolemia leads to a fall in mean arterial pressure, which in turn triggers a series of neural and humoral responses that include activation of the sympathetic nervous and renin-angiotensin-aldosterone systems and ADH release. [GFR](#) is maintained by prostaglandin-mediated relaxation of afferent arterioles and angiotensin II-mediated constriction of efferent arterioles. Once the mean arterial pressure falls below 80 mmHg, there is a steep decline in GFR.

Blockade of prostaglandin production by [NSAIDs](#) can result in severe vasoconstriction and acute renal failure under these circumstances. Angiotensin-converting enzyme (ACE) inhibitors decrease efferent arteriolar tone and can decrease glomerular capillary perfusion pressure. Patients on NSAIDs and/or ACE inhibitors are most susceptible to hemodynamically mediated acute renal failure when blood volume is reduced for any reason. Patients with renal artery stenosis are dependent upon efferent arteriolar vasoconstriction for maintenance of glomerular filtration pressure and are particularly susceptible to precipitous decline in [GFR](#) when given ACE inhibitors.

Prolonged renal hypoperfusion can lead to acute tubular necrosis (ATN; an intrinsic renal disease discussed below). The urinalysis and urinary electrolytes can be useful in distinguishing prerenal azotemia from ATN ([Table 47-2](#)). The urine of patients with prerenal azotemia can be predicted from the stimulatory actions of norepinephrine, angiotensin II, ADH, and low tubule fluid flow on salt and water reabsorption from the urine. In prerenal conditions the tubules are intact, leading to a concentrated urine (>500 mosm), avid Na retention (urine Na concentration <20 mM/L; fractional excretion of Na <1%), and $U_{Cr}/P_{Cr} > 40$ ([Table 47-2](#)). The prerenal urine sediment is usually normal or has occasional hyaline and granular casts, while the sediment of ATN is usually filled with cellular debris and muddy brown granular casts.

Intrinsic Renal Disease When prerenal and postrenal azotemia have been excluded as etiologies of renal failure, an intrinsic parenchymal renal disease is present. Intrinsic renal disease can arise from processes involving large renal vessels, microvasculature and glomeruli, or tubulointerstitium. Ischemic and toxic [ATN](#) account for about 90% of acute intrinsic renal failure. As outlined in [Fig. 47-1](#), the clinical setting and urinalysis are

helpful in separating the possible etiologies of acute intrinsic renal failure. Prerenal azotemia and ATN are part of a spectrum of renal hypoperfusion; evidence of structural tubule injury is present in ATN, whereas prompt reversibility occurs with prerenal azotemia upon restoration of adequate renal perfusion. Thus, ATN can often be distinguished from prerenal azotemia by urinalysis and urine electrolyte composition ([Table 47-2](#) and [Fig. 47-1](#)). Ischemic ATN is observed most frequently in patients who have undergone major surgery, trauma, severe hypovolemia, overwhelming sepsis, or extensive burns. Nephrotoxic ATN complicates the administration of many common medications, usually by inducing a combination of intrarenal vasoconstriction, direct tubule toxicity, and/or tubular obstruction. The kidney is vulnerable to toxic injury by virtue of its rich blood supply (25% of cardiac output) and its ability to concentrate and metabolize toxins. A diligent search for hypotension and nephrotoxins will usually uncover the specific etiology of ATN. Discontinuation of nephrotoxins and stabilizing blood pressure will often suffice without the need for dialysis while the tubules recover. **An extensive list of potential drugs and toxins implicated in ATN can be found in [Chap. 269](#).*

Processes that involve the tubules and interstitium can lead to acute renal failure. These include drug-induced interstitial nephritis (especially antibiotics, [NSAIDs](#), and diuretics), severe infections (both bacterial and viral), systemic diseases (e.g., systemic lupus erythematosus), or infiltrative disorders (e.g., sarcoid, lymphoma, or leukemia). A list of drugs associated with allergic interstitial nephritis can be found in [Chap. 277](#). The urinalysis usually shows mild to moderate proteinuria, hematuria, and pyuria (approximately 75% of cases) and occasionally white blood cell casts. The finding of RBC casts in interstitial nephritis has been reported but should prompt a search for glomerular diseases. Occasionally renal biopsy will be needed to distinguish among these possibilities. The finding of eosinophils in the urine is suggestive of allergic interstitial nephritis and is optimally observed by using a Hansel stain. The absence of eosinophiluria, however, does not exclude the possibility of acute interstitial nephritis.

Occlusion of large renal vessels including arteries and veins is an uncommon cause of acute renal failure. A significant reduction in [GFR](#) by this mechanism suggests bilateral processes or a unilateral process in a patient with a single functioning kidney. Renal arteries can be occluded with atheroemboli, thromboemboli, in situ thrombosis, aortic dissection, or vasculitis. Atheroembolic renal failure can occur spontaneously but is most often associated with recent aortic instrumentation. The emboli are cholesterol-rich and lodge in medium and small renal arteries leading to an eosinophil-rich inflammatory reaction. Atheroembolic acute renal failure often has a normal urinalysis but may contain eosinophils and casts. The diagnosis can be confirmed by renal biopsy, but this is often unnecessary when other stigmata of atheroemboli are present (livedo reticularis, distal peripheral infarcts, eosinophilia). Renal artery thrombosis may lead to mild proteinuria and hematuria, whereas renal vein thrombosis typically induces heavy proteinuria and hematuria. **These vascular catastrophes often require angiography for confirmation and are discussed in [Chap. 278](#).*

Diseases of glomeruli (glomerulonephritis or vasculitis) and the renal microvasculature (hemolytic uremic syndromes, thrombotic thrombocytopenic purpura, or malignant hypertension) usually present with various combinations of glomerular injury: proteinuria, hematuria, reduced [GFR](#), and alterations of Na excretion leading to

hypertension, edema, and circulatory congestion (acute nephritic syndrome). These findings may occur as primary renal diseases or as renal manifestations of systemic diseases. The clinical setting and other laboratory data will help distinguish primary renal from systemic diseases. The finding of [RBC casts](#) in the urine is an indication for early renal biopsy ([Fig. 47-1](#)) as the pathologic pattern has important implications for diagnosis, prognosis, and treatment. Hematuria without RBC casts can also be an indication of glomerular disease, and this evaluation is summarized in [Fig. 47-2](#).**A detailed discussion of glomerulonephritis and diseases of the microvasculature can be found in Chap. 274.*

Postrenal Azotemia Urinary tract obstruction accounts for fewer than 5% of cases of acute renal failure, but it is usually reversible and must be ruled out early in the evaluation ([Fig. 47-1](#)). Since a single kidney is capable of adequate clearance, acute renal failure from obstruction requires obstruction at the urethra or bladder outlet, bilateral ureteral obstruction, or unilateral obstruction in a patient with a single functioning kidney. Obstruction is usually diagnosed by the presence of ureteral dilatation on renal ultrasound. However, early in the course of obstruction or if the ureters are unable to dilate (such as encasement by pelvic tumors), the ultrasound examination may be negative.**The specific urologic conditions that cause obstruction are discussed in Chap. 281.*

Oliguria and Anuria *Oliguria* refers to a 24-h urine output of <500 mL, and *anuria* is the complete absence of urine formation. Anuria can be caused by total urinary tract obstruction, total renal artery or vein occlusion, and shock (manifested by severe hypotension and intense renal vasoconstriction). Cortical necrosis, [ATN](#), and rapidly progressive glomerulonephritis can occasionally cause anuria. Oliguria can accompany any cause of acute renal failure and carries a more serious prognosis for renal recovery in all conditions except prerenal azotemia. *Nonoliguria* refers to urine output in excess of 500 mL/day in patients with acute or chronic azotemia. With nonoliguric ATN, disturbances of potassium and hydrogen balance are less severe than in oliguric patients and recovery to normal renal function is usually more rapid.

ABNORMALITIES OF THE URINE

PROTEINURIA

The evaluation of proteinuria is shown schematically in [Fig. 47-3](#) and is typically initiated after colorimetric detection of proteinuria by dipstick examination. Current methods for measuring proteinuria vary significantly. The dipstick measurement detects mostly albumin and gives false-positive results when pH > 7.0 and the urine is very concentrated or contaminated with blood. A very dilute urine may obscure significant proteinuria on dipstick examination, and proteinuria that is not predominantly albumin will be missed. This is particularly important for the detection of Bence Jones proteins in the urine of patients with multiple myeloma. Tests to measure total urine concentration accurately rely on precipitation with sulfosalicylic or trichloroacetic acids. Currently, ultrasensitive dipsticks are available to measure microalbuminuria (30 to 300 mg/d), an early marker of glomerular disease that has been shown to predict glomerular injury in early diabetic nephropathy ([Fig. 47-3](#)).

The magnitude of proteinuria and the protein composition in the urine depend upon the mechanism of renal injury leading to protein losses. Large amounts of plasma proteins normally course through the glomerular capillaries but do not enter the urinary space. Both charge and size selectivity prevent virtually all of albumin, globulin, and other large-molecular-weight proteins from crossing the glomerular wall. However, if this barrier is disrupted, there can be leakage of plasma proteins into the urine (glomerular proteinuria; [Fig. 47-3](#)). Smaller proteins (<20 kDa) are freely filtered but are readily reabsorbed by the proximal tubule. Normal individuals excrete less than 150 mg/d of total protein and only about 30 mg/d of albumin. The remainder of the protein in the urine is secreted by the tubules (Tamm-Horsfall, IgA, and urokinase) or represents small amounts of filtered β_2 -microglobulin, apoproteins, enzymes, and peptide hormones. Another mechanism of proteinuria occurs when there is excessive production of an abnormal protein that exceeds the capacity of the tubule for reabsorption. This most commonly occurs with plasma cell dyscrasias such as multiple myeloma and lymphomas that are associated with monoclonal production of immunoglobulin light chains.

The normal glomerular endothelial cell forms a barrier penetrated by pores of about 100 nm that holds back cells and other particles but offers little impediment to passage of most proteins. The glomerular basement membrane traps most large proteins (>100 kDa), while the foot processes of epithelial cells (podocytes) cover the urinary side of the glomerular basement membrane and produce a series of narrow channels (slit diaphragms) to allow molecular passage of small solutes and water ([Fig. 47-4](#)). The channels are coated with anionic glycoproteins that are rich in glutamate, aspartate, and sialic acid, which are negatively charged at physiologic pH. This negatively charged barrier impedes the passage of anionic molecules such as albumin. Some glomerular diseases, such as minimal change disease, cause fusion of glomerular epithelial cell foot processes, resulting in predominantly "selective" ([Fig. 47-3](#)) loss of albumin. Other glomerular diseases can present with disruption of the basement membrane and slit diaphragms (e.g., by immune complex deposition), resulting in large amounts of protein losses that include albumin and other plasma proteins. The fusion of foot processes causes increased pressure across the capillary basement membrane, resulting in areas with larger pore sizes. The combination of increased pressure and larger pores results in significant proteinuria ("nonselective"; [Fig. 47-3](#)).

When the total daily excretion of protein exceeds 3.5 g, there is often associated hypoalbuminemia, hyperlipidemia, and edema (nephrotic syndrome; [Table 47-1](#)). However, total daily urinary protein excretion greater than 3.5 g can occur without the other features of the nephrotic syndrome in a variety of other renal diseases ([Fig. 47-3](#)). Plasma cell dyscrasias (multiple myeloma) can be associated with large amounts of excreted light chains in the urine, which may not be detected by dipstick (which detects mostly albumin). The light chains produced from these disorders are filtered by the glomerulus and overwhelm the reabsorptive capacity of the proximal tubule. A sulfosalicylic acid precipitate that is out of proportion to the dipstick estimate is suggestive of light chains (Bence Jones protein), and light chains typically redissolve upon warming of the precipitate. Renal failure from these disorders occurs through a variety of mechanisms including tubule obstruction (cast nephropathy) and light chain deposition ([Chap. 275](#)).

Hypoalbuminemia in nephrotic syndrome occurs through excessive urinary losses, increased renal catabolism, and inadequate hepatic synthesis. The resulting decrease in plasma oncotic pressure contributes to edema formation by altering the Starling forces and favoring fluid movement from capillaries to interstitium. The resulting homeostatic mechanisms designed to correct the decrease in effective intravascular volume contribute to edema formation in some patients. These mechanisms include activation of the renin-angiotensin system, antidiuretic hormone, and the sympathetic nervous system, which contribute to excessive renal salt and water reabsorption and can contribute to unrelenting edema.

The severity of edema correlates with the degree of hypoalbuminemia and is modified by other factors such as heart disease or peripheral vascular disease. The diminished plasma oncotic pressure and urinary losses of regulatory proteins appear to stimulate hepatic lipoprotein synthesis. The resulting hyperlipidemia results in lipid bodies (fatty casts, oval fat bodies) in the urine. Other proteins are lost in the urine, leading to a variety of metabolic disturbances. These include thyroxine-binding globulin, cholecalciferol-binding protein, transferrin, and metal-binding proteins. A hypercoagulable state frequently accompanies severe nephrotic syndrome due to urinary losses of antithrombin III, reduced serum levels of proteins S and C, hyperfibrinogenemia, and enhanced platelet aggregation. Some patients develop severe IgG deficiency with resulting defects in immunity. Many diseases (some listed in [Fig. 47-3](#)) and drugs can cause the nephrotic syndrome, and a complete list can be found in [Chap. 274](#).

HEMATURIA, PYURIA, AND CASTS

Isolated hematuria without proteinuria, other cells, or casts is often indicative of bleeding from the urinary tract. Normal red blood cell excretion is up to 2 million [RBCs](#) per day. Hematuria is defined as two to five RBCs per high-power field (HPF) and can be detected by dipstick. Common causes of isolated hematuria include stones, neoplasms, tuberculosis, trauma, and prostatitis. Gross hematuria with blood clots is almost never indicative of glomerular bleeding; rather, it suggests a postrenal source in the urinary collecting system. Evaluation of patients presenting with microscopic hematuria is outlined in [Fig. 47-2](#). A single urinalysis with hematuria is common and can result from menstruation, viral illness, allergy, exercise, or mild trauma. Annual urinalysis of servicemen over a 10-year period showed an incidence of 38%. However, persistent or significant hematuria (>three RBCs/HPF on three urinalyses, or single urinalysis with >100 RBCs, or gross hematuria) identified significant renal or urologic lesions in 9.1% of over 1000 patients. Even patients who are chronically anticoagulated should be investigated as outlined in [Fig. 47-2](#). The suspicion for urogenital neoplasms in patients with isolated painless hematuria (nondysmorphic RBCs) increases with age. Neoplasms are rare in the pediatric population, and isolated hematuria is more likely to be "idiopathic" or associated with a congenital anomaly. Hematuria with pyuria and bacteriuria is typical of infection and should be treated with antibiotics after appropriate cultures. Acute cystitis or urethritis in women can cause gross hematuria. Hypercalciuria and hyperuricosuria are also risk factors for unexplained isolated hematuria in both children and adults. In some of these patients (50 to 60%), reducing calcium and uric acid excretion through dietary interventions can eliminate the microscopic hematuria.

Isolated microscopic hematuria can be a manifestation of glomerular diseases. The [RBCs](#) of glomerular origin are often dysmorphic when examined by phase-contrast microscopy. Irregular shapes of RBCs may also occur due to pH and osmolarity changes found in the distal tubule. There is, however, significant observer variability in detecting dysmorphic RBCs, especially if a phase-contrast microscope is not available. The most common etiologies of isolated glomerular hematuria are IgA nephropathy, hereditary nephritis, and thin basement membrane disease. IgA nephropathy and hereditary nephritis can have episodic gross hematuria. A family history of renal failure is often present in patients with hereditary nephritis, and patients with thin basement membrane disease often have other family members with microscopic hematuria. A renal biopsy is needed for the definitive diagnosis of these disorders, which are discussed in more detail in [Chap. 275](#). Hematuria with dysmorphic RBCs, RBC casts, and protein excretion >500 mg/d is virtually diagnostic of glomerulonephritis. RBC casts form as RBCs that enter the tubular fluid become trapped in a cylindrical mold of gelled Tamm-Horsfall protein. Even in the absence of azotemia, these patients should undergo serologic evaluation and renal biopsy as outlined in [Fig. 47-2](#).

Isolated pyuria is unusual since inflammatory reactions in the kidney or collecting system are also associated with hematuria. The presence of bacteria suggests infection, and white blood cell casts with bacteria are indicative of pyelonephritis. White blood cells and/or white blood cell casts may also be seen in tubulointerstitial processes such as interstitial nephritis, systemic lupus erythematosus, and transplant rejection. In chronic renal diseases, degenerated cellular casts called *waxy casts* can be seen in the urine. *Broad casts* are thought to arise in the dilated tubules of enlarged nephrons that have undergone compensatory hypertrophy in response to reduced renal mass (i.e., chronic renal failure). A mixture of broad casts typically seen with chronic renal failure together with cellular casts and [RBCs](#) may be seen in smoldering processes such as chronic glomerulonephritis with active glomerulitis.

ABNORMALITIES OF URINE VOLUME

The volume of urine produced varies depending upon the fluid intake, renal function, and physiologic demands of the individual. See "Azotemia," above, for discussion of decreased (oliguria) or absent urine production (anuria). **The physiology of water formation and renal water conservation are discussed in [Chap. 268](#).*

POLYURIA

By history, it is often difficult for patients to distinguish urinary frequency (often of small volumes) from polyuria, and a 24-h urine collection is needed for evaluation ([Fig. 47-5](#)). It is necessary to determine if the polyuria represents a solute or water diuresis and if the diuresis is appropriate for the clinical circumstances. The average person excretes between 600 and 800 mosmol of solutes per day, primarily as urea and electrolytes. The urine osmolality can help distinguish a solute from water diuresis. If the urine output is >3 L/d (arbitrarily defined as polyuria) and the urine is dilute (<250 mosmol/L), then total mosmol excretion is normal and a water diuresis is present. This circumstance could arise from polydipsia, inadequate secretion of vasopressin (central diabetes insipidus), or failure of renal tubules to respond to vasopressin (nephrogenic diabetes insipidus). If the urine volume is >3 L/d and urine osmolality is >300 mosmol/L, then a

solute diuresis is clearly present and a search for the responsible solute(s) is mandatory.

Excessive filtration of a poorly reabsorbed solute such as glucose, mannitol, or urea can depress reabsorption of NaCl and water in the proximal tubule and lead to enhanced excretion in the urine. Poorly controlled diabetes mellitus is the most common cause of a solute diuresis, leading to volume depletion and serum hypertonicity. Since the urine Na concentration is less than that of blood, more water than Na is lost, causing hypernatremia and hypertonicity. Common iatrogenic solute diuresis occurs from mannitol administration, radiocontrast media, and high-protein feedings (enterally or parenterally), leading to increased urea production and excretion. Less commonly, excessive Na loss may occur from cystic renal diseases, Bartter's syndrome, or during the course of a tubulointerstitial process (such as resolving [ATN](#)). In these so-called salt-wasting disorders, the tubule damage results in direct impairment of Na reabsorption and indirectly reduces the responsiveness of the tubule to aldosterone. Usually, the Na losses are mild, and the obligatory urine output is less than 2 L/d (resolving ATN and postobstructive diuresis are exceptions and may be associated with significant natriuresis and polyuria.)

Formation of large volumes of dilute urine represent polydipsic states or diabetes insipidus. Primary polydipsia can result from habit, psychiatric disorders, neurologic lesions, or medications. During deliberate polydipsia, extracellular fluid volume is normal or expanded and vasopressin levels are reduced because serum osmolality tends to be near the lower limits of normal.

Central diabetes insipidus may be idiopathic in origin or secondary to a variety of hypothalamic conditions including posthypophysectomy or trauma or neoplastic, inflammatory, vascular, or infectious hypothalamic diseases. Idiopathic central diabetes insipidus is associated with selective destruction of the vasopressin-secreting neurons in the supraoptic and paraventricular nuclei and can be inherited as an autosomal dominant trait or occur spontaneously. Nephrogenic diabetes insipidus can occur in a variety of clinical situations as summarized in [Fig. 47-5](#).

A plasma vasopressin level is recommended as the best method for distinguishing between central and nephrogenic diabetes insipidus. Alternatively, a water deprivation test plus exogenous vasopressin may also distinguish primary polydipsia from central and nephrogenic diabetes insipidus. **For a detailed discussion, see [Chap. 329](#).*

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48. INCONTINENCE AND LOWER URINARY TRACT SYMPTOMS - Philippe E. Zimmern, John D. McConnell

PHYSIOLOGY OF VOIDING

Normal bladder filling depends on unique elastic properties of the bladder wall that allow it to increase in volume at a pressure lower than that of the bladder neck and urethra (otherwise incontinence would occur). Despite provocative maneuvers such as coughing, voluntary bladder contractions do not occur. Emptying is dependent on the integrity of a complex neuromuscular network that causes relaxation of the urethral sphincter a few milliseconds before the onset of the detrusor (bladder muscle) contraction. With normal, sustained detrusor contraction, the bladder empties completely. A bladder that can fill and empty in this manner has a normal detrusor muscle and is described as *stable* according to conventional terminology.

Since the voluntary control of micturition depends on the neural connections between the cerebral cortex and the brainstem, disruption of these pathways (brain tumor, stroke, head trauma, Parkinson's disease) impairs the ability to suppress and control bladder contractions. A bladder contraction without voluntary effort characterizes an unstable bladder. Bladder or detrusor instability of neurologic origin is termed *detrusor hyperreflexia*. Conversely, the detrusor muscle that cannot contract during voiding is called *noncontractile*; underactivity of the detrusor due to a lesion of the sacral cord or pelvic nerves is termed *detrusor areflexia*.

Contrary to common belief, the center that controls normal micturition is not in the spinal cord but in the brainstem. Proper coordination (*synergia*) between the detrusor and urethral sphincters requires an intact neural (autonomic and somatic nervous systems) communication between bladder and urethra. Injury to the upper spinal cord, for example, can cause dyssynergia between bladder and urethra that results in urge incontinence, residual urine retention, bladder wall changes (trabeculation and fibrosis), and possibly renal insufficiency.

A simple way to classify voiding dysfunction is to determine whether it is primarily a *storage failure* or an *emptying failure* by asking two questions:

Is the voiding dysfunction due to the bladder or outlet (bladder neck or urethra) (failure to store)?

Is there neurologic dysfunction (failure to empty)?

Bladder storage and emptying problems may coexist in the same individual and can cause similar lower urinary tract symptoms (LUTS).

LOWER URINARY TRACT SYMPTOMS IN MEN

The most common cause of [LUTS](#) in men of middle age and older is prostatic hyperplasia, which causes obstruction to urine flow by encroachment on the urethral lumen ([Chap. 95](#)). Histologically, 50 to 80% of the prostatic volume is composed of stromal tissue (smooth muscle), while the remainder is glandular. The transitional zone,

which is responsible for benign prostatic growth, comprises 10 to 15% of the prostate at the end of puberty but increases in volume after age 40. However, prostatic enlargement is not always accompanied by symptoms because the direction of growth can be outward, so that little change may occur in urine flow. Alternatively, men with early histologic evidence of prostatic hyperplasia can experience significant voiding symptoms. In this circumstance, increased tone of the prostatic smooth muscle and enhanced prostatic tension within a nondistensible capsule can cause obstruction.

In response to obstruction, the bladder smooth-muscle cells hypertrophy to generate the higher pressures necessary for voiding, and the increase in bladder muscle mass leads to reduced elasticity, or compliance, and decreased bladder capacity. Detrusor dysfunction from bladder outlet obstruction can cause any combination of the [LUTS](#) described above. When the obstruction progresses, infiltration of extracellular matrix between the smooth-muscle bundles of the bladder wall can result in a hypocontractile or acontractile bladder (bladder failure).

Other complications such as urinary tract infections or bladder stones secondary to the large postvoid residuals (stasis) and upper tract damage (hydronephrosis, reflux) can develop during the course of the obstructive process. Although prostatic hyperplasia is the most common cause of bladder outlet obstruction in men, other sources of obstruction include prostate cancer, urethral stricture, and lack of proper sphincteric relaxation (neurologic cause). Nonobstructive causes of [LUTS](#) include diabetic neuropathy, which can affect the parasympathetic nerves of the bladder. Decreased sensation of bladder fullness leads to incomplete emptying and overdistention of the bladder and, in turn, to increased frequency and nocturia due to bladder overflow; these symptoms are frequently made worse by the polydipsia/polyuria of diabetes mellitus. At times, storage symptoms can be caused by other neurologic causes such as stroke, multiple sclerosis, or Parkinson's disease.

The International Prostate Symptom Score (IPSS) is used to assess the severity of [LUTS](#):

Decreased force of stream -- over the past month how often have you had a weak urinary stream?

Intermittency -- over the past month how often have you found you stopped and started again several times when urinating?

Incomplete emptying -- over the past month how often have you had a sensation of not emptying your bladder completely after finishing urination?

Straining -- over the past month how often have you had to push or strain to begin urination? The [IPSS](#) also assesses the impact of storage symptoms:

Frequency -- over the past month how often have you had to urinate again within 2 h after urinating?

Urgency -- over the past month how often have you found it difficult to postpone urination?

Nocturia -- over the past month how many times did you typically get up to urinate between going to bed and getting up in the morning? (Range: none to five or more times.) Except for nocturia, the answers range from 0 (not at all) to 5 (almost always). A total score of <8 indicates minimal voiding dysfunction; a total score of ≥ 13 is usually required to enroll patients in drug studies for the management of benign prostatic hyperplasia (BPH); and symptom scores >23 suggest significant bladder outlet obstruction. Because similar symptoms can result from neurologic causes, the [IPSS](#) questionnaire cannot be used to make the diagnosis of prostatic hyperplasia but is useful only as an index of severity and of the response to treatment.

LOWER URINARY TRACT SYMPTOMS IN WOMEN

Urethral obstruction is an uncommon cause of [LUTS](#) in women. A careful bimanual examination and passage of a urethral catheter are sufficient to exclude urethral stenosis, which is usually secondary to prior instrumentation or operative procedures, and urethral cancer. Urinary tract infection (cystitis) is more prevalent in women and must be excluded by urinalysis. Multiple sclerosis should be considered in middle-aged women presenting with frequency, urgency, or incontinence. In addition to many of the same disorders that produce voiding symptoms in men, estrogen deficiency, frequency-urgency syndrome, and interstitial cystitis (IC) with minimal pain must be considered. Cystocele and pelvic prolapse can cause urinary frequency secondary to impairment of bladder emptying.

EVALUATION

Men and women with [LUTS](#) and concomitant neurologic disease should undergo a complete urodynamic evaluation. In the absence of neurologic disease, men with LUTS most commonly have prostatic hyperplasia. However, it is necessary to exclude prostate cancer, especially if there is a positive family history, an abnormal prostate examination, or an elevated level of prostate-specific antigen (PSA). In both sexes bladder cancer can also cause storage symptoms and is suggested by microscopic hematuria and/or abnormal urine cytology. Usually, a detailed genitourinary history, a symptom assessment, a careful neurologic examination including rectal examination and assessment of the bulbocavernosus reflex, measurements of urine flow and postvoid residual urine volume (by bladder ultrasound), and limited laboratory evaluation (urinalysis, urine culture, PSA levels, urine cytology, urea/creatinine levels, as indicated) should be sufficient to direct therapy. More complex investigations of the lower urinary tract (cystoscopy, voiding cystography, urodynamics) and upper urinary tract (pyelogram or ultrasonography) are sometimes indicated. **For therapy of BPH, see [Chap. 95](#).*

INCONTINENCE

Incontinence is a condition where involuntary loss of urine is objectively demonstrated and is a social or hygienic problem. A common variant, *stress incontinence*, denotes involuntary loss of urine with physical exercise (coughing, sneezing, sports, sexual activity). *Urge incontinence* is an involuntary loss of urine associated with a strong desire to void, and *overflow incontinence* is an involuntary loss of urine when the

elevation of intravesical pressure with bladder overfilling or distention exceeds the maximal urethral pressure. Loss of urine through channels other than the urethra is rare (ectopic ureter, fistulae) but causes total or continuous incontinence.

INCONTINENCE IN WOMEN

Among noninstitutionalized women 60 years of age and older, 25 to 30% have urinary incontinence daily or weekly, and approximately half of institutionalized women are incontinent more than once a day. The annual cost of caring for incontinent persons is very high and, if not well managed, can be associated with complications such as decubitus ulcers.

Stress urinary incontinence (SUI) is secondary to urethral hypermobility or, less commonly (<10%), to intrinsic sphincteric deficiency (ISD). In the continent woman the bladder neck and proximal urethra are supported by the anterior vaginal wall and its lateral attachment to the levator muscles. Anterior vaginal wall relaxation causes urethral hypermobility, usually due to aging and/or estrogen deficiency or a prior traumatic delivery or pelvic surgery. Paradoxically, women can have clinical evidence of urethral hypermobility but no stress urinary incontinence.

Some women have an anatomically normal urethra and bladder neck but still have [SUI](#) due to damage to the internal sphincter (fixed, rigid, or "pipestem" urethra), due to prior anti-incontinence surgery, pelvic radiation or trauma, or neurologic disorders that cause denervation of the urethra. Urethral hypermobility and [ISD](#) can coexist in some patients and cause persistence (or rapid recurrence) of incontinence after a simple bladder neck suspension procedure that fixes the hypermobility but leaves the sphincter untreated.

Urge incontinence can be present alone or in association with [SUI](#) (mixed incontinence). The cause of the unsuppressible or uninhibited bladder contractions is usually idiopathic, but bacterial cystitis, bladder tumor, bladder outlet obstruction, and neurogenic bladder must be excluded. Overflow incontinence is due either to bladder outlet obstruction (rare in women), an acontractile bladder (diabetic neuropathy, multiple sclerosis), excessive smooth-muscle relaxation from drugs (anticholinergic medications), or psychogenic retention.

INCONTINENCE IN MEN

In men incontinence is less common than obstruction, but urgency and urge incontinence can occur as the result of bladder outlet obstruction (as from prostatic hyperplasia) that impairs detrusor smooth-muscle function and leads to detrusor instability. Men with neurogenic bladders (diabetic neuropathy, multiple sclerosis, Parkinson's disease, stroke) can develop urge incontinence. Other causes such as bacterial cystitis or bladder tumor must be excluded. [SUI](#) in men is usually the result of distal sphincteric damage, for example, as the result of radical prostatectomy for prostate cancer.

INCONTINENCE IN THE ELDERLY

Transient urinary incontinence is common in the elderly. A mnemonic devised by

Resnick delineates its numerous causes, namely *delirium*, *infection*, *atrophic urethritis*, *pharmacologic*, *psychological*, *excessive urine output* (hyperglycemia, congestive heart failure), *restricted mobility*, and *stool impaction* (DIAPPERS). Urge incontinence is the next most common disorder in this age group and is attributed to the progressive loss of the modulating influence of the frontal lobes of the cortex on the micturition center in the brainstem.

EVALUATION

The evaluation of urinary incontinence in women should include history and quality-of-life assessment, voiding diary, physical examination including pelvic examination, urinalysis and urine culture, and measurement of postvoid residual urine volume. For patients with an unclear history or after prior pelvic or anti-incontinence procedures, evaluation may include cystoscopy, urodynamic evaluation, and imaging studies (lower and/or upper urinary tract). The history should define the onset, duration, evolution, and triggering events of leakage. Prior treatments with medications, frequent voiding schedules, and exercise regimens should be noted. Severity of incontinence is denoted by recording the type and number of pads used per day or at night and how the incontinence affects daily activities (incontinence-impact questionnaire). The amount and type of fluid consumed, sexual history (hormonal status, deliveries, venereal diseases), gastrointestinal function (fecal incontinence, constipation), and past urologic history (bed-wetting, surgeries) must also be documented. The physical examination should place special emphasis on the abdominal, genital, pelvic (associated prolapses), and neurologic systems. [SUI](#) must be demonstrated by asking the patient to cough, strain, or even stand or squat. While leakage during a cough confirms SUI, leakage after a cough is due to bladder instability (stress-induced instability). SUI in the absence of urethral hypermobility raises the suspicion of a sphincteric defect. More complex testing is needed to determine whether the urethral anatomy is normal (evaluation of urethral mobility, lateral view of the urethra on the voiding cystourethrogram, cystoscopy), whether urethral function is normal with adequate closure (leak point pressure, urethral profilometry, videourodynamics), or whether bladder function is normal (bladder volume based on home diary, filling cystometrogram).

TREATMENT

Mild stress incontinence can be treated nonoperatively with medications, estrogen replacement, or biofeedback techniques. Modalities such as urethral plugs and anterior vaginal wall prostheses are under investigation. Moderate to severe stress incontinence responds to surgical procedures aimed at supporting the anterior vaginal wall (vaginal, laparoscopic, or abdominal operations) or enhancing urethral closure when stress incontinence is secondary to internal sphincter deficiency (periurethral injection of fat or collagen, autologous or cadaveric fascial sling, synthetic sling, or insertion of an artificial urinary sphincter).

Urge incontinence responds to the management of its cause. When it is due to neurogenic or idiopathic causes, anticholinergic agents are partially effective, although side effects such as mouth dryness, blurring of vision, or constipation can limit their usefulness. Better tolerated medications are now available including slow-release oxybutinin (Ditropan XL), which is administered as 5- to 10-mg tablets once daily, and

the more specific antimuscarinic agent, tolterodine (Detrol), which is usually given as 2 mg orally twice daily. Fluid restriction (which must be undertaken only with great caution) and bladder retraining with biofeedback may also be helpful. More aggressive intervention with bladder augmentation or urinary diversion are seldom necessary in the absence of neurologic disease.

BLADDER PAIN

Painful bladder disease is a general term for any bladder pathology that causes suprapubic, urethral, or pelvic pain. [IC](#) is the most common cause of bladder pain, but endometriosis, bacterial cystitis, and outlet obstruction that causes bladder instability can mimic the symptoms of IC.

INTERSTITIAL CYSTITIS

[IC](#) is a severe, chronic bladder disorder that causes frequency, nocturia, and suprapubic pain. The disorder usually affects women and is rare in blacks. Routine urine culture is uniformly negative, and the symptoms do not respond to antibiotic therapy. The etiology is probably multifactorial ([Table 48-1](#)). Current hypotheses as to etiology include autoimmune reaction against bladder antigens, deficiency in the glycosaminoglycan layer of the bladder surface allowing presumed toxins to penetrate the mucosa, mast cell infiltration and activation leading to the histamine release, and local bladder wall damage from bacteria.

The National Institutes of Health has established a series of criteria to define [IC](#) clinically ([Table 48-2](#)). The diagnosis is one of exclusion -- infection, radiation cystitis, urethral diverticula, herpes simplex, and malignancy must be excluded. Cystoscopy under anesthesia may be used for the following: (1) reveal glomerulations (submucosal vascular anomalies) or the infrequent Hunner's ulcer suggestive of IC; (2) make it possible to estimate bladder capacity (an important guide to treatment); (3) allow biopsy of the bladder wall when indicated; and (4) by bladder filling, sometimes provide therapeutic benefit with a reduction in pain level and urinary frequency up to 6 months or rarely longer.

EVALUATION

Chronic urinary frequency and bladder pain affect the quality of life to an extreme degree, though most patients experience a waxing and waning evolution; only 10% of patients have a consistent progression in symptoms. Evaluation should include a detailed history; physical examination designed to exclude neurologic and gynecologic pathology; voiding cystogram to exclude urethral defects; and urodynamic testing to eliminate a neurogenic bladder, bladder instability, or outlet obstruction and to document sensory instability. Referral to specialists may be indicated to exclude adnexal pathology, endometriosis, or bowel dysfunction or to utilize modern pain management techniques to prevent drug addiction.

TREATMENT

Empirical treatments that have been used include oral medications (amitriptyline,

hydroxyzine, pentosanpolysulfate) and intravesical agents (dimethyl sulfoxide, chlorpactin, heparin). These measures may improve the urinary symptoms and occasionally reduce pain but do not modify the long-term course. Surgical intervention (augmentation cystoplasty, urinary diversion) is indicated in fewer than 5% of cases because this is a non-life-threatening, chronic disease with occasional spontaneous remissions. "Last-resort" interventions such as removal of the bladder and urethra are not a guarantee of success because some patients continue to experience pelvic pain afterwards.

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49. FLUID AND ELECTROLYTE DISTURBANCES - Gary G. Singer, Barry M. Brenner

SODIUM AND WATER

COMPOSITION OF BODY FLUIDS

Water is the most abundant constituent in the body, comprising approximately 50% of body weight in women and 60% in men. This difference is attributable to differences in the relative proportions of adipose tissue in men and women. Total body water is distributed in two major compartments -- 55 to 75% is intracellular [intracellular fluid (ICF)], and 25 to 45% is extracellular [extracellular fluid (ECF)]. The [ECF](#) is further subdivided into intravascular (plasma water) and extravascular (interstitial) spaces in a ratio of 1:3.

The solute or particle concentration of a fluid is known as its *osmolality* and is expressed as milliosmoles per kilogram of water (mosmol/kg). Water crosses cell membranes to achieve osmotic equilibrium ([ECF](#) osmolality = [ICF](#) osmolality). The extracellular and intracellular solutes or osmoles are markedly different due to disparities in permeability and the presence of transporters and active pumps. The major ECF particles are Na^+ and its accompanying anions Cl^- and HCO_3^- , whereas K^+ and organic phosphate esters (ATP, creatine phosphate, and phospholipids) are the predominant ICF osmoles. Solute that are restricted to the ECF or the ICF determine the *effective osmolality* (or *tonicity*) of that compartment. Since Na^+ is largely restricted to the extracellular compartment, total body Na^+ content is a reflection of ECF volume. Likewise, K^+ and its attendant anions are predominantly limited to the ICF and are necessary for normal cell function. Therefore, the number of intracellular particles is relatively constant, and a change in ICF osmolality is usually due to a change in ICF water content. However, in certain situations, brain cells can vary the number of intracellular solutes in order to defend against large water shifts. This process of *osmotic adaptation* is important in the defense of cell volume and occurs in chronic hyponatremia and hypernatremia. This response is mediated initially by transcellular shifts of K^+ and Na^+ , followed by synthesis, import, or export of organic solutes (so-called osmolytes) such as inositol, betaine, and glutamine. During chronic hyponatremia, brain cells lose solutes, thereby defending cell volume and diminishing neurologic symptoms. The converse occurs during chronic hypernatremia. Certain solutes, such as urea, do not contribute to water shift across cell membranes and are known as *ineffective osmoles*.

Fluid movement between the intravascular and interstitial spaces occurs across the capillary wall and is determined by the Starling forces -- capillary hydraulic pressure and colloid osmotic pressure. The transcapillary hydraulic pressure gradient exceeds the corresponding oncotic pressure gradient, thereby favoring the movement of plasma ultrafiltrate into the extravascular space. The return of fluid into the intravascular compartment occurs via lymphatic flow.

WATER BALANCE (See also [Chap. 268](#))

The normal plasma osmolality is 275 to 290 mosmol/kg and is kept within a narrow range by mechanisms capable of sensing a 1 to 2% change in tonicity. To maintain a

steady state, water intake must equal water excretion. Disorders of water homeostasis result in hypo- or hypernatremia. Normal individuals have an obligate water loss consisting of urine, stool, and evaporation from the skin and respiratory tract. Gastrointestinal excretion is usually a minor component of total water output, except in patients with vomiting, diarrhea, or high enterostomy output states. Evaporative or insensitive water losses are important in the regulation of core body temperature. Obligatory renal water loss is mandated by the minimum solute excretion required to maintain a steady state. Normally, about 600 mosmol must be excreted per day, and since the maximal urine osmolality is 1200 mosmol/kg a minimum urine output of 500 mL/d is required for neutral solute balance.

Water Intake The primary stimulus for water ingestion is *thirst*, mediated either by an increase in effective osmolality or a decrease in [ECF](#) volume or blood pressure. *Osmoreceptors*, located in the anterolateral hypothalamus, are stimulated by a rise in tonicity. Ineffective osmoles, such as urea and glucose, do not play a role in stimulating thirst. The average osmotic threshold for thirst is approximately 295 mosmol/kg and varies among individuals. Under normal circumstances, daily water intake exceeds physiologic requirements.

Water Excretion In contrast to the ingestion of water, its excretion is tightly regulated by physiologic factors. The principal determinant of renal water excretion is *arginine vasopressin* (AVP; formerly antidiuretic hormone), a polypeptide synthesized in the supraoptic and paraventricular nuclei of the hypothalamus and secreted by the posterior pituitary gland. The binding of AVP to V₂ receptors on the basolateral membrane of principal cells in the collecting duct activates adenylyl cyclase and initiates a sequence of events that leads to the insertion of water channels into the luminal membrane. These water channels that are specifically activated by AVP are encoded by the *aquaporin-2* gene ([Chap. 329](#)). The net effect is passive water reabsorption along an osmotic gradient from the lumen of the collecting duct to the hypertonic medullary interstitium. The major stimulus for AVP secretion is hypertonicity. Since the major [ECF](#) solutes are Na⁺ salts, effective osmolality is primarily determined by the plasma Na⁺ concentration. An increase or decrease in tonicity is sensed by hypothalamic osmoreceptors as a decrease or increase in cell volume, respectively, leading to enhancement or suppression of AVP secretion. The osmotic threshold for AVP release is 280 to 290 mosmol/kg, and the system is sufficiently sensitive that plasma osmolality varies by no more than 1 to 2%.

Nonosmotic factors that regulate [AVP](#) secretion include *effective circulating* (arterial) *volume*, nausea, pain, stress, hypoglycemia, pregnancy, and numerous drugs. The hemodynamic response is mediated by baroreceptors in the carotid sinus. The sensitivity of these receptors is significantly lower than that of the osmoreceptors. In fact, depletion of blood volume sufficient to result in a decreased mean arterial pressure is necessary to stimulate AVP release, whereas small changes in effective circulating volume have little effect. In the setting of hypovolemia, the osmotic regulation of AVP remains intact. However, the osmotic threshold, or set point, for AVP release is decreased, and the sensitivity is increased.

To maintain homeostasis and a normal plasma Na⁺ concentration, the ingestion of solute-free water must eventually lead to the loss of the same volume of electrolyte-free

water. Three steps are required for the kidney to excrete a water load: (1) filtration and delivery of water (and electrolytes) to the diluting sites of the nephron; (2) active reabsorption of Na^+ and Cl^- without water in the thick ascending limb of the loop of Henle and, to a lesser extent, in the distal nephron; and (3) maintenance of a dilute urine due to impermeability of the collecting duct to water in the absence of [AVP](#). Abnormalities of any of these steps can result in impaired free water excretion, and eventual hyponatremia.

SODIUM BALANCE

Sodium is actively pumped out of cells by the Na^+, K^+ -ATPase pump. As a result, 85 to 90% of all Na^+ is extracellular, and the [ECF](#) volume is a reflection of total body Na^+ content. Normal volume regulatory mechanisms ensure that Na^+ loss balances Na^+ gain. If this does not occur, conditions of Na^+ excess or deficit ensue and are manifest as edematous or hypovolemic states, respectively. It is important to distinguish between disorders of osmoregulation and disorders of volume regulation since water and Na^+ balance are regulated independently. Changes in Na^+ concentration generally reflect disturbed water homeostasis, whereas alterations in Na^+ content are manifest as ECF volume contraction or expansion and imply abnormal Na^+ balance.

Sodium Intake Individuals eating a typical western diet consume approximately 150 mmol of NaCl daily. This normally exceeds basal requirements. As noted above, sodium is the principal extracellular cation. Therefore, dietary intake of Na^+ results in [ECF](#) volume expansion, which in turn promotes enhanced renal Na^+ excretion to maintain steady state Na^+ balance.

Sodium Excretion (See also [Chap. 268](#)) The regulation of Na^+ excretion is multifactorial and is the major determinant of Na^+ balance. A Na^+ deficit or excess is manifest as a decreased or increased effective circulating volume, respectively. Changes in effective circulating volume tend to lead to parallel changes in glomerular filtration rate (GFR). However, tubule Na^+ reabsorption, and not GFR, is the major regulatory mechanism controlling Na^+ excretion. Almost two-thirds of filtered Na^+ is reabsorbed in the proximal convoluted tubule -- this process is electroneutral and isoosmotic. Further reabsorption (25 to 30%) occurs in the thick ascending limb of the loop of Henle via the apical *$\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ cotransporter* -- this is an active process and is also electroneutral. Distal convoluted tubule reabsorption of Na^+ (5%) is mediated by the *thiazide-sensitive $\text{Na}^+ - \text{Cl}^-$ cotransporter*. Final Na^+ reabsorption occurs in the cortical and medullary collecting ducts, the amount excreted being reasonably equivalent to the amount ingested per day ([Chap. 268](#)).

HYPOVOLEMIA

ETIOLOGY

True volume depletion, or hypovolemia, generally refers to a state of combined salt and water loss exceeding intake, leading to [ECF](#) volume contraction. The loss of Na^+ may be renal or extrarenal ([Table 49-1](#)).

Renal Many conditions are associated with excessive urinary NaCl and water losses,

including diuretics. Pharmacologic diuretics inhibit specific pathways of Na⁺-reabsorption along the nephron with a consequent increase in urinary Na⁺-excretion. Enhanced filtration of non-reabsorbed solutes, such as glucose or urea, can also impair tubular reabsorption of Na⁺ and water, leading to an osmotic or solute diuresis. This often occurs in poorly controlled diabetes mellitus and in patients receiving high-protein hyperalimentation. Mannitol is a diuretic that produces an osmotic diuresis because the renal tubule is impermeable to mannitol. Many tubule and interstitial renal disorders are associated with Na⁺-wasting. Excessive renal losses of Na⁺ and water may also occur during the diuretic phase of acute tubular necrosis ([Chap. 269](#)) and following the relief of bilateral urinary tract obstruction. The natriuresis and water diuresis associated with these two conditions are often short-lived and an appropriate response to a state of [ECF](#) volume expansion that ensued as a result of prior oliguria. However, ongoing losses in the absence of adequate replacement fluids may eventually lead to a state of hypovolemia. Chronic renal insufficiency is associated with a diminished ability to regulate renal salt and water excretion appropriately ([Chap. 270](#)). Therefore, patients with a [GFR](#) of less than 25 mL/min have an obligatory renal Na⁺ loss that may result in progressive ECF volume depletion if Na⁺-intake is restricted. Finally, mineralocorticoid deficiency (hypoaldosteronism) causes salt wasting in the presence of normal intrinsic renal function.

Massive renal water excretion can also lead to hypovolemia. The [ECF](#) volume contraction is usually less severe since two-thirds of the volume lost is intracellular. Conditions associated with excessive urinary water loss include *central diabetes insipidus* (CDI) and *nephrogenic diabetes insipidus* (NDI). These two disorders are due to impaired secretion of and renal unresponsiveness to [AVP](#), respectively, and are discussed below.

Extrarenal Nonrenal causes of hypovolemia include fluid loss from the gastrointestinal tract, skin, and respiratory system and third space accumulations (burns, pancreatitis, peritonitis). Approximately 9 L of fluid enters the gastrointestinal tract daily, 2 L by ingestion and 7 L by secretion. Almost 98% of this volume is reabsorbed so that fecal fluid loss is only 100 to 200 mL/d. Impaired gastrointestinal reabsorption or enhanced secretion leads to volume depletion. Since gastric secretions have a low pH (high H⁺-concentration) and biliary, pancreatic, and intestinal secretions are alkaline (high HCO₃⁻-concentration), vomiting and diarrhea are often accompanied by metabolic alkalosis and acidosis, respectively.

Water evaporation from the skin and respiratory tract contributes to thermoregulation. These *insensible losses* amount to 500 mL/d. During febrile illnesses, prolonged heat exposure, or exercise, increased salt and water loss from skin, in the form of sweat, can be significant and lead to volume depletion. The Na⁺-concentration of sweat is normally 20 to 50 mmol/L and decreases with profuse sweating due to the action of aldosterone. Since sweat is hypotonic, the loss of water exceeds that of Na⁺. The water deficit is minimized by enhanced thirst. Nevertheless, ongoing Na⁺ loss is manifest as hypovolemia. Enhanced evaporative water loss from the respiratory tract may be associated with hyperventilation, especially in mechanically ventilated febrile patients.

Certain conditions lead to fluid sequestration in a *third space*. This compartment is extracellular but is not in equilibrium with either the [ECF](#) or the [ICF](#). The fluid is

effectively lost from the ECF and can result in hypovolemia. Examples include the bowel lumen in gastrointestinal obstruction, subcutaneous tissues in severe burns, retroperitoneal space in acute pancreatitis, and peritoneal cavity in peritonitis. Finally, severe hemorrhage from any source can result in volume depletion.

PATHOPHYSIOLOGY

[ECF](#) volume contraction is manifest as a decreased plasma volume and hypotension. Hypotension is due to decreased venous return (preload) and diminished cardiac output; it triggers baroreceptors in the carotid sinus and aortic arch and leads to activation of the sympathetic nervous system and the renin-angiotensin system. The net effect is to maintain mean arterial pressure and cerebral and coronary perfusion. In contrast to the cardiovascular response, the renal response is aimed at restoring the ECF volume by decreasing the [GFR](#) and filtered load of Na⁺ and, most importantly, by promoting tubular reabsorption of Na⁺. Increased sympathetic tone increases proximal tubular Na⁺ reabsorption and decreases GFR by causing preferential afferent arteriolar vasoconstriction. Sodium is also reabsorbed in the proximal convoluted tubule in response to increased angiotensin II and altered peritubular capillary hemodynamics (decreased hydraulic and increased oncotic pressure). Enhanced reabsorption of Na⁺ by the collecting duct is an important component of the renal adaptation to ECF volume contraction. This occurs in response to increased *aldosterone* and [AVP](#) secretion, and suppressed *atrial natriuretic peptide* secretion.

CLINICAL FEATURES

A careful history is often helpful in determining the etiology of [ECF](#) volume contraction (e.g., vomiting, diarrhea, polyuria, diaphoresis). Most symptoms are nonspecific and secondary to electrolyte imbalances and tissue hypoperfusion and include fatigue, weakness, muscle cramps, thirst, and postural dizziness. More severe degrees of volume contraction can lead to end-organ ischemia manifest as oliguria, cyanosis, abdominal and chest pain, and confusion or obtundation. Diminished skin turgor and dry oral mucous membranes are poor markers of decreased interstitial fluid. Signs of intravascular volume contraction include decreased jugular venous pressure, postural hypotension, and postural tachycardia. Larger and more acute fluid losses lead to hypovolemic shock, manifest as hypotension, tachycardia, peripheral vasoconstriction, and hypoperfusion -- cyanosis, cold and clammy extremities, oliguria, and altered mental status.

DIAGNOSIS

A thorough history and physical examination are generally sufficient to diagnose the etiology of hypovolemia. Laboratory data usually confirm and support the clinical diagnosis. The blood urea nitrogen (BUN) and plasma creatinine concentrations tend to be elevated, reflecting a decreased [GFR](#). Normally, the BUN:creatinine ratio is about 10:1. However, in *prerenal azotemia*, hypovolemia leads to increased urea reabsorption and a proportionately greater elevation in BUN than plasma creatinine, and a BUN:creatinine ratio of 20:1 or higher. An increased BUN (relative to creatinine) may also be due to increased urea production that occurs with hyperalimentation (high-protein), glucocorticoid therapy, and gastrointestinal bleeding.

Volume depletion may be associated with hyponatremia, hypernatremia, or a normal plasma Na⁺ concentration, depending on the tonicity of the fluid lost, the presence of thirst, and the access to water. Hypokalemia is common in settings of increased renal or gastrointestinal K⁺ loss, and hyperkalemia occurs in renal failure, adrenal insufficiency, and certain types of metabolic acidosis. Metabolic alkalosis occurs with diuretic-induced hypovolemia and in cases of vomiting or nasogastric suction. In contrast, metabolic acidosis is associated with renal failure, tubulointerstitial disorders, adrenal insufficiency, diarrhea, diabetic ketoacidosis, and lactic acidosis. Since albumin and erythrocytes are confined to the intravascular compartment, [ECF](#) volume contraction often leads to a relative elevation in hematocrit (hemoconcentration) and plasma albumin concentration.

The appropriate response to hypovolemia is enhanced renal Na⁺ and water reabsorption, which is reflected in the urine composition. Therefore, the urine Na⁺ concentration should usually be less than 20 mmol/L except in conditions associated with impaired Na⁺ reabsorption, as in acute tubular necrosis ([Chap. 269](#)). Another exception is hypovolemia due to vomiting, since the associated metabolic alkalosis and increased filtered HCO₃⁻ impair proximal Na⁺ reabsorption. In this case, the urine Cl⁻ is low (<20 mmol/L). The urine osmolality and specific gravity in hypovolemic subjects are generally greater than 450 mosmol/kg and 1.015, respectively, reflecting the presence of enhanced [AVP](#) secretion. However, in hypovolemia due to diabetes insipidus, urine osmolality and specific gravity are indicative of inappropriately dilute urine.

TREATMENT

The therapeutic goals are to restore normovolemia with fluid similar in composition to that lost and to replace ongoing losses. Symptoms and signs, including weight loss, can help estimate the degree of volume contraction and should also be monitored to assess response to treatment. Mild volume contraction can usually be corrected via the oral route. More severe hypovolemia requires intravenous therapy. Isotonic or normal saline (0.9% NaCl or 154 mmol/L Na⁺) is the solution of choice in normonatremic and mildly hyponatremic individuals and should be administered initially in patients with hypotension or shock. Severe hyponatremia may require hypertonic saline (3.0% NaCl or 513 mmol/L Na⁺). Hypernatremia reflects a proportionally greater deficit of water than Na⁺, and its correction will therefore require a hypotonic solution such as half-normal saline (0.45% NaCl or 77 mmol/L Na⁺) or 5% dextrose in water. Patients with significant hemorrhage, anemia, or intravascular volume depletion may require blood transfusion or colloid-containing solutions (albumin, dextran). Hypokalemia may be present initially or may ensue as a result of increased urinary K⁺ excretion; it should be corrected by adding appropriate amounts of KCl to replacement solutions.

HYPONATREMIA

ETIOLOGY

A plasma Na⁺ concentration less than 135 mmol/L usually reflects a hypotonic state. However, plasma osmolality may be normal or increased in some cases of hyponatremia, referred to as *pseudohyponatremia*. Plasma is 93% water, the remaining 7% consisting of plasma proteins and lipids. Since Na⁺ ions are dissolved in plasma

water, increasing the nonaqueous phase artificially lowers the Na⁺-concentration measured per liter of plasma (except when Na⁺-sensitive glass electrodes are used). The plasma osmolality and the Na⁺-concentration remain normal. This type of hyponatremia has little clinical significance, except to ascertain the cause of the hyperproteinemia or hyperlipidemia. Isotonic or slightly hypotonic hyponatremia may complicate transurethral resection of the prostate or bladder because large volumes of isoosmotic (mannitol) or hypoosmotic (sorbital or glycine) bladder irrigation solution can be absorbed and result in a dilutional hyponatremia. The metabolism of sorbitol and glycine to CO₂ and water may lead to hypotonicity if the accumulated fluid and solutes are not rapidly excreted. Hypertonic hyponatremia is usually due to hyperglycemia or, occasionally, intravenous administration of mannitol. Relative insulin deficiency causes myocytes to become impermeable to glucose. Therefore, during poorly controlled diabetes mellitus, glucose is an effective osmole and draws water from muscle cells, resulting in hyponatremia. Plasma Na⁺-concentration falls by 1.4 mmol/L for every 100 mg/dL rise in the plasma glucose concentration.

Most causes of hyponatremia are associated with a low plasma osmolality ([Table 49-2](#)). In general, hypotonic hyponatremia is due either to a primary water gain (and secondary Na⁺-loss) or a primary Na⁺ loss (and secondary water gain). In the absence of water intake or hypotonic fluid replacement, hyponatremia is usually associated with hypovolemic shock due to a profound sodium deficit and transcellular water shift. Contraction of the [ECF](#) volume stimulates thirst and [AVP](#) secretion. The increased water ingestion and impaired renal excretion result in hyponatremia. It is important to note that *diuretic-induced hyponatremia* is almost always due to thiazide diuretics. Loop diuretics decrease the tonicity of the medullary interstitium and impair maximal urinary concentrating capacity. This limits the ability of AVP to promote water retention. In contrast, thiazide diuretics lead to Na⁺ and K⁺-depletion, and AVP-mediated water retention. In the presence of a large K⁺-deficit, transcellular ion exchange (K⁺-exits and Na⁺-enters cells) may contribute to hyponatremia. Hyponatremia can also occur by a process of *desalination*. This occurs when the urine tonicity (the sum of the concentrations of Na⁺ and K⁺) exceeds that of administered intravenous fluids (including isotonic saline). This accounts for some cases of acute postoperative hyponatremia and cerebral salt wasting after neurosurgery.

Hyponatremia in the setting of [ECF](#) volume expansion is usually associated with edematous states, such as congestive heart failure, hepatic cirrhosis, and the nephrotic syndrome. These disorders all have in common a decreased effective circulating arterial volume, leading to increased thirst and increased [AVP](#) levels. Additional factors impairing the excretion of solute-free water include a reduced [GFR](#), decreased delivery of ultrafiltrate to the diluting site (due to increased proximal fractional reabsorption of Na⁺ and water), and diuretic therapy. The degree of hyponatremia often correlates with the severity of the underlying condition and is an important prognostic factor. Oliguric acute and chronic renal failure may be associated with hyponatremia if water intake exceeds the ability to excrete equivalent volumes.

Hyponatremia in the absence of [ECF](#) volume contraction, decreased effective circulating arterial volume, or renal insufficiency is usually due to increased [AVP](#) secretion resulting in impaired water excretion. Ingestion or administration of water is also required since high levels of AVP alone are usually insufficient to produce hyponatremia. This disorder,

commonly termed the *syndrome of inappropriate antidiuretic hormone secretion* (SIADH), is the most common cause of normovolemic hyponatremia and is due to the nonphysiologic release of AVP from the posterior pituitary or an ectopic source ([Chap. 329](#)). Renal free water excretion is impaired while the regulation of Na⁺ balance is unaffected. The most common causes of SIADH include neuropsychiatric and pulmonary diseases, malignant tumors, major surgery (postoperative pain), and pharmacologic agents. Severe pain and nausea are physiologic stimuli of AVP secretion; these stimuli are inappropriate in the absence of hypovolemia or hyperosmolality. A variety of central nervous system disorders may be associated with SIADH, such as meningitis, encephalitis, hemorrhage, stroke, psychosis, primary and metastatic tumors, and acute porphyria. Pneumonia, empyema, tuberculosis, and acute respiratory failure can be complicated by hyponatremia secondary to SIADH. Hypoxemia, hypercarbia, and positive-pressure ventilation are all nonosmotic stimuli for AVP release. Various tumors, notably oat cell carcinoma of the lung, have been demonstrated to secrete AVP ectopically. Many drugs either stimulate AVP release or potentiate its actions on the kidney. The pattern of AVP secretion can be used to classify SIADH into four subtypes: (1) erratic autonomous AVP secretion (ectopic production); (2) normal regulation of AVP release around a lower osmolality set point or *reset osmostat* (cachexia, malnutrition); (3) normal AVP response to hypertonicity with failure to suppress completely at low osmolality (incomplete pituitary stalk section); and (4) normal AVP secretion with increased sensitivity to its actions or secretion of some other antidiuretic factor (rare).

Hormonal excess or deficiency may cause hyponatremia. Adrenal insufficiency ([Chap. 331](#)) and hypothyroidism ([Chap. 330](#)) may present with hyponatremia and should not be confused with [SIADH](#). Although decreased mineralocorticoids may contribute to the hyponatremia of adrenal insufficiency, it is the cortisol deficiency that leads to hypersecretion of [AVP](#) both indirectly (secondary to volume depletion) and directly (cosecreted with corticotropin-releasing factor). The mechanisms by which hypothyroidism leads to hyponatremia include decreased cardiac output and [GFR](#) and increased AVP secretion in response to hemodynamic stimuli.

Finally, hyponatremia may occur in the absence of [AVP](#) or renal failure if the kidney is unable to excrete the dietary water load. In psychogenic or primary polydipsia, compulsive water consumption may overwhelm the normally large renal excretory capacity of 12 L/d ([Chap. 329](#)). These patients often have psychiatric illnesses and may be taking medications, such as phenothiazines, that enhance the sensation of thirst by causing a dry mouth. The maximal urine output is a function of the minimum urine osmolality achievable and the mandatory solute excretion. Metabolism of a normal diet generates about 600 mosmol/d, and the minimum urine osmolality in humans is 50 mosmol/kg. Therefore, the maximum daily urine output will be about 12 L ($600 \div 50 = 12$). A solute excretion rate of greater than ~750 mosmol/d is, by definition, an *osmotic diuresis*. A low-protein diet may yield as few as 250 mosmol/d, which translates into a maximal urine output of 5 L/d at a minimum urine tonicity of 50 mosmol/kg. Beer drinkers typically have a poor dietary intake of protein and electrolytes and consume large volumes (of beer), which may exceed the renal excretory capacity and result in hyponatremia. This phenomenon is referred to as *beer potomania*.

CLINICAL FEATURES

The clinical manifestations of hyponatremia are related to osmotic water shift leading to increased [ICF](#) volume, specifically brain cell swelling or cerebral edema. Therefore, the symptoms are primarily neurologic, and their severity is dependent on the rapidity of onset and absolute decrease in plasma Na^+ concentration. Patients may be asymptomatic or complain of nausea and malaise. As the plasma Na^+ concentration falls, the symptoms progress to include headache, lethargy, confusion, and obtundation. Stupor, seizures, and coma do not usually occur unless the plasma Na^+ concentration falls acutely below 120 mmol/L or decreases rapidly. As described above, adaptive mechanisms designed to protect cell volume occur in chronic hyponatremia. Loss of Na^+ and K^+ , followed by organic osmolytes, from brain cells decreases brain swelling due to secondary transcellular water shifts (from ICF to [ECF](#)). The net effect is to minimize cerebral edema and its symptoms. Hospitalized patients with hyponatremia have an increased mortality rate compared to normonatremic control subjects. However, the excess mortality is usually attributed to the underlying disorder rather than the electrolyte disturbance.

DIAGNOSIS

Hyponatremia is not a disease but a manifestation of a variety of disorders. The underlying cause can often be ascertained from an accurate history and physical examination, including an assessment of [ECF](#) volume status and effective circulating arterial volume. The differential diagnosis of hyponatremia, an expanded ECF volume, and decreased effective circulating volume includes congestive heart failure, hepatic cirrhosis, and the nephrotic syndrome. Hypothyroidism and adrenal insufficiency tend to present with a near-normal ECF volume and decreased effective circulating arterial volume. All of these diseases have characteristic signs and symptoms. Patients with [SIADH](#) are usually euvoletic.

Four laboratory findings often provide useful information and can narrow the differential diagnosis of hyponatremia: (1) the plasma osmolality, (2) the urine osmolality, (3) the urine Na^+ concentration, and (4) the urine K^+ concentration. Since [ECF](#) tonicity is determined primarily by the Na^+ concentration, most patients with hyponatremia have a decreased plasma osmolality. If the plasma osmolality is not low, pseudohyponatremia must be ruled out. The appropriate renal response to hypoosmolality is to excrete the maximum volume of dilute urine, i.e., urine osmolality and specific gravity of less than 100 mosmol/kg and 1.003, respectively. This occurs in patients with primary polydipsia. If this is not present, it suggests impaired free water excretion due to the action of [AVP](#) on the kidney. The secretion of AVP may be a physiologic response to hemodynamic stimuli or it may be inappropriate in the presence of hyponatremia and euvoletic. Since Na^+ is the major ECF cation and is largely restricted to this compartment, ECF volume contraction represents a deficit in total body Na^+ content. Therefore, volume depletion in patients with normal underlying renal function results in enhanced tubule Na^+ reabsorption and a urine Na^+ concentration less than 20 mmol/L. The finding of a urine Na^+ concentration greater than 20 mmol/L in hypovolemic hyponatremia implies a salt-wasting nephropathy, diuretic therapy, hypoaldosteronism, or occasionally vomiting. Both the urine osmolality and the urine Na^+ concentration can be followed serially when assessing response to therapy.

[SIADH](#) is characterized by hypoosmotic hyponatremia in the setting of an inappropriately concentrated urine (urine osmolality greater than 100 mosmol/kg). Patients are typically normovolemic and have normal Na⁺ balance. They tend to be mildly volume expanded secondary to water retention and have a urine Na⁺ excretion rate equal to intake (urine Na⁺ concentration usually greater than 40 mmol/L). By definition, they have normal renal, adrenal, and thyroid function and usually have normal K⁺ and acid-base balance. SIADH is often associated with hypouricemia due to the uricosuric state induced by volume expansion. In contrast, hypovolemic patients tend to be hyperuricemic secondary to increased proximal urate reabsorption.

CLINICAL APPROACH

See [Fig. 49-1](#).

TREATMENT

The goals of therapy are twofold: (1) to raise the plasma Na⁺ concentration by restricting water intake and promoting water loss; and (2) to correct the underlying disorder. Mild asymptomatic hyponatremia is generally of little clinical significance and requires no treatment. The management of asymptomatic hyponatremia associated with [ECF](#) volume contraction should include Na⁺ repletion, generally in the form of isotonic saline. The direct effect of the administered NaCl on the plasma Na⁺ concentration is trivial. However, restoration of euvolemia removes the hemodynamic stimulus for [AVP](#) release, allowing the excess free water to be excreted. The hyponatremia associated with edematous states tends to reflect the severity of the underlying disease and is usually asymptomatic. These patients have increased total body water that exceeds the increase in total body Na⁺ content. Treatment should include restriction of Na⁺ and water intake, correction of hypokalemia, and promotion of water loss in excess of Na⁺. The latter may require the use of loop diuretics with replacement of a proportion of the urinary Na⁺ loss to ensure net free water excretion. Dietary water restriction should be less than the urine output. Correction of the K⁺ deficit may raise the plasma Na⁺ concentration by favoring a shift of Na⁺ out of cells as K⁺ moves in. Water restriction is also a component of the therapeutic approach to hyponatremia associated with primary polydipsia, renal failure, and [SIADH](#) ([Chap. 329](#)).

The rate of correction of hyponatremia depends on the absence or presence of neurologic dysfunction. This, in turn, is related to the rapidity of onset and magnitude of the fall in plasma Na⁺ concentration. In asymptomatic patients, the plasma Na⁺ concentration should be raised by no more than 0.5 to 1.0 mmol/L per hour and by less than 10 to 12 mmol/L over the first 24 h. Acute or severe hyponatremia (plasma Na⁺ concentration <110 to 115 mmol/L) tends to present with altered mental status and/or seizures and requires more rapid correction. Severe symptomatic hyponatremia should be treated with hypertonic saline, and the plasma Na⁺ concentration should be raised by 1 to 2 mmol/L per hour for the first 3 to 4 h or until the seizures subside. Once again, the plasma Na⁺ concentration should probably be raised by no more than 12 mmol/L during the first 24 h. The quantity of Na⁺ required to increase the plasma Na⁺ concentration by a given amount can be estimated by multiplying the deficit in plasma Na⁺ concentration by the total body water. Under normal conditions, total body water is 50 or 60% of lean body weight in women or men, respectively. Therefore, to

raise the plasma Na⁺ concentration from 105 to 115 mmol/L in a 70-kg man requires 420 mmol $[(115 - 105) \times 70 \times 0.6]$ of Na⁺. The risk of correcting hyponatremia too rapidly is the development of the *osmotic demyelination syndrome* (ODS). This is a neurologic disorder characterized by flaccid paralysis, dysarthria, and dysphagia. The diagnosis is usually suspected clinically and can be confirmed by appropriate neuroimaging studies. There is no specific treatment for the disorder, which is associated with significant morbidity and mortality. Patients with chronic hyponatremia are most susceptible to the development of ODS, since their brain cell volume has returned to near normal as a result of the osmotic adaptive mechanisms described above. Therefore, administration of hypertonic saline to these individuals can cause sudden osmotic shrinkage of brain cells. In addition to rapid or overcorrection of hyponatremia, risk factors for ODS include prior cerebral anoxic injury, hypokalemia, and malnutrition, especially secondary to alcoholism. Water restriction in primary polydipsia and intravenous saline therapy in [ECF](#) volume-contracted patients may also lead to overly rapid correction of hyponatremia as a result of AVP suppression and a brisk water diuresis. This can be prevented by administration of water or use of an AVP analogue to slow down the rate of free water excretion. **For further discussion, see [Chap. 329](#).*

HYPERNATREMIA

ETIOLOGY

Hypernatremia is defined as a plasma Na⁺ concentration greater than 145 mmol/L. Since Na⁺ and its accompanying anions are the major effective [ECF](#) osmoles, hypernatremia is a state of hyperosmolality. As a result of the fixed number of [ICF](#) particles, maintenance of osmotic equilibrium in hypernatremia results in ICF volume contraction. Hypernatremia may be due to primary Na⁺ gain or water deficit. The two components of an appropriate response to hypernatremia are increased water intake stimulated by thirst and the excretion of the minimum volume of maximally concentrated urine reflecting [AVP](#) secretion in response to an osmotic stimulus.

In practice, the majority of cases of hypernatremia result from the loss of water. Since water is distributed between the [ICF](#) and the [ECF](#) in a 2:1 ratio, a given amount of solute-free water loss will result in a twofold greater reduction in the ICF compartment than the ECF compartment. For example, consider three scenarios: the loss of 1 L of water, isotonic NaCl, or half-isotonic NaCl. If 1 L of water is lost, the ICF volume will decrease by 667 mL, whereas the ECF volume will fall by only 333 mL. Due to the fact that Na⁺ is largely restricted to the ECF, this compartment will decrease by 1 L if the fluid lost is isoosmotic. One liter of half-isotonic NaCl is equivalent to 500 mL of water (one-third ECF, two-thirds ICF) plus 500 mL of isotonic saline (all ECF). Therefore, the loss of 1 L of half-isotonic saline decreases the ECF and ICF volumes by 667 mL and 333 mL, respectively.

The degree of hyperosmolality is typically mild unless the thirst mechanism is abnormal or access to water is limited. The latter occurs in infants, the physically handicapped, patients with impaired mental status, in the postoperative state, and in intubated patients in the intensive care unit. On rare occasions, impaired thirst may be due to *primary hypodipsia*. This usually occurs as a result of damage to the hypothalamic osmoreceptors that control thirst and tends to be associated with abnormal osmotic

regulation of [AVP](#) secretion. Primary hypodipsia may be due to a variety of pathologic changes including granulomatous disease, vascular occlusion, and tumors. A subset of hypodipsic hypernatremia, referred to as *essential hypernatremia*, does not respond to forced water intake. This appears to be due to a specific osmoreceptor defect resulting in nonosmotic regulation of AVP release. Thus, the hemodynamic effects of water loading lead to AVP suppression and excretion of dilute urine.

The source of free water loss is either renal or extrarenal. Nonrenal loss of water may be due to evaporation from the skin and respiratory tract (insensible losses) or loss from the gastrointestinal tract. Insensible losses are increased with fever, exercise, heat exposure, and severe burns and in mechanically ventilated patients. Furthermore, the Na^+ -concentration of sweat decreases with profuse perspiration, thereby increasing solute-free water loss. Diarrhea is the most common gastrointestinal cause of hypernatremia. Specifically, osmotic diarrheas (induced by lactulose, sorbitol, or malabsorption of carbohydrate) and viral gastroenteritides result in water loss exceeding that of Na^+ and K^+ . In contrast, secretory diarrheas (e.g., cholera, carcinoid, VIPoma) have a fecal osmolality (twice the sum of the concentrations of Na^+ and K^+) similar to that of plasma and present with [ECF](#) volume contraction and a normal plasma Na^+ -concentration or hyponatremia.

Renal water loss is the most common cause of hypernatremia and is due to drug-induced or osmotic diuresis or diabetes insipidus ([Chap. 329](#)). Loop diuretics interfere with the countercurrent mechanism and produce an isoosmotic solute diuresis. This results in a decreased medullary interstitial tonicity and impaired renal concentrating ability. The presence of non-reabsorbed organic solutes in the tubule lumen impairs the osmotic reabsorption of water. This leads to water loss in excess of Na^+ and K^+ , known as an osmotic diuresis. The most frequent cause of an osmotic diuresis is hyperglycemia and glucosuria in poorly controlled diabetes mellitus. Intravenous administration of mannitol and increased endogenous production of urea (high-protein diet) can also result in an osmotic diuresis. Hypernatremia secondary to nonosmotic urinary water loss is usually due to: (1) [CDI](#) or neurogenic diabetes insipidus characterized by impaired [AVP](#) secretion, or (2) [NDI](#) resulting from end-organ (renal) resistance to the actions of AVP. The most common cause of CDI is destruction of the neurohypophysis. This may occur as a result of trauma, neurosurgery, granulomatous disease, neoplasms, vascular accidents, or infection. In many cases, CDI is idiopathic and may occasionally be hereditary. The familial form of the disease is inherited in an autosomal dominant fashion and has been attributed to mutations in the propressophysin (AVP precursor) gene. NDI may be either inherited or acquired. Congenital NDI is an X-linked recessive trait due to mutations in the V_2 receptor gene. Mutations in the autosomal aquaporin-2 gene may also result in NDI. The aquaporin-2 gene encodes the water channel protein whose membrane insertion is stimulated by AVP. The causes of sporadic NDI are numerous and include drugs (especially lithium), hypercalcemia, hypokalemia, and conditions that impair medullary hypertonicity (e.g., papillary necrosis or osmotic diuresis). Pregnant women, in the second or third trimester, may develop NDI as a result of excessive elaboration of vasopressinase by the placenta.

Finally, although infrequent, a primary Na^+ gain may cause hypernatremia. For example, inadvertent administration of hypertonic NaCl or NaHCO_3 or replacing sugar with salt in

infant formula can produce this complication.

CLINICAL FEATURES

As a consequence of hypertonicity, water shifts out of cells, leading to a contracted [ICF](#) volume. A decreased brain cell volume is associated with an increased risk of subarachnoid or intracerebral hemorrhage. Hence, the major symptoms of hypernatremia are neurologic and include altered mental status, weakness, neuromuscular irritability, focal neurologic deficits, and occasionally coma or seizures. Patients may also complain of polyuria or thirst. For unknown reasons, patients with polydipsia from [CDI](#) tend to prefer ice-cold water. The signs and symptoms of volume depletion are often present in patients with a history of excessive sweating, diarrhea, or an osmotic diuresis. The mortality rate associated with a plasma Na^+ concentration greater than 180 mmol/L is very high. As with hyponatremia, the severity of the clinical manifestations is related to the acuity and magnitude of the rise in plasma Na^+ concentration. Chronic hypernatremia is generally less symptomatic as a result of adaptive mechanisms designed to defend cell volume. Brain cells initially take up Na^+ and K^+ salts, later followed by accumulation of organic osmolytes such as inositol. This serves to restore the brain ICF volume towards normal.

DIAGNOSIS

A complete history and physical examination will often provide clues as to the underlying cause of hypernatremia. Relevant symptoms and signs include the absence or presence of thirst, diaphoresis, diarrhea, polyuria, and the features of [ECF](#) volume contraction. The history should include a list of current and recent medications, and the physical examination is incomplete without a thorough mental status and neurologic assessment. Measurement of urine volume and osmolality are essential in the evaluation of hyperosmolality. The appropriate renal response to hypernatremia is the excretion of the minimum volume (500 mL/d) of maximally concentrated urine (urine osmolality >800 mosmol/kg). These findings suggest extrarenal or remote renal water loss or administration of hypertonic Na^+ salt solutions. The presence of a primary Na^+ excess can be confirmed by the presence of ECF volume expansion and natriuresis (urine Na^+ concentration usually >100 mmol/L). Many causes of hypernatremia are associated with polyuria and a submaximal urine osmolality. The product of the urine volume and osmolality, i.e., the solute excretion rate, is helpful in determining the basis of the polyuria (see above). To maintain a steady state, total solute excretion must equal solute production. As stated above, individuals eating a normal diet generate ~ 600 mosmol/d. Therefore, daily solute excretion in excess of 750 mosmol defines an osmotic diuresis. This can be confirmed by measuring the urine glucose and urea. In general, both [CDI](#) and [NDI](#) present with polyuria and hypotonic urine (urine osmolality <250 mosmol/kg). The degree of hypernatremia is usually mild unless there is an associated thirst abnormality. The clinical history, physical examination, and pertinent laboratory data can often rule out causes of acquired NDI. CDI and NDI can generally be distinguished by administering the [AVP](#) analogue desmopressin (10 μg intranasally) after careful water restriction. The urine osmolality should increase by at least 50% in CDI and will not change in NDI. Unfortunately, the diagnosis may sometimes be difficult due to partial defects in AVP secretion and action.

CLINICAL APPROACH

See [Fig. 49-2](#).

TREATMENT

The therapeutic goals are to stop ongoing water loss by treating the underlying cause and to correct the water deficit. The [ECF](#) volume should be restored in hypovolemic patients. The quantity of water required to correct the deficit can be calculated from the following equation:

In hypernatremia due to water loss, total body water is approximately 50 and 40% of lean body weight in men and women, respectively. For example, a 50-kg woman with a plasma Na^+ concentration of 160 mmol/L has an estimated free water deficit of 2.9 L $\text{icub}; [(160 - 140) \cdot 140] \cdot (0.4 \cdot 50) \text{rcub}$. As in hyponatremia, rapid correction of hypernatremia is potentially dangerous. In this case, a sudden decrease in osmolality could potentially cause a rapid shift of water into cells that have undergone osmotic adaptation. This would result in swollen brain cells and increase the risk of seizures or permanent neurologic damage. Therefore, the water deficit should be corrected slowly over at least 48 to 72 h. When calculating the rate of water replacement, ongoing losses should be taken into account, and the plasma Na^+ concentration should be lowered by 0.5 mmol/L per hour and by no more than 12 mmol/L over the first 24 h. The safest route of administration of water is by mouth or via a nasogastric tube (or other feeding tube). Alternatively, 5% dextrose in water or half-isotonic saline can be given intravenously. The appropriate treatment of [CDI](#) consists of administering desmopressin intranasally ([Chap. 329](#)). Other options for decreasing urine output include a low-salt diet in combination with low-dose thiazide diuretic therapy. In some patients with partial CDI, drugs that either stimulate [AVP](#) secretion or enhance its action on the kidney have been useful. These include chlorpropamide, clofibrate, carbamazepine, and nonsteroidal anti-inflammatory drugs (NSAIDs). The concentrating defect in [NDI](#) may be reversible by treating the underlying disorder or eliminating the offending drug. Symptomatic polyuria due to NDI can be treated with a low- Na^+ diet and thiazide diuretics as described above. This induces mild volume depletion, which leads to enhanced proximal reabsorption of salt and water and decreased delivery to the site of action of AVP, the collecting duct. By impairing renal prostaglandin synthesis, NSAIDs potentiate AVP action and thereby increase urine osmolality and decrease urine volume. Amiloride may be useful in patients with NDI who need to be on lithium. The nephrotoxicity of lithium requires the drug to be taken up into collecting duct cells via the amiloride-sensitive Na^+ channel.

POTASSIUM

POTASSIUM BALANCE

Potassium is the major intracellular cation. The normal plasma K^+ concentration is 3.5 to 5.0 mmol/L, whereas that inside cells is about 150 mmol/L. Therefore, the amount of K^+ in the [ECF](#) (30 to 70 mmol) constitutes less than 2% of the total body K^+ content (2500 to 4500 mmol). The ratio of [ICF](#) to ECF K^+ concentration (normally 38:1) is the principal

result of the resting membrane potential and is crucial for normal neuromuscular function. The basolateral Na⁺, K⁺-ATPase pump actively transports K⁺ in and Na⁺ out of the cell in a 2:3 ratio, and the passive outward diffusion of K⁺ is quantitatively the most important factor that generates the resting membrane potential. The activity of the electrogenic Na⁺, K⁺-ATPase pump may be stimulated as a result of an increased intracellular Na⁺ concentration and inhibited in the setting of digoxin toxicity or chronic illness such as heart failure or renal failure.

The distribution of K⁺ is also affected by several other factors, including hormones, acid-base balance, osmolality, and cell turnover. Insulin increases Na⁺, K⁺-ATPase activity indirectly and independent of its effect on glucose transport, leading to K⁺ shift into muscle and liver cells. Conversely, insulin deficiency results in K⁺ movement from the [ICF](#) to the [ECF](#) compartment. Catecholamines have variable effects on K⁺ distribution -- β_2 -adrenergic agonists promote whereas α -adrenergic agonists impair K⁺ uptake by cells. The Na⁺, K⁺-ATPase pump as well as insulin secretion are stimulated by β_2 -adrenergic agonists. In contrast, α -adrenergic agonists have the opposite effect. The major action of aldosterone is to increase K⁺ excretion (see below). The role of extracellular pH in K⁺ balance relates to the underlying acid-base disorder. In metabolic acidosis, 60% of the H⁺ load is buffered inside cells. To maintain electroneutrality, the H⁺ ion must either be accompanied by an anion or exchanged for intracellular K⁺ (leading to hyperkalemia). Organic acidoses are not usually associated with a pH-related K⁺ shift, since anions such as lactate and β -hydroxybutyrate can be readily taken up by the cell. The converse, movement of K⁺ into cells, may be seen with metabolic alkalosis. However, this is less important due to diminished intracellular buffering. Primary respiratory disturbances in acid-base balance result in minimal transcellular K⁺ shifts. In hyperosmolal states, K⁺ diffuses out of cells along with water due to *solvent drag*. The concentration gradient favoring K⁺ movement out of cells is also increased as a result of ICF water loss. Tissue destruction or breakdown results in the release of intracellular K⁺, whereas the production of new cells shifts K⁺ out of the ECF. Finally, moderate to severe exercise may be associated with K⁺ release from muscle, leading to glycogenolysis and local vasodilatation. This is usually transient but may affect the plasma K⁺ concentration if patients repeatedly clench and unclench their fist prior to venipuncture.

The K⁺ intake of individuals on an average western diet is 40 to 120 mmol/d or approximately 1 mmol/kg per day, 90% of which is absorbed by the gastrointestinal tract. Maintenance of the steady state necessitates matching K⁺ ingestion with excretion. Initially, extrarenal adaptive mechanisms, followed later by urinary excretion, prevent a doubling of the plasma K⁺ concentration that would occur if the dietary K⁺ load remained in the [ECF](#) compartment. Immediately following a meal, most of the absorbed K⁺ enters cells as a result of the initial elevation in the plasma K⁺ concentration and facilitated by insulin release and basal catecholamine levels. Eventually, however, the excess K⁺ is excreted in the urine (see below). The regulation of gastrointestinal K⁺ handling is not well understood. The amount of K⁺ lost in the stool can increase from 10 to 50 or 60% (of dietary intake) in chronic renal insufficiency. In addition, colonic secretion of K⁺ is stimulated in patients with large volumes of diarrhea, resulting in potentially severe K⁺ depletion.

POTASSIUM EXCRETION (See also [Chap. 268](#))

Renal excretion is the major route of elimination of dietary and other sources of excess K⁺. The filtered load of K⁺ ($\text{GFR} \times \text{plasma K}^+ \text{concentration} = 180 \text{ L/d} \times 4 \text{ mmol/L} = 720 \text{ mmol/d}$) is 10- to 20-fold greater than the [ECF](#) K⁺ content. Some 90% of filtered K⁺ is reabsorbed by the proximal convoluted tubule and loop of Henle. Proximally, K⁺ is reabsorbed passively with Na⁺ and water, whereas the luminal Na⁺-K⁺-2Cl⁻ cotransporter mediates K⁺ uptake in the thick ascending limb of the loop of Henle. Therefore, K⁺ delivery to the distal nephron [distal convoluted tubule and cortical collecting duct (CCD)] approximates dietary intake. Net distal K⁺ secretion or reabsorption occurs in the setting of K⁺ excess or depletion, respectively. The cell responsible for K⁺ secretion in the late distal convoluted tubule (or connecting tubule) and CCD is the principal cell. Virtually all regulation of renal K⁺ excretion and total body K⁺ balance occurs in the distal nephron. The driving force for K⁺ secretion is a favorable electrochemical gradient across the luminal membrane of the principal cell. As a result of the action of the basolateral Na⁺, K⁺-ATPase pump, the intracellular K⁺ concentration far exceeds that of the fluid in the lumen of the CCD. The electrical gradient is created by electrogenic Na⁺ reabsorption leading to a lumen-negative transepithelial potential difference (TEPD), favoring K⁺ secretion. The generation of a lumen-negative TEPD depends on the relative rates of reabsorption of Na⁺ and its accompanying anion (primarily Cl⁻). Equimolar reabsorption of Na⁺ and Cl⁻ at equivalent rates is electroneutral, whereas reabsorption of Na⁺ in excess of Cl⁻ is electrogenic. The cellular uptake of Na⁺ by the principal cell occurs via an apical Na⁺ channel and is driven by a low intracellular Na⁺ concentration relative to that in the lumen of the CCD. The mechanism and regulation of distal nephron Cl⁻ transport is less clear. Obviously, factors that impact on either Na⁺ or Cl⁻ reabsorption by the principal cell will influence the TEPD. Potassium secretion is regulated by two physiologic stimuli -- aldosterone and hyperkalemia. Aldosterone is secreted by the zona glomerulosa cells of the adrenal cortex in response to high renin and angiotensin II or hyperkalemia. The actions of aldosterone on the principal cell include enhanced apical membrane Na⁺ conductivity, stimulation of the basolateral Na⁺, K⁺-ATPase, and increased luminal K⁺ channels. The plasma K⁺ concentration, independent of aldosterone, can directly affect K⁺ secretion. In addition to the K⁺ concentration in the lumen of the CCD, renal K⁺ loss depends on the urine flow rate, a function of daily solute excretion (see above). Since excretion is equal to the product of concentration and volume, increased distal flow rate can significantly enhance urinary K⁺ output. Finally, in severe K⁺ depletion, secretion of K⁺ is reduced and reabsorption, via apical H⁺, K⁺-ATPase pumps in cortical and medullary collecting ducts, is upregulated.

HYPOKALEMIA

ETIOLOGY (See [Table 49-3](#))

Hypokalemia, defined as a plasma K⁺ concentration < 3.5 mmol/L, may result from one (or more) of the following: decreased net intake, shift into cells, or increased net loss. Diminished intake is seldom the sole cause of K⁺ depletion since urinary excretion can be effectively decreased to less than 15 mmol/d as a result of net K⁺ reabsorption in the distal nephron. With the exception of the urban poor and certain cultural groups, the amount of K⁺ in the diet almost always exceeds that excreted in the urine. However, dietary K⁺ restriction may exacerbate the hypokalemia secondary to increased gastrointestinal or renal loss. An unusual cause of decreased K⁺ intake is ingestion of

clay (geophagia), which binds dietary K^+ and iron. This custom was previously common among African Americans in the American South.

Redistribution into Cells Movement of K^+ into cells may transiently decrease the plasma K^+ concentration without altering total body K^+ content. For any given cause, the magnitude of the change is relatively small, often less than 1 mmol/L. However, a combination of factors may lead to a significant fall in the plasma K^+ concentration and may amplify the hypokalemia due to K^+ wasting. Alkalosis, especially that due to a primary increase in plasma HCO_3^- (metabolic alkalosis), is often associated with hypokalemia. This occurs as a result of K^+ redistribution as well as excessive renal K^+ loss. Treatment of diabetic ketoacidosis with insulin may lead to hypokalemia due to stimulation of the Na^+ - H^+ antiporter and (secondarily) the Na^+ , K^+ -ATPase pump. Furthermore, uncontrolled hyperglycemia often leads to K^+ depletion from an osmotic diuresis (see below). Stress-induced catecholamine release and administration of β_2 -adrenergic agonists directly induce cellular uptake of K^+ and promote insulin secretion by pancreatic islet β cells. *Hypokalemic periodic paralysis* is a rare condition characterized by recurrent episodic weakness or paralysis ([Chap. 381](#)). Since K^+ is the major [ICF](#) cation, anabolic states can potentially result in hypokalemia due to a K^+ shift into cells. This may occur following rapid cell growth seen in patients with pernicious anemia treated with vitamin B_{12} or with neutropenia after treatment with granulocyte-macrophage colony stimulating factor. Massive transfusion with thawed washed red blood cells (RBCs) could cause hypokalemia since frozen RBCs lose up to half of their K^+ during storage.

Nonrenal Loss of Potassium Excessive sweating may result in K^+ depletion from increased integumentary and renal K^+ loss. Hyperaldosteronism, secondary to [ECF](#) volume contraction, enhances K^+ excretion in the urine ([Chap. 331](#)). Normally, K^+ lost in the stool amounts to 5 to 10 mmol/d in a volume of 100 to 200 mL. Hypokalemia subsequent to increased gastrointestinal loss can occur in patients with profuse diarrhea (usually secretory), villous adenomas, VIPomas, or laxative abuse. However, the loss of gastric secretions does not account for the moderate to severe K^+ depletion often associated with vomiting or nasogastric suction. Since the K^+ concentration of gastric fluid is 5 to 10 mmol/L, it would take 30 to 80 L of vomitus to achieve a K^+ deficit of 300 to 400 mmol typically seen in these patients. In fact, the hypokalemia is primarily due to increased renal K^+ excretion. Loss of gastric contents results in volume depletion and metabolic alkalosis, both of which promote kaliuresis. Hypovolemia stimulates aldosterone release, which augments K^+ secretion by the principal cells. In addition, the filtered load of HCO_3^- exceeds the reabsorptive capacity of the proximal convoluted tubule, thereby increasing distal delivery of $NaHCO_3$, which enhances the electrochemical gradient favoring K^+ loss in the urine.

Renal Loss of Potassium In general, most cases of chronic hypokalemia are due to renal K^+ wasting. This may be due to factors that increase the K^+ concentration in the lumen of the [CCD](#) or augment distal flow rate. As described above, distal nephron K^+ secretion is driven by a lumen-negative [TEPD](#), affected by aldosterone and the relative rates of reabsorption of Na^+ and its accompanying anion(s). Mineralocorticoid excess commonly results in hypokalemia ([Chap. 331](#)). *Primary hyperaldosteronism* is due to dysregulated aldosterone secretion by an adrenal adenoma (Conn's syndrome) or carcinoma or to adrenocortical hyperplasia. In a rare subset of patients, the disorder

is familial (autosomal dominant) and aldosterone levels can be suppressed by administering low doses of exogenous glucocorticoid. The molecular defect responsible for *glucocorticoid-remediable hyperaldosteronism* is a rearranged gene (due to a chromosomal crossover), containing the 5'-regulatory region of the 11 β -hydroxylase gene and the coding sequence of the aldosterone synthase gene. Consequently, mineralocorticoid is synthesized in the zona fasciculata and regulated by corticotropin. A number of conditions associated with hyperreninemia result in secondary hyperaldosteronism and renal K⁺wasting. High renin levels are commonly seen in both renovascular and malignant hypertension. Renin-secreting tumors of the juxtaglomerular apparatus are a rare cause of hypokalemia. Other tumors that have been reported to produce renin include renal cell carcinoma, ovarian carcinoma, and Wilms' tumor. Hyperreninemia may also occur secondary to decreased effective circulating arterial volume.

In the absence of elevated renin or aldosterone levels, enhanced distal nephron secretion of K⁺ may result from increased production of non-aldosterone mineralocorticoids in *congenital adrenal hyperplasia* ([Chap. 331](#)).

Glucocorticoid-stimulated kaliuresis does not normally occur due to the conversion of cortisol to cortisone by 11 β -hydroxysteroid dehydrogenase (11 β -HSDH). Therefore, 11 β -HSDH deficiency or suppression allows cortisol to bind to the aldosterone receptor and leads to the *syndrome of apparent mineralocorticoid excess*. Drugs that inhibit the activity of 11 β -HSDH include glycyrrhetic acid, present in licorice, chewing tobacco, and carbenoxolone. The presentation of Cushing's syndrome may include hypokalemia if the capacity of 11 β -HSDH to inactivate cortisol is overwhelmed by persistently elevated glucocorticoid levels.

Liddle's syndrome is a rare familial (autosomal dominant) disease characterized by hypertension, hypokalemic metabolic alkalosis, renal K⁺wasting, and suppressed renin and aldosterone secretion ([Chap. 331](#)). Increased distal delivery of Na⁺ with a non-reabsorbable anion (not Cl⁻) enhances the lumen-negative [TEPD](#) and K⁺secretion. Classically, this is seen with *proximal (type 2) renal tubular acidosis* (RTA) and vomiting, associated with bicarbonaturia. Diabetic ketoacidosis and toluene abuse (glue-sniffing) can lead to increased delivery of β -hydroxybutyrate and hippurate, respectively, to the [CCD](#) and to renal K⁺ loss. High doses of penicillin derivatives administered to volume-depleted patients may likewise promote renal K⁺secretion as well as an osmotic diuresis. *Classic distal (type 1) RTA* is associated with hypokalemia due to increased renal K⁺ loss, the mechanism of which is uncertain. Amphotericin B causes hypokalemia due to increased distal nephron permeability to Na⁺ and K⁺ and to renal K⁺wasting.

Bartter's syndrome is a disorder characterized by hypokalemia, metabolic alkalosis, hyperreninemic hyperaldosteronism secondary to [ECF](#) volume contraction, and juxtaglomerular apparatus hyperplasia ([Chap. 331](#)). Finally, diuretic use and abuse are common causes of K⁺depletion. Carbonic anhydrase inhibitors, loop diuretics, and thiazides are all kaliuretic. The degree of hypokalemia tends to be greater with long-acting agents and is dose-dependent. Increased renal K⁺excretion is due primarily to increased distal solute delivery and secondary hyperaldosteronism (due to volume depletion).

CLINICAL FEATURES

The clinical manifestations of K⁺depletion vary greatly between individual patients, and their severity depends on the degree of hypokalemia. Symptoms seldom occur unless the plasma K⁺concentration is less than 3 mmol/L. Fatigue, myalgia, and muscular weakness of the lower extremities are common complaints and are due to a lower (more negative) resting membrane potential. More severe hypokalemia may lead to progressive weakness, hypoventilation (due to respiratory muscle involvement), and eventually complete paralysis. Impaired muscle metabolism and the blunted hyperemic response to exercise associated with profound K⁺depletion increase the risk of rhabdomyolysis. Smooth-muscle function may also be affected and manifest as paralytic ileus.

The electrocardiographic changes of hypokalemia ([Fig. 226-19](#)) are due to delayed ventricular repolarization and do not correlate well with the plasma K⁺concentration. Early changes include flattening or inversion of the T wave, a prominent U wave, ST-segment depression, and a prolonged QU interval. Severe K⁺depletion may result in a prolonged PR interval, decreased voltage and widening of the QRS complex, and an increased risk of ventricular arrhythmias, especially in patients with myocardial ischemia or left ventricular hypertrophy. Hypokalemia may also predispose to digitalis toxicity. Epidemiologic studies have linked a low-K⁺diet with an increased prevalence of hypertension, particularly among African Americans. Furthermore, in patients with essential hypertension, systemic blood pressure may be lowered by K⁺supplementation. The mechanism of the hypertensive effect of K⁺depletion is not certain but may relate to enhanced distal NaCl reabsorption.

Hypokalemia is often associated with acid-base disturbances related to the underlying disorder. In addition, K⁺depletion results in intracellular acidification and an increase in net acid excretion or new HCO₃⁻production. This is a consequence of enhanced proximal HCO₃⁻reabsorption, increased renal ammoniagenesis, and increased distal H⁺secretion. This contributes to the generation of metabolic alkalosis frequently present in hypokalemic patients. [NDI](#) (see above) is not uncommonly seen in K⁺depletion and is manifest as polydipsia and polyuria. Glucose intolerance may also occur with hypokalemia and has been attributed to either impaired insulin secretion or peripheral insulin resistance.

DIAGNOSIS

In most cases, the etiology of K⁺depletion can be determined by a careful history. Diuretic and laxative abuse as well as surreptitious vomiting may be difficult to identify but should be excluded. Rarely, patients with a marked leukocytosis (e.g., acute myeloid leukemia) and normokalemia may have a low measured plasma K⁺concentration due to white blood cell uptake of K⁺ at room temperature. This *pseudohypokalemia* can be avoided by storing the blood sample on ice or rapidly separating the plasma (or serum) from the cells. After eliminating decreased intake and intracellular shift as potential causes of hypokalemia, examination of the renal response can help to clarify the source of K⁺loss. The appropriate response to K⁺depletion is to excrete less than 15 mmol/d of K⁺ in the urine, due to increased reabsorption and decreased distal secretion. Hypokalemia with minimal renal K⁺excretion suggests that K⁺ was lost via the skin or gastrointestinal tract or that there is a remote history of vomiting or diuretic use. As

described above, renal K⁺ wasting may be due to factors that either increase the K⁺ concentration in the [CCD](#) or increase the distal flow rate (or both). The [ECF](#) volume status, blood pressure, and associated acid-base disorder may help to differentiate the causes of excessive renal K⁺ loss. A rapid and simple test designed to evaluate the driving force for net K⁺ secretion is the *transtubular K⁺ concentration gradient* (TTKG). The TTKG is the ratio of the K⁺ concentration in the lumen of the CCD ($[K^+]_{\text{CCD}}$) to that in peritubular capillaries or plasma ($[K^+]_{\text{P}}$). The validity of this measurement depends on three assumptions: (1) few solutes are reabsorbed in the medullary collecting duct (MCD), (2) K⁺ is neither secreted nor reabsorbed in the MCD, and (3) the osmolality of the fluid in the terminal CCD is known. Significant reabsorption or secretion of K⁺ in the MCD seldom occurs, except in profound K⁺ depletion or excess, respectively. When [AVP](#) is acting ($OSM_U \approx OSM_P$), the osmolality in the terminal CCD is the same as that of plasma, and the K⁺ concentration in the lumen of the distal nephron can be estimated by dividing the urine K⁺ concentration ($[K^+]_U$) by the ratio of the urine to plasma osmolality (OSM_U/OSM_P):

Hypokalemia with a [TTKG](#) greater than 4 suggests renal K⁺ loss due to increased distal K⁺ secretion. Plasma renin and aldosterone levels are often helpful in differentiating the various causes of hyperaldosteronism. Bicarbonaturia and the presence of other non-reabsorbed anions also increase the TTKG and lead to renal K⁺ wasting.

CLINICAL APPROACH

See [Fig. 49-3](#).

TREATMENT

The therapeutic goals are to correct the K⁺ deficit and to minimize ongoing losses. With the exception of periodic paralysis, hypokalemia resulting from transcellular shifts rarely requires intravenous K⁺ supplementation, which can lead to rebound hyperkalemia. It is generally safer to correct hypokalemia via the oral route. The degree of K⁺ depletion does not correlate well with the plasma K⁺ concentration. A decrement of 1 mmol/L in the plasma K⁺ concentration (from 4.0 to 3.0 mmol/L) may represent a total body K⁺ deficit of 200 to 400 mmol, and patients with plasma levels under 3.0 mmol/L often require in excess of 600 mmol of K⁺ to correct the deficit. Furthermore, factors promoting K⁺ shift out of cells (e.g., insulin deficiency in diabetic ketoacidosis) may result in underestimation of the K⁺ deficit. Therefore, the plasma K⁺ concentration should be monitored frequently when assessing the response to treatment. Potassium chloride is usually the preparation of choice and will promote more rapid correction of hypokalemia and metabolic alkalosis. Potassium bicarbonate and citrate (metabolized to HCO₃⁻) tend to alkalinize the patient and would be more appropriate for hypokalemia associated with chronic diarrhea or [RTA](#).

Patients with severe hypokalemia or those unable to take anything by mouth require intravenous replacement therapy with KCl. The maximum concentration of administered K⁺ should be no more than 40 mmol/L via a peripheral vein or 60 mmol/L via a central vein. The rate of infusion should not exceed 20 mmol/h unless paralysis or malignant

ventricular arrhythmias are present. Ideally, KCl should be mixed in normal saline since dextrose solutions may initially exacerbate hypokalemia due to insulin-mediated movement of K⁺ into cells. Rapid intravenous administration of K⁺ should be used judiciously and requires close observation of the clinical manifestations of hypokalemia (electrocardiogram and neuromuscular examination).

HYPERKALEMIA

ETIOLOGY

Hyperkalemia, defined as a plasma K⁺ concentration >5.0 mmol/L, occurs as a result of either K⁺ release from cells or decreased renal loss. Increased K⁺ intake is rarely the sole cause of hyperkalemia since the phenomenon of *potassium adaptation* ensures rapid K⁺ excretion in response to increases in dietary consumption. Iatrogenic hyperkalemia may result from overzealous parenteral K⁺ replacement or in patients with renal insufficiency. *Pseudohyperkalemia* represents an artificially elevated plasma K⁺ concentration due to K⁺ movement out of cells immediately prior to or following venipuncture. Contributing factors include prolonged use of a tourniquet with or without repeated fist clenching, hemolysis, and marked leukocytosis or thrombocytosis. The latter two result in an elevated serum K⁺ concentration due to release of intracellular K⁺ following clot formation. Pseudohyperkalemia should be suspected in an otherwise asymptomatic patient with no obvious underlying cause. If proper venipuncture technique is used and a plasma (not serum) K⁺ concentration is measured, it should be normal. Intravascular hemolysis, tumor lysis syndrome, and rhabdomyolysis all lead to K⁺ release from cells as a result of tissue breakdown. Metabolic acidoses, with the exception of those due to the accumulation of organic anions, can be associated with mild hyperkalemia resulting from intracellular buffering of H⁺ (see above). As previously described (p. 278), insulin deficiency and hypertonicity (e.g., hyperglycemia) promote K⁺ shift from the [ICF](#) to the [ECF](#). The severity of exercise-induced hyperkalemia is related to the degree of exertion. It is due to release of K⁺ from muscles and is usually rapidly reversible, often associated with rebound hypokalemia. Treatment with beta blockers rarely causes hyperkalemia but may contribute to the elevation in plasma K⁺ concentration seen with other conditions. *Hyperkalemic periodic paralysis* ([Chap. 381](#)) is a rare autosomal dominant disorder characterized by episodic weakness or paralysis, precipitated by stimuli that normally lead to mild hyperkalemia (e.g., exercise). The genetic defect appears to be a single amino acid substitution due to a mutation in the gene for the skeletal muscle Na⁺ channel. Hyperkalemia may occur with severe digitalis toxicity due to inhibition of the Na⁺, K⁺-ATPase pump. Depolarizing muscle relaxants such as succinylcholine can increase the plasma K⁺ concentration, especially in patients with massive trauma, burns, or neuromuscular disease.

Chronic hyperkalemia is virtually always associated with decreased renal K⁺ excretion due to either impaired secretion or diminished distal solute delivery ([Table 49-4](#)). The latter is seldom the only cause of impaired K⁺ excretion but may significantly contribute to hyperkalemia in protein-malnourished (low urea excretion) and [ECF](#) volume-contracted (decreased distal NaCl delivery) patients. Decreased K⁺ secretion by the principal cells results from either impaired Na⁺ reabsorption or increased Cl⁻ reabsorption, both of which give rise to a diminished (less lumen-negative) [TEPD](#) in the [CCD](#). *Hyporeninemic hypoaldosteronism* is a syndrome characterized by euvolemia or ECF volume

expansion and suppressed renin and aldosterone levels ([Chaps. 331](#) and [333](#)). This disorder is commonly seen in mild renal insufficiency, diabetic nephropathy, or chronic tubulointerstitial disease. Patients frequently have an impaired kaliuretic response to exogenous mineralocorticoid administration, suggesting that enhanced distal Cl⁻ reabsorption (electroneutral Na⁺ reabsorption) may account for many of the findings of hyporeninemic hypoaldosteronism. [NSAIDs](#) inhibit renin secretion and the synthesis of vasodilatory renal prostaglandins. The resultant decrease in [GFR](#) and K⁺ secretion is often manifest as hyperkalemia. As a rule, the degree of hyperkalemia due to hypoaldosteronism is mild in the absence of increased K⁺ intake or renal dysfunction. Angiotensin-converting enzyme (ACE) inhibitors block the conversion of angiotensin I to angiotensin II, resulting in impaired aldosterone release. Patients at increased risk of ACE inhibitor-induced hyperkalemia include those with diabetes mellitus, renal insufficiency, decreased effective circulating arterial volume, bilateral renal artery stenosis, or concurrent use of K⁺-sparing diuretics or NSAIDs.

Decreased aldosterone synthesis may be due to *primary adrenal insufficiency* (Addison's disease) or congenital adrenal enzyme deficiency ([Chap. 331](#)). Heparin (including low-molecular-weight heparin) inhibits production of aldosterone by the cells of the zona glomerulosa and can lead to severe hyperkalemia in a subset of patients with underlying renal disease; diabetes mellitus; or those receiving K⁺-sparing diuretics, [ACE](#) inhibitors, or [NSAIDs](#). *Pseudohypoaldosteronism* is a rare familial disorder characterized by hyperkalemia, metabolic acidosis, renal Na⁺ wasting, hypotension, high renin and aldosterone levels, and end-organ resistance to aldosterone. The gene encoding the mineralocorticoid receptor is normal in these patients, and the electrolyte abnormalities can be reversed with suprapharmacologic doses of an exogenous mineralocorticoid (e.g., 9α-fludrocortisone) or an inhibitor of [11β-HSDH](#) (e.g., carbenoxolone). The kaliuretic response to aldosterone is impaired by K⁺-sparing diuretics. Spironolactone is a competitive mineralocorticoid antagonist, whereas amiloride and triamterene block the apical Na⁺ channel of the principal cell. Two other drugs that impair K⁺ secretion by blocking distal nephron Na⁺ reabsorption are trimethoprim and pentamidine. These antimicrobial agents may contribute to the hyperkalemia often seen in patients infected with HIV who are being treated for *Pneumocystis carinii* pneumonia.

Hyperkalemia frequently complicates acute oliguric renal failure due to increased K⁺ release from cells (acidosis, catabolism) and decreased excretion. Increased distal flow rate and K⁺ secretion per nephron compensate for decreased renal mass in chronic renal insufficiency. However, these adaptive mechanisms eventually fail to maintain K⁺ balance when the [GFR](#) falls below 10 to 15 mL/min or oliguria ensues. Otherwise asymptomatic urinary tract obstruction is an often overlooked cause of hyperkalemia. Other nephropathies associated with impaired K⁺ excretion include drug-induced interstitial nephritis, lupus nephritis, sickle cell disease, and diabetic nephropathy.

Gordon's syndrome is a rare condition characterized by hyperkalemia, metabolic acidosis, and a normal [GFR](#). These patients are usually volume-expanded with suppressed renin and aldosterone levels as well as refractory to the kaliuretic effect of exogenous mineralocorticoids. It has been suggested that these findings could all be accounted for by increased distal Cl⁻ reabsorption (electroneutral Na⁺ reabsorption), also referred to as a *Cl⁻ shunt*. A similar mechanism may be partially responsible for the

hyperkalemia associated with cyclosporine nephrotoxicity. *Hyperkalemic distal (type 4) RTA* may be due to either hypoaldosteronism or a Cl⁻ shunt (aldosterone-resistant).

CLINICAL FEATURES

Since the resting membrane potential is related to the ratio of the [ICF](#) to [ECF](#) K⁺ concentration, hyperkalemia partially depolarizes the cell membrane. Prolonged depolarization impairs membrane excitability and is manifest as weakness, which may progress to flaccid paralysis and hypoventilation if the respiratory muscles are involved. Hyperkalemia also inhibits renal ammoniogenesis and reabsorption of NH₄⁺ in the thick ascending limb of the loop of Henle. Thus, net acid excretion is impaired and results in metabolic acidosis, which may further exacerbate the hyperkalemia due to K⁺ movement out of cells.

The most serious effect of hyperkalemia is cardiac toxicity, which does not correlate well with the plasma K⁺ concentration. The earliest electrocardiographic changes include increased T-wave amplitude, or peaked T waves. More severe degrees of hyperkalemia result in a prolonged PR interval and QRS duration, atrioventricular conduction delay, and loss of P waves. Progressive widening of the QRS complex and merging with the T wave produces a sinewave pattern. The terminal event is usually ventricular fibrillation or asystole.

DIAGNOSIS

With rare exceptions, chronic hyperkalemia is always due to impaired K⁺ excretion. If the etiology is not readily apparent and the patient is asymptomatic, pseudohyperkalemia should be excluded, as described above. Oliguric acute renal failure and severe chronic renal insufficiency should also be ruled out. The history should focus on medications that impair K⁺ handling and potential sources of K⁺ intake. Evaluation of the [ECF](#) compartment, effective circulating volume, and urine output are essential components of the physical examination. The severity of hyperkalemia is determined by the symptoms, plasma K⁺ concentration, and electrocardiographic abnormalities.

The appropriate renal response to hyperkalemia is to excrete at least 200 mmol of K⁺ daily. In most cases, diminished renal K⁺ loss is due to impaired K⁺ secretion, which can be assessed by measuring the [TTKG](#) (see above). A TTKG <10 implies a decreased driving force for K⁺ secretion due to either hypoaldosteronism or resistance to the renal effects of mineralocorticoid. This can be determined by evaluating the kaliuretic response to administration of mineralocorticoid (e.g., 9α-fludrocortisone). Primary adrenal insufficiency can be differentiated from hyporeninemic hypoaldosteronism by examining the renin-aldosterone axis. Renin and aldosterone levels should be measured in the supine and upright positions, following three days of Na⁺ restriction (Na⁺ intake <10 mmol/d) in combination with a loop diuretic to induce mild volume contraction. Aldosterone-resistant hyperkalemia can result from the various causes of impaired distal Na⁺ reabsorption or from a Cl⁻ shunt. The former leads to salt wasting, [ECF](#) volume contraction, and high renin and aldosterone levels. In contrast, enhanced distal Cl⁻ reabsorption is associated with volume expansion and suppressed renin and aldosterone secretion. As mentioned above, hypoaldosteronism seldom causes severe hypokalemia in the absence of increased dietary K⁺ intake, renal

insufficiency, transcellular K⁺-shifts, or antikaliuretic drugs.

CLINICAL APPROACH

See [Fig. 49-4](#).

TREATMENT

The approach to therapy depends on the degree of hyperkalemia as determined by the plasma K⁺-concentration, associated muscular weakness, and changes on the electrocardiogram. Potentially fatal hyperkalemia rarely occurs unless the plasma K⁺-concentration exceeds 7.5 mmol/L and is usually associated with profound weakness and absent P waves, QRS widening, or ventricular arrhythmias on the electrocardiogram.

Severe hyperkalemia requires emergent treatment directed at minimizing membrane depolarization, shifting K⁺ into cells, and promoting K⁺ loss. In addition, exogenous K⁺-intake and antikaliuretic drugs should be discontinued. Administration of calcium gluconate decreases membrane excitability. The usual dose is 10 mL of a 10% solution infused over 2 to 3 min. The effect begins within minutes but is short-lived (30 to 60 min), and the dose can be repeated if no change in the electrocardiogram is seen after 5 to 10 min. Insulin causes K⁺ to shift into cells by mechanisms described previously and will temporarily lower the plasma K⁺-concentration. Although glucose alone will stimulate insulin release from normal pancreatic β cells, a more rapid response generally occurs when exogenous insulin is administered (with glucose to prevent hypoglycemia). A commonly recommended combination is 10 to 20 units of regular insulin and 25 to 50 g of glucose. Obviously, hyperglycemic patients should not be given glucose. If effective, the plasma K⁺-concentration will fall by 0.5 to 1.5 mmol/L in 15 to 30 min and the effect will last for several hours. Alkali therapy with intravenous NaHCO₃ can also shift K⁺ into cells. This is safest when administered as an isotonic solution of 3 ampules per liter (134 mmol/L NaHCO₃) and ideally should be reserved for severe hyperkalemia associated with metabolic acidosis. Patients with end-stage renal disease seldom respond to this intervention and may not tolerate the Na⁺ load and resultant volume expansion. When administered parenterally or in nebulized form, β_2 -adrenergic agonists promote cellular uptake of K⁺ (see above). The onset of action is 30 min, lowering the plasma K⁺-concentration by 0.5 to 1.5 mmol/L, and the effect lasts 2 to 4 h.

Removal of K⁺ can be achieved using diuretics, cation-exchange resin, or dialysis. Loop and thiazide diuretics, often in combination, may enhance K⁺-excretion if renal function is adequate. Sodium polystyrene sulfonate is a cation-exchange resin that promotes the exchange of Na⁺ for K⁺ in the gastrointestinal tract. Each gram binds 1 mmol of K⁺ and releases 2 to 3 mmol of Na⁺. When given by mouth, the usual dose is 25 to 50 g mixed with 100 mL of 20% sorbitol to prevent constipation. This will generally lower the plasma K⁺-concentration by 0.5 to 1.0 mmol/L within 1 to 2 h and last for 4 to 6 h. Sodium polystyrene sulfonate can also be administered as a retention enema consisting of 50 g of resin and 50 mL of 70% sorbitol mixed in 150 mL of tap water. The sorbitol should be omitted from the enema in postoperative patients due to the increased incidence of sorbitol-induced colonic necrosis, especially following renal transplantation. The most rapid and effective way of lowering the plasma K⁺-concentration is hemodialysis. This

should be reserved for patients with renal failure and those with severe life-threatening hyperkalemia unresponsive to more conservative measures. Peritoneal dialysis also removes K⁺ but is only 15 to 20% as effective as hemodialysis. Finally, the underlying cause of the hyperkalemia should be treated. This may involve dietary modification, correction of metabolic acidosis, cautious volume expansion, and administration of exogenous mineralocorticoid.

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50. ACIDOSIS AND ALKALOSIS - Thomas D. DuBose, Jr.

NORMAL ACID-BASE HOMEOSTASIS

Systemic arterial pH is maintained between 7.35 and 7.45 by extracellular and intracellular chemical buffering together with respiratory and renal regulatory mechanisms. The control of arterial CO₂ tension (P_{aCO_2}) by the central nervous system and respiratory systems and the control of the plasma bicarbonate by the kidneys stabilize the arterial pH by excretion or retention of acid or alkali. The metabolic and respiratory components that regulate systemic pH are described by the Henderson-Hasselbalch equation:

Under most circumstances, CO₂ production and excretion are matched, and the usual steady-state P_{aCO_2} is maintained at 40 mmHg. Underexcretion of CO₂ produces hypercapnia, and overexcretion causes hypocapnia. Nevertheless, production and excretion are again matched at a new steady-state P_{aCO_2} . Therefore, the P_{aCO_2} is regulated primarily by neural respiratory factors ([Chap. 263](#)) and is not subject to regulation by the rate of CO₂ production. Hypercapnia is usually the result of hypoventilation rather than of increased CO₂ production. Increases or decreases in P_{aCO_2} represent derangements of neural respiratory control or are due to compensatory changes in response to a primary alteration in the plasma $[HCO_3^-]$.

Primary changes in P_{aCO_2} can cause acidosis or alkalosis, depending on whether P_{aCO_2} is above or below the normal value of 40 mmHg (respiratory acidosis or alkalosis, respectively). Primary alteration of P_{aCO_2} evokes cellular buffering and renal adaptation, a slow process that becomes more efficient with time. A primary change in the plasma $[HCO_3^-]$ as a result of metabolic or renal factors results in compensatory changes in ventilation that blunt the changes in blood pH that would occur otherwise. Such respiratory alterations are referred to as *secondary*, or compensatory, changes, since they occur in response to primary metabolic changes.

The kidneys regulate plasma $[HCO_3^-]$ through three main processes: (1) "reabsorption" of filtered HCO_3^- , (2) formation of titratable acid, and (3) excretion of NH_4^+ in the urine. The kidney filters approximately 4000 mmol of HCO_3^- per day. To reabsorb the filtered load of HCO_3^- , the renal tubules must therefore secrete 4000 mmol of hydrogen ions. Between 80 and 90% of HCO_3^- is reabsorbed in the proximal tubule. The distal nephron reabsorbs the remainder and secretes protons, as generated from metabolism, to defend systemic pH. While this quantity of protons, 40 to 60 mmol/d, is small, it must be secreted to prevent chronic positive H⁺ balance and metabolic acidosis. This quantity of secreted protons is represented in the urine as titratable acid and NH_4^+ . Metabolic acidosis in the face of normal renal function increases NH_4^+ production and excretion. NH_4^+ production and excretion are impaired in chronic renal failure, hyperkalemia, and renal tubular acidosis.

In sum, these regulatory responses, including chemical buffering, the regulation of P_{aCO_2} by the respiratory system, and of $[HCO_3^-]$ by the kidneys, act in concert to maintain a systemic arterial pH between 7.35 and 7.45.

DIAGNOSIS OF GENERAL TYPES OF DISTURBANCES

The most common clinical disturbances are simple acid-base disorders, i.e., metabolic acidosis or alkalosis or respiratory acidosis or alkalosis. Since compensation is not complete, the pH is abnormal in simple disturbances. More complicated clinical situations can give rise to mixed acid-base disturbances.

SIMPLE ACID-BASE DISORDERS

Primary respiratory disturbances (primary changes in P_{aCO_2}) invoke compensatory metabolic responses (secondary changes in $[HCO_3^-]$), and primary metabolic disturbances elicit predictable compensatory respiratory responses. Physiologic compensation can be predicted from the relationships displayed in [Table 50-1](#). Primary changes in P_{aCO_2} or $[HCO_3^-]$ alter systemic pH and cause acidosis or alkalosis. To illustrate, metabolic acidosis due to an increase in endogenous acids (e.g., ketoacidosis) lowers extracellular fluid $[HCO_3^-]$ and decreases extracellular pH. This stimulates the medullary chemoreceptors to increase ventilation and to return the ratio of $[HCO_3^-]$ to P_{aCO_2} , and thus pH, toward normal, although not to normal. The degree of respiratory compensation expected in a simple form of metabolic acidosis can be predicted from the relationship: $P_{aCO_2} = (1.5 \times [HCO_3^-]) + 8$, i.e., the P_{aCO_2} is expected to decrease 1.25 mmHg for each mmol per liter decrease in $[HCO_3^-]$. Thus, a patient with metabolic acidosis and $[HCO_3^-]$ of 12 mmol/L would be expected to have a P_{aCO_2} between 24 and 28 mmHg. Values for P_{aCO_2} below 24 or greater than 28 mmHg define a mixed disturbance (metabolic acidosis and respiratory alkalosis or metabolic alkalosis and respiratory acidosis, respectively). Another way to judge the appropriateness of the response in $[HCO_3^-]$ or P_{aCO_2} is to use an acid-base nomogram ([Fig. 50-1](#)). While the shaded areas of the nomogram show the 95% confidence limits for normal compensation in simple disturbances, finding acid-base values within the shaded area does not necessarily rule out a mixed disturbance. Imposition of one disorder over another may result in values lying within the area of a third. Thus, the nomogram, while convenient, is not a substitute for the equations in [Table 50-1](#).

MIXED ACID-BASE DISORDERS

Mixed acid-base disorders -- defined as independently coexisting disorders, not merely compensatory responses -- are often seen in patients in critical care units and can lead to dangerous extremes of pH. A patient with diabetic ketoacidosis (metabolic acidosis) may develop an independent respiratory problem leading to respiratory acidosis or alkalosis. Patients with underlying pulmonary disease may not respond to metabolic acidosis with an appropriate ventilatory response because of insufficient respiratory reserve. Such imposition of respiratory acidosis on metabolic acidosis can lead to severe acidemia and a poor outcome. When metabolic acidosis and metabolic alkalosis coexist in the same patient, the pH may be normal or near normal. When the pH is normal, an elevated anion gap (see below) denotes the presence of a metabolic acidosis. A diabetic patient with ketoacidosis may have renal dysfunction resulting in simultaneous metabolic acidosis. Patients who have ingested an overdose of drug combinations such as sedatives and salicylates may have mixed disturbances as a result of the acid-base response to the individual drugs (metabolic acidosis mixed with

respiratory acidosis or respiratory alkalosis, respectively). Even more complex are triple acid-base disturbances. For example, patients with metabolic acidosis due to alcoholic ketoacidosis may develop metabolic alkalosis due to vomiting and superimposed respiratory alkalosis due to the hyperventilation of hepatic dysfunction or alcohol withdrawal.

DIAGNOSIS OF ACID-BASE DISORDERS

Care should be taken when measuring blood gases to obtain the arterial blood sample without using excessive heparin. In the determination of arterial blood gases by the clinical laboratory, both pH and P_{aCO_2} are measured, and the $[HCO_3^-]$ is calculated from the Henderson-Hasselbalch equation. This calculated value should be compared with the measured $[HCO_3^-]$ (total CO_2) on the electrolyte panel. These two values should agree within 2 mmol/L. If they do not, the values may not have been drawn simultaneously, a laboratory error may be present, or an error could have been made in calculating the $[HCO_3^-]$. After verifying the blood acid-base values, one can then identify the precise acid-base disorder.

The most common causes of acid-base disorders should be kept in mind while probing the history for clues about the etiology. For example, established chronic renal failure is expected to cause a metabolic acidosis, and chronic vomiting frequently causes metabolic alkalosis. Patients with pneumonia, sepsis, or cardiac failure frequently have respiratory alkalosis, and patients with chronic obstructive pulmonary disease or a sedative drug overdose often display a respiratory acidosis. The drug history is important since loop or thiazide diuretics may cause metabolic alkalosis, and the carbonic anhydrase inhibitor, acetazolamide, can result in metabolic acidosis.

Blood for electrolytes and arterial blood gases should be drawn simultaneously prior to therapy, since an increase in $[HCO_3^-]$ occurs with metabolic alkalosis and respiratory acidosis. Conversely, a decrease in $[HCO_3^-]$ occurs in metabolic acidosis and respiratory alkalosis.

Metabolic acidosis leads to hyperkalemia as a result of cellular shifts in which H^+ is exchanged for K^+ or Na^+ . For each decrease in blood pH of 0.10, the plasma $[K^+]$ should rise by 0.6 mmol/L. This relationship is not invariable. Diabetic ketoacidosis, lactic acidosis, diarrhea, and renal tubular acidosis (RTA) are often associated with potassium depletion because of urinary K^+ wasting.

Anion Gap All evaluations of acid-base disorders should include a simple calculation of the anion gap (AG); it represents those unmeasured anions in plasma (normally 10 to 12 mmol/L) and is calculated as follows: $AG = Na^+ - (Cl^- + HCO_3^-)$. The unmeasured anions include anionic proteins, phosphate, sulfate, and organic anions. When acid anions, such as acetoacetate and lactate, accumulate in extracellular fluid, the AG increases, causing a high-AG acidosis. An increase in the AG is most often due to an increase in unmeasured anions and less commonly is due to a decrease in unmeasured cations (calcium, magnesium, potassium). In addition, the AG may increase with an increase in anionic albumin, either because of increased albumin concentration or alkalosis, which alters albumin charge. A decrease in the AG can be due to: (1) an increase in unmeasured cations; (2) the addition to the blood of abnormal cations, such

as lithium (lithium intoxication) or cationic immunoglobulins (plasma cell dyscrasias); (3) a reduction in the major plasma anion albumin concentration (nephrotic syndrome); (4) a decrease in the effective anionic charge on albumin by acidosis; or (5) hyperviscosity and severe hyperlipidemia, which can lead to an underestimation of sodium and chloride concentrations.

In the face of a normal serum albumin, a high [AG](#) is usually due to non-chloride-containing acids that contain inorganic (phosphate, sulfate), organic (ketoacids, lactate, uremic organic anions), exogenous (salicylate or ingested toxins with organic acid production), or unidentified anions. By definition, therefore, a high-AG acidosis has two identifying features: a low $[\text{HCO}_3^-]$ and an elevated AG. The latter is present even if an additional acid-base disorder is superimposed to modify the $[\text{HCO}_3^-]$ independently. Simultaneous metabolic acidosis of the high-AG variety plus either chronic respiratory acidosis or metabolic alkalosis represents such a situation in which $[\text{HCO}_3^-]$ may be normal or even high. However, the AG is elevated, and $[\text{Cl}^-]$ is depressed.

Similarly, normal values for $[\text{HCO}_3^-]$, PaCO_2 , and pH do not ensure the absence of an acid-base disturbance. For instance, an alcoholic who has been vomiting may develop a metabolic alkalosis with a pH of 7.55, PaCO_2 of 48 mmHg, $[\text{HCO}_3^-]$ of 40 mmol/L, $[\text{Na}^+]$ of 135, $[\text{Cl}^-]$ of 80, and $[\text{K}^+]$ of 2.8. If such a patient were then to develop a superimposed alcoholic ketoacidosis with a β -hydroxybutyrate concentration of 15 mM, arterial pH would fall to 7.40, $[\text{HCO}_3^-]$ to 25 mmol/L, and the PaCO_2 to 40 mmHg. Although these blood gases are normal, the [AG](#) is elevated at 30 mmol/L, indicating a mixed metabolic alkalosis and metabolic acidosis.

METABOLIC ACIDOSIS

Metabolic acidosis can occur because of an increase in endogenous acid production (such as lactate and ketoacids), loss of bicarbonate (as in diarrhea), or accumulation of endogenous acids (as in renal failure). Metabolic acidosis has profound effects on the respiratory, cardiac, and nervous systems. The fall in blood pH is accompanied by a characteristic increase in ventilation, especially the tidal volume (Kussmaul respiration). Intrinsic cardiac contractility may be depressed, but inotropic function can be normal because of catecholamine release. Both peripheral arterial vasodilation and central venoconstriction can be present; the decrease in central and pulmonary vascular compliance predisposes to pulmonary edema with even minimal volume overload. Central nervous system function is depressed, with headache, lethargy, stupor, and, in some cases, even coma. Glucose intolerance may also occur.

There are two major categories of clinical metabolic acidosis: high-[AG](#) and normal-AG, or hyperchloremic acidosis ([Table 50-2](#) and [Table 50-3](#)).

TREATMENT

Treatment of metabolic acidosis with alkali should be reserved for severe acidemia except when the patient has no "potential HCO_3^- " in plasma. Potential $[\text{HCO}_3^-]$ can be estimated from the increment (D) in the anion gap ($\text{DAG} = \text{patient's AG} - 10$). It must be determined if the acid anion in plasma is metabolizable (i.e., β -hydroxybutyrate,

acetoacetate, and lactate) or nonmetabolizable (anions that accumulate in chronic renal failure and after toxin ingestion). The latter requires return of renal function to replenish the $[\text{HCO}_3^-]$ deficit, a slow and often unpredictable process. Consequently, patients with a normal AG acidosis (hyperchloremic acidosis), a slightly elevated AG (mixed hyperchloremic and AG acidosis), or an AG attributable to a nonmetabolizable anion in the face of renal failure should receive alkali therapy, either orally (NaHCO_3 or Shohl's solution) or intravenously (NaHCO_3), in an amount necessary to slowly increase the plasma $[\text{HCO}_3^-]$ into the 20 to 22 mmol/L range.

Controversy exists, however, in regard to the use of alkali in patients with a pure [AG](#) acidosis owing to accumulation of a metabolizable organic acid anion (ketoacidosis or lactic acidosis). In general, severe acidosis ($\text{pH} < 7.20$) warrants the intravenous administration of 50 to 100 meq of NaHCO_3 , over 30 to 45 min, during the initial 1 to 2 h of therapy. Provision of such modest quantities of alkali in this situation seems to provide an added measure of safety, but it is essential to monitor plasma electrolytes during the course of therapy, since the $[\text{K}^+]$ may decline as pH rises. The goal is to increase the $[\text{HCO}_3^-]$ to 10 meq/L and the pH to 7.25, not to increase these values to normal.

HIGH-ANION-GAP ACIDOSES

There are four principal causes of a high-[AG](#) acidosis: (1) lactic acidosis, (2) ketoacidosis, (3) ingested toxins ([Table 50-2](#)), and (4) acute and chronic renal failure. Initial screening to differentiate the high-AG acidoses should include: (1) a probe of the history for evidence of drug and toxin ingestion and measurement of arterial blood gas to detect coexistent respiratory alkalosis (salicylates); (2) determination of whether diabetes mellitus is present (diabetic ketoacidosis); (3) a search for evidence of alcoholism or increased levels of β -hydroxybutyrate (alcoholic ketoacidosis); (4) observation for clinical signs of uremia and determination of the blood urea nitrogen (BUN) and creatinine (uremic acidosis); (5) inspection of the urine for oxalate crystals (ethylene glycol); and (6) recognition of the numerous clinical settings in which lactate levels may be increased (hypotension, shock, cardiac failure, leukemia, cancer, and drug or toxin ingestion).

Lactic Acidosis An increase in plasma L-lactate may be secondary to poor tissue perfusion (type A) -- circulatory insufficiency (shock, circulatory failure), severe anemia, mitochondrial enzyme defects, and inhibitors (carbon monoxide, cyanide) -- or to aerobic disorders (type B) -- malignancies, diabetes mellitus, renal or hepatic failure, severe infections (cholera, malaria), seizures, AIDS, or drugs/toxins (biguanides, ethanol, methanol, isoniazid, AZT analogues, and fructose). Unrecognized bowel ischemia or infarction in a patient with severe atherosclerosis or cardiac decompensation receiving vasopressors is a common cause of lactic acidosis. D-Lactic acid acidosis, which may be associated with jejunoileal bypass or intestinal obstruction and is due to formation of D-lactate by gut bacteria, may cause both an increased [AG](#) and hyperchloremia.

TREATMENT

The underlying condition that disrupts lactate metabolism must first be corrected; tissue

perfusion must be restored when it is inadequate. Vasoconstrictors should be avoided, if possible, since they may worsen tissue perfusion. Alkali therapy is generally advocated for acute, severe acidemia ($\text{pH} < 7.1$) to improve cardiac function and lactate utilization. However, NaHCO_3 therapy may paradoxically depress cardiac performance and exacerbate acidosis by enhancing lactate production (HCO_3^- stimulates phosphofructokinase). While the use of alkali in moderate lactic acidosis is controversial, it is generally agreed that attempts to return the pH or $[\text{HCO}_3^-]$ to normal by administration of exogenous NaHCO_3 are deleterious. A reasonable approach is to infuse sufficient NaHCO_3 to raise the arterial pH to no more than 7.2 over 30 to 40 min.

NaHCO_3 therapy can cause fluid overload and hypertension because the amount required can be massive when accumulation of lactic acid is relentless. Fluid administration is poorly tolerated because of central venoconstriction, especially in the oliguric patient. If the underlying cause of the lactic acidosis can be remedied, blood lactate will be converted to HCO_3^- and may result in an overshoot alkalosis.

Ketoacidosis

Diabetic Ketoacidosis This condition is caused by increased fatty acid metabolism and the accumulation of ketoacids (acetoacetate and β -hydroxybutyrate). Diabetic ketoacidosis usually occurs in insulin-dependent diabetes mellitus in association with cessation of insulin or an intercurrent illness, such as an infection, gastroenteritis, pancreatitis, or myocardial infarction, which increases insulin requirements temporarily and acutely. The accumulation of ketoacids accounts for the increment in the [AG](#) and is accompanied most often by hyperglycemia [$\text{glucose} > 17 \text{ mmol/L}$ (300 mg/dL)]. It should be noted that since insulin prevents production of ketones, bicarbonate therapy is rarely needed except with extreme acidemia ($\text{pH} < 7.1$), and then in only limited amounts (see "Treatment" for lactic acidosis). **The management of this condition is described in [Chap. 333](#).*

Alcoholic Ketoacidosis Chronic alcoholics can develop ketoacidosis when alcohol consumption is abruptly curtailed; it is usually associated with binge drinking, vomiting, abdominal pain, starvation, and volume depletion. The glucose concentration is low or normal, and acidosis may be severe because of elevated ketones, predominantly β -hydroxybutyrate. Mild lactic acidosis may coexist because of alteration in the redox state. The nitroprusside ketone reaction (Acetest) can detect acetoacetic acid but not β -hydroxybutyrate, so that the degree of ketosis and ketonuria can be underestimated. Typically, insulin levels are low, and concentrations of triglyceride, cortisol, glucagon, and growth hormone are increased.

TREATMENT

Extracellular fluid deficits should be repleted by intravenous administration of saline and glucose (5% dextrose in 0.9% NaCl). Hypophosphatemia, hypokalemia, and hypomagnesemia may coexist and should be corrected. Hypophosphatemia usually emerges 12 to 24 h after admission, may be exacerbated by glucose infusion, and, if severe, may induce rhabdomyolysis. Upper gastrointestinal hemorrhage, pancreatitis, and pneumonia may accompany this disorder.

Drug- and Toxin-Induced Acidosis

Salicylates (See also [Chap. 396](#)) Salicylate intoxication in adults usually causes respiratory alkalosis, mixed metabolic acidosis-respiratory alkalosis, or a pure high-[AG](#) metabolic acidosis. In the latter example, which is less common, only a portion of the AG is due to the salicylates. Lactic acid production is also often increased.

TREATMENT

This should begin with vigorous gastric lavage with isotonic saline (not NaHCO_3) followed by administration of activated charcoal. In the acidotic patient, to facilitate removal of salicylate, intravenous NaHCO_3 is administered in amounts adequate to alkalinize the urine and to maintain urine output (urine pH > 7.5). While this form of therapy is straightforward in acidotic patients, a coexisting respiratory alkalosis may make this approach hazardous. Acetazolamide may be administered when an alkaline diuresis cannot be achieved, but this drug can cause systemic metabolic acidosis if HCO_3^- is not replaced. Hypokalemia may occur with an alkaline diuresis from NaHCO_3 and should be treated promptly and aggressively. Glucose-containing fluids should be administered because of the danger of hypoglycemia. Excessive insensible fluid losses may cause severe volume depletion and hypernatremia. If renal failure prevents rapid clearance of salicylate, hemodialysis can be performed against a bicarbonate dialysate.

Alcohols Under most physiologic conditions, sodium, urea, and glucose generate the osmotic pressure of blood. Plasma osmolality is calculated according to the following expression: $P_{\text{osm}} = 2\text{Na}^+ + \text{Glu} + \text{BUN}$ (all in mmol/L), or, using conventional laboratory values in which glucose and BUN are expressed in milligrams per deciliter: $P_{\text{osm}} = 2\text{Na}^+ + \text{Glu}/18 + \text{BUN}/2.8$. The calculated and determined osmolality should agree within 10 to 15 mmol/kg H_2O . When the measured osmolality exceeds the calculated osmolality by more than 15 to 20 mmol/kg H_2O , one of two circumstances prevails. Either the serum sodium is spuriously low, as with hyperlipidemia or hyperproteinemia (pseudohyponatremia), or osmolytes other than sodium salts, glucose, or urea have accumulated in plasma. Examples include mannitol, radiocontrast media, isopropyl alcohol, ethylene glycol, ethanol, methanol, and acetone. In this situation, the difference between the calculated osmolality and the measured osmolality (*osmolar gap*) is proportional to the concentration of the unmeasured solute. With an appropriate clinical history and index of suspicion, identification of an osmolar gap is helpful in identifying the presence of poison-associated [AG](#) acidosis.

Ethylene Glycol (See also [Chap. 396](#)) Ingestion of ethylene glycol (commonly used in antifreeze) leads to a metabolic acidosis and severe damage to the central nervous system, heart, lungs, and kidneys. The increased [AG](#) and osmolar gap are attributable to ethylene glycol and its metabolites, oxalic acid, glycolic acid, and other organic acids. Lactic acid production increases secondary to inhibition of the tricarboxylic acid cycle and altered intracellular redox state. Diagnosis is facilitated by recognizing oxalate crystals in the urine, the presence of an osmolar gap in serum, and a high-AG acidosis. Treatment should not be delayed while awaiting measurement of ethylene glycol levels in this setting.

TREATMENT

This includes the prompt institution of a saline or osmotic diuresis, thiamine and pyridoxine supplements, fomepizole or ethanol, and hemodialysis. The intravenous administration of the new alcohol dehydrogenase inhibitor, fomepizole (4-methylpyrazole; 7 mg/kg as a loading dose), or ethanol intravenously to achieve a level of 22 mmol/L (100 mg/dL) serves to lessen toxicity because they compete with ethylene glycol for metabolism by alcohol dehydrogenase. Fomepizole, although expensive, offers the advantages of a predictable decline in ethylene glycol levels without the adverse effects, such as excessive obtundation, associated with ethyl alcohol infusion.

Methanol (See also [Chap. 396](#)) The ingestion of methanol (wood alcohol) causes metabolic acidosis, and its metabolites formaldehyde and formic acid cause severe optic nerve and central nervous system damage. Lactic acid, ketoacids, and other unidentified organic acids may contribute to the acidosis. Due to its low molecular weight (32 Da), an osmolar gap is usually present.

TREATMENT

This is similar to that for ethylene glycol intoxication, including general supportive measures, fomepizole or ethanol administration, and hemodialysis.

Renal Failure (See also [Chaps. 269](#) and [270](#)) The hyperchloremic acidosis of moderate renal insufficiency is eventually converted to the high-[AG](#) acidosis of advanced renal failure. Poor filtration and reabsorption of organic anions contribute to the pathogenesis. As renal disease progresses, the number of functioning nephrons eventually becomes insufficient to keep pace with net acid production. Uremic acidosis is characterized, therefore, by a reduced rate of NH_4^+ production and excretion, primarily due to decreased renal mass. $[\text{HCO}_3^-]$ rarely falls below 15 mmol/L, and the AG rarely exceeds 20 mmol/L. The acid retained in chronic renal disease is buffered by alkaline salts from bone. Despite significant retention of acid (up to 20 mmol/d), the serum $[\text{HCO}_3^-]$ does not decrease further, indicating participation of buffers outside the extracellular compartment. Chronic metabolic acidosis results in significant loss of bone mass due to reduction in bone calcium carbonate. Chronic acidosis also increases urinary calcium excretion, proportional to cumulative acid retention.

TREATMENT

Both uremic acidosis and the hyperchloremic acidosis of renal failure require oral alkali replacement to maintain the $[\text{HCO}_3^-]$ between 20 and 24 mmol/L. This can be accomplished with relatively modest amounts of alkali (1.0 to 1.5 mmol/kg body weight per day). It is assumed that alkali replacement prevents the harmful effects of H^+ -balance on bone and prevents or retards muscle catabolism. Sodium citrate (Shohl's solution) or NaHCO_3 tablets are equally effective alkalinizing salts. Citrate enhances the absorption of aluminum from the gastrointestinal tract and should never be given together with aluminum-containing antacids because of the risk of aluminum intoxication. When hyperkalemia is present, furosemide (60 to 80 mg/d) should be added.

HYPERCHLOREMIC METABOLIC ACIDOSES

Alkali can be lost from the gastrointestinal tract in diarrhea or from the kidneys (renal tubular acidosis, [RTA](#)). In these disorders ([Table 50-3](#)), reciprocal changes in $[\text{Cl}^-]$ and $[\text{HCO}_3^-]$ result in a normal [AG](#). In pure hyperchloremic acidosis, therefore, the increase in $[\text{Cl}^-]$ above the normal value approximates the decrease in $[\text{HCO}_3^-]$. The absence of such a relationship suggests a mixed disturbance.

In diarrhea, stools contain a higher $[\text{HCO}_3^-]$ and decomposed HCO_3^- than plasma so that metabolic acidosis develops along with volume depletion. Instead of an acid urine pH (as anticipated with systemic acidosis), urine pH is usually around 6 because metabolic acidosis and hypokalemia increase renal synthesis and excretion of NH_4^+ , thus providing a urinary buffer that increases urine pH. Metabolic acidosis due to gastrointestinal losses with a high urine pH can be differentiated from [RTA](#) ([Chap. 276](#)) because urinary NH_4^+ excretion is typically low in RTA and high with diarrhea. Urinary NH_4^+ levels can be estimated by calculating the urine anion gap (UAG): $\text{UAG} = [\text{Na}^+ + \text{K}^+]_{\text{u}} - [\text{Cl}^-]_{\text{u}}$. When $[\text{Cl}^-]_{\text{u}} > [\text{Na}^+ + \text{K}^+]_{\text{u}}$, the urine ammonium level is appropriately increased, suggesting an extrarenal cause of the acidosis.

Loss of functioning renal parenchyma by progressive renal disease leads to hyperchloremic acidosis when the glomerular filtration rate (GFR) is between 20 and 50 mL/min and to uremic acidosis with a high [AG](#) when the GFR falls to <20 mL/min. Such a progression occurs commonly with tubulointerstitial forms of renal disease, but hyperchloremic metabolic acidosis can persist with advanced glomerular disease. In advanced renal failure, ammoniogenesis is reduced in proportion to the loss of functional renal mass, and ammonium accumulation and trapping in the outer medullary collecting tubule may also be impaired. Because of adaptive increases in K^+ secretion by the collecting duct and colon, the acidosis of chronic renal insufficiency is typically normokalemic.

Proximal [RTA](#) (type 2 RTA) is most often due to generalized proximal tubular dysfunction manifested by glycosuria, generalized aminoaciduria, and phosphaturia (Fanconi syndrome). With a low plasma $[\text{HCO}_3^-]$, the urine pH is acid ($\text{pH} < 5.5$). The fractional excretion of $[\text{HCO}_3^-]$ may exceed 10 to 15% when the serum $\text{HCO}_3^- > 20$ mmol/L. Since HCO_3^- is not reabsorbed normally in the proximal tubule, therapy with NaHCO_3 will enhance renal potassium wasting and hypokalemia.

The typical findings in classic distal [RTA](#) (type 1 RTA) ([Chap. 276](#)) include hypokalemia, hyperchloremic acidosis, low urinary NH_4^+ excretion (positive [UAG](#), low urine $[\text{NH}_4^+]$), and inappropriately high urine pH ($\text{pH} > 5.5$). Such patients are unable to acidify the urine below a pH of 5.5. Most patients have hypocitraturia and hypercalciuria, so that nephrolithiasis, nephrocalcinosis, and bone disease are common. In type 4 RTA, hyperkalemia is disproportionate to the reduction in [GFR](#) because of coexisting dysfunction of potassium and acid secretion. Urinary ammonium excretion is invariably depressed, and renal function may be compromised, for example, due to diabetic nephropathy, amyloidosis, or tubulointerstitial disease. **See [Chap. 276](#) for the pathophysiology, diagnosis, and treatment of RTA.*

Hyporeninemic Hypoaldosteronism (See also [Chap. 331](#)) This condition typically causes hyperchloremic metabolic acidosis, most commonly in older adults with diabetes mellitus or tubulointerstitial disease and renal insufficiency. Patients usually have mild to moderate renal insufficiency and acidosis, with elevation in serum $[K^+]$ (5.2 to 6.0 mmol/L), concurrent hypertension, and congestive heart failure. Both the metabolic acidosis and the hyperkalemia are out of proportion to impairment in [GFR](#). Nonsteroidal anti-inflammatory drugs -- trimethoprim, pentamidine, and ACE-inhibitors -- can also cause hyperkalemia with hyperchloremic metabolic acidosis in patients with renal insufficiency ([Table 50-3](#)).

METABOLIC ALKALOSIS

Metabolic alkalosis is manifested by an elevated arterial pH, an increase in the serum $[HCO_3^-]$, and an increase in P_{aCO_2} as a result of compensatory alveolar hypoventilation. It is often accompanied by hypochloremia and hypokalemia. The patient with a high $[HCO_3^-]$ and a low $[Cl^-]$ has either metabolic alkalosis or chronic respiratory acidosis. As shown in [Table 50-1](#), the P_{aCO_2} increases 6 mmHg for each 10-mmol/L increase in the $[HCO_3^-]$ above normal. Stated differently, in the range of $[HCO_3^-]$ from 10 to 40 mmol/L, the predicted P_{aCO_2} is approximately equal to the $[HCO_3^-] + 15$. The arterial pH establishes the diagnosis, since it is increased in metabolic alkalosis and decreased or normal in respiratory acidosis. Metabolic alkalosis frequently occurs in association with other disorders such as respiratory acidosis or alkalosis or metabolic acidosis.

PATHOGENESIS

Metabolic alkalosis occurs as a result of net gain of $[HCO_3^-]$ or loss of nonvolatile acid (usually HCl by vomiting) from the extracellular fluid. Since it is unusual for alkali to be added to the body, the disorder involves a generative stage, in which the loss of acid usually causes alkalosis, and a maintenance stage, in which the kidneys fail to compensate by excreting HCO_3^- because of volume contraction, a low [GFR](#), or depletion of Cl^- or K^+ .

Under normal circumstances, the kidneys have an impressive capacity to excrete HCO_3^- . Continuation of metabolic alkalosis represents a failure of the kidneys to eliminate HCO_3^- in the usual manner. For HCO_3^- to be added to the extracellular fluid, it must be administered exogenously or synthesized endogenously, in part or entirely by the kidneys. The kidneys will retain, rather than excrete, the excess alkali and maintain the alkalosis if (1) volume deficiency, chloride deficiency, and K^+ deficiency exist in combination with a reduced [GFR](#), which augments distal tubule H^+ secretion; or (2) hypokalemia exists because of autonomous hyperaldosteronism. In the first example, alkalosis is corrected by administration of NaCl and KCl, while in the latter it is necessary to repair the alkalosis by pharmacologic or surgical intervention, not with saline administration.

DIFFERENTIAL DIAGNOSIS

To establish the cause of metabolic alkalosis ([Table 50-4](#)), it is necessary to assess the status of the extracellular fluid volume (ECFV), the recumbent and upright blood pressure, the serum $[K^+]$, and the renin-aldosterone system. For example, the presence

of chronic hypertension and chronic hypokalemia in an alkalotic patient suggests either mineralocorticoid excess or that the hypertensive patient is receiving diuretics. Low plasma renin activity and normal urine $[\text{Na}^+]$ and $[\text{Cl}^-]$ in a patient who is not taking diuretics indicate a primary mineralocorticoid excess syndrome. The combination of hypokalemia and alkalosis in a normotensive, nonedematous patient can be due to Bartter's or Gitelman's syndrome, magnesium deficiency, vomiting, exogenous alkali, or diuretic ingestion. Determination of urine electrolytes (especially the urine $[\text{Cl}^-]$) and screening of the urine for diuretics may be helpful. If the urine is alkaline, with an elevated $[\text{Na}^+]$ and $[\text{K}^+]$ but low $[\text{Cl}^-]$, the diagnosis is usually either vomiting (overt or surreptitious) or alkali ingestion. If the urine is relatively acid and has low concentrations of Na^+ , K^+ , and Cl^- , the most likely possibilities are prior vomiting, the posthypercapnic state, or prior diuretic ingestion. If, on the other hand, neither the urine sodium, potassium, nor chloride concentrations are depressed, magnesium deficiency, Bartter's or Gitelman's syndrome, or current diuretic ingestion should be considered. Bartter's syndrome is distinguished from Gitelman's syndrome because of hypocalciuria and hypomagnesemia in the latter disorder. The genetic and molecular basis of these two disorders has been elucidated recently ([Chap. 276](#)).

Alkali Administration Chronic administration of alkali to individuals with normal renal function rarely, if ever causes alkalosis. However, in patients with coexistent hemodynamic disturbances, alkalosis can develop because the normal capacity to excrete HCO_3^- may be exceeded or there may be enhanced reabsorption of HCO_3^- . Such patients include those who receive oral or intravenous HCO_3^- , acetate loads (parenteral hyperalimentation solutions), citrate loads (transfusions), or antacids plus cation-exchange resins (aluminum hydroxide and sodium polystyrene sulfonate).

METABOLIC ALKALOSIS ASSOCIATED WITH [ECFV](#) CONTRACTION, K^+ DEPLETION, AND SECONDARY HYPERRENINEMIC HYPERALDOSTERONISM

Gastrointestinal Origin Gastrointestinal loss of H^+ from vomiting or gastric aspiration results in retention of HCO_3^- . The loss of fluid and NaCl in vomitus or nasogastric suction results in contraction of the [ECFV](#) and an increase in the secretion of renin and aldosterone. Volume contraction causes a reduction in [GFR](#) and an enhanced capacity of the renal tubule to reabsorb HCO_3^- . During active vomiting, there is continued addition of HCO_3^- to plasma in exchange for Cl^- , and the plasma $[\text{HCO}_3^-]$ exceeds the reabsorptive capacity of the proximal tubule. The excess NaHCO_3 reaches the distal tubule, where secretion is enhanced by an aldosterone and the delivery of the poorly reabsorbed anion, HCO_3^- . Because of contraction of the ECFV and hypochloremia, Cl^- is avidly conserved by the kidney. Correction of the contracted ECFV with NaCl and repair of K^+ deficits corrects the acid-base disorder.

Renal Origin

Diuretics (See also [Chap. 232](#)) Drugs that induce chloruresis, such as thiazides and loop diuretics (furosemide, bumetanide, torsemide, and ethracrynic acid), acutely diminish the [ECFV](#) without altering the total body bicarbonate content. The serum $[\text{HCO}_3^-]$ increases. The chronic administration of diuretics tends to generate an alkalosis by increasing distal salt delivery, so that K^+ and H^+ secretion are stimulated. The alkalosis is maintained by persistence of the contraction of the ECFV, secondary

hyperaldosteronism, K⁺ deficiency, and the direct effect of the diuretic (as long as diuretic administration continues). Repair of the alkalosis is achieved by providing isotonic saline to correct the ECFV deficit.

Bartter's Syndrome and Gitelman's Syndrome See [Chap. 276](#).

Nonreabsorbable Anions and Magnesium Deficiency Administration of large quantities of nonreabsorbable anions, such as penicillin or carbenicillin, can enhance distal acidification and K⁺ secretion by increasing the transepithelial potential difference (lumen negative). Mg²⁺ deficiency results in hypokalemic alkalosis by enhancing distal acidification through stimulation of renin and hence aldosterone secretion.

Potassium Depletion Chronic K⁺ depletion may cause metabolic alkalosis by increasing urinary acid excretion. Both NH₄⁺ production and absorption are enhanced and HCO₃⁻ reabsorption is stimulated. Chronic K⁺ deficiency upregulates the renal H⁺, K⁺-ATPase to increase K⁺ absorption at the expense of enhanced H⁺ secretion. Alkalosis associated with severe K⁺ depletion is resistant to salt administration, but repair of the K⁺ deficiency corrects the alkalosis.

After Treatment of Lactic Acidosis or Ketoacidosis When an underlying stimulus for the generation of lactic acid or ketoacid is removed rapidly, as with repair of circulatory insufficiency or with insulin therapy, the lactate or ketones are metabolized to yield an equivalent amount of HCO₃⁻. Other sources of new HCO₃⁻ are additive with the original amount generated by organic anion metabolism to create a surfeit of HCO₃⁻. Such sources include (1) new HCO₃⁻ added to the blood by the kidneys as a result of enhanced acid excretion during the preexisting period of acidosis, and (2) alkali therapy during the treatment phase of the acidosis. Acidosis-induced contraction of the [ECFV](#) and K⁺ deficiency act to sustain the alkalosis.

Posthypercapnia Prolonged CO₂ retention with chronic respiratory acidosis enhances renal HCO₃⁻ absorption and the generation of new HCO₃⁻ (increased net acid excretion). If the PaCO₂ is returned to normal, metabolic alkalosis results from the persistently elevated [HCO₃⁻]. Alkalosis develops if the elevated PaCO₂ is abruptly returned toward normal by a change in mechanically controlled ventilation. Associated [ECFV](#) contraction does not allow complete repair of the alkalosis by correction of the PaCO₂ alone, and alkalosis persists until Cl⁻ supplementation is provided.

METABOLIC ALKALOSIS ASSOCIATED WITH ECFV EXPANSION, HYPERTENSION, AND HYPERALDOSTERONISM

Mineralocorticoid administration or excess production [primary aldosteronism of Cushing's syndrome and adrenal cortical enzyme defects ([Chap. 331](#))] increases net acid excretion and may result in metabolic alkalosis, which may be worsened by associated K⁺ deficiency. [ECFV](#) expansion from salt retention causes hypertension and antagonizes the reduction in [GFR](#) and/or increases tubule acidification induced by aldosterone and by K⁺ deficiency. The kaliuresis persists and causes continued K⁺ depletion with polydipsia, inability to concentrate the urine, and polyuria. Increased aldosterone levels may be the result of autonomous primary adrenal overproduction or of secondary aldosterone release due to renal overproduction of renin. In both

situations, the normal feedback of ECFV on net aldosterone production is disrupted, and hypertension from volume retention can result.

Liddle's syndrome ([Chap. 276](#)) results from increased activity of collecting duct Na⁺-channel (ENaC) and is a rare inherited disorder associated with hypertension due to volume expansion manifested as hypokalemic alkalosis and normal aldosterone levels.

Symptoms With metabolic alkalosis, changes in central and peripheral nervous system function are similar to those of hypocalcemia ([Chap. 340](#)); symptoms include mental confusion, obtundation, and a predisposition to seizures, paresthesia, muscular cramping, tetany, aggravation of arrhythmias, and hypoxemia in chronic obstructive pulmonary disease. Related electrolyte abnormalities include hypokalemia and hypophosphatemia.

TREATMENT

This is primarily directed at correcting the underlying stimulus for HCO₃⁻-generation. If primary aldosteronism is present, correction of the underlying cause will reverse the alkalosis. [H⁺] loss by the stomach or kidneys can be mitigated by the use of H₂receptor blockers, H⁺, K⁺-ATPase inhibitors, or the discontinuation of diuretics. The second aspect of treatment is to remove the factors that sustain HCO₃⁻-reabsorption, such as [ECFV](#) contraction or K⁺-deficiency. Although K⁺-deficits should be repaired, NaCl therapy is usually sufficient to reverse the alkalosis if ECFV contraction is present, as indicated by a low urine [Cl⁻].

If associated conditions preclude infusion of saline, renal HCO₃⁻-loss can be accelerated by administration of acetazolamide, a carbonic anhydrase inhibitor, which is usually effective in patients with adequate renal function but can worsen K⁺-losses. Dilute hydrochloric acid (0.1 N HCl) is also effective but can cause hemolysis. Alternatively, acidification can also be achieved with oral NH₄Cl, which should be avoided in the presence of liver disease. Hemodialysis against a dialysate low in [HCO₃⁻] and high in [Cl⁻] can be effective when renal function is impaired.

RESPIRATORY ACIDOSIS

Respiratory acidosis can be due to severe pulmonary disease, respiratory muscle fatigue, or abnormalities in ventilatory control and is recognized by an increase in PaCO₂ and decrease in pH ([Table 50-5](#)). In acute respiratory acidosis, there is an immediate compensatory elevation (due to cellular buffering mechanisms) in HCO₃⁻, which increases 1 mmol/L for every 10-mmHg increase in PaCO₂. In chronic respiratory acidosis (>24 h), renal adaptation increases the [HCO₃⁻] by 4 mmol/L for every 10-mmHg increase in PaCO₂. The serum HCO₃⁻ usually does not increase above 38 mmol/L.

The clinical features vary according to the severity and duration of the respiratory acidosis, the underlying disease, and whether there is accompanying hypoxemia. A rapid increase in PaCO₂ may cause anxiety, dyspnea, confusion, psychosis, and hallucinations and may progress to coma. Lesser degrees of dysfunction in chronic hypercapnia include sleep disturbances, loss of memory, daytime somnolence,

personality changes, impairment of coordination, and motor disturbances such as tremor, myoclonic jerks, and asterixis. Headaches and other signs that mimic raised intracranial pressure, such as papilledema, abnormal reflexes, and focal muscle weakness, are due to vasoconstriction secondary to loss of the vasodilator effects of CO_2 .

Depression of the respiratory center by a variety of drugs, injury, or disease can produce respiratory acidosis. This may occur acutely with general anesthetics, sedatives, and head trauma or chronically with sedatives, alcohol, intracranial tumors, and the syndromes of sleep-disordered breathing, including the primary alveolar and obesity-hypoventilation syndromes ([Chaps. 263](#) and [264](#)). Abnormalities or disease in the motor neurons, neuromuscular junction, and skeletal muscle can cause hypoventilation via respiratory muscle fatigue. Mechanical ventilation, when not properly adjusted and supervised, may result in respiratory acidosis, particularly if CO_2 production suddenly rises (because of fever, agitation, sepsis, or overfeeding) or alveolar ventilation falls because of worsening pulmonary function. High levels of positive end-expiratory pressure in the presence of reduced cardiac output may cause hypercapnia as a result of large increases in alveolar dead space ([Chap. 266](#)). Permissive hypercapnia is being used with increasing frequency because of studies suggesting lower mortality rates than with conventional mechanical ventilation, especially with severe central nervous system or heart disease. Although the potential beneficial effects of permissive hypercapnia may be mitigated by correction of the acidemia, it seems prudent, nevertheless, to keep the pH in the range of 7.2 to 7.3 by administration of NaHCO_3 .

Acute hypercapnia follows sudden occlusion of the upper airway or generalized bronchospasm as in severe asthma, anaphylaxis, inhalational burn, or toxin injury. Chronic hypercapnia and respiratory acidosis occur in end-stage obstructive lung disease. Restrictive disorders involving both the chest wall and the lungs can cause respiratory acidosis because the high metabolic cost of respiration causes ventilatory muscle fatigue. Advanced stages of intrapulmonary and extrapulmonary restrictive defects present as chronic respiratory acidosis.

The diagnosis of respiratory acidosis requires, by definition, the measurement of PaCO_2 and arterial pH. A detailed history and physical examination often indicate the cause. Pulmonary function studies ([Chap. 250](#)), including spirometry, diffusion capacity for carbon monoxide, lung volumes, and arterial PaCO_2 and O_2 saturation, usually make it possible to determine if respiratory acidosis is secondary to lung disease. The workup for nonpulmonary causes should include a detailed drug history, measurement of hematocrit, and assessment of upper airway, chest wall, pleura, and neuromuscular function.

TREATMENT

The management of respiratory acidosis depends on its severity and rate of onset. Acute respiratory acidosis can be life-threatening, and measures to reverse the underlying cause should be undertaken simultaneously with restoration of adequate alveolar ventilation. This may necessitate tracheal intubation and assisted mechanical ventilation. Oxygen administration should be titrated carefully in patients with severe

obstructive pulmonary disease and chronic CO₂ retention who are breathing spontaneously ([Chap. 258](#)). When oxygen is used in judiciously, these patients may experience progression of the respiratory acidosis. Aggressive and rapid correction of hypercapnia should be avoided, because the falling PaCO₂ may provoke the same complications noted with acute respiratory alkalosis (i.e., cardiac arrhythmias, reduced cerebral perfusion, and seizures). The PaCO₂ should be lowered gradually in chronic respiratory acidosis, aiming to restore the PaCO₂ to baseline levels and to provide sufficient Cl⁻ and K⁺ to enhance the renal excretion of HCO₃⁻.

Chronic respiratory acidosis is frequently difficult to correct, but measures aimed at improving lung function ([Chap. 258](#)) can help some patients and forestall further deterioration in most.

RESPIRATORY ALKALOSIS

Alveolar hyperventilation decreases PaCO₂ and increases the HCO₃⁻/PaCO₂ ratio, thus increasing pH ([Table 50-5](#)). Nonbicarbonate cellular buffers respond by consuming HCO₃⁻. Hypocapnia develops when a sufficiently strong ventilatory stimulus causes CO₂ output in the lungs to exceed its metabolic production by tissues. Plasma pH and [HCO₃⁻] appear to vary proportionately with PaCO₂ over a range from 40 to 15 mmHg. The relationship between arterial [H⁺] concentration and PaCO₂ is about 0.7 mmol/L per mmHg (or 0.01 pH unit/mmHg), and that for plasma [HCO₃⁻] is 0.2 mmol/L per mmHg. Hypocapnia sustained longer than 2 to 6 h is further compensated by a decrease in renal ammonium and titrable acid excretion and a reduction in filtered HCO₃⁻ reabsorption. Full renal adaptation to respiratory alkalosis may take several days and requires normal volume status and renal function. The kidneys appear to respond directly to the lowered PaCO₂ rather than to alkalosis per se. In chronic respiratory alkalosis a 1-mmHg fall in PaCO₂ causes a 0.4- to 0.5-mmol/L drop in [HCO₃⁻] and a 0.3-mmol/L fall (or 0.003 rise in pH) in [H⁺].

The effects of respiratory alkalosis vary according to duration and severity but are primarily those of the underlying disease. Reduced cerebral blood flow as a consequence of a rapid decline in PaCO₂ may cause dizziness, mental confusion, and seizures, even in the absence of hypoxemia. The cardiovascular effects of acute hypocapnia in the conscious human are generally minimal, but in the anesthetized or mechanically ventilated patient, cardiac output and blood pressure may fall because of the depressant effects of anesthesia and positive-pressure ventilation on heart rate, systemic resistance, and venous return. Cardiac arrhythmias may occur in patients with heart disease as a result of changes in oxygen unloading by blood from a left shift in the hemoglobin-oxygen dissociation curve (Bohr effect). Acute respiratory alkalosis causes intracellular shifts of Na⁺, K⁺, and PO₄⁻ and reduces free [Ca²⁺] by increasing the protein-bound fraction. Hypocapnia-induced hypokalemia is usually minor.

Chronic respiratory alkalosis is the most common acid-base disturbance in critically ill patients and, when severe, portends a poor prognosis. Many cardiopulmonary disorders manifest respiratory alkalosis in their early to intermediate stages, and the finding of normocapnia and hypoxemia in a patient with hyperventilation may herald the onset of rapid respiratory failure and should prompt an assessment to determine if the patient is becoming fatigued. Respiratory alkalosis is common during mechanical ventilation.

The hyperventilation syndrome may be disabling. Paresthesia, circumoral numbness, chest wall tightness or pain, dizziness, inability to take an adequate breath, and, rarely, tetany may themselves be sufficiently stressful to perpetuate the disorder. Arterial blood-gas analysis demonstrates an acute or chronic respiratory alkalosis, often with hypocapnia in the range of 15 to 30 mmHg and no hypoxemia. Central nervous system diseases or injury can produce several patterns of hyperventilation and sustained P_{aCO_2} levels of 20 to 30 mmHg. Hyperthyroidism, high caloric loads, and exercise raise the basal metabolic rate, but ventilation usually rises in proportion so that arterial blood gases are unchanged and respiratory alkalosis does not develop. Salicylates are the most common cause of drug-induced respiratory alkalosis as a result of direct stimulation of the medullary chemoreceptor ([Chap. 396](#)). The methylxanthines, theophylline, and aminophylline stimulate ventilation and increase the ventilatory response to CO_2 . Progesterone increases ventilation and lowers arterial P_{aCO_2} by as much as 5 to 10 mmHg. Therefore, chronic respiratory alkalosis is a common feature of pregnancy. Respiratory alkalosis is also prominent in liver failure, and the severity correlates with the degree of hepatic insufficiency. Respiratory alkalosis is often an early finding in gram-negative septicemia, before fever, hypoxemia, or hypotension develop.

The diagnosis of respiratory alkalosis depends on measurement of arterial pH and P_{aCO_2} . The plasma $[K^+]$ is often reduced and the $[Cl^-]$ increased. In the acute phase, respiratory alkalosis is not associated with increased renal HCO_3^- excretion, but within hours net acid excretion is reduced. In general, the HCO_3^- concentration falls by 2.0 mmol/L for each 10-mmHg decrease in P_{aCO_2} . Chronic hypocapnia reduces the serum $[HCO_3^-]$ by 5.0 mmol/L for each 10-mmHg decrease in P_{aCO_2} . It is unusual to observe a plasma $HCO_3^- < 12$ mmol/L as a result of a pure respiratory alkalosis.

When a diagnosis of respiratory alkalosis is made, its cause should be investigated. The diagnosis of hyperventilation syndrome is made by exclusion. In difficult cases, it may be important to rule out other conditions such as pulmonary embolism, coronary artery disease, and hyperthyroidism.

TREATMENT

The management of respiratory alkalosis is directed toward alleviation of the underlying disorder. If respiratory alkalosis complicates ventilator management, changes in dead space, tidal volume, and frequency can minimize the hypocapnia. Patients with the hyperventilation syndrome may benefit from reassurance, rebreathing from a paper bag during symptomatic attacks, and attention to underlying psychological stress. Antidepressants and sedatives are not recommended. β -Adrenergic blockers may ameliorate peripheral manifestations of the hyperadrenergic state.

(Bibliography omitted in Palm version)

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SECTION 8 -ALTERATIONS IN SEXUAL FUNCTION AND REPRODUCTION

51. ERECTILE DYSFUNCTION - Kevin T. McVary

Erectile dysfunction (ED) affects 10 to 25% of middle-aged and elderly men. Demographic changes, the popularity of newer treatments, and greater acceptance of ED by patients and society have led to increased diagnosis and associated health care expenditures for the management of this common disorder. Impairment of erectile function has a profound impact on the well-being of affected men. Because many patients are reluctant to initiate discussion of sexual function, the physician should address this topic directly to elicit a history of ED.

PHYSIOLOGIC CONTROL OF ERECTION AND MALE SEXUAL FUNCTION

Normal male sexual function requires (1) an intact libido, (2) the ability to achieve and maintain penile erection, (3) ejaculation, and (4) detumescence. *Libido* refers to sexual desire and is influenced by a variety of visual, olfactory, tactile, auditory, imaginative and hormonal stimuli. Sex steroids, particularly testosterone, act to increase libido. Libido can be diminished by hormonal or psychiatric disorders or by medications.

The major anatomic structures of the penis that are involved in erectile function include the three corpora, which consist of the paired cavernosa and a single spongiosum that encloses the urethra. A collagenous sheath, called the *tunica albuginea*, individually surrounds each corpora. The micro-architecture of the corpora is composed of a mass of smooth muscle (trabecula) which contains a network of endothelial-lined vessels (lacunar spaces).

Penile tumescence leading to erection depends on the increased flow of blood into the lacunar network after complete relaxation of the arteries and corporal smooth muscle. Subsequent compression of the trabecular smooth muscle against the fibroelastic tunica albuginea causes a passive closure of the emissary veins and accumulation of blood in the corpora. In the presence of a full erection and a competent valve mechanism, the corpora become noncompressible cylinders from which blood does not escape.

The central nervous system exerts an important influence by either stimulating or antagonizing spinal pathways that mediate erectile function and ejaculation. The erectile response is mediated by a combination of central (psychogenic) and peripheral (reflexogenic) innervation. Sensory nerves that originate from receptors in the penile skin and glans converge to form the dorsal nerve of the penis, which travels to the S2-S4 dorsal root ganglia via the pudendal nerve. Parasympathetic nerve fibers to the penis arise from neurons in the intermediolateral columns of S2-S4 sacral spinal segments. Sympathetic innervation originates from the T-11 to the L-2 spinal segments and descends through the hypogastric plexus.

Neural input to smooth muscle tone is crucial to the initiation and maintenance of an erection. There is also an intricate interaction between the corporal smooth muscle cell and its overlying endothelial cell lining ([Fig. 51-1A](#)). Nitric oxide, which induces vascular relaxation, promotes erection and is opposed by endothelin-1 (ET-1), which mediates vascular contraction. Nitric oxide is synthesized from L-arginine by nitric oxide synthase,

and is released from the nonadrenergic, noncholinergic (NANC) autonomic nerve supply to act postjunctionally on smooth muscle cells. Nitric oxide increases the production of cyclic 3',5'-guanosine monophosphate (cyclic GMP), which interacts with protein kinase G and decreases intracellular calcium, causing relaxation of the smooth muscle ([Fig. 51-1B](#)). Cyclic GMP is gradually broken down by phosphodiesterase type 5 (PDE-5). Inhibitors of PDE-5, such as the oral medication sildenafil, maintain erections by reducing the breakdown of cyclic GMP. However, if nitric oxide is not produced at some level, the addition of PDE-5 inhibitor is not effective, as the drug facilitates but does not initiate the initial enzyme cascade. In addition to nitric oxide, vasoactive prostaglandins (PGE₁, PGF_{2a}) are synthesized within the cavernosal tissue and increase cyclic AMP levels, also leading to relaxation of cavernosal smooth muscle cells.

Ejaculation is stimulated by the sympathetic nervous system, which results in contraction of the epididymis, vas deferens, seminal vesicles, and prostate, causing seminal fluid to enter the urethra. Seminal fluid emission is followed by rhythmic contractions of the bulbocavernosus and ischiocavernosus muscles, leading to ejaculation. *Premature ejaculation* is usually related to anxiety or a learned behavior and is amenable to behavioral therapy or treatment with medications such as selective serotonin reuptake inhibitors (SSRIs). *Retrograde ejaculation* results when the internal urethral sphincter does not close, and it may occur in men with diabetes or after surgery involving the bladder neck.

Detumescence is mediated by released norepinephrine from the sympathetic nerves, release of endothelin from the vascular surface, and contraction of smooth muscle induced by activation of postsynaptic α -adrenergic receptors. These events increase venous outflow and restore the flaccid state. Venous leak can cause premature detumescence and is thought to be caused by insufficient relaxation of the corporal smooth muscle rather than a specific anatomic defect. *Priapism* refers to a persistent and painful erection and may be associated with sickle cell anemia, hypercoagulable states, spinal cord injury, or injection of vasodilator agents into the penis.

ERECTILE DYSFUNCTION

EPIDEMIOLOGY

In the Massachusetts Male Aging Study (MMAS), a community-based survey of men between the ages of 40 and 70, 52% of responders reported some degree of [ED](#). Complete ED occurred in 10% of respondents, moderate ED occurred in 25%, and minimal ED in 17%. The incidence of moderate or severe ED more than doubled between the ages of 40 and 70. In the National Health and Social Life Survey (NHSLS), which was a nationally representative sample of men and women age 18 to 59 years, 10% of men reported being unable to maintain an erection (corresponding to the proportion of men in the MMAS reporting severe ED). Incidence was highest among men in the 50 to 59 age group (21%) and among men who were poor (14%), divorced (14%), and less educated (13%).

The incidence of [ED](#) is also higher among men with certain medical disorders. In the [MMAS](#), ED correlated with the presence of diabetes mellitus, heart disease, hypertension, and decreased HDL levels. Medications used to treat diabetes or

cardiovascular disease are additional risk factors (see below). There is a higher incidence of ED among men who have undergone radiation or surgery for cancer of the prostate and in those with a lower spinal cord injury. Psychological causes of ED include depression and anger. The [NHSLS](#) found a higher incidence of ED among men who reported fair-to-poor health or experienced stress from unemployment or other causes. ED is not considered a normal part of the aging process. Nonetheless, it is associated with certain physiologic and psychological changes related to age.

PATHOPHYSIOLOGY

[ED](#) may result from three basic mechanisms: (1) failure to initiate (psychogenic, endocrinologic, or neurogenic); (2) failure to fill (arteriogenic); or (3) failure to store (venoocclusive dysfunction) adequate blood volume within the lacunar network. The inability to initiate an erection may have psychogenic, endocrinologic, or neurogenic etiologies. These categories are not mutually exclusive, and multiple factors contribute to ED in many patients. For example, diminished filling pressure can lead secondarily to venous leak. Psychogenic factor frequently co-exist with other etiologic factors and should be considered in all cases. Diabetic, atherosclerotic, and drug-related causes account for >80% of cases of ED in older men.

Vasculogenic The most frequent organic cause of [ED](#) is a disturbance of blood flow to and from the penis. Atherosclerotic or traumatic arterial disease can decrease flow to the lacunar spaces, resulting in decreased rigidity and an increased time to full erection. Excessive outflow through the veins, despite adequate inflow, may also contribute to ED. In this case, the achieved perfusion pressures cannot compensate for the unrestricted outflow needed to ensure adequate erection. This situation may be due to insufficient relaxation of trabecular smooth muscle and may occur in anxious individuals with excessive adrenergic tone or in those with damaged parasympathetic outflow. Structural alterations to the fibroelastic components of the corpora may cause a loss of compliance and an inability to compress the tunical veins. This condition may result from aging, increased cross-leaking of collagen fibers induced by nonenzymatic glycosylation, hypoxia, or altered synthesis of collagen associated with hypercholesterolemia. Fibroelastic structures can also be damaged by surgery, radiation, or trauma to the penis.

Neurogenic Disorders that affect the sacral spinal cord or the autonomic fibers to the penis preclude nervous system relaxation of penile smooth muscle, thus leading to [ED](#). In patients with spinal cord injury, the degree of ED depends on the completeness and level of the lesion. Patients with incomplete lesions or injuries to the upper part of the spinal cord are more likely to retain erectile capabilities than those with complete lesions or injuries to the lower part. Although 75% of patients with spinal cord injuries have some erectile capability, only 25% have erections sufficient for penetration. Other neurologic disorders commonly associated with ED include multiple sclerosis and peripheral neuropathy. The latter is often due to either diabetes or alcoholism. Pelvic surgery may cause ED through disruption of the autonomic nerve supply.

Endocrinologic Androgens increase libido, but their exact role in erectile function remains unclear. Individuals with castrate levels of testosterone can achieve erections from visual or sexual stimuli. Nonetheless, normal levels of testosterone appear to be

important for erectile function, particularly in older males. Androgen replacement therapy can improve depressed erectile function when it is secondary to hypogonadism; it is not useful for ED when endogenous testosterone levels are normal. Increased prolactin may decrease libido by suppressing gonadotropin-releasing hormone (GnRH), and it also leads to decreased testosterone levels. Treatment of hyperprolactinemia with dopamine agonists can restore libido and testosterone.

Diabetic ED occurs in 35 to 75% of men with diabetes mellitus. Pathologic mechanisms are primarily related to diabetes-associated vascular and neurologic complications. Diabetic macrovascular complications are mainly related to age, whereas microvascular complications correlate with the duration of diabetes and the degree of glycemic control ([Chap. 333](#)). Individuals with diabetes also have reduced amounts of nitric oxide synthase in both endothelial and neural tissues.

Psychogenic Two mechanisms contribute to the inhibition of erections in psychogenic [ED](#). First, psychogenic stimuli to the sacral cord may inhibit reflexogenic responses, thereby blocking activation of vasodilator outflow to the penis. Second, excess sympathetic stimulation in an anxious man may increase penile smooth muscle tone. The most common causes of psychogenic ED are performance anxiety, depression, relationship conflict, loss of attraction, sexual inhibition, conflicts over sexual preference, sexual abuse in childhood, and fear of pregnancy or sexually transmitted disease. Almost all patients with ED, even when it has a clear-cut organic basis, develop a psychogenic component as a reaction to ED.

Medication-Related Medication-induced [ED](#) ([Table 51-1](#)) is estimated to occur in 25% of men seen in general medical outpatient clinics. Among the antihypertensive agents, the thiazide diuretics and beta blockers have been implicated most frequently. Calcium channel blockers and angiotensin-converting enzyme inhibitors are less frequently cited. These drugs may act directly at the corporal level (e.g., calcium channel blockers) or indirectly by reducing pelvic blood pressure, which is important in the development of penile rigidity. Alpha adrenergic blockers are less likely to cause ED. Estrogens, [GnRH](#) agonists, H_2 antagonists, and spironolactone cause ED by suppressing gonadotropin production or by blocking androgen action. Antidepressant and antipsychotic agents -- particularly neuroleptics, tricyclics, and [SSRIs](#) -- are associated with erectile, ejaculatory, orgasmic, and sexual desire difficulties. Digoxin induces ED via blockade of the Na^+, K^+ -ATPase pump, resulting in a net increase in intracellular calcium and increased corporal smooth muscle tone.

Although many medications can cause [ED](#), patients frequently have concomitant risk factors that confound the clinical picture. If there is a strong association between the institution of a drug and the onset of ED, alternative medications should be considered. Otherwise, it is often practical to treat the ED without attempting multiple changes in medications, as it may be difficult to establish a causal role for the drug.

CLINICAL EVALUATION

A good physician-patient relationship helps to unravel the possible causes of [ED](#), many of which require discussion of personal and sometimes embarrassing topics. For this reason, a primary care provider is often ideally suited to initiate the evaluation. A

complete medical and sexual history should be taken in an effort to assess whether the cause of ED is organic, psychogenic, or multifactorial ([Fig. 51-2](#)). Initial questions should focus on the onset of symptoms, the presence and duration of partial erections, and the progression of ED. A history of nocturnal or early morning erections is useful for distinguishing physiologic from psychogenic ED. Nocturnal erections occur during rapid eye movement (REM) sleep and require intact neurologic and circulatory systems. Organic causes of ED are generally characterized by a gradual and persistent change in rigidity or the inability to sustain nocturnal, coital, or self-stimulated erections. The patient should also be questioned about the presence of penile curvature or pain with coitus. It is also important to address libido, as decreased sexual drive and ED are sometimes the earliest signs of endocrine abnormalities (e.g., increased prolactin, decreased testosterone levels). It is useful to ask whether the problem is confined to coitus with one or other partners; ED arises not uncommonly in association with new or extramarital sexual relationships. Situational ED, as opposed to consistent ED, suggests psychogenic causes. Ejaculation is much less commonly affected than erection, but questions should be asked about whether ejaculation is normal, premature, delayed, or absent. Relevant risk factors should be identified, such as diabetes mellitus, coronary artery disease, lipid disorders, hypertension, peripheral vascular disease, smoking, alcoholism, and endocrine or neurologic disorders. The patient's surgical history should be explored with an emphasis on bowel, bladder, prostate, or vascular procedures. A complete drug history is also important, as medications constitute a major source of reversible ED. Social changes that may precipitate ED are also crucial to the evaluation, including health worries, spousal death, divorce, relationship difficulties, and financial concerns.

The physical examination is an essential element in the assessment of [ED](#). Signs of hypertension as well as evidence of thyroid, hepatic, hematologic, cardiovascular, or renal diseases should be sought. An assessment should be made of the endocrine and vascular systems, the external genitalia, and the prostate gland. The penis should be carefully palpated along the corpora to detect fibrotic plaques. Reduced testicular size and loss of secondary sexual characteristics are suggestive of hypogonadism. Neurologic examination should include assessment of anal sphincter tone, the bulbocavernosus reflex, and testing for peripheral neuropathy.

Selected laboratory testing is recommended in all cases. Although hyperprolactinemia is uncommon, a serum prolactin level should be measured, as decreased libido and/or erectile dysfunction may be the presenting symptoms of a prolactinoma or other mass lesions of the sella ([Chap. 328](#)). The serum testosterone level should be measured and, if low, gonadotropins should be measured to determine whether hypogonadism is primary (testicular) or secondary (hypothalamic-pituitary) in origin ([Chap. 335](#)). Serum chemistries, CBC, and lipid profiles may be of value, if not performed recently, as they can yield evidence of anemia, diabetes, hyperlipidemia, or other systemic diseases associated with [ED](#). Determination of serum PSA should be conducted according to recommended clinical guidelines ([Chap. 95](#)).

Additional diagnostic testing is rarely necessary in the evaluation of ED. However, in selected patients, specialized testing may provide insight into pathologic mechanisms of ED and aid in the selection of treatment options. Optional specialized testing includes: (1) studies of nocturnal penile tumescence and rigidity; (2) vascular testing (in-office

injection of vasoactive substances, penile Doppler ultrasound, penile angiography, dynamic infusion cavernosography/cavernosometry); (3) neurologic testing (biothesiometry-graded vibratory perception; somatosensory evoked potentials); and (4) psychological diagnostic tests. The information potentially gained from these procedures must be balanced against their invasiveness and cost.

TREATMENT

Patient Education Patient and partner education is essential in the treatment of [ED](#). In goal-directed therapy, education facilitates understanding of the disease, results of the tests, and selection of treatment. Discussion of treatment options helps to clarify how treatment is best offered, and to stratify first- and second-line therapies. Patients with high-risk lifestyle issues, such as smoking, alcohol abuse, or recreational drug use, should be counseled on the role these factors play in the development of ED.

Oral Agents Sildenafil is the only approved and effective oral agent for the treatment of [ED](#). Sildenafil has markedly improved the management of ED because it is effective for the treatment of a broad range of causes of ED, including psychogenic, diabetic, vasculogenic, post-radical prostatectomy (nerve-sparing procedures), and spinal cord injury. Sildenafil is a selective and potent inhibitor of PDE-5, the predominant phosphodiesterase isoform found in the penis. It is administered in doses of 25, 50, or 100 mg, and enhances erections after sexual stimulation. The onset of action is approximately 60 to 90 min. Reduced initial doses should be considered for patients who are elderly, have renal insufficiency, or are taking medications that inhibit the CYP3A4 metabolic pathway in the liver (e.g., erythromycin, cimetidine, ketoconazole, and, possibly, itraconazole and mibefradil), as they may increase the serum concentration of sildenafil. The drug does not affect ejaculation, orgasm, or sexual drive. Side effects associated with sildenafil include headaches (19%), facial flushing (9%), dyspepsia (6%) and nasal congestion (4%). Approximately 7% of men may experience transient altered color vision (blue halo effect). Sildenafil is contraindicated in men receiving nitrate therapy for cardiovascular disease, including agents delivered by oral, sublingual, transnasal, or topical routes. These agents can potentiate its hypotensive effect and may result in profound shock. Likewise, amyl/butyl nitrates (poppers) may have a fatal synergistic effect on blood pressure. Sildenafil should also be avoided in patients with congestive heart failure and cardiomyopathy because of the risk of vascular collapse. Because sexual activity leads to an increase in physiologic expenditure [5 to 6 metabolic equivalents (METS)], physicians have been advised to exercise caution in prescribing any drug for sexual activity to those with active coronary disease, heart failure, borderline hypotension, hypovolemia, and to those on complex antihypertensive regimens.

Androgen Therapy Testosterone replacement is used to treat both primary and secondary causes of hypogonadism ([Chap. 335](#)). Androgen supplementation in the setting of normal testosterone is rarely efficacious and is discouraged. Methods of androgen replacement include parenteral administration of long-acting testosterone esters (enanthate and cypionate), oral preparations (17 α -alkylated derivatives), and transdermal patches ([Chap. 335](#)). The long-acting 17 β -hydroxy esters of testosterone are the safest, most cost-effective, and practical preparations available. The administration of 200 to 300 mg intramuscularly every 2 to 3 weeks provides a practical

option but is far from an ideal physiologic replacement. Oral androgen preparations have the potential for hepatotoxicity and should be avoided. Transdermal delivery of testosterone more closely mimics physiologic testosterone levels, but it is unclear whether this translates into improved sexual function. Because testosterone gradually decreases into the hypogonadal range by 24 hours, patches need to be replaced daily. Testosterone therapy is contraindicated in men with androgen-sensitive cancers and may be inappropriate for men with bladder neck obstruction. It is generally advisable to measure PSA before giving androgen. Hepatic function should be tested before and during testosterone therapy.

Vacuum Constriction Devices Vacuum constriction devices (VCD) are a well-established, noninvasive therapy. They are a reasonable treatment alternative for select patients who cannot take sildenafil or do not desire other interventions. VCD draw venous blood into the penis and use a constriction ring to restrict venous return and maintain tumescence. Adverse events with VCD include pain, numbness, bruising, and altered ejaculation. Additionally, many patients complain that the devices are cumbersome and that the induced erections have a non-physiologic appearance.

Intraurethral Alprostadil If a patient fails to respond to oral agents, a reasonable next choice is intraurethral or self-injection or vasoactive substances. Intraurethral prostaglandin E₁ (alprostadil), in the form of a semisolid pellet (doses of 125 to 1000 ug), is delivered with an applicator. Approximately 65% of men receiving intraurethral alprostadil respond with an erection when tested in the office, but only 50% of those achieve successful coitus at home. Intraurethral insertion is associated with a markedly reduced incidence of priapism in comparison to intracavernosal injection.

Intracavernosal Self-Injection Injection of synthetic formulations of alprostadil is effective in 70 to 80% of patients with [ED](#), but discontinuation rates are high because of the invasive nature of administration. Doses range between 1 and 40 ug. Injection therapy is contraindicated in men with a history of hypersensitivity to the drug and in men at risk for priapism (hypercoagulable states, sickle cell disease). Side effects include local adverse events, prolonged erections, pain, and fibrosis with chronic use. Various combinations of alprostadil, phentolamine, and/or papaverine are sometimes used.

Surgery A less frequently used form of therapy for [ED](#) involves the surgical implantation of a semi-rigid or inflatable penile prosthesis. These surgical treatments are invasive, associated with potential complications, and generally reserved for treatment of refractory ED. Despite their high cost and invasiveness, penile prostheses are associated with high rates of patient satisfaction.

Sex Therapy A course of sex therapy may be useful for addressing specific interpersonal factors that may affect sexual functioning. Sex therapy generally consists of in-session discussion and at-home exercises specific to the person and the relationship. It is preferable if therapy includes both partners, provided the patient is involved in an ongoing relationship.

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52. DISTURBANCES OF MENSTRUATION AND OTHER COMMON GYNECOLOGIC COMPLAINTS IN WOMEN - *Bruce R. Carr, Karen D. Bradshaw*

Complaints related to the female reproductive tract can be categorized as disorders of menstruation, pelvic pain, disturbances in sexual function, or infertility. However, a single disorder, e.g., leiomyoma of the uterus, can present with symptoms referable to any one or more of these categories. Furthermore, sexual dysfunction can interdigitate with other problems in several ways. On the one hand, in women with complaints related to other reproductive tract functions, the underlying problem may actually be severe sexual dysfunction or marital conflict. Alternatively, women with severe organic disorders of the pelvis, e.g., pelvic inflammatory disease or endometriosis, may present with sexual dysfunction such as dyspareunia that in fact is only a minor manifestation of the underlying disease.

Since normal reproductive function depends on the integrated action of the central nervous system, the endocrine glands, and the reproductive organs, menstrual cycle abnormalities, sexual dysfunction, and infertility may be the result of systemic and psychological disorders as well as of primary defects in the endocrine and reproductive organs. The endocrine and physiologic control -- normal and abnormal -- of puberty, reproductive life, and menopause are discussed in [Chap. 336](#). The focus of this chapter is on the initial evaluation of women with disturbances of the reproductive tract.

DISTURBANCES IN MENSTRUATION

Disorders of menstruation can be divided into abnormal uterine bleeding and amenorrhea.

Abnormal Uterine Bleeding The menstrual cycle is defined as the interval between the onset of one bleeding episode and the onset of the next. In normal women the cycle averages 28 ± 3 days, the mean duration of menstrual flow is 4 ± 2 days, and the average blood loss is 35 to 80 mL. Between menarche and menopause most women experience one or more episodes of abnormal uterine bleeding, here defined as any bleeding pattern outside the normal ranges of frequency, duration, and/or amount of blood loss. The decision to evaluate a patient depends on the severity and frequency of the abnormal bleeding pattern.

When vaginal bleeding occurs, it should first be determined whether the blood is derived from the uterine endometrium. Rectal, bladder, cervical, and vaginal sources of bleeding must be excluded. Once the bleeding is established to be uterine in origin, a pregnancy-related disorder (such as threatened or incomplete abortion or ectopic pregnancy) must be ruled out by physical examination and appropriate laboratory tests. It should also be remembered that uterine bleeding may also be the initial or principal manifestation of a generalized bleeding diathesis. The remaining causes of abnormal uterine bleeding can be divided into those associated with ovulatory or anovulatory cycles.

Ovulatory Cycles Menstrual bleeding with ovulatory cycles is spontaneous, regular in onset, predictable in duration and amount of flow, and frequently associated with discomfort; it is the consequence of progesterone withdrawal at the end of the luteal

(postovulatory) phase and requires prior estrogen priming of the endometrium during the follicular (preovulatory) phase of the cycle. When deviations from an established pattern of menstrual flow occur but the cycles are still regular, the usual cause is disease of the outflow tract. For example, regular, prolonged, excessive bleeding episodes can result from abnormalities of the uterus such as submucous leiomyomas, adenomyosis, or endometrial polyps. On the other hand, cyclic, predictable menstruation characterized by spotting or light bleeding suggests obstruction of the outflow tract as with uterine synechiae or scarring of the cervix. Intermittent bleeding between cyclic ovulatory menses is often due to cervical or endometrial lesions.

Anovulatory Cycles Uterine bleeding that is irregular in occurrence, unpredictable as to amount and duration of flow, and usually painless is called *dysfunctional or anovulatory uterine bleeding*. This type of bleeding is the result of a failure of normal follicular maturation with consequent anovulation and may be either transient or chronic. Transient disruption of ovulatory cycles occurs most often in the early menarcheal years, during the perimenopausal period, or as the consequence of a variety of stresses and intercurrent illnesses. Persistent dysfunctional uterine bleeding during the reproductive years can occur in several organic diseases that affect ovarian function and is most often due to estrogen breakthrough bleeding. Estrogen breakthrough bleeding occurs when estrogen stimulation of the endometrium is continuous and is not interrupted by cyclic progesterone withdrawal, as can occur in polycystic ovarian disease.

Amenorrhea *Amenorrhea* is defined either as failure of menarche by age 16, regardless of the presence or absence of secondary sexual characteristics, or as the absence of menstruation for 6 months in a woman with previous periodic menses. Amenorrhea in a woman who has never menstruated is termed *primary amenorrhea*; cessation of menses is termed *secondary amenorrhea*. Because some disorders can cause both primary and secondary amenorrhea, we prefer a functional classification based on the nature of the underlying defect, namely, anatomic defects of the outflow tract (uterus, cervix, or vagina), ovarian failure, and chronic anovulation.

Anatomic defects of the outflow tract include congenital defects of the vagina, imperforate hymen, transverse vaginal septa, cervical stenosis, intrauterine adhesions (synechiae), absence of the vagina or uterus, and uterine maldevelopment. The diagnosis of an anatomic defect is usually made by physical examination but may be confirmed by demonstrating failure of bleeding following administration of estrogen plus a progestogen for 21 days. Pelvic ultrasonography, magnetic resonance imaging, hysterosalpingogram, or hysteroscopy may be helpful in defining the defect.

Causes of *ovarian failure* include gonadal dysgenesis, deficiency of 17 α -hydroxylase, resistant ovary syndrome, and premature ovarian failure. Ovarian failure encompasses disorders in which the ovary is deficient in germ cells and those in which the germ cells are resistant to follicle-stimulating hormone (FSH). The diagnosis of ovarian failure as the cause of amenorrhea is confirmed by an elevated plasma FSH level.

Women with *chronic anovulation* fail to ovulate spontaneously but have the capability of ovulating with appropriate therapy. In some women with chronic anovulation, total estrogen production is adequate, but it is not secreted in a cyclic fashion. In others,

estrogen production is deficient.

Women who have adequate estrogen production and demonstrate withdrawal bleeding after progestogen challenge often have polycystic ovarian disease (see [Fig. 336-8](#)). Other causes include hormone-secreting ovarian and adrenal tumors. Women with deficient or absent estrogen production, and therefore with absence of withdrawal bleeding after progestogen administration, usually have hypogonadotropic hypogonadism due to organic or functional disorders of the pituitary or central nervous system such as brain tumors, pituitary tumors (especially prolactin-secreting adenomas), primary hypopituitarism, or Sheehan's syndrome.

PELVIC PAIN

Pelvic pain may originate in the pelvis or be referred from another region of the body. A pelvic source is suggested by the history (e.g., dysmenorrhea and dyspareunia) and physical findings, but a high index of suspicion must be entertained for extrapelvic disorders that refer to the pelvis, such as appendicitis, diverticulitis, cholecystitis, intestinal obstruction, and urinary tract infections ([Chap. 14](#)).

"Physiologic" Pelvic Pain

Pain Associated with Ovulation ("Mittelschmerz") Many women experience low abdominal discomfort with ovulation, typically a dull aching pain at midcycle in one lower quadrant lasting from minutes to hours. It is rarely severe or incapacitating. The pain may result from peritoneal irritation by follicular fluid released into the peritoneal cavity at ovulation. The onset at midcycle and short duration of pain suggest this diagnosis.

Premenstrual or Menstrual Pain In normal ovulatory women, somatic symptoms during the few days prior to menses may be insignificant or disabling. Such symptoms include edema, breast engorgement, and abdominal bloating or discomfort. A symptom complex of cyclic irritability, depression, and lethargy is known as the *premenstrual syndrome* (PMS). PMS appears to be caused by changes in gonadal steroid levels. Although there is no consensus about therapy, randomized, controlled trials suggest significant improvement with the daily use of serotonin-reuptake inhibitors.

Severe or incapacitating uterine cramping during ovulatory menses and in the absence of demonstrable disorders of the pelvis is termed *primary dysmenorrhea*. Primary dysmenorrhea is caused by prostaglandin-induced uterine ischemia and is treated with prostaglandin synthetase inhibitors and/or oral contraceptive agents.

Pelvic Pain due to Organic Causes Severe dysmenorrhea associated with disease of the pelvis is termed *secondary dysmenorrhea*. Organic causes of pelvic pain can be classified as (1) uterine, (2) adnexal, (3) vulvar or vaginal, and (4) pregnancy-associated.

Uterine Pain Pain of uterine etiology is often chronic and continuous and increases in intensity during menstruation and intercourse. Causes include leiomyomas of the uterus (particularly submucous and degenerating leiomyomas), adenomyosis, and cervical stenosis. Infections of the uterus associated with intrauterine manipulation following

dilatation and curettage or with the insertion of intrauterine devices can also cause pelvic pain ([Chap. 336](#)). Pelvic pain due to endometrial or cervical cancer is usually a late manifestation ([Chap. 336](#)).

Adnexal Pain The most common cause of pain in the adnexae (fallopian tubes and ovaries) is infection ([Chap. 133](#)). Acute salpingo-oophoritis presents as low abdominal pain, fever, and chills; begins a few days after a menstrual period; and is usually due to chlamydial or gonococcal disease with or without a superimposed pyogenic infection. Chronic pelvic inflammatory disease results from either a single episode or multiple episodes of infection and may present as infertility associated with chronic pelvic pain that increases in intensity with menses and intercourse. On physical examination, cervical motion tenderness, adnexal tenderness, and adnexal thickening and/or masses may be present. Pelvic inflammatory disease may become a surgical emergency if peritonitis results from rupture of a tuboovarian abscess. Ovarian cysts or neoplasms may cause pelvic pain that becomes more severe with torsion or rupture of the mass, and ectopic pregnancy must be considered in the differential diagnosis (see below). Endometriosis involving fallopian tubes, ovaries, or peritoneum may cause both chronic low abdominal pain and infertility; the magnitude of tissue involvement does not always correlate with the severity of symptoms. Endometriosis pain typically increases with menstruation and, if the posterior ligaments of the uterus are involved, with intercourse.

Vulvar or Vaginal Pain Pain in these areas is most often due to infectious vaginitis caused by *Monilia*, *Trichomonas*, or bacteria and is characteristically associated with vaginal discharge and pruritus. Herpetic vulvitis, other dermatologic conditions of the vulva, condyloma acuminatum, and cysts or abscesses of Bartholin's glands may also cause vulvar pain.

Pregnancy-Associated Disorders Pregnancy must be considered in the differential diagnosis of pelvic pain during the reproductive years. Threatened abortion or incomplete abortion often presents with uterine cramping, bleeding, or passage of tissue following a period of amenorrhea. Ectopic pregnancy may be insidious in presentation or result in abrupt intraperitoneal hemorrhage and maternal death.

Evaluation of Pelvic Pain The evaluation of pelvic pain requires a careful history and pelvic examination. This often leads to the correct diagnosis and institution of appropriate treatment. If the pain is severe and the diagnosis is unclear, the workup should follow that outlined for the acute abdomen ([Chap. 14](#)). A culdocentesis may be indicated if a ruptured ectopic pregnancy is suspected. If there is a question of an adnexal mass or if the patient is so obese as to preclude a thorough pelvic examination, abdominal or vaginal sonography may be useful. Serial human chorionic gonadotropin (hCG) measurements may help in establishing a diagnosis of tubal pregnancy and are useful in determining if an intrauterine pregnancy is viable. Finally, diagnostic laparoscopy and laparotomy may be indicated with pain of undetermined etiology.

SEXUAL DYSFUNCTION

Some women with sexual dysfunction describe minor complaints related to the reproductive tract as a means of bringing sexual problems to the attention of the physician. Alternatively, sexual dysfunction may be thought to be the cause of low

abdominal discomfort or dyspareunia when the actual etiology is organic. However, more and more women seek medical advice because of sexual problems that interface in provenance between medicine, psychiatry, and sociology.

The normal sexual response begins with sexual arousal, which causes genital vasocongestion that results in vaginal lubrication in preparation for intromission. The lubrication is due to the formation of a transudate in the vagina and in conjunction with genital congestion produces the so-called orgasmic platform prior to orgasm. Sexual stimuli (visual, tactile, auditory, and olfactory) as well as healthy vaginal tissue are prerequisites for genital vasocongestion and vaginal lubrication. During the second stage of the sexual response, involuntary contractions of the muscles of the pelvis result in a pleasurable cortical sensory phenomenon known as orgasm. Direct or indirect stimulation of the clitoris is important in the production of the female orgasm. In simple terms, sexual dysfunction can be due to interference with the arousal or orgasmic phases of the sexual response. Either disorder can be due to an organic or functional cause or both.

Illnesses that impair neurologic function such as diabetes mellitus or multiple sclerosis can prevent normal sexual arousal. Local pelvic diseases such as vaginitis, endometriosis, and salpingo-oophoritis may preclude normal sexual response because of resulting dyspareunia. Debilitating systemic diseases such as cancer and cardiovascular diseases may inhibit normal sexual response indirectly.

More commonly, failure of a normal sexual response is due to psychological factors that impair sexual arousal. Such problems include misinformation, e.g., the perception of sexual satisfaction as bad, or feelings of guilt about previous psychologically traumatic events such as incest, rape, or unwanted pregnancy. In addition, women who have had previous hysterectomy or mastectomy may perceive themselves as "incomplete." Stresses such as anxiety, depression, fatigue, and marital or interpersonal conflicts may lead to failure of the vasocongestive response and prevent normal vaginal lubrication. Women with such experiences may be unable to achieve normal sexual response unless they receive professional counseling. Such problems are approached by attempting to identify and reduce the causative stresses.

Failure to achieve orgasm is a specific form of sexual dysfunction. In the absence of orgasm many women enjoy sexual encounters to variable degrees because of the pleasure derived from closeness in a cherished relationship, particularly with a loving partner. However, for other women sexual relations with rare or absent orgasms are frustrating and unsatisfying. In many instances, failure of orgasm is due to insufficient clitoral stimulation and may be rectified by appropriate counseling and patient education.

A specific entity, "vaginismus," painful, involuntary contractions of the musculature surrounding the entrance to the vagina, is a rare cause of dyspareunia. It is a conditioned response to a previous real or imagined frightening or traumatic sexual experience. Treatment is directed to elimination of the conditioned response by progressive vaginal dilation by the patient in conjunction with marital therapy.

REPRODUCTION

Infertility is discussed in detail in [Chap. 54](#). The approach to infertile couples always involves evaluation of both the man and woman. The history should address the frequency of intercourse, the sexual responses of both, the use of contraceptives or lubricants, prior pregnancies, interval to conception and outcome of pregnancy, previous or past medical illnesses, and all medications taken.

Male-associated factors account for a third of infertility problems. Therefore, one of the first procedures in the workup of infertile couples should be a semen analysis. The initial evaluation of the woman includes documentation of normal ovulatory cycles. A history of regular, cyclic, predictable, spontaneous menses usually indicates ovulatory cycles, which may be confirmed by basal body temperature graphs, properly timed endometrial biopsies, or plasma progesterone measurements during the luteal phase of the cycle. Also, the diagnosis of luteal-phase dysfunction (low progesterone secretion during the luteal phase) can be established by these methods. Transvaginal ultrasonography is useful for evaluating follicular development.

The most common cause of infertility in women is tubal disease, usually due to infection (pelvic inflammatory disease) or endometriosis. Tubal disease can be evaluated by obtaining a hysterosalpingogram or by diagnostic laparoscopy. Tubal diseases can usually be treated by laparoscopic tuboplasty and lysis of adhesions.

In many instances of infertility, it is now possible to use assisted reproductive technologies including in vitro fertilization and embryo transfer, gamete intrafallopian tube transfer, transfer of cryopreserved ova and embryos, donor oocytes or donor sperm, and ovarian hyperstimulation with clomiphene citrate or gonadotropins followed by intrauterine insemination.

The desire for contraception is also a frequent cause for women to seek medical treatment or evaluation. The most widely used methods for fertility control include (1) rhythm and withdrawal techniques, (2) barrier methods, (3) intrauterine devices, (4) oral steroid contraceptives, (5) sterilization, and (6) abortion. **These methods and their complications are discussed in [Chap. 54](#).*

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53. HIRSUTISM AND VIRILIZATION - David A. Ehrmann

Hirsutism, defined as excessive male-pattern hair growth, affects approximately 10% of women of reproductive age. Hirsutism may be mild, essentially representing a variation of normal hair growth, or rarely it may be the harbinger of a serious underlying condition. It is often idiopathic but may be caused by several conditions associated with androgen excess, such as polycystic ovarian syndrome (PCOS) or congenital adrenal hyperplasia (CAH) ([Table 53-1](#)). Cutaneous manifestations commonly associated with hirsutism include acne and male-pattern balding (androgenic alopecia). *Virilization*, on the other hand, refers to the state in which androgen levels are sufficiently high to cause additional signs and symptoms such as deepening of the voice, breast atrophy, increased muscle bulk, clitoromegaly, and increased libido; virilization is an ominous sign that suggests the possibility of an ovarian or adrenal neoplasm.

HAIR FOLLICLE GROWTH AND DIFFERENTIATION

Hair can be categorized as either *vellus* (fine, soft, and not pigmented) or *terminal* (long, coarse, and pigmented). The number of hair follicles does not change over an individual's lifetime, but the follicle size and type of hair can change in response to numerous factors, particularly androgens. Androgens are necessary for terminal hair and sebaceous gland development and mediate differentiation of pilosebaceous units (PSUs) into either a terminal hair follicle or a sebaceous gland. In the former case, androgens transform the vellus hair into a terminal hair; in the latter, the sebaceous component proliferates and the hair remains vellus.

There are three phases in the cycle of hair growth: (1) *anagen* (growth phase), (2) *catagen* (involution phase), and (3) *telogen* (rest phase). Depending on the body site, hormonal regulation may play an important role in the hair growth cycle. For example, the eyebrows, eyelashes, and vellus hairs are androgen-insensitive, whereas the axillary and pubic areas are sensitive to low doses of androgens. Hair growth on the face, chest, upper abdomen, and back requires greater levels of androgens and is therefore more characteristic of the pattern typically seen in males. Androgen excess in women leads to increased hair growth in most androgen-sensitive sites but will manifest with loss of hair in the scalp region, in part by reducing the time hairs spend in anagen phase.

Although androgen excess underlies most cases of hirsutism, there is only a modest correlation between androgen levels and the quantity of hair growth. This is due to the fact that hair growth from the follicle depends on local factors and variability in end-organ sensitivity, as well as circulating androgen concentrations. Genetic factors and ethnic background also influence hair growth. In general, dark-haired individuals tend to be more hirsute than blonde or fair individuals. Asians and Native Americans have relatively sparse hair in regions sensitive to high androgen levels, whereas people of Mediterranean descent are more hirsute. For these reasons, family history and ethnic background are important considerations when assessing the etiology and severity of hirsutism.

CLINICAL ASSESSMENT

Historic elements relevant to the assessment of hirsutism include the age of onset and rate of progression of hair growth and associated symptoms or signs (e.g., acne). Depending on the cause, excess hair growth is typically first noted during the second and third decades. The growth is usually slow but progressive. Sudden development and rapid progression of hirsutism suggests the possibility of an androgen-secreting neoplasm, in which case findings of virilization may also be present.

The age of onset of menstrual cycles (menarche) and the pattern of the menstrual cycle should be ascertained; irregular cycles from the time of menarche onward are more likely to result from ovarian rather than adrenal androgen excess. Associated symptoms such as galactorrhea should prompt evaluation for hyperprolactinemia ([Chap. 328](#)) and possibly hypothyroidism ([Chap. 330](#)). Hypertension, striae, easy bruising, centripetal weight gain, and weakness suggest hypercortisolism (Cushing's syndrome; [Chap. 331](#)). Rarely, patients with growth hormone excess (i.e., acromegaly) will present with hirsutism. Use of medications such as phenytoin, minoxidil, or cyclosporine may be associated with androgen-independent causes of excess hair growth (i.e., hypertrichosis). A family history of infertility and/or hirsutism may indicate disorders such as nonclassic congenital adrenal hyperplasia ([CAH](#)), a disorder particularly common in Ashkenazi Jews, among others ([Chap. 331](#)).

Physical examination should include measurement of height, weight, and calculation of body mass index (BMI). A BMI $>25 \text{ kg/m}^2$ is indicative of excess weight for height, and values $>30 \text{ kg/m}^2$ are often seen in association with hirsutism. Notation should be made of blood pressure. Cutaneous signs sometimes associated with androgen excess and insulin resistance include acanthosis nigricans and skin tags.

An objective clinical assessment of hair distribution and quantity is central to the evaluation in any woman presenting with hirsutism. This assessment permits the distinction between hirsutism and hypertrichosis and provides a baseline reference point to gauge the response to treatment. *Hypertrichosis* refers to the excessive growth of androgen-independent hair which is vellus, prominent in nonsexual areas, and most commonly familial or caused by metabolic disorders (e.g., thyroid disturbances, anorexia nervosa) or medications (e.g., phenytoin, minoxidil or cyclosporine).

A simple and commonly used method to grade hair growth is the modified scale of Ferriman and Gallwey ([Fig. 53-1](#)), where each of nine androgen-sensitive sites is graded from 0 to 4. Approximately 95% of Caucasian women have a score below 8 on this scale; thus, it is normal for most women to have some hair growth in androgen-sensitive sites. Scores above 8 suggest an excess of androgen-mediated hair growth, a finding that should be assessed further by hormonal evaluation (see below). In racial/ethnic groups that are less likely to manifest hirsutism (e.g., Asian women), additional cutaneous evidence of androgen excess should be sought, including pustular acne or thinning hair.

HORMONAL EVALUATION

Androgens are secreted by both the ovaries and adrenal glands in response to their respective tropic hormones, luteinizing hormone (LH) and adrenocorticotrophic hormone (ACTH). The principal circulating steroids involved in the etiology of hirsutism are

androstenedione, dehydroepiandrosterone (DHEA) and its sulfated form (DHEAS), and testosterone. The ovaries and adrenal glands normally contribute about equally to testosterone production. Further, approximately half of the total testosterone originates from direct glandular secretion, and the remainder is derived from the peripheral conversion of androstenedione and DHEA ([Chap. 335](#)).

Although it is the most important circulating androgen, testosterone is, in effect, the penultimate androgen in mediating hirsutism; it is converted to the more potent dihydrotestosterone (DHT) by the enzyme 5 α -reductase, which is located in the pilosebaceous unit. DHT has a higher affinity for, and slower dissociation from, the androgen receptor. The local production of DHT allows it to serve as the primary mediator of androgen action at the level of the pilosebaceous unit. There are two isoenzymes of 5 α -reductase: type 2 is found in the prostate gland and in hair follicles, whereas type 1 is primarily found in sebaceous glands.

One approach to testing for hyperandrogenemia is depicted in [Fig. 53-2](#). This involves measuring blood levels of testosterone and [DHEAS](#). It is also important to measure the level of free (or unbound) testosterone, because it is the fraction of testosterone that is not bound to its carrier protein, sex-hormone binding globulin (SHBG), that is biologically available. Hyperinsulinemia and/or androgen excess decrease hepatic production of SHBG, often resulting in levels of total testosterone within the high-normal range at a time when the free hormone is substantially elevated. Because adrenal androgens are readily suppressed by low doses of glucocorticoids, the dexamethasone androgen-suppression test may broadly distinguish ovarian from adrenal androgen overproduction. A blood sample is obtained before and after administering dexamethasone (0.5 mg orally every 6 h for 4 days). An adrenal source is suggested by suppression of plasma free testosterone into the normal range; incomplete suppression suggests ovarian androgen excess.

A baseline plasma total testosterone level >12 nmol/L (>3.5 ng/mL) usually indicates a virilizing tumor, whereas a level >7 nmol/L (>2 ng/mL) is suggestive. A basal [DHEAS](#) level >18.5 μ mol/L (>7000 μ g/L) suggests an adrenal tumor. Although DHEAS has been proposed as a "marker" of predominant adrenal androgen excess, it is not unusual to find modest elevations in DHEAS among women with [PCOS](#). Computed tomography (CT) or magnetic resonance imaging (MRI) should be used to localize an adrenal mass, and ultrasound will usually suffice to identify an ovarian mass, if clinical evaluation and hormonal levels suggest these possibilities.

[PCOS](#) is the most common cause of ovarian androgen excess ([Chap. 336](#)). However, the increased ratio of [LH](#) to follicle-stimulating hormone that is often seen in carefully studied patients with PCOS may not be exhibited in up to half of these women due to the pulsatility of gonadotropins. If performed, ultrasound shows enlarged ovaries and/or increased stroma in many women with PCOS. However, polycystic ovaries may also be found in women without clinical or laboratory features of PCOS. Therefore, polycystic ovaries are a relatively insensitive and nonspecific finding for the diagnosis of ovarian hyperandrogenism. Though it is not widely used, gonadotropin-releasing hormone agonist testing can be used to make a specific diagnosis of ovarian hyperandrogenism. A peak 17-hydroxyprogesterone level $\geq$ 7.8 nmol/L ($\geq$ 2.6 μ g/L), after the administration of 100 μ g nafarelin (or 10 μ g/kg leuprolide) subcutaneously, is virtually diagnostic of

ovarian hyperandrogenism.

Nonclassic [CAH](#) is most commonly due to 21-hydroxylase deficiency but can also be caused by autosomal recessive defects in other steroidogenic enzymes necessary for adrenal corticosteroid synthesis ([Chap. 331](#)). Because of the enzyme defect, the adrenal gland cannot secrete glucocorticoids efficiently (especially cortisol). This results in diminished negative feedback inhibition of [ACTH](#), leading to compensatory hyperplasia of the adrenal cortex and accumulation of steroid precursors proximal to the enzyme defect. These precursors are subsequently converted to androgen.

Deficiency of 21-hydroxylase can be reliably excluded by determining a morning 17-hydroxyprogesterone level <6 nmol/L (<2 ug/L) (drawn in the follicular phase). Alternatively, 21-hydroxylase deficiency can be diagnosed by measurement of 17-hydroxyprogesterone 1 h after administration of 250 ug of synthetic [ACTH](#) (cosyntropin) intravenously. Measurement after ACTH is slightly more cumbersome, though the results obtained in this manner are highly reproducible and can be compared to published nomograms.

TREATMENT

Treatment of hirsutism may be accomplished pharmacologically and by mechanical means of hair removal. Nonpharmacologic treatments should be considered in all patients, either as the only treatment or as an adjunct to drug therapy.

Nonpharmacologic treatments include (1) bleaching; (2) depilatory (removal from the skin surface) such as shaving and chemical treatments; or (3) epilatory (removal of the hair including the root) such as plucking, waxing, electrolysis, and laser therapy. Despite perceptions to the contrary, shaving does not increase the rate or density of hair growth. Chemical depilatory treatments may be useful for mild hirsutism that affects only limited skin areas, though they can cause skin irritation. Wax treatment removes hair temporarily but is uncomfortable. Electrolysis is effective for more permanent hair removal, particularly in the hands of a skilled electrologist. Laser phototherapy appears to be efficacious for hair removal. It delays hair regrowth and causes permanent hair removal in some patients. The long-term effects and complications associated with laser treatment are being evaluated.

Pharmacologic therapy for androgen excess is directed at interrupting one or more of the steps in the pathway leading to its expression: (1) suppression of adrenal and/or ovarian androgen production; (2) enhancement of androgen-binding to plasma-binding proteins, particularly [SHBG](#); (3) impairment of the peripheral conversion of androgen precursors to active androgen; and (4) inhibition of androgen action at the target tissue level. Attenuation of hair growth is typically not evident until 4 to 6 months after initiation of medical treatment and, in most cases, leads to a modest reduction in hair growth.

Combination estrogen-progestin therapy, in the form of an oral contraceptive, is usually the first-line endocrine treatment for hirsutism and acne, after cosmetic and dermatologic management. The estrogenic component of most oral contraceptives currently in use is either ethinyl estradiol or mestranol. The suppression of [LH](#) leads to reduced production of ovarian androgens. The reduced androgen levels also result in a

dose-related increase in [SHBG](#), thereby lowering the fraction of unbound plasma testosterone. Combination therapy has also been demonstrated to decrease [DHEAS](#), perhaps by reducing [ACTH](#) levels. Estrogens also have a direct, dose-dependent suppressive effect on sebaceous cell function.

The choice of a specific oral contraceptive should be predicated on the progestational component, as progestins vary in their suppressive effect on [SHBG](#) levels and in their androgenic potential. Ethynodiol diacetate has relatively low androgenic potential, whereas progestins such as norgestrel and levonorgestrel are particularly androgenic, as judged from their attenuation of the estrogen-induced increase in SHBG. Norgestimate exemplifies the newer generation of progestins that are virtually nonandrogenic. Oral contraceptives are contraindicated in women with a history of thromboembolic disease or in women with breast cancer or other estrogen-dependent cancers ([Chap. 336](#)). There is a relative contraindication to the use of oral contraceptives in smokers or in those with hypertension or a history of migraine headaches. In most trials, estrogen-progestin therapy alone improves the extent of acne by a maximum of 50 to 70%. In contrast, the effect on hair growth may not be evident for 6 months, and the maximum effect may require 9 to 12 months owing to the length of the hair growth cycle. Improvements in hirsutism are typically in the range of 20%, and often there is little more than arrest of further progression of hair growth.

Adrenal androgens are more sensitive than cortisol to the suppressive effects of glucocorticoids. Therefore, glucocorticoids are the mainstay of treatment in patients with [CAH](#). Although glucocorticoids have been reported to restore ovulatory function in some women with [PCOS](#), this effect is highly variable. Because of side effects from excessive glucocorticoids, low doses should be used. Dexamethasone (0.2 to 0.5 mg) or prednisone (5 to 10 mg) should be given at bedtime to achieve maximal suppression by inhibiting the nocturnal surge of [ACTH](#).

Cyproterone acetate is the prototypic antiandrogen. It acts mainly by competitive inhibition of the binding of testosterone and [DHT](#) to the androgen receptor. In addition, it may act to enhance the metabolic clearance of testosterone by inducing hepatic enzymes. Although not available for use in the United States, cyproterone acetate is widely used in Canada, Mexico, and Europe. Cyproterone (50 to 100 mg) is given on days 1 to 15 and ethinyl estradiol (50 ug) is given on days 5 to 26 of the menstrual cycle. Side effects of cyproterone acetate include irregular uterine bleeding, nausea, headache, fatigue, weight gain, and decreased libido.

Spirolactone, usually used as a mineralocorticoid antagonist, is also a weak antiandrogen. It is almost as effective as cyproterone acetate when used at high enough doses (100 to 200 mg daily). Patients should be monitored intermittently for hyperkalemia or hypotension, though these side effects are uncommon. Pregnancy should be avoided because of the risk of feminization of a male fetus. Spirolactone can also cause menstrual irregularity. It is often used in combination with an oral contraceptive, which helps in prevention of pregnancy and suppression of ovarian androgen production.

Flutamide is a potent nonsteroidal antiandrogen that is effective in treating hirsutism, but concerns about the induction of hepatocellular dysfunction have limited its use.

Finasteride is a competitive inhibitor of 5 α -reductase type 2. Beneficial effects on hirsutism have been reported, but the prominence of 5 α -reductase type 1 in the pilosebaceous unit appears to account for its limited efficacy. Finasteride would also be expected to impair sexual differentiation in a male fetus, and thus it should not be used in women who may become pregnant.

A prospective, randomized trial comparing low-dose flutamide, finasteride, and combination cyproterone acetate-ethinyl estradiol demonstrated relative superiority of flutamide and cyproterone acetate-ethinyl estradiol in the treatment of hirsutism. Ultimately, the choice of any specific agent(s) must be tailored to the unique needs of the patient being treated. As noted previously, pharmacologic treatments for hirsutism should be used in conjunction with nonpharmacologic approaches. Patients should be reminded about the relatively slow and usually modest responses to pharmacologic treatment. It is also helpful to review the pattern of female hair distribution in the normal population to dispel unrealistic expectations.

(Bibliography omitted in Palm version)

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54. INFERTILITY AND FERTILITY CONTROL - Janet E. Hall

The concept of reproductive choice is now firmly entrenched in developed countries and has dramatically altered reproductive behavior. The availability of effective contraceptive methods prevents unintended pregnancies and gives women the option of pursuing educational and career opportunities without interruption. Population control also has important economic and social implications. Infertility, on the other hand, can be accompanied by substantial stress and disappointment. Fortunately, the ability to diagnose and to treat various causes of infertility now provides an array of effective new approaches to this condition.

INFERTILITY

DEFINITION AND PREVALENCE

Infertility is defined as the inability to conceive after 12 months of unprotected sexual intercourse. In a study of 5574 English and American women who ultimately conceived, pregnancy occurred in 50% within 3 months, 72% within 6 months, and 85% within 12 months. These findings are consistent with predictions based on *fecundability*, the probability of achieving pregnancy in one menstrual cycle (approximately 20 to 25% in healthy young couples). Assuming a fecundability of 0.25, 98% of couples should conceive within 13 months. Based on this definition, the National Survey of Family Growth reports a 14% rate of infertility in the United States in married women aged 15 to 44. The infertility rate has remained relatively stable over the past 30 years, although the proportion of couples without children has risen, reflecting a trend to delay childbearing. This trend has important implications because of an age-related decrease in fecundability, which begins at age 35, and decreases markedly after age 40.

CAUSES OF INFERTILITY

There is a spectrum of infertility, ranging from reduced conception rates or the need for medical intervention to irreversible causes of infertility (*sterility*). Infertility can be attributed primarily to male factors in 25%, female factors in 58%, and is unexplained in about 17% of couples ([Fig. 54-1](#)). Not uncommonly, both male and female factors contribute to infertility.

Approach to the Patient

Initial Evaluation In all couples presenting with infertility, the initial evaluation includes discussion of the appropriate timing of intercourse and a description of the range of investigations that may be required. A brief description of infertility treatment options, including adoption, should be reviewed. Initial investigations are focused on determining whether the primary cause of the infertility is male, female, or both. These investigations include a semen analysis in the male, confirmation of ovulation in the female, and, in the majority of situations, documentation of tubal patency in the female. Although frequently used in the past, recent studies have not supported the efficacy of postcoital testing of sperm interaction with cervical mucus as a routine component of initial testing. Strategies for further evaluation are described below and in [Chaps. 335](#) and [336](#). In some cases, after an extensive workup excluding all male and female factors, a specific

cause cannot be identified and infertility may ultimately be classified as unexplained.

Psychological Aspects of Infertility Infertility is invariably associated with psychological stress related not only to the diagnostic and therapeutic procedures themselves but also to repeated cycles of hope and loss associated with each new procedure or cycle of treatment that does not result in the birth of a child. These feelings are often combined with a sense of isolation from friends and family. Counseling and stress-management techniques should be introduced early in the evaluation of infertility. In addition to the psychological benefits of stress management, it is possible that stress contributes to infertility in some couples (e.g., impaired ovulation). Importantly, infertility and its treatment do not appear to be associated with long-term psychological sequelae.

Female Causes Abnormalities in menstrual function constitute the most common cause of female infertility. These disorders, which include ovulatory dysfunction and abnormalities of the uterus or outflow tract, may present as amenorrhea (absence of menses) or as irregular or short menstrual cycles. A careful history and physical examination and a limited number of laboratory tests will help to determine whether the abnormality is: (1) hypothalamic or pituitary [low follicle-stimulating hormone (FSH), luteinizing hormone (LH), and estradiol with or without an increase in prolactin]; (2) polycystic ovarian syndrome (PCOS; irregular cycles and hyperandrogenism in the absence of other causes of androgen excess); (3) ovarian (low estradiol with increased FSH); or (4) uterine or outflow tract abnormality. The frequency of these diagnoses depends on whether the amenorrhea is primary or occurs after normal puberty and menarche ([Fig. 54-1](#)). **The approach to further evaluation of these disorders is described in detail in Chap. 52.*

OVULATORY DYSFUNCTION In women with a history of regular menstrual cycles, *evidence of ovulation* should be sought by using urinary ovulation predictor kits (they reflect the preovulatory gonadotropin surge but do not confirm ovulation), basal body temperature charts, or a mid-luteal phase progesterone level. The mid-luteal phase progesterone increase (usually >3 ng/mL) confirms ovulation and corpus luteum function and is responsible for the rise in basal body temperature [$>0.3^{\circ}\text{C}$ ($>0.6^{\circ}\text{F}$) for 10 days]. An endometrial biopsy to exclude luteal phase insufficiency is no longer considered an essential part of the infertility workup for most patients. Even in the presence of ovulatory cycles, evaluation of *ovarian reserve* is recommended for women over 35 by measurement of [FSH](#) on day 3 of the cycle or in response to clomiphene, an estrogen antagonist (see below). An FSH level <10 IU/mL on cycle day 3 predicts adequate ovarian oocyte reserve. Inhibin B, an ovarian hormone that selectively suppresses FSH, is being investigated as an additional marker of ovarian reserve.

TUBAL DISEASE This may result from pelvic inflammatory disease (PID), appendicitis, endometriosis, pelvic adhesions, tubal surgery, and previous use of an intrauterine device (IUD). However, a cause is not identified in up to 50% of patients with documented tubal factor infertility. Because of the high prevalence of tubal disease, testing should occur early in the majority of couples with infertility. Subclinical infections with *Chlamydia trachomatis* may be an underdiagnosed cause of tubal infertility and requires the treatment of both partners. A hysterosalpingogram (HSG) is the most common screening test and will determine the presence of tubal patency and identify potential abnormalities of the uterine cavity.

ENDOMETRIOSIS *Endometriosis* is defined as the presence of endometrial glands or stroma outside the endometrial cavity and uterine musculature. Its presence is suggested by a history of dyspareunia (painful intercourse), worsening dysmenorrhea that often begins before menses, or by a thickened rectovaginal septum or deviation of the cervix on pelvic examination. The pathogenesis of the infertility associated with endometriosis is unclear but may involve indirect effects on the normal endometrium as well as the direct effects of adhesions in advanced disease. Endometriosis is often clinically silent, however, and can only be excluded definitively by laparoscopy.

Male Causes Known causes of male infertility include primary testicular disease, disorders of sperm transport, and hypothalamic-pituitary disease resulting in secondary hypogonadism. However, the etiology is not ascertained in up to half of men with suspected male factor infertility ([Fig. 54-1](#)). The key initial diagnostic test is a *semen analysis*. Although 95% confidence limits can be used to define normal semen parameters, data relating sperm counts to fecundability are more useful. Such studies suggest that sperm counts of <20 million/mL, with a motility of less than 40%, are associated with an increased risk of infertility. Analysis of sperm morphology is less well validated, but >40% normal forms are usually present in fertile men. Successful in vitro fertilization (IVF) can usually be accomplished with >14% normal forms (using strict Kruger criteria), whereas low fertilization is seen with <4% normal forms. Other tests such as the hamster egg penetration test and the zona-binding assay are not of proven value.

Testosterone levels should be measured if the sperm count is low on repeated examination or if there is clinical evidence of hypogonadism. A low testosterone level may result from *primary gonadal deficiency*; in this condition, levels of [LH](#) and [FSH](#) will be elevated. Less commonly, low testosterone and decreased spermatogenesis result from hypothalamic or pituitary disease, in which case the LH and FSH levels will be low ([Chap. 335](#)).

Abnormalities of spermatogenesis may have a genetic component. Y chromosome microdeletions and substitutions are increasingly recognized as a cause of *azoospermia* (absence of sperm) or *oligospermia* (low sperm count). Microdeletions (Yq6 region) have also been identified in a subset of men with elevated [FSH](#) levels or otherwise idiopathic infertility. Several candidate genes have been identified including *DAZ* (deleted in azoospermia) and *YRRM* (Y chromosome RNA recognition motif).

Acquired disorders of the testes are often associated with impaired spermatogenesis with relatively preserved Leydig cell function; thus, testosterone levels may be normal. Such abnormalities include viral orchitis (especially mumps) and other infectious causes such as tuberculosis or sexually transmitted diseases (STDs), chemotherapy (especially the alkylating agents cyclophosphamide and chlorambucil), ionizing radiation, and drugs that may impair fertility directly or through inhibition of testicular androgen production or action. Anabolic androgen abuse should be considered in a well-androgenized man with low gonadotropins and testosterone but a suppressed sperm count. Prolonged elevation of testicular temperature may impair spermatogenesis, e.g., after an acute febrile illness or in association with varicocele. A potential role for environmental toxins as a cause of impaired spermatogenesis has been suggested based on an apparent decrease in

sperm counts over the past several decades, but a direct cause-and-effect relationship has not been established.

SECONDARY HYPOGONADISM Low gonadotropin levels, associated with low testosterone, may signal the presence of a pituitary macroadenoma or hypothalamic tumor (in both cases prolactin levels may be elevated; [Chap. 328](#)) or may be the first presentation of hemochromatosis ([Chap. 345](#)) or other systemic illness. Recent studies have identified several genetic causes of gonadotropin-releasing hormone (GnRH) deficiency (*KAL* and *DAX-1*), as well as mutations that lead to isolated gonadotropin deficiency (GnRH receptor, [LHb](#), [FSHb](#) mutations) ([Chap. 328](#)).

DISORDERED SPERM TRANSPORT Patients with low sperm counts and normal hormonal levels may be found to have obstructive abnormalities of the vas deferens or epididymus. The most common causes of vas deferens obstruction are previous vasectomy or accidental ligation during inguinal surgery. Patency rates with microsurgical reversal techniques are high in the first 3 years after vasectomy but decrease markedly thereafter. Congenital absence of the vas deferens can be diagnosed by a deficiency of fructose in the ejaculate and is often associated with an abnormality of the cystic fibrosis transmembrane regulator (*CFTR*) gene. Young's syndrome, characterized by inspissated secretions, can also preclude normal sperm transport.

TREATMENT

The treatment of infertility should be tailored to the problems unique to each couple ([Table 54-1](#)). In many situations, including unexplained infertility, mild to moderate endometriosis, and/or borderline semen parameters, a stepwise approach to infertility is optimal, beginning with low-risk interventions and moving to more invasive, higher risk interventions only if necessary. After determination of all infertility factors and their correction, if possible, this approach might include, in increasing order of complexity: (1) expectant management, (2) clomiphene citrate (see below) with or without intrauterine insemination (IUI), (3) gonadotropins with or without IUI, and (4) [IVF](#). The time used to complete the evaluation, correction, and expectant management can be longer in women <30, but this process should be advanced rapidly in women >35. In some situations expectant management will not be appropriate.

Ovulatory Dysfunction Treatment of ovulatory dysfunction should first be directed at identification of the etiology of the disorder to allow specific management when possible. Dopamine agonists, for example, may be indicated in patients with hyperprolactinemia ([Chap. 328](#)); lifestyle modification may be successful in women with low body weight or a history of intensive exercise ([Chap. 78](#)).

Pulsatile GnRH is highly effective for restoring ovulation in patients with hypothalamic amenorrhea. When administered subcutaneously by an automated pump at a physiologic dose and frequency, pulsatile GnRH induces normal [LH](#) and [FSH](#) dynamics. Direct comparisons between pulsatile GnRH and gonadotropin treatment for ovulation induction indicate similar pregnancy rates; pulsatile GnRH is associated with lower rates of multiple gestation and virtually no risk of ovarian hyperstimulation.

Clomiphene citrate is a nonsteroidal estrogen antagonist that increases [FSH](#) and [LH](#) levels by blocking estrogen negative feedback at the hypothalamus. The efficacy of clomiphene for ovulation induction is highly dependent on patient selection. It induces ovulation in ~60% of women with [PCOS](#) and is the initial treatment of choice in these patients. The starting dose is 50 mg daily for 5 days beginning on day 5 of a spontaneous cycle or after a progestin-induced withdrawal bleed. The dose can be increased to 150 mg, if necessary, in subsequent cycles, and human chorionic gonadotropin (hCG) can be added as the ovulatory stimulus. In women with PCOS, the use of insulin-sensitizing agents, such as metformin appears to be particularly effective in combination with clomiphene.

Gonadotropins are highly effective for ovulation induction in women with hypogonadotropic hypogonadism and [PCOS](#). Gonadotropins are also used to induce multiple follicular recruitment in unexplained infertility and in older reproductive-aged women, particularly in conjunction with [IUI](#). Disadvantages include a significant risk of multiple gestation and the risk of ovarian hyperstimulation, a side effect that is more common in women with PCOS. However, careful monitoring and a conservative approach to ovarian stimulation reduce these risks; gonadotropin stimulation is an effective and safe treatment when applied by experienced practitioners. Currently available gonadotropins include urinary preparations of [LH](#) and [FSH](#), highly purified FSH, and recombinant FSH. Though FSH is the key component, there is growing data that the addition of some LH (or [hCG](#)) may improve results, particularly in hypogonadotropic patients.

None of these methods are effective in women with premature ovarian failure in whom donor oocyte or adoption are the methods of choice.

Tubal Disease If hysterosalpingography suggests a tubal or uterine cavity abnormality, or if a patient is ³⁵ at the time of initial evaluation, laparoscopy with tubal lavage is recommended, often with a hysteroscopy. Although tubal reconstruction may be attempted if tubal disease is identified, it is generally being replaced by the use of [IVF](#), as these patients are at increased risk of developing an ectopic pregnancy.

Endometriosis Though 60% of women with minimal or mild endometriosis may conceive within 1 year without treatment, laparoscopic resection or ablation appear to improve conception rates. Medical management of advanced stages of endometriosis is widely used for symptom control but has not been shown to enhance fertility ([Chap. 336](#)). In moderate to severe endometriosis, conservative surgery is associated with pregnancy rates of 50 and 39% respectively, compared with rates of 25 and 5% with expectant management alone. In some patients, [IVF](#) may be the treatment of choice.

Male Factor Infertility The treatment options for male factor infertility have expanded greatly in recent years. Secondary hypogonadism is highly amenable to treatment with pulsatile [GnRH](#) or gonadotropins ([Chap. 335](#)). In vitro techniques have provided new opportunities for patients with primary testicular failure and disorders of sperm transport. Choice of initial treatment options depends on sperm concentration and motility. Expectant management should be attempted initially in men with mild male factor infertility (sperm count of 15 to 20 × 10⁶/mL and normal motility). Moderate male factor infertility (10 to 15 × 10⁶/mL and 20 to 40% motility) should begin with [IUI](#) alone or in

combination with treatment of the female partner with clomiphene or gonadotropins, but it may require [IVF](#) with or without intracytoplasmic sperm injection (ICSI). For men with a severe defect (sperm count of $<10^6$ /mL, 10% motility), IVF with ICSI or donor sperm should be used.

Assisted Reproductive Technologies The development of assisted reproductive technologies (ART) has dramatically altered the treatment of male and female infertility. [IVF](#) is indicated for patients with many causes of infertility that have not been successfully managed with more conservative approaches. IVF or [ICSI](#) is often the treatment of choice in couples with a significant male factor or tubal disease, whereas IVF using donor oocytes is used in patients with premature ovarian failure and in women of advanced reproductive age. Success rates depend on the age of the woman and the cause of the infertility and are generally 18 to 24% per cycle when initiated in women <40 . In women >40 , there is a marked decrease in both the number of oocytes retrieved and their ability to be fertilized. Though often effective, IVF is expensive and requires careful monitoring of ovulation induction and invasive techniques including the aspiration of multiple follicles. IVF is associated with a significant risk of multiple gestation (29% twins, 7% triplets, and 0.6% higher order multiples). More recently developed blastocyst transfer protocols decrease the number of transfers but increase pregnancy rates.

CONTRACEPTION

Though various forms of contraception are widely available, approximately 30% of births in the United States are the result of unintended pregnancy. Teenage pregnancies continue to represent a serious public health problem in the United States, with >1 million unintended pregnancies each year -- a significantly greater incidence than in other industrialized nations ([Chap. 8](#)).

Contraceptive methods are widely used ([Table 54-2](#)). Only 15% of couples report having unprotected sexual intercourse in the past 3 months. A reversible form of contraception is used by $>50\%$ of couples. Sterilization (in either the male or female) has been employed as a permanent form of contraception by about 25% of couples. Pregnancy termination is relatively safe when directed by health care professionals but is rarely the option of choice.

No single contraceptive method is ideal, although all are safer than carrying a pregnancy to term. The effectiveness of a given method of contraception is dependent on the efficacy of the method itself, compliance, and appropriate use. Knowledge of the advantages and disadvantages of each contraceptive is essential for counseling an individual about the methods that are safest and most consistent with his or her lifestyle. Discrepancies between theoretical and actual effectiveness emphasize the importance of patient education and compliance when considering various forms of contraception ([Table 54-2](#)).

BARRIER METHODS

Barrier contraceptives, such as condoms, diaphragms, cervical caps, and spermicides, are easily available, reversible, and have fewer side effects than hormonal methods.

However, their effectiveness is highly dependent on compliance and proper use ([Table 54-2](#)). A major advantage of barrier contraceptives is the protection provided against [STDs](#) ([Chap. 132](#)). Consistent use is associated with a decreased risk of gonorrhea, nongonococcal urethritis, and genital herpes, probably due in part to the concomitant use of spermicides. Condom use also reduces the transmission of HIV infection. Natural membrane condoms may be less effective than latex condoms, and petroleum-based lubricants can degrade condoms and decrease their efficacy for preventing HIV infection. A highly effective female condom, which also provides protection against STDs, was approved in 1994 but has not achieved widespread use.

STERILIZATION

Sterilization is the method of birth control most frequently chosen by fertile men and multiparous women >30 ([Table 54-2](#)). Sterilization refers to a procedure that prevents fertilization by surgical interruption of the fallopian tubes in women or the vas deferens in men. Although tubal ligation and vasectomy are potentially reversible, these procedures should be considered permanent and should not be undertaken without careful patient counseling.

Several methods of *tubal ligation* have been developed, all of which are highly effective with a 10-year cumulative pregnancy rate of 1.85 per 100 women. However, when pregnancy does occur, the risk of ectopic pregnancy may be as high as 30%. The success rate of tubal reanastomosis depends on the method used -- the clip, silastic band, and modified Pomeroy procedures are easier to reverse than the Irving, Uchida, and electrocoagulation methods. Even after successful reversal, the risk of ectopic pregnancy remains great. In addition to prevention of pregnancy, tubal ligation reduces the risk of ovarian cancer, possibly by limiting the upward migration of potential carcinogens.

Vasectomy is an outpatient surgical procedure that has little risk and is highly effective. The development of azoospermia may be delayed for 2 to 6 months, and other forms of contraception must be used until two sperm-free ejaculations provide proof of sterility. Reanastomosis may restore fertility in 30 to 50% of men, but the success rate appears to decline with time after vasectomy and may be influenced by nonmechanical factors such as the development of anti-sperm antibodies.

INTRAUTERINE DEVICES

[IUDs](#) inhibit pregnancy primarily through a spermicidal effect caused by a sterile inflammatory reaction produced by the presence of a foreign body in the uterine cavity. There may also be effects on cervical mucus sperm transport through the oviduct. IUDs provide a high level of efficacy in the absence of systemic metabolic effects. An additional advantage is that ongoing motivation is not required to ensure efficacy once the device has been placed. However, only 1% of women in the United States use this method compared to a utilization rate of 15 to 30% in much of Europe and Canada. This relatively low utilization rate continues despite evidence that the newer devices are not associated with increased rates of pelvic infection and infertility, as occurred with earlier devices. Screening for [STD](#) should be performed prior to insertion, and an IUD should not be used in women at high risk for development of STD or in women at high risk for

bacterial endocarditis. In addition, the IUD may not be effective in women with uterine leiomyomas because they alter the size or shape of the uterine cavity. IUD use is associated with increased menstrual blood flow, although this is less pronounced with the progesterone-releasing IUD than the copper-containing device.

HORMONAL METHODS

No male hormonal contraceptive methods are currently approved in the United States. However, hormonal methods of male contraception, including [GnRH](#)-mediated suppression of the hypothalamic-pituitary-gonadal axis in combination with testosterone replacement, are under investigation.

Oral Contraceptive Pills Because of their ease of use and efficacy, oral contraceptive pills are the most widely used form of hormonal contraception. They act by suppressing ovulation, changing cervical mucus, and altering the endometrium. The current formulations are made from synthetic estrogens and progestins. The estrogen component of the pill consists of ethinyl estradiol or mestranol, which is metabolized to ethinyl estradiol. Multiple synthetic progestins are used. Norethindrone and its derivatives are used in many formulations. Low-dose norgestimate and third-generation progestins (desogestrel, gestodene) have a less androgenic profile; levonorgestrel appears to be the most androgenic of the progestins and should be avoided in patients with hyperandrogenic symptoms. The three major formulations of oral contraceptives include: (1) fixed-dose estrogen-progestin combination, (2) phasic estrogen-progestin combination, and (3) progestin only. Each of these formulations is administered daily for 3 weeks followed by a week of no medication during which menstrual bleeding generally occurs.

Current doses of ethinyl estradiol range from 20 to 50 ug. However, indications for the 50-ug dose are rare, and the majority of formulations contain 35 ug of ethinyl estradiol. The reduced estrogen and progesterone content in the second- and third-generation pills has decreased both side effects and risks associated with oral contraceptive use. At the currently used doses, patients must be cautioned not to miss pills due to the potential for ovulation. Side effects, including break-through bleeding, amenorrhea, and weight gain, are often responsive to a change in formulation. There is no evidence that low-dose oral contraceptives increase the risk of cardiovascular disease in women <30 or in nonsmoking women without additional risk factors. However, the risk of myocardial infarction and stroke in women who smoke is increased by the use of oral contraceptives. The risk of developing hypertension is increased somewhat, even with the low-dose preparations. An increased risk of venous thromboembolism occurs with all oral contraceptives and may be even greater with the third-generation preparations. The factor V Leiden mutation and other thrombophilic disorders ([Chap. 117](#)) are important risk factors for venous thrombosis during oral contraceptive therapy. However, biochemical or genetic screening for these disorders before starting oral contraceptives is not cost-effective at present. In most studies, oral contraceptive use has not been shown to increase the risk of breast cancer, but there is a slight increase in the risk of cervical cancer. Risks for endometrial and ovarian cancer are decreased in oral contraceptive users.

Previous thromboembolic events or stroke are absolute contraindications for the use of

oral contraceptive pills. A history of hormone-dependent tumors and liver disease are also contraindications. Oral contraceptive pills should not be given in pregnancy or in women with undiagnosed uterine bleeding or amenorrhea.

The microdose progestin-only minipill is less effective as a contraceptive, having a pregnancy rate of 2 to 7 per 100 women-years. However, it may be appropriate for women with cardiovascular disease or for women who cannot tolerate synthetic estrogens.

Injectable Contraceptives Depot medroxyprogesterone acetate (Depo-Provera) and Norplant ([Table 54-2](#)) act primarily by inhibiting ovulation and causing changes in the endometrium and cervical mucus that result in decreased implantation and sperm transport. Depo-Provera is effective for 3 months, but return of fertility after discontinuation may be delayed for up to 12 to 18 months. Norplant requires surgical insertion but is effective for up to 5 years after insertion; fertility is possible shortly after its removal. The U.S. Food and Drug Administration (FDA) has recently approved the use of covered rods in addition to the capsules. Amenorrhea, irregular bleeding, and weight gain are the most common adverse effects associated with both injectable forms of contraception. An injectable progestin/estrogen combination contraceptive will be available soon. It requires monthly injection, but irregular bleeding and weight gain are less common. A major advantage of the injectable progestin-based contraceptives is the apparent lack of increased arterial and venous thromboembolic events.

POSTCOITAL CONTRACEPTION

Postcoital contraceptive methods prevent implantation or cause regression of the corpus luteum and are highly efficacious if used appropriately. Although postcoital contraception is not specifically licensed for use in the United States, an FDA notice published in 1997 indicated that certain oral contraceptive pills could be used within 72 h of unprotected intercourse [Ovral (2 tablets 12 h apart) and Lo/Ovral (4 tablets 12 h apart)]. The Preven Emergency Contraceptive Kit contains four combination tablets (50 mg ethinyl estradiol and 0.25 mg levonorgestrel) and a pregnancy kit to rule out pregnancy before taking the pills. Side effects are common with these high doses of hormones and include nausea, vomiting, and breast soreness. Recent studies suggest that 600 mg mifepristone (RU486), a progesterone receptor antagonist, may be equally as effective or more effective than hormonal regimens, with fewer side effects. Mifepristone is not currently available in the United States.

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SECTION 9 -ALTERATIONS IN THE SKIN

55. APPROACH TO THE PATIENT WITH A SKIN DISORDER - *Thomas J. Lawley, Kim B. Yancey*

The challenge of examining the skin lies in distinguishing normal from abnormal, significant findings from trivial ones, and in integrating pertinent signs and symptoms into an appropriate differential diagnosis. The fact that the largest organ in the body is visible is both an advantage and a disadvantage to those who examine it. It is advantageous because no special instrumentation, other than a magnifying glass, is necessary and because the skin can be biopsied with little morbidity. However, the casual observer can be overwhelmed by a variety of stimuli and overlook important, subtle signs of skin or systemic disease. For instance, the sometimes minor differences in color and shape that distinguish a malignant melanoma (see [Plate IIC-30](#)) from a benign pigmented nevus (see [Plate IIC-28](#)) can be difficult to recognize. To aid in the interpretation of skin lesions, a variety of descriptive terms have been developed to characterize cutaneous lesions ([Tables 55-1](#) and [55-2](#) and [Fig. 55-1](#)) and to formulate a differential diagnosis ([Table 55-3](#)). For instance, the finding of large numbers of scaling papules, usually indicative of a primary skin disease, places the patient in a different diagnostic category than would hemorrhagic papules, which may indicate vasculitis or sepsis (see [Plates IIE-71](#) and [IID-44](#), respectively). It is important to differentiate primary skin lesions from secondary skin changes. If the examiner focuses on linear erosions overlying an area of erythema and scaling, he or she may incorrectly assume that the erosion is the primary lesion and the redness and scale are secondary, while the correct interpretation would be that the patient has a pruritic eczematous dermatitis and the erosions have been caused by scratching.

Approach to the Patient

In examining the skin it is usually advisable to assess the patient before taking a history. This way, the entire cutaneous surface is sure to be evaluated, and objective findings can be integrated with relevant historic data. Four basic features of any cutaneous lesion must be noted and considered in the examination of skin: the distribution of the eruption, the type(s) of primary lesion, the shape of individual lesions, and the arrangement of the lesions. In the initial examination it is important that the patient be disrobed as completely as possible. This will minimize chances of missing important individual skin lesions and make it possible to assess the distribution of the eruption accurately. The patient should first be viewed from a distance of about 1.5 to 2 m (4 to 6 ft) so that the general character of the skin and the distribution of lesions can be evaluated. Indeed, distribution of lesions often correlates highly with diagnosis ([Fig. 55-2](#)). For example, a hospitalized patient with a generalized erythematous exanthem is more likely to have a drug eruption than is a patient with a similar rash limited to the sun-exposed portions of the face. The presence or absence of lesions on mucosal surfaces should also be determined. Once the distribution of the lesions has been established, the nature of the primary lesion must be determined. Thus, when lesions are distributed on elbows, knees, and scalp, the most likely possibility based solely on distribution is psoriasis or dermatitis herpetiformis (see [Plates IIA-3](#) and [IIE-68](#), respectively). The primary lesion in psoriasis is a scaly papule that soon forms erythematous plaques covered with a white scale, whereas that of dermatitis

herpetiformis is an urticarial papule that quickly becomes a small vesicle. In this manner, identification of the primary lesion directs the examiner toward the proper diagnosis. Secondary changes in skin can also be quite helpful. For example, scale represents excessive epidermis, while crust is the result of an inadequate or discontinuous epithelial cell layer. Palpation of skin lesions can also yield insight into the character of an eruption. Thus red papules on the lower extremities that blanch with pressure can be a manifestation of many different diseases, but hemorrhagic red papules that do not blanch with pressure indicate palpable purpura characteristic of necrotizing vasculitis (see [Plate IIE-71](#)).

The shape of lesions is also an important feature. Flat, round, erythematous papules and plaques are common in many cutaneous diseases. However, target-shaped lesions that consist in part of erythematous plaques are specific for erythema multiforme (see [Plate IIE-67](#)). In the same way, the arrangement of individual lesions is important. Erythematous papules and vesicles can occur in many conditions, but their arrangement in a specific linear array suggests an external etiology such as allergic contact (see Plate IIA-8) or primary irritant dermatitis. In contrast, lesions with a generalized arrangement are common and suggest a systemic etiology.

As in other branches of medicine, a complete history should be obtained to emphasize the following features:

1. Evolution of lesions
 - a. Site of onset
 - b. Manner in which eruption progressed or spread
 - c. Duration
 - d. Periods of resolution or improvement in chronic eruptions
2. Symptoms associated with the eruption
 - a. Itching, burning, pain, numbness
 - b. What, if anything, has relieved symptoms
 - c. Time of day when symptoms are most severe
3. Current or recent medications (prescribed as well as over-the-counter)
4. Associated systemic symptoms (e.g., malaise, fever, arthralgias)
5. Ongoing or previous illnesses
6. History of allergies
7. Presence of photosensitivity

8. Review of systems

DIAGNOSTIC TECHNIQUES

Many skin diseases can be diagnosed on gross clinical appearance, but sometimes relatively simple diagnostic procedures can yield valuable information. In most instances, they can be performed at the bedside with a minimum of equipment.

Skin Biopsy A skin biopsy is a straightforward minor surgical procedure; however, it is important to biopsy the anatomic site most likely to yield diagnostic findings. This decision may require expertise in skin diseases and knowledge of superficial anatomic structures in selected areas of the body. In this procedure, a small area of skin is anesthetized with 1% lidocaine with or without epinephrine. The skin lesion in question can be excised with a scalpel or removed by punch biopsy. In the latter technique, a punch is pressed against the surface of the skin and rotated with downward pressure until it penetrates to the subcutaneous tissue. The circular biopsy is then lifted with forceps, and the bottom is cut with iris scissors. Biopsy sites may or may not need suture closure, depending on size and location.

KOH Preparation A potassium hydroxide (KOH) preparation is performed on scaling skin lesions when a fungal etiology is suspected. The edge of such a lesion is scraped gently with a scalpel blade, and the removed scale is collected on a glass microscope slide and treated with 1 to 2 drops of a solution of 10 to 20% KOH. KOH dissolves keratin and allows easier visualization of fungal elements. Brief heating of the slide accelerates dissolution of keratin. When the preparation is viewed under the microscope, the refractile hyphae will be seen more easily when the light intensity is reduced. This technique can be utilized to identify hyphae in dermatophyte infections (see [Plate IID-51](#)), pseudohyphae and budding yeast in *Candida* infections (see [Plate IID-43](#)), and fragmented hyphae and spores in tinea versicolor. The same sampling technique can be used to obtain scale for culture of selected pathogenic organisms.

Tzanck Smear A Tzanck smear is a cytologic technique most often used in the diagnosis of herpesvirus infections [simplex or varicella-zoster (see [Plates IID-36](#) and [IID-37](#))]. An early vesicle, not a pustule or crusted lesion, is unroofed, and the base of the lesion is scraped gently with a scalpel blade. The material is placed on a glass slide, air-dried, and stained with Giemsa or Wright's stain. Multinucleated giant cells suggest the presence of herpes, but culture or immunofluorescence testing must be performed to identify the specific virus.

Diascopy Diascopy is designed to assess whether a skin lesion will blanch with pressure as, for example, in determining whether a red lesion is hemorrhagic or simply blood-filled. For instance, a hemangioma (see [Plate IIA-17](#)) will blanch with pressure, whereas a purpuric lesion caused by necrotizing vasculitis (see [Plate IIE-71](#)) will not. Diascopy is performed by pressing a microscope slide or magnifying lens against a specified lesion and noting the amount of blanching that occurs. Granulomas often have an "apple jelly" appearance on diascopy.

Wood's Light A Wood's lamp generates 360-nm ultraviolet (or "black") light that can be

used to aid the evaluation of certain skin disorders. For example, a Wood's lamp will cause erythrasma (a superficial, intertriginous infection caused by *Corynebacterium minutissimum*) to show a characteristic coral red color, and wounds colonized by *Pseudomonas* to appear pale blue. Tinea capitis caused by certain dermatophytes such as *Microsporum canis* or *M. audouini* exhibits a yellow fluorescence. Pigmented lesions of the epidermis such as freckles are accentuated, while dermal pigment such as postinflammatory hyperpigmentation fades under a Wood's light. Vitiligo (see [Plate IIA-11](#)) appears totally white under a Wood's lamp, and previously unsuspected areas of involvement often become apparent. A Wood's lamp may also aid in the demonstration of tinea versicolor and in recognition of ash leaf spots in patients with tuberous sclerosis.

Patch Tests Patch testing is designed to document sensitivity to a specific antigen. In this procedure, a battery of suspected allergens is applied to the patient's back under occlusive dressings and allowed to remain in contact with the skin for 48 h. The dressings are removed, and the area is examined for evidence of delayed hypersensitivity reactions (e.g., erythema, edema, or papulovesicles). This test is best performed by physicians with special expertise in patch testing and is often helpful in the evaluation of patients with chronic dermatitis.

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56. ECZEMA, PSORIASIS, CUTANEOUS INFECTIONS, ACNE, AND OTHER COMMON SKIN DISORDERS - Robert A. Swerlick, Thomas J. Lawley

ECZEMA AND DERMATITIS

Eczema, or dermatitis, is a reaction pattern that presents with variable clinical and histologic findings and is the final common expression for a number of disorders, including atopic dermatitis, allergic contact and irritant contact dermatitis, dyshidrotic eczema, nummular eczema, lichen simplex chronicus, asteatotic eczema, and seborrheic dermatitis. Primary lesions may include papules, erythematous macules, and vesicles, which can coalesce to form patches and plaques. In severe eczema, secondary lesions from infection or excoriation, marked by weeping and crusting, may predominate. Long-standing dermatitis is often dry and is characterized by thickened, scaling skin (*lichenification*).

ATOPIC DERMATITIS

Atopic dermatitis (AD) is the cutaneous expression of the atopic state, characterized by a family history of asthma, hay fever, or dermatitis in up to 70% of patients. The criteria for the diagnosis of atopic eczema are shown in [Table 56-1](#). The prevalence of atopic dermatitis is increasing worldwide, with a point prevalence in Norwegian school children as high as 23%.

The etiology of [AD](#) is only partially defined. There is a clear genetic predisposition. When both parents are affected by AD, over 80% of their children manifest the disease. When only one parent is affected, the prevalence drops to slightly over 50%. A number of genes have been tentatively linked to AD including genes coding for IgE, the high-affinity IgE receptor, mast cell tryptase, and interleukin (IL) 4. Patients with AD may display a variety of immunoregulatory abnormalities including increased IgE synthesis; increased specific IgE to foods, aeroallergens, bacteria, and bacterial products; increased expression of CD23 (low-affinity IgE receptor) on monocytes and B cells; impaired delayed type hypersensitivity reactions; and increased type II and decreased type I cytokine responses.

The clinical presentation often varies with age. Half of patients with [AD](#) present within the first year of life, and 80% present by 5 years of age. Some 80% ultimately coexpress allergic rhinitis or asthma later in life. The infantile pattern is characterized by weeping inflammatory patches and crusted plaques that occur on the face, neck, extensor surfaces, and groin. The childhood and adolescent pattern is marked by dermatitis of flexural skin, particularly in the antecubital and popliteal fossae (see [Plate IIA-4](#)). AD may resolve spontaneously in adults, but the dermatitis will persist into adult life in over half of individuals affected as children. The distribution of lesions may be similar to those seen in childhood. However, adults affected with AD frequently have localized disease, manifesting as hand eczema or lichen simplex chronicus (see below).

Pruritus is a prominent characteristic of [AD](#), and many of the cutaneous findings in affected patients are secondary to rubbing and scratching. Other cutaneous stigmata of AD are perioral pallor, an extra fold of skin beneath the lower eyelid (Dennie's line), increased palmar markings, and increased incidence of cutaneous infections,

particularly with *Staphylococcus aureus*. Atopic individuals often have dry itchy skin, abnormalities in cutaneous vascular responses, and, in some instances, elevations in serum IgE.

Histologic examination of the skin affected by AD may demonstrate features of acute or chronic dermatitis. Immunopathology shows activated, memory T helper cells, which express the cutaneous lymphocyte antigen, the ligand for the inducible endothelial cell adhesion molecule E-selectin. AD skin lesions may also demonstrate IgE-bearing CD1a+ positive Langerhans cells, and these cells have been implicated in AD disease pathophysiology through mediation of hypersensitivity responses to environmental antigens.

TREATMENT

Therapy of AD should be based on avoidance of cutaneous irritants, adequate cutaneous hydration, judicious use of low- or midpotency topical glucocorticoids, and prompt treatment of secondarily infected skin lesions. Patients should be instructed to bathe using warm, but not hot, water and to limit their use of soap. Immediately after bathing while the skin is still moist, the skin should be lubricated with a low- or midpotency topical glucocorticoid in a cream or ointment base. Potent fluorinated topical glucocorticoids should not be used on the face or intertriginous areas. It takes a minimum of 30 g of glucocorticoid ointment to cover the entire body surface of an average adult.

Crusted and weeping skin lesions should be treated with systemic antibiotics with activity against *S. aureus* since secondary infection often exacerbates eczema. The frequency of macrolide-resistant organisms makes the use of penicillinase-resistant penicillins or cephalosporins preferable. Dicloxacillin or cephalexin (250 mg four times daily for 7 to 10 days) is generally adequate to decrease heavy colonization. As an adjunct, the use of triclosan-containing antibacterial washes and intermittent nasal mupirocin may be useful as prophylactic measures. The role of dietary allergens in atopic dermatitis is controversial, and there is little evidence that they play any role outside of infancy.

Control of pruritus is essential for treatment, since AD often represents "an itch that rashes." Antihistamines are useful to control the pruritus, but sedation may limit their usefulness. Unlike their effects in urticaria, nonsedating antihistamines are of little use since the effectiveness of antihistamines in the treatment of pruritus associated with AD is primarily related to their sedative effects as opposed to any specific action on histamine-mediated pathways.

Treatment with systemic glucocorticoids should be limited to severe exacerbations unresponsive to conservative topical therapy. In the patient with chronic AD, therapy with systemic glucocorticoids will generally clear the skin only briefly, but cessation of the systemic therapy will invariably be accompanied by return, if not worsening, of the dermatitis. Patients who do not respond to conventional therapies should be considered for patch testing to rule out allergic contact dermatitis. Immunotherapy with aeroallergens has not proven useful in AD, unlike its effect in allergic rhinitis and extrinsic asthma.

CONTACT DERMATITIS

Contact dermatitis is an inflammatory process in skin caused by an exogenous agent or agents that directly or indirectly injure the skin. This injury may be caused by an inherent characteristic of a compound -- irritant contact dermatitis (ICD). An example of ICD would be dermatitis induced by a concentrated acid or base. Agents that cause allergic contact dermatitis (ACD) induce an antigen-specific immune response. The clinical lesions of contact dermatitis may be acute (wet and edematous) or chronic (dry, thickened, and scaly), depending on the persistence of the insult (see Plate IIA-8). The most common presentation of contact dermatitis is hand eczema, and it is frequently related to occupational exposures. Occupation-related contact dermatitis represents a significant proportion of occupation-induced injury, affecting over 60,000 persons annually.

[ICD](#) is generally strictly demarcated and often localized to areas of thin skin (eyelids, intertriginous areas) or to areas where the irritant was occluded. Lesions may range from minimal skin erythema to areas of marked edema, vesicles, and ulcers. Chronic low-grade irritant dermatitis is the most common type of ICD and the most common area of involvement is the hands (see below). The most common irritants encountered are chronic wet work, soaps, and detergents. Treatment should be directed to avoidance of irritants and use of protective gloves or clothing.

[ACD](#) is a manifestation of delayed type hypersensitivity mediated by memory T lymphocytes in the skin. The most common cause of ACD is exposure to plants, specifically to members of the family Anacardiaceae; including the genera *Toxicodendron*, *Anacardium*, *Gluta*, *Mangifera*, and *Semecarpus*. Poison ivy, poison oak, and poison sumac are members of the genus *Toxicodendron* and cause an allergic reaction marked by erythema, vesiculation, and severe pruritus. The eruption is often linear, corresponding to areas where plants have touched the skin. However, other allergens may be more difficult to identify, especially if the exposure is chronic and the skin becomes thickened and scaly. The sensitizing antigen common to these plants is urushiol, an oleoresin containing the active ingredient pentadecylcatechol. The oleoresin may adhere to skin, clothing, tools, and pets, and contaminated articles may cause dermatitis even after prolonged storage. Blister fluid does not contain urushiol and is not capable of inducing skin eruption in exposed subjects.

TREATMENT

If [ACD](#) is suspected and an offending agent is identified and removed, the eruption will resolve. Usually, treatment with high-potency fluorinated topical glucocorticoids is enough to relieve symptoms while the ACD runs its course. For those patients who require systemic therapy, a tapering course over 2 to 3 weeks given as single morning doses is the preferred method.

Identification of a contact allergen can be a difficult and time-consuming task. Patients with dermatitis unresponsive to conventional therapy or with an unusual and patterned distribution should be suspected of having [ACD](#). They should be questioned carefully regarding occupational exposures, topical medicaments, and oral medications.

Common sensitizers include preservatives in topical preparations, nickel sulfate, potassium dichromate, thimerosal in ocular preparations, neomycin sulfate, fragrances, formaldehyde, and rubber-curing agents. Patch testing is helpful in identifying these agents, but should not be attempted on patients with widespread active dermatitis or on those taking systemic glucocorticoids.

HAND ECZEMA

Hand eczema is a very common, chronic skin disorder. It represents a large proportion of occupation-associated skin disease. It may be associated with other cutaneous disorders such as atopic dermatitis or may occur by itself. Similar to other forms of dermatitis, both exogenous and endogenous factors play important roles in the expression of hand dermatitis. Chronic, excessive exposure to water and detergents may initiate or aggravate this disorder. It may present with dryness and cracking of the skin of the hands as well as with variable amounts of erythema and edema. Often, the dermatitis will begin under rings where water and irritants are trapped. A variant of hand dermatitis, dyshidrotic eczema, presents with multiple, intensely pruritic, small papules and vesicles occurring on the thenar and hypothenar eminences and the sides of the fingers (see Plate IA-5). Lesions tend to occur in crops that slowly form crusts and heal.

The evaluation of a patient with hand eczema should include an assessment of potential occupation-associated exposures. Predominant involvement of the dorsal surface of the hands with sparing of the palmar surface suggests a possible contact dermatitis. The history should be directed to identifying possible irritant or allergen exposures. The use of rubber gloves to protect dermatitic skin is sometimes associated with the development of delayed type hypersensitivity reactions to agents used for cross-linking rubber. Such reactions can be detected by patch testing. Less commonly, patients may manifest hand dermatitis as a consequence of developing immediate type hypersensitivity reactions to latex. These are of particular concern since these patients are at risk for anaphylactic reactions. The most sensitive method of detection is the use of scratch testing with latex extract. However, this should be done with extreme caution only in a setting where an anaphylactic reaction can be treated. A latex radioallergosorbent test is available but is only about 60% sensitive.

TREATMENT

Therapy of hand dermatitis is directed toward avoidance of irritants, identification of possible contact allergens, treatment of coexistent infection, and application of topical glucocorticoids. Whenever possible, the hands should be protected by gloves, preferably vinyl. Most patients can be treated with cool moist compresses (dressings) to dry and debride acute inflammatory lesions and to decrease swelling, followed by application of a mid- to high-potency topical glucocorticoid in a cream or ointment base. As with atopic dermatitis, treatment of secondary infection by staphylococci or streptococci is essential for good control. Additionally, patients with hand dermatitis should be examined for dermatophyte infection by KOH preparation and culture (see below).

NUMMULAR ECZEMA

Nummular eczema is characterized by circular or oval "coinlike" lesions. Initially, this eruption consists of small edematous papules that become crusted and scaly. The most common locations are on the trunk or the extensor surfaces of the extremities, particularly on the pretibial areas or dorsum of the hands. It occurs more frequently in men and is most commonly seen in middle age. The etiology of nummular eczema is unknown. Whether nummular eczema represents a variant of atopic eczema is controversial. The treatment of nummular eczema is similar to that for other forms of dermatitis.

LICHEN SIMPLEX CHRONICUS

Lichen simplex chronicus may represent the end stage of a variety of pruritic and eczematous disorders. It consists of a well-circumscribed plaque or plaques with lichenified or thickened skin due to chronic scratching or rubbing. Common areas involved include the posterior nuchal region, dorsum of the feet, or ankles. Treatment of lichen simplex chronicus centers around breaking the cycle of chronic itching and scratching, which often occur during sleep. High-potency topical glucocorticoids are helpful in alleviating pruritus in most cases, but in recalcitrant cases, application of topical glucocorticoids under occlusion or intralesional injection of glucocorticoids may be required. Oral antihistamines such as hydroxyzine (10 to 50 mg every 6 h) or tricyclic antidepressants with antihistaminic activity such as doxepin (10 to 25 mg at bedtime) are useful as antipruritics primarily due to their sedating action, and are particularly useful at bedtime (see above). Patients need to be counseled regarding driving or operating heavy equipment after taking these medications due to their potentially potent sedative activity.

ASTEATOTIC ECZEMA

Asteatotic eczema, also known as xerotic eczema or "winter itch," is a mildly inflammatory variant of dermatitis that develops most commonly on the lower legs of elderly individuals during dry times of year. Fine cracks, with or without erythema, characteristically develop on the anterior surface of the lower extremities. Pruritus is variable. Asteatotic eczema responds well to avoidance of irritants, rehydration of the skin, and application of topical emollients.

STASIS DERMATITIS AND STASIS ULCERATION

Stasis dermatitis develops on the lower extremities secondary to venous incompetence and chronic edema. Early findings in stasis dermatitis consist of mild erythema and scaling associated with pruritus. The typical initial site of involvement is the medial aspect of the ankle, often over a distended vein (see [Plate IIA-7](#)). As the disorder progresses, the dermatitis becomes progressively pigmented, due to chronic erythrocyte extravasation leading to cutaneous hemosiderin deposition. As with other forms of dermatitis, stasis dermatitis may become acutely inflamed, with crusting and exudate. Chronic stasis dermatitis is often associated with dermal fibrosis that is recognized clinically as brawny edema of the skin. Stasis dermatitis is often complicated by secondary infection and contact dermatitis. Severe stasis dermatitis may precede the development of stasis ulcers.

TREATMENT

Avoidance of irritants and use of emollients and/or midpotency topical glucocorticoids are the cornerstones of therapy for stasis dermatitis. Control of chronic edema is important to prevent leg ulcers. Patients should be encouraged to elevate the affected extremity when sitting. A compression stocking with a gradient of at least 30 to 40 mmHg is most effective for edema control and is much more effective for preventing chronic edema than is antiembolism hose.

Stasis ulcers are difficult to treat, and resolution of these lesions is slow even under the best of circumstances. It is extremely important to elevate the affected limb as much as possible. The ulcer should be kept clear of necrotic material by gentle debridement and covered with a semipermeable dressing under pressure. Glucocorticoids should not be applied to ulcers, since they may retard healing. Secondarily infected lesions should be treated with appropriate oral antibiotics, but it should be noted that all ulcers will become colonized with bacteria, and the purpose of antibiotic therapy should not be to clear all bacterial growth. Some ulcers may take months to heal or require skin grafting.

SEBORRHEIC DERMATITIS

Seborrheic dermatitis is a common, chronic disorder, characterized by greasy scales overlying erythematous patches or plaques. The most common location is in the scalp where it may be recognized as severe dandruff. On the face, seborrheic dermatitis affects the eyebrows, eyelids, glabella, nasolabial fold, or ears (see [Plate IIA-6](#)). Scaling within the external ear is often mistaken for a chronic fungal infection (otomycosis), and postauricular dermatitis often becomes macerated and tender. Additionally, seborrheic dermatitis may develop in the central chest, axilla, groin, submammary folds, and gluteal cleft. Rarely, it may cause a widespread generalized dermatitis. Seborrheic dermatitis is usually symptomatic, with patients complaining of itching or burning.

Seborrheic dermatitis may be evident within the first few weeks of life, and within this context it occurs in the scalp ("cradle cap"), face, or groin. It is rarely seen in children beyond infancy but becomes evident again during adult life. Although it is frequently seen in patients with Parkinson's disease, in those who have had cerebrovascular accidents, and in those with human immunodeficiency virus (HIV) infection, the overwhelming majority of individuals with seborrheic dermatitis have no underlying disorder.

TREATMENT

Treatment with low-potency topical glucocorticoids in conjunction with shampoos containing coal tar and/or salicylic acid is generally sufficient to control activity of this disorder. High-potency topical glucocorticoid solutions (betamethasone or fluocinonide) are effective for control of scalp involvement. Fluorinated topical glucocorticoids should not be used on the face since this is often associated with the development of rebound worsening and steroid-induced rosacea or atrophy.

PAPULOSQUAMOUS DISORDERS ([Table 56-2](#))

PSORIASIS

Psoriasis is one of the most common dermatologic diseases, affecting up to 1 to 2% of the world's population. It is a chronic inflammatory skin disorder clinically characterized by erythematous, sharply demarcated papules and rounded plaques, covered by silvery micaceous scale. The skin lesions of psoriasis are variably pruritic. Traumatized areas often develop lesions of psoriasis (Koebner or isomorphic phenomenon). Additionally, other external factors may exacerbate psoriasis including infections, stress, and medications (lithium, beta blockers, and antimalarials).

The most common variety of psoriasis is called *plaque type*. Patients with plaque-type psoriasis will have stable, slowly growing plaques, which remain basically unchanged for long periods of time. The most common areas for plaque psoriasis to occur are the elbows, knees, gluteal cleft, and the scalp. Involvement tends to be symmetric. *Inverse psoriasis* affects the intertriginous regions including the axilla, groin, submammary region, and navel, it also tends to affect the scalp, palms, and soles. The individual lesions are sharply demarcated plaques (see [Plate IIA-3](#)) but may be moist due to their location. Plaque psoriasis generally develops slowly and runs an indolent course. It rarely remits spontaneously.

Eruptive psoriasis (guttate psoriasis) is most common in children and young adults. It develops acutely in individuals without psoriasis or in those with chronic plaque psoriasis. Patients present with many small erythematous, scaling papules, frequently after upper respiratory tract infection with *β*-hemolytic streptococci. The differential diagnosis should include pityriasis rosea and secondary syphilis. Patients with psoriasis may also develop pustular lesions. These may be localized to the palms and soles or may be generalized and associated with fever, malaise, diarrhea, and arthralgias.

About half of all patients with psoriasis have fingernail involvement, appearing as punctate pitting, nail thickening, or subungual hyperkeratosis. About 5 to 10% of patients with psoriasis have associated joint complaints, and these are most often found in patients with fingernail involvement. Although some have the coincident occurrence of classic rheumatoid arthritis ([Chap. 312](#)), many have joint disease that falls into one of three types associated with psoriasis: (1) asymmetric inflammatory arthritis most commonly involving the distal and proximal interphalangeal joints and less commonly the knees, hips, ankles, and wrists; (2) a seronegative rheumatoid arthritis-like disease; a significant portion of these patients go on to develop a severe destructive arthritis; or (3) disease limited to the spine (psoriatic spondylitis).

The etiology of psoriasis is still poorly understood. There is clearly a genetic component to psoriasis. Over 50% of patients with psoriasis report a positive family history, and a 65 to 72% concordance among monozygotic twins has been reported in twin studies. Psoriasis has been linked to HLA-Cw6 and, to a lesser extent, to HLA-DR7. Evidence has accumulated clearly indicating a role for T cells in the pathophysiology of psoriasis. Stimulation of immune function with cytokines such as [IL-2](#) has been associated with abrupt worsening of preexisting psoriasis, and bone marrow transplantation has resulted in clearance of disease. Psoriatic lesions are characterized by infiltration of skin with activated memory T cells, with CD8+ cells predominating in the epidermis. Agents that inhibit activated T cell function are often effective for the treatment of severe psoriasis.

Presumably, cytokines from activated T cells elaborate growth factors that stimulate keratinocyte hyperproliferation.

TREATMENT

Treatment of psoriasis depends on the type, location, and extent of disease. All patients should be instructed to avoid excess drying or irritation of their skin and to maintain adequate cutaneous hydration. Most patients with localized plaque-type psoriasis can be managed with midpotency topical glucocorticoids, although their long-term use is often accompanied by loss of effectiveness (tachyphylaxis). Crude coal tar (1 to 5% in an ointment base) is an old but useful method of treatment in conjunction with ultraviolet light therapy. A topical vitamin D analogue (calcipotriol) is also efficacious in the treatment of psoriasis.

Ultraviolet light is an effective therapy for patients with widespread psoriasis. The ultraviolet B (UV-B) spectrum is effective alone, or may be combined with coal tar (Goeckerman regimen) or anthralin (Ingram regimen). Natural sunlight or an artificial light source can be used. The combination of the ultraviolet A (UV-A) spectrum with either oral or topical psoralens (PUVA) is also extremely effective for the treatment of psoriasis, but long-term use may be associated with an increased incidence of squamous cell cancer and melanoma of the skin.

Various other agents can be used for widespread psoriatic disease. Methotrexate is an effective agent, especially in patients with associated psoriatic arthritis. Liver toxicity from long-term use limits its use to patients with widespread disease not responsive to less aggressive modalities. The synthetic retinoid, acetrein, has been shown to be effective in some patients with severe psoriasis but is a potent teratogen, thus limiting its use in women with childbearing potential. The evidence implicating psoriasis as a T cell-mediated disorder has created a new perspective relating to the treatment of psoriasis. Based on this presumed disease mechanism, immunomodulatory therapy utilizing cyclosporine has proven to be highly effective in selected patients with severe, crippling, and potentially life-threatening disease.

LICHEN PLANUS

Lichen planus (LP) is a papulosquamous disorder in which the primary lesions are pruritic, polygonal, flat-topped, violaceous papules. Close examination of the surface of these papules often reveals a network of gray lines (Wickham's striae). The skin lesions may occur anywhere but have a predilection for the wrists, shins, lower back, and genitalia (see [Plate IIA-9](#)). Involvement of the scalp may lead to hair loss. LP commonly involves mucous membranes, particularly the buccal mucosa, where it can present as a white netlike eruption. Its etiology is unknown, but cutaneous eruptions clinically resembling LP have been observed after administration of numerous drugs, including diuretics, gold, antimalarials, penicillamine, and phenothiazines, and in patients with skin lesions of chronic graft-versus-host disease. Additionally, LP associated with abnormal liver function has been correlated with viral hepatitis, particularly hepatitis C infection. The course of LP is variable, but most patients have spontaneous remissions 6 months to 2 years after the onset of disease. Topical glucocorticoids are the mainstay of therapy.

PITYRIASIS ROSEA

Pityriasis rosea (PR) is a papulosquamous eruption of unknown etiology that occurs more commonly in the spring and fall. Its first manifestation is the development of a 2- to 6-cm annular lesion (the herald patch). This is followed in a few days to a few weeks by the appearance of many smaller annular or papular lesions with a predilection to occur on the trunk (see [Plate IIA-13](#)). The lesions are generally oval, with their long axis parallel to the skin-fold lines. Individual lesions may range in color from red to brown and have a trailing scale. PR shares many clinical features with the eruption of secondary syphilis, but palm and sole lesions are extremely rare in PR and common in secondary syphilis. The eruption tends to be moderately pruritic and lasts 3 to 8 weeks. Treatment is generally directed at alleviating pruritus and consists of oral antihistamines, midpotency topical glucocorticoids, and, in some cases, the use of [UV-B](#) phototherapy.

CUTANEOUS INFECTIONS ([Table 56-3](#))

IMPETIGO AND ECTHYMA

Impetigo is a common superficial bacterial infection of skin caused by group Ab-hemolytic streptococci ([Chap. 140](#)) or *S. aureus* ([Chap. 139](#)). The primary lesion is a superficial pustule that ruptures and forms a characteristic yellow-brown honey-colored crust (see [Plate IID-38](#)). Lesions caused by staphylococci may be tense, clear bullae, and this less common form of the disease is called *bullous impetigo*. Lesions may occur on normal skin or in areas already affected by another skin disease. Ecthyma is a variant of impetigo that generally occurs on the lower extremities and causes punched-out ulcerative lesions. Treatment of both ecthyma and impetigo involves gentle debridement of adherent crusts, which is facilitated by the use of soaks and topical antibiotics, in conjunction with appropriate oral antibiotics.

ERYSIPELAS AND CELLULITIS

See [Chap. 128](#)

DERMATOPHYTOSIS

Dermatophytes are fungi that infect skin, hair, and nails and include members of the genera *Trichophyton*, *Microsporum*, and *Epidermophyton*. Infection of the foot (tinea pedis) is most common and is often chronic; it is characterized by variable erythema and edema, scaling, pruritus, and occasionally vesiculation. Involvement may be widespread or localized, but almost invariably the web space between the fourth and fifth toes is affected. Infection of the nails (tinea unguium) occurs in many patients with tinea pedis and is characterized by opacified, thickened nails and subungual debris. The groin is the next most commonly involved area (tinea cruris), with males affected much more often than females. It presents as a scaling erythematous eruption that spares the scrotum. Microscopic examination of either untreated tinea pedis or tinea cruris scale after digestion with KOH preparation will generally demonstrate hyphae.

Dermatophyte infection of the scalp (tinea capitis) has returned in epidemic proportions,

particularly affecting inner city children. The predominant organism is *T. tonsurans*. This organism can produce an inflammatory or relatively noninflammatory infection that may present with either well-defined or irregular, diffuse areas of mild scaling and hair loss. Tinea corporis, or infection on non-hair-bearing skin, may have a variable appearance, depending on the extent of the associated inflammatory reaction (see [Plate IID-51](#)). It may have the typical annular appearance of "ringworm" or appear as deep inflammatory nodules (on the scalp known as a *kerion*) or granulomas. KOH examination of scale or hair from patients with tinea capitis or inflammatory tinea corporis often does not reveal hyphae, and diagnosis may require culture or biopsy.

TREATMENT

Both topical and systemic therapies may be used to treat dermatophyte infection. Treatment depends on the site involved and the type of infection. Topical therapy is generally effective for uncomplicated tinea corporis, tinea cruris, and limited tinea pedis. It is not effective as monotherapy for tinea capitis or tinea unguium. Topical imidazoles (miconazole, ketoconazole, econazole, clotrimazole, oxiconazole, and sulconazole), triazoles (terconazole), and allylamines (terbinafine and naftifine) may all be effective topical therapies for dermatophyte infections. Haloprogin, undecylic acid, ciclopirox-olamine, and tolnaftate are also effective, but nystatin is not active against dermatophytes. Treatment should continue until the patient is clear of infection by clinical examination and culture. Tinea pedis often requires longer treatment courses and is associated with a high relapse rate.

Griseofulvin is the drug of choice for dermatophyte infections requiring systemic therapy. A daily dose of 500 mg of microsize or 350 mg of ultramicrosize griseofulvin administered with a fatty meal is an adequate dose for most dermatophyte infections. The duration of therapy may be as short as 2 weeks for uncomplicated tinea corporis but may be as long as 6 to 12 months for nail infections. The most common side effects of griseofulvin are gastrointestinal distress and headache. Dermatophyte infection of hair-bearing areas (e.g., tinea capitis) requires systemic antifungal therapy. The usual adult dose of griseofulvin is 1 g of microsize or 0.5 g of ultramicrosize given daily, and treatment should be continued for 6 to 8 weeks. Children should be treated with 15 to 20 mg/kg as a single daily dose given with a fatty meal. The adjunctive use of topical antifungal agents in addition to systemic therapy may be useful, but topical therapy alone is not adequate. Markedly inflammatory tinea capitis may result in scarring and hair loss, and systemic or topical glucocorticoids may be helpful in preventing this sequela. Recent studies in children have also suggested that both itraconazole (3 to 5 mg/kg for 6 to 10 weeks) and terbinafine (125 mg/d for 6 weeks) may be effective treatments for tinea capitis.

Until recently, griseofulvin was the recommended therapy for dermatophyte infection of the nails. However, despite prolonged treatment, cure rates were poor. Itraconazole given as either continuous daily therapy (200 mg/d for 3 months) or pulses (200 mg twice daily for 1 week per month for 3 consecutive months) has been shown to be a safe and effective therapy. Itraconazole has the potential for interactions with other drugs requiring the P450 enzyme system for metabolism. Similarly, terbinafine (250 mg/d for 3 months) has shown similar cure rates. Only limited data are available on the dosing and effectiveness of the newer antifungal agents in tinea corporis, tinea cruris,

and uncomplicated tinea pedis.

TINEA VERSICOLOR

Tinea versicolor is caused by a nondermatophyte dimorphic fungus that is a normal inhabitant of the skin. As the yeast form *Pityrosporum orbiculare*, it generally does not cause disease (except for folliculitis in certain individuals). However, in some individuals, it converts to the hyphal form and causes characteristic lesions. The expression of infection is promoted by heat and humidity. The typical lesions consist of oval scaly macules, papules, and patches concentrated on the chest, shoulders, and back but only rarely on the face or distal extremities. On dark skin, they often appear as hypopigmented areas, while on light skin, they are slightly hyperpigmented. In some darkly pigmented individuals, they may only appear as scaling patches. A KOH preparation from scaling lesions will demonstrate a confluence of short hyphae and round spores (so-called spaghetti and meatballs). Solutions containing sulfur, salicylic acid, or selenium sulfide will clear the infection if used daily for a week and then intermittently thereafter. Treatment with a single 400-mg dose of ketoconazole is also effective.

CANDIDIASIS

Candidiasis is a fungal infection caused by a related group of yeasts, whose manifestations may be localized to the skin, or rarely, may be systemic and life-threatening. The causative organism is usually *Candida albicans*, but may also be *C. tropicalis*, *C. parapsilosis*, or *C. krusei*. These organisms are normal saprophytic inhabitants of the gastrointestinal tract but may overgrow (usually due to broad-spectrum antibiotic therapy) and cause disease at a number of cutaneous sites. Other predisposing factors include diabetes mellitus, chronic intertrigo, oral contraceptive use, and cellular immune deficiency. Candidiasis is a very common infection in HIV-infected individuals ([Chap. 309](#)). The oral cavity is commonly involved. Lesions may occur on the tongue or buccal mucosa (thrush) and appear as white plaques (see [Plate IID-43](#)). Microscopic examination of scrapings demonstrate both pseudohyphae and yeast forms. Fissured, macerated lesions at the corners of the mouth (perleche) are often seen in individuals with poorly fitting dentures and may also be associated with candidal infection. Additionally, candidal infections have an affinity for sites that are chronically wet and macerated and may occur around nails (onycholysis and paronychia) and in intertriginous areas. Intertriginous lesions are characteristically edematous, erythematous, and scaly, with scattered "satellite pustules." In males, there is often involvement of the penis and scrotum as well as the inner aspect of the thighs. In contrast to dermatophyte infections, candidal infections are frequently accompanied by a marked inflammatory response. Diagnosis of candidal infection is based upon the clinical pattern and demonstration of yeast on KOH preparation, or culture.

TREATMENT

Treatment routinely involves removing any predisposing factors such as antibiotic therapy or chronic wetness and the use of appropriate topical or systemic antifungal therapy. Effective topical agents include nystatin or topical azoles (miconazole,

clotrimazole, econazole, or ketoconazole). These agents are generally effective in clearing mucous membrane or glabrous skin involvement in nonimmunosuppressed patients. The associated inflammatory response that often accompanies candidal infection on glabrous skin should be treated with a mild glucocorticoid lotion or cream (2.5% hydrocortisone). Systemic therapy is generally reserved for immunosuppressed patients or individuals with chronic or recurrent disease who fail to respond to or tolerate appropriate topical therapy. Vulvovaginal candidiasis may respond to treatment with a single dose of fluconazole (150 mg). Chronic recurrent oral or vaginal candidiasis may be treated with weekly to monthly oral fluconazole (150 to 200 mg) in conjunction with topical therapy.

WARTS

Warts are cutaneous neoplasms that are caused by papilloma viruses. Over 50 different human papilloma viruses (HPV) have been described, and this number will almost certainly continue to grow. Typical verruca vulgaris lesions are sessile, dome-shaped, usually about a centimeter in diameter, and their surface is made up of many small filamentous projections. The HPV that cause typical verruca vulgaris also cause typical plantar warts, flat warts (or verruca plana), and filiform warts in intertriginous areas. Plantar warts are endophytic and are covered by thick keratin. Paring of the wart will generally demonstrate a central core of keratinized debris and punctate bleeding points. Filiform warts are most commonly seen on the face, neck, and skin folds and present as papillomatous lesions on a narrow base. Flat warts are only slightly elevated and have a velvety, nonverrucous surface. They have a propensity for the face, arms, and legs and are often spread by shaving.

Multiple [HPV](#) types have been associated with genital tract lesions. They generally begin as small papillomas that may grow to form large fungating lesions. In women, they may involve either the labia, perineum, or perianal skin. Additionally, the mucosa of the vagina, urethra, and anus can be involved, as well as the cervical epithelium. In men, the lesions often occur initially in the coronal sulcus, but may be seen on the shaft of the penis, the scrotum, perianal skin, or in the urethra.

Within the past decade, appreciable evidence has accumulated that suggests [HPV](#) plays a role in the development of neoplasia of the uterine cervix and external genitalia ([Chap. 97](#)). HPV types 16 and 18 have been most intensely studied, while recent evidence also implicates other types. Lesions may initially appear as small, flat, velvety, hyperpigmented papules occurring on the genitalia or perianal skin. Histologic examination of biopsies from affected sites may reveal changes associated with typical warts and/or features typical of intraepidermal carcinoma (Bowen's disease). Squamous cell carcinomas associated with HPV infections have also been observed in extragenital skin ([Chap. 86](#)). This is most commonly seen in patients immunosuppressed after organ transplantation.

TREATMENT

There are many modalities available to treat warts, but no single therapy is universally effective. Factors that influence the choice of therapy include the location of the wart, extent of disease, the age and immunologic status of the patient, and the patient's

desire for therapy. Perhaps the most useful and convenient method for treating warts in almost any location is cryotherapy with liquid nitrogen. Equally effective, but requiring much more patient compliance, is the use of keratolytic agents such as salicylic acid plasters or combinations of lactic acid and salicylic acid. For genital warts, application of podophyllin solution is moderately effective but may be associated with marked local reactions in certain individuals. Dilute preparations of purified podophyllin permit physician-directed by patient-applied use, facilitating treatment of mucosal warts. Topical imiquimod, a potent inducer of local cytokine release, has also been approved for use in genital warts. Other topical agents that are used include trichloroacetic acid or cantharidin. Electrodesiccation and curettage or CO₂ laser excision are also effective therapies but require local anesthesia. Recurrence of warts appears to be common to all these modalities because viral genomic material is present in normal-appearing skin adjacent to the clinical lesions.

Treatment of warts should be tempered by the observation that an overwhelming majority of warts in normal individuals resolve spontaneously within 1 to 2 years. Also, only an extremely small proportion of warts is associated with neoplasia, and those are almost exclusively located on the genitalia or perianal skin.

HERPES SIMPLEX

[*See Chap. 182](#)

HERPES ZOSTER

[*See Chap. 183](#)

ACNE

ACNE VULGARIS

Acne vulgaris is a self-limited disorder primarily of teenagers and young adults, although perhaps 10 to 20% of adults may continue to experience some form of the disorder. The permissive factor for the expression of the disease in adolescence is the increase in sebum release by sebaceous glands after puberty. Small cysts, called *comedones*, form in hair follicles due to blockage of the follicular orifice by retention of sebum and keratinous material. The activity of lipophilic yeast (*Pityrosporum orbiculare*) and bacteria (*Propionibacterium acnes*) within the comedones releases free fatty acids from sebum, causes inflammation within the cyst, and results in rupture of the cyst wall. An inflammatory foreign-body reaction develops as a result of extrusion of oily and keratinous debris from the cyst.

The clinical hallmark of acne vulgaris is the comedone, which may be closed (whitehead) or open (blackhead). Closed comedones appear as 1- to 2-mm pebbly white papules, which are accentuated when the skin is stretched. They are the precursors of inflammatory lesions of acne vulgaris. The contents of closed comedones are not easily expressed. Open comedones, which rarely result in inflammatory acne lesions, have a large dilated follicular orifice and are filled with easily expressible oxidized, darkened, oily debris. Comedones are usually accompanied by inflammatory

lesions: papules, pustules, or nodules.

The earliest lesions seen in early adolescence are generally mildly inflamed or noninflammatory comedones on the forehead. Subsequently, more typical inflammatory lesions develop on the cheeks, nose, and chin (see [Plate IIA-1](#)). The most common location for acne is the face, but involvement of the chest and back is not uncommon. Most disease remains mild and does not lead to scarring. However, a small number of patients develop large inflammatory cysts and nodules, which may drain and result in significant scarring.

Exogenous and endogenous factors can alter the expression of acne vulgaris. Friction and trauma may rupture preexisting microcomedones and elicit inflammatory acne lesions. This is commonly seen with headbands or chin straps of athletic helmets. Application of comedogenic topical agents in cosmetics or hair preparations or chronic topical exposure to certain industrial compounds that are comedogenic may elicit or aggravate acne. Glucocorticoids, applied topically or administered systemically in high doses, may also elicit acne. Other systemic medications such as lithium, isoniazid, halogens, phenytoin, and phenobarbital may produce acneiform eruptions, or aggravate preexisting acne.

TREATMENT

Treatment of acne vulgaris is directed toward elimination of comedones by normalization of follicular keratinization, decreasing sebaceous gland activity, decreasing the population of lipophilic bacteria and yeast, and decreasing inflammation. Acne vulgaris may be treated with either local or systemic medications. Minimal to moderate, pauci-inflammatory disease may respond adequately to local therapy alone. Although areas affected with acne should be kept clean, there is little evidence to suggest that removal of surface oils plays an important role in therapy. Overly vigorous scrubbing may aggravate acne due to mechanical rupture of comedones. Topical agents such as retinoic acid, benzoyl peroxide, or salicylic acid may alter the pattern of epidermal desquamation, preventing the formation of comedones and aiding in the resolution of preexisting cysts. Topical antibacterial agents such as benzoyl peroxide, azelaic acid, topical erythromycin (with or without zinc), clindamycin, or tetracycline are also useful adjuncts to therapy.

Patients with moderate to severe acne with a prominent inflammatory component will benefit from the addition of systemic therapy. Oral tetracyclines or erythromycin in doses of 250 to 1000 mg/d will decrease follicular colonization with some of the lipophilic organisms. They also appear to have an anti-inflammatory effect independent of their antibacterial effect. Female patients who do not respond to oral antibiotics may benefit from hormonal therapy. Women placed on oral contraceptives containing ethinyl estradiol and norgestimate have demonstrated improvement in their acne when compared to a placebo control.

Severe nodulocystic acne not responsive to oral antibiotics, hormonal therapy, or topical therapy may be treated with the synthetic retinoid isotretinoin. It is used at doses of 0.5 to 2.0 mg/kg as a single daily dose for 15 to 20 weeks. The use of this drug is limited by its teratogenicity, and female patients must be screened for pregnancy prior to initiating

therapy, maintain a method of birth control during therapy, and be screened for pregnancy during treatment. Patients receiving this medication develop extremely dry skin and cheilitis and must be followed for development of hypertriglyceridemia.

ACNE ROSACEA

Acne rosacea is an inflammatory disorder predominantly affecting the central face. It is seen almost exclusively in adults, only rarely affecting patients under 30 years of age. Rosacea is seen more often in women, but those most severely affected are men. It is characterized by the presence of erythema, telangiectases, and superficial pustules (see [Plate IIA-2](#)), but is not associated with the presence of comedones. Rosacea only rarely involves the chest or back.

There is a relationship between the tendency for pronounced facial flushing and the subsequent development of acne rosacea. Often, individuals with rosacea initially demonstrate a pronounced flushing reaction. This may be in response to heat, emotional stimuli, alcohol, hot drinks, or spicy foods. As the disease progresses, the flush persists longer and longer and may eventually become permanent. Papules, pustules, and telangiectases can become superimposed on the persistent flush. Rosacea of very long standing may lead to connective tissue overgrowth, particularly of the nose (rhinophyma). Rosacea may also be complicated by various inflammatory disorders of the eye, including keratitis, blepharitis, iritis, and recurrent chalazion. These ocular problems are potentially sight-threatening and warrant ophthalmologic evaluation.

TREATMENT

Acne rosacea can generally be treated effectively with oral tetracycline in doses ranging from 250 to 1000 mg/d. Topical metronidazole or sodium sulfacetamide has also been shown to be effective. In addition, the use of low-potency, nonfluorinated topical glucocorticoids, particularly after cool soaks, is helpful in alleviating facial erythema. Fluorinated topical glucocorticoids should be avoided since chronic use of these preparations may actually elicit rosacea. Topical therapy is not effective treatment for ocular disease.

(Bibliography omitted in Palm version)

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57. SKIN MANIFESTATIONS OF INTERNAL DISEASE - Jean L. Bologna, Irwin M. Braverman

It is now a generally accepted concept in medicine that the skin can show signs of internal disease. Therefore, in textbooks of medicine one finds a chapter describing in detail the major systemic disorders that can be identified by cutaneous signs. The underlying assumption of such a chapter is that the clinician has been able to identify the disorder in the patient and needs only to read about it in the textbook. In reality, concise differential diagnoses and the identification of these disorders are actually difficult for the nondermatologist because he or she is not well versed in the recognition of cutaneous lesions or their spectrum of presentations. Therefore, the authors of this chapter have decided to cover this particular topic of cutaneous medicine not by discussing individual disorders but by describing and discussing the various presenting clinical signs and symptoms that indicate the presence of these disorders. Concise differential diagnoses will be generated in which the significant diseases will be briefly discussed and distinguished from the more common disorders that have no significance for internal diseases. The latter disorders are reviewed in table form and always need to be excluded when considering the former. For a detailed description of individual diseases, the reader should consult a dermatologic text.

PAPULOSQUAMOUS SKIN LESIONS ([Table 57-1](#))

When an eruption is characterized by elevated lesions, papules (<1 cm) or plaques (>1 cm), in association with scale, it is referred to as *papulosquamous*. The most common papulosquamous diseases -- *psoriasis*, *tinea*, *pityriasis rosea*, and *lichen planus* -- are primary cutaneous disorders ([Chap. 56](#)). When psoriatic lesions are accompanied by arthritis, the possibility of psoriatic arthritis or *Reiter's disease* should be considered. A history of oral ulcers, conjunctivitis, uveitis, and/or urethritis points to the latter diagnosis. In *guttate psoriasis* there is an acute onset of small, widely scattered, uniform lesions, often in association with a streptococcal infection. Lithium, beta blockers, HIV infection, and a rapid taper of systemic glucocorticoids are also known to exacerbate psoriasis.

Whenever the diagnosis of pityriasis rosea or lichen planus is made, it is important to review the patient's medications because the eruption can be treated by simply discontinuing the offending agent. Pityriasis rosea-like drug eruptions are seen most commonly with beta blockers, angiotensin-converting enzyme (ACE) inhibitors, gold, and metronidazole, while the drugs that can produce a lichenoid eruption include gold, antimalarials, thiazides, quinidine, phenothiazines, sulfonyleureas, and ACE inhibitors. Lichen planus-like lesions are also observed in chronic graft-versus-host disease.

Parapsoriasis is an intermediate disease, for it can remain solely as a primary cutaneous disease or it can progress to cutaneous T cell lymphoma (CTCL) after a latency period of as long as 40 years. There are several forms of parapsoriasis, including small plaque (0.5 to 5 cm), large plaque (>6 cm), and retiform. The lesions of both small plaque and large plaque parapsoriasis are thin and salmon-pink in color with fine white scale. In small plaque forms, the lesions are commonly on the trunk but can be widely scattered. In large plaque forms, the most common location is the "girdle" area, and fine wrinkling secondary to epidermal atrophy is often seen. Retiform

parapsoriasis forms a netlike pattern, and the individual papules are red-brown and flat-topped. The latter two forms of parapsoriasis, large plaque and retiform, can progress to CTCL.

A clue to the development of [CTCL](#) within lesions of large plaque or retiform parapsoriasis is an increase in the palpable component of the plaque (increased infiltration). In its early stages, CTCL may be confused with eczema or psoriasis, but it often fails to respond to the appropriate therapy for those inflammatory diseases. The diagnosis of CTCL is established by skin biopsy in which collections of atypical T lymphocytes are found in the epidermis and dermis. As the disease progresses, cutaneous tumors and lymph node involvement may appear.

In *secondary syphilis* there are scattered red-brown papules with thin scale. The eruption often involves the palms and soles and can resemble pityriasis rosea. Associated findings are helpful in making the diagnosis and include annular plaques on the face, nonscarring alopecia, condyloma lata (broad-based and moist), and mucous patches as well as lymphadenopathy, malaise, fever, headache, and myalgias. The interval between the primary chancre and the secondary stage is usually 4 to 8 weeks, and spontaneous resolution without appropriate therapy is seen.

ERYTHRODERMA ([Table 57-2](#))

Erythroderma is the term used when the majority of the skin surface is erythematous (red in color). There may be associated scale, erosions, or pustules as well as shedding of the hair and nails. Potential systemic manifestations include fever, chills, hypothermia, reactive lymphadenopathy, peripheral edema, hypoalbuminemia, and high-output cardiac failure. The major etiologies of erythroderma are (1) *cutaneous diseases* such as psoriasis and dermatitis ([Table 57-3](#)); (2) *drugs*; (3) *systemic diseases*, most commonly [CTCL](#); and (4) *idiopathic*. In the first three groups, the location and description of the initial lesions, prior to the development of the erythroderma, aid in the diagnosis. For example, a history of red scaly plaques on the elbows and knees would point to psoriasis. It is also important to examine the skin carefully for a migration of the erythema and associated secondary changes such as pustules or erosions. Migratory waves of erythema studded with superficial pustules are seen in *pustular psoriasis*.

Drug-induced erythroderma (exfoliative dermatitis) may begin as a morbilliform eruption ([Chap. 59](#)) or may arise as diffuse erythema. Fever and peripheral eosinophilia often accompany the eruption, and occasionally there is an associated allergic interstitial nephritis. A number of drugs can produce an erythroderma, including penicillins, sulfonamides, carbamazepine, phenytoin, gold, allopurinol, and captopril. While reactions to anticonvulsants can lead to a pseudolymphoma syndrome, reactions to allopurinol may be accompanied by hepatitis, gastrointestinal bleeding, and nephropathy.

The most common malignancy that is associated with erythroderma is [CTCL](#); in some series, up to 25% of the cases of erythroderma were due to CTCL. The patient may progress from isolated plaques and tumors, but more commonly the erythroderma is present throughout the course of the disease (Sezary syndrome). In the Sezary

syndrome, there are circulating atypical T lymphocytes, pruritus, and lymphadenopathy. In cases of erythroderma where there is no apparent cause (idiopathic), longitudinal follow-up is mandatory to monitor for the possible development of CTCL. There have been isolated case reports of erythroderma secondary to some solid tumors -- lung, liver, prostate, thyroid, and colon -- but it is usually in a late stage of the disease.

ALOPECIA ([Table 57-4](#))

The two major forms of alopecia are scarring and nonscarring. In *scarring alopecia* there is associated fibrosis, inflammation, and loss of hair follicles. A smooth scalp with a decreased number of follicular openings is usually observed clinically, but in some cases the changes are seen only in biopsy specimens from the affected areas. In *nonscarring alopecia* the hair shafts are gone, but the hair follicles are preserved, explaining the reversible nature of nonscarring alopecia.

Primary cutaneous disorders are the most common causes of nonscarring alopecia and they include *telogen effluvium*, *androgenetic alopecia*, *alopecia areata*, *tinea capitis*, and *traumatic alopecia* ([Table 57-5](#)). In women with androgenetic alopecia, an elevation in circulating levels of androgens may be seen as a result of ovarian or adrenal gland dysfunction. When there are signs of virilization, such as a deepened voice and enlarged clitoris, the possibility of an ovarian or adrenal gland tumor should be considered.

Exposure to various *drugs* can also cause diffuse hair loss, usually by inducing a telogen effluvium. An exception is the anagen effluvium observed with antimitotic agents such as daunorubicin. Alopecia is a side effect of the following drugs: warfarin, heparin, propylthiouracil, carbimazole, vitamin A, isotretinoin, acetrein, lithium, beta blockers, colchicine, and amphetamines. Fortunately, spontaneous regrowth usually follows discontinuation of the offending agent.

Less commonly, nonscarring alopecia is associated with *lupus erythematosus* and *secondary syphilis*. In systemic lupus there are two forms of alopecia -- one is scarring secondary to discoid lesions (see below) and the other is nonscarring. The latter form may be diffuse and involve the entire scalp, or it may be localized to the frontal scalp and result in multiple short hairs ("lupus hairs"). Scattered, poorly circumscribed patches of alopecia with a "moth-eaten" appearance are a manifestation of the secondary stage of syphilis. Diffuse thinning of the hair is also associated with hypothyroidism, hyperthyroidism, and HIV infection ([Table 57-4](#)).

Scarring alopecia is more frequently the result of a primary cutaneous disorder such as *lichen planus*, *folliculitis decalvans*, *cutaneous lupus*, or *linear scleroderma (morphea)* than it is a sign of systemic disease. Although the scarring lesions of *discoid lupus* can be seen in patients with systemic lupus, in the majority of cases the disease process is limited to the skin. Less common causes of scarring alopecia include *sarcoidosis* (see "Papulonodular Skin Lesions") and cutaneous *metastases*.

In the early phases of discoid lupus, lichen planus, and folliculitis decalvans, there are circumscribed areas of alopecia. Fibrosis and subsequent loss of follicles are observed primarily in the center of the individual lesions, while the inflammatory process is most

prominent at the periphery. The areas of active inflammation in discoid lupus are erythematous with scale, whereas the areas of previous inflammation are often hypopigmented with a rim of hyperpigmentation. In lichen planus the peripheral perifollicular macules are usually violet-colored. Complete examination of the skin and oral mucosa combined with a biopsy and direct immunofluorescence microscopy will aid in distinguishing these two entities. The peripheral active lesions in folliculitis decalvans are perifollicular pustules; these patients can develop a reactive arthritis.

FIGURATE SKIN LESIONS ([Table 57-6](#))

In *figurate* eruptions, the lesions form rings and arcs that are usually erythematous but can be flesh-colored to brown. Most commonly, they are due to primary cutaneous diseases such as *tinea*, *urticaria*, *erythema annulare centrifugum*, and *granuloma annulare* ([Chaps. 56](#) and [58](#)). An underlying systemic illness is found in a second, less common group of migratory annular erythemas. It includes *erythema gyratum repens*, *erythema migrans*, *erythema marginatum*, and *necrolytic migratory erythema*.

In *erythema gyratum repens*, one sees hundreds of mobile concentric arcs and wavefronts that resemble the grain in wood. A search for an underlying malignancy is mandatory in a patient with this eruption. *Erythema migrans* is the cutaneous manifestation of Lyme disease, which is caused by the spirochete *Borrelia burgdorferi*. In the initial stage (3 to 30 days after tick bite), a single annular lesion is usually seen, which can expand to 3–10 cm in diameter. Within several days, approximately half the patients develop multiple smaller erythematous lesions at sites distant from the bite. Associated symptoms include fever, headache, photophobia, myalgias, arthralgias, and malar rash. *Erythema marginatum* is seen in patients with rheumatic fever, primarily on the trunk. Lesions are pink-red in color, flat to mildly elevated, and transient.

There are additional cutaneous diseases that present as annular eruptions but lack an obvious migratory component. Examples include [CTCL](#), *annular cutaneous lupus*, also referred to as *subacute lupus*, *secondary syphilis*, and *sarcoidosis* (see "Papulonodular Skin Lesions").

ACNE ([Table 57-7](#))

Acne vulgaris and *acne rosacea* are the two major forms of acne ([Chap. 56](#)). Estrogens decrease sebaceous gland activity, whereas androgens enhance sebum production. Therefore, acne vulgaris in an adult, especially if it is of recent onset, may be a reflection of increased levels of circulating *androgens*. Dysfunction of the ovary or adrenal gland, e.g., polycystic ovary disease or Cushing's syndrome, can lead to the hormonal imbalance. Examination of the patient for signs such as hirsutism, androgenetic alopecia, hypertension, and redistribution of subcutaneous fat will aid in the diagnosis.

Exacerbations of acne vulgaris follow the ingestion of several *drugs*, such as anabolic steroids, glucocorticoids, lithium, and iodides as well as the application of oil-containing compounds. Acne-like lesions can be seen in patients with Behcet's disease (see "Ulcers"), and in immunocompromised hosts, disseminated fungal infections (e.g., cryptococcosis) may present as an acneiform eruption.

Patients with the carcinoid syndrome have episodes of flushing of the head, neck, and sometimes the trunk. Resultant skin changes of the face, in particular telangiectasias, may mimic the clinical appearance of acne rosacea.

PUSTULAR LESIONS

Acneiform eruptions (see "Acne") and *folliculitis* represent the most common pustular dermatoses. An important consideration in the evaluation of perifollicular pustules is a determination of the associated pathogen, e.g., normal flora, *Staphylococcus aureus*, *Pityrosporum*. Noninfectious forms of folliculitis include HIV-associated eosinophilic folliculitis and folliculitis secondary to drugs such as glucocorticoids and lithium. Administration of high-dose oral glucocorticoids can result in a widespread eruption of perifollicular pustules on the trunk, characterized by lesions in the same stage of development. With regard to underlying systemic diseases, pustules are a characteristic component of pustular psoriasis and can be seen in septic emboli of bacterial or fungal origin (see "Purpura").

TELANGIECTASIAS ([Table 57-8](#))

In order to distinguish the various types of telangiectasias, it is important to examine the shape and configuration of the dilated blood vessels. *Linear telangiectasias* are seen on the face of patients with *actinically damaged skin* and *acne rosacea* and they are found on the legs of patients with *venous hypertension* and *essential telangiectasia*. Patients with an unusual form of *mastocytosis* (telangiectasia macularis eruptiva perstans), the *carcinoid* syndrome (see "Acne"), and *ataxia-telangiectasia* also have linear telangiectasias. In ataxia-telangiectasia, linear telangiectasias appear on the bulbar conjunctiva during childhood. Eventually, there is involvement of the ears, eyelids, cheeks, and/or flexural areas such as the antecubital and popliteal fossae. Lastly, linear telangiectasias are found in areas of cutaneous inflammation. For example, lesions of discoid lupus frequently have telangiectasias within them.

Poikiloderma is a term used to describe a patch of skin with (1) reticulated hypo- and hyperpigmentation, (2) wrinkling secondary to epidermal atrophy, and (3) telangiectasias. Poikiloderma does not imply a single disease entity -- it is seen in skin damaged by *ionizing radiation*, in the disorders *poikiloderma vasculare atrophicans* (PVA) and *xeroderma pigmentosum*, as well as in patients with connective-tissue diseases, primarily *dermatomyositis* (DM). [PVA](#) is a precursor lesion of [CTCL](#), and the areas of poikiloderma usually begin in the flexural areas of the axillae and groin.

In *scleroderma*, the dilated blood vessels have a unique configuration and are known as *mat telangiectasias*. The lesions are broad macules that usually measure 2 to 7 mm in diameter but occasionally are larger. Mats have a polygonal or oval shape, and their erythematous color may be uniform or the result of delicate telangiectasias. The most common locations for mat telangiectasias are the face, oral mucosa, and hands -- peripheral sites that are prone to intermittent ischemia. The CREST (calcinosis cutis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia) variant of scleroderma ([Chap. 313](#)) is associated with a chronic course and anticentromere antibodies. Mat telangiectasias are an important clue to the diagnosis of

the CREST syndrome as well as systemic scleroderma, for they may be the only cutaneous finding.

Periungual telangiectasias are pathognomonic signs of the three major connective tissue diseases -- *lupus erythematosus*, *scleroderma*, and [DM](#). They are easily visualized by the naked eye and occur in at least two-thirds of these patients. In both DM and lupus there is associated nailfold erythema, and in DM the erythema is often accompanied by "ragged" cuticles and fingertip tenderness. Under 10^x magnification, the blood vessels in the nailfolds of lupus patients are tortuous and resemble "glomeruli," whereas in scleroderma and DM there is a loss of capillary loops and those that remain are markedly dilated.

In *hereditary hemorrhagic telangiectasia* (Osler-Rendu-Weber disease), the lesions usually appear during adulthood and are most commonly seen on the mucous membranes, face, and distal extremities, including under the nails. They represent arteriovenous (AV) malformations of the dermal microvasculature, are dark red in color, and are usually slightly elevated. When the skin is stretched over an individual lesion, an eccentric punctum with radiating legs is seen. Although the degree of systemic involvement varies in this autosomal dominant disease (due to mutations in either the endoglin or activin receptor-like kinase gene), the major symptoms are recurrent epistaxis and gastrointestinal bleeding. The fact that these mucosal telangiectasias are actually AV communications helps to explain their tendency to bleed.

HYPOPIGMENTATION ([Table 57-9](#))

Disorders of hypopigmentation are classified as either diffuse or localized. The classic example of *diffuse hypopigmentation* is *oculocutaneous albinism* (OCA). The most common forms are due to mutations in the tyrosinase gene (type I) or the *P* gene (type II); patients with type IA OCA have a total lack of enzyme activity. At birth, different forms of OCA can appear similar -- white hair, gray-blue eyes, and pink-white skin. However, the patients with no tyrosinase activity maintain this phenotype, whereas those with decreased activity or *P* gene mutations will acquire some pigmentation of the eyes, hair, and skin as they age. The degree of pigment formation is also a function of racial background, and the pigmentary dilution is readily apparent when patients are compared to their first-degree relatives.

The ocular findings in [OCA](#) correlate with the degree of hypopigmentation and include decreased visual acuity, nystagmus, photophobia, and monocular vision. Generalized vitiligo, phenylketonuria, and homocystinuria are other unusual causes of diffuse pigmentary dilution. In generalized vitiligo, melanocytes are not found in affected skin, whereas in OCA they are present but have decreased activity. Appropriate laboratory tests exclude the other disorders of metabolism.

The differential diagnosis of *localized hypomelanosis* includes the following primary cutaneous disorders: *idiopathic guttate hypomelanosis*, *postinflammatory hypopigmentation*, *tinea (pityriasis) versicolor*, *vitiligo*, *chemical leukoderma*, *nevus depigmentosus* (see below), and *piebaldism* ([Table 57-9](#)). In this group of diseases, the areas of involvement are macules or patches with a decrease or absence of pigmentation. Patients with vitiligo also have an increased incidence of several

autoimmune disorders, including hypothyroidism, Graves' disease, pernicious anemia, Addison's disease, uveitis, alopecia areata, chronic mucocutaneous candidiasis, and the polyglandular autoimmune syndromes (types I and II). Diseases of the thyroid gland are the most frequently associated disorders, occurring in up to 30% of patients with vitiligo. Circulating autoantibodies are often found, and the most common ones are antithyroglobulin, antimicrosomal, and antiparietal cell antibodies.

There are three systemic diseases that should be considered in a patient with skin findings suggestive of vitiligo -- *Vogt-Koyanagi-Harada syndrome*, *scleroderma*, and *melanoma-associated leukoderma*. A history of aseptic meningitis, nontraumatic uveitis, tinnitus, hearing loss, and/or dysacusis points to the diagnosis of the Vogt-Koyanagi-Harada syndrome. In these patients, the face and scalp are the most common locations of pigment loss. The vitiligo-like leukoderma seen in patients with scleroderma has a clinical resemblance to idiopathic vitiligo that has begun to repigment as a result of treatment; that is, perifollicular macules of normal pigmentation are seen within areas of depigmentation. The basis of this leukoderma is unknown; there is no evidence of inflammation in areas of involvement, but it can resolve if the underlying connective tissue disease becomes inactive. In contrast to idiopathic vitiligo, melanoma-associated leukoderma often begins on the trunk, and its appearance should prompt a search for metastatic disease. The possibility exists that the destruction of normal melanocytes is the result of an immune response against malignant melanocytes.

There are two systemic disorders that may have the cutaneous findings of piebaldism ([Table 57-10](#)). They are *Hirschsprung's disease* and *Waardenburg's syndrome*. A possible explanation for both disorders is an abnormal embryonic migration or survival of two neural crest-derived elements, one of them being melanocytes and the other myenteric ganglion cells (Hirschsprung's disease) or auditory nerve cells (Waardenburg's syndrome). The latter syndrome is characterized by congenital sensorineural hearing loss, dystopia canthorum (lateral displacement of the inner canthi but normal interpupillary distance), heterochromic irises, and a broad nasal root, in addition to the piebaldism. Patients with Waardenburg's syndrome have been shown to have mutations in two genes that encode DNA-binding proteins, *PAX-3* and *MITF*, while patients with Hirschsprung's disease and white spotting have mutations in one of three genes -- endothelin 3, endothelin B receptor, and *SOX-10*.

In *tuberous sclerosis*, the earliest cutaneous sign is an ash leaf spot. These lesions are often present at birth and are usually multiple; however, detection may require Wood's lamp examination, especially in fair-skinned individuals. The pigment within them is reduced but not absent. The average size is 1 to 3 cm, and the common shapes are polygonal and lance-ovate. Examination of the patient for additional cutaneous signs such as adenoma sebaceum (multiple angiofibromas of the face), ungual and gingival fibromas, fibrous plaques of the forehead, and connective tissue nevi (shagreen patches) is recommended. It is important to remember that an ash leaf spot on the scalp will result in *poliosis*, which is a circumscribed patch of gray-white hair. Internal manifestations include seizures, mental retardation, central nervous system (CNS) and retinal hamartomas, renal angiomyolipomas, and cardiac rhabdomyomas. The latter can be detected in up to 60% of children (<18 years) with tuberous sclerosis by echocardiography.

Nevus depigmentosus is a stable, well-circumscribed hypomelanosis that is present at birth. There is usually a single circular or rectangular lesion, but occasionally the nevus has a dermatomal or whorled pattern. It is important to distinguish this more common entity from ash leaf spots especially when there are multiple lesions. In *hypomelanosis of Ito*, swirls and streaks of hypopigmentation run parallel to one another in a pattern that resembles a marble cake. Lesions may progress or regress with time, and in up to a third of patients, associated abnormalities are found including in the musculoskeletal system (asymmetry), the [CNS](#) (seizures and mental retardation), and the eyes (strabismus and hypertelorism). Chromosomal mosaicism has been detected in these patients; this lends support to the hypothesis that the pattern is the result of the migration of two clones of primordial melanocytes, each with a different pigment potential.

Localized areas of decreased pigmentation are commonly seen as a result of cutaneous inflammation ([Table 57-10](#)) and have been observed in the skin overlying active lesions of sarcoidosis (see "Papulonodular Skin Lesions") as well as in [CTCL](#). Cutaneous infections also present as disorders of hypopigmentation, and in *tuberculoid leprosy* there are a few asymmetric patches of hypomelanosis that have associated anesthesia, anhidrosis, and alopecia. Biopsy specimens of the palpable border show dermal granulomas that lack *Mycobacterium leprae* organisms.

HYPERPIGMENTATION ([Table 57-11](#))

Disorders of hyperpigmentation are also divided into two groups -- localized and diffuse. The *localized* forms are due to an epidermal alteration, a proliferation of melanocytes, or an increase in pigment production. Both seborrheic keratoses and acanthosis nigricans belong to the first group. *Seborrheic keratoses* are common lesions, but in one clinical setting they are a sign of systemic disease, and that setting is the sudden appearance of multiple lesions, often with an inflammatory base and in association with acrochordons (skin tags) and acanthosis nigricans. This is termed the *sign of Leser-Trelat* and signifies an internal malignancy. *Acanthosis nigricans* can also be a reflection of an internal malignancy, most commonly of the gastrointestinal tract, and it appears as velvety hyperpigmentation, primarily in flexural areas. In the majority of patients, acanthosis nigricans is associated with obesity, but it may be a reflection of an endocrinopathy such as acromegaly, Cushing's syndrome, the Stein-Leventhal syndrome, or insulin-resistant diabetes mellitus (type A, type B, and lipodystrophic forms).

A proliferation of melanocytes results in the following pigmented lesions: *lentigo*, *melanocytic nevus*, and *melanoma* ([Chap. 86](#)). In an adult, the majority of lentigines are related to sun exposure, which explains their distribution. However, in the Peutz-Jeghers and LEOPARD [lentigines; ECG abnormalities, primarily conduction defects, ocular hypertelorism; pulmonary stenosis and subaortic valvular stenosis; abnormal genitalia (cryptorchidism, hypospadias); retardation of growth; and deafness (sensorineural)] syndromes, lentigines do serve as a clue to systemic disease. In the multiple lentigines or *LEOPARD syndrome*, hundreds of lentigines develop during childhood and are scattered over the entire surface of the body. The lentigines in patients with *Peutz-Jeghers syndrome* are located primarily around the nose and mouth, on the hands and feet, and within the oral cavity. While the pigmented macules on the

face may fade with age, the oral lesions persist. However, similar intraoral lesions are also seen in Addison's disease and as a normal finding in darkly pigmented individuals. Patients with this autosomal dominant syndrome (due to mutations in a novel serine threonine kinase gene) have multiple benign polyps of the gastrointestinal tract, testicular tumors, and an increased risk of developing gastrointestinal (primarily colon), breast, and gynecologic cancers.

Lentigines are also seen in association with cardiac myxomas and have been described in two syndromes whose findings overlap: *LAMB* (lentigines, atrial myxomas, mucocutaneous myxomas, and blue nevi) syndrome and *NAME* [nevus, atrial myxoma, myxoid neurofibroma, and ephelides (freckles)] syndrome. These patients can also have evidence of endocrine overactivity in the form of Cushing's syndrome, acromegaly, or sexual precocity.

The third type of localized hyperpigmentation is due to a local increase in pigment production, and it includes *ephelides* and cafe au lait macules (CALM). The latter are most commonly associated with two disorders -- neurofibromatosis (NF) and McCune-Albright syndrome. CALM are flat, uniformly light brown in color, and can vary in size from 0.5 to 12 cm. Approximately 80% of adult patients with *type I NF* will have six or more CALM measuring 1.5 cm or greater in diameter. Additional findings are discussed in the section on neurofibromas (see "Papulonodular Skin Lesions"). In comparison with NF, the CALM in patients with *McCune-Albright syndrome* [polyostotic fibrous dysplasia with precocious puberty in females due to mosaicism for an activating mutation in a G protein (G_{sa}) gene] are usually larger, more irregular in outline, and tend to respect the midline. CALM have also been associated with pulmonary stenosis (Watson syndrome), tuberous sclerosis, the [LEOPARD](#) syndrome, and ataxia telangiectasia, but a few such lesions can be found in normal individuals.

In *incontinentia pigmenti*, dyskeratosis congenita, and bleomycin pigmentation, the areas of localized hyperpigmentation form a pattern -- swirled in the first, reticulated in the second, and flagellate in the third. Patients with the X-linked dominant disorder *incontinentia pigmenti* can have linear blisters and verrucous papules during infancy. During childhood, parallel swirls and streaks of hyperpigmentation appear on the trunk, and occasionally streaks of hypopigmentation appear on the extremities. Associated findings include seizures, mental retardation, retinal vascular abnormalities, and delayed or impaired dentition. Biopsy specimens of the streaks will show pigment within dermal macrophages ("incontinent pigment"). In *dyskeratosis congenita*, atrophic reticulated hyperpigmentation is seen on the neck, thighs, and trunk and is accompanied by nail dystrophy, pancytopenia, and leukoplakia of the oral and anal mucosa. The latter often develops into squamous cell carcinoma. In addition to the flagellate pigmentation (linear streaks) on the trunk, patients receiving bleomycin often have hyperpigmentation on the elbows, knees, and small joints of the hand.

Localized hyperpigmentation is seen as a side effect of several other *systemic medications*, including those that produce fixed drug reactions [phenolphthalein, nonsteroidal anti-inflammatory drugs (NSAIDs), sulfonamides, and barbiturates] and those that can complex with melanin (antimalarials). Fixed drug eruptions recur in the same location as circular areas of erythema that can become bullous and then resolve as brown macules. The eruption usually appears within hours of administration of the

offending agent, and common locations include the genitalia, extremities, and perioral region. Chloroquine and hydroxychloroquine produce gray-brown to blue-black discoloration of the shins, hard palate, and face, while blue macules can be seen on the lower extremities and in sites of inflammation with prolonged minocycline administration. Estrogen in oral contraceptives can induce melasma -- symmetric brown patches on the face, especially the cheeks, upper lip, and forehead. Similar changes are seen in pregnancy, in patients receiving hydantoin, and in the adult form of Gaucher's disease. In the latter group there is also hyperpigmentation of the distal lower extremities.

In the *diffuse* forms of hyperpigmentation, the darkening of the skin may be of equal intensity over the entire body or may be accentuated in sun-exposed areas. The causes of diffuse hyperpigmentation can be divided into four groups -- endocrine, metabolic, autoimmune, and drugs. The endocrinopathies that frequently have associated hyperpigmentation include *Addison's disease*, *Nelson syndrome*, and *ectopic ACTH syndrome*. In these diseases, the increased pigmentation is diffuse but is accentuated in the palmar creases, sites of friction, scars, and the oral mucosa. An overproduction of the pituitary hormones α -MSH (melanocyte-stimulating hormone) and ACTH can lead to an increase in melanocyte activity. These peptides are products of the proopiomelanocortin gene and exhibit homology; e.g., α -MSH and ACTH share 13 amino acids. A minority of the patients with Cushing's disease or hyperthyroidism have generalized hyperpigmentation.

The metabolic causes of hyperpigmentation include *porphyria cutanea tarda* (PCT), *hemochromatosis*, *vitamin B₁₂ deficiency*, *folic acid deficiency*, *pellagra*, *malabsorption*, and *Whipple's disease*. In patients with PCT (see "Vesicles/Bullae"), the skin darkening is seen in sun-exposed areas and is a reflection of the photoreactive properties of porphyrins. The increased level of iron in the skin of patients with hemochromatosis stimulates melanin pigment production and leads to the classic bronze color. Patients with pellagra have a brown discoloration of the skin, especially in sun-exposed areas, as a result of nicotinic acid (niacin) deficiency. In the areas of increased pigmentation, there is a thin varnish-like scale. These changes are also seen in patients who are vitamin B₆ deficient, have functioning carcinoid tumors (increased consumption of niacin), or take isoniazid. Approximately 50% of the patients with Whipple's disease have an associated generalized hyperpigmentation in association with diarrhea, weight loss, arthritis, and lymphadenopathy. A diffuse slate-blue color is seen in patients with melanosis secondary to metastatic melanoma. Although there is a debate as to whether the color is due to single-cell metastases in the dermis or to a widespread deposition of melanin resulting from the high concentration of circulating melanin precursors, there is more evidence to support the latter.

Of the autoimmune diseases associated with diffuse hyperpigmentation, *biliary cirrhosis* and *scleroderma* are the most common, and occasionally, both disorders are seen in the same patient. The skin is dark brown in color, especially in sun-exposed areas. In biliary cirrhosis the hyperpigmentation is accompanied by pruritus, jaundice, and xanthomas, whereas in scleroderma it is accompanied by sclerosis of the extremities, face, and, less commonly, the trunk. Additional clues to the diagnosis of scleroderma are telangiectasias, calcinosis cutis, Raynaud's phenomenon, and distal ulcerations (see "Telangiectasias"). The differential diagnosis of cutaneous sclerosis with hyperpigmentation includes the POEMS [polyneuropathy; organomegaly (liver, spleen,

lymph nodes); endocrinopathies (impotence, gynecomastia); *M*-protein; and skin changes] syndrome. The skin changes include hyperpigmentation, skin thickening, hypertrichosis, and angiomas.

In the late 1980s, an epidemic of the eosinophilia-myalgia syndrome was described that was presumably due to contaminated L-tryptophan preparations. In addition to maculopapular eruptions and alopecia, large areas of scleroderma-like induration were observed with overlying hyperpigmentation.

Diffuse hyperpigmentation that is due to *drugs* or *metals* can result from one of several mechanisms -- induction of melanin pigment formation, complexing of the drug or its metabolites to melanin, and deposits of the drug in the dermis. Busulfan, cyclophosphamide, long-term, high-dose ACTH, and inorganic arsenic induce pigment production. Complexes containing melanin or hemosiderin plus the drug or its metabolites are seen in patients receiving chlorpromazine and minocycline. The sun-exposed skin as well as the conjunctivae of patients on long-term, high-dose chlorpromazine can become blue-gray in color. Patients taking minocycline may develop a diffuse blue-gray, muddy appearance in sun-exposed areas in addition to pigmentation of the mucous membranes, teeth, nails, bones, and thyroid. Administration of amiodarone can result in both a phototoxic eruption (exaggerated sunburn) and/or a brown or blue-gray discoloration of sun-exposed skin. Biopsy specimens of the latter show yellow-brown granules in dermal macrophages, which represent intralysosomal accumulations of lipids, amiodarone, and its metabolites. Actual deposits of a particular drug or metal in the skin are seen with silver (argyria), where the skin appears blue-gray in color; gold (chrysiasis), where the skin has a brown to blue-gray color; and clofazimine, where the skin appears reddish brown. The associated hyperpigmentation is accentuated in sun-exposed areas, and discoloration of the eye is seen with gold (sclerae) and clofazimine (conjunctivae).

VESICLES/BULLAE ([Table 57-12](#))

Depending on their size, cutaneous blisters are referred to as *vesicles* (<0.5 cm) or *bullae* (>0.5 cm). The primary blistering disorders include *pemphigus vulgaris*, *pemphigus foliaceus*, *pemphigus erythematosus*, *paraneoplastic pemphigus*, *bullous pemphigoid*, *herpes gestationis*, *cicatricial pemphigoid*, *epidermolysis bullosa acquisita*, *linear IgA disease*, and *dermatitis herpetiformis* ([Chap. 58](#)).

Vesicles and bullae are also seen in *contact dermatitis*, both allergic and irritant forms ([Chap. 56](#)). When there is a linear arrangement of vesicular lesions, an exogenous cause should be suspected. Bullous disease secondary to the ingestion of drugs can take one of several forms, including phototoxic eruptions, isolated bullae, toxic epidermal necrolysis, and erythema multiforme ([Chap. 59](#)). Clinically, phototoxic eruptions resemble an exaggerated sunburn with diffuse erythema and bullae in sun-exposed areas. The most commonly associated drugs are thiazides, doxycycline, sulfonamides, [NSAIDs](#), and psoralens. The development of a phototoxic eruption is dependent on the doses of both the drug and UV-A irradiation.

Toxic epidermal necrolysis (TEN) is characterized by bullae that arise on widespread areas of erythema and then slough. This results in large areas of denuded skin. The

associated morbidity, such as sepsis, and mortality are relatively high and are a function of the extent of epidermal necrosis. In addition, these patients may also have involvement of the mucous membranes and intestinal tract. Drugs are the primary cause of TEN, and the most common offenders are phenytoin, barbiturates, sulfonamides, penicillins, and [NSAIDs](#). Severe acute graft-versus-host disease (grade 4) also can resemble TEN.

In *erythema multiforme* (EM), the primary lesions are pink-red macules and edematous papules, the centers of which may become vesicular. The clue to the diagnosis of EM, as opposed to a drug-induced morbilliform exanthem, is the development of a "dusky" violet color or petechiae in the center of the lesions. Target or iris lesions are also characteristic of EM and arise as a result of active centers and borders in combination with centrifugal spread. However, iris lesions need not be present to make the diagnosis of EM. Preferred sites of involvement include the distal extremities and mucous membranes (oral, nasal, ocular, and genital). Hemorrhagic crusts of the lips are characteristic of EM as well as herpes simplex, pemphigus vulgaris, and paraneoplastic pemphigus. Fever, malaise, myalgias, sore throat, and cough may precede or accompany the eruption. The lesions of EM usually resolve over 3 to 6 weeks but may be recurrent.

Drugs can induce [EM](#), in particular sulfonamides, phenytoin, barbiturates, penicillins, and carbamazepine, but they do not cause the majority of cases, especially in young adults. Infections with herpes simplex are the most common cause of EM in this age group, and the lesions appear 7 to 12 days after the viral eruption. Other infectious agents associated with EM include *Mycoplasma pneumoniae*, dimorphic fungi, and several viruses (echovirus, coxsackievirus, Epstein-Barr, and influenza). EM can also follow vaccinations with BCG, poliomyelitis, or vaccinia viruses; radiation therapy; and exposure to environmental toxins.

In addition to primary blistering disorders and hypersensitivity reactions, bacterial and viral infections can lead to vesicles and bullae. The most common infectious agents are herpes simplex ([Chap. 182](#)), herpes varicella-zoster ([Chap. 183](#)), and staphylococci ([Chap. 139](#)).

Staphylococcal scalded-skin syndrome (SSSS) and *bullous impetigo* are two blistering disorders associated with staphylococcal (phage group II) infection. In SSSS, the initial findings are redness and tenderness of the central face, neck, trunk, and intertriginous zones. This is followed by short-lived flaccid bullae and a slough or exfoliation of the superficial epidermis. Crusted areas then develop, characteristically around the mouth. SSSS is distinguished from [TEN](#) by the following features: younger age group, more superficial site of blister formation, no oral lesions, shorter course, less morbidity and mortality, and an association with staphylococcal exfoliative toxin ("exfoliatin"), not drugs. A rapid diagnosis of SSSS versus TEN can be made by a frozen section of the blister roof or exfoliative cytology of the blister contents. In SSSS the site of staphylococcal infection is usually extracutaneous (conjunctivitis, rhinorrhea, otitis media, pharyngitis, tonsillitis), and the cutaneous lesions are sterile, whereas in bullous impetigo the skin lesions are the site of infection. Impetigo is more localized than SSSS and usually presents with honey-colored crusts. Occasionally, superficial purulent blisters also form. *Cutaneous emboli* from gram-negative infections may present as

isolated bullae, but the base of the lesion is purpuric or necrotic, and it may develop into an ulcer (see "Purpura").

Several metabolic disorders are associated with blister formation, including diabetes mellitus, renal failure, and porphyria. Local hypoxia secondary to decreased cutaneous blood flow can also produce blisters, which explains the presence of bullae over pressure points in comatose patients (coma bullae). In *diabetes mellitus*, tense bullae with clear viscous fluid arise on normal skin. The lesions can be as large as 6 cm in diameter and are located on the distal extremities. There are several types of porphyria, but the most common form with cutaneous findings is [PCT](#). In sun-exposed areas (primarily the face and hands), the skin is very fragile, and trauma leads to erosions and tense vesicles. These lesions then heal with scarring and formation of milia; the latter are firm, 2- to 3-mm white or yellow papules that represent epidermoid inclusion cysts. Associated findings can include hypertrichosis of the lateral malar region (males) or face (females) and, in sun-exposed areas, hyperpigmentation and firm sclerotic plaques. An elevated level of urinary uroporphyrins confirms the diagnosis and is due to a decrease in uroporphyrinogen decarboxylase activity. Precipitating agents include alcohol, iron, chlorinated hydrocarbons, and hepatitis C infection.

The differential diagnosis of [PCT](#) includes (1) *porphyria variegata* -- the skin signs of PCT plus the systemic findings of acute intermittent porphyria; it has a diagnostic plasma porphyrin fluorescence emission at 626 nm; (2) *drug-induced bullous photosensitivity* (pseudoporphyria) -- the clinical and histologic findings are similar to PCT, but porphyrins are normal; etiologic agents include naproxen, furosemide, tetracycline, and nalidixic acid; (3) *bullous dermatosis of hemodialysis* -- the same appearance as PCT, but porphyrins are usually normal or occasionally borderline elevated; patients have chronic renal failure and are on hemodialysis; (4) PCT associated with hepatomas, hepatic carcinomas, and hemodialysis; and (5) *epidermolysis bullosa acquisita* ([Chap. 58](#)).

EXANTHEMS ([Table 57-13](#))

Exanthems are characterized by an acute generalized eruption. The two most common presentations are erythematous macules and papules (morbilliform) and confluent blanching erythema (scarlatiniform). *Morbilliform* eruptions are usually due to either *drugs* or *viral infections*. For example, up to 5% of the patients receiving penicillins, sulfonamides, phenytoin, or gold will develop a maculopapular eruption. Accompanying signs may include pruritus, fever, eosinophilia, and transient lymphadenopathy. Similar maculopapular eruptions are seen in the classic childhood viral exanthems, including (1) *rubeola* (measles) -- a prodrome of coryza, cough, and conjunctivitis followed by Koplik's spots on the buccal mucosa; the eruption begins behind the ears, at the hairline, and on the forehead and then spreads down the body, often becoming confluent; (2) *rubella* -- it begins on the forehead and face and then spreads down the body; it resolves in the same order and is associated with retroauricular and suboccipital lymphadenopathy; and (3) *erythema infectiosum* (fifth disease) -- erythema of the cheeks is followed by a reticulated pattern on extremities; it is secondary to a parvovirus B19 infection, and an associated arthritis is seen in adults.

Both measles and rubella are seen in unvaccinated young adults, and an atypical form

of measles is seen in adults immunized with either killed measles vaccine or killed vaccine followed in time by live vaccine. In contrast to classic measles, the eruption of atypical measles begins on the palms, soles, wrists, and knuckles, and the lesions may become purpuric. The patient with atypical measles can have pulmonary involvement and be quite ill. Rubelliform and roseoliform eruptions are also associated with *Epstein-Barr virus* (5 to 15% of patients), *echovirus*, *coxsackievirus*, and *adenovirus* infections. Detection of specific IgM antibodies or fourfold elevations in IgG antibodies allows the proper diagnosis. Occasionally, a maculopapular eruption is the result of a drug-viral interaction. For example, about 95% of the patients with infectious mononucleosis who are given ampicillin will develop a rash.

Of note, early in the course of infections with *Rickettsia* and *meningococcus*, prior to the development of purpura, the lesions may be erythematous macules and papules. This is also the case in chickenpox prior to the development of vesicles. Maculopapular eruptions are associated with early *HIV infection*, early secondary *syphilis*, *typhoid fever*, and *acute graft-versus-host disease*. In the last, lesions frequently begin on the palms and soles; the macular rose spots of typhoid fever involve primarily the anterior trunk.

The prototypic *scarlatiniform* eruption is seen in *scarlet fever* and is due to an erythrogenic toxin produced by group A β -hemolytic streptococcal infections, most commonly pharyngitis. This eruption is characterized by diffuse erythema, which begins on the neck and upper trunk, and red perifollicular puncta. Additional findings include a white strawberry tongue (white coating with red papillae) followed by a red strawberry tongue (red tongue with red papillae); petechiae of the palate; a facial flush with circumoral pallor; linear petechiae in the antecubital fossae; and desquamation of the involved skin, palms, and soles 5 to 20 days after onset of the eruption. A similar desquamation of the palms and soles is seen in toxic shock syndrome, Kawasaki's disease, and after severe febrile illnesses. Certain strains of staphylococci also produce an erythrogenic toxin that leads to the same clinical findings as in streptococcal scarlet fever, except that the antistreptolysin O titers are not elevated.

In *toxic shock syndrome* (TSS), staphylococcal (phage group I) infections produce an exotoxin (TSST-1) that causes the fever and rash, as well as enterotoxins. Initially, the majority of cases were reported in menstruating women who were using tampons. However, other sites of infection, including wounds and vaginitis, may produce TSS. The diagnosis of TSS is based on clinical criteria ([Chap. 139](#)), and three of these involve mucocutaneous sites. The clinical criteria are (1) fever; (2) diffuse erythema of the skin; (3) desquamation of the palms and soles 1 to 2 weeks after onset of illness; (4) hypotension; and (5) involvement of three or more organ systems, including the gastrointestinal tract, muscles, kidney, liver, [CNS](#), hematologic (thrombocytopenia), and mucous membranes. The latter is characterized as hyperemia of the vagina, oropharynx, or conjunctivae. Similar systemic findings have been described in *streptococcal toxic shock-like syndrome* ([Chap. 140](#)), and although an exanthem is seen less often than in TSS due to a staphylococcal infection, the underlying infection is often in the soft tissue.

The cutaneous eruption in *Kawasaki's disease* (mucocutaneous lymph node syndrome) ([Chap. 317](#)) is polymorphous, but the two most common forms are morbilliform and

scarlatiniform. The majority of cases are seen in children less than 5 years of age, but adult cases have been reported. The diagnosis is based on a fever lasting more than 5 days plus four of the five following criteria: (1) bilateral conjunctival injection; (2) exanthem; (3) cervical lymphadenopathy, usually unilateral; (4) erythema and edema of the hands and feet followed by desquamation; and (5) diffuse erythema of the oropharynx, red strawberry tongue, and erosions with crusting on the lips. This clinical picture can resemble [TSS](#) and scarlet fever, but clues to the diagnosis of Kawasaki's disease are the cervical lymphadenopathy, lip erosions, and increased platelets. The most serious associated systemic finding in this disease is coronary aneurysm secondary to arteritis. Aneurysms may lead to sudden death, primarily within the first 30 days of the illness. Scarlatiniform eruptions are also seen in the early phase of [SSSS](#) (see "Vesicles/Bullae") and as reactions to drugs.

URTICARIA ([Table 57-14](#))

Urticaria (hives) are transient lesions that are composed of a central wheal surrounded by an erythematous halo. Individual lesions are round, oval, or figurate and are often pruritic. *Acute* and *chronic* urticaria have a wide variety of allergic etiologies. Less common systemic causes of urticaria are mastocytosis (urticaria pigmentosa), hyperthyroidism, malignancy, and juvenile rheumatoid arthritis (JRA). In JRA, the lesions coincide with the fever spike and are transient but not migratory as in erythema marginatum.

The common *physical urticarias* include dermographism, solar urticaria, cold urticaria, and cholinergic urticaria. Patients with *dermographism* exhibit linear wheals following minor pressure or scratching of the skin. It is a common disorder, affecting approximately 5% of the population. *Solar urticaria* characteristically occur within minutes of sun exposure and are a skin sign of one systemic disease -- erythropoietic protoporphyria. In addition to the urticaria, these patients have subtle pitted scarring of the nose and hands. *Cold urticaria* are precipitated by exposure to the cold, and therefore exposed areas are usually affected. In some cases, the disease is associated with abnormal circulating proteins -- more commonly cryoglobulins and less commonly cryofibrinogens and cold agglutinins. Additional systemic symptoms include wheezing and syncope, thus explaining the need for these patients to avoid swimming in cold water. *Cholinergic urticaria* are precipitated by heat, exercise, or emotion and are characterized by small wheals with relatively large flares. They are occasionally associated with wheezing.

Whereas urticaria are the result of dermal edema, subcutaneous edema leads to the clinical picture of *angioedema*. Sites of involvement include the eyelids, lips, tongue, larynx, and gastrointestinal tract as well as the subcutaneous tissue. Angioedema occurs alone or in combination with urticaria, including urticarial vasculitis and the physical urticarias. Both acquired and hereditary (autosomal dominant) forms of angioedema occur ([Chap. 310](#)), and in the latter, urticaria is rarely seen.

Urticarial vasculitis is an immune complex disease that may be confused with simple urticaria. In contrast to simple urticaria, individual lesions tend to last longer than 24 h and usually develop central petechiae that can be observed even after the urticarial phase has resolved. The patient may also complain of burning rather than pruritus. On

biopsy, there is a leukocytoclastic vasculitis of the small blood vessels. Although many cases of urticarial vasculitis are idiopathic in origin, it can be a reflection of an underlying systemic illness such as lupus erythematosus, Sjogren's syndrome, or hereditary complement deficiency. There is a spectrum of urticarial vasculitis that ranges from purely cutaneous to multisystem involvement. The most common systemic signs and symptoms are arthralgias and/or arthritis, nephritis, and crampy abdominal pain, with asthma and chronic obstructive lung disease seen less often. Hypocomplementemia occurs in one- to two-thirds of patients, even in the idiopathic cases. Urticarial vasculitis can also be seen in patients with *hepatitis B* and *hepatitis C* infections, *serum sickness*, and *serum sickness-like illnesses*.

PAPULONODULAR SKIN LESIONS ([Table 57-15](#))

In the *papulonodular diseases*, the lesions are elevated above the surface of the skin and may coalesce to form plaques. The location, consistency, and color of the lesions are the keys to their diagnosis; this section is organized on the basis of color.

White Lesions In *calcinosis cutis* there are firm white to white-yellow papules with an irregular surface. When the contents are discharged, a chalky white material is seen. *Dystrophic* calcification is seen at sites of previous inflammation or damage to the skin. It develops in acne scars as well as on the distal extremities of patients with scleroderma and in the subcutaneous tissue and intermuscular fascial planes in [DM](#). The latter is more extensive and is more commonly seen in children. An elevated calcium phosphate product, as in secondary hyperparathyroidism, can lead to nodules of *metastatic* calcinosis cutis, which tend to be subcutaneous and periarticular. This form is often accompanied by calcification of muscular arteries and subsequent ischemic necrosis (calciophylaxis).

Skin-Colored Lesions There are several types of skin-colored lesions, including epidermoid inclusion cysts, lipomas, rheumatoid nodules, neurofibromas, angiofibromas, neuromas, and adnexal tumors such as tricholemmomas. Both *epidermoid inclusion cysts* and *lipomas* are very common mobile subcutaneous nodules -- the former are rubbery and compressible and drain cheeselike material (sebum and keratin) if incised. Lipomas are firm and somewhat lobulated on palpation. When extensive facial epidermoid inclusion cysts develop in childhood or there is a family history of such lesions, the patient should be examined for other signs of Gardner syndrome, including osteomas and desmoid tumors. *Rheumatoid nodules* are firm, 0.5- to 4-cm nodules that tend to localize around pressure points, especially the elbows. They are seen in approximately 20% of patients with rheumatoid arthritis and 6% of patients with Still's disease. Biopsies of the nodules show palisading granulomas. Similar lesions that are smaller and shorter-lived are seen in rheumatic fever.

Neurofibromas (benign Schwann cell tumors) are soft papules or nodules that exhibit the "button-hole" sign, that is, they invaginate into the skin with pressure in a manner similar to a hernia. Single lesions are seen in normal individuals, but multiple neurofibromas, usually in combination with six or more [CALM](#) measuring >1.5 cm (see "Hyperpigmentation") and multiple Lisch nodules, are seen in von Recklinghausen's disease ([NF](#) type I). Lisch nodules are 1-mm yellow-brown spots within the iris that are best observed with slit-lamp examination. Additional manifestations include axillary

freckling and peripheral and [CNS tumors \(Chap. 370\)](#). In some patients the neurofibromas are localized and unilateral, whereas in others they are limited to the CNS.

Angiofibromas are firm, pink to skin-colored papules that measure from 3 mm to several centimeters in diameter. When they are located on the central cheeks (adenoma sebaceum) or multiple fibromas are seen around the nails, the patient has tuberous sclerosis. It is an autosomal disorder due to mutations in two different genes, and the associated findings are discussed in the section on ash leaf spots as well as in [Chap. 370](#). Multiple facial angiofibromas have also been observed in patients with multiple endocrine neoplasia (MEN) syndrome, type 1.

Neuromas (benign proliferations of nerve fibers) are also firm, skin-colored papules. They are more commonly found at sites of amputation and as rudimentary supernumerary digits. However, when there are multiple neuromas on the eyelids, lips, distal tongue, and/or oral mucosa, the patient should be investigated for other signs of the [MEN syndrome, type 2b](#). Associated findings include marfanoid habitus, protuberant lips, intestinal ganglioneuromas, and medullary thyroid carcinoma (>75% of patients; [Chap. 339](#)).

Adnexal tumors are derived from pluripotential cells of the epidermis that can differentiate toward hair, sebaceous, apocrine, or eccrine glands or remain undifferentiated. *Basal cell epitheliomas* (BCEs) are examples of adnexal tumors that have little or no evidence of differentiation. Clinically, they are translucent papules with rolled borders, telangiectasias, and central erosion. BCEs commonly arise in sun-damaged skin of the head and neck. When a patient has multiple BCEs, especially prior to age 30, the possibility of the basal cell nevus syndrome should be raised. It is inherited as an autosomal dominant trait and is associated with jaw cysts, palmar and plantar pits, frontal bossing, medulloblastomas and calcification of the falx cerebri and diaphragma sellae. *Tricholemmomas* are also skin-colored adnexal tumors but differentiate toward hair follicles and can have a wartlike appearance. The presence of multiple tricholemmomas on the face and cobblestoning of the oral mucosa points to the diagnosis of Cowden's disease (multiple hamartoma syndrome) due to mutations in the *PTEN* gene. Internal organ involvement (in decreasing order of frequency) includes fibrocystic disease and carcinoma of the breast, adenomas and carcinomas of the thyroid, and gastrointestinal polyposis. Keratoses of the palms, soles, and dorsa of the hands are also seen.

Pink Lesions The cutaneous lesions associated with primary systemic *amyloidosis* are pink in color and translucent. Common locations are the face, especially the periorbital and perioral regions, and flexural areas. On biopsy, homogeneous deposits of amyloid are seen in the dermis and in the walls of blood vessels; the latter lead to an increase in vessel wall fragility. As a result, petechiae and purpura develop in clinically normal skin as well as in lesional skin following minor trauma, hence the term "pinch purpura." Amyloid deposits are also seen in the striated muscle of the tongue and result in macroglossia.

Even though specific mucocutaneous lesions are rarely seen in secondary amyloidosis and are present in only about 30% of the patients with primary amyloidosis, a rapid

diagnosis of systemic amyloidosis can be made by an examination of abdominal subcutaneous fat. By special staining, deposits are seen around blood vessels or individual fat cells in 40 to 50% of patients. There are also three forms of amyloidosis that are limited to the skin and that should not be construed as cutaneous lesions of systemic amyloidosis. They are macular amyloidosis (upper back), lichenoid amyloidosis (usually lower extremities), and nodular amyloidosis. In macular and lichenoid amyloidosis, the deposits are composed of altered epidermal keratin. Recently, macular and lichenoid amyloidosis have been associated with [MEN](#) syndrome, type 2a.

Patients with *multicentric reticulohistiocytosis* also have pink-colored papules and nodules on the face and mucous membranes as well as on the extensor surface of the hands and forearms. They have a polyarthritis that can mimic rheumatoid arthritis clinically. On histologic examination, the papules have characteristic giant cells that are not seen in biopsies of rheumatoid nodules. Pink to skin-colored papules that are firm, 2 to 5 mm in diameter, and often in a linear arrangement are seen in patients with *papular mucinosis*. This disease is also referred to as *lichen myxedematosus* or *scleromyxedema*. The latter name comes from the brawny induration of the face and extremities that may accompany the papular eruption. Biopsy specimens of the papules show localized mucin deposition, and serum protein electrophoresis demonstrates a monoclonal spike of IgG, usually with a λ light chain.

Yellow Lesions Several systemic disorders are characterized by yellow-colored cutaneous papules or plaques -- hyperlipidemia (xanthomas), gout (tophi), diabetes (necrobiosis lipoidica), pseudoxanthoma elasticum, and Torre syndrome (sebaceous tumors). Eruptive xanthomas are the most common form of *xanthomas*, and are associated with hypertriglyceridemia (types I, III, IV, and V). Crops of yellow papules with erythematous halos occur primarily on the extensor surfaces of the extremities and the buttocks, and they spontaneously involute with a fall in serum triglycerides. Increased β -lipoproteins (primarily types II and III) result in one or more of the following types of xanthoma: xanthelasma, tendon xanthomas, and plane xanthomas. Xanthelasma are found on the eyelids, whereas tendon xanthomas are frequently associated with the Achilles and extensor finger tendons; plane xanthomas are flat and favor the palmar creases, face, upper trunk, and scars. Tuberous xanthomas are frequently associated with hypertriglyceridemia, but they are also seen in patients with hypercholesterolemia (type II) and are found most frequently over the large joints or hand. Biopsy specimens of xanthomas show collections of lipid-containing macrophages (foam cells).

Patients with several disorders, including biliary cirrhosis, can have a secondary form of hyperlipidemia with associated tuberous and planar xanthomas. However, patients with myeloma have *normolipemic* flat xanthomas. This latter form of xanthoma may be 3-12 cm in diameter and is most frequently seen on the upper trunk or side of the neck. It is also important to note that the most common setting for eruptive xanthomas is uncontrolled diabetes mellitus. The least specific sign for hyperlipidemia is xanthelasma, because at least 50% of the patients with this finding have normal lipid profiles.

In *tophaceous gout* there are deposits of monosodium urate in the skin around the joints, particularly those of the hands and feet. Additional sites of *tophi* formation include

the helix of the ear and the olecranon and prepatellar bursae. The lesions are firm, yellow in color, and occasionally discharge a chalky material. Their size varies from 1 mm to 7 cm, and the diagnosis can be established by polarization of the aspirated contents of a lesion. Lesions of *necrobiosis lipoidica* are found primarily on the shins (90%), and patients can have diabetes mellitus or develop it subsequently. Characteristic findings include a central yellow color, atrophy (transparency), telangiectasias, and an erythematous border. Ulcerations can also develop within the plaques. Biopsy specimens show necrobiosis of collagen, granulomatous inflammation, and obliterative endarteritis.

In *pseudoxanthoma elasticum* (PXE) there is an abnormal deposition of calcium on the elastic fibers of the skin, eye, and blood vessels. In the skin, the flexural areas such as the neck, axillae, antecubital fossae, and inguinal area are the primary sites of involvement. Yellow papules coalesce to form reticulated plaques that have an appearance similar to that of plucked chicken skin. In severely affected skin, hanging, redundant folds develop. Some patients have a more subtle macular form of the disease, and careful inspection is required. Biopsy specimens of involved skin show swollen and irregularly clumped elastic fibers with deposits of calcium. In the eye, the calcium deposits in Bruch's membrane lead to angioid streaks and choroiditis; in the arteries of the heart, kidney, gastrointestinal tract, and extremities, the deposits lead to angina, hypertension, gastrointestinal bleeding, and claudication, respectively. Long-term administration of D-penicillamine can lead to PXE-like skin changes as well as elastic fiber alterations in internal organs.

Adnexal tumors that have differentiated toward sebaceous glands include sebaceous adenoma, sebaceous epithelioma, sebaceous carcinoma, and sebaceous hyperplasia. Except for sebaceous hyperplasia, which is commonly seen on the face, these tumors are fairly rare. Patients with Torre syndrome have *sebaceous adenomas*, and in the majority of cases there are multiple such tumors. These patients can also have sebaceous carcinomas and sebaceous hyperplasia as well as keratoacanthomas. The internal manifestations of Torre syndrome include *multiple* carcinomas of the gastrointestinal tract (primarily colon) as well as cancers of the larynx, genitourinary tract, and endometrium. Some patients also have a strong family history of cancer.

Red Lesions Cutaneous lesions that are red in color have a wide variety of etiologies; in an attempt to simplify their identification, they will be subdivided into papules, papules/plaques, and subcutaneous nodules. Common red papules include *arthropod bites* and *cherry hemangiomas*; the latter are small, bright-red, dome-shaped papules that represent benign proliferation of capillaries. In patients with AIDS, the development of multiple red hemangioma-like lesions points to bacillary angiomatosis, and biopsy specimens show clusters of bacilli that stain positive with the Warthin-Starry stain; the pathogens have been identified as *Bartonella henselae* and *B. quintana*. Disseminated visceral disease is seen primarily in immunocompromised hosts but can occur in immunocompetent individuals.

Multiple *angiokeratomas* are seen in Fabry's disease, an X-linked recessive lysosomal storage disease that is due to a deficiency of α -galactosidase A. The lesions are red to red-blue in color and can be quite small in size (1 to 3 mm), with the most common location being the lower trunk. Associated findings include chronic renal failure,

peripheral neuropathy, and corneal opacities (cornea verticillata). Electron photomicrographs of angiokeratomas and clinically normal skin demonstrate lamellar lipid deposits in fibroblasts, pericytes, and endothelial cells that are diagnostic of this disease. Widespread acute eruptions of erythematous papules are discussed in the section on exanthems.

There are several infectious diseases that present as erythematous papules or nodules in a sporotrichoid pattern, that is, in a linear arrangement along the lymphatic channels. The two most common etiologies are *Sporothrix schenckii* (sporotrichosis) and *M. marinum* (atypical mycobacteria). The organisms are introduced as a result of trauma, and a primary inoculation site is often seen in addition to the lymphatic nodules. Additional causes include *Nocardia*, *Leishmania*, and other dimorphic fungi; culture of lesional tissue will aid in the diagnosis.

The diseases that are characterized by erythematous plaques with scale are reviewed in the papulosquamous section, and the various forms of dermatitis are discussed in the section on erythroderma. Additional disorders in the differential diagnosis of red papules/plaques include *erysipelas*, *polymorphous light eruption* (PMLE), *lymphocytoma cutis*, *cutaneous lupus*, *lymphoma cutis*, and *leukemia cutis*. The first three diseases represent primary cutaneous disorders. PMLE is characterized by erythematous papules and plaques in a primarily sun-exposed distribution -- dorsum of the hand, extensor forearm, and face. Lesions follow exposure to UV-B and/or UV-A, and in northern latitudes PMLE is most severe in the late spring and early summer. A process referred to as "hardening" occurs with continued UV exposure, and the eruption fades, but in temperate climates it will recur in the spring. PMLE must be differentiated from cutaneous lupus, and this is accomplished by histologic examination and direct immunofluorescence of the lesions. Lymphocytoma cutis (pseudolymphoma) is a *benign* polyclonal proliferation of lymphocytes in the skin that presents as infiltrated pink-red to red-purple papules and plaques; it must be distinguished from lymphoma cutis.

Several types of red plaques are seen in patients with systemic *lupus*, including (1) erythematous urticarial plaques across the cheeks and nose in the classic butterfly rash; (2) erythematous discoid lesions with fine or "carpet-tack" scale, telangiectasias, central hypopigmentation, peripheral hyperpigmentation, follicular plugging, and atrophy located on the face, scalp, external ears, arms, and upper trunk; and (3) psoriasiform or annular lesions of subacute lupus with hypopigmented centers located on the face, extensor arms, and upper trunk. Additional cutaneous findings include (1) a violaceous flush on the face and V of the neck; (2) urticarial vasculitis (see "Urticaria"); (3) lupus panniculitis (see below); (4) diffuse alopecia; (5) alopecia secondary to discoid lesions; (6) periungual telangiectasias and erythema; (7) erythema multiforme-like lesions that may become bullous; and (8) distal ulcerations secondary to Raynaud's phenomenon, vasculitis, or livedoid vasculitis. Patients with only discoid lesions usually have the form of lupus that is limited to the skin. However, 2 to 10% of these patients eventually develop systemic lupus. Direct immunofluorescence of involved skin shows deposits of IgG or IgM and C3 in a granular distribution along the dermal-epidermal junction.

In *lymphoma cutis* there is a proliferation of malignant lymphocytes or histiocytes in the skin, and the clinical appearance resembles that of lymphocytoma cutis -- infiltrated pink-red to red-purple papules and plaques. Lymphoma cutis can occur anywhere on

the surface of the skin, whereas the sites of predilection for lymphocytomas include the malar ridge, tip of the nose, and earlobes. Patients with non-Hodgkin's lymphomas have specific cutaneous lesions more often than those with Hodgkin's disease, and occasionally, the skin nodules precede the development of extracutaneous non-Hodgkin's lymphoma or represent the only site of involvement. Arcuate lesions are sometimes seen in lymphoma and lymphocytoma cutis as well as in [CTCL](#). *Leukemia cutis* has the same appearance as lymphoma cutis, and specific lesions are seen more commonly in monocytic leukemias than in lymphocytic or granulocytic leukemias. Cutaneous chloromas (granulocytic sarcomas) may precede the appearance of circulating blasts in acute nonlymphocytic leukemia and, as such, represent a form of aleukemic leukemia cutis.

Common causes of erythematous subcutaneous nodules include inflamed epidermoid inclusion cysts, acne cysts, and furuncles. *Panniculitis*, an inflammation of the fat, also presents as subcutaneous nodules and is frequently a sign of systemic disease. There are several forms of panniculitis, including erythema nodosum, erythema induratum, lupus profundus, lipomembranous lipodermatosclerosis, α_1 -antitrypsin deficiency, factitial, and fat necrosis secondary to pancreatic disease. Except for erythema nodosum, these lesions may break down and ulcerate or heal with a scar. The shin is the most common location for the nodules of erythema nodosum, whereas the calf is the most common location for lesions of erythema induratum. In erythema nodosum the nodules are initially red but then develop a blue color as they resolve. Patients with erythema nodosum but no underlying systemic illness can still have fever, malaise, leukocytosis, arthralgias, and/or arthritis. However, the possibility of an underlying illness should be excluded, and the most common associations are streptococcal infections, upper respiratory infections, sarcoidosis, and inflammatory bowel disease. The less common associations include tuberculosis, histoplasmosis, coccidioidomycosis, psittacosis, drugs (oral contraceptives, sulfonamides, aspartame, bromides, iodides), cat-scratch fever, and infections with *Yersinia*, *Salmonella*, and *Chlamydia*.

In some patients, erythema induratum/nodular vasculitis is an idiopathic disease; however, in approximately 25 to 70% of patients, polymerase chain reaction (PCR) analysis will demonstrate *M. tuberculosis* complex DNA. The lesions of lupus profundus are found primarily on the upper arms and buttocks (sites of abundant fat) and are seen in both the cutaneous and systemic forms of lupus. The overlying skin may be normal, erythematous, or have the changes of discoid lupus. The subcutaneous fat necrosis that is associated with pancreatic disease is presumably secondary to circulating lipases and is seen in patients with pancreatic carcinoma as well as in patients with acute and chronic pancreatitis. In this disorder there may be an associated arthritis, fever, and inflammation of visceral fat. Histologic examination of deep incisional biopsy specimens will aid in the diagnosis of the particular type of panniculitis.

Subcutaneous erythematous nodules are also seen in *cutaneous polyarteritis nodosa* (PAN) and as a manifestation of *systemic vasculitis*, e.g., systemic PAN, allergic granulomatosis, or Wegener's granulomatosis ([Chap. 317](#)). Cutaneous PAN presents with painful subcutaneous nodules and ulcers within a red-purple, netlike pattern of livedo reticularis. The latter is due to slowed blood flow through the superficial horizontal venous plexus. The majority of lesions are found on the lower extremity, and while

arthralgias and myalgias may accompany cutaneous PAN, there is no evidence of systemic involvement. In both the cutaneous and systemic forms of vasculitis, skin biopsy specimens of the associated nodules will show the changes characteristic of a vasculitis; the size of the vessel involved will depend on the particular disease.

Red-Brown Lesions The cutaneous lesions in *sarcoidosis* ([Chap. 318](#)) are classically red to red-brown in color, and with diascopy (pressure with a glass slide) a yellow-brown residual color is observed that is secondary to the granulomatous infiltrate. The waxy papules and plaques may be found anywhere on the skin, but the face is the most common location. Usually there are no surface changes, but occasionally the lesions will have scale. Biopsy specimens of the papules show "naked" granulomas in the dermis, i.e., granulomas surrounded by a minimal number of lymphocytes. Other cutaneous findings in *sarcoidosis* include annular lesions with an atrophic or scaly center, papules within scars, hypopigmented macules and papules, alopecia, acquired ichthyosis, erythema nodosum, and lupus pernio (see below). Additional physical findings are peripheral lymphadenopathy and parotid and lacrimal gland enlargement. When there is cutaneous involvement of the hands, radiographs will often show lytic lesions in the underlying bone.

The differential diagnosis of *sarcoidosis* includes foreign-body granulomas produced by chemicals such as beryllium and zirconium, late secondary syphilis, and *lupus vulgaris*. *Lupus vulgaris* is a form of cutaneous tuberculosis that is seen in previously infected and sensitized individuals. There is often underlying active tuberculosis elsewhere, usually in the lungs or lymph nodes. At least 90% of the lesions occur in the head and neck area and are red-brown plaques with a yellow-brown color on diascopy. Secondary scarring and squamous cell carcinomas can develop within the plaques. Cultures or [PCR](#) analysis of the lesions should be done because it is rare for the acid-fast stain to show bacilli within the dermal granulomas.

Sweet's syndrome is characterized by red to red-brown plaques and nodules that are frequently painful and occur primarily on the head, neck, and upper extremities. The patients also have fever, neutrophilia, and a dense dermal infiltrate of neutrophils in the lesions. In approximately 10% of the patients there is an associated malignancy, most commonly acute nonlymphocytic leukemia. *Sweet's syndrome* has also been reported with lymphoma, chronic leukemia, myeloma, myelodysplastic syndromes, and solid tumors (primarily of the genitourinary tract). The differential diagnosis includes neutrophilic eccrine hidradenitis and atypical forms of pyoderma gangrenosum. Extracutaneous sites of involvement include joints, muscles, eye, kidney (proteinuria, occasionally glomerulonephritis), and lung (neutrophilic infiltrates). The idiopathic form of *Sweet's syndrome* is seen more often in women, following a respiratory tract infection.

A generalized distribution of red-brown macules and papules is seen in the form of mastocytosis known as *urticaria pigmentosa* ([Chap. 310](#)). Each lesion represents a collection of mast cells in the dermis, with hyperpigmentation of the overlying epidermis. Stimuli such as rubbing cause these mast cells to degranulate, and this leads to the formation of localized urticaria (Darier's sign). Additional symptoms can result from mast cell degranulation and include headache, flushing, diarrhea, and pruritus. Mast cells also infiltrate various organs such as the liver, spleen, and gastrointestinal tract in up to

30 to 50% of patients with urticaria pigmentosa, and accumulations of mast cells in the bones may produce either osteosclerotic or osteolytic shadows on radiographs. In the majority of these patients, however, the internal involvement remains fairly static. A subtype of chronic leukocytoclastic vasculitis, *erythema elevatum diutinum* (EED), also presents with papules that are red-brown in color. The papules coalesce into plaques on the extensor surfaces of knees, elbows, and the small joints of the hand. Flares of EED have been associated with streptococcal infections.

Blue Lesions Lesions that are blue in color are the result of either vascular ectasias and tumors or melanin pigment in the dermis. *Venous lakes* (ectasias) are compressible dark-blue lesions that are found commonly in the head and neck region. *Venous malformations* are also compressible blue papules and nodules that can occur anywhere on the body, including the oral mucosa. When they are multiple rather than single congenital lesions, the patient may have the blue rubber bleb syndrome or Mafucci's syndrome. Patients with the blue rubber bleb syndrome also have vascular anomalies of the gastrointestinal tract that may bleed, whereas patients with Mafucci's syndrome have associated dyschondroplasia and osteochondromas. *Blue nevi* (moles) are seen when there are collections of pigment-producing nevus cells in the dermis. These benign papular lesions are dome-shaped and occur most commonly on the dorsum of the hand or foot.

Violaceous Lesions Violaceous papules and plaques are seen in *lupus pernio*, *lymphoma cutis*, and *cutaneous lupus*. Lupus pernio is a particular type of sarcoidosis that involves the tip of the nose and the earlobes, with lesions that are violaceous in color rather than red-brown. This form of sarcoidosis is associated with involvement of the upper respiratory tract. The plaques of lymphoma cutis and cutaneous lupus may be red or violaceous in color and were discussed above.

Purple Lesions Purple-colored papules and plaques are seen in vascular tumors, such as *Kaposi's sarcoma* ([Chap. 309](#)) and *angiosarcoma*, and when there is extravasation of red blood cells into the skin in association with inflammation, as in *palpable purpura* (see "Purpura"). Patients with congenital or acquired [AV](#)fistulas and venous hypertension can develop purple papules on the lower extremities that can resemble Kaposi's sarcoma clinically and histologically; this condition is referred to as pseudo-Kaposi sarcoma (acral angiodermatitis). Angiosarcoma is found most commonly on the scalp and face of elderly patients or within areas of chronic lymphedema and presents as purple papules and plaques. In the head and neck region the tumor often extends beyond the clinically defined borders and may be accompanied by facial edema.

Brown and Black Lesions Brown- and black-colored papules are reviewed in "Hyperpigmentation."

Cutaneous Metastases These are discussed last because they can have a wide range of colors. Most commonly they present as either firm, skin-colored subcutaneous nodules or firm, red to red-brown papulonodules. The lesions of lymphoma cutis range from pink-red to plum in color, whereas metastatic melanoma can be pink, blue, or black in color. Cutaneous metastases develop from hematogenous or lymphatic spread and are most often due to the following primary carcinomas: in men, lung, colon, melanoma,

and oral cavity; and in women, breast, colon, and lung. These metastatic lesions may be the initial presentation of the carcinoma, especially when the primary site is the lung, kidney, or ovary.

PURPURA ([Table 57-16](#))

Purpura are seen when there is an extravasation of red blood cells into the dermis, and as a result, the lesions do not blanch with pressure. This is in contrast to those erythematous or violet-colored lesions that are due to localized vasodilatation -- they do blanch with pressure. Purpura (≥ 3 mm) and petechiae (≤ 2 mm) are divided into two major groups, palpable and nonpalpable. The most frequent causes of *nonpalpable* petechiae and purpura are primary cutaneous disorders such as *trauma*, *solar purpura*, and *capillaritis*. Less common causes are *steroid purpura* and *livedoid vasculitis* (see "Ulcers"). Solar purpura are seen primarily on the extensor forearms, while glucocorticoid purpura secondary to potent topical steroids or endogenous or exogenous Cushing's syndrome can be more widespread. In both cases there is alteration of the supporting connective tissue that surrounds the dermal blood vessels. In contrast, the petechiae that result from capillaritis are found primarily on the lower extremities. In capillaritis there is an extravasation of erythrocytes as a result of perivascular lymphocytic inflammation. The petechiae are bright red, 1 to 2 mm in size, and scattered within annular or coin-shaped yellow-brown macules. The yellow-brown color is caused by hemosiderin deposits within the dermis.

Systemic causes of nonpalpable purpura fall into several categories, and those secondary to clotting disturbances and vascular fragility will be discussed first. The former group includes *thrombocytopenia* ([Chap. 116](#)), *abnormal platelet function* as is seen in uremia, and *clotting factor defects*. The initial site of presentation for thrombocytopenia-induced petechiae is the distal lower extremity. Capillary fragility leads to nonpalpable purpura in patients with systemic *amyloidosis* (see "Papulonodular Skin Lesions"), disorders of collagen production such as *Ehlers-Danlos syndrome*, and *scurvy*. In scurvy there are flattened corkscrew hairs with surrounding hemorrhage on the lower extremities, in addition to gingivitis. Vitamin C is a cofactor for lysyl hydroxylase, an enzyme involved in the posttranslational modification of procollagen that is necessary for cross-link formation.

In contrast to the previous group of disorders, the purpura seen in the following group of diseases are associated with thrombi formation within vessels. It is important to note that these thrombi are demonstrable in skin biopsy specimens. This group of disorders includes disseminated intravascular coagulation (DIC), monoclonal cryoglobulinemia, thrombotic thrombocytopenic purpura, and reactions to warfarin. DIC is triggered by several types of infection (gram-negative, gram-positive, viral, and rickettsial) as well as by tissue injury and neoplasms. Widespread purpura and hemorrhagic infarcts of the distal extremities are seen. Similar lesions are found in purpura fulminans, which is a form of DIC associated with fever and hypotension that occurs more commonly in children following an infectious illness such as varicella, scarlet fever, or an upper respiratory tract infection. In both disorders, hemorrhagic bullae can develop in involved skin.

Monoclonal cryoglobulinemia is associated with multiple myeloma, Waldenstrom's

macroglobulinemia, lymphocytic leukemia, and lymphoma. Purpura, primarily of the lower extremities, and hemorrhagic infarcts of the fingers and toes are seen in these patients. Exacerbations of disease activity can follow cold exposure or an increase in serum viscosity. Biopsy specimens show precipitates of the cryoglobulin within dermal vessels. Similar deposits have been found in the lung, brain, and renal glomeruli. Patients with *thrombotic thrombocytopenic purpura* can also have hemorrhagic infarcts as a result of intravascular thromboses. Additional signs include thrombocytopenic purpura, fever, and microangiopathic hemolytic anemia ([Chap. 108](#)).

Administration of *warfarin* can result in painful areas of erythema that become purpuric and then necrotic with an adherent black eschar. This reaction is seen more often in women and in areas with abundant subcutaneous fat -- breasts, abdomen, buttocks, thighs, and calves. The erythema and purpura develop between the third and tenth day of therapy, most likely as a result of a transient imbalance in the levels of anticoagulant and procoagulant vitamin K-dependent factors. Continued therapy does not exacerbate preexisting lesions, and patients with an inherited or acquired deficiency of protein C are at increased risk for this particular reaction as well as for purpura fulminans.

Purpura secondary to *cholesterol emboli* are usually seen on the lower extremities of patients with atherosclerotic vascular disease. They often follow anticoagulant therapy or an invasive vascular procedure such as an arteriogram but also occur spontaneously from disintegration of atheromatous plaques. Associated findings include livedo reticularis, gangrene, cyanosis, subcutaneous nodules, and ischemic ulcerations. Multiple step sections of the biopsy specimen may be necessary to demonstrate the cholesterol clefts with the vessels. Petechiae are also an important sign of *fat embolism* and occur primarily on the upper body 2 to 3 days after a major injury. By using special fixatives, the emboli can be demonstrated in biopsy specimens of the petechiae. Emboli of tumor or thrombus are seen in patients with atrial myxomas and marantic endocarditis.

In the *Gardner-Diamond syndrome* (autoerythrocyte sensitivity), female patients develop large ecchymoses within areas of painful, warm erythema. An episode of significant trauma frequently precedes the onset of this syndrome. Intradermal injections of autologous erythrocytes or phosphatidyl serine derived from the red cell membrane can reproduce the lesions in some patients; however, there are instances where a reaction is seen at an injection site of the forearm but not in the midback region. The latter has led some observers to view Gardner-Diamond syndrome as a cutaneous manifestation of severe emotional stress. *Waldenstrom's hypergammaglobulinemic purpura* is a chronic disorder characterized by petechiae on the lower extremities. There are circulating complexes of IgG-anti-IgG molecules, and exacerbations are associated with prolonged standing or walking.

Palpable purpura are further subdivided into vasculitic and embolic. In the group of vasculitic disorders, *leukocytoclastic vasculitis* (LCV), also known as *allergic vasculitis*, is the one most commonly associated with palpable purpura ([Chap. 317](#)). *Henoch-Schonlein purpura* is a subtype of acute LCV that is seen primarily in children and adolescents following an upper respiratory infection. The majority of lesions are found on the lower extremities and buttocks. Systemic manifestations include fever, arthralgias (primarily of the knees and ankles), abdominal pain, gastrointestinal

bleeding, and nephritis. Direct immunofluorescence examination shows deposits of IgA within dermal blood vessel walls. In *polyarteritis nodosa*, specific cutaneous lesions result from a vasculitis of arterial vessels rather than postcapillary venules as in LCV. The arteritis leads to ischemia of the skin, and this explains the irregular outline of the purpura (see below).

Several types of infectious emboli can give rise to palpable purpura. These embolic lesions are usually *irregular* in outline as opposed to the lesions of leukocytoclastic vasculitis, which are *circular* in outline. The irregular outline is indicative of a cutaneous infarct, and the size corresponds to the area of skin that received its blood supply from that particular arteriole or artery. The palpable purpura in [LCV](#) are circular because the erythrocytes simply diffuse out evenly from the postcapillary venules as a result of inflammation. Infectious emboli are most commonly due to gram-negative cocci (meningococcus, gonococcus), gram-negative rods (Enterobacteriaceae), and gram-positive cocci (staphylococcus). Additional causes include *Rickettsia* and, in immunocompromised patients, *Candida* and opportunistic fungi.

The embolic lesions in *acute meningococcemia* are found primarily on the trunk, lower extremities, and sites of pressure, and a gunmetal-gray color often develops within them. Their size varies from 1 mm to several centimeters, and the organisms can be cultured from the lesions. Associated findings include a preceding upper respiratory tract infection, fever, meningitis, [DIC](#), and, in some patients, a deficiency of the terminal components of complement. In *disseminated gonococcal infection* (arthritis-dermatitis syndrome), a small number of papules and vesicopustules with central purpura or hemorrhagic necrosis are found over the joints of the distal extremities. Additional symptoms include arthralgias, tenosynovitis, and fever. To establish the diagnosis, a Gram stain of these lesions should be performed. *Rocky mountain spotted fever* is a tick-borne disease that is caused by *R. rickettsii*. A several-day history of fever, chills, severe headache, and photophobia precedes the onset of the cutaneous eruption. The initial lesions are erythematous macules and papules on the wrists, ankles, palms, and soles. With time, the lesions spread centripetally and become purpuric.

Lesions of *ecthyma gangrenosum* begin as edematous, erythematous papules or plaques and then develop central purpura and necrosis. Bullae formation also occurs in these lesions, and they are frequently found in the girdle region. The organism that is classically associated with ecthyma gangrenosum is *Pseudomonas aeruginosa*, but other gram-negative rods such as *Klebsiella*, *Escherichia coli*, and *Serratia* can produce similar lesions. In immunocompromised hosts, the list of potential pathogens is expanded to include *Candida* and opportunistic fungi.

ULCERS

The approach to the patient with a cutaneous ulcer, is outlined in [Table 57-17](#). **Peripheral vascular diseases of the extremities are reviewed in Chap. 248, as is Raynaud's phenomenon.*

Livedoid vasculitis (atrophie blanche) represents a combination of a vasculopathy with intravascular thrombosis. Purpuric lesions and livedo reticularis are found in association with painful ulcerations of the lower extremities. These ulcers are often slow to heal, but

when they do, irregularly shaped white scars are formed. The majority of cases are secondary to venous hypertension, but possible underlying illnesses include cryofibrinogenemia and disorders of hypercoagulability, e.g., the antiphospholipid syndrome ([Chap. 117](#)).

In *pyoderma gangrenosum*, the border of the ulcers has a characteristic appearance of an undermined necrotic bluish edge and a peripheral erythematous halo. The ulcers often begin as pustules that then expand rather rapidly to a size as large as 20 cm. Although these lesions are most commonly found on the lower extremities, they can arise anywhere on the surface of the body, including sites of trauma (pathergy). An estimated 30 to 50% of cases are idiopathic, and the most common associated disorders are ulcerative colitis and Crohn's disease. Less commonly, it is associated with chronic active hepatitis, seropositive rheumatoid arthritis, acute and chronic granulocytic leukemia, polycythemia vera, and myeloma. Additional findings in these patients, even those with idiopathic disease, are cutaneous anergy and a benign monoclonal gammopathy. Because the histology of pyoderma gangrenosum is nonspecific, the diagnosis is made clinically by excluding less common causes of similar-appearing ulcers such as necrotizing vasculitis, Meleney's ulcer (synergistic infection at a site of trauma or surgery), dimorphic fungi, cutaneous amebiasis, spider bites, and factitial. In the myeloproliferative disorders, the ulcers may be more superficial with a pustulobullous border, and these lesions provide a connection between classic pyoderma gangrenosum and acute febrile neutrophilic dermatosis (Sweet's syndrome).

FEVER AND RASH

The major considerations in a patient with a fever and a rash are inflammatory diseases versus infectious diseases. In the hospital setting, the most common scenario is a patient who has a drug rash plus a fever secondary to an underlying infection. However, it should be emphasized that a drug reaction can lead to both a cutaneous eruption and a fever ("drug fever"). Additional inflammatory diseases that are often associated with a fever include pustular psoriasis, erythroderma, and Sweet's syndrome. Lyme disease, secondary syphilis, and viral and bacterial exanthems (see "Exanthems") are examples of infectious diseases that produce a rash and a fever. Lastly, it is important to determine whether or not the cutaneous lesions represent septic emboli (see "Purpura"). Such lesions usually have evidence of ischemia in the form of purpura, necrosis, or impending necrosis (gunmetal-gray color). In the patient with thrombocytopenia, however, purpura can be seen in inflammatory reactions such as morbilliform drug eruptions and infectious lesions.

(Bibliography omitted in Palm version)

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58. IMMUNOLOGICALLY MEDIATED SKIN DISEASES - Kim B. Yancey, Thomas J. Lawley

A number of immunologically mediated skin diseases and cutaneous manifestations of immunologically mediated systemic disorders are now recognized as distinct entities with relatively consistent clinical, histologic, and immunopathologic findings. Many of these disorders are due to autoimmune mechanisms. Clinically, they are characterized by morbidity (pain, pruritus, disfigurement) and in some instances by mortality (largely due to loss of epidermal barrier function and/or secondary infection). The major features of the more common immunologically mediated skin diseases are summarized in this chapter ([Table 58-1](#)).

PEMPHIGUS VULGARIS

Pemphigus vulgaris (PV) is a blistering skin disease seen predominantly in elderly patients. Patients with PV have an increased incidence of the HLA-DR4 and -DRw6 serologically defined haplotypes. This disorder is characterized by the loss of cohesion between epidermal cells (a process termed *acantholysis*) with the resultant formation of intraepidermal blisters. Clinical lesions of PV typically consist of flaccid blisters on either normal-appearing or erythematous skin. These blisters rupture easily, leaving denuded areas that may crust and enlarge peripherally (Plate IIE-69). Substantial portions of the body surface may be denuded in severe cases. Manual pressure to the skin of these patients may elicit the separation of the epidermis (Nikolsky's sign). This finding, while characteristic of PV, is not specific to this disorder and is also seen in toxic epidermal necrolysis, Stevens-Johnson syndrome, and a few other skin diseases. Lesions in PV typically present on the oral mucosa, scalp, face, neck, axilla, and trunk. In half or more of patients, lesions begin in the mouth; approximately 90% of patients have oromucosal involvement at some time during the course of their disease. Involvement of other mucosal surfaces (e.g., pharyngeal, laryngeal, esophageal, conjunctival, vulval, or rectal) can occur in severe disease. Pruritus may be a feature of early pemphigus lesions; extensive denudation may be associated with severe pain. Lesions usually heal without scarring, except at sites complicated by secondary infection or mechanically induced dermal wounds. Nonetheless, postinflammatory hyperpigmentation is usually present at sites of healed lesions for some time.

Biopsies of early lesions demonstrate intraepidermal vesicle formation secondary to loss of cohesion between epidermal cells (i.e., acantholytic blisters). Blister cavities contain acantholytic epidermal cells, which appear as round homogeneous cells containing hyperchromatic nuclei. Basal keratinocytes remain attached to the epidermal basement membrane, hence blister formation is within the suprabasal portion of the epidermis. Lesional skin may contain focal collections of intraepidermal eosinophils within blister cavities; dermal alterations are slight, often limited to an eosinophil-predominant leukocytic infiltrate. Direct immunofluorescence microscopy of lesional or intact patient skin shows deposits of IgG on the surface of keratinocytes; in contrast, deposits of complement components are typically found in lesional but not uninvolved skin. Deposits of IgG on keratinocytes are derived from circulating autoantibodies directed against cell-surface antigens. Circulating autoantibodies can be demonstrated in 80 to 90% of [PV](#) patients by indirect immunofluorescence microscopy; monkey esophagus is the optimal substrate for demonstration of these autoantibodies. Patients with PV have

IgG autoantibodies directed against desmogleins (Dsgs), transmembrane desmosomal glycoproteins that belong to the cadherin supergene family of calcium-dependent adhesion molecules. While Dsg3 is specifically recognized by PV autoantibodies, approximately 50% of PV sera also contain IgG against Dsg1. Most patients with early PV and only mucosal involvement have only anti-Dsg3 autoantibodies, whereas most patients with advanced disease (i.e., involvement of skin and mucosa) have both anti-Dsg3 and anti-Dsg1 autoantibodies. Recent studies have shown that the anti-Dsg autoantibody profile in these patients' sera as well as the tissue distribution of Dsg3 and Dsg1 determine the site of blister formation in patients with pemphigus. Experimental studies have also shown that these autoantibodies are pathogenic (i.e., responsible for blister formation) and that their titer correlates with disease activity.

[PV](#) can be life-threatening. Prior to the availability of glucocorticoids, the mortality ranged from 60 to 90%; the current mortality is approximately 5%. Common causes of morbidity and mortality are infection and complications of treatment with glucocorticoids. Bad prognostic factors include advanced age, widespread involvement, and the requirement for high doses of glucocorticoids (with or without other immunosuppressive agents) for control of disease. The course of PV in individual patients is variable and difficult to predict. Some patients achieve remission (40% of patients in some series), but others may require long-term treatment or succumb to complications of their disease or its treatment. The mainstay of treatment is systemic glucocorticoids. Patients with moderate to severe disease are usually started on prednisone, 60 to 80 mg/d. If new lesions continue to appear after 1 to 2 weeks of treatment, the dose should be increased. Many regimens combine an immunosuppressive agent with systemic glucocorticoids for control of PV. The two most frequently used are either azathioprine (1 mg/kg per day) or cyclophosphamide (1 mg/kg per day). It is important to bring severe or progressive disease under control quickly to lessen the severity and/or duration of this disorder.

PEMPHIGUS FOLIACEUS

Pemphigus foliaceus (PF) is distinguished from [PV](#) by several features. In PF, acantholytic blisters are located high within the epidermis, usually just beneath the stratum corneum. Hence PF is a more superficial blistering disease than PV. The distribution of lesions in the two disorders is much the same, except that in PF mucous membrane lesions are very rare. Patients with PF rarely demonstrate intact blisters but rather exhibit shallow erosions associated with erythema, scale, and crust formation. Mild cases of PF resemble severe seborrheic dermatitis; severe PF may cause extensive exfoliation. Sun exposure (ultraviolet irradiation) may be an aggravating factor. A blistering skin disease endemic to south central Brazil known as *fogo selvagem*, or *Brazilian pemphigus*, is clinically, histologically, and immunopathologically indistinguishable from PF.

Patients with [PF](#) have immunopathologic features in common with [PV](#). Specifically, direct immunofluorescence microscopy of perilesional skin demonstrates IgG on the surface of keratinocytes. As in PV, patients with PF frequently have circulating IgG autoantibodies against keratinocyte cell surface antigens. Guinea pig esophagus is the optimal substrate for indirect immunofluorescence microscopy studies of sera from patients with PF. In PF, autoantibodies are directed against Dsg1, a 160-kDa desmosomal cadherin.

As noted for PV, the autoantibody profile in patients with PF (i.e., anti-Dsg1) and the normal tissue distribution of this autoantigen (i.e., low expression in oral mucosa) is thought to account for the distribution of lesions in this disease.

Although pemphigus has been associated with several autoimmune diseases, its association with thymoma and/or myasthenia gravis is particularly notable. To date, more than 30 cases of thymoma and/or myasthenia gravis have been reported in association with pemphigus, usually with [PF](#). Patients may also develop pemphigus as a consequence of drug exposure. The most frequently implicated agent is penicillamine; other offenders include captopril, rifampin, piroxicam, penicillin, and phenobarbital. Drug-induced pemphigus usually resembles PF rather than [PV](#); autoantibodies in these patients have the same antigenic specificity as they do in other pemphigus patients. In most patients, lesions resolve following discontinuation of the drug; however, some patients require treatment with systemic glucocorticoids and/or immunosuppressive agents.

[PF](#) is generally a far less severe disease than [PV](#) and carries a better prognosis. Localized disease can be treated conservatively with topical or intralesional glucocorticoids; more active cases can usually be controlled with systemic glucocorticoids.

PARANEOPLASTIC PEMPHIGUS

Paraneoplastic pemphigus (PNP) is an autoimmune acantholytic mucocutaneous disease associated with an occult or confirmed neoplasm. Patients with PNP typically show painful mucosal erosive lesions in association with pruritic papulosquamous eruptions that often progress to blisters. Palm and sole involvement is common in these patients and raises the possibility that prior reports of neoplasia-associated erythema multiforme actually may have represented unrecognized cases of PNP. Biopsies of lesional skin from these patients show varying combinations of acantholysis, keratinocyte necrosis, and vacuolar-interface dermatitis. Direct immunofluorescence microscopy of patient skin shows deposits of IgG and complement on the surface of keratinocytes and (variably) similar immunoreactants in the epidermal basement membrane zone. Patients with PNP have IgG autoantibodies against cytoplasmic proteins that are members of the plakin family (e.g., desmoplakins I and II, bullous pemphigoid antigen 1, envoplakin, periplakin, and plectin) and cell-surface proteins that are members of the cadherin family (e.g., Dsg3 and Dsg1). Because immunoadsorption of anti-Dsg3 IgG is sufficient to eliminate the ability of PNP sera to induce blisters in an experimental passive transfer animal model, these particular autoantibodies are thought to play a key pathogenic role in blister formation in these patients.

Although PNP is generally resistant to conventional therapies (i.e., those used to treat PV), patients may improve (or even remit) following resection of underlying neoplasms. The predominant neoplasms associated with this disorder are non-Hodgkin's lymphoma, chronic lymphocytic leukemia, Castleman's disease, thymoma, and spindle cell tumors.

BULLOUS PEMPHIGOID

Bullous pemphigoid (BP) is an autoimmune subepidermal blistering disease usually

seen in the elderly. Lesions typically consist of tense blisters on either normal-appearing or erythematous skin ([Plate IIE-72](#)). The lesions are usually distributed over the lower abdomen, groin, and flexor surface of the extremities; oral mucosal lesions are found in 10 to 40% of patients. Pruritus may be nonexistent or severe. As lesions evolve, tense blisters tend to rupture and be replaced by flaccid lesions or erosions with or without surmounting crust. Nontraumatized blisters heal without scarring. The major histocompatibility complex class II allele HLA-DQb1*0301 is prevalent in patients with BP. Despite isolated reports, several studies have shown that patients with BP do not have an increased incidence of malignancy in comparison with appropriately age- and sex-matched controls.

While biopsies of early lesional skin demonstrate subepidermal blisters, the histologic features depend on the character of the particular lesion. Lesions on normal-appearing skin generally show a sparse perivascular leukocytic infiltrate with some eosinophils; conversely, biopsies of inflammatory lesions typically show an eosinophil-rich infiltrate within the papillary dermis at sites of vesicle formation and in perivascular areas. In addition to eosinophils, cell-rich lesions also contain mononuclear cells and neutrophils. It is not always possible to distinguish [BP](#) from other subepidermal blistering diseases by routine histologic techniques.

Immunopathologic studies have broadened our understanding of this disease and aided its diagnosis. Direct immunofluorescence microscopy of normal-appearing perilesional skin shows linear deposits of IgG and/or C3 in the epidermal basement membrane. The sera of approximately 70% of these patients contain circulating IgG autoantibodies that bind the epidermal basement membrane of normal human skin in indirect immunofluorescence microscopy. An even higher percentage of patients shows reactivity to the epidermal side of 1 M NaCl split skin [an alternative immunofluorescence microscopy test substrate that is commonly used to distinguish circulating IgG anti-basement membrane autoantibodies in patients with [BP](#) from those in patients with similar, yet different, subepidermal blistering diseases (e.g., epidermolysis bullosa acquisita, see below)]. No correlation exists between the titer of these autoantibodies and disease activity. In BP, circulating autoantibodies recognize 230- and (in approximately 70% of BP patients) 180-kDa hemidesmosome-associated proteins in basal keratinocytes [i.e., bullous pemphigoid antigen (BPAG)1 and BPAG2, respectively]. Autoantibodies are thought to develop against these antigens (more specifically, initially against BPAG2), deposit in situ, and activate complement that subsequently produces dermal mast cell degranulation and granulocyte-rich infiltrates that cause tissue damage and blister formation.

[BP](#) may persist for months to years, with exacerbations or remissions. Although extensive involvement may result in widespread erosions and compromise cutaneous integrity, the mortality rate is low even in the absence of treatment. Nonetheless, deaths may occur in elderly and/or debilitated patients. The mainstay of treatment is systemic glucocorticoids. Patients with local or minimal disease can sometimes be controlled with topical glucocorticoids alone; patients with more extensive lesions generally respond to systemic glucocorticoids either alone or in combination with immunosuppressive agents. Patients will usually respond to prednisone, 40 to 60 mg/d. In some instances, azathioprine (1 mg/kg per day) or cyclophosphamide (1 mg/kg per day) are necessary adjuncts.

PEMPHIGOID GESTATIONIS

Pemphigoid gestationis (PG), also known as herpes gestationis, is a rare, nonviral, subepidermal blistering disease of pregnancy and the puerperium. PG may begin during any trimester of pregnancy or present shortly after delivery. Lesions are usually distributed over the abdomen, trunk, and extremities; mucous membrane lesions are rare. Skin lesions in these patients may be quite polymorphic and consist of erythematous urticarial papules and plaques, vesiculopapules, and/or frank bullae. Lesions are almost always very pruritic. Severe exacerbations of PG frequently occur after delivery, typically within 24 to 48 h. PG tends to recur in subsequent pregnancies, often beginning earlier during such gestations. Brief flare-ups of disease may occur with resumption of menses and may develop in patients later exposed to oral contraceptives. Occasionally, infants of affected mothers demonstrate transient skin lesions.

Biopsies of early lesional skin show teardrop-shaped subepidermal vesicles forming in dermal papillae in association with an eosinophil-rich leukocytic infiltrate. Differentiation of [PG](#) from other subepidermal bullous diseases by light microscopy is often difficult. However, direct immunofluorescence microscopy of perilesional skin from PG patients reveals the immunopathologic hallmark of this disorder -- linear deposits of C3 in the epidermal basement membrane zone. These deposits develop as a consequence of complement activation produced by low titer IgG anti-basement membrane zone autoantibodies. Recent studies have shown that the majority of PG sera contain autoantibodies that recognize [BPAG2](#), the same 180-kDa hemidesmosome-associated protein that is targeted by autoantibodies in roughly 70% of patients with [BP](#) -- a subepidermal bullous disease that resembles PG morphologically, histologically, and immunopathologically.

The goals of therapy in patients with [PG](#) are to prevent the development of new lesions, relieve intense pruritus, and care for erosions at sites of blister formation. Most patients require treatment with moderate doses of daily glucocorticoids (i.e., 20 to 40 mg of prednisone) at some point in their course. Mild cases (or brief flare-ups) may be controlled by vigorous use of potent topical glucocorticoids. Although PG was once thought to be associated with an increased risk of fetal morbidity and mortality, the best evidence now suggests that these infants may only be at increased risk of being slightly premature or "small for dates." Current evidence suggests that there is no difference in the incidence of uncomplicated live births in PG patients treated with systemic glucocorticoids and in those managed more conservatively. If systemic glucocorticoids are administered, newborns are at risk for development of reversible adrenal insufficiency.

DERMATITIS HERPETIFORMIS

Dermatitis herpetiformis (DH) is an intensely pruritic, papulovesicular skin disease characterized by lesions symmetrically distributed over extensor surfaces (i.e., elbows, knees, buttocks, back, scalp, and posterior neck) ([Plate IIE-68](#)). The primary lesion in this disorder is a papule, papulovesicle, or urticarial plaque. Because pruritus is prominent, patients may present with excoriations and crusted papules but no observable primary lesions. Patients sometimes report that their pruritus has a

distinctive burning or stinging component; the onset of such local symptoms reliably heralds the development of distinct clinical lesions 12 to 24 h later. Almost all DH patients have an associated, usually subclinical, gluten-sensitive enteropathy ([Chap. 286](#)), and more than 90% express the HLA-B8/DRw3 and HLA-DQw2 haplotypes. DH may present at any age, including childhood; onset in the second to fourth decades is most common. The disease is typically chronic.

Biopsy of early lesional skin reveals neutrophil-rich infiltrates within dermal papillae. Neutrophils, fibrin, edema, and microvesicle formation at these sites are characteristic of early disease. Older lesions may demonstrate nonspecific features of a subepidermal bulla or an excoriated papule. Because the clinical and histologic features of this disease can be variable and resemble other subepidermal blistering disorders, the diagnosis is confirmed by direct immunofluorescence microscopy of normal-appearing perilesional skin. Such studies demonstrate granular deposits of IgA (with or without complement components) in the papillary dermis and along the epidermal basement membrane zone. IgA deposits in the skin are unaffected by control of disease with medication; however, these immunoreactants may diminish in intensity or disappear in patients maintained for long periods on a strict gluten-free diet (see below). Patients with granular deposits of IgA in their epidermal basement membrane zone typically do not have circulating IgA anti-basement membrane autoantibodies and should be distinguished from individuals with linear IgA deposits at this site (see below).

Although most [DH](#) patients do not report overt gastrointestinal symptoms or laboratory evidence of malabsorption, biopsies of small bowel usually reveal blunting of intestinal villi and a lymphocytic infiltrate in the lamina propria. As is true for patients with celiac disease, this gastrointestinal abnormality can be reversed by a gluten-free diet. Moreover, if maintained, this diet alone may control the skin disease and eventuate in clearance of IgA deposits from these patients' epidermal basement membrane zone. Subsequent gluten exposure in such patients alters the morphology of their small bowel, elicits a flare-up of their skin disease, and is associated with the reappearance of IgA in their epidermal basement membrane zone. Additional evidence that DH develops as a consequence of dietary gluten exposure is the demonstration of IgA anti-endomysial antibodies in these patients' sera (as found in the sera of patients with ordinary gluten-sensitive enteropathy). Recent studies have shown that such autoantibodies are directed against tissue transglutaminase. Patients with DH also have an increased incidence of thyroid abnormalities, achlorhydria, atrophic gastritis, and antigastric parietal cell antibodies. These associations likely relate to the high frequency of the HLA-B8/DRw3 haplotype in these patients, since this marker is commonly linked to autoimmune disorders. The mainstay of treatment of DH is dapsone, a sulfone. Patients respond rapidly (24 to 48 h) to dapsone but require careful pretreatment evaluation and close follow-up to ensure that complications are avoided or controlled. All patients on more than 100 mg/d dapsone will have some hemolysis and methemoglobinemia. These are expected pharmacologic side effects of this agent. Gluten restriction can control DH and lessen dapsone requirements; this diet must rigidly exclude gluten to be of maximal benefit. Many months of dietary restriction may be necessary before a beneficial result is achieved. Good dietary counselling by a trained dietitian is essential.

LINEAR IGA DISEASE

Linear IgA disease, once considered a variant form of dermatitis herpetiformis, is actually a separate and distinct entity. Clinically, these patients may resemble patients with typical cases of [DH](#), [BP](#), or other subepidermal blistering diseases. Lesions typically consist of papulovesicles, bullae, and/or urticarial plaques, predominantly on extensor (as seen in "classic" DH), central, or flexural sites. Oral mucosal involvement occurs in some patients. Severe pruritus resembles that in patients with DH. Patients with linear IgA disease do not have an increased frequency of the HLA-B8/DRw3 haplotype or an associated enteropathy and hence are not candidates for a gluten-free diet.

The histologic alterations in early lesions may be virtually indistinguishable from those in [DH](#). However, direct immunofluorescence microscopy of normal-appearing perilesional skin reveals linear deposits of IgA (and often C3) in the epidermal basement membrane zone. Most patients with linear IgA disease demonstrate circulating IgA anti-basement membrane autoantibodies against epitopes in the extracellular domain of [BPAG2](#), a transmembrane protein found in hemidesmosomes of basal keratinocytes. These patients generally respond to treatment with dapsone, 50 to 150 mg/d.

EPIDERMOLYSIS BULLOSA ACQUISITA

[EBA](#) is a rare, noninherited, polymorphic, subepidermal blistering disease. (The inherited form is discussed in [Chap. 351](#).) Patients with classic or noninflammatory EBA have blisters on noninflamed skin, atrophic scars, milia, nail dystrophy, and oral lesions. Because lesions generally occur at sites exposed to minor trauma, classic EBA is considered to be a mechanobullous disease. Other patients with EBA have widespread inflammatory, scarring, bullous lesions and oromucosal involvement that resembles severe [BP](#). Some patients present with an inflammatory bullous disease that evolves into the classic noninflammatory form of this disorder. In general, EBA is chronic; associations with multiple myeloma, amyloidosis, inflammatory bowel disease, and diabetes mellitus have been reported. The HLA-DR2 haplotype is found with increased frequency in these patients.

The histology of lesional skin varies depending on the character of the lesion being studied. Noninflammatory bullae show subepidermal blisters with a sparse leukocytic infiltrate and resemble those in patients with porphyria cutanea tarda. Inflammatory lesions consist of a subepidermal blister and neutrophil-rich leukocytic infiltrates in the superficial dermis. [EBA](#) patients have continuous deposits of IgG (and frequently C3 as well as other complement components) in a linear pattern within the epidermal basement membrane zone. Ultrastructurally, these immunoreactants are found in the sublamina densa region in association with anchoring fibrils, wheat stack-like structures that extend from the lamina densa into the underlying papillary dermis. Approximately 25 to 50% of EBA patients have circulating IgG anti-basement membrane autoantibodies directed against type VII collagen -- the collagen species that comprises anchoring fibrils. Such IgG autoantibodies bind the dermal side of 1 M NaCl split skin (in contrast to IgG autoantibodies in patients with [BP](#) that bind either epidermal or both sides of this indirect immunofluorescence microscopy test substrate).

Treatment of [EBA](#) is generally unsatisfactory. Some patients with inflammatory EBA may respond to systemic glucocorticoids, either alone or in combination with immunosuppressive agents. Other patients (especially those with neutrophil-rich

inflammatory lesions) may respond to dapsone. The chronic, noninflammatory form of this disease is largely resistant to treatment, although some patients may respond to cyclosporine.

CICATRICAL PEMPHIGOID

Cicatricial pemphigoid (CP) is a rare, acquired, subepithelial blistering disease characterized by erosive lesions of mucous membranes and skin that result in scarring of at least some sites of involvement. Immunopathologically, perilesional mucosa and skin of patients with CP demonstrate in situ deposits of immunoreactants in epithelial basement membranes. Common sites of involvement include the oral mucosa (especially the gingiva) and conjunctiva; other sites that may be affected include the nasopharyngeal, laryngeal, esophageal, urogenital, and rectal mucosa. Skin lesions (present in about one-third of patients) tend to predominate on the scalp, face, and upper trunk and generally consist of a few scattered erosions or tense blisters on an erythematous or urticarial base. CP is typically a chronic and progressive disorder. Serious complications may arise as a consequence of ocular, laryngeal, esophageal, or urogenital lesions. Erosive conjunctivitis may result in shortened fornices, symblephara, ankyloblepharon, entropion, corneal opacities, and (in severe cases) blindness. Similarly, erosive lesions of the larynx may cause hoarseness, pain, and tissue loss that if unrecognized and untreated may eventuate in complete destruction of the airway. Esophageal lesions may result in stenosis and/or strictures that may place patients at risk for aspiration. Strictures may also complicate urogenital involvement.

Biopsies of lesional tissue generally demonstrate subepithelial vesiculobullae and a mononuclear leukocytic infiltrate. Neutrophils and eosinophils may be seen in biopsies of early lesions; older lesions may demonstrate a scant leukocytic infiltrate and fibrosis. Direct immunofluorescence microscopy of perilesional tissue typically demonstrates deposits of IgG, IgA, and/or C3 in these patients' epithelial basement membranes. Because many of these patients show no evidence of circulating anti-basement membrane autoantibodies, testing of perilesional skin is important diagnostically. Although [CP](#) was once thought to be a single nosologic entity, it is now largely regarded as a disease phenotype that may develop as a consequence of an autoimmune reaction against a variety of different molecules in epithelial basement membranes (e.g., [BPAG2](#), laminin 5, type VII collagen, and other antigens yet to be completely defined). Treatment of CP is largely dependent upon sites of involvement. Due to potentially severe complications, ocular, laryngeal, esophageal, and/or urogenital involvement require aggressive systemic treatment with dapsone, prednisone, or the latter in combination with another immunosuppressive agent (e.g., azathioprine or cyclophosphamide). Less threatening forms of the disease may be managed with topical or intralesional glucocorticoids.

AUTOIMMUNE SYSTEMIC DISEASES WITH PROMINENT CUTANEOUS FEATURES

DERMATOMYOSITIS

The cutaneous manifestations of dermatomyositis ([Chap. 382](#)) are often distinctive but at times may resemble those of systemic lupus erythematosus (SLE) ([Chap. 311](#)), scleroderma ([Chap. 313](#)), or other overlapping connective tissue diseases ([Chap. 313](#)).

The extent and severity of cutaneous disease may or may not correlate with the extent and severity of the myositis. Patients with severe muscle involvement may have relatively minor skin changes, whereas patients with marked skin involvement may have mild muscle disease. The cutaneous manifestations of dermatomyositis are similar whether the disease appears in childhood or old age, except that calcification of subcutaneous tissue is a common late sequela in childhood dermatomyositis.

The cutaneous signs of dermatomyositis may precede or follow the development of myositis by weeks to years. Cases lacking muscle involvement (i.e., dermatomyositis sine myositis) have also been reported. The most common manifestation is a purple-red discoloration of the upper eyelids, sometimes associated with scaling ("heliotrope" erythema; [Plate IIE-63](#)) and periorbital edema. Erythema on the cheeks and nose in a "butterfly" distribution may resemble the eruption in [SLE](#). Erythematous or violaceous scaling patches are common on the upper anterior chest, posterior neck, scalp, and the extensor surfaces of the arms, legs, and hands. Erythema and scaling may be particularly prominent over the elbows, knees, and the dorsal interphalangeal joints. Approximately one-third of patients have violaceous, flat-topped papules over the dorsal interphalangeal joints that are pathognomonic of dermatomyositis (Gottron's sign or Gottron's papules; [Plate IIE-65](#)). These lesions can be contrasted with the erythema and scaling on the dorsum of the fingers in some patients with SLE, which spares the skin over the interphalangeal joints. Periungual telangiectasia may be prominent, and a lacy or reticulated erythema may be associated with fine scaling on the extensor surfaces of the thighs and upper arms. Other patients, particularly those with long-standing disease, develop areas of hypopigmentation, hyperpigmentation, mild atrophy, and telangiectasia known as *poikiloderma vasculare atrophicans*. Poikiloderma is rare in both SLE and scleroderma and thus can serve as a clinical sign that distinguishes dermatomyositis from these two diseases. Cutaneous changes may be similar in scleroderma and dermatomyositis and may include thickening and binding down of the skin of the hands (sclerodactyly) as well as Raynaud's phenomenon. However, the presence of severe muscle disease, Gottron's papules, heliotrope erythema, and poikiloderma serve to distinguish patients with dermatomyositis. Skin biopsy of erythematous, scaling lesions of dermatomyositis may reveal only mild nonspecific inflammation but sometimes may show changes indistinguishable from those found in SLE, including epidermal atrophy, hydropic degeneration of basal keratinocytes, edema of the upper dermis, and a mild mononuclear cell infiltrate. Direct immunofluorescence microscopy of lesional skin is usually negative, although granular deposits of immunoglobulin(s) and complement in the epidermal basement membrane zone have been described in some patients. Treatment should be directed at the systemic disease. In the few instances where adjunctive cutaneous therapy is desirable, topical glucocorticoids are sometimes useful. These patients should avoid exposure to ultraviolet irradiation and use photoprotective measures such as sunscreens.

LUPUS ERYTHEMATOSUS

The cutaneous manifestations of lupus erythematosus (LE) ([Chap. 311](#)) can be divided into acute, subacute, and chronic (i.e., discoid LE) types. *Acute cutaneous LE* is characterized by erythema of the nose and malar eminences in a "butterfly" distribution ([Plate IIE-61](#)). The erythema is often sudden in onset, accompanied by edema and fine scale, and correlated with systemic involvement. Patients may have widespread

involvement of the face as well as erythema and scaling of the extensor surfaces of the extremities and upper chest. These acute lesions, while sometimes evanescent, usually last for days and are often associated with exacerbations of systemic disease. Skin biopsy of acute lesions may show only a sparse dermal infiltrate of mononuclear cells and dermal edema. In some instances, cellular infiltrates around blood vessels and hair follicles are notable, as is hydropic degeneration of basal cells of the epidermis. Direct immunofluorescence microscopy of lesional skin frequently reveals deposits of immunoglobulin(s) and complement in the epidermal basement membrane zone. Treatment is aimed at control of systemic disease; photoprotection in this, as well as in other forms of LE, is very important.

Subacute cutaneous lupus erythematosus (SCLE) is characterized by a widespread photosensitive, nonscarring eruption. About half of these patients have [SLE](#) in which severe renal and central nervous system involvement is uncommon. SCLE may present as a papulosquamous eruption that resembles psoriasis or annular lesions that resemble those seen in erythema multiforme. In the papulosquamous form, discrete erythematous papules arise on the back, chest, shoulders, extensor surfaces of the arms, and the dorsum of the hands; lesions are uncommon on the face, flexor surfaces of the arms, and below the waist. The slightly scaling papules tend to merge into large plaques, some with a reticulate appearance. The annular form involves the same areas and presents with erythematous papules that evolve into oval, circular, or polycyclic lesions. The lesions of SCLE are more widespread but have less tendency for scarring than do lesions of discoid [LE](#). Skin biopsy reveals a dense mononuclear cell infiltrate around hair follicles and blood vessels in the superficial dermis, combined with hydropic degeneration of basal cells in the epidermis. Direct immunofluorescence microscopy of lesional skin reveals deposits of immunoglobulin(s) in the epidermal basement membrane zone in about half these cases. A particulate pattern of IgG deposition around basal keratinocytes has recently been associated with SCLE. Most SCLE patients have anti-Ro antibodies. Local therapy is usually unsuccessful, and most patients require treatment with aminoquinoline antimalarials. Low-dose therapy with oral glucocorticoids is sometimes necessary; photoprotective measures against both ultraviolet B and A wavelengths are very important.

Discoid lupus erythematosus (DLE) is characterized by discrete lesions, most often on the face, scalp, or external ears. The lesions are erythematous papules or plaques with a thick, adherent scale that occludes hair follicles (follicular plugging). When the scale is removed, its underside will show small excrescences that correlate with the openings of hair follicles and is termed a "carpet tack" appearance. This finding is relatively specific for DLE. Long-standing lesions develop central atrophy, scarring, and hypopigmentation but frequently have erythematous, sometimes raised borders at the periphery ([Plate IIE-62](#)). These lesions persist for years and tend to expand slowly. Only 5 to 10% of patients with DLE meet the American Rheumatism Association criteria for [SLE](#). However, typical discoid lesions are frequently seen in patients with SLE. Biopsy of DLE lesions shows hyperkeratosis, follicular plugging, and atrophy of the epidermis. The dermal-epidermal junction reveals hydropic degeneration of basal keratinocytes, and a mononuclear cell infiltrate surrounding hair follicles and blood vessels. Direct immunofluorescence microscopy demonstrates immunoglobulin(s) and complement deposits at the basement membrane zone in about 90% of cases. Treatment is focused on control of local cutaneous disease and consists mainly of photoprotection and topical

or intralesional glucocorticoids. If local therapy is ineffective, use of aminoquinoline antimalarials may be indicated.

SCLERODERMA AND MORPHEA

The skin changes of scleroderma ([Chap. 313](#)) usually begin on the hands, feet, and face, with episodes of recurrent nonpitting edema. Sclerosis of the skin begins distally on the fingers (sclerodactyly) and spreads proximally, usually accompanied by resorption of bone of the fingertips, which may have punched out ulcers, stellate scars, or areas of hemorrhage ([Plate IIE-66](#)). The fingers may actually shrink in size and become sausage-shaped, and since the fingernails are usually unaffected, the nails may curve over the end of the fingertips. Periungual telangiectasias are usually present, but periungual erythema is rare. In advanced cases, the extremities show contractures and calcinosis cutis. Face involvement includes a smooth, unwrinkled brow, taut skin over the nose, shrinkage of tissue around the mouth, and perioral radial furrowing ([Plate IIE-64](#)). Matlike telangiectasias are often present, particularly on the face and hands. Involved skin feels indurated, smooth, and bound to underlying structures; hyperpigmentation and hypopigmentation are also often present. Raynaud's phenomenon, i.e., cold-induced blanching, cyanosis, and reactive hyperemia, is present in almost all patients and can precede development of scleroderma by many years. The combination of calcinosis cutis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia has been termed the *CREST syndrome*. Anticentromere antibodies have been reported in a very high percentage of patients with the CREST syndrome but in only a small minority of patients with scleroderma. Skin biopsy reveals thickening of the dermis and homogenization of collagen bundles. Direct immunofluorescence microscopy of lesional skin is usually negative.

Morphea, which has been called *localized scleroderma*, is characterized by localized thickening and sclerosis of skin, usually affecting young adults or children. Morphea begins as erythematous or flesh-colored plaques that become sclerotic, develop central hypopigmentation, and demonstrate an erythematous border. In most cases, patients have one or a few lesions, and the disease is termed *localized morphea*. In some patients, widespread cutaneous lesions may occur, without systemic involvement. This form is called *generalized morphea*. Most patients with morphea do not have autoantibodies. Skin biopsy of morphea is indistinguishable from that of scleroderma. Linear scleroderma is a limited form of disease that presents in a linear, bandlike distribution and tends to involve deep as well as superficial layers of skin. Scleroderma and morphea are usually quite resistant to therapy. For this reason, physical therapy to prevent joint contractures and to maintain function is employed and is often helpful.

Diffuse fasciitis with eosinophilia is a clinical entity that can sometimes be confused with scleroderma. There is usually the sudden onset of swelling, induration, and erythema of the extremities frequently following significant physical exertion. The proximal portions of extremities (arms, forearms, thighs, legs) are more often involved than are the hands and feet. While the skin is indurated, it is usually not bound down as in scleroderma; contractures may occur early secondary to fascial involvement. The latter may also cause muscle groups to be separated (i.e., the "groove sign") and veins to appear depressed (i.e., sunken veins). These skin findings are accompanied by peripheral blood eosinophilia, increased erythrocyte sedimentation rate, and sometimes

hypergammaglobulinemia. Deep biopsy of affected areas of skin reveals inflammation and thickening of the deep fascia overlying muscle. An inflammatory infiltrate composed of eosinophils and mononuclear cells is usually found. Patients with eosinophilic fasciitis appear to be at increased risk to develop bone marrow failure or other hematologic abnormalities. While the ultimate course of eosinophilic fasciitis is uncertain, many patients respond favorably to treatment with prednisone in doses ranging from 40 to 60 mg/d.

The *eosinophilia-myalgia syndrome*, a disorder reported in epidemic numbers in 1989 and linked to ingestion of L-tryptophan manufactured by a single company in Japan, is a multisystem disorder characterized by debilitating myalgias and absolute eosinophilia in association with varying combinations of arthralgias, pulmonary symptoms, and peripheral edema. In a later phase (i.e., 3 to 6 months after initial symptoms), these patients often develop localized sclerodermatous skin changes, weight loss, and/or neuropathy ([Chap. 313](#)). The precise cause of this syndrome, which may resemble other sclerotic skin conditions, is unknown. However, the implicated lots of L-tryptophan contained the contaminant 1,1-ethylidene bis[tryptophan]. This contaminant may be pathogenic or a marker for another substance that provokes the disorder.

(Bibliography omitted in Palm version)

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59. CUTANEOUS DRUG REACTIONS - Robert S. Stern, Olivier M. Chosidow, Bruce U. Wintroub

Cutaneous reactions are among the most frequent adverse reactions to drugs. Prompt recognition of these reactions, drug withdrawal, and appropriate therapeutic interventions can minimize toxicity. This chapter focuses on adverse cutaneous reactions to drugs other than topical agents and reviews the incidence, patterns, and pathogenesis of cutaneous reactions to drugs and other therapeutic agents.

USE OF PRESCRIPTION DRUGS IN THE UNITED STATES

More than 1.5 billion prescriptions for 60,000 drug products, which include over 2000 different active agents, are dispensed each year in the United States. Hospital inpatients alone annually receive about 120 million courses of drug therapy, and half of adult Americans receive prescription drugs on a regular outpatient basis. Many additional patients use over-the-counter medicines that may cause adverse cutaneous reactions.

INCIDENCE OF CUTANEOUS REACTIONS

Although adverse drug reactions are common, it is difficult to ascertain their incidence, seriousness, and ultimate health effects. Available information comes from evaluations of hospitalized patients, epidemiologic surveys, premarketing studies, and voluntary reporting, most notably to the U.S. Food and Drug Administration's Medwatch System. None of these efforts provides comprehensive comparable data on the risk of cutaneous reactions associated with most medicines.

In one study about 2% of medical inpatients had skin reactions consisting of rash, urticaria, or pruritus during hospitalization. The overall reaction rate per course of drug therapy was about 3:1000. Among inpatients, penicillins, sulfonamides, and blood products accounted for two-thirds of cutaneous reactions. Among outpatients, reaction rates for many antibiotics were comparable to those observed in inpatients. Fluoroquinolones are notable causes of cutaneous reactions not observed in earlier studies. Reaction rates for selected commonly used antibiotics are summarized in [Table 59-1](#). Most cutaneous reactions occur within 2 weeks of exposure to a drug. The risk of allergic reactions does not vary greatly with age or sex. Among outpatients, the risk of a reaction to an antibiotic was comparable for first and subsequent courses of a given drug.

The distribution of morphologic patterns of drug eruptions cared for within a Finnish hospital dermatology department with a special interest in fixed drug eruptions included exanthematous reactions (32%), urticaria and/or angioedema (20%), fixed drug eruptions (34%), erythema multiforme (2%), Stevens-Johnson syndrome (SJS; 1%), exfoliative dermatitis (1%), and photosensitivity reactions (3%). Other studies suggest that about 80% of all cutaneous reactions are morbilliform or erythematous, 10 to 15% are urticaria or angioedema, and all other types of reactions are relatively rare.

The relative risk of [SJS](#) and toxic epidermal necrolysis (TEN), perhaps the most important severe cutaneous reactions, has been quantified in an international case control study and case series. Sulfonamide antibiotics, allopurinol, amine antiepileptic

drugs (phenytoin and carbamazepine), and lamotrigine (a new antiepileptic) are associated with the highest risk of these reactions.

PATHOGENESIS OF DRUG REACTIONS

Untoward cutaneous responses to drugs can arise as a result of immunologic or nonimmunologic mechanisms. Immunologic reactions require activation of host immunologic pathways and are designated *drug allergy*. Drug reactions occurring through nonimmunologic mechanisms may be due to activation of effector pathways, overdosage, cumulative toxicity, side effects, ecologic disturbance, interactions between drugs, metabolic alterations, exacerbation of preexisting dermatologic conditions, or inherited protein or enzyme deficiencies. It is often not possible to specify the responsible drug or pathogenic mechanism because the skin responds to a variety of stimuli through a limited number of reaction patterns. The mechanism of many drug reactions is unknown.

IMMUNOLOGIC DRUG REACTIONS

Drugs frequently elicit an immune response, but only a small number of individuals experience clinical hypersensitivity reactions. For example, most patients exposed to penicillin develop demonstrable antibodies to penicillin but do not manifest drug reactions when exposed to penicillin. Multiple factors determine the capacity of a drug to elicit an immune response, including the molecular characteristics of the drug and host effects.

Increases in *molecular* size and complexity are associated with increased immunogenicity, and macromolecular drugs such as protein or peptide hormones are highly antigenic. Most drugs are small organic molecules <1000 Da in size, and the capacity of such small molecules to elicit an immune response depends on their ability to act as haptens, i.e., to form stable, usually covalent, bonds with tissue macromolecules, an extremely rare event.

Route of administration of a drug or simple chemical can influence the nature of the *host* immune response. For example, topical application of antigens tends to induce delayed hypersensitivity, and exposure to antigens via oral or nasal cavities stimulates production of secretory immunoglobins, IgA and IgE, and occasionally IgM. Frequency of sensitization through intravenous administration of drugs varies, but anaphylaxis is a more likely consequence with this route of exposure than following oral administration.

The degree of drug exposure and individual variability in absorption and metabolism of a given agent may alter immunogenic load. The variable degree of *in vivo* acetylation of hydralazine provides a clinical example of this phenomenon. Hydralazine produces a lupus-like syndrome associated with antinuclear antibody formation more frequently in patients who acetylate the drug slowly. Frequent high-dose and interrupted courses of therapy are also important risk factors for development of drug allergy.

Pathogenesis of Allergic Drug Reactions> >

> >*IgE-Dependent Reactions* IgE-dependent drug reactions are usually manifest in the

skin and gastrointestinal, respiratory, and cardiovascular systems ([Chap. 310](#)). Primary symptoms and signs include pruritus, urticaria, nausea, vomiting, cramps, bronchospasm, and laryngeal edema and, on occasion, anaphylactic shock with hypotension and death. Immediate reactions may occur within minutes of drug exposure, and accelerated reactions occur hours or days after drug administration. Accelerated reactions are usually urticarial and may include laryngeal edema. Penicillin and related drugs are the most frequent causes of IgE-dependent reactions. Release of chemical mediators such as histamine, adenosine, leukotrienes, prostaglandins, platelet-activating factor, enzymes, and proteoglycans from sensitized tissue, mast cells, or circulating basophilic leukocytes results in vasodilation and edema. Release is triggered when polyvalent drug protein conjugates cross-link IgE molecules fixed to sensitized cells. The clinical manifestations are determined by interaction of the released chemical mediator with its target organ, i.e., skin, respiratory, gastrointestinal, and/or cardiovascular systems. Certain routes of administration favor different clinical patterns (i.e., oral route: gastrointestinal effects; intravenous route: circulatory effects).

Immune-Complex-Dependent Reactions Serum sickness is produced by circulating immune complexes and is characterized by fever, arthritis, nephritis, neuritis, edema, and an urticarial, papular, or purpuric rash ([Chap. 317](#)). The syndrome requires an antigen that remains in the circulation for prolonged periods so that when antibody is synthesized, circulating antigen-antibody complexes are formed. Serum sickness was first described following administration of foreign sera, but drugs are now the usual cause. Drugs that produce serum sickness include the penicillins, sulfonamides, thiouracils, cholecystographic dyes, phenytoin, aminosalicic acid, heparin, and antilymphocyte globulin. Cephalosporin administration in febrile children is associated with a high risk of a clinically similar reaction, but the mechanism of this reaction is unknown. In classic serum sickness, symptoms develop 6 days or more after exposure to a drug, the latent period representing the time needed to synthesize antibody. The antibodies responsible for immune-complex-dependent drug reactions are largely of the IgG or IgM class. Vasculitis, a relatively rare cutaneous complication of drugs, may also be a result of immune complex deposition ([Chap. 317](#)).

Cytotoxicity and Delayed Hypersensitivity Cytotoxicity and delayed hypersensitivity mechanisms may be important in the etiology of morbilliform exanthema, hypersensitivity syndrome, [SJS](#), or [TEN](#), but this is not proven. Systemic manifestations occur frequently. The nature of the antigen leading to cytotoxic reactions is unknown, but it is likely that different T lymphocyte populations are activated. T_H1 type cells will lead to the production of interleukin (IL)-2 and interferon (IFN)- γ and subsequent activation of cytotoxic T cells. In early lesions of morbilliform exanthema or TEN, histopathologic studies have shown expression of HLA-DR and intercellular adhesion molecule (ICAM-1) by keratinocytes, CD4 cells (in the dermis), and CD8 T cells (in the epidermis) and apoptosis of keratinocytes (facilitated by tumor necrosis factor α secretion and *fas*-ligand expression). T_H2 type cells produce cytokines such as IL-5, which may be involved in hypersensitivity syndrome (see below).

NONIMMUNOLOGIC DRUG REACTIONS

Nonimmunologic mechanisms are responsible for the majority of drug reactions; however, only the most important mechanisms will be discussed.

Nonimmunologic Activation of Effector Pathways Drug reactions may result from nonimmunologic activation of effector pathways by three mechanisms: First, drugs may release mediators directly from mast cells and basophils and present as anaphylaxis, urticaria, and/or angioedema. Urticarial anaphylactic reactions induced by opiates, polymyxin B, tubocurarine, radiocontrast media, and dextrans may occur by this mechanism. Second, drugs may activate complement in the absence of antibody. This is an additional mechanism through which radiocontrast media may act. Third, drugs such as aspirin and other nonsteroidal anti-inflammatory agents (NSAIDs) may alter pathways of arachidonic acid metabolism and induce urticaria.

Phototoxicity Phototoxic reactions may be drug-induced or may occur in metabolic disorders in which a photosensitizing chemical is overproduced. A phototoxic reaction occurs when enough chromophore (drug or metabolic product) absorbs sufficient radiation to cause a reaction or interaction with target tissue. Drug-induced phototoxic reactions can occur on first exposure. The incidence of phototoxicity is a direct function of the concentration of sensitizer and the amount of light of the appropriate wavelengths. At least three distinct photochemical mechanisms have been described: (1) the reaction between the excited state of a phototoxic molecule and a biologic target may cause formation of a covalent photoaddition product, (2) the phototoxic molecule may form stable photoproducts that are toxic to biologic substrates, and (3) radiation of a phototoxic molecule may result in transfer of energy to oxygen molecules and cause formation of toxic oxygen species, such as singlet oxygen superoxide anion, or hydroxyl radicals. Interaction of these reactive species with biologic targets produces photooxidized molecules. Phototoxic injury is usually manifest as a sunburn-like reaction.

Exacerbation of Preexisting Diseases A variety of agents can exacerbate preexisting diseases. For example, lithium can exacerbate acne and psoriasis in a dose-dependent manner. Beta-blocking agents and [IFN- \$\alpha\$](#) may induce psoriasis. Withdrawal of glucocorticoids can exacerbate psoriasis or atopic dermatitis.

Inherited Enzyme or Protein Deficiencies Specific genetically determined defects in the ability of an individual to detoxify toxic reactive drug metabolites may predispose such individuals to the development of severe drug reactions, especially hypersensitivity syndrome, and perhaps [TEN](#) associated with use of sulfonamides and anticonvulsants.

Alterations of Immunologic Status Alterations in patients' immunologic status may also modify the risk of cutaneous reactions. Bone marrow transplant patients, HIV-infected persons, and persons with Epstein-Barr virus infection are at higher risk of developing cutaneous reactions to drugs. Skin reactions to trimethoprim-sulfamethoxazole are seen in about a third of HIV-infected users of this drug, but desensitization can be accomplished. Dapsone, trimethoprim alone, and amoxicillin-clavulanate are also frequent causes of drug eruptions in HIV-infected patients. The advent of highly active antiretroviral therapy (HAART) may have decreased the risk of cutaneous reactions in HIV patients ([Chap. 309](#)).

[A CLINICAL CLASSIFICATION OF CUTANEOUS DRUG REACTIONS](#)

URTICARIA/ANGIOEDEMA

Urticaria is a skin reaction characterized by pruritic, red wheals. Lesions may vary from a small point to a large area. Individual lesions rarely last more than 24 h. When deep dermal and subcutaneous tissues are also swollen, this reaction is known as *angioedema*. Angioedema may involve mucous membranes and may be part of a life-threatening anaphylactic reaction. Urticarial lesions, along with pruritus and morbilliform (or maculopapular) eruptions, are among the most frequent types of cutaneous reactions to drugs.

Drug-induced urticaria may be caused by three mechanisms: an IgE-dependent mechanism, circulating immune complexes (serum sickness), and nonimmunologic activation of effector pathways. IgE-dependent urticarial reactions usually occur within 36 h but can occur within minutes. Reactions occurring within minutes to hours of drug exposure are termed *immediate reactions*, whereas those that occur 12 to 36 h after drug exposure are designated *accelerated reactions*. Immune-complex-induced urticaria associated with serum sickness usually occurs from 6 to 12 days after first exposure. In this syndrome, the urticarial eruption may be accompanied by fever, hematuria, arthralgias, hepatic dysfunction, and neurologic symptoms.

Certain drugs, such as [NSAIDs](#), angiotensin-converting enzyme (ACE) inhibitors, and radiographic dyes, may induce urticarial reactions, angioedema, and anaphylaxis in the absence of drug-specific antibody. Although ACE inhibitors, aspirin, penicillin, and blood products are the most frequent causes of urticarial eruptions, urticaria has been observed in association with nearly all drugs. Drugs also may cause chronic urticaria, which lasts more than 6 weeks. Aspirin frequently exacerbates this problem.

The treatment of urticaria or angioedema depends on the severity of the reaction and the rate at which it is evolving. In severe cases, especially with respiratory or cardiovascular compromise, epinephrine is the mainstay of therapy, but its effect is reduced in patients using beta blockers. For more seriously affected patients, treatment with systemic glucocorticoids, sometimes intravenously administered, are helpful. In addition to drug withdrawal, for patients with only cutaneous symptoms and without symptoms of angioedema or anaphylaxis, oral antihistamines are usually sufficient.

PHOTOSENSITIVITY ERUPTIONS

Photosensitivity eruptions are usually most marked in sun-exposed areas but may extend to sun-protected areas. Phototoxic reactions are more common with some drugs. Photoallergic reactions to systemically administered drugs are very rare. Phototoxic reactions usually resemble sunburn and can occur with the first exposure to a drug. Their severity depends on the tissue level of the drug, the extent of exposure to light, and the efficiency of the photosensitizer ([Chap. 60](#)).

Orally administered phototoxic drugs include many fluoroquinolones, chlorpromazine, tetracycline, thiazides, and at least two [NSAIDs](#) (benoxaprofen and piroxicam). The majority of the common phototoxic drugs have action spectrums in the long-wave ultraviolet A (UV-A) range. Phototoxic reactions abate with removal of either the drug or ultraviolet radiation. Because UV-A and visible light, which trigger these reactions, are

not easily absorbed by nonopaque sunscreens and are transmitted through window glass, these reactions may be difficult to block.

Photosensitivity reactions are treated by avoiding exposure to ultraviolet light (sunlight) and treating the reaction as one would a sunburn. Rarely, individuals develop persistent reactivity to light, necessitating long-term avoidance of sun exposure.

PIGMENTATION CHANGES

Drugs may cause a variety of pigmentary changes in the skin. Some drugs stimulate melanocytic activity and increase pigmentation. Drug deposition can also lead to pigmentation; this phenomenon occurs with heavy metals. Phenothiazines may be deposited in the skin and cause a slate-gray color. Antimalarial drugs may cause a slate-gray or yellow pigmentation. Long term minocycline use may cause slate-gray hyperpigmentation, especially in areas of chronic inflammation. Inorganic arsenic, once used to treat psoriasis, is associated with diffuse macular pigmentation. Other heavy metals that cause pigmentary changes include silver, gold, bismuth, and mercury. Long-term use of phenytoin can produce a chloasma-like pigmentation in women. Certain cytostatic agents can also cause pigmentary changes. Histologic examination is often diagnostic for drug deposition diseases.

Zidovudine (AZT) is a frequent cause of pigmentation, especially of the nails ([Chap. 309](#)). Nicotinic acid in large doses may cause brown pigmentation, and oral contraceptives may produce chloasma. In addition, amiodarone may cause violaceous hyperpigmentation that is increased in sun-exposed skin. Drugs such as heavy metals, copper antimalarial and arsenical agents, and ACTH also may discolor oral mucosa.

VASCULITIS

Cutaneous necrotizing vasculitis often presents as palpable purpuric lesions that may be generalized or limited to the lower extremities or other dependent areas ([Chap. 317](#)). Urticarial lesions, ulcers, and hemorrhagic blisters also occur. Vasculitis may involve other organs, including the liver, kidney, brain, and joints. Drugs are only one cause of vasculitis, with infection and collagen vascular disease responsible for the majority of cases.

Propylthiouracil induces a cutaneous vasculitis that is accompanied by leukopenia and splenomegaly. Direct immunofluorescent changes in these lesions suggest immune-complex deposition. Drugs implicated in vasculitic eruptions include allopurinol, thiazides, sulfonamides, penicillin, and some [NSAIDs](#).

HYPERSENSITIVITY SYNDROME

Initially described with phenytoin, hypersensitivity syndrome presents as an erythematous eruption that may become purpuric and is accompanied by many of the following features: fever, facial and periorbital edema, tender generalized lymphadenopathy, leukocytosis (often with atypical lymphocytes and eosinophils), hepatitis, and sometimes nephritis or pneumonitis. The cutaneous reaction usually begins 1 to 6 weeks after phenytoin is begun and usually resolves with drug cessation,

but symptoms, especially hepatitis, may persist. The eruption recurs with rechallenge, and cross-reactions among aromatic anticonvulsants, including phenytoin, carbamazepine, and barbiturates, are frequent. With phenytoin, an increased risk of this syndrome is associated with an inherited deficiency of epoxide hydrolase, an enzyme required for metabolism of a toxic intermediate arene oxide that is formed during metabolism of phenytoin by the cytochrome P450 system. Other drugs causing this syndrome include lamotrigine, dapsone, allopurinol, sulfonamides, minocycline, and sulfones. Systemic glucocorticoids (prednisone, 0.5 to 1.0 mg/kg) seem to reduce symptoms. Mortality as high as 10% has been reported.

WARFARIN NECROSIS OF THE SKIN

This rare reaction occurs usually between the third and tenth days of therapy with warfarin derivatives, usually in women. Lesions are sharply demarcated, erythematous, indurated, and purpuric and may resolve or progress to form large, irregular, hemorrhagic bullae with eventual necrosis and slow-healing eschar formation.

Development of the syndrome is unrelated to drug dose or underlying condition. Favored sites are breasts, thighs, and buttocks. The course is not altered by discontinuation of the drug after onset of the eruption. Similar reactions have been associated with heparin. Warfarin reactions are associated with protein C deficiency. Protein C is a vitamin K-dependent protein with a shorter half-life than other clotting proteins and is in part responsible for control of fibrinolysis. Since warfarin inhibits synthesis of vitamin K-dependent coagulation factors, warfarin anticoagulation in heterozygotes for protein C deficiency causes a precipitous fall in circulating levels of protein C, permitting hypercoagulability and thrombosis in the cutaneous microvasculature, with consequent areas of necrosis. Heparin-induced necrosis may have clinically similar features but is probably due to heparin-induced platelet aggregation with subsequent occlusion of blood vessels.

Warfarin-induced cutaneous necrosis is treated with vitamin K and heparin. Vitamin K reverses the effects of warfarin, and heparin acts as an anticoagulant. Treatment with protein C concentrates may also be helpful in individuals with deficiencies of protein C, the predisposing factor for development of these reactions.

MORBILLIFORM REACTIONS

Morbilliform or maculopapular eruptions are the most common of all drug-induced reactions, often start on the trunk or areas of pressure or trauma, and consist of erythematous macules and papules that are frequently symmetric and may become confluent. Involvement of mucous membranes, palms, and soles is variable; the eruption may be associated with moderate to severe pruritus and fever.

The pathogenesis is unclear. A hypersensitivity mechanism has been suggested, although these reactions do not always recur following drug rechallenge. Diagnosis is rarely assisted by laboratory or patch testing; differentiation from viral exanthem is the principal differential diagnostic consideration. Unless the suspect drug is essential it should be discontinued. Occasionally these eruptions may decrease or fade with continued use of the responsible drug.

Morbilloform reactions usually develop within 1 week of initiation of therapy and last 1 to 2 weeks; however, reactions to some drugs, especially penicillin and drugs with long half-lives, may begin more than 2 weeks after therapy has begun and last as long as 2 weeks after therapy has ceased.

Morbilloform eruptions are usually treated by discontinuing the suspect medications symptomatically. Oral antihistamines, emollients, and soothing baths are useful for treatment of pruritus. Short courses of potent topical glucocorticoids can reduce inflammation and symptoms and are probably helpful. The beneficial effect of systemic glucocorticoids relative to risk is less clear.

FIXED DRUG REACTIONS

These reactions are characterized by one or more sharply demarcated, erythematous lesions in which hyperpigmentation results after resolution of the acute inflammation; with rechallenge, the lesion recurs in the same (i.e., "fixed") location. Lesions often involve the lips, hands, legs, face, genitalia, and oral mucosa and cause burning. Most patients have multiple lesions. Patch testing is useful to establish the etiology. Fixed drug eruptions have been associated with phenolphthalein, sulfonamides, tetracyclines, phenylbutazone, [NSAIDs](#), and barbiturates. Although cross-sensitivity appears to occur between different tetracycline compounds, cross-sensitivity was not elicited when different sulfonamide compounds were administered to patients as part of provocation testing.

LICHENOID DRUG ERUPTIONS

A lichenoid cutaneous reaction, clinically and morphologically indistinguishable from lichen planus, is associated with a variety of drugs and chemicals. Eosinophils are more common when the reaction is drug-induced. Gold and antimalarials are most often associated with this eruption. Antihypertensive agents, including beta blockers and captopril, have also been reported to cause lichenoid reactions.

BULLOUS ERUPTIONS

Blisters accompany a wide variety of cutaneous reactions, including fixed drug eruptions, severe morbilliform eruptions in dependent areas of the body, and phototoxic reactions. [SJS](#) and [TEN](#) are the most serious and important bullous reactions to drugs. Nalidixic acid and furosemide cause blistering eruptions indistinguishable from the primary bullous diseases. A pemphigus foliaceus-like eruption is seen with penicillamine.

PUSTULAR ERUPTIONS

Acute generalized exanthematous pustulosis is often associated with exposure to drugs, most notably antibiotics. Usually beginning on the face or intertriginous areas, small nonfollicular pustules overlying erythematous and edematous skin may coalesce and lead to superficial ulceration. Fever is present and differentiating this eruption from [TEN](#) in its initial stages may be difficult. Acute generalized exanthematous pustulosis often

begins within a few days of initiating drug treatment.

ERYTHEMA MULTIFORME

Erythema multiforme is an acute, self-limited inflammatory disorder of skin and mucous membranes characterized by distinctive iris or target lesions, usually acraly distributed and often associated with sore throat, mucosal lesions, and malaise. Classic erythema multiforme usually has nondrug causes, most commonly herpes simplex infection, and must be differentiated from true [SJS](#), which is usually drug related.

STEVENS-JOHNSON SYNDROME

[SJS](#) is a blistering disorder that is usually more severe than erythema multiforme. Initial presentation is often a sore throat, malaise, and fever. Within a few days, in addition to erosions of multiple mucous membranes, small blisters developing on dusky or purpuric macules or atypical target lesions characterize this eruption. Total percent of body surface area blistering and eventual detachment is less than 10%. Overlap [SJS/TEN](#) shares characteristics of both SJS and TEN, with 10 to 30% of body surface area exhibiting epidermal detachment.

TOXIC EPIDERMAL NECROLYSIS

[TEN](#) is the most serious cutaneous drug reaction and may be fatal. Drugs are usually the cause of TEN. Onset is generally acute and is characterized by fever $>39^{\circ}\text{C}$ (102.2°F), blisters or ulcers of multiple mucous membranes, malaise, and epidermal necrosis involving $>30\%$ of body surface area. Intestinal and pulmonary involvement is associated with a poor prognosis, as is a greater extent of epidermal detachment and older age. About 30% of affected persons die. Many treatments affecting immune response or cytokines (thalidomide) or apoptosis (intravenous immunoglobulin) have been advocated, but none have been shown to be efficacious in well-controlled trials. In spite of its theoretical potential benefits, thalidomide therapy increases TEN-associated mortality. Supportive treatment in burn units is helpful in reducing morbidity and mortality.

DRUGS OF SPECIAL INTEREST

PENICILLIN

The incidence of cutaneous reactions to penicillin is about 1%. About 85% of cutaneous reactions to penicillin are morbilliform, and about 10% are urticaria or angioedema.

IgG, IgM, and IgE antibodies can be produced; IgG and IgM anti-penicillin antibodies play a role in the development of hemolytic anemia, whereas anaphylaxis and serum sickness appear to be due to IgE antibodies in serum.

In patients with suspected IgE-mediated reactions to penicillin for whom future treatment is anticipated, accurate tests for sensitization are available. Current practice is to perform skin testing with a commercially available penicilloyl determinant preparation (Pre-pen, Kremers-Urban) and with fresh penicillin and, if possible, with another source

of minor (nonpenicilloyl) determinants such as aged or base-treated penicillin. Antibodies to minor determinants are common in patients experiencing anaphylaxis, but testing with major determinants alone detects most patients at risk for anaphylaxis.

About one-fourth of patients with positive history of penicillin allergy have a positive skin test, while 6% (3 to 10%) with no history of penicillin sensitivity demonstrate a positive skin response to penicillin. Administering penicillin to those patients with a positive skin test produces reactions in a high proportion (50 to 100%); conversely, only a few patients (0.5%) with a negative skin test react to the drug, and reactions tend to be mild and to occur late. Since a false-negative skin test may occur during or just after an acute reaction, testing should be performed either prospectively or several months after a suspected reaction. As many as 80% of patients lose anaphylactic sensitivity and IgE antibody after several years. Radioallergosorbent tests and other in vitro tests offer no advantage over properly performed skin testing. Some cross-reactivity between penicillin and nonpenicillin b-lactam antibiotics (e.g., cephalosporins) occurs, but the majority of penicillin-allergic patients will tolerate cephalosporins. Persons who have negative skin tests to penicillin rarely develop reactions to cephalosporins.

In the face of a positive clinical history of penicillin reaction, another drug should be chosen. If this is not feasible or prudent (e.g., in a pregnant patient with syphilis or with enterococcal endocarditis), skin testing with penicillin is warranted. If skin tests are negative, cautious administration of penicillin is acceptable, although some recommend desensitization of such patients if the reaction was likely to be IgE-mediated. In those with positive skin tests, desensitization is mandatory if therapeutic use of b-lactam antibiotics is to be undertaken. Various protocols are available, including oral and parenteral approaches. Oral desensitization appears to have lower risk of serious anaphylactic reactions during desensitization. However, desensitization carries the risk of anaphylaxis regardless of how it is performed. After desensitization, many patients experience non-life-threatening IgE-mediated untoward reactions to penicillin during their course of therapy. Desensitization is not effective in those with exfoliative dermatitis or morbilliform reactions due to penicillin.

NONSTEROIDAL ANTI-INFLAMMATORY DRUGS

[NSAIDs](#), including aspirin and indomethacin (indometacin), cause two broad categories of allergic-like symptoms in susceptible individuals: (1) approximately 1% of persons experience urticaria or angioedema, and (2) about half as many (0.5%) experience rhinosinusitis and asthma; however, about 10% of adults with asthma and one-third of individuals with nasal polyposis and sinusitis may respond adversely to aspirin.

Urticaria/angioedema may be delayed up to 24 h and may occur at any age. The rhinosinusitis-asthma syndrome generally develops within 1 h of drug administration. In young patients, the reaction pattern often begins as watery rhinorrhea, which can be complicated by nasal and sinus infection, and polyposis, bloody discharge, and nasal eosinophilia. In many individuals with this syndrome, asthma that can be life-threatening eventually ensues whenever [NSAIDs](#) are subsequently ingested, and symptoms may persist despite avoidance of these drugs. Proof of the association of symptoms and NSAID use requires either clear-cut history of symptoms following drug ingestion or an oral challenge. For the latter to be performed with relative safety, (1) asthma must be

under good control, (2) the procedure must be conducted in a hospital setting by experienced personnel capable of recognizing and treating acute respiratory responses, and (3) the challenge should begin with very low doses (i.e., not >30 mg) of aspirin and increase every 1 to 2 h in doubling doses as tolerated to 650 mg.

While cross-reactivity between [NSAIDs](#) is common, it is not immunologic, and patients who are sensitive to NSAIDs cannot be identified by assessment of IgE antibody to aspirin, lymphocyte sensitization, or in vitro immunologic testing.

RADIOCONTRAST MEDIA

Large numbers of patients are exposed to radiocontrast agents. High-osmolality radiocontrast media are about five times more likely to induce urticaria (1%) or anaphylaxis than newer low-osmolality media. Severe reactions are rare with either type of contrast media. About one-third of those with mild reactions to previous exposure rereact on reexposure. In most cases, these reactions are probably not immunologic. Pretreatment with prednisone and diphenhydramine reduces reaction rates. Persons with a reaction to a high-osmolality contrast media should be given low-osmolality media if later contrast studies are required.

ANTICONVULSANTS

Of the anticonvulsants, the single orally administered agent with the highest risk of severe adverse cutaneous reactions is the antiseizure medicine lamotrigine. Older anticonvulsants, including phenytoin and carbamazepine, are also associated with many types of severe reactions and a high incidence of less severe reactions, particularly in children. In addition to [SJS](#), [TEN](#), and the hypersensitivity syndrome discussed above, the aromatic anticonvulsants can induce a pseudolymphoma syndrome and induce gingival hyperplasia.

SULFONAMIDES

Sulfonamides have perhaps the highest risk of causing cutaneous eruptions and are the drugs most frequently implicated in [SJS](#) and [TEN](#). The combination of sulfamethoxazole and trimethoprim frequently induces adverse cutaneous reactions in patients with AIDS ([Chap. 309](#)). Desensitization is often successful in AIDS patients with morbilliform eruptions but is a high-risk procedure in AIDS patients who manifest erythroderma, fever, or a bullous reaction in response to their earlier sulfonamide exposure.

AGENTS USED IN CANCER CHEMOTHERAPY

Since many agents used in cancer chemotherapy inhibit cell division, rapidly proliferating elements of the skin, including hair, mucous membranes, and appendages, are sensitive to their effects; as a result, stomatitis and alopecia are among the most frequent dose-dependent side effects of chemotherapy. Onychodystrophy (dystrophic changes in nails) is also seen with bleomycin, hydroxyurea (hydroxycarbamide), and 5-fluorouracil. Sterile cellulitis and phlebitis and ulceration of pressure areas occur with many of these agents. Urticaria, angioedema, exfoliative dermatitis, and erythema of the palms and soles have also been seen, as has local and diffuse hyperpigmentation.

GLUCOCORTICOIDS

Both systemic and topical glucocorticoids cause a variety of skin changes, including acneiform eruptions, atrophy, striae, and other stigmata of Cushing's syndrome, and in sufficiently high doses can retard wound healing. Patients using glucocorticoids are at higher risk for bacterial, yeast, and fungal skin infections that may be misinterpreted as drug eruptions but are instead drug side effects.

CYTOKINE THERAPY

Alopecia is a common complication of [IFN-a](#). Induction or exacerbation of various immune-mediated disorders (psoriasis, lichen planus, lupus erythematosus) has been also reported with this agent. IFN- β injection has been associated with local necrosis of the skin. Granulocyte colony stimulating factor may induce various neutrophilic dermatosis, including Sweet's syndrome, pyoderma gangrenosum, neutrophilic eccrine hidradenitis, and vasculitis, and can exacerbate psoriasis.

[IL-2](#) is associated with frequent cutaneous reactions including exanthema, facial edema, xerosis, and pruritus. Cases of pemphigus vulgaris, linear IgA disease, psoriasis, and vitiligo have also been described in association with this drug.

ANTIMALARIAL AGENTS

Antimalarial agents are used as therapy for several skin diseases, including the skin manifestations of lupus and polymorphous light eruption, but they can also induce cutaneous reactions. Although also used to treat porphyria cutanea tarda at low doses, in patients with asymptomatic porphyria cutanea tarda, higher doses of chloroquine increase porphyrin levels to such an extent that they may exacerbate the disease.

Pigmentation disturbances, including black pigmentation of the face, mucous membranes, and pretibial and subungual areas, occur with antimalarials. Quinacrine (mepacrine) causes generalized, cutaneous yellow discoloration.

GOLD

Chrysotherapy has been associated with a variety of dose-related dermatologic reactions (including maculopapular eruptions), which can develop as long as 2 years after initiation of therapy and require months to resolve. Erythema nodosum, psoriasiform dermatitis, vaginal pruritus, eruptions similar to those of pityriasis rosea, hyperpigmentation, and lichenoid eruptions resembling those seen with antimalarial agents have been reported. After a cutaneous reaction, it is sometimes possible to reinstitute gold therapy at lower doses without recurrence of the dermatitis.

DIAGNOSIS OF DRUG REACTIONS

Possible causes of an adverse reaction can be assessed as definite, probable, possible, or unlikely based on six variables: (1) previous experience with the drug in the general population, (2) alternative etiologic candidates, (3) timing of events, (4) drug levels or

evidence of overdose, (5) patient reaction to drug discontinuation, and (6) patient reaction to rechallenge.

PREVIOUS EXPERIENCE

Tables of relative reaction rates are available and are useful to assess the likelihood that a given drug is responsible for a given cutaneous reaction. The specific morphologic pattern of a drug reaction, however, may modify these reaction rates by increasing or decreasing the likelihood that a given drug is responsible for a given reaction. For example, since fixed eruptions due to drugs are more often seen with barbiturates than with penicillin, a fixed drug reaction in a patient taking both types of agents is more likely to be due to the barbiturate, even though penicillins have a higher overall drug reaction rate.

ALTERNATIVE ETIOLOGIC CANDIDATES

A cutaneous eruption may be due to exacerbation of preexisting disease or to development of new disease unrelated to drugs. For example, a patient with psoriasis may have a flare-up of disease coincidental with administration of penicillin for streptococcal infection; in this case, infection is a more likely cause for the flare-up than drug reaction.

TIMING OF EVENTS

Most drug reactions of the skin occur within 1 to 2 weeks of initiation of therapy. Hypersensitivity syndrome may occur later (up to 8 weeks) after initiating drug therapy. Fixed drug reactions and generalized exanthematous pustulosis often occur earlier (within 48 h), as do reactions of all types in persons with prior sensitization to that drug or a cross-sensitizing agent.

DRUG LEVELS

Some cutaneous reactions are dependent on dosage or cumulative toxicity. For example, lichenoid dermatoses due to gold administration appear more often in patients taking high doses.

DISCONTINUATION

Most adverse cutaneous reactions to drugs remit with discontinuation of the suspected agent. A reaction is considered unlikely to be drug-related if improvement occurs while the drug is continued or if a patient fails to improve after stopping the drug and appropriate therapy.

RECHALLENGE

Rechallenge provides the most definitive information concerning adverse cutaneous reactions to drugs, since a reaction failing to recur on rechallenge with a drug is unlikely to be due to that agent. Rechallenge is usually impractical, however, because the need to ensure patient safety and comfort outweighs the value of the possible information

derived from rechallenge.

Of special importance is the rapid recognition of reactions that may become serious or life-threatening. [Table 59-2](#) lists clinical and laboratory features that, if present, suggest the reaction may be serious. [Table 59-3](#) provides key features of the most serious adverse cutaneous reactions.

DIAGNOSIS OF DRUG ALLERGY

Tests for IgE responses include in vivo and in vitro methods, but such tests are available for only a limited number of drugs, including penicillins and cephalosporins, some peptide and protein drugs (insulin, xenogeneic sera), and some agents used for general anesthesia. In vivo testing is accomplished by prick puncture and/or by intradermal skin testing. A wheal-and-flare response 2 × 2 mm greater than that seen with a saline control within 20 min is considered indicative of IgE-mediated mast cell degranulation, provided (1) the patient is not dermographic, (2) the drug does not nonspecifically degranulate mast cells, (3) the drug concentration is not high enough to be irritating, and (4) the buffer itself does not cause wheal-and-flare responses.

Skin testing with major and minor determinants of penicillins or cephalosporins has proved useful for identifying patients at risk of anaphylactic reactions to these agents. However, skin tests themselves carry a small risk of anaphylaxis. Negative skin tests do not rule out IgE-mediated reactivity, and the risk of anaphylaxis in response to penicillin administration in patients with negative skin tests is about 1%; about two-thirds of patients with a positive skin test and history of a previous adverse reaction to penicillin experience an allergic response on rechallenge. Skin tests may be negative in allergic patients receiving antihistamines or in those whose allergy is to determinants not present in the test reagent. Although less well studied, similar techniques can identify patients who are sensitive to protein drugs and to agents such as gallamine and succinylcholine. Most other drugs are small molecules, and skin testing with them is unreliable.

There are no generally available and reliable tests for assessing causality of non-IgE-mediated reactions, except possibly patch tests for assessment of fixed drug reactions. Therefore, diagnosis usually relies on clinical factors rather than test results.

(Bibliography omitted in Palm version)

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60. PHOTSENSITIVITY AND OTHER REACTIONS TO LIGHT - *David R. Bickers*

SOLAR RADIATION

Sunlight is the most visible and obvious source of comfort in the environment. This natural proclivity for the sun has the beneficial results of warmth and vitamin D synthesis but also can produce pathologic consequences. Few effects of sun exposure beyond those affecting the skin have been identified, but cutaneous exposure to sunlight can evoke immunosuppressive responses and genetic changes that may be relevant to the pathogenesis of nonmelanoma skin cancer and perhaps infections such as herpes simplex.

The sun's energy encompasses a broad range from ultrashort highly energetic ionizing radiation (10⁻²μm) to ultralong radiowaves of very low photon energy (10⁷μm). Thus, the emission spectrum ranges over nine orders of magnitude, but that reaching the earth's surface is narrow and is limited to components of the ultraviolet (UV), visible light, and portions of the infrared. The cutoff at the short end of the UV is at approximately 290 nm, because stratospheric ozone is formed by ionizing radiation of wavelengths less than 100 nm and absorbs solar energy between 120 and 310 nm, thereby preventing penetration to the earth's surface of the shorter, more energetic, potentially more harmful wavelengths of solar radiation. Indeed, concern about destruction of the ozone layer by chlorofluorocarbons released into the atmosphere has led to international agreements to reduce production of these chemicals.

Measurements of solar flux indicate that there is a twentyfold regional variation in the amount of energy at 300 nm that reaches the earth's surface. This variability relates to seasonal effects, the path of sunlight transmission through ozone and air, the altitude (4% increase for each 300 m of elevation), the latitude (increasing intensity with decreasing latitude), and the amount of cloud cover, fog, and pollution.

The major components of the photobiologic action spectrum include the [UV](#) and visible wavelengths between 290 and 700 nm. In addition, the wavelengths beyond 700 nm in the infrared primarily evoke heat, but warming of the skin may enhance biologic responses to wavelengths in the UV and visible spectrum.

The [UV](#) spectrum is arbitrarily divided into three major segments: C, B, and A. This includes the wavelengths between 10 and 400 nm. Ultraviolet C (UV-C) consists of wavelengths between 10 and 290 nm and does not reach the earth because of its absorption by stratospheric ozone. These wavelengths are not a cause of photosensitivity except in occupational settings where artificial sources of this energy are employed -- e.g., for germicidal effects. Ultraviolet B (UV-B) consists of wavelengths between 290 and 320 nm. This portion of the photobiologic action spectrum is the most efficient in producing redness or erythema in human skin and hence is sometimes known as the "sunburn spectrum." Ultraviolet A (UV-A) represents those wavelengths between 320 and 400 nm and is approximately 1000-fold less efficient in producing skin hyperemia than is UV-B. The UV-A has also been divided into two parts known as UV-A 1 (340 to 400 nm) and UV-A 2 (320 to 340 nm).

The visible wavelengths between 400 and 700 nm include the familiar white light which

when directed through a prism can be shown to consist of various colors including violet, indigo, blue, green, yellow, orange, and red. The energy possessed by photons in the visible spectrum is not capable of damaging human skin in the absence of a photosensitizing chemical. The absorption of energy is critical to the development of photosensitivity. Thus the *absorption spectrum* of a molecule is defined as the range of wavelengths absorbed by it, whereas the *action spectrum* for an effect of incident radiation is defined as the range of wavelengths that evoke the response.

Photosensitivity occurs when a photon-absorbing chemical (chromophore) present in the skin absorbs incident energy, becomes excited, and transfers the absorbed energy to various structures or to oxygen. The absorbed energy must be dissipated by processes including heat, fluorescence, and phosphorescence. It is important to emphasize that absorption spectra and action spectra need not be superimposable, but there must be overlap at some point to produce photosensitization.

STRUCTURE AND FUNCTION OF SKIN

The skin's exposure to sunlight permits the absorption of some wavelengths and the transmission of others. Essentially, human skin is a sandwich of two distinctive compartments, the epidermis and dermis, separated by a basement membrane. The outer epidermis is a stratified squamous epithelium comprising the surface stratum corneum (a protein- and lipid-rich compact membrane), the stratum granulosum, stratum spinosum, and the basal cell layer. The basal cell layer contains a heterogeneous population of cells, a subset of which migrate upward in the process of terminal differentiation that results in the expression of specific keratin genes and the formation of the stratum corneum. Epidermal cells include resident keratinocytes and melanocytes and immigrant cells, including the immunologically active Langerhans cells, lymphocytes, polymorphonuclear leukocytes, monocytes, and macrophages, making the epidermis a major component of the immune system. Branches of sensory nerve endings also reach into this compartment.

The second major component of skin is the dermis, which is relatively large and less densely populated with cells that include fibroblasts, endothelial cells within dermal vessels, and mast cells. Tissue macrophages and sparsely distributed inflammatory cells are also present. All these cells exist within an extracellular matrix of collagen, elastin, and glycosaminoglycans. In contrast to the epidermis, rich vascularization of the dermis allows it to play an important role in temperature regulation and in inflammatory responses to skin injury.

UV RADIATION (UVR) AND SKIN

The epidermis and the dermis contain several chromophores capable of interacting with incident solar energy. These interactions include reflection, refraction, absorption, and transmission. The stratum corneum is a major impediment to the transmission of [UV-B](#), and less than 10% of incident wavelengths in this region penetrate the basement membrane. Approximately 3% of radiation below 300 nm, 20% of radiation below 360 nm, and 33% of short visible radiation reaches the basal cell layer in untanned human skin. Proteins and nucleic acids absorb intensely in the short UV-B. In contrast, UV-A 1 and 2 penetrate the epidermis efficiently to reach the dermis, where they likely produce

changes in structural and matrix proteins that contribute to the aged appearance of chronically sun-exposed skin, particularly in individuals of light complexion.

One of the consequences of [UV-B](#) absorption by DNA is the production of pyrimidine dimers. These structural changes can be repaired by mechanisms that result in their recognition and excision, and the reestablishment of normal base sequences. The efficient repair of these structural aberrations is crucial, since individuals with defective DNA repair are at high risk for the development of cutaneous cancer. For example, patients with xeroderma pigmentosum, an autosomal recessive disorder, are characterized by variably decreased repair of UV-induced photoproducts, and their skin may develop the xerotic appearance of photoaging as well as basal cell and squamous cell carcinomas and melanoma in the first two decades of life. Studies in mice using knockout gene technology have verified the importance of genes regulating these repair pathways in preventing the development of UV-induced cancer.

Cutaneous Optics and Chromophores Chromophores are endogenous or exogenous chemical components that can absorb physical energy. Endogenous chromophores of skin are of two types: (1) chemicals that are normally present, including nucleic acids, proteins, lipids, and 7-dehydrocholesterol, the precursor of vitamin D; and (2) chemicals, such as porphyrins, synthesized elsewhere in the body that circulate in the bloodstream and diffuse into the skin. Normally, only trace amounts of porphyrins are present in the skin, but in the diseases known as the porphyrias, increased amounts are released into the circulation and are transported to the skin, where they absorb incident energy both in the Soret band around 400 nm (short visible) and to a lesser extent in the red portion of the visible spectrum (580 to 660 nm). This results in structural damage to the skin that may be manifest as erythema, edema, urticaria, or blister formation ([Chap. 346](#)).

Acute Effects of Sun Exposure The immediate cutaneous consequences of sun exposure include sunburn and vitamin D synthesis.

Sunburn This very common affliction of human skin is caused by exposure to [UVR](#). Generally speaking, the individual's ability to tolerate sunlight is inversely proportional to his or her melanin pigmentation. Melanin is a complex polymer of tyrosine that functions as an efficient neutral-density filter with broad absorbance within the [UV](#) portion of the solar spectrum. Melanin is synthesized in specialized epidermal dendritic cells termed *melanocytes* and is packaged into *melanosomes* that are transferred via dendritic processes into *keratinocytes*, where they provide photoprotection. Sun-induced melanogenesis is a consequence of increased tyrosinase activity in melanocytes that in turn may be due to a combination of eicosanoid and endothelin-1 release. Tolerance of sun exposure is a function of the efficiency of the epidermal-melanin unit and can usually be ascertained by asking an individual two questions: (1) Do you burn after sun exposure? and (2) Do you tan after sun exposure? By the answers to these questions, it is usually possible to divide the population into six skin types varying from type I (always burn, never tan) to type VI (never burn, always tan) ([Table 60-1](#)).

There are two general theories about the pathogenesis of the sunburn response. First, the lag phase in time between skin exposure and the development of visible redness (usually 4 to 12 h) suggests an epidermal chromophore that causes delayed production and/or release of vasoactive mediator(s), or cytokines, that diffuse to the dermal

vasculature to evoke vasodilatation. Indeed, [UVR](#) stimulates the release of numerous proinflammatory cytokines and nitric oxide by keratinocytes. Second, it is possible that the small amount of incident [UV-B](#) radiation (10% or less) that penetrates to the dermis can be absorbed directly by endothelial cells in the vasculature, thereby resulting in vasodilatation. The issue remains unresolved.

The action spectrum for sunburn erythema includes the [UV-B](#) and UV-A regions. Photons in the shorter UV-B are at least 1000-fold more efficient than photons in the longer UV-B and the UV-A in evoking the response. However, UV-A may contribute to sunburn erythema at midday when much more UV-A than UV-B is present.

The mechanism of injury remains poorly defined, but the action spectrum for [UV-B](#) erythema closely resembles the absorption spectrum for DNA after adjusting for the absorbance of incident energy by the stratum corneum. Apoptotic keratinocytes (so-called sunburn cells) are visible histologically within an hour of exposure and are maximal within 24 h. UV-A is less effective than UV-B in producing sunburn cells. Mast cells may release inflammatory mediators after exposure to UV-B and UV-A. For example, erythema doses of both UV-B and UV-A increase histamine levels in experimentally induced suction blisters of human skin that return to normal after 24 h (before visible erythema has subsided). Prostaglandin E_2 increases to approximately 150% of control levels after 24 h and then diminishes. Since prostaglandins evoke both pain and redness when injected intradermally, their presence in suction blisters after UV-B exposure suggests a role in UV-B erythema. Age-related declines occur in the amount of inflammatory mediators detectable in human skin after UV-B irradiation. UV-A erythema results in few epidermal sunburn cells, but vascular endothelial injury is greater than with UV-B. In addition, there are increased levels of arachidonic acid and of prostaglandins D_2 , E_2 , and I_2 that peak within 5 to 9 h and then subside before peak redness occurs. Despite evidence for the role of prostaglandins in both UV-B- and UV-A-irradiated skin, administration of nonsteroidal anti-inflammatory drugs is more effective in reducing erythema evoked by UV-B than by UV-A. UV-B also induces cutaneous matrix-degrading metalloproteinases within hours of exposure.

Vitamin D Photochemistry Cutaneous exposure to [UV-B](#) causes photolysis of epidermal previtamin D_3 (7-dehydrocholesterol) to previtamin D_3 , which then undergoes a temperature-dependent isomerization to form the stable hormone vitamin D_3 . This compound then diffuses to the dermal vasculature and circulates systemically where it is converted to the functional hormone 1,25-dihydroxy vitamin D_3 [$1,25(OH)_2D_3$]. Vitamin D metabolites from the circulation or those produced in the skin itself can augment epidermal differentiation signaling. Aging substantially decreases the ability of human skin to produce vitamin D_3 . This, coupled with the widespread use of sunscreens that filter out [UV-B](#), has led to concern that vitamin D deficiency may become a significant clinical problem in the elderly. Indeed, studies have shown that the use of sunscreens can diminish the production of vitamin D_3 in human skin.

Chronic Effects of Sun Exposure: Nonmalignant The clinical features of photodamaged sun-exposed skin consist of wrinkling, blotchiness, telangiectasia, and a roughened, irregular, "weather-beaten" appearance. Whether these changes, which some refer to as *photoaging* or *dermatoheliosis*, represent accelerated chronologic aging or a separate and distinct process is not clear.

Within chronically sun-exposed epidermis, there is thickening (acanthosis) and morphologic heterogeneity within the basal cell layer. Higher but irregular melanosome content may be present in some keratinocytes, indicating prolonged residence of the cells in the basal cell layer. These structural changes may help to explain the leathery texture and the blotchy discoloration of sun-damaged skin.

The dermis is the major site for sun-associated chronic damage, manifest as a massive increase in thickened irregular masses of tangled elastic fibers resulting from enhanced expression of elastin genes. Collagen fibers are also abnormally clumped in the deeper dermis. Fibroblasts are increased in number and show morphologic signs suggesting activation. Degraded mast cells may be present in the dermis, the relevance of which remains unclear.

These morphologic changes, both gross and microscopic, are features of chronically sun-exposed skin. The chromophore(s), the action spectra, and the specific biochemical events orchestrating these changes are unknown.

Chronic Effects of Sun Exposure: Malignant One of the major known consequences of chronic skin exposure to sunlight is nonmelanoma skin cancer. The two types of nonmelanoma skin cancer are basal cell and squamous cell carcinoma ([Chap. 86](#)). There are three major steps for cancer induction: initiation, promotion, and progression. Chronic exposure of animal skin to artificial light sources that mimic solar [UVR](#) results in *initiation*, a step whereby structural (mutagenic) changes in DNA evoke an irreversible alteration in the target cell (keratinocyte) that begins the tumorigenic process. Exposure to a tumor initiator is believed to be a necessary but not sufficient step in the malignant process, since initiated skin cells not exposed to tumor promoters do not generally develop tumors. The second stage in tumor development is *promotion*, a multistep process whereby initiated cells are exposed to chemical and physical agents that evoke epigenetic changes that culminate in the clonal expansion of initiated cells and cause the development, over a period of weeks to months, of benign growths known as *papillomas*. Again, using transgenic animals, the importance of [UV](#) effects on the expression of additional oncogenes such as *fos* and *jun* in developing papillomas has been demonstrated. UV-B is a *complete carcinogen*, meaning that it can function as both an initiator and a promoter, leading to tumor induction. *Incomplete carcinogens* can initiate tumorigenesis but require additional skin exposure to tumor promoters to elicit tumors. The prototype tumor promoter is the phorbol ester 12-*O*-tetradecanoyl phorbol-13-acetate. Tumor promotion usually requires multiple exposures over time to evoke a neoplasm.

The final step in the malignant process is the conversion of benign precursors into malignant lesions, a process thought to require additional genetic alterations in already transformed cells. Indeed, *ras* gene mutations have been detected in a minority of human nonmelanoma skin cancers. Mutations of the tumor suppressor gene p53 also occur in sun-damaged human skin.

Sun exposure causes nonmelanoma and melanoma cancers of the skin, although the evidence is far more direct for its role in nonmelanoma (basal cell and squamous cell carcinoma) than in melanoma. Approximately 80% of nonmelanoma skin cancers

develop on exposed body area, including the face, the neck, and the hands. Men of fair complexion who work outdoors are twice as likely as women to develop these types of cancers. Whites of darker complexions (e.g., Hispanics) have one-tenth the risk of developing such cancers as do light-skinned individuals. Blacks are at lowest risk for all forms of skin cancer. Between 600,000 and 800,000 individuals in the United States develop nonmelanoma skin cancer annually, and the lifetime risk for a white individual to develop such a neoplasm is estimated at approximately 15%. A consensus exists that the incidence of nonmelanoma skin cancer in the population is rising, for reasons that are unclear.

The relationship of sun exposure to melanoma is less clear-cut, but suggestive evidence supports an association. Melanomas occasionally develop by the teenage years, indicating that the latent period for tumor growth is less than that of nonmelanoma skin cancer. Melanomas are among the most rapidly increasing of all human malignancies ([Chap. 86](#)). Epidemiologic studies of immigrants of similar ethnic stock indicate that individuals born in one area or who migrated to the same locale before age 10 have higher age-specific melanoma rates than individuals arriving later. It is thus reasonable to conclude that life in a sunny climate from birth or early childhood increases the risk of melanoma. In general, risk does not correlate with cumulative sun exposure but may relate to sequelae of sun exposure in childhood. Thus, a blistering sunburn is associated with a doubling of melanoma risk at the site of the reaction.

Immunologic Effects Exposure to solar radiation influences both local and systemic immune responses. [UV-B](#) appears to be most efficient in altering immune responses, likely related to the capacity of such energy to affect antigen presentation in skin by interacting with epidermal Langerhans cells. These bone marrow-derived dendritic cells possess surface markers characteristic of monocytes and macrophages. Following skin exposure to erythema doses of UV-B, Langerhans cells undergo both morphologic and functional changes that result in decreased contact allergic responses when haptens are applied to the irradiated site. This diminished capacity for sensitization is due to the induction of antigen-specific suppressor T lymphocytes. Indeed, while the immunosuppressive effect of irradiation is limited to haptens applied to the irradiated site, the net result is systemic immune suppression to that antigen because of the induction of suppressor T cells.

Higher doses of radiation evoke diminished immunologic responses to antigens introduced either epicutaneously or intracutaneously at sites distant from the irradiated site. These suppressed responses are also associated with the induction of antigen-specific suppressor T lymphocytes and may be mediated by as yet undefined factors that are released from epidermal cells at the irradiated site. The implications of this generalized immune suppression in terms of altered susceptibility to cutaneous cancer or to infection remain to be defined.

It is known that [UV](#)-induced tumors in murine skin are antigenic and are rapidly rejected when transplanted into normal syngeneic animals. If the tumors are transplanted into animals previously exposed to subcarcinogenic doses of UV-B, they are not rejected and instead grow progressively in the recipients. This failure of irradiated animals to reject the transplanted tumors is due to the development of T suppressor cells that prevent the rejection response. While the mechanism of suppression of tumor rejection

is unknown, such a response might be a critical determinant of cancer risk in human skin.

PHOTOSENSITIVITY DISEASES

The diagnosis of photosensitivity requires a careful history to define the duration of the signs and symptoms, the length of time between exposure to sunlight and the development of subjective complaints, and visible changes in the skin. The age of onset also can be a helpful clue; for example, the acute photosensitivity of erythropoietic protoporphyria almost always begins in childhood, whereas the chronic photosensitivity of porphyria cutanea tarda typically begins in the fourth and fifth decades. A history of exposure to topical and systemic drugs and chemicals may provide important information. Many classes of drugs can cause photosensitivity on the basis of either phototoxicity or photoallergy. Fragrances such as musk ambrette that were previously present in numerous cosmetic products are also potent photosensitizers.

Examination of the skin may also offer important clues. Anatomic areas that are naturally protected from direct sunlight such as the hairy scalp, the upper eyelids, the retroauricular areas, and the infranasal and submental regions may be spared, whereas exposed areas show characteristic features of the pathologic process. These anatomic localization patterns are often helpful but not infallible in making the diagnosis. For example, airborne contact sensitizers that are blown onto the skin may produce dermatitis that can be difficult to distinguish from photosensitivity, despite the fact that such material may trigger skin reactivity in areas shielded from direct sunlight.

Many dermatologic conditions may be caused or aggravated by light ([Table 60-2](#)). The role of light in evoking these responses may be dependent on genetic abnormalities ranging from well-described defects in DNA repair that occur in xeroderma pigmentosum to the inherited abnormalities in heme synthesis that characterize the porphyrias. In certain photosensitivity diseases, the chromophore has been identified, whereas in the majority, the energy-absorbing agent is unknown.

Polymorphous Light Eruption After sunburn, the most common type of photosensitivity disease is *polymorphous light eruption*, the mechanism of which is unknown. Many affected individuals never seek medical attention because the condition is often transient, becoming manifest each spring with initial sun exposure but then subsiding spontaneously with continuing exposure, a phenomenon known as "hardening." The major manifestations of polymorphous light eruption include pruritic (often intensely so) erythematous papules that may coalesce into plaques on exposed areas of the face and arms or other areas as well, making the distribution spotty and uneven.

The diagnosis can be confirmed by skin biopsy and by performing phototest procedures in which skin is exposed to multiple erythema doses of [UV](#)-A and UV-B. The action spectrum for polymorphous light eruption is usually within these portions of the solar spectrum.

Treatment of this disease includes the induction of hardening by the cautious administration of [UV](#)light, either alone or in combination with photosensitizers such as

the psoralens (see below).

Phototoxicity and Photoallergy These photosensitivity disorders are related to the topical or systemic administration of drugs and other chemicals. Both reactions require the absorption of energy by a drug or chemical resulting in the production of an excited-state photosensitizer that can transfer its absorbed energy to a bystander molecule or to molecular oxygen, thereby generating tissue-destructive chemical species.

Phototoxicity is a nonimmunologic reaction caused by drugs and chemicals, a few of which are listed in [Table 60-3](#). The usual clinical manifestations include erythema resembling a sunburn that quickly desquamates or "peels" within several days. In addition, edema, vesicles, and bullae may occur.

Photoallergy is distinct in that the immune system participates in the pathologic process. The excited-state photosensitizer may create highly unstable haptenic free radicals that bind covalently to macromolecules to form a functional antigen capable of evoking a delayed hypersensitivity response. Some of the drugs and chemicals that produce photoallergy are listed in [Table 60-4](#). The clinical manifestations typically differ from those of phototoxicity in that an intensely pruritic eczematous dermatitis tends to predominate and evolves into lichenified, thickened, "leathery" changes in sun-exposed areas. A small subset (perhaps 5 to 10%) of patients with photoallergy may develop a persistent exquisite hypersensitivity to light even when the offending drug or chemical is identified and eliminated. Known as *persistent light reaction*, this may be incapacitating for years. Some have used the term *chronic actinic dermatitis* to encompass these chronic hyperresponsive states.

Diagnostic confirmation of phototoxicity and photoallergy often can be obtained using phototest procedures. In patients with suspected phototoxicity, determination of the minimal erythema dose (MED) while the patient is exposed to a suspected agent and then repeating the MED after discontinuation of the agent may provide a clue to the causative drug or chemical. Photopatch testing can be performed to confirm the diagnosis of photoallergy. This is a simple variant of ordinary patch testing in which a series of known photoallergens is applied to the skin in duplicate and one set is irradiated with a suberythema dose of UV-A. Development of eczematous changes at sites exposed to sensitizer and light is a positive result. The characteristic abnormality in patients with persistent light reaction is a diminished threshold to erythema evoked by UV-B. Patients with chronic actinic dermatitis may have a broad spectrum of UV hyperresponsiveness.

The management of drug photosensitivity is first and foremost to eliminate exposure to the chemical agents responsible for the reaction and to minimize sun exposure. The acute symptoms of phototoxicity may be ameliorated by cool, moist compresses, topical glucocorticoids, and systemically administered nonsteroidal antiinflammatory agents. In severely affected individuals, a rapidly tapered course of systemic glucocorticoids may be useful. Judicious use of analgesics may be necessary.

Photoallergic reactions require a similar management approach. Furthermore, individuals suffering from persistent light reactivity must be meticulously protected

against light exposure. In selected patients in whom chronic systemic high-dose glucocorticoids pose unacceptable risks, it may be necessary to employ cytotoxic agents such as azathioprine or cyclophosphamide.

Porphyria The porphyrias ([Chap. 346](#)) are a group of diseases that have in common various derangements in the synthesis of heme. Heme is an iron-chelated tetrapyrrole or porphyrin, and the nonmetal chelated porphyrins are potent photosensitizers that absorb light intensely in both the short (400 to 410 nm) and the long (580 to 650 nm) portions of the visible spectrum.

Heme cannot be reutilized and must be continuously synthesized, and the two body compartments with the largest capacity for its production are the bone marrow and the liver. Accordingly, the porphyrias originate in one or the other of these organs, with the end result of excessive endogenous production of potent photosensitizing porphyrins. The porphyrins circulate in the bloodstream and diffuse into the skin, where they absorb solar energy, become photoexcited, and evoke cutaneous photosensitivity. The mechanism of porphyrin photosensitization is known to be photodynamic or oxygen-dependent and is mediated by reactive oxygen species such as superoxide anions.

Porphyria cutanea tarda is the most common type of human porphyria and is associated with decreased activity of the enzyme uroporphyrinogen decarboxylase associated with a number of gene mutations. There are two basic types of porphyria cutanea tarda: the sporadic or acquired type, generally seen in individuals ingesting ethanol or receiving estrogens; and the inherited type, in which there is autosomal dominant transmission of deficient enzyme activity. Both forms are associated with increased hepatic iron stores.

In both types of porphyria cutanea tarda, the predominant feature is a chronic photosensitivity characterized by increased fragility of sun-exposed skin, particularly areas subject to repeated trauma such as the dorsa of the hands, the forearms, the face, and the ears. The predominant skin lesions are vesicles and bullae that rupture, producing moist erosions, often with a hemorrhagic base, that heal slowly with crusting and purplish discoloration of the affected skin. Hypertrichosis, mottled pigmentary change, and scleroderma-like induration are associated features. Biochemical confirmation of the diagnosis can be obtained by measurement of urinary porphyrin excretion, plasma porphyrin assay, and by assay of erythrocyte and/or hepatic uroporphyrinogen decarboxylase. Multiple mutations of the uroporphyrinogen decarboxylase gene have been identified in human populations, including exon skipping and base substitutions.

Treatment consists of repeated phlebotomies to diminish the excessive hepatic iron stores and/or intermittent low doses of the antimalarial drugs chloroquine and hydroxychloroquine. Long-term remission of the disease can be achieved if the patient eliminates exposure to porphyrinogenic agents.

Erythropoietic protoporphyria originates in the bone marrow and is due to a decrease in the mitochondrial enzyme ferrochelatase secondary to numerous gene mutations. The major clinical features include an acute photosensitivity characterized by subjective burning and stinging of exposed skin that often develops during or just after exposure.

There may be associated skin swelling and, after repeated episodes, a waxlike scarring.

The diagnosis is confirmed by demonstration of measurement of free elevated erythrocyte protoporphyrin. Detection of increased plasma protoporphyrin helps to differentiate lead poisoning and iron-deficiency anemia, in both of which elevated erythrocyte protoporphyrin occurs in the absence of cutaneous photosensitivity and of elevated plasma protoporphyrin.

Treatment consists of reducing sun exposure and the oral administration of the carotenoid β -carotene, which is an effective scavenger of free radicals. This drug increases tolerance to sun exposure in many affected individuals, although it has no effect on deficient ferrochelatase.

An algorithm for the approach to a patient with photosensitivity is illustrated in [Fig. 60-1](#).

PHOTOPROTECTION

Since photosensitivity of the skin results from exposure to sunlight, it follows that avoidance of the sun would eliminate these disorders. Unfortunately, social pressures make this an impractical alternative for most individuals, and this has led to a search for better approaches to photoprotection.

Natural photoprotection is provided by structural proteins in the epidermis, particularly keratins and melanin. The amount of melanin and its distribution in cells is genetically regulated, and individuals of darker complexion (skin types IV to VI) are at decreased risk for the development of cutaneous malignancy.

Other forms of photoprotection include clothing and sunscreens. Clothing constructed of tightly woven sun-protective fabrics, irrespective of color, affords substantial protection. Wide-brimmed hats, long sleeves, and trousers all reduce direct exposure. Sunscreens are of two major types -- chemical and physical. Chemical sunscreens are chromophores that absorb energy in the [UV-B](#) and/or [UV-A](#) regions, thereby diminishing photon absorption by the skin ([Table 60-5](#)). Sunscreens are rated for their photoprotective effect by their *sun protective factor* (SPF). The SPF is simply a ratio of the time required to produce sunburn erythema with and without sunscreen application. SPF ratings of 15 or higher provide effective protection against [UV-B](#) and, to a lesser extent, [UV-A](#). The major categories of chemical sunscreens include *p*-aminobenzoic acid and its esters, benzophenones, anthranilates, cinnamates, and salicylates. Physical sunscreens are light-opaque mixtures containing metal particles such as titanium oxide and zinc oxide that scatter light, thereby reducing photon absorption by the skin.

In addition to light absorption, a critical determinant of the photoprotective effect of sunscreens is their ability to remain on the skin, a property known as *substantivity*. In general, the *p*-aminobenzoic acid esters formulated in moisturizing vehicles provide the greatest substantivity.

Photoprotection can also be achieved by limiting the time of exposure during the day. Since the majority of an individual's total lifetime sun exposure may occur by the age of

18, it is important to educate parents and young children about the hazards of sunlight. Simply eliminating exposure at midday will substantially reduce lifetime [UV](#)-B exposure.

PHOTOTHERAPY AND PHOTOCHEMOTHERAPY

[UVR](#) can also be used therapeutically. The administration of [UV](#)-B alone or in combination with topically applied agents can induce remissions of psoriasis and atopic dermatitis.

Photochemotherapy in which topically applied or systemically administered *psoralens* are combined with [UV](#)-A (PUVA) is also effective in treating psoriasis and in the early stages of cutaneous T cell lymphoma and vitiligo. Psoralens are tricyclic furocoumarins that, when intercalated into DNA and exposed to UV-A, form adducts with pyrimidine bases and eventually form DNA cross-links. These structural changes are thought to decrease DNA synthesis and relate to improvement that occurs in psoriasis. The reason that PUVA photochemotherapy is effective in cutaneous T cell lymphoma is not clear.

In addition to its effects on DNA, PUVA photochemotherapy also stimulates melanin synthesis, and this provides the rationale for its use in the depigmenting disease vitiligo. Oral 8-methoxypsoralen and [UV](#)-A appear to be most effective in this regard, but as many as 100 treatments extending over 12 to 18 months may be required to promote satisfactory repigmentation.

The major side effects of [UV](#)-B phototherapy and PUVA photochemotherapy are due to the cumulative effects of photon absorption and include skin dryness, actinic keratoses, and an increased risk of nonmelanoma skin cancer. Despite these risks, the therapeutic index of these modalities is quite acceptable.

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SECTION 10 -HEMATOLOGIC ALTERATIONS

61. ANEMIA AND POLYCYTHEMIA - John W. Adamson, Dan L. Longo

HEMATOPOIESIS AND THE PHYSIOLOGIC BASIS OF RED CELL PRODUCTION

Hematopoiesis is the process by which the formed elements of the blood are produced. The process is regulated through a series of steps beginning with the pluripotent hematopoietic stem cell. Stem cells are capable of producing red cells, all classes of granulocytes, monocytes, platelets, and the cells of the immune system. Commitment of the stem cell to the specific cell lineages appears not to be regulated by known exogenous growth factors or cytokines. Rather, stem cells develop into differentiated cell types through incompletely defined molecular events that are intrinsic to the stem cell itself ([Chap. 104](#)). Following lineage commitment (or differentiation), hematopoietic progenitor and precursor cells come increasingly under the regulatory influence of growth factors and hormones, such as erythropoietin (EPO) for red cell production. EPO is required for the maintenance of committed erythroid progenitor cells which, in the absence of the hormone, undergo programmed cell death (*apoptosis*). The regulated process of red cell production is *erythropoiesis*, and its key elements are illustrated in [Fig. 61-1](#).

In the bone marrow, the first morphologically recognizable erythroid precursor is the pronormoblast. This cell can undergo 4 to 5 cell divisions that result in the production of 16 to 32 mature red cells. With increased [EPO](#) production, or the administration of EPO as a drug, early progenitor cell numbers are amplified and, in turn, give rise to increased numbers of erythrocytes. The regulation of EPO production itself is linked to O₂ transport.

In mammals, O₂ is transported to tissues bound to the hemoglobin contained within circulating red cells. The mature red cell is 8µm in diameter, anucleate, discoid in shape, and extremely pliable in order for it to traverse the microcirculation successfully; its membrane integrity is maintained by the intracellular generation of ATP. Normal red cell production results in the daily replacement of 0.8 to 1% of all circulating red cells in the body. The average red cell lives 100 to 120 days. The machinery responsible for red cell production is called the *erythron*. The erythron is a dynamic organ made up of a rapidly proliferating pool of marrow erythroid precursor cells and a large mass of mature circulating red blood cells. The size of the red cell mass reflects the balance of red cell production and destruction. The physiologic basis of red cell production and destruction provides an understanding of the mechanisms that can lead to anemia.

The physiologic regulator of red cell production, the glycoprotein hormone [EPO](#), is produced and released by peritubular capillary lining cells within the kidney. These cells are highly specialized epithelial-like cells. A small amount of EPO is produced by hepatocytes. The fundamental stimulus for EPO production is the availability of O₂ for tissue metabolic needs. Impaired O₂ delivery to the kidney can result from a decreased red cell mass (*anemia*), impaired O₂ loading of the hemoglobin molecule (*hypoxemia*), or, rarely, impaired blood flow to the kidney (renal artery stenosis). EPO governs the day-to-day production of red cells, and ambient levels of the hormone can be measured in the plasma by sensitive immunoassays -- the normal level being 10 to 25 U/L. When

the hemoglobin concentration falls below 100 to 120 g/L (10 to 12 g/dL), plasma EPO levels increase logarithmically in inverse proportion to the severity of the anemia. In circulation, EPO has a half-clearance time of 6 to 9 h. EPO acts by binding to specific receptors on the surface of marrow erythroid precursors, inducing them to proliferate and to mature. Under the stimulus of EPO, red blood cell production can increase four- to fivefold within a 1- to 2-week period but only in the presence of adequate nutrients, especially iron. The functional capacity of the erythron, therefore, requires normal renal production of EPO, a functioning erythroid marrow, and an adequate supply of substrates for hemoglobin synthesis. A defect in any of these key components can lead to anemia. Generally, anemia is recognized in the laboratory when a patient's hemoglobin level or hematocrit is reduced below an expected value (the normal range). The likelihood and severity of anemia are defined based on the deviation of the patient's hemoglobin/hematocrit from values expected for age- and sex-matched normal subjects. The lower ranges of distribution of hemoglobin/hematocrit values for adult males and females are shown in [Fig. 61-2](#). The hemoglobin concentration in adults has a Gaussian distribution. The mean hematocrit value for adult males is 47% ($\pm$ SD 7) and that for adult females is 42% ($\pm$ 5). Any individual hematocrit or hemoglobin value carries with it a likelihood of associated anemia. Thus, a hematocrit of $\leq$ 39% in an adult male or $\leq$ 35% in an adult female has only about a 25% chance of being normal. Suspected low hemoglobin or hematocrit values are more easily interpreted if there are historic values for the same patient for comparison.

The critical elements of erythropoiesis -- [EPO](#) production, iron availability, the proliferative capacity of the bone marrow, and effective maturation of red cell precursors -- are used for the initial classification of anemia (see "Definition and Classification, below).

ANEMIA

CLINICAL PRESENTATION OF ANEMIA

Signs and Symptoms Anemia is most often recognized by abnormal screening laboratory tests. Patients only occasionally present with advanced anemia and its attendant signs and symptoms. Acute anemia is nearly always due to blood loss or hemolysis. In fact, with acute blood loss, hypovolemia dominates the clinical picture and the hematocrit and hemoglobin levels do not reflect the volume of blood lost. Signs of vascular instability dominate with acute losses of 10 to 15% of the total blood volume. In such patients, the issue is not anemia but hypotension and decreased organ perfusion. When $>30\%$ of the blood volume is lost suddenly, patients are unable to compensate with the usual mechanisms of vascular contraction and changes in regional blood flow. The patient prefers to remain supine and will show postural hypotension and tachycardia if upright. If the volume of blood lost is $>40\%$ (i.e., >2 L in the average-sized adult), signs of hypovolemic shock including confusion, air hunger, diaphoresis, hypotension, and tachycardia appear ([Chap. 108](#)). Such patients have significant deficits in vital organ perfusion and require immediate volume replacement. With mild blood loss, enhanced O_2 delivery is achieved through changes in the O_2 -hemoglobin dissociation curve mediated by a decreased pH or increased CO_2 (*Bohr effect*).

With acute hemolytic disease, the signs and symptoms depend on the mechanism that

leads to red cell destruction. Intravascular hemolysis with release of free hemoglobin may be associated with acute back pain, free hemoglobin in the plasma and urine, and renal failure. Symptoms associated with more chronic or progressive anemia depend on the age of the patient and the adequacy of blood supply to critical organs. Symptoms associated with moderate anemia include fatigue, loss of stamina, breathlessness, and tachycardia (particularly with physical exertion). However, because of the intrinsic compensatory mechanisms that govern the O₂-hemoglobin dissociation curve, the gradual onset of anemia -- particularly in young patients -- may not be associated with signs or symptoms until the anemia is severe [hemoglobin <70 to 80 g/L (7 to 8 g/dL)]. When anemia develops over a period of days or weeks, the total blood volume is normal to slightly increased and changes in cardiac output and regional blood flow help compensate for the overall loss in O₂-carrying capacity. Changes in the position of the O₂-hemoglobin dissociation curve account for some of the compensatory response to anemia. With chronic anemia, intracellular levels of 2,3-bisphosphoglycerate (BPG) rise, shifting the dissociation curve to the right and facilitating O₂ unloading. This compensatory mechanism can only maintain normal tissue O₂ delivery in the face of a 20 to 30 g/L (2 to 3 g/dL) deficit in hemoglobin concentration. Finally, further protection of O₂ delivery to vital organs is achieved by the shunting of blood away from organs that are relatively rich in blood supply, particularly the kidney, gut, and skin.

Certain disorders are commonly associated with anemia. Chronic inflammatory states (e.g., infection, rheumatoid arthritis) are associated with mild to moderate anemia, whereas lymphoproliferative disorders, such as chronic lymphocytic leukemia and certain other B cell neoplasms, may be associated with autoimmune hemolysis.

Approach to the Patient

The evaluation of the patient with anemia requires a careful history and physical examination. Historic information that may be useful includes exposure to certain toxic agents or drugs and symptoms related to other disorders commonly associated with anemia. These include symptoms and signs such as bleeding, fatigue, malaise, fever, weight loss, night sweats, and other systemic symptoms. Clues to the mechanisms of anemia may be provided on physical examination by findings of infection, blood in the stool, lymphadenopathy, splenomegaly, or petechiae. Splenomegaly and lymphadenopathy suggest an underlying lymphoproliferative disease, while petechiae suggest platelet dysfunction. If it is uncertain whether a mild anemia represents an extreme normal value or an abnormal finding, past laboratory measurements may be helpful. Nutritional history related to drugs or alcohol intake and family history of anemia should always be assessed. Certain geographic backgrounds and ethnic origins are associated with an increased likelihood of an inherited disorder of the hemoglobin molecule or intermediary metabolism. Glucose-6-phosphate dehydrogenase deficiency and certain hemoglobinopathies are seen more commonly in those of middle-Eastern or African origin.

In the anemic patient, physical examination may demonstrate a forceful heartbeat, strong peripheral pulses, and a systolic "flow" murmur. The skin and mucous membranes may be pale if the hemoglobin is <80 to 100 g/L (8 to 10 g/dL). This part of the physical examination should focus on areas where vessels are close to the surface such as the mucous membranes, nail beds, and palmar creases. If the palmar creases

are lighter in color than the surrounding skin when the hand is hyperextended, the hemoglobin level is usually <80 g/L (8 g/dL).

Laboratory Evaluation [Table 61-1](#) lists the tests used in the initial workup of anemia. A routine complete blood count (CBC) is required as part of the evaluation and includes the hemoglobin, hematocrit, and red cell indices: the mean cell volume (MCV) in femtoliters, mean cell hemoglobin (MCH) in picograms per cell, and mean concentration of hemoglobin per volume of red cells (MCHC) in grams per liter (non-SI: grams per deciliter). The red cell indices are calculated as shown in [Table 61-2](#), and the normal variations in the CBC with age are shown in [Table A-7](#). A number of physiologic factors affect the normal CBC values including age, gender, pregnancy, smoking, and altitude. High-normal hemoglobin values may be seen in men and women who live at altitude or smoke heavily. The elevations in smokers reflect normal compensation due to the displacement of O₂ by CO in hemoglobin binding. Other important information is provided by the reticulocyte count and measurements of iron supply including the *serum iron*, the *total iron-binding capacity* (TIBC; an indirect measure of the transferrin level), and *serum ferritin*. Marked alterations in the red cell indices usually reflect disorders of maturation or iron deficiency. Clinical laboratories also provide a description of both the red and white cells, a white cell differential count, and the platelet count. In patients with severe anemia and abnormalities in red blood cell morphology, a bone marrow aspirate or biopsy may be important to assist in the diagnosis. Other tests of value in the diagnosis of specific anemias are discussed in chapters on specific disease states.

The components of the [CBC](#) also help in the classification of anemia. *Microcytosis* is reflected by a lower than normal [MCV](#) (<80), whereas high values (>100) reflect *macrocytosis*. The [MCH](#) and [MCHC](#) reflect defects in hemoglobin synthesis (*hypochromia*). Automated cell counters describe the red cell volume distribution width (RDW). The MCV (representing the peak of the distribution curve) is insensitive to the appearance of small populations of macrocytes or microcytes. An experienced laboratory technician will be able to identify minor populations of large or small cells or hypochromic cells before the red cell indices change.

PERIPHERAL BLOOD SMEAR The peripheral blood smear provides important information about defects in red cell production. As a complement to the red cell indices, the blood smear also reveals variations in cell size (*anisocytosis*) and shape (*poikilocytosis*). The degree of anisocytosis usually correlates with increases in the [RDW](#) or the range of cell sizes. Poikilocytosis suggests a defect in the maturation of red cell precursors in the bone marrow or fragmentation of circulating red cells. The blood smear may also reveal *polychromasia* -- red cells that are slightly larger than normal and grayish blue in color on the Wright-Giemsa stain. These cells are reticulocytes that have been prematurely released from the bone marrow, and their color represents residual amounts of ribosomal RNA. These cells appear in circulation in response to [EPO](#) stimulation or to architectural damage of the bone marrow (fibrosis, infiltration of the marrow by malignant cells, etc.) that results in their disordered release from the marrow. The appearance of nucleated red cells, Howell-Jolly bodies, target cells, sickle cells, and others may provide clues to specific disorders (see [Plates V-2, V-3, V-8, V-9, V-16, V-21, V-24, V-26, V-27, V-28, and V-39](#)).

RETICULOCYTE COUNT An accurate reticulocyte count is key to the initial

classification of anemia. Normally, reticulocytes are red cells that have been recently released from the bone marrow. They are identified by staining with a supravital dye that precipitates the residual ribosomal RNA. These precipitates appear as blue or black punctate spots. This residual RNA is metabolized over the first 24 to 36 h of the reticulocyte's lifespan in circulation. Normally, the reticulocyte count ranges from 1 to 2% and reflects the daily replacement of 0.8 to 1.0% of the circulating red cell population. A correctly interpreted reticulocyte count provides a reliable measure of red cell production.

In the initial classification of anemia, the patient's reticulocyte count is compared with the expected reticulocyte response. In general, if the [EPO](#) and erythroid marrow responses to moderate anemia [hemoglobin < 100 g/L (10 g/dL)] are intact, the red cell production rate increases to two to three times normal within 10 days following the onset of anemia. In the face of established anemia, a reticulocyte response less than two to three times normal indicates an inadequate marrow response.

In order to use the reticulocyte count to estimate marrow response, two corrections are necessary. The first correction adjusts the reticulocyte count based on the reduced number of circulating red cells. With anemia, the percentage of reticulocytes may be increased while the absolute number is unchanged. To correct for this effect, the reticulocyte percentage is multiplied by the ratio of the patient's hemoglobin or hematocrit to the expected hemoglobin/hematocrit for the age and gender of the patient ([Table 61-3](#)). This provides an estimate of the absolute reticulocyte count. In order to convert the corrected reticulocyte count to an index of marrow production, a further correction is required, depending on whether some of the reticulocytes in circulation have been released from the marrow prematurely. For this second correction, the peripheral blood smear is examined to see if there are polychromatophilic macrocytes present. These cells, representing prematurely released reticulocytes, are referred to as "shift" cells, and the relationship between the degree of shift (and the necessary shift correction factor) is shown in [Fig. 61-3](#). The correction is necessary because these prematurely released cells survive as reticulocytes in circulation for >1 day, thereby providing a falsely high estimate of daily red cell production. If polychromasia is increased, the reticulocyte count, already corrected for anemia, should be divided again by a factor of 2 to account for the prolonged reticulocyte maturation time. The second correction factor varies from 1 to 3 depending upon the severity of anemia. In general, a correction of 2 is commonly used. An appropriate correction is shown in [Table 61-3](#). If polychromatophilic cells are not seen on the blood smear, the second correction is not required. The now doubly corrected reticulocyte count is the *reticulocyte production index*, and it provides an estimate of marrow production relative to normal.

Premature release of reticulocytes is normally due to increased [EPO](#) stimulation. However, if the integrity of the bone marrow release process is lost through tumor infiltration, fibrosis, or other disorders, the appearance of nucleated red cells or polychromatophilic macrocytes should still invoke the second reticulocyte correction. The shift correction should always be applied to a patient with anemia and a very high reticulocyte count to provide a true index of effective red cell production. Patients with severe chronic hemolytic anemia may increase red cell production as much as six- to sevenfold. This measure alone, therefore, confirms the fact that the patient has an appropriate EPO response, a normally functioning bone marrow, and sufficient iron

available to meet the demands for new red cell formation. If the reticulocyte production index is <2 in the face of established anemia, a defect in erythroid marrow proliferation or maturation must be present.

TESTS OF IRON SUPPLY AND STORAGE The laboratory measurements that reflect the availability of iron for hemoglobin synthesis include the serum iron, the [TIBC](#), and the percent transferrin saturation. The percent transferrin saturation is derived by dividing the serum iron level ($\times 100$) by the TIBC. The normal serum iron ranges from 9 to 27 $\mu\text{mol/L}$ (50 to 150 $\mu\text{g/dL}$), while the normal TIBC is 54 to 64 $\mu\text{mol/L}$ (300 to 360 $\mu\text{g/dL}$); the transferrin saturation ranges from 25 to 50%. A diurnal variation in the serum iron leads to a variation in the percent transferrin saturation. The serum ferritin is used to evaluate total-body iron stores. Adult males have serum ferritin levels that average about 100 $\mu\text{g/L}$, corresponding to iron stores of about 1 g. Adult females have lower serum ferritin levels averaging 30 $\mu\text{g/L}$, reflecting lower iron stores. A serum ferritin level of 10 to 15 $\mu\text{g/L}$ represents depletion of body iron stores. However, ferritin is also an acute-phase reactant and, in the presence of acute or chronic inflammation, may rise severalfold above baseline levels. As a rule, a serum ferritin $>200 \mu\text{g/L}$ means there is at least some iron in tissue stores.

BONE MARROW EXAMINATION A bone marrow aspirate and smear or a needle biopsy may be useful in the diagnosis of a marrow disorder such as myelofibrosis, a red cell maturation defect, or an infiltrative disease ([Plates V-5, V-13, V-14, V-15, V-19, V-29, and V-33](#)). The increase or decrease of one cell lineage (myeloid vs. erythroid) compared to another is obtained by a differential count of nucleated cells in a bone marrow smear [the erythroid/granulocytic (E/G) ratio]. A patient with a hypoproliferative anemia (see below) and a reticulocyte production index <2 will demonstrate an E/G ratio of 1:2 or 1:3. In contrast, patients with hemolytic disease and a production index >3 will have an E/G ratio of at least 1:1. Maturation disorders are identified from the discrepancy between a high E/G ratio and a low reticulocyte production index (see below). Either the marrow smear or biopsy can be stained for the presence of iron stores or iron in developing red cells. The storage iron is in the form of *ferritin* or *hemosiderin*. On carefully prepared bone marrow smears, small ferritin granules can normally be seen in 10 to 20% of developing erythroblasts. Such cells are called *sideroblasts*.

OTHER LABORATORY MEASUREMENTS

Additional laboratory tests may be of value in confirming specific diagnoses. **For details of these tests and how they are applied in individual disorders, see [Chaps. 105 to 109](#).*

DEFINITION AND CLASSIFICATION OF ANEMIA

Initial Classification of Anemia Classifying an anemia according to the functional defect in red cell production helps organize the subsequent use of laboratory studies. The three major classes of anemia are: (1) marrow production defects (*hypoproliferation*), (2) red cell maturation defects (*ineffective erythropoiesis*), and (3) decreased red cell survival (*blood loss/hemolysis*). This functional classification of anemia then guides the selection of specific clinical and laboratory studies designed to complete the differential diagnosis and to plan appropriate therapy. The classification is

shown in [Fig. 61-4](#). A hypoproliferative anemia is typically seen with a low reticulocyte production index together with little or no change in red cell morphology (a normocytic, normochromic anemia) ([Chap. 105](#)). Maturation disorders typically have a slight to moderately elevated reticulocyte production index that is accompanied by either macrocytic ([Chap. 107](#)) or microcytic ([Chaps. 105,106](#)) red cell indices. Increased red blood cell destruction secondary to hemolysis results in an increase in the reticulocyte production index to at least three times normal ([Chap. 108](#)), provided sufficient iron is available for hemoglobin synthesis. Hemorrhagic anemia does not typically result in production indices of more than 2.5 times normal because of the limitations placed on expansion of the erythroid marrow by iron availability.

In the first branch point of the classification of anemia, a reticulocyte production index >2.5 indicates that hemolysis is most likely. A reticulocyte production index of <2 indicates either a hypoproliferative anemia or maturation disorder. The latter two possibilities can often be distinguished by the red cell indices, by examination of the peripheral blood smear, or by a marrow examination. If the red cell indices are normal, the anemia is almost certainly hypoproliferative in nature. Maturation disorders are characterized by ineffective red cell production and a low reticulocyte production index with bizarre red cell shapes -- macrocytes or hypochromic microcytes on the peripheral blood smear. With a hypoproliferative anemia, no erythroid hyperplasia is noted in the marrow, whereas patients with ineffective red cell production have erythroid hyperplasia and an E/G ratio³1:1.

Hypoproliferative Anemias At least 75% of all cases of anemia are hypoproliferative in nature. A hypoproliferative anemia reflects absolute or relative marrow failure in which the erythroid marrow has not proliferated appropriately for the degree of anemia. The majority of hypoproliferative anemias are due to mild to moderate iron deficiency or inflammation. A hypoproliferative anemia can result from marrow damage, iron deficiency, or inadequate [EPO](#) stimulation. The last may reflect impaired renal function, suppression of EPO production by inflammatory cytokines such as interleukin 1, or reduced tissue needs for O₂ from metabolic disease such as hypothyroidism. Only occasionally is the marrow unable to produce red cells at a normal rate, and this is most prevalent in patients with renal failure. In general, hypoproliferative anemias are characterized by normocytic, normochromic red cells, although microcytic, hypochromic cells may be observed with mild iron deficiency or long-standing chronic inflammatory disease. The key laboratory tests in distinguishing between the various forms of hypoproliferative anemia include the serum iron and iron-binding capacity, evaluation of renal and thyroid function, a marrow biopsy or aspirate to detect marrow damage or infiltrative disease, and serum ferritin to assess iron stores. Occasionally, an iron stain of the marrow will be needed to determine the pattern of iron distribution. Patients with the anemia of acute or chronic inflammation show a distinctive pattern of serum iron (low), [TIBC](#) (normal or low), percent transferrin saturation (low), and serum ferritin (normal or high). A distinct pattern of results is noted in mild to moderate iron deficiency (low serum iron, high TIBC, low percent transferrin saturation, low serum ferritin) ([Chap. 105](#)). Marrow damage by a drug, infiltrative disease such as leukemia or lymphoma, or marrow aplasia can usually be diagnosed from the peripheral blood and bone marrow morphology. With infiltrative disease or fibrosis, a marrow biopsy will likely be required.

Maturation Disorders The presence of anemia with an inappropriately low reticulocyte

production index, macro- or microcytosis on smear, and abnormal red cell indices suggests a maturation disorder. Maturation disorders are divided into two categories: nuclear maturation defects, associated with macrocytosis and abnormal marrow development, and cytoplasmic maturation defects, associated with microcytosis and hypochromia usually from defects in hemoglobin production. The low reticulocyte production index is a reflection of the ineffective erythropoiesis that results from the destruction within the marrow of developing erythroblasts. Marrow morphology shows an E/G ratio of $\approx 1:1$, diagnostic of erythroid hyperplasia.

Nuclear maturation defects result from vitamin B₁₂ or folic acid deficiency, drug damage, or myelodysplasia. Drugs that interfere with cellular DNA metabolism, such as methotrexate or alkylating agents, can produce a nuclear maturation defect. Alcohol, alone, is also capable of producing macrocytosis and a variable degree of anemia, but this is usually associated with coincident folic acid deficiency. Measurements of folic acid and vitamin B₁₂ are key not only in identifying the specific vitamin deficiency but also because they reflect different pathogenetic mechanisms.

Cytoplasmic maturation defects result from severe iron deficiency or abnormalities in globin or heme synthesis. Iron deficiency occupies an unusual position in the classification of anemia. If the iron-deficiency anemia is mild to moderate, erythroid marrow proliferation is decreased and the anemia is classified as hypoproliferative. However, if the anemia is severe and prolonged, the erythroid marrow will become hyperplastic despite the inadequate iron supply, and the anemia will be classified as ineffective erythropoiesis with a cytoplasmic maturation defect. In either case, a reduced reticulocyte production index, microcytosis, and a classic pattern of iron values make the diagnosis clear and easily distinguish iron deficiency from other cytoplasmic maturation defects such as the thalassemias. Defects in heme synthesis, in contrast to globin synthesis, are less common and may be acquired or inherited ([Chap. 346](#)). Acquired abnormalities are usually associated with myelodysplasia, may lead to either a macro- or microcytic anemia, and are frequently associated with mitochondrial iron loading. In these cases, iron is taken up by the mitochondria of the developing erythroid cell but not incorporated into heme. The iron-encrusted mitochondria surround the nucleus of the erythroid cell, forming a ring. Based on the distinctive finding of so-called ringed sideroblasts on the marrow iron stain (see [Plate V-37](#)), patients are diagnosed as having a sideroblastic anemia -- almost always reflecting myelodysplasia. Again, studies of iron parameters are helpful in the differential diagnosis and management of these patients.

Blood Loss/Hemolytic Anemia In contrast to anemias associated with an inappropriately low reticulocyte production index, blood loss or hemolysis is associated with red cell production indices of ≥ 2.5 times normal. The stimulated erythropoiesis is reflected in the blood smear by the appearance of increased numbers of polychromatophilic macrocytes. A marrow examination is rarely indicated if the reticulocyte production index is increased appropriately. The red cell indices are typically normocytic or slightly macrocytic, reflecting the increased number of reticulocytes. Acute blood loss is not associated with an increased reticulocyte production index because of the time required to increase [EPO](#) production and, subsequently, marrow proliferation. Subacute blood loss may be associated with modest reticulocytosis because iron is lost along with the red cells. Anemia from chronic

blood loss more often presents as iron deficiency than with the picture of increased red cell production.

The evaluation of blood loss anemia is usually not difficult. Most problems arise when a patient presents with an increased red cell production index from an episode of acute blood loss that went unrecognized. The cause of the anemia and increased red cell production may not be obvious. The confirmation of a recovering state may require observations over a period of 2 to 3 weeks, during which the hemoglobin concentration will be seen to rise and the reticulocyte production index fall.

Hemolytic disease, while dramatic, is among the least common forms of anemia. The ability to sustain a high reticulocyte production index reflects the ability of the erythroid marrow to compensate for hemolysis and the efficient recycling of iron from the destroyed red cells to support new hemoglobin synthesis. The level of response will depend on the severity of the anemia and the nature of the underlying disease process.

Hemolytic anemias present in different ways. Some appear suddenly as an acute, self-limited episode of intravascular or extravascular hemolysis, a presentation pattern often seen in patients with autoimmune hemolysis or with inherited defects of the Embden-Myerhof pathway or the glutathione reductase pathway. Patients with inherited disorders of the hemoglobin molecule or red cell membrane generally have a lifelong clinical history typical of the disease process. Those with chronic hemolytic disease, such as hereditary spherocytosis, may actually present not with anemia but with a complication stemming from the prolonged increase in red cell destruction such as aplastic crisis, symptomatic bilirubin gallstones, or splenomegaly.

The differential diagnosis of an acute or chronic hemolytic event requires the careful integration of family history, pattern of clinical presentation, and a number of highly specific laboratory studies ([Chap. 108](#)). Some of the more common congenital hemolytic anemias may be identified from the red cell morphology, a routine laboratory test such as hemoglobin electrophoresis, or a screen for red cell enzymes. Acquired defects in red cell survival are often immunologically mediated and require the immunoglobulin test or a cold agglutinin titer to detect the presence of hemolytic antibodies or complement-mediated red cell destruction.

TREATMENT

An overriding principle is to not initiate treatment of mild to moderate anemia without a specific diagnosis. Rarely, in the acute setting, anemia may be so severe that red cell transfusions are required before a specific diagnosis is made. Whether the anemia is of acute or gradual onset, the selection of the appropriate treatment is determined by the documented cause(s) of the anemia. Often, the cause of the anemia may be multifactorial. For example, a patient with severe rheumatoid arthritis who has been taking anti-inflammatory drugs may have a hypoproliferative anemia associated with chronic inflammation as well as chronic blood loss associated with intermittent gastrointestinal bleeding. In every circumstance, it is important to evaluate the patient's iron status fully before and during the treatment of any anemia. **Transfusion is discussed in [Chap. 114](#); iron therapy is discussed in [Chap. 105](#); treatment of megaloblastic anemia is discussed in [Chap. 107](#); treatment of other entities is discussed*

in their respective chapters (sickle cell anemia, [Chap. 106](#); hemolytic anemias, [Chap. 108](#); aplastic anemia and myelodysplasia, [Chap. 109](#)).

Therapeutic options for the treatment of anemias have expanded dramatically during the past 25 years. Blood component therapy is available and safe. Recombinant [EPO](#) as an adjunct to anemia management has transformed the lives of patients with chronic renal failure on dialysis. Improvements in the management of sickle cell crises and sickle cell anemia have also occurred. Eventually, patients with inherited disorders of globin synthesis or mutations in the globin gene, such as sickle cell disease, may benefit from the successful introduction of targeted genetic therapy ([Chap. 69](#)).

POLYCYTHEMIA

Polycythemia is defined as an increase in circulating red blood cells above normal. This increase may be real or only apparent (spurious or relative) because of a decrease in plasma volume. The term *erythrocytosis* may be used interchangeably with polycythemia, but some draw a distinction between them; erythrocytosis implies documentation of increased red cell mass, whereas polycythemia refers to any increase in red cells. Often patients with polycythemia are detected through an incidental finding of elevated hemoglobin or hematocrit levels. Concern that the hemoglobin level may be abnormally high is usually triggered at 170 g/L (17 g/dL) for men and 150 g/L (15 g/dL) for women. Hematocrit levels >50% in men or >45% in women may be abnormal. Hematocrits >60% in men and >55% in women are almost invariably associated with increased red cell mass.

Historic features useful in the differential diagnosis include smoking history; living at high altitude; or a history of congenital heart disease, peptic ulcer disease, sleep-apnea, chronic lung disease, or renal disease.

Patients with polycythemia may be asymptomatic or experience symptoms related to the increased red cell mass or an underlying disease process that leads to increased red cell production. The dominant symptoms from increased red cell mass are thrombotic (both venous and arterial), because the blood viscosity increases logarithmically at hematocrits >55%. Manifestations range from digital ischemia to Budd-Chiari syndrome with hepatic vein thrombosis. Abdominal thromboses are particularly common. Neurologic symptoms such as vertigo, tinnitus, headache, and visual disturbances may occur. Hypertension is often present. Patients with *polycythemia vera* may have aquagenic pruritus and symptoms related to hepatosplenomegaly. Patients may have easy bruising, epistaxis, or bleeding from the gastrointestinal tract. Patients with hypoxemia may develop cyanosis on minimal exertion or have headache, impaired mental acuity, and fatigue.

The physical examination usually reveals a ruddy complexion. Splenomegaly favors polycythemia vera as the diagnosis ([Chap. 110](#)). The presence of cyanosis or evidence of a right-to-left shunt suggests congenital heart disease presenting in the adult, particularly tetralogy of Fallot or Eisenmenger syndrome ([Chap. 234](#)). Increased blood viscosity raises pulmonary artery pressure; hypoxemia can lead to increased pulmonary vascular resistance. Together these factors can produce cor pulmonale.

Polycythemia can be spurious (related to a decrease in plasma volume; Gaisbock's syndrome), primary, or secondary in origin. The secondary causes are all associated with increases in [EPO](#) levels: either a physiologically adapted appropriate elevation based upon tissue hypoxia (lung disease, high altitude, CO poisoning, high-affinity hemoglobinopathy) or an abnormal overproduction (renal disease, tumors with ectopic EPO production). A rare familial form of polycythemia is associated with normal EPO levels but mutations producing hyperresponsive EPO receptors.

Approach to the Patient

As shown in [Fig. 61-5](#), the first step is to document the presence of an increased red cell mass using the principle of isotope dilution by administering ^{51}Cr -labeled autologous red blood cells to the patient and sampling blood radioactivity over a 2-h period. If the red cell mass is normal (<36 mL/kg in men, <32 mL/kg in women), the patient has spurious polycythemia. If the red cell mass is increased (>36 mL/kg in men, >32 mL/kg in women), serum [EPO](#) levels should be measured. If EPO levels are low or absent, the patient most likely has polycythemia vera. Ancillary tests that support this diagnosis include elevated white blood cell count, increased absolute basophil count, thrombocytosis, elevated leukocyte alkaline phosphatase levels, and elevated serum vitamin B₁₂ and vitamin B₁₂-binding protein levels.

If serum [EPO](#) levels are elevated, one attempts to distinguish whether the elevation is a physiologic response to hypoxia or is related to autonomous production. Patients with low arterial O₂ saturation ($<92\%$) should be further evaluated for the presence of heart or lung disease, if they are not living at high altitude. Patients with normal O₂ saturation who are smokers may have elevated EPO levels because of CO displacement of O₂. If carboxyhemoglobin (COHb) levels are high, the diagnosis is smoker's polycythemia. Such patients should be urged to stop smoking. Those who cannot stop smoking require phlebotomy to control their polycythemia. Patients with normal O₂ saturation who do not smoke either have an abnormal hemoglobin that does not deliver O₂ to the tissues (evaluated by finding elevated O₂-hemoglobin affinity) or have a source of EPO production that is not responding to the normal feedback inhibition. Further workup is dictated by the differential diagnosis of EPO-producing neoplasms. Hepatoma, uterine leiomyoma, and renal disease or cysts are all detectable with abdominopelvic computed tomography scans. Cerebellar hemangiomas may produce EPO, but they nearly always present with localizing neurologic signs and symptoms rather than polycythemia-related symptoms.

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Dr. Robert S. Hillman wrote this chapter in the 14th edition, and elements of his chapter were retained here.

(Bibliography omitted in Palm version)

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62. BLEEDING AND THROMBOSIS - Robert I. Handin

Hemorrhage, intravascular thrombosis, and embolism are common clinical manifestations of many diseases. The normal hemostatic system limits blood loss by precisely regulated interactions between components of the vessel wall, blood platelets, and plasma proteins. However, when disease or trauma damages large arteries and veins, excessive bleeding may occur, despite a normal hemostatic system. Less frequently, hemorrhage is caused by an inherited or acquired disorder of the hemostatic machinery itself. A large number of such bleeding disorders have been identified.

In addition, unregulated activation of the hemostatic system may cause thrombosis and embolism, which can reduce blood flow to critical organs such as the brain and myocardium. Although we understand less about the pathophysiology of thrombosis than of hemostatic failure, certain patient groups have been identified that are particularly prone to thrombosis and embolism. These include patients who (1) are immobilized after surgery, (2) have chronic congestive heart failure, (3) have atherosclerotic vascular disease, (4) have a malignancy, or (5) are pregnant. Most of these "thrombosis-prone" patients have inherited or acquired "hypercoagulable" or "prethrombotic" disorders.

Certain information in the patient's history, such as the mode of onset and sites of bleeding, a family bleeding tendency, and a record of drug ingestion, helps establish the correct diagnosis. Physical examination can identify bleeding in the skin or joint deformities due to previous hemarthroses. Ultimately, however, bleeding disorders are diagnosed by laboratory tests. General screening tests are used first, to document a systemic disorder, and are then supplemented by specific tests of coagulation protein or platelet function to arrive at an accurate diagnosis.

The hypercoagulable or prethrombotic patient can also be identified by a careful history. Three important clues to this diagnosis are: (1) repeated episodes of thromboembolism without an obvious predisposing condition, (2) a family history of thrombosis, and (3) well-documented thromboembolism in adolescents and young adults. All of the known inherited prethrombotic disorders can be diagnosed with specific immunologic, functional, and, in some cases, genetic tests.

NORMAL HEMOSTASIS

Accurate diagnosis and treatment of patients with either bleeding or thrombosis require knowledge of the pathophysiology of hemostasis. The process can be divided into primary and secondary components and is initiated when trauma, surgery, or disease disrupts the vascular endothelial lining and blood is exposed to subendothelial connective tissue. *Primary hemostasis* is the name given to the process of platelet plug formation at sites of injury. It occurs within seconds of injury and is of prime importance in stopping blood loss from capillaries, small arterioles, and venules ([Fig. 62-1](#)).

Secondary hemostasis consists of the reactions of the plasma coagulation system that result in fibrin formation. It requires several minutes for completion. The fibrin strands that are produced strengthen the primary hemostatic plug. This reaction is particularly important in larger vessels and prevents bleeding from recurring hours or days after the injury. Although presented here as separate events, primary and secondary hemostasis

are closely linked. For example, activated platelets accelerate plasma coagulation, and products of the plasma coagulation reaction, such as thrombin, induce platelet activation.

Effective primary hemostasis requires three critical events -- platelet adhesion, granule release, and platelet aggregation. Within a few seconds of injury, platelets adhere to collagen fibrils in vascular subendothelium via a specific platelet collagen receptor, glycoprotein Ia/IIa, which is a member of the integrin family. As shown in [Fig. 62-2](#), this interaction is stabilized by the von Willebrand factor, an adhesive glycoprotein that allows platelets to remain attached to the vessel wall despite the high shear forces generated within the vascular lumen. The von Willebrand factor accomplishes this task by forming a link between a platelet receptor site on glycoprotein Ib/IX and subendothelial collagen fibrils. The adherent platelets then release preformed granule constituents and generate de novo mediators like those depicted in [Fig. 62-1](#).

As in other cells, platelet activation and secretion are regulated by changes in the level of cyclic nucleotides, the influx of calcium, hydrolysis of membrane phospholipids, and phosphorylation of critical intracellular proteins. The relevant pathways are depicted in [Figs. 62-3](#) and [62-4](#). The binding of agonists such as epinephrine, collagen, or thrombin to platelet surface receptors activates two membrane enzymes -- phospholipase C and phospholipase A₂. These enzymes catalyze the release of arachidonic acid from two of the major membrane phospholipids, phosphatidylinositol and phosphatidylcholine. Initially, a small quantity of the released arachidonic acid is converted to thromboxane A₂ (TXA₂), which, in turn, can activate phospholipase C. The formation of TXA₂ from arachidonic acid is mediated by the enzyme cyclooxygenase ([Fig. 62-3](#)). This enzyme is inhibited by aspirin and nonsteroidal anti-inflammatory drugs. Inhibition of TXA₂ synthesis is a cause of mild bleeding in some patients and is the same way some antithrombotic drugs work.

Hydrolysis of the membrane phospholipid phosphatidylinositol 4,5-bisphosphate produces diacylglycerol (DAG) and inositol triphosphate (IP₃), both of which play critical roles in platelet metabolism. IP₃ mediates the movement of calcium into the platelet cytosol and stimulates the phosphorylation of myosin light chains. The latter interact with actin to facilitate granule movement and platelet shape change. DAG activates protein kinase C, which, in turn, phosphorylates several substrates, including myosin light chain kinase and a 47-kDa protein (plekstrin). Phosphorylation of these or other proteins may regulate platelet granule secretion.

A finely balanced mechanism controls the rate and extent of platelet activation ([Fig. 62-3](#)). TXA₂, a platelet product of arachidonic acid, stimulates platelet activation and secretion. In contrast, prostacyclin, an endothelial cell product of arachidonic acid metabolism, inhibits platelet activation by raising intraplatelet levels of cyclic adenosine monophosphate.

Following activation, platelets secrete their granule contents into plasma. Endoglycosidases and a heparin-cleaving enzyme are released from lysosomes; calcium, serotonin, and adenosine diphosphate (ADP) are released from the dense granules; and several proteins, including the von Willebrand factor, fibronectin, thrombospondin, the platelet-derived growth factor (PDGF), and a heparin-neutralizing

protein (platelet factor 4), are released from granules. Released ADP binds to purinergic receptors, which, when activated, change the conformation of the glycoprotein IIb/IIIa complex so that it binds fibrinogen, linking adjacent platelets into a hemostatic plug ([Fig. 62-2](#)). Released PDGF stimulates the growth and migration of fibroblasts and smooth-muscle cells within the vessel wall, an important part of the repair process.

As the primary hemostatic plug is being formed, plasma coagulation proteins are activated to initiate secondary hemostasis. An overall picture of the coagulation scheme, including the role of various inhibitors, is shown in [Fig. 62-5](#). The coagulation pathway can be broken down into a series of reactions ([Fig. 62-6](#)) that culminate in the production of enough thrombin to convert a small amount of plasma fibrinogen to fibrin. Each of the reactions requires the formation of a surface-bound complex and the conversion of inactive precursor proteins into active proteases by limited proteolysis, and each is regulated by both plasma and cellular cofactors and calcium.

In *reaction 1*, the intrinsic or contact phase of coagulation, three plasma proteins, Hageman factor (factor XII), high-molecular-weight kininogen (HMWK), and prekallikrein (PK), form a complex on vascular subendothelial collagen. After binding to HMWK, factor XII is slowly converted to an active protease (XIIa), which then activates PK to kallikrein and factor XI to XIa. Kallikrein accelerates the conversion of XII to XIIa, while XIa participates in subsequent coagulation reactions. An alternative mechanism for the activation of factor XI may exist, as patients who are deficient in either factor XII, HMWK, or PK have apparently normal hemostasis and no clinical bleeding.

Reaction 2 provides a second pathway to initiate coagulation by activating factor VII to a protease. In this extrinsic or tissue-factor-dependent pathway, a complex is formed between factor VII, calcium, and tissue factor, a ubiquitous lipoprotein present in cellular membranes and exposed by cellular injury. The tissue factor-VII pathway is continuously active and makes a major contribution to basal coagulation. Factor VII and three other coagulation proteins -- factors II (prothrombin), IX, and X -- require calcium and vitamin K for biologic activity. These proteins are synthesized in the liver, where a vitamin K-dependent carboxylase catalyzes a unique posttranslational modification that adds a second carboxyl group to certain glutamic acid residues. Pairs of these di-g-carboxyglutamic acid (Gla) residues bind calcium, which alters protein conformation for binding to phospholipid surfaces and confers biologic activity. Inhibition of this modification by vitamin K antagonists (e.g., warfarin) is the basis of one of the most common forms of anticoagulant therapy.

In *reaction 3*, factor X is activated by the proteases generated in the two previous reactions. In one reaction, a calcium- and lipid-dependent complex is formed between factors VIII, IX, and X. Within this complex, factor IX is first converted to IXa by factor XIa that was generated within the intrinsic pathway (*reaction 1*). Factor X is then activated by factor IXa in concert with factor VIII. Alternatively, both factors IX and X can be activated more directly by factor VIIa, generated via the extrinsic pathway (*reaction 2*). Activation of factors IX and X provides a link between the intrinsic and extrinsic coagulation pathways ([Fig. 62-5](#)).

Reaction 4, the final step, converts prothrombin to thrombin in the presence of factor V,

calcium, and phospholipid. Although prothrombin conversion can take place on various natural and artificial phospholipid-rich surfaces, it proceeds several thousand times faster on the surface of activated platelets or endothelial cells. Thrombin has multiple functions in hemostasis. Although its principal role in hemostasis is the conversion of fibrinogen to fibrin, it also activates factors V, VIII, and XIII and stimulates platelet aggregation and secretion. Following the release of fibrinopeptides A and B from the α and β chains of fibrinogen, the modified molecule, now called *fibrin monomer*, polymerizes into an insoluble gel. The fibrin polymer is then stabilized by the cross-linking of individual chains by factor XIIIa, a plasma transglutaminase ([Fig. 62-5](#)).

Although the classic view of coagulation had clinical utility, it left several important questions unanswered: (1) Why does factor XII deficiency dramatically prolong partial thromboplastin time (PTT) but not cause bleeding? (2) Why is there heterogeneity in the bleeding symptoms of patients with factor XI deficiency? (3) Why do deficiencies in factors VIII or IX produce such dramatic bleeding even though the "extrinsic" pathway remains intact? Activation of factors IX and X by the tissue factor-VIIa complex is thought to play a major role in the initiation of hemostasis. Once coagulation is initiated by this interaction, the tissue factor pathway inhibitor (TFPI) blocks the pathway, and elements of the intrinsic pathway, particularly factors VIII and IX, become the dominant regulators of thrombin generation. This step in the pathway explains why factor XII-deficient patients are asymptomatic and why factor XI-deficient patients have only a mild to moderate bleeding diathesis ([Fig. 62-7](#)).

Clot lysis and vessel repair begin immediately after the formation of the definitive hemostatic plug. Three potential activators of the fibrinolytic system are: Hageman factor fragments, urinary plasminogen activator (uPA) or urokinase, and tissue plasminogen activator (tPA). The principal physiologic activators, tPA and uPA, diffuse from endothelial cells and convert plasminogen, adsorbed to the fibrin clot, into plasmin ([Fig. 62-8](#)). Plasmin then degrades fibrin polymer into small fragments, which are cleared by the monocyte-macrophage scavenger system. Although plasmin can also degrade fibrinogen, the reaction remains localized because (1) tPA and some forms of uPA activate plasminogen more effectively when it is adsorbed to fibrin clots; (2) any plasmin that enters the circulation is rapidly bound and neutralized by the α_2 plasmin inhibitor (patients who lack this factor have unchecked fibrinolysis and bleed); and (3) endothelial cells release a plasminogen activator inhibitor (PAI-1), which blocks the action of tPA.

Only a small quantity of each coagulation enzyme is converted to its active form. As a consequence, the hemostatic plug does not propagate beyond the site of injury. Precise regulation is important, since each milliliter of blood contains enough clotting potential to clot all the fibrinogen in the body in 10 to 15 s. Blood fluidity is maintained by the flow of blood, the adsorption of coagulation factors to surfaces and their trapping in the emerging clot, and by multiple inhibitors in plasma. Antithrombin, proteins C and S, and [TFPI](#) are important inhibitors that maintain blood fluidity.

These inhibitors have distinct modes of action. Antithrombin forms complexes with all serine protease coagulation factors except factor VII ([Fig. 62-5](#)). Rates of complex formation are accelerated by heparin and heparin-like molecules on the surface of the endothelial cells. Heparin's ability to accelerate antithrombin activity is the basis for its

anticoagulant action. Protein C is converted to an active protease by thrombin after it is bound to an endothelial cell protein called *thrombomodulin*. Activated protein C then inactivates the two plasma cofactors V and VIII by limited proteolysis, which slows down two critical coagulation reactions. Protein C may also stimulate the release of [tPA](#) from endothelial cells. The inhibitory function of protein C is enhanced by protein S. Reduced levels of antithrombin or proteins C and S, or dysfunctional forms of these molecules, result in a hypercoagulable or prethrombotic state. In addition, a particularly common heritable defect associated with a hypercoagulable state is the presence of a form of factor V (factor V Leiden) that is resistant to protein C inhibition. Between 20 and 50% of patients with unexplained venous thromboembolism have this defect.

Blood coagulation is not uniform throughout the body. The composition of the blood clot varies with the site of injury. Hemostatic plugs or thrombi that form in veins where blood flow is slow are rich in fibrin and trapped red blood cells and contain relatively few platelets. They are often called *red thrombi* because of their appearance in surgical and pathologic specimens. The friable ends of these red thrombi, which most often form in leg veins, can break off and embolize to the pulmonary circulation. Conversely, clots that form in arteries under conditions of high flow are predominantly composed of platelets and have little fibrin. These *white thrombi* may readily dislodge from the arterial wall and embolize to distant sites, causing temporary or permanent ischemia. These clots are a particularly common cause of embolism in the cerebral and retinal circulation, where they may lead to transient neurologic dysfunction (transient ischemic attacks), including temporary monocular blindness (amaurosis fugax), or to strokes. In addition, most episodes of myocardial infarction are due to thrombi that form after the rupture of atherosclerotic plaques within diseased coronary arteries. Hemostatic plugs, which are a physiologic response to injury, are very similar to pathologic thrombi. Thrombosis has been described as coagulation occurring in the wrong place or at the wrong time.

CLINICAL EVALUATION

HISTORY

Certain elements of the history are particularly useful in determining whether bleeding is caused by an underlying hemostatic disorder or by a local anatomic defect. One clue is a history of bleeding following common hemostatic stresses such as dental extraction, childbirth, or minor surgery. Bleeding that is sufficiently severe to require a blood transfusion merits special attention. A family history of bleeding and bleeding from multiple sites that cannot be linked to trauma or surgery also suggest a systemic disorder. Since bleeding can be mild, lack of a family history of bleeding does not exclude an inherited hemostatic disorder.

Bleeding from a platelet disorder is usually localized to superficial sites such as the skin and mucous membranes, comes on immediately after trauma or surgery, and is readily controlled by local measures ([Table 62-1](#)). In contrast, bleeding from secondary hemostatic or plasma coagulation defects occurs hours or days after injury and is unaffected by local therapy. Such bleeding most often occurs in deep subcutaneous tissues, muscles, joints, or body cavities. A careful and thorough history may establish the presence of a hemostatic disorder and guide initial laboratory testing.

PHYSICAL EXAMINATION

The most common site to observe bleeding is in the skin and mucous membranes. Collections of blood in the skin are called *purpura* and may be subdivided on the basis of the site of bleeding in the skin. Small pinpoint hemorrhages into the dermis due to the leakage of red cells through capillaries are called *petechiae* and are characteristic of platelet disorders -- in particular, severe thrombocytopenia. Larger subcutaneous collections of blood due to leakage of blood from small arterioles and venules are called *ecchymoses* (common bruises) or, if somewhat deeper and palpable, *hematomas*. They are also common in patients with platelet defects and result from minor trauma. Dilated capillaries, or *telangiectasia*, may cause bleeding without any hemostatic defect. In addition, the loss of connective tissue support for capillaries and small veins that accompanies aging increases the fragility of superficial vessels, such as those on the dorsum of the hand, leading to extravasation of blood into subcutaneous tissue -- *senile purpura*. Menorrhagia is sometimes a serious problem in women with severe thrombocytopenia or platelet dysfunction. Some patients with primary hemostatic defects, especially von Willebrand's disease, may have recurrent gastrointestinal hemorrhage, often associated with angiodysplasia, a common vascular malformation in the gastrointestinal tract.

Bleeding into body cavities, the retroperitoneum, or joints is a common manifestation of plasma coagulation defects. Repeated joint bleeding may cause synovial thickening, chronic inflammation, and fluid collections and may erode articular cartilage and lead to chronic joint deformity and limited mobility. Such deformities are particularly common in deficiencies of factors VIII and IX, the two sex-linked coagulation disorders referred to as the *hemophilias*. For unclear reasons, hemarthroses are much less common in patients with other plasma coagulation defects. Blood collections in various body cavities or soft tissues can cause secondary necrosis of tissues or nerve compression. Retroperitoneal hematomas can cause femoral nerve compression, and large collections of poorly coagulated blood in soft tissues occasionally mimic malignant growths -- the pseudotumor syndrome. Two of the most life-threatening sites of bleeding are in the oropharynx, where bleeding can compromise the airway, and in the central nervous system. Intracerebral hemorrhage is one of the leading causes of death in patients with severe coagulation disorders. Because of their need for plasma and factor concentrates derived from multiple donors, many patients with hemophilia were infected with HIV before effective testing of donors was in place.

LABORATORY TESTS

The most important screening tests of the primary hemostatic system are (1) a *bleeding time* (a sensitive measure of platelet function), and (2) a *platelet count*. The latter correlates well with the propensity to bleed. The normal platelet count is 150,000 to 450,000/uL of blood. As long as the count is >100,000/uL, patients are usually not symptomatic and the bleeding time remains normal. Platelet counts of 50,000 to 100,000/uL cause mild prolongation of the bleeding time; bleeding occurs only from severe trauma or other stress. Patients with platelet counts <50,000/uL have easy bruising, manifested by skin purpura after minor trauma and bleeding after mucous membrane surgery. Patients with a platelet count <20,000/uL have an appreciable

incidence of spontaneous bleeding, usually have petechiae, and may have intracranial or other spontaneous internal bleeding. The major causes of thrombocytopenia are outlined in [Table 62-2](#).

Patients with qualitative platelet abnormalities have a normal platelet count and a prolonged bleeding time ([Table 62-3](#)). The bleeding time is ascertained by making a small, superficial skin incision and timing the duration of blood flow from the wounded area. By careful standardization, bleeding time is a reliable and sensitive test of platelet function. A template or an automated scalpel controls the length and depth of the incision (usually 1 mm deep by 9 mm long), and a sphygmomanometer inflated to 40 mmHg distends the capillary bed of the forearm uniformly. The bleeding time test must be performed by an experienced technician, as small differences in technique have a big effect on outcome. Any patient with a bleeding time >10 min has an increased risk of bleeding, but the risk does not become great until the bleeding time >15 or 20 min. As shown in [Fig. 62-9](#), the relationship between the platelet count and the bleeding time is roughly linear. When a defect in primary hemostasis is uncovered, specialized testing is needed to determine the cause of the platelet dysfunction ([Table 62-3](#)). A precise diagnosis is important in determining the proper treatment. Occasional patients with a strong history of bleeding, particularly those with mild von Willebrand's disease, may have a normal bleeding time when initially tested, owing to cyclical variations in the level of the von Willebrand factor. Repeated testing may be necessary to establish an accurate diagnosis. Bleeding time is not an effective screening test for preoperative patients.

Plasma coagulation function is readily assessed with the [PTT](#), prothrombin time (PT), thrombin time (TT), and quantitative fibrinogen determination ([Fig. 62-5, Table 62-4](#)). The PTT screens the intrinsic limb of the coagulation system and tests for the adequacy of factors XII, [HMWK, PK](#), XI, IX, and VIII. The PT screens the extrinsic or tissue factor-dependent pathway. Both tests also evaluate the common coagulation pathway involving all the reactions that occur after the activation of factor X. Prolongation of the PT and PTT that does not resolve after the addition of normal plasma suggests a coagulation inhibitor. A specific test for the conversion of fibrinogen to fibrin is needed when both the PTT and PT are prolonged -- either a TT or a clottable fibrinogen level can be employed. When abnormalities are noted in any of the screening tests, more specific coagulation factor assays can be ordered to determine the nature of the defect.

Several rare coagulation abnormalities that may be missed as they do not affect these screening tests: factor XIII deficiency, α_2 plasmin inhibitor deficiency, [PAI-1](#) deficiency (PAI-1 is the major inhibitor of plasminogen activators), and Scott's syndrome, a platelet coagulant defect. A test for factor XIII-dependent fibrin cross-linking, such as clot solubility in 5 M urea, should be ordered when the [PT](#) and [PTT](#) are both normal but the history of bleeding is strong. The fibrinolytic system can be assessed by measuring the rate of clot lysis with the euglobulin lysis or whole blood clot lysis tests and by measuring the levels of α_2 plasmin inhibitor and PAI-1. Scott's syndrome can be detected by measuring the serum PT, which assesses the amount of residual prothrombin.

Conditions associated with thrombosis are listed in [Table 62-5](#). Patients suspected of having a hypercoagulable or prethrombotic disorder on the basis of clinical information should be tested with specific assays to screen for the known defects. Currently

available tests can identify 50 to 60% of the cases of familial or recurrent venous thrombosis.

Inhibitor syndromes or circulating anticoagulants are usually due to antibodies that impair coagulation factor activity. They are an infrequent cause of bleeding and require specialized diagnostic testing. Inhibitors are likely when screening test abnormalities cannot be reversed by adding normal plasma to patient plasma. Antibodies against specific coagulation factors may develop in (1) postpartum women, (2) patients with autoimmune disorders such as systemic lupus erythematosus, (3) patients taking drugs such as penicillin and streptomycin, and (4) otherwise healthy elderly individuals. In addition, between 10 to 20% of patients with severe hemophilia who have received multiple plasma infusions develop inhibitory antibodies. Some patients, especially those with systemic lupus erythematosus, may also have a nonspecific form of anticoagulant antibody that interferes with phospholipid binding of coagulation factors and prolongs the [PT](#) and [PTT](#) but does not cause clinical bleeding. The presence of the lupus anticoagulant may increase the risk of thromboembolism and may cause placental infarction, recurrent midtrimester abortion, and venous and arterial thrombosis. The lupus-like anticoagulant is one manifestation of the anticardiolipin antibody syndrome. Patients may have anticardiolipin antibodies that do not prolong the PTT, but patients are still at risk from thrombosis. Occasionally, patients develop inhibitors that are not antibodies. For example, several patients with clinical bleeding have been found to have circulating mucopolysaccharides that have heparin-like activity.

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63. ENLARGEMENT OF LYMPH NODES AND SPLEEN - Patrick H. Henry, Dan L. Longo

This chapter is intended to serve as a guide to the evaluation of patients who present with enlargement of the lymph nodes (*lymphadenopathy*) or the spleen (*splenomegaly*). Lymphadenopathy is a rather common clinical finding in primary care settings, whereas palpable splenomegaly is less so.

LYMPHADENOPATHY

Lymphadenopathy may be an incidental finding in patients being examined for various reasons or it may be a presenting sign or symptom of the patient's illness. The physician must eventually decide whether the lymphadenopathy is a normal finding or one that requires further study, up to and including biopsy. Soft, flat, submandibular nodes (<1 cm) are often palpable in healthy children and young adults, and healthy adults may have palpable inguinal nodes of up to 2 cm, which are considered normal. Further evaluation of these normal nodes is not warranted. In contrast, if the physician believes the node(s) to be abnormal, then pursuit of a more precise diagnosis is needed.

Approach to the Patient

Lymphadenopathy may be a primary or secondary manifestation of numerous disorders, as shown in [Table 63-1](#). Many of these disorders are infrequent causes of lymphadenopathy. Analysis of lymphadenopathy in primary care practice has shown that more than two-thirds of patients have nonspecific causes or upper respiratory illnesses (viral or bacterial), and fewer than 1% have a malignancy. In one study, researchers reported that 186 of 220 patients (84%) referred for evaluation of lymphadenopathy had a "benign" diagnosis. The remaining 34 patients (16%) had a malignancy (lymphoma or metastatic adenocarcinoma). Sixty-three percent (112) of the 186 patients with benign lymphadenopathy had a nonspecific or reactive etiology (no causative agent found), and the remainder had a specific cause demonstrated, most commonly infectious mononucleosis, toxoplasmosis, or tuberculosis. Thus, the vast majority of patients with lymphadenopathy will have a nonspecific etiology requiring few diagnostic tests.

Clinical Assessment The physician will be aided in the pursuit of an explanation for the lymphadenopathy by a careful medical history, physical examination, selected laboratory tests, and perhaps an excisional lymph node biopsy.

The *medical history* should reveal the setting in which lymphadenopathy is occurring. Symptoms such as sore throat, cough, fever, night sweats, fatigue, weight loss, or pain in the nodes should be sought. The patient's age, sex, occupation, exposure to pets, sexual behavior, and use of drugs such as diphenylhydantoin are other important historic points. For example, children and young adults usually have benign (i.e., nonmalignant) disorders, such as viral or bacterial upper respiratory infections, infectious mononucleosis, toxoplasmosis, and, in some countries, tuberculosis, which account for the observed lymphadenopathy. In contrast, after age 50 the incidence of malignant disorders increases and that of benign disorders decreases.

The *physical examination* can provide useful clues such as the extent of lymphadenopathy (localized or generalized), size of nodes, texture, presence or absence of nodal tenderness, signs of inflammation over the node, skin lesions, and splenomegaly. A thorough ear, nose, and throat (ENT) examination is indicated in adult patients with cervical adenopathy and a history of tobacco use. Localized or regional adenopathy implies involvement of a single anatomic area. Generalized adenopathy has been defined as involvement of three or more noncontiguous lymph node areas. Many of the causes of lymphadenopathy ([Table 63-1](#)) can produce localized or generalized adenopathy, so this distinction is of limited utility in the differential diagnosis. Nevertheless, generalized lymphadenopathy is frequently associated with nonmalignant disorders such as infectious mononucleosis [Epstein-Barr virus (EBV) or cytomegalovirus (CMV)], toxoplasmosis, AIDS, other viral infections, systemic lupus erythematosus (SLE), and mixed connective tissue disease. Acute and chronic lymphocytic leukemias and malignant lymphomas also produce generalized adenopathy in adults.

The site of localized or regional adenopathy may provide a useful clue about the cause. Occipital adenopathy often reflects an infection of the scalp, and preauricular adenopathy accompanies conjunctival infections and cat-scratch disease. The most frequent site of regional adenopathy is the neck, and most of the causes are benign -- upper respiratory infections, oral and dental lesions, infectious mononucleosis, other viral illnesses. The chief malignant causes include metastatic cancer from head and neck, breast, lung, and thyroid primaries. Enlargement of supraclavicular and scalene nodes is always abnormal. Because these nodes drain regions of the lung and retroperitoneal space, they can reflect either lymphomas, other cancers, or infectious processes arising in these areas. Virchow's node is an enlarged left supraclavicular node infiltrated with metastatic cancer from a gastrointestinal primary. Metastases to supraclavicular nodes also occur from lung, breast, testis, or ovarian cancers. Tuberculosis, sarcoidosis, and toxoplasmosis are nonneoplastic causes of supraclavicular adenopathy. Axillary adenopathy is usually due to injuries or localized infections of the ipsilateral upper extremity. Malignant causes include melanoma or lymphoma and, in women, breast cancer. Inguinal lymphadenopathy is usually secondary to infections or trauma of the lower extremities and may accompany sexually transmitted diseases such as lymphogranuloma venereum, primary syphilis, genital herpes, or chancroid. These nodes may also be involved by lymphomas and metastatic cancer from primary lesions of the rectum, genitalia, or lower extremities (melanoma).

The size and texture of the lymph node(s) and the presence of pain are useful parameters in evaluating a patient with lymphadenopathy. Nodes $<1.0 \text{ cm}^2$ in area ($1.0 \times 1.0 \text{ cm}$ or less) are almost always secondary to benign, nonspecific reactive causes. In one retrospective analysis of younger patients (9 to 25 years) who had a lymph node biopsy, a maximum diameter of $>2 \text{ cm}$ served as one discriminant for predicting that the biopsy would reveal malignant or granulomatous disease. Another study showed that a lymph node size of 2.25 cm^2 ($1.5 \text{ cm} \times 1.5 \text{ cm}$) was the best discriminating limit for distinguishing malignant or granulomatous lymphadenopathy from other causes of lymphadenopathy. Patients with node(s) $\geq 1.0 \text{ cm}^2$ should be observed after excluding infectious mononucleosis and/or toxoplasmosis unless there are symptoms and signs of an underlying systemic illness.

The texture of lymph nodes may be described as soft, firm, rubbery, hard, discrete, matted, tender, movable, or fixed. Tenderness is found when the capsule is stretched during rapid enlargement, usually secondary to an inflammatory process. Some malignant diseases such as acute leukemia may produce rapid enlargement and pain in the nodes. Nodes involved by lymphoma tend to be large, discrete, symmetric, rubbery, firm, mobile, and nontender. Nodes containing metastatic cancer are often hard, nontender, and nonmovable because of fixation to surrounding tissues. The coexistence of splenomegaly in the patient with lymphadenopathy implies a systemic illness such as infectious mononucleosis, lymphoma, acute or chronic leukemia, [SLE](#), sarcoidosis, toxoplasmosis, cat-scratch disease, or other less common hematologic disorders. The patient's story should provide helpful clues about the underlying systemic illness.

Nonsuperficial presentations (thoracic or abdominal) of adenopathy are usually detected as the result of a symptom-directed diagnostic workup. Thoracic adenopathy may be detected by routine chest roentgenography or during the workup for superficial adenopathy. It may also be found because the patient complains of a cough or wheezing from airway compression; hoarseness from recurrent laryngeal nerve involvement; dysphagia from esophageal compression; or swelling of the neck, face, or arms secondary to compression of the superior vena cava or subclavian vein. The differential diagnosis of mediastinal and hilar adenopathy includes primary lung disorders and systemic illnesses that characteristically involve mediastinal or hilar nodes. In the young, mediastinal adenopathy is associated with infectious mononucleosis and sarcoidosis. In endemic regions, histoplasmosis can cause unilateral paratracheal lymph node involvement that mimics lymphoma. Tuberculosis can also cause unilateral adenopathy. In older patients, the differential diagnosis includes primary lung cancer (especially among smokers), lymphomas, metastatic carcinoma (usually lung), tuberculosis, fungal infection, and sarcoidosis.

Enlarged intraabdominal or retroperitoneal nodes are usually malignant. Although tuberculosis may present as mesenteric lymphadenitis, these masses usually contain lymphomas or, in young men, germ cell tumors.

Laboratory Investigation The laboratory investigation of patients with lymphadenopathy must be tailored to elucidate the etiology suspected from the patient's history and physical findings. One study from a family practice clinic evaluated 249 younger patients with "enlarged lymph nodes, not infected" or "lymphadenitis." No laboratory studies were obtained in 51%. When studies were performed, the most common were a complete blood count (33%), throat culture (16%), chest x-ray (12%), or monospot test (10%). Only eight patients (3%) had a node biopsy, and half of those were normal or reactive. The complete blood count can provide useful data for the diagnosis of acute or chronic leukemias, [EBV](#) or [CMV](#) mononucleosis, lymphoma with a leukemic component, pyogenic infections, or immune cytopenias in illnesses such as [SLE](#). Serologic studies may demonstrate antibodies specific to components of EBV, CMV, HIV, and other viruses; *Toxoplasma gondii*; *Brucella*; etc. If SLE is suspected, then antinuclear and anti-DNA antibody studies are warranted.

The chest x-ray is usually negative, but the presence of a pulmonary infiltrate or mediastinal lymphadenopathy would suggest tuberculosis, histoplasmosis, sarcoidosis, lymphoma, primary lung cancer, or metastatic cancer and demands further

investigation.

A variety of imaging techniques [computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, color Doppler ultrasonography] have been employed to differentiate benign from malignant lymph nodes, especially in patients with head and neck cancer. CT and MRI are comparably accurate (65 to 90%) in the diagnosis of metastases to cervical lymph nodes. Ultrasonography has been used to determine the long (L) axis, short (S) axis, and a ratio of long to short axis in cervical nodes. An L/S ratio of <2.0 has a sensitivity and a specificity of 95% for distinguishing benign and malignant nodes in patients with head and neck cancer. This ratio has greater specificity and sensitivity than palpation or measurement of either the long or the short axis alone.

The indications for lymph node biopsy are imprecise, yet it is a valuable diagnostic tool. The decision to biopsy may be made early in a patient's evaluation or delayed for up to 2 weeks. Prompt biopsy should occur if the patient's history and physical findings suggest a malignancy; examples include a solitary, hard, nontender cervical node in an older patient who is a chronic user of tobacco; supraclavicular adenopathy; and solitary or generalized adenopathy that is firm, movable, and suggestive of lymphoma. If a primary head and neck cancer is suspected as the basis of a solitary, hard cervical node, then a careful [ENT](#) examination should be performed. Any mucosal lesion that is suspicious for a primary neoplastic process should be biopsied first. If no mucosal lesion is detected, an excisional biopsy of the largest node should be performed. Fine-needle aspiration should not be performed as the first diagnostic procedure. Most diagnoses require more tissue than such aspiration can provide and it often delays a definitive diagnosis. Fine-needle aspiration should be reserved for thyroid nodules and for confirmation of relapse in patients whose primary diagnosis is known. If the primary physician is uncertain about whether to proceed to biopsy, consultation with a hematologist or medical oncologist should be helpful. In primary care practices, fewer than 5% of lymphadenopathy patients will require a biopsy. That percentage will be considerably larger in referral practices, i.e., hematology, oncology, or otolaryngology (ENT).

Two groups have reported algorithms that they claim will identify more precisely those lymphadenopathy patients who should have a biopsy. Both reports were retrospective analyses in referral practices. The first study involved patients 9 to 25 years of age who had a node biopsy performed. Three variables were identified that predicted those young patients with peripheral lymphadenopathy who should undergo biopsy; lymph node size >2 cm in diameter and abnormal chest x-ray had positive predictive value, whereas recent [ENT](#) symptoms had negative predictive values. The second study evaluated 220 lymphadenopathy patients in a hematology unit and identified five variables [lymph node size, location (supraclavicular or non-supraclavicular), age (>40 years or <40 years), texture (nonhard or hard), and tenderness] that were utilized in a mathematical model to identify those patients requiring a biopsy. Positive predictive value was found for age >40 years, supraclavicular location, node size >2.25 cm², hard texture, and lack of pain or tenderness. Negative predictive value was evident for age <40 years, node size <1.0 cm², nonhard texture, and tender or painful nodes. Ninety-one percent of those who required biopsy were correctly classified by this model. Since both of these studies were retrospective analyses and one was limited to young patients, it is not known how useful these models would be if applied prospectively in a primary care

setting.

Most lymphadenopathy patients do not require a biopsy, and at least half require no laboratory studies. If the patient's history and physical findings point to a benign cause for lymphadenopathy, then careful follow-up at a 2- to 4-week interval can be employed. The patient should be instructed to return for reevaluation if the node(s) increase in size. Antibiotics are not indicated for lymphadenopathy unless there is strong evidence of a bacterial infection. Glucocorticoids should not be used to treat lymphadenopathy because their lympholytic effect obscures some diagnoses (lymphoma, leukemia, Castleman's disease) and they contribute to delayed healing or activation of underlying infections. An exception to this statement is the life-threatening pharyngeal obstruction by enlarged lymphoid tissue in Waldeyer's ring that is occasionally seen in infectious mononucleosis.

SPLENOMEGALY

STRUCTURE AND FUNCTION OF THE SPLEEN

The spleen is a reticuloendothelial organ that has its embryologic origin in the dorsal mesogastrium at about 5 weeks' gestation. It arises in a series of hillocks, migrates to its normal adult location in the left upper quadrant (LUQ), and is attached to the stomach via the gastrosplenic ligament and to the kidney via the lienorenal ligament. When the hillocks fail to unify into a single tissue mass, accessory spleens may develop in around 20% of persons. The function of the spleen has been elusive. Galen believed it was the source of "black bile" or melancholia, and the word *hypochondria* (literally, beneath the ribs) and the idiom "to vent one's spleen" attest to the beliefs that the spleen had an important influence on the psyche and emotions. In humans, its normal physiologic roles seem to be the following:

1. Maintenance of quality control over erythrocytes in the red pulp by removal of senescent and defective red blood cells. The spleen accomplishes this function through a unique organization of its parenchyma and vasculature ([Fig. 63-1](#)).
2. Synthesis of antibodies in the white pulp.
3. The removal of antibody-coated bacteria and antibody-coated blood cells from the circulation.

An increase in these normal functions may result in splenomegaly.

The spleen is composed of red pulp and white pulp, which are Malpighi's terms for the red blood-filled sinuses and reticuloendothelial cell-lined cords and the white lymphoid follicles arrayed within the red pulp matrix. The spleen is in the portal circulation. The reason for this is unknown but may relate to the fact that lower blood pressure allows less rapid flow and minimizes damage to normal erythrocytes. Blood flows into the spleen at a rate of about 150 mL/min through the splenic artery, which ultimately ramifies into central arterioles. Some blood goes from the arterioles to capillaries and then to splenic veins and out of the spleen, but the majority of blood from central arterioles flows into the macrophage-lined sinuses and cords. The blood entering the

sinuses reenters the circulation through the splenic venules, but the blood entering the cords is subjected to an inspection of sorts. In order to return to the circulation, the blood cells in the cords must squeeze through slits in the cord lining to enter the sinuses that lead to the venules. Old and damaged erythrocytes are less deformable and are retained in the cords, where they are destroyed and their components recycled. Red cell inclusion bodies such as parasites, nuclear residua (Howell-Jolly bodies), or denatured hemoglobin (Heinz bodies) are pinched off in the process of passing through the slits, a process called *pitting*. The culling of dead and damaged cells and the pitting of cells with inclusions appear to occur without significant delay since the blood transit time through the spleen is only slightly slower than in other organs.

The spleen is also capable of assisting the host in adapting to its hostile environment. It has at least three adaptational functions: (1) clearance of bacteria and particulates from the blood, (2) the generation of immune responses to certain invading pathogens, and (3) the generation of cellular components of the blood under circumstances in which the marrow is unable to meet the needs (i.e., extramedullary hematopoiesis). The latter adaptation is a recapitulation of the blood-forming function the spleen plays during gestation. In some animals, the spleen also serves a role in the vascular adaptation to stress because it stores red blood cells (often hemoconcentrated to higher hematocrits than normal) under normal circumstances and contracts under the influence of α -adrenergic stimulation to provide the animal with an autotransfusion and improved oxygen-carrying capacity. However, the normal human spleen does not sequester or store red blood cells and does not contract in response to sympathetic stimuli. The normal human spleen contains approximately one-third of the total body platelets and a significant number of marginated neutrophils. These sequestered cells are available when needed to respond to bleeding or infection.

Approach to the Patient

Clinical Assessment The most common *symptoms* produced by diseases involving the spleen are pain and a heavy sensation in the [LUQ](#). Massive splenomegaly may cause early satiety. Pain may result from acute swelling of the spleen with stretching of the capsule, infarction, or inflammation of the capsule. For many years, it was believed that splenic infarction was clinically silent, which at times is true. However, Soma Weiss, in his classic 1942 report of the self-observations by a Harvard medical student on the clinical course of subacute bacterial endocarditis, documented that severe LUQ and pleuritic chest pain may accompany thromboembolic occlusion of splenic blood flow. Vascular occlusion, with infarction and pain, is commonly seen in children with sickle cell crises. Rupture of the spleen, either from trauma or infiltrative disease that breaks the capsule, may result in intraperitoneal bleeding, shock, and death. The rupture itself may be painless.

A palpable spleen is the major *physical sign* produced by diseases affecting the spleen and suggests enlargement of the organ. The normal spleen is said to weigh less than 250 g, decreases in size with age, normally lies entirely within the rib cage, has a maximum cephalocaudad diameter of 13 cm by ultrasonography or maximum length of 12 cm and/or width of 7 cm by radionuclide scan, and is usually not palpable. However, a palpable spleen was found in 3% of 2200 asymptomatic, male, freshman college students. Follow-up at 3 years revealed that 30% of those students still had a palpable

spleen without any increase in disease prevalence. Ten-year follow-up found no evidence for lymphoid malignancies. Furthermore, in some tropical countries (e.g., New Guinea) the incidence of splenomegaly may reach 60%. Thus, the presence of a palpable spleen does not always equate with presence of disease. Even when disease is present, splenomegaly may not reflect the primary disease, but rather a reaction to it. For example, in patients with Hodgkin's disease, only two-thirds of the palpable spleens show involvement by the cancer.

Physical examination of the spleen utilizes primarily the techniques of palpation and percussion. Inspection may reveal a fullness in the [LUQ](#) that descends on inspiration, a finding associated with a massively enlarged spleen. Auscultation may reveal a venous hum or a friction rub.

Palpation can be accomplished by bimanual palpation, ballotment, and palpation from above (Middleton maneuver). For bimanual palpation, which is at least as reliable as the other techniques, the patient is supine with flexed knees. The examiner's left hand is placed on the lower rib cage and pulls the skin toward the costal margin, allowing the fingertips of the right hand to feel the tip of the spleen as it descends while the patient inspires slowly, smoothly, and deeply. Palpation is begun with the right hand in the left lower quadrant with gradual movement toward the left costal margin, thereby identifying the lower edge of a massively enlarged spleen. When the spleen tip is felt, the finding is recorded as centimeters below the left costal margin at some arbitrary point, i.e., 10 to 15 cm, from the midpoint of the umbilicus or the xiphisternal junction. This allows other examiners to compare findings or the initial examiner to determine changes in size over time. Bimanual palpation in the right lateral decubitus position adds nothing to the supine examination.

Percussion for splenic dullness is accomplished with any of three techniques described by Nixon, Castell, or Barkun:

1. *Nixon's method*: The patient is placed on the right side so that the spleen lies above the colon and stomach. Percussion begins at the lower level of pulmonary resonance in the posterior axillary line and proceeds diagonally along a perpendicular line toward the lower midanterior costal margin. The upper border of dullness is normally 6 to 8 cm above the costal margin. Dullness greater than 8 cm in an adult is presumed to indicate splenic enlargement.

2. *Castell's method*: With the patient supine, percussion in the lowest intercostal space in the anterior axillary line (8th or 9th) produces a resonant note if the spleen is normal in size. This is true during expiration or full inspiration. A dull percussion note on full inspiration suggests splenomegaly.

3. *Percussion of Traube's semilunar space*: The borders of Traube's space are the sixth rib superiorly, the left midaxillary line laterally, and the left costal margin inferiorly. The patient is supine with the left arm slightly abducted. During normal breathing, this space is percussed from medial to lateral margins, yielding a normal resonant sound. A dull percussion note suggests splenomegaly.

Studies comparing methods of percussion and palpation with a standard of

ultrasonography or scintigraphy have revealed sensitivity of 56 to 71% for palpation and 59 to 82% for percussion. Reproducibility among examiners is better for palpation than percussion. Both techniques are less reliable in obese patients or patients who have just eaten. Thus, the physical examination techniques of palpation and percussion are imprecise at best. It has been suggested that the examiner perform percussion first and, if positive, proceed to palpation; if the spleen is palpable, then one can be reasonably confident that splenomegaly exists. However, not all [LUQ](#) masses are enlarged spleens; gastric or colon tumors and pancreatic or renal cysts or tumors can mimic splenomegaly.

The presence of an enlarged spleen can be more precisely determined, if necessary, by liver-spleen radionuclide scan, [CT](#), [MRI](#), or ultrasonography. The latter technique is the current procedure of choice for routine assessment of spleen size (normal = a maximum cephalocaudad diameter of 13 cm) because it has high sensitivity and specificity and is safe, noninvasive, quick, mobile, and less costly. Nuclear medicine scans are accurate, sensitive, and reliable but are costly, require greater time to generate data, and utilize immobile equipment. They have the advantage of demonstrating accessory splenic tissue. CT and MRI provide accurate determination of spleen size, but the equipment is immobile and the procedures are expensive. MRI appears to offer no advantage over CT. Changes in spleen structure such as mass lesions, infarcts, inhomogeneous infiltrates, and cysts are more readily assessed by CT, MRI, or ultrasonography. None of these techniques is very reliable in the detection of patchy infiltration (e.g., Hodgkin's disease).

Differential Diagnosis Many of the diseases associated with splenomegaly are listed in [Table 63-2](#). They are grouped according to the presumed basic mechanisms responsible for organ enlargement:

1. Hyperplasia or hypertrophy related to a particular splenic function such as reticuloendothelial hyperplasia (work hypertrophy) in diseases such as hereditary spherocytosis or thalassemia syndromes that require removal of large numbers of defective red blood cells; immune hyperplasia in response to systemic infection (infectious mononucleosis, subacute bacterial endocarditis) or to immunologic diseases (immune thrombocytopenia, [SLE](#), Felty's syndrome).
2. Passive congestion due to decreased blood flow from the spleen in conditions that produce portal hypertension (cirrhosis, Budd-Chiari syndrome, congestive heart failure).
3. Infiltrative diseases of the spleen (lymphomas, metastatic cancer, amyloidosis, Gaucher's disease, myeloproliferative disorders with extramedullary hematopoiesis).

The differential diagnostic possibilities are much fewer when the spleen is "massively enlarged," that is, it is palpable more than 8 cm below the left costal margin or its drained weight is ≥ 1000 g ([Table 63-3](#)). The vast majority of such patients will have non-Hodgkin's lymphoma, chronic lymphocytic leukemia, hairy cell leukemia, chronic myelogenous leukemia, myelofibrosis with myeloid metaplasia, or polycythemia vera.

Laboratory Assessment The major laboratory abnormalities accompanying splenomegaly are determined by the underlying systemic illness. Erythrocyte counts

may be normal, decreased (thalassemia major syndromes, [SLE](#), cirrhosis with portal hypertension), or increased (polycythemia vera). Granulocyte counts may be normal, decreased (Feltz's syndrome, congestive splenomegaly, leukemias), or increased (infections or inflammatory disease, myeloproliferative disorders). Similarly, the platelet count may be normal, decreased when there is enhanced sequestration or destruction of platelets in an enlarged spleen (congestive splenomegaly, Gaucher's disease, immune thrombocytopenia), or increased in the myeloproliferative disorders such as polycythemia vera.

The complete blood count may reveal cytopenia of one or more blood cell types, which should suggest *hypersplenism*. This condition is characterized by splenomegaly, cytopenia(s), normal or hyperplastic bone marrow, and a response to splenectomy. The latter characteristic is less precise because reversal of cytopenia, particularly granulocytopenia, is sometimes not sustained after splenectomy. The cytopenias result from increased destruction of the cellular elements secondary to reduced flow of blood through enlarged and congested cords (congestive splenomegaly) or to immune-mediated mechanisms. In hypersplenism, various cell types usually have normal morphology on the peripheral blood smear, although the red cells may be spherocytic due to loss of surface area during their longer transit through the enlarged spleen. The increased marrow production of red cells should be reflected as an increased reticulocyte production index, although the value may be less than expected due to increased sequestration of reticulocytes in the spleen.

The need for additional laboratory studies is dictated by the differential diagnosis of the underlying illness of which splenomegaly is a manifestation.

SPLENECTOMY

Splenectomy is infrequently performed for diagnostic purposes, especially in the absence of clinical illness or other diagnostic tests that suggest underlying disease. More often splenectomy is performed for staging the extent of disease in patients with Hodgkin's disease, for symptom control in patients with massive splenomegaly, for disease control in patients with traumatic splenic rupture, or for correction of cytopenias in patients with hypersplenism or immune-mediated destruction of one or more cellular blood elements. Splenectomy is necessary for routine staging of patients with Hodgkin's disease only in those with clinical stage I or II disease in whom radiation therapy alone is contemplated as the treatment. Noninvasive staging of the spleen in Hodgkin's disease is not a sufficiently reliable basis for treatment decisions because one-third of normal-sized spleens will be involved with Hodgkin's disease and one-third of enlarged spleens will be tumor-free. Although splenectomy in chronic myelogenous leukemia does not affect the natural history of disease, removal of the massive spleen usually makes patients significantly more comfortable and simplifies their management by significantly reducing transfusion requirements. Splenectomy is an effective secondary or tertiary treatment for two chronic B cell leukemias, hairy cell leukemia and prolymphocytic leukemia, and for the very rare splenic mantle cell or marginal zone lymphoma. Splenectomy in these diseases may be associated with significant tumor regression in bone marrow and other sites of disease. Similar regressions of systemic disease have been noted after splenic irradiation in some types of lymphoproliferative disease, especially chronic lymphocytic leukemia and prolymphocytic leukemia. This

has been termed the *abscopal effect*. Such systemic tumor responses to local therapy directed at the spleen suggest that there may be some hormone or growth factor produced by the spleen that affects tumor cell proliferation, but this conjecture is not yet substantiated. The most common indication for splenectomy is traumatic or iatrogenic splenic rupture. In a fraction of patients with splenic rupture, peritoneal seeding of splenic fragments can lead to splenosis -- the presence of multiple rests of spleen tissue not connected to the portal circulation. This ectopic spleen tissue may cause pain or gastrointestinal obstruction, as in endometriosis. A large number of hematologic, immunologic, and congestive causes of splenomegaly can lead to destruction of one or more cellular blood elements. In the majority of such cases, splenectomy can correct the cytopenias, particularly anemia and thrombocytopenia. Perhaps the only contraindication to splenectomy is the presence of marrow failure, in which the enlarged spleen is the only source of hematopoietic tissue.

The absence of the spleen has minimal long-term effects on the hematologic profile. In the immediate postsplenectomy period, there may be some leukocytosis (up to 25,000/uL) and thrombocytosis (up to 1×10^6 /uL), but within 2 to 3 weeks, blood cell counts and survival of each cell lineage are usually normal. The chronic manifestations of splenectomy are marked variation in size and shape of erythrocytes (anisocytosis, poikilocytosis) and the presence of Howell-Jolly bodies (nuclear remnants), Heinz bodies (denatured hemoglobin), basophilic stippling, and an occasional nucleated erythrocyte in the peripheral blood. When such erythrocyte abnormalities appear in a patient whose spleen has not been removed, one should suspect splenic infiltration by tumor that has interfered with its normal culling and pitting function.

The most serious consequence of splenectomy is increased susceptibility to bacterial infections, particularly those with capsules such as *Streptococcus pneumoniae*, *Haemophilus influenzae*, and some gram-negative enteric organisms. Patients under age 20 years are particularly susceptible to overwhelming sepsis with *S. pneumoniae*, and the overall actuarial risk of sepsis in patients who have had their spleens removed is about 7% in 10 years. The case-fatality rate for pneumococcal sepsis in splenectomized patients is 50 to 80%. About 25% of patients without spleens will develop a serious infection at some time in their life. The frequency is highest within the first 3 years after splenectomy. About 15% of the infections are polymicrobial, and lung, skin, and blood are the most common sites. No increased risk of viral infection has been noted in patients who have no spleen. The susceptibility to bacterial infections relates to the inability to remove opsonized bacteria from the bloodstream and a defect in making antibodies to T cell-independent antigens such as the polysaccharide components of bacterial capsules. Pneumococcal vaccine (23-valent polysaccharide vaccine) should be administered to all patients 2 weeks before elective splenectomy. The Advisory Committee on Immunization Practices recommends that even splenectomized patients receive pneumococcal vaccine with a repeat vaccination 5 years later. Efficacy has not been proven in this setting, and the recommendation discounts the possibility that administration of the vaccine may actually lower the titer of specific pneumococcal antibodies. A more effective pneumococcal vaccine that involves T cells in the response is in development. The vaccine to *H. influenzae* should also be given to patients in whom elective splenectomy is planned. No other vaccines are routinely recommended in this setting.

Splenectomized patients should be educated to consider any unexplained fever as a medical emergency. Prompt medical attention with evaluation and treatment of suspected bacteremia may be life-saving. Routine chemoprophylaxis with oral penicillin can result in the emergence of drug-resistant strains and is not recommended.

In addition to an increased susceptibility to bacterial infections, splenectomized patients are also more susceptible to the parasitic disease babesiosis. The splenectomized patient should avoid areas where the parasite *Babesia* is endemic (e.g., Cape Cod, MA).

Surgical removal of the spleen is an obvious cause of *hyposplenism*. Patients with sickle cell disease often suffer from autosplenectomy as a result of splenic destruction by the numerous infarcts associated with sickle cell crises during childhood. Indeed, the presence of a palpable spleen in a patient with sickle cell disease after age 5 suggests a coexisting hemoglobinopathy, e.g., thalassemia or hemoglobin C. In addition, patients who receive splenic irradiation for a neoplastic or autoimmune disease are also functionally hyposplenic. The term *hyposplenism* is preferred to *asplenism* in referring to the physiologic consequences of splenectomy because asplenia is a rare, specific, and fatal congenital abnormality in which there is a failure of the left side of the coelomic cavity (which includes the splenic anlagen) to develop normally. Infants with asplenia have no spleens, but that is the least of their problems. The right side of the developing embryo is duplicated on the left so there is liver where the spleen should be, there are two right lungs, and the heart comprises two right atria and two right ventricles.

(Bibliography omitted in Palm version)

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64. DISORDERS OF GRANULOCYTES AND MONOCYTES - Steven M. Holland, John I. Gallin

Leukocytes are the major cells comprising inflammatory and immune responses and include neutrophils, T and B lymphocytes, natural killer (NK) cells, monocytes, eosinophils, and basophils. These cells have specific functions, such as antibody production by B lymphocytes or destruction of bacteria by neutrophils, but in no single infectious disease is the exact role of the cell types completely established. Thus, whereas neutrophils are classically thought to be critical to host defense against bacteria, they may also play important roles in defense against viral infections.

The blood delivers leukocytes to the various tissues from the bone marrow, where they are produced. Normal blood leukocyte counts are given in the Appendix (Tables A-7 and A-8). The various leukocytes are derived from a common stem cell in the bone marrow. Three-fourths of the nucleated cells of bone marrow are committed to the production of leukocytes. Leukocyte maturation in the marrow is under the regulatory control of a number of different factors, known as colony stimulating factors and interleukins ([Chap. 104](#)). Because an alteration in the number and type of leukocytes is often associated with disease processes, total white blood count (WBC) (cells per microliter) and differential counts are informative. The lymphocytes and basophils are discussed in [Chaps. 305](#) and [310](#), respectively. This chapter focuses on the neutrophils, monocytes, and eosinophils.

NEUTROPHILS

MATURATION

Important events in neutrophil life are summarized in [Fig. 64-1](#). In normal humans, neutrophils are produced only in the bone marrow. The minimum number of stem cells necessary to support hematopoiesis is estimated to be 400 to 500. Human blood monocytes, tissue macrophages, and stromal cells produce colony stimulating factors, hormones required for the growth of monocytes and neutrophils in the bone marrow. The hematopoietic system not only produces enough neutrophils ($\sim 1.3 \times 10^{11}$ cells per 80-kg person per day) to carry out physiologic functions but also has a large reserve stored in the marrow which can be mobilized in response to inflammation or infection. An increase in the number of blood neutrophils is called neutrophilia, and the presence of immature cells is termed a shift to the left. A decrease in the number of blood neutrophils is called neutropenia.

Neutrophils and monocytes evolve from pluripotent stem cells under the influence of cytokines and colony stimulating factors ([Fig. 64-2](#)). The proliferation phase through the metamyelocyte takes about 1 week, while the maturation phase from metamyelocyte to mature neutrophil takes another week. The myeloblast is the first recognizable precursor cell and is followed by the *promyelocyte* ([Plate V-23](#)). The promyelocyte evolves when the classic lysosomal granules, called the *primary* or *azurophil granules*, are produced. The primary granules contain hydrolases, elastase, myeloperoxidase, cationic proteins, and bactericidal/permeability-increasing protein, which is important for killing gram-negative bacteria. Azurophil granules also contain *defensins*, a family of cysteine-rich polypeptides with broad antimicrobial activity against bacteria, fungi, and

certain enveloped viruses. The promyelocyte divides to produce the *myelocyte*, a cell responsible for the synthesis of the *specific* or *secondary granules* which contain unique (specific) constituents such as lactoferrin, vitamin B₁₂-binding proteins, membrane components of the nicotinamide-adenine dinucleotide phosphate (NADPH) oxidase required for hydrogen peroxide production, histaminase, and receptors for certain chemoattractants and adherence-promoting factors (CR3) as well as receptors for the basement membrane component, laminin. The secondary granules do not contain acid hydrolases and therefore are not classic lysosomes. Packaging of secondary granule contents during myelopoiesis is controlled by CCAAT/enhancer binding protein-ε. Secondary granule contents are readily released extracellularly, and their mobilization is important in modulating inflammation. During the final stages of maturation no cell division occurs, and the cell passes through the *metamyelocyte* stage and then to the *band* neutrophil with a sausage-shaped nucleus ([Plate V-35](#)). As the band cell matures, the nucleus assumes a lobulated configuration. The nucleus of neutrophils normally contains up to four segments. Excessive segmentation (more than five nuclear lobes) may be a manifestation of folate or vitamin B₁₂ deficiency ([Plate V-38](#)). The Pelger-Huet anomaly ([Plate V-34B](#)), an infrequent dominant benign inherited trait, results in neutrophils with distinctive bilobed nuclei that must be distinguished from band forms. The physiologic role of the multilobed nucleus of neutrophils is unknown, but it may allow great deformation of neutrophils during migration into tissues at sites of inflammation.

In severe acute bacterial infection, prominent neutrophil cytoplasmic granules called *toxic granulations* are occasionally seen ([Plate V-11](#)). Toxic granulations are immature or abnormally staining azurophil granules. Cytoplasmic inclusions, also called *Dohle bodies* ([Plate V-35](#)), can be seen during infection and are fragments of ribosome-rich endoplasmic reticulum. Large neutrophil vacuoles are often present in acute bacterial infection and probably represent pinocytosed (internalized) membrane.

Neutrophils are heterogeneous in function. Monoclonal antibodies have been developed that recognize only a subset of mature neutrophils. The meaning of neutrophil heterogeneity is not known.

MARROW RELEASE AND CIRCULATING COMPARTMENTS

Specific signals, including interleukin (IL) 1, tumor necrosis factor-α (TNF-α), the colony stimulating factors, complement fragment C3e, and perhaps other cytokines mobilize leukocytes from the bone marrow and deliver them to the blood in an unstimulated state. Under normal conditions, about 90% of the neutrophil pool is in the bone marrow, 2 to 3% in the circulation, and the remainder in the tissues ([Fig. 64-3](#)).

The circulating pool exists in two dynamic compartments: one freely flowing and one marginated. The freely flowing pool is about one-half the neutrophils in the basal state and is composed of those cells that are in the blood and not in contact with the endothelium. Marginated leukocytes are those that are in close physical contact with the endothelium ([Fig. 64-4](#)). In the pulmonary circulation, where an extensive capillary bed (~1000 capillaries per alveolus) exists, margination occurs because the capillaries are about the same size as a mature neutrophil. Therefore, neutrophil fluidity and deformability are necessary to make the transit through the pulmonary bed. Increased

neutrophil rigidity and decreased deformability lead to augmented neutrophil trapping and margination in the lung. In contrast, in the systemic postcapillary venules, margination is mediated by the interaction of specific cell-surface molecules. *Selectins* are glycoproteins expressed on neutrophils and endothelial cells, among others, that cause a low-affinity interaction, resulting in "rolling" of the neutrophil along the endothelial surface. On neutrophils, the molecule L-selectin [cluster determinant (CD) 62L] binds to glycosylated proteins on endothelial cells [e.g., glycosylation-dependent cell adhesion molecule (GlyCAM1) and CD34]. Glycoproteins on neutrophils, most importantly sialyl-Lewis_x (SLe_x, CD15s), are targets for binding of selectins expressed on endothelial cells [E-selectin (CD62E) and P-selectin (CD62P)] and other leukocytes. In response to chemotactic stimuli from injured tissues (e.g., complement product C5a, leukotriene B₄, [IL-8](#)) or bacterial products [e.g., *N*-formylmethionylleucylphenylalanine (f-metleuphe)], neutrophil adhesiveness increases, and the cells "stick" to the endothelium through *integrins*. The integrins are leukocyte glycoproteins that exist as complexes of a common CD18 b-chain with CD11a (LFA-1), CD11b (also called either Mac-1, CR3, or the C3b receptor), and CD11c (p150,95). CD11a/CD18 and CD11b/CD18 bind to specific endothelial receptors [intercellular adhesion molecules (ICAM) 1 and 2].

On cell stimulation, L-selectin is shed; receptors for chemoattractants and opsonins are mobilized; the phagocytes orient toward the chemoattractant source in the extravascular space, increase their motile activity (chemokinesis), and migrate directionally (chemotaxis) into tissues. The process of migration into tissues is called *diapedesis* and involves the crawling of neutrophils between postcapillary endothelial cells that open junctions between adjacent cells to permit leukocyte passage. Diapedesis involves platelet/endothelial cell adhesion molecule (PECAM) 1 (CD31), which is expressed on both the emigrating leukocyte and the endothelial cells. The endothelial responses (increased blood flow from increased vasodilation and permeability) are mediated by anaphylatoxins (e.g., C3a and C5a) as well as vasodilators such as histamine, bradykinin, serotonin, nitric oxide, vascular endothelial growth factor (VEGF), and prostaglandins E and I. Cytokines regulate some of these processes [e.g., [TNF-α](#) induction of VEGF, interferon (IFN) γ inhibition of prostaglandin E].

In the healthy adult, most neutrophils leave the body by migration through the mucous membrane of the gastrointestinal tract. Normally, neutrophils spend a short time in the circulation (half-life, 6 to 7 h). Senescent neutrophils are cleared from the circulation by macrophages in the lung and spleen. Once in the tissues, neutrophils release enzymes, such as collagenase and elastase, that help establish abscess cavities. Neutrophils ingest pathogenic materials that have been opsonized by IgG and C3b. Fibronectin and the tetrapeptide tuftsin facilitate phagocytosis.

With phagocytosis comes a burst of oxygen consumption and activation of the hexose-monophosphate shunt. A membrane-associated [NADPH](#) oxidase, consisting of membrane and cytosolic components, is assembled and catalyzes the reduction of oxygen to superoxide anion, which is then converted to hydrogen peroxide and other toxic oxygen products (e.g., hydroxyl radical). Hydrogen peroxide + chloride + neutrophil myeloperoxidase generates hypochlorous acid (bleach), hypochlorite, and chlorine. These products oxidize and halogenate microorganisms and tumor cells and, when uncontrolled, can damage host tissue. Strongly cationic proteins, defensins, and

probably nitric oxide also participate in microbial killing. Other enzymes, such as lysozyme and acid proteases, help digest microbial debris. After 1 to 4 days in tissues neutrophils die. The apoptosis of neutrophils is also cytokine regulated; granulocyte colony stimulating factor (G-CSF) and [IFN- \$\gamma\$](#) prevent their death. Under certain conditions, such as in delayed-type hypersensitivity, monocyte accumulation occurs within 6 to 12 h of initiation of inflammation. Neutrophils, monocytes, microorganisms in various states of digestion, and altered local tissue cells make up the inflammatory exudate, pus. Myeloperoxidase confers the characteristic green color to pus and may participate in turning off the inflammatory process by inactivating chemoattractants and immobilizing phagocytic cells.

Neutrophils respond to certain cytokines [[IFN- \$\gamma\$](#) , granulocyte-macrophage colony stimulating factor (GM-CSF), [IL-8](#)] and produce cytokines and chemotactic signals [[TNF- \$\alpha\$](#) , IL-8, macrophage inflammatory protein (MIP) 1] that modulate the inflammatory response. In the presence of fibrinogen, f-metleuphe or leukotriene B₄ induces IL-8 production by neutrophils, providing autocrine amplification of inflammation. *Chemokines (chemoattractant cytokines)* are small proteins produced by many different cell types, including endothelial cells, fibroblasts, epithelial cells, neutrophils, and monocytes, that regulate neutrophil and monocyte recruitment and activation. The chemokines transduce their signals through heterotrimeric G protein-linked receptors that have seven cell membrane-spanning domains, the same type of cell-surface receptor that mediates the response to the classical chemoattractants *N*-f-metleuphe and C5a. Four major groups of chemokines are recognized based on the cysteine structure near the N terminus: C, CC, CXC, and CXXXC. The CXC cytokines like IL-8 mainly attract neutrophils; CC chemokines like [MIP-1 \$\alpha\$](#) attract lymphocytes, monocytes, eosinophils, and basophils; the C chemokine lymphotactin is T cell tropic; the CXXXC chemokine fractalkine attracts neutrophils, monocytes, and T cells. These molecules and their receptors not only regulate the trafficking and activation of inflammatory cells, but chemokine receptors serve as co-receptors for HIV infection ([Chap. 309](#)).

NEUTROPHIL ABNORMALITIES

A defect in the neutrophil life cycle can lead to dysfunction and compromised host defenses. Inflammation is often depressed, and the clinical result is often recurrent and severe bacterial and fungal infections. Aphthous ulcers of mucous membranes (gray ulcers without pus) and gingivitis and periodontal disease suggest a phagocytic cell disorder. Patients with congenital phagocyte defects can have infections within the first few days of life. Skin, ear, upper and lower respiratory tract, and bone infections are common. Sepsis and meningitis are rare. In some disorders the frequency of infection is variable, and patients can go for months or even years without major infection. Aggressive management of these congenital diseases has extended the life span of patients beyond 30 years.

Neutropenia The consequences of absent neutrophils are dramatic. Susceptibility to infectious diseases increases sharply when neutrophil counts fall below 1000 cells/uL. When the absolute neutrophil count (ANC; band forms and mature neutrophils combined) falls below 500 cells/uL, control of endogenous microbial flora (e.g., mouth, gut) is impaired; when the ANC is < 200/uL, the inflammatory process is absent. Neutropenia can be due to depressed production, increased peripheral destruction, or

excessive peripheral pooling. A falling neutrophil count or a significant decrease in the number of neutrophils below steady state levels, together with a failure to increase neutrophil counts in the setting of infection or other challenge, requires investigation. Acute neutropenia, such as that caused by cancer chemotherapy, is more likely to be associated with increased risk of infection than neutropenia of long duration (months to years) that reverses in response to infection or carefully controlled administration of endotoxin (see "Laboratory Diagnosis," below).

Some causes of inherited and acquired neutropenia are listed in [Table 64-1](#). The most common neutropenias are iatrogenic, resulting from the use of cytotoxic or immunosuppressive therapies for malignancy or control of autoimmune disorders. These drugs cause neutropenia because they result in decreased production of rapidly growing progenitor (stem) cells of the marrow. Certain antibiotics such as chloramphenicol, trimethoprim-sulfamethoxazole, flucytosine, vidarabine, and the antiretroviral drug zidovudine may cause neutropenia by inhibiting proliferation of myeloid precursors. The marrow suppression is generally dose-related and dependent on continued administration of the drug. Recombinant human [G-CSF](#) reverses this form of neutropenia.

Another important mechanism for iatrogenic neutropenia is the effect of drugs that serve as immune haptens and sensitize neutrophils or neutrophil precursors to immune-mediated peripheral destruction. This form of drug-induced neutropenia can be seen within 7 days of exposure to the drug; with previous drug exposure, resulting in preexisting antibodies, neutropenia may occur a few hours after administration of the drug. Although any drug can cause this form of neutropenia, the most frequent causes are commonly used antibiotics, such as sulfa-containing compounds, penicillins, and cephalosporins. Fever and eosinophilia also may be associated drug reactions, but often these signs are not present. Drug-induced neutropenia can be severe, but discontinuation of the sensitizing drug is sufficient for recovery, which is usually seen within 5 to 7 days and is complete by 10 days. Readministration of the sensitizing drug should be avoided, since abrupt neutropenia often will result. For this reason, diagnostic challenge should be avoided.

Autoimmune neutropenias caused by circulating antineutrophil antibodies are another form of acquired neutropenia that results in increased destruction of neutrophils. Acquired neutropenia also may be seen with viral infections, including infection with HIV. Acquired neutropenia may be cyclic in nature, occurring at intervals of several weeks. Acquired cyclic or stable neutropenia may be associated with an expansion of large granular lymphocytes (LGL), which may be T cells, [NK](#) cells, or NK-like cells. Patients with LGL lymphocytosis may have moderate blood and bone marrow lymphocytosis, neutropenia, polyclonal hypergammaglobulinemia, splenomegaly, rheumatoid arthritis, and absence of lymphadenopathy. Such patients may have a chronic and relatively stable course. Recurrent bacterial infections are frequent. Benign and malignant forms of this syndrome occur. In some patients, a spontaneous regression has occurred even after 11 years, suggesting an immunoregulatory defect as the basis for at least one form of the disorder. Glucocorticoids, cyclosporine, [IFN- \$\alpha\$](#) , and nucleosides such as 2-chlorodeoxyadenosine each have induced remission.

Hereditary Neutropenias Hereditary neutropenias are rare and may manifest in early

childhood as a profound constant neutropenia or agranulocytosis. Congenital forms of neutropenia include Kostmann's syndrome (neutrophil count <100/uL), which is often fatal; more benign chronic idiopathic neutropenia (neutrophil count of 300 to 1500/uL); the cartilage-hair hypoplasia syndrome; Shwachman's syndrome associated with pancreatic insufficiency; myelokathexis, a congenital disorder characterized by neutrophil degeneration, hypersegmentation, and myeloid hyperplasia in the marrow associated with decreased expression of bcl-X_L in myeloid precursors and accelerated apoptosis; and neutropenias associated with other immune defects (X-linked agammaglobulinemia, ataxia telangiectasia, IgA deficiency). Mutations in the [G-CSF](#) receptor on chromosome 1 associated with poor response to G-CSF can occur with severe congenital neutropenia and predispose to myeloid malignancy. Hereditary cyclic neutropenia, an autosomal dominant trait, may occur in infancy and is characterized by a remarkably regular 3-week cycle. Hereditary cyclic neutropenia actually is cyclic hematopoiesis, due to mutations in the neutrophil elastase gene. Glucocorticoids and G-CSF blunt the cycling in some patients.

Maternal factors can be associated with neutropenia in the newborn. Transplacental transfer of IgG directed against antigens on fetal neutrophils can result in peripheral destruction. Drugs (e.g., thiazides) ingested during pregnancy can cause neutropenia in the newborn by either depressed production or peripheral destruction.

The presence of immunoglobulin directed toward neutrophils is seen in Felty's syndrome -- a triad of rheumatoid arthritis, splenomegaly, and neutropenia ([Chap. 312](#)). Patients with Felty's syndrome who respond to splenectomy with an increase in their neutrophil count also have lower postoperative serum neutrophil-binding IgG. Some of these patients have neutropenia associated with an increased number of [LGL](#). Splenomegaly with peripheral trapping and destruction of neutrophils is also seen in lysosomal storage diseases and in portal hypertension.

Neutrophilia Neutrophilia results from increased neutrophil production, increased marrow release, or defective margination ([Table 64-2](#)). The most important acute cause of neutrophilia is infection. Neutrophilia from acute infection represents both increased production and increased marrow release. Increased production is also associated with chronic inflammation and certain myeloproliferative diseases. Increased marrow release and mobilization of the marginated leukocyte pool are induced by glucocorticoids. Release of epinephrine, as with vigorous exercise, excitement, or stress, will demarginate neutrophils in the spleen and lungs and double the neutrophil count in minutes. Leukocytosis with cell counts of 10,000 to 25,000/uL occurs in response to infection and other forms of acute inflammation and results from both release of the marginated pool and mobilization of marrow reserves. Persistent neutrophilia with cell counts of 30,000 to 50,000/uL or higher is called a *leukemoid reaction*, a term often used to distinguish this degree of neutrophilia from leukemia. In a leukemoid reaction, the circulating neutrophils are usually mature and not clonally derived.

Abnormal Neutrophil Function Inherited and acquired abnormalities of phagocyte function are listed in [Table 64-3](#). The resulting diseases are best considered in terms of the functional defects of adherence, chemotaxis, and microbicidal activity. The distinguishing features of the important inherited disorders of phagocyte function are shown in [Table 64-4](#).

Disorders of Adhesion Two types of leukocyte adhesion deficiency (LAD) have been described. Both are autosomal recessive traits and result in the inability of neutrophils to exit the circulation to sites of infection, leading to leukocytosis and increased susceptibility to infection ([Fig. 64-4](#)). Patients with LAD 1 have mutations in CD18, the common component of the integrins LFA-1, Mac-1, and p150,95, leading to a defect in tight adhesion between neutrophils and the endothelium. The heterodimer formed by CD18/CD11b (Mac-1) is also the receptor for the complement-derived opsonin C3bi (CR3). The CD18 gene is located on distal chromosome 21q. Variable expression of the defect determines the severity of clinical disease. Complete lack of expression of the leukocyte adhesion proteins results in the severe phenotype in which inflammatory cytokines do not increase the expression of leukocyte adhesion proteins on neutrophils or activated T and B cells. Neutrophils (and monocytes) from patients with LAD 1 adhere poorly to endothelial cells and protein-coated surfaces and exhibit defective spreading, aggregation, and chemotaxis. Patients with LAD 1 have recurrent bacterial and fungal infections involving skin, oral and genital mucosa, and respiratory and intestinal tracts; persistent leukocytosis (neutrophil counts of 15,000 to 20,000/uL) because cells do not marginate; and, in severe cases, a history of delayed separation of the umbilical stump. Infections, especially of the skin, may become necrotic with progressively enlarging borders, slow healing, and development of dysplastic scars. The most common bacteria are *Staphylococcus aureus* and enteric gram-negative bacteria. LAD 2 is caused by an abnormality of SLe^x(CD15s), the ligand on neutrophils that interacts with selectins on endothelial cells.

Disorders of Neutrophil Granules The most common neutrophil defect is *myeloperoxidase deficiency*, a primary granule defect inherited as an autosomal recessive trait; the incidence is ~1 in 2000 persons. Isolated myeloperoxidase deficiency is not associated with clinically compromised defenses, because other defense systems such as hydrogen peroxide generation are amplified. Microbicidal activity of neutrophils is delayed but not absent. Myeloperoxidase deficiency may make other acquired host defense defects more serious. An acquired form of myeloperoxidase deficiency occurs in myelomonocytic leukemia and acute myeloid leukemia.

Chediak-Higashi syndrome (CHS) is a rare disease with autosomal recessive inheritance due to defects in the lysosomal transport protein LYST, encoded by the gene *CHS1* at 1q42. This protein is required for normal packaging and disbursement of granules. Neutrophils (and all cells containing lysosomes) from patients with CHS characteristically have large granules ([Plate V-34A](#)). Patients with CHS have an increased number of infections resulting from many agents. CHS neutrophils and monocytes have impaired chemotaxis and abnormal rates of microbial killing due to slow rates of fusion of the lysosomal granules with phagosomes. [NK](#) cell function is also impaired.

Specific granule deficiency is a rare autosomal recessive disease in which the production of secondary granules and their contents, as well as primary granule component defensins, is defective. The defect in bacterial killing leads to severe bacterial infections. One type of specific granule deficiency is due to a mutation in the CCAAT/enhancer binding protein-ε, a regulator of expression of granule components.

Chronic granulomatous disease Chronic granulomatous disease (CGD) is a group of disorders of granulocyte and monocyte oxidative metabolism. Although CGD is rare, with an incidence of 1 in 200,000 individuals, it is an important model of defective neutrophil oxidative metabolism. Most often CGD is inherited as an X-linked recessive trait; 30% of patients inherit the disease in an autosomal recessive pattern. Mutations in the genes for the four proteins that assemble at the plasma membrane account for all patients with CGD. Two proteins (a 91-kDa protein, abnormal in X-linked CGD, and a 22-kDa protein, absent in one form of autosomal recessive CGD) form the heterodimer cytochrome b-558 in the plasma membrane. Two other proteins (47 and 67 kDa, abnormal in the other autosomal recessive forms of CGD) are cytoplasmic in origin and interact with the cytochrome after cell activation to form [NADPH](#)oxidase, required for hydrogen peroxide production. Leukocytes from patients with CGD have severely diminished hydrogen peroxide production. The genes involved in each of the defects have been cloned and sequenced and the chromosome locations identified. Patients with CGD characteristically have increased numbers of infections due to catalase-positive microorganisms (organisms that destroy their own hydrogen peroxide). When patients with CGD become infected, they often have extensive inflammatory reactions, and lymph node suppuration is common despite the administration of appropriate antibiotics. Aphthous ulcers and chronic inflammation of the nares are often present. Granulomas are frequent and can obstruct the gastrointestinal or genitourinary tracts. The excessive inflammation probably reflects failure to degrade chemoattractants and antigens, leading to persistent neutrophil accumulation. Impaired killing of intracellular microorganisms by macrophages may lead to persistent cell-mediated immunity and granuloma formation.

MONONUCLEAR PHAGOCYTES

The mononuclear phagocyte system is composed of monoblasts, promonocytes, and monocytes in addition to the structurally diverse tissue macrophages that make up what was previously referred to as the reticuloendothelial system. Macrophages are long-lived phagocytic cells capable of many of the functions of neutrophils. They are also secretory cells that participate in many immunologic and inflammatory processes distinct from neutrophils. Monocytes leave the circulation by diapedesis more slowly than neutrophils and have a half-life in the blood of 12 to 24 h.

After blood monocytes arrive in the tissues, they differentiate into macrophages ("big eaters") with specialized functions suited for specific anatomic locations. Macrophages are particularly abundant in capillary walls of the lung, spleen, liver, and bone marrow, where they function to remove microorganisms and other noxious elements from the blood. Alveolar macrophages, liver Kupffer cells, splenic macrophages, peritoneal macrophages, bone marrow macrophages, lymphatic macrophages, brain microglial cells, and dendritic macrophages all have specialized functions. Macrophage-secreted products include lysozyme, neutral proteases, acid hydrolases, arginase, complement components, enzyme inhibitors (plasmin, α_2 -macroglobulin), binding proteins (transferrin, fibronectin, transcobalamin II), nucleosides, and cytokines ([TNF- \$\alpha\$](#) ; [IL-1](#), -8, -12, and -18). [IL-1](#) ([Chaps. 17](#) and [305](#)) has many functions, including initiating fever in the hypothalamus, mobilizing leukocytes from the bone marrow, activating lymphocytes and neutrophils. [TNF- \$\alpha\$](#) is a pyrogen that duplicates many of the actions of [IL-1](#) and plays an important role in the pathogenesis of gram-negative shock ([Chap. 124](#)). [TNF- \$\alpha\$](#)

stimulates production of hydrogen peroxide and related toxic oxygen species by macrophages and neutrophils. In addition, TNF- α induces catabolic changes that contribute to the profound wasting (cachexia) associated with many chronic diseases.

Other macrophage-secreted products include reactive oxygen and nitrogen metabolites, bioactive lipids (arachidonic acid metabolites and platelet-activating factors), chemokines, colony stimulating factors, and factors stimulating fibroblast and vessel proliferation. Macrophages help regulate the replication of lymphocytes and participate in the killing of tumors, viruses, and certain bacteria (*Mycobacterium tuberculosis* and *Listeria monocytogenes*). Macrophages are key effector cells in the elimination of intracellular microorganisms. Their ability to fuse to form giant cells that coalesce into granulomas in response to some inflammatory stimuli is important in the elimination of intracellular microbes and is under the control of IFN- γ . Nitric oxide induced by IFN- γ is an important effector against intracellular parasites including tuberculosis and *Leishmania*.

Macrophages play an important role in the immune response ([Chap. 305](#)). They process and present antigen to lymphocytes and secrete cytokines that modulate and direct lymphocyte development and function. Macrophages participate in autoimmune phenomena by removing immune complexes and other substances from the circulation. Polymorphisms in macrophage receptors for immunoglobulin (Fc γ RII) determine susceptibility to some infections and autoimmune diseases. In wound healing, they dispose of senescent cells, and they contribute to atheroma development. Macrophage elastase mediates development of emphysema from cigarette smoking.

DISORDERS OF THE MONONUCLEAR PHAGOCYTE SYSTEM

Many disorders of neutrophils extend to mononuclear phagocytes. Thus, drugs that suppress neutrophil production in the bone marrow can cause monocytopenia. Transient monocytopenia occurs after stress or glucocorticoid administration. Monocytosis is associated with tuberculosis, brucellosis, subacute bacterial endocarditis, Rocky Mountain spotted fever, malaria, and visceral leishmaniasis (kala azar). Monocytosis also occurs with malignancies, leukemias, myeloproliferative syndromes, hemolytic anemias, chronic idiopathic neutropenias, and granulomatous diseases such as sarcoidosis, regional enteritis, and some collagen vascular diseases. Patients with [LAD](#), hyperimmunoglobulin E-recurrent infection (Job's) syndrome, [CHS](#), and [CGD](#) all have defects in the mononuclear phagocyte system.

Monocyte cytokine production is impaired in some patients with disseminated nontuberculous mycobacterial infection who are not infected with HIV. Genetic defects in IFN- γ receptors 1 and 2 impair monocyte killing of intracellular parasites, as do lesions in the potent IFN- γ inducer, IL-12 and its receptor ([Fig. 64-5](#)).

Certain viral infections impair mononuclear phagocyte function. For example, influenza virus infection causes abnormal monocyte chemotaxis. Mononuclear phagocytes can be infected by HIV using CCR5, the chemokine receptor that acts as a coreceptor with CD4 for HIV. T lymphocytes produce IFN- γ , which induces FcR expression and phagocytosis and stimulates hydrogen peroxide production by mononuclear phagocytes and neutrophils. In certain diseases, such as AIDS, IFN- γ production may be deficient, while

in other diseases, such as T cell lymphomas, excessive release of IFN- γ may be associated with erythrophagocytosis by splenic macrophages.

Monocytopenia occurs with acute infections, with stress, and after treatment with glucocorticoids. Monocytopenia also occurs in aplastic anemia, hairy cell leukemia, acute myeloid leukemia, and as a direct result of myelotoxic drugs.

EOSINOPHILS

Eosinophils and neutrophils share similar morphology, many lysosomal constituents, phagocytic capacity, and oxidative metabolism. Eosinophils express a specific chemoattractant receptor and respond to a specific chemokine, eotaxin. Little is known about the role of eosinophils. Eosinophils are much longer lived than neutrophils, and unlike neutrophils, tissue eosinophils can recirculate. During most infections, eosinophils are not important. However, in invasive helminthic infections, such as hookworm, schistosomiasis, strongyloidiasis, toxocariasis, trichinosis, filariasis, echinococcosis, and cysticercosis, the eosinophil plays a central role in host defense. Eosinophils are associated with bronchial asthma, cutaneous allergic reactions, and other hypersensitivity states.

The distinctive feature of the red-staining (Wright's stain) eosinophil granules is its crystalline core consisting of an arginine-rich protein (major basic protein) with histaminase activity, important in host defense against parasites. Eosinophil granules also contain a unique eosinophil peroxidase that catalyzes the oxidation of many substances by hydrogen peroxide and may facilitate killing of microorganisms.

Eosinophil peroxidase, in the presence of hydrogen peroxide and halide, initiates mast cell secretion in vitro and thereby promotes inflammation. Eosinophils contain cationic proteins, some of which bind to heparin and reduce its anticoagulant activity. Eosinophil-derived neurotoxin and eosinophil cationic protein are ribonucleases that can kill respiratory syncytial virus. Eosinophil cytoplasm contains Charcot-Leyden crystal protein, a hexagonal bipyramidal crystal first observed in a patient with leukemia and then in sputum of patients with asthma; this protein is lysophospholipase and may function to detoxify certain lysophospholipids.

Several factors enhance the eosinophil's function in host defense. T cell-derived factors enhance the ability of eosinophils to kill parasites. Mast cell-derived eosinophil chemotactic factor of anaphylaxis (ECFa) increases the number of eosinophil complement receptors and enhances eosinophil killing of parasites. Eosinophil colony stimulating factors (e.g., IL-5) produced by macrophages increase eosinophil production in the bone marrow and activate eosinophils to kill parasites.

EOSINOPHILIA

Eosinophilia is the presence of >500 eosinophils per microliter of blood and is common in many settings besides parasite infection. Significant tissue eosinophilia can occur without an elevated blood count. The most common cause of eosinophilia is allergic reactions to drugs (iodides, aspirin, sulfonamides, nitrofurantoin, penicillins, and cephalosporins). Allergies such as hay fever, asthma, eczema, serum sickness, allergic

vasculitis, and pemphigus are associated with eosinophilia. Eosinophilia also occurs in collagen vascular diseases (e.g., rheumatoid arthritis, eosinophilic fasciitis, allergic angitis, and periarteritis nodosa) and malignancies (e.g., Hodgkin's disease; mycosis fungoides; chronic myelogenous leukemia; and cancer of the lung, stomach, pancreas, ovary, or uterus), as well as in Job's syndrome and [CGD](#). Eosinophilia commonly is present in the helminthic infections. [IL-5](#) is the dominant eosinophil growth factor. Therapeutic administration of the cytokines IL-2 and [GM-CSF](#) frequently leads to transient eosinophilia. The most dramatic hypereosinophilic syndromes are Loeffler's syndrome, tropical pulmonary eosinophilia, Loeffler's endocarditis, eosinophilic leukemia, and idiopathic hypereosinophilic syndrome (50,000 to 100,000/uL).

The idiopathic hypereosinophilic syndrome represents a heterogeneous group of disorders with the common feature of prolonged eosinophilia of unknown cause and organ system dysfunction, including the heart, central nervous system, kidneys, lungs, gastrointestinal tract, and skin. The bone marrow is involved in all affected individuals, but the most severe complications involve the heart and central nervous system. Clinical manifestations and organ dysfunction are highly variable. Eosinophils are found in the involved tissues and likely cause tissue damage by local deposition of toxic eosinophil proteins such as eosinophil cationic protein and major basic protein. In the heart, the pathologic changes lead to thrombosis, endocardial fibrosis, and restrictive endomyocardialopathy. The damage to tissues in other organ systems is similar. The mechanism for the hypereosinophilia is not known. Glucocorticoids usually induce remission. In patients who do not respond to glucocorticoids, a cytotoxic agent such as hydroxyurea has been used successfully to lower the peripheral blood eosinophil counts and to improve markedly the prognosis. [IFN- \$\alpha\$](#) also is effective in some patients, including those unresponsive to hydroxyurea. Aggressive medical and surgical approaches are used to manage patients with cardiovascular complications.

The *eosinophilia-myalgia syndrome* is a multisystem disease with prominent cutaneous, hematologic, and visceral manifestations that frequently evolves into a chronic course and can occasionally be fatal. The syndrome is characterized by eosinophilia (eosinophil count >1000/uL) and generalized disabling myalgias without other recognized causes. Eosinophil fasciitis, pneumonitis, and myocarditis; neuropathy culminating in respiratory failure; and encephalopathy may occur. The disease is caused by ingesting contaminants in L-tryptophan-containing products. Eosinophils, lymphocytes, macrophages, and fibroblasts accumulate in the affected tissues, but their role in pathogenesis is unclear. Activation of eosinophils and fibroblasts and the deposition of eosinophil-derived toxic proteins in affected tissues may contribute. [IL-5](#) and transforming growth factor β have been implicated as potential mediators. Treatment is withdrawal of L-tryptophan-containing products and the administration of glucocorticoids. Most patients recover fully, remain stable, or show slow recovery; but the disease can be fatal in up to 5% of patients.

EOSINOPENIA

Eosinopenia occurs with stress, such as acute bacterial infection, and after treatment with glucocorticoids. The mechanism of eosinopenia of acute bacterial infection is unknown but is independent of endogenous glucocorticoids, since it occurs in animals after total adrenalectomy. There is no known adverse effect of eosinopenia.

HYPERIMMUNOGLOBULIN E-RECURRENT INFECTION SYNDROME

The hyperimmunoglobulin E-recurrent infection (HIE) syndrome or *Job's syndrome* is a rare multisystem disease in which the immune system, bone, teeth, lung and skin are affected. Abnormal chemotaxis is a variable feature. The molecular basis for this syndrome is not known, but some cases show autosomal dominant transmission with linkage to 4q. Patients with this syndrome have characteristic facies with broad nose, kyphoscoliosis and osteoporosis, and eczema. The primary teeth erupt normally but do not deciduate, often requiring extraction. Patients develop recurrent sinopulmonary and cutaneous infections that tend to be much less inflamed than appropriate for the degree of infection and have been referred to as "cold abscesses." A high degree of suspicion is required to diagnose infections in these patients, who may appear well despite extensive disease. The cold abscesses have been considered a reflection of impaired chemotaxis with too few phagocytes arriving too late, perhaps due to a lymphocyte factor inhibiting chemotaxis. However, the chemotactic defect in these patients is variable, and the fundamental basis for the impaired defenses is complex and poorly defined.

LABORATORY DIAGNOSIS AND MANAGEMENT

Initial studies of [WBC](#) and differential and often a bone marrow examination are followed by assessment of bone marrow reserves (steroid challenge test), marginated circulating pool of cells (epinephrine challenge test), and marginating ability (endotoxin challenge test) ([Fig. 64-3](#)). In vivo assessment of inflammation is possible with a Rebuck skin window test or an in vivo blister assay, which measures the ability of leukocytes and inflammatory mediators to accumulate locally in the skin. In vitro tests of phagocyte aggregation, adherence, chemotaxis, phagocytosis, degranulation, and microbicidal activity (for *S. aureus*) may help pinpoint cellular or humoral lesions. Deficiencies of oxidative metabolism are detected with the nitroblue tetrazolium (NBT) dye test, which is based on the ability of products of oxidative metabolism to reduce yellow, soluble NBT to blue-black formazan, an insoluble material that can be seen microscopically. Studies of superoxide and hydrogen peroxide production may further define neutrophil oxidative function.

Patients with leukopenias or leukocyte dysfunction often have delayed inflammatory responses. Therefore, clinical manifestations may be minimal despite overwhelming infection, and unusual infections must always be suspected. Early signs of infection demand prompt, aggressive culturing for microorganisms, use of antibiotics, and surgical drainage of abscesses. Prolonged antibiotics are often required. In patients with [CGD](#), prophylactic antibiotics (trimethoprim-sulfamethoxazole) diminish the frequency of life-threatening infections. Short courses of glucocorticoids may relieve gastrointestinal or genitourinary tract obstruction by granulomas in patients with CGD. Recombinant human [IFN-g](#), which nonspecifically stimulates phagocytic cell function, reduces the frequency of infections in patients with CGD by 70% and reduces the severity of infection. This effect of IFN-g in CGD is additive to the effect of prophylactic antibiotics. The recommended dose is 50 ug/m² subcutaneously three times weekly. IFN-g also has been used successfully in the treatment of leprosy, nontuberculous mycobacteria, and visceral leishmaniasis.

Rigorous oral hygiene reduces but does not eliminate the discomfort of gingivitis, periodontal disease, and aphthous ulcers; chlorhexidine mouthwash and tooth brushing with a hydrogen peroxide-sodium bicarbonate paste helps many patients. Oral antifungal agents (fluconazole) have reduced mucocutaneous candidiasis in patients with Job's syndrome. Androgens, glucocorticoids, lithium, and immunosuppressive therapy have been used to restore myelopoiesis in patients with neutropenia due to impaired production. Recombinant [G-CSF](#) is useful in the management of certain forms of neutropenia due to depressed neutrophil production, especially that related to cancer chemotherapy. Patients with chronic neutropenia with evidence of a good bone marrow reserve need not receive prophylactic antibiotics.

Patients with constant or cyclic neutrophil counts <500/uL may benefit from prophylactic antibiotics and [G-CSF](#) during periods of neutropenia. Oral trimethoprim-sulfamethoxazole (160/800 mg) twice daily can prevent infection. Increased numbers of fungal infections are not seen in patients with [CGD](#) on this regimen. Oral quinolones such as norfloxacin and ciprofloxacin are alternatives.

In the setting of cytotoxic chemotherapy with severe, persistent neutropenia, trimethoprim-sulfamethoxazole prevents *Pneumocystis carinii* pneumonia. These patients, and patients with phagocytic cell dysfunction, should avoid heavy exposure to airborne soil, dust, or decaying matter (mulch, manure), which are often rich in spores of *Aspergillus* or other fungi. Restriction of activities or social contact has no proven role in reducing risk of infection.

Cure of some congenital phagocyte defects is possible by bone marrow transplantation ([Chap. 115](#)). However, complications of bone marrow transplantation are still serious, and with rigorous medical care many patients with phagocytic disorders can go for years without a life-threatening infection. The identification of specific gene defects in patients with [LAD 1](#) and [CGD](#) has led to gene therapy trials in a number of genetic white cell disorders.

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PART THREE -GENETICS AND DISEASE

65. PRINCIPLES OF HUMAN GENETICS - *J. Larry Jameson, Peter Kopp*

IMPACT OF GENETICS ON MEDICAL PRACTICE

New insights into the genetic basis of disease are being generated at an ever-increasing rate. This explosion of information was ignited by technological advances, such as the polymerase chain reaction (PCR) and automated DNA sequencing, and is fueled by rapid progress in the Human Genome Project (HGP). Although its promise is great, the integration of genetics into the everyday practice of medicine remains challenging. To date, the most significant impact of genetics has been to enhance our understanding of disease etiology and pathogenesis. In the near term, we can expect an even greater role for genetics in the diagnosis, prevention, and treatment of disease ([Chaps. 68](#) and [69](#)).

Genetic disorders are more common than generally appreciated. It is estimated, for example, that 3% of pregnancies result in a child with a genetic disease or birth defect. About 10% of all pediatric and adult hospitalization admissions involve genetic diseases. This number would increase substantially if one included complex, multifactorial genetic diseases, such as diabetes or cardiovascular disease. The prevalence of genetic diseases, combined with their severity and chronic nature, imposes a great financial, social, and emotional burden on society.

Genetics has historically focused on chromosomal and metabolic disorders, reflecting the long-standing availability of techniques to diagnose these conditions. For example, conditions such as trisomy 21 (Down syndrome) or monosomy X (Turner syndrome) can be diagnosed using cytogenetics ([Chap. 66](#)). Likewise, many metabolic disorders (e.g., phenylketonuria, familial hypercholesterolemia) have been diagnosed using biochemical analyses. Recent advances in DNA diagnostics have extended the field of genetics to include virtually all medical specialties. In cardiology, for example, the molecular basis of inherited cardiomyopathies and ion channel defects that predispose to arrhythmias is being defined ([Chaps. 230](#) and [238](#)). In neurology, genetics has unmasked the pathophysiology of a startling number of neurodegenerative disorders ([Chap. 359](#)). Hematology has evolved dramatically, from its incipient genetic descriptions of hemoglobinopathies to the current understanding of the molecular basis of red cell membrane defects, clotting disorders, and thrombotic disorders ([Chaps. 106](#) and [117](#)). It is now abundantly clear that neoplasia and the acquisition of metastatic potential can be described in genetic terms ([Chaps. 81, 82](#), and [83](#)).

New concepts derived from genetic studies can sometimes clarify topics that were previously opaque. For example, although many different genetic defects can cause peripheral neuropathies, disruption of the normal folding of the myelin sheaths is frequently a common final pathway ([Chap. 379](#)). Several genetic causes of obesity appear to converge on a physiologic pathway that involves products of the proopiomelanocortin polypeptide and the MC4R receptor, thus identifying a key mechanism for appetite control ([Chap. 77](#)). A similar situation is emerging for genetically distinct forms of Alzheimer disease, several of which lead to the formation of neurofibrillary tangles ([Chap. 362](#)). Increasingly, the identification of defective genes can

pinpoint cellular pathways involved in key physiologic processes. Examples include identification of the cystic fibrosis conductance regulator (*CFTR*) gene, the Duchenne's muscular dystrophy (*DMD*) gene, which encodes dystrophin, and the fibroblast growth factor receptor-3 (*FGFR3*) gene, which is responsible for achondroplastic dwarfism. Similarly, transgenic and gene "knockout" models can help to unravel the physiologic function of genes. Genetic approaches have proven invaluable for the detection of infectious pathogens and are used clinically to identify agents that are difficult to culture such as mycobacteria, viruses, and parasites ([Chap. 121](#)). In many cases, molecular genetics has improved the feasibility and accuracy of diagnostic testing, enhanced our understanding of pathophysiology, and is beginning to open new avenues for therapy, including gene therapy ([Chap. 69](#)).

It is increasingly apparent that genetic background plays some role in virtually every medical condition. This is particularly true when one considers disease susceptibility, the interaction of genetic background with the environment, host responses to illness and to pharmaceutical agents, or the metabolism of drugs. Although genetics has traditionally been viewed through the window of relatively rare single-gene diseases, many disorders such as hypertension, asthma, diabetes, susceptibility to cardiovascular disease, and mental illness are also affected by genetic background, as often evident from a patient's family history. These complex genetic traits involve the contributions of many different genes, as well as environmental factors that can modify disease risk ([Chap. 68](#)).

The astounding rate at which new genetic information is being generated creates a major challenge for physicians and other health care providers. The terminology and techniques used for discovery evolve continuously. Much genetic information presently resides in computer databases or is being published in basic science journals. The ongoing development of bioinformatics promises to simplify this seemingly daunting onslaught of new information. It is now possible, for example, to search for genetic testing centers through a web site (<http://www.genlink.wustl.edu>) that can be accessed conveniently by organ system, disease state, or gene. Monogenic disorders are summarized in a large, continuously evolving compendium, referred to as the *Online Mendelian Inheritance in Man* (OMIM; <http://www.ncbi.nlm.nih.gov/omim/>). These and other databases (<http://www.genbank.com>) will expand rapidly in conjunction with advances in the [HGP](#).

CHROMOSOMES AND DNA REPLICATION

ORGANIZATION OF DNA INTO CHROMOSOMES

Size of the Human Genome The human genome is divided into 23 different chromosomes, including 22 autosomes (numbered 1 to 22) and the X and Y sex chromosomes. Adult cells are diploid, meaning they contain two homologous sets of 22 autosomes and a pair of sex chromosomes. Females have two X chromosomes (XX), whereas males have one X and one Y chromosome (XY). As a consequence of meiosis, germ cells (sperm or oocytes) are haploid and contain one set of 22 autosomes and one of the sex chromosomes. At the time of fertilization, the diploid genome is reconstituted by pairing of the homologous chromosomes from the mother and father. With each cell division (mitosis), chromosomes are replicated, paired, segregated, and

divided into two daughter cells ([Chap. 66](#)).

The genome is estimated to contain about 100,000 genes that are divided among the 23 chromosomes. A *gene* is a functional unit that is regulated by transcription (see below) and encodes a product, either RNA or protein, that exerts activity within the cell. Historically, genes were identified because they conferred specific traits that are transmitted from one generation to the next.

Human DNA is estimated to consist of about 3 billion base pairs (bp) of DNA per haploid genome. DNA length is normally measured in units of 1000 bp (kilobases, kb) or 1,000,000 bp (megabases, Mb). Not all DNA encodes genes. In fact, genes account for only about 10 to 15% of DNA. Much of the remaining DNA consists of highly repetitive sequences, the function of which is poorly understood. These repetitive DNA regions, along with nonrepetitive sequences that do not encode genes, may serve a structural role in the packaging of DNA into chromatin (DNA bound to histone proteins) and chromosomes ([Fig. 65-1](#)). If only 10% of DNA is expressed and there are 100,000 genes, the average gene would be about 3 kb in length. Although many genes are about this size, the range is quite broad. For example, some genes are only a few hundred bp, whereas others, like the *DMD* gene, are extraordinarily large (2 million bp).

Structure of DNA Each gene is composed of a linear polymer of DNA. DNA is a double-stranded helix composed of four different bases: adenine (A), thymidine (T), guanine (G), and cytosine (C). Adenine is paired to thymidine, and guanine is paired to cytosine, by hydrogen bond interactions that span the double helix. DNA has several remarkable features that make it ideal for the transmission of genetic information. It is relatively stable, at least in comparison to RNA or proteins. The double-stranded nature of DNA and its feature of strict base-pair complementarity permit faithful replication during cell division. As described below, complementarity also allows the transmission of genetic information from DNA → RNA → protein ([Fig. 65-2](#)). Messenger RNA (mRNA) is encoded by the so-called sense strand of the DNA double helix and is translated into proteins by ribosomes.

The presence of four different bases provides surprising genetic diversity. In the protein-coding regions of genes, the DNA bases are arranged into codons, a triplet of bases that specifies a particular amino acid. It is possible to arrange the four bases into 64 different triplet codons (4^3). Each codon specifies 1 of the 20 different amino acids, or a regulatory signal, such as stop translation. Because there are more codons than amino acids, the genetic code is degenerate; that is, most amino acids can be specified by several different codons. By arranging the codons in different combinations and in various lengths, it is possible to generate the tremendous diversity of primary protein structure.

REPLICATION OF DNA AND MITOSIS

Genetic information in DNA is transmitted to daughter cells under two different circumstances: (1) somatic cells divide by mitosis, allowing the diploid ($2n$) genome to replicate itself completely in conjunction with cell division; and (2) germ cells (sperm and ova) undergo meiosis, a process that enables the reduction of the diploid ($2n$) set of chromosomes to the haploid state ($1n$) ([Chap. 66](#)).

Prior to mitosis, cells exit the resting, or G₀ state, and enter the cell cycle ([Chap. 82](#)). After traversing a critical checkpoint in G₁, cells undergo DNA synthesis (S phase), during which the DNA in each chromosome is replicated, yielding two pairs of sister chromatids ($2n \rightarrow 4n$). The process of DNA synthesis requires stringent fidelity in order to avoid transmitting errors to subsequent generations of cells. Genetic abnormalities of DNA mismatch/repair include xeroderma pigmentosum, Bloom syndrome, ataxia telangiectasia, and hereditary nonpolyposis colon cancer (HNPCC), among others. Many of these disorders strongly predispose to neoplasia because of the rapid acquisition of additional mutations ([Chap. 81](#)). After completion of DNA synthesis, cells enter G₂ and progress through a second checkpoint before entering mitosis. At this stage, the chromosomes condense and are aligned along the equatorial plate at metaphase. The two identical sister chromatids, held together at the centromere, divide and migrate to opposite poles of the cell ([Fig. 66-3](#)). After formation of a nuclear membrane around the two separated sets of chromatids, the cell divides and two daughter cells are formed, thus restoring the diploid ($2n$) state.

ASSORTMENT AND SEGREGATION OF GENES DURING MEIOSIS

Meiosis occurs only in germ cells of the gonads. It shares certain features with mitosis but involves two distinct steps of cell division that reduce the chromosome number to the haploid state. In addition, there is active recombination that generates genetic diversity. During the first cell division, two sister chromatids ($2n \rightarrow 4n$) are formed for each chromosome pair and there is an exchange of DNA between homologous paternal and maternal chromosomes. This process involves the formation of *chiasmata*, structures that correspond to the DNA segments that cross over between the maternal and paternal homologues ([Fig. 65-3](#)). Usually there is at least one crossover on each chromosomal arm; recombination occurs more frequently in female meiosis than in male meiosis. Subsequently, the chromosomes segregate randomly. Because there are 23 chromosomes, there exist 2^{23} (>8 million) possible combinations of chromosomes. Together with the genetic exchanges that occur during recombination, chromosomal segregation generates tremendous diversity, and each gamete is genetically unique. The process of recombination, and the independent segregation of chromosomes, provide the foundation for performing linkage analyses, whereby one attempts to correlate the inheritance of certain chromosomal regions (or linked genes) with the presence of a disease or genetic trait (see below).

After the first meiotic division, which results in two daughter cells ($2n$), the two chromatids of each chromosome separate during a second meiotic division to yield four gametes with a haploid state ($1n$). When the egg is fertilized by sperm, the two haploid sets are combined, thereby restoring the diploid state ($2n$) in the zygote.

REGULATION OF GENE EXPRESSION

Mechanisms that regulate gene expression play a critical role in the function of genes. The new field of *functional genomics* is based on the concept that understanding gene regulation and function will provide a better understanding of physiology and offer novel therapeutic opportunities. The transcription of genes is controlled primarily by *transcription factors* that bind to DNA sequences in the regulatory regions of genes. As

described below, mutations in transcription factors cause an unexpectedly large number of genetic disorders. Gene expression is also influenced by *epigenetic events*, such as X-inactivation and imprinting, processes in which DNA methylation is associated with the silencing (i.e., suppression) of expression. Several genetic disorders, such as Prader-Willi syndrome (neonatal hypotonia, developmental delay, obesity, short stature, and hypogonadism) and Albright hereditary osteodystrophy (resistance to parathyroid hormone, short stature, brachydactyly, resistance to other hormones in certain subtypes), exhibit the consequences of genomic imprinting. Most studies of gene expression have focused on the regulatory DNA elements of genes that control transcription. However, it should be emphasized that gene expression requires a series of steps including mRNA processing, protein translation, and posttranslational modifications, all of which are actively regulated ([Fig. 65-2](#)).

STRUCTURE OF GENES

A gene product is usually a protein but can occasionally consist of RNA that is not translated. *Exons* refer to the portion of genes that are eventually spliced together to form mRNA. *Introns* refer to the spacing regions between the exons that are spliced out of precursor RNAs during RNA processing ([Fig. 65-2](#)).

The gene locus also includes regions that are necessary to control its expression. The regulatory regions most commonly involve sequences upstream (5') of the transcription start site, although there are also examples of control elements within introns or downstream of the coding regions of a gene. The upstream regulatory regions are also referred to as the *promoter*. The minimal promoter usually consists of a TATA box (which binds TATA-binding protein, TBP) and initiator sequences that enhance the formation of an active transcription complex. Transcriptional termination signals reside downstream, or 3', of a gene. Specific sequences, such as the AAUAAA sequence at the 3' end of the mRNA, designate the site for polyadenylation (poly-A tail), a process that influences mRNA transport to the cytoplasm, stability, and translation efficiency. A rigorous test of the regulatory region boundaries involves expressing a gene in a transgenic animal to determine whether the isolated DNA flanking sequences are sufficient to recapitulate the normal developmental, tissue-specific, and signal-responsive features of the endogenous gene. This has been accomplished for only a few genes; there are many examples in which large genomic fragments only partially reconstitute normal gene regulation in vivo, implying the presence of distant regulatory sequences. This approach is critical to our understanding of mechanisms that regulate genes and is also relevant for gene therapy strategies that require normal gene regulation ([Chap. 69](#)).

As genes are dissected with greater resolution, the number of DNA sequences and transcription factors that regulate transcription is much greater than originally anticipated. Most genes contain at least 15 to 20 discrete regulatory elements within 300 bp of the transcription start site. This densely packed promoter region often contains binding sites for ubiquitous transcription factors such as CAAT box/enhancer binding protein (C/EBP), cyclic AMP response element binding (CREB) protein, selective promoter factor 1 (Sp-1), or activator protein 1 (AP-1). However, factors involved in cell-specific expression may also bind to these sequences. For example, basic helix-loop-helix (bHLH) proteins bind to E-boxes in the promoters of myogenic

genes, and steroidogenic factor 1 (SF-1) binds to a specific recognition site in the regulatory region of multiple steroidogenic enzyme genes. Key regulatory elements may also reside at some distance from the proximal promoter. The globin and the immunoglobulin genes, for example, contain *locus control regions* that are several kilobases away from the structural sequences of the gene. Specific groups of transcription factors that bind to these promoter and enhancer sequences provide a combinatorial code for regulating transcription. In this manner, relatively ubiquitous factors interact with more restricted factors to allow each gene to be expressed and regulated in a unique manner. As described below, the transcription factors that bind to DNA actually represent only the first level of regulatory control. Other proteins -- *coactivators* and *corepressors* -- interact with the DNA-binding transcription factors to generate large regulatory complexes. These complexes are subject to control by numerous cell-signaling pathways, including phosphorylation and acetylation. Ultimately, the recruited transcription factors interact with, and stabilize, components of the basal transcription complex that assembles at the site of the TATA box and initiator region. This basal transcription factor complex consists of >30 different proteins. Gene transcription occurs when RNA polymerase begins to synthesize RNA from the DNA template.

TRANSCRIPTIONAL ACTIVATION AND REPRESSION

Every gene is controlled uniquely, whether in its spatial or temporal pattern of expression or in its response to extracellular signals. It is estimated that transcription factors account for about 30% of expressed genes. A growing number of identified genetic diseases involve transcription factors ([Table 65-1](#)). The MODY (maturity-onset diabetes of the young) disorders are representative of this group of diseases; mutations in several different islet cell-specific transcription factors cause various forms of MODY ([Chap. 333](#)).

Transcriptional activation can be divided into three main mechanisms:

1. Events that alter chromatin structure can enhance the access of transcription factors to DNA. For example, histone acetylation opens chromatin structure and is correlated with transcriptional activation.
2. Posttranslational modifications of transcription factors, such as phosphorylation, can induce the assembly of active transcription complexes. As an example, phosphorylation of [CREB](#) protein on serine 133 induces a conformational change that allows the recruitment of CREB-binding protein (CBP), a factor that integrates the actions of many transcription factors, including proteins, with histone acetyltransferase activity.
3. Transcriptional activators can displace a repressor protein. This mechanism is particularly common during development when the pattern of transcription factor expression changes dynamically.

Of course, these mechanisms are not mutually exclusive, and most genes are activated by some combination of these events.

In general, mechanisms of transcriptional repression have not been studied to the same

extent as mechanisms of transcriptional activation. Nonetheless, suppression of gene expression is as important as gene activation. Some mechanisms of repression are the corollary of activation. For example, repression is often associated with histone deacetylation or protein dephosphorylation. For nuclear hormone receptors, transcriptional silencing involves the recruitment of repression complexes that contain histone deacetylase activity. Aberrant expression of repressor proteins is sometimes associated with neoplasia. The t(15;17) chromosomal translocation that occurs in promyelocytic leukemia fuses the *PML* gene to a portion of the retinoic acid receptor α (*RAR* α) gene ([Table 65-1](#)). This event causes unregulated transcriptional repression in a manner that precludes normal cellular differentiation. The addition of the *RAR* ligand, retinoic acid, activates the receptor, thereby relieving repression and allowing cells to differentiate and ultimately undergo apoptosis. This mechanism has therapeutic importance as the addition of retinoic acid to treatment regimens induces a higher remission rate in patients with promyelocytic leukemia ([Chap. 111](#)).

CLONING AND SEQUENCING DNA

Since the mid-1970s, eight Nobel prizes have been awarded for research that led, directly or indirectly, to major methodological advances as well as to profound insights into genetics. Examples include the discoveries of reverse transcriptase, restriction enzymes, plasmid cloning vectors, DNA sequencing, and [PCR](#). A description of recombinant DNA techniques, the methodology used for the manipulation, analysis, and characterization of DNA segments, is beyond the scope of this chapter. As these methods are widely used in genetics and molecular diagnostics, however, it is useful to review briefly some of the fundamental principles of cloning and DNA sequencing.

CLONING OF GENES

Cloning refers to the creation of a recombinant DNA molecule that can be propagated indefinitely. The ability to clone genes and cDNAs therefore provides a permanent and renewable source of these reagents. Cloning is essential for DNA sequencing, nucleic acid hybridization studies, expression of recombinant proteins, and other recombinant DNA procedures.

The cloning of DNA involves the insertion of a DNA fragment into a cloning vector, followed by the propagation of the recombinant DNA in a host cell. The most straightforward cloning strategy involves inserting a DNA fragment into bacterial plasmids. Plasmids are small, autonomously replicating, circular DNA molecules that propagate separately from the chromosome in bacterial cells. The process of DNA insertion relies heavily on the use of restriction enzymes, which cleave DNA at highly specific sequences (usually 4 to 6 bp in length). Restriction enzymes generate complementary, cohesive sequences at the ends of the DNA fragment, which allow them to be efficiently ligated to the plasmid vector. Because plasmids contain genes that confer resistance to antibiotics, their presence in the host cell can be used for selection and DNA amplification.

A variety of vectors and appropriate hosts are now used for cloning ([Table 65-2](#)). Many of these are used for creating *libraries*, a term that refers to a collection of DNA clones. A genomic library represents an array of clones derived from genomic DNA. These

overlapping DNA fragments represent the entire genome and can ultimately be arranged according to their linear order. Genomic libraries are propagated using a variety of vectors, such as lambda (l) phage, cosmids, bacterial artificial chromosomes (BACs), and yeast artificial chromosomes (YACs). Phage libraries have been used extensively to isolate specific genes. Cosmids, BACs, and YACs are particularly useful for studying large genes and for defining the order of genes along the chromosomes (Fig. 65-4). cDNA libraries reflect clones derived from mRNA, typically from a particular tissue source. Thus, a cDNA library from the heart contains copies of mRNA expressed specifically in cardiac myocytes, in addition to those that are expressed ubiquitously. For this reason, a heart cDNA library will be enriched with cardiac-specific gene products and will differ from cDNA libraries generated from liver or pituitary mRNAs. As an example of the complexity of a genomic library, consider that the human genome contains 3×10^9 bp and the average genomic insert in a l phage library is about 10^4 bp. Therefore, it requires at least 3×10^5 clones to represent all of the genomic DNA. Specific clones are isolated from the several hundred thousand clones by using DNA hybridization.

With completion of the HGP, all human genes have been cloned and sequenced. As a result, many of these cloning procedures will be unnecessary or greatly facilitated by the extensive information concerning DNA markers and the sequence of DNA (see below).

NUCLEIC ACID HYBRIDIZATION

Nucleic acid *hybridization* is a fundamental principle in molecular biology that takes advantage of the fact that the two complementary strands of nucleic acids bind, or *hybridize*, to one another with very high specificity. The goal of hybridization is to detect specific nucleic acid (DNA or RNA) sequences in a complex background of other sequences. This technique is used for Southern blotting, northern blotting, and for screening libraries (see above). Further adaptation of hybridization techniques has led to the development of microarray DNA chips.

Southern Blot Southern blotting is used to analyze whether genes have been deleted or rearranged. It is also used to detect restriction fragment length polymorphisms (RFLPs). Genomic DNA is digested with restriction endonucleases and separated by gel electrophoresis. Individual fragments can then be transferred to a membrane and detected after hybridization with specific radioactive DNA probes. Because single base-pair mismatches can disrupt the hybridization of short DNA probes (oligonucleotides), a variation of the Southern blot, termed *oligonucleotide-specific hybridization* (OSH), uses short oligonucleotides to distinguish normal from mutant genes.

Northern Blot Northern blots are used to analyze patterns and levels of gene expression in different tissues. In a northern blot, mRNA is separated on a gel and transferred to a membrane, and specific transcripts are detected using radiolabeled DNA as a probe. This technique is rapidly being supplanted by more sensitive and comprehensive methods such as reverse transcriptase (RT)-PCR and gene expression arrays on DNA chips (see below).

Microarray Technology A rapidly evolving approach to genome-scale studies consists

of *microarrays*, or *DNA chips*. These approaches consist of thousands of synthetic nucleic acid sequences aligned on thin glass or silicon surfaces. Fluorescently labeled test sample DNA or RNA is hybridized to the chip, and a computerized scanner detects sequence matches. Microarrays allow the detection of variations in DNA sequence and are used for mutational analysis and genotyping. Alternatively, the expression pattern of large numbers of mRNA transcripts can be determined by hybridization of RNA samples to cDNA or genomic microarrays. This method has tremendous potential in the era of functional genomics. As one example, microarrays can be used to develop genetic fingerprints of different types of lymphomas, providing information useful for classification, pathophysiology, prognosis, and treatment.

THE POLYMERASE CHAIN REACTION

The [PCR](#), introduced in 1985, has revolutionized the way DNA analyses are performed and has become a cornerstone of molecular biology and genetic analysis. In essence, PCR provides a rapid way of cloning (amplifying) specific DNA fragments in vitro ([Fig. 65-5](#)). Exquisite specificity is conferred by the use of PCR primers, which are designed for a given DNA sequence. The geometric amplification of the DNA after multiple cycles yields remarkable sensitivity. As a result, PCR can be used to amplify DNA from very small samples, including single cells. These properties also allow DNA amplification from a variety of tissue sources including blood samples, biopsies, surgical or autopsy specimens, or cells from hair or saliva. PCR can also be used to study mRNA. In this case, the enzyme [RT](#) is first used to convert the RNA to DNA, which can then be amplified by PCR. This procedure, commonly known as *RT-PCR*, is useful as a quantitative measure of gene expression.

[PCR](#) provides a key component of molecular diagnostics. It provides a strategy for the rapid amplification of DNA (or mRNA) to search for mutations by a wide array of techniques, including DNA sequencing. PCR is also used for the amplification of highly polymorphic di- or trinucleotide repeat sequences, which allow various polymorphic alleles to be traced in genetic linkage or association studies. PCR is increasingly used to diagnose various microbial pathogens ([Chap. 121](#)).

DNA SEQUENCING

DNA sequencing is now an automated procedure. Although many protocols exist, the most commonly used strategy is based on the Sanger method in which dideoxynucleotides are used to randomly terminate DNA polymerization at each of the four bases (A,G,T,C). After separating the array of terminated DNA fragments using high-resolution gel or capillary electrophoresis, it is possible to deduce the DNA sequence by examining the progression of fragment lengths generated in each of the four nucleotide reactions. The use of fluorescently labeled dideoxynucleotides allows automated detection of the different bases and direct computer analysis of the DNA sequence. Efforts are underway to develop faster, more cost-effective DNA sequencing technologies. These include the use of mass spectrometry; detection of fluorescently labeled bases in flow cytometry; direct reading of the DNA sequence by scanning, tunneling, or atomic force microscopy; and sequence analysis using DNA chips.

TRANSGENIC MICE AS MODELS OF GENETIC DISEASE

Several organisms have been studied extensively as genetic models, including *Mus musculus* (mouse), *Drosophila melanogaster* (fruit fly), *Caenorhabditis elegans* (nematode), *Saccharomyces cerevisiae* (baker's yeast), and *Escherichia coli* (colonic bacterium). The ability to use these evolutionarily distant organisms as genetic models that are relevant to human physiology reflects a surprising conservation of genetic pathways and gene function. Transgenic mouse models have been particularly valuable, because many human and mouse genes exhibit similar structure and function and because manipulation of the mouse genome is relatively straightforward compared to those of other mammalian species.

Transgenic strategies in mice can be divided into two main approaches: (1) overexpression of a gene by random insertion into the genome, and (2) deletion or targeted mutagenesis of a gene by homologous recombination with the native endogenous gene ([Fig. 65-6](#)). Many variations of these basic approaches now exist that allow genes to be expressed in specific cell types, at different times during development, or at varying levels. Consequently, transgenic technology has emerged as a powerful strategy for defining the physiologic effects of deleting or overexpressing a gene, as well as providing unique genetic models for dissecting pathophysiology or testing therapies.

Examples of transgenic models relevant to human genetic disorders are listed in [Table 65-3](#). Transgenic overexpression of genes is useful for studying disorders that are sensitive to gene dosage. Overexpression of *PMP22*, for example, mimics a common duplication of this gene in type 1A Charcot-Marie-Tooth disease ([Chap. 379](#)). Duplication of the *PMP22* gene results in high levels of expression of peripheral myelin protein 22, and this dosage effect is responsible for the demyelinating neuropathy. Expression of the Y chromosome-specific gene, *SRY*, in XX females demonstrates that *SRY* is sufficient to induce the formation of testes. This finding confirms the pathogenic role of *SRY* translocations to the X chromosome in sex-reversed XX females. Huntington disease is an autosomal dominant disorder caused by expansion of a CAG trinucleotide repeat that encodes a polyglutamine tract. Targeted deletion of the Huntington disease (*HD*) gene does not induce the neurologic disorder. On the other hand, transgenic expression of the entire gene or of the first exon containing the expanded polyGlu repeat is sufficient to cause many features of the neurologic disorder, indicating a gain-of-function property for the expanded polyGlu-containing protein. Transgenic strategies can also be used as a precursor to gene therapy. Expression of dystrophin, the protein that is deleted in Duchenne muscular dystrophy, partially corrects the disorder in a mouse model of Duchenne's. Targeted expression of oncogenes has been valuable to study mechanisms of neoplasia and to generate immortalized cell lines. For example, expression of the simian virus 40 (SV40) large T antigen under the direction of the insulin promoter induces the formation of islet cell tumors.

Targeted deletion mutagenesis, commonly known as *gene knockout*, is performed using a targeting vector that carries a mutant version of the gene. After homologous recombination in embryonic stem (ES) cells, chimeric animals are produced by injection of ES cells into blastocysts. Subsequently, animals are bred to be heterozygous or homozygous for the mutation. In addition to their use in examining gene function, gene knockouts provide valuable animal models for many loss-of-function mutations. A

variation of this strategy is to use cre recombinase to induce genetic recombination in vivo. Cre recombinase will delete genes that have been flanked by its recognition sequences, called *loxP* sites. One advantage to this approach is that transgenic expression of cre in specific tissues can be used to delete a gene in a tissue-specific or developmentally staged manner. This is particularly useful for genes that would be lethal if deleted universally or during early development.

The list of genes that have been knocked out in mice is already very large. Many of these knockouts do not have an apparent phenotype, either because of redundant functions of the other genes or because the phenotype is subtle. For example, deletion of the hypoxanthine phosphoribosyltransferase (HPRT) gene (*Hprt*) does not cause characteristic features of Lesch-Nyhan syndrome in mice because of their reliance on adenine phosphoribosyltransferase (APRT) in the purine salvage pathway. The administration of an APRT inhibitor to HPRT-deficient mice, however, results in the typical self-injurious behavior seen in patients with Lesch-Nyhan syndrome. The phenotypes of some knockouts are quite different from their human disease counterparts. For example, deletion of the retinoblastoma (*Rb*) gene does not lead to retinoblastoma or other tumors that characterize the human syndrome. These examples underscore the fact that the functions of genes, and their interactions with genetic background and the environment, cannot be assumed to be identical in mice and humans. On the other hand, the deletion of many genes provides a remarkably faithful model of human disorders ([Table 65-3](#)). In addition to clarifying pathophysiology, these models facilitate the development of therapies, both genetic and pharmaceutical.

In addition to transgenic animal models, naturally occurring mutations in mice and other species continue to provide fundamental insights into human disease. A compendium of natural and transgenic animal models is provided in continuously evolving databases (Online Mendelian Inheritance in Animals OMIA: <http://www.angis.su.oz.au/Databases/BIRX/omia/>; The Jackson Library: <http://www.jax.org/>).

Human pluripotent *stem cells* have recently been developed, and, consistent with their potential for self-renewal, these cell lines express high levels of telomerase, an enzyme that is essential for allowing repeated replication of the ends of eukaryotic chromosomes. Although much remains to be learned about the properties of pluripotent stem cells, they may prove useful for transplantation, drug testing, or for other purposes.

THE HUMAN GENOME PROJECT

The [HGP](#) was initiated in the mid-1980s as an ambitious effort to characterize the human genome, culminating in a complete DNA sequence. In the United States, the National Institutes of Health (NIH) and the Department of Energy (DOE) officially launched the genome project in 1990; the project has evolved as an international effort and has also included important contributions from the private sector. The main goals include: (1) creation of genetic maps, (2) development of physical maps, and (3) determination of the complete human DNA sequence.

Some analogies help in appreciating the scope of the [HGP](#). The 23 pairs of human

chromosomes are thought to encode approximately 100,000 genes. The total length of DNA is about 3 billion bp, which is nearly 1000-fold greater than the *E. coli* genome. If the human DNA sequence were printed out, it would correspond to about 120 volumes of *Harrison's Principles of Internal Medicine*.

THE GENETIC MAP

Given the size and complexity of the human genome, genetic maps have been developed to provide orientation and to delimit where a gene of interest may be located. A *genetic map* describes the order of genes and defines the position of a gene relative to other loci on the same chromosome. It is constructed by assessing how frequently two markers are inherited together by linkage studies. Distances of the genetic map are expressed in recombination units, or centimorgans (cM). One cM corresponds to a recombination frequency of 1% between two polymorphic markers; 1 cM corresponds to approximately 1 Mb of DNA ([Fig. 65-3](#)). Any polymorphic sequence variation can be useful for mapping purposes. Examples of polymorphic markers include variable number of tandem repeats (VNTRs), [RFLPs](#), microsatellite repeats, and single nucleotide polymorphisms (SNPs); the latter two methods are now used predominantly because of the high density of markers and because they are amenable to automated procedures.

The current genetic map exists at about 1 cM resolution. A goal for the near term is to add about 100,000 [SNPs](#) to these maps. This would provide 1 SNP approximately every 100,000 bp. The addition of SNPs will facilitate automation using DNA chips and will enhance the ability to perform linkage studies of complex genetic diseases.

THE PHYSICAL MAP

Cytogenetics and chromosomal banding techniques provide a relatively low-resolution microscopic view of genetic loci. Physical maps indicate the position of a locus or gene in absolute values. Sequence-tagged sites (STSs) are used as a standard unit for physical mapping and serve as sequence-specific landmarks for arranging overlapping cloned fragments in the same order as they occur in the genome. These overlapping clones, usually in [YACs](#) or [BACs](#), allow the characterization of contiguous DNA sequences, commonly referred to as *contigs* ([Fig. 65-4](#)). The STSs consist of 200 to 500 bp, which can be retrieved from computer databases; >50,000 STSs have been mapped. The goal of achieving a high-resolution physical map of the human genome has essentially been achieved as all of the genome has been cloned into overlapping fragments. The highest resolution physical map will provide the complete DNA sequence of each chromosome in the human genome.

STATUS OF DNA SEQUENCING

The primary focus of the genome project is to obtain DNA sequence for the entire human genome as well as model organisms. The sequences of *E. coli* and many other bacteria, *S. cerevisiae*, *C. elegans*, and *D. melanogaster* have already been completed. Sequencing of the laboratory mouse genome is in progress. Although the prospect of determining the complete sequence of the human genome was a daunting prospect several years ago, technical advances in DNA sequencing and bioinformatics have led

to the completion of a draft human sequence in June 2000, well in advance of the original goal of the year 2003. The current standard is to achieve 99.99% (1 error in 10,000 bp) accuracy. This level of accuracy is important for many reasons, including efforts to determine the degree of DNA sequence variation in the population. Comparisons of the DNA sequence from multiple individuals or populations will allow assessments of genetic variance in the human population. Another goal is to develop a complete set of full-length human cDNAs and to define their locations on the physical map.

ETHICAL ISSUES

Implicit in the [HGP](#) is the idea and hope that identifying disease-causing genes can lead to improvements in diagnosis, prognosis, and treatment. It is estimated that most individuals harbor several serious recessive genes. However, completion of the human genome sequence, determination of the association of genetic defects with disease, and studies of genetic variation raise many new issues with implications for the individual and mankind. The controversies concerning the cloning of mammals and the establishment of human embryonic stem cells underscore the relevance of these questions. Moreover, the information gleaned from genotypic results can have quite different impacts, depending on the availability of strategies to modify the course of disease. For example, the identification of mutations that cause multiple endocrine neoplasia (MEN) type 2 or hemochromatosis allows specific interventions for affected family members. On the other hand, at present the identification of an Alzheimer or Huntington disease gene does not alter therapy. Genetic test results can generate anxiety in affected individuals and family members, and there is the possibility of discrimination on the basis of the test results. Most genetic disorders are likely to fall into an intermediate category where the opportunity for prevention or treatment is significant but limited ([Chap. 68](#)). For these reasons, the scientific components of the HGP have been paralleled by efforts to examine ethical and legal implications as new issues arise.

Many issues raised by the genome project are familiar, in principle, to medical practitioners. For example, an asymptomatic patient with increased low-density lipoprotein (LDL) cholesterol, high blood pressure, or a strong family history of early myocardial infarction, is known to be at increased risk of coronary heart disease. In such cases, it is clear that the identification of risk factors and an appropriate intervention are beneficial. Likewise, patients with phenylketonuria, cystic fibrosis, or sickle cell anemia are often identified as having a genetic disease early in life. These precedents can be helpful for adapting policies that relate to genetic information. We can anticipate similar efforts, whether based on genotypes or other markers of genetic predisposition, to be applied to many disorders. One confounding aspect of the rapid expansion of information is that our ability to make clinical predictions often lags behind genetic advances. For example, when genes that predispose to breast cancer, such as *BRCA1*, are described, they generate tremendous public interest in the potential to predict disease, but many years of clinical research are still required to rigorously establish genotype and phenotype correlations.

Whether related to informed consent, participation in research, or the management of a genetic disorder that affects an individual or their families, there is a great need for more

information about fundamental principles of genetics. The pervasive nature of the role of genetics in medicine makes it imperative for physicians and other health care professionals to become more informed about genetics and to provide advice and counseling in conjunction with trained genetic counselors ([Chap. 68](#)). The application of screening and prevention strategies will therefore require intensive patient and physician education, changes in health care financing, and legislation to protect patient's rights.

TRANSMISSION OF GENETIC DISEASE

ORIGINS AND TYPES OF MUTATIONS

A *mutation* can be defined as any change in the primary nucleotide sequence of DNA regardless of its functional consequences. Some mutations may be lethal, others are less deleterious, and some may confer an evolutionary advantage. Mutations can occur in the germline (sperm or oocytes); these can be transmitted to progeny. Alternatively, mutations can occur during embryogenesis or in somatic tissues. Mutations that occur during development lead to *mosaicism*, a situation in which tissues are composed of cells with different genetic constitutions. If the germline is mosaic, a mutation can be transmitted to some progeny but not others, which sometimes leads to confusion in assessing the pattern of inheritance. Somatic mutations that do not affect cell survival can sometimes be detected because of variable phenotypic effects in tissues (e.g., pigmented lesions in McCune-Albright syndrome). Other somatic mutations are associated with neoplasia because they confer a growth advantage to cells. Epigenetic events such as altered DNA methylation may also influence gene expression. With the exception of triplet nucleotide repeats, which can expand (see below), mutations are usually stable.

Mutations are structurally diverse -- they can involve the entire genome, as in triploidy (one extra set of chromosomes), or gross numerical or structural alterations in chromosomes or individual genes ([Chap. 66](#)). Large deletions may affect a portion of a gene or an entire gene, or, if several genes are involved, they may lead to a *contiguous gene syndrome*. Unequal crossing-over between homologous genes can result in fusion gene mutations, as illustrated by color blindness ([Chap. 28](#)). Mutations involving single nucleotides are referred to as *point mutations*. Substitutions are called *transitions* if a purine is replaced by another purine base (A $\leftrightarrow$ G) or if a pyrimidine is replaced by another pyrimidine (C $\leftrightarrow$ T). Changes from a purine to a pyrimidine, or vice versa, are referred to as *transversions*. If the DNA sequence change occurs in a coding region and alters an amino acid, it is called a *missense mutation*. Depending on the functional consequences of such a missense mutation, amino acid substitutions in different regions of the protein can lead to distinct phenotypes. *Polymorphisms* are sequence variations that have a frequency of at least 1%. Usually, they do not result in a perceptible phenotype. Often they consist of single base-pair substitutions that do not alter the protein coding sequence because of the degenerate nature of the genetic code, although it is possible that some might alter mRNA stability, translation, or the amino acid sequence. These types of silent base substitutions and [SNPs](#) are encountered frequently during genetic testing and must be distinguished from true mutations that alter protein expression or function. Small nucleotide deletions or insertions cause a shift of the codon reading frame. Most commonly, reading frame

alterations result in an abnormal protein segment of variable length before termination of translation occurs at a stop codon (*nonsense mutation*). Mutations in intronic sequences or in exon junctions may destroy or create splice donor or splice acceptor sites. Mutations may also be found in the regulatory sequences of genes, resulting in reduced gene transcription.

Mutation Rates As noted before, mutations represent an important cause of genetic diversity as well as disease. Mutation rates are difficult to determine in humans because many mutations are silent and because testing is often not adequate to detect the phenotypic consequences. Mutation rates vary in different genes but are estimated to occur at a rate of about 10^{-10} /bp per cell division. Germline mutation rates (as opposed to somatic mutations) are relevant in the transmission of genetic disease. Because the population of oocytes is established very early in development, only about 20 cell divisions are required for completed oogenesis, whereas spermatogenesis involves about 30 divisions by the time of puberty and 20 cell divisions each year thereafter. Consequently, the probability of acquiring new point mutations is much greater in the male germline than the female germline, in which rates of aneuploidy are increased ([Chap. 66](#)). Thus, the incidence of new point mutations in spermatogonia increases with paternal age (e.g., achondrodysplasia, Marfan syndrome, neurofibromatosis). It is estimated that about 1 in 10 sperm carries a new deleterious mutation. The rates for new mutations are calculated most readily for autosomal dominant and X-linked disorders and are $\sim 10^{-5}$ to 10^{-6} /locus per generation. Because most monogenic diseases are relatively rare, new mutations account for a significant fraction of cases. This is important in the context of genetic counseling, as a new mutation can be transmitted to the affected individual but does not necessarily imply that the parents are at risk to transmit the disease to other children. An exception to this is when the new mutation occurs early in germline development, leading to *gonadal mosaicism*.

Unequal Crossing-Over Normally, DNA recombination in germ cells occurs with remarkable fidelity to maintain the precise junction sites for the exchanged DNA sequences ([Fig. 65-2](#)). However, mispairing of homologous sequences leads to unequal crossover, with gene duplication on one of the chromosomes and gene deletion on the other chromosome. A significant fraction of growth hormone (*GH*) gene deletions, for example, involve unequal crossing-over ([Chap. 328](#)). The *GH* gene is a member of a large gene cluster that includes a growth hormone variant gene as well as several structurally related chorionic somatomammotropin genes and pseudogenes (highly homologous but functionally inactive relatives of a normal gene). Because such gene clusters contain multiple homologous DNA sequences arranged in tandem, they are particularly prone to undergo recombination and, consequently, gene duplication or deletion. On the other hand, duplication of the *PMP22* gene as a result of unequal crossing-over results in increased gene dosage and type 1A Charcot-Marie-Tooth disease ([Chap. 379](#)). Unequal crossing-over resulting in deletion of *PMP22* results in a distinct neuropathy called *hereditary liability to pressure palsy* ([Chap. 379](#)).

Glucocorticoid-remediable aldosteronism (GRA) is caused by a rearrangement involving the genes that encode aldosterone synthase (*CYP11B2*) and steroid 11 β -hydroxylase (*CYP11B1*), normally arranged in tandem on chromosome 8q. These two genes are 95% identical, predisposing to gene duplication and deletion by unequal crossing-over. The rearranged gene product contains the regulatory regions of 11 β -hydroxylase fused

to the coding sequence of aldosterone synthetase. Consequently, the latter enzyme is expressed in the adrenocorticotrophic hormone (ACTH)-dependent zone of the adrenal gland, resulting in overproduction of mineralocorticoids and hypertension ([Chap. 331](#)).

Gene conversion refers to a nonreciprocal exchange of homologous genetic information; it is probably more common than generally recognized. In human genetics, gene conversion has been used to explain how an internal portion of a gene is replaced by a homologous segment copied from another allele or locus; these genetic alterations may range from a few nucleotides to a few thousand nucleotides. As a result of gene conversion, it is possible for short DNA segments of two chromosomes to be identical, even though these sequences are distinct in the parents. A practical consequence of this phenomenon is that nucleotide substitutions can occur during gene conversion between related genes, often altering the function of the gene. In disease states, gene conversion often involves intergenic exchange of DNA between a gene and a related pseudogene. For example, the 21-hydroxylase gene (*CYP21A*) is adjacent to a nonfunctional pseudogene. Many of the nucleotide substitutions that are found in the *CYP21A* gene in patients with congenital adrenal hyperplasia correspond to sequences that are present in the pseudogene, suggesting gene conversion as a mechanism of mutagenesis. In addition, mitotic gene conversion has been suggested as a mechanism to explain revertant mosaicism in which an inherited mutation is "corrected" in certain cells. For example, patients with autosomal recessive generalized atrophic benign epidermolysis bullosa have acquired reverse mutations in one of the two mutated *COL17A1* alleles, leading to clinically unaffected patches of skin.

Insertions and Deletions Though many instances of insertions and deletions occur as a consequence of unequal crossing-over, there is also evidence for internal duplication, inversion, or deletion of DNA sequences. The fact that certain deletions or insertions appear to occur repeatedly as independent events suggests that specific regions within the DNA sequence predispose to these errors. For example, certain regions of the *DMD* gene appear to be hot spots for deletions.

Errors in DNA Repair Because mutations caused by defects in DNA repair accumulate as somatic cells divide, these types of mutations are particularly important in the context of neoplastic disorders ([Chap. 82](#)). Several genetic disorders involving DNA repair enzymes underscore their importance. Patients with xeroderma pigmentosum have defects in DNA damage recognition or in the nucleotide excision and repair pathway ([Chap. 86](#)). Exposed skin is dry and pigmented and is extraordinarily sensitive to the mutagenic effects of ultraviolet irradiation. More than 10 different genes have been shown to cause the different forms of xeroderma pigmentosum. This finding is consistent with the earlier classification of this disease into different complementation groups ([Table 65-4](#)) in which normal function is rescued by the fusion of cells derived from two different forms of xeroderma pigmentosum.

Ataxia telangiectasia causes large telangiectatic lesions of the face, cerebellar ataxia, immunologic defects, and hypersensitivity to ionizing radiation ([Chap. 364](#)). The discovery of the ataxia telangiectasia mutated (*ATM*) gene reveals that it is homologous to genes involved in DNA repair and control of cell cycle checkpoints. Mutations in the *ATM* gene give rise to defects in meiosis as well as increasing susceptibility to damage from ionizing radiation. Fanconi's anemia is also associated with an increased risk of

multiple acquired genetic abnormalities. It is characterized by diverse congenital anomalies and a strong predisposition to develop aplastic anemia and acute myelogenous leukemia ([Chap. 111](#)). Cells from these patients are susceptible to chromosomal breaks caused by a defect in genetic recombination. At least eight different complementation groups have been identified, and several loci and genes associated with Fanconi's anemia have been mapped or cloned ([Table 65-4](#)).

[HNPCC](#) is caused by mutations in one of several different mismatch repair (MMR) genes including MutS homologue 2 (*MSH2*) and MutL homologue 1 (*MLH1*) ([Chap. 90](#)). These enzymes are involved in the detection of nucleotide mismatches and in the recognition of slipped-strand trinucleotide repeats. Germline mutations in these genes lead to microsatellite instability and a high mutation rate in colon cancer. This syndrome is characterized by autosomal dominant transmission of colon cancer, young age (<50 years) of presentation, predisposition to lesions in the proximal large bowel, and associated malignancies such as uterine cancer and ovarian cancer. Genetic screening tests for this disorder are now being used for families considered to be at risk ([Chap. 68](#)). Recognition of HNPCC allows early screening with colonoscopy and the implementation of prevention strategies using nonsteroidal anti-inflammatory drugs.

CpG and Dipyrimidine Sequences Certain DNA sequences are particularly susceptible to mutagenesis. Successive pyrimidine residues (e.g., T-T or C-C) are subject to the formation of ultraviolet light-induced photoadducts. If these pyrimidine dimers are not repaired by the nucleotide excision repair pathway, mutations will be introduced after DNA synthesis. The dinucleotide C-G, or CpG, is also a hot spot for a specific type of mutation. In this case, methylation of the cytosine is associated with an enhanced rate of deamination to uracil, which is then replaced with thymine. This C → T transition (or G → A on the opposite strand) accounts for at least one-third of point mutations associated with polymorphisms and mutations. Many of the *MSH2* mutations in [HNPCC](#), for example, involve CpG sequences.

Certain types of mutations (C → T or G → A) are relatively common. Moreover, the redundant nature of the genetic code results in overrepresentation of certain amino acid substitutions. For example, arginine codons are most likely to be converted to cysteine, tryptophan, or a stop codon when a C → T transition occurs, and to histidine or glutamine when a G → A transition occurs.

Unstable DNA Sequences *Trinucleotide repeats* may be unstable and expand beyond a critical number. Mechanistically, the expansion is thought to be caused by unequal recombination and slipped mispairing. A premutation represents a small increase in trinucleotide copy number. In subsequent generations, the expanded repeat may increase further in length and result in an increasingly severe phenotype, a process called *dynamic mutation* (see below for discussion of anticipation). Trinucleotide expansion was first recognized as a cause of the fragile X syndrome, one of the most common causes of mental retardation ([Chap. 359](#)). Other disorders arising from a similar mechanism include Huntington disease ([Chap. 362](#)), X-linked spinobulbar muscular atrophy ([Chap. 365](#)), and myotonic dystrophy ([Chap. 383](#)) ([Tables 65-5](#) and [65-6](#)). Malignant cells are also characterized by genetic instability, indicating a breakdown in mechanisms that regulate DNA repair and the cell cycle.

FUNCTIONAL CONSEQUENCES OF MUTATIONS

Functionally, mutations can be broadly classified as gain-of-function and loss-of-function mutations. Gain-of-function mutations are typically dominant; that is, they result in phenotypic alterations when a single allele is affected. Inactivating mutations are usually recessive, and an affected individual is homozygous or compound heterozygous (i.e., carrying two different mutant alleles) for the disease-causing mutations. Alternatively, mutation in a single allele can result in *haploinsufficiency*, a situation in which one normal allele is not sufficient for a normal phenotype. This phenomenon applies, for example, to expression of rate-limiting enzymes in heme synthesis that cause porphyrias ([Chap. 346](#)). An increase in dosage of a gene product may also result in disease, as illustrated by the duplication of the *DAX1* gene in dosage-sensitive sex-reversal ([Chap. 338](#)). Mutation in a single allele can also result in loss of function due to a dominant-negative effect. In this case, the mutated allele interferes with the function of the normal gene product by one of several different mechanisms: (1) a mutant protein may interfere with the function of a multimeric protein complex, as illustrated by mutations in type 1 collagen (*COL1A1*, *COL1A2*) genes in osteogenesis imperfecta ([Chap. 351](#)); (2) a mutant protein may occupy binding sites on proteins or promoter response elements, as illustrated by thyroid hormone resistance, a disorder in which inactivated thyroid hormone receptor binds to target genes and functions as an antagonist of normal receptors ([Chap. 330](#)); or (3) a mutant protein can be cytotoxic as in α_1 -antitrypsin deficiency ([Chap. 258](#)) or autosomal dominant neurohypophyseal diabetes insipidus ([Chap. 329](#)), in which the abnormally folded proteins are trapped within the endoplasmic reticulum and ultimately cause cellular damage.

GENOTYPE AND PHENOTYPE

Alleles, Genotypes, and Haplotypes An observed trait is referred to as a *phenotype*; the genetic information defining the phenotype is called the *genotype*. Alternative forms of a gene or a genetic marker are referred to as alleles. Alleles may be polymorphic variants of nucleic acids that have no apparent effect on gene expression or function. In other instances, these variants may have subtle effects on gene expression, thereby conferring the adaptive advantages associated with genetic diversity. On the other hand, allelic variants may reflect mutations in a gene that clearly alter its function. The common Glu⁶ Val sickle cell mutation (E6V) in the *β-globin* gene and the ΔF508 deletion of phenylalanine (F) in the *CFTR* gene are examples of allelic variants of these genes. Because each individual has two copies of each chromosome (one inherited from the mother and one inherited from the father), he or she can only have two alleles at a given locus. However, there can be many different alleles in the population. The normal or common allele is usually referred to as *wild type*. When alleles at a given locus are identical, the individual is *homozygous*. Inheriting such identical copies of a mutant allele occurs in many autosomal recessive disorders, particularly in circumstances of consanguinity. If the alleles are different, the individual is *heterozygous* at this locus. If two different mutant alleles are inherited at a given locus, the individual is said to be a *compound heterozygote*. *Hemizygous* is used to describe males with a mutation in an X chromosomal gene, or a female with a loss of one X chromosomal locus.

Genotypes describe the specific alleles at a particular locus. For example, there are

three common alleles (E2, E3, E4) of the apolipoprotein E (*APOE*) gene. The genotype of an individual can therefore be described as *APOE*3/4 or *APOE*4/4 or any other variant. These designations indicate which alleles are present on the two chromosomes in the *APOE* gene at locus 19q13.2. In other cases, the genotype might be assigned arbitrary numbers (e.g., 1/2) or letters (e.g., B/b) to distinguish different alleles.

A *haplotype* refers to a group of alleles that are closely linked together at a genomic locus. Haplotypes are useful for tracking the transmission of genomic segments within families and for detecting evidence of genetic recombination, if the crossover event occurs between the alleles ([Fig. 65-3](#)). As an example, various alleles at the histocompatibility locus antigen (HLA) on chromosome 6p are used to establish haplotypes associated with certain disease states. For example, 21-hydroxylase deficiency, complement deficiency, and hemochromatosis are each associated with specific HLA haplotypes. It is now recognized that these genes lie in close vicinity to the HLA locus, which explains why HLA associations were identified even before the disease genes were cloned and localized. In other cases, specific HLA associations with diseases such as ankylosing spondylitis (HLA-B27) or type 1 diabetes mellitus (HLA-DR4) reflect the role of specific HLA allelic variants in susceptibility to these autoimmune diseases.

Allelic Heterogeneity *Allelic heterogeneity* refers to the fact that different mutations in the same genetic locus can cause an identical or similar phenotype. For example, many different mutations of the β -globin locus can cause β -thalassemia ([Fig. 65-7](#)). In essence, allelic heterogeneity reflects the fact that many different mutations are capable of altering protein structure and function. For this reason, maps of inactivating mutations in genes usually show a near-random distribution. Exceptions include: (1) a founder effect, in which a particular mutation that does not affect reproductive capacity can be traced to a single individual; (2) "hot spots" for mutations, in which the nature of the DNA sequence predisposes to a recurring mutation; and (3) localization of mutations to certain domains that are particularly critical for protein function. Allelic heterogeneity creates a practical problem for genetic testing because one must often examine the entire genetic locus for mutations, as these can differ in each patient.

Phenotypic Heterogeneity *Phenotypic heterogeneity* occurs when more than one phenotype is caused by allelic mutations (e.g., different mutations in the same gene). For example, mutations in the *myosin VIIA* gene can result in four distinct clinical disorders: (1) autosomal recessive deafness DFNB2, (2) autosomal dominant nonsyndromic deafness DFNA11, (3) Usher 1B syndrome [congenital deafness, retinitis pigmentosa ([Plate IV-14](#))], and (4) an atypical variant of Usher's syndrome. Similarly, identical mutations in the *FGFR2* gene can result in very distinct phenotypes: Crouzon syndrome (craniofacial synostosis), or Pfeiffer syndrome (acrocephalopolysyndactyly).

Locus or Nonallelic Heterogeneity and Phenocopies *Nonallelic or locus heterogeneity* refers to the situation in which a similar disease phenotype results from mutations at different genetic loci ([Table 65-4](#)). This often occurs when more than one gene product produces different subunits of an interacting complex or when different genes are involved in the same genetic cascade or physiologic pathway. For example, osteogenesis imperfecta can arise from mutations in two different procollagen genes (*COL1A1* or *COL1A2*) that are located on different chromosomes ([Chap. 351](#)). The

effects of inactivating mutations in these two genes are similar because the protein products comprise different subunits of the helical collagen fiber. Similarly, muscular dystrophy syndromes can be caused by mutations in various genes, consistent with the fact that it can be transmitted in an X-linked (Duchenne or Becker), autosomal dominant (limb-girdle muscular dystrophy type 1), or autosomal recessive (limb-girdle muscular dystrophy type 2) manner ([Chap. 383](#)). Mutations in the X-linked *DMD* gene, which encodes dystrophin, are the most common cause of muscular dystrophy. This feature reflects the large size of the gene as well as the fact that the phenotype is expressed in hemizygous males because they only have a single copy of the X chromosome. Dystrophin is associated with a large group of additional proteins that form the membrane-associated cytoskeleton in muscle. Mutations in several components of this protein complex can also cause muscular dystrophy syndromes. Although the phenotypic features of some of these disorders are distinct, the phenotypic spectrum caused by mutations in different genes overlaps, thereby leading to nonallelic heterogeneity. It should be noted that mutations in dystrophin also cause allelic heterogeneity. For example, mutations in the *DMD* gene can cause either Duchenne or the less severe Becker muscular dystrophy, depending on the severity of the protein defect.

Recognition of nonallelic heterogeneity is important for several reasons: (1) the ability to identify disease loci in linkage studies is reduced by including patients with similar phenotypes but different genetic disorders; (2) genetic testing is more complex because several different genes need to be considered along with the possibility of different mutations in each of the candidate genes; and (3) novel information is gained about how genes or proteins interact, providing unique insights into molecular physiology.

Phenocopies refer to circumstances in which nongenetic conditions mimic a genetic disorder. For example, features of toxin- or drug-induced neurologic syndromes can resemble those seen in Huntington disease, and vascular causes of dementia share phenotypic features with familial forms of Alzheimer dementia ([Chap. 362](#)). Children born with activating mutations of the thyroid-stimulating hormone receptor (TSH-R) exhibit goiter and thyrotoxicosis similar to that seen in neonatal Graves' disease, which is caused by the transfer of maternal autoantibodies to the fetus ([Chap. 330](#)). As in nonallelic heterogeneity, the presence of phenocopies has the potential to confound linkage studies and genetic testing. Patient history and subtle differences in phenotype can often provide clues that distinguish these disorders from related genetic conditions.

Variable Expressivity and Incomplete Penetrance It is not uncommon for the same genetic mutation to cause a phenotypic spectrum illustrating the phenomenon of *variable expressivity*. This may include different manifestations of a complex disorder (e.g., [MEN](#)), the severity of the disorder (e.g., sickle cell anemia), or the age of disease onset (e.g., Alzheimer dementia). MEN-1 illustrates several of these features. Families with this autosomal dominant disorder develop tumors of the parathyroid gland, endocrine pancreas, and the pituitary gland ([Chap. 339](#)). However, the pattern of tumors in the different glands, the age at which tumors develop, and the types of hormones produced vary among affected individuals, even within a given family. In this example, the phenotypic variability arises, in part, because of the requirement for a second mutation in the normal copy of the *MEN1* gene, as well as the large array of different cell types that are susceptible to the effects of *MEN1* gene mutations. In part, variable

expression reflects the influence of other genes, or genetic background, on the effects of a particular mutation. Even in identical twins, in whom the genetic constitution is the same, one can occasionally see variable expression of a genetic disease.

Interactions with the environment can also influence the course of a disease. For example, the manifestations and severity of hemochromatosis can be influenced by iron intake ([Chap. 345](#)), and the course of phenylketonuria is affected by exposure to phenylalanine in the diet ([Chap. 352](#)). Other metabolic disorders, such as hyperlipidemias and porphyria, also fall into this category. Many mechanisms, including genetic effects and environmental influences, can therefore lead to variable expressivity. In genetic counseling, it is particularly important to recognize this variability, as one cannot always predict the course of disease, even when the mutation is known.

Penetrance is the probability of expressing the phenotype given a defined genotype; it can be complete or incomplete. For example, hypertrophic obstructive cardiomyopathy (HOCM) caused by mutations in the *myosin heavy chain* gene is a dominant disorder with clinical features in only a subset of patients who carry the mutation ([Chap. 238](#)). Patients who have the mutation but no evidence of the disease can still transmit the disorder to subsequent generations. In this situation, the disorder is said to be *nonpenetrant* or *incompletely penetrant*. This classification depends to some degree on the criteria and techniques used for diagnosis. For disorders such as Huntington disease or familial amyotrophic lateral sclerosis, which present late in life, the rate of penetrance is influenced by the age at which the clinical assessment is performed. *Imprinting* can also modify the penetrance of a disease (see below). For example, in patients with Albright hereditary osteodystrophy, mutations in the Gsa subunit (*GNAS1* gene) are expressed clinically only in individuals who inherit the mutation from their mother ([Chap. 343](#)).

Sex-Influenced Phenotypes Certain mutations affect males and females quite differently. In some instances, this is because the gene resides on the X or Y sex chromosomes (X-linked disorders and Y-linked disorders). As a result, the phenotype of mutated X-linked genes will be expressed fully in males but variably in heterozygous females, depending on the degree of X-inactivation and the function of the gene. For example, most heterozygous female carriers of factor VIII deficiency (hemophilia A) are asymptomatic because sufficient factor VIII is produced to prevent a defect in coagulation ([Chap. 117](#)). On the other hand, some females heterozygous for the X-linked lipid storage defect caused by α -galactosidase A deficiency (Fabry disease) experience mild manifestations of painful neuropathy, as well as other features of the disease ([Chap. 349](#)). Because only males have a Y chromosome, mutations in genes such as *SRY* (which causes male-to-female sex-reversal) or *DAZ* (which causes abnormalities of spermatogenesis) are unique to males ([Chap. 338](#)).

Other diseases are expressed in a sex-limited manner because of the differential function of the gene product in males and females. Activating mutations in the luteinizing hormone receptor cause dominant male-limited precocious puberty in boys ([Chap. 335](#)). The phenotype is unique to males because activation of the receptor induces testosterone production in the testis, whereas it is functionally silent in the immature ovary. Homozygous inactivating mutations of the follicle-stimulating hormone (FSH) receptor cause primary ovarian failure in females because the follicles do not

develop in the absence of FSH action. In contrast, affected males have a more subtle phenotype, because testosterone production is preserved (allowing sexual maturation) and spermatogenesis is only partially impaired ([Chap. 335](#)). In congenital adrenal hyperplasia, most commonly caused by 21-hydroxylase deficiency, cortisol production is impaired and [ACTH](#) stimulation of the adrenal gland leads to increased production of androgenic precursors ([Chap. 331](#)). In females, the increased androgen level causes ambiguous genitalia, which can be recognized at the time of birth. In males, the diagnosis may be made on the basis of adrenal insufficiency at birth, because the increased adrenal androgen level does not alter sexual differentiation, or later in childhood, because of the development of precocious puberty. Hemochromatosis is more common in males than in females, presumably because of differences in dietary iron intake and losses associated with menstruation and pregnancy in females ([Chap. 345](#)).

GENETIC LINKAGE

Genetic linkage refers to the fact that genes are physically connected, or linked, to one another along the chromosomes. Two fundamental principles are essential for understanding the concept of a genetic linkage: (1) When two genes are close together on a chromosome, they are usually transmitted together, unless a recombination event separates them ([Fig. 65-3](#)); and (2) the odds of a crossover, or recombination event, between two linked genes is proportional to the distance that separates them. Thus, genes that are further apart are more likely to undergo a recombination event than genes that are very close together. Linkage is used in genetic counseling to predict the odds of disease gene transmission.

Polymorphisms are essential for linkage studies because they provide a means to distinguish the maternal and paternal chromosomes in an individual. On average, 1 out of every 1000 bp varies from one person to the next. Although this degree of variation seems low (99.9% identical), it means that >3 million sequence differences exist between any two unrelated individuals. This sequence variation usually has no significant functional consequence and provides much of the basis for variation in genetic traits. Although many of these sequence variations are [SNPs](#), other variants include [VNTRs](#) or short tandem repeats (STRs). In VNTRs and STRs, the number of times a sequence is repeated is highly variable in the population. Consequently, the probability that sequences will differ on the two homologous chromosomes is high (often >70 to 90%). Most STRs, also called *polymorphic microsatellite markers*, consist of di-, tri-, or tetranucleotide repeats that can be measured readily using [PCR](#) and primers that reside on either side of the repeat sequences ([Fig. 65-8](#)). Many other methods for analyzing polymorphic variation are also available. Historically, [RFLPs](#) were used to detect sequence variations that caused changes in the recognition sites for restriction enzymes. This procedure has been largely replaced by the use of STRs. Analyses of SNPs, using DNA chips, provide a promising means for rapid analysis of genetic variation and linkage.

In order to identify a chromosomal locus that segregates with a disease, it is necessary to determine the genotype or haplotype of DNA samples from one or several pedigrees. One can then assess whether certain marker alleles cosegregate with the disease. Markers that are closest to the disease gene are less likely to undergo recombination

events and therefore receive a higher linkage score. Linkage is expressed as a lod (logarithm of odds) score -- the ratio of the probability that the disease and marker loci are linked rather than unlinked. Lod scores of +3 (1000:1) are generally accepted as supporting linkage, whereas a score of -2 is consistent with the absence of linkage.

An example of the use of linkage analysis is shown in [Fig. 65-8](#). In this case, the gene for the autosomal dominant disorder, [MEN-1](#), is known to be located on chromosome 11q13. Using positional cloning, the *MEN1* gene was identified and shown to encode menin, the function of which is poorly understood. However, the transmission of the disorder suggests that menin acts like a tumor-suppressor gene. Affected individuals inherit a mutant form of the *MEN1* gene, predisposing them to certain types of tumors (parathyroid, pituitary, pancreatic islet) ([Chap. 339](#)). In the tissues that develop a tumor, a "second hit" occurs in the normal copy of the *MEN1* gene. This somatic mutation may be a point mutation, a microdeletion, or loss of a chromosomal fragment (detected as loss of heterozygosity, LOH). Within a given family, linkage to the *MEN1* gene locus can be assessed without necessarily knowing the specific mutation in the *MEN1* gene. Using polymorphic [STRs](#) that are close to the *MEN1* gene, one can assess transmission of the different *MEN1* alleles and compare this pattern to development of the disorder to determine which allele is associated with risk of MEN-1. In the pedigree shown, the affected grandfather in generation I carries alleles 3 and 4 on the chromosome with the mutated *MEN1* gene and alleles 2 and 2 on his other chromosome 11. Consistent with linkage of the 3/4 genotype to the *MEN1* locus, his son in generation II is affected, whereas his daughter (who inherits the 2/2 genotype from her father) is unaffected. In the third generation, transmission of the 3/4 genotype indicates risk of developing MEN-1, assuming that no genetic recombination between the 3/4 alleles and the *MEN1* gene has occurred. After a specific mutation in the *MEN1* gene is identified within a family, it is possible to track transmission of the mutation itself, thereby eliminating uncertainty caused by recombination.

CHROMOSOMAL DISORDERS

Chromosomal or cytogenetic disorders are caused by numerical or structural aberrations in chromosomes. Deviations in chromosome number are common causes of abortions, developmental disorders, and malformations. **For discussion of disorders of chromosome number and structure, see [Chap. 66](#).*

Contiguous Gene Syndromes Large deletions or duplications may affect a portion of a gene, an entire gene, or, if several genes are involved, cause a *contiguous gene syndrome*. Syndromes associated with chromosomal deletions or duplications have a wide phenotypic spectrum that is dependent on the number of involved gene loci. For example, the cri-du-chat syndrome, one of the most common deletion disorders, is associated with deletions on the short arm of chromosome 5 that vary in size from extremely small deletions within 5p15.2 to the loss of the entire short arm. Because of the variable size of the involved deletions, the phenotype encompasses a spectrum that ranges from severe mental retardation and microcephaly to an isolated catlike cry without morphologic or mental abnormalities.

Contiguous gene syndromes have been useful for identifying the location of new disease-causing genes. Because of the variable size of gene deletions in different

patients, a systemic comparison of phenotypes and locations of deletion breakpoints allows positions of particular genes to be mapped within the critical genomic region.

MONOGENIC MENDELIAN DISORDERS

Monogenic human diseases are frequently referred to as *Mendelian disorders* because they obey the principles of genetic transmission originally set forth in Gregor Mendel's classic work. The mode of inheritance for a given phenotypic trait or disease is determined by pedigree analysis. All affected and unaffected individuals in the family are recorded in a pedigree using standard symbols ([Fig. 65-9](#)). The principles of allelic segregation, and the transmission of alleles from parents to children, are illustrated in [Fig. 65-10](#). One dominant (A) allele and one recessive (a) allele can display three Mendelian modes of inheritance: autosomal dominant, autosomal recessive, and X-chromosomal. About 65% of human monogenic disorders are autosomal dominant, 25% are autosomal recessive, and 5% are X-linked ([Table 65-5](#)). Genetic testing is now available for many of these disorders and plays an increasingly important role in clinical medicine.

Autosomal Dominant Disorders Autosomal dominant disorders assume particular relevance because mutations in a single allele are sufficient to cause the disease. In contrast to recessive disorders, in which disease pathogenesis is relatively straightforward because there is loss of gene function, in dominant disorders there are various disease mechanisms, many of which are unique to the function of the genetic pathway involved.

In autosomal dominant disorders, individuals are affected in successive generations; the disease does not occur in the offspring of unaffected individuals. Males and females are affected with equal frequency because the defective gene resides on one of the 22 autosomes ([Fig. 65-11A](#)). Autosomal dominant mutations alter one of the two alleles at a given locus. Because the alleles segregate randomly at meiosis, the probability that an offspring will be affected is 50%. Unless there is a new germline mutation, an affected individual has an affected parent. Children with a normal genotype do not transmit the disorder. Due to differences in penetrance or expressivity (see above), the clinical manifestations of autosomal dominant disorders may be variable. Because of these variations, it is sometimes challenging to determine the pattern of inheritance.

It should be recognized, however, that some individuals acquire a mutated gene from an unaffected parent. De novo germline mutations occur more frequently during later cell divisions in gametogenesis, explaining why siblings are rarely affected. As noted before, new germline mutations occur more frequently in fathers of advanced age. For example, the average age of fathers with new germline mutations that cause Marfan's syndrome is approximately 37 years, whereas fathers who transmit the disease by inheritance have an average age of about 30 years.

Autosomal Recessive Disorders The clinical expression of autosomal recessive disorders is more uniform than in autosomal dominant disorders. Most mutated alleles lead to a complete or partial loss of function. They frequently involve enzymes in metabolic pathways, receptors, or proteins in signaling cascades. Though most recessive disorders are rare, the relatively high frequency of certain recessive disorders,

such as sickle cell anemia, cystic fibrosis, and thalassemia, is partially explained by a selective biologic advantage for the heterozygous state (see below).

In an autosomal recessive disease, the affected individual, who can be of either sex, is a homozygote or compound heterozygote for a single-gene defect. With a few important exceptions, autosomal recessive diseases are rare and often occur in the context of parental consanguinity. Though heterozygous carriers of a defective allele are usually clinically normal, they may display subtle differences in phenotype that only become apparent with more precise testing or in the context of certain environmental influences. In sickle cell anemia, for example, heterozygotes are normally asymptomatic. However, in situations of dehydration or diminished oxygen pressure, sickle cell crises can also occur in heterozygotes ([Chap. 106](#)).

In most instances, an affected individual is the offspring of heterozygous parents. In this situation, there is a 25% chance that the offspring will have a normal genotype, a 50% probability of a heterozygous state, and a 25% risk of homozygosity for the recessive alleles ([Fig. 65-11B](#)). In the case of one unaffected heterozygous and one affected homozygous parent, the probability of disease increases to 50% for each child. In this instance, the pedigree analysis mimics an autosomal dominant mode of inheritance (*pseudodominance*). In contrast to autosomal dominant disorders, new mutations in recessive alleles are rarely manifest because they usually result in an asymptomatic carrier state.

X-Linked Disorders Males have only one X chromosome; consequently, a daughter always inherits her father's X chromosome in addition to one of her mother's two X chromosomes. A son inherits the Y chromosome from his father and one maternal X chromosome. Thus, the characteristic features of X-linked inheritance are (1) the absence of father-to-son transmission, and (2) the fact that all daughters of an affected male are obligate carriers of the mutant allele ([Fig. 65-11C](#)). The risk of developing disease due to a mutant X-chromosomal gene differs in the two sexes. Because males have only one X chromosome, they are hemizygous for the mutant allele; thus, they are more likely to develop the mutant phenotype, regardless of whether the mutation is dominant or recessive. A female may be either heterozygous or homozygous for the mutant allele, which may be dominant or recessive. The terms *X-linked dominant* or *X-linked recessive* are therefore only applicable to expression of the mutant phenotype in women. In addition, the expression of X-chromosomal genes is influenced by X chromosome inactivation (see below).

Y-Linked Disorders Only a few genes are known on the Y chromosome. One such gene, the sex-region determining Y factor (*SRY*), or testis-determining factor (*TDF*), is crucial for normal male development. Normally there is infrequent exchange of sequences on the Y chromosome with the X chromosome. Because the *SRY* region is closely adjacent to the pseudoautosomal region, a chromosomal segment on the X and Y chromosomes with a high degree of homology, a crossing-over occasionally involves the *SRY* region. Translocations can result in XY females with the Y chromosome lacking the *SRY* gene or XX males harboring the *SRY* gene on one of the X chromosomes ([Chap. 338](#)). Point mutations in the *SRY* gene may also result in individuals with an XY genotype and an incomplete female phenotype. Most of these mutations occur de novo. Men with oligospermia/azoospermia frequently have microdeletions on the long arm of

the Y chromosome that involve one or more of the azoospermia factor (AZF) genes.

EXCEPTIONS TO SIMPLE MENDELIAN INHERITANCE PATTERNS

Mitochondrial Disorders Each mitochondrion contains several copies of a circular chromosome. Mitochondrial DNA predominantly encodes transfer RNAs and proteins that are components of the respiratory chain involved in oxidative phosphorylation and ATP generation. The mitochondrial genome is inherited through the maternal line because sperm does not contribute significant cytoplasmic components to the zygote. All children from an affected mother will inherit the disease, but it will not be transmitted from an affected father to his children. During cell replication, the proportion of wild-type and mutant mitochondria can drift; differences in the fraction of defective mitochondria are referred to as *heteroplasmy* and explain, in part, the phenotypic variability that is common in mitochondrial diseases. **For detailed discussion of mitochondrial disorders, see Chap. 67.*

Mosaicism Mosaicism refers to the presence of two or more genetically distinct cell lines in the tissues of an individual. It results from a mutation that occurs during embryonic, fetal, or extrauterine development. The developmental stage at which the mutation arises will determine whether germ cells and/or somatic cells are involved. Chromosomal mosaicism results from non-disjunction at an early embryonic mitotic division, leading to the persistence of more than one cell line, as exemplified by some patients with Turner syndrome ([Chap. 338](#)). Somatic mosaicism is characterized by a patchy distribution of genetically altered somatic cells. The McCune-Albright syndrome, for example, is caused by activating mutations in the stimulatory G protein $\alpha(G_{sa})$ that occur early in development ([Chap. 343](#)). The clinical phenotype varies depending on the tissue distribution of the mutation; manifestations include ovarian cysts that secrete sex steroids and cause precocious puberty, polyostotic fibrous dysplasia, cafe-au-lait skin pigmentation, growth hormone-secreting pituitary adenomas, and hypersecreting autonomous thyroid nodules ([Chap. 336](#)).

X-Inactivation, Imprinting, and Uniparental Disomy According to traditional Mendelian principles, the parental origin of a mutant gene is irrelevant for the expression of the phenotype. Nonetheless, there are important exceptions to this rule. X-inactivation prevents the expression of most genes on one of the two X-chromosomes in every cell of a female. Gene inactivation also occurs on selected chromosomal regions of autosomes. This phenomenon, referred to as *genomic imprinting*, leads to preferential expression of an allele depending on its parental origin. It is of pathophysiologic importance in disorders where the transmission of disease is dependent on the sex of the transmitting parent and, thus, plays an important role in the expression of certain genetic disorders. Two classic examples are the Prader-Willi syndrome and Angelman syndrome ([Chap. 66](#)). Prader-Willi syndrome is characterized by diminished fetal activity, obesity, hypotonia, mental retardation, short stature, and hypogonadotropic hypogonadism. Deletions in the Prader-Willi syndrome occur exclusively on the paternal chromosome 15. In contrast, patients with Angelman syndrome, characterized by mental retardation, seizures, ataxia, and hypotonia, have deletions at the same site of chromosome 15; however, they are located on the maternal chromosome 15. These two syndromes may also result from *uniparental disomy*. In this case, the syndromes are not caused by deletions on chromosome 15 but

by the inheritance of either two paternal chromosomes (Prader-Willi syndrome), or two maternal chromosomes (Angelman syndrome).

Imprinting and the related phenomenon of allelic exclusion may be more common than currently documented, as it is difficult to examine levels of mRNA expression from the maternal and paternal alleles in specific tissues or in individual cells. Genomic imprinting, or uniparental disomy, is involved in the pathogenesis of several other disorders and malignancies ([Chap. 66](#)). Hydatidiform mole contains a normal number of diploid chromosomes, but they are all of paternal origin. The opposite situation occurs in ovarian teratomata, with 46 chromosomes of maternal origin. Expression of the imprinted gene for insulin-like growth factor II (IGF-II) is involved in the pathogenesis of the cancer-predisposing Beckwith-Wiedemann syndrome (BWS) ([Chap. 81](#)). These children show somatic overgrowth with organomegalies and hemihypertrophy, and they have an increased risk of embryonal malignancies such as Wilm's tumor. Normally only the paternally derived copy of the *IGF-II* gene is active and the maternal copy is inactive. Imprinting of the *IGF-II* gene is regulated by *H19*, which encodes an RNA transcript that is not translated into protein. Disruption or lack of *H19* methylation leads to a relaxation of *IGF-II* imprinting and expression of both alleles. Heritable changes in gene expression not associated with DNA sequence alterations are referred to as *epigenetic effects*; these changes are increasingly recognized to play a role in human diseases and possibly in aging as well ([Chap. 9](#)).

Somatic Mutations In many cancer syndromes, there is an inherited predisposition to tumor formation. However, the neoplastic process requires the acquisition of additional somatic mutations ([Chap. 81](#)). In retinoblastoma, the tumor develops when both copies of the retinoblastoma (*RB*) gene are inactivated through two somatic events (sporadic retinoblastoma) or through a somatic loss of the normal allele in an individual with a hereditary defect in the other allele (hereditary retinoblastoma). This "two-hit" model applies to other inherited cancer syndromes such as *MEN-1* ([Chap. 339](#)) and neurofibromatosis type 2 ([Chap. 370](#)). The defective allele is transmitted in a dominant pattern, though tumorigenesis results from a recessive loss of the tumor suppressor gene in an affected tissue. In other instances, the development of cancer typically requires somatic defects in multiple genes, a process termed *multistep carcinogenesis* ([Chap. 82](#)).

Nucleotide Repeat Expansion Disorders Several diseases are associated with an increase in the number of nucleotide repeats above a certain threshold ([Table 65-6](#)). The repeats are sometimes located within the coding region of the genes, as in Huntington disease or the X-linked form of spinal and bulbar muscular atrophy (SBMA, Kennedy syndrome). In other instances, the repeats probably alter gene regulatory sequences. If an expansion is present, the DNA fragment is unstable and tends to expand further during cell division. The length of the nucleotide repeat often correlates with the severity of the disease. When repeat length increases from one generation to the next, disease manifestations may worsen or be observed at an earlier age; this phenomenon is referred to as *anticipation*. In Huntington disease, for example, there is a correlation between age of onset and length of the triplet codon expansion ([Chap. 362](#)). Anticipation has also been documented in other diseases caused by dynamic mutations in trinucleotide repeats ([Table 65-6](#)). The repeat number may also vary in a tissue-specific manner. In myotonic dystrophy, the CTG repeat may be tenfold greater in

muscle tissue than in lymphocytes ([Chap. 383](#)).

POPULATION GENETICS AND ASSOCIATION STUDIES

Overview of Population Genetics In population genetics, the focus changes from alterations in an individual's genome to the distribution pattern of different genotypes of alleles in the population. In a case where there are only two alleles, A and a, the frequency of the genotypes will be $p^2 + 2pq + q^2 = 1$, with p^2 corresponding to the frequency of AA, $2pq$ to the frequency of Aa, and q^2 to aa. When the frequency of an allele is known, the frequency of the genotype can be calculated. Alternatively, one can determine an allele frequency, if the genotype frequency has been determined.

Allele frequencies vary among ethnic groups and geographical regions. For example, heterozygous mutations in the *CFTR* gene are relatively common in populations of European origin but are rare in the African population. Allele frequencies may vary because certain allelic variants confer a selective advantage. For example, heterozygotes for the sickle cell mutation, which is particularly common in West Africa, are more resistant to malarial infection because the erythrocytes of heterozygotes provide a less favorable environment for *Plasmodium* parasites. Though homozygosity for the sickle cell gene is associated with severe anemia and sickle crises ([Chap. 106](#)), heterozygotes have a higher probability of survival because of the reduced morbidity and mortality from malaria; this phenomenon has led to an increased frequency of the mutant allele. Recessive conditions are more prevalent in geographically isolated populations because of the more restricted gene pool.

Allelic Association and Linkage Disequilibrium There are two primary strategies for mapping genes that cause or increase susceptibility to human disease: (1) classic linkage can be performed based on a known genetic model (see above) or, when the model is unknown, by studying pairs of affected relatives; or (2) disease genes can be mapped using allelic association studies ([Table 65-7](#)). *Allelic association* refers to a situation in which the frequency of an allele is significantly increased or decreased in a particular disease. Linkage and association differ in several aspects. Genetic linkage is demonstrable in families or sibships. Association studies, on the other hand, compare a population of affected individuals with a control population. Association studies can be performed as case-control studies that include unrelated affected individuals and matched controls, or as family-based studies that compare the frequencies of alleles transmitted or not transmitted to affected children.

Allelic association studies are particularly useful for identifying susceptibility genes in complex diseases. When alleles at two loci occur more frequently in combination than would be predicted (based on known allele frequencies and recombination fractions), they are said to be in *linkage disequilibrium*. In [Fig. 65-12](#), a mutation, Z, has occurred at a susceptibility locus where the normal allele is Y. The mutation is in close proximity to a genetic polymorphism with allele A or B. With time, the chromosomes carrying the A and Z alleles accumulate and represent 10% of the chromosomes in the population. The fact that the disease susceptibility gene, Z, is found preferentially, or exclusively, in association with the A allele illustrates linkage disequilibrium. Though not all chromosomes carrying the A allele carry the disease gene, the A allele is associated with an increased risk because of its possible association with the Z allele. This model

implies that it may be possible in the future to identify Z directly to provide a more accurate prediction of disease susceptibility. Evidence for linkage disequilibrium can be helpful in mapping disease genes because it suggests that the two loci, in this case A and Z, are tightly linked.

POLYGENIC DISEASE AND COMPLEX GENETIC TRAITS

Approach to Polygenic and Multifactorial Disease The expression of many common diseases such as cardiovascular disease, hypertension, diabetes, asthma, psychiatric disorders, and certain cancers is determined by genetic background, environmental factors, and lifestyle ([Table 65-8](#)). A trait is called *polygenic* if multiple genes are thought to contribute to the phenotype or *multifactorial* if multiple genes are assumed to interact with environmental factors. Genetic models for complex traits need to account for genetic heterogeneity and interactions with other genes and the environment. Complex genetic traits may be influenced by modifying genes that are not linked to the main gene involved in the pathogenesis of the trait. This type of gene-gene interaction, or *epistasis*, plays an important role in polygenic traits that require the simultaneous presence of variations in multiple genes in order to result in a pathologic phenotype.

Gene-environment interactions are relevant for many monogenic and polygenic disorders. In phenylketonuria, the phenotypic expression of the disease depends not only on the presence of the mutation in the phenylalanine hydroxylase gene but also on the exposure to the amino acid phenylalanine ([Chap. 352](#)). Another example is type 2 diabetes mellitus, in which genetic, nutritional, and lifestyle factors are intimately interrelated in disease pathogenesis ([Chap. 333](#)). The identification of genetic variations and environmental factors that either predispose or protect against disease is essential for predicting disease risk, designing preventive strategies, and developing novel therapeutic approaches ([Chap. 68](#)). The study of rare monogenic diseases may provide insights into genetic and molecular mechanisms that are subsequently of importance for the understanding of complex diseases. For example, the identification of the insulin promoter factor 1 in maturity-onset of diabetes type 4 was followed by the observation that it also plays a role in the pathogenesis of diabetes mellitus type 2 ([Tables 65-1 and 65-8](#)).

Approach to the Patient

Identifying the Disease-Causing Gene *Genomic medicine* aims to enhance the quality of medical care through the use of genotypic analysis (DNA testing) to identify genetic predisposition to disease, to select more specific pharmacotherapy, and to design individualized medical care based on genotype. Genotype can be deduced by analysis of protein (e.g., hemoglobin, apoprotein E), mRNA, or DNA. However, technological advances have made DNA analysis particularly useful because it can be readily applied to all but the largest genes ([Fig. 65-13](#)).

DNA testing is performed by mutational analysis or linkage studies in individuals at risk for a genetic disorder known to be present in a family. Mass screening programs require tests of high sensitivity and specificity to be cost-effective. Prerequisites for the success of genetic screening programs include the following: that the disorder is potentially serious; that it can be influenced at a presymptomatic stage by changes in behavior, diet, and/or pharmaceutical manipulations; and that the screening does not result in any

harm or discrimination. Screening in Jewish populations for the autosomal recessive neurodegenerative storage disease Tay-Sachs has reduced the number of affected individuals. In contrast, screening for sickle cell trait/disease in African Americans has led to unanticipated problems of discrimination by health insurers and employers. Mass screening programs harbor additional potential problems. For example, screening for the most common genetic alteration in cystic fibrosis, the $\Delta F508$ mutation with a frequency of ~70% in northern Europe, is feasible and seems to be effective. One has to keep in mind, however, that there is pronounced allelic heterogeneity and that the disease can be caused by >600 other mutations. The search for these less common mutations would substantially increase costs but not the effectiveness of the screening program as a whole. Occupational screening programs aim to detect individuals with increased risk for certain professional activities (e.g., α_1 -antitrypsin deficiency and smoke or dust exposure).

MUTATIONAL ANALYSES DNA sequence analysis is increasingly used as a diagnostic tool and significantly enhanced diagnostic accuracy. It is used for determining carrier status and for prenatal testing in monogenic disorders ([Table 65-5](#)). Certain cancer susceptibility genes, such as *BRCA1* and *BRCA2*, may identify individuals with an increased risk for the development of malignancies. The detection of mutations is an important diagnostic and prognostic tool in leukemias and lymphomas. The demonstration of the presence or absence of mutations is also relevant for the rapidly evolving field of pharmacogenetics, including the identification of differences in drug treatment response or metabolism as a function of genetic background.

A general algorithm for the approach to mutational analysis is outlined in [Fig. 65-13](#). The importance of a detailed clinical phenotype cannot be overemphasized. This is the step where one should also consider the possibility of genetic heterogeneity and phenocopies. If obvious candidate genes are suggested by the phenotype, they can be analyzed directly. After identification of a mutation, it is essential to demonstrate that it segregates with the phenotype. The functional characterization of novel mutations is labor-intensive and may require analyses in vitro or in transgenic models in order to document the relevance of the genetic alteration.

Numerous techniques are available for the detection of mutations ([Table 65-9](#)). In a very broad sense, one can distinguish between techniques that allow for screening the absence or presence of known mutations (screening mode) or techniques that definitively characterize mutations. Analyses of large alterations in the genome are possible using cytogenetics, fluorescent in situ hybridization (FISH), and Southern blotting ([Chap. 66](#)).

More discrete sequence alterations rely heavily on the use of the [PCR](#), which allows rapid gene amplification and analysis. Moreover, PCR makes it possible to perform genetic testing and mutational analysis with small amounts of DNA extracted from leukocytes or even from single cells, buccal cells, or hair roots. Screening for point mutations can be performed by numerous methods ([Table 65-9](#)); most are based on the recognition of mismatches between nucleic acid duplexes, electrophoretic separation of single- or double-stranded DNA, or sequencing of DNA fragments amplified by PCR. DNA sequencing can be performed directly on PCR products or on fragments cloned into plasmid vectors amplified in bacterial host cells.

[RT-PCR](#) may be useful to detect absent or reduced levels of mRNA expression due to a mutated allele. Protein truncation tests (PTT) can be used to detect the broad array of mutations that result in premature termination of a polypeptide during its synthesis. The isolated cDNA is transcribed and translated in vitro, and the proteins are analyzed by gel electrophoresis. Comparison of electrophoretic mobility with the wild-type protein allows detection of truncated mutants.

The majority of traditional diagnostic methods are gel-based. Novel technologies for the analysis of mutations, genetic mapping, and mRNA expression profiles are in rapid development. DNA chip technologies allow hybridization of DNA or RNA to hundreds of thousands of probes simultaneously. Microarrays are being used clinically for mutational analysis of several human disease genes, as well as for the identification of viral sequence variations. Together with the knowledge gained from the [HGP](#), these technologies provide the foundation to expand from a focus on single genes to analyses at the scale of the genome.

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(Bibliography omitted in Palm version)

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66. CHROMOSOME DISORDERS - Terry Hassold, Stuart Schwartz

In humans, the normal diploid number of chromosomes is 46, consisting of 22 pairs of autosomal chromosomes (numbered 1 to 22 in decreasing size) and one pair of sex chromosomes (XX in females and XY in males). The genome is estimated to contain between 80,000 and 100,000 genes, with the smallest autosome housing between 500 and 1000 genes. Not surprisingly, duplications or deletions of even small chromosome segments have profound consequences on normal gene expression.

Deviations in the number or structure of the 46 human chromosomes are astonishingly common, despite severe deleterious consequences. Chromosomal disorders occur in an estimated 10 to 25% of all pregnancies. They are the leading cause of fetal loss and, among pregnancies surviving to term, the leading known cause of birth defects and mental retardation.

In recent years, the practice of cytogenetics has shifted from conventional cytogenetic methodology to a union of cytogenetic and molecular techniques. Formerly the province of research laboratories, *fluorescence in situ hybridization* (FISH) and related molecular cytogenetic technologies have been incorporated into everyday practice in clinical laboratories. As a result, there is an increased appreciation of the importance of "subtle" constitutional cytogenetic abnormalities, such as microdeletions and imprinting disorders, as well as previously recognized translocations and disorders of chromosome number.

VISUALIZING CHROMOSOMES

CONVENTIONAL CYTOGENETIC ANALYSIS

In theory, chromosome preparations can be obtained from any actively dividing tissue by causing the cells to arrest in metaphase, the stage of the cell cycle at which chromosomes are maximally condensed. In practice, only a small number of tissues are used for routine chromosome analysis: amniocytes or chorionic villi for prenatal testing; and blood, bone marrow, or skin fibroblasts for postnatal studies. Samples of blood, bone marrow, and chorionic villi can be processed using short-term culture techniques that yield results in 1 to 3 days. Analysis of other tissue types typically involves long-term tissue culture, requiring 1 to 3 weeks of processing before cytogenetic analysis is possible.

Regardless of the culturing technique, cells are processed to recover chromosomes at metaphase or prometaphase and treated chemically or enzymatically to reveal chromosome "bands" ([Fig. 66-1](#)). Analysis of the number of chromosomes in the cell, and the distribution of bands on individual chromosomes, allows the identification of numerical or structural abnormalities. This strategy is useful for characterizing the normal chromosome complement and determining the incidence and types of major chromosome abnormalities.

Chromosomes are complex structures, consisting of the DNA double helix and chromosome-associated proteins. As for virtually all organisms, each human chromosome contains two specialized structures: a centromere and two telomeres. The

centromere, or primary constriction, divides the chromosome into short (p) and long (q) arms and is responsible for the segregation of chromosomes during cell division. The *telomeres*, or chromosome ends, "cap" the p and q arms and are important for allowing DNA replication at the ends of the chromosomes. Prior to DNA replication, each chromosome consists of a single chromatid copy of the DNA double helix. After DNA replication and continuing until the time of cell division (including metaphase, when chromosomes are typically visualized), each chromosome consists of two identical sister chromatids ([Fig. 66-1](#)).

MOLECULAR CYTOGENETICS

The introduction of [FISH](#) methodologies in the late 1980s revolutionized the field of cytogenetics. In principle, FISH is similar to other DNA-DNA hybridization methodologies. The labeled probe DNA and the target DNA (usually metaphase chromosomes) are denatured to become single-stranded and are hybridized together. The probe is labeled with a hapten, such as biotin or digoxigenin, to allow detection with a fluorophore (e.g., FITC or rhodamine). Alternatively, many probes are already labeled with fluorophores and thus can be detected directly. After the hybridization step, the specimen is counter-stained and the preparations are visualized with a fluorescence microscope.

Types of FISH Probes A variety of probes are available for use with FISH, including chromosome-specific paints (chromosome libraries), repetitive probes, and single-copy probes. Chromosome libraries were developed initially from flow-sorted individual chromosomes and more recently from monochromosomal human-rodent hybrids. These probes hybridize to sequences that span the entirety of the chromosome from which they are derived and, as a result, they can be used to "paint" individual chromosomes ([Fig. 66-2](#)).

Repetitive probes recognize amplified DNA sequences present in chromosomes. The most common are α -satellite DNA probes that are complementary to DNA sequences found at the centromeric regions of all human chromosomes. There are also α -satellite probes that hybridize to the centromeric regions of specific chromosomes ([Fig. 66-2](#)).

A vast number of *single-copy probes* are now available, both commercially and as a result of the human genome project. These probes can be as small as 1 kb, though normally they are packaged in cosmids (40 kb), bacterial artificial chromosomes (BACs) or P1 clones (100 to 200 kb), or yeast artificial chromosomes (YACs) (1 to 2 Mb). With the advent of the National Cancer Institute BAC initiative, these large DNA fragments will be placed at 1-Mb intervals on every chromosome, each of which can be used for [FISH](#) hybridization. Probes for a variety of microdeletion syndromes and for subtelomeric regions of individual chromosomes are commercially available ([Fig. 66-2](#)).

Applications of FISH The majority of FISH applications involve hybridization of one or two probes of interest as an adjunctive procedure to conventional chromosomal banding techniques. In this regard, FISH can be utilized to identify specific chromosomes, characterize de novo duplications or deletions, and identify and clarify subtle chromosomal rearrangements. Its greatest utilization, however, is in the detection of microdeletions (see below), including those associated with Prader-Willi syndrome

(PWS), Angelman syndrome (AS), William syndrome, velocardiofacial (VCF) and DiGeorge syndromes, Smith-Magenis syndrome, and Miller-Dieker syndrome (MDS) (see below). Though conventional cytogenetic studies can detect some of these microdeletions, initial detection and/or confirmation with FISH is essential. In fact, since appropriate FISH probes have become available, detection of the aforementioned syndromes has increased significantly.

In addition to metaphase [FISH](#), cells can be analyzed at a variety of stages. *Interphase analysis*, for example, can be used to make a rapid diagnosis in instances where metaphase chromosome preparations are not yet available (e.g., amniotic fluid interphase analysis). Interphase analysis also increases the number of cells available for examination, allows for investigation of nuclear organization, and provides results when cells do not progress to metaphase. One specialized type of interphase analysis involves the application of FISH to paraffin-embedded sections, thereby preserving the architecture of the tissue.

The use of interphase [FISH](#) has increased recently, especially for analyses of amniocentesis samples. These studies are performed on uncultured amniotic fluid, typically using DNA probes specific for the chromosomes most commonly identified in trisomies (chromosomes 13, 18, 21, and the X and Y). These studies can be performed rapidly (24 to 72 h) and will ascertain about 60% of the abnormalities detected prenatally. Nevertheless, guidelines from the American College of Medical Genetics suggest that standard cytogenetic analysis be conducted on all specimens following FISH analysis.

Another area in which interphase analysis is routinely utilized is cancer cytogenetics ([Chap. 81](#)). Many site-specific translocations are associated with specific types of malignancies. For example, there are probes available for both the Abelson (Abl) oncogene and breakpoint cluster region (bcr) involved in chronic myelogenous leukemia (CML). These probes are labeled with rhodamine and FITC, respectively; the fusion of these genes in CML combines the fluorescent colors and appears as a yellow hybridization signal.

In addition to standard metaphase and interphase [FISH](#) analyses, a number of enhanced techniques have been developed for specific types of analysis, including multicolor FISH techniques, reverse painting, comparative genomic hybridization, and fiber FISH. *Spectral karyotyping* (SKY) and *multicolor FISH* (m-FISH) techniques use combinatorially labeled probes that create a unique color for individual chromosomes. In this manner, all of the chromosomes are studied simultaneously, and computer software is used to generate "pseudo-colors" for the individual chromosomes. This technology is useful in the identification of unknown chromosome material (such as markers of duplications) but is most commonly used with the complex rearrangements seen in cancer specimens.

Reverse painting is accomplished by either flow-sorting a chromosome of interest or scraping the chromosome off a slide. The DNA from this chromosome (or portion of a chromosome) is extracted, amplified, labeled, and used as a [FISH](#) probe. This probe is then hybridized to a normal metaphase chromosome to identify the origin of the DNA of interest. It is also utilized to identify marker chromosomes or chromosome duplications

of unknown origin.

Comparative genomic hybridization (CGH) is a method that can be used when only DNA is available from a specimen of interest. The entire DNA specimen from the sample of interest is labeled in one color (e.g., green), and the normal control DNA specimen in another color (e.g., red). These are mixed in equal amounts and hybridized to normal metaphase chromosomes. The red-to-green ratio is analyzed by a computer program, which determines where the DNA of interest may have gains or loss of material. This technique is useful in the analysis of tumors, particularly in those cases where cytogenetic analysis is not possible.

Fiber FISH is a technique in which chromosomes are mechanically stretched, using one of a variety of different methods. FiberFISH provides a higher resolution of analysis than conventional FISH and more precise information on the chromosomal localization of a specific probe.

CYTOGENETIC TESTING IN PRENATAL DIAGNOSIS (See also [Chap. 68](#))

The vast majority of prenatal diagnostic studies are performed to rule out a chromosomal abnormality, but cells may also be propagated for biochemical studies or molecular analyses of DNA. Three procedures are used to obtain samples for prenatal diagnosis: amniocentesis, chorionic villus sampling (CVS), and fetal blood sampling. *Amniocentesis* is the most commonly used procedure and is routinely performed at 15 to 17 weeks of gestation. On some occasions, early amniocentesis at 12 to 14 weeks is done to expedite results, though less fluid is obtained at this time. Early amniocentesis carries a greater risk of spontaneous abortion or fetal injury but provides results at an earlier stage of pregnancy.

The vast majority of amniocentesis are performed in the context of advanced maternal age, the best-known correlate of trisomy (see below). Additional reasons for referral for amniocentesis include an abnormal "triple-marker assay" and/or detection of ultrasound abnormalities. In the triple-marker assay, levels of human chorionic gonadotropin, a fetoprotein, and unconjugated estriol in the maternal serum are quantified and used to adjust the maternal age-predicted risk of a trisomy 21 or trisomy 18 fetus. Specific ultrasound abnormalities, when detected at mid-trimester, can also be associated with chromosomal defects. When a nonspecific ultrasound abnormality is present, the estimated risk of a chromosomal defect is approximately 16%. Associations of chromosomal abnormalities and specific types of abnormal ultrasound findings are listed in [Table 66-1](#).

Chorionic villus sampling is the second most common procedure for genetic prenatal diagnosis. Because this procedure is routinely performed at about 8 to 10 weeks of gestation, it allows for an earlier detection of abnormalities and a safer pregnancy termination, if desired. CVS is a relatively safe procedure (spontaneous abortions <0.5 to 1%). Because there is an increased association of limb defects when the procedure is performed later (³11 weeks of gestation), CVS is applicable during a very narrow window of time of gestation. CVS involves the use of a catheter inserted transvaginally; approximately 25 mg of villi are aspirated from the chorion frondosum (the fetal portion of the placenta). Care must be taken not to obtain villi from the maternal portion from

the placenta to avoid compromising the analysis. The majority of the sample (the mesenchymal cells from the CVS sample) is enzymatically digested and cultured in a fashion similar to amniotic fluid cells. However, cells in the outer layer of the villi -- the cytotrophoblasts -- are actively dividing and can be analyzed directly. Therefore, by adding colchicine directly to these cells, a result can be obtained within 24 to 48 h. Findings from these procedures should be confirmed by analyses of cultured mesenchymal cells, as they are more reliably derived from the fetus.

Percutaneous umbilical blood sampling (PUBS) is a method for obtaining fetal blood during the second and third trimesters of pregnancy. It allows for acquisition of a blood sample, which can be used for cytogenetic studies; results can be obtained within 48 h of sampling. PUBS is carried out under ultrasound guidance. It is usually performed when ultrasound abnormalities are detected late in the second trimester. PUBS is also used when cytogenetic results from amniocentesis need clarification, such as the detection of mosaicism.

CHROMOSOME ABNORMALITIES

CHROMOSOMES IN CELL DIVISION

To understand the etiology of chromosome abnormalities, it is important to review the movement of chromosomes during cell division. In somatic tissues, chromosomes are replicated during the S-phase of the cell cycle, so that each replicated chromosome consists of two identical sister chromatids ([Fig. 66-1](#)). When the cell enters mitosis, each of the 46 chromosomes align on the metaphase plate, with the centromeres cooriented toward opposite spindle poles ([Fig. 66-3](#)). At anaphase the sister chromatids separate, with each of the daughter cells receiving one sister chromatid from each of the 46 chromosomes.

Chromosome segregation is more complicated in germ cell division, since the number of chromosomes must be reduced from 46 to 23 in the mature sperm and eggs. This is accomplished by two rounds of division -- meiosis I and meiosis II ([Fig. 66-3](#)). In meiosis I, homologous chromosomes become paired and exchange genetic material, then align on the metaphase plate, and finally separate from one another. Thus, by the end of meiosis I, only 23 of the original 46 chromosomes are represented in each of the two daughter cells. Meiosis II quickly follows meiosis I and is essentially a "haploid mitosis," involving separation of the sister chromatids in each of the 23 chromosomes.

Although the fundamentals of meiosis are the same in males and females, there are important distinctions, particularly in the timing of the meiotic divisions. In males, meiosis begins with puberty and continues throughout the individual's lifetime. In females, meiosis begins prenatally, with oocytes proceeding through the first stages of meiosis I but arresting at mid-prophase. At the time of birth, the first meiotic division is suspended in oocytes. Only after ovulation many years later do oocytes complete meiosis I and proceed to the metaphase stage of meiosis II; if fertilized, the oocyte then completes the second meiotic division. Thus, in females, the first meiotic division takes at least 10 to 15 years, and possibly as many as 40 to 45 years, to complete. Maternal age-related increases in the incidence of trisomy are likely the consequence of this protracted process of cell division.

INCIDENCE AND TYPES OF CHROMOSOME ABNORMALITIES

Errors in meiosis, or in early cleavage divisions, occur with extraordinary frequency. At least 10 to 25% of all pregnancies, for example, involve chromosomally abnormal conceptions. A large proportion of these terminate in the earliest stages of pregnancy. Nevertheless, even among clinically recognized pregnancies, nearly 10% of fetuses are chromosomally unbalanced. The occurrence of different classes of chromosome abnormalities are summarized in [Table 66-2](#) for the three types of clinically recognized pregnancies: spontaneous abortions, stillbirths, and livebirths. The commonest abnormalities are numerical, involving fetuses with additional (*trisomy*) or missing (*monosomy*) chromosomes or those with one (*triploidy*) or two (*tetraploidy*) additional sets of chromosomes. Structural chromosome abnormalities are much less common, although several of the most important clinical chromosomal disorders involve structural rearrangements (see below).

By far the most common abnormality is trisomy, which is identified in approximately 25% of spontaneous abortions and 0.3% of newborns. Trisomies for all chromosomes have now been identified in embryos or fetuses, but there is considerable variation in frequency for various chromosomes. For example, trisomy 16 is extraordinarily common, accounting for about one-third of all trisomies in spontaneous abortions, whereas trisomies 1, 5, 11, and 19 have been identified less often. Available evidence suggests two reasons for this variation: (1) some chromosomes (e.g., chromosome 16) are more likely to segregate abnormally or undergo nondisjunction during meiosis than are others; and (2) the potential for development varies widely among different trisomic conditions, with some being eliminated very early in gestation, others surviving to the time of clinical pregnancy recognition, and some (e.g., trisomies 13, 18, and 21 and sex chromosome trisomies) being compatible with survival to term.

CHROMOSOMAL SYNDROMES

While most chromosomally abnormal conceptions perish in utero ([Table 66-2](#)), several conditions are compatible with survival to term. The best-characterized of these are numerical abnormalities involving loss or gain of individual chromosomes and abnormalities resulting from unbalanced translocations. [FISH](#) and other molecular studies have led to the identification of two "new" types of chromosome abnormalities, commonly referred to as *microdeletion syndromes* and *imprinting syndromes*.

Numerical Abnormalities Virtually all types of numerical abnormalities are eliminated prenatally, so that only those involving small, gene-poor autosomes or the sex chromosomes are identified with any frequency among live-borns. Clinically, the most important of these is *trisomy 21*, the most frequent cause of Down syndrome. Depending on the maternal age structure of the population and the utilization of prenatal testing, the incidence of trisomy 21 ranges from 1/600 to 1/1000 live births, making it the most common chromosome abnormality in live-born individuals. Like most trisomies, the incidence of trisomy 21 is highly correlated with maternal age, increasing from about 1/1500 live births for women 20 years of age to 1/30 for women 45 years of age and older.

In addition to trisomy 21, only two other autosomal trisomies, 13 and 18, occur with any frequency in livebirths. Incidence rates for trisomies 13 and 18 in live births are 1/20,000 and 1/10,000 respectively. Unlike trisomy 21, which is associated with near-normal life expectancy, both trisomies 13 and 18 are associated with death in infancy, typically occurring during the first year of life.

Three sex chromosome trisomies -- the 47,XXX, 47,XXY (*Klinefelter syndrome*), and 47,XYY conditions -- are quite common, with each occurring in about 1/2000 newborns. Of all the trisomic conditions, these three have the fewest phenotypic complications. In fact, with the exception of infertility in Klinefelter syndrome ([Chap. 335](#)), it is likely that most individuals with such trisomic conditions would go undetected. The additional chromosome in the 47,XYY condition is small and contains only a few genes. Most Y-linked genes are involved in testicular development or spermatogenesis. Thus, dosage imbalance of Y-linked genes has relatively little effect on other developmental processes. The 47,XYY genotype is associated with increased height. Its role in antisocial behavior, postulated initially because of an increased prevalence among some penalized populations, is unclear.

For the 47,XXX and 47,XXY conditions, the situation is different -- the X chromosome contains over 1000 genes, many of them essential for normal development. How, then, are 47,XXX and 47,XXY individuals spared from the catastrophic consequences of dosage imbalance? The answer lies in the biology of X chromosome gene expression. In normal females, one of the X chromosomes is inactivated in somatic cells. The inactivation of the paternal or maternal X chromosome occurs randomly in each somatic cell and thereby serves as a mechanism of dosage compensation, ensuring that males and females have equal expression of most X-linked genes. The inactivation process occurs at the blastocyst stage of development; prior to this time both X chromosomes are active. In addition, the rules for inactivation are different for germ cells than for somatic cells: in female germ cells both X chromosomes remain active, whereas in male germ cells the X chromosome is inactivated. In addition, not all X-linked genes are inactivated. Some genes on the X chromosome "escape" the inactivating mechanism and are expressed from both X chromosomes. In disorders such as Klinefelter syndrome, some genes may be expressed from both X chromosomes, resulting in phenotypic abnormalities. Individuals with Klinefelter syndrome have small testes, hyalinized seminiferous tubules, and azoospermia or severe oligospermia ([Chap. 335](#)). Testosterone levels are variably reduced, and often there is gynecomastia and eunuchoidal body proportions. Antisocial behavior and mild mental deficiency are seen in some individuals. Females with the 47,XXX genotype are more likely to have mild mental deficiency and may be subfertile. Despite these features, sex chromosome trisomies impart relatively minor phenotypic complications in comparison to aneuploidies that involve autosomal chromosomes.

As a rule, monosomic conditions are incompatible with fetal development and, consequently, autosomal monosomies are only rarely identified in spontaneous abortions and are not found among live-born individuals. In fact, the only monosomy compatible with live birth is the 45,X condition, which causes *Turner syndrome*. The 45,X chromosome constitution occurs with surprisingly high frequency, being present in at least 1 to 2% of all pregnancies. More than 99% of all 45,X conceptions are spontaneously aborted. Thus, live-born individuals with a 45,X chromosome constitution

represent a rare group of survivors. The 45,X phenotype is mild, presumably because the second copy of many X chromosomal genes is normally inactivated. Nonetheless, Turner syndrome causes gonadal dysgenesis, resulting in infertility and failure to undergo secondary sexual development. Other prominent features are more variable and include short stature, webbing of the neck, and shield-shaped chest; lymphedema; increased carrying angle at the elbow; cardiovascular and renal abnormalities; and a propensity to hypertension, glucose intolerance, and autoimmune thyroid disease ([Chap. 336](#)). Several other structural abnormalities of the X chromosome such as partial deletions, isochromosome X, or ring chromosomes can cause Turner syndrome. Mosaicism, including 45,X/45,XX, 45X/45,XXX, 45,X/45,XY, and others, also occurs (see below) and contributes to the phenotypic spectrum seen in Turner syndrome.

Because numerical abnormalities originate in meiosis ([Table 66-3](#)), affected individuals have missing or extra chromosomes in all cells. In a small proportion of cases, though, a mitotic nondisjunctional event occurs at an early stage in an individual with an initially normal chromosome constitution. Alternatively, a "normalizing" mitotic nondisjunctional event may result in a normal chromosome complement in some cells of an embryo. In either case, the embryo is a *mosaic*, with some cells bearing a normal chromosome constitution and others an aneuploid number of chromosomes. The phenotypic consequences are difficult to predict because they depend on the timing of nondisjunction and the distribution of normal and abnormal cells in different tissues. Nevertheless, mosaicism may lead to clinical abnormalities indistinguishable from those of nonmosaic individuals; for example, nearly 5% of all cases of Down syndrome involve individuals with mosaic trisomy 21, and about 15% of individuals with Turner syndrome are mosaic for various sex chromosomal constitutions as described above.

The Origin and Etiology of Numerical Abnormalities Over the past decade, a number of studies have used DNA polymorphisms to investigate the origin of different types of chromosome abnormalities ([Fig. 66-4](#)). The most thoroughly investigated types have been numerical abnormalities ([Table 66-3](#)). Sex chromosome monosomy usually results from loss of the paternal sex chromosome. This is the case regardless of whether the conception is live-born or spontaneously aborted, indicating that the parental origin of the abnormality does not affect its likelihood of surviving to term.

Trisomies show remarkable variation in parental origin. For example, paternal nondisjunction is responsible for nearly 50% of 47,XXY but only 5 to 10% of cases of trisomies 13, 14, 15, 21, and 22; it is rarely, if ever, the source of the additional chromosome in trisomy 16. Similarly, there is considerable variability in the meiotic stage of origin. For example, among maternally derived trisomies, all cases of trisomy 16 may be due to meiosis I errors, whereas for trisomy 21, one-third of cases are associated with meiosis II errors, and for trisomy 18, the majority of cases are apparently due to meiosis II nondisjunction. In spite of this variation in parental and meiotic origin, nondisjunction at maternal meiosis I appears to be the most common source of trisomy.

Molecular studies have also begun to shed light on the molecular mechanisms underlying nondisjunction, the source of trisomy and monosomy. Most, if not all, trisomies are associated with alterations in genetic recombination. This is the process by which chromosomes exchange genetic material during the first of the two meiotic

divisions. In other organisms, the physical connections, or chiasmata, associated with recombination are known to hold chromosomes together at meiosis I. This mechanism is now known to be true for humans as well. Nondisjunction at meiosis I is linked to a reduced extent of crossing-over, with some cases involving outright failure of recombination between the homologous chromosomes, and others associated with distally placed exchanges. Unexpectedly -- since recombination occurs at meiosis I -- maternal meiosis II errors involving chromosome 21 may also be linked to altered recombination. In this instance, though, the effect involves increased -- not decreased -- recombination, especially in proximal 21q. Presumably, this indicates that errors scored as arising at meiosis II are, in fact, precipitated by events occurring at meiosis I.

Maternal Age and Trisomy The association between increasing maternal age and trisomy is arguably the most important etiologic factor in congenital chromosomal disorders. Among women under the age of 25, approximately 2% of all clinically recognized pregnancies are trisomic; by the age of 36, however, this figure increases to 10% and by the age of 42, to >33% ([Fig. 66-5](#)). This association between maternal age and trisomy is exerted without respect to race, geography, or socioeconomic factors and likely affects segregation of all chromosomes.

Despite the importance of increasing age, almost nothing is known about the mechanism by which aging leads to abnormal chromosomal segregation. As noted above, it is thought to originate in maternal meiosis I owing to the protracted time to completion (often³40 years) in females. As noted above, alterations in genetic recombination may explain age-related trisomy. In trisomy 21, for example, recombination patterns appear to be similarly altered in younger and older mothers of trisomic conceptions. With this in mind, it has been suggested that two distinct steps, or "hits," may be involved in maternal age-related nondisjunction. The first hit, which is age-independent, involves the establishment of a "vulnerable" recombination configuration in the fetal oocyte; the second hit, which is age-dependent, involves abnormal processing of the vulnerable bivalent structure at metaphase I. If this model is correct, it means that the nondisjunctional process is the same in younger and older women, but it occurs more frequently with aging, possibly because of age-dependent degradation of cell cycle proteins or meiotic proteins responsible for maintaining sister chromatid cohesion.

Structural Chromosome Abnormalities Structural rearrangements involve breakage and reunion of chromosomes. Although less common than numerical abnormalities, they present additional challenges from a genetic counseling standpoint. This is because structural abnormalities, unlike numerical abnormalities, can be present in "balanced" form in clinically normal individuals but transmitted in "unbalanced" form to progeny, thereby resulting in a hereditary form of chromosome abnormality.

Rearrangements may involve exchanges of material between different chromosomes (*translocations*) or loss, gain, or rearrangements of individual chromosomes (e.g., *deletions*, *duplications*, *inversions*, *rings* or *isochromosomes*). Of particular clinical importance are translocations, which involve two basic types: Robertsonian and reciprocal. *Robertsonian rearrangements* are a special class of translocation, in which the long arms of two acrocentric chromosomes (chromosomes 13, 14, 15, 21, and 22) join together, generating a fusion chromosome that contains virtually all of the genetic

material of the original two chromosomes. If the Robertsonian translocation is present in unbalanced form, a monosomic or trisomic conception ensues. For example, approximately 3% of cases of Down syndrome are attributable to unbalanced Robertsonian translocations, most often involving chromosomes 14 and 21. In this instance, the affected individual has 46 chromosomes, including one structurally normal chromosome 14, two structurally normal chromosomes 21, and one fusion 14/21 chromosome. This effect leads to a normal diploid dosage for chromosome 14 and to a triplication of chromosome 21, thus resulting in Down syndrome. Similarly, a small proportion of individuals with trisomy 13 syndrome are clinically affected because of an unbalanced Robertsonian translocation.

Reciprocal translocations involve exchanges between any two chromosomes. In this circumstance, the phenotypic consequences associated with unbalanced translocations depend on the location of the breakpoints, which dictate the amount of material that has been "exchanged" between the two chromosomes. Because most reciprocal translocations involve unique sets of breakpoints, it is difficult to predict the phenotypic consequences in any one situation. In general, severity is determined by the amount of excess or missing chromosome material in individuals with unbalanced translocations.

In addition to rearrangements between chromosomes, there are several examples of intrachromosome structural abnormalities. The most common and deleterious of these involve loss of chromosome material due to deletions. The two best-characterized deletion syndromes, *Wolf-Hirschhorn syndrome* and *cri-du-chat syndrome*, result from loss of relatively small chromosomal segments on chromosomes 4p and 5p, respectively. Nonetheless, each is associated with multiple congenital anomalies, developmental delays, profound retardation, and reduced lifespan.

Microdeletion Syndromes The term *contiguous gene syndromes* refers to genetic disorders that mimic a combination of single gene disorders. They result from the deletion of a small number of tightly clustered genes. Because they are usually too small to be detected cytogenetically, they are termed *microdeletions*. The application of molecular techniques has led to the identification of at least 18 of these microdeletion syndromes ([Table 66-4](#)). Some of the more common ones include the Wilms' tumor-aniridia complex (WAGR), [MDS](#), and [VCF](#) syndrome. *WAGR* is characterized by mental retardation and involvement of multiple organs, including kidney (Wilm's tumor), eye (aniridia), and the genitourinary system. The cytogenetic abnormality involves a deletion of part of the short arm of chromosome 11 (11p13), which typically is detectable on well-banded chromosome preparations. In *MDS*, a disorder characterized by mental retardation, dysmorphic faces, and lissencephaly, the deletion involves chromosome 17 (17p13). Using [FISH](#), 17p deletions have been detected in >90% of *MDS* patients as well as in 20% of cases of isolated lissencephaly.

Deletions involving the long arm of chromosome 22 (22q11) are the most common microdeletions identified to date, present in approximately 1/3000 newborns. [VCF](#) syndrome, the most commonly associated syndrome, consists of learning disabilities or mild mental retardation, palatal defects, a hypoplastic alae nasi and long nose, and congenital heart defects (conotruncal defect). Some individuals with 22q11 deletion are more severely affected and present with *DiGeorge syndrome*, which involves abnormalities in the development of the third and fourth branchial arches

leading to thymic hypoplasia, parathyroid hypoplasia, and conotruncal heart defects. In approximately 30% of these cases, a deletion at 22q11 can be detected with high-resolution banding; by combining conventional cytogenetics, [FISH](#), and molecular detection techniques (i.e., Southern blotting or polymerase chain reaction analyses), these rates improve to >90%. Additional studies have demonstrated a surprisingly high frequency of 22q11 deletions in individuals with nonsyndromic conotruncal defects. Approximately 10% of individuals with a 22q11 deletion inherited it from a parent with a similar deletion.

Smith-Magenis syndrome involves a microdeletion localized to the short arm of chromosome 17 (17p11.2). Affected individuals have mental retardation, dysmorphic facial features, delayed speech, peripheral neuropathy, and behavior abnormalities. Most of these deletions can be detected with cytogenetic analysis, although [FISH](#) is available to confirm these findings. In contrast, *William syndrome*, a chromosome 7 (7q11.23) microdeletion, cannot be diagnosed with standard or high-resolution analysis; it is only detectable utilizing FISH or other molecular methods. William syndrome involves a deletion of the elastin gene and is characterized by mental retardation, dysmorphic features, a gregarious personality, premature aging, and congenital heart disease (usually supravulvar aortic stenosis).

In addition to microdeletion syndromes, there is now at least one well-described microduplication syndrome, Charcot-Marie-Tooth type 1A (CMT1A). This is a nerve conduction disease previously thought to be transmitted as a simple autosomal dominant disorder. Recent molecular studies have demonstrated that affected individuals are heterozygous for duplication of a small region of chromosome 17 (17p11.2-12). Although it is not yet clear why increased gene dosage would result in CMT1A, the inheritance pattern is explained by the fact that one-half of the offspring of affected individuals inherit the duplication-carrying chromosome.

Imprinting Disorders Two other microdeletion syndromes, [PWS](#) and [AS](#), exhibit parent-of-origin, or "imprinting," effects. For many years, it has been known that cytogenetically detectable deletions of chromosome 15 occur in a proportion of patients with PWS, as well as in those with AS. This seemed curious, as the clinical manifestations of the two syndromes are very dissimilar. PWS is characterized by obesity, hypogonadism, and mild to moderate mental retardation, whereas AS is associated with microcephaly, ataxic gait, seizures, inappropriate laughter, and severe mental retardation. New insight into the pathogenesis of these disorders has been provided by the recognition that parental origin of the deletion determines which phenotype ensues: if the deletion is paternal, the result is PWS, whereas if the deletion is maternal, the result is AS ([Fig. 66-2](#)).

This scenario is complicated further by the recognition that not all individuals with [PWS](#) or [AS](#) carry the chromosome 15 deletion. For such individuals, it turns out that the parental origin of the chromosome 15 region is again the important determinant. In PWS, for example, nondeletion patients invariably have two maternal and no paternal chromosomes 15 [*maternal uniparental disomy* (UPD)], whereas for some nondeletion AS patients the reverse is true (*paternal UPD*). This indicates that at least some genes on chromosome 15 are differently expressed, depending on which parent contributed the chromosome. Additionally, this means that normal fetal development requires the

presence of one maternal and one paternal copy of chromosome 15.

Approximately 70% of [PWS](#) cases are due to paternal deletions of 15q11-q13, whereas 25% are due to maternal uniparental disomy, and about 5% are caused by mutations in a chromosome 15 imprinting center. Though 75% of the [AS](#) cases are due to maternal deletions, only 2% are due to paternal uniparental disomy. The rest of the cases are presumed to be caused by imprinting mutations (5%) or mutations in the UBE3A gene, which is one of the genes associated with AS. The [UPD](#) cases are mostly caused by meiotic nondisjunction resulting in trisomy 15, subsequently followed by a normalizing mitotic nondisjunction event ("trisomy rescue") resulting in two normal chromosomes 15, both from the same parent. UBE3A is the only maternally imprinted gene known in the critical region of chromosome 15. However, several paternally imprinted genes, or expressed-sequence tags (ESTs), have been identified, including ZNF127, IPW, SNRPN, SNURF, PAR1, and PAR5.

Chromosomal regions that behave in the manner observed in [PWS](#) and [AS](#) are said to be *imprinted*. This phenomenon is involved in differential expression of certain genes on different chromosomes. Chromosome 11 must be one of these with an imprinted region, since it is known that a small proportion of individuals with the *Beckwith-Wiedemann overgrowth syndrome* have two paternal but no maternal copies of this chromosome.

ACQUIRED CHROMOSOME ABNORMALITIES IN CANCER

In addition to the constitutional cytogenetic chromosomal abnormalities that are present at birth, somatic chromosomal changes can be acquired later in life and are often associated with malignant conditions. As with constitutional abnormalities, somatic changes can include the net loss of chromosomal material (due to a deletion or loss of a chromosome), net gain of material (duplication or gain of a chromosome), and relocation of DNA sequences (translocation). These chromosomal changes are intertwined with the three major categories of cancer genes: (1) *tumor-suppressor genes* that inhibit cell proliferation may be deleted; (2) *oncogenes* that activate cell proliferation may be activated by duplication, amplification, or translocation; and (3) *DNA repair genes* may also be deleted from somatic cells, thereby predisposing to the accumulation of additional DNA damage. Cytogenetic changes have been particularly well studied in (1) leukemias, e.g., Philadelphia chromosome translocation in [CML](#) [t(9;22)(q34.1;q11.2)]; and (2) lymphomas, e.g., translocations of MYC in Burkitt's [t(8;14)(q24;q32)]. These and other translocations are useful for diagnosis, classification, and prognosis. Analyses of cytogenetic changes are also proving useful in certain solid tumors. For example, a complex karyotype with Wilms' tumor, diploidy in medulloblastoma, and Her-2/neu amplification in breast cancer are poor prognostic signs. The genetic basis of malignancy, including the role of germline and somatic chromosomal alterations, is discussed further in [Chap. 81](#).

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67. DISEASES CAUSED BY GENETIC DEFECTS OF MITOCHONDRIA - *Donald R. Johns*

Mitochondrial defects play a role in several metabolic and neurodegenerative diseases as well as aging. Mitochondrial disorders have protean clinical manifestations, reflecting the fact that nearly all organ systems utilize oxidative metabolism. Clinical features often involve tissues with high energy requirements such as the central and peripheral nervous systems, the eye, muscle, kidney, and endocrine organs. The age and mode of onset and clinical course of mitochondrial diseases range widely as a result of the unusual mechanisms of mitochondrial DNA (mtDNA) replication, which is distinct from that of the nuclear genome. A maternal mode of inheritance is characteristic of many mitochondrial diseases because mtDNA is transmitted by the oocyte. Hundreds of different mtDNA mutations have been described since the first mutation was described in 1988.

STRUCTURE AND FUNCTION OF MITOCHONDRIAL DNA

Most cells contain several hundred mitochondria, though the number varies depending on the energy requirements and function of a tissue. Mitochondria are the only cellular organelles that contain their own extrachromosomal DNA. Human [mtDNA](#) is a small (16,569 bp) double-stranded, circular molecule that encodes 13 protein subunits of 4 different oxidative phosphorylation biochemical complexes. mtDNA also encodes the 24 structural RNAs (2 ribosomal RNAs and 22 transfer RNAs) required for the intramitochondrial translation of these proteins. The noncoding D (displacement)-loop is a regulatory region that controls transcription and replication. mtDNA mutations are found in each type of mitochondrial gene ([Fig. 67-1](#)).

Mitochondria probably evolved from independent organisms that became endosymbiotically incorporated into the cell. As a result, mitochondria replicate, transcribe, and translate their DNA independently of nuclear DNA. However, cellular and mitochondrial function are interdependent. Nuclear DNA-encoded proteins are also involved in oxidative phosphorylation, and the myriad macromolecular compounds required for mitochondrial structure and function (e.g., [mtDNA](#) replication, transcription, and translation) are imported from the cytoplasm into the mitochondria.

Oxidative phosphorylation and the generation of adenosine triphosphate (ATP) for energy-requiring cellular processes are the central functions of mitochondria. Alterations in mitochondrial function lead to disease pathogenesis by three main mechanisms: (1) reduction of ATP supply when mutations impair oxidative phosphorylation; (2) generation of reactive oxygen species such as H_2O_2 and $\text{OH}^\times$ free radicals that can damage DNA, proteins, or lipids; and (3) execution of the apoptosis pathway when mitochondria release cell death-promoting factors including caspases, cytochrome c, and apoptosis-inducing factor.

Several unique features of [mtDNA](#) render it vulnerable to mutations and contribute to its role in disease. For example, mtDNA has no introns (a random mutation will therefore usually strike a coding DNA sequence) or protective histones, and it has an imperfect DNA repair system and is exposed to oxygen free radicals generated by oxidative phosphorylation. mtDNA is estimated to mutate 10 times more rapidly than nuclear

DNA. Importantly, mtDNA is strictly *maternally inherited* and does not recombine, and mtDNA mutations sequentially accumulate along maternal lineages. These properties have made mtDNA sequence variation an invaluable tool for evolutionary biologists and forensic scientists.

Each mitochondrion contains 2 to 10 [mtDNA](#) molecules, and each cell contains multiple mitochondria (*polyplasm*y). Population genetic principles, as opposed to Mendelian genetics, govern mitochondrial genetics. When a new mtDNA mutation arises, cells initially harbor copies of both normal and mutant DNA sequences -- a condition known as *heteroplasm*y -- which allows an otherwise lethal mutation to persist and cause disease. The presence of either completely normal or completely mutant mtDNA is known as *homoplasm*y. During cell division, mitochondria are unevenly partitioned to the daughter cells through the process of *replicative segregation*; consequently, the proportion of mutant and normal mtDNA molecules can drift. Selection pressures apply at the molecular, cellular, and organismal levels. The critical proportion of mutant mtDNA required for deleterious phenotypic expression is known as the *threshold effect*. It varies among individuals, among organ systems, and within a given tissue, depending on the delicate balance of oxidative supply and demand. These features of mtDNA segregation, combined with the uneven transmission of mitochondria to daughter cells during cell division, form much of the basis for the phenotypic diversity seen in mitochondrial diseases.

MITOCHONDRIAL DNA MUTATIONS

[mtDNA](#) mutations that cause a severe, lethal impairment of oxidative phosphorylation -- such as gross structural defects or point mutations in structural RNAs -- are only viable if heteroplasmic. In contrast, the majority of the milder, missense mtDNA mutations in protein-coding genes are homoplasmic. mtDNA point mutations have been found in each type of mtDNA gene, but tRNA mutations predominate in severe, multisystemic mitochondrial encephalomyopathy phenotypes; protein-coding gene mutations predominate in Leber's hereditary optic neuropathy (LHON). A point mutation in the 12S ribosomal RNA gene is associated with both spontaneous and aminoglycoside-associated sensorineural deafness ([Fig. 67-1](#)).

A definitive cause-and-effect relationship between a [mtDNA](#) mutation and a clinical phenotype can be difficult to establish for several reasons: (1) mtDNA is highly polymorphic, (2) different mutations can be associated with the same phenotype or the same mutation can be associated with different phenotypes, and (3) epigenetic factors can affect clinical manifestations.

CLASSIC MITOCHONDRIAL ENCEPHALOMYOPATHY PHENOTYPES

Many diseases were provisionally classified as mitochondrial disorders on the basis of abnormal mitochondrial morphology, biochemistry, or a pattern of maternal inheritance. Disease-associated [mtDNA](#) mutations are now an important diagnostic criterion. Though each classic mitochondrial encephalomyopathy phenotype has distinctive clinical features ([Table 67-1](#)), each also shares many clinical and laboratory features ([Tables 67-2](#) and [67-3](#)).

Mitochondrial Myopathy Mitochondrial myopathy is characterized by fixed proximal weakness with marked exercise intolerance. Fatigability and poor stamina are prominent clinical features, but a mitochondrial etiology is often considered in the context of other neurologic, somatic, or laboratory features. Frank rhabdomyolysis is rare. Electromyography documents a nonirritative myopathy, and serum creatine kinase is usually normal or slightly elevated. Skeletal muscle biopsy shows abnormal proliferating mitochondria and "ragged red fibers," a histologic hallmark of the severe biochemical defects in oxidative phosphorylation. Large [mtDNA](#) deletions and a variety of mtDNA point mutations occur in mitochondrial myopathy.

Chronic Progressive External Ophthalmoplegia (CPEO) Ptosis, ophthalmoplegia, and limb myopathy characterize CPEO. Additional clinical features ([Table 67-2](#)) may also occur along with the laboratory abnormalities characteristic of mitochondrial disorders ([Table 67-3](#)). Patients with CPEO have abnormal skeletal muscle biopsies, with ragged red fibers and ultrastructural changes. Additional somatic and central nervous system findings in conjunction with CPEO are known as the "CPEO-plus" syndromes. Kearns-Sayre syndrome is a subset of CPEO-plus that begins before age 20 and is characterized by CPEO and atypical pigmentary retinopathy; ancillary features include elevated cerebrospinal fluid protein, ataxia, or heart block.

Most patients with [CPEO](#) have large, single deletions in [mtDNA](#) that can be reliably detected by molecular genetic methods. However, skeletal muscle is required as the source of DNA because almost all single mtDNA deletions occur sporadically. The presence of pigmentary retinopathy is strongly predictive of a deletion. The mechanism of DNA deletion is unknown, though recombination or slippage during replication is plausible. The junctions of most deletions contain directly repeated sequences, including a 13-nucleotide direct repeat "hot spot" that accounts for about 25% of all deletions. Approximately half of all deletions are bound by other direct repeats; one-quarter have no apparent direct repeat. A few patients have partially duplicated mtDNA molecules. A point mutation at nucleotide position 3243 has been found in many patients who lack a mtDNA deletion or duplication.

Autosomally Transmitted Multiple Mitochondrial DNA Deletions Several families with clinical variants of [CPEO](#) have been found to harbor multiple [mtDNA](#) deletions in skeletal muscle. The autosomal inheritance of multiple mtDNA deletions implies a primary defect in a nuclear DNA gene that has secondary qualitative effects on mtDNA. Multiple mtDNA deletions can be transmitted in either an autosomal dominant or autosomal recessive mode or can occur as somatic mutations.

Thymidine phosphorylase was the first nuclear-encoded gene shown to influence the regulation and function of normal [mtDNA](#) in a trans-acting manner. Mutations in thymidine phosphorylase cause an autosomal recessive disease known as *myoneurogastrointestinal encephalomyopathy* (MNGIE). Several different nuclear loci have been linked to the autosomal dominant forms of [CPEO](#). Tissue-specific, autosomally transmitted depletion of mtDNA represents a quantitative mtDNA defect caused by an intergenomic communication error.

Mitochondrial Encephalomyopathy, Lactic Acidosis, and Strokelike Episodes (MELAS) This syndrome is characterized by strokelike events that cause subacute

brain dysfunction, cerebral structural changes, seizures, and several other common clinical and laboratory features ([Tables 67-2,67-3](#)). Maternal inheritance of the MELAS syndrome may be obscured because of mild clinical features in relatives. A point mutation at nucleotide 3243 in the tRNA^{Leu}(UUR) gene accounts for 80% of MELAS cases. However, the clinical features of the 3243 mtDNA mutation are pleiomorphic; it is also associated with nondeletion [CPEO](#), myopathy, deafness, diabetes, and dystonia.

Myoclonic Epilepsy with Ragged Red Fibers (MERRF) Syndrome The MERRF syndrome consists of myoclonus, seizures, cerebellar ataxia, and mitochondrial myopathy. Pathogenetic mutations have been demonstrated at nucleotide positions 8344 and 8356 in the tRNA^{Lys} gene. Neurologic and laboratory features common to other mitochondrial encephalomyopathies are seen. Maternal relatives may be asymptomatic or may have partial clinical syndromes, including lipomas in a characteristic "horse-collar" distribution, and hypertension.

Neuropathy, Ataxia, and Retinitis Pigmentosa (NARP)/Maternally Inherited Leigh's Disease NARP is characterized by proximal weakness, sensory neuropathy, developmental delay, ataxia, seizures, dementia, and retinal pigmentary degeneration. This maternally inherited disorder is associated with two different heteroplasmic missense mutations at nucleotide position 8993 in the ATPase 6 gene. High proportions of the same mutations are also seen in maternally inherited Leigh's disease. Autosomal recessive Leigh's disease is associated with cytochrome c oxidase deficiency and is caused by a deficiency of the nuclear-encoded protein, SURF1, which is required for biogenesis of the cytochrome c oxidase complex (complex IV). The [mtDNA](#) point mutations associated with NARP, [MELAS](#), [MERRF](#), and other mitochondrial disorders are readily detected by molecular genetic analysis of mtDNA extracted from muscle or blood.

Leber's Hereditary Optic Neuropathy [LHON](#) typically presents with painless, subacute, bilateral visual loss with central scotomas and dyschromatopsia. The mean age of onset is 23 years, and males are affected three to four times more commonly than females. LHON bears little clinical resemblance to the other mitochondrial disease phenotypes. It was first classified in this group on the basis of the maternal inheritance pattern. The pathophysiology of visual loss appears to involve both genetic and epigenetic (tobacco and alcohol) factors. The [mtDNA](#) mutations exhibit a high degree of genetic heterogeneity. Primary LHON-associated mtDNA mutations predominantly affect complex I genes [at nucleotide positions 11778 (ND-4 gene), 3460 (ND-1 gene), and 14484 (ND-6 gene)] ([Fig. 67-1](#)). Several other mtDNA mutations may have secondary pathogenetic roles in LHON, including a mutation at nucleotide position 13708 (ND-5 gene) of Caucasian haplogroup J mtDNA.

Genotype-phenotype correlations are beginning to emerge for the primary [LHON](#)-associated [mtDNA](#) mutations. The prognosis for visual recovery, for example, varies nearly ten-fold depending on the mutation. mtDNA mutations that cause LHON plus dystonia have also been described.

ORGAN SYSTEM MANIFESTATIONS OF MITOCHONDRIAL DISEASE

Because virtually all tissues of the body depend, to some extent, on oxidative

metabolism, patients with mitochondrial disease can present to many specialists in medicine. The somatic manifestations listed in [Table 67-2](#), first noted in association with the classic mitochondrial diseases, may be the dominant or initial clinical symptom or may be important comorbid features.

The ophthalmologic manifestations of [mtDNA](#) mutations are prominent, with involvement of virtually the entire visual axis from the lids, cornea, and extraocular muscles to the occipital cortex. The cardinal eye findings include ophthalmoplegia, optic neuropathy, and pigmentary retinopathy. Cardiovascular manifestations include dilated and hypertrophic cardiomyopathy, conduction disease and heart block, Wolff-Parkinson-White syndrome, and hypertension. The prevalence of diabetes mellitus is higher than expected in patients with mitochondrial encephalomyopathies, and it occurs in association with a variety of mtDNA mutations. Diabetes mellitus has been linked with the 3243 mtDNA point mutation, usually, but not exclusively, in association with sensorineural hearing loss.

ROLE OF MITOCHONDRIAL DNA MUTATIONS IN PREVALENT DISEASES

The role of [mtDNA](#) mutations in common, socioeconomically significant diseases is under active investigation. The genetic basis of many prevalent diseases is complex and does not follow simple, single-gene Mendelian inheritance ([Chap. 65](#)). Mitochondrial diseases, such as [LHON](#) and aminoglycoside-induced deafness, illustrate the potential for complex pathophysiologic interactions between genetic and epigenetic factors. As a result of these interactions, mtDNA mutations may be involved in subsets of common diseases, such as diabetes mellitus, in which the maternal inheritance pattern is not obvious.

The tissue-specific accumulation of somatic (noninherited) [mtDNA](#) mutations is likely relevant to some late-onset degenerative disorders, such as Alzheimer's disease and Parkinson's disease. It has been shown, for example, that as people age, mtDNA mutations accumulate in tissues, including some postmitotic tissues such as the basal ganglia and cerebral cortex. The high mutations rate and poor repair capacity of mtDNA contribute to the buildup of mtDNA mutations in postmitotic tissues or those with a slower turnover rate. Oxidative damage, as occurs with repeated episodes of ischemia and reperfusion, markedly increases the accumulation of mtDNA mutations. Environmental factors may also affect mtDNA. The anti-retroviral drug azidothymidine depletes muscle mtDNA and causes an acquired mitochondrial myopathy. Cumulative, age-dependent mitochondrial dysfunction, mediated to a significant degree by oxidative damage to mtDNA and other mitochondrial macromolecules, may be a major contributor to aging.

The unequivocal establishment of the diagnosis of a mitochondrial disease by molecular genetic methods is a prerequisite for proper genetic counseling and ultimately treatment.

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68. SCREENING, COUNSELING, AND PREVENTION OF GENETIC DISORDERS - **Susan Miesfeldt, J. Larry Jameson**

IMPLICATIONS OF MOLECULAR GENETICS FOR INTERNAL MEDICINE

Approximately 1 in 50 children is born with a serious congenital abnormality or mental handicap. It is known that each of us harbors mutations in several genes that can potentially lead to serious diseases. The field of medical genetics has traditionally focused on chromosomal abnormalities ([Chap. 66](#)) and Mendelian disorders ([Chap. 65](#)). However, there is genetic susceptibility to many common adult-onset diseases including atherosclerosis, hypertension, autoimmune diseases, diabetes mellitus, Alzheimer disease, psychiatric disorders, and many forms of cancer. Genetic contributions to these common disorders involve more than the ultimate expression of a disease; these genes can also influence the severity of infirmity, response to treatment, and progression of disease.

The primary care clinician is now faced with the role of recognizing and counseling patients at risk for a large number of genetically influenced illnesses. This role reflects the advances in genetic medicine that are changing the way diseases are classified, enhancing our understanding of pathophysiology, providing practical information concerning drug metabolism and therapeutic responses, and promising to allow individualized screening and health care management programs. In view of these changes, the physician must integrate personal medical history, family history, and diagnostic molecular testing into the overall care of individual patients and their families. In addition, the internist has an important role in educating patients about the indications, benefits, risks, and limitations of genetic testing in the management of diverse diseases. This is a formidable task as scientific advances in genetic medicine, and media attention to these advances, have outpaced the translation into standards of clinical care.

PRENATAL AND NEWBORN GENETIC SCREENING AND TESTING

During pregnancy and in the newborn period, genetic screening and testing are both used to detect genetic disorders ([Table 68-1](#)). *Genetic screening* refers to the search for a genetic disorder in the entire population or in a high-risk population. Screening techniques must be cost-effective, have a high positive predictive value, and should yield information that leads to disease prevention or a useful therapeutic intervention. Examples of screening tests include maternal serum markers used to detect increased risk of Down syndrome, postnatal tests for phenylketonuria, and cholesterol levels in children used to identify individuals at risk for familial hyperlipidemias. Thus, genetic screening tests do not necessarily imply DNA- or chromosome-based tests. Instead, many such tests use a surrogate biochemical marker or phenotypic feature of the underlying genetic disorder. Determination of which genetic disorders should be screened routinely depends on disease frequency, the severity of the disorder, cost of screening, and whether treatment interventions can alter the course of the disease.

Genetic testing, on the other hand, is used in an individual suspected to have a disease based on physical features, family history, or biochemical findings. Genetic testing can include the following: (1) diagnostic testing as for hemochromatosis, (2) predictive

testing as for breast cancer predisposition, (3) carrier testing as for muscular dystrophy, and (4) prenatal testing as for β -thalassemia when both parents are carriers.

Depending on the disorder(s) under consideration, several different techniques are currently used for genetic screening during pregnancy. For instance, first-trimester screening for Down syndrome is performed using measurements of maternal serum pregnancy-associated plasma protein A and the free β subunit of human chorionic gonadotropin in combination with ultrasonographic measurement of nuchal translucency (at 10 to 14 weeks of gestation). Second-trimester maternal serum screening includes measurements of human chorionic gonadotropin, unconjugated estriol, α -fetoprotein, and inhibin that, in combination, increase the sensitivity of Down syndrome detection ([Chap. 66](#)). Increased α -fetoprotein levels are also associated with open neural tube defects.

Amniocentesis or chorionic villus sampling (CVS) is used to isolate fetal cells for chromosomal analysis or to test for specific genetic abnormalities. Amniocentesis is typically performed during the early second trimester (14 to 16 weeks' gestational age). CVS can be performed earlier (8 to 10 weeks' gestational age) and involves transcervical or transabdominal biopsy of fetal trophoblastic tissue. CVS is associated with a somewhat greater risk of spontaneous abortion (<0.5 to 1%) than amniocentesis (<0.5%) but allows for elective abortion earlier during pregnancy. Ultrasonography is used to analyze the fetus directly at different stages of development. Preimplantation molecular diagnostic testing is now possible by isolating single cells from the 8- to 10-cell embryo. The polymerase chain reaction (PCR) is then used to test for selected single-gene disorders such as Tay-Sachs disease, cystic fibrosis, or sickle cell anemia. This testing strategy requires in vitro fertilization but has the advantage that affected embryos are not implanted.

It should be remembered that prenatal genetic testing focuses on chromosomal abnormalities ([Chap. 66](#)), along with specific genetic disorders for which there is increased risk of parental transmission. There is no guarantee that a child will be free of birth defects or other genetic disorders not included among the diagnostic tests.

The genetic counseling process should begin before prenatal testing and should include: (1) a description of the test, (2) the types of disorders that will be screened, (3) the limitations of the screening, (4) an exploration of what the parents will do with the information, and (5) an indication of when the results will be available. If a genetic disorder is identified, the full repertoire of genetic counseling skills is required (see below). The nature of the genetic disease needs to be reviewed with the parents, often on separate occasions. The counselor should also discuss the kinds of physical and emotional challenges that a genetic disease might pose for the affected individual and the family. Written information should be provided, if available. Ultimately, the parents must reach a decision to continue or terminate the pregnancy. If pregnancy is terminated, counseling should continue after the procedure. If pregnancy is continued, counseling should continue to address the medical needs of the affected fetus or child, as well as to help the family meet any challenges presented by the genetic disorder.

At the time of birth, all newborns undergo a complete physical examination to detect gross developmental abnormalities. Within 24 to 72 h of birth, blood samples from

newborns are sent to a state-designated laboratory to screen for selected diseases, such as congenital hypothyroidism and a variety of inherited metabolic disorders ([Table 68-1](#)). This program represents a clear example of the benefits of selected genetic screening. The early diagnosis of phenylketonuria, for example, permits parents to introduce a phenylalanine-free diet before the development of severe neurologic sequelae ([Chap. 352](#)).

COMMON ADULT-ONSET GENETIC DISORDERS

MULTIFACTORIAL INHERITANCE

The risk for many adult-onset disorders reflects the additive effects of genetic factors at multiple loci that may function independently -- or in combination -- with other genes or environmental factors. Our understanding of the genetic basis of these disorders is incomplete, despite the clear recognition of genetic susceptibility. In type 2 diabetes mellitus, for example, the concordance rate in monozygotic twins ranges between 50 to 90%. Diabetes or impaired glucose tolerance occurs in 40% of siblings and in 30% of the offspring of an affected individual. Despite the fact that diabetes affects 5% of the population and exhibits a high degree of heritability, there are only a few examples of genetic mutations that might account for the familial nature of the disease (most of which are rare). They include certain mitochondrial DNA disorders ([Chap. 67](#)), mutations in a cascade of genes that control pancreatic islet cell development and function (*HNF4a*, *HNF1a*, *IPF1*), insulin receptor mutations, and others ([Chap. 333](#)). Obesity and other factors that contribute to insulin resistance also represent important risk factors for type 2 diabetes. Current models for the genetic basis of type 2 diabetes propose the involvement of more than a dozen genes: some genes influence pancreatic islet development or function; others likely modulate glucose-sensing; and an important group determine insulin sensitivity, either directly by affecting insulin signaling or indirectly by regulating body weight or composition. Superimposed on this genetic background are environmental influences such as diet, exercise, pregnancy, and medications.

Identifying these susceptibility genes is a formidable task. Nonetheless, a reasonable goal for this type of disease is to identify genes that increase (or decrease) disease risk by a factor of two or more. For common diseases such as diabetes or heart disease, this level of risk has important implications for health. Much the same way that cholesterol is currently used as a biochemical marker of cardiovascular risk, we can anticipate the development of genetic panels with similar predictive power. Genetic tests for a large number of genetic disorders are now available; a web site (<http://www.genetests.org>) lists various laboratories that perform specific tests. The advent of DNA-sequencing chips represents an important technical advance that promises to make large-scale testing more feasible ([Chap. 65](#)). The decision whether or not to perform a genetic test for a particular inherited adult-onset disorder, such as hemochromatosis, multiple endocrine neoplasia (MEN) type 1, prolonged QT syndrome, or Huntington disease, is complex; it depends on the clinical features of the disorder, the desires of the patient and family, and whether the results of genetic testing will alter medical decision-making or treatment (see below).

THE FAMILY HISTORY

Pending additional advances in genetic testing, the key to determining the inherited risk for common adult-onset diseases still rests in the collection and interpretation of a detailed personal and family medical history in conjunction with a directed physical examination. For example, a history of multiple family members with early-onset coronary artery diseases, glucose intolerance, and hypertension should suggest increased risk for genetic, and perhaps environmental, predisposition to insulin resistance ([Chap. 333](#)). Individual patients with this family history should be monitored for the possible development of hypertension, diabetes, and hyperlipidemia. They should be counseled about the importance of avoiding additional risk factors such as obesity and cigarette smoking.

Family history, recorded in the form of a pedigree, greatly assists the assessment of risk in the individual patient. At a minimum, pedigrees should convey health-related data on all first-degree relatives and selected second-degree relatives, including grandparents. When pedigrees appear to suggest an inherited disease, they should be extended to include additional family members. The determination of risk for an asymptomatic individual will vary depending on the size of the pedigree, the number of unaffected relatives, and the types of diagnoses within the family. For example, a woman with two first-degree relatives with breast cancer is more at risk if she has a total of three female first-degree relatives than if she has a total of eight female first-degree relatives. Additional variables that factor into the assessment of risk -- and should be documented in the pedigree -- include the age at diagnosis of each affected family member, present age of all family members, and the presence or absence of nonhereditary risk factors among those affected with diseases.

When assessing the personal and family history, the physician should be alert to a younger age of disease onset than is usually seen in the general population. A 30-year-old with acute myocardial infarction should be considered at risk for a hereditary trait, even if there is no family history of premature coronary artery disease ([Chap. 241](#)). The absence of the nonhereditary risk factors typically associated with a disease also raises the prospect of genetic risk factors. A personal or family history of deep vein thrombosis, in the absence of known nongenetic risk factors, might suggest a hereditary thrombotic disorder ([Chap. 117](#)). The physical examination may also provide important clues concerning the risk for a specific inherited disorder. A patient with xanthomas at a young age should prompt consideration of familial hypercholesterolemia. Some adult-onset disease-causing mutations are more prevalent in certain ethnic groups. For instance, >2% of the Ashkenazi population carry one of three specific mutations in the *BRCA1* or *BRCA2* genes. The prevalence of the factor V Leiden allele ranges from 3 to 7% in Caucasians but is much less common in Africans or Asians.

Recall of family history is often inaccurate. This is especially so when the history is remote and as families become more dispersed. It can be helpful to ask patients to fill out family history forms before or after their visits, as this provides them with an opportunity to contact relatives. Attempts should be made to confirm the illnesses reported in the family history before making management decisions. This process is often labor-intensive and ideally involves interviews of additional family members or reviewing medical records, autopsy reports, and death certificates.

Nongenetic factors associated with disease risk should also be reviewed in full, including occupation, diet, living conditions, and social habits. For example, patients at hereditary risk for heart disease should be questioned about tobacco use, diet, exercise, and lipid levels. Patients should also be asked about their health screening and prevention behaviors. These nonhereditary factors contribute to the assessment of overall risk and represent an important focus for disease prevention.

Although many inherited disorders will be suggested by the clustering of relatives with the same or related conditions, it is important to note that *disease penetrance* is incomplete for most multifactorial genetic disorders. As a result, the pedigree obtained in such families may not exhibit a clear Mendelian pattern of inheritance, as not all family members carrying the disease-associated alleles will manifest disease. Furthermore, genes associated with some of these disorders often exhibit *variable expression* of disease. For example, the breast cancer-associated gene, *BRCA1*, can predispose to several different malignancies in the same family, including cancers of the breast, ovary, and prostate ([Chap. 81](#)). For common diseases such as breast cancer, some family members without the disease-causing mutation may also develop breast cancer, representing another confounding variable in the pedigree analysis.

Some of the aforementioned features of the family history are illustrated in [Fig. 68-1](#). The proband, a 36-year-old woman, has a strong history of breast and ovarian cancer on the paternal side of her family. The early age of onset, as well as the co-occurrence of breast and ovarian cancer in this family, suggests the possibility of an inherited mutation in *BRCA1* or *BRCA2*. It is unclear though -- without genetic testing -- whether her father inherited such a mutation and transmitted it to her. After appropriate genetic counseling of the proband and her family, one approach to DNA analysis in this family is to test the potentially affected 42-year-old living cousin (IV-4) for the presence of a *BRCA1* or *BRCA2* mutation. If a mutation is found, then it is possible to test for this particular alteration in the proband and other family members, if they so desire. In the example shown, if the proband's father has inherited the *BRCA1* mutation, there is a 50:50 probability that the mutation has been transmitted to her. Genetic testing can be used to establish the absence or presence of this particular risk factor.

GENETIC TESTING FOR ADULT-ONSET DISORDERS

A critical first step before initiating genetic testing is to assure that the correct clinical diagnosis has been made, whether based on family history, characteristic physical findings, or biochemical testing. Careful clinical assessment will prevent unnecessary testing and will direct testing towards the most probable candidate genes. Many disorders exhibit the feature of *locus heterogeneity*, which refers to the fact that mutations in different genes can cause phenotypically similar disorders. For example, osteogenesis imperfecta ([Chap. 351](#)), muscular dystrophy ([Chap. 383](#)), homocystinuria ([Chap. 352](#)), and hereditary predisposition to colon cancer ([Chap. 90](#)) or breast cancer ([Chap. 89](#)) can each be caused by mutations in distinct genes. The pattern of disease transmission, clinical course, and treatment may differ significantly, depending on which gene is affected. In these cases, the choice of which genes to test is often determined by unique clinical features, the relative prevalence of mutations in various genes, or test availability.

Like all laboratory tests, there are limitations to the accuracy and interpretation of genetic tests. In addition to technical errors, genetic tests are often designed to detect only the most common mutations. In this case, a negative result must be qualified by the possibility that the individual may have a mutation that is not included in the test.

In addition to molecular testing for established disease, presymptomatic testing for susceptibility to chronic disease is being increasingly integrated into the practice of medicine. In most cases, however, the discovery of disease-associated genes has greatly outpaced studies that assess clinical outcomes and the impact of interventions. Until such outcomes-based studies are available, predictive molecular testing must be approached with caution and should be offered only to patients who have been adequately counseled and have provided informed consent ([Fig. 68-2](#)). In the majority of cases, presymptomatic testing should be offered only to individuals with a suggestive personal or family medical history or in the context of a clinical trial.

Molecular analysis is generally more informative if testing is initiated in a symptomatic family member, since the identification of a mutation can direct the testing of other at-risk family members (whether they are symptomatic or not). In the absence of additional familial or environmental risk factors, individuals who test negative for the mutation found in the affected family member can be informed that they are at general population risk for that particular disease. Furthermore, they can be reassured that they are not at risk for passing on the mutation to their children. On the other hand, asymptomatic family members who test positive for the known mutation must be informed that they are at increased risk for disease development and for transmitting the mutation to their children. Nevertheless, for most multifactorial genetic disorders, the test results cannot predict with confidence whether, or when, the disease will develop. For example, not everyone with the apolipoprotein E allele (e4) will develop Alzheimer disease, and many individuals without this susceptibility gene can still develop the disorder ([Chap. 362](#)).

A negative test result is interpreted differently when no genetic mutation is found in a symptomatic family member. In this difficult circumstance, the test performed on a given gene may not detect all mutations in that gene (false negative) or the individual may have a mutation in a different disease-associated gene that was not tested.

Clinicians providing pretest counseling and education should assess the patient's emotional ability to cope with test results. Individuals who demonstrate signs and symptoms of psychiatric illness should have their emotional needs addressed before proceeding with molecular testing. Generally, genetic testing should not be offered at a time of personal crisis or acute illness within the family. Patients will derive more benefit from test results if they are emotionally able to comprehend and absorb the information. It is important to assess patients' preconceived notions of their personal likelihood of disease in preparing pretest educational strategies. Often, patients harbor unwarranted fear or denial of their likelihood of genetic risk.

Genetic testing has the potential of affecting the way individual family members relate to one another, both negatively and positively. As a result, patients addressing the option of molecular testing must consider how test results might impact their relationships with relatives, spouses, and friends. In families with a known genetic mutation, those who

test positive must address the impact of the disease on their present and future lifestyles; those who test negative may manifest survivor guilt. Family members are likely to differ in their emotional and social responses to the same information. Counseling should also address the potential consequences of test results on relationships with a spouse or child. Parents who are found to have a disease-associated mutation often express considerable anxiety and despair as they address the issue of risk to their children.

When a condition does not manifest until adulthood, clinicians will be faced with the question of whether at-risk children should be offered molecular testing and, if so, at what age. Several professional organizations have cautioned that genetic testing for adult-onset disorders should not be offered to children. Many of these conditions are not preventable and, consequently, such information can pose significant psychosocial risk. In addition, there is concern that testing during childhood violates a child's right to make an informed decision regarding testing upon reaching adulthood. On the other hand, testing should be offered in childhood for disorders that may be manifest early in life, especially when management options are available. For example, children at risk for familial adenomatous polyposis (FAP) may develop polyps as early as their teens, and progression to an invasive cancer can occur by their twenties. Likewise, children at risk for [MEN](#) type 2, which is caused by mutations in the *RET* proto-oncogene, may develop medullary thyroid cancer as early as 6 years of age, and the issue of prophylactic thyroidectomy should be addressed with the parents of children with documented mutations ([Chap. 339](#)).

INFORMED CONSENT

When the issue of testing is addressed, patients should be strongly encouraged to involve other relatives in the decision-making process, as molecular diagnostics will likely have an impact on the entire family. Informed consent for molecular testing begins with detailed education and counseling. The patient must fully understand the risks, benefits, and limitations of undergoing the analysis. Informed consent should be in the form of a written document, drafted clearly and concisely in a language and format that is comprehensible to the patient, who should be made aware of the disposition of test results. Informed consent should also include a discussion of the mechanics of testing. Most molecular testing for hereditary disease involves DNA-based analysis of peripheral blood. In the majority of circumstances, test results should be given only to the individual, in person, and with a support person in the room.

Because molecular testing of an asymptomatic individual often allows prediction of future risk, the patient should understand any potential long-term medical, psychological, and social implications of this decision. In the United States, legislation affecting this area is still evolving, and it is important to explore with the patient the potential impact that test results may have on employment, future health, and life insurance coverage.

Patients should understand that alternatives to molecular analysis remain available if they decide not to proceed with this option. They should also be notified that testing is available in the future if they are not prepared to undergo analysis immediately. The option of DNA banking should be presented so that samples are readily available for

future use by family members, if needed.

FOLLOW-UP CARE AFTER TESTING

Depending on the nature of the genetic disorder, posttest interventions may include (1) cautious surveillance and appropriate health care screening, (2) specific medical interventions, (3) chemoprevention, (4) risk avoidance, and (5) referral to support services. For example, patients with known pathologic mutations in *BRCA1* or *BRCA2* are offered intensive screening as well as the option of prophylactic mastectomy and/or oophorectomy. In addition, such women may be eligible for preventive treatment with agents such as tamoxifen, raloxifene, or retinoids. In contrast, those at known risk for Huntington disease are offered continued follow-up and supportive services, including physical and occupational therapy, and social services and support groups as indicated. Specific interventions will change as translational research enhances our understanding of these genetic diseases and as more is learned about the functions of the genes involved.

Individuals who test negative for a mutation in a disease-associated gene identified in an affected family member must be reminded that they may still be at risk for the disease. This is of particular importance for common diseases such as diabetes mellitus, cancer, and coronary artery disease. For example, a woman who finds that she does not carry the disease-associated mutation in *BRCA2* previously discovered in her family must be reminded that she still requires the same breast cancer screening used in the general population.

GENETIC COUNSELING AND EDUCATION

Genetic counseling should be distinguished from genetic testing and screening, even though genetic counselors are often involved in issues related to testing. Genetic counseling refers to *a communication process that deals with human problems associated with the occurrence or risk of a genetic disorder in a family*. Genetic risk assessment can be complex and often involves elements of uncertainty. Counseling therefore includes genetic education as well as psychosocial counseling. Genetic counselors may be called upon by other health care professionals (or by individual patients and families) to address a broad range of issues directly and indirectly involved with genetic disease ([Table 68-2](#)). The genetic counselor will do the following:

- Gather and document a detailed family history
- Educate the patient about general genetic principles related to disease risk, both for themselves and others in the family
- Assess and enhance the patient's ability to cope with the genetic information offered
- Discuss how nongenetic factors may relate to the ultimate expression of disease
- Address medical management issues
- Assist in determining the role of genetic testing for the individual and family

- Ensure that the patient is aware of the risks, benefits, and limitations of the various genetic testing options
- Refer the patient and other at-risk family members for additional medical and support services, if necessary.

The complexity of genetic counseling and the broad scope of genetic diseases are leading to the development of specialized, multidisciplinary clinics designed to provide broad-based support and medical care for those at risk and their family members. Such multidisciplinary teams are often composed of medical geneticists, specialist physicians, genetic counselors, nurses, psychologists, social workers, and biomedical ethicists who work together to consider difficult diagnostic, treatment, and testing decisions. Such a format also provides primary care physicians with invaluable support and assistance as they follow and treat at-risk patients.

The approach to genetic counseling has important ethical, social, and financial implications. Philosophies related to genetic counseling vary widely by country and center. In North American centers, for example, counseling is generally offered in a nondirective manner, wherein patients learn to understand how their values factor into a particular medical decision. Nondirective counseling is particularly appropriate when there are no data demonstrating a clear benefit associated with a particular intervention or when an intervention is considered experimental. For example, nondirective genetic counseling is employed when a person is deciding whether or not to undergo genetic testing for Huntington disease ([Chap. 362](#)). At this time, there is no clear benefit (in terms of medical outcome) to an at-risk individual undergoing genetic testing for this disease, as its course cannot be altered by therapeutic interventions. However, testing can have an important impact on such a person's perception of the future and his or her interpersonal relationships and plans for reproduction. Therefore, the decision to pursue testing rests on the individual's belief system and values. On the other hand, a more directive approach is appropriate when a condition can be treated. In a family with [FAP](#) (associated with mutations in the *APC* gene), colon cancer screening and prophylactic colectomy should be recommended for known *APC* mutation carriers. The counselor and clinician following this family must ensure that the at-risk individuals have access to the resources necessary to adhere to these recommendations.

Genetic education is central to an individual's ability to make an informed decision regarding testing options and treatment. Although genetic counselors represent one source of genetic education, other health care providers also need to contribute to patient education. Patients at risk for genetic disease should understand fundamental medical genetic principles and terminology relevant to their situation. This includes the concept of genes, how they are transmitted, and how they confer hereditary disease risk. An adequate knowledge of patterns of inheritance will allow patients to understand the probability of disease risk for themselves and other family members. It is also important to impart the concepts of disease penetrance and expression. For complex genetic disorders, asymptomatic patients should be advised that a positive test result does not always translate into future disease development. In addition, the role of nongenetic factors, such as environmental exposures, must be discussed in the context of multifactorial disease risk and disease prevention. Finally, patients should understand

the natural history of the disease as well as the potential options for intervention, including screening, prevention, and -- in certain circumstances -- pharmacologic treatment or prophylactic surgery.

THERAPEUTIC INTERVENTIONS BASED ON GENETIC SUSCEPTIBILITY TO DISEASE

Specific treatments are now available for an increasing number of genetic disorders, whether identified through population-based screening or directed testing ([Table 68-3](#)). A number of metabolic disorders fall into this group. For example, the complications of phenylketonuria can be mitigated by recognizing the disease early and avoiding foods that contain phenylalanine ([Chap. 352](#)). Similar principles apply to maple syrup urine disease ([Chap. 352](#)) and galactosemia ([Chap. 350](#)). Children with 21-hydroxylase deficiency present with adrenal insufficiency, usually within the first few weeks of life ([Chap. 331](#)). Because of the block in cortisol synthesis, the adrenal steroid precursors are shunted into the androgen pathway, causing ambiguous genitalia in females and premature virilization in males. In this disorder, treatment with glucocorticoid and mineralocorticoid not only corrects the hormone deficits but is also required to suppress ACTH overproduction, which otherwise worsens virilization.

Although the strategies for therapeutic interventions are best developed for childhood genetic diseases, these principles are gradually making their way into the diagnosis and management of adult-onset disorders. Hereditary hemochromatosis illustrates many of the issues raised by the potential availability of genetic screening in the adult population. For instance, it is relatively common (approximately 1 in 200 individuals of northern European descent are homozygous), and its complications are potentially preventable through phlebotomy ([Chap. 345](#)). The recent identification of the *HFE* gene, mutations of which are associated with this syndrome, has sparked interest in the use of DNA-based testing for presymptomatic diagnosis of the disorder. However, up to one-third of individuals who are homozygous for the *HFE* mutation do not have evidence of iron overload. Consequently, in the absence of a positive family history, current recommendations are phenotypic screening for evidence of iron overload followed by genetic testing. Whether genetic screening for hemochromatosis will someday be coupled to assessment of phenotypic expression awaits further studies. In contrast to the issue of population screening, it is important to test and counsel other family members when the diagnosis of hemochromatosis has been made in a proband. Testing allows the physician to exclude family members who are not at risk. It also permits presymptomatic detection of iron overload and the institution of treatment (phlebotomy) before the development of organ damage.

Preventive measures and therapeutic interventions are not restricted to metabolic disorders. Identification of familial forms of long QT syndrome, associated with ventricular arrhythmias, allows early electrocardiographic testing and the use of prophylactic antiarrhythmic therapy ([Chap. 230](#)). Individuals with familial hypertrophic cardiomyopathy can be screened by ultrasound, treated with beta blockers or other drugs, and counseled about the importance of avoiding strenuous exercise and dehydration ([Chap. 238](#)). Likewise, individuals with Marfan syndrome can be treated with beta blockers and monitored for the development of aortic aneurysms ([Chap. 247](#)). Individuals with antitrypsin deficiency can be strongly counseled to avoid cigarette

smoking and exposure to environmental pulmonary and hepatotoxins. Various host genes influence the pathogenesis of certain infectious diseases in humans, including HIV ([Chap. 309](#)). The factor V Leiden allele increases risk of thrombosis ([Chap. 62](#)). Approximately 3% of the worldwide population is heterozygous for this mutation. Moreover, it is found in up to 25% of patients with recurrent deep venous thrombosis or pulmonary embolism. Women who are heterozygous or homozygous for this allele should therefore avoid the use of oral contraceptives and receive heparin prophylaxis after surgery or trauma.

The field of pharmacogenetics seeks to identify genes that alter drug metabolism or confer susceptibility to toxic drug reactions ([Chap. 71](#)). Examples include succinylcholine sensitivity, malignant hyperthermia, the porphyrias, and glucose-6-phosphate dehydrogenase (G6PD) deficiency.

As noted above, the identification of genes that increase the risk of specific types of neoplasia is rapidly changing the management of many cancers. Identifying family members with mutations that predispose to [FAP](#) or hereditary nonpolyposis colon cancer (HNPCC) can lead to recommendations of early screening by colonoscopy or prophylactic surgery ([Chap. 90](#)). Similar principles apply to familial forms of melanoma, basal cell carcinoma, and breast cancer. It should be recognized, however, that most cancers harbor several distinct genetic abnormalities by the time they acquire invasive or metastatic potential ([Chaps. 81](#) and [82](#)). Consequently, the major impact of genetic testing in these cases is to allow more intensive clinical screening, as it remains very challenging to predict disease penetrance or the clinical course of these diseases.

Although genetic diagnosis of these and other disorders is only beginning to be used in the clinical setting, susceptibility testing holds the promise of allowing earlier and more targeted interventions that can reduce the morbidity and mortality associated with these disorders. We can expect the availability of genetic tests to expand rapidly. A critical challenge for physicians and other health care providers is to keep pace with these advances in genetic medicine and to implement testing judiciously. Meeting this goal will enhance patient care through adequate counseling, directed testing, and appropriate interventions, with the ultimate objective being the reduction of morbidity and mortality from genetic diseases.

(Bibliography omitted in Palm version)

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69. GENE THERAPY - Mark A. Kay, David W. Russell

Gene therapy is generally defined as *the delivery of nucleic acids to alter or prevent a pathologic process*. Although initially considered primarily in the context of inherited monogenic disorders, gene therapy is now recognized as a potential treatment strategy for a wide range of acquired disorders, such as cancer, neurodegenerative diseases, and infections. Gene therapy has been used in several hundred protocols. Despite early hopes that gene therapy might be quickly incorporated into medical practice, it has yielded limited success to date and remains an investigational treatment. The technical challenges associated with gene therapy are formidable, but with steady progress in vector development, definitive therapeutic milestones may soon be realized for selected disorders.

GENERAL APPROACHES TO GENE THERAPY

FORMS OF GENE THERAPY

Gene therapy can be used, in principle, to modify all cells in the body, including the germ line. *Germ-line gene therapy* would allow transmission of the modified genetic information to the next generation and is not currently accepted as an appropriate therapeutic approach. For ethical reasons, opposition to germ-line gene therapy is likely to continue in the foreseeable future. *Somatic gene therapy* refers to modification of the somatic, differentiated cells of the body. It is used in an effort to correct inherited diseases such as cystic fibrosis and acquired disorders such as rheumatoid arthritis or malignancies.

Whole-organ transplantation (e.g., bone marrow, liver, and kidney) has been used for years as a strategy to replace the function of defective genes. The idea of using nonautologous cell transplantation has been revisited in the past few years, and with the advent of human embryonic stem cells, similar transplantation strategies are being considered for a variety of disorders once considered prime targets for gene therapy. Thus, the interface between gene therapy and *cell transplantation* is complementary.

The production of *genetically engineered proteins* is another area closely aligned with gene therapy. The cloning of human genes allows proteins to be produced in unlimited quantities and free of the potential contaminants associated with their purification from natural sources such as plasma. Moreover, recombinant DNA technology allows these proteins to be modified in ways that can enhance their therapeutic benefit. Examples of recombinant proteins are listed in [Table 69-1](#). These include: (1) hormones such as insulin, growth hormone, and gonadotropins; (2) factors used to enhance blood cell production including erythropoietin, granulocyte colony stimulating factor (CSF), granulocyte-macrophage CSF, and thrombopoietin; (3) interferons (IFNs) and interleukins (ILs) used to treat a variety of autoimmune, infectious, and neoplastic diseases; (4) clotting factors VIII and IX; (5) thrombolytic agents such as tissue plasminogen activator or the antithrombotic agent hirudin; (6) recombinant antigens used for hepatitis B vaccines; and (7) humanized monoclonal antibodies used for immunosuppression or to treat specific types of malignancy. Although these recombinant proteins are an indirect form of gene therapy, they represent an important outgrowth of genetic medicine.

Long-Term versus Transient Gene Delivery Strategies for Gene Therapy The goals of gene therapy vary depending on the nature of the disease being treated. For an inherited, monogenic disorder such as hemophilia, the goal is lifelong replacement of the missing gene product. Expression of the missing secreted clotting factor, even at modest levels, might ameliorate the disease or reduce the need for exogenous treatment. For other inherited disorders, such as sickle cell anemia, strategies for gene replacement are more demanding. In this case, there is a requirement for cell-specific and exquisitely regulated expression of the transferred gene, and one is still faced with endogenous expression of the mutant form of β -globin. Long-term gene delivery strategies typically involve direct modification of host chromosomal sequences, which allows normal inheritance and stability of the delivered gene.

A growing use of gene therapy involves transient gene expression to treat a variety of diseases ([Table 69-2](#)). Gene therapy is now being considered or used for (1) killing cancer cells; (2) providing chemoprotection to normal cells; (3) preventing coronary restenosis or enhancing vascularization; (4) providing DNA-based immunization (e.g., viral DNA), in which the injection of DNA results in antigen expression and generation of an immune response; and (5) impairing viral replication. In cancer therapy, for example, the requirements for long-term gene expression and the level of gene expression are not as stringent as for gene replacement. Rather, the challenge of many cancer-based gene therapies is to achieve highly efficient gene transfer into cancer cells without expression in surrounding normal cells.

EX VIVO VERSUS IN VIVO ADMINISTRATION OF GENE THERAPY

Gene therapy has been administered ex vivo for conditions in which cells can be readily harvested, manipulated in tissue culture, and then reintroduced into the patient. The ex vivo approach is potentially applicable to a variety of hematologic or immune deficiency disorders. For example, in adenosine deaminase (ADA) deficiency, the missing ADA gene has been integrated into the patient's T lymphocytes, which are then reintroduced after genetic manipulation. In familial hypercholesterolemia, an analogous approach has been used for ex vivo treatment of hepatocytes. After surgical resection of liver tissue, the low-density lipoprotein (LDL) receptor gene is inserted into hepatocytes in cell culture, and the modified cells are infused back through the portal vein. Another form of ex vivo gene therapy involves the treatment of saphenous veins with oligonucleotides designed to block vascular smooth-muscle cell proliferation before using the tissue for coronary artery bypass graft.

In vivo approaches to gene therapy vary depending on the nature of the disease. In cystic fibrosis, an aerosol has been used to administer viral vectors to the lung. Factor IX expression has been achieved by introduction of the gene into muscle. Tumors have been injected directly with viral vectors expressing cytotoxic genes or immunotherapies. A major advantage of in vivo approaches is the potential to target cells that cannot easily be removed from the patient.

GENE TRANSFER VECTORS

A vector, or vehicle, is required to transfer a gene to an appropriate cell. When the goal

is to add a gene (often called a *transgene*) to supply a function not present in the recipient cell, the vector typically contains DNA sequences encoding a therapeutic protein under the control of transcriptional regulatory elements necessary for gene expression ([Fig. 69-1](#)). The gene is expressed from an ectopic location as opposed to its normal chromosomal position. An alternative strategy is to correct mutant genes at their normal chromosomal location through *gene targeting*. Although this represents an ideal approach for many genetic diseases, it is far more technically demanding, and therapeutic gene-targeting efficiencies are difficult to achieve currently.

Two major classes of vectors are used for transferring nucleic acids into cells for the purposes of gene therapy: viral and nonviral vectors. *Viral vectors* have been genetically engineered so that the viruses transfer exogenous (therapeutic) nucleic acids into cells through a process called transduction. *Nonviral vectors* consist of nucleic acids that are typically complexed with other chemicals to facilitate gene transfer. Although nonviral vectors offer improved safety by avoiding viral components, their gene transfer efficiencies are generally much lower than those of viral vectors. The major vector systems currently used in clinical trials or under development are summarized in [Table 69-3](#).

Vectors that integrate into host chromosomes are considered ideal for lifelong gene expression, whereas episomal (unintegrated) vectors are preferred for transient gene delivery. The host range of a particular vector, in combination with the mode of delivery, will determine which cell types can be genetically modified. Also, it may be possible to alter the natural tropism of viral capsids or envelopes, as well as of nonviral vector complexes, to limit transduction to particular cell types. Transcriptional control elements, such as promoters and enhancers, can be used to regulate expression of the transgene. In some cases, the inclusion of special regulatory elements may allow gene expression to be controlled by the administration of pharmacologic agents that specifically switch the regulatory elements on or off. Many types of vectors, each exhibiting a unique set of properties, will ultimately be needed for safe and effective gene therapy of different diseases.

RETROVIRAL VECTORS

Retroviral vectors based on murine leukemia viruses (MLVs) were the first vectors used in clinical gene transfer protocols and illustrate many of the principles of viral-mediated gene therapy. Typically, the vector genome consists of a transgene cassette placed between the two *cis*-acting long terminal repeats (LTRs) of the viral genome. Viral coding regions are removed from the vector genome to allow the insertion of the therapeutic gene and to prevent viral replication and toxicity to the host. The packaging capacity of MLV vectors is approximately 8 kb (including the LTRs), enough to accommodate most therapeutic cDNAs. The viral gene products needed for vector production and packaging are supplied by helper virus expression constructs.

The envelope protein of the virus interacts with specific cellular receptors to allow cellular entry of the core proteins and vector genome. After transduction, the vector gene integrates at random locations in host chromosomes and replicates with the host chromosome during cell proliferation, offering the potential for lifelong transgene expression. *Chromosomal integration* may also cause insertional mutagenesis, but no

clinically relevant consequences of such events have been observed to date with replication-incompetent retroviral vectors.

Although [MLV](#) vectors are widely used, primarily because of their ability to integrate and simple production requirements, they still have significant disadvantages. Complement-mediated vector particle inactivation has largely limited their use to ex vivo applications. MLV vectors also require cell division for transduction because the vector genome can only enter the cell nucleus during mitosis, when the nuclear membrane breaks down. Since the majority of cells present in the human body are nondividing, this severely limits the potential uses of MLV-based vectors.

Promising new retroviral vectors are based on complex retroviruses (rather than the relatively simple oncoviruses such as [MLV](#)). Both lentiviral vectors and spumaviral vectors have broad host ranges, improved transduction of nondividing cells, and may function well in vivo. Although lentiviral vector production systems can be designed to eliminate the potential for replication-competent retrovirus contamination, and these vectors appear safe from a virologic standpoint, the stigma associated with vectors based on human pathogens such as HIV may limit their acceptance.

ADENOVIRAL VECTORS

Adenoviral-mediated gene expression is not lifelong, making it well suited for applications that require transient gene expression. In contrast to retroviruses, which integrate into the host genome, the adenoviral vector genome remains *episomal*, or extrachromosomal. Adenoviral vectors typically are derived from serotypes 2 or 5 and contain double-stranded DNA genomes. Wild-type adenovirus encodes over 50 peptides on overlapping gene fragments from both DNA strands. Nonessential viral genes have been removed to make room for up to 8 kb of exogenous DNA. The vector containing the therapeutic transgene is amplified in a cell line that supplies viral proteins needed for replication and packaging. Recombinant adenoviruses can be generated at high titers in the range of 10^{13} to 10^{14} particles per milliliter, a feature that is important for efficient in vivo gene therapy.

In clinical practice, adenoviral vectors are limited by relatively short durations of expression (usually several weeks) and by the synthesis of remaining cytotoxic or antigenic viral proteins that can cause an acute inflammatory response and/or a robust cellular immune response. These features of the virus appear to have contributed to hepatic failure and death in an individual receiving adenoviral-based gene therapy for ornithine transcarbamylase deficiency. On the other hand, the inflammatory properties of adenoviruses may enhance their efficacy in cancer trials, as discussed below. Several methods that eliminate the remaining viral genes have been devised recently, and these "gutted" vectors appear to exhibit much reduced toxicity and inflammatory properties in comparison to previous generations of adenoviral vectors.

ADENO-ASSOCIATED VIRUS VECTORS

Adeno-associated viruses (AAV) are parvoviruses that normally require a helper virus, such as adenovirus, to mediate a productive infection. A major limitation of the AAV vector is its relatively small packaging capacity, which restricts the size of exogenous

DNA to about 4.5 kb. AAV vectors have been shown to transduce cells both through episomal transgene expression and by chromosomal integration. They can also be used to modify homologous chromosomal sequences through gene targeting, which is an important strategy for the correction of genetic mutations.

Since the AAV vector genome lacks viral coding sequences, the vector itself causes little immune or inflammatory response (except for the generation of neutralizing antibodies that may limit readministration). The vector particle can be delivered to many different organs [e.g., the central nervous system (CNS), liver, lung, and muscle] by in vivo administration, and AAV vectors have been found to transduce nondividing cells efficiently. Clinical trials using AAV for the treatment of cystic fibrosis and hemophilia are underway.

OTHER VIRAL VECTORS

Many other viruses are under development as vector systems, such as herpesviruses, double-stranded RNA viruses, autonomous parvoviruses, and papovaviruses. Hybrid viral vectors that use components from more than one virus are also under development, offering the potential to combine the most desirable properties of different vectors. Ultimately, vectors will be developed that utilize both viral components and synthetic functions, allowing vectors to be custom designed.

NONVIRAL VECTORS

Nonviral vectors usually consist of DNA complexes with lipids, carbohydrates, proteins, and/or other synthetic chemicals to facilitate delivery or increase vector stability. Gene delivery with naked DNA is also possible but is relatively inefficient. Nonviral vectors are desirable because they eliminate the risk of viral contamination and can be produced under more controlled conditions. Their major disadvantage is low gene transfer rates, at least in relation to most viral vector systems. Because the vectors do not integrate and the majority of DNA molecules that enter the cell are rapidly degraded, only transient gene expression is possible. Examples of applications that may be well suited to gene delivery with nonviral vectors include vaccination with specific antigen genes, ex vivo treatment of vessels used for coronary artery bypass graft, and perhaps transient immune modulation for the treatment of cancer.

Nonviral vectors composed of RNA or modified nucleotides may also prove useful, if problems associated with effective delivery can be addressed. For example, antisense oligonucleotides can be used to inhibit gene expression by pairing with complementary mRNA molecules. Antisense oligonucleotides might be used, for example, to block the expression of cell cycle proteins or cytokines. Oligonucleotides with binding sites for specific DNA- or RNA-binding proteins may be designed to function as decoys that inhibit protein function. Chimeric RNA/DNA oligonucleotides have been used to introduce specific genetic modifications through gene targeting. In animal studies, this novel approach has been shown to work efficiently in the liver, and it may be used soon for clinical trials of uridine diphosphate-glucuronosyltransferase deficiency.

GENE THERAPY FOR SELECTED DISEASE CATEGORIES

LIVER AND GASTROINTESTINAL TRACT

The liver has been studied intensively as a target organ for gene therapy, in part because of the many genetic diseases amenable to gene replacement in hepatocytes. In addition, because of the regenerative capacity of the hepatocyte, integration of a vector offers the possibility of lifelong gene expression. The first hepatic gene therapy attempted in humans involved the ex vivo transplant of autologous hepatocytes transduced in culture by retroviral vectors encoding the [LDL](#) receptor as a treatment for familial hypercholesterolemia. Due to the low levels (nontherapeutic) of gene expression observed and the labor-intensive nature of the ex vivo transduction process, current strategies are geared towards in vivo gene transfer. [MLV](#)-based retroviral vectors require cell division for transduction of host cells. The use of growth factors (rather than partial hepatectomy) to stimulate hepatocellular replication can enhance the efficacy of MLV vectors. Alternative retroviral vectors based on lentiviruses may overcome these obstacles.

Adenovirus vectors are very effective at gene transfer into the liver, thus allowing for the transduction of a majority of hepatocytes. However, the early generations of these vectors exhibited dose-dependent toxicity, inflammation, and immunogenicity that limited the persistence of transgene expression. Adenoviral vectors devoid of all viral genes have been shown to transduce hepatocytes efficiently with reduced toxicity or immunogenicity. In rodents, transgene expression appears to be lifelong, but in nonhuman primates, expression declines by 90% over a 2-year period.

[AAV](#) vectors stably integrate their proviral DNA into hepatocytes in vivo with no apparent toxicity. This vector has been used in preclinical liver gene therapy studies to cure mice with hemophilia B and to partially correct the defect in dogs with hemophilia B. Hepatocyte gene transfer by AAV vectors is likely to be useful for treating a variety of metabolic diseases, such as urea cycle disorders, aminoacidopathies, disorders of carbohydrate metabolism, and lysosomal storage diseases. This strategy is particularly applicable when expression of the transgene in a relatively small percentage of hepatocytes is of therapeutic benefit.

The intestinal tract offers a large cell mass for gene therapy and is accessible by oral administration of vectors. Although a number of vectors have been tested in the gastrointestinal tract, major limitations still exist, such as the rapid turnover of the intestinal epithelial cells, the inability to target the stem or early progenitor cells, and the effects of digestive secretions on vector delivery. If these obstacles can be overcome, many diseases may be amenable to treatment by intestinal gene transfer.

THE HEMATOPOIETIC SYSTEM

Clinical trials targeting hematopoietic cells have almost exclusively taken an ex vivo approach, with gene transfer occurring outside the body, followed by reinfusion of transduced cells. The most common cellular targets are lymphocytes and hematopoietic stem cells. Because stem cells can reconstitute the entire hematopoietic system, ex vivo genetic manipulation and autologous transplantation of bone marrow or peripheral blood stem cells represents a powerful therapeutic paradigm. Long-term gene transfer to hematopoietic stem cells requires the use of integrating viral vectors to ensure that the

transgene replicates with the chromosomes during the extensive cellular proliferation that occurs during hematopoiesis. Thus, most stem cell gene therapy trials have used retroviral vectors. In part, because stem cells are not actively dividing, transduction rates have been too low to be therapeutic. Despite this, recent success in treating X-linked severe combined immunodeficiency with MLV vectors encoding the cytokine receptor gamma common chain demonstrates the feasibility of this approach. The use of different envelope proteins and vectors based on lentiviruses or spumaviruses may yield greater success in stem cell transduction.

Gene transfer to hematopoietic stem cells could be used to treat diseases of erythroid, myeloid, and lymphoid cells as well as platelet disorders (since the stem cell ultimately differentiates into all these cell types). An important aspect of this approach is the level of myeloablation required to allow engraftment of ex vivo-modified stem cells. An optimal therapy would minimize myeloablation and its associated toxicity, perhaps by delivering a gene that confers a selective advantage to transduced stem cells along with the therapeutic transgene. Many of the clinical trials to date have involved genes that confer increased resistance to chemotherapeutic agents used in treating cancer. An alternative strategy is to apply gene therapy to disorders in which expression is not required in all cells to prevent clinical disease. In chronic granulomatous disease, in which there is failure of granulocytes and monocytes to generate the hydrogen peroxide needed for bacterial killing, it may be feasible to achieve a threshold level of normal cells to respond to infections. Another major area of stem cell gene therapy research is the treatment of hemoglobinopathies with vectors expressing globin genes. Although clinical trials have not begun, these efforts have helped define the problems that must be solved for successful stem cell gene therapy, especially with regard to regulating transgene expression levels. Globin gene replacement is a daunting task, since the gene must be expressed at high levels in a select subset of erythroid cells, thus requiring the inclusion of a variety of transcriptional control elements in the vector (many of which have resulted in vector instability and low viral titers).

Lymphocyte gene transfer has the potential to treat genetic immunodeficiencies and to modulate immune functions. [ADA](#) deficiency was one of the first diseases to be treated with gene therapy and employed ex vivo transduction of lymphocytes. Although this early clinical trial represented an important milestone in gene therapy, the assessment of a therapeutic effect has been complicated by concomitant treatment with the ADA protein. Replacement of the ADA gene also confers a selective advantage to ADA-deficient lymphocytes, a property that would not be generally applicable to lymphocyte gene transfer. Still, as lymphocyte transduction protocols improve, many treatments may be developed with this cellular target, especially for acquired diseases. Genetic manipulations with antibody or cytokine genes offer a multitude of possible treatments for infectious and autoimmune diseases, as well as for cancer.

PULMONARY SYSTEM

Gene therapy of the lung has concentrated primarily on treatments for cystic fibrosis. The gene for cystic fibrosis, *CFTR*, was cloned in 1989; by 1993 the first trials using adenovirus vectors were attempted in the nasal epithelium and airway. Though no clinical efficacy was demonstrated, these studies underscored the need to develop defined clinical endpoints for future trials. In addition to the adenovirus trials, clinical

trials with nonviral liposome and [AAV](#) vectors also failed to demonstrate therapeutic effects. Difficulties encountered in pulmonary gene therapy include poor vector delivery due to respiratory secretions, lack of accessible vector receptors on the exposed cellular surface, transient transgene expression due to turnover of the respiratory epithelium, and vector-induced inflammation and pneumonitis. It also remains unclear which type of lung cells should be targeted with the *CFTR* gene to result in clinical benefit. As efficient and safe vectors for lung gene transfer are developed, treatments for acquired disorders such as chronic obstructive pulmonary disease may also be possible.

NERVOUS SYSTEM AND RETINA

Gene transfer to the nervous system will be important for the treatment of many inherited and acquired neurologic diseases. Depending on the disorder, both glial cells and neurons may be appropriate cellular targets. Several vectors have been shown to transduce cells (including neurons) in animals after in vivo injection into the brain. Gene transfer is typically localized near the site of injection, a feature that is desirable for disorders such as Parkinson's disease, where transduction in the striatum could allow selective expression of genes involved in dopamine synthesis. Vector delivery throughout the nervous system will be more difficult to achieve, making gene therapy of global neurologic disorders problematic. The possibility of delivering neurotrophic factors to the [CNS](#) is a potential strategy for the treatment of neurodegenerative diseases, such as amyotrophic lateral sclerosis, and for facilitating recovery after spinal cord injury. Genetically modified stem cells have also been suggested as a strategy for the treatment of neurodegenerative disorders, but studies of these cells are at very early stages.

Retinitis pigmentosa (RP), a common cause of blindness from retinal degeneration, represents one class of disorders that stands to benefit from gene therapy. Though many different gene defects may cause RP, treatment of a form caused by mutation in the rod photoreceptor cGMP phosphodiesterase b-subunit gene has been studied in animal models by intraocular gene addition with adenovirus, lentivirus, and [AAV](#) vectors. In a different type of RP, progression of an autosomal dominant form of the disease was significantly slowed in a transgenic rat model using AAV vectors that express *ribozymes* (RNA enzymes) against the mutant mRNA. Ribozymes are catalytically active RNA molecules that specifically anneal to, and cleave, other RNA sequences, resulting in their selective degradation. This strategy may be useful for the treatment of other genetic diseases caused by dominant mutations. Viral vectors that express cytotoxic genes have been used to treat [CNS](#) tumors (see below).

MUSCULOSKELETAL SYSTEM

Efficient gene transfer to myotubes through intramuscular injection has been demonstrated with adenoviral vectors, [AAV](#) vectors, and lentiviral vectors. In the case of AAV vectors, long-term transgene expression in muscles has been observed in several animal species. The treatment of the muscular dystrophies will require efficient delivery to multiple muscle groups throughout the body, and vector delivery to all the necessary locations remains a formidable task. In the case of Duchenne muscular dystrophy, the large size of the dystrophin cDNA also poses technical challenges in vector design. In contrast to primary diseases of muscle, success is more likely in muscle gene therapy

trials aimed at the synthesis of secreted proteins, such as clotting factors. The large tissue mass and accessibility of muscle make it particularly attractive for these applications.

CARDIOVASCULAR SYSTEM

The cardiovascular system (including the peripheral vasculature) has become an important target for gene therapy. Vascular wall gene delivery is being studied as a way to inhibit smooth-muscle cell proliferation and prevent restenosis. The transgenes used in this application are designed to interfere with the cell cycle or induce apoptosis in smooth-muscle cells. This approach is especially attractive if the vector can be delivered during angioplasty.

Other gene therapy strategies are used to promote the vascularization of tissues. There is some evidence for a therapeutic response in patients with critical limb ischemia due to poor peripheral vascularization who received intramuscular injection of naked DNA vectors encoding the vascular endothelial growth factor (VEGF) gene. Small amounts of the protein are secreted from the muscle, resulting in collateral vessel development that can reverse the ischemia. Patients who were expected to require limb amputation were spared this procedure after gene therapy. Similar clinical trials with the VEGF gene using nonviral and viral vectors are being studied in the context of myocardial ischemia.

CANCER

Most of the gene therapy clinical trials to date have been aimed at the treatment of cancer. One approach uses gene therapy with cytokine or neoantigen genes to increase tumor immunogenicity. The vector is usually injected directly into the tumor, and there is some evidence that once the immune system is stimulated, nontransduced tumor cells may also be eliminated by the immune system. In melanoma, for example, cells have been genetically altered to express mismatched histocompatibility antigens or cytokines such as tumor necrosis factor, [IFN- \$\gamma\$](#) , or [IL-2](#) in an effort to stimulate an immune response. Another approach involves the delivery of genes to tumor cells that convert a prodrug into a cytotoxic compound. Although several strategies are being developed, the most common involves the transfer of the herpes thymidine kinase (TK) gene. The TK enzyme converts gancyclovir into a thymidine analogue that interferes with DNA synthesis and causes cell death. The toxic effects also occur in adjacent nontransduced cells due to the uptake of the toxic analogue; this process is referred to as the *bystander effect*. Vectors designed to replace defective tumor-suppressor proteins with normal versions are being considered for cancer gene therapy, but it is difficult to envision success with this approach unless virtually all the tumor cells are genetically modified. Genes that control tumor growth when expressed in nontumor cells may also be effective in cancer gene therapy. The delivery of gene products that interfere with tumor angiogenesis best exemplifies this approach. Finally, lytic viral vectors that selectively replicate and kill malignant cells are being developed. One example is an adenovirus designed to replicate in cells deficient in p53, a tumor-suppressor protein that is mutated in many different cancers.

These and other related strategies have shown efficacy in animal models when tumor cells are transplanted into various anatomic locations. These models do not always

emulate the natural processes that result in bonafide cancers, in part due to the tumor cells not being truly autologous in origin. Although there have been encouraging results in some of the clinical trials performed to date, efficacy has not been demonstrated definitively. In many cases the patient populations studied had advanced malignancies, and more informative results might be obtained at earlier disease stages. Still, cancer gene therapy has a promising future, and ongoing research involving tumor-specific antigens, angiogenesis, cell cycle control, and apoptosis are all likely to lead to new gene therapy approaches. Improvements in vector targeting of tumor cells and in the development of tumor-specific gene expression will also enhance future therapeutic approaches.

COAGULOPATHIES

Inherited coagulopathies, especially hemophilia A and B, are promising areas of gene therapy research because even a low level of coagulation factor reconstitution can potentially benefit a patient with a severe phenotype. Although the liver is the major site of synthesis for factors VIII and IX, several other tissues may support factor synthesis and secretion into the bloodstream. [AAV](#) vectors, in particular, have shown prolonged therapeutic effects in animal models of hemophilia B when delivered to muscle or liver. Hemophilia A has been more difficult to treat, due to the larger cDNA (which approaches the packaging capacity of AAV vectors) and the requirement that expressing cells deliver the protein directly into the intravascular space. Other coagulopathies, such as factor X deficiency, are also good candidates for gene therapy. Various approaches to the treatment of hemophilia are currently in the early stages of clinical trials.

INFECTIOUS DISEASES

Most infectious pathogens studied as targets for gene therapy are viral (particularly HIV), in part because they replicate inside human cells. One approach involves introducing inhibitory versions of essential viral proteins that disrupt the viral life cycle even in the presence of their normal counterparts (e.g., by disrupting the packaging of virions). Another strategy involves the expression of proteins, peptides, or even RNA transcripts that function as decoys by binding to other proteins required for viral replication and preventing them from acting on their normal viral target sites. The cellular proteins required for a pathogen's life cycle, such as receptor molecules used for viral entry, or specific proteins used for adherence of the pathogen, can also be manipulated through gene therapy; this tactic could also be applied to nonviral pathogens. Ribozymes can be engineered to cleave specific viral transcripts, thereby blocking expression of viral gene products or destroying viral genomes (in the case of RNA viruses). Because ribozymes can, in principle, be engineered to attack any RNA transcript, they are being considered as treatments for many different viral diseases. Clinical trials have begun using ribozymes to block HIV infection. Antisense oligonucleotides can also be used to interfere with viral or cellular nucleic acid sequences. Other gene therapy approaches that may be applied to infectious diseases include vaccination with specific antigens and manipulation of the immune system to enhance the clearance of pathogens.

SUMMARY

The gene can be thought of as a new pharmaceutical agent in the armamentarium used to treat disease. The availability of cloned genes has already yielded a large array of recombinant proteins for clinical use. The limited success of gene therapy to date is due primarily to difficulties inherent in the efficient and safe delivery of genes to their appropriate target cells. The field is still in its infancy, and many of the technical problems are likely to be solved by advances in vector design.

(Bibliography omitted in Palm version)

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PART FOUR -CLINICAL PHARMACOLOGY

70. PRINCIPLES OF DRUG THERAPY - John A. Oates, Dan M. Roden, Grant R. Wilkinson

Safe and effective drug therapy requires that drugs be delivered to their molecular targets in tissues at concentrations within the range that yields efficacy without toxicity. Variability in drug effects between individual patients can thus be attributed, in part, to differences in the processes that determine these concentrations, i.e., drug disposition involving absorption, distribution, and elimination. *Pharmacokinetics* provides a quantitative description of these processes, i.e., what the body does to the drug. In addition, variability in response may also reflect differences in the drug-target interaction that affect the concentration-related effects of the drug on the body, termed *pharmacodynamics*. This chapter reviews the principles of pharmacokinetics and pharmacodynamics and their application to optimizing therapeutic regimens.

CLINICAL PHARMACOKINETICS

PLASMA LEVELS AFTER A SINGLE DOSE

The levels of lidocaine in plasma following intravenous administration decline in two phases, as illustrated in [Fig. 70-1](#); such a biphasic decline is typical for many drugs. Immediately after rapid injection, essentially all of the drug is in the plasma or central compartment, and the high initial plasma level reflects the confinement of the drug to this small volume. The drug is then transferred into an extravascular or peripheral compartment during a period called the *distribution phase*. For lidocaine, the distribution phase is virtually complete within 30 min. A phase of slower decline, the *equilibrium phase*, then occurs. During this phase, the drug levels in the plasma and in tissues change in parallel. Although this disposition profile is common to many drugs given intravenously, the characteristic parameters vary among drugs.

Distribution Phase The pharmacologic effects during the distribution phase depend on whether the concentration of drug at the receptor site is similar to that in the plasma. If that is the case, the pharmacologic effects may be intense during this period because of the high initial levels in the plasma. For example, after a small bolus dose (50 mg) of lidocaine, antiarrhythmic effects may be evident during the early distribution phase but will disappear as levels fall below those that are minimally effective, even before equilibrium between plasma and tissue is reached. Thus, a larger single dose or multiple small doses must be administered to achieve an effect that is sustained into the equilibrium phase. Toxicity resulting from high levels of some drugs during the distribution phase precludes administration of a single intravenous loading dose that will yield therapeutic levels during the equilibrium phase. For example, the administration of the entire loading dose of phenytoin as a single intravenous bolus can cause cardiovascular collapse due to the high levels during the distribution phase. If a loading dose of phenytoin is administered intravenously, it must be given slowly, e.g., at infusion rates of 25 to 50 mg/min. For similar reasons, the intravenous loading dose of many potent drugs that equilibrate rapidly with their receptors is either divided into fractional doses given at intervals or administered by infusion over a similar period.

A dose given orally results in lower plasma levels during the initial period than an intravenous bolus dose that delivers the same amount of drug to the systemic circulation. Because the drug is not absorbed instantly after oral administration and is delivered into the systemic circulation more slowly, much of it has been distributed by the time absorption is complete. Thus, procainamide, which is almost totally absorbed after oral administration, can be given orally as a single 750-mg loading dose with little risk of hypotension; in contrast, loading of this drug by the intravenous route is more safely accomplished by giving the dose in fractions of about 100 mg at 5-min intervals or by slow infusion to avoid hypotension during the distribution phase.

Some drugs are so predictably lethal when infused too rapidly that special precautions should be taken to prevent even the inadvertent occurrence. For example, solutions of potassium for intravenous administration in excess of 20 meq/L should be avoided in all but the most exceptional and carefully monitored circumstances. This minimizes the possibility of cardiac arrests, which can occur as a result of accidental increases in infusion rates of more concentrated solutions.

As these examples illustrate, excessively rapid administration of many drugs can lead to catastrophic consequences that result from high concentrations in the blood during the distribution phase.

In contrast, for some centrally active drugs, the higher concentration of drug during the distribution phase after intravenous administration is used to advantage. The use of midazolam for "IV sedation," for example, depends upon its rapid uptake by the brain during the distribution phase to produce sedation quickly, with subsequent egress from the brain during the redistribution of the drug as equilibrium is achieved.

Some drugs are distributed slowly to their sites of action during the distribution phase, i.e., concentration at the relevant receptor does not parallel that in plasma early after drug administration. For example, the level of digoxin at the receptor site (and the drug's pharmacologic effect) do not reflect plasma levels during the distribution phase. Digoxin is transported (or bound) to its cardiac receptors slowly by a process that proceeds throughout distribution. Thus, over the distribution phase of several hours, plasma levels fall while the level at the site of action and the pharmacologic effect increase. Only at the end of the distribution phase, when the drug has reached equilibrium with the receptor, does the concentration of digoxin in plasma reflect pharmacologic effect. For this reason, there should be a 6- to 8-h wait after administration before plasma levels of digoxin are measured as a guide to therapy.

Equilibrium Phase After the concentration of drug in plasma has reached a dynamic equilibrium with that in the tissues, the levels in plasma and tissues fall in parallel as the drug is eliminated from the body. Thus, the equilibrium phase is also called the *elimination phase*. During this phase, drug concentrations measured in plasma can provide a useful index of drug levels in tissues.

Most drugs are eliminated as a first-order process. This means that the time required for the plasma level of the drug to fall to one-half the original value (the half-life, $t_{1/2}$) is the same regardless of the point on the plasma level curve at which measurement begins. Another characteristic of the first-order process is that a plot of the logarithm of the

plasma concentration versus time is linear. From such a plot ([Fig. 70-1](#)), it can be seen that the half-life of lidocaine is 108 min. If the half-life is known, the amount of a dose remaining in the body at any time following administration of a single dose can be calculated. [Table 70-1](#) shows how this amount changes over five successive half-lives.

From a clinical standpoint, elimination is essentially complete when it has reached about 90%. Therefore, for practical purposes, *a first-order elimination process reaches completion after three to four half-lives.*

DRUG ACCUMULATION -- LOADING AND MAINTENANCE DOSES

If a drug is given repeatedly at intervals shorter than the time required to eliminate a dose, both the amount of drug in the body and its pharmacologic effect increase with successive doses until they reach a plateau. [Figure 70-2](#) shows the accumulation of digoxin administered in repeated maintenance doses (without a loading dose). Since the half-life of digoxin is about 1.6 days in a patient with normal renal function, 65% of a digoxin dose remains in the body at the end of 1 day. Thus, the second dose will raise the amount of digoxin in the body (and the average plasma level) to 165% of the level produced by the first dose. Each subsequent dose causes a further increase until a *steady state* is achieved. At that point, the drug dosing rate (bioavailable dose/dosage interval) is equal to the rate of elimination, with the fluctuation between peak and trough plasma levels remaining constant. If the rate of drug delivery is then altered, a new steady state will be attained. Continuous infusion of a drug at constant rate also results in progressive accumulation to a predictable steady state ([Fig. 70-2](#)). In this case, the steady-state plasma level (C_{ss}) is equivalent to the average between the peak and trough levels ($C_{avg,ss}$) produced by intermittent administration of the same amount of drug over the same period. For *all* drugs with first-order kinetics, the time required to achieve steady-state levels can be predicted from the half-life, because accumulation is a first-order process with a half-life identical to that for elimination. Thus, accumulation reaches 90% of steady-state levels at the end of three to four half-lives. This is true for either intermittent or continuous dosing ([Fig. 70-2](#)).

When a therapeutic effect is required urgently, simply administering the maintenance dose of a drug with a long half-life results in an unacceptable delay in reaching steady-state levels of the drug and its intended effect. The time required to achieve the desired pharmacologic effect may be shortened by the administration of a *loading dose* -- the amount of drug that will bring the plasma concentration rapidly to the steady-state level. Therefore, if treatment with lidocaine ($t_{1/2} \sim 108$ min) were initiated by infusion at only the maintenance dose level, it would take about 4 to 8 h before the drug's maximal effect was achieved. Because ventricular arrhythmias may be life-threatening, it is not reasonable to wait that long to achieve an effective steady-state drug level. Accordingly, it would be appropriate to administer one or more loading doses at the onset of therapy, together with infusion at the rate of the maintenance dose.

Loading may be accomplished by the administration of a single loading dose or, if that would create a risk of toxicity, by the administration of the loading dose in a series of fractions of that dose over time. The latter approach is particularly appropriate for most drugs that have a low therapeutic index. A divided loading dose strategy or the administration of the loading dose in a slow intravenous infusion are particularly

advisable when the drug is given intravenously. Thus, in the case of lidocaine, a common regimen is to administer an initial intravenous bolus of 1 mg/kg followed by up to three additional bolus injections of 0.5 mg/kg every 8 to 10 min as necessary, and a maintenance infusion of 2 mg/min.

Regardless of the size of the loading dose, *after maintenance therapy has been given for three to four half-lives, the amount of drug in the body is determined by the maintenance dose.* The independence of the steady state plasma levels from the loading dose is illustrated in [Fig. 70-2](#), which shows that the elimination of the loading dose would be practically complete after three to four half-lives.

DETERMINANTS OF DRUG DISPOSITION

A number of physiologic and pathophysiologic factors determine the disposition of a drug and hence its pharmacokinetics in an individual patient. The three most important are *clearance*, a measure of the body's ability to eliminate drug; *volume of distribution*, an indication of the extent to which the drug is distributed outside of the blood compartment; and *bioavailability*, the fraction of the administered dose that reaches the systemic circulation. The elimination half-life ($t_{1/2}$), which measures the rate of drug removal from the body, is determined by the relationship between the physiologically determined clearance and volume of distribution.

Clearance The majority of drugs are given over a prolonged period of time according to a multiple dosage regimen, e.g., x mg every y h. The clinical goal is to maintain the drug's steady-state concentration within the therapeutic range for the individual patient; if the level is too low, reduced efficacy results, whereas if it is too high, the likelihood of adverse effects increases.

Steady state is achieved when the rate of drug elimination equals the rate of drug delivery into the systemic circulation, which, if bioavailability is complete, corresponds to the rate at which the drug dose is administered:

where Cl is clearance and C_{ss} is the steady-state concentration in plasma. Therefore, *at a given dosing rate the concentration of drug in plasma is completely dependent on its clearance.* If the drug is infused, the concentration remains constant so long as drug delivery continues; however, when the drug is given intermittently, the above relationship is expressed as:

where $C_{avg,ss}$ is the *average* value during the dosage interval and is equal to C_{ss} although the *actual* concentrations will be higher or lower at various points during this period. Thus, clearance determines the rate at which a drug should be administered in order to obtain a desired steady-state concentration; stated in a different fashion, steady-state drug levels can be modified either up or down by changing the dosage rate. *The clearance of the vast majority of drugs is constant over the therapeutic range of concentrations.*

Clearance is a measure of the rate at which the organs that eliminate drug from the body remove drug from the blood.

Accordingly, clearance reflects the volume of blood (or plasma) from which the drug would have to be removed per unit time to account for the elimination; it can be related to either total or unbound drug.

Drug elimination generally occurs as a result of metabolism and/or excretion in the liver, kidney, and possibly other organs. Clearance of drug from the body, therefore, reflects the overall contribution of each of these organs, as indicated by their individual rates of elimination normalized to the concentration of drug, and is additive.

Clearance of a drug is usually estimated following administration of an intravenous dose (dose_{iv}) and measurement of the resulting total area under the curve (AUC) for the blood or plasma concentration-time curve (AUC_{iv}) from zero to infinite time.

[Table 70-2](#) indicates the marked differences in plasma clearance for some commonly used drugs. Some drugs such as phenobarbital and valproic acid have relatively low values (<10 mL/min), whereas others such as procainamide and lidocaine have much larger clearances (>500 mL/min). Such differences mainly reflect different rate-limiting determinants such as blood flow through the organ(s) of clearance, the extent of binding of the drug to plasma proteins, and the efficiency of the clearance process to remove drug from tissue water by metabolism and/or excretion. The data in [Table 70-2](#) also demonstrate that the relative contributions of the two major routes of elimination, i.e., renal and nonrenal, also vary according to the individual drug. In some cases, such as amikacin, digoxin, gentamicin, lithium, and tobramycin, excretion by the kidneys is predominant. However, with many other drugs (e.g., carbamezipine, lidocaine), nonrenal elimination, which usually reflects metabolism in the liver, is more important.

Volume of Distribution The relationship between the amount of drug in the body and the concentration of drug in the plasma provides a measure of the apparent volume of distribution:

This volume does not, except in a limited number of special cases, reflect an identifiable physiologic volume but corresponds to the virtual volume of fluid that would be required to contain all of the drug in the body at the same concentration as in the plasma. In a typical 70-kg human, plasma volume is about 3 L, blood volume around 5.5 L, and extracellular water outside the vasculature is approximately 42 L. The volume of distribution of drugs that are extensively bound to plasma proteins but are not bound to tissue components approaches plasma volume. However, for most drugs, the volume of

distribution is far greater than any physiologic space. For example, the volume of distribution of digoxin is about 700 L, which obviously exceeds total body volume. This simply indicates that digoxin is largely distributed outside the vascular system and hence the proportion of the drug present in the plasma compartment is low.

The volume of distribution may be estimated by back-extrapolation of the plasma concentration-time curve to zero time (C_0) ([Fig. 70-1](#)) and dividing this into the dose of drug administered intravenously.

When distribution of the drug is not instantaneous, a more useful volume estimate is based on the area under the plasma-concentration time curve (AUC) and the terminal elimination half-life of the drug ($t_{1/2}$).

Extent and Rate of Bioavailability After intravenous administration, all of the administered dose reaches the systemic circulation. In contrast, with all other routes of administration, such as oral, intramuscular, and subcutaneous, there is the potential for only a part of the dose to be absorbed; this fraction (F) is termed the drug's *bioavailability*. Lack of complete bioavailability may reflect the inability of the drug to be completely released from the dosage form or vehicle, chemical destruction at the site of administration, incomplete absorption into the vascular system, and metabolism and/or excretion during translocation of the drug from its site of administration to the systemic circulation. In the case of oral dosing, this would include the intestinal epithelium and the liver and lungs. Metabolism and/or excretion by the gastrointestinal tract (F_G) and liver (F_L) are collectively referred to as the *first-pass effect*, since the resulting drug elimination only occurs during drug delivery to the systemic circulation. Loss of drug because of a first-pass effect thus requires that the dosing rate appropriately take into account bioavailability. For example, after oral drug administration:

Bioavailability and the first-pass effect are important with respect to possible differences in drug responsiveness dependent on the route of administration. For example, glyceryl trinitrate has such a large oral first-pass effect (>99%) that systemic concentrations after an oral dose are negligible and no antianginal effect is present. Giving the drug by the sublingual or transdermal routes bypasses the splanchnic organs and allows essentially all of the drug to reach the systemic circulation. For drugs that are efficiently metabolized by the intestinal epithelium and/or liver, i.e., drugs with high extraction ratios in either of these organs, differences in the extent of the first-pass effect between individuals frequently explain variability in drug response. With propranolol, for example, the 15-fold variability in plasma concentrations after the same oral dose results from differences in the individual hepatic extraction ratios reflective of different levels of drug metabolizing activity.

Drug administration by nonintravenous routes involves an absorption process characterized by the plasma level increasing to a maximum value at some time after

administration and then declining as the rate of drug elimination exceeds the rate of absorption. Thus, the peak concentration is lower and occurs later than after the same dose given by rapid intravenous injection. The rate of absorption can be an important consideration during the initial period after drug administration, especially for drugs with a narrow *therapeutic index* -- the ratio of the toxic dose to the therapeutic dose. If absorption is too rapid, then the resulting high concentration may cause adverse effects not observed with a more slowly available formulation. At the other extreme, slow absorption is deliberately designed into "slow-release" or "sustained-release" drug formulations in order to maintain plasma concentrations essentially constant during the dosage interval, because the drug's rate of elimination is offset by an equivalent rate of absorption controlled by formulation factors.

Half-Life The organs of elimination can only clear drug from the blood. Thus, the rate at which drug is eliminated from the body is a function of both clearance and the extent to which drug is distributed outside of the vascular compartment. The fraction of total drug in the body that is eliminated in a given time is designated the *fractional elimination constant* (k).

For example, if the volume of distribution is 10 L and clearance is 1 L/min, then one-tenth of the drug is eliminated per minute. If k is multiplied by the total amount of drug in the body, the actual rate of elimination at any given time can be determined:

This relationship, indicating that the rate of drug elimination is proportional to the drug concentration, describes a first-order, or monoexponential, process. With a few notable exceptions, the elimination of drugs used clinically is first-order.

Half-life ($t_{1/2}$) is the time that it takes for the plasma concentration or amount of drug in the body to decline by 50%. This parameter is related to k as follows:

where 0.693 is the natural logarithm of $2(C_0/0.5 C_0)$.

Because

then

This is an important relationship since it indicates that the rate of drug elimination, reflected by $t_{1/2}$, is dependent on both the efficiency of drug removal (Cl) and the drug's volume of distribution (V). When V remains constant, $t_{1/2}$ is a reflection of clearance. Thus, $t_{1/2}$ is shortened when rifampin induces the enzymes responsible for a drug's

hepatic clearance and is lengthened when a drug's renal clearance is impaired in renal failure. However, when there are concomitant alterations in V , as occurs for some drugs in cardiac failure, $t_{1/2}$ is not an accurate measure of Cl or drug dose.

DESIGNING DOSAGE REGIMENS

Most drugs are administered as part of long-term therapy involving multiple dosing, and it is critical that the dosage regimen be optimized to the individual patient. With some drugs, the desired response, e.g., coagulation or blood pressure, is readily measurable and an individualized dosage regimen can be developed with dosage titration. However, dosage changes should be conservative (<50% for drugs with a low therapeutic index) and not more frequent than every three to four half-lives. Other drugs have little dose-related toxicity so the therapeutic window is large, e.g., penicillins and β -adrenoceptor antagonists. In these situations, effective and prolonged drug effects may be obtained by a "maximal dose" strategy. It is also possible to use this strategy to extend the duration of action of a drug, especially one that is eliminated rapidly from the body. Thus, 75 mg of captopril will result in reduced blood pressure for up to 12 h, even though the elimination half-life of this angiotensin-converting enzyme (ACE) inhibitor is about 2 h; this is because the dose raises the concentration of drug in plasma many times higher than the threshold for its pharmacologic effect.

Determination of the Maintenance Dose The relationship between the maintenance dose and the final steady-state concentration is

Thus, steady-state concentrations can be predictably increased or decreased by appropriate modification of the maintenance dosing rate to achieve a desired target value:

In most cases, this is best achieved by changing the drug dose but not the dosing interval, e.g., by giving 250 mg every 8 h instead of 200 mg every 8 h. However, this approach is acceptable only if the resulting maximum concentration is not toxic and the trough value does not fall below the minimum effective concentration for an undesirable period of time. Alternatively, the steady state may be changed by altering the frequency of intermittent dosing but not the size of each dose. In this case, the magnitude of the fluctuations around the average steady-state level will change -- the shorter the dosing interval, the smaller the difference between peak and trough levels ([Fig. 70-3](#)).

The extent of fluctuation is determined by the relationship between the dosing interval and the drug's half-life. For example, if the dosing interval is equal to the drug's half-life, then the fluctuation would be twofold, which is usually a tolerable variation. If a longer dosing interval is used, then the difference between the maximum and minimum plasma levels will be greater ([Fig. 70-3](#)). Marked fluctuations increase the likelihood of increased concentration-dependent drug effects early during the dosing interval and possible ineffectiveness at the end of the period, even though the average steady-state drug concentration is the same as that following administration at the same dosing rate but at

shorter intervals.

Determination of the Loading Dose The loading dose can be estimated if both the desired plasma level (C) and the apparent volume of distribution (V) are known:

The loading amount required to achieve steady-state plasma levels can also be determined from the fraction of drug eliminated during the dosing interval and the maintenance dose. For example, if the fraction of digoxin eliminated daily is 35% and the planned maintenance dose is 0.25 mg daily, then the loading dose to achieve steady-state levels would be $(100/35)$ times the maintenance dose, or approximately 0.75 mg. Thus,

NONLINEAR DRUG ELIMINATION

The elimination of some drugs (e.g., phenytoin, salicylate, propafenone, and theophylline) does not follow first-order kinetics because the clearance of these drugs changes as levels in the body fall during elimination or change after alterations in dose. Such elimination is called *concentration-dependent* or *dose-dependent*. Accordingly, the time for the concentration to fall to one-half becomes less as plasma levels fall. (This halving time is not truly a half-life, because the term *half-life* applies to first-order kinetics and is a constant.) When a drug is eliminated by first-order kinetics, the plasma level at steady state is directly related to the amount of the maintenance dose, and a doubling of the dose should lead to doubling of the steady-state plasma level. However, for drugs with dose-dependent kinetics, an increase in the dose may be accompanied by a disproportionate increase in the plasma level. For example, a threefold increase in the dose of propafenone (from 300 mg to 900 mg daily) leads to a tenfold increase in the concentration of propafenone in plasma. Changes in dosage regimens for drugs with dose-dependent kinetics should always be accompanied by surveillance for adverse effects and by measurement of the concentration of the drug in plasma during the time of transition to the new steady-state, if this is feasible.

INDIVIDUALIZATION OF DRUG THERAPY

EFFECTS OF RENAL DISEASE

Whether a drug's dosing rate needs to be modified in patients with renal dysfunction depends on whether the drug is primarily excreted through the kidneys and whether increased drug levels, secondary to impaired renal clearance, will be associated with adverse effects. If both of these factors are present, it is likely that with decreased renal clearance the drug will accumulate to a greater extent than in patients with normal renal function and toxicity will result. This is especially true for drugs with long half-lives and narrow therapeutic indexes (e.g., digoxin). In general, over 60 to 70% of the drug must be renally excreted for dosage modification to be necessary and then only when renal function is less than about 30 to 50% of normal.

The goal of any dosing rate adjustment is to modify the dosing schedule so that the drug's plasma concentration-time profile is as similar to the desired one as possible and that the steady state is reached in about the same time as in a patient with normal renal function. To obtain the desired profile, a modification may be made by decreasing the dose while maintaining the dosage interval, keeping the dose the same but increasing the dosing interval, or a combination of these two approaches.

A drug's renal clearance is proportional to creatinine's clearance (Cl_{CR}), which may be measured directly or estimated from the serum creatinine level (C_{CR}). In men:

For women, the estimate by the above equation should be multiplied by 0.85 to reflect their smaller muscle mass. It should also be noted that this equation is not valid for patients with severe renal insufficiency ($C_{CR} < 5$ mg/dL) or when renal function is changing rapidly. For simplicity, normal creatinine clearance is conveniently considered to be 100 mL/min. Thus, if the relative contributions of renal and nonrenal elimination to systemic clearance are known, an appropriate modification of the dose in a patient with a given level of insufficiency can be estimated. For example, if the fraction of drug excreted unchanged is 0.9 and creatinine clearance is reduced to 10% of normal, the dosing rate should be reduced to 19% of normal. This modification, which in practice would be rounded to 20%, is based on the fact that nonrenal clearance is unchanged (10% of normal clearance); renal clearance is reduced from 90% to 9% of normal Cl ; thus systemic clearance is reduced to $10 + 9 = 19\%$ of normal Cl .

In clinical practice today, most decisions involving dosing adjustment in patients with renal failure use published tables of recommended dosage reduction or dosing interval lengthening based on the level of renal function indicated by Cl_{CR} , or similar information provided in the drug "label" ([Table 70-3](#)). Such modifications are, however, rigorously based on pharmacokinetic principles and are best used when resulting plasma concentration data and clinical observation are used, as necessary, to further optimize therapy for the individual patient.

Often metabolites of the drug are pharmacologically active or cause toxicity, and renal insufficiency may result in their unanticipated accumulation. Meperidine, for example, is extensively metabolized, and renal failure has little effect on its plasma concentration; however, its metabolite, normeperidine, accumulates above its usual level when renal function is impaired. Because normeperidine has greater convulsant activity than meperidine, this accumulation probably accounts for the signs of central nervous system (CNS) excitation, such as irritability, twitching, and seizures, that appear when multiple doses of meperidine are administered to patients with renal disease.

EFFECTS OF LIVER DISEASE

In contrast to the predictable decline in renal clearance of drugs in renal insufficiency, it is not possible to make a general prediction of the effect of liver disease on hepatic biotransformation of drugs ([Chap. 292](#)). Rather, the possible effects of hepatitis or cirrhosis range from impaired to increased drug clearance. Even in advanced hepatocellular disease, drug clearance is usually impaired only about two- to fivefold.

The extent of such changes, however, cannot be predicted by the common tests of liver function. Consequently, even when it is suspected that drug elimination is altered in liver disease, there is no quantitative basis on which to adjust the dosage regimen other than assessment of clinical response and the concentration of the drug in plasma.

A drug's oral bioavailability may markedly increase in patients with liver disease. This is particularly the case for those drugs that normally are very well extracted by the liver and thus have a high first-pass effect. In addition, the presence of portacaval shunts may further reduce first-pass elimination and lead to higher drug concentrations reaching the systemic circulation, with the increased risk of adverse effects. For example, the oral availability for high first-pass drugs such as morphine, meperidine, midazolam, and nifedipine is almost doubled in patients with cirrhosis, compared to those with normal liver function. The size of the oral dose of such drugs should, therefore, be reduced in such patients.

EFFECTS OF CIRCULATORY INSUFFICIENCY -- CARDIAC FAILURE AND SHOCK

Under conditions of decreased tissue perfusion, the cardiac output is redistributed to preserve blood flow to the heart and brain at the expense of other tissues ([Chap. 38](#)). As a result, the drug may be distributed into a smaller volume of distribution, higher drug concentrations will be present in the plasma, and the tissues that are best perfused will be exposed to these higher concentrations. If either the brain or heart is sensitive to the drug, an alteration in response will occur.

Furthermore, the decreased perfusion of the kidney and liver may impair drug clearance by these organs, directly or indirectly. Thus, in severe congestive heart failure, in hemorrhagic shock, and in cardiogenic shock, the response to the usual dose of drug may be excessive, and dosage modification may be necessary. For example, the clearance of lidocaine is reduced by about 50% in cardiac failure, and therapeutic plasma levels are achieved at infusion rates only about half those usually required. The volume of distribution of lidocaine is also reduced, meaning that the correct loading dose will be smaller than usual. Similar situations are thought to exist for procainamide, theophylline, and possibly quinidine. Unfortunately, predictors of these types of pharmacokinetic alterations are unavailable. Therefore, loading doses should be conservative, and continued therapy should be monitored closely, following clinical indicators of toxicity and plasma levels.

DISEASE-INDUCED CHANGES IN PLASMA BINDING

Many drugs circulate in the plasma partly bound to plasma proteins. Since only the unbound (free) drug can distribute to the site of pharmacologic action, the therapeutic response should be related to the free rather than the total circulating plasma drug concentration. In most cases, the degree of binding is fairly constant across the therapeutic concentration range, so that the total drug levels in plasma can be used as a basis for adjusting dosage without resulting in significant error. However, conditions such as hypoalbuminemia, liver disease, and renal disease can decrease the extent of drug binding, particularly of acidic and neutral drugs so that at any total plasma level there is a greater concentration of free drug than usual and thus a risk of increased response and toxicity. By contrast, conditions that lead to an increased plasma

concentration of the acute-phase reactant α_1 -acid glycoprotein -- such as myocardial infarction, surgery, neoplastic disease, rheumatoid arthritis, and burns -- cause an increase in drug binding for the basic drugs, e.g., lidocaine and quinidine, that bind to this macromolecule, resulting in an opposite set of effects. The drugs for which changes in binding are important are those that are normally highly bound to plasma proteins (>90%), because a small alteration in the extent of binding produces a large change in the amount of unbound drug.

For many drugs, elimination and distribution are restricted largely to the unbound fraction, and so a decrease in binding leads to an increase in the clearance and distribution of the drug. The relative magnitudes of these changes are such that the net effect is a shortened half-life.

GENETIC DETERMINANTS OF THE RESPONSE TO DRUGS

Knowledge of the enzyme that catalyzes the predominant pathway of metabolism of a drug provides a basis for understanding the therapeutic consequences of variations in the genotype of that enzyme. For a number of the enzymes that metabolize drugs, there are differences (polymorphisms) in catalytic function that are genetically determined ([Chap. 65](#)). *A phenotypic trait or its corresponding gene is said to be polymorphic if there is more than one form of the trait or gene in the population.* Polymorphisms in the function of an enzyme are determined by allelic variants in its gene. Increasingly it is possible to individualize treatment based on analysis of the phenotype and/or genotype of the relevant drug-metabolizing enzyme.

Similarly, polymorphism in the receptor for a drug can determine variability in its pharmacologic effect. Genotyping those drug receptors for which polymorphisms influence response may also assist in individualizing drug therapy.

THIOPURINE S-METHYLTRANSFERASE (TPMT)

The metabolism of azathioprine provides an example of the importance of genetic polymorphisms of enzymes. Azathioprine exerts its immunosuppressive action via an active metabolite, 6-mercaptopurine. Within target cells, the major pathway of inactivation of 6-mercaptopurine is by TPMT. Genetic polymorphisms in this enzyme lead to differences in inactivation of 6-mercaptopurine, with corresponding vast differences in the sensitivity of patients to the toxic and therapeutic effects of azathioprine. Homozygotes for alleles encoding inactive TPMT (0.3 to 1% of the population) predictably exhibit severe pancytopenia on standard doses of azathioprine. Heterozygotes for alleles encoding enzymes with deficient TPMT activity also experience more bone marrow suppression on "usual" doses. From a therapeutic standpoint it is likely that the bone marrow suppression in the heterozygotes has influenced the empiric determination of the "usual" dose range of azathioprine and that this results in undertreatment of some of the patients homozygous for the allele encoding a TPMT with full catalytic activity. To detect the TPMT deficient phenotype in patients anticipating therapy with azathioprine, the catalytic function of TPMT may be measured in red blood cells (if no blood or red cell transfusions have been given within 2 months). A high concordance of genotype with phenotype (~95%) suggests that analysis of genotype may be used to individualize dosing and therefore improve

treatment with azathioprine (and 6-mercaptopurine) in the future.

ACETYLATION

Isoniazid, hydralazine, sulfonamides, procainamide, and a number of other drugs are metabolized by acetylation of a hydrazino or amino group. This reaction is catalyzed by *N*-acetyl transferase-2 (NAT-2), an enzyme in the liver cytosol that transfers an acetyl group from acetyl coenzyme A to the drug. Individuals differ markedly in the rate at which drugs are acetylated, because of polymorphisms in the NAT-2 gene, resulting in a bimodal distribution of the population into "rapid acetylators" and "slow acetylators."

Acetylation phenotype can be determined by measuring the ratio of acetylated to nonacetylated forms of the probe drugs, dapsone, caffeine, or sulfamethazine, in plasma or urine following administration of a test dose of these acetylation substrates. Slow, intermediate, and rapid acetylators may be identified by these methods for phenotyping. It is also possible to identify slow acetylators by genotyping, using genomic DNA obtained from blood leukocytes.

METABOLISM BY CYTOCHROME P450 MONOOXYGENASES

In healthy individuals taking no other medications, the major determinant of the rate of metabolism of drugs by the cytochrome P450 monooxygenases is genetic. Hepatic endoplasmic reticulum contains a family of cytochrome P450 (CYP) isoforms with different substrate specificities. Many drugs undergo oxidative metabolism by more than one isoform, and the steady-state concentrations of such drugs in the plasma is a function of the sum of the activities of these and other metabolizing enzymes. When a drug is metabolized by multiple pathways, the catalytic activities of the participating enzymes are regulated by a number of genes, so that the clearance rates and steady-state concentrations of the drug tend to distribute unimodally within the population. The range of activity may differ markedly (3tenfold) between different individuals, as is the case for chlorpromazine, and there is no way to predict the rate before beginning therapy.

Certain metabolic pathways show a bimodal or trimodal distribution of activity, suggesting control by a single gene, and polymorphisms in these genes have been identified. Most individuals are extensive metabolizers (EM phenotype); a smaller group have a lower ability to metabolize the drug (or no ability at all) and are called poor metabolizers (PM phenotype). Heterozygotes for genes encoding the enzymes lacking catalytic activity may be intermediate metabolizers (IM phenotype). And patients with duplicate or multiple copies of the gene may exhibit ultrarapid metabolism. These polymorphisms are of greatest clinical relevance during administration of substrate drugs for which there are no major alternative routes of elimination. The clinical consequences of the PM phenotype will then depend on the resultant accumulation of the drug or occasionally on the absence of generation of active metabolites.

The cytochrome P450 isoform CYP2D6 is polymorphically distributed in the population, and about 8 to 10% of Caucasians are deficient in this enzyme. CYP2D6 represents the main metabolic pathway for a number of drugs, including antiarrhythmic agents (propafenone, flecainide), β -adrenoceptor blockers (timolol, metoprolol, and alprenolol),

tricyclic antidepressants (nortriptyline, desipramine, imipramine, clomipramine), neuroleptic drugs (perphenazine, thioridazine, and possibly haloperidol), selective serotonin reuptake inhibitors (fluoxetine and paroxetine), and certain opiates, such as codeine and dextromethorphan. Thus, codeine has a much lower analgesic effect in [PM](#) patients because of impaired production of the active metabolite, morphine. Conversely, a patient with duplicate or multiple copies of CYP2D6 will exhibit an exaggerated response to codeine. Patients with the PM phenotype experience more pronounced systemic β -adrenoceptor blockade after the administration of timolol ophthalmic solution. The catalytic activity of CYP2D6 in humans may be assessed by using a test drug, debrisoquin, which is eliminated almost entirely via hydroxylation by CYP2D6. Individuals with the PM phenotype can be identified by genotyping for the alleles that encode proteins with loss of catalytic function. There are ethnic variations in the frequency of the PM phenotype, which occurs in 5 to 10% of Caucasians but with a lesser frequency in Asians (1 to 2%).

The isoform CYP2C19 also exhibits polymorphism; it was initially detected with the hydroxylation of mephenytoin, which is used as a probe drug for the function of this P450 isoform. This enzyme catalyzes the major metabolic pathway of omeprazole, proguanil, diazepam, and citalopram. The impact of the polymorphism in CYP2C19 on treatment outcome is clearly illustrated by omeprazole. The efficacy of omeprazole (20 mg in combination with amoxicillin) in eradicating *Helicobacter pylori* is markedly reduced in persons with the homozygous [EM](#) genotype (29% cured) as compared with 100% cure in those with homozygous [PM](#) genotype. This reflects the relative lack of effect of the recommended dose of omeprazole (20 mg) on gastric acid secretion and ulcer healing in patients with the CYP2C19 EM genotype. Certainly knowledge of a patient's CYP2C19 genotype would improve therapy with this proton pump inhibitor. Impaired hydroxylation of mephenytoin is present in only 3 to 5 percent of Caucasians, but the incidence is about 20 percent in individuals of Japanese and Chinese descent.

CYP2C9 catalyzes the major pathways of metabolism of warfarin and phenytoin. There are allelic variants of the gene for this enzyme that encode proteins with loss of catalytic function. These variant alleles are associated with requirement for a very low dose of warfarin, difficulties in initiating warfarin therapy, and an increased risk of bleeding complications. Similarly, high concentrations of phenytoin in plasma and resulting adverse effects of phenytoin occur in patients with loss of function alleles for CYP2C9.

Polymorphisms in drug-metabolizing ability may be associated with large differences in the disposition of a drug among individuals, especially when the involved pathway makes a major contribution to the elimination of the drug. For example, the clearance of mephenytoin given orally differs 100- to 200-fold between individuals of the [EM](#) and [PM](#) phenotypes. As a result, the peak plasma concentrations and bioavailability after oral administration are much higher, and the rate of drug elimination much lower, in PM than in EM individuals. In PM individuals, the result is excessive drug accumulation and exaggerated pharmacologic responses, including toxicity, when usual drug dosages are administered. Individualization of drug therapy is especially critical for drugs that exhibit polymorphic drug metabolism. The increasing availability of laboratory methods to identify the PM phenotype for NAT-2, CYP2D6, and CYP2C19 by genotyping should be useful for this purpose.

INTERINDIVIDUAL VARIABILITY IN THE MOLECULAR TARGETS WITH WHICH DRUGS INTERACT

The increasing emphasis on identifying molecular mechanisms of disease ([Chap. 65](#)) has important consequences for further understanding a genetic basis for individual variability in drug actions. As molecular approaches identify the role of specific gene products in human physiology, polymorphisms that alter expression or function of those gene products are being recognized; it is estimated that such polymorphisms occur in 1 in 1000 bp in the human genome. These genes in turn, are now being recognized as the molecular targets with which available and new drugs interact to produce beneficial and adverse effects.

Genome-wide searches in families with premature Alzheimer's disease identified the *APOE* locus as linked to the disease ([Chap. 362](#)). Specifically, the *E4* allele of the *APOE* gene appears associated with a worse prognosis, and this is thought to relate to reduced expression of choline acetyl transferase. Further, a therapeutic response to the choline acetyl transferase inhibitor, tacrine, appears to be more common with the prognostically more benign *APOE2* or *APOE3* alleles. Multiple polymorphisms identified in the β_2 -adrenergic receptor appear to be linked to specific phenotypes in asthma and congestive heart failure, diseases in which β_2 -receptor function might be expected to determine prognosis. It has been suggested that polymorphisms in the β_2 -receptor may be a determinant of response to inhaled β_2 -receptor agonists.

The development of marked QT prolongation and the polymorphic ventricular tachycardia, *torsade de pointes* ([Chap. 230](#)), in response to certain action potential-prolonging drugs such as quinidine used to be characterized as an "idiosyncratic" response. Advances in understanding the molecular basis of normal cardiac repolarization have resulted in identification of genes encoding ion channel proteins, the molecules whose normal function results in physiologic cardiac repolarization. Mutations in these genes cause congenital arrhythmia syndromes, such as the long QT syndrome, and block of ion channels is a common mechanism whereby drugs prolong QT intervals. Patients with mutations in these genes that remain subclinical until challenge with drugs are now recognized. In summary, continuing efforts to unravel the molecular basis of disease are likely also to provide insights into determinants of the response to drug therapy.

DRUG USE IN THE ELDERLY (See also [Chap. 9](#))

Aging results in changes in organ function, especially of the organs involved in drug disposition, as well as alterations in body size and composition. Not surprisingly, therefore, pharmacokinetics are often different in elderly individuals than in younger adults. Also, elderly patients often have multiple diseases and may therefore be taking a large number of drugs. Consequently, drug interactions, as well as an increased vulnerability to morbidity and mortality, contribute to the higher incidence of adverse drug reactions in elderly patients. Increased sensitivity of target organs and impairment of physiologic control systems, such as those involved in the regulation of the circulation, may also be a factor. Accordingly, optimization of drug therapy in the elderly, particularly in frail patients, is often difficult, as a variety of factors (often poorly defined) accentuate the usual interindividual variability in drug response.

Although many individuals preserve good renal function into old age, elderly patients as a group have an increased likelihood of impaired renal excretion of drugs. Even in the absence of kidney disease, renal clearance is generally reduced by about 35 to 50% in elderly patients. Dosage adjustments analogous to those in patients with renal dysfunction (see above) are therefore necessary for drugs that are eliminated mainly by the kidneys, such as digoxin, aminoglycosides, lithium, and other drugs listed in [Table 70-3](#). In this regard, it is important to recognize that the reduced muscle mass of older individuals results in a reduced rate of creatinine production; thus, a normal serum creatinine concentration can be present even though creatinine clearance is impaired.

Aging also results in a decrease in the size of and blood flow to the liver and possibly in the activity of hepatic drug-metabolizing enzymes; accordingly, the hepatic clearance of some drugs is impaired in the elderly. Unfortunately, no consistent pattern of clinical application appears to be present. Moreover, the changes are often modest relative to other causes for interindividual variability in these patients. However, even a small reduction in hepatic extraction may significantly increase the oral bioavailability of drugs with a high first-pass effect, such as propranolol and labetalol.

Impaired clearance and/or increased distribution may cause the elimination half-life of a drug to increase with aging. Thus, if a dosage modification in an elderly patient is required, it is often possible to accomplish it by decreasing the frequency of drug administration, possibly along with a reduction in dose.

Even if the pharmacokinetics of a drug are not altered, an elderly patient may require a smaller dosage because of an increase in pharmacodynamic sensitivity. Examples include increased analgesic effects of opioids, increased sedation from benzodiazepines and other [CNS](#) depressants, and increased risk of bleeding while receiving anticoagulant therapy, even when clotting parameters are well controlled. Exaggerated responses to cardiovascular drugs are also common because of the impaired responsiveness of normal homeostatic mechanisms. Such age-related changes require close monitoring of the patient's clinical response and appropriate dosage titration. Accordingly, in the elderly, initial doses should be less than the usual adult dosage and should be increased slowly. The final therapeutic regimen should be as simple as possible, and the number of different drugs used should be kept as low as possible. Also, because interindividual variability in drug responsiveness is greater in geriatric patients than in younger adults, individualization of therapy is even more critical.

INTERACTIONS BETWEEN DRUGS

The effect of some drugs can be altered markedly by the administration of other agents. Such interactions can complicate therapy by adversely increasing or decreasing the action of a drug. Drug interactions must be considered in the differential diagnosis of unexpected responses to drugs, and it should be recognized that patients often come to the physician with a legacy of drugs acquired during previous medical experiences. A meticulous drug history will minimize such unknown elements. It should include examination of the patient's medications and, if necessary, calls to the pharmacist to identify prescriptions. It should also address the use agents not often volunteered on

initial questioning, such as over-the-counter drugs, health food supplements, and topical agents such as eye drops.

There are two principal types of interactions between drugs. In *pharmacokinetic interactions*, the delivery of a drug to its site of action is altered, whereas in *pharmacodynamic interactions*, the responsiveness of the target organ or system is modified.

An index of the drug interactions discussed in this chapter is provided in [Table 70-4](#). The table includes interactions that have verified significance in patients, plus a few that are so potentially dangerous that cognizance should be taken of the experimental data or case reports suggesting they occur.

I. PHARMACOKINETIC INTERACTIONS CAUSING DIMINISHED DRUG DELIVERY

A. Impaired Gastrointestinal Absorption Examples include aluminum ions, present in antacids, which form insoluble chelates with the tetracyclines, preventing absorption of these drugs. Ferrous ions similarly block tetracycline absorption. Kaolin-pectin suspensions bind digoxin, and when these substances are administered together, digoxin absorption is reduced by about one-half. However, when kaolin-pectin is administered 2 h after digoxin, digoxin absorption is unaffected.

Ketoconazole is a weak base that dissolves well only at acidic pH. Histamine H₂receptor antagonists, such as ranitidine and cimetidine, reduce gastric acidity and thus impair the dissolution and absorption of ketoconazole. By contrast, the absorption of fluconazole is not impaired by an increase in gastric pH.

B. Induction of Hepatic Drug-Metabolizing Enzymes When a drug is eliminated largely by metabolism, an increase in the rate of its metabolism reduces its availability to sites of action. Most drugs are metabolized largely in the liver because of this organ's large mass, high blood flow, and high concentration of drug-metabolizing enzymes. The first step in the metabolism of many drugs is catalyzed by a group of cytochrome P450 mixed-function oxidases located in the endoplasmic reticulum (see "Metabolism by Cytochrome P450 Monooxygenases," above). These enzyme systems oxidize drug molecules by a variety of reactions, including aromatic hydroxylations, *N*-demethylations, *O*-demethylations, and sulfoxidations. The products of these reactions are usually more polar than the parent compound (and more readily excreted by the kidney).

The expression of some of the mixed-function oxidase (CYP) isoforms is regulated, and their content in the liver can be increased, by a number of drugs. Phenobarbital is the prototype of these inducers, and all the barbiturates in clinical use increase CYP enzyme activity. Induction with phenobarbital can occur with doses of as little as 60 mg daily. Mixed-function oxidases are also induced by rifampin, carbamazepine, phenytoin, and glutethimide and by smoking, exposure to chlorinated insecticides such as DDT, and chronic alcohol ingestion.

Phenobarbital, rifampin, and other inducers lower plasma levels of many drugs, including warfarin, quinidine, mexiletine, verapamil, ketoconazole, itraconazole,

cyclosporine, dexamethasone, methylprednisolone, prednisolone (the active metabolite of prednisone), oral contraceptive steroids, methadone, metronidazole, and metyrapone. These interactions all have obvious clinical significance. In the case of the coumarin anticoagulants, the patient is placed at major risk if the appropriate level of anticoagulation is achieved when an inducer is also being administered and the inducer is later discontinued (for example, at discharge from the hospital). The plasma levels of the coumarin anticoagulant will rise as the induction effect wears off, leading to excessive anticoagulation. There is considerable variation among individuals in the extent to which drug metabolism can be induced.

C. Inhibition of Cellular Uptake or Binding The guanidinium antihypertensive agents guanethidine and guanadrel are transported to their site of action in adrenergic neurons by an energy-requiring membrane transport system for biogenic monoamines; the physiologic function of this system is reuptake of the adrenergic neurotransmitter. Inhibitors of norepinephrine uptake prevent the uptake of the guanidinium antihypertensive agents into adrenergic neurons and thereby block their pharmacologic effects. The tricyclic antidepressants are potent inhibitors of norepinephrine uptake. Consequently, concomitant administration of clinical doses of tricyclic antidepressants, including desipramine, protriptyline, nortriptyline, and amitriptyline, almost totally abolishes the antihypertensive effects of guanethidine and guanadrel. Although they are less potent inhibitors of norepinephrine uptake, doxepin and chlorpromazine produce dose-related antagonism of the action of the guanidinium antihypertensives.

The antihypertensive effect of clonidine is partially antagonized by tricyclic antidepressants. Clonidine lowers arterial pressure by reducing sympathetic outflow from the blood pressure-regulating centers in the hindbrain ([Chap. 246](#)). This central hypotensive action is antagonized by the tricyclic antidepressants.

II. PHARMACOKINETIC INTERACTIONS CAUSING INCREASED DRUG DELIVERY

A. Inhibition of Drug Metabolism If the active form of a drug is eliminated largely by biotransformation, inhibition of its metabolism leads to reduced clearance, prolonged half-life, and accumulation of the drug during maintenance therapy. Excessive accumulation due to inhibited metabolism can lead to adverse effects.

Cimetidine is a potent inhibitor of the oxidative metabolism of many drugs, including warfarin, quinidine, nifedipine, lidocaine, theophylline, and phenytoin. Adverse reactions, many of them severe, have resulted from the administration of these drugs in conjunction with cimetidine. Cimetidine is a more potent inhibitor of mixed-function oxidases than ranitidine, whereas ranitidine is more potent as a histamine H₂receptor antagonist. Famotidine and nizatidine are not known to produce clinically appreciable inhibition of drug metabolism.

Knowledge of the CYP isoforms that catalyze the main pathway of metabolism of a drug provides a basis for predicting and understanding drug interactions. For example, the CYP3A subfamily of isoforms catalyzes the metabolism of many drugs for which blockage of metabolism results in toxicity. Drugs that depend on CYP3A as a major route of metabolism include cyclosporine, quinidine, lovastatin, simvastatin, atorvastatin, nifedipine, lidocaine, cisapride, erythromycin, methylprednisolone, carbamazepine,

midazolam, and triazolam.

The antifungal agents ketoconazole and itraconazole are potent inhibitors of enzymes in the CYP3A family. When fluconazole levels are elevated as a result of higher doses and/or renal insufficiency, this drug can also inhibit CYP3A. The macrolide antibiotics erythromycin and clarithromycin inhibit CYP3A4 to a clinically significant extent, but azithromycin does not inhibit this enzyme. Some of the calcium antagonists, diltiazem, nicardipine, and verapamil can also inhibit CYP3A, as can some of its other substrates, such as cyclosporine.

Cyclosporine can cause serious toxicity when its metabolism is inhibited by erythromycin, ketoconazole, diltiazem, nicardipine, or verapamil. A serious complication of HMG-CoA reductase inhibitors is myopathy. Fortunately, this is infrequent except in the context of interactions of a subset of the HMG-CoA reductase inhibitors with other drugs, particularly those that inhibit CYP3A4. The disposition of lovastatin is reduced markedly by drugs that inhibit CYP3A4, causing increases in plasma levels by more than tenfold. As a consequence, lovastatin has produced severe myopathy with rhabdomyolysis when administered together with erythromycin or cyclosporine. Not all of the HMG-CoA reductase inhibitors are as dependent on CYP3A4 for disposition as is lovastatin. Blocking CYP3A4 causes moderate elevations of plasma levels of simvastatin and atorvastatin (increases of severalfold), whereas elevations of the levels of fluvastatin and cerivastatin are only slight. Pravastatin disposition and plasma levels are not altered by inhibitors of CYP3A4. Cisapride can cause polymorphic ventricular tachycardia (torsade de pointes) when its metabolism is blocked by inhibitors of CYP3A, such as ketoconazole, itraconazole, clarithromycin, and erythromycin.

Whenever an inhibitor of CYP3A4 is administered to a patient, the physician should be alert to the possibility of serious interactions with drugs that are metabolized by CYP3A.

The CYP2D6 isoform that catalyzes the polymorphic metabolism of debrisoquin is markedly inhibited by quinidine and is also blocked by a number of neuroleptic drugs, such as chlorpromazine and haloperidol, and by fluoxetine. The analgesic effect of codeine depends on its metabolism to morphine via CYP2D6 in individuals with the [EM](#) phenotype. Thus, quinidine reduces the analgesic efficacy of codeine in EMs. Since desipramine is cleared largely by metabolism via CYP2D6 in EMs, its levels are increased substantially by concurrent administration of quinidine, fluoxetine, or the neuroleptic drugs that inhibit CYP2D6.

Some drugs are inactivated by mechanisms other than the hepatic drug-metabolizing enzymes. Azathioprine is converted in the body to an active metabolite, 6-mercaptopurine, which in turn is oxidized by xanthine oxidase to 6-thiouric acid. When allopurinol, a potent inhibitor of xanthine oxidase, is administered concurrently with standard doses of azathioprine or 6-mercaptopurine, life-threatening toxicity (bone marrow suppression) can result.

Other drugs that inhibit biotransformation of pharmacologic compounds (with examples of drugs whose metabolism is blocked by the inhibitor listed in parenthesis) include:

- Amiodarone (warfarin, quinidine)

- Clofibrate (phenytoin, tolbutamide)
- Excessive ingestion of ethanol (warfarin)
- Isoniazid (phenytoin)
- Metronidazole, cotrimoxazole (warfarin)
- Phenylbutazone (warfarin, phenytoin, tolbutamide)

B. Inhibition of Drug Transport Specific molecules that transport drugs into and out of cells are increasingly recognized, and inhibition of their function can be a major cause of clinically important drug interactions. The best studied to date is P-glycoprotein, originally isolated from tumor cells displaying resistance to multiple, structurally unrelated anticancer agents. The mechanism underlying this "multidrug resistance" phenomenon is P-glycoprotein-mediated pumping of anticancer agents out of cells, thereby inhibiting their anticancer effects. P-glycoprotein is also expressed in normal tissues (the luminal aspect of intestinal and renal tubular cells, the canalicular aspect of hepatocytes, the capillary endothelium of the blood-brain barrier), where it is responsible for efflux of not only antineoplastics but also digoxin and HIV protease inhibitors. Quinidine inhibits P-glycoprotein function in vitro, and it now seems apparent that the widely recognized doubling of plasma digoxin when quinidine is coadministered reflects this action in vivo, particularly since the effects of quinidine (increased digoxin bioavailability and reduced renal and hepatic secretion) occur at the sites of P-glycoprotein expression. Many other drugs also elevate digoxin concentrations (e.g., amiodarone, verapamil, cyclosporine, itraconazole, and erythromycin), and a similar mechanism seems likely. Reduced CNS penetration of multiple HIV protease inhibitors (with the attendant risk of facilitating a sanctuary site for the virus) appears attributable to P-glycoprotein-mediated exclusion of the drug from the CNS.

A number of drugs are secreted by the renal tubular transport systems for organic anions. Inhibition of this tubular transport system can cause excessive accumulation of a drug. Phenylbutazone, probenecid, and salicylates competitively inhibit this transport system. Salicylate, for example, reduces the renal clearance of methotrexate, an interaction that may lead to methotrexate toxicity. Renal tubular secretion contributes substantially to the elimination of penicillin, which can be inhibited by probenecid.

Inhibition of the tubular cation transport system by cimetidine impedes the renal clearance of procainamide and its active metabolite *N*-acetylprocainamide.

III. PHARMACODYNAMIC AND OTHER INTERACTIONS BETWEEN DRUGS

Therapeutically useful interactions occur in which the effect of two drugs in combination is greater than the sum of their effects when used individually. Favorable drug combinations are described in specific therapeutic sections in this text, and this section focuses on interactions that create unwanted effects. Two drugs may act on separate components of a common process to yield effects greater than either has alone. For example, although small doses of aspirin (<1 g daily) do not alter the prothrombin time

appreciably in patients who are receiving warfarin therapy, aspirin nevertheless increases the risk of bleeding in these patients because it inhibits platelet aggregation. Thus the combination of impaired functions of platelets and the clotting system, while useful for some therapeutic purposes, also increases the potential for hemorrhagic complications in patients receiving warfarin therapy.

Nonsteroidal antiinflammatory drugs (NSAIDs) cause gastric and duodenal ulcers, and, in patients treated with warfarin, the risk of bleeding from a peptic ulcer is increased almost threefold by concomitant use of a NSAID. This clearly is a serious drug interaction.

Indomethacin, piroxicam, and probably other NSAIDs antagonize the antihypertensive effects of β -adrenergic receptor blockers, diuretics, ACE inhibitors, and other drugs. The resulting elevation in blood pressure ranges from trivial to severe. Aspirin and sulindac, however, do not elevate the blood pressure in treated hypertensive patients.

Polymorphic ventricular tachycardia (torsade de pointes) during quinidine administration occurs much more frequently in patients receiving diuretics, probably owing to potassium and/or magnesium depletion.

The administration of supplemental potassium leads to more frequent and more severe hyperkalemia when potassium elimination is reduced by concurrent treatment with ACE inhibitors, spironolactone, amiloride, or triamterene.

The pharmacologic effects of sildenafil result from inhibition of the phosphodiesterase type 5 isoform that inactivates cyclic GMP in the vasculature. Nitroglycerin and related nitrates produce vasodilation by elevating cyclic GMP. Thus, coadministration of these nitrates with sildenafil will cause profound and potentially catastrophic hypotension.

CONCENTRATION OF DRUGS IN PLASMA AS A GUIDE TO THERAPY

In many cases, the plasma concentration of a drug is measured as a guide in the individualization of therapy. Genetic variation in elimination rates, interactions with other drugs, disease-induced alterations in elimination and distribution, and other factors combine to yield a wide range of plasma levels in patients given the same dose. Furthermore, the problem of noncompliance with prescribed regimens during continuing therapy is an endemic and elusive cause of therapeutic failure (see below). Clinical indicators assist in the titration of some drugs into the desired range, but no chemical determination is a substitute for careful observations of the response to treatment. However, the therapeutic and adverse effects are not precisely quantifiable for all drugs, and, in complex clinical situations, estimates of the action of a drug may be misleading. For example, previously existing neurologic disease may obscure the neurologic consequences of intoxication with phenytoin. Because clearance, half-life, accumulation, and steady-state plasma levels are difficult to predict, the measurement of plasma levels is often useful as a guide to the optimal dose. This is particularly true when there is a narrow range between the plasma levels yielding therapeutic and adverse effects. For drugs having such characteristics -- e.g., digoxin, theophylline, lidocaine, aminoglycosides, cyclosporine, and anticonvulsants -- dose optimization should involve modification of the standard dose on the basis of the pharmacokinetic

principles described above. In certain instances, predictive nomograms and algorithms have been developed to facilitate the necessary modifications. However, the most flexible and accurate method for individualizing drug dosage appears to be a feedback approach using a small number of previously obtained plasma levels and Bayesian forecasting. In controlled studies, this type of computer-assisted dosing has been shown to improve patient care. However, the overall cost/benefit ratio of such methods in routine management still remains to be conclusively demonstrated.

For drugs with a narrow therapeutic window that exhibit first-order elimination, then, dosage adjustments may be made on the assumption that the average, maximum, and minimum steady-state concentrations are related linearly to the dosing rate. Accordingly, the dose may be adjusted on the basis of the ratio between the desired and measured concentrations:

For drugs that have dose-dependent kinetics (e.g., phenytoin and theophylline), plasma concentrations change disproportionately more than the alteration in the dosing rate. Not only should changes in dose be small to minimize the degree of unpredictability, but plasma concentration monitoring is also critical to ensure appropriate modification.

The variability among individual responses to given plasma levels must be recognized. This is illustrated by a hypothetical population concentration-response curve ([Fig. 70-4](#)) and its relationship to the therapeutic range or therapeutic window of desired plasma levels. The defined therapeutic window should include the levels at which the intended pharmacologic effect is achieved in most patients. However, a few persons, who are sensitive to the therapeutic effects, respond to lower levels, whereas others are refractory enough to require levels that may cause adverse effects. For example, a few patients with strong seizure foci require plasma levels of phenytoin exceeding 20 ug/mL to control seizures. Dosages to achieve this effect may be appropriate if tolerated.

As also illustrated in [Fig. 70-4](#), some patients are prone to adverse effects at levels that are tolerated by most of the population. Therefore, raising the plasma concentration of a drug to a level that has a high probability of being therapeutically effective may bring on unwanted actions in an occasional patient. [Table 70-2](#) presents for a number of drugs the plasma concentrations that are associated with adverse and therapeutic effects in most patients. Use of this information according to the guidelines discussed should permit more effective and safer therapy for those patients who are not "average."

EFFECTIVE PARTICIPATION OF THE PATIENT IN THERAPY

Measurement of the concentration of a drug in plasma is the most effective way to detect failure to take a drug. Such "noncompliance" is a frequent problem in the long-term treatment of diseases such as hypertension and epilepsy, occurring in 25% or more of patients in therapeutic environments in which no special effort is made to involve patients in the responsibility for their own health. Occasionally, noncompliance can be uncovered by sympathetic, nonincriminating questioning, but more often it is recognized only after determining that the concentration of drug in plasma is nil or is recurrently low. Because other factors can cause plasma levels to be lower than

expected, comparison with levels obtained during inpatient treatment may be required to confirm that noncompliance has occurred. Once the physician is certain of noncompliance, a nonaccusatory discussion of the problem with the patient may clarify the reason for the noncompliance and serve as a basis for more effective cooperation on the part of the patient. Many approaches have been tried to help patients exercise more responsibility for their own treatment, most based on better communication regarding the nature of the disease and the chances of success or failure of the treatment. The patient is given a chance to discuss problems associated with treatment. The process may be improved by the involvement of nurses and other paramedical personnel. Minimizing the complexity of the regimen is helpful in terms of both the number of drugs and the frequency of administration. Educating patients to assume the principal role in their own health care requires a blend of the art and science of medicine.

(Bibliography omitted in Palm version)

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71. ADVERSE REACTIONS TO DRUGS - *Alastair J. J. Wood*

The beneficial effects of drugs are coupled with the inescapable risk of untoward effects. The morbidity and mortality from these untoward effects often present diagnostic problems because they can involve every organ and system of the body and are frequently mistaken for signs of underlying disease. Major advances in the investigation, development, and regulation of drugs ensure in most instances that they are uniform, effective, and relatively safe and that their recognized hazards are publicized. However, prior to regulatory approval and marketing, new drugs are tested in relatively few patients who tend to be less sick and to have fewer concomitant diseases than those patients who subsequently receive the drug therapeutically. Because of the relatively small number of patients studied in clinical trials, and the selected nature of these patients, rare adverse effects may not be detected prior to a drug's approval, and physicians therefore need to be cautious in the prescription of new drugs and alert for the appearance of previously unrecognized adverse events.

The large number and variety of drugs available over the counter (OTC), herbal preparations, and by prescription, make it impossible for patient or physician to obtain or retain the knowledge necessary to use all drugs well. It is understandable, therefore, that many OTC drugs are used unwisely by the public and that restricted drugs may be prescribed incorrectly by physicians.

Most physicians use no more than 50 drug products in their practice, gaining familiarity with their effectiveness and safety. Most patients probably use only a limited number of [OTC](#) drugs. Nevertheless, many patients receive care and drug prescriptions from more than one physician, and in any 30-day period, many patients consume more than three different OTC drug products containing nine or more different chemical agents.

Some 25 to 50% of patients make errors in self-administration of prescribed medicines, and these errors can be responsible for adverse drug effects. Elderly patients are the group most likely to commit such errors, perhaps in part because they consume more medicines. One-third or more of patients also may not take their prescribed medications. Similarly, patients commit errors in taking [OTC](#) drugs by not reading or following the directions on the containers. Physicians must recognize that providing directions with prescriptions does not always guarantee compliance.

Every drug can produce untoward consequences, even when used according to standard or recommended methods of administration. When used incorrectly, the effectiveness may be reduced, and adverse reactions can be expected to occur more frequently. Also, the administration of several drugs concurrently may result in adverse drug interactions ([Chap. 70](#)). In the hospital, all drugs a patient is given should be under the control of a physician, and patient compliance is, in general, ensured. Errors may occur nevertheless -- the wrong drug or dose may be given, or the drug may be given to the wrong patient -- although improved drug distribution and administration systems have reduced this problem. On the other hand, there are no easy means for controlling how ambulatory patients take prescription or [OTC](#) drugs.

EPIDEMIOLOGY

Epidemiologic studies of adverse drug reactions have been helpful in evaluating the magnitude of the overall problem, in calculating the rate of reactions to individual drugs, and in characterizing some of the determinants of adverse drug effects.

Patients receive, on average, 10 different drugs during each hospitalization. The sicker the patient, the more drugs are given, and there is a corresponding increase in the likelihood of adverse drug reactions. When fewer than 6 different drugs are given to hospitalized patients, the probability of an adverse reaction is about 5%, but if more than 15 drugs are given, the probability is over 40%. Retrospective analyses of ambulatory patients have revealed adverse drug effects in 20%.

Thus, the magnitude of drug-induced disease is large. Of patients admitted to the medical and pediatric services of general hospitals, 2 to 5% are admitted because of illnesses attributed to drugs. The case/ fatality ratio from drug-induced disease in hospitalized patients varies from 2 to 12%. Furthermore, some fetal or neonatal abnormalities are due to medicines taken by the mother during pregnancy or parturition.

A small group of widely used drugs accounts for a disproportionate number of reactions. Aspirin and other nonsteroidal anti-inflammatory drugs, analgesics, digoxin, anticoagulants, diuretics, antimicrobials, glucocorticoids, antineoplastics, and hypoglycemic agents account for 90% of reactions, although the drugs involved differ between ambulatory and hospitalized patients. Estimates of the cost of drug-related morbidity and mortality in the ambulatory setting range from \$30 billion to \$130 billion.

ADVERSE DRUG REACTIONS IN THE ELDERLY (See also [Chap. 9](#))

The elderly as a group have a greater burden of disease and receive a greater number of medications than other persons. Thus, it is not surprising that adverse drug reactions occur frequently in elderly patients. The issue of whether an elderly individual is more likely to develop an adverse drug reaction than a young person with a similar number of concurrent diseases and taking the same number of drugs has not been answered unequivocally. However, in population surveys of the noninstitutionalized elderly, as many as 10% report having had at least one adverse drug reaction in the last year. The incidence appears to be even greater in hospitalized elderly patients. Although it is widely believed that the elderly are more sensitive to drugs than the young, that is not true for all drugs. For example, a consistent decrease in sensitivity to drugs acting at the α -adrenergic receptor has been demonstrated in the elderly. The consequences of adverse drug effects may differ in the elderly because of their greater likelihood of other disease. For example, use of long-half-life benzodiazepines is linked to the occurrence of hip fractures in elderly patients, perhaps reflecting both a risk of falls from these drugs and the increased incidence of osteoporosis in elderly patients. Even when a drug impairs function similarly in patients of different age groups, the poorer baseline function in elderly persons may put them at greater risk for an adverse drug reaction. When prescribing for an elderly patient, the possibility that hepatic or renal mechanisms of drug excretion may be impaired should be taken into account. Adverse drug effects in the elderly may be subtle and, as in all populations, the physician must be alert to the possibility that a patient's signs and symptoms reflect an adverse effect of medication.

ETIOLOGY

Most adverse drug reactions are preventable, and recent studies using a systems analysis approach suggest that the most common system failure associated with an adverse drug reaction is the failure to disseminate knowledge about drugs to individuals involved in prescribing and administering them. Most adverse reactions can be classified into two groups. The most frequent ones result from an exaggeration of a predicted pharmacologic action of the drug. Other adverse reactions ensue from toxic effects unrelated to the intended pharmacologic actions. The latter effects are often unpredictable, are frequently severe, and result from recognized as well as undiscovered mechanisms. Some mechanisms unrelated to the drug's primary pharmacologic activity may include direct cytotoxicity, initiation of abnormal immune responses, and perturbation of metabolic processes in individuals with genetic enzymatic defects. Further understanding of interindividual differences in the expression of the enzymes responsible for drug metabolism has contributed to the understanding of adverse drug reactions that previously were thought to be idiosyncratic (see below). Prior consideration of the factors known to modify drug action often make it possible to prevent adverse reactions of this type.

Genetic Variations in Drug Oxidation by Cytochromes There is considerable interindividual variability in drug metabolism, resulting in variability in drug concentrations ([Chap. 70](#)). The majority of drugs are oxidized by cytochrome P450s (CYP) in the liver and gut. Some of these enzymes exhibit genetic polymorphisms resulting in enzymes with absent or reduced drug metabolizing activity, which may result in concentration-dependent toxicity. Conversely, where toxicity or pharmacologic effect is produced by a metabolite, individuals with low enzyme activity may have reduced drug effect whereas those with genetically determined increased enzyme activity will have increased drug effect. Examples of such polymorphically distributed oxidation enzymes include CYP2D6, CYP2C9, and CYP2C19.

The clinical consequences of the poor metabolizer phenotype are now becoming clearer and depend on the specific consequences of excessive drug concentrations. For example, the more potent (S) isomer of warfarin is metabolized by the polymorphically distributed enzyme [CYP2C9](#), resulting in lower (S) warfarin clearance and higher concentrations in both heterozygotes and homozygotes for the allelic variants associated with reduced enzyme activity. Recently, the clinical consequences of this polymorphism have been demonstrated in patients followed in an anticoagulant clinic. Patients who were stabilized on warfarin doses of 1.5 mg/d or less had an increased frequency of the genotypes associated with low warfarin metabolism compared to either community controls or patients requiring higher doses of warfarin ([Table 71-1](#)). In addition, the group stabilized on low-dose warfarin had a greater incidence of initial over-anticoagulation and hemorrhage than the group on the higher dose. This serves as an example of how genetic variations of a cytochrome P450 enzyme alter the response to a drug.

The oral hypoglycemic glipizide is also metabolized by [CYP2C9](#), and excessively low blood glucose concentrations occur after usual doses in genetic CYP2C9 poor metabolizers. Another oral hypoglycemic, phenformin, produced lactic acidosis in some patients. It is now recognized that phenformin is metabolized by another polymorphically distributed oxidative enzyme, CYP2D6. Patients who have genetically determined low

activity of CYP2D6 may be at particular risk from phenformin-induced lactic acidosis.

Genotypically determined variability in drug toxicity may also occur with drugs metabolized by enzymes other than cytochrome P450s. Such toxicity can be severe with the clinical use of the antimetabolites azathioprine and 6-mercaptopurine (to which it is converted in vivo). The cytotoxic thioguanine nucleotides produced in vivo are detoxified by further metabolism by xanthine oxidase and thiopurine methyltransferase (TPMT). The latter enzyme shows a trimodal distribution ([Fig. 71-1](#)). In children receiving mercaptopurine for treatment of leukemia, low activity of this enzyme is associated with excessive myelosuppression, whereas children with high TPMT levels have a poor antileukemic response. Azathioprine is currently used as a disease-modifying agent in the treatment of rheumatoid arthritis. Individuals with mutant TPMT alleles that result in impaired metabolism developed toxicity rapidly after beginning azathioprine and were uniformly unable to take the drug chronically. Thus the genotypic basis for drug toxicity is beginning to emerge.

Pharmacokinetic Bases for Adverse Reactions *An abnormally high drug concentration at the receptor site* (site of action) owing to pharmacokinetic variability is the usual cause of these reactions ([Chap. 70](#)). For example, a reduction in the volume of distribution, in the rate of metabolism, or in the rate of excretion all result in higher than expected concentration of drug at the receptor site, with a consequent increase in the pharmacologic effect.

Alteration in the dose-response curve due to increased receptor sensitivity results in an increase in drug effect at a given drug concentration. An example is the excessive response of elderly persons to the anticoagulant warfarin at normal or lower than normal blood levels. Such alterations in the dose-response curve may reflect altered drug sensitivity due to receptor polymorphisms, which are now being recognized. One such example is the prolonged QT syndrome, which has both a genetic basis, in individuals with abnormal potassium channels involved in cardiac repolarization, and a pharmacologic basis in individuals who receive drugs known to prolong the QT interval. Such individuals may develop torsade de pointes ([Chap. 230](#)). A large number of drugs have now been identified that can produce this potentially lethal effect ([Table 71-2](#); also <http://www.dml.georgetown.edu/depts/pharmacology/torsades.html>).

The shape of the dose-response curve also determines the likelihood of adverse drug reactions. Drugs with a steep dose-response curve or a narrow therapeutic index ([Chap. 70](#)) are more likely to cause dose-related toxicity because a small increase in dose produces a large change in pharmacologic effect. An increase in the dose of drugs that exhibit nonlinear kinetics, such as phenytoin ([Chap. 70](#)), may produce a proportionately greater increase in the blood level, resulting in toxicity.

Concurrent administration of other drugs may affect pharmacokinetics or pharmacodynamics. Pharmacokinetics may be affected by alterations in bioavailability, protein binding, or the rate of metabolism or excretion. Pharmacodynamics may be altered by another drug that competes for the same receptor sites, that prevents the drug from reaching its site of action, or that antagonizes or enhances the drug's pharmacologic effect. Inhibition of the metabolism of one drug by another may occur when both drugs bind to the same [CYP](#). Therefore, as the specific CYPs responsible for

the metabolism of individual drugs become known, prediction of drug interactions is put on a more rational scientific basis. An important example of such a mechanism is the inhibition of terfenadine's metabolism by inhibitors of CYP3A, such as erythromycin and systemic antimycotics. Such inhibition has resulted in torsade de pointes and lethal cardiac arrhythmia ([Chap. 70](#)).

TOXICITY UNRELATED TO A DRUG'S PRIMARY PHARMACOLOGIC ACTIVITY

Cytotoxic Reactions The understanding of so-called idiosyncratic reactions has greatly improved with the recognition that many of them are due to irreversible binding of a drug or its metabolites to tissue macromolecules by covalent bonds. Some chemical carcinogens, such as the alkylating agents, combine directly with DNA. Usually, it is only after metabolic activation of a drug to reactive metabolites that covalent binding occurs. This activation usually occurs in the microsomal mixed-function oxidase system, the hepatic enzyme system responsible for the metabolism of many drugs ([Chap. 70](#)). During the course of drug metabolism, reactive metabolites may covalently bind to tissue macromolecules, causing tissue damage. Because of the reactive nature of these metabolites, covalent binding often occurs close to the site of production. Typically that is the liver, but the mixed-function oxidase system is found in other tissues as well.

An example of this type of adverse drug reaction is the hepatotoxicity associated with isoniazid. This drug is metabolized principally by acetylation to acetylisoniazid, which is then hydrolyzed to acetylhydrazine. The further metabolism of acetylhydrazine by the mixed-function oxidase system liberates reactive metabolites that covalently bind to hepatic macromolecules, causing hepatic necrosis. The administration of drugs known to increase the activity of the mixed-function oxidase system, such as phenobarbital or rifampin, together with isoniazid results in the production of increased amounts of reactive metabolites, increased covalent binding, and a greater risk of hepatic damage.

The hepatic necrosis produced by overdosage of acetaminophen is also caused by reactive metabolites. Normally these metabolites are detoxified by combining with hepatic glutathione. When glutathione becomes exhausted, the metabolites bind instead to hepatic protein, with resultant hepatocyte damage. The hepatic necrosis produced by the ingestion of acetaminophen can be prevented, or at least attenuated, by the administration of substances such as *N*-acetylcysteine that reduce the binding of electrophilic metabolites to hepatic proteins. The risk of hepatic necrosis is increased in patients receiving drugs such as phenobarbital that increase the rate of drug metabolism and the rate of production of toxic metabolite(s).

It is likely, though as yet not proved, that other idiosyncratic reactions are caused by the covalent binding of reactive metabolites to tissue macromolecules, resulting either in direct cytotoxicity or in the initiation of an immune response.

Immunologic Mechanisms Most pharmacologic agents are poor immunogens because they are small molecules with molecular weights of less than 2000. Stimulation of antibody synthesis or sensitization of lymphocytes by a drug or one of its metabolites usually requires in vivo activation and covalent linkage to protein, carbohydrate, or nucleic acid.

Drug stimulation of antibody production may mediate tissue injury by one of several mechanisms. The antibody may attack the drug when the drug is covalently attached to a cell, and thereby destroy the cell. This mechanism occurs in penicillin-induced hemolytic anemia. Antibody-drug-antigen complexes may be passively adsorbed by a bystander cell, which is then destroyed by activation of complement; this occurs in quinine- and quinidine-induced thrombocytopenia. Drugs or their reactive metabolites may alter a host tissue, rendering it antigenic and eliciting autoantibodies. For example, hydralazine and procainamide can chemically alter nuclear material, stimulating the formation of anti-nuclear antibodies and occasionally causing lupus erythematosus. Autoantibodies can be elicited by drugs that neither interact with the host antigen nor have any chemical similarity to the host tissue; for example, α-methyldopa frequently stimulates the formation of antibodies to host erythrocytes, yet the drug neither attaches to the erythrocyte nor shares any chemical similarities with the antigenic determinants on the erythrocyte.

Drug-induced pure red cell aplasia ([Chap. 109](#)) is due to an immune-based drug reaction. Red cell formation in bone marrow cultures can be inhibited by phenytoin and purified IgG obtained from a patient with pure red cell aplasia associated with phenytoin.

Serum sickness ([Chap. 310](#)) results from the deposition of circulating drug-antibody complexes on endothelial surfaces. Complement activation occurs, chemotactic factors are generated locally, and an inflammatory response develops at the site of complex entrapment. Arthralgias, urticaria, lymphadenopathy, glomerulonephritis, or cerebritis may result. Penicillin is the most common cause of serum sickness today. Many drugs, particularly antimicrobial agents, induce production of IgE, which binds to mast cell membranes. Contact with a drug antigen initiates a series of biochemical events in the mast cell and results in the release of mediators that can produce the urticaria, wheezing, flushing, rhinorrhea, and (occasionally) hypotension characteristic of anaphylaxis.

Drugs may also excite cell-mediated immune responses. Topically administered substances may interact with sulfhydryl or amino groups in the skin and react with sensitized lymphocytes to produce the rash characteristic of contact dermatitis. Other types of rashes may also result from the interaction of serum factors, drugs, and sensitized lymphocytes. The role of drug-activated lymphocytes in the immune mechanisms governing destruction of visceral tissue is unknown.

Toxicity Associated with Genetically Determined Enzymatic Defects In the porphyrias, drugs that increase the activity of enzymes proximal to the deficient enzyme in the biosynthetic pathway of porphyrins can increase the quantity of porphyrin precursors that accumulate proximal to the deficient enzyme ([Chap. 346](#)). These drugs are listed in [Table 71-1](#).

Patients with a deficiency of glucose-6-phosphate dehydrogenase (G6PD) develop hemolytic anemia in response to primaquine and a number of other drugs ([Table 71-1](#)) that do not cause hemolysis in patients with adequate quantities of this enzyme ([Chap. 108](#)).

DIAGNOSIS

The manifestations of drug-induced diseases frequently resemble those of other diseases, and a given set of manifestations may be produced by different and dissimilar drugs. Recognition of the role of a drug or drugs in an illness depends on appreciation of the possible adverse reactions to drugs in any disease, on identification of the temporal relationship between drug administration and development of the illness, and on familiarity with the common manifestations of the drugs. Many associations between particular drugs and specific reactions have been described, but there is always a "first time" for a novel association, and any drug should be suspected of causing an adverse effect if the clinical setting is appropriate.

Illness related to a drug's pharmacologic action is often more easily recognized than illness attributable to immune or other mechanisms. For example, side effects such as cardiac arrhythmias in patients receiving digitalis, hypoglycemia in patients given insulin, and bleeding in patients receiving anticoagulants are more readily related to a specific drug than are symptoms such as fever or rash, which may be caused by many drugs or by other factors.

Once an adverse reaction is suspected, discontinuance of the suspected drug followed by disappearance of the reaction is presumptive evidence of a drug-induced illness. Confirming evidence may be sought by cautiously reintroducing the drug and seeing if the reaction reappears. However, that should be done only if confirmation would be useful in the future management of the patient and if the attempt would not entail undue risk. With concentration-dependent adverse reactions, lowering the dosage may cause the reaction to disappear, and raising it may cause the reaction to reappear. When the reaction is thought to be allergic, however, readministration of the drug may be hazardous, since anaphylaxis may develop. Readministration is unwise under these conditions unless no alternative drugs are available and treatment is necessary.

If the patient is receiving many drugs when an adverse reaction is suspected, the drugs likeliest to be responsible can usually be identified. All drugs may be discontinued at once, or, if that is not practical, they should be discontinued one at a time, starting with the one that is most suspect, and the patient observed for signs of improvement. The time needed for a concentration-dependent adverse effect to disappear depends on the time required for the concentration to fall below the range associated with the adverse effect, and that, in turn, depends on the initial blood level and on the rate of elimination or metabolism of the drug. Adverse effects of drugs with long half-lives, such as phenobarbital, take a considerable time to disappear.

Drugs recognized as producing a number of reactions are listed in [Table 71-1](#). This table includes both well-documented and some less well-documented reactions, focusing on those that are sufficiently important to require consideration. This information should be used to suggest the drug likely to be causing a reaction; the absence of a drug from the table does not mean that it cannot be responsible for the reaction, however.

Serum antibody has been demonstrated in some persons with drug allergies involving cellular blood elements, as in agranulocytosis, hemolytic anemia, and thrombocytopenia. For example, both quinine and quinidine can produce platelet agglutination in vitro in the presence of complement and the serum from a patient who

has developed thrombocytopenia following use of this drug.

Eliciting a drug history from patients is important for diagnosis. Attention must be directed to [OTC](#) drugs and herbal preparations as well as to prescription drugs. Each type can be responsible for adverse drug effects, and adverse interactions may occur between OTC drugs and prescribed drugs. In addition, it is common for patients to be cared for by several physicians, and duplicative, additive, counteractive, or synergistic drug combinations may therefore be administered if the physicians are not aware of the patients' drug histories. Every physician should determine what drugs a patient has been taking, at least during the preceding 30 days, before prescribing any medications. A frequently overlooked source of additional drug exposure is topical therapy; for example, a patient complaining of bronchospasm may not mention that an ophthalmic beta blocker is being used unless specifically asked. A history of previous adverse drug effects in patients is common. Since these patients have shown a predisposition to drug-induced illnesses, such a history should dictate added caution in prescribing drugs.

Patients with biochemical abnormalities such as erythrocyte [G6PD](#) deficiency can be identified. Most patients with the G6PD defect are of African or Mediterranean descent. Drug-induced hemolytic crisis can be avoided by testing for the enzyme defect before administering drugs that could cause the reaction. Similarly, persons with an abnormal serum pseudocholinesterase level may have abnormally prolonged apnea when given succinylcholine.

GENERAL COMMENTS

No drug is completely without side effects, and a side effect in one patient may be the desired pharmacologic effect in another. Current drug regulations allow physicians to have considerable confidence in the purity, bioavailability, and effectiveness of the drugs they prescribe. However, physicians have to weigh potential toxicity against possible benefits. Toxicity that would be acceptable for an effective antineoplastic agent would not be permitted in an oral contraceptive, for example. Because of the necessarily small number of patients treated in premarketing studies, rare adverse reactions may not be identified, so the first responsibility for identifying and reporting these effects must rest with the practicing clinician through the use of the various national adverse reaction reporting systems, such as those operated by the Food and Drug Administration in the United States and the Committee on Safety of Medicines in Great Britain. The publication of a newly recognized adverse reaction can in a short time stimulate many similar such reports of reactions that previously had gone unrecognized.

The prevention of adverse drug reactions first involves a high index of suspicion that the development of a new symptom or sign may be drug-related. Reduction of the dose or discontinuation of the suspected agent usually clarifies the issue in concentration-dependent toxic reactions. Physicians should be familiar with the common adverse effects of the drugs they use and, when in doubt, should consult the literature.

(Bibliography omitted in Palm version)

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72. PHYSIOLOGY AND PHARMACOLOGY OF THE AUTONOMIC NERVOUS SYSTEM - *Lewis Landsberg, James B. Young*

FUNCTIONAL ORGANIZATION OF THE AUTONOMIC NERVOUS SYSTEM

The autonomic nervous system innervates vascular and visceral smooth muscle, exocrine and endocrine glands, and parenchymal cells throughout the various organ systems. Functioning below the conscious level, the autonomic nervous system responds rapidly and continuously to perturbations that threaten the constancy of the internal environment. The many functions governed by this system include the distribution of blood flow and the maintenance of tissue perfusion, the regulation of blood pressure, the regulation of the volume and composition of the extracellular fluid, the expenditure of metabolic energy and supply of substrate, and the control of visceral smooth muscle and glands.

ANATOMIC ORGANIZATION

The autonomic neurons, located in ganglia outside the central nervous system (CNS), give rise to the postganglionic autonomic nerves that innervate organs and tissues throughout the body ([Fig. 72-1](#)). The activity of autonomic nerves is regulated by central neurons responsive to diverse afferent inputs. After central integration of afferent information, autonomic outflow is adjusted to permit the functioning of the major organ systems in accordance with the needs of the organism as a whole. Connections between the cerebral cortex and the autonomic centers in the brainstem coordinate autonomic outflow with higher mental functions.

The Sympathetic and Parasympathetic Divisions The preganglionic neurons of the parasympathetic nervous system leave the [CNS](#) in the third, seventh, ninth, and tenth cranial nerves and in the second and third sacral nerves, while the preganglionic neurons of the sympathetic nervous system exit the spinal cord between the first thoracic and the second lumbar segments ([Fig. 72-1](#)). Responses to sympathetic and parasympathetic stimulation are frequently antagonistic, as exemplified by their opposing effects on heart rate and gut motility. This antagonism reflects highly coordinated interactions within the CNS; the resultant changes in parasympathetic and sympathetic activity, often reciprocal, provide more precise control of autonomic responses than could be achieved by the modulation of a single system. Moreover, both sympathetic and parasympathetic portions of the autonomic nervous system are composed of multiple function-specific subdivisions. Neurons with the various subdivisions differ neurochemically and neurophysiologically and are controlled by distinct regions within the CNS. This specialization within sympathetic and parasympathetic divisions contributes to the precision and specificity of autonomic regulation.

Neurotransmitters *Acetylcholine* (ACh) is the preganglionic neurotransmitter for both divisions of the autonomic nervous system as well as the postganglionic neurotransmitter of the parasympathetic neurons. Nerves that release ACh are said to be cholinergic. *Norepinephrine* (NE) is the neurotransmitter of the postganglionic sympathetic neurons; these nerves are said to be adrenergic. Within the sympathetic outflow, postganglionic neurons innervating the eccrine sweat glands (and perhaps

some blood vessels supplying skeletal muscle) are of the cholinergic type.

THE SYMPATHETIC NERVOUS SYSTEM AND ADRENAL MEDULLA

CATECHOLAMINES

All three of the naturally occurring catecholamines, [NE](#), *epinephrine* (E), and *dopamine*, function as neurotransmitters within the [CNS](#). NE, the neurotransmitter of postganglionic sympathetic nerve endings, exerts its effects locally, in the immediate vicinity of its release. Epinephrine, the circulating hormone of the adrenal medulla, influences processes throughout the body. A peripheral dopaminergic system also exists but has not been characterized in detail.

Biosynthesis (Fig. 72-2) Catecholamines are synthesized from the amino acid tyrosine, which is sequentially hydroxylated to form dihydroxyphenylalanine (dopa), decarboxylated to form dopamine, and hydroxylated on the β position of the side chain to form [NE](#). The initial step, the hydroxylation of tyrosine, is rate-limiting and is regulated so that synthesis of dopa is coupled to NE release. This regulation is achieved by alterations in both the activity and the amount of tyrosine hydroxylase. In the adrenal medulla and in those central neurons utilizing epinephrine as neurotransmitter, NE is *N*-methylated to epinephrine by the enzyme phenylethanolamine-*N*-methyltransferase (PNMT).

Catecholamine Metabolism The major metabolic transformations of catecholamines involve *O*-methylation at the meta-hydroxyl group and oxidative deamination. *O*-Methylation is catalyzed by the enzyme catechol-*O*-methyltransferase (COMT), and oxidative deamination is promoted by monoamine oxidase (MAO). COMT in liver and kidney is important in the metabolism of circulating catecholamines. MAO, a mitochondrial enzyme present in most tissues, including nerve endings, has a lesser role in the metabolism of circulating catecholamines but is important in regulating the catecholamine stores within the peripheral sympathetic nerve endings. The metanephrines and 3-methoxy-4-hydroxymandelic acid (vanilmandelic acid, VMA) are the major end products of E and [NE](#) metabolism. Homovanillic acid (HVA) is the end product of dopamine metabolism.

STORAGE AND RELEASE OF CATECHOLAMINES

In both the adrenal medulla and sympathetic nerve endings catecholamines are stored in subcellular vesicles and released by exocytosis. The large stores of catecholamines in these tissues provide an important physiologic reserve that maintains an adequate supply of catecholamines in the face of intense stimulation. A variety of substances may be stored along with catecholamines in sympathetic nerve endings and adrenal medulla and released with catecholamines during exocytosis. These substances, which may function as cotransmitters or neuromodulators, include peptides such as neuropeptide Y, substance P, and enkephalins; purines such as ATP and adenosine; and other amines such as serotonin. At the neuroeffector junction, coreleased neuromodulators modify the response to [NE](#), while cotransmitters exert physiologic effects independent of those induced by NE.

Adrenal Medulla The adrenal medullary chromaffin tissue in a pair of normal human adrenal glands weighs about 1 g and contains approximately 6 mg catecholamines, 85% of which is epinephrine.

Catecholamine secretion, stimulated by [ACh](#) from the preganglionic sympathetic nerves, occurs after calcium influx triggers fusion of the chromaffin granule membrane and cell membrane; obliteration of the cell membrane at the point of fusion and extrusion of the entire soluble contents of the granule into the extracellular space complete the process of exocytosis ([Fig. 72-2](#)). Although the molecular mechanisms involved in the exocytotic process are only partially understood, evidence has accumulated that specific calcium-binding proteins are involved. Once bound, calcium induces a conformational change in these proteins that induces fusion of granules and docking of granules at the cell membrane.

Peripheral Sympathetic Nerve Endings The peripheral sympathetic nerve endings form a reticulum or ground plexus that brings the terminal fibers into close contact with effector cells. All the [NE](#) in peripheral tissues is in the sympathetic nerve endings, and heavily innervated tissues contain as much as 1 to 2 ug/g of tissue. NE stored in the nerve endings is in discrete subcellular particles analogous to the adrenal medullary chromaffin granules. [MAO](#) in the mitochondria of the nerve endings plays an important role in regulating the local concentration of NE ([Fig. 72-2](#)). Amines in storage vesicles are protected from oxidative deamination; amines within the cytoplasm, however, are deaminated to inactive metabolites. Release from the nerve ending occurs in response to action potentials propagated in terminal sympathetic fibers.

THE PERIPHERAL ADRENERGIC NEUROEFFECTOR JUNCTION

The peripheral sympathetic nerve endings possess an amine transport system that actively takes up amines from the extracellular fluid. Neuronal uptake or recapture of locally released [NE](#) terminates the action of the transmitter and contributes to the constancy of the NE stores.

A variety of factors alter the relationship between neuronal impulse traffic and [NE](#) release. Diminished temperature and acidosis, for example, both decrease the amount of NE released in response to sympathetic impulses. Several chemical mediators operate at the peripheral sympathetic nerve ending (referred to as *prejunctional* or *presynaptic sites*) to modify sympathetic neurotransmission by influencing the amount of NE released in response to nerve impulses. Prejunctional modulation may be either inhibitory or facilitatory. Certain modulators, such as catecholamines and [ACh](#), may either inhibit or facilitate NE release, antagonistic effects that are mediated by different adrenergic or cholinergic receptors, respectively. Those compounds exerting an *inhibitory* effect on NE release at the prejunctional nerve ending include the following: catecholamines (α_2 receptor), ACh (muscarinic receptor), dopamine (D_2 receptor), histamine (H_2 receptor), serotonin, adenosine, enkephalins, and prostaglandins.

Catecholamines reduce [NE](#) release via prejunctional α_2 receptors in a classic negative-feedback system. Feedback regulation is complicated by the fact that β -receptor activation facilitates NE release.

Though both inhibitory and facilitatory effects of [ACh](#) on [NE](#) release have been described, the inhibitory effect of ACh, mediated by the muscarinic cholinergic receptor, occurs at lower ACh concentrations and is probably of greater physiologic significance.

CENTRAL REGULATION OF SYMPATHOADRENAL OUTFLOW

Brainstem Sympathetic Centers Sympathetic outflow is initiated from the reticular formation of the medulla oblongata and pons and from centers in the hypothalamus. The rostral ventral portion of the medulla, particularly the area designated the rostral ventrolateral medulla (RVLM), appears to contain especially important sympathoexcitatory areas. Descending fibers originating from all these centers synapse in the intermediolateral cell column of the spinal cord with the preganglionic sympathetic neurons. Changes in the physical and chemical properties of the extracellular fluid, including the circulating levels of hormones and substrates, also affect sympathetic nervous system outflow. The area postrema, in the floor of the fourth ventricle, along with other circumventricular organs lie outside the blood-brain barrier and may play an important role in this regard. Although the hallmark of intense sympathoadrenal stimulation is a global response (the fight-or-flight reaction of Cannon), discrete changes in sympathetic outflow to different organ systems continuously regulate many autonomic functions.

Relationship Between the Sympathetic Nervous System and the Adrenal Medulla

Sympathetic nervous system activity and adrenal medullary secretion are coordinated but not always congruent. During periods of intense sympathetic stimulation, such as cold exposure and exhaustive exercise, the adrenal medulla is progressively recruited, and circulating epinephrine reinforces the physiologic effects of sympathetic stimulation. In other situations, the sympathetic nervous system and the adrenal medulla are stimulated independently. The response to upright posture, for example, involves predominantly the sympathetic nervous system, while hypoglycemia stimulates only the adrenal medulla.

Sympathetic Regulation of the Cardiovascular System Stretch receptors in the systemic and pulmonary arteries and veins continuously monitor intravascular pressures; the resulting afferent impulses, after relay and integration in the brainstem, alter sympathetic activity in defense of blood pressure and blood flow to critical areas ([Fig. 72-3](#)).

Arterial Baroreceptors An increase in blood pressure stimulates receptors in the carotid sinus and aortic arch. The ensuing afferent impulses, after relay within the nucleus of the solitary tract (NTS) in the brainstem, suppress the brainstem sympathetic centers ([Fig. 72-3](#)). This baroreceptor reflex arc forms a negative-feedback loop in which a rise in arterial pressure results in the inhibition of central sympathetic outflow. A brainstem noradrenergic pathway interacts with the NTS to participate in suppression of sympathetic outflow. This noradrenergic inhibitory pathway is stimulated by centrally acting adrenergic agonists and may be involved in the action of certain antihypertensive drugs, such as clonidine, that potentiate the baroreceptor-mediated vasodepressor response ([Chap. 246](#)). In the opposite manner, when the blood pressure falls, decreased afferent impulses diminish central inhibition, resulting in an increase in

sympathetic outflow and a rise in arterial pressure.

Central Venous Pressure Receptors in the walls of the great veins and within the atria are also involved in the regulation of sympathetic outflow. Stimulation of these receptors by high venous pressure suppresses the brainstem sympathetic centers; when central venous pressure is low, sympathetic outflow increases. The central connections are poorly understood, but the afferent impulses are carried in the vagus ([Fig. 72-3](#)).

ASSESSMENT OF SYMPATHOADRENAL ACTIVITY

The clinical assessment of sympathoadrenal activity involves the measurement of catecholamines in plasma and of catecholamines and catecholamine metabolites in urine, and the assessment of sympathetic nerve impulse traffic by microneurography. Microneurography, utilizing microelectrodes implanted in nerves supplying skeletal muscle (such as the peroneal nerve), is primarily a research tool. Quantitation of urinary catecholamines and metabolites is useful in the diagnosis of pheochromocytoma ([Chap. 332](#)).

Plasma Catecholamines Catecholamines in human plasma may be measured by radioenzymatic isotope derivative techniques or by high-performance liquid chromatography in conjunction with electrochemical detection. Plasma catecholamine measurements provide an index of sympathetic nervous system and adrenal medullary activity and have been widely used to assess sympathoadrenal activity in clinical investigation in human subjects. The usefulness of plasma catecholamine measurements, however, is compromised by factors that alter the relationship between the plasma concentration of catecholamines and the functional state of the sympathoadrenal system, and also by important regional differences in sympathetic outflow. Techniques utilizing tracer infusions of tritiated [NE](#), which correct for changes in NE clearance when applied across a particular anatomic region, estimate regional sympathetic outflow with some precision and have helped to define differentiated sympathetic nervous system activity in the investigational setting. The clinical usefulness of plasma catecholamine levels remains limited to the evaluation of patients with autonomic insufficiency and, on occasion, patients with suspected pheochromocytoma ([Chap. 332](#)).

Basal plasma [NE](#) concentrations are in the range of 0.09 to 1.8 nmol/L (150 to 350 pg/mL); basal E levels are about 135 to 270 pmol/L (25 to 50 pg/mL). The half-time of disappearance of NE from the circulation is approximately 2 min. The plasma NE level is markedly affected by a variety of factors, including posture; accordingly, the conditions under which blood is obtained for assay must be controlled. By convention, basal plasma NE levels are those obtained through an indwelling intravenous line after the patient has rested supine in a relaxed environment for 30 min.

Plasma NE response to upright posture The predictable increase in circulating NE concentration during upright posture provides a convenient test of sympathetic nervous system function. Five minutes of quiet standing results in a two- to threefold increase in plasma NE level. A normal response requires an intact afferent system, appropriate [CNS](#) relays, and an intact peripheral sympathetic nervous system; a defect of any of these components reduces the increment in circulating NE.

Plasma E levels are also dependent on the physical and mental state of the subject. Change in plasma E with upright posture is usually small. Hypoglycemia, strenuous exercise, and various types of mental stress, however, can cause large increases in the plasma E level.

PERIPHERAL DOPAMINERGIC SYSTEM

In addition to its role as neurotransmitter in the [CNS](#), dopamine functions as an inhibitory transmitter in the carotid body and the sympathetic ganglia. A distinct peripheral dopaminergic system is also believed to exist. Dopamine elicits a variety of responses not attributable to stimulation of classic adrenergic receptors; it relaxes the lower esophageal sphincter, delays gastric emptying, causes vasodilation in the renal and mesenteric arterial circulation, suppresses aldosterone secretion, directly stimulates renal sodium excretion, and suppresses [NE](#) release at sympathetic nerve terminals by a presynaptic inhibitory mechanism. The mediation of these dopaminergic effects in vivo is poorly understood. Dopamine does not appear to be a circulating hormone.

ADRENERGIC RECEPTORS

Catecholamines influence effector cells by interacting with specific surface *receptors* coupled to G proteins. Two major categories of response to catecholamines reflect the activation of two populations of adrenergic receptors, designated α and β . Both α and β receptors have been further divided into subtypes that serve different functions and are susceptible to differential stimulation and blockade.

α -ADRENERGIC RECEPTORS

α -Adrenergic receptors mediate vasoconstriction, intestinal relaxation, and pupillary dilatation. Epinephrine and [NE](#) are approximately equipotent as α -receptor agonists. Distinct α_1 - and α_2 -receptor subtypes are also recognized. Originally the postsynaptic or postjunctional α -adrenergic receptors on effector cells were designated α_1 , while the prejunctional α -adrenergic receptors on the sympathetic nerve endings were designated α_2 . It is now recognized that nonneuronal (postsynaptic) processes are mediated by the α_2 receptor as well. The α_1 receptor mediates the classic α effects, including vasoconstriction; phenylephrine and methoxamine are selective α_1 agonists, and prazosin is a selective α_1 antagonist. The α_2 receptor mediates presynaptic inhibition of NE release from adrenergic nerves and other responses, including inhibition of [ACh](#) release from cholinergic nerves, inhibition of lipolysis in adipocytes, inhibition of insulin secretion, stimulation of platelet aggregation, and vasoconstriction in some vascular beds. Specific α_2 agonists include clonidine and α -methylnorepinephrine; these agents, the latter derived from α -methyldopa in vivo, exert an antihypertensive effect by interacting with α_2 receptors within the brainstem sympathetic centers that regulate blood pressure. Yohimbine is a specific α_2 antagonist.

β -ADRENERGIC RECEPTORS

Physiologic events associated with β -adrenergic receptor responses include stimulation of heart rate and contractility, vasodilation, bronchodilation, and lipolysis. β -Receptor

responses can also be divided into two types. The β_1 receptor responds equally to E and NE and mediates cardiac stimulation and lipolysis. The β_2 receptor is more responsive to E than to NE and mediates responses such as vasodilation and bronchodilation. Isoproterenol stimulates and propranolol blocks both β_1 and β_2 receptors. Other agonists and antagonists that have partial selectivity for the β_1 or β_2 receptors have been used therapeutically where the desired response involves predominantly one of the two subtypes.

Both pharmacologic and molecular genetic studies have demonstrated an additional distinct β_3 -adrenergic receptor that subserves lipolysis in white and brown adipose tissue as well as the stimulation of heat production in brown adipose tissue. The human β_3 -adrenergic receptor has been cloned, and a distinct polymorphism noted that may, in some populations, be associated with weight gain, insulin resistance, and type 2 diabetes mellitus. The β_3 -adrenergic receptor has a much greater affinity for NE than E and, unlike the β_1 and β_2 receptors, does not undergo desensitization. Synthetic agonists for the β_3 receptor, currently under development, have a potential role in the treatment of obesity by increasing metabolic rate.

DOPAMINERGIC RECEPTORS

Specific dopaminergic receptors, distinct from the classic α - and β -adrenergic receptors, are present in the CNS and peripheral nervous system and in several nonneural tissues. Two types of dopaminergic receptors serve different functions and have different second messengers. Dopamine is a potent agonist of both types of receptors; the action of dopamine is antagonized by phenothiazines and thioxanthenes. The D_1 receptor mediates vasodilation in the renal, mesenteric, coronary, and cerebral vascular beds. Fenoldopam is an agonist selective for the D_1 receptor. The D_2 receptor inhibits transmission in the sympathetic ganglia, inhibits NE release from sympathetic nerve endings by an effect on the presynaptic membrane (Fig. 72-2), inhibits prolactin release from the pituitary, and causes vomiting. Selective agonists of the D_2 receptor include bromocriptine, cabergoline, and apomorphine, while butyrophenones such as haloperidol (active within the CNS), domperidone (does not cross blood-brain barrier readily), and the benzamide sulpiride are relatively selective D_2 antagonists.

STRUCTURE AND FUNCTION OF ADRENERGIC RECEPTORS

Adrenergic receptors belong to a superfamily of related membrane proteins, including the visual protein rhodopsin and the muscarinic acetylcholine receptors, that interacts with G proteins. These proteins share significant sequence homologies and, as deduced from the properties of the constituent amino acids, a similar topographic structure in the cell membrane. The postulated structure of this family of receptor proteins is shown schematically in Fig. 72-4. The characteristic features include seven membrane-spanning hydrophobic domains containing 20 to 28 amino acids each. The membrane-spanning domains, particularly M-7 (Fig. 72-4), appear to be important in determining the characteristic agonist binding.

Coupling of Receptor Occupancy with Cellular Response The major mediators of adrenergic (as well as many other) cellular responses are a family of regulatory proteins termed *G proteins* that, when activated, bind the nucleotide guanosine triphosphate

(GTP). The best-characterized G proteins are those that stimulate or inhibit adenylyl cyclase, designated G_s or G_i , respectively (Fig. 72-5). The β_1 , β_2 , and D_1 receptors are coupled to G_s ; receptor occupancy is therefore associated with stimulation of adenylyl cyclase and results in an increase in intracellular cyclic adenosine monophosphate (AMP), which in turn results in activation of protein kinase A and other cyclic AMP-dependent protein kinases. The resultant protein phosphorylation alters the activity of enzymes and the function of other proteins, culminating in a cellular response that is characteristic of the tissue being stimulated. The α_2 , M_2 subtype of the muscarinic acetylcholine receptor and the D_2 receptor are coupled to G_i , resulting in diminished adenylyl cyclase activity and a fall in cyclic AMP. The subsequent alterations in enzyme activity and function of other proteins produce an alternate, frequently opposite, series of cellular responses. Although many α_2 responses can be explained by inhibition of adenylyl cyclase, other mechanisms may be involved as well.

The α_1 -adrenergic receptor (as well as the M_1 subtype of the acetylcholine receptor) is coupled to a different G protein that activates phospholipase C; this G protein has not been as well characterized but is sometimes designated G_q . Receptor occupancy in this system stimulates phospholipase C, which catalyzes the breakdown of membrane-bound phospholipids, particularly phosphatidylinositol-4,5-bisphosphate (PIP_2) with the production of inositol-1,4,5-trisphosphate (IP_3) and 1,2-diacylglycerol (DAG), both of which act as second messengers (Fig. 72-5). IP_3 rapidly mobilizes calcium from intracellular stores within the endoplasmic reticulum, producing an increase in free cytoplasmic calcium which by itself and via calcium-calmodulin-dependent protein kinases influences cellular processes appropriate to the stimulated cell. The transient rise in calcium induced by IP_3 from the intracellular stores is reinforced in the presence of continued agonist stimulation by alterations in membrane calcium flux that result eventually in net calcium uptake from the extracellular fluid by mechanisms that have been incompletely defined.

DAG, the other second messenger produced by the action of phospholipase C on PIP_2 (as well as other membrane phospholipids), remains associated with the cell membrane and activates protein kinase C, which has different substrates than the calcium-calmodulin kinases stimulated by IP_3 . Protein phosphorylation stimulated by protein kinase C contributes to the tissue-specific response in ways that remain poorly understood. Increases in intracellular calcium also potentiate the activation of protein kinase C (Fig. 72-5).

REGULATION OF ADRENERGIC RECEPTORS

Prolonged exposure to α - or β -adrenergic agonists decreases the number of corresponding adrenergic receptors on effector cells. Although the biochemical mechanisms involved are obscure, internalization of the β -adrenergic receptor within the cell occurs during agonist exposure in some systems, suggesting that internal translocation contributes to the decrease in receptor number under these circumstances.

Alteration in agonist concentration may also affect the affinity of the receptor for the agonist. Adrenergic receptors that utilize adenylyl cyclase for the second messenger (β receptors, α_2 receptors) exist in high- and low-affinity states; exposure to agonist

diminishes the proportion of receptors in the high-affinity state. Such alterations in adrenergic receptors induced by adrenergic agonists are termed *homologous regulation*. Agonist-induced alterations in adrenergic-receptor density and affinity are believed to contribute to the diminished physiologic response that occurs after prolonged exposure of an effector tissue to adrenergic agonist, a phenomenon known as *tachyphylaxis* or *desensitization*.

Adrenergic receptors are also influenced by factors other than adrenergic agonists, so-called *heterologous regulation*. Enhanced α -adrenergic-receptor affinity, for example, may underlie the potentiation of α -adrenergic responses that occur in response to lowered environmental temperatures. Thyroid hormones potentiate β -receptor responses by alterations in β -receptor number and in the efficiency of coupling receptor occupancy with physiologic response. Estrogen and progesterone alter the sensitivity of the myometrium to catecholamines by effects on α -adrenergic receptors. Glucocorticoids may influence adrenergic function by antagonizing agonist-induced decreases in adrenergic receptors, thereby counteracting tachyphylaxis in response to intense adrenergic stimulation.

Alterations in sensitivity to catecholamines also occur as a consequence of postreceptor changes, although the latter remain poorly characterized.

PHYSIOLOGY OF THE SYMPATHOADRENAL SYSTEM

Catecholamines influence all of the major organ systems. The effects take place in seconds and may occur in anticipation of physiologic requirement. An increase in sympathoadrenal activity prior to strenuous exercise, for example, lessens the impact of exercise on the internal environment.

DIRECT EFFECTS OF CATECHOLAMINES

Cardiovascular System Catecholamines stimulate vasoconstriction in the subcutaneous, mucosal, splanchnic, and renal vascular beds by α -receptor-mediated mechanisms. Although vasoconstriction was originally considered an α_1 -receptor response, vascular tone appears to be more complexly regulated and, in many areas, involves α_2 -mediated responses as well. The venous portion of the circulation, in particular, is endowed with α_2 receptors. Differential regulation of the two types of α receptors, under certain circumstances, contributes to an integrated physiologic response. Since vasoconstriction in the coronary and cerebral circulations is minimal, flow to these areas is maintained during sympathetic stimulation. Skeletal muscle vasculature contains β receptors sensitive to low circulating levels of epinephrine so that skeletal muscle blood flow is augmented during adrenal medullary activation.

The effects of catecholamines on the heart are mediated by β_1 receptors and include increase in heart rate, enhancement of cardiac contractility, and increase in conduction velocity. The increase in myocardial contractility is illustrated by a leftward and upward shift of the ventricular function curve (see [Fig. 231-6](#)) that relates cardiac work to ventricular diastolic fiber length; at any initial fiber length, catecholamines increase cardiac work. Catecholamines also enhance cardiac output by stimulating venoconstriction, enhancing venous return, and increasing the force of atrial contraction,

thereby augmenting diastolic volume and hence fiber length. The acceleration of conduction in the junctional tissues results in a more synchronous, and hence more effective, ventricular contraction. Cardiac stimulation increases myocardial oxygen consumption, a major factor in the pathogenesis and treatment of myocardial ischemia.

Metabolism Catecholamines increase metabolic rate. In small mammals, mitochondrial respiration in brown adipose tissue is functionally uncoupled by [NE](#). In a reaction unique to brown adipose tissue, NE stimulates the β_3 -adrenergic receptor that activates a specific mitochondrial uncoupling protein that dissipates the proton gradient between the inner mitochondrial matrix and the cytoplasm, thereby uncoupling substrate utilization and ATP synthesis. In humans, a functional role for brown adipose tissue has not been established with certainty.

Substrate Mobilization In a variety of tissues, catecholamines stimulate the breakdown of stored fuel with the production of substrate for local consumption; glycogenolysis in the heart, for example, provides substrate for immediate metabolism by the myocardium. Catecholamines also accelerate fuel mobilization in liver, adipose tissue, and skeletal muscle, liberating substrates (glucose, free fatty acids, lactate) into the circulation for use throughout the body.

Fluids and Electrolytes By a direct action on the renal tubule, [NE](#) stimulates sodium reabsorption, thereby defending extracellular fluid volume. Dopamine, in contrast, promotes sodium excretion. NE and E also promote cellular uptake of potassium.

Viscera Catecholamines affect visceral function by actions on smooth muscle and glandular epithelium. Urinary bladder and intestinal smooth muscle are relaxed while the corresponding sphincters are stimulated. Gallbladder emptying also involves sympathetic mechanisms. Catecholamine-mediated smooth-muscle contraction in the female aids ovulation and ovum transport along the fallopian tubes, and in the male provides propulsive force for the seminal fluid during ejaculation. Inhibitory α_2 -receptors on cholinergic neurons within the gut contribute to intestinal relaxation. Catecholamines induce bronchodilation by β_2 -receptor mechanism.

INDIRECT EFFECTS OF CATECHOLAMINES

The ultimate physiologic response induced by catecholamines involves changes in hormone secretion and in blood flow distribution, both of which support and amplify the direct effects of catecholamines.

Endocrine System Catecholamines influence the secretion of renin, insulin, glucagon, calcitonin, parathormone, thyroxine, gastrin, erythropoietin, progesterone, and, possibly, testosterone. Secretion of each of these hormones is governed by complex feedback loops. With the exception of thyroxine and the gonadal steroids, each is a polypeptide not under the direct control of the pituitary gland. Sympathoadrenal input into the secretion of these hormones provides a mechanism for regulation by the [CNS](#) and ensures a coordinated hormonal response in accord with the homeostatic needs of the organism.

Renin (See also [Chap. 246](#)) Sympathetic stimulation increases renin release by a direct

b-receptor effect independent of vascular changes within the kidney. The renin response to volume depletion is sympathetically mediated and is initiated by a fall in central venous pressure. Since renin secretion activates the angiotensin-aldosterone system, angiotensin-induced vasoconstriction supports the direct effects of catecholamines on blood vessels, while aldosterone-mediated sodium reabsorption complements the direct increase in sodium reabsorption induced by sympathetic stimulation. b-receptor blocking agents suppress renin secretion.

Insulin and Glucagon Stimulation of pancreatic sympathetic nerves or an elevation in circulating catecholamines suppresses insulin and increases glucagon release. Inhibition of insulin secretion is mediated by the α_2 receptor, and stimulation of glucagon is mediated by the β receptor. This combination of effects supports substrate mobilization, reinforcing the direct effects of catecholamines on hepatic glucose output and lipolysis. Although α -receptor-mediated suppression of insulin release usually predominates, a b-receptor mechanism may augment insulin secretion under some circumstances.

SYMPATHOADRENAL FUNCTION IN SELECTED PHYSIOLOGIC AND PATHOPHYSIOLOGIC STATES

Support of the Circulation The sympathetic nervous system functions to maintain an adequate circulation. During upright posture and volume depletion, reduction of afferent venous and arterial baroreceptor impulse traffic diminishes an inhibitory input to the vasomotor center, thereby increasing sympathetic activity ([Fig. 72-3](#)) and reducing efferent vagal tone. As a result, heart rate is increased, and cardiac output is diverted from the skin, subcutaneous tissues, mucosa, and viscera. Sympathetic stimulation of the kidney increases sodium reabsorption, and sympathetically mediated venoconstriction enhances venous return ([Fig. 72-6](#)). With pronounced hypotension, the adrenal medulla is recruited and epinephrine reinforces the effects of the sympathetic nervous system.

The intense sympathoadrenal stimulation that accompanies severe volume depletion may contribute to the development of ketoacidosis in alcoholics as well as to the ketoacidosis sometimes seen in association with hyperemesis gravidarum. Catecholamine-mediated suppression of insulin and stimulation of glucagon markedly potentiate ketogenesis in these disease states. Volume resuscitation and provision of adequate glucose promptly reverse the ketoacidosis in most cases.

Congestive Heart Failure The sympathetic nervous system also provides circulatory support during congestive heart failure ([Chap. 232](#)). Venoconstriction and sympathetic stimulation of the heart increase cardiac output while peripheral vasoconstriction directs blood flow to the heart and brain. The afferent signals are less clear than in simple volume depletion because the venous pressure is usually elevated. In severe heart failure, depletion of cardiac [NE](#) may impair the effectiveness of sympathetic circulatory support. On the other hand, the possibility has been raised that intense sympathetic stimulation may further impair cardiac function, suggesting possible benefit from β -adrenergic blockade. The use of beta blockers in the treatment of congestive heart failure, in fact, has increased in recent years.

Trauma and Shock In acute traumatic injury or shock, adrenal catecholamines support the circulation and mobilize substrates. In the chronic, reparative phase following injury, catecholamines also contribute to substrate mobilization and to the elevation in metabolic rate.

Exercise Sympathetic activation during exercise increases cardiac output and ensures sufficient substrate to meet the increased metabolic needs. Central neural factors, such as anticipation, and circulatory factors, such as fall in venous pressure, trigger the sympathetic response. Mild degrees of exercise stimulate the sympathetic nervous system alone; during more severe exertion the adrenal medulla is activated as well. Cardiovascular conditioning is associated with a decrease in sympathetic nervous system activity both at rest and during exercise, in comparison with the untrained state.

Hypoglycemia (See also [Chap. 334](#)) Hypoglycemia causes a marked increase in adrenal medullary epinephrine secretion. When glucose concentrations fall below overnight fasting levels, regulatory glucose-sensitive neurons in the [CNS](#) initiate a prompt increase in adrenal medullary secretion. The increase is especially intense at plasma glucose levels below 2.8 mmol/L (50 mg/dL), when plasma E levels increase 25 to 50 times above baseline, thereby increasing hepatic glucose output, providing alternative substrate in the form of free fatty acids, suppressing endogenous insulin release, and inhibiting insulin-mediated glucose utilization in muscle. Many clinical manifestations of hypoglycemia, such as tachycardia, palpitations, nervousness, tremor, and widened pulse pressure, are secondary to increased E secretion, and these manifestations constitute an "early warning" system in insulin-requiring diabetics. In patients with long-standing diabetes mellitus, however, the E response to hypoglycemia may be diminished or absent, leaving affected patients at greater risk to develop severe hypoglycemia.

Cold Exposure The sympathetic nervous system plays a critical role in the maintenance of normal body temperature during exposure to a cold environment. Receptors in the skin and [CNS](#) respond to a fall in temperature by activating hypothalamic and brainstem centers that increase sympathetic activity. Sympathetic stimulation leads to vasoconstriction in the superficial vascular beds, thereby diminishing heat loss. The sympathetic response involves a complex interaction between lowered environmental temperatures and α_2 -adrenergic receptors. Acclimatization during chronic cold exposure increases the capacity for metabolic heat production in response to sympathetic stimulation.

Dietary Intake Fasting suppresses and overfeeding stimulates the sympathetic nervous system. The reduction in sympathetic activity during fasting or starvation contributes to the decrease in metabolic rate, bradycardia, and hypotension in these states. Enhanced sympathetic activity during periods of increased caloric intake contributes to the elevation in metabolic rate associated with a chronic increase in dietary intake.

Hypoxia Chronic hypoxia is associated with stimulation of the sympathoadrenal system, and some of the cardiovascular changes attendant to hypoxia are dependent on catecholamines.

Orthostatic Hypotension The maintenance of arterial pressure during upright posture

depends on an adequate blood volume, an unimpaired venous return, and an intact sympathetic nervous system. Significant postural hypotension, therefore, often reflects extracellular fluid volume depletion or dysfunction of the circulatory reflexes. Diseases of the nervous system, such as tabes dorsalis, syringomyelia, or diabetes mellitus, may disrupt these sympathetic reflexes with resultant orthostatic hypotension. Although any antiadrenergic agent may impair the postural sympathetic response, orthostatic hypotension is most prominent with drugs that block neurotransmission within the ganglia or adrenergic neurons.

The term *idiopathic orthostatic hypotension* refers to a group of degenerative diseases involving either the pre- or postganglionic sympathetic neurons ([Chaps. 21](#) and [365](#)).

Treatment of orthostatic hypotension is usually unsatisfactory except in the mildest cases. There is no way of reestablishing the normal relationship between fall in venous return and sympathetic neuronal activation. Volume expansion with fludrocortisone and a liberal salt diet in conjunction with fitted stockings to the waist, as well as elevation of the head of the bed to avoid recumbency, will maintain plasma volume and venous return and frequently provide symptomatic improvement.

PHARMACOLOGY OF THE SYMPATHOADRENAL SYSTEM

A variety of therapeutic agents affect sympathetic nervous system function or interact with adrenergic receptors, making it possible to stimulate or suppress effects mediated by catecholamines with some degree of specificity ([Table 72-1](#)).

SYMPATHOMIMETIC AMINES

Sympathomimetic amines may directly activate adrenergic receptors (direct acting) or release [NE](#) from the sympathetic nerve endings (indirect acting). Many agents have both direct and indirect effects.

Epinephrine and Norepinephrine The naturally occurring catecholamines act predominantly by the direct stimulation of adrenergic receptors. [NE](#) is employed to support the circulation and elevate the blood pressure in hypotensive states ([Chap. 38](#)). Peripheral vasoconstriction is the major effect, although cardiac stimulation occurs as well. Epinephrine, also employed as a pressor, has special usefulness in the treatment of allergic reactions, especially those associated with anaphylaxis. Epinephrine antagonizes the effects of histamine and other mediators on vascular and visceral smooth muscle and is useful in the treatment of bronchospasm.

Dopamine *Dopamine* is used in treating hypotension, shock ([Chap. 38](#)), and certain forms of heart failure ([Chap. 232](#)). At low infusion rates it exerts a positive inotropic effect both by a direct action on the cardiac β_1 receptors and by the indirect release of [NE](#) from sympathetic nerve endings in the heart. At low doses direct stimulation of dopaminergic receptors in the renal and mesenteric vasculature also results in vasodilation in the gut and kidney and facilitates sodium excretion. At higher infusion rates interaction with α -adrenergic receptors results in vasoconstriction, an increase in peripheral resistance, and an elevation of blood pressure.

b-Receptor Agonists *Isoproterenol*, a direct-acting b-receptor agonist, stimulates the heart, decreases peripheral resistance, and relaxes bronchial smooth muscle. It raises the cardiac output and accelerates atrioventricular conduction while increasing the automaticity of ventricular pacemakers. *Isoproterenol* was formerly used in the treatment of heart block and bronchoconstriction. *Dobutamine*, a congener of dopamine with relative selectivity for the β_1 receptor and with a greater effect on myocardial contractility than on heart rate, is also used in the treatment of congestive heart failure, often in combination with vasodilators ([Chap. 232](#)). In conjunction with radionuclide imaging or echocardiographic assessment of wall motion, *dobutamine*, as well as other investigational congeners that have a relatively greater effect on heart rate, is used in the diagnosis of demand-induced myocardial ischemia.

Selective β_2 -Receptor Agonists The cardiac stimulation caused by nonselective b agonists, such as *isoproterenol* or *epinephrine*, is occasionally dangerous when these agents are used in the treatment of bronchoconstriction ([Chap. 252](#)). Selective β_2 agonists, administered by inhalation for bronchoconstriction, include agents with an intermediate duration of action (*metaproterenol*, *albuterol*, *terbutaline*, *pirbuterol*, *isoetharine*, and *bitolterol*) and the newer long-acting agents (*salmeterol* and *formoterol*); these drugs improve the therapeutic ratio by achieving bronchial dilatation with less activation of the cardiovascular system ([Chaps. 252](#) and [258](#)). Although selectivity is relative and cardiac stimulation may occur with these agents at higher dose levels, inhaled agonists at the usual doses result in relatively little cardiac stimulation. Oral administration, which is no longer preferred, is associated with more systemic b-agonist effects. *Ritodrine*, another selective β_2 agonist, is used as a tocolytic agent (as is *terbutaline*) to relax the uterus and antagonize premature labor.

a-Adrenergic Agonists *Phenylephrine* and *methoxamine* are direct-acting a agonists that elevate blood pressure by increasing peripheral vasoconstriction. They are used primarily in the treatment of hypotension and paroxysmal supraventricular tachycardia ([Chap. 230](#)), in the latter case by increasing cardiac vagal tone through reflex baroreceptor stimulation. *Phenylephrine* and a related proprietary compound, *phenylpropanolamine*, are common constituents of decongestant medications (often combined with antihistamines) for the treatment of allergic rhinitis and upper respiratory infections.

Miscellaneous Sympathomimetic Amines with Mixed Actions *Ephedrine* has both direct b-receptor agonist properties and an indirect effect on sympathetic nerve endings, from which it releases [NE](#), and is used primarily as a bronchodilator. *Sudaphedrine*, a congener of *ephedrine*, is less potent at dilating bronchi and serves as a nasal decongestant. *Metaraminol* has both direct and indirect effects on sympathetic nerve endings and is employed in the treatment of hypotensive states.

Dopaminergic Agonists The D_2 -receptor agonists, *bromocriptine* and *cabergoline*, are used to suppress prolactin secretion ([Chap. 328](#)). *Apomorphine*, another D_2 -receptor agonist, is used to induce emesis. The D_1 -receptor agonist, *fenoldopam*, has recently been approved for the short-term in-hospital treatment of severe hypertension.

ANTIADRENERGIC OR SYMPATHOLYTIC AGENTS (See also [Chap. 246](#))

Agents Inhibiting Central Sympathetic Outflow The antihypertensive agents *methyldopa*, *clonidine*, *guanabenz*, and *guanfacine* diminish central sympathetic outflow by stimulating a central adrenergic pathway (α_2 receptor) that diminishes vasomotor outflow. CNS side effects such as sedation are common. When administration of clonidine is stopped abruptly, a withdrawal syndrome characterized by rebound hyperactivity of the sympathetic nervous system can produce a state resembling the crises of patients with pheochromocytoma. *Opiates* also may exert a central sympatholytic effect; the sympathetic excitation of morphine withdrawal responds to clonidine and vice versa. *Propranolol* and *reserpine* may exert some sympatholytic effects at the level of the CNS.

Ganglionic Blocking Agents Ganglionic transmission may be antagonized by drugs that block the (nicotinic) cholinergic synapse between the pre- and postganglionic autonomic nerves. These agents inhibit the parasympathetic as well as the sympathetic nervous system. Only *trimethaphan* is in general clinical use; its major application is in the treatment of hypertensive crises, particularly aortic dissection, when controlled hypotension and decreased myocardial contractility are desirable ([Chap. 246](#)).

Agents Acting at the Peripheral Sympathetic Nerve Endings Adrenergic neuron-blocking agents depress the function of the peripheral sympathetic nerves by decreasing the amount of neurotransmitter released. *Guanethidine*, the prototype of this class of drugs, is concentrated in the sympathetic nerve endings by the amine-uptake mechanism. Within the terminal it blocks the release of NE in response to nerve impulses and eventually depletes the nerve of NE by displacing it from the intraneuronal storage granules. The drug was formerly useful in the management of severe hypertension, although orthostatic hypotension was a limiting side effect. *Bretylium*, an agent whose effects are similar to those of guanethidine, is employed in the treatment of ventricular fibrillation ([Chap. 230](#)). Both guanethidine and bretylium are antagonized by agents that affect the amine-uptake transport process such as sympathomimetic amines, tricyclic antidepressants, phenoxybenzamine, and phenothiazines. The antihypertensive action of guanethidine may be rapidly reversed by these drugs.

Reserpine depletes catecholamines from the peripheral sympathetic nerve endings, the brain, and the adrenal medulla. Its antihypertensive effect in humans is usually attributed to depletion of peripheral NE stores within sympathetic nerve endings. The sedation and occasionally morbid depression attending its use result from NE depletion within the CNS.

Adrenergic-Receptor Blocking Agents Adrenergic blocking agents antagonize the effects of catecholamines at the level of the peripheral tissue.

α -Adrenergic-receptor blocking agents *Phenoxybenzamine* and *phentolamine* are utilized principally in treating pheochromocytoma ([Chap. 332](#)). Phenoxybenzamine produces prolonged, noncompetitive α blockade, while phentolamine leads to reversible, competitive blockade. Because of its rapid action and short duration, phentolamine is commonly used in the treatment of acute hypertensive paroxysms secondary to catecholamine excess, such as occur with pheochromocytoma. Both phentolamine and phenoxybenzamine antagonize α_1 and α_2 receptors, although phenoxybenzamine is more potent at the α_1 -receptor site. *Prazosin*, an α -adrenergic

blocking agent with selectivity for the α_1 -receptor, possesses properties that resemble those of primary vasodilators and has been used in the treatment of essential hypertension, as an afterload-reducing agent in congestive heart failure, and as an adjunct in the treatment of pheochromocytoma ([Chap. 332](#)). *Doxazosin* and *terazosin*, long-acting selective α_1 -blockers, are more useful in the treatment of essential hypertension because of better dosing schedule and less orthostatic hypotension. These agents also lower triglyceride levels and raise high-density lipoprotein (HDL) cholesterol levels. These selective α_1 -blockers, along with *tamsulosin* are useful in the symptomatic treatment of urinary outflow track obstruction and prostatism because they antagonize contraction of the sphincter at the bladder trigone and the prostate smooth muscle.

b-Adrenergic-receptor blocking agents These drugs antagonize the effects of catecholamines on the heart and are useful in the treatment of angina pectoris, hypertension, and cardiac arrhythmias. The benefit of beta blockade in angina derives from the decrease in myocardial oxygen consumption following reduction in heart rate and myocardial contractility ([Chap. 244](#)). The hypotensive effect of beta blockade is not clearly understood ([Chap. 246](#)). Diminished cardiac output, decreased NE release at postganglionic sympathetic nerve endings, reduced renin secretion, and suppressed central sympathetic outflow are possible mechanisms. The efficacy of b-blocking agents in the treatment of arrhythmias depends on reduction of the rate of spontaneous depolarization of pacemaker cells in the sinus node and junctional pacemakers and on slowing conduction within the atria and atrioventricular node. Beta blockade is also effective in the symptomatic management of hyperthyroidism and the control of tachycardia and arrhythmias in patients with pheochromocytoma. b-adrenergic blocking agents are also useful in the treatment of migraine, essential tremor, idiopathic hypertrophic subaortic stenosis, and aortic dissection. Several trials have demonstrated that b-receptor blocking agents, administered long-term, diminish mortality following acute myocardial infarction. The mechanism of this cardioprotective effect may involve antiarrhythmic action, prevention of reinfarction, and reduction in infarct size ([Chap. 243](#)).

Pharmacologic Properties of b-Receptor Blocking Agents Fourteen beta-blocking agents (atenolol, acebutolol, betaxolol, bisoprolol, carvedilol, carteolol, esmolol, metoprolol, nadolol, pindolol, penbutolol, propranolol, sotalol, and timolol) are available for use in the United States. Other agents (alprenolol, bevantolol, dilevalol, oxprenolol, etc.) are in use in other countries and investigational within the United States. The utility of these agents is derived predominantly from blockade of b-adrenergic receptors.

Although much has been written about other pharmacologic properties, including cardioselectivity, membrane-stabilizing (local anesthetic) effects, intrinsic sympathomimetic (partial-agonist) activity, and lipid solubility, the clinical significance of these additional properties is small. Local anesthetic properties are most prominent with propranolol; however, membrane stabilization probably does not contribute substantially to the clinical utility. The various beta blockers do differ in their water and lipid solubility. The lipophilic agents (propranolol, metoprolol, oxprenolol, bisoprolol, carvedilol) are readily absorbed from the gastrointestinal tract, metabolized by the liver, have large volumes of distribution, and penetrate the CNS well; the hydrophilic agents (acebutolol, atenolol, betaxolol, carteolol, nadolol, sotalol) are less readily absorbed, not extensively

metabolized, and have relatively long plasma half-lives. As a consequence, the hydrophilic agents may be administered once per day. Hepatic failure may prolong the plasma half-life of the lipophilic agents, whereas renal failure may prolong the action of the hydrophilic group. The degree of lipid solubility, therefore, provides a basis for choice of a particular agent in patients with hepatic or renal insufficiency. Although the hydrophilic agents penetrate the CNS less well, CNS side effects (sedation, depression, hallucinations) are well described with the hydrophilic as well as with the lipophilic agents.

Some β -adrenergic blocking agents possess β -agonist activity. This has been referred to as "intrinsic sympathomimetic activity" (ISA). Agents with partial agonist activity (pindolol, alprenolol, acebutolol, carteolol, dilevalol, oxprenolol) cause little or no depression of resting heart rate (partial agonist effect) while blocking the increase in heart rate that occurs in response to exercise or the administration of a β agonist such as isoproterenol. The presence of partial agonist activity may be useful when bradycardia limits treatment in patients with slow resting heart rates. Pindolol also produces mild vasodilation, perhaps in part related to peripheral α_2 stimulation. Agents with partial agonist activity cause less change in blood lipid levels than agents without agonist properties. On theoretical grounds, intrinsic sympathomimetic activity would be undesirable in the treatment of thyrotoxicosis, idiopathic hypertrophic subaortic stenosis, aortic dissection, and tachyarrhythmias.

Cardioselective (β_1) Adrenergic-Receptor Blocking Agents Propranolol, the prototype of the nonselective β -adrenergic blocking agent, induces a competitive blockade of both β_1 and β_2 receptors. Other nonselective β -blocking agents include alprenolol, carteolol, dilevalol, nadolol, oxprenolol, penbutolol, pindolol, sotalol, timolol, and carvedilol. Metoprolol, esmolol, acebutolol, atenolol, and betaxolol possess relative selectivity for the β_1 receptor. Although β_1 -(cardio-) selective agents have the theoretical advantage of producing less bronchoconstriction and less peripheral vasoconstriction, a clear-cut clinical advantage of the cardioselective agents has not been decisively demonstrated, since the β_1 selectivity is only relative. Bronchoconstriction may occur when β_1 -selective agents are administered in full therapeutic doses.

Adverse Effects of β -Receptor Blocking Agents Aside from the effects on the [CNS](#), most adverse reactions to β -blocking agents are consequences of β -adrenergic blockade. These include the precipitation of heart failure in patients in whom cardiac compensation depends on enhanced sympathetic drive; the aggravation of bronchospasm in patients with asthma; predisposition to the development of hypoglycemia in insulin-requiring diabetics (blockade of catecholamine-mediated counterregulation and antagonism of the adrenergic warning signs of hypoglycemia); the development of hyperkalemia in diabetic or uremic patients with impaired potassium tolerance; the enhancement of coronary or peripheral arterial vasospasm; and elevation in triglycerides and depression of [HDL](#) levels. The lipid (and perhaps the peripheral vascular) effects are less (or absent) in agents with partial (β_2) agonist activity or α -blocking properties (carvedilol).

Miscellaneous Adrenergic Blocking Agents Labetalol, approved for use in the United States as an antihypertensive agent, is a competitive antagonist of both α - and β -adrenergic receptors. Although labetalol induces relatively more β - than α -receptor

blockade, fall in peripheral resistance may be marked following acute administration of the drug. Vasodilation may be mediated in part by a partial agonist effect on the β_2 -adrenergic receptor; labetalol does not possess partial agonist activity for the β_1 (cardiac) receptor.

Metoclopramide is a dopaminergic antagonist with cholinergic agonist properties. It enhances gastric emptying, increases the tone of the lower esophageal sphincter, increases prolactin and aldosterone secretion, and antagonizes emesis induced by apomorphine. It is useful clinically in enhancing gastric emptying (in the absence of organic obstruction such as in diabetic gastroparesis), in antagonizing gastroesophageal reflux, and as an antiemetic during cancer chemotherapy.

THE PARASYMPATHETIC NERVOUS SYSTEM

ACETYLCHOLINE

[ACh](#) serves as the neurotransmitter at all autonomic ganglia, at the postganglionic parasympathetic nerve endings, at the postganglionic sympathetic nerve endings innervating the eccrine sweat glands, and at the skeletal muscle end plate (neuromuscular junction). The enzyme choline acetyltransferase catalyzes the synthesis of ACh from acetyl coenzyme A (CoA) produced within the nerve ending and from choline, actively taken up from the extracellular fluid. Within the cholinergic nerve endings, ACh is stored in discrete synaptic vesicles and released in response to nerve impulses that depolarize the nerve terminals and increase calcium influx.

Cholinergic Receptors Different receptors for [ACh](#) exist on the postganglionic neurons within the autonomic ganglia and at the postjunctional autonomic effector sites. Those within the autonomic ganglia and adrenal medulla are stimulated predominantly by nicotine (*nicotinic receptors*) and those on autonomic effector cells by the alkaloid muscarine (*muscarinic receptors*). Ganglionic blocking agents antagonize the nicotinic receptors, while atropine blocks the muscarinic receptors. The muscarinic (M) receptor, furthermore, has been recently subdivided into additional types. The M_1 receptor is localized to the [CNS](#) and perhaps parasympathetic ganglia; the M_2 receptor is the nonneuronal muscarinic receptor on smooth muscle, cardiac muscle, and glandular epithelium. Bethanechol is a selective agonist of the M_3 receptor; pirenzepine, an investigational agent, is a selective antagonist of the M_1 receptor that markedly reduces gastric acid secretion. The M_2 receptor inhibits adenylyl cyclase and utilizes the regulatory G protein; the M_1 receptor interacts with G_q and stimulates phospholipase C ([Fig. 72-5](#)). The M_3 receptor, present on smooth muscle and secretory glands, is antagonized by atropine and utilizes phospholipase C, [IP₃](#), and [DAG](#) as second messengers. Other subtypes have been identified by molecular biologic techniques but have not yet been fully characterized.

Acetylcholinesterase Hydrolysis of [ACh](#) by acetylcholinesterase inactivates the neurotransmitter at cholinergic synapses. This enzyme (also known as specific or true cholinesterase) is present within neurons and is distinct from butyrylcholinesterase (serum cholinesterase or pseudocholinesterase). The latter enzyme is present in plasma and nonneuronal tissues and is not primarily involved in the termination of the effects of ACh at autonomic effector sites. The pharmacologic effects of

anticholinesterase agents are due to inhibition of neuronal (true) acetylcholinesterase.

PHYSIOLOGY OF THE PARASYMPATHETIC NERVOUS SYSTEM

The parasympathetic nervous system participates in the regulation of the cardiovascular system, the gastrointestinal tract, and the genitourinary system. Tissues such as liver, kidney, pancreas, and thyroid also receive parasympathetic innervation, suggesting a role for the parasympathetic nervous system in metabolic regulation as well, although cholinergic effects on metabolism are not well characterized.

Cardiovascular System Parasympathetic effects on the heart are mediated by the vagus nerve. [ACh](#) reduces the rate of spontaneous depolarization of the sinoatrial node and decreases heart rate. ACh also delays impulse conduction within the atrial musculature while shortening the effective refractory period, a combination of factors that may initiate or perpetuate atrial arrhythmias. At the atrioventricular node, ACh reduces conduction velocity, increases the effective refractory period, and thus diminishes the ventricular response during atrial flutter or fibrillation ([Chap. 230](#)). The decrease in inotropy induced by ACh is related to a prejunctional inhibitory effect on sympathetic nerve endings as well as to a direct inhibitory effect on the atrial myocardium. The ventricular myocardium is not much affected since innervation by cholinergic fibers is minimal. A direct cholinergic contribution to the regulation of peripheral resistance appears unlikely since parasympathetic innervation of the vasculature is not extensive. The parasympathetic nervous system, however, may influence peripheral resistance indirectly by inhibiting [NE](#) release from sympathetic nerves.

Gastrointestinal Tract Parasympathetic innervation of the gut is via the vagus nerve and the pelvic sacral nerves. The parasympathetic nervous system increases the tone of gastrointestinal smooth muscle, enhances peristaltic activity, and relaxes the gastrointestinal sphincters. [ACh](#) stimulates exocrine secretion from the glandular epithelium and enhances the secretion of gastrin, secretin, and insulin.

Genitourinary and Respiratory Systems Sacral parasympathetic nerves supply the urinary bladder and genitalia. [ACh](#) increases ureteral peristalsis, contracts the urinary detrusor muscle, and relaxes the trigone and sphincter, thereby playing a critical role in the coordination of urination. The respiratory tract is innervated with parasympathetic fibers derived from the vagus nerve. ACh increases tracheobronchial secretions and stimulates bronchial constriction.

PHARMACOLOGY OF THE PARASYMPATHETIC NERVOUS SYSTEM

Cholinergic Agonists [ACh](#) itself has no therapeutic role because of its widespread effects and short duration of action. Congeners of ACh are less susceptible to hydrolysis by cholinesterase and have a narrower range of physiologic effects. Bethanechol, the only systemic cholinergic agonist in general use, stimulates gastrointestinal and genitourinary smooth muscle with minimal effect on the cardiovascular system. It is used in the treatment of urinary retention in the absence of outflow tract obstruction and, less commonly, in gastrointestinal disorders such as postvagotomy gastric atony. Pilocarpine and carbachol are topical cholinergic agonists used in the treatment of

glaucoma.

Acetylcholinesterase Inhibitors Cholinesterase inhibitors enhance the effects of parasympathetic stimulation by diminishing the inactivation of [ACh](#). The therapeutic application of reversible cholinesterase inhibitors depends on the role of ACh as neurotransmitter at the skeletal muscle neuroeffector junction and within the [CNS](#) and includes the treatment of myasthenia gravis ([Chap. 380](#)), the termination of neuromuscular blockade following general anesthesia, and the reversal of intoxication by agents with a central anticholinergic action. Physostigmine, a tertiary amine, penetrates the CNS well, while related quaternary amines (neostigmine, pyridostigmine, ambenonium, and edrophonium) do not. Organophosphorous cholinesterase inhibitors produce irreversible cholinesterase blockade; these agents are used principally as insecticides and are primarily of toxicologic interest. With regard to the autonomic nervous system, cholinesterase inhibitors are of limited use in the treatment of intestinal and bladder smooth-muscle dysfunction such as occurs in paralytic ileus and atonic urinary bladder. Cholinesterase inhibitors induce a vagotonic response in the heart and may be useful in terminating attacks of paroxysmal supraventricular tachycardia ([Chap. 230](#)).

Cholinergic-Receptor Blocking Agents *Atropine* blocks muscarinic cholinergic receptors, with little effect on cholinergic transmission at the autonomic ganglia and the neuromuscular junctions. Many of the [CNS](#) actions of atropine and atropine-like drugs are attributable to blockade of central muscarinic synapses. The related alkaloid, *scopolamine*, is similar to atropine but causes drowsiness, euphoria, and amnesia, effects that make it suitable as a preanesthetic medication.

Atropine increases heart rate and enhances atrioventricular conduction, actions that may be useful in combating the bradycardia or heart block associated with heightened vagal tone. In addition, atropine reverses cholinergically mediated bronchoconstriction and diminishes respiratory tract secretions. These effects contribute to its utility as a preanesthetic medication.

Atropine also decreases gastrointestinal tract motility and secretion. Although various derivatives and congeners of atropine (such as *propantheline*, *isopropamide*, and *glycopyrrolate*) have been advocated in patients with peptic ulcer or with diarrheal syndromes, the chronic use of such agents is limited by other manifestations of parasympathetic inhibition such as dry mouth and urinary retention. The investigational selective M₁ inhibitor pirenzepine inhibits gastric secretion at doses that have minimal anticholinergic effects at other sites; this agent may be useful in the treatment of peptic ulcer. Atropine and its congener *ipratropium*, when given by inhalation, cause bronchodilation and have been used experimentally in the treatment of asthma.

(Bibliography omitted in Palm version)

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PART FIVE -NUTRITION

73. NUTRITIONAL REQUIREMENTS AND DIETARY ASSESSMENT - *Johanna Dwyer*

Nutrients are substances that are not synthesized in the body in sufficient amounts and therefore must be supplied by the diet. Nutrient requirements for groups of healthy persons have been thoroughly defined on the basis of experimental evidence. For good health we require energy-providing nutrients (protein, fat, and carbohydrate), vitamins, minerals, and water. Specific nutrient requirements include 9 essential amino acids, several fatty acids, 4 fat-soluble vitamins, 10 water-soluble vitamins, and choline. Several inorganic substances, including four minerals, seven trace minerals, three electrolytes, and the ultratrace elements, must also be supplied in the diet ([Chap. 75](#)).

The required amounts of the essential nutrients differ by age and physiologic state. Conditionally essential nutrients are not required in the diet but must be supplied to individuals who do not synthesize them in adequate amounts, such as those with genetic defects, those having pathologic states with nutritional implications, and developmentally immature infants ([Chap. 74](#)). Many organic phytochemicals and zoochemicals present in foods have various health effects. For example, dietary fiber has been shown to have beneficial effects on gastrointestinal function.

ESSENTIAL NUTRIENT REQUIREMENTS

Energy For weight to remain stable, energy intake must match energy output ([Chap. 77](#)). The major categories of energy output are resting energy expenditure (REE) and physical activity; minor sources include the energy cost of metabolizing food (thermic effect of food or specific dynamic action) and shivering thermogenesis (e.g., cold-induced thermogenesis). The average energy intake is about 2800 kcal/d for American men and about 1800 kcal/d for American women, though these estimates vary with body size and activity level. Formulas for estimating REE are useful for assessing the energy needs of an individual whose weight is stable. Thus, for males, $REE = 900 + 10w$, and for females, $REE = 700 + 7w$, where w is weight in kg. The calculated REE is then adjusted for physical activity level by multiplying by 1.2 for sedentary, 1.4 for moderately active, or 1.8 for very active individuals. The final figure provides an estimate of total caloric needs in a state of energy balance.

Illness often alters energy needs. Unstressed hospitalized patients at bed rest usually require 1.2 times their [REE](#), whereas those who are stressed, febrile, and catabolic require 1.5 to 2 times their REE ([Chap. 74](#)). Intestinal malabsorption may decrease net utilizable energy to as little as 25% of ingested energy and may necessitate feeding by parenteral routes ([Chap. 76](#)). Fever increases energy expenditure by 10 to 13% per degree Celsius above normal. Other diseases increase energy needs by varying amounts, such as burns (40 to 100%), trauma (40 to 100%), and hyperthyroidism (10 to 100%). Hypothyroidism and adrenal insufficiency decrease resting energy needs, but these alterations are corrected after adequate hormone replacement. In obese patients, weight reduction can be accomplished by reducing energy intakes by approximately 500 kcal/d to achieve a loss of 0.5 kg of fat per week, or 1000 kcal/d to lose 1 kg per week ([Chap. 77](#)).

Protein Dietary protein consists of both essential and nonessential amino acids that are required for protein synthesis, whereas certain amino acids can also be used for energy and gluconeogenesis ([Chap. 334](#)). The nine essential amino acids are histidine, isoleucine, leucine, lysine, methionine/cystine, phenylalanine/tyrosine, threonine, tryptophan, and valine. When energy intake is inadequate, protein intake must be increased, since ingested amino acids are diverted into pathways of glucose synthesis and oxidation. In extreme energy deprivation, protein-calorie malnutrition may ensue ([Chaps. 74](#) and [76](#)).

For adults, the recommended dietary allowance (RDA) for protein is about 0.6 g/kg desirable body weight per day, assuming that energy needs are met and that the protein is of relatively high biologic value. Current recommendations for a healthy diet call for at least 10 to 14% of calories from protein. Biologic value tends to be highest for animal proteins, followed by proteins from legumes (beans), cereals (rice, wheat, corn), and roots. Combinations of plant proteins that complement one another in biologic value or combinations of animal and plant proteins can increase biologic value and lower total protein requirements.

Protein needs increase during growth, pregnancy, lactation, and rehabilitation during treatment of malnutrition. The tolerance of dietary protein is decreased in renal insufficiency and liver failure. Normal protein intake can precipitate encephalopathy in patients with cirrhosis of the liver ([Chap. 299](#)) or worsen uremia in those with renal failure ([Chap. 270](#)).

Fat and Carbohydrate Fats are a concentrated source of energy and constitute on average 34% of calories in U.S. diets. However, for optimal health, fat intake should total no more than 30% of calories. Saturated fat and trans-fat should be limited to <10% of calories, and polyunsaturated fats to <10% of calories, with monounsaturated fats comprising the remainder of fat intake. At least 55% of total calories should be derived from carbohydrates. The brain requires about 100 g/d of glucose for fuel; other tissues use about 50 g/d. Over time, adaptations in carbohydrate needs are possible in hypocaloric states. For example, reduced insulin levels lead to adipose tissue breakdown and cause the body to burn more fatty acids. However, some tissues (e.g., brain and red blood cells) rely on glucose supplied either exogenously or from muscle proteolysis ([Chap. 334](#)).

Water For adults, 1 to 1.5 mL water per kcal of energy expenditure is sufficient under usual conditions to allow for normal variations in physical activity levels, sweating, and solute load of the diet. Water losses include 50 to 100 mL/d in the feces, 500 to 1000 mL/d by evaporation or exhalation, and, depending on the renal solute load, ³1000 mL/d in the urine. If external losses increase, intakes must increase accordingly to avoid underhydration. Fever increases water losses by approximately 200 mL/d per°C; diarrheal losses vary but may be as great as 5 L/d in severe diarrhea. Heavy sweating and vomiting also increase water losses. When renal function is normal and solute intakes are adequate, the kidneys can adjust to increased water intake by excreting up to 18 L/d of excess water ([Chap. 329](#)). However, obligatory urine outputs can compromise hydration status when there is inadequate intake or when losses increase in disease or kidney damage.

Infants have high requirements for water because of their large ratio of surface area to volume, the limited capacity of the immature kidney to handle high renal solute loads, and their inability to communicate their thirst. Increased water needs during pregnancy are low, perhaps an additional 30 mL/d; but during lactation, milk production increases water requirements so that approximately 1000 mL/d of additional water is needed, or 1 mL for each mL of milk produced. Special attention must be paid to the water needs of the elderly, who have reduced total body water and blunted thirst sensation and may be taking diuretics.

Other Nutrients The vitamins and minerals required for health and the clinical disorders caused by vitamin deficiency or excess are discussed in [Chap. 75](#).

DIETARY REFERENCE INTAKES, RECOMMENDED ALLOWANCES, AND TOLERANCES

Fortunately, human life and well-being can be maintained within a fairly wide range for most nutrients. However, the capacity for adaptation is not infinite -- too much, as well as too little, intake of a nutrient may have adverse effects or alter the health benefits conferred by another nutrient ([Chap. 75](#)). Therefore, benchmark recommendations on nutrient intakes have been developed to guide clinical practice. These quantitative estimates of nutrient intakes are collectively referred to as the *dietary reference intakes* (DRIs). The DRIs supplant the [RDAs](#), the single reference values used in the United States since 1989. DRIs include the estimated average requirement (EAR) for nutrients, as well as three other reference values used for dietary planning for individuals: the RDAs, the adequate intake (AI), and the safe upper level (UL). The current RDAs and AIs are provided in [Tables 73-1](#) and [73-2](#), respectively.

Estimated Average Requirement When florid dietary deficiency diseases such as rickets, scurvy, xerophthalmia, and protein-calorie malnutrition were common, nutrient adequacy was assumed by the absence of clinical signs of a dietary deficiency disease. Later, it was determined that biochemical and other changes were evident long before the clinical deficiency became apparent. Consequently, criteria of adequacy are chosen using such biologic markers when they are available. Current efforts focus on the amount of a nutrient that reduces the risk of chronic degenerative diseases. Priority is given to sensitive biochemical, physiologic, or behavioral tests that reflect early changes in regulatory processes or maintenance of body stores of nutrients.

The [EAR](#) is the amount of a nutrient estimated to be adequate for half of the healthy individuals of a specific age and sex. The types of evidence and criteria used to establish nutrient requirements vary by nutrient, age, and physiologic group. The EAR is not an effective estimate of nutrient adequacy in individuals because it is a median requirement for a group, and the variation around this number is considerable. As the EAR specifies, 50% of individuals in a group fall below the requirement and 50% fall above it. Thus, a person with a usual intake at the EAR has a 50% risk of an inadequate intake during the reporting period. For these reasons, other standards, described below, are more useful for clinical purposes.

Recommended Dietary Allowances The [RDA](#) is the average daily dietary intake level

that meets the nutrient requirements of nearly all healthy persons of a specific sex, age, life stage, or physiologic condition (such as pregnancy or lactation). The RDA is commonly used as a nutrient-intake goal for planning diets of individuals.

The [RDA](#) is defined statistically as 2 standard deviations (SD) above the [EAR](#) to ensure that the needs of any given individual are met. The RDAs are used to formulate food guides such as the U.S. Department of Agriculture (USDA) Food Guide Pyramid for individuals, food exchange lists for therapeutic diet planning, and as a standard for describing the nutritional content of processed foods and nutrient supplements. The nutrient content in a food is stated by weight or as a percent of the daily value (DV), a variant of the RDA which, for an adult, represents the highest RDA for an adult consuming 2000 kcal/d.

The risk of dietary inadequacy increases as intakes fall further below the [RDA](#). However, the RDA is an overly generous criterion for evaluating nutrient adequacy. For example, by definition the RDA exceeds the actual requirements of all but about 2 to 3% of the population. Therefore, many people whose intakes fall below the RDA may still be getting enough of the nutrient.

Adequate Intake It is not possible to set an [RDA](#) for some nutrients that do not have an established [EAR](#). In this circumstance, the [AI](#) is based on observed, or experimentally determined, approximations of nutrient intakes in healthy people. In the [DRIs](#) established to date, AIs rather than RDAs are proposed for infants up to age 1, as well as for calcium, vitamin D, fluoride, pantothenic acid, biotin, and choline for persons of all ages.

Tolerable Upper Levels of Nutrient Intake Excessive nutrient intake can disturb body functions and cause acute, progressive, or permanent disabilities ([Chap. 75](#)). Some diseases of nutritional excess include fluorosis, hypervitaminosis A, hypervitaminosis D, and obesity. The tolerable [UL](#) is the highest level of chronic nutrient intake (usually daily) that is unlikely to pose a risk of adverse health effects for most of the population. An uncertainty factor is applied to ensure that even very sensitive persons would not experience adverse effects at the UL dose chosen. For many nutrients, data on the adverse effects of large amounts of the nutrient are unavailable or too limited to establish a UL. Therefore, the lack of a UL does *not* mean that the risk of adverse effects from high intakes is nonexistent; caution is warranted in those who consume large amounts of such nutrients. Healthy individuals derive no established benefit from consuming nutrient levels above the [RDA](#) or [AI](#). Individual nutrients in foods that most people eat rarely reach levels that exceed the UL. However, nutritional supplements provide more concentrated amounts of nutrients per dose and, as a result, pose a potential risk of toxicity. Nutrient supplements are labeled with "supplement facts" that express the amount of nutrient in absolute units or as the percent of the [DV](#) provided per recommended serving size. Those who use supplements should be advised that total nutrient consumption, including both food and supplements, should not exceed RDA levels.

FACTORS ALTERING NUTRIENT NEEDS

The [DRIs](#) are affected by age, sex, rate of growth, pregnancy, lactation, physical activity, composition of diet, concomitant diseases, and drugs. When only slight differences exist

between the requirements for nutrient sufficiency and excess, dietary planning becomes more difficult. Renal insufficiency provides one example in which protein intakes must be sufficient to maintain protein nutritional status, while avoiding exacerbation of uremic symptoms because of protein excess.

Physiologic Factors Growth, strenuous physical activity, pregnancy, and lactation increase needs for energy and several essential nutrients. Energy needs rise during pregnancy, due to the demands of fetal growth, and during lactation, because of the increased energy required for milk production. Energy needs decrease with loss of lean body mass, the major determinant of [REE](#). Because both health and physical activity tend to decline with age, energy needs in older persons, especially those over 70, tend to be less than those of younger persons.

Dietary Composition Dietary composition affects the biologic availability and utilization of nutrients. For example, the absorption of iron may be impaired by high amounts of calcium or lead; non-heme iron uptake may be impaired by the lack of ascorbic acid and amino acids in the meal. The absorption of calcium and magnesium is decreased by large amounts of phytates in the diet. Protein utilization by the body may be decreased when essential amino acids are not present in sufficient amounts. Animal foods, such as milk, eggs, and meat, have high biologic values with most of the needed amino acids present in adequate amounts. Plant proteins in corn (maize), soy, and wheat have lower biologic values and must be combined with other plant or animal proteins to achieve optimal utilization by the body.

Route of Administration The [RDAs](#) apply only to oral intakes. When nutrients are administered parenterally, similar values can sometimes be used for amino acids, carbohydrates, fats, sodium, chloride, potassium, and most of the vitamins, since their intestinal absorption is nearly 100% ([Chap. 75](#)). However, the oral bioavailability of most mineral elements may be only half that obtained by parenteral administration. For some nutrients that are not readily stored in the body, or cannot be stored in large amounts, timing of administration may also be important. For example, amino acids cannot be used for protein synthesis if they are not supplied together; instead they will be used for energy production.

Disease Specific dietary deficiency diseases include protein-calorie malnutrition; iron, iodine, and vitamin A deficiency; megaloblastic anemia due to vitamin B₁₂ or folic acid deficiency; vitamin D deficiency rickets; and scurvy, beriberi, and pellagra ([Chaps. 74 and 75](#)). Each deficiency disease is characterized by imbalances at the cellular level between the supply of nutrients or energy and the body's nutritional needs for growth, maintenance, and other functions. Imbalances in nutrient intakes are recognized as risk factors for certain chronic degenerative diseases, such as saturated fat and cholesterol in coronary artery disease; sodium in hypertension; obesity in hormone-dependent endometrial, breast, and prostate cancers; and ethanol in alcoholism. Since the etiology and pathogenesis of these disorders are multifactorial, diet is only one of many risk factors. Osteoporosis, for example, is associated with calcium deficiency, as well as risk factors related to environment (e.g., smoking, sedentary lifestyle), physiology (e.g., estrogen deficiency), genetic determinants (e.g., defects in collagen metabolism), and drug use (chronic steroids) ([Chap. 342](#)).

DIETARY ASSESSMENT

In clinical situations, nutritional assessment is an iterative process that involves: (1) screening for malnutrition, (2) assessing the diet and other data to establish either the absence or presence of malnutrition and its possible causes, and (3) planning for the most appropriate nutritional therapy. Some disease states affect the bioavailability, requirements, utilization, or excretion of specific nutrients. In these circumstances, specific measurements of various nutrients may be required to assure adequate replacement ([Chap. 75](#)).

Most health care facilities have a nutrition screening process in place for identifying possible malnutrition after hospital admission. Nutritional screening is required by the Joint Commission on Accreditation of Healthcare Organizations (JCAHO), but there are no universally recognized or validated standards, so techniques vary. The factors that are usually assessed include: abnormal weight for height or body mass index (e.g., BMI <19 or >25); reported weight change (involuntary loss or gain of >5 kg in past 6 months) ([Chap. 43](#)); diagnoses with known nutritional implications (metabolic disease, any disease affecting the gastrointestinal tract, alcoholism, and others); present therapeutic dietary prescription; chronic poor appetite; presence of chewing and swallowing problems or major food intolerances; need for assistance with preparing or shopping for food, eating, or other aspects of self care; and social isolation. Reassessment of nutrition status should occur periodically in hospitalized patients -- at least once every week.

A more complete dietary assessment is indicated for patients who exhibit a high risk of malnutrition on nutrition screening. The type of assessment varies based on the clinical setting, severity of the patient's illness, and stability of his or her condition.

Acute Care Settings In acute care settings, anorexia, various diseases, test procedures, and medications can compromise dietary intake. Under such circumstances, the goal is to identify and avoid inadequate intake and assure appropriate alimentation. Dietary assessment in acute care situations focuses on what patients are currently eating, whether they are able and willing to eat, and whether they experience any problems with eating. Dietary intake assessment is based on information from observed intakes; medical record; history; clinical examination; and anthropometric, biochemical, and functional status. The objective is to gather enough information to establish the likelihood of malnutrition due to poor dietary intake or other causes in order to determine whether nutritional therapy is indicated.

Simple observations may suffice to suggest inadequate oral intake. These include dietitians' and nurses' notes, the amount of food eaten on trays, frequent tests and procedures that are likely to cause meals to be skipped, nutritionally inadequate diet orders such as clear liquids or full liquids for more than a few days, fever, gastrointestinal distress, vomiting, diarrhea, or a comatose state. Patients with diseases or treatments that involve any part of the alimentary tract are at high nutritional risk. Acutely ill patients with diet-related diseases such as diabetes need assessment because an inappropriate diet may exacerbate these conditions and adversely affect other therapies. Abnormal biochemical values [serum albumin levels <35 g/L (<3.5 mg/dL); serum cholesterol levels <3.9 mmol/L (<150 mg/dL)] are nonspecific but may

also indicate a need for further nutritional assessment.

Most therapeutic diets offered in hospitals are calculated to meet individual nutrient requirements and the [RDA](#). Exceptions include clear liquids, some full liquid diets, and test diets, which are inadequate for several nutrients and should not be used, if possible, for more than 24 h. As much as half of the food served to hospitalized patients is not eaten, and so it cannot be assumed that the intakes of hospitalized patients are adequate. The dietary assessment should therefore compare how much and what food the patient has consumed with the diet that has been provided in the hospital. Major deviations in intakes of energy, protein, fluids, or other nutrients of special concern for the patient's illness should be noted and corrected.

Nutritional monitoring is especially important for patients who are very ill and who have extended lengths of stay. Patients who are fed by special enteral and parenteral routes also require special nutritional assessment and monitoring by physicians with training in nutrition support and/or dietitians with certification in nutrition support ([Chap. 76](#)).

Ambulatory Settings The aim of dietary assessment in the outpatient setting is to determine whether the patient's usual diet is a health risk in itself or if it contributes to existing chronic disease-related problems. It also provides the basis for planning a diet that fulfills therapeutic goals while ensuring patient compliance. The outpatient dietary assessment should review the adequacy of present and usual food intakes, including vitamin and mineral supplements, medications, and alcohol, as all of these may affect the patient's nutritional status. The dietary assessment should focus on the dietary constituents that are most likely to be involved or compromised by a specific diagnosis, as well as any comorbidities that are present. More than one day's intake should be reviewed to provide a better representation of the usual diet.

There are many ways to assess the adequacy of the patient's habitual diet. These include a food guide, a food exchange list, a diet history, or a food frequency questionnaire. A commonly used food guide for healthy persons is the [USDA's](#) food pyramid, which is useful as a basis for identifying inadequate intakes of essential nutrients, as well as likely excesses in fat, saturated fat, sodium, sugar, and alcohol ([Table 73-3](#)). The guide is calculated to provide approximately 1600 kcal for sedentary women and some older adults; 2200 kcal for most children, teenage girls, active women, and many sedentary men (women who are pregnant or breastfeeding may need somewhat more); and 2800 kcal for teenage boys, most active men, and some very active women. Results provide a rough guide to food groups that may be eaten either in excess of recommendations or in insufficient quantities. Respondents who follow ethnic or unusual dietary patterns may need extra instruction on how foods should be categorized, as well as the appropriate portion sizes that constitute a serving. The process of reviewing the guide with patients helps them transition to healthier dietary patterns. For those on therapeutic diets, assessment against food exchange lists may be useful. These include, for example, the American Diabetes Association food exchange lists for diabetes, or the American Dietetic Association food exchange lists for renal disease.

Nutritional Status Assessment Full nutritional status assessment is a complex, time-consuming, and expensive process that requires considerable expertise.

Candidates include seriously ill patients and those at very high nutritional risk when the cause of malnutrition is still uncertain after initial clinical evaluation and dietary assessment. Full nutritional status assessment involves multiple dimensions, including documentation of dietary intake, anthropometric measurements, biochemical measurements of blood and urine, clinical examination, health history, and functional. **For further discussion of Nutritional Assessment, see [Chap. 74](#).*

(Bibliography omitted in Palm version)

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74. MALNUTRITION AND NUTRITIONAL ASSESSMENT - Charles H. Halsted

Malnutrition is a frequent and integral component of acute and chronic illness. When recognized by appropriate clinical assessment, malnutrition is found in >50% of all hospitalized adults. It contributes to increased in-hospital morbidity and mortality in both medical and surgical patients, and leads to more frequent hospital admissions among the elderly. Malnutrition results from various combinations of starvation, including inadequate intake or abnormal gastrointestinal assimilation of the diet, the stress response to acute injury or chronic inflammation, and abnormal nutrient metabolism. Nutritional assessment should be considered an integral part of the clinical evaluation and be used as a basis for nutritional support in the overall therapeutic plan.

DEFINITIONS OF MALNUTRITION

In the strict sense, the term *malnutrition* includes extremes of underweight and overweight. The current chapter, however, focuses on the evaluation of the undernourished patient who presents with diminished body protein and energy stores and micronutrient deficiencies.

To the practicing physician, both outpatients and inpatients should be considered at risk for malnutrition if they meet one or more of the following criteria: (1) unintentional loss of >10% of usual body weight in the preceding 3 months, (2) body weight <90% of ideal for height, or (3) body mass index (BMI; the weight in kilograms divided by the height in square meters) <18.5. With regard to varying levels of severity, body weight <90% of ideal for height represents risk of malnutrition, body weight <85% of ideal constitutes *malnutrition*, <70% of ideal represents *severe malnutrition*, and <60% of ideal is usually incompatible with survival.

Malnutrition may be endemic in regions of famine, and two forms of severe malnutrition are recognized under conditions of inadequate food supply or distribution: *marasmus* refers to generalized starvation with loss of body fat and protein, whereas *kwashiorkor* refers to selective protein malnutrition with edema and fatty liver. The latter form occurs following restriction of dietary protein among children in settings of recurrent diarrheal illness. These distinctions, however, seldom apply to malnourished patients in more developed societies. In this setting, features of combined protein-calorie malnutrition (PCM) are more commonly seen in the context of a wide variety of acute and chronic illnesses that lead to depletion of body fat, muscle wasting, multiple signs of micronutrient deficiencies, decubitus ulcers, and life-threatening infections. An overview of the evaluation of malnutrition in the sick adult is depicted in [Fig. 74-1](#).

PATHOPHYSIOLOGY AND ETIOLOGIES OF MALNUTRITION

In simple terms, patients lose weight when: (1) the intake or gastrointestinal assimilation of dietary calories is insufficient to meet normal energy expenditure; (2) the expenditure of body energy stores is greater than energy normally consumed and assimilated by the body; or (3) the metabolism of energy supplies, protein, and other nutrients is significantly impaired by the intrinsic disease process.

Body Composition As depicted in [Fig. 74-2](#), the human body stores between 15 and

25% of its energy as fat (greater in women than men), which is available for the metabolism of endogenous fatty acids during starvation. The remaining fat-free mass (FFM) is composed of extracellular and intracellular water, the bony skeleton, glycogen, and skeletal and visceral protein. Aside from body fat, energy reserves are also provided by intracellular glycogen and protein, which, together with intracellular water, constitute the body cell mass (BCM). Thus, in addition to the enzymes that support the normal metabolic machinery of the body, the BCM provides reserve protein for energy production by gluconeogenesis during the stress response.

The Metabolic Response to Starvation and Stress The expenditure of body stores of energy (as fat, glycogen, and protein) is different during *starvation* (due to decreased intake and/or assimilation of the diet) and *stress* (due to excessive expenditure of energy and body protein). Consequently, these events affect body compartments differently. Starvation decreases the size of all body compartments, whereas stress reduces [BCM](#), increases extracellular water, and has variable effects on body fat.

A normal 70-kg man stores fuel at about 15 kg as fat, 6 kg as protein, and 0.4 kg as glycogen. During a 24-h fast, energy needs are met by the consumption of liver glycogen stores and the conversion of up to 75 g of body protein to glucose (by gluconeogenesis). During prolonged starvation, metabolism is supported by stores of body fat (about 150 g/d), which provides fatty acid-derived ketones, and muscle protein (about 20 g/d), which is used for gluconeogenesis. Under these conditions, total energy expenditure is decreased in order to conserve energy. While normal-weight individuals can sustain total fasting for about 2 months, obese individuals can fast for periods >12 months, depending on their fat stores.

The metabolic responses to the stress of acute critical illness (e.g., following accidental or surgical trauma or sepsis) significantly modify this sequence of events. In contrast to the hypometabolism, protein conservation, and reliance on body fat stores for energy needs during starvation, the acute stress response is characterized by hypermetabolism, in which the demands of accelerated energy expenditure are met by skeletal and visceral proteolysis to provide amino acid substrate for gluconeogenesis. Muscle proteolysis and gluconeogenesis are promoted by high levels of circulating catecholamines, glucagon, cortisol, and cytokines, including tumor necrosis factor (TNF) α and interleukins 1 and 6, in the setting of insulin resistance. When untreated, body protein catabolism is accelerated to 240 g/d, which is sufficient to deplete 50% of body protein stores within 3 weeks.

A more common clinical situation is the malnourished patient with chronic illness in whom acute trauma or sepsis superimposes cytokine-mediated proteolysis with increased metabolic demands. If unchecked by appropriate therapy, the process of progressive [PCM](#) in such patients is associated with decreased cardiac and renal function, fluid retention, intestinal mucosal atrophy, loss of intracellular minerals (zinc, magnesium, and phosphorus), diminished cell-mediated immune functions, increased risk of infection, and eventual death ([Fig. 74-3](#)).

Etiologies of Malnutrition The causes of decreased dietary intake are diverse and include social and economic conditions, psychiatric diseases, neurodegenerative dementias, cytokine-mediated appetite suppression in chronic infections such as AIDS

or in disseminated cancer, and self-limited food intake in abdominal pain syndromes ([Table 74-1;Chap. 43](#)). Given the central role of the gastrointestinal tract in the assimilation of nutrients,[PCM](#) is a predictable component of many chronic gastrointestinal diseases. These diseases promote starvation through decreased assimilation of the diet by: (1) blocking the transit of dietary constituents to the intestinal absorbing surface, (2) impairing normal processes of pancreatic or biliary digestion, or (3) preventing the intestinal mucosal transport of dietary constituents. Diseases that are characterized by increased catabolism of stored energy and protein include acute surgical or medical critical illness and acute or chronic inflammatory or infectious disorders affecting diverse organ systems. Other chronic diseases promote malnutrition through mixed mechanisms that contribute to abnormal nutrient metabolism. Both AIDS and disseminated malignancy, for example, cause progressive malnutrition through combinations of anorexia and futile cycles of fatty acid and glucose metabolism. Chronic obstructive pulmonary disease increases risk of malnutrition through the increased energy expenditure of labored respiration, chronic indolent bronchial infection, and the anorexic side effects of many bronchodilating drugs. Chronic liver disease is often associated with PCM caused by the cumulative effects of anorexia; decreased biliary circulation; and abnormal lipid, carbohydrate, and protein metabolism. The chronic intestinal inflammation of Crohn's disease or ulcerative colitis accelerates fecal losses of protein, electrolytes, and zinc.

CLINICAL EVALUATION OF THE MALNOURISHED PATIENT

THE PATIENT HISTORY

The clinical nutritional history should include diet and weight change, socioeconomic conditions, and symptoms unique to each clinical setting ([Table 74-2](#)). Social and economic conditions that may lead to poverty include inadequate income, homelessness, and activities that restrict real income and promote involuntary diet restriction, such as drug abuse or chronic alcoholism. Anorexia, or loss of appetite, is a feature of psychiatric disorders, such as anorexia nervosa and neurodegenerative dementia in the elderly. Many self-selected, inadequate diets may promote malnutrition. During binge drinking, chronic alcoholics typically substitute more than half their daily food calories with excessive amounts of ethanol, the metabolism of which consumes energy and promotes unbalanced metabolism of fat and carbohydrates. Other inadequate diets include unbalanced and commercially promoted formulas for rapid weight loss and strict vegetarianism, which may lead to selective deficiencies of specific nutrients such as vitamin B₁₂ and iron.

Digestive diseases are major causes of malnutrition, both in the inpatient and outpatient settings. The malnourished patient with digestive disease may present with symptoms of: (1) dysphagia or recurrent vomiting due to benign or malignant esophageal or gastrointestinal obstruction; (2) chronic diarrhea due to abnormal pancreatic or biliary digestion, intestinal mucosal malabsorption, or protein-losing enteropathy; or (3) recurrent abdominal pain exacerbated by eating, as occurs in patients with chronic pancreatitis, inflammatory bowel disease, or intestinal ischemia.

On the general medical service,[PCM](#) is prevalent in patients with multiple chronic illnesses that are associated with anorexia, recurrent stress, and abnormal nutrient

metabolism. In addition, PCM is comorbid with chronic recurrent pancreatitis, renal failure, chronic liver disease, chronic obstructive pulmonary disease, disseminated cancer, and chronic infections such as AIDS and tuberculosis. Depending on the severity of injury or illness, critically ill surgical and medical patients predictably develop stress-related PCM if increased nutritional needs are not met after 5 to 10 days.

THE PHYSICAL EXAMINATION

A careful physical examination can both characterize and define the extent of malnutrition. Measurements of unclothed weight and height are essential for establishing the severity of malnutrition in all patients but may be confounded by the effects of fluid overload as a result of edema and ascites. The normal values for weight (in kg) and height (in cm) in men and women are provided in [Table 74-3](#). These values can be adjusted by $\pm 10\%$ to account for variability in body frame.

Anthropometry Measurements of subcutaneous fat and skeletal muscle are important to determine the severity of [PCM](#). Using specialized calipers and a tape measure, anthropometry estimates body fat from the thickness of the skin-fold of the posterior mid-upper arm. Anthropometric measurements in healthy and malnourished adults are shown in [Table 74-4](#). Mid-arm muscle circumference is estimated from the equation:

The use of anthropometry is limited by the requirement for specialized calipers, the experience of the observer, and potential confounding effects of edema or dehydration.

Specific Physical Findings of Malnutrition During the conventional physical examination, the observant and experienced clinician can identify multiple and specific findings of [PCM](#) and its associated micronutrient deficiencies ([Chap. 75](#)). A variety of nutritional deficiencies can be identified by examination of the patient's general appearance, including skin, hair, nails, mucus membranes, and neurologic system ([Table 74-5](#)). Initially, a pinch of the posterior upper arm may reveal loss of subcutaneous fat in the malnourished patient. Hollowing of the temporal muscles, wasting of upper arms and thigh muscles, easily plucked hair, and peripheral edema are all consistent with protein deficiency. Examination of the skin may reveal the papular keratitis ("goose bump rash") of vitamin A deficiency, perifollicular hemorrhages of vitamin C deficiency, ecchymoses of vitamin K deficiency, the "flaky paint" lower extremity rash of zinc deficiency, hyperpigmentation of skin-exposed areas from niacin deficiency, seborrhea of essential fatty acid deficiency, spooning of nails in iron deficiency, and transverse nail pigmentation in protein deficiency. The eye examination yields conjunctival pallor of anemia, pericorneal and corneal opacities of severe vitamin A deficiency ("Bitot spots"), and nystagmus and isolated ocular muscle paresis of thiamine deficiency. The oral examination may reveal angular stomatitis and cheilosis of either riboflavin or niacin deficiency; glossitis with smooth and red tongue of riboflavin, niacin, vitamin B₁₂, or pyridoxine deficiency; and hypertrophied bleeding gums of vitamin C deficiency. Examination of the neurologic system, particularly in the setting of chronic alcohol abuse, may detect memory loss with confabulation, a wide-based gait, and past pointing, which, together with ophthalmoplegia and peripheral neuropathy, constitute the Wernicke-Korsakoff syndrome of thiamine deficiency. Other neurologic causes of

dementia include pellagra due to niacin and/or tryptophan deficiency. Additional causes of peripheral neuropathy include deficiencies of pyridoxine or vitamin E; loss of distal vibratory and position sense is characteristic of the subacute combined degeneration of vitamin B₁₂ deficiency.

LABORATORY ASSESSMENT

Selected use of laboratory tests, most of which are widely available, is essential for characterizing and quantifying malnutrition. Laboratory findings that are often attributed to chronic disease may, in actuality, reflect the response to [PCM](#) or selected micronutrient deficiencies in the setting of chronic illness.

Serum Visceral Proteins Serum albumin, which has a 2- to 3-week half-life, is a highly sensitive but nonspecific measure of [PCM](#). A normal serum albumin level in a well-hydrated patient is inconsistent with PCM. On the other hand, a low serum albumin level must be interpreted in its clinical context, since the concentration of albumin is decreased in the setting of increased plasma volume (as seen in acute trauma or sepsis and in chronic liver, renal, or cardiopulmonary failure). The acute stress of surgery, sepsis, or other acute inflammatory illness lowers the serum albumin level because of a combination of increased circulating extracellular volume and [TNF](#)- α -mediated inhibition of albumin synthesis. Hepatic albumin synthesis is inhibited in the setting of liver cirrhosis, AIDS, and disseminated cancer, whereas albumin loss from the body is accelerated in inflammatory bowel diseases, including ulcerative colitis, Crohn's disease, and radiation enteritis. Several shorter-lived visceral proteins can also be measured for estimation of the severity of PCM. These include transferrin (1-week half-life), prealbumin or retinol-binding protein complex (2-day half-life), and fibronectin (1-day half-life). However, like the serum albumin level, the circulating level of each of these proteins is affected by the changes in extracellular volume that occur in acute and chronic illnesses.

Vitamin and Mineral Assays Specific micronutrient deficiencies can be measured by a variety of serum and red blood cell assays, often utilizing high-performance liquid chromatography or enzyme or microbiologic assays ([Chap. 75](#)). Commonly available assays and their interpretations are listed in [Table 74-6](#). [PCM](#) is typically associated with low serum levels of vitamin A, zinc, and magnesium. Abnormal digestion and absorption of dietary fat are associated with deficiencies of fat-soluble vitamins A, D, and E, whereas intestinal mucosal malabsorption (as in celiac disease) is commonly associated with additional deficiencies of iron and folic acid. Chronic alcoholism is frequently associated with thiamine, folate, vitamin A, and zinc deficiencies. Vitamin B₁₂ deficiency due to achlorhydria occurs in up to 15% of elderly individuals as well as in those with pernicious anemia or with diseases involving the terminal ileum. As described in [Chap. 105](#), both folate and vitamin B₁₂ deficiencies are associated with elevations in plasma homocysteine; vitamin B₁₂ deficiency can also elevate the plasma level of methylmalonic acid.

Assessment of Immune Function [PCM](#) is associated with atrophy of thymic-dependent lymphoid structures and reduced T cell-mediated immunity. Conversely, B cell-mediated production of immunoglobulins is usually unaffected. Total lymphocyte count (total white cell count $\times$ fraction as lymphocytes) is often $<1000/\mu\text{L}$ in

PCM and may be accompanied by anergy to common skin test antigens. While sensitive for PCM, these measures of cell-mediated immunity are nonspecific and can be affected by other disorders such as acute or chronic infections, uremia, or immunosuppressive therapy.

SPECIALIZED PROCEDURES FOR NUTRITIONAL ASSESSMENT

Several specialized procedures are used to assess energy and protein stores and energy expenditure in malnourished patients. These procedures may be employed during the initial nutritional assessment or may serve as an index of the efficacy of nutritional support during the treatment of malnourished patients.

Bioelectric Impedance Analysis Bioelectric impedance analysis (BIA) is a simplified and portable method for measurement of body fat, [FFM](#), and total-body water. BIA is based on differences in the electric conductivity of a weak current between electrodes placed on the dorsal surfaces of the hands and feet. The measurement reflects differences in the impedance to electric current, which is greatest through fat and least through water. Lean body mass can be calculated as the difference between fat mass and body weight or as total-body water divided by 0.73.

Overall, [BIA](#) is most useful in assessing body fat and [FFM](#) in stable patients and in those who suffer from conditions leading to relative starvation. However, BIA can also be used to assess critically ill patients with decreased intracellular water space and [BCM](#) and expanded extracellular compartment size. Reduced BCM correlates inversely with increased metabolic rate. BIA may be confounded in AIDS patients receiving protease-inhibitor therapy, if they exhibit lipodystrophy with associated redistribution of interscapular, abdominal, and breast fat ([Chap. 309](#)).

Energy Expenditure Body weight and energy balance are sustained in health by the consumption of dietary calories in an amount equal to the daily expenditure of energy. Therefore, caloric needs can be determined from the estimated daily total energy expenditure (TEE), which is composed of basal or resting energy expenditure (REE, about 75% of total), the thermic expenditures of digestion (about 10% of total), and modest physical activity (about 15% of total). The REE is directly proportional to both the [FFM](#) and [BCM](#) and can be estimated in healthy people using the Harris and Benedict formula on the basis of weight in kg (W), height in cm (H), and age in years (A):

A simplified bedside estimation for [TEE](#) in sick patients is 25 kcal/kg of body weight, to which is added 10% for digestion or metabolism of intravenous or enteral nutrition. In the acutely ill patient, one should include an additional 12.5% for each degree of fever over 37°C, as well as an additional multiplier commensurate with the severity of illness (e.g., 25% for general surgery, 50% for sepsis, and 100% for extensive third-degree burns).

While [REE](#) can be predicted by the Harris-Benedict equations in healthy persons, it is

decreased in starvation because of hypometabolism. In contrast, REE is increased in the hypermetabolic stress that accompanies critical illness. REE and caloric requirements cannot be predicted in certain clinical conditions. These include the relatively starved, chronically ill patient admitted with a critical illness, the obese patient who develops a critical illness on the background of both increased body fat and [FFM](#), or the patient with chronic liver disease accompanied by combinations of anorexia and ongoing hepatic inflammation. In these situations, REE can be measured accurately by the gas-exchange method of indirect calorimetry. In practice, indirect calorimetry is performed at the bedside using a mobile metabolic cart. This procedure is applicable to ventilator-independent and -dependent patients whose fractional intake of oxygen is less than 0.45. Because the goal is to reach an accurate approximation of the 24-h energy requirement, measurements must be taken at intervals during the day and must account for several variables, including food intake and activity. To calculate the energy cost of metabolism by indirect calorimetry, the volumes (V) of oxygen consumed and carbon dioxide produced are measured over a given period of time, according to the modified Weir equation where

Indirect calorimetry also provides the respiratory quotient (RQ), which is the ratio of carbon dioxide produced to oxygen consumed during the process of gas collection. The RQ decreases when fat is the predominant substrate for metabolism (as in starvation) and increases when the contribution of carbohydrate increases (as during stress with gluconeogenesis). In healthy individuals, the RQ usually falls between 0.80 and 0.90. A RQ <0.7 is consistent with active ketogenesis from endogenous fatty acid metabolism with limited generation of carbon dioxide. An RQ >1.0 indicates net lipogenesis, or the conversion of substrate carbohydrate to fat, a situation that occurs with overfeeding. Values that fall outside the range of 0.65 to 1.25 suggest an error in measurement technique.

Creatinine Excretion in the 24-h Urine Creatinine, the metabolic product of skeletal muscle creatine, is produced at a constant rate and in an amount directly proportional to skeletal muscle mass. With steady-state day-to-day renal function, each gram of creatinine in the 24-h urine collection represents 18.5 g of fat-free skeletal muscle. Since skeletal muscle is the major component of [FFM](#) and [BCM](#), measurement of creatinine in the 24-h urine collection can be used as a relative measure of these body compartments during the initial assessment and/or during the course of nutritional support. The *creatinine coefficient* represents the amount of creatinine excreted per kilogram of body weight; it is equal to 23 mg/kg of ideal body weight in men and 18 mg/kg of ideal body weight in women. The *creatinine-height index* represents the ratio of the measured 24-h urine creatinine excretion to the value predicted by the creatinine coefficient for the patient's ideal body weight. These values can be calculated from estimation of the patient's ideal body weight ([Table 74-3](#)) or from tables that relate creatinine excretion to height in men and women ([Table 74-7](#)). In practice, the accuracy of the 24-h urine creatinine depends primarily on completeness of the urine collection. Together with variations due to fever and fluctuations in dietary intake, inaccuracies of urine collections may result in as much as 10% error in the quantitative 24-h urine creatinine measurement. The constancy of creatinine excretion depends on steady-state renal function, and unpredictable creatinine excretion may occur through

feces or skin in patients with serum creatinine levels $>530 \text{ } \mu\text{mol/L}$ ($>6 \text{ mg/dL}$). The presence of ascites, however, apparently does not compromise the accuracy of the 24-h urine creatinine as a reflection of FFM or BCM in patients with chronic liver disease.

Urine Nitrogen Excretion and Nitrogen Balance Nitrogen balance provides an index of protein gain or loss: 1 g nitrogen is equivalent to 6.25 g protein. Nitrogen balance can be assessed by measuring the difference between nitrogen consumed through the mouth, enteral tube, or intravenous sources and nitrogen excreted in the urine, feces, and other intestinal sources. Protein requirements to achieve zero or positive balance are less in starvation states, where daily protein losses are minimized because of hypometabolism, than in clinical states of stress, where the catabolism of skeletal muscle is accelerated for gluconeogenesis. Accurate measurement of nitrogen balance requires complete measurement of nitrogen losses from all possible excretory routes. In most cases, total urine nitrogen can be calculated by dividing 24-h urinary urea nitrogen by 0.85 and assuming approximately 2 g/d for nitrogen losses in feces and sweat. On the other hand, when the clinical condition includes extensive diarrhea and/or protein losses from pancreatic or enterocutaneous fistulas, the accuracy of nitrogen balance requires measurement of total nitrogen by the modified Kjeldahl technique in both urine and enteric sources. Total nitrogen measurements are also advisable in patients with liver failure, where urinary ammonia becomes a major and alternative source of nitrogen.

INTEGRATED BEDSIDE NUTRITIONAL ASSESSMENT

Several different approaches have been developed in order to simplify the process of nutritional assessment by using selective measurements that relate malnutrition to the specific medical condition and the severity of the underlying disease process.

Subjective Global Assessment This approach incorporates historic and physical findings as a basis for nutrition assessment by the trained physician. Major components in the history include evaluation of the extent of recent weight loss, changes in dietary intake, presence of significant gastrointestinal symptoms persisting more than 2 weeks, alterations in functional status, and the metabolic demand of the patient's underlying disease. Emphasis in the physical examination is placed on findings of depletion of subcutaneous body fat; skeletal muscle wasting; typical changes in skin, mucus membranes, and neurologic examination; as well as the presence of edema. Integration of the historic and physical data permits ranking of patients according to the following categories: adequate nutrition, moderate malnutrition, or severe malnutrition. Though the developers of the subjective global assessment have reported good sensitivity and specificity, the approach is still quite dependent on the training and experience of the clinician.

Prognostic Nutritional Assessment Several paradigms have been developed to link different parameters of nutritional assessment with clinical prognosis. Each approach links specific features of malnutrition with certain measurements of cell-mediated immunity, since abnormal immune function is a common pathway for increased risk in the malnourished patient ([Fig. 74-3](#)). A surgical prognostic nutritional index predicts morbidity based on preoperative measurements of serum albumin, transferrin, triceps skin-fold thickness, and delayed hypersensitivity to skin-test antigens.

Another [PCM](#) score was developed to link survival in alcoholic liver disease to both skin-fold and mid-arm muscle measurements; the creatinine-height index; values for serum albumin, transferrin, prealbumin, and retinol-binding protein; the total lymphocyte count; and the skin-test response to a series of antigens. The Maastricht index predicts survival in patients with serious gastrointestinal diseases on the basis of factors related to serum albumin, retinol-binding protein, lymphocyte count, and deviation from the patient's ideal body weight.

(Bibliography omitted in Palm version)

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75. VITAMIN AND TRACE MINERAL DEFICIENCY AND EXCESS- *Robert M. Russell*

Vitamins and trace minerals are required constituents of the human diet since they are either inadequately synthesized or not synthesized in the human body. Only small amounts of these substances are needed for carrying out essential biochemical reactions (e.g., acting as coenzymes or prosthetic groups). Overt vitamin or trace mineral deficiencies are rare in western countries due to a plentiful, varied, and inexpensive food supply; however, multiple nutrient deficiencies may appear together in persons who are ill or alcoholic. Moreover, subclinical vitamin and trace mineral deficiencies, as diagnosed by laboratory testing, are quite common in the normal population -- especially the geriatric population.

Body stores of vitamins and minerals vary tremendously. For example, vitamin B₁₂ and vitamin A stores are large, and an adult may not become deficient for 1 or more years after being on a depleted diet. However, folate and thiamine may become depleted within weeks when eating a deficient diet. Therapeutic modalities can deplete essential nutrients from the body; for example, hemodialysis removes water-soluble vitamins, which must be replaced by supplementation.

There are several roles for vitamins and trace minerals in diseases: (1) deficiencies of vitamins and minerals may be caused by disease states such as malabsorption; (2) both deficiency and excess of vitamins and minerals can cause disease in and of themselves (e.g., vitamin A intoxication and liver disease); and (3) vitamins and minerals in high doses may be used as drugs (e.g., niacin for hypercholesterolemia). The hematologic-related vitamins and minerals ([Chaps. 105,107](#)) are not considered in this chapter, nor are the bone-related vitamins and minerals (vitamin D, calcium, phosphorus; [Chap. 340](#)), as they are covered elsewhere.

VITAMINS

THIAMINE (VITAMIN B₁)

Thiamine was the first B vitamin to be identified and is therefore also referred to as vitamin B₁. Thiamine pyrophosphate, the coenzyme form of thiamine, is required for branched-chain amino acid metabolism and carbohydrate metabolism ([Fig. 75-1](#)).

Thiamine functions in the decarboxylation of α -ketoacids, such as pyruvate and α -ketoglutarate, and branched-chain amino acids and thus is a source of energy generation. In addition, thiamine pyrophosphate acts as a coenzyme for a transketolase reaction that mediates the conversion of hexose and pentose phosphates. It has also been postulated that thiamine plays a role in peripheral nerve conduction, although the exact chemical reactions underlying this function are unknown.

Absorption and Requirements At high doses, thiamine is absorbed by a passive mechanism; at low doses, it is absorbed by a carrier-mediated, active transport system and becomes phosphorylated in the process. Once absorbed, thiamine circulates bound to plasma proteins (mainly albumin) and erythrocytes. Storage sites for thiamine include muscle, heart, liver, kidney, and brain, although muscle is the principal storage site. The total body store of thiamine, mainly in the form of thiamine pyrophosphate, is approximately 30 mg, and its biologic half-life ranges between 9 and 18 days.

Given the fact that thiamine is involved in carbohydrate metabolism and energy generation, the Recommended Dietary Allowance (RDA) for males has been adjusted upward to account for increased energy utilization. Experiments have shown that heavy athletic training also increases thiamine utilization slightly. The RDA for thiamine is 1.2 mg/d for males and 1.1 mg/d for females. The median intake of thiamine in the United States from food alone is 2 mg/d. There is a 10% increase in the need for thiamine in pregnancy, and a small further increase in lactating females.

Primary food sources for thiamine include yeast, pork, legumes, beef, whole grains, and nuts. Milled and polished rice contain little, if any, thiamine. Thiamine deficiency is therefore more common in cultures that rely heavily on a rice-based diet. The molecule is heat-sensitive and is destroyed at pH > 8. Tea, coffee (caffeinated and decaffeinated), raw fish, and shellfish contain thiamineases, which can destroy the vitamin. Thus, drinking large amounts of tea or coffee can theoretically lower thiamine body stores.

Deficiency Most dietary deficiency of thiamine worldwide is the result of poor dietary intake. In western countries, the primary causes of thiamine deficiency are alcoholism and chronic illness, such as cancer. Alcohol is known to interfere directly with the absorption of thiamine and with the synthesis of thiamine pyrophosphate. Malnourished individuals with alcoholic liver disease are also at increased risk of thiamine deficiency because of diminished storage sites in liver and muscle. Thiamine should always be replenished when refeeding a patient with alcoholism, as carbohydrate repletion without adequate thiamine can precipitate acute thiamine deficiency.

Thiamine deficiency in its early stage induces anorexia, irritability, apathy, and generalized weakness. Prolonged thiamine deficiency causes beriberi, which is classically categorized as wet or dry, although there is considerable overlap. In either form of beriberi, patients may complain of pain and paresthesia. *Wet beriberi* presents primarily with cardiovascular symptoms, due to impaired myocardial energy metabolism and dysautonomia, and can occur after 3 months of a thiamine-deficient diet. Patients present with an enlarged heart, tachycardia, high-output congestive heart failure, peripheral edema, and peripheral neuritis. Patients with *dry beriberi* present with a symmetric peripheral neuropathy of the motor and sensory systems with diminished reflexes. The neuropathy affects the legs most markedly, and patients have difficulty rising from a squatting position.

Alcoholic patients with chronic thiamine deficiency may also have central nervous system manifestations known as *Wernicke's encephalopathy*, consisting of horizontal nystagmus, ophthalmoplegia (due to weakness of one or more extraocular muscles), cerebellar ataxia, and mental impairment ([Chap. 387](#)). When there is an additional loss of memory and a confabulatory psychosis, the syndrome is known as *Wernicke-Korsakoff syndrome*. Although this syndrome is generally described in alcoholic patients, there may be a genetic predisposition to Wernicke-Korsakoff that involves a variant transketolase isozyme.

In severely malnourished infants 2 to 3 months old, thiamine deficiency may occur precipitously with sudden cardiovascular failure and collapse, resulting in death within

hours. In addition, infants with thiamine deficiency may present with features suggesting meningitis, including vomiting, nystagmus, and convulsions. An aphonic presentation has also been described in which there is extreme irritability and either a very hoarse cry or total inability to emit any noise whatsoever (a silent scream).

The laboratory diagnosis of thiamine deficiency is usually made by a functional enzymatic assay of transketolase activity measured before and after the addition of thiamine pyrophosphate. A >25% stimulation by the addition of thiamine pyrophosphate (an activity coefficient of 1.25) is taken as abnormal. Thiamine or the phosphorylated esters of thiamine in serum or blood can also be measured by high-performance liquid chromatography (HPLC) to detect deficiency. Moreover, a urinary level of thiamine <27 ug per gram of creatinine per day is abnormal. In measuring urinary excretion of thiamine, one should make sure the patient is not taking diuretics, which increase thiamine excretion.

TREATMENT

In acute thiamine deficiency with either cardiovascular or neurologic signs, 100 mg/d of thiamine should be given parenterally for 7 days, followed by 10 mg/d orally until there is complete recovery. Cardiovascular improvement occurs in $\leq$ 12 h, and ophthalmoplegic improvement occurs within 24 h. Other manifestations gradually clear, although psychosis in the Wernicke-Korsakoff syndrome may be permanent or persist for several months. Consistent with this, pathologic changes occur in the cortex, cerebellum, and mammillary bodies of the thalamus. Parenteral thiamine should be given prophylactically to all chronic alcoholic patients in the emergency room, or as soon as they are admitted, to prevent precipitation of thiamine deficiency after the provision of glucose-containing solutions.

Thiamine-responsive conditions requiring pharmacologic doses of thiamine include branched-chain ketoaciduria (maple sugar urine disease), subacute necrotizing encephalopathy due to thiamine triphosphate deficiency in the brain (Leigh syndrome), thiamine-responsive lactic acidosis, and thiamine-responsive megaloblastic anemia associated with diabetes mellitus and deafness ([Chap. 353](#)). The gene for this recessive disorder, *SLC19A2*, encodes a thiamine transporter.

Toxicity Although anaphylaxis has been reported after high doses of thiamine, no adverse effects have been recorded from either food or supplements at high doses. Thiamine supplements may be bought over the counter in doses of up to 50 mg/d.

RIBOFLAVIN (VITAMIN B₂)

Riboflavin is important for the metabolism of fat, carbohydrate, and protein, reflecting its role as a respiratory coenzyme and an electron donor. Riboflavin is esterified with phosphoric acid in the body to form two coenzymes, flavin-mononucleotide (FMN) and flavin adenine dinucleotide (FAD) which are involved in a variety of cellular oxidation-reduction processes. Enzymes that contain FAD or FMN as prosthetic groups are known as *flavoenzymes* (e.g., succinic acid dehydrogenase, monoamine oxidase, glutathione reductase). Riboflavin plays an important role in niacin metabolism, since flavoenzymes act as intermediaries in the oxidation of the reduced forms of

nicotinamide adenine dinucleotide (NAD) and NAD phosphate (NADP).

Riboflavin is normally absorbed by active and carrier-mediated saturable mechanisms, whereas diffusion is the principal mechanism of absorption at high concentrations. Riboflavin phosphorylation takes place mainly in the wall of the small intestine. Both [FMN](#) and [FAD](#) are bound to immunoglobulins and albumin in the circulation, and both forms are stored to some degree in liver and muscle.

Although much is known about the chemical and enzymatic reactions of riboflavin, the clinical manifestations of riboflavin deficiency are nonspecific and similar to those of other B vitamin deficiencies. Further, riboflavin deficiency usually occurs in combination with other water-soluble vitamin deficiencies ([Chap. 74](#)). Riboflavin deficiency is manifested principally by lesions of the mucocutaneous surfaces of the mouth (angular stomatitis, cheilosis, atrophic glossitis, magenta tongue, pharyngitis) and skin (seborrhea, genital dermatitis). In addition to the mucocutaneous lesions, corneal vascularization, anemia, and personality changes have been described with riboflavin deficiency.

Deficiency and Excess Riboflavin deficiency is almost always due to dietary deficiency. The requirement for riboflavin is increased during pregnancy and lactation and possibly by heavy exercise. The use of phenothiazines and antibiotics also appears to increase the need for riboflavin. Milk, other dairy products, and enriched breads and cereals are the most important dietary sources of riboflavin in the United States, although lean meat, fish, eggs, broccoli, and legumes are also good sources. Riboflavin is extremely sensitive to light, and milk should be stored in containers that protect against photodegradation. In non-milk-drinking societies (e.g., Central America), the laboratory diagnosis of riboflavin deficiency is common. Laboratory diagnosis of riboflavin deficiency can be made by measurement of red blood cell or urinary riboflavin concentrations or by measurement of erythrocyte glutathione reductase activity, with and without added [FAD](#). A stimulation (activity coefficient) of >1.4 is diagnostic of a deficient state. The [RDA](#) for riboflavin is 1.1 to 1.3 mg/d in adults, with slightly higher recommendations for lactating and pregnant women. Rare genetic defects of flavoprotein synthesis may require pharmacologic doses of riboflavin for treatment. Because the capacity of the gastrointestinal tract to absorb riboflavin is limited (~20 mg if given in one oral dose), riboflavin toxicity has not been described. Thus, the most recent revision of the RDAs did not set an upper limit for this nutrient.

NIACIN (VITAMIN B₃)

The term *niacin* refers to nicotinic acid and nicotinamide and their biologically active derivatives. Nicotinic acid and nicotinamide serve as precursors of two coenzymes, [NAD](#) and [NADP](#). These coenzymes are important in numerous oxidation and reduction reactions in the body. NAD and NADP serve as cofactors for dehydrogenases and are involved in the transfer of the hydride ion in many redox reactions. Thus, niacin is important in pentose, steroid, and fatty acid biosynthesis; glycolysis; protein metabolism; and the oxidation of fuels such as lactate, pyruvate, and alcohol. In addition, NAD and NADP are active in adenine diphosphate-ribose transfer reactions involved in DNA repair and calcium mobilization.

Absorption, Metabolism, and Requirements Nicotinic acid and nicotinamide are absorbed well from the stomach and small intestine. Both forms of niacin are absorbed by a sodium-dependent, facilitated diffusion mechanism at low doses, whereas passive diffusion occurs at high doses. Some storage of [NAD](#) takes place in the liver. The amino acid tryptophan can be converted to niacin with an efficiency of 60:1 by weight. Thus, the [RDA](#) for niacin is expressed in niacin equivalents. Greater conversion of tryptophan to niacin occurs in niacin-deficient states, pregnancy, and in women using oral contraceptives. However, a lower conversion efficiency occurs if a patient is vitamin B₆- or riboflavin-deficient. The drug isoniazid inhibits the conversion of tryptophan to niacin. The urinary excretion products of niacin include nicotinic acid and niacin oxide; however, the major urinary metabolites are 2-pyridone and 2-methyl nicotinamide, measurements of which are used in diagnosis of niacin deficiency.

The [RDA](#) for niacin is 16 niacin equivalents per day for men and 14 niacin equivalents for women. Median intakes of niacin in the United States considerably exceed these values. Diets that are corn-based can predispose to niacin deficiency due to the low tryptophan and niacin content. Niacin bioavailability is high from beans, milk, meat, and eggs; bioavailability from cereal grains is lower. Since flour is enriched with the "free" niacin (i.e., non-coenzyme form), bioavailability is excellent.

Deficiency Niacin deficiency causes *pellagra*, which is mostly found among people eating corn-based diets in parts of China, Africa, and India. Pellagra in North America is found mainly among alcoholics; in patients with congenital defects of intestinal and kidney absorption of tryptophan (Hartnup's disease; [Chap. 352](#)); and in patients with carcinoid syndrome ([Chap. 93](#)), in which there is increased conversion of tryptophan to serotonin. The early symptoms of pellagra include loss of appetite, generalized weakness and irritability, abdominal pain, and vomiting. Epithelial cell changes then ensue with stomatitis and bright-red glossitis, followed by a characteristic skin rash that is pigmented and scaling, particularly in skin areas exposed to sunlight. This rash is known as "Casal's necklace," when it rings the neck, and is seen in advanced cases. Vaginitis and esophagitis may also occur. Diarrhea (in part due to proctitis and in part due to malabsorption), depression, seizures, and dementia are also part of the pellagra syndrome -- the four D's: dermatitis, diarrhea, and dementia leading to death.

The diagnosis of niacin deficiency is based on low levels of the urinary metabolites 2-methyl nicotinamide and 2-pyridone. Treatment of pellagra consists of oral supplementation of 100 to 200 mg of nicotinamide or nicotinic acid three times daily for 5 days. High doses of nicotinic acid (3 g nicotinic acid per day) are used for the treatment of elevated cholesterol levels and in the treatment of types 2, 4, and 5 hyperlipidemias ([Chap. 344](#)).

Toxicity Prostaglandin-mediated flushing has been observed at daily doses as low as 50 mg of niacin when taken as a supplement or as therapy for hypertriglyceridemia. No toxicity has been seen from niacin derived from food sources. Flushing may be accompanied by skin dryness, itching, and headache. Premedication with aspirin may alleviate these symptoms. Nausea, vomiting, and abdominal pain also occur at similar doses of niacin. Hepatic toxicity is the most serious toxic reaction due to niacin and may present as jaundice with elevated AST and ALT levels. A few cases of fulminant hepatitis requiring liver transplantation have been reported at doses of 3 to 9 g/d. Other

toxic reactions include glucose intolerance, macular edema, and macular cysts. It is not clear whether sustained-release forms of nicotinic acid are more toxic than regular forms. The upper limit for daily niacin intake has been set at 35 mg. However, this upper limit does not pertain to the therapeutic use of niacin.

PYRIDOXINE (VITAMIN B₆)

Vitamin B₆ refers to a family of compounds including pyridoxine, pyridoxal, pyridoxamine, and their 5¢-phosphate derivatives. 5¢-Pyridoxal phosphate (PLP) is a cofactor for more than 100 enzymes involved in amino acid metabolism (e.g., 5¢-PLP is a cofactor for the transulfuration enzymes involved in the conversion of homocysteine to cystathionine; [Chap. 352](#)). Vitamin B₆ is also involved in heme and neurotransmitter synthesis and in the metabolism of glycogen, lipids, steroids, sphingoid bases, and several vitamins, including the conversion of tryptophan to niacin.

Absorption and Metabolism Approximately 75% of vitamin B₆ is absorbed from a mixed diet by a nonsaturable, passive process. Much of dietary vitamin B₆ is in the phosphorylated form, and the phosphate must be removed by intestinal alkaline phosphatase before absorption takes place. Once absorbed, the vitamin becomes rephosphorylated in the liver, where the various forms can be interconverted. In the liver, [PLP](#) binds avidly to cellular proteins and albumin. Since these binding proteins protect it from phosphatase activity, tissue levels of PLP can become quite high with continuous supplementation. Sixty mg of vitamin B₆ is stored in the body, and much is in the form of PLP bound to phosphorylase A in muscle. The biologic half-life of vitamin B₆ is 25 days.

Dietary Sources Plants contain vitamin B₆ in the form of pyridoxine, whereas animal tissues contain [PLP](#) and pyridoxamine phosphate. The vitamin B₆ contained in plants is less bioavailable than that from animal tissues. All forms of vitamin B₆ are labile in alkaline conditions. Rich food sources of vitamin B₆ include legumes, nuts, wheat bran, and meat, although the vitamin is present in all food groups. The [RDA](#) for young adults (both males and females) has been set at 1.3 mg/d. For older adults, the RDA is slightly higher (1.5 mg/d for women, 1.7 mg/d for men).

Deficiency Symptoms of vitamin B₆ deficiency include seborrheic dermatitis, glossitis, stomatitis, and cheilosis, as frequently seen with other B vitamin deficiencies ([Chap. 74](#)). In addition, severe vitamin B₆ deficiency can lead to generalized weakness, irritability, peripheral neuropathy, abnormal electroencephalograms, and personality changes including depression and confusion. In infants, diarrhea, seizures, and anemia have been reported. Microcytic, hypochromic anemia is due to diminished hemoglobin synthesis, since the first enzyme involved in heme biosynthesis (amino-levulinate synthase) requires [PLP](#) as a cofactor ([Chap. 104](#)). In some case reports, platelet dysfunction has also been reported. Since vitamin B₆ is necessary for the conversion of homocysteine to cystathionine, it is possible that chronic low-grade vitamin B₆ deficiency may result in hyperhomocystinemia and increased risk of cardiovascular disease ([Chaps. 242](#) and [352](#)).

Certain medications such as isoniazid, L-dopa, penicillamine, and cycloserine interact with [PLP](#) due to a reaction with carbonyl groups. Oral contraceptives have been reported

to decrease vitamin B₆ status indicators, although the mechanism for this is uncertain. Alcoholism also decreases vitamin B₆ status due to poor diet, liver disease, and the fact that acetaldehyde can compete with PLP for protein binding, leading to increased degradation and excretion. The increased ratio of aspartate aminotransferase (AST or SGOT) to alanine aminotransferase (ALT or SGPT) seen in alcoholic liver disease reflects the relative vitamin B₆ dependence of ALT. Vitamin B₆ requirements are higher in preeclampsia, eclampsia, and hemodialysis. Vitamin B₆ dependency syndromes that require pharmacologic doses of vitamin B₆ are rare, but include cystathionine b-synthase deficiency, pyridoxine-responsive (primarily sideroblastic) anemias, and gyrate atrophy with chorioretinal degeneration due to decreased activity of the mitochondrial enzyme ornithine aminotransferase. In these situations, 100 to 200 mg/d of oral vitamin B₆ are required for treatment.

High doses of vitamin B₆ have been used to treat carpal tunnel syndrome, premenstrual tension, schizophrenia, autism, and diabetic neuropathy but have not been found to be effective.

The laboratory diagnosis of vitamin B₆ deficiency is generally made on the basis of low plasma [PLP](#) values (<20 nmol/L). Other measures of vitamin B₆ deficiency include low erythrocyte levels of PLP, low plasma pyridoxal, and low urinary levels of 4-pyridoxic acid. Treatment of vitamin B₆ deficiency is 50 mg/d; higher doses of 100 to 200 mg/d are given if vitamin B₆ deficiency is related to medication use. Vitamin B₆ should not be given with L-dopa, since the vitamin interferes with the action of this drug.

Toxicity The safe upper limit for vitamin B₆ has been set at 100 mg/d, although the lowest dose at which toxicity (sensory neuropathy) has been seen is 500 mg/d. No adverse effects have been associated with high intakes of vitamin B₆ from food sources only. When toxicity occurs, it causes a severe sensory neuropathy, leaving patients unable to walk. Some cases of photosensitivity and dermatitis have also been reported.

VITAMIN C

Both ascorbic acid and its oxidized product dehydroascorbic acid are biologically active. Vitamin C participates in oxidation-reduction reactions and hydrogen ion transfer reactions. As an antioxidant, vitamin C donates electrons to quench reactive free radical and oxygen species. It also acts to regenerate other antioxidants such as vitamin E, flavonoids, and glutathione. Other actions of vitamin C include promotion of nonheme iron absorption, carnitine biosynthesis, and the conversion of dopamine to norepinephrine. Vitamin C is also important for connective tissue metabolism and cross-linking and is a component of many drug-metabolizing enzyme systems, particularly the mixed-function oxidase systems. As such, the vitamin participates in the synthesis of corticosteroids, aldosterone, and the metabolism of cholesterol. Vitamin C also participates in enzymatic reactions requiring a reduced metal, although the exact molecular basis for this role has not been delineated.

Absorption and Physiology Vitamin C is absorbed by an energy-dependent, saturable transport system, and a progressively smaller proportion of the vitamin is absorbed with increasing dose. Almost complete absorption of the vitamin occurs if <100 mg is administered in a single dose; however, only 50% or less is absorbed at doses >1 g.

Enhanced degradation and fecal and urinary excretion of vitamin C occur at higher intake levels. High levels of the reduced form of vitamin C are contained in white blood cells, lens tissue, and brain. The maximum body pool in adult males is approximately 1500 mg, and 3% of this body pool is turned over each day, resulting in a half-life of approximately 18 days.

Dietary Sources and Requirements Good dietary sources of vitamin C include citrus fruits, green vegetables (especially broccoli), tomatoes, and potatoes. Appreciable amounts of vitamin C may be consumed as an antioxidant food additive, and the consumption of five servings of fruits and vegetables a day provides vitamin C in excess of the [RDA](#) of 60 mg/d for males and females. Moreover, approximately 40% of the U.S. population takes vitamin C as a dietary supplement. Vitamin C requirements are increased slightly to 70 mg in pregnancy and are increased further to 90 to 95 mg/d during lactation. Smoking, hemodialysis, and stress (e.g., infection, trauma) appear to increase vitamin C requirements. "Natural forms" of vitamin C are no more bioavailable than synthetic forms.

Deficiency Vitamin C deficiency causes scurvy; in the United States, this is seen primarily among poor and elderly people and alcoholics who consume <10 mg/d of vitamin C. Vitamin C deficiency has also been described among individuals consuming macrobiotic diets. Scurvy occurs when the body pool for vitamin C drops to <300 mg/d and plasma levels drop to <11 $\mu\text{mol/L}$. Symptoms of scurvy primarily reflect impaired formation of mature connective tissue and include bleeding into skin (petechiae, ecchymoses, perifollicular hemorrhages); inflamed and bleeding gums; and manifestations of bleeding into joints, the peritoneal cavity, pericardium, and the adrenal glands. Other generalized symptoms include weakness, fatigue, and depression. In children, vitamin C deficiency may cause impaired bone growth. Laboratory diagnosis of vitamin C deficiency is made on the basis of low plasma or leukocyte levels.

Administration of vitamin C (200 mg/d) results in marked improvement in the symptoms of scurvy in a matter of several days. High-dose vitamin C supplementation (e.g., 1 to 2 g/d) has been shown to slightly decrease the symptoms and duration of upper respiratory tract infections and to improve glycemic control. Vitamin C supplementation has also been reported to be useful in Chediak-Higashi syndrome ([Chap. 64](#)) and osteogenesis imperfecta ([Chap. 351](#)). It has been claimed that foods high in vitamin C may lower the incidence of certain cancers, particularly esophageal and gastric cancers. If proven, this effect may be due to the fact that vitamin C can prevent the conversion of nitrites and secondary amines to carcinogenic nitrosamines. However, one intervention study from China did not show vitamin C to be protective. Other chronic diseases for which diets high in vitamin C have been reported to be protective include cardiovascular disease, stroke, and cataracts. However, these studies are correlational, and no large-scale intervention studies have been reported.

Toxicity Taking >2 g of vitamin C in a single dose may result in abdominal pain, diarrhea, and nausea; doses >3 g have been reported to elevate blood levels of alanine aminotransferase, lactic acid dehydrogenase, and uric acid. Since vitamin C may be metabolized to oxalate, it has been feared that chronic, high-dose vitamin C supplementation could result in an increased prevalence of kidney stones. However, this has not been borne out in several trials, except in individual patients with preexisting

renal disease. Thus, it is reasonable to advise patients with a past history of kidney stones not to take large doses of vitamin C. There is also an unproven, but possible risk that chronic high doses of vitamin C could promote iron overload in patients taking supplemental iron. High doses of vitamin C can induce hemolysis in patients with glucose-6-phosphate dehydrogenase deficiency, and doses >1 g/d can cause false-negative guaiac reactions as well as interfering with tests for urinary glucose.

BIOTIN

Biotin is a water-soluble vitamin with a bicyclic structure. The vitamin plays an important role in gluconeogenesis and fatty acid synthesis and serves as a CO₂carrier on the surface of both cytosolic and mitochondrial carboxylase enzymes. The vitamin also functions in the catabolism of specific amino acids (e.g., leucine).

Biotin in food sources is bound to protein from which it must be cleaved in order to be absorbed. The enzyme biotinidase dissociates the vitamin and facilitates its subsequent transport. Excellent food sources of biotin include liver, soy, beans, yeast, and egg yolks, although egg white contains the protein avidin that strongly binds the vitamin and reduces its bioavailability. Biotin is contained in moderate amounts in legumes, nuts, mushrooms, cauliflower, and certain cereals. Although biotin is synthesized by intestinal bacteria, the relative importance of this source in humans is uncertain. The recommended intake of biotin for adults is 30 ug/d and 35 ug/d in lactating women.

Biotin deficiency has been induced by experimental feeding of egg white diets and in patients with short bowels who received biotin-free parenteral nutrition. In the adult, biotin deficiency results in mental changes (depression, hallucinations), paresthesia, anorexia, and nausea. A scaling, seborrheic, and erythematous rash may occur around the eyes, nose, and mouth as well as on the extremities. In infants, biotin deficiency presents as hypotonia, lethargy, and apathy. In addition, the infant may develop alopecia and a characteristic rash that includes the ears. Two types of inherited infantile biotin deficiency states have been described. Multiple carboxylase deficiency syndrome is an autosomal recessive disorder that is expressed during the first week of life and is characterized by severe metabolic ketoacidosis and dermatitis. Treatment requires pharmacologic doses of biotin, using up to 10 mg/d. Late-onset infantile biotin deficiency due to absorptive and transport defects occurs between 3 and 6 months with dermatitis, seizures, ataxia, hypotonia, and variable metabolic acidosis. The laboratory diagnosis of biotin deficiency can be established based on a decreased urinary concentration.

PANTOTHENIC ACID

Pantothenic acid is a component of coenzyme A and phosphopantetheine, which are involved in fatty acid metabolism and the synthesis of cholesterol, steroid hormones, and all compounds formed from isoprenoid units. In addition, pantothenic acid is involved in the acetylation of proteins. Pantothenic acid is actively transported when given at low doses, but it is passively absorbed when given at high doses. The vitamin is excreted in the urine, and the laboratory diagnosis of deficiency is made on the basis of low urinary vitamin levels.

The vitamin is ubiquitous in the food supply. Liver, yeast, egg yolks, and vegetables are

particularly good sources. The recommended adequate intake for adults is 5 mg/d. Human pantothenic acid deficiency has only been demonstrated in experimental feeding of diets low in pantothenic acid or by giving a specific pantothenic acid antagonist. The symptoms of pantothenic acid deficiency are nonspecific and include gastrointestinal disturbance, depression, muscle cramps, paresthesia, ataxia, and hypoglycemia. Pantothenic acid deficiency was thought to cause the burning feet syndrome seen in prisoners of war during World War II. No toxicity of this vitamin has been reported.

CHOLINE

Choline is a precursor for acetylcholine, phospholipids, and betaine. Choline is necessary for the structural integrity of cell membranes, cholinergic neurotransmission, lipid and cholesterol metabolism, and transmembrane signaling. Recently, a recommended adequate intake was set at 550 mg/d for adult males and 425 mg/d for adult females. Choline is thought to be a "conditionally essential" nutrient, in that de novo synthesis occurs in the liver and is less than the vitamin's utilization only under certain stress conditions. Choline deficiency has occurred in patients receiving parenteral nutrition devoid of choline. Deficiency results in fatty liver and elevated transaminase levels. The diagnosis of choline deficiency is made on the basis of low plasma levels.

Toxicity from choline results in hypotension, cholinergic sweating, diarrhea, salivation, and a fishy body odor. The upper limit for choline has been set at 3.5 g/d. Therapeutically, choline has been suggested for patients with dementia and for patients at high risk of cardiovascular disease, due to its ability to lower cholesterol and homocysteine levels. However, such benefits have yet to be documented.

VITAMIN A

Vitamin A, in the strictest sense, refers to retinol. However, the oxidized metabolites, retinaldehyde and retinoic acid, are also biologically active compounds. The term *retinoids* includes synthetic molecules that are chemically related to retinol. Retinaldehyde is the essential form of vitamin A that is required for normal vision, whereas retinoic acid is necessary for normal morphogenesis, growth, and cell differentiation. Retinoic acid does not function in vision and, in contrast to retinol, is not involved in reproduction. Vitamin A also plays a role in iron utilization, humoral immunity, T cell-mediated immunity, natural killer cell activity, and phagocytosis. Vitamin A is commercially available in esterified forms (e.g., acetate, palmitate) since it is more stable as an ester.

There are over 600 carotenoids in nature, and approximately 50 of these can be metabolized to vitamin A. β -Carotene is the most prevalent carotenoid in the food supply that has provitamin A activity. Although the breakdown of β -carotene should theoretically yield two molecules of vitamin A, the conversion of carotenoids to vitamin A, in fact, is much less efficient. It is estimated that 6 μ g or greater of dietary β -carotene is equivalent to 1 μ g of retinol, whereas 12 μ g or greater of other dietary provitamin A carotenoids (e.g., cryptoxanthin, α -carotene) is equivalent to 1 μ g of retinol.

Absorption and Metabolism Approximately 80% of preformed vitamin A is absorbed

from food, and absorption is via a carrier-mediated mechanism at low concentrations and passive diffusion at high concentrations. Approximately 15 to 30% of provitamin A carotenoids are absorbed passively from the diet, and the absorption becomes much less efficient at high dosage. The absorption of both vitamin A and carotenoids are partially dependent on an adequate bile concentration within the intestinal lumen for the formation of micelles. Once a provitamin A carotene is absorbed into the epithelial cell, a small proportion of it is split to form vitamin A. At higher doses of *b*-carotene, the conversion to vitamin A is less efficient, thereby preventing vitamin A toxicity. The absorption of both vitamin A and intact *b*-carotene is via the lymphatics after chylomicron formation.

Hepatic clearance of vitamin A in chylomicrons is efficient, and the liver contains approximately 90% of the vitamin A reserves. Approximately 10 to 40% of a vitamin A dose is oxidized or conjugated in the liver and excreted in urine or bile. Of a given dose of vitamin A, approximately 50% enters the liver storage pool. Storage of vitamin A takes place in the lipid storage (Ito) cell of the liver, which is also a collagen-producing cell. The liver secretes vitamin A in the form of retinol, which is bound to retinol-binding protein. Once this has occurred, the retinol-binding protein complex interacts with a second protein, transthyretin. This trimolecular complex functions to prevent vitamin A from being filtered by the kidney glomerulus, to protect the body against the toxicity of retinol and to allow retinol to be taken up by specific cell-surface receptors that recognize retinol-binding protein. A certain amount of vitamin A enters peripheral cells even if it is not bound to retinol-binding protein. After retinol is internalized by the cell, it becomes bound to a series of cellular retinol-binding proteins, which function as sequestering and transporting agents as well as cofactors for enzymatic reactions. Certain cells also contain retinoic acid-binding proteins, which have the same sequestering functions as well as enabling retinoic acid metabolism.

11-*cis*-Retinaldehyde functions as a visual pigment chromophore to capture light. Rhodopsin is composed of the protein opsin and retinaldehyde and is contained in the rod cells, whereas iodopsin is contained in cones. When the dark-adapted retina is exposed to light, the 11-*cis*-retinaldehyde contained in rhodopsin isomerizes to an all-*trans* form. This conformational change causes dissociation from the opsin, resulting in a nerve impulse and a visual response. Once the retina returns to dim light conditions, rhodopsin is regenerated.

Retinoic acid is a ligand for certain nuclear receptors that act as transcription factors. Two families of receptors (RAR and RXR receptors) are active in retinoid-mediated gene transcription. Retinoid receptors regulate transcription by binding as dimeric complexes to specific DNA sites, the retinoic acid response elements, in target genes ([Chap. 327](#)). The receptors can either stimulate or repress gene expression in response to their ligands. RAR binds all-*trans* retinoic acid and 9-*cis* retinoic acid, whereas RXR binds only 9-*cis* retinoic acid.

The retinoid receptors play an important role in controlling cell proliferation and differentiation. Retinoic acid is useful in the treatment of promyelocytic leukemia ([Chap. 111](#)). In this case, a gene rearrangement fuses the RAR to one of several other genes [e.g., t(15;17)], causing an apparent block in cell differentiation. Treatment with retinoic acid activates the RAR, dissociating repressor complexes and leading to cell

differentiation and more normal cell turnover. Retinoic acid is also used in the treatment of cystic acne because it inhibits keratinization, decreases sebum secretion, and possibly alters the inflammatory reaction ([Chap. 56](#)). RXRs dimerize with other nuclear receptors to function as coregulators of genes responsive to retinoids, thyroid hormone, and calcitriol. RXR agonists induce insulin sensitivity experimentally, perhaps because RXR is a cofactor for the peroxisome-proliferator-activated receptors (PPARs), which are targets for the thiazolidinedione drugs such as rosiglitazone and troglitazone ([Chap. 333](#)).

Dietary Sources The retinol equivalent (RE) is used to express the vitamin A value of food. One RE is defined as 1 μg of retinol (0.003491 mmol). In the past, 1 RE was considered to be equal to 6 μg of β -carotene, but additional studies indicate that 1 RE may, in fact, be equal to 12 to 20 μg of β -carotene from a dietary source. In older literature, vitamin A was often expressed in international units (IU), with 1 RE being equal to 3.33 IU of retinol and 12 IU of β -carotene, but these units are no longer in current medical or scientific use. The RDA for vitamin A is set at 1000 RE for adult males and 800 RE for adult females.

Liver and fish are excellent food sources for preformed vitamin A; vegetable sources of provitamin A carotenoids include dark-green and -colored fruits and vegetables. Diets consisting mainly of rice, wheat, maize, and tubers can produce vitamin A deficiency, as few carotenoids are contained in these foods. In areas where these foods are staples, children are particularly susceptible to vitamin A deficiency because neither breast nor cow's milk supplies enough vitamin A to prevent deficiency. Areas of the world where vitamin A deficiency is particularly prevalent include parts of Africa, South America, and Southeast Asia. Vitamin A deficiency occurs in more than 250,000 children each year, resulting in blindness and a 50% mortality rate within the year. In western countries, vitamin A deficiency is seen primarily among patients with diseases associated with fat malabsorption (e.g., celiac sprue, short-bowel syndrome). Concurrent zinc deficiency can interfere with the mobilization of vitamin A from liver stores as well as the synthesis of rhodopsin in the eye; thus vitamin A deficiency is exacerbated by concurrent zinc deficiency. Alcohol also interferes with the conversion of retinol to retinaldehyde in the eye by competing for alcohol (retinol) dehydrogenase. Drugs that interfere with the absorption of vitamin A include mineral oil, neomycin, and cholestyramine.

Deficiency Symptoms of vitamin A deficiency include hyperkeratotic skin lesions, night blindness (inability to see in dim light), dryness of the eyes, xerosis, and Bitot spots, which are white patches of keratinized epithelium appearing on the sclera ([Fig. 75-CD1](#)). Aggressive xerophthalmia can result in corneal ulceration. If untreated, proteolytic destruction and rupture of the cornea ensues with permanent blindness, although vitamin A treatment of patients with corneal ulcers can also result in blindness due to permanent corneal scarring. Children with vitamin A deficiency have increased mortality, primarily from infectious diseases, measles, respiratory diseases, and diarrhea. Extremely low birth weight infants (<1000 g) should be treated parenterally with 5000 IU (1500 μg or RE) of vitamin A three times a week for 4 weeks.

There are no specific deficiency signs or symptoms that result from carotenoid deficiency. However, dietary carotenoids have been suggested to protect against cataract formation, low-density lipoprotein (LDL) oxidation, and certain cancers. It was

hoped that β -carotene would be an effective chemopreventive for cancer because numerous epidemiologic studies had shown that diets high in β -carotene were associated with lower incidences of cancers of the respiratory and digestive system. However, intervention studies using high doses of β -carotene actually resulted in more lung cancers than in placebo-treated groups. Non-provitamin A carotenoids, such as lutein and zeaxanthin, have been suggested to protect against macular degeneration. The non-provitamin A carotenoid lycopene has been suggested to protect against prostate cancer. However, the effectiveness of these agents has not been proven by intervention studies, and the mechanisms underlying these purported biologic actions are unknown.

The diagnosis of vitamin A deficiency is made by measurement of serum retinol (normal range, 30 to 65 $\mu\text{g/dL}$), tests of dark adaptation, impression cytology of the conjunctiva (decreased numbers of mucous-secreting cells), or measurement of body storage pools, either directly by liver biopsy or by isotopic dilution after administering a stable isotope of vitamin A.

Vitamin A deficiency with ocular changes should be treated by administering 100,000 IU (30 mg) of vitamin A intramuscularly, or 200,000 IU (60 mg) orally. In areas of endemic vitamin A deficiency, this is followed by vitamin A capsules of 200,000 IU at 6-month intervals. Vitamin A deficiency in patients with malabsorptive diseases, who have abnormal dark adaptation or symptoms of night blindness without ocular changes, should be treated for 1 month with 50,000 IU/d (15 mg/d) orally of a water micelle preparation of vitamin A. This is followed by lower maintenance doses with the exact amount determined by monitoring serum retinol.

Toxicity Acute toxicity of vitamin A was first noted in Arctic explorers after eating polar bear liver and has been seen after administration of 150 mg in adults or 100 mg in children. Acute toxicity is manifest by increased intracranial pressure, vertigo, diplopia, bulging fontanels in children, seizures, and exfoliative dermatitis; it may result in death. Chronic vitamin A intoxication has been seen in normal adults who ingest 50,000 IU/d (15 mg/d) of vitamin A for a period of several months and in children who ingest 20,000 IU/d (6 mg/d). Manifestations include dry skin, cheilosis, glossitis, vomiting, alopecia, bone pain, hypercalcemia, lymph node enlargement, hyperlipidemia, amenorrhea, and features of pseudotumor cerebri with increased intracranial pressure and papilledema. Liver fibrosis with portal hypertension and bone demineralization may also result from chronic vitamin A intoxication. When vitamin A is provided in excess of pregnant women, congenital malformations have included spontaneous abortions, craniofacial abnormalities, and valvular heart disease. In pregnancy, the daily dose of vitamin A should not exceed 10,000 IU (3 mg). Elderly individuals appear to be more prone to vitamin A intoxication, as are alcoholics and patients with liver disease. In fact acute hepatitis may precipitate vitamin A intoxication in patients who have extremely high vitamin A stores in the liver. It should be noted that the commercially available retinoid derivatives are also toxic, including 13-*cis*-retinoic acid, which has been associated with birth defects. As a result, contraception should be continued for at least 1 year, and possibly longer, in women who have taken 13-*cis* retinoic acid.

High doses of carotenoids do not result in toxic symptoms. However, carotenemia, which is characterized by a yellowing of the skin (creases of the palms and soles) but

not the sclerae, may be seen after ingestion of >30 mg of β -carotene on a daily basis. Hypothyroid patients are particularly susceptible to the development of carotenemia due to impaired breakdown of carotene to vitamin A. Reduction of carotenes from the diet results in the disappearance of skin yellowing and carotenemia over a period of 30 to 60 days.

VITAMIN D (See [Chap. 340](#)).

VITAMIN E

Vitamin E is a collective name for a group of tocopherols and tocotrienols, the latter having an unsaturated sidechain. There are eight naturally occurring plant compounds with vitamin E activity. RRR- α tocopherol is the most active, while synthetic stereoisomers of vitamin E are less biologically active. Vitamin E acts as a chain-breaking antioxidant and is an efficient peroxyl radical scavenger, which protects [LDLs](#) and polyunsaturated fats in membranes from oxidation. A network of other antioxidants (e.g., vitamin C, glutathione) and enzymes maintains vitamin E in a reduced state. Vitamin E also inhibits prostaglandin synthesis and the activities of protein kinase C and phospholipase A₂.

Absorption and Metabolism Vitamin E is a fat-soluble vitamin and requires all the processes needed for micelle formation to be absorbed. About 15 to 40% is absorbed passively from a single physiologic dose, and there is less efficient absorption at high doses. Polyunsaturated fat may inhibit absorption. Vitamin E is taken up from chylomicrons by the liver, and an hepatic tocopherol transport protein is involved in intracellular vitamin E transport and incorporation into very low density lipoprotein (VLDL). The transport protein has particular affinity for the RRR isomeric form of a tocopherol; thus this natural isomer has the most biologic activity. In the circulation, vitamin E is bound to all lipoprotein classes and becomes widely distributed in tissues, with fat and muscle being the most important storage depots. Vitamin E metabolites are mainly excreted in feces, although some are also excreted in urine.

Requirement The [RDA](#) for vitamin E is currently 10 mg for adults. Additional vitamin E is recommended during pregnancy (12 mg/d) and lactation (14 mg/d). Vitamin E is widely distributed in the food supply. The RRR- α isomers are particularly high in sunflower oil, safflower oil, and wheat germ oil; γ tocotrienols are notably present in soybean and corn oils. Vitamin E is also found in meats, nuts, and cereal grains, and small amounts are present in fruits and vegetables. Vitamin E pills containing doses of 50 to 1000 mg are ingested by a large fraction of the U.S. population. In the older literature, 1 IU of vitamin E is equal to 1 mg *all-racemic* tocopherol acetate. Diets high in polyunsaturated fats may necessitate a slightly higher requirement for vitamin E.

Dietary deficiency of vitamin E does not exist. Vitamin E deficiency is seen only in severe and prolonged malabsorptive diseases, such as celiac disease, or after small-intestinal resection, leading to short-bowel syndrome. Children with cystic fibrosis or prolonged cholestasis may develop vitamin E deficiency characterized by areflexia and hemolytic anemia. Children with abetalipoproteinemia cannot absorb or transport vitamin E and become deficient quite rapidly. A familial form of isolated vitamin E deficiency also exists, which is due to a defect in the tocopherol transport protein.

Vitamin E deficiency causes axonal degeneration of the large myelinated axons and results in posterior column and spinocerebellar symptoms. Peripheral neuropathy is initially characterized by areflexia, with progression to an ataxic gait, and by decreased vibration and position sensations. Ophthalmoplegia, skeletal myopathy, and pigmented retinopathy may also be features of vitamin E deficiency. The laboratory diagnosis of vitamin E deficiency is made on the basis of low blood levels of a tocopherol (<5 ug/mL, or <0.8 mg of a tocopherol per gram of total lipids).

TREATMENT

Symptomatic vitamin E deficiency should be treated with 800 to 1200 mg of a tocopherol per day. Patients with abetalipoproteinemia may need as much as 5000 to 7000 mg/d. Children with symptomatic vitamin E deficiency should be treated with 400 mg/d orally of water-soluble esters; alternatively, 2 mg/kg per day may be administered intramuscularly. Vitamin E in high doses may protect against oxygen-induced retrolental fibroplasia and bronchopulmonary dysplasia in prematurity, as well as intraventricular hemorrhage of prematurity. Vitamin E has been suggested to increase sexual performance, to treat intermittent claudication, and to slow the aging process, but evidence for these properties is lacking. High doses (60 to 800 mg/d) of vitamin E have been shown in controlled trials to improve parameters of immune function, and there are two intervention studies showing that vitamin E at 400 to 800 mg/d may be protective against cardiovascular disease, possibly by inhibiting [LDL](#) oxidation. Also, supplemental intake of vitamin E (100 to 200 mg/d) has been associated with a decreased risk of cataracts.

Toxicity High doses of vitamin E (>800 mg/d) may reduce platelet aggregation and interfere with vitamin K metabolism and are therefore contraindicated in patients taking coumadin. Nausea, flatulence, and diarrhea have been reported at doses >1 g/d.

VITAMIN K

There are two natural forms of vitamin K: vitamin K I, also known as *phylloquinone*, from vegetable and animal sources, and vitamin K II, or *menaquinone*, which is synthesized by bacterial flora and found in hepatic tissue. *Menadione*, or vitamin K III, is a chemically synthesized pro-vitamin that can be converted to menaquinone by the liver.

Phylloquinone and menaquinones differ only in their lipophilic sidechains, and both are destroyed in an alkaline pH and by ultraviolet light.

Absorption and Physiology As with other fat-soluble vitamins, vitamin K absorption is dependent on normal pancreatic function and the presence of bile salts. Phylloquinones are absorbed by a saturable energy-dependent mechanism in the proximal small intestine, whereas menaquinones are absorbed by passive diffusion in the small intestine and colon. Approximately 100 ug of vitamin K is stored in the liver as well as in lung, bone marrow, kidneys, and adrenal glands. Most vitamin K circulates bound to [VLDL](#), although it is also carried by [LDL](#) and high-density lipoprotein (HDL). The half-life of vitamin K is only 1½ days, despite the presence of a vitamin K regeneration cycle.

Vitamin K is necessary for the posttranslational carboxylation of glutamic acid, which is

necessary for calcium binding to γ -carboxylated proteins such as prothrombin (factor II); factors VII, IX, and X; protein C; protein S; and proteins found in bone (bone gla, matrix gla protein, and osteocalcin). The importance of vitamin K for bone mineralization is not known. Warfarin-type drugs inhibit γ -carboxylation by preventing the conversion of vitamin K to its active hydroquinone form. Vitamin E, at high doses, may act as a vitamin K antagonist.

Dietary Sources Vitamin K is found in green leafy vegetables such as kale and spinach, but appreciable amounts are also present in butter, margarine, liver, milk, ground beef, coffee, and pears. Vitamin K is present in vegetable oils and is particularly rich in olive oil and soybean oil. The recommended intake of vitamin K is 70 $\mu\text{g}/\text{d}$ in adults. The average daily intake by Americans is estimated to be approximately 100 $\mu\text{g}/\text{d}$.

Deficiency The symptoms of vitamin K deficiency are due to hemorrhage, and newborns are particularly susceptible because of low fat stores, low breast milk levels of vitamin K, sterility of the infantile intestinal tract, liver immaturity, and poor placental transport. Intracranial bleeding, as well as gastrointestinal and skin bleeding, can be seen in vitamin K-deficient infants 1 to 7 days after birth. Thus, vitamin K (1 mg intramuscularly) is given prophylactically at the time of delivery.

Vitamin K deficiency in adults may be seen in patients with chronic small-intestinal disease (e.g., celiac disease, Crohn's disease), obstructed biliary tracts, or after small-bowel resection. Broad-spectrum antibiotic treatment can precipitate vitamin K deficiency by reducing gut bacteria, which synthesize menaquinones, as well as by inhibiting the metabolism of vitamin K. The diagnosis of vitamin K deficiency is usually made on the basis of an elevated prothrombin time or reduced clotting factors. Vitamin K may also be measured directly by [HPLC](#). In addition, undercarboxylated prothrombin and low gla levels in urine are indicative of vitamin K deficiency. Vitamin K deficiency is treated using a parenteral dose of 10 mg. For patients with chronic malabsorption, 1 to 2 mg/d of vitamin K may be given orally, or 1 to 2 mg/week can be taken parenterally. Patients with liver disease may have an elevated prothrombin time because of liver cell destruction as well as vitamin K deficiency. If an elevated prothrombin time does not improve on vitamin K therapy, it can be assumed that it is not the result of vitamin K deficiency.

Toxicity Parenteral doses of the water-soluble vitamin K derivative (menadione) have been reported to cause hemolytic anemia and hypobilirubinemia in infants. Toxicity from dietary phyloquinones and menaquinones has not been described. High doses of vitamin K can impair the actions of oral anticoagulants.

TRACE MINERALS (See [Table 75-1](#))

ZINC

Zinc is an integral component of many metalloenzymes in the body; it is involved in the synthesis and stabilization of proteins, DNA, and RNA and plays a structural role in ribosomes and membranes. Zinc is necessary for the binding of steroid hormone receptors and several other transcription factors to DNA and thereby plays an important

role in the regulation of gene transcription. Zinc is absolutely required for normal spermatogenesis, fetal growth, and embryonic development.

Absorption and Physiology Zinc is absorbed in the small intestine by a carrier-mediated mechanism. The absorption of zinc from the diet is inhibited by dietary phytate, fiber, oxalate, iron, and copper, as well as by certain drugs including penicillamine, sodium valproate, and ethambutol. The [RDA](#) for zinc is 15 mg in males and 12 mg in females, with an additional 3 mg in pregnancy and 4 to 7 mg during lactation. Supplemental zinc is recommended for women taking ³60 mg/d of iron during pregnancy.

Meat, shellfish, nuts, and legumes are good sources of bioavailable zinc, whereas zinc in grains is less available for absorption. Zinc is excreted mainly in the feces but also in urine and sweat. The body contains approximately 2 g of zinc, and high concentrations are found in liver, prostate, pancreas, bone, and brain (hippocampus and cerebral cortex), where the metal may function in neural transmission.

Deficiency Mild zinc deficiency has been described in many diseases including diabetes mellitus, AIDS, cirrhosis, alcoholism, inflammatory bowel disease, malabsorption syndromes, and sickle cell anemia ([Figs. 75-CD2](#) and [75-CD3](#)). In these diseases, mild chronic zinc deficiency can cause stunted growth in children, decreased taste sensation (hypogusia), impaired immune function, and night blindness due to impaired conversion of retinol to retinaldehyde. Severe chronic zinc deficiency has been described as a cause of hypogonadism and dwarfism in several Middle Eastern countries. In these children, hypopigmented hair is also part of the syndrome. Acrodermatitis enteropathica is a rare autosomal recessive disorder characterized by abnormalities in zinc absorption. Clinical manifestations include diarrhea, alopecia, muscle wasting, depression, irritability, and a rash involving the extremities, face, and perineum. The rash is characterized by vesicular and pustular crusting with scaling and erythema. In addition, hypopigmentation and corneal edema have been described in these patients. Occasional patients with Wilson's disease have developed zinc deficiency as a consequence of penicillamine therapy. Patients on long-term parenteral nutrition have developed deficiency when zinc has been omitted from the total parenteral nutrition (TPN) solution.

The diagnosis of zinc deficiency is usually made by a serum zinc level of <12 $\mu\text{mol/L}$ (<70 $\mu\text{g/dL}$). Pregnancy and birth control pills may cause a slight depression in serum zinc levels, and hypoalbuminemia from any cause can result in hypozincemia. In acute stress situations, zinc may be redistributed from serum into tissues. Zinc deficiency may be treated with 60 mg elemental zinc, given orally twice a day. Zinc gluconate lozenges (13 mg elemental Zn every 2 h while awake) have been reported to reduce the duration and symptoms of the common cold in adults, but these studies are conflicting.

Toxicity Acute zinc toxicity after oral ingestion causes nausea and vomiting, fever, and respiratory distress. Zinc fumes from welding may also be toxic and cause fever, chills, excessive salivation, sweating, and headache. Chronic large doses of zinc may depress immune function and cause hypochromic anemia as a result of copper deficiency.

COPPER

Copper is an integral part of numerous enzyme systems including amine oxidases, ferroxidase (ceruloplasmin), cytochrome-c oxidase, superoxide dismutase, and dopamine hydroxylase. As such, copper plays a role in iron metabolism, melanin synthesis, and central nervous system function; the synthesis and cross-linking of elastin and collagen; and the scavenging of superoxide radicals.

Copper is absorbed in the proximal small intestine, and 90% of circulating copper is bound to ceruloplasmin. The body contains 50 to 120 mg of copper, and high concentrations are found in liver, brain, heart, spleen, kidney, and blood. The U.S. [RDA](#) is 1.5 to 3 mg of copper intake per day, although World Health Organization recommendations are somewhat lower. Dietary sources of copper include shellfish, liver, nuts, legumes, bran, and organ meats, whereas milk is a very poor source. Copper is primarily excreted in the feces, and small amounts are also excreted in urine.

Deficiency Dietary copper deficiency is relatively rare, although it has been described in premature infants fed milk diets and in infants with malabsorption. Signs and symptoms of copper deficiency include a hypochromic-normocytic anemia, osteopenia, depigmentation, mental retardation, and psychomotor abnormalities. Copper deficiency anemia has been reported in patients with malabsorptive diseases and nephrotic syndrome and in patients treated for Wilson's disease with chronic high doses of oral zinc, which can interfere with copper absorption. Menkes kinky hair syndrome is an X-linked metabolic disturbance of copper metabolism characterized by mental retardation, hypocupremia, and decreased circulating ceruloplasmin ([Chap. 351](#)). It is caused by mutations in a copper-transporting *ATP7A* gene. Children with this disease often die within 5 years due to dissecting aneurysms or cardiac rupture.

The diagnosis of copper deficiency is usually made on the basis of low serum levels of copper (<65 ug/dL) and low ceruloplasmin levels (<18 mg/dL). Serum levels of copper may be elevated in pregnancy or stress conditions since ceruloplasmin is an acute-phase reactant.

Toxicity Toxicity due to copper is usually accidental and may include nausea, vomiting, diarrhea, and hemolytic anemia. In severe cases, kidney failure, liver failure, and coma may ensue. In Wilson's disease, mutations in the copper-transporting *ATP7B* gene lead to accumulation of copper in the liver and brain, with low blood levels due to decreased ceruloplasmin ([Chap. 348](#)). Indian childhood cirrhosis is another hereditary disease characterized by extremely high copper levels in the liver. The World Health Organization recommends that adult females should not ingest >10 mg/d and males should not take in >12 mg/d of copper.

SELENIUM

Selenium, in the form of selenocysteine, is a component of the enzyme glutathione peroxidase, which serves to protect proteins, cell membranes, lipids, and nucleic acids from oxidant molecules. Selenocysteine is also found in the deiodinase enzymes, which mediate the deiodination of thyroxine to the more active triiodothyronine ([Chap. 330](#)). Rich sources of selenium include seafood, muscle meat, and cereals, although the selenium content of cereal is determined by the soil concentration. Countries with low

soil concentrations include parts of Scandinavia, China, and New Zealand. *Keshan disease* is an endemic cardiomyopathy found in children and young women residing in regions of China where dietary intake of selenium is low (<20 ug/d). Concomitant deficiencies of iodine and selenium may worsen the clinical manifestations of cretinism. The adult [RDAs](#) for selenium in the United States are 55 and 70 ug/d for females and males, respectively. Low blood levels of selenium in various populations have been correlated with an increase in coronary artery disease and certain cancers, although the data are not consistent. Selenosis occurs at intakes of ³400 ug/d and can result in nausea, vomiting, loss of hair, nail changes, peripheral neuropathy, and fatigue.

CHROMIUM

Chromium potentiates the action of insulin in patients with impaired glucose tolerance, presumably by increasing insulin receptor-mediated signaling. In addition, in some patients, improvement in blood lipid profiles has been reported. The usefulness of chromium supplements in muscle building are not substantiated. Rich food sources of chromium include yeast, meat, and grain products. Chromium deficiency has been reported to cause glucose intolerance, peripheral neuropathy, and confusion. The suggested intake of chromium for adults is 50 to 200 ug/d. Chromium in the trivalent state is found in supplements and is largely nontoxic; however, chromium-6 is a product of stainless steel welding and is a known pulmonary carcinogen, as well as causing liver, kidney, and central nervous system.

MAGNESIUM See [Chap. 340](#)

FLUORIDE, MANGANESE, AND ULTRATRACE ELEMENTS

An essential function for *fluoride* in humans has not been described, although it is useful for the maintenance of structure in teeth and bone. An adequate intake for fluoride (on the basis of protection against dental caries) has been set at 3.1 and 3.8 mg/d in adult females and males, respectively. Adult fluorosis can occur at an intake of 10 mg/d for prolonged periods and results in mottled and pitted defects in tooth enamel as well as brittle bone (skeletal fluorosis). Much lower doses of fluoride (0.7 to 2 mg) can cause dental fluorosis or mottled enamel in infants and children.

Manganese and molybdenum deficiencies have been reported in patients with rare genetic abnormalities as well as in a few patients receiving prolonged [TPN](#). Several manganese-specific enzymes have been identified (e.g., manganese superoxide dismutase). The estimated adequate daily dietary *manganese* intake for adults is 2 to 3 mg/d. Deficiencies of manganese have been reported to result in bone demineralization, poor growth, ataxia, and convulsions.

Ultratrace elements are those for which the need is <1 mg/d. Essentiality has not been established for most ultratrace elements, although *iodine* is clearly essential ([Chap. 330](#)). *Molybdenum* is necessary for the activity of sulfite and xanthine oxidase, and molybdenum deficiency may result in skeletal and brain lesions. The minimum required daily molybdenum intake is estimated to be ~25 ug/d. There is circumstantial evidence to suggest that *arsenic* (impaired growth, infertility), *boron* (impaired energy metabolism, impaired brain function), *nickel* (impaired-growth and reproduction), *silicon* (impaired

growth) and *vanadium* (impaired skeletal formation) might also be essential.

(Bibliography omitted in Palm version)

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76. ENTERAL AND PARENTERAL NUTRITION THERAPY - Lyn Howard

Parenteral and enteral nutrition provide life-sustaining therapy for patients who cannot take adequate food by mouth and who consequently are at risk for malnutrition and its effects, including susceptibility to infection, weakness and immobility; these features predispose the patient to aspiration pneumonia, pulmonary embolism, and pressure sores, all of which delay recovery from illness and increase mortality.

The term *enteral* refers to feeding via the gut and hence includes normal eating, but in the present context implies the infusion of formulas via a tube into the upper gastrointestinal tract. *Parenteral* refers to the infusion of nutrient solutions into the bloodstream. While these are different approaches to nutritional support, their goals are the same. Where feasible, enteral nutrition is the preferred route because it sustains the digestive, absorptive, and immunologic barrier functions of the gastrointestinal tract. The cost of enteral tube feeding is about one-tenth the cost of parenteral feeding.

Several developments have made tube feeding easier and more acceptable to patients. Small-bore pliable tubes have largely replaced large-bore rubber tubes, and double-lumen tubes are now available for simultaneous gastric suction and jejunal feeding when there is concern about gastric retention and aspiration. Enteral tubes can be inserted into the stomach or jejunum through the nose or, for long-term use, directly through the abdominal wall, using endoscopic, radiologic, or surgical techniques. Once the enterocutaneous tract is established, the protruding tube can be replaced by a "button" entry port, flush with the abdominal wall.

Complete nutrition by vein with sufficient calories, amino acids, minerals, and vitamins to permit wound healing, restoration of normal body composition of a cachectic patient, or growth in children became feasible in the 1960s with the development of high-flow central vein catheters. Parental nutrition is now available in all large hospitals and for some patients at home. Adequate calories and other nutrients can be delivered in the form of high-energy, isotonic intravenous fat solutions via a peripheral vein. However, peripheral veins usually cannot sustain such infusions indefinitely, and long-term support requires central venous access.

THE DECISION PROCESS FOR USING PARENTERAL OR ENTERAL NUTRITION

The decision to use specialized nutrition support should be based on the likelihood that averting or redressing malnutrition will improve the quality of life or the ability to recover from a serious illness.

Approximately 15 to 20% of hospitalized patients have evidence of malnutrition. Some malnourished patients benefit from specialized nutrition support; for others, wasting is an inevitable component of a terminal disease. Selecting the appropriate form of nutritional support for the patient requires knowledge of the potential benefits and risks of nutritional support, and the physician must inform the patient and family of these issues. A flow diagram of the steps involved in deciding whether specialized nutrition support should be undertaken and, if so, how, is depicted in [Fig. 76-1](#). Like all life-support measures, these therapies are difficult to withdraw once started.

The first step requires consideration of the nutritional implications of the disease process. Is the condition or its treatment likely to impair appetite or food ingestion and absorption for a prolonged period of time? Because prevention of malnutrition is easier than repleting a cachectic patient, this issue must be considered in the initial evaluation ([Chap. 74](#)). The second step is to determine whether the patient is already sufficiently malnourished that lean body mass is decreased and critical functions such as healing and ventilation are impaired. The presence or absence of metabolic stress should be noted, since injury or infection can evoke the secretion of hormonal and cytokine factors that reduce the efficiency of nutrition repletion.

Weight loss without physiologic impairment is probably of no consequence. Physiologic impairment usually develops when more than 20% of body protein is lost and is more likely if key organ systems, such as the gut or liver, are directly affected by disease. Once it is recognized that the patient is malnourished or at major risk, the next question is whether specialized nutritional support will impact positively on the patient's response to the disease, improving the quality of life. While the provision of food and water is part of basic medical care, nutrition support by enteral or parenteral means is associated with risk and discomfort and should be recommended only when potential benefit exceeds risk and undertaken only with the consent of the patient.

If it is decided that preventing or treating malnutrition with specialized nutrition support would improve the prognosis and quality of life, the nutritional requirements must be determined and the route of nutrient delivery must be selected.

RISKS AND BENEFITS OF NUTRITION SUPPORT

The risks are determined primarily by the route required to deliver nutrition support. Providing nutritional requirements by special attention to oral intake of food, or by adding oral liquid supplements and monitoring food intake with frequent calorie counts, is the safest and least costly approach. It is also the most metabolically efficient since normal eating initiates the cephalic phase of digestion. Tube-fed infants grow better if the cephalic phase is stimulated by having the infant suck on a pacifier.

Anorexia, impairment of swallowing, or bowel disease may limit oral intake or the absorption of oral nutrients, in which case tube enteral nutrition is the next consideration. The bowel and its associated digestive organs derive 70% of their required nutrients directly from food in the lumen. In addition, glutamine, short-chain fatty acids, and nucleotides may have particular importance in maintaining gut integrity. Enteral feeding also supports gut function by stimulating splanchnic blood flow, neuronal activity, IgA antibody release, and secretion of gastrointestinal hormones such as epidermal growth factor that stimulate gut trophic activity. All these factors support the gut as an immunologic barrier against enteric pathogens, reducing the likelihood of bacterial overgrowth. For these reasons, some enteral nutrition should always be provided if possible, even when parenteral nutrition is required to provide most of the support. In the past, bowel rest through parenteral nutrition was thought to be the cornerstone of treatment of many severe gastrointestinal disorders, but the value of some enteral nutrition is now widely accepted, and strict bowel rest is rarely appropriate. Parenteral nutrition alone is necessary in severe hemorrhagic pancreatitis, necrotizing enterocolitis, prolonged ileus, and distal bowel obstruction.

Specialized nutrition support is expensive, accounting for >1% of all health care dollars. Consequently, hard clinical endpoints such as mortality rate, incidence of major complications, and duration of hospital stay are required of risk-benefit studies. Better nitrogen balance, increased levels of serum albumin, and improved delayed hypersensitivity are softer endpoints. [Table 76-1](#) summarizes clinical trials that evaluate the use of specialized nutrition support in different disease states.

Perioperative Nutrition There is a clear-cut association between preoperative malnutrition and poor surgical outcome, but it has been difficult to demonstrate the benefit of preoperative parenteral nutrition on the outcome of surgery in malnourished patients. However, a meta-analysis of several small studies and a large cooperative Veterans Administration study indicates that preoperative parenteral nutrition does improve the outcome of severely malnourished surgical patients. In treated patients, noninfectious complications (e.g., pulmonary emboli and delayed wound healing) are reduced in the postoperative period. Effective preoperative restoration of nutrition by the parenteral route requires at least 7 to 14 days. If feasible, a safer and less costly approach is preoperative enteral nutrition, especially if provided at home.

Immediate postoperative nutritional support is appropriate for patients who received preoperative support and for patients unlikely to resume oral feeding within 10 days. The parenteral route is commonly used because of postoperative ileus or concern about disrupting a new bowel anastomosis. However, cautious jejunal feeding is often tolerated. Specialized enteral formulas supplemented with conditionally essential nutrients may be particularly beneficial in debilitated and immunosuppressed postoperative patients ([Table 76-2](#)).

Critical Illness Very early nutrition support (within the first 48 h) improves survival and reduces infections and length of hospital stay in patients with severe head injuries, burns, and major abdominal trauma. Enteral therapy, where feasible, is superior to parenteral therapy in several randomized trials. Enteral nutrition equally benefits malnourished and well-nourished injured patients. Animal studies show that enteral feeding reduces translocation of gut bacteria and the systemic catabolic response; however, these phenomena have not been substantiated in humans. Early enteral feeding may prevent bacterial overgrowth and decrease aspiration pneumonia.

The practical issue is obtaining jejunal access in a critically ill patient, who is not easily transferred out of the intensive care unit for endoscopic or radiologic tube placement. Sometimes a nasal or percutaneous combined gastric suction and jejunal feeding tube can be inserted at the bedside. If a surgical laparotomy is indicated, a feeding tube can be placed simultaneously.

Most studies of enteral feeding in critically ill patients used either a general polymeric formula or one with hydrolyzed protein. Formulas supplemented with conditionally essential nutrients reduce infections and length of hospital stay. Parenteral formulas enriched with large amounts of branched-chain amino acids (BCAA) improve nitrogen balance but do not appear to affect clinical outcome.

Cancer Cachexia Early nonrandomized studies suggested that patients with cancer

cachexia benefitted from parenteral nutrition, but randomized trials demonstrated more risk than benefit for patients receiving chemotherapy or radiation. Severely malnourished cancer patients undergoing surgery benefit from preoperative parenteral nutrition, as do other malnourished patients.

In two randomized, prospective trials, patients undergoing bone marrow transplantation had better long-term survival after parenteral or enteral nutrition in the cytoreduction phase; nutrition support did not influence the initial infection rate or the frequency of graft rejection or graft-versus-host disease. Immediate morbidity is reduced if glutamine supplements are added to the parenteral nutrition solution. Parenteral nutrition continued at home delays return to oral feeding. For cancer patients with unresectable upper gastrointestinal cancer, enteral feeding is usually justified if it is desired by the patient and family. Parenteral feeding should be provided only if clinical improvement can be expected and when quality survival at home for several months is predicted.

Liver Failure Malnutrition is common in advanced liver disease. Patients with acute or chronic liver failure have decreased levels of [BCAA](#) and elevated levels of aromatic amino acids (AAA) in plasma and cerebrospinal fluid. Randomized, prospective trials with parenteral and enteral formulas high in BCAA and low in AAA have demonstrated better nitrogen balance and less risk of encephalopathy. One large multicenter study also reported improved survival. The BCAA-enriched formulas are expensive and should be used only in patients who have encephalopathy or who develop encephalopathy when fed a standard protein formula providing 0.8 g protein/kg per day.

Renal Failure Since renal failure is associated with impaired nitrogen excretion, it is rational to assume that protein restriction might benefit patients with both acute and chronic renal failure. Patients with acute renal failure given parenteral calories and amino acids have fewer infectious complications and a better chance of leaving the hospital than similar patients given only calories. An early randomized study showed benefit when essential amino acids were the sole source of nitrogen, but in other studies, standard solutions supplying both essential and nonessential amino acids provide similar advantage. Thus, the benefit of using expensive formulas containing only essential amino acids or their keto analogues is not established. A large national study failed to show any benefit of a low-protein diet on slowing progression of renal impairment in patients on chronic dialysis. Some 15 to 20% of patients on chronic dialysis have significant nutritional impairment, usually due to profound anorexia. The anorexia may improve with stepped-up dialysis or treatment of gastritis but usually persists. The resulting growth impairment in younger patients has been treated with supplemental tube enteral nutrition. This approach has not been widely used in adult patients. Limited parenteral calories and amino acids can be provided in the last 90 min of hemodialysis treatment (intradialytic parenteral nutrition). This may improve appetite, serum protein levels, and body weight. No randomized studies have documented better survival, so the appropriateness of this regimen is not established. Standard dialysis uses glucose to provide an osmotic load, and some glucose calories are absorbed. During continuous ambulatory peritoneal dialysis, amino acids can be substituted for glucose and are also partly absorbed, offsetting the loss of endogenous amino acids into the dialysate. This approach to nutrition repletion is expensive and also awaits a randomized study.

Pancreatitis Parenteral nutrition does not improve the outcome of patients with mild or moderate pancreatitis. However, in severe pancreatitis, survival decreases as malnutrition becomes more severe. When parenteral nutrition support was delayed beyond 72 h, patients with severe pancreatitis had a threefold higher complication and mortality rate, compared to similar patients treated earlier. In the absence of severe hyperlipidemia or thrombocytopenia, intravenous lipids appear safe and are especially useful if glucose intolerance is present. Several studies report successful enteral jejunal feeding in acute pancreatitis and, compared to parenteral nutrition, the inflammatory response and infectious complications are less.

Inflammatory Bowel Disease Evidence of nutritional deficiencies such as weight loss, growth failure, anemia, and hypoalbuminemia are common in inflammatory bowel disease (IBD), more so in Crohn's disease than in ulcerative colitis ([Chap. 287](#)). Nutrition support plays a role in correcting these nutritional deficiencies, particularly prior to elective surgery. Since IBD often improves with diversion of the fecal stream, the question is whether bowel rest and parenteral nutrition have a role as primary treatment. However, randomized, prospective studies have shown no special benefit from bowel rest. Elemental diets are not quite as effective as glucocorticoids for inducing remission in acute Crohn's disease but may be preferable in children to avoid growth impairment. Relapse is common when a regular diet is resumed. In controlled studies, remissions are prolonged if the Crohn's patient does not return to a regular diet but instead eliminates from the diet those foods that induce gastrointestinal symptoms. For the majority of Crohn's patients this leads to avoidance of cereals, yeast, green vegetables, and, early on, dairy products. Because of the possibility that diets high in omega-3 fatty acids have a beneficial effect in immune disorders by altering prostaglandin synthesis, their value in IBD is under investigation. Some studies suggest that high-fiber diets benefit IBD patients, but fiber can also cause obstruction in patients with bowel strictures.

Short Bowel Syndrome Before the advent of parenteral nutrition, patients with acute short bowel syndrome from mesenteric vascular infarction or massive small bowel surgical resection seldom survived. Parenteral nutrition has allowed many patients to survive indefinitely with only a foot or two of small intestine. In some, the remaining bowel eventually adapts and allows the absorption of adequate calories and protein. This is especially true of patients who retain their ileocecal valve and colon. However, fluid and electrolyte imbalance may persist, necessitating some parenteral fluid and electrolyte support. A gradual switch to overnight tube enteral hydration or constant sipping of an electrolyte solution may allow discontinuation of all parenteral support.

Pulmonary Disease Weight loss in patients with advanced pulmonary disease is due to increased work of breathing and poor food intake. Patients with chronic pulmonary disease who are <90% of their ideal weight have a higher 5-year mortality, independent of pulmonary status. The recommended energy intake for these patients is 1.7 times their resting energy expenditure. The use of a low-carbohydrate formula is beneficial in the weaning of patients from ventilators, but the superiority of such formulas in ambulatory patients with chronic lung disease is not established. In cystic fibrosis, malnutrition may hasten pulmonary deterioration, and enteral tube feeding enhances growth and stabilizes or improves pulmonary function, particularly in young children. Tube feeding is safest when delivered into the jejunum. Postpyloric feeding is no safer

than gastric feeding.

HIV Disease Specialized nutrition support can replete body cell mass if the weight loss is due to inadequate oral intake caused by oral or esophageal disease or to inadequate intestinal absorption, which is common in HIV patients with cryptosporidiosis or microsporidiosis infections ([Chap. 309](#)). The route of nutrition support has usually been parenteral, but patients respond equally well with an isocaloric semi-elemental oral diet. Patients using the oral supplement experience a better quality of life, and their medical costs are significantly lower. Wasting due to systemic infection and increased cytokine secretion is not redressed by specialized nutrition support. Survival, CD4+ counts, and intestinal function also are not improved by specialized nutrition support.

Pregnancy Severe hyperemesis gravidarum can make any oral or tube enteral nutrition impossible, and profound weight loss and ketosis may harm the developing fetus. The underlying mechanism of the disorder is not understood, but it is cured by abortion or delivery. Temporary parenteral nutrition usually results in a successful outcome, but nausea and vomiting tend to persist, despite bowel rest.

Home Parenteral and Enteral Nutrition Some patients require long-term nutrition support, and for many this can be administered at home. Clinical outcomes of patients with severe intestinal disorders that used either parenteral or enteral nutrition are summarized in [Table 76-3](#). Nutrition support is not usually appropriate in terminally ill patients but is an option if the patient is expected to survive for several months. Such therapy must make sense to the patient, and sufficient help must be available so the treatment can be given at home without undue hazard. Both home therapies are relatively safe, with <5% therapy-related mortality.

THE DESIGN OF INDIVIDUAL REGIMENS

Fluid Requirements These can be estimated by adding the normal daily requirement (120 mL/kg of body weight for infants, 35 mL/kg of body weight for adults) to any abnormal loss. If the patient is on parenteral therapy, any enteral intake should be subtracted from the estimate ([Table 76-4](#)). Since abnormal loss of enteric fluid implies significant mineral losses, extra amounts of these nutrients, as well as fluid ([Table 76-5](#)), must be added to the parenteral formula.

Energy Requirements These can be determined as outlined in [Chaps. 73](#) and [74](#). In the long run, energy expenditure dictates energy requirements, but in the early phase of nutrition repletion, requirements may not reflect expenditure. For example, malnourished patients are hypometabolic and may expend only 85 kJ/kg (20 kcal/kg) per day, but more calories are needed both for tissue repletion and because the metabolic rate increases with refeeding. Conversely, a highly stressed patient (sepsis, trauma) may expend 165 kJ/kg (40 kcal/kg) per day with a significant proportion of the calories coming from protein breakdown and gluconeogenesis and from catecholamine-induced lipolysis. Oxidation of exogenous glucose plateaus at 100 kJ/kg (25 kcal/kg) per day, and administering additional glucose induces hepatic steatosis. Providing such patients with additional calories as exogenous fat does not suppress endogenous lipolysis. Furthermore, lipid solutions are made from vegetable oil and egg phospholipid and lack apoproteins, which they acquire from endogenous lipoproteins.

Initially, the artificial chylomicron may be taken up by the reticuloendothelial system enhancing its blockade. For all these reasons, modest hypocaloric glucose feeding with minimal parenteral fat is safer in the acutely stressed subject.

Parenteral lipid solutions are available as 10 or 20% isotonic solutions and are infused separately from amino acids and glucose or as a combined "three-in-one solution," obviating the need for an extra pump. Three-in-one solutions are less stable than the glucose and amino acid mix, and destabilized fat particles have the potential to coalesce into larger droplets, becoming fat emboli. For this reason, three-in-one solutions have a shorter storage life and must be mixed by a pharmacist knowledgeable about the correct mixing sequence and safe levels of electrolytes and trace elements. Iron, for example, cannot be added to this solution.

Polyunsaturated vegetable oils are used in most enteral formulas because they are absorbed better than animal fat by a diseased gastrointestinal tract. Fat must supply the essential fatty acid requirement (1 to 4% of energy from linoleic and linolenic acid) ([Table 76-6](#)). Larger amounts (30% of energy) are safe in relatively stable patients and avoid the problems of providing large amounts of glucose (e.g., hyperglycemia and hepatic steatosis). Substituting omega-3 polyunsaturated fish oils for polyunsaturated vegetable fats may reduce the catabolic response to burn injury, trauma, and radiation by reducing the synthesis of prostaglandins that enhance the inflammatory response ([Table 76-2](#)). Such fats are available in enteral formulas and are currently being tested in parenteral formulas.

Protein or Amino Acid Requirements The recommended dietary protein allowance of 0.8 g/kg per day is adequate for nonstressed patients, such as a starved patient with a high-grade esophageal stricture. Catabolic patients, in contrast, may require up to 1.5 g/kg per day of protein to induce positive nitrogen balance and reconstitute normal body mass. Early studies showed that recombinant human growth hormone (rHGH) increases lean body mass. However, subsequent trials have shown that it is associated with increased mortality in critically ill patients, and it should not be used in this setting.

In a stable patient the adequacy of protein support can be assessed by analyzing protein balance:

where protein loss = $[(24\text{-h urine urea nitrogen (g)} + 4) \times 6.25]$. Over a long period, protein balance is assessed by documenting wound healing, restoration of normal body composition, or resumption of longitudinal growth. In states of disturbed protein utilization (e.g., renal and hepatic failure), azotemia and abnormal plasma amino acid patterns develop. The benefit of special enteral and parenteral solutions that correct these aberrations is only established in hepatic encephalopathy (see "Risks and Benefits of Nutrition Support").

Certain nutrients that can normally be synthesized endogenously become essential in severely ill patients when endogenous production or salvage pathways are impaired. This is true of glutamine, nucleotides, and the products of methionine metabolism ([Table 76-2](#)). Glutamine, an important fuel for the enterocyte and lymphocyte, is fairly insoluble

and is absent from standard parenteral formulas and present in low concentrations in most enteral formulas. Soluble glutamine-containing dipeptides are under investigation.

Nucleotides and their related metabolic products have beneficial effects on the immune system, growth of the small intestine, lipid metabolism, and hepatic function. Nucleotides can be synthesized *de novo* in all cells only in small amounts, and the body therefore depends on dietary nucleotides or on salvage pathways that recycle nucleotides from purine and pyrimidine turnover. Nutritionally depleted patients benefit from formulas enriched in nucleotides.

When amino acids are infused systemically, rather than via the more physiologic portal vein, methionine, the only sulfur-containing amino acid in most parenteral solutions, is transaminated in peripheral tissues rather than transsulfurated in the liver. As a result, downstream sulfur products such as carnitine, taurine, and glutathione become relatively deficient ([Chap. 352](#)). Preliminary studies suggest that the addition of an intermediate compound, S-adenosyl methionine, to parenteral solutions results in less cholestasis.

Mineral and Vitamin Requirements Parenteral and enteral mineral and vitamin requirements are summarized in [Table 76-6](#). Electrolyte modifications are necessary if the patient has significant gastrointestinal losses ([Table 76-5](#)) or renal failure. Requirements of some minerals and vitamins are higher when administered parenterally for several reasons: (1) many micronutrients delivered into the systemic rather than the portal circulation are not captured by the liver and instead pass directly into the urine; (2) patients with bowel disease may have enteric loss of sodium, potassium, chloride, and bicarbonate and malabsorption of divalent cations, fat-soluble vitamins, and vitamin B₁₂; and (3) nutrients may adhere to the tubing and delivery bags, and exposure to oxygen and light may destroy vitamins (particularly vitamin A).

PARENTERAL NUTRITION

Infusion Technique and Patient Monitoring Partial and short-term total parenteral nutrition can be provided via a peripheral vein if the majority of the energy is supplied by isotonic fat solutions; long-term total parenteral nutrition using glucose as the chief energy source requires administration via a central vein catheter so the hypertonic solution can be rapidly diluted in a high-flow system. The preferred site for central vein infusion is the superior vena cava. Access sites and catheter choices are summarized in [Table 76-7](#). Peripherally inserted central catheters are the most economical option for short-term parenteral nutrition. In one randomized study, the number of catheter-related infections was the same with peripherally and centrally inserted catheters. Tunneled catheters and implanted subcutaneous ports require operating room insertion and are more stable for long-term use. Central catheters should be changed when clinically indicated; routine changes are costly and hazardous. Chlorhexidine solution is a more effective local antiseptic than iodophor or alcohol. Although transparent dressings are helpful in stabilizing catheters and allow easy observation of the skin site, the incidence of catheter-related sepsis is higher than with traditional dry gauze dressings; newer transparent dressings that trap less moisture are under investigation. Catheters made from Silastic material or polyurethane are associated with lower complications than polyvinylchloride catheters. Several types of needleless systems use hub valves, and

contamination rates are higher with these devices when used for long-term parenteral nutrition. Appropriate clinical and laboratory monitoring for patients on parenteral nutrition are summarized in [Table 76-8](#).

Complications (See [Table 76-9](#))

Mechanical The insertion of a central venous catheter should be done only by trained personnel under aseptic techniques. Major mechanical complications include pneumothorax; hemothorax from laceration of the subclavian artery or vein; brachial plexus injury; and malpositioning of the catheter in a cerebral vein, the azygos vein, or the right ventricle. The correct catheter position must be confirmed by x-ray before hypertonic nutrient solution is infused. Catheters can subsequently dislocate, develop leaks, or become detached from the hub and embolize into the heart or pulmonary artery. Catheter thrombosis may occur, especially if the catheter is used for withdrawing blood samples, and extension of the thrombosis to the central vein is frequently coincident with infection. Thrombosed catheters can sometimes be unblocked by urokinase treatment. The addition of low-dose heparin (1000 U/L) to limit thrombosis in parenteral catheters is controversial; no randomized, controlled studies demonstrate benefit, and heparin can contribute to loss of bone mineral, which is already a problem with long-term parenteral nutrition.

Metabolic Fluid overload can cause congestive heart failure, particularly in elderly and debilitated patients. Glucose overload can cause an osmotic diuresis or stimulate insulin secretion, which in turn promotes extracellular to intracellular shifts of potassium and phosphorous. Such shifts are most dangerous in cachectic patients with depletion of potassium and phosphorus stores and can cause arrhythmias, cardiopulmonary dysfunction, and neurologic symptoms. To avoid these problems, parenteral nutrition should be started slowly and monitored carefully. Glucose content is increased gradually as the patient demonstrates tolerance of the high glucose load. Late metabolic complications include cholestatic liver disease with bile sludging and gallstone formation. The exact cause of the liver disease is not understood, but lack of enteral stimulation to bile flow and defective sulfur amino acid metabolism and cholesterol solubilization appear to play a role. Cholestasis is less likely to occur if some enteral feeding is maintained. Parenteral nutrition induces hypercalciuria, which can result in negative calcium balance and osteopenia. Hypercalciuria may have several causes, including the high fixed-acid load of infused amino acids and the bisulfite preservative in parenteral solutions. Earlier, protein hydrolysates were used as an amino acid source and were contaminated with aluminum, which blocked bone mineralization. Aluminum is still a contaminant of some additives such as calcium gluconate. Once patients on long-term parenteral nutrition change from catabolic breakdown to sustained anabolism, deficiencies of micronutrients such as essential fatty acids, trace minerals, and vitamins may develop unless they are supplied in adequate amounts ([Table 76-6](#)).

Infectious Infection of the access line rarely occurs in the first 72 h, and fever during this period is usually due to infection elsewhere or some other cause. Infection of the access line is likely if the fever defervesces when the infusion of the parenteral formula is tapered.

Positive central line cultures suggest catheter sepsis, especially if no other infectious

source is identified and if the organism is *Staphylococcus* or *Candida*. Although removal of the central catheter may allow fungemia to clear spontaneously, antibiotic therapy is recommended for bacterial infections and the more invasive fungi. Catheter sepsis rates are similar in single lumen central lines dedicated to parenteral nutrition whether inserted peripherally via the subclavian vein or tunneled; multiple-lumen catheters are associated with a greater incidence of sepsis. While there is no evidence to support the use of prophylactic antibiotics, recurrent catheter sepsis may be avoided if cuffs are used around the catheter exit site or small amounts of an antibiotic solution are left in the line along with a heparin lock.

ENTERAL NUTRITION

Tube Placement and Patient Monitoring The types of enteral feeding tubes, methods of insertion, their clinical uses, and potential complications are outlined in [Table 76-10](#). The different types of enteral formulas are listed in [Table 76-11](#). Patients on enteral feeding are at risk for many of the same metabolic complications as those receiving parenteral nutrition and should be monitored in the same way ([Table 76-8](#)). Since small-bore tubes are easily displaced, tube position should be checked at intervals by aspirating and measuring the pH of the gut fluid (<4 in stomach, >6 in jejunum).

Complications

Aspiration The debilitated patient with poor gastric emptying and impairment of swallowing and cough mechanisms is at risk for aspiration; this is particularly so for those on respirators. Tracheal suctioning induces coughing and gastric regurgitation, and cuffs on endotracheal or tracheostomy tubes seldom provide protection against aspiration. Under these circumstances, it may be safer to use a large-bore feeding tube to allow for temporary removal of gastric contents during tracheal suction or to use jejunal feeding. Constant gastric infusion of an enteral formula is better tolerated in sick patients than intermittent bolus feeding. A continuous infusion is best achieved with a pump, especially when using fine-bore tubes that have a greater potential to clog. If long-term feeding is anticipated, endoscopic, radiologic, or surgical placement of a gastric tube is preferred by most patients. For long-term ambulatory patients, a gastrostomy tube can be converted to a gastric "button," an access device that is flush with the skin. A nasojejunal tube reduces the risk of aspiration. However, fluoroscopically guided placement of fine-bore tubes through the pylorus is time-consuming, and such tubes frequently pull back into the stomach. A percutaneous combined gastric-suction and jejunal-feeding tube is more reliable. This can be placed radiologically, endoscopically, or surgically.

Diarrhea Enteral feeding often causes diarrhea, especially if bowel function is compromised by bowel disease or drugs. The diarrhea may be controlled by the use of continuous feeding of fiber-containing formulas or by adding an anticholinergic medication to the formula. Diarrhea associated with enteral feeding does not necessarily imply inadequate absorption of nutrients, other than water and electrolytes. Furthermore, since luminal nutrients exert trophic effects on the gut mucosa and enhance the enteric immunologic barrier, it is often appropriate to persist with tube feeding, despite the diarrhea, even when this necessitates supplemental parenteral fluid support.

THE SCOPE AND COST OF NUTRITION SUPPORT

As many as 25% of patients entering tertiary care hospitals have central catheters placed, and 20 to 30% of these catheters are used for parenteral nutrition. The incidence of catheter-related infection reflects the severity of the underlying medical condition and varies from 2 to 30 per thousand catheter days, depending on the type of patients involved. In critically ill patients, catheter sepsis is associated with a 35% mortality rate and a high cost per survivor. Most catheter-related complications derive from faulty insertion and management of the catheter rather than defects in the device. In large tertiary care hospitals, the insertion and management of these lines by specially trained teams can reduce complications by 80%, impacting significantly on outcome and costs. A growing shift from parenteral to enteral nutrition also promises significant cost savings. Home parenteral nutrition costs approximately half as much as similar treatment in the hospital, and home enteral nutrition costs much less.

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77. OBESITY - Jeffrey S. Flier

In a world where food supplies are intermittent, the ability to store energy in excess of what is required for immediate use is essential for survival. Fat cells, residing within widely distributed adipose tissue depots, are adapted to store excess energy efficiently as triglyceride and, when needed, to release stored energy as free fatty acids for use at other sites. This physiologic system, orchestrated through endocrine and neural pathways, permits humans to survive starvation for as long as several months. However, in the presence of nutritional abundance and a sedentary lifestyle, and influenced importantly by genetic endowment, this system increases adipose energy stores and produces adverse health consequences.

DEFINITION AND MEASUREMENT

Obesity is a state of excess adipose tissue mass. Although often viewed as equivalent to increased body weight, this need not be the case -- lean but very muscular individuals may be overweight by arbitrary standards without having increased adiposity. Body weights are distributed continuously in populations, so that a medically meaningful distinction between lean and obese is somewhat arbitrary. Obesity is therefore more effectively defined by assessing its linkage to morbidity or mortality.

Although not a direct measure of adiposity, the most widely used method to gauge obesity is the *body mass index* (BMI), which is equal to weight/height^2 (in kg/m^2) ([Fig. 77-1](#)). Other approaches to quantifying obesity include anthropometry (skin-fold thickness), densitometry (underwater weighing), computed tomography (CT) or magnetic resonance imaging (MRI), and electrical impedance. Using data from the Metropolitan Life Tables, BMIs for the midpoint of all heights and frames among both men and women range from 19 to 26 kg/m^2 ; at a similar BMI, women have more body fat than men. Based on unequivocal data of substantial morbidity, a BMI of 30 is most commonly used as a threshold for obesity in both men and women. Large-scale epidemiologic studies suggest that all-cause, metabolic, and cardiovascular morbidity begin to rise (albeit at a slow rate) when BMIs are ³ 25, suggesting that the cut-off for obesity should be lowered. Some authorities use the term *overweight* (rather than obese) to describe individuals with BMIs between 25 or 27 and 30. A BMI between 25 and 30 should be viewed as medically significant and worthy of therapeutic intervention, especially in the presence of risk factors that are influenced by adiposity, such as hypertension and glucose intolerance.

The distribution of adipose tissue in different anatomic depots also has substantial implications for morbidity. Specifically, intraabdominal and abdominal subcutaneous fat have more significance than subcutaneous fat present in the buttocks and lower extremities. This distinction is most easily made by determining the waist-to-hip ratio, with a ratio >0.9 in women and >1.0 in men being abnormal. Many of the most important complications of obesity, such as insulin resistance, diabetes, hypertension, and hyperlipidemia, and hyperandrogenism in women, are linked more strongly to intraabdominal and/or upper body fat than to overall adiposity. The mechanism underlying this association is unknown but may relate to the fact that intraabdominal adipocytes are more lipolytically active than those from other depots. Release of free fatty acids into the portal circulation has adverse metabolic actions, especially on the

liver.

PREVALENCE

Recent data from the National Health and Nutrition Examination Surveys (NHANES) show that the percent of the American adult population with obesity ([BMI](#) > 30) has increased from 14.5% (between 1976 and 1980) to 22.5% (between 1998 and 1994). As many as 50% of U.S. adults 20 years of age were overweight (defined as BMI > 25) between the years of 1998 and 1991. Because substantial health risks exist in many individuals with BMI between 25 and 30, the increasing prevalence of medically significant obesity raises great concern. Obesity is more common among women and in the poor; the prevalence in children is also rising at a worrisome rate.

PHYSIOLOGIC REGULATION OF ENERGY BALANCE

Substantial evidence suggests that body weight is regulated by both endocrine and neural components that ultimately influence the effector arms of energy intake and expenditure. This complex regulatory system is necessary because even small imbalances between energy intake and expenditure will ultimately have large effects on body weight. For example, a 0.3% positive imbalance over 30 years would result in a 9-kg (20-lb) weight gain. Alterations in stable weight by forced overfeeding or food deprivation induce physiologic changes that resist these perturbations: with weight loss, appetite increases and energy expenditure falls; with overfeeding, appetite falls and energy expenditure increases. This latter compensatory mechanism frequently fails, however, permitting obesity to develop when food is abundant and physical activity is limited. A major regulator of these adaptative responses is the adipocyte-derived hormone leptin, which acts through brain circuits (predominantly in the hypothalamus) to influence appetite, energy expenditure, and neuroendocrine function (see below).

Appetite is influenced by many factors that are integrated by the brain, most importantly within the hypothalamus ([Fig. 77-2](#)). Signals that impinge on the hypothalamic center include neural afferents, hormones, and metabolites. Vagal inputs are particularly important, bringing information from viscera, such as gut distention. Hormonal signals include leptin, insulin, cortisol, and gut peptides such as cholecystokinin, which signals to the brain through the vagus nerve. Metabolites, including glucose, can influence appetite, as seen by the effect of hypoglycemia to induce hunger; however, glucose is not normally a major regulator of appetite. These diverse hormonal, metabolic, and neural signals act by influencing the expression and release of various hypothalamic peptides [e.g., neuropeptide Y (NPY), Agouti-related peptide (AgRP), melanocyte-stimulating hormone (MSH), and melanin concentrating hormone (MCH)] that are integrated with serotonergic, catecholaminergic, and opioid signaling pathways (see below). Psychological and cultural factors also appear to play a role in the final expression of appetite. Apart from rare syndromes involving leptin, its receptor, and the melanocortin system (see below), the defects in this complex appetite control network that account for common causes of obesity are not well understood.

Energy expenditure includes the following components: (1) resting or basal metabolic rate; (2) the energy cost of metabolizing and storing food; (3) the thermic effect of exercise; and (4) adaptive thermogenesis, which varies in response to chronic caloric

intake (rising with increased intake). Basal metabolic rate accounts for about 70% of daily energy expenditure, whereas active physical activity contributes 5 to 10%. Thus, a significant component of daily energy consumption is fixed.

Adaptive thermogenesis occurs in *brown adipose tissue* (BAT), which plays an important role in energy metabolism in many mammals. In contrast to white adipose tissue, which is used to store energy in the form of lipids, BAT expends stored energy as heat. A mitochondrial *uncoupling protein* (UCP-1) in BAT dissipates the hydrogen ion gradient in the oxidative respiration chain and releases energy as heat. The metabolic activity of BAT is increased by a central action of leptin, acting through the sympathetic nervous system, which heavily innervates this tissue. In rodents, BAT deficiency causes obesity and diabetes; stimulation of BAT with a specific adrenergic agonist (β_3 agonist) protects against diabetes and obesity. Although BAT exists in humans (especially neonates), its physiologic role is not yet established. Homologues of UCP-1 may mediate uncoupled mitochondrial respiration in other tissues.

THE ADIPOCYTE AND ADIPOSE TISSUE

Adipose tissue is composed of the lipid-storing adipose cell and a stromal/vascular compartment in which preadipocytes reside. Adipose mass increases by enlargement of adipose cells through lipid deposition, as well as by an increase in the number of adipocytes. The process by which adipose cells are derived from a mesenchymal preadipocyte involves an orchestrated series of differentiation steps mediated by a cascade of specific transcription factors. One of the key transcription factors is *peroxisome proliferator-activated receptor γ* (PPAR γ), a nuclear receptor that binds the thiazolidinedione class of insulin-sensitizing drugs used in the treatment of type 2 diabetes ([Chap. 333](#)).

Although the adipocyte has generally been regarded as a storage depot for fat, it is also an endocrine cell that releases numerous molecules in a regulated fashion ([Fig. 77-3](#)). These include the energy balance-regulating hormone leptin, cytokines such as tumor necrosis factor (TNF) α , complement factors such as factor D (also known as adipsin), prothrombotic agents such as plasminogen activator inhibitor 1, and a component of the blood pressure regulating system, angiotensinogen. These factors, and others not yet identified, play a role in the physiology of lipid homeostasis, insulin sensitivity, blood pressure control, and coagulation and are likely to contribute to obesity-related pathologies.

ETIOLOGY OF OBESITY

Though the molecular pathways regulating energy balance are beginning to be illuminated, the causes of obesity remain elusive. In part, this reflects the fact that obesity is a heterogeneous group of disorders. At one level, the pathophysiology of obesity seems simple: a chronic excess of nutrient intake relative to the level of energy expenditure. However, due to the complexity of the neuroendocrine and metabolic systems that regulate energy intake, storage, and expenditure, it has been difficult to quantitate all the relevant parameters (e.g., food intake and energy expenditure) over time in human subjects.

Role of Genes vs. Environment Obesity is commonly seen in families. Inheritance is usually not Mendelian, however, and it is difficult to distinguish the role of genes and environmental factors. Adoptees usually resemble their biologic rather than adoptive parents with respect to obesity, providing strong support for genetic influences. Likewise, identical twins have very similar [BMIs](#) whether reared together or apart, and their BMIs are much more strongly correlated than those of dizygotic twins. These genetic effects appear to relate to both energy intake and expenditure.

Whatever the role of genes, it is clear that the environment plays a key role in obesity, as evidenced by the fact that famine prevents obesity in even the most obesity-prone individual. In addition, the recent increase in the prevalence of obesity in the United States is too rapid to be due to changes in the gene pool. Cultural factors are also important -- these relate to both availability and composition of the diet and to changes in the level of physical activity. In industrial societies, obesity is more common among poor women, whereas in underdeveloped countries, wealthier women are more often obese. In children, obesity correlates to some degree with time spent watching television. High-fat diets may promote obesity, as may diets rich in simple (as opposed to complex) carbohydrates.

Specific Genetic Syndromes Obesity in rodents has been known for many years to be caused by a number of distinct mutations distributed through the genome. Most of these single-gene mutations cause both hyperphagia and diminished energy expenditure, suggesting a link between these two parameters of energy homeostasis. Identification of the *ob* gene mutation in genetically obese (*ob/ob*) mice represents a major breakthrough in the field. The *ob/ob* mouse develops severe obesity, insulin resistance, and hyperphagia, as well as efficient metabolism (e.g., it gets fat even when given the same number of calories as lean littermates). The product of the *ob* gene is the peptide leptin, a name derived from the Greek root *leptos*, meaning thin. Leptin is secreted by adipose cells and acts through the hypothalamus. Its level of production provides an index of adipose energy stores ([Fig. 77-4](#)). High leptin levels decrease food intake and increase energy expenditure. Another mouse mutant, *db/db*, which is resistant to leptin, has a mutation in the leptin receptor and develops a similar syndrome. The *ob* gene is present in humans and expressed in fat. Several families with morbid, early-onset obesity due to inactivating mutations in either leptin or the leptin receptor have been described, thus demonstrating the biologic relevance of leptin in humans. The obesity in these individuals begins shortly after birth, is severe, and is accompanied by neuroendocrine abnormalities. The most prominent of these is hypogonadotropic hypogonadism, which is reversed by leptin replacement. Central hypothyroidism and growth retardation are seen in the mouse model, but their occurrence in leptin-deficient humans is less clear. To date, there is no evidence to suggest that mutations or polymorphisms in the leptin or leptin receptor genes play a prominent role in common forms of obesity.

Mutations in several other genes cause severe obesity in humans ([Table 77-1](#)), each of these syndromes is rare. Mutations in the gene encoding proopiomelanocortin (POMC) cause severe obesity through failure to synthesize α -[MSH](#), a key neuropeptide that inhibits appetite in the hypothalamus. The absence of POMC also causes secondary adrenal insufficiency due to absence of adrenocorticotrophic hormone (ACTH), as well as pale skin and red hair due to absence of MSH. Proenzyme convertase 1 (PC-1)

mutations are thought to cause obesity by preventing synthesis of α -MSH from its precursor peptide, POMC. α -MSH binds to the type 4 melanocortin receptor (MC4R), a key hypothalamic receptor that inhibits eating; mutations of this receptor also cause obesity. These three genetic defects, although rare, define a pathway through which leptin (by stimulating POMC and increasing MSH) restricts food intake and limits weight ([Fig. 77-5](#)).

In addition to these human obesity genes, studies in rodents reveal several other molecular candidates for hypothalamic mediators of human obesity or leanness. The *tub* gene encodes a hypothalamic peptide of unknown function; mutation of this gene causes late-onset obesity. The *fat* gene encodes carboxypeptidase E, a peptide-processing enzyme; mutation of this gene is thought to cause obesity by disrupting production of one or more neuropeptides. [AgRP](#) is coexpressed with [NPY](#) in arcuate nucleus neurons. AgRP antagonizes [MSH](#) action at MC4 receptors, and its overexpression induces obesity. A putative activating mutation in the gene encoding PPAR γ , the adipocyte transcription factor required for adipogenesis, has been linked to obesity in a group of German subjects.

A number of complex human syndromes with defined inheritance are associated with obesity ([Table 77-2](#)). Although specific genes are undefined at present, their identification will likely enhance our understanding of more common forms of human obesity. In the Prader-Willi syndrome, obesity coexists with short stature, mental retardation, hypogonadotropic hypogonadism, hypotonia, small hands and feet, fish-shaped mouth, and hyperphagia. Most patients have a chromosome 15 deletion ([Chap. 66](#)). Laurence-Moon-Biedl syndrome involves obesity, mental retardation, retinitis pigmentosa, polydactyly, and hypogonadotropic hypogonadism.

Other Specific Syndromes Associated with Obesity

Cushing's Syndrome Although obese patients commonly have central obesity, hypertension, and glucose intolerance, they lack other specific stigmata of Cushing's syndrome ([Chap. 331](#)). Nonetheless, a potential diagnosis of Cushing's syndrome is often entertained. Cortisol production and urinary metabolites (17OH steroids) may be increased in simple obesity. Unlike in Cushing's syndrome, however, cortisol levels in blood and urine in the basal state and in response to CRH or [ACTH](#) are normal; the overnight 1-mg dexamethasone suppression test is normal in 90%, with the remainder being normal on a standard 2-day low-dose dexamethasone suppression test.

Hypothyroidism The possibility of hypothyroidism should be considered when evaluating obesity, but it is an uncommon cause of obesity; hypothyroidism is easily ruled out by measuring thyroid stimulating hormone (TSH). Much of the weight gain that occurs in hypothyroidism is due to myxedema ([Chap. 330](#)).

Insulinoma Patients with insulinoma often gain weight as a result of overeating to avoid hypoglycemia symptoms ([Chap. 334](#)). The increased substrate plus high insulin levels promotes energy storage in fat. This can be marked in some individuals but is modest in most.

Craniopharyngioma and Other Disorders Involving the Hypothalamus Whether through

tumors, trauma, or inflammation, hypothalamic dysfunction of systems controlling satiety, hunger, and energy expenditure can cause varying degrees of obesity ([Chap. 328](#)). It is uncommon to identify a discrete anatomic basis for these disorders. Subtle hypothalamic dysfunction is probably a more common cause of obesity than can be documented using currently available techniques. Growth hormone (GH), which exerts lipolytic activity, is diminished in obesity and increases with weight loss. Despite low growth hormone levels, insulin-like growth factor (IGF) I (somatomedin) production is normal, suggesting that GH suppression is a compensatory response to increased nutritional supply.

Pathogenesis of Common Obesity Obesity can result from increased energy intake, decreased energy expenditure, or a combination of the two. Thus, identifying the etiology of obesity should involve measurements of both parameters. However, it is nearly impossible to perform direct and accurate measurements of energy intake in free-living individuals. Obese people, in particular, appear to underreport intake. Measurements of chronic energy expenditure have only recently become available using doubly-labeled water or metabolic chamber/rooms. In subjects at stable weight and body composition, energy intake equals expenditure. Consequently, these techniques allow determination of energy intake in free-living individuals. The level of energy expenditure differs in established obesity, during periods of weight gain or loss, and in the pre- or postobese state. Studies that fail to take note of this phenomenon are not easily interpreted.

There is increased interest in the concept of a body weight "set point." This idea is supported by physiologic mechanisms centered around a sensing system in adipose tissue that reflects fat stores, and a receptor, or "adipostat," that is in the hypothalamic centers. When fat stores are depleted, the adipostat signal is low, and the hypothalamus responds by stimulating hunger and decreasing energy expenditure to conserve energy. Conversely, when fat stores are abundant, the signal is increased, and the hypothalamus responds by decreasing hunger and increasing energy expenditure. The recent discovery of the *ob* gene, and its product leptin, provides a molecular basis for this physiologic concept (see above).

What Is the Status of Food Intake in Obesity (Do the Obese Eat More Than the Lean?) This question has stimulated much debate, due in part to the methodologic difficulties inherent in determining food intake. Many obese individuals believe that they eat small quantities of food, and this claim has often been supported by the results of food intake questionnaires. However, it is now established that average energy expenditure increases as people get more obese, due primarily to the fact that metabolically active lean tissue mass increases with obesity. Given the laws of thermodynamics, the obese person must therefore eat more than the average lean person to maintain their increased weight. It may be the case, however, that a subset of individuals who are predisposed to obesity have the capacity to become obese initially without an absolute increase in caloric consumption.

What Is the State of Energy Expenditure in Obesity? The average total daily energy expenditure is higher in obese than lean individuals when measured at stable weight. However, energy expenditure falls as weight is lost, due in part to loss of lean body mass and to decreased sympathetic nerve activity. When reduced to near-normal

weight and maintained there for a while, (some) obese individuals have lower energy expenditure than (some) lean individuals. There is also a tendency for those who develop obesity as infants or children to have lower resting energy expenditure rates than those who remain lean.

The physiologic basis for variable rates of energy expenditure (at a given body weight and level of energy intake) is essentially unknown. A mutation in the human β_3 adrenergic receptor may be associated with increased risk of obesity and/or insulin resistance in certain (but not all) populations. Homologues of the [BAT](#) uncoupling protein, named [UCP](#)-2 and UCP-3, have been identified in both rodents and humans. UCP-2 is expressed widely, whereas UCP-3 is primarily expressed in skeletal muscle. These proteins may play a role in disordered energy balance.

One newly described component of thermogenesis, called *nonexercise activity thermogenesis* (NEAT), has been linked to obesity. It is the thermogenesis that accompanies physical activities other than volitional exercise, such as the activities of daily living, fidgeting, spontaneous muscle contraction, and maintaining posture. NEAT accounts for about two-thirds of the increased daily energy expenditure induced by overfeeding. The wide variation in fat storage seen in overfed individuals is predicted by the degree to which NEAT is induced. The molecular basis for NEAT and its regulation are unknown.

Leptin in Typical Obesity The vast majority of obese people have increased leptin levels but do not have mutations of either leptin or its receptor. They appear, therefore, to have a form of functional "leptin resistance." Data suggesting that some individuals produce less leptin per unit fat mass than others or have a form of relative leptin deficiency that predisposes to obesity are at present contradictory and unsettled. The mechanism for leptin resistance, and whether it can be overcome by raising leptin levels, is not yet established. Some data suggest that leptin may not effectively cross the blood-brain barrier as levels rise. It is also possible that leptin signaling inhibitors are involved in the leptin-resistant state.

PATHOLOGIC CONSEQUENCES OF OBESITY

Obesity has major adverse effects on health. Morbidly obese individuals (>200% ideal body weight) have as much as a twelvefold increase in mortality. Mortality rates rise as obesity increases, particularly when obesity is associated with increased intraabdominal fat (see above). It is also apparent that the degree to which obesity affects particular organ systems is influenced by susceptibility genes that vary in the population.

Insulin Resistance and Type 2 Diabetes Mellitus Hyperinsulinemia and insulin resistance are pervasive features of obesity, increasing with weight gain and diminishing with weight loss. Insulin resistance is more strongly linked to intraabdominal fat than to fat in other depots. The molecular link between obesity and insulin resistance has been sought for many years, with the major factors under investigation being: (1) insulin itself, by inducing receptor downregulation; (2) free fatty acids, known to be increased and capable of impairing insulin action; and (3) the cytokine [TNF](#)- α , which is produced by adipocytes, overexpressed in obese adipocytes, and capable of inhibiting insulin action. Despite insulin resistance, most obese individuals do not develop diabetes, suggesting

that the onset of diabetes requires an interaction between obesity-induced insulin resistance and other factors that predispose to diabetes, such as impaired insulin secretion ([Chap. 333](#)). Obesity, however, is a major risk factor for diabetes, and as many as 80% of patients with type 2 diabetes mellitus are obese. Weight loss, even of modest degree, is associated with increased insulin sensitivity and often improves glucose control in diabetes.

Reproductive Disorders Disorders that affect the reproductive axis are associated with obesity in both men and women. Male hypogonadism is associated with increased adipose tissue, often distributed in a pattern more typical of females. In men >160% ideal body weight, plasma testosterone and sex hormone-binding globulin (SHBG) are often reduced, and estrogen levels (derived from conversion of adrenal androgens in adipose tissue) are increased ([Chap. 335](#)). Gynecomastia may be seen. However, masculinization, libido, potency, and spermatogenesis are preserved in most of these individuals. Free testosterone may be decreased in morbidly obese men whose weight exceeds 200% ideal body weight.

Obesity has long been associated with menstrual abnormalities in women, particularly in women with upper body obesity ([Chaps. 52 and 336](#)). Common findings are increased androgen production, decreased SHBG, and increased peripheral conversion of androgen to estrogen. Most obese women with oligomenorrhea have the polycystic ovarian syndrome (PCOS), with its associated anovulation and ovarian hyperandrogenism; 40% of women with PCOS are obese. Interestingly, most nonobese women with PCOS are also insulin-resistant, suggesting that insulin resistance, hyperinsulinemia, or the combination of the two are causative or contribute to the ovarian pathophysiology in PCOS in both obese and lean individuals. In obese women with PCOS, weight loss or treatment with insulin-sensitizing drugs often restores normal menses, along with a fall in estrone levels and normalized gonadotropin secretion. The increased conversion of androstenedione to estrogen, which occurs to a greater degree in women with lower body obesity, may contribute to the increased incidence of uterine cancer in postmenopausal women with obesity.

Cardiovascular Disease The Framingham Study revealed that obesity was an independent risk factor for the 26-year incidence of cardiovascular disease in men and women [including coronary disease, stroke, and congestive heart failure (CHF)]. The waist/hip ratio may be the best predictor of these risks. When the additional effects of hypertension and glucose intolerance associated with obesity are included, the adverse impact of obesity is even more evident. The effect of obesity on cardiovascular mortality in women may be seen at BMIs as low as 25. Obesity, especially abdominal obesity, is associated with an atherogenic lipid profile, with increased low-density lipoprotein (LDL) cholesterol, very low density lipoprotein and triglyceride, and decreased high-density lipoprotein cholesterol ([Chap. 344](#)). Obesity is also associated with hypertension. Measurement of blood pressure in the obese requires use of a larger cuff size to avoid artifactual increases. Obesity-induced hypertension is associated with increased peripheral resistance and cardiac output, increased sympathetic nervous system tone, increased salt sensitivity, and insulin-mediated salt retention; it is often responsive to modest weight loss.

Pulmonary Disease Obesity may be associated with a number of pulmonary

abnormalities. These include reduced chest wall compliance, increased work of breathing, increased minute ventilation due to increased metabolic rate, and decreased total lung capacity and functional residual capacity ([Chap. 250](#)). Severe obesity may be associated with obstructive sleep apnea and the "obesity hypoventilation syndrome" ([Chap. 263](#)). Sleep apnea can be obstructive (most common), central, or mixed. Weight loss (10 to 20 kg) can bring substantial improvement, as can major weight loss following gastric bypass or restrictive surgery. Continuous positive airway pressure has been used with some success.

Gallstones Obesity is associated with enhanced biliary secretion of cholesterol, supersaturation of bile, and a higher incidence of gallstones, particularly cholesterol gallstones ([Chap. 302](#)). A person 50% above ideal body weight has about a sixfold increased incidence of symptomatic gallstones. Paradoxically, fasting increases supersaturation of bile by decreasing the phospholipid component. Fasting-induced cholecystitis is a complication of extreme diets.

Cancer Obesity in males is associated with higher mortality from cancer of the colon, rectum, and prostate; obesity in females is associated with higher mortality from cancer of the gallbladder, bile ducts, breasts, endometrium, cervix, and ovaries. Some of the latter may be due to increased rates of conversion of androstenedione to estrone in adipose tissue of obese individuals.

Bone, Joint, and Cutaneous Disease Obesity is associated with an increased risk of osteoarthritis, no doubt partly due to the trauma of added weight bearing. The prevalence of gout may also be increased ([Chap. 322](#)). Among the skin problems associated with obesity is acanthosis nigricans, manifested by darkening and thickening of the skin folds on the neck, elbows, and dorsal interphalangeal spaces. Acanthosis reflects the severity of underlying insulin resistance and diminishes with weight loss. Friability of skin may be increased, especially in skin folds, enhancing the risk of fungal and yeast infections. Finally, venous stasis is increased in the obese.

TREATMENT

Obesity is a chronic medical condition. Successful treatment, defined as the sustained attainment of normal body weight without producing unacceptable treatment-induced morbidity, is rarely achieved in clinical practice. Many approaches produce short-term weight loss, and this has clear benefits for associated morbidities such as hypertension and diabetes. Despite the fact that sustained weight loss is uncommon, enormous resources are expended in pursuit of this goal.

Treatment goals should be guided by the health risks of obesity in any given individual ([Fig. 77-6](#)). The clinician should always consider the possibility that an individual has an identified cause of obesity, such as hypothyroidism, hypercortisolism, male hypogonadism, insulinoma, or central nervous system disease that affects hypothalamic function. Although they are infrequent causes of obesity, specific therapy may be available.

Behavior Modification The principles of behavior modification provide the underpinnings for many current programs of weight reduction. Typically, the patient is

requested to monitor and record the circumstances related to eating, and rewards are designed to modify maladaptive behaviors. Patients may benefit from counseling offered in a stable group setting for extended periods of time, including after weight loss.

Diet Reduced caloric intake is the cornerstone of obesity treatment. The fundamental goal is the sustained reduction of energy intake below that of energy expenditure. The difficulty in achieving this goal has led to a wide array of suggested diets that vary in recommended calorie content (from total fasting to mild reductions), as well as specific food content and form (e.g., liquid vs. solid). There is no scientific evidence to validate the utility of specific "fad diets." The main diet regimens in use follow several general facts relevant to food intake and weight loss. First, a deficit of 7500 kcal will produce a weight loss of approximately 1 kg. Therefore, eating 100 kcal/d less for a year should cause a 5-kg weight loss, and a deficit of 1000 kcal/d should cause a loss of approximately 1 kg per week. The rate of weight loss on a given caloric intake is related to the rate of energy expenditure. Because obese individuals have a higher metabolic rate than lean individuals, and because men have a higher metabolic rate than women (due to their greater lean body mass), the rate of weight loss is greater among the more obese and among men (relative to women). With chronic caloric restriction, metabolic rate diminishes, but because of reduced lean body mass (along with much greater loss in fat mass) and possibly because of other adaptations. This fall in metabolic rate with food restriction slows the rate of weight loss on a constant diet. With total starvation or diets restricted to <600 kcal/d, initial weight loss over the first week results predominantly from natriuresis and the loss of fluids.

Very low energy diets (e.g., 400 to 600 kcal/d) are widely used. The liquid protein diets popularized in the 1970s were proved to be unsafe, causing >60 deaths. Life-threatening arrhythmias were documented in the clinical research setting, a consequence of both low-quality protein and deficiencies of vitamins, minerals, and trace elements. These types of diets have now been substantially modified. A very low energy diet consisting of 45 to 70 g high-quality protein, 30 to 50 g carbohydrate, and approximately 2 g fat per day, as well as supplements of vitamins, minerals, and trace elements, appears to be safe in selected patients under medical supervision. Patients should not be started on such diets unless they are >130% of their ideal body weight. Contraindications include pregnancy, cancer, recent myocardial infarction, cerebrovascular disease, hepatic disease, or untreated psychiatric disease. When used in patients with diabetes who are receiving insulin or oral agent therapy, close supervision is required and diabetic treatment may need to be adjusted. Whenever possible, exercise regimens and behavioral modification approaches should be used in conjunction with the diet.

Advantages of very low calorie diets are the greater rate of weight loss compared to less restrictive diets, as well as the possible beneficial effect of hunger suppression brought about by the production of ketones. In patients on such diets, blood pressure, blood glucose, cholesterol, and triglyceride levels fall, and pulmonary function and exercise tolerance improve. Sleep apnea may improve within a few weeks. Complications of these very low energy diets are usually minor and include fatigue, constipation or diarrhea, dry skin, hair loss, menstrual irregularities, orthostatic dizziness, and difficulty concentrating. Cholelithiasis and pancreatitis may occur when such diets are interrupted by binge eating; gallstones have been shown to develop in as many as 25% of patients

while on the diet.

Low-calorie diets, >800 kcal/d, are applicable to most patients and have fewer restrictions than the very low calorie diets. Considerable controversy has attended the question of which diet composition is most appropriate for promoting weight loss. Though commonly recommended, benefits resulting from very low fat diets are modest at best. Nonetheless, the health effects of low-fat diets -- apart from curbing obesity -- may be important. A diet rich in fruits, vegetables, and whole grains may promote weight loss and is preferable to low-fat diets in which large amounts of simple carbohydrates are substituted for fats. The latter may actually promote obesity. Some have advocated diets with protein replacement of simple carbohydrates in an effort to minimize insulin production. The efficacy of this strategy, aside from overall calorie reduction, is unknown.

Exercise Exercise is an important component of the overall approach to treating obesity. Increased energy expenditure is the most obvious mechanism for an effect of exercise. The impact of an exercise regimen as a sole therapy of obesity has been difficult to document. On the other hand, exercise appears to be a valuable means to sustain diet therapy ([Fig. 77-7](#)). Even if exercise had no such salutary effect, it would be valuable in the obese individual for its effects on cardiovascular tone and blood pressure. Because many obese individuals have not engaged in exercise on a regular basis and may have cardiovascular risk factors, it should be introduced gradually and under medical supervision, especially in the most obese individuals.

Drugs Unfortunately, drug treatment of obesity is rarely efficacious. Despite short-term benefits, medication-induced weight loss is often associated with rebound weight gain after the cessation of drug use, side effects from the medications, and the potential for drug abuse. Given the need for effective therapies, many possible compounds have been evaluated. On the basis of placebo-controlled trials, the U.S. Food and Drug Administration (FDA) approved several amphetamine-like agents for short-term use. Phentermine is an amphetamine-like drug with low addictive potential that has shown modest efficacy (10 vs. 4.4 kg of weight loss over a 24-week period in well-controlled study). This class of drugs is thought to act centrally by reducing appetite. Effects on energy expenditure are less clear. Over-the-counter drugs, such as phenylpropanolamine HCl, have similar efficacy to prescription appetite suppressants in short-term studies. Drugs that promote serotonin release or inhibit serotonin reuptake, such as fenfluramine, also have modest efficacy. When fenfluramine was administered together with phentermine, as "fen-phen," the combination was widely used based on controlled trials that demonstrated modest but definite efficacy. However, the risk of primary pulmonary hypertension was increased up to 20-fold in association with this treatment. The FDA withdrew approval of the fen-phen combination in 1997 when reports suggested an association with right- and left-sided valvular heart disease. The histopathologic features of the valvular disease are similar to those seen in carcinoid syndrome and are thought to result from fenfluramine. Though the true incidence and long-term effects of these valvular lesions are currently unknown, the occurrence of this complication has been verified in multiple studies.

Sibutramine is a novel central reuptake inhibitor of both norepinephrine and serotonin. Using a once-daily dose over 24 weeks, it produced a 7% weight loss in a double-blind,

placebo-controlled trial. It lowered cholesterol and triglyceride levels and exhibited similar clinical efficacy to fenfluramine. Sibutramine increases pulse and blood pressure in some patients, and long-term safety is not established. Orlistat is an inhibitor of intestinal lipase that causes modest weight loss due to drug-induced fat malabsorption. A randomized, double-blind trial over 2 years revealed modest weight loss (8.7 kg for 120 mg orlistat versus 5.8 kg from diet alone) during the first year and better maintenance of weight loss in a second year compared to the placebo-treated group (3.2 kg regained versus 5.6 kg regained for placebo). [LDL](#) cholesterol and insulin levels were also reduced. In patients with obesity and type 2 diabetes mellitus, the antidiabetic medication metformin tends to decrease body weight. The mechanism appears to involve inhibition of appetite. Thyroid hormone has little place in the treatment of obesity, as the vast majority of obese individuals are euthyroid. It promotes loss of lean body mass and raises the risk of complications from the hyperthyroid state.

β_3 -Adrenergic receptor agonists may provide a new treatment approach for obesity. Drugs of this class are in clinical trials. In animals, β_3 agonists promote leanness by stimulating thermogenesis in [BAT](#); they also stimulate lipolysis in white adipose tissue. These drugs also reduce insulin resistance and lower blood glucose in animal models by a mechanism that is not yet defined. Recombinant human leptin is also in clinical trials. In the rare cases of leptin deficiency caused by mutations of the leptin gene, the administration of recombinant leptin is highly effective. Preliminary reports suggest that the response to leptin is limited or absent in common causes of obesity (which are associated with hyperleptinemia and leptin resistance). New drugs are also being developed based on insights into central pathways that regulate body weight. These include antagonists for [NPY](#) receptors (subtypes Y1, Y5) and agonists for melanocortin 4 receptors.

Surgery Morbid obesity, commonly defined as either 45 kg (100 lb) or 100% above ideal body weight, is estimated to increase mortality by as much as twelvefold in men between 25 and 34 years of age and sixfold between 35 and 45 years of age. Deaths from cardiovascular disease, diabetes, and accidents have been documented. In response to ineffective treatment using diet, exercise, and available drugs, surgical approaches have been tried. The potential benefits of surgery include major weight loss and improvement in hypertension, diabetes, sleep apnea, [CHF](#), angina, hyperlipidemia, and venous disease. Many different approaches have been used, often without adequate assessment of efficacy and complications. Jejunoileal bypass surgery has largely been abandoned because of complications, which have included electrolyte disturbances, nephrolithiasis, gallstones, gastric ulcers, arthritis, and hepatic dysfunction, with cirrhosis occurring in as many as 7% of patients. Two procedures in common use today are the vertical-banded gastroplasty and the Roux-en-Y gastric bypass ([Fig. 77-8](#)).

Following the National Institutes of Health Consensus Conference on Gastrointestinal Surgery for Severe Obesity in 1991, it was recommended that suitable patients be selected using the following criteria: (1) the presence of 45 kg (100 lb) or 100% above ideal body weight, or one or more severe medical conditions related to refractory obesity; (2) repeated failures of other therapeutic approaches; (3) at eligible weight for 3 to 5 years; (4) capability of tolerating surgery; (5) absence of alcoholism, other addictions, or major psychopathology; and (6) prior clearance by a psychiatrist. It is

recommended that an appropriately experienced surgeon work together with nutritionists and other support personnel; evaluation and follow-up programs should be monitored closely.

ACKNOWLEDGEMENT

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78. EATING DISORDERS - B. Timothy Walsh

Anorexia nervosa and bulimia nervosa are characterized by severe disturbances of eating behavior. The salient feature of *anorexia nervosa* is a refusal to maintain a minimally normal body weight. *Bulimia nervosa* is characterized by recurrent episodes of binge eating followed by abnormal compensatory behaviors, such as self-induced vomiting. Anorexia nervosa and bulimia nervosa are closely related. Both occur primarily among previously healthy young women who become overly concerned with body shape and weight. Many patients with bulimia nervosa have past histories of anorexia nervosa, and many patients with anorexia nervosa engage in binge eating and purging behavior. In the current diagnostic system, the critical distinction between anorexia nervosa and bulimia nervosa depends on body weight: patients with anorexia nervosa are, by definition, significantly underweight, whereas the weights of patients with bulimia nervosa are in the normal range or above.

Another syndrome of disturbed eating behavior has been described recently: *Binge eating disorder* is characterized by repeated episodes of binge eating, similar to those of bulimia nervosa, in the absence of inappropriate compensatory behavior. Patients with binge eating disorder are typically middle-aged men or women with significant obesity. They have an increased frequency of anxiety and depression compared to similarly obese patients without binge eating disorder. It is not known whether patients with binge eating disorder are at increased risk for medical complications or what treatments are most useful.

EPIDEMIOLOGY

In women, the full syndrome of anorexia nervosa occurs with a lifetime prevalence of approximately 0.5%; bulimia nervosa occurs with a lifetime prevalence of 1 to 3%. Variants of these eating disorders with only some features of anorexia nervosa or bulimia nervosa are much more common and occur in 5 to 10% of young women. Both anorexia nervosa and bulimia nervosa also occur in males but at frequencies approximately one-tenth of those in females.

Anorexia nervosa and bulimia nervosa are more prevalent in cultures where food is plentiful and in which being thin is associated with attractiveness. These disorders are more frequent among young women who place a premium on thinness, such as ballet dancers and models. The incidence of anorexia nervosa appears to have increased in recent decades. The frequency of bulimia nervosa increased dramatically in the early 1970s and 1980s but may have declined somewhat in recent years.

DIAGNOSIS

The diagnosis of eating disorders is based on the presence of characteristic behavioral, psychological, and physical attributes ([Tables 78-1](#) and [78-2](#)). Widely accepted diagnostic criteria are provided by the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders* (DSM-IV).

ANOREXIA NERVOSA

For anorexia nervosa, these criteria include weight <85% of the expected weight for age and height, which is roughly equivalent to a body mass index (BMI) of 18.5 kg/m² for adult women. This weight criterion is somewhat arbitrary, so that a patient who meets all other diagnostic criteria but weighs between 85 and 90% of expected would still merit the diagnosis of anorexia nervosa. Despite being underweight, patients with anorexia nervosa are irrationally afraid of gaining weight, often out of a concern that weight gain will get "out of control." They also exhibit a distortion of body image (criterion 3, [Table 78-1](#)), which may express itself in several ways. For example, despite being emaciated, patients with anorexia nervosa may believe that their body as a whole, or some part of their body, is too fat and experience additional weight loss as a highly rewarding achievement. The current diagnostic criteria require that women with anorexia nervosa not have spontaneous menses, but occasional patients with the characteristics and complications of anorexia nervosa describe regular menses. Two mutually exclusive subtypes of anorexia nervosa are specified in [DSM-IV](#). Patients whose weight loss is maintained primarily by caloric restriction, perhaps augmented by excessive exercise, are considered to have the "restricting" subtype of anorexia nervosa. The "binge eating/purging" subtype is characterized by self-induced vomiting or laxative abuse. Patients with the binge/purge subtype are more prone to develop electrolyte imbalances, are more emotionally labile, and are more likely to have other problems with impulse control, such as drug abuse.

The diagnosis of anorexia nervosa can usually be made confidently on the basis of history when significant weight loss is accomplished by restrictive dieting and excessive exercise and is accompanied by a marked reluctance to gain weight. Patients with anorexia nervosa often deny that they have a serious problem and may be brought to medical attention by concerned family or friends. In atypical presentations, other causes of significant weight loss in previously healthy young people should be considered, including inflammatory bowel disease, gastric outlet obstruction, central nervous system (CNS) tumors, and neoplasm ([Chap. 43](#)).

BULIMIA NERVOSA

The critical diagnostic features of bulimia nervosa are repeated episodes of binge eating followed by inappropriate and abnormal behaviors aimed at avoiding weight gain. During binges, patients with this disorder tend to consume large amounts of sweet foods with a high fat content, such as dessert items. The most frequent compensatory behaviors are self-induced vomiting and laxative abuse, but a wide variety of techniques have been described, including the omission of insulin injections by diabetics. Typically, patients with bulimia nervosa are ashamed of their behavior and endeavor to keep their disorder hidden from family and friends. Like patients with anorexia nervosa, those with bulimia nervosa place an unusual emphasis on weight and shape as a basis for their self-esteem.

As in anorexia nervosa, there are two mutually exclusive subtypes of bulimia nervosa. Patients with the "purging" subtype utilize compensatory behaviors that directly rid the body of calories or fluids (e.g., self-induced vomiting, laxative or diuretic abuse), whereas those with the "nonpurging" subtype attempt to compensate for binges by fasting or by excessive exercise. Patients with the nonpurging subtype tend to be heavier and are less prone to fluid and electrolyte disturbances.

The diagnosis of bulimia nervosa requires a candid history from the patient detailing recurrent, large eating binges followed by the purposeful use of inappropriate mechanisms to avoid weight gain. Most patients with bulimia nervosa who present for treatment are distressed by their inability to control their eating behavior but are able to provide such details if queried in a supportive and nonjudgmental fashion.

ETIOLOGY

The fundamental etiology of the eating disorders is unknown but is believed to involve a combination of psychological, biologic, and cultural risk factors. Many of these risk factors, such as sexual or physical abuse and a family history of mood disturbance or substance abuse, are best viewed as nonspecific risk factors that increase vulnerability to a range of psychiatric disorders. Other factors appear to be more specific to the development of an eating disorder.

Patients who develop anorexia nervosa are inclined to be more obsessional and perfectionist than their peers. The disorder often begins as a diet not distinguishable at the outset from those undertaken by many adolescents and young women. As weight loss progresses, the fear of gaining weight grows; dieting becomes stricter; and psychological, behavioral, and medical aberrations increase. The fact that most cases are reported from countries where food is plentiful and where thinness, especially among women, is highly valued suggests that cultural factors play a significant role in the development of anorexia nervosa. However, it is notable that the clinical syndrome was well described over a century ago, when the cultural pressures were quite different.

Numerous physiologic disturbances, including abnormalities in a variety of neurotransmitter systems, have been described in anorexia nervosa (see below). It is difficult to distinguish neurochemical, metabolic, and hormonal changes that may have a role in the initiation or perpetuation of the syndrome from those that are secondary to the disorder. The resolution of most of these abnormalities with weight restoration argues against their having a critical etiologic role.

Bulimia nervosa typically begins during or following an episode of dieting, often in association with depressed mood. Patients who develop bulimia nervosa describe a higher-than-expected prevalence of childhood and parental obesity, suggesting that a predisposition towards obesity may increase vulnerability to this eating disorder. The marked increase in the number of cases of bulimia nervosa during the past 25 years and the rarity of bulimia nervosa in underdeveloped countries suggest that cultural factors are important. Several biologic abnormalities in patients with bulimia nervosa may perpetuate this disorder once it has begun. These include abnormalities of [CNS](#) serotonergic function, which is involved in the regulation of eating behavior, and disruption of peripheral satiety mechanisms, including the release of cholecystokinin (CCK) from the small intestine.

Genetic factors probably contribute to the risk of development of eating disorders, as the incidence of these disorders is greater in families with one affected member and the concordance in monozygotic twins is greater than in dizygotic twins. However, specific genes have not been identified, and the range of the estimates of heritability is large.

CLINICAL FEATURES ([Table 78-3](#))

ANOREXIA NERVOSA

Anorexia nervosa typically begins in mid to late adolescence, sometimes in association with a stressful life event such as leaving home for school. The disorder occasionally develops in early puberty, before menarche, but seldom begins after age 40. Despite being underweight, patients with anorexia nervosa rarely complain of hunger or fatigue and often exercise extensively. Further weight loss is viewed by the patient as a fulfilling accomplishment, while weight gain is seen as a personal failure. Patients tend to become socially withdrawn and increasingly committed to work or study, dieting, and exercise. As weight loss progresses, thoughts of food dominate mental life and idiosyncratic rules develop around eating. Patients with anorexia nervosa may obsessively collect cookbooks and recipes and be drawn to food-related occupations. Despite the denial of hunger, one-quarter to one-half of patients with anorexia nervosa engage in eating binges.

Physical Features Patients with anorexia nervosa typically have few physical complaints but may note cold intolerance and constipation. Some women who develop anorexia nervosa after menarche report that their menses ceased before significant weight loss occurred. Weight and height should be measured to allow calculation of [BMI](#) (kg/m^2). Vital signs may reveal bradycardia, hypotension, and hypothermia. Soft, downy hair growth (lanugo) sometimes occurs, especially on the back, and alopecia may be seen. Salivary gland enlargement, which is associated with starvation as well as with binge eating and vomiting, may make the face appear surprisingly full in contrast to the marked general wasting. Acrocyanosis of the digits is common, and peripheral edema can be seen in the absence of hypoalbuminemia, particularly when the patient begins to regain weight. Some patients who consume large amounts of vegetables containing vitamin A develop a yellow tint to the skin (*hypercarotenemia*), which is especially notable on the palms.

Laboratory Abnormalities Mild normochromic, normocytic anemia is frequent, as is mild to moderate leukopenia, with a disproportionate reduction of polymorphonuclear leukocytes. Dehydration may result in slightly increased levels of blood urea nitrogen and creatinine. Serum liver enzyme levels may increase, especially during the early phases of refeeding. The level of serum proteins is usually normal. Blood sugar is often low and serum cholesterol may be moderately elevated. Gastrointestinal motility is diminished, leading to reduced gastric emptying and constipation. A range of electrolyte disturbances may develop, reflecting the degree to which the patient restricts or overconsumes fluids and whether the patient engages in purging behavior. Hypokalemic alkalosis suggests self-induced vomiting or the use of diuretics. Hyponatremia is common and may result from excess fluid intake and disturbances in the secretion of antidiuretic hormone.

Endocrine Abnormalities The regulation of virtually every endocrine system is altered in anorexia nervosa, but the most striking changes occur in the reproductive system. Amenorrhea is hypothalamic in origin and reflects diminished production of gonadotropin-releasing hormone (GnRH). When exogenous GnRH is administered in a

physiologic pulsatile manner, pituitary responses of luteinizing hormone (LH) and follicle stimulating hormone (FSH) are normalized, indicating the absence of a primary pituitary abnormality. The resulting gonadotropin deficiency causes low plasma estrogen in women and reduced testosterone in men. The hypothalamic GnRH pulse generator is exquisitely sensitive, particularly in women, to body weight, stress, and exercise, each of which may contribute to *hypothalamic amenorrhea* in anorexia nervosa ([Chap. 336](#)). Although the mechanisms underlying these effects are unknown, the decreased adipose tissue associated with weight loss leads to a marked reduction in leptin, a hormone that plays a permissive role in GnRH production ([Chap. 77](#)). In many patients, weight gain to a specific threshold triggers restoration of the GnRH pulse generator, initially recapitulating the pubertal pattern of nocturnal gonadotropin secretion before returning to the normal adult pattern.

Serum cortisol and 24-h urine free cortisol levels are generally elevated but without characteristic clinical signs of cortisol excess. Thyroid function tests resemble the pattern seen in euthyroid sick syndrome ([Chap. 330](#)). Thyroxine (T₄) and free T₄ levels are usually in the low-normal range, triiodothyronine (T₃) levels are reduced, and reverse T₃(rT₃) is elevated. The level of thyroid stimulating hormone (TSH) is normally or partially suppressed. Growth hormone is increased, but insulin-like growth factor 1 (IGF-1), which is produced mainly by the liver, is reduced, as it is in other conditions of starvation. Diminished bone density is routinely observed in anorexia nervosa and reflects the effects of multiple nutritional deficiencies, reduced gonadal steroids, and increased cortisol. The degree of bone density reduction is proportional to the length of the illness, and patients are at risk for the development of symptomatic fractures. The occurrence of anorexia nervosa during adolescence may lead to the premature cessation of linear bone growth and a failure to achieve expected adult height.

Cardiac Abnormalities Cardiac output is reduced, and congestive heart failure occasionally occurs during rapid refeeding. The electrocardiogram usually shows sinus bradycardia, reduced QRS voltage, and nonspecific ST-T-wave abnormalities. Some patients develop a prolonged QT_c interval, which may predispose to serious arrhythmias.

BULIMIA NERVOSA

The typical patient presenting for treatment of bulimia nervosa is a woman of normal weight in her mid-twenties who reports binge eating and purging 5 to 10 times a week for 5 to 10 years. The disorder usually begins in late adolescence or early adulthood during or following a diet. The self-imposed caloric restriction leads to increased hunger and to overeating. In an attempt to avoid weight gain, the patient induces vomiting, takes laxatives or diuretics, or engages in some other form of compensatory behavior. Initially, patients may experience a sense of satisfaction that appealing food can be eaten without weight gain. However, as the disorder progresses, patients perceive diminished control over eating. Binges increase in size and frequency and are provoked by a variety of stimuli, such as transient depression, anxiety, or a sense that too much food has been consumed in a normal meal. Between binges, patients attempt to restrict caloric intake, which increases hunger and sets the stage for the next binge.

Although vomiting may be triggered initially by manual stimulation of the gag reflex, most patients with bulimia nervosa develop the ability to induce vomiting at will. Rarely,

patients resort to the regular use of syrup of ipecac. Laxatives and diuretics are frequently taken in impressive quantities, such as 30 or 60 laxative pills on a single occasion. The resulting fluid loss produces dehydration and a feeling of emptiness but has little impact on caloric balance.

The physical abnormalities associated with bulimia nervosa primarily result from the purging behavior. Painless bilateral salivary gland hypertrophy (sialadenosis) may be noted. A scar or callus on the dorsum of the hand may develop due to repeated trauma from the teeth among patients who manually stimulate the gag reflex. Recurrent vomiting and the exposure of the lingual surfaces of the teeth to stomach acid leads to loss of dental enamel and eventually to chipping and erosion of the front teeth. Laboratory abnormalities are surprisingly infrequent, but hypokalemia, hypochloremia, and hyponatremia are observed occasionally. Repeated vomiting may lead to alkalosis, whereas repeated laxative abuse may produce a mild metabolic acidosis. Serum amylase may be mildly elevated due to an increase in the salivary isoenzyme.

Serious physical complications resulting from bulimia nervosa are rare. Oligomenorrhea and amenorrhea are more frequent than in women without eating disorders. Arrhythmias occasionally occur secondary to electrolyte disturbances. Tearing of the esophagus and rupture of the stomach, which constitute life-threatening events, have been reported. Some patients who have chronically abused laxatives or diuretics develop transient peripheral edema when this behavior ceases, presumably due to high levels of aldosterone resulting from persistent fluid and electrolyte depletion.

PROGNOSIS

The course and outcome of anorexia nervosa are highly variable. One-quarter to one-half of patients eventually recover fully, with few psychological or physical sequelae. However, many patients have persistent difficulties with weight maintenance, depression, and eating disturbances, including bulimia nervosa. The development of obesity following anorexia nervosa is rare. The long-term mortality of anorexia nervosa is among the highest associated with any psychiatric disorder. Approximately 5% of patients die per decade of follow-up, primarily due to the physical effects of chronic starvation or by suicide.

Virtually all of the physiologic abnormalities associated with anorexia nervosa are observed in other forms of starvation and markedly improve or disappear with weight gain. A worrisome exception is the reduction in bone mass, which may not recover fully, particularly when anorexia nervosa occurs during adolescence when peak bone mass is normally achieved.

The prognosis of bulimia nervosa is much more favorable. Mortality is low, and full recovery occurs in approximately 50% of patients within 10 years. Approximately 25% of patients have persistent symptoms of bulimia nervosa over many years. Few patients progress from bulimia nervosa to anorexia nervosa.

TREATMENT

Anorexia Nervosa Because of the profound physiological and psychological effects of

starvation, there is a broad consensus that weight restoration to 90% of predicted weight is the primary goal in the treatment of anorexia nervosa. Unfortunately, because most patients resist this goal, its accomplishment is often accompanied by frustration for the patient, the family, and the physician. In attempting to engage the patient in treatment, it may be useful for the physician to elicit the patient's physical concerns (e.g., about osteoporosis, weakness, or fertility) and, if possible, educate the patient regarding the importance of normalizing nutritional status in order to address those concerns. The physician should attempt to reassure the patient that weight gain will not be permitted to get "out of control" but simultaneously emphasize that weight restoration is medically and psychologically imperative.

The intensity of the initial treatment, including the need for hospitalization, is determined by the patient's current weight, the rapidity of recent weight loss, and the severity of medical and psychological complications ([Fig. 78-1](#)). Hospitalization should be strongly considered for patients weighing <75% of expected, even if the results of routine blood studies are within normal limits. Acute medical problems, such as severe electrolyte imbalances, should be identified and addressed. Nutritional restoration can almost always be successfully accomplished by oral feeding, and parenteral methods are rarely required. For severely underweight patients, sufficient calories (approximately 1500 to 1800 kcal/d) should be provided initially in divided meals as food or liquid supplements to maintain weight and to permit stabilization of fluid and electrolyte balance. Calories can then be gradually increased to achieve a weight gain of 1 to 2 kg (2 to 4 lb) per week, typically requiring an intake of 3000 to 4000 kcal/d. Meals must be supervised, ideally by personnel who are firm regarding the necessity of food consumption, empathic regarding the challenges entailed, and reassuring regarding the patient's eventual recovery. Patients have great psychological difficulty complying with the need for increased caloric consumption, and the assistance of psychiatrists or psychologists experienced in the treatment of anorexia nervosa is usually necessary.

Psychiatric treatment focuses primarily on two issues. First, patients require much emotional support during the period of weight gain. Second, patients must learn to base their self-esteem, not on the achievement of an inappropriately low weight, but on the development of satisfying personal relationships and the attainment of reasonable academic and occupational goals. For younger patients, the active involvement of the family in treatment is crucial.

Less severely affected patients may be treated in a partial hospitalization program where medical and psychiatric supervision is available and several meals can be monitored each day. Outpatient treatment may suffice for mildly ill patients. Weight must be monitored at frequent intervals, and explicit goals agreed on for weight gain, with the understanding that more intensive treatment will be required if the level of care initially employed is not successful.

Medical complications occasionally occur during refeeding. Most patients transiently retain excess fluid, occasionally resulting in peripheral edema. Fluid retention occurs during recovery from other forms of malnutrition and generally does not require specific treatment in the absence of cardiac, renal, or hepatic dysfunction. Congestive heart failure and acute gastric dilatation have been described when refeeding has been rapid. Transient modest elevations in serum levels of liver enzymes occasionally occur. Low

levels of magnesium and phosphate should be repleted. Multivitamins should be given, and it is important to ensure adequate intake of vitamin D (400 IU/d) and calcium (1500 mg/d) to minimize bone loss.

No psychotropic medications are of established value in the treatment of anorexia nervosa; tricyclic antidepressants are contraindicated when there is prolongation of the QT_c interval. The alterations of cortisol and thyroid hormone metabolism do not require specific treatment and are corrected by weight gain. Estrogen treatment appears to have minimal impact on bone density in underweight patients but may be helpful to relieve symptoms of estrogen deficiency.

Bulimia Nervosa Bulimia nervosa can usually be treated on an outpatient basis. Cognitive behavioral therapy (CBT) is a short-term (4 to 6 months) psychological treatment that focuses on the intense concern with shape and weight, the persistent dieting, and the binge eating and purging that characterize this disorder. Patients are directed to monitor the circumstances, thoughts, and emotions associated with binge/purge episodes, to eat regularly, and to challenge their assumptions linking weight to self-esteem. CBT produces symptomatic remission in 25 to 50% of patients.

Numerous double-blind, placebo-controlled trials have documented that antidepressant medications are useful in the treatment of bulimia nervosa but are probably somewhat less effective than CBT. Although efficacy has been established for virtually all chemical classes of antidepressants, only the selective serotonin reuptake inhibitor fluoxetine (Prozac) has been approved for use in bulimia nervosa by the U.S. Food and Drug Administration. Antidepressant medications are helpful even for patients with bulimia nervosa who are not depressed, and the dose of fluoxetine recommended for bulimia nervosa (60 mg/d) is higher than that typically used to treat depression. These observations suggest that different mechanisms may underlie the utility of these medications in bulimia nervosa and in depression.

A substantial minority of patients with bulimia nervosa do not respond adequately to CBT, antidepressant medication, or their combination. More intensive forms of treatment, including hospitalization, may be required for such patients.

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PART SIX -ONCOLOGY AND HEMATOLOGY

SECTION 1 -NEOPLASTIC DISORDERS

79. APPROACH TO THE PATIENT WITH CANCER - *Dan L. Longo*

The application of current treatment techniques (surgery, radiation therapy, chemotherapy, and biological therapy) results in the cure of >50% of patients diagnosed with cancer. Nevertheless, patients experience the diagnosis of cancer as one of the most traumatic and revolutionary events that has ever happened to them. Independent of prognosis, the diagnosis brings with it a change in a person's self-image and in his or her role in the home and workplace. The prognosis of a person who has just been found to have pancreatic cancer is the same as the prognosis of the person with aortic stenosis who develops the first symptoms of congestive heart failure (median survival, about 8 months). However, the patient with heart disease may remain functional and maintain a view of him- or herself as a fully intact person with just a malfunctioning part, a diseased organ ("a bum ticker"). By contrast, the patient with pancreatic cancer has a completely altered self-image and is viewed differently by family and anyone who knows the diagnosis. He or she is being attacked and invaded by a disease that could be anywhere in the body. Every ache or pain takes on desperate significance. Cancer is an exception to the coordinated interaction among cells and organs. In general, the cells of a multicellular organism are programmed for collaboration. Many diseases occur because the specialized cells fail to perform their assigned task. Cancer takes this malfunction one step further. Not only is there a failure of the cancer cell to maintain its specialized function, but it also strikes out on its own; the cancer cell competes to survive using natural mutability and natural selection to seek advantage over normal cells in a recapitulation of evolution. One consequence of the traitorous behavior of cancer cells is that the patient feels betrayed by his or her body. The cancer patient feels that he or she, and not just a body part, is diseased.

THE MAGNITUDE OF THE PROBLEM

There is no nationwide cancer registry; therefore, the incidence of cancer is estimated on the basis of the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) database, which tabulates cancer incidence and death figures from nine sites, accounting for about 10% of the U.S. population, and from population data from the Bureau of the Census. In 2000, 1.22 million new cases of invasive cancer (619,700 men, 600,400 women) were diagnosed and 552,200 people (284,100 men, 268,100 women) died from cancer. The percent distribution of new cancer cases and cancer deaths by site for men and women are shown in [Table 79-1](#). Cancer incidence has been declining by about 2% each year since 1992.

The most significant risk factor for cancer overall is age; two-thirds of all cases were in people over age 65. Cancer incidence increases as the third, fourth, or fifth power of age in different sites. For the interval between birth and age 39, 1 in 62 men and 1 in 52 women will develop cancer; for the interval between ages 40 and 59, 1 in 12 men and 1 in 11 women will develop cancer; and for the interval between ages 60 and 79, 1 in 3 men and 1 in 4 women will develop cancer.

Cancer is the second leading cause of death behind heart disease. Deaths from heart disease have declined 45% in the United States since 1950 and continue to decline. After a 70-year period of increases, cancer deaths began to decline in 1997 ([Fig. 79-1](#)). The five leading causes of cancer deaths are shown for various populations in [Table 79-2](#). Along with the decrease in incidence has come an increase in survival for cancer patients. The 5-year survival for white patients was 39% in 1960-1963 and 61% in 1989-1995. Cancers are more often deadly in blacks; the 5-year survival was 48% for the 1989-1995 interval. Incidence and mortality vary among racial and ethnic groups ([Table 79-3](#)). The basis for these differences is unclear.

PATIENT MANAGEMENT

Important information is obtained from every portion of the routine history and physical examination. The duration of symptoms may reveal the chronicity of disease. The past medical history may alert the physician to the presence of underlying diseases that may affect the choice of therapy or the side effects of treatment. The social history may reveal occupational exposure to carcinogens or habits, such as smoking or alcohol consumption, that may influence the course of disease and its treatment. The family history may suggest an underlying familial cancer predisposition and point out the need to begin surveillance or other preventive therapy for unaffected siblings of the patient. The review of systems may suggest early symptoms of metastatic disease or a paraneoplastic syndrome.

DIAGNOSIS

The diagnosis of cancer relies most heavily on invasive tissue biopsy and should never be made without obtaining tissue; no noninvasive diagnostic test is sufficient to define a disease process as cancer. Although in rare clinical settings (e.g., thyroid nodules) fine-needle aspiration is an acceptable diagnostic procedure, the diagnosis generally depends on obtaining adequate tissue to permit careful evaluation of the histology of the tumor, its grade, and its invasiveness and to yield further molecular diagnostic information, such as the expression of cell-surface markers or intracellular proteins that typify a particular cancer, or the presence of a molecular marker, such as the t(8;14) translocation of Burkitt's lymphoma. Increasing evidence links the expression of certain genes with the prognosis and response to therapy ([Chaps. 81](#) and [82](#)).

Occasionally a patient will present with a metastatic disease process that is defined as cancer on biopsy but has no apparent primary site of disease. Efforts should be made to define the primary site based on age, sex, sites of involvement, histology and tumor markers, and personal and family history. Particular attention should be focused on ruling out the most treatable causes ([Chap. 99](#)).

Once the diagnosis of cancer is made, the management of the patient is best undertaken as a multidisciplinary collaboration among the primary care physician, medical oncologists, surgical oncologists, radiation oncologists, oncology nurse specialists, pharmacists, social workers, rehabilitation medicine specialists, and a number of other consulting professionals working closely with each other and with the patient and family.

DEFINING THE EXTENT OF DISEASE AND THE PROGNOSIS

The first priority in patient management after the diagnosis of cancer is established and shared with the patient is to determine the extent of disease. The curability of a tumor usually is inversely proportional to the tumor burden. Ideally, the tumor will be diagnosed before symptoms develop or as a consequence of screening efforts ([Chap. 80](#)). A very high proportion of such patients can be cured. However, most patients with cancer present with symptoms related to the cancer, caused either by mass effects of the tumor or by alterations associated with the production of cytokines or hormones by the tumor.

For most cancers, the extent of disease is evaluated by a variety of noninvasive and invasive diagnostic tests and procedures. This process is called *staging*. There are two types. *Clinical staging* is based on physical examination, radiographs, isotopic scans, computed tomography, and other imaging procedures; *pathologic staging* takes into account information obtained during a surgical procedure, which might include intraoperative palpation, resection of regional lymph nodes and/or tissue adjacent to the tumor, and inspection and biopsy of organs commonly involved in disease spread. Pathologic staging includes histologic examination of all tissues removed during the surgical procedure. Surgical procedures performed may include a simple lymph node biopsy or more extensive procedures such as thoracotomy, mediastinoscopy, or laparotomy. Surgical staging may occur in a separate procedure or may be done at the time of definitive surgical resection of the primary tumor.

Knowledge of the predilection of particular tumors for spread to adjacent or distant organs helps direct the staging evaluation.

Information obtained from staging is used to define the extent of disease either as localized, as exhibiting spread outside of the organ of origin to regional but not distant sites, or as metastatic to distant sites. The most widely used system of staging is the TNM (tumor, node, metastasis) system codified by the International Union Against Cancer and the American Joint Committee on Cancer (AJCC).

¹The AJCC *Manual for Staging Cancer*, 5th edition, can be obtained from the AJCC at 55 East Erie Street, Chicago, IL, 60611.

The TNM classification is an anatomically based system that categorizes the tumor on the basis of the size of the primary tumor lesion (T1-4, where a higher number indicates a tumor of larger size), the presence of nodal involvement (usually N0 and N1 for the absence and presence, respectively, of involved nodes, although some tumors have more elaborate systems of nodal grading), and the presence of metastatic disease (M0 and M1 for the absence and presence, respectively, of metastases). The various permutations of T, N, and M scores are then broken into stages, usually designated by the roman numerals I through IV. Tumor burden increases and curability decreases with increasing stage. Other anatomic staging systems are used for some tumors, e.g., the Dukes classification for colorectal cancers, the International Federation of Gynecologists and Obstetricians (FIGO) classification for gynecologic cancers, and the Ann Arbor classification for Hodgkin's disease.

Certain tumors cannot be grouped appropriately on the basis of anatomic considerations. For example, hematopoietic tumors such as leukemia, myeloma, and lymphoma are often disseminated at presentation and do not spread in the fashion typical of solid tumors. For these tumors, other prognostic factors have been identified ([Chaps. 111,112, and 113](#)).

In addition to tumor burden, a second major determinant of treatment outcome is the physiologic reserve of the patient. Patients who are bedridden before developing cancer are likely to fare worse, stage for stage, than fully active patients. Physiologic reserve is a determinant of how a patient is likely to cope with the physiologic stresses imposed by the cancer and its treatment. This factor is difficult to assess directly. Instead, surrogate markers for physiologic reserve are used, such as the patient's age or Karnofsky performance status ([Table 79-4](#)). Older patients and those with a Karnofsky performance status <70 have a poor prognosis unless the poor performance is a reversible consequence of the tumor.

Increasingly, biologic features of the tumor are being related to prognosis. The expression of particular oncogenes, drug-resistance genes, apoptosis-related genes, and genes involved in metastasis are being found to influence response to therapy and prognosis. The presence of selected cytogenetic abnormalities may influence survival. Tumors with higher growth fractions, as assessed by expression of proliferation-related markers such as proliferating cell nuclear antigen (PCNA), behave more aggressively than tumors with lower growth fractions. Information obtained from studying the tumor itself will increasingly be used to influence treatment decisions.

MAKING A TREATMENT PLAN

From information on the extent of disease and the prognosis and in conjunction with the patient's wishes, it is determined whether the treatment approach should be curative or palliative in intent. Cooperation among the various professionals involved in cancer treatment is of the utmost importance in treatment planning. For some cancers, chemotherapy or chemotherapy plus radiation therapy delivered before the use of definitive surgical treatment (so-called neoadjuvant therapy) may improve the outcome, as seems to be the case for locally advanced breast cancer and head and neck cancers. In certain settings in which combined modality therapy is intended, coordination among the medical oncologist, radiation oncologist, and surgeon is crucial to achieving optimal results. Sometimes the chemotherapy and radiation therapy need to be delivered sequentially, and other times concurrently. Surgical procedures may precede or follow other treatment approaches. It is best for the treatment plan either to follow a standard protocol precisely or else to be part of an ongoing clinical research protocol evaluating new treatments. Ad hoc modifications of standard protocols are likely to compromise treatment results.

The choice of treatment approaches was formerly dominated by the local culture in both the university and the practice settings. However, it is now possible to gain access electronically to standard treatment protocols and to every approved clinical research study in North America through a personal computer interface with the Internet.

²The National Cancer Institute maintains a database called PDQ (Physician Data Query)

that is accessible on the Internet under the name CancerNet at www.wicic.nci.nih.gov/health.htm. Information can be obtained through a facsimile machine using CancerFax by dialing 301-402-5874. Patient information is also provided by the National Cancer Institute in at least three formats: on the Internet via CancerNet at www.wicic.nci.nih.gov/patient.htm, through the CancerFax number listed above, or by calling 1-800-4-CANCER. The quality control for the information provided through these services is rigorous.

The skilled physician also has much to offer the patient for whom curative therapy is no longer an option. Often a combination of guilt and frustration over the inability to cure the patient and the pressure of a busy schedule greatly limit the time a physician spends with a patient who is receiving only palliative care. Resist these forces. In addition to the medicines administered to alleviate symptoms (see below), it is important to remember the comfort that is provided by holding the patient's hand, continuing regular examinations, and taking time to talk.

MANAGEMENT OF DISEASE AND TREATMENT COMPLICATIONS

Because cancer therapies are toxic ([Chap. 84](#)), patient management involves addressing complications of both the disease and its treatment as well as the complex psychosocial problems associated with cancer. In the short term during a course of curative therapy, the patient's functional status may decline. Treatment-induced toxicity is less acceptable if the goal of therapy is palliation. The most common side effects of treatment are nausea and vomiting (see below), febrile neutropenia ([Chap. 85](#)), and myelosuppression ([Chap. 104](#)). Therapeutic tools are now available to minimize the acute toxicity of cancer treatment.

New symptoms developing in the course of cancer treatment should always be assumed to be reversible until proven otherwise. The fatalistic attribution of anorexia, weight loss, and jaundice to recurrent or progressive tumor could result in a patient dying from a reversible intercurrent cholecystitis. Intestinal obstruction may be due to reversible adhesions rather than progressive tumor. Systemic infections, sometimes with unusual pathogens, may be a consequence of the immunosuppression associated with cancer therapy. Some drugs used to treat cancer or its complications (e.g., nausea) may produce central nervous system symptoms that look like metastatic disease or may mimic paraneoplastic syndromes such as the syndrome of inappropriate antidiuretic hormone. A definitive diagnosis should be pursued and may even require a repeat biopsy.

A critical component of cancer management is assessing the response to treatment. In addition to a careful physical examination in which all sites of disease are physically measured and recorded in a flow chart by date, response assessment usually requires periodic repeating of imaging tests that were abnormal at the time of staging. If imaging tests have become normal, repeat biopsy of previously involved tissue is performed to document complete response by pathologic criteria. Biopsies are not usually required if there is macroscopic residual disease. A *complete response* is defined as disappearance of all evidence of disease, and a *partial response* as >50% reduction in the sum of the products of the perpendicular diameters of all measureable lesions. *Progressive disease* is defined as the appearance of any new lesion or an increase of

>25% in the sum of the products of the perpendicular diameters of all measurable lesions. Tumor shrinkage or growth that does not meet any of these criteria is considered *stable disease*. Some sites of involvement (e.g., bone) or patterns of involvement (e.g., lymphangitic lung or diffuse pulmonary infiltrates) are considered unmeasurable. No response is complete without biopsy documentation of their resolution but partial responses may exclude their assessment unless clear objective (though unmeasurable) progression has occurred.

Tumor markers may be useful in patient management in certain tumors. Response to therapy may be difficult to gauge with certainty. However, some tumors produce or elicit the production of markers that can be measured in the serum or urine and, in a particular patient, rising and falling levels of the marker are usually associated with increasing or decreasing tumor burden, respectively. Some clinically useful tumor markers are shown in [Table 79-5](#). Tumor markers are not in themselves specific enough to permit a diagnosis of malignancy to be made, but once a malignancy has been diagnosed and shown to be associated with elevated levels of a tumor marker, the marker can be used to assess response to treatment.

The recognition and treatment of depression are important components of management. The incidence of depression in cancer patients is ~25% overall and may be greater in patients with greater debility. This diagnosis is likely in a patient with a depressed mood (dysphoria) and/or a loss of interest in pleasure (anhedonia) for at least 2 weeks. In addition, three or more of the following symptoms are usually present: appetite change, sleep problems, psychomotor retardation or agitation, fatigue, feelings of guilt or worthlessness, inability to concentrate, and suicidal ideation. Patients with these symptoms should receive therapy. Medical therapy with a serotonin reuptake inhibitor such as fluoxetine (10 to 20 mg/d), sertraline (50 to 150 mg/d), or paroxetine (10 to 20 mg/d) or a tricyclic antidepressant such as amitriptyline (50 to 100 mg/d) or desipramine (75 to 150 mg/d) should be tried, allowing 4 to 6 weeks for response. Effective therapy should be continued at least 6 months after resolution of symptoms. If therapy is unsuccessful, other classes of antidepressants may be used. In addition to medication, psychosocial interventions such as support groups, psychotherapy, and guided imagery may be of benefit.

Many patients opt for unproven or unsound approaches to treatment when it appears that conventional medicine is unlikely to be curative. Those seeking such alternatives are often well educated and may be early in the course of their disease. Unsound approaches are usually hawked on the basis of unsubstantiated anecdotes and not only cannot help the patient but may be harmful. Physicians should strive to keep communications open and nonjudgmental, so that patients are more likely to discuss with the physician what they are actually doing. The appearance of unexpected toxicity may be an indication that a supplemental therapy is being taken.

Information about unsound methods may be obtained from the National Council Against Health Fraud, Box 1276, Loma Linda, CA 92354, or from the Center for Medical Consumers and Health Care Information, 237 Thompson Street, New York, NY 10012.

LONG-TERM FOLLOW-UP/LATE COMPLICATIONS

At the completion of treatment, sites originally involved with tumor are reassessed, usually by radiography or imaging techniques, and any persistent abnormality is biopsied. If disease persists, the multidisciplinary team discusses a new salvage treatment plan. If the patient has been rendered disease-free by the original treatment, the patient is followed regularly for disease recurrence. The optimal guidelines for follow-up care are not known. For many years, a routine practice has been to follow the patient monthly for 6 to 12 months, then every other month for a year, every 3 months for a year, every 4 months for a year, every 6 months for a year, and then annually. At each visit, a battery of laboratory and radiographic and imaging tests were obtained on the assumption that it is best to detect recurrent disease before it becomes symptomatic. However, where follow-up procedures have been examined, this assumption has been found to be untrue. Studies of breast cancer, melanoma, lung cancer, colon cancer, and lymphoma have all failed to support the notion that asymptomatic relapses are more readily cured by salvage therapy than symptomatic relapses. In view of the enormous cost of a full battery of diagnostic tests and their manifest lack of impact on survival, new guidelines are emerging for less frequent follow-up visits during which the history and physical examination are the major investigations performed.

As time passes, the likelihood of recurrence of the primary cancer diminishes. For many types of cancer, survival for 5 years without recurrence is tantamount to cure. However, important medical problems can occur in patients treated for cancer and must be examined ([Chap 103](#)). Some problems emerge as a consequence of the disease and some as a consequence of the treatment. An understanding of these disease- and treatment-related problems may help in their detection and management.

Despite these concerns, most patients who are cured of cancer return to normal lives.

SUPPORTIVE CARE

In many ways, the success of cancer therapy depends on the success of the supportive care. Failure to control the symptoms of cancer and its treatment may lead patients to abandon curative therapy. Of equal importance, supportive care is a major determinant of quality of life. Even when life cannot be prolonged, the physician must strive to preserve its quality. Quality-of-life measurements have become common end-points of clinical research studies. Furthermore, palliative care has been shown to be cost-effective when approached in an organized fashion. A credo for oncology could be to cure sometimes, to extend life often, and to comfort always.

Pain Pain occurs with variable frequency in the cancer patient: 25 to 50% of patients present with pain at diagnosis, 33% have pain associated with treatment, and 75% have pain with progressive disease. The pain may have several causes. In about 70% of cases, pain is caused by the tumor itself -- by invasion of bone, nerves, blood vessels, or mucous membranes or obstruction of a hollow viscus or duct. In about 20% of cases, pain is related to a surgical or invasive medical procedure, to radiation injury (mucositis, enteritis, or plexus or spinal cord injury), or to chemotherapy injury (mucositis, peripheral neuropathy, phlebitis, steroid-induced aseptic necrosis of the femoral head). In 10% of cases, pain is unrelated to cancer or its treatment.

Assessment of pain requires the methodical investigation of the history of the pain, its location, character, temporal features, provocative and palliative factors, and intensity ([Chap. 12](#)); a review of the oncologic history and past medical history as well as personal and social history; and a thorough physical examination. The patient should be given a 10-division visual analogue scale on which to indicate the severity of the pain. The clinical condition is often dynamic, making it necessary to reassess the patient frequently. Pain therapy should not be withheld while the cause of pain is being sought.

A variety of tools are available with which to address cancer pain. About 85% of patients will have pain relief from pharmacologic intervention. However, other modalities, including antitumor therapy (such as surgical relief of obstruction, radiation therapy, and strontium-89 or samarium-153 treatment for bone pain), neurostimulatory techniques, regional analgesia, or neuroablative procedures are effective in an additional 12% or so. Thus, very few patients will have inadequate pain relief if appropriate measures are taken.

The World Health Organization (WHO) has devised a simple and effective method for the rational titration of oral analgesia, called the *WHO ladder*. The ladder has the following three steps. (1) For mild to moderate pain, one begins with acetaminophen (650 mg every 4 h or 975 mg every 6 h), aspirin (650 mg every 4 h or 975 mg every 6 h), or a nonsteroidal anti-inflammatory agent (NSAID; e.g., ketoprofen, 25 to 60 mg every 6 h) with or without an adjuvant such as a glucocorticoid (dexamethasone) or an antidepressant (amitriptyline). (2) When pain persists or increases, an opioid such as codeine or hydrocodone (30 mg every 3 to 4 h is roughly equivalent to 10 mg of intravenous morphine) should be added (not substituted); fixed combinations such as oxycodone/acetaminophen (Percocet) or oxycodone/aspirin (Percodan) are worth testing. (3) Pain that is persistent or that is moderate to severe at the outset should be treated by increasing the potency of the opioid or using higher dosages (e.g., morphine, 15 to 30 mg every 3 to 4 h, or controlled-release morphine, 90 to 120 mg bid), and fixed opioid/NSAID combinations should be abandoned. Adjuvants may be used at all steps. The critical features of this approach are that the treatment is oral, should be given around the clock with supplemental doses as needed to control pain, and is tailored to the individual patient. Transmucosal fentanyl (in lollipop form) may aid in control of breakthrough pain. Records of pain control should be a prominent component of the medical record. When opioids are used, the patient should be placed on a prophylactic regimen to prevent constipation.

Nausea Emesis in the cancer patient is usually caused by chemotherapy ([Chap. 84](#)). Its severity can be predicted from the drugs used to treat the cancer. Three forms of emesis are recognized on the basis of their timing with regard to the noxious insult. *Acute emesis*, the most common variety, occurs within 24 h of treatment. *Delayed emesis* occurs 1 to 7 days after treatment; it is rare, but, when present, usually follows cisplatin administration. *Anticipatory emesis* occurs before administration of chemotherapy and represents a conditioned response to visual and olfactory stimuli previously associated with chemotherapy delivery.

Acute emesis is the best understood form. Stimuli that activate signals in the chemoreceptor trigger zone in the medulla, the cerebral cortex, and peripherally in the intestinal tract lead to stimulation of the vomiting center in the medulla, the motor center

responsible for coordinating the secretory and muscle contraction activity that leads to emesis. Diverse receptor types participate in the process, including dopamine, serotonin, histamine, opioid, and acetylcholine receptors. The serotonin receptor antagonists ondansetron and granisetron are the most effective drugs against highly emetogenic agents, but they are expensive.

As with the analgesia ladder, emesis therapy should be tailored to the situation. For mildly and moderately emetogenic agents, prochlorperazine, 5 to 10 mg orally or 25 mg rectally, is effective. Its efficacy may be enhanced by administering the drug before the chemotherapy is delivered. Dexamethasone, 10 to 20 mg intravenously, is also effective and may enhance the efficacy of prochlorperazine. For highly emetogenic agents such as cisplatin, mechlorethamine, dacarbazine, and streptozocin, combinations of agents work best and administration should begin 6 to 24 h before treatment. Ondansetron, 8 mg orally every 6 h the day before therapy and intravenously on the day of therapy, plus dexamethasone, 20 mg intravenously before treatment, is an effective regimen. Like pain, emesis is easier to prevent than to alleviate.

Delayed emesis may be related to bowel inflammation from the therapy and can be controlled with oral dexamethasone and oral metoclopramide, a dopamine receptor antagonist that also blocks serotonin receptors at high dosages. The best strategy for preventing anticipatory emesis is to control emesis in the early cycles of therapy to prevent the conditioning from taking place. If this is unsuccessful, prophylactic antiemetics the day before treatment may help. Experimental studies are evaluating behavior modification.

Effusions Fluid may accumulate abnormally in the pleural cavity, pericardium, or peritoneum. Asymptomatic malignant effusions may not require treatment. Symptomatic effusions occurring in tumors responsive to systemic therapy usually do not require local treatment but respond to the treatment for the underlying tumor. Symptomatic effusions occurring in tumors unresponsive to systemic therapy may require local treatment in patients with a life expectancy of at least 6 months.

Pleural effusions due to tumors may or may not contain malignant cells. Lung cancer, breast cancer, and lymphomas account for about 75% of malignant pleural effusions. Their exudative nature is usually gauged by an effusion/serum protein ratio of 0.5 or an effusion/serum lactate dehydrogenase ratio of 0.6. When the condition is symptomatic, thoracentesis is usually performed first. In most cases, symptomatic improvement occurs for <1 month. Chest tube drainage is required if symptoms recur within 2 weeks. Fluid is aspirated until the flow rate is <100 mL in 24 h. Then either 60 units of bleomycin or 1 g of doxycycline is infused into the chest tube in 50 mL of 5% dextrose in water; the tube is clamped; the patient is rotated on four sides, spending 15 min in each position; and, after 1 to 2 h, the tube is again attached to suction for another 24 h. The tube is then disconnected from suction and allowed to drain by gravity. If <100 mL drains over the next 24 h, the chest tube is pulled, and a radiograph taken 24 h later. If the chest tube continues to drain fluid at an unacceptably high rate, sclerosis can be repeated. Bleomycin may be somewhat more effective than doxycycline but is very expensive. Doxycycline is usually the drug of first choice. If neither doxycycline nor bleomycin is effective, talc can be used.

Symptomatic pericardial effusions are usually treated by creating a pericardial window or by stripping the pericardium. If the patient's condition does not permit a surgical procedure, sclerosis can be attempted with doxycycline and/or bleomycin.

Malignant ascites is usually treated with repeated paracentesis of small volumes of fluid. If the underlying malignancy is unresponsive to systemic therapy, peritoneovenous shunts may be inserted. Despite the fear of disseminating tumor cells into the circulation, widespread metastases are an unusual complication. The major complications are occlusion, leakage, and fluid overload. Patients with severe liver disease may develop disseminated intravascular coagulation.

Nutrition Cancer and its treatment may lead to a decrease in nutrient intake of sufficient magnitude to cause weight loss and alteration of intermediary metabolism. The prevalence of this problem is difficult to estimate because of variations in the definition of cancer cachexia, but most patients with advanced cancer experience weight loss and decreased appetite. A variety of both tumor-derived factors (e.g., bombesin, adrenocorticotrophic hormone) and host-derived factors (e.g., tumor necrosis factor, interleukins 1 and 6, growth hormone) contribute to the altered metabolism, and a vicious cycle is established in which protein catabolism, glucose intolerance, and lipolysis cannot be reversed by the provision of calories.

It remains controversial how to assess nutritional status and when and how to intervene. Efforts to make the assessment objective have included the use of a prognostic nutritional index based on albumin levels, triceps skin fold thickness, transferrin levels, and delayed-type hypersensitivity skin testing. However, a simpler approach has been to define the threshold for nutritional intervention as >10% unexplained body weight loss, serum transferrin level <1500 mg/L (150 mg/dL), and serum albumin <34 g/L (3.4 g/dL).

The decision is important, because it appears that cancer therapy is substantially more toxic and less effective in the face of malnutrition. Nevertheless, it remains unclear whether nutritional intervention can alter the natural history. Unless some pathology is affecting the absorptive function of the gastrointestinal tract, enteral nutrition provided orally or by tube feeding is preferred over parenteral supplementation. However, the risks associated with the tube may outweigh the benefits. Megestrol acetate, a progestational agent, has been advocated as a pharmacologic intervention to improve nutritional status. Research in this area may provide more tools in the future as cytokine-mediated mechanisms are further elucidated.

Psychosocial Support The psychosocial needs of patients vary with their situation. Patients undergoing treatment experience fear, anxiety, and depression. Self-image is often seriously compromised by deforming surgery and loss of hair. Women who receive cosmetic advice that enables them to look better also feel better. Loss of control over how one spends time can contribute to the sense of vulnerability. Juggling the demands of work and family with the demands of treatment may create enormous stresses. Sexual dysfunction is highly prevalent and needs to be discussed openly with the patient. An empathetic health care team is sensitive to the individual patient's needs and permits negotiation where such flexibility will not adversely affect the course of treatment.

Cancer survivors have other sets of difficulties. Patients may have fears associated with the termination of a treatment they associate with their continued survival. Adjustments are required to physical losses and handicaps, real and perceived. Patients may be preoccupied with minor physical problems. They perceive a decline in their job mobility and view themselves as less desirable workers. They may be victims of job and/or insurance discrimination. Patients may experience difficulty reentering their normal past life. They may feel guilty for having survived and may carry a sense of vulnerability to colds and other illnesses. Perhaps the most pervasive and threatening concern is the ever-present fear of relapse (the Damocles syndrome).

Patients in whom therapy has been unsuccessful have other problems related to the end of life.

Death and Dying The most common causes of death in patients with cancer are infection (leading to circulatory failure), respiratory failure, hepatic failure, and renal failure. Intestinal blockage may lead to inanition and starvation. Central nervous system disease may lead to seizures, coma, and central hypoventilation. About 70% of patients develop dyspnea preterminally. However, many months usually pass between the diagnosis of cancer and the occurrence of these complications, and during this period the patient is severely affected by the possibility of death. The path of unsuccessful cancer treatment usually occurs in three phases. First, there is optimism at the hope of cure; when the tumor recurs, there is the acknowledgment of an incurable disease, and the goal of palliative therapy is embraced in the hope of being able to live with disease; finally, at the disclosure of imminent death, another adjustment in outlook takes place. The patient imagines the worst in preparation for the end of life and may go through stages of adjustment to the diagnosis. These stages include denial, isolation, anger, bargaining, depression, acceptance, and hope. Of course, patients do not all progress through all the stages or proceed through them in the same order or at the same rate. Nevertheless, developing an understanding of how the patient has been affected by the diagnosis and is coping with it is an important goal of patient management.

It is best to speak frankly with the patient and the family regarding the likely course of disease. These discussions can be difficult for the physician as well as for the patient and family. The critical features of the interaction are to reassure the patient and family that everything that can be done to provide comfort will be done. They will not be abandoned. Many patients prefer to be cared for in their homes or in a hospice setting rather than a hospital. The American College of Physicians has published a book called *Home Care Guide for Cancer: How to Care for Family and Friends at Home* that teaches an approach to successful problem-solving in home care. With appropriate planning, it should be possible to provide the patient with the necessary medical care as well as the psychological and spiritual support that will prevent the isolation and depersonalization that can attend in-hospital death.

The care of dying patients may take a toll on the physician. A "burnout" syndrome has been described that is characterized by fatigue, disengagement from patients and colleagues, and a loss of self-fulfillment. Efforts at stress reduction, maintenance of a balanced life, and setting realistic goals may combat this disorder.

End-of-Life Decisions Unfortunately, a smooth transition in treatment goals from

curative to palliative may not be possible in all cases because of the occurrence of serious treatment-related complications or rapid disease progression. Vigorous and invasive medical support for a reversible disease or treatment complication is assumed to be justified. However, if the reversibility of the condition is in doubt, the patient's wishes determine the level of medical care. These wishes should be elicited before the terminal phase of illness and reviewed periodically. This information can guide the physician should the patient be unable to speak for him- or herself. The family cannot be expected to make such decisions without guidance from the patient and support from the physician when surrogate decisions are required. Advance directives such as a living will or a durable power of attorney for health care provide guidance for the health care team and the family regarding the patient's wishes and may protect the patient's assets from depletion on expensive but unwanted care.

Only about 15% of the population has implemented an advance directive. Physicians should take the initiative to speak with patients and family members about advance directives.

4Information about advance directives can be obtained from the American Association of Retired Persons, 601 E Street, NW, Washington, DC 20049, 202-434-2277 or Choice in Dying, 250 West 57th Street, New York, NY 10107, 212-366-5540.

(Bibliography omitted in Palm version)

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80. PREVENTION AND EARLY DETECTION OF CANCER - Otis W. Brawley, Barnett S. Kramer

The prevention and control of cancer is a burgeoning field because of advances in understanding the biology of carcinogenesis. The field has expanded beyond the identification and avoidance of carcinogens to include studies of specific interventions to lower cancer risk, as well as screening for early detection of cancer.

Central to the prevention and control of cancer is the concept that carcinogenesis is not an event but a process, a series of discrete cellular changes that result in progressively more autonomous cellular processes. *Primary prevention* concerns the identification and manipulation of the genetic, biologic, and environmental factors in the causal pathway. Smoking cessation, diet modification, and chemoprevention are primary prevention activities. *Secondary prevention* concerns the identification of asymptomatic neoplastic lesions combined with effective therapy. Screening is a form of secondary prevention. Screening may also be a form of primary prevention of invasive cancer; screening Pap smears are used to identify and treat preinvasive lesions of the cervix.

EDUCATION AND HEALTHFUL HABITS

Public education on the avoidance of identified risk factors for cancer and encouraging healthy habits were among early efforts in cancer prevention and control. Many educational messages have come to the public through commercials in the print and electronic media and through school health courses. The physician is a potentially powerful messenger in this education campaign about the hazards of smoking, the benefits of a healthful diet, and sun avoidance.

Smoking Cessation Tobacco use through cigarettes and other means is the most avoidable risk factor for cardiovascular disease and cancer. Lung cancer mortality rates correlate with the number of cigarettes smoked per day as well as the degree of inhalation of cigarette smoke. Those who stop smoking have a lower lung cancer mortality rate than those who continue smoking, despite the persistence for years of some carcinogen-induced genetic mutations. In addition to lung cancer, cigarette smoking is a causative agent in cancers of the larynx, oropharynx, esophagus, bladder, and pancreas. Smoking cessation and avoidance have the potential to save and extend more lives than any other public health activity. About 400,000 Americans die prematurely every year because of cigarette smoking. A smoker has a one in three lifetime risk of dying prematurely of a cancer or cardiovascular or pulmonary disease caused by cigarette smoking. Indeed, more human lives are lost due to cardiovascular disease caused by smoking than from smoking-related cancer. The risk of tobacco smoke is not necessarily limited to the smoker. Epidemiologic studies suggest that environmental tobacco smoke may cause lung cancer and other pulmonary diseases in nonsmokers.

Nonsmoking persons should be encouraged not to start smoking, and persons who smoke should be encouraged to stop. Tobacco prevention is a pediatric issue. Over 80% of American smokers begin smoking before the age of 18. Nearly 20% of Americans aged 12 to 18 have smoked a cigarette in the past month. Counseling of adolescents and young adults is critical to prevent smoking. A physician's simple advice

to not start smoking or to quit smoking can be of benefit. The U.S. Agency for Health Care Research and Quality recommends that physicians query patients on tobacco use on every office visit, record the answer with the vital signs, and ask smokers if they would like assistance in quitting.

Current approaches to smoking cessation recognize that smoking is an addiction ([Chap. 390](#)). The smoker who is quitting goes through a process with identifiable stages that include contemplation of quitting, an action phase in which the smoker quits, and a maintenance phase. Smokers who quit completely are more likely to be successful than those who gradually reduce the number of cigarettes smoked or change to cigarettes lower in tar or nicotine. More than 90% of the Americans who have successfully quit smoking did so on their own without participation in an organized cessation program, but cessation programs are helpful for some smokers. The Community Intervention Trial for Smoking Cessation (COMMIT) was a community-based 4-year program. One community of each of 11 matched community pairs was randomly assigned to intervention. The intervention included public education through the media and community-wide events, health care providers, worksites and other organizations, and cessation resources. COMMIT demonstrated that light smokers can benefit from simple cessation messages and cessation programs. The quit rate (fraction of the subjects followed who achieved and maintained cessation at the end of the trial) was 30.6% in the intervention communities and 27.5% in the control communities. This finding is statistically significant, but modest. The control communities enjoyed a substantial decrease in smoking through study participation. The COMMIT interventions were not successful for heavy smokers (>25 cigarettes per day). Heavy smokers need an intensive, broad-based cessation program that includes counseling, behavioral strategies, and pharmacologic adjuncts such as nicotine gum and nicotine patches.

Cigar and pipe smoking carry the same risks as tobacco smoke including lung cancer. Smokeless tobacco is the fastest growing part of the tobacco industry and represents a significant health risk. Chewing tobacco is a carcinogen linked to dental caries, gingivitis, oral leukoplakia, and oral cancer. The systemic effects of smokeless tobacco may increase risks for other cancers. Nitrosamines found in smokeless tobacco cause lung cancer in laboratory animals.

Diet Modification Dietary modification may have significant potential for lowering cancer risk in western culture. Studies of international dietary patterns and animal studies suggest that diets high in fat increase the risk for cancers of the breast, colon, prostate, and endometrium. These cancers have their highest incidence and mortalities in western countries, where fat comprises an average of 40 to 45% of the total calories consumed. In populations at low risk for these cancers, fat accounts for <20% of dietary calories.

Nonetheless, dietary fat has not been accepted by all as important in the etiology of cancers. Case-control and cohort epidemiologic studies give conflicting results. In addition, diet is a highly complex exposure to many nutrients and chemicals. Low-fat diets may render some protection through anticarcinogens found in vegetables, fruits, legumes, nuts, and grains. Substances found in these foods that may be protective include phenols, sulfur-containing compounds, flavones, and fiber.

In observational studies, dietary fiber appears protective against colonic polyps and invasive cancer of the colon. The mechanisms involved are complex and speculative. They involve binding of oxidized bile acids, a decrease in bowel transit time, and generation of soluble fiber products, such as butyrate, that may have differentiating properties. High-fiber diets may also protect against breast and prostate cancer by absorbing and inactivating dietary estrogenic and androgenic cancer promoters. Protective effects of fiber have not been proved in a prospective clinical trial.

The U.S. National Institutes of Health (NIH) Women's Health Initiative, launched in 1994, is a long-term clinical trial enrolling more than 100,000 women aged 45 to 69. It studies the potential cancer-preventing effects of a low-fat diet and vitamin supplementation. It must be stressed that the scientific evidence does not currently establish the anticarcinogenic value of vitamin, mineral, or nutritional supplements in amounts greater than that provided by a good diet.

The Polyp Prevention Trial studied 2000 elderly persons randomly assigned to a low-fat, high-fiber diet or a routine diet followed for 4 years. No significant differences in polyp formation were noted.

A simple way to decrease dietary fat and increase fiber is to consume at least 5 to 9 servings of fruits and vegetables a day. Such a diet may lower the risk of cardiac disease as well as cancer.

Sun Avoidance Nonmelanoma skin cancers (basal cell and squamous cell) are induced by cumulative exposure to ultraviolet radiation. Intermittent acute sun exposure and sun damage have been linked to melanoma. Sunburns, especially in childhood and adolescence, are associated with an increased risk of melanoma in adulthood. Reduction of sun exposure through use of protective clothing and changes in the pattern of outdoor activities can reduce skin cancer risk. Sunscreens decrease the risk of actinic keratoses, the precursor to squamous cell skin cancer, but melanoma risk may be increased. Sunscreens prevent burning and may encourage more prolonged exposure to the sun; yet they may not filter out wavelengths of energy that cause melanoma.

Educational interventions to help people assess their risk of developing skin cancer accurately have some impact. Self-examination for skin pigment characteristics associated with melanoma, such as freckling, may be useful in identifying people at high risk. People who recognize themselves as being at risk tend to be more compliant with sun-avoidance recommendations. Possible risk factors for melanoma include a propensity to sunburn, a large number of benign melanocytic nevi, and atypical nevi.

CANCER CHEMOPREVENTION

Chemoprevention of cancer is a relatively new concept. It involves the use of specific natural or synthetic chemical agents to reverse, suppress, or prevent carcinogenesis before the development of invasive malignancy. While the concept that pharmacologic agents can prevent a cancer is relatively new, the idea that a compound can prevent chronic disease is not. Clinicians routinely prevent heart disease, kidney disease, and stroke by treating hypertension with pharmacologic agents. Lipid-lowering drugs are used to prevent coronary artery disease.

Improved understanding of the biology of cancer makes chemoprevention a real possibility. Cancer develops through an accumulation of genetic changes that are potential points of intervention to prevent cancer. The initial genetic changes are termed *initiation*. The alteration can be inherited or acquired through the action of physical, infectious, or chemical carcinogens. Like most human diseases, cancer arises through an interaction between genetics and environmental exposures ([Table 80-1](#)). Influences that cause the initiated cell to progress through the carcinogenic process and to change phenotypically are termed *promoters*. Promoters include hormones such as androgens, linked to prostate cancer, and estrogen, linked to breast and endometrial cancer. The distinction between an initiator and a promoter is sometimes arbitrary; some components of cigarette smoke are "complete carcinogens," acting as both initiators and promoters. Cancer can be prevented or controlled through interference with the factors that cause initiation, promotion, or progression. Compounds of interest in chemoprevention often have antimutagenic, antioxidant, or antiproliferative activity.

Before a chemoprevention strategy can become standard practice, evidence of benefit must be gathered from clinical trials. These trials are usually large, long-term, randomized, placebo-controlled, and double-blinded. They often allow for the study of drugs for prevention of multiple cancers and the study of end-points beyond cancer, such as other chronic diseases. Several large clinical trials have been completed, and a number are continuing in the twenty-first century. Only tamoxifen has been approved by the U.S. Food and Drug Administration for prevention; it lowers risk of breast cancer in high-risk women.

Multiple Cancer Site Prevention Trials The Physicians' Health Trial involves 22,071 American male physicians. Participants were randomly assigned to receive b-carotene, aspirin, and/or placebo in a 2³ factorial design. All major medical events were recorded. In 1988, the aspirin arm was unblinded after the trial demonstrated that aspirin therapy causes a significant reduction in cardiovascular mortality. The b-carotene arm of the study stopped in 1998, and data analysis is proceeding.

The Women's Health Study, launched in 1992, is a 10-year trial involving 44,000 female nurses. Subjects are randomly assigned to b-carotene, a-tocopherol, aspirin, and/or placebo in a factorial design yielding eight different treatment groups. The end-points are total epithelial cancers, breast cancer, lung cancer, colon cancer, and vascular disease.

The Women's Health Initiative uses a partial factorial design that places women in 22 intervention groups. Participants can receive calcium and vitamin D supplementation, hormone replacement therapy, and counseling to increase exercise and cease smoking. Prevention of a number of cancers, cardiovascular disease, osteoporosis, and other diseases will be assessed.

Prevention of Hormonally Driven Cancers Hormonal manipulation is being tested in the primary prevention of breast and prostate cancer. Tamoxifen is an antiestrogen with partial estrogen agonistic activity in some tissues, such as endometrium and bone. One of its actions is to upregulate transforming growth factor- β , which decreases breast cell proliferation. In randomized placebo-controlled trials to assess tamoxifen as an adjuvant

in breast cancer treatment, this drug reduced the number of new breast cancers in the uninvolved breast by more than a third. In a randomized placebo-controlled trial involving >13,000 women at high risk, tamoxifen decreased the risk of developing cancer by 49% compared to placebo. Tamoxifen also reduced the risk of bone fractures; a small increase in risk of endometrial cancer, stroke, pulmonary emboli, and deep vein thrombosis was noted. A trial to compare tamoxifen with another selective estrogen receptor modulator, raloxifene, is ongoing.

Finasteride is a 5 α -reductase inhibitor. It inhibits the conversion of testosterone to dihydrotestosterone, a more potent stimulator of prostate cell proliferation than testosterone. In an F344 rat model of carcinogen-induced prostate cancer, finasteride decreased the incidence of cancers. Finasteride is being tested as a preventive agent for prostate cancer in a 10-year study involving 18,000 men age 55 and older.

Chemoprevention of Cancers of the Upper Aerodigestive Tract Smoking causes diffuse epithelial injury in the head, neck, esophagus, and lung. Patients cured of squamous cell cancers of the lung, esophagus, head, and neck are at risk (as high as 5% per year) of developing a second cancer of the upper aerodigestive tract. Cessation of cigarette smoking does not markedly decrease the cured cancer patient's risk of second malignancy, even though it does lower the cancer risk in those who have never developed a malignancy. Smoking cessation may halt the early stages of the carcinogenic process (such as metaplasia), but it may have no effect on late stages of carcinogenesis. This "field carcinogenesis" hypothesis for cancer of the upper aerodigestive tract has made "cured" patients an important population for chemoprevention of second malignancies. A randomized, placebo-controlled clinical trial has demonstrated that adjuvant isotretinoin (13-*cis*-retinoic acid) can reduce the incidence of second primary tumors in patients treated with local therapy for head and neck cancer. However, overall survival was not improved due to mortality from recurrences of the primary tumor.

Oral leukoplakia, a premalignant lesion commonly found in smokers, has been used as an intermediate marker allowing the demonstration of chemopreventive activity in smaller, shorter-duration, randomized, placebo-controlled trials. Response was associated with upregulation of retinoic acid receptor β . Therapy with isotretinoin causes regression of oral leukoplakia. However, the lesions recur when the agent is withdrawn, suggesting the need for chronic administration of retinoids. Premalignant lesions in the oropharyngeal area have also responded to β -carotene, retinol, α -tocopherol (vitamin E), and selenium. Further study to improve the definition of the activity of these drugs is ongoing. The ability of isotretinoin to prevent second malignancies in patients cured of early-stage non-small cell lung cancer is also being assessed.

Several large-scale trials have assessed agents in the chemoprevention of lung cancer in patients at high risk. In the Alpha-Tocopherol/Beta-Carotene (ATBC) Lung Cancer Prevention Trial, participants were male smokers, aged 50 to 69 at entry. At entry, participants had smoked an average of one pack of cigarettes per day for 35.9 years. Participants received α -tocopherol, β -carotene, and/or placebo in a randomized, 2 $\times$ 2 factorial design. After a median follow-up of 6.1 years, lung cancer incidence and mortality were statistically significantly *increased* in those receiving β -carotene. α -Tocopherol had no significant impact on lung cancer mortality, and no

evidence suggested interaction between the two drugs. Patients receiving a-tocopherol had a higher incidence of hemorrhagic stroke.

The Beta-Carotene and Retinol Efficacy Trial (CARET) involved 17,000 American smokers and workers with asbestos exposure. Entrants were randomly assigned to one of four arms and received b-carotene, retinol, and/or placebo in a 2'2 factorial design. This trial demonstrated harm from b-carotene: a lung cancer rate of 5 per 1000 subjects per year for those taking placebo and of 6 per 1000 subjects per year for those taking b-carotene. The difference was statistically significant.

These [ATBC](#) and [CARET](#) results demonstrate the importance of testing chemoprevention hypotheses before implementing them widely, because the results stand in contrast to a number of observational epidemiologic studies. In the ATBC trial, participants taking a-tocopherol had a one-third reduction in the incidence of prostate cancer, compared to those not taking a-tocopherol. Assessment of these findings continues. The Physicians' Health Trial showed neither an increased nor a decreased risk of lung cancer in those using b-carotene; fewer of its participants were smokers than those in the ATBC and CARET studies.

Chemoprevention of Colon Cancer Many of the current colon cancer prevention trials are based on the premise that most colorectal cancers develop from adenomatous polyps. These trials use adenoma recurrence or disappearance as a surrogate end-point to assess colon cancer prevention. Early clinical trial results suggest that nonsteroidal anti-inflammatory drugs (NSAIDs), such as piroxicam, sulindac, and aspirin, may prevent adenoma formation or cause regression of adenomatous polyps. The mechanism of action of NSAIDs is unknown, but they are presumed to work through the cyclooxygenase pathway. In the Physicians' Health Trial, aspirin had no effect on colon cancer incidence, although the 6-year assessment period may not have been long enough to evaluate definitively this end-point.

Epidemiologic studies suggest that diets high in calcium lower colon cancer risk. Calcium binds bile and fatty acids, which cause hyperproliferation of colonic epithelium. It is hypothesized that this effect reduces intraluminal exposure to these compounds. Early data from randomized studies suggest that calcium supplementation decreases the risk of adenomatous polyp recurrence by about 20%, even though it does not decrease the proliferative rate of the colonic epithelium. Epithelial proliferative rate may not be an adequate surrogate marker in colon cancer prevention trials.

Cyclooxygenase II inhibitors may be even more effective at colon cancer prevention.

Vaccines and Cancer Prevention A number of infectious agents have been linked to the development of cancer, leading to interest in developing vaccines to protect against these agents. The hepatitis B vaccine is quite effective in preventing hepatitis and hepatomas due to chronic hepatitis B infection. Public health officials are encouraging widespread administration of this vaccine, especially in Asia, where the disease is epidemic. In the future, human papilloma virus (HPV) vaccines could be developed to prevent cervical cancer, and *Helicobacter pylori* vaccines may be developed to prevent gastric cancer.

CANCER SCREENING

Screening is a means of detecting disease early in asymptomatic individuals with the goal of decreasing morbidity and mortality. Screening for cancer is intuitively appealing and has attracted great public interest as technology has generated a number of diagnostic tests and procedures that are safe, quick, and inexpensive. While screening can potentially save lives, and has been shown clearly to do so in the case of breast, cervical, and colon cancer, it is also subject to a number of biases, which can suggest a benefit when actually there is none. Bias can even mask net harm. Early detection does not in itself confer benefit. To be of value, screening must detect disease earlier, and treatment of earlier disease must yield a better outcome than treatment at the onset of symptoms. Cause-specific mortality, rather than survival after diagnosis, is the preferred end point (see below).

Because screening is done on asymptomatic, healthy persons, it should offer substantial likelihood of benefit. A critical approach to screening is necessary to ensure that benefit results. Screening tests and their appropriate use should be carefully evaluated before their use is widely encouraged in screening programs as a matter of public policy.

Screening examinations, tests, or procedures are usually not diagnostic of cancer but instead indicate that a cancer may be present. The diagnosis is then made following a workup that includes a biopsy and pathologic confirmation.

A number of genes have been identified that predispose for a disease, and many more will be identified in the near future. Testing for these genes can define a high-risk population. The ability to predict the development of a particular cancer may some day present therapeutic options as well as ethical dilemmas. It may eventually allow for early intervention to prevent a cancer or limit its severity. People at high risk will be ideal candidates for chemoprevention and screening; however, the efficacy of these interventions in the high-risk population should be investigated. Currently, persons at high risk for a particular cancer can engage in intensive screening. While this course is clinically prudent, it is not known if it saves lives in these populations.

The Accuracy of Screening A screening test's accuracy or ability to discriminate disease is described by four indices: sensitivity, specificity, positive predictive value, and negative predictive value ([Table 80-2](#)). *Sensitivity* is the proportion of persons with the disease who test positive in the screen (i.e., the ability of the test to detect disease when it is present). *Specificity* is the proportion of persons who do not have the disease and test negative in the screening test (i.e., the ability of a test to tell that the disease is not present). The *positive predictive value* is the proportion of persons who test positive who actually have the disease. Similarly, *negative predictive value* is the proportion of who test negative and do not have the disease. The sensitivity and specificity of a test are relatively independent of the underlying prevalence (or risk) of the disease in the population screened, but the predictive values depend strongly on the prevalence of the disease ([Table 80-3](#)).

Screening is most beneficial, efficient, and economical when the target disease is common in the population being screened. To be valuable, the screening test should

have a high specificity; sensitivity need not be very high, as demonstrated in [Table 80-3](#).

Potential Biases of Screening Tests The common biases of screening are lead time, length, and selection. These biases can make a screening test seem beneficial when actually it is not (or even causes net harm). Whether beneficial or not, screening can create the false impression of an epidemic by increasing the number of cancers diagnosed. It can also give the appearance of a shift in stage, thus improving survival statistics without reducing mortality (i.e., the number of deaths from a given cancer relative to the number of people at risk for the cancer). In such a case, the *apparent* duration of survival increases without lives being saved or life expectancy changed.

Lead-time bias occurs when a test does not influence the natural history of the disease; the patient is merely diagnosed at an earlier date. When lead-time bias occurs, survival *appears* increased, but life is not really prolonged. The screening test only prolongs the time the subject is aware of the disease and spends as a patient.

Length bias occurs when slow-growing, less aggressive cancers are detected during screening. Cancers diagnosed owing to the onset of symptoms between scheduled screenings are on average more aggressive, and treatment outcomes are not as favorable. An extreme form of length bias is termed *overdiagnosis*, the detection of "pseudodisease." The reservoir of some undetected slow-growing tumors is large. Many of these tumors fulfill the histologic criteria of cancer but will never become clinically significant or cause death. This problem is compounded by the fact that the most common cancers appear most frequently at ages when competing causes of death are more frequent.

Selection bias must be considered in assessing the results of any screening effort. The population most likely to seek screening may differ from the general population to which the screening test might be applied. The individuals screened may have volunteered because of a particular risk factor not found in the general population, such as a strong family history. In general, volunteers for studies may be more health conscious and thus likely to have a better prognosis or lower mortality rate, irrespective of the screening result. This is termed the *healthy volunteer effect*.

Potential Drawbacks of Screening Risks associated with screening include harm caused by the screening intervention itself, harm due to the further investigation of persons with positive test results (both true and false positives), and harm from the treatment of persons with a true-positive result, even if life is extended by treatment. The diagnosis and treatment of cancers that would never have caused medical problems can lead to the harm of unnecessary treatment and give patients the anxiety of a cancer diagnosis. The psychosocial impact of cancer screening, whether the result is positive or negative, can also be substantial when applied to the entire population.

Assessment of Screening Tests Good clinical trial design can offset some biases of screening and demonstrate the relative risks and benefits of a screening test. A randomized, controlled screening trial with cause-specific mortality as the end-point provides the strongest support for a screening intervention. In a randomized trial, two like populations are randomly established. One is given the medical standard of care (which may be no screening at all), and the other receives the screening intervention

being assessed. The two populations are compared over time. Efficacy for the population studied is established when the group receiving the screening test has a better cause-specific mortality rate than the control group. Studies showing a reduction in the incidence of advanced-stage disease, an improved survival, or a stage shift are weaker evidence of benefit. These latter criteria are necessary but not sufficient to establish the value of a screening test.

Although a randomized, controlled screening trial provides the strongest evidence to support the usefulness of a screening test, it is not perfect. Unless the trial is population-based, it does not remove the issue of generalizability to the target population. Screening trials generally involve thousands of persons and last for years. Less definitive study designs are therefore often used to estimate the effectiveness of screening practices. After a randomized controlled clinical trial, in descending order of strength, evidence may be derived from:

- The findings of internally controlled trials using intervention allocation methods other than randomization (e.g., allocation determined by birth date, date of clinic visit);
- The findings of cohort or case-control analytic observational studies;
- The results of multiple time series studies with or without the intervention;
- The opinions of respected authorities based on clinical experience, descriptive studies, or consensus reports of experts (the weakest evidence because even experts can be misled by the biases described above).

Screening for Specific Cancers Widespread screening for breast, cervical, and colon cancer is beneficial for certain age groups. Special surveillance of those at high risk for a specific cancer because of a family history or a genetic risk factor may be prudent, but few studies have been carried out to assess the impact of this practice on mortality in specific high-risk populations. A number of organizations have considered whether or not to endorse routine use of certain screening tests. Because these groups have not used the same criteria to judge whether a screening test should be endorsed, they have arrived at different recommendations. The screening guidelines of the U.S. Preventive Services Task Force, the Canadian Task Force on Preventive Health Care, and the American Cancer Society are often quoted and show a range of recommendations ([Table 80-4](#)).

Breast Cancer Breast self-examination, clinical breast examination by a care giver, and mammography have been advocated as useful screening tools. Only screening mammography alone and screening mammography with clinical examination have been evaluated in randomized controlled trials. A number of well-designed trials have demonstrated that annual or biennial screening with mammography or mammography plus clinical breast examination in women over the age of 50 saves lives. In these trials, the breast cancer mortality rate is decreased by about a third. Experts disagree on whether average-risk women aged 40 to 49 should receive regular screening ([Table 80-4](#)). The statistical significance of the screening effect in women aged 40 to 49 depends on the statistical test used. An analysis of eight large randomized trials showed no benefit from mammographic screening for women aged 40 to 49 when assessed 5 to

7 years after trial entry. However, a small benefit emerged 10 to 12 years after study entry. What proportion of this possible benefit is due to screening after these women turned 50 is not known. In randomized screening studies of women aged 50 to 69, the decline in mortality begins about 5 years after initiation of screening. Nearly half of women aged 40 to 49 years screened annually will have false-positive mammograms necessitating further evaluation, often including biopsy. The risk of false-positive testing should be discussed with the patient.

While no study has shown breast self-examination to decrease mortality, it is recommended as prudent by many organizations. A substantial fraction of breast cancers are first detected by the patient, even with widespread mammographic screening.

Cervical Cancer Screening with Papanicolaou smears decreases cervical cancer mortality. The cervical cancer mortality rate has fallen significantly since the widespread use of the Pap smear, although this trend actually began earlier. Most screening guidelines recommend regular Pap testing for all women who are or have been sexually active or have reached the age of 18. With the onset of sexual activity comes the risk of sexual transmission of [HPV](#), the most common etiologic factor for cervical cancer. The recommended interval for Pap screening varies from 1 to 3 years. An upper age limit at which screening ceases to be effective is not known.

Colorectal Cancer Fecal occult blood testing, digital rectal examination, rigid and flexible sigmoidoscopy, radiographic barium contrast studies, and colonoscopy have been considered for colorectal cancer screening. Annual fecal occult blood testing using hydrated specimens could reduce colorectal cancer mortality by a third. The sensitivity for fecal occult blood is increased if specimens are rehydrated before testing, but at the cost of lower specificity. The false-positive rate for rehydrated fecal occult blood testing is high; 1 to 5% of persons tested have a positive result. About 2 to 10% of those with occult blood in the stool have cancer, and 20 to 30% have adenomas. The high false-positive rate of fecal occult blood testing dramatically increases the number of colonoscopies performed.

Two case-control studies suggest that regular screening of people over 50 with sigmoidoscopy decreases mortality. These types of studies are prone to selection biases. A quarter to a third of polyps can be discovered with the rigid sigmoidoscope; half are found with a 35-cm flexible scope, and two-thirds to three-quarters are found with a 60-cm scope. Diagnosis of polyposis by sigmoidoscopy should lead to evaluation of the entire colon with colonoscopy and/or barium enema. The most efficient interval for screening sigmoidoscopy is unknown. Case-control studies suggest that testing at intervals of up to 9 years may confer benefit. Most authorities feel that full colonoscopy is too cumbersome and invasive for widespread use as a screening tool in standard-risk populations. It may be suitable for subjects at extremely high risk, such as members of families with a genetic predisposition to colorectal cancer. Colonoscopy is accepted in screening persons with inflammatory bowel disease. Data are not available on digital rectal examination or barium enema as colon cancer screening tools, but both are insensitive.

Lung Cancer Screening chest radiographs and sputum cytology have been evaluated

as methods for lung cancer screening. No reduction in lung cancer mortality has been found in these studies, although all the controlled trials performed have had low statistical power. Even screening of high-risk subjects (smokers) has not been proved to be beneficial. Spiral computed tomography (CT) can diagnose lung cancers at early stages; however, false-positive rates are high. Ongoing studies are evaluating spiral CT screening.

Ovarian Cancer Adnexal palpation, transvaginal ultrasound, and serum CA-125 determination have been considered for ovarian cancer screening. Adnexal palpation is too insensitive to detect ovarian cancer at an early enough stage to affect mortality substantially. Neither transvaginal ultrasound nor CA-125 screening has been tested in a completed randomized prospective trial. Ovarian cancer screening can lead to an invasive diagnostic workup, which may include laparotomy. In a clinical study, 0.6% of 900 adult women had a serum CA-125 level >35 U/mL. Thus, if 100,000 adult women were screened, 600 would be identified as having a high CA-125. The prevalence of ovarian cancer in the female adult population is approximately 20 per 100,000. Thus, the screening test would identify 600 women who would undergo further evaluation to identify 20 cases of ovarian cancer. Some of these 600 would only be inconvenienced by an ultrasound examination. Others would undergo an exploratory laparotomy. A large proportion of the 20 women identified as having ovarian cancer would have advanced, incurable disease and thus not benefit from screening. An [NIH](#) consensus conference in 1994 concluded that routine screening for ovarian cancer was not indicated for standard-risk women or those with a single affected family member, but that it might be worthwhile in families with genetic ovarian cancer syndromes.

Prostate Cancer The most common prostate cancer screening modalities are digital rectal examination and assays for serum prostate-specific antigen (PSA). Newer serum tests, such as measurement of the ratio of bound to free serum PSA, have yet to be fully evaluated. An emphasis on PSA screening has caused prostate cancer to become the most common non-skin cancer diagnosed in American males. Screening for this disease is very prone to lead-time bias, length bias, and overdiagnosis, and substantial debate rages among experts on whether it is effective. Some experts are concerned that prostate cancer screening, more than screening for other cancers, may cause net harm. Prostate cancer screening clearly detects many asymptomatic cancers, but the ability to distinguish tumors that are lethal but still curable from those that pose little or no threat to health is limited. Men over age 50 have a very high prevalence of indolent, clinically insignificant prostate cancers. No well-designed trial has demonstrated the true benefit of prostate cancer screening and treatment, but trials are in progress.

The effectiveness of radical prostatectomy, radiation therapy, and other treatments for low-stage prostate cancer is also under study in randomized trials. Definitive treatment of cancers detected by screening may cause morbidity for some men, such as impotence and urinary incontinence, and carries a low but finite risk of death. Pending the completion of ongoing randomized trials comparing usual care to prostate screening and comparing definitive therapy to "watchful waiting," organizations have provided conflicting recommendations on prostate cancer screening ([Table 80-4](#)). After a thorough review of the literature, the American Cancer Society and the American Urologic Association changed their guidelines from a recommendation for screening to a recommendation that men be offered screening after being informed of the potential

risks and benefits. A man should have a life expectancy of at least 10 years to be eligible for screening.

Endometrial Cancer Transvaginal ultrasound and endometrial sampling have been advocated as screening tests for endometrial cancer. Benefit from routine screening has not been shown. Transvaginal ultrasound and endometrial sampling are indicated for workup of vaginal bleeding in postmenopausal women but are not considered as screening tests in symptomatic women.

Skin Cancer Visual examination of all skin surfaces by the patient or by a health care provider is used in screening for basal and squamous cell cancers and melanoma. No prospective randomized study has been performed to look for a mortality decrease. Observational epidemiologic evidence from Scotland and Australia suggests that screening programs have caused a stage shift in melanomas diagnosed. Screening may reinforce sun avoidance and other skin cancer prevention behaviors.

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81. CANCER GENETICS - Francis S. Collins, Jeffrey M. Trent

THE CLONAL NATURE OF CANCER

Nearly all cancers originate from a single cell. While multiple cumulative events are invariably required to move a cell from normal to the transformed phenotype (see below and [Chap. 82](#)), the origin of tumors from a single clone of cells is a critical discriminating feature between neoplasia and hyperplasia.

CANCER IS A GENETIC DISEASE

Cancer arises because of alterations in DNA that result in unrestrained cellular proliferation. Most of these alterations involve actual sequence changes in the DNA (i.e., mutation). They may arise as a consequence of random replication errors, exposure to carcinogens (e.g., radiation), or faulty DNA repair processes.

While virtually all cancer is genetic, most cancer is not inherited. Certain individuals with cancer have inherited a germline mutation that predisposes them to the cancer, but even in that situation additional somatic mutations are required for a tumor to develop. In a truly sporadic cancer, *all* of the mutations responsible for the malignant phenotype arise somatically. Such a cancer is caused by genetic alterations but has no hereditary implications.

RNA AND RNA TUMOR VIRUSES

Many malignancies in animals are transmissible, and the etiologic agent is frequently a retrovirus, which possesses a single-stranded RNA genome. During the life cycle of the virus, the single-stranded RNA is converted to double-stranded DNA and is inserted at random into the host chromosome. On rare occasions the virus can be remobilized, carrying along with it an adjacent segment of host DNA. Should this host DNA contain a growth-promoting gene, then the retrovirus is potentially transforming. Although efforts to identify retroviruses in human malignancies have mostly been fruitless, retroviruses are implicated in at least one human malignancy. Human T cell lymphotropic virus (HTLV) type I causes adult T cell lymphoma/leukemia, particularly in Japan and the Caribbean ([Chap. 191](#)). Unlike animal retroviruses that induce neoplasia, HTLV-I does not contain a growth-promoting transforming oncogene. The tax protein, a 40-kDa molecule encoded in the pX region of the viral genome, induces the activation of a number of genes (including some promoting growth) through interactions with *rel* family and CREB (cyclic AMP response element binding protein) family transcription factors.

DNA tumor viruses are more commonly involved in human malignancy. Human papilloma viruses (especially types 16 and 18) cause cervical cancer ([Chap. 188](#)), and both hepatitis B and hepatitis C viruses have been implicated in hepatocellular carcinoma ([Chap. 297](#)). In addition, the Epstein-Barr virus, a herpesvirus that causes a mild illness in children but infectious mononucleosis in nonimmune adolescents and adults, causes Burkitt's lymphoma in Africa, nasopharyngeal carcinoma in Asia, and lymphomas in the setting of immune deficiency ([Chap. 184](#)).

GENERAL CLASSES OF CANCER GENES

In 1914 Boveri hypothesized that cells become malignant either because of overactivation of a gene that promotes cell division or because of loss of function of a gene that normally restrains growth. This hypothesis is largely correct, although defects in DNA repair genes are also involved. Genes that promote normal cell growth are referred to as *protooncogenes*, and activation of such genes by point mutation, amplification, or dysregulation converts them to *oncogenes*.

Genes that normally restrain growth are called *tumor suppressors* (use of the alternative designation of anti-oncogenes is to be discouraged), and unregulated cell growth arises if their function is lost. The diploid nature of mammalian cells allows certain predictions about the consequences of somatic mutations of tumor suppressor genes. Loss of one allele is unlikely to have significant consequences in most instances, as the remaining normal allele is usually sufficient for normal function. Thus, most cells of an individual with an inherited loss of function of one tumor suppressor allele are functionally normal. Only the rare cell that loses or develops a mutation in the remaining normal copy will exhibit uncontrolled growth. This model correctly predicts that the inheritance pattern of cancer in a family with a tumor suppressor gene mutation will be expressed as an autosomal dominant trait, though the cellular mechanism is recessive.

The third category of genes that contribute to malignancy is the DNA repair genes. Every cell division involves the copying of 6 billion base pairs (bp) of DNA. DNA polymerase has a finite error rate, and many environmental influences can damage DNA. As a consequence, repair systems are essential to protect the integrity of the genome. When the repair systems themselves are faulty, either on the basis of inherited or acquired mutation, the rate of accumulation of mutations throughout the genome rises as cell divisions occur. To the extent that these mutations involve oncogenes and tumor suppressor genes, the likelihood of developing malignancy increases.

MENDELIAN CANCER SYNDROMES

Roughly 100 syndromes of familial cancer have been reported, though many are rare. The majority are inherited as autosomal dominant traits, although some of those associated with DNA repair abnormalities (xeroderma pigmentosum, Fanconi anemia, ataxia telangiectasia) are autosomal recessives. Most of the genes responsible for the dominantly inherited cancer syndromes are tumor suppressor genes ([Table 81-1](#)). The hallmarks of a tumor suppressor gene are as follows: (1) the germline mutation that affects one allele generally causes a loss of function; (2) tumors also show loss of the second normal allele as a result of a somatic mutation; and (3) often the *normal* function of the gene is to suppress unrestrained cellular growth or to promote differentiation.

The retinoblastoma gene (*RB*) is a paradigm of such a tumor suppressor gene. In a pedigree showing dominant inheritance of susceptibility to retinoblastoma, a loss-of-function germline mutation occurs in one allele of the *RB* gene on chromosome 13. Analysis of the DNA from the tumors invariably shows that the wild-type allele has also been lost by one of several possible mechanisms ([Fig. 81-1](#)). However, not all retinoblastoma tumors arise in the context of a strong family history. Sporadic retinoblastoma, which is usually unilateral and on average occurs at a slightly older age than familial retinoblastoma, is usually a consequence of somatic mutation in both

alleles of the *RB* gene without any germline predisposition.

Another tumor suppressor gene is the p53 gene on chromosome 17p, which is frequently altered in solid tumors. p53 is somewhat unusual for a tumor suppressor gene in that missense mutations that produce a dominant negative protein product may also be growth-promoting, so that not all alterations obliterate function. Mutations in p53 are found in nearly half of human tumors. Germline mutations in p53 have dramatic consequences, resulting in a phenotype known as the *Li-Fraumeni syndrome*, where affected individuals may develop a variety of sarcomas, brain tumors, and leukemia. [Figure 81-2](#) illustrates a typical pedigree of this devastating disorder.

In many instances the discovery of genes responsible for familial cancer syndromes has provided insight into the normal control of cell growth. For instance, in type I neurofibromatosis -- one of the more common dominant disorders of humans -- positional cloning efforts uncovered a previously unknown gene on chromosome 17q that, when mutated, produces a clinical phenotype of cafe au lait spots, neurofibromas, Lisch nodules of the iris, and a predisposition to neurofibrosarcoma and glioma. The responsible gene, which (like many of the genes in [Table 81-1](#)) has a close homologue in yeast, is neurofibromin (*NF1*), a critical participant in the regulation of the protooncogene *ras*. As shown in [Fig. 81-3](#), the NF1 protein is a GTPase-activating protein (GAP) that normally acts to convert *ras* from its active, growth-promoting, GTP-bound form to its inactive, GDP-bound form. Loss of both copies of *NF1* (one copy by inheritance, one by somatic mutation) thus renders a cell vulnerable to overgrowth, since *ras* is left in the "on" position.

While most autosomal dominant inherited cancer syndromes are due to mutations in tumor suppressor genes, there are a few interesting exceptions. Multiple endocrine neoplasia type II -- a dominant disorder characterized by pituitary adenomas, medullary carcinoma of the thyroid, and (in some pedigrees) pheochromocytoma -- is due to gain-of-function mutations in the protooncogene *ret* on chromosome 10. Interestingly, loss-of-function mutations in *ret* cause a totally different phenotype, Hirschsprung's disease (aganglionic megacolon) ([Chaps. 339](#) and [289](#)).

Dominantly inherited colon cancer is sometimes associated with familial polyposis, which is usually due to mutations in the adenomatous polyposis coli (*APC*) tumor suppressor gene on chromosome 5 ([Table 81-1](#)). However, in most colon cancer families affected individuals do not have familial polyposis, but instead the cancer arises from normal-appearing epithelium. Hereditary nonpolyposis colon cancer (HNPCC, or Lynch's syndrome) is commonly defined as the occurrence of colon cancer in at least three individuals over at least two generations and with at least one individual diagnosed under the age of 50. As many as 1 in 200 individuals in the general population may have HNPCC, although this number is somewhat controversial. Most HNPCC is due to mutations in one of four DNA mismatch repair genes ([Table 81-2](#)). All four of these genes are components of a repair system that is normally responsible for correcting errors in freshly copied DNA. Tumors in patients with HNPCC are characterized by profound genomic instability, especially for short repeated sequences called *microsatellites*. [Figure 81-4](#) shows an example of the instability in allele sizes for dinucleotide repeats in the cancers in HNPCC. The unstable phenotype [sometimes referred to as the "mutator" phenotype, or the "RER+" (replication error) phenotype]

probably requires loss of both copies of the particular mismatch repair gene (one inherited, one somatic), so that the mechanism is similar to that typical of a tumor suppressor gene.

MORE COMPLEX INHERITED FORMS OF CANCER

While the Mendelian forms of cancer described above have taught us much about mechanisms of cellular growth control, most forms of cancer do not follow such simple patterns of inheritance. In many instances (e.g., lung cancer), a strong environmental contribution is at work, but even in such circumstances some individuals may be genetically more susceptible to developing cancer given the appropriate exposure.

In the case of breast and ovarian cancer, circumstantial evidence indicates that a subset of affected individuals (5 to 10%) might be accounted for by dominantly inherited high-penetrance susceptibility genes; two of these genes have been identified by positional cloning. *BRCA1*, located on chromosome 17, is capable when mutated of producing a high risk (up to 85% lifelong) of breast cancer and also of ovarian cancer (50% lifelong risk). Roughly 1 in 500 women carries a germline *BRCA1* mutation, often giving rise to a strong family history. Men with *BRCA1* mutations may have a modestly increased risk of prostate cancer. An array of mutational heterogeneity has been described for *BRCA1* ([Fig. 81-5](#)), as is often the case for genetic disorders. An exception is the Ashkenazi Jewish population, where 1 in 100 individuals carries a particular 2-bp deletion (denoted 185delAG) of *BRCA1*, apparently as a consequence of descent from a common ancestor.

Mutations in another gene on chromosome 13, *BRCA2*, also confer a high risk of breast cancer (and a somewhat lower risk of ovarian cancer); men with *BRCA2* mutations are prone to develop breast cancer. The frequency of *BRCA2* mutations is estimated to be about half that of *BRCA1*. About 1% of Ashkenazi Jews again have a common mutation: 6174delT.

What then of the 90 to 95% of breast cancers that arise in individuals without germline alterations in *BRCA1* or *BRCA2*? Hereditary factors may still contribute to a significant fraction of these, but those factors must be weaker and therefore more difficult to discern.

GENETIC TESTING AND COUNSELING FOR CANCER SUSCEPTIBILITY

The discovery of genes like *RB*, *p53*, *NF1*, *ret*, the [HNPCC](#) mismatch repair genes, *BRCA1*, and *BRCA2* raises the possibility of DNA analysis to predict risk of cancer. There are many complexities associated with such testing. First, one must know the sensitivity and specificity of the test; the mutational heterogeneity for each of these genes constitutes a considerable technical challenge, as it is often necessary to sample every nucleotide of the coding region, the splice junctions, and the promoter to identify most mutations. False-positive results -- i.e., sequence alterations that turn out to be benign polymorphic variants (allelic variations) rather than disease-causing mutations -- can present a thorny problem. Unless proven interventions are available and the test is sensitive, specific, and relatively inexpensive, it will be inappropriate to offer such tests to the general population; the number of false-positive tests will exceed the number of

true positives and a great deal of anxiety and expense will be incurred evaluating persons who are not at an increased risk. Generally, therefore, such testing is not considered except for individuals of higher-than-normal risk, usually on the basis of their family history. In deciding whether to offer such testing, it is critical to determine whether evidence exists for effective interventions to reduce the risk of cancer in those found to be at high risk. If such interventions do not exist (as is the case for Li-Fraumeni syndrome), then the value of the information is limited, and the major negative psychological consequences of this information must be seriously considered.

For conditions such as colon and breast cancer, prophylactic measures exist (total colectomy and bilateral mastectomy, respectively), but these prophylactic measures are more radical and potentially disfiguring than the surgical procedures that would be used to treat the patient if the malignancies actually occurred (segmental bowel resection and lumpectomy, respectively). Other potential negative consequences of a positive genetic test include insurance and employment discrimination. One can still argue, of course, that a close relative of an individual known to carry a mutation in a cancer-causing gene is already sensitized to his or her personal risk of cancer, and that a test establishing that an individual at risk does *not* harbor the mutation can be quite useful. Testing should never be undertaken, however, without a full consideration of how the individual will handle a positive as well as a negative result.

Despite these caveats, genetic testing for some cancer syndromes already appears to have greater benefits than risks, and in those situations it is reasonable to offer testing to individuals at high risk. This would include conditions such as multiple endocrine neoplasia type 2 ([Chap. 339](#)) and von Hippel-Lindau disease ([Chap. 370](#)). More in the gray zone, although potentially applicable to much larger numbers of individuals, are tests for *BRCA1*, *BRCA2*, and the [HNPCC](#) genes. More research is urgently needed in those situations to determine the effectiveness of various interventions (life-style, diet, surveillance, or surgery). Until those answers are available, such testing should be offered only as part of a research protocol. As more susceptibility genes are identified, better answers become available about the effectiveness of interventions, and health insurance discrimination is legislatively prohibited, genetic testing will move into the mainstream of medicine. Every physician of the future will need to have the skills of a genetic counselor.

ACQUIRED MUTATIONS IN CANCER

The identification of mutations in the germline of patients with heritable cancers means that the alteration is present in every cell of the body. However, in most cancers a normal cell becomes a malignant cell by a series of mutations that arise not in the germline but in somatic cells. Usually mutations must occur in several genes to give rise to neoplasia. The underlying questions are "how many mutations cause a cancer?" and "what specific genes are affected?" rather than whether or not mutational events cause cancer.

While answers to these questions are not available for every human malignancy, advances in molecular and cellular biology and epidemiologic analyses of human and experimental cancers are providing insights in cancer causation. [Table 81-3](#) summarizes evidence from several lines of investigation pointing to a mutational basis for cancer

causation. One particularly striking feature is the fact that the overall incidence of cancer increases as the fourth to sixth power of age for most malignancies ([Fig. 81-6A](#)). For some tumors, the shape of the age-incidence curve suggests heterogeneity in molecular mechanisms. For example, Hodgkin's disease has a bimodal age distribution, suggesting that two etiologically (and therefore mutationally) distinct forms of this disease may exist ([Fig. 81-6B](#)).

MULTISTEP BASIS OF CANCER

From 5 to 10 accumulated mutations are thought to be necessary for a cell to move from the normal to the fully malignant phenotype. At each step the mutated cell may gain a slight growth advantage, so that it is increased in its representation relative to its neighbors. [Figure 81-7](#), a representation of a lineage diagram hypothesized by Peter Nowell, illustrates how a single cell, afflicted with progressive alterations in tumor suppressor genes and protooncogenes, can develop into a clonal malignancy.

We are beginning to understand the precise nature of the genetic alterations responsible for some malignancies and to get a sense of the order in which they occur. Perhaps the best studied example is colon cancer, where an analysis of DNA from tissues extending from normal epithelium through adenoma to carcinoma have identified some of the genes mutated along the way ([Fig. 81-8](#)). However, the order of mutational events is far from uniform, and the diagram in [Fig. 81-8](#) should be considered a generalization and not a defined pathway. Similar data are being accumulated for other malignancies.

MECHANISMS OF SOMATIC MUTATION OF ONCOGENES IN MALIGNANCY

Cellular protooncogenes, their necessity and importance in normal cell growth, and their responsibility for transformation-associated change after removal of normal growth controls are discussed in [Chap. 82](#). Mechanisms that upregulate (or activate) cellular protooncogenes can be grouped into three broad areas: point mutations, DNA amplification, and chromosome rearrangements.

Point Mutation One protooncogene that is commonly activated in solid tumors by point mutation is a member of the *ras* family of oncogenes; these were initially cloned from human bladder carcinoma cells and are critical regulators of normal and aberrant cell growth ([Fig. 81-3](#)). Mutations in one of the *ras* genes (*H-ras*, *K-ras*, or *N-ras*) are present in up to 85% of pancreatic cancers and 15% of all human cancers. In studies of *K-ras* (particularly in lung and colon cancer), the mutational spectrum of this gene has been identified. Remarkably, and in contrast to the diversity of mutations observed in the *BRCA1* gene ([Fig. 81-5](#)), most of these activated genes contain point mutations in codons 12 or 61 (which convey resistance to the inactivating action of [GAP](#)). The specificity of this pattern of mutation means that it has potential value in diagnostic or prognostic studies of cancer. For *K-ras*, mutations may be a useful prognostic marker in lung cancer, but for most other cancers (including pancreas and colon cancer) no prognostic utility has been demonstrated. This is in part because *ras* mutations occur early in colon cancer ([Fig. 81-8](#)), being common in precancerous lesions of the bowel.

DNA Amplification The second mechanism for activation of oncogene overexpression

is DNA sequence amplification. This increase in DNA sequence copy number may cause cytologically recognizable chromosome alterations referred to as *homogeneous staining regions* (HSRs), if integrated within chromosomes, or *double minutes* (dmins), if extrachromosomal in nature ([Fig. 81-9](#)).

The recognition of DNA amplification was greatly facilitated by the development of a procedure based on dual-color fluorescence in situ hybridization (FISH) called *comparative genomic hybridization* (CGH). DNA from tumor and normal cells is labeled with different fluorescent reporter molecules and then hybridized to normal metaphase chromosomes. Regions of duplications and deletions within tumor DNA are then demonstrated as quantifiable alterations in signal intensity at particular sites. With this technique the entire genome can be surveyed for gains and losses of DNA sequences, thus pinpointing chromosomal regions likely to contain genes important in the development or progression of cancer.

Numerous genes are known to be amplified in human malignancies. Several genes, including *N-myc* were identified because they were present within the amplified DNA sequences of a tumor and had homology to known oncogenes. Because the region amplified often extends to hundreds of thousands of base pairs, more than one oncogene may be amplified in some cancers (particularly sarcomas). Genes simultaneously amplified in many cases include *MDM2*, *GLI*, *CDK4*, *SAS*, and others implicated in cellular growth control. The clinical implications of gene amplification have been explored for some cancers [most notably *ERBB2* (*HER-2/neu*) in breast cancer and *N-myc* in neuroblastoma]; demonstration of amplification of a cellular gene is usually a predictor of poor prognosis. Once a patient has been exposed to the selective effects of chemotherapy, gene amplification may lead to drug resistance. Amplification of the dihydrofolate reductase gene may follow clinical exposure to methotrexate, a drug that inhibits the activity of the enzyme.

Chromosomal Alterations in Human Cancer Chromosomal alterations provide important clues to the genetic changes in cancers. To date, most chromosome analyses have been performed on hematopoietic cancers, although solid tumors may also have translocations. The breakpoints of several recurring chromosome abnormalities often occur at the sites of cellular protooncogenes. Translocations are particularly common in lymphoid tumors, probably because these cell types normally rearrange DNA to generate antigen receptors. Indeed, antigen receptor genes are commonly involved in the translocations, implying that an imperfect regulation of receptor gene rearrangement may be involved in the pathogenesis. An example is Burkitt's lymphoma, a B-cell tumor characterized by a reciprocal translocation between chromosomes 8 and 14. Molecular analysis of Burkitt's lymphomas demonstrated that the breakpoints occurred within or near the *myc* locus on chromosome 8 and within the immunoglobulin heavy chain locus on chromosome 14, resulting in the transcriptional activation of *myc*. Enhancer activation by translocation, although not universal, appears to play an important role in malignant progression.

Chromosome rearrangements can lead to the abnormal overexpression of a transcription factor that performs its normal function and turns on growth-related genes. The translocation may create a chimeric transcription factor that has altered function. For example, the t(15;17) of acute promyelocytic leukemia produces a retinoic acid

receptor with an abnormal cell distribution that inhibits differentiation. Gene rearrangements most commonly involve transcription factors, but other components of signaling pathways may also be involved.

The first reproducible chromosome abnormality detected in human malignancy was the Philadelphia chromosome in chronic myelogenous leukemia (CML). This cytogenetic abnormality is generated by reciprocal translocation involving the *ABL* oncogene, a tyrosine kinase on chromosome 9 being placed in proximity to the *BCR* (breakpoint cluster region) on chromosome 22. [Figure 81-10](#) illustrates the generation of the translocation and its protein product. The consequence of expression of the *BCR-ABL* gene product is the activation of signal transduction pathways, leading to cell growth independent of normal external growth factor signals.

In addition to transcription factors and signal transduction molecules, translocations may involve the overexpression of cell cycle regulatory proteins, such as cyclins, and of proteins that regulate cell death, such as *bcl2*. Altering control of expression of cell cycle regulatory proteins can lead to aberrant cell cycle control. The overexpression of *bcl-2* can prevent the death of a cell that has endured enough genetic damage to cause its death. If such a cell survives to receive additional genetic damage, a tumor can develop. [Table 81-4](#) lists representative examples of recurring chromosome alterations in malignancy and the associated gene(s) rearranged or dysregulated by the chromosomal change.

Technical obstacles have slowed the identification of recurring chromosome abnormalities in human solid tumors (particularly carcinomas) because of the complexity of chromosome alterations in such tumors, in contrast to the solitary, often reciprocal, nature of chromosome rearrangements in hematopoietic malignancies.

EPIGENETIC REGULATION OF GENE EXPRESSION AND CANCER

The term *epigenetic* refers to mechanisms of gene regulation independent of DNA sequence. The inactivation of the second X chromosome in female cells is an example of an epigenetic mechanism that prevents gene expression from the inactivated chromosome. During embryologic development, entire regions of chromosomes from one parent are silenced and gene expression proceeds from the chromosome of the other parent. For most genes, expression occurs from both parental alleles or randomly from one parent or the other. The preferential expression of a particular gene exclusively from the allele contributed by one parent is called *parental imprinting* and is thought to be regulated by the methylation of the silenced allele.

The role of epigenetic control mechanisms in the development of human cancer is unclear. However, a general decrease in the level of DNA methylation has been noted as a common change in cancers. In addition, the loss of imprinting of the normally silent maternal allele of the insulin-like growth factor II gene at chromosome 11p15.5 has been implicated in some cases of the rare pediatric malignancy Wilms' tumor. The loss of imprinting may result in the overexpression of the growth factor and a predisposition to malignant transformation.

THE FUTURE

The real challenge in oncology is to convert the growing molecular understanding of cancer into clinical advances, particularly the development of new therapies. One can anticipate that in the coming years the molecular analysis of mutations in tumors will allow stratification of malignancies into more precise subgroups than is currently possible by histologic classification, including subgroups with particularly good or bad prognoses or that have a lower or higher likelihood of responding to a particular therapy. Some of this information is already accumulating, but usually only one or two genes are assessed; the promise of the future is to obtain a detailed molecular "fingerprint" of every tumor in order to provide the maximum information about its biology and response to therapy. The application of techniques for assessing global gene expression (cDNA microarrays, serial analysis of gene expression, or SAGE, and others) is leading to novel ways of looking at cancer that are considerably more discriminating than light microscopy, the gold standard of medical practice. The National Cancer Institute in conjunction with the National Center for Biotechnology Information have undertaken the Cancer Genome Anatomy Project (CGAP) (<http://www.ncbi.nlm.nih.gov/ncicgap/>) to collect data on the differences in gene expression between normal and malignant tissues and make it available on the Internet.

Genetics will also influence cancer prevention and early detection. The ability to identify cancer susceptibility genes presages a new era of cancer prevention, if the potential risks of such testing can be surmounted. Currently, most cancer early detection strategies (such as mammography, stool occult blood testing, or digital rectal examination) are applied to population groups. The ability to identify the individuals at highest risk and to focus preventive medicine efforts accordingly may be both better received by patients and more cost effective. Early detection strategies will be even more effective if we can develop the ability to identify very small numbers of malignant or premalignant cells at a time when the risk of metastasis is still very low.

More importantly, detailed molecular information about the regulation of the cell cycle and the interplay of tumor suppressor genes and proto-oncogenes that control it may lead to new effective therapies, based on pathophysiology rather than empiricism. Whether such strategies will rely on drugs of the traditional types or will be based on more novel strategies such as gene therapy or immunotherapy is hard to predict.

(Bibliography omitted in Palm version)

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82. CELL BIOLOGY OF CANCER - Robert G. Fenton, Dan L. Longo

Two characteristic features define a cancer: cell growth not regulated by external signals (i.e., autonomous) and the capacity to invade tissues and metastasize to and colonize distant sites ([Chap. 83](#)). The first of these features, the uncontrolled growth of abnormal cells, is a property of all neoplasms, or new growths. A neoplasm may be benign or malignant. If invasion, the second cardinal feature of cancer, is present, the neoplasm is malignant. Cancer is a synonym for *malignant neoplasm*. Cancers of epithelial tissues are called *carcinomas*; cancers of nonepithelial (mesenchymal) tissues are called *sarcomas*.

Cancer is a genetic disease, but the level of its expression is the single cell. Although some forms of cancer are heritable, most mutations occur in somatic cells and are caused by intrinsic errors in DNA replication or are induced by carcinogen exposure. A single genetic lesion is usually not sufficient to induce neoplastic transformation of a cell. The malignant phenotype is acquired only after several (5 to 10) mutations lead to derangements in a variety of gene products. Each genetic alteration may cause phenotypic changes typified by the progression in epithelial tissues from hyperplasia to adenoma to dysplasia to carcinoma in situ to invasive carcinoma. Cells have evolved mechanisms to resist neoplastic transformation (see below).

The >200 discrete cell types in the body are not equally susceptible to developing cancer. Some cells, such as cardiac myocytes, sensory receptor cells for light and sound, and lens fibers, persist throughout life without dividing or being replaced. Neoplasia in such tissues is exceedingly rare. Most differentiated tissues undergo constant renewal characterized by cell death and replacement.

In tissues with rapid turnover, such as skin, bone marrow, and gut, an individual cell is on one of two largely mutually exclusive paths: division or differentiation. Cells capable of dividing are undifferentiated (stem cells), whereas terminally differentiated cells are unable to divide. Stem cells produce daughter cells that can either become new stem cells (thus replenishing the stem cell compartment) or undergo terminal differentiation, depending on the circumstances and the environmental signals. Stem cells are distinguished from differentiating cells by different patterns of gene expression. Gene expression is the product of the tissue-specific programming interacting with environmental factors such as cell-to-cell contact; interactions with extracellular matrix; endocrine hormones; paracrine growth and differentiation factors; and stresses such as heat, oxidation, irradiation, and physical distortion or traction.

Cancer is most common in tissues with rapid turnover, especially those exposed to environmental carcinogens and whose proliferation is regulated by hormones. The most common genetic changes involve the activation of proto-oncogenes or the inactivation of tumor suppressor genes ([Chap. 81](#)). Although genetic damage is nearly universal in human cancer, cells with neoplastic features can be generated in vitro without genetic damage. Removal and in vitro culture of cells from the epiblast of a murine embryo lead to the uncontrolled proliferation of the cells and the generation of a teratocarcinoma cell line capable of producing tumors when inoculated into animals. The removal of these normal embryonic cells from their normal environment leads to uncontrolled growth. However, if the teratocarcinoma cells are reinjected into an early embryo, under the

inductive influence of their normal neighbors they can differentiate into normal organs and tissues appropriate for the location where they are injected.

Thus, environmental factors exert potent effects on the gene expression of target cells. The panoply of signals received by a particular cell leads to the activation of particular sets of transcription factors. The pattern of gene expression determines whether a cell will divide, differentiate, or die.

PRINCIPLES OF CELL CYCLE REGULATION

The mechanism of cell division is substantially the same in all dividing cells and has been conserved throughout evolution. The process assures that the cell accurately duplicates its contents, especially its chromosomes. The cell cycle is divided into four phases. During M phase, the replicated chromosomes are separated and packaged into two new nuclei by mitosis and the cytoplasm is divided between the two daughter cells by cytokinesis. The other three phases of the cell cycle are called *interphase*: G1 (gap 1), a period of growth during which the cell determines its readiness to commit to DNA synthesis; S (DNA synthesis), during which the genetic material is replicated and no re-replication is permitted; and G2 (gap 2), during which the fidelity of DNA replication is determined and errors are corrected.

During S phase, DNA synthesis begins with the unfolding of chromatin and nucleosome complexes rendering DNA accessible for the addition of DNA helicase and single-strand binding proteins that help open the double helix. Replication origins are spaced roughly 100,000 nucleotide pairs apart throughout the genome. DNA polymerase and DNA primase attach to these sites and catalyze the polymerization of the DNA at a rate of about 50 nucleotides per second. DNA polymerase catalyzes leading-strand synthesis, while DNA polymerase α uses DNA primase-generated Okazaki fragments for lagging-strand synthesis. Topoisomerase I nicks DNA, relieving torsional tension of the replicating helix; topoisomerase II introduces double strand breaks to avoid DNA tangles. Topoisomerases are targets of many chemotherapeutic drugs. Once a DNA segment is replicated, chromatin is reassembled, and replication origins are relicensed by binding of specific proteins that prevent re-replication until the next S phase. Although this system for replication is efficient and accurate, occasional mistakes are made, and these are repaired by a variety of mechanisms. In some cancers, the mismatch repair mechanisms are defective and errors increase by 3 to 4 logs, greatly increasing the likelihood of mutations in growth regulatory genes in daughter cells.

DNA polymerase is unable to replicate the end of a DNA chain completely, resulting in loss of DNA with each replication. This problem has been solved by a mechanism that replicates tandem repeats of a six-nucleotide sequence (GGGTTA) to the ends of each chromosome. These repeated sequences are called *telomeres* and are replicated through an RNA-dependent DNA polymerase called *telomerase*. Normal somatic cells do not express telomerase, and the replicative lifetime of such cells is limited to approximately 30 cell divisions due to the progressive loss of telomere repeats; the limit imposed on somatic cell division is called the *Hayflick limit*, at which time replicative senescence occurs. Germ cells do express telomerase and have a long (possibly unlimited) replicative lifetime. The aberrant expression of telomerase in cancer cells is thought to be a component of the neoplastic process, assuring that the cell will be able

to undergo many divisions without inducing senescence or genetic catastrophe. Inhibition of telomerase activity in cancer cells could have antitumor effects.

The cell cycle transitions between G1 and S and between G2 and M are tightly regulated to ensure that cells are prepared to divide and to minimize errors in the replication process. Checkpoints in G1 and G2 determine whether a cell will enter S or M phase, respectively; these checkpoints are regulated by serine/threonine protein kinases (cyclin-dependent kinases, or cdk) and kinase-associated proteins called *cyclins*. The enzymatic activity of each cdk is determined by its association with a cyclin and its phosphorylation state. There are at least seven cdk family members, and a like number of cyclins, which generate a group of cdk/cyclin complexes with differing substrate specificities and times of action during the cell cycle. Cyclin expression varies with the cell cycle, and the synthesis of these proteins is transcriptionally regulated and their degradation is mediated by ubiquitin conjugation and destruction in proteasomes.

The cyclin B/cdc2 complex (also called *mitosis promoting factor*, or MPF) is the primary regulator of transition from G2 to M phase. It is activated by a [cdk](#)-activating kinase (CAK) and a phosphatase (cdc25c) that removes inhibitory phosphates. The cdc25C is the target of a DNA damage-induced kinase that inhibits its activity. DNA damage leads to phosphorylation of cdc25C and its transport out of the nucleus, away from cyclin B/cdc2. This prevents entry into M phase until DNA damage is repaired. The regulated movement of molecules into and out of the nucleus is a common control mechanism in signal transduction. Some of the substrates of cyclin B/cdc2 are defined; its phosphorylation of histone H1, nuclear lamins, and microtubule-associated proteins facilitates chromosome condensation, nuclear membrane breakdown, and spindle formation, respectively.

The checkpoint regulating transition from G1 to S is frequently disrupted in cancer. The product of the retinoblastoma tumor suppressor gene, the nuclear phosphoprotein Rb, governs the key transition referred to as the *restriction point*. A second pathway regulated by the p53 tumor suppressor interacts with the Rb pathway to ensure that cell proliferation can safely take place. Rb and p53 are inactivated by products encoded by DNA tumor viruses, including SV40 large T antigen, adenovirus E1A and E1B, and human papillomavirus E6 and E7. The Rb and p53 pathways each include other oncogenes and tumor suppressors that are frequently disrupted in cancer ([Table 82-1](#)).

Regulation of passage through the restriction point is complex. In early G1, Rb is hypophosphorylated and in a complex with the E2F transcription factor. This nuclear complex binds to the promoters of genes required for cell cycle progression and inhibits their expression. However, in mid and late G1, Rb becomes phosphorylated (at ~10 sites) by the sequential activity of the cyclin D/[cdk](#)4 and cyclin E/cdk2 complexes. Hyperphosphorylated Rb releases E2F, thus relieving transcriptional repression, and heterodimers of E2F and DP1 transcription factor family members activate several genes required for S phase progression, including dihydrofolate reductase, thymidine kinase, DNA polymerase, and cdc2 ([Fig. 82-1](#)). In addition to its role in growth regulation, Rb is required for the *in vitro* differentiation of muscle cells and adipocytes. Cell cycle control and induction of differentiation are functions of Rb that contribute to tumor suppression.

The activity of [cdk](#) is also regulated by cdk inhibitors (cdki). These low-molecular-weight proteins are divided into two families: the Cip/Kip family encoding p21^{Cip1/Waf1}, p27^{Kip1}, p57^{Kip2}, which inhibit cdk activity broadly, and the Ink4 family encoding p16^{INK4a}, p15^{INK4b}, p18^{INK4c}, and p19^{INK4d}, which block cyclin D/cdk4 activity and inhibit Rb phosphorylation and G1/S transition. p21 is induced by p53 in response to DNA damage, causing G1 arrest to permit DNA repair. If DNA damage is too great, a cell suicide pathway is induced to eliminate cells that may be dysfunctional (see below).

The [cdki](#) can be induced by growth inhibitors such as transforming growth factor (TGF) β and can be inhibited by growth factors such as interleukin (IL) 2. Genetic alterations in cdki, especially p16 and p15, occur with high frequency in certain tumors. Alterations at the p16 locus on chromosome 9p21 have been detected in 75% of pancreatic cancers; 40 to 70% of glioblastomas; 50% of esophageal cancers; and about 20% of non-small cell lung cancers, soft tissue sarcomas, and bladder cancers. Mutations in p16 account for half of familial melanomas. Some tumors fail to express cdki because the genes are methylated, an epigenetic mechanism for blocking transcription. Rb pathway regulation is also circumvented by overexpression of cyclin D1 [breast cancer and the t(11;14) in mantle cell lymphoma] and by mutations in cdk4 that abrogate p16 binding.

Whereas Rb, cyclin D, [cdk4](#), and p16 are commonly altered in cancer, E2F overexpression or p21 mutations have not yet been seen. Additional study may reveal why some components of the system are susceptible to alterations and other components are not.

p53, the "guardian of the genome," is a transcription factor that is not usually called upon to act in the course of normal replication. Levels of p53 are normally kept low by its association with mdm2, which binds p53 and shuttles it out of the nucleus for degradation. However, with DNA damage, p53 is phosphorylated by the ataxia telangiectasia gene product ATM, it dissociates from mouse double minute 2 (mdm2), and its destruction is slowed, leading to increased levels. Also, p53 influences transcription to either halt cell cycle progression (e.g., through induction of p21 expression to inhibit [cdk](#) activity) to permit repair of the DNA or, if the damage is too great, to initiate cell suicide (*apoptosis*). p53-induced genes involved in apoptosis include death receptors (DR5) and death-inducing members of the Bcl-2 family. p53 also induces expression of mdm2, thus down-regulating its own activity.

Inducers of p53 include hypoxia, DNA damage, ribonucleotide depletion, and telomere shortening. Dysregulated activity of oncogenes such as *myc*, which promote aberrant G1/S transition, induces p53-mediated apoptosis. A second product of the Ink4a locus is p14^{ARF}, encoded by an alternative reading frame (ARF) from p16. Levels of ARF are upregulated by *myc* and E2F. ARF binds to mdm2/p53 complexes and rescues p53 from the inhibitory effects of mdm2 with subsequent activation of p53-induced genes. This oncogene checkpoint leads to the death of renegade cells that attempt to traverse the restriction point without the right signals.

Mutation in p53 is the most common genetic alteration found in human cancer (>50%) and is the causative lesion in Li-Fraumeni familial cancer syndrome. In tumors, usually one p53 allele on chromosome 17p is deleted and the other is mutated. The mutations often involve the region between codons 120 and 290, the portion of the gene specifying

the site of p53 involved in transcription. Some environmental agents cause mutations at specific sites. In 81% of hepatomas in persons from developing countries, codon 249 is mutated due to exposure to the carcinogen, aflatoxin. Codon 249 mutations occur in only 11% of hepatomas in persons from industrialized countries where aflatoxin exposure is low. Inactivation of the p53 pathway compromises cell cycle arrest, inhibits apoptosis induced by DNA damage or oncogene activation, and predisposes cells to chromosome instability.

Regardless of the pathogenesis of the tumor, most have some mechanism(s) to bypass the G1 checkpoint, avoid activation of cell suicide pathways, and propagate cells with damaged DNA. [Table 82-1](#) summarizes some of the changes in cell cycle regulators detected in human cancers.

SIGNALING FROM OUTSIDE THE CELL TO THE NUCLEUS

The behavior of cells in the body is tightly regulated by environmental signals. The ability of a cell to respond to a specific set of signals determines whether the cell will live or die, differentiate, proliferate, or remain quiescent. In normal cells and tissues, coordinated action such as wound healing or the inflammatory response is regulated by signaling pathways that convert extracellular signals into the performance of specialized action in the responding cells. In cancer cells, the process of invading and metastasizing is influenced by signal transduction pathways activated by paracrine and autocrine factors.

The coupling of extracellular signals to cell response varies for different receptor and signaling systems. The binding of a growth factor [e.g., epidermal growth factor (EGF)] to its receptor on the cell surface produces measurable changes in the cell within seconds and elicits a sequence of events that may last for days. Rapid responses are elicited by changes in ion flux, phosphorylation events, lipid metabolism, and production of second messengers. Long-term responses are mediated by the transfer of signaling information from the receptor to the nucleus, where alterations in the pattern of gene expression result in phenotypic change. Signal transduction comprises the mechanisms by which information received at the plasma membrane is imparted to the nuclear transcriptional machinery and other cell functions. Many signal transduction pathways are perturbed in cancer cells. There are three families of cell surface receptors: ion channel-linked receptors, G protein-linked receptors, and enzyme-linked receptors. Although ion channel-linked receptors are a component of growth-related activation in many cell types, they are primarily involved in neurotransmitter signal transduction and are somewhat less important in the pathogenesis of neoplasia than the other two types and will not be discussed further.

G PROTEIN-LINKED RECEPTORS

The G protein-linked receptors traverse the plasma membrane seven times (serpentine receptors). They do not induce covalent modification of their substrates, as do enzyme-linked receptors, but generate second messenger molecules such as cyclic AMP, cyclic GMP, and calcium to activate downstream processes. Upon ligand binding, these receptors activate trimeric G proteins inducing the release of G_α and $G_\beta\gamma$ subunits, each of which elicits downstream signals. The process is terminated by GTP

hydrolysis by G α subunits. The roles of G protein signaling pathways in human cancer include certain endocrine tumors whose cells of origin depend on cyclic AMP for growth. About half of growth hormone-secreting pituitary tumors encode mutant G α subunits that are defective in GTPase activity and are constitutively activated even in the absence of ligand. These G α subunits bind to and stimulate the activity of adenylyl cyclase, leading to unregulated synthesis of cyclic AMP. Cyclic AMP binds to the repressor subunit of protein kinase A thus activating the kinase, which enters the nucleus and phosphorylates CREB (cyclic AMP response element binding protein), a transcription factor that activates genes required for proliferation of the cancer cells. Growth stimulatory G α mutations have also been described in adrenal cortical tumors and endocrine tumors of the ovary. Factors involved in normal cell and tumor cell migration also stimulate cells through G protein-coupled receptors.

ENZYME-LINKED RECEPTORS

There are at least five classes of enzyme-linked receptors: receptor guanylyl cyclases, receptor tyrosine kinases, tyrosine kinase-associated receptors, receptor tyrosine phosphatases, and receptor serine/threonine kinases. The atrial natriuretic peptide receptor is a receptor guanylyl cyclase. Some disease manifestations in cancer may be related to atrial natriuretic peptide activity (such as hyponatremia), but little is known about this receptor class. Receptor phosphatases are not known to be involved in cancer. The other classes of enzyme-linked receptors are better defined and play a more important role in cancer.

Receptor Tyrosine Kinases The receptors for most growth factors are transmembrane tyrosine kinase receptors, including platelet-derived growth factor (PDGF), fibroblast growth factors (FGFs), [EGF](#), heregulin, insulin, insulin-like growth factors (IGF) I and II, nerve growth factor (NGF), stem cell factor, vascular endothelial growth factor, macrophage colony stimulating factors (CSF), and others. Much of what we know about receptor tyrosine kinases and the events that follow their ligation emerged from the study of the proliferation-inducing altered forms of the normal cellular genes (proto-oncogenes) that are the cancer-causing genes (oncogenes) in animal retroviruses. Although downstream events vary with the receptor/ligand combinations, the activation of the receptor follows a typical pattern. Ligand binding induces dimerization or oligomerization of receptor subunits, which activates tyrosine kinase activity and causes autophosphorylation of specific tyrosine residues in the cytoplasmic domain of the receptor. Phosphorylated tyrosine residues on the receptors or on associated adaptor proteins form docking sights for other signal transduction molecules that contain one or more *src-homology region 2*, or SH2, domains, named because the sequence was first identified in the *src* nonreceptor tyrosine kinase. Phosphorylation of tyrosine residues provides a unique amplification signal because of the rapid and specific binding of SH2 domains to p-Tyr, although p-Tyr comprises only 0.05% of total cell phosphoamino acid. These associations via SH2 domains trigger subsequent events ([Fig. 82-2](#)). Signaling is terminated by the action of p-Tyr-specific phosphatases.

Protein domain interactions between evolutionarily conserved motifs play critical roles in all forms of signal transduction, ranging from tyrosine kinase pathways, death-inducing molecules, and the association of transcription complexes on gene regulatory regions. The most common docking mechanisms are based on recognition of particular protein

sequences; the SH2 domains recognize p-Tyr-containing sequences with specificity conferred by surrounding amino acid residues, the SH3 domains dock with proline-rich sequences, and the pleckstrin homology domains [pleckstrin is a major protein kinase C (PKC) substrate in platelets] lead to associations with phosphatidylinositol lipids phosphorylated in the 3 position by phosphatidylinositol 3-kinase (PI3K; see below). Some molecules that do not have docking domains are brought into association with the receptors through the activity of adaptor proteins that are composed of docking domains only. Thus, the nucleotide exchange factor son of sevenless (SOS; named for its role in *Drosophila* eye development) is brought close to the membrane to activate Ras through its association with the adaptor protein grb2 (identified because it "grabbed" p-Tyr-containing proteins).

Receptor tyrosine kinases activate many signaling pathways including phospholipase C- α , which hydrolyzes phosphoinositide 4,5-bisphosphate (PIP₂) into diacylglycerol (DAG) and inositol triphosphate (IP₃). DAG together with calcium ion activates [PKC](#), a family of serine/threonine kinases with different activation requirements, subcellular locations, and substrates in different cell types. PKC is the target of tumor-promoting phorbol esters, and its activation can influence cell proliferation, differentiation, and tumorigenesis. IP₃ induces the release of intracellular calcium, which binds to calmodulin, a protein that regulates the activity of many enzymes, including phosphatases. Calcium fluxes within cells can be short or prolonged, and the duration has profound physiologic effects. PI3K is a lipid kinase that generates PI(3,4)P₂ and PI(3,4,5)P₃, membrane lipids that act as binding sites for proteins containing PH motifs. Such proteins include Akt serine/threonine kinase, which is implicated in activating survival pathways in many cells. Src family tyrosine kinases bind to p-Tyr on activated receptors and amplify signaling information by phosphorylation of distinct protein substrates within the cell. Src activity is required for G₁ progression in some cells through its induction of the transcription factor c-myc. Another consequence of receptor tyrosine kinase activation is stimulation of the Ras/MAP (mitogen-activated protein) kinase pathway that leads to activation of a number of transcription factors that regulate proliferation, differentiation, and cell survival. This pathway is frequently abnormal in cancer cells.

Ras is a 21-kDa member of a large family of proteins, including rho, rac, rab, and others, that regulate cytoskeletal changes, vesicular and nuclear transport, and proliferation and that share sequence homology with the G α subunit of G protein-linked receptors. Ras is attached to the inner cell membrane through an isoprenyl lipid group added after translation by the enzyme farnesyl transferase. If the lipid group is not added, Ras does not localize to the membrane and cannot function normally. In unstimulated cells, Ras is bound to GDP and is inactive. Following receptor tyrosine kinase activation, the guanine nucleotide exchange factor [SOS](#) is brought to the membrane by its association with grb2. SOS removes GDP from Ras and adds GTP. GTP-bound Ras then activates a cascade ending with [MAP](#) kinase, which migrates to the nucleus and phosphorylates (activates) a number of transcription factors ([Fig. 82-3](#)). The kinetics of MAP kinase activity are critical: in PC12 rat pheochromocytoma cells, stimulation with [EGF](#) results in transient stimulation of MAP kinase activity, retention of MAP kinase in the cytoplasm, and cell proliferation; stimulation of PC12 cells with NGF induces sustained activation of MAP kinase, nuclear translocation of MAP kinase, and neuronal differentiation.

Genetic defects leading to increased signaling from receptor tyrosine kinase-linked pathways are important in the etiology and progression of human cancer. About 30% of human cancers (especially pancreatic, lung, and colon adenocarcinomas) have mutated *Ras*. The mutations usually involve codons 12, 13, or 61 and result in a *Ras* protein that fails to hydrolyze its bound GTP and is thus constitutively active. In the hereditary disorder, neurofibromatosis, a mutation in the gene that encodes neurofibromin, a GTPase activating protein (GAP), inhibits its ability to inactivate *Ras* by converting the GTP to GDP. Constitutively activated *Ras* results in the unregulated activity of the signaling pathways downstream of *Ras*, including the MAP kinase pathway and activation of PI3K. Some epithelial cancers overexpress one or more members of the receptor tyrosine kinase family. [EGF](#) receptors, [IGF-I](#) receptors, and *HER-2/neu* are overexpressed in lung, bladder, breast, head and neck, and ovarian cancers. Mutations within the *Ret* tyrosine kinase receptor lead to constitutive receptor dimerization and kinase activation and are responsible for the dominant inherited cancer syndromes multiple endocrine neoplasia (MEN) type 2A and type 2B and familial medullary thyroid carcinoma. Autocrine and paracrine sources of the relevant growth factors have been noted in some cases.

Tyrosine Kinase-Associated Receptors The receptors for growth hormone, prolactin, erythropoietin, thrombopoietin, most interleukins, granulocyte [CSF](#), granulocyte-macrophage CSF, interferon- α , interferon- γ , and many other cytokines are members of the tyrosine kinase-associated receptor family. These single-transmembrane receptors contain ligand-specific subunits and shared signaling subunits. Ligand binding induces the activation of receptor-associated tyrosine kinases. Three families of kinases are known to be associated with this class of receptors: *src* family (*src*, *yes*, *fgr*, *fyn*, *lck*, *lyn*, *hck*, *blk*, and counting), *syk* family (*syk*, ZAP-70), and Janus family (JAK1, JAK2, JAK3, Tyk2). The Janus family kinases have receptor sites that act as docking sites for SH2-containing transcription factors called STATs (signal transducers and activators of transcription). Tyrosine phosphorylation of STATs induces their dimerization by SH2-p-Tyr association followed by translocation to target genes in the nucleus. A unique feature of JAK/STAT signaling is that the pathway from cell membrane to nucleus is traversed by a single dimeric molecule, as opposed to the cascade of kinase and adaptor molecules associated with membrane tyrosine kinases. The *src* family kinases can associate with receptor tyrosine kinases as well as tyrosine kinase-associated receptors, and, not surprisingly, signal transduction through either receptor class leads to the activation of similar signaling cascades. As a consequence of *src* family activation, *myc* is one of the transcription factors activated. The *syk* family usually activates the *src* family member in the receptor complex.

These receptors are often overexpressed on tumors of hematopoietic origin, and, similar to receptor tyrosine kinases, autocrine or paracrine stimulation may contribute to the neoplastic state of the tumor cell.

Serine/Threonine Kinase Receptors These receptors recognize [TGF- \$\beta\$](#) , bone morphogenetic factors, and other activins as ligands. Ligand binding leads to activation of the receptor kinase activity, but downstream events are not well defined. Bone morphogenetic factors are important in bone formation and in determining ventral vs. dorsal orientation in the developing embryo. TGF- β induces transformation of

mesenchymal cells but inhibits the proliferation of most cell types through the induction of [cdki](#), which block G1 progression in an Rb-dependent manner. Activation of TGF- β receptors leads to the phosphorylation of transcription factors Smad2 and Smad3, which then associate with their obligate partner Smad4. This complex, probably a heterotrimer, translocates into the nucleus where specific genes are activated. The direct path to the nucleus is analogous to the JAK/[STAT](#) signaling pathway. Smads bind to specific DNA sequences adjacent to other transcription factor sites in the promoters of target genes, leading to cooperative interaction for gene induction. TGF- β -induced genes include plasminogen activator inhibitor-1 (PAI-1), collagenase I, and p15^{INK4b}.

Many cancers are resistant to growth inhibition by TGF- β , including leukemias, lymphomas, melanomas, and breast and colon cancers. Colon cancers harboring defects in DNA mismatch repair develop inactivating mutations in the extracellular domain of the TGF- β receptor. In pancreatic and colon cancers, missense mutations and loss of heterozygosity at the DPC4 locus on chromosome 18q21 are frequent. This locus has been found to encode Smad4, and its inactivation blocks TGF- β signaling. Loss of expression or loss of function of TGF- β receptors occurs in several tumor types including colon cancer and lymphomas.

Nuclear Hormone Receptors Steroids, retinoids, thyroid hormone, vitamin D, and other lipid-soluble hormones diffuse through the plasma membrane and bind to members of the nuclear hormone receptor family. Receptors for these ligands are transcription factors that reside in the nucleus. The hydrophobic nature of the ligands obviates the need for machinery to transduce signals from the cell surface to the cell interior. Steroid hormone receptors are bound as heterodimers to promoter/enhancer elements of genes; in the absence of ligand, these complexes act as transcriptional repressors. Upon ligand binding, conformational changes are induced and the active transcription factor binds to coactivating factors and transcription is induced. Coactivators tend to open chromatin structure by adding acetyl groups to histones. Histone acetylation in nucleosomal complexes permits access of promoter regions to RNA polymerase II. Transcriptional repressor complexes associate with histone deacetylases (HDAC; co-repressor complexes), and chromatin remains condensed. Nuclear hormone-induced pathways affect virtually all biologic processes. Retinoic acid receptors (RAR) provide a clear link to cancer. Retinoids bind receptors composed of an RAR subunit (α , β , γ) dimerized with a retinoid-X receptor (RXR) and activate genes that influence differentiation in many cell types.

Acute promyelocytic leukemia (APL) is associated with the t(15;17) translocation, which fuses sequences from a novel gene PML (promyelocytic leukemia) to the [RAR](#) gene, resulting in expression of a PML-RAR α fusion protein. PML-RAR α binds to and represses RAR α -inducible genes required for myeloid differentiation. Repression is mediated by [HDAC](#) binding to PML-RAR α . The developmental arrest at the promyelocyte stage of differentiation is associated with unregulated proliferation in these cells. Patients with APL can achieve complete remission with pharmacologic doses of all-trans retinoic acid (tretinoin), the ligand for RAR α . Tretinoin induces the release of HDAC, permitting coactivator binding. Drugs that inhibit HDAC activity may provide a therapeutic benefit by activating genes required for the differentiation of cancer cells.

Cell-Cell and Cell-Extracellular Matrix (ECM) Communication In addition to

information conveyed by soluble mediators, cell surface receptors relay important signals between cells, such as contact inhibition, and cell-ECM signals, such as anchorage-dependent growth. In cancer, these highly organized mechanisms of intercellular interaction often become disrupted or are subsumed for the purpose of metastasizing ([Chap. 83](#)). Individual cells no longer respond to signals from their neighbors, actin filaments are highly disorganized, and adherens junctions are lost. Normal patterns of growth and differentiation are disrupted, and the potential for metastasis increases.

E-cadherins are integral membrane glycoproteins that mediate calcium-dependent homophilic adhesion as the major component of adherens junctions between epithelial cells. E-cadherin cytoplasmic domains bind complexes containing α - and β -catenins, which are structurally linked to the cytoskeleton (actin cables and intermediate filaments). β -Catenin that is not sequestered in E-cadherin complexes is rapidly phosphorylated by glycogen synthase kinase (GSK) 3 β in a complex with the APC (adenomatous polyposis coli) gene product (maps to chromosome 5, mutated in familial polyposis) and is degraded by the ubiquitin/proteasome pathway. Degradation of β -catenin can be blocked by several mechanisms, including mutations that inactivate APC and mutations in serine phosphorylation sites within β -catenin that target it for degradation. Such mutations result in increased free β -catenin, which translocates into the nucleus and binds to members of the T cell factor (TCF) family of transcription factors, influencing the expression of genes such as *c-myc* and cyclin D1 that promote progression through G1. Excess free β -catenin has been implicated in hereditary and sporadic forms of colon cancer and melanoma. Decreased expression of E-cadherin has been noted in breast, colon, prostate, gastric, and other cancers and is a marker of poor prognosis.

Epithelial cell growth and survival require attachment of cells to components of the [ECM](#) that compose basement membranes, including collagen, fibronectin, vitronectin, and laminin. The integrin family of transmembrane receptors is composed of α and β subunits that adhere to the ECM and convey information to cytoplasmic membrane-associated structures called *focal adhesions*. The complexes, whose assembly is mediated by the Rho and Rac GTPases, are sites of attachment of actin cables but are also active in cell signaling through their association with focal adhesion kinase (FAK) and Src tyrosine kinases. Integrin-ECM interactions lead to activation of the Ras/[MAP](#)kinase, PI3K, and phospholipase C- γ pathways. Detachment of epithelial and endothelial cells from ECM induces their death by a form of programmed cell death called *anoikis* (Greek, "homeless"). This molecular safeguard prevents abnormal spread of cells. Invasive cancers often avoid anoikis by activating Ras or Src, which allow anchorage-independent growth of cells by activation of Akt kinase.

REGULATION OF GENE TRANSCRIPTION

One consequence of signal transduction is the activation of sequence-specific transcription factors that regulate gene expression. Whether a cell proliferates, differentiates, or undergoes apoptosis is regulated by gene products made in response to physiologic stimuli. For some transcription factors, the ligand goes directly to the nucleus where they reside (nuclear hormone receptors). For others, activated kinases enter the nucleus and phosphorylate factors already bound to DNA ([MAP](#)kinase and

AP-1 transcription factor). Some transcription factors are activated in the cytoplasm and translocate to the nucleus ([STATs](#)). NF- κ B is held in the cytoplasm by the negative regulator I κ B, which is phosphorylated and degraded as a consequence of signal transduction. NF- κ B is then released from I κ B, and NF- κ B translocates to the nucleus.

Transcription factors recognize short stretches of DNA of a defined nucleotide sequence 6 to 12 base pairs in length. These recognition sites may be upstream or downstream of the transcription start site [the TATA box where the first subunit of the transcription machinery, transcription factor IID (TFIID), binds]. Transcription factors may affect transcription at sites remote from the start site by looping out large intervening DNA sequences.

Transcription factors contain specific amino acid sequences capable of recognizing the DNA sequence and usually form one of several structural motifs: helix-turn-helix, homeodomain, zinc finger, leucine zipper, and helix-loop-helix are all used as DNA-binding motifs or mediate dimerization of factors required for DNA binding. Transcription factors function in one or more of several ways. They can physically bend the DNA to permit the ordered addition of the components of the transcription machinery. Activated transcription factors bind to coactivator proteins in complexes that encode enzymatic activity leading to the acetylation of histones. This alters nucleosomal conformation and increases accessibility of DNA to transcription proteins. Transcription factors can inhibit transcription by blocking binding of a positive transcription factor or preventing the assembly of a transcription complex. They can form complexes with co-repressors and deacetylate histones. Promoters can also be made inaccessible by methylation of cytosine- and guanosine-rich sequences near promoters. The complex interaction between positive and negative transcription factors dictates the level of gene transcription. Individual genes may have 20 sites for transcription factor binding. The pattern of gene expression is determined by which factors are expressed in a given cell type.

Most genes are regulated at multiple levels, though transcription initiation is the dominant control point. The von Hippel-Lindau gene on chromosome 3p (a tumor suppressor gene involved in the pathogenesis of renal cell cancer) appears to act by inhibiting the elongation of an RNA chain after transcription initiation. A message may be spliced alternatively and encode different proteins in different cells. Transport of the message from the nucleus to the cytoplasm may be altered. Messenger RNA turnover may be accelerated. Some proteins, such as apoferritin and thymidylate synthase, regulate the translation of their own messages (and perhaps other messages) by binding to mRNA and preventing initiation of protein synthesis. Thus, there are many levels at which gene expression may be influenced.

Some transcription factors were identified because of their transforming effects when their genes, usually in mutated form, were incorporated into animal retroviral genomes; *myc*, *rel*, *fos*, *jun*, and others are examples of proto-oncogene transcription factors that are overexpressed in certain cancers and that contribute to the malignant phenotype of tumor cells. The mutated oncogenes are often more resistant to protein degradation and have a longer half-life than the normal cellular counterpart. Transcription factors with unusual properties may be generated by chromosome translocation that produces a chimeric protein. Novel genes may be activated that promote proliferation and inhibit

apoptosis. Usually the genetic changes lead to inhibition of normal lineage-specific differentiation, resistance to apoptosis, and proliferation.

REGULATION OF CELL DEATH

The homeostasis of adult organisms requires a balance between the generation of new cells and the death of old cells. Some cells die when their telomeres no longer protect the integrity of DNA replication. Some cells die when they have sustained sufficient hypoxic, heat, oxidative, or ultraviolet irradiation damage that cannot be repaired. A cell can be killed if it becomes infected with a virus or other intracellular pathogen that destroys the cell or is recognized by the host's lymphocytes, which kill the infected cell. Multicellular organisms are models of cellular cooperation; some cells die to preserve the rest of the organism.

Genetic damage to growth-regulating genes of stem cells could be catastrophic; however, single genetic events such as activation of *myc* expression or loss of the Rb checkpoint often lead to the death of the cell by apoptosis. Apoptosis is a form of cell death initiated by extracellular or intracellular signals in which enzymes are activated to degrade nuclear DNA by making intranucleosomal cuts, causing the cell to shrink and finally break up. The core apoptosis machinery consists of a family of specialized proteases called *caspases* (they contain a cysteine at their active site and cleave substrates after *aspartic acid* residues). Like coagulation and complement systems, caspases exist as proenzymes with minimal enzymatic activity that can be rapidly induced by activators, and each enzyme acts to activate the next enzyme in the cascade. Key targets include DNA (chromatin degraded into nucleosomal multimers), nuclear lamins (nucleus shrinks and fragments), cytoskeletal regulatory proteins, DNA repair enzymes, and others. The cell shrinks, its chromatin fragments, and the cell breaks apart forming apoptotic bodies.

The latent activity of caspases is tightly regulated to prevent the death of normal cells. Assembly of initiator caspases into active complexes occurs by two main mechanisms. Members of the tumor necrosis factor (TNF) receptor superfamily, including Fas (CD95), and DR4 and DR5 death receptors encode transmembrane proteins whose cytoplasmic domains encode protein association or docking domains, called *death domains* and *death effector domains* (DED). Ligand binding induces dimerization of death domains with recruitment to the membrane of an adaptor signaling protein called *Fas-associated death domain* (FADD). FADD forms a complex with procaspase 8 mediated by DED interactions; caspase 8 is activated by self-cleavage. Caspase 8 then cleaves effector caspase 3 into active heterodimeric subunits, and death ensues ([Fig. 82-4](#)). Regulation of this pathway occurs at the level of expression of Fas receptor and ligand and the secretion of death-inducing cytokines, TNF and tumor necrosis factor-related apoptosis-inducing ligand (TRAIL) (ligand for DR4 and -5).

The second pathway of caspase activation encompasses responses to a greater variety of noxious signals including DNA damage, growth factor deprivation, reactive oxygen damage, and heat stress. The mitochondrion plays a key role in this pathway as the storehouse of protein cofactors required for the activation of caspases. Damage within the cell is detected by the mitochondria by unknown mechanisms. The mitochondria then lose membrane potential and release cytochrome c, which forms a complex with

apoptosis-activating factor (Apaf) 1. This complex binds to procaspase 9 via a caspase recruitment domain (CARD), and caspase 9 is activated. Caspase 9 then cleaves effector caspases and induces cell death. The release of cytochrome c from the mitochondria appears to be regulated by *bcl2*.

The *bcl2* gene was discovered as the chromosome 18q contribution to the t(14;18) translocation in follicular lymphoma. The gene did not transform cells but prolonged the life of cells destined to die, greatly increasing a pool of cells available for subsequent genetic mutations. Members of the *bcl2* family fall into two groups: *bcl2* and *bcl-X_L* prevent cell death, whereas *bax*, *bad*, *bak*, and others promote cell death. The *bcl2* family members associate as homodimers or heterodimers; the combinatorial effects of the various dimers allow a fine level of control over cell survival. When any of the death promoters exist as homo- or heterodimers, the cell dies by apoptosis. When a death promoter heterodimerizes with either *bcl2* or *bcl-X_L*, cell death is prevented. Thus, the relative amounts of different *bcl2* family members determine whether a cell will survive potentially damaging insults. Furthermore, phosphorylation of *bcl2* family members by cellular kinases can alter the biologic activity of individual members, altering the balance in favor of death or survival.

In addition to its presumed role in the etiology of follicular lymphoma, *bcl2* is expressed in a number of cancers. It prevents the normal p53-mediated destruction of cells with damaged DNA and also appears to prevent the death of cells severely damaged by cancer chemotherapy. In addition, *bcl2* mediates drug resistance and contributes to neoplasia in a novel way, preventing the death that would normally eliminate the damaged cell, rather than promoting aberrant cell growth. Strategies to overcome *bcl2* function might well make available therapies more effective.

The apoptotic machinery is subject to regulation by multiple signal transduction pathways, and many of these are subverted in cancer cells to shift the balance toward survival of the malignant clone. In addition to *bcl2*, two other important links have been established between growth factor signaling and survival pathways. Activation of P13K by tyrosine kinases leads to activation of the serine/threonine kinase Akt. Akt directly promotes cell survival by phosphorylation of Bad and procaspase 9, inhibiting their apoptotic functions. Cancer cells can usurp the activity of Akt; in some cases, cells expressing a mutated Ras oncogene or increased levels of tyrosine kinase receptors (e.g., HER-2/*neu*) have upregulated the Akt pathway. An alternative genetic lesion leading to increased Akt kinase activity results in the loss of the PTEN tumor suppressor, a lipid phosphatase that normally downregulates the P13K pathway by dephosphorylating lipid second messengers. Another important pathway usurped by cancer cells involves NF- κ B activation. NF- κ B induces expression of the inhibitor of apoptosis (IAP) family of genes whose products inhibit caspase activity; one such, survivin, is expressed in lymphomas and other tumors.

Thus, cancer becomes more adaptive to its host from genetic events that alter apoptosis. Stimulation of proliferation or prevention of death can be complementary targets for cell transformation. Apoptosis pathways are important targets for treatment. Paclitaxel and other microtubule inhibitors induce the phosphorylation and inactivation of *bcl2*. One could make bone marrow cells highly resistant to chemotherapy-induced death by expressing a form of *bcl2* that lacks the loop domain in the BH1 region, as this

is the site that is phosphorylated to inhibit *bcl2* function. Tumor cells and normal cells express DR4 and -5 receptors; however, tumor cells fail to express decoy receptors that protect normal cells from the DR4 and -5 ligand, TRAIL. Thus, therapies directed at DR4 or -5 may be tumor selective.

CELL BIOLOGY AND CANCER

For a cancer to arise, mutations must occur that affect a variety of pathways. Often the G1 cell cycle checkpoint is affected. Apoptosis is averted by mutations in the p53 pathway or by other mechanisms. The expression of telomerase is a common feature in cancers. Overexpression of growth factors and their receptors is frequently detected. Activation of the *Ras* proto-oncogene or other changes leading to a constitutively active [MAP](#)kinase cascade are common. Changes in cytoskeleton and responsiveness to contact-mediated growth inhibition are frequent in cancer cells. Usually when a mutation occurs in one component of a signaling pathway, other mutations are seen in other pathways rather than in another component of the same pathway. The high level of mutability of cancer cells facilitates adaptation to the environment, including the development of resistance to anticancer drugs. As tumors progress, they acquire the ability to secrete proteases that aid in the escape from local barriers so that they may metastasize ([Chap. 83](#)). Discrete steps in tumor progression lead to the production of factors by the tumor cells that permit neovascularization to supply nutrients to the growing tumor. Other mutations allow the tumor to escape immune surveillance mechanisms; for example, some tumors downregulate expression of class I major histocompatibility complex antigens so that they become invisible to T cells. The wide range of changes that must occur in a single cell to permit the behavior associated with a malignant neoplasm makes it clear why carcinogenesis is a multistep process and why human cancers may have 10 or more genetic lesions that account for the biology.

The characteristic of cancer cells that has dominated clinical thinking is their uncontrolled proliferation. However, the growth fraction of most human cancers is usually not higher than the growth fraction of normal gut epithelia or normal bone marrow, and most human tumor explants are difficult to propagate for long periods of time in culture. Cancer cell lines immortal in vitro may have additional genetic lesions that permit their growth in vitro. Naturally occurring tumors growing in vivo show a Gompertzian or exponential decline in their growth fraction because the daughter cells of a division are not uniformly capable of further division. The accumulation of genetic damage, poor oxygen or nutrient supply, and other unknown factors contribute to the senescence of some tumor cells, so that by the time a tumor becomes clinically apparent at a tumor burden of 10^8 to 10^9 cells, most of the proliferative capability of the tumor is finished. Often by this time, more malignant and highly selected clonal derivatives of the tumor have metastasized to other sites where new tumor deposits with more aggressive characteristics are formed. Thus, cancer cells can be viewed as having lost the altruism that usually characterizes cell behavior in multicellular organisms. Cancer cells operate under natural selection imposed by a hostile environment. Ironically, the more successful they are at achieving independence from environmental influences, the more assured is the destruction of their host and ultimately themselves.

Many potential therapeutic agents are in clinical development based on our concepts of tumor cell biology. They include the development of growth factor and growth factor

receptor antagonists; inhibitors of phosphoryl transfer to block key kinases; selective inhibitors of [PKC](#), P13K, phospholipase C, and other targets; farnesyl transferase inhibitors that block the insertion of *Ras* into the membrane; mutant versions of proteins such as *Ras* and p53 that may make the cell vulnerable to immunologic attack if employed as a vaccine; and inhibitors of angiogenesis or the steps in metastasis that may limit tumor growth and prevent its spread. However, it seems unlikely that a single target will be the highly sought-after point of vulnerability. More likely, combinations of inhibitors will be required to improve antitumor effects. For example, the combination of chemotherapy and antibody to [EGF](#) receptors appears to produce greater antitumor effects than the sum of the effects produced individually.

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83. ANGIOGENESIS - Judah Folkman

Virtually every cell in the body lives adjacent to a capillary blood vessel, or at least no further than the mean oxygen diffusion distance of 100 to 200 μm . Some cell types, such as beta cells in the pancreatic islets, fat cells, and skeletal muscle cells, are surrounded by at least two capillaries ([Fig. 83-1](#)). Capillaries of 8 to 20 μm diameter are lined by a single layer of endothelial cells. These cells cover $\sim 1000 \text{ m}^2$, an area the size of a tennis court. The length of capillary tubing in 1 mm^3 of human heart muscle is $\sim 2500 \text{ mm}$, and 1 kg of fat contains $\sim 3500 \text{ m}$ of capillaries. During normal conditions vascular endothelial cell proliferation is barely detectable -- $<0.01\%$ of endothelial cells are in cycle. Endothelial cell turnover is >1000 days and in retinal vasculature may be >5000 days. In contrast, in the normal adult bone marrow, ~ 6 billion cell divisions occur per hour and the turnover time is ~ 5 days, (i.e., the time during which bone marrow is completely replaced). Endothelial cells can emerge from their resting state and proliferate as rapidly as bone marrow cells during formation of new capillaries. This process is called *angiogenesis* and leads to *neovascularization*.

Physiologic angiogenesis is tightly regulated and of limited duration. It is essential to reproduction and embryonic development. During postnatal and adult life, angiogenesis in wound repair and in exercised muscle is restricted to days or weeks.

Pathologic angiogenesis, in contrast, is usually persistent and unabated. Angiogenesis that continues for months or years supports the growth and progression of solid tumors and leukemias, provides a conduit for the entry of inflammatory cells into sites of chronic inflammation (e.g., Crohn's disease and chronic cystitis), is the most common cause of blindness, destroys cartilage in rheumatoid arthritis, contributes to growth and hemorrhage of atherosclerotic plaques, leads to intraperitoneal bleeding in endometriosis, is the basis of life-threatening hemangiomas of infancy, and permits prostate growth in benign prostatic hypertrophy. These are just a few of the "angiogenic disease processes," which are found in almost all specialties of medicine. Angiogenesis inhibitors are a new *class* of drugs that suppress or reverse the pathologic neovascularization upon which these diseases are dependent.

NEOPLASTIC DISEASE

HISTORIC BACKGROUND

Tumor hyperemia, observed during surgery since the 1870s, was for the next 100 years attributed to simple dilation of existing host vessels. Two reports, in 1939 and 1945, suggested that tumor vascularity was due to the induction of new blood vessels. This idea was dismissed by most investigators. The few who accepted it believed that new vessels were an inflammatory side effect of tumor growth.

In 1971, based on experiments carried out in the 1960s, a hypothesis was proposed that tumor growth could be angiogenesis-dependent, i.e., tumors could recruit their own private blood supply by releasing a diffusible chemical signal that stimulated angiogenesis. Tumor angiogenesis could then be a novel second target for anticancer therapy. These concepts were not accepted at the time. The conventional wisdom was that tumor neovascularization was (1) an inflammatory host response to necrotic tumor

cells, (2) a host response detrimental to the tumor, or (3) "established" vasculature that could not regress. From these assumptions most scientists concluded that it was fruitless to attempt to discover an angiogenesis stimulator, to say nothing of discovering angiogenesis inhibitors. Eventual acceptance of the 1971 hypothesis was slow because it would be 2 more years before the first vascular endothelial cells were successfully cultured in vitro, 8 more years before *capillary* endothelial cells could be cultured in vitro, 11 years before the discovery of the first angiogenesis inhibitor, and 13 years before the purification of the first angiogenic protein. By the mid-1980s, after a series of reports from several laboratories demonstrating that tumor growth was angiogenesis-dependent, this hypothesis had been confirmed by genetic methods.

THE ANGIOGENIC SWITCH

The Prevascular Phase Most human tumors arise without angiogenic activity and exist in situ as microscopic-sized lesions of 0.2 to 2 mm diameter for months to years, after which a small percentage may switch to the angiogenic phenotype. Autopsy studies of people who died of trauma but who never had cancer during their lifetime reveal that in women from 40 to 50 years of age, 39% had in situ carcinomas in their breast, but breast cancer is diagnosed in only 1% of women in this age range. In men from age 50 to 70, 46% had in situ prostate cancers at the time of death, but only 1% are diagnosed in this age range during life. In people from age 50 to 70, >98% had small carcinomas of the thyroid ([Fig. 83-2](#)), but thyroid cancer is diagnosed in only 0.1% in this age range. In the majority of human tumors, the angiogenic phenotype appears after the malignant phenotype is recognized histologically. However, for certain human tumors (e.g., carcinoma of the cervix), the preneoplastic stage of dysplasia becomes angiogenic before the malignant phenotype is recognized histologically. When a nonangiogenic in situ carcinoma emerges in avascular epidermis or mucosa (e.g., melanoma or breast cancer), it is separated from host vessels by a basement membrane ([Fig. 83-3](#)). If a nonangiogenic tumor emerges in the midst of a vascularized tissue (e.g., an islet cell carcinoma), it may form an in situ microcylinder of tumor cells around capillary vessels (called *cooption*).

At the clinical level the angiogenic switch is recognized by expansion of tumor mass to a detectable size, local bleeding, and metastasis. For example, a positive mammogram usually represents a neovascularized tumor -- a non-neovascularized in situ carcinoma is below the detectable limits of mammography. Hematuria in bladder cancer, melena in colorectal cancer, and hemoptysis in lung cancer all result from neovascularized tumors. Tumor cells are not usually shed into the circulation until after neovascularization has occurred. Furthermore, distant metastases themselves cannot be detected until they have "turned on" the angiogenic switch.

At the cellular level at least four mechanisms of the angiogenic switch have been identified in human and mouse tumors: (1) avascular in situ carcinomas can recruit their own blood supply by stimulating neovascularization in an adjacent host vascular bed -- the most common process in human tumors; (2) circulating precursor endothelial cells from bone marrow may incorporate into an angiogenic focus; (3) tumors may induce host fibroblasts and/or macrophages in the tumor bed to overexpress an angiogenic factor [e.g., vascular endothelial growth factor (VEGF)]; and (4) preexisting vessels can be coopted by tumor cells. The angiogenic switch may also include combinations of

these mechanisms. Once tumors have switched on angiogenesis, they rarely revert to the nonangiogenic phenotype. Neuroblastoma and retinoblastoma may be exceptions, but spontaneous loss of angiogenic activity (and tumor regression) is rare even in these two tumors. After the angiogenic switch, new microvessels converge on the tiny in situ tumor. Tumor cells grow as microcylinders, or "perivascular cuffs," around each new vessel. One endothelial cell can support from 5 to 100 tumor cells.

At the molecular level, the angiogenic switch operates as a shift in the balance of production by tumor cells of molecules that positively or negatively regulate angiogenesis. The overexpression of positive regulators of angiogenesis and the downregulation of inhibitors of angiogenesis during early tumor development are generally triggered by genetic mutations that control angiogenesis. For example, overexpression of the *ras* oncogene increases production of the angiogenic protein [VEGF](#), while a mutation in the p53 tumor-suppressor gene or its deletion decreases production of the angiogenesis inhibitor protein, thrombospondin-1. In the normal cell wild-type p53 upregulates thrombospondin-1 and downregulates VEGF.

The angiogenic switch can be further modified by environmental conditions such as hypoxia, endogenous angiogenesis inhibitors, and genetic background of the host.

1. *Hypoxia*: After a tumor has become neovascularized, its continued expansion may lead to increased tissue pressure. This increased interstitial pressure is caused mainly by plasma that leaks from new vessels but is slow to efflux from the tumor because of a dearth of intratumoral lymphatics. Microvessels in the center of the tumor are the first to be compressed, which leads to central necrosis. Hypoxia activates an hypoxia-inducible factor (HIF-1) binding sequence in the [VEGF](#) promoter. This leads to transcription of VEGF mRNA, increased stability of VEGF message, and increased production of VEGF protein beyond what may have been triggered genetically. Tumors, therefore, do not "outgrow their blood supply" but compress it. A counterintuitive lesson is that in situ tumors arising in an avascular compartment (e.g., epidermis) are *not* hypoxic, but larger neovascularized tumors become hypoxic after compressing their blood supply. Low pH and low glucose in a tumor may also upregulate production of angiogenic factors, especially VEGF.

2. *Endogenous angiogenesis inhibitors*: At the molecular level, the angiogenic switch can also be modified by endothelial inhibitors that either circulate [e.g., interferon (IFN) β , platelet factor 4, angiostatin] or are releasable from extracellular matrix [e.g., endostatin, thrombospondin-1, and tissue inhibitors of metalloproteinases (TIMPs)]. Therapeutic administration of an endogenous angiogenesis inhibitor, such as angiostatin or endostatin, can tip the balance of the angiogenic switch so that angiogenic output of a tumor is opposed or abrogated.

3. *Genetic control of host response*: Just as the angiogenic output of a given tumor is governed by oncogenes and tumor-suppressor genes, the angiogenic response of the host is genetically regulated. It is known that hemangiomas predominate in white infants and that ocular neovascularization in macular degeneration is almost never found in black patients. The genes that regulate these effects are not yet known.

Endogenous Angiogenesis Promoters The known endogenous angiogenic promoters

are listed in [Table 83-1](#). Virtually all of these proteins are produced by different types of tumors. However, acidic FGF (aFGF), basic FGF (bFGF), [VEGF](#), and angiopoietin-1 and -2 are the most well studied and have been found in a wide variety of human tumors.

Fibroblast Growth Factors [aFGF](#) and [bFGF](#) stimulate endothelial cell mitosis and migration in vitro and are among the most potent angiogenic proteins in vivo. They have high affinity for heparin and heparan sulfate. They lack a signal sequence for secretion but are stored in extracellular matrix. An unsolved problem is how bFGF is exported from tumor cells in the absence of a signal sequence. Many different cells synthesize bFGF, including tumor cells of the central nervous system, sarcomas, genitourinary tumors, and even endothelial cells in the tumor vasculature. Proteinases and heparanases are thought to mobilize bFGF from the extracellular matrix. Furthermore, some tumors recruit macrophages and activate them to secrete bFGF, while others attract mast cells, which, because of their high heparin content, sequester bFGF. bFGF is not a specific endothelial mitogen but has several cell targets including fibroblasts, smooth-muscle cells, and neurons. However, experimental tumors transfected with bFGF containing an engineered signal sequence stimulate endothelial proliferation almost to the exclusion of smooth-muscle and fibroblast proliferation. This is similar to the process in human tumors. This selective attraction of vascular endothelial cells by bFGF released from a tumor may be explained by the smooth-muscle repellent activity of angiopoietin-2 elaborated from proliferating endothelial cells in a tumor bed (see below). bFGF interferes with adhesion of leukocytes to endothelium; thus, tumors that elaborate bFGF may produce a form of local immunologic tolerance.

Abnormally elevated levels of [bFGF](#) are found in the serum and urine of cancer patients and in the cerebrospinal fluid of patients with different types of brain tumors. High bFGF levels in renal carcinoma correlate with poor outcome. Also, bFGF levels in the urine of children with Wilms' tumor correlate with stage of disease and tumor grade.

Vascular Endothelial Growth Factor/Vascular Permeability Factor The first proposal that tumor angiogenesis is associated with increased microvascular permeability led to the identification of vascular permeability factor (VPF). VPF was subsequently sequenced and shown to be a specific inducer of angiogenesis; it was called *vascular endothelial growth factor*. [VEGF](#) is an endothelial cell mitogen and motogen that is angiogenic in vivo. Its permeability effect on capillaries is more potent than histamine and contributes to ascites in ovarian cancer and to edema in brain tumors. Its expression correlates with blood vessel growth during embryogenesis and with angiogenesis in the female reproductive tract and in tumors. VEGF is a 40- to 45-kDa homodimeric protein with a signal sequence secreted by a wide variety of cells and by the majority of human tumor cells. For example, >60% of breast cancers overexpress VEGF. VEGF exists as five different isoforms of 121, 145, 165, 189, and 206 amino acids, of which VEGF₁₆₅ is the predominant molecular species produced by a variety of normal and neoplastic cells. Two receptors for VEGF are found mainly on vascular endothelial cells, the 180-kDa fms-like tyrosine kinase (Flt-1) and the 200-kDa human kinase insert domain-containing receptor (KDR) and its mouse homologue, Flk-1. VEGF binds to both receptors, but KDR/Flk-1 transduces the signals for endothelial proliferation and chemotaxis. Other structural homologues of the VEGF family have recently been identified, including VEGF-B, -C, -D, and -E. VEGF-C stimulates lymphatic growth and binds to Flt4, which is preferentially expressed on lymphatic endothelium. Neuropilin-1, a neuronal guidance

molecule, is a recently discovered receptor for VEGF₁₆₅. Neuropilin is not a tyrosine kinase receptor and is expressed on nonendothelial cells including tumor cells. This allows VEGF that is synthesized by tumor cells to bind to their surface. Surface-bound VEGF could make endothelial cells chemotactic to tumor cells or it could act in a paracrine manner to mediate cooption of tumor cells around microvessels ([Fig. 83-3](#)).

[VEGF](#) expression is upregulated by the *ras* oncogene. The farnesyl transferase inhibitors inhibit *ras* expression. One mechanism of their antitumor effect is to inhibit angiogenesis by inhibiting VEGF expression.

[VEGF](#) expression is inhibited by the von Hippel-Lindau (VHL) protein. The VHL-tumor suppressor gene is inactivated in patients with VHL disease and in most sporadic clear-cell renal carcinomas, which leads to VEGF-mediated angiogenesis. The VHL gene normally suppresses hypoxia-inducible genes including erythropoietin. When VHL is mutated or deleted, these genes are overexpressed even under normoxic conditions. This explains why renal cell carcinomas driven by mutant VHL are associated with a high hematocrit.

Experimental evidence indicates that [bFGF](#) function may be in part dependent upon [VEGF](#). bFGF induces the expression of VEGF. The two endothelial mitogens act synergistically to stimulate capillary tube formation in vitro. Systemic administration of a soluble receptor for VEGF (flk-1) completely blocks cornea angiogenesis induced by implanted bFGF. An important implication of these studies is that angiogenesis inhibitors that block VEGF (currently in clinical trial) may also inhibit bFGF.

Angiopoietins Tie2 is a receptor found only on vascular endothelial cells. It is a specific tyrosine kinase whose ligand is angiopoietin-1. Angiopoietin-1 induces endothelial cells to recruit pericytes and smooth-muscle cells [mainly by producing platelet-derived growth factor (PDGF) BB] to become incorporated in the vessel wall. Vessels stimulated by angiopoietin-1 are not leaky and are analogous to new vessels in a healing wound. Angiopoietin-2 blocks the Tie-2 receptor and acts to repel pericytes and smooth muscle. It is produced by vascular endothelium in a tumor bed, but it is unclear how tumor cells mediate this. Nevertheless, tumor vessels remain thin "endothelial-lined tubes" even though some of these microvessels reach the diameter of venules ([Fig. 83-3](#)). A key point is that angiopoietin-2 and [VEGF](#) together increase angiogenesis. However, if VEGF is neutralized or withdrawn, endothelial cells in the absence of perivascular smooth muscle and pericytes undergo apoptosis and new microvessels regress. These differences indicate that angiogenesis in tumors may be more vulnerable to certain angiogenesis inhibitors than angiogenesis in healing wounds. Endostatin inhibits tumor growth in mice without delaying wound healing.

Endogenous Angiogenesis Inhibitors Certain endogenous inhibitors of angiogenesis are known to play a role in the angiogenic switch, including: [IFN- \$\alpha\$](#) and platelet factor 4, and the class of angiostatic steroids typified by tetrahydrocortisol ([Table 83-2](#)).

Thrombospondin-1 The production of thrombospondin-1 has been shown to be inversely related to the ability of a cell line to produce a tumor and vessels in vivo; loss of thrombospondin-1 production allowed non-tumorigenic cells to become tumorigenic. Thrombospondin-1 is regulated by wild-type p53. Loss of p53 function in tumor cells

dramatically decreased the level of angiogenesis inhibitor. Restoration of p53 increased the inhibitor and suppressed the angiogenic activity of the tumor cells. The angiogenic switch was controlled by a negative regulator of angiogenesis generated by the tumor. The switch itself was viewed as a result of a shift in the "net balance" of angiogenesis stimulators and inhibitors. This led to the discovery of angiostatin, a second inhibitor found to be involved in the angiogenic switch.

Angiostatin, Endostatin, and Antiangiogenic Antithrombin III Several clinical and experimental observations suggested that certain tumors may produce angiogenesis inhibitors. The removal of certain tumors (e.g., breast carcinomas, colon carcinomas, and osteogenic sarcomas) can be followed by rapid growth of distant metastases. A primary tumor can suppress metastases from a different type of tumor, e.g., a breast cancer can inhibit melanoma metastases. In melanoma, partial spontaneous regression of the primary tumor may be followed by rapid growth of metastases. Regression of small cell lung cancer by ionizing radiation may be followed by rapid growth of distant metastases. If one portion of a primary tumor is removed (e.g., cytoreductive surgery for testicular cancer), the residual tumor increases its rate of expansion. A similar phenomenon is observed in animal tumors, i.e., certain primary tumors inhibit the growth, but not the number, of their own metastases. Surgical removal of a primary tumor increases growth rate of the residual tumors. Many primary tumors can suppress the growth of a second tumor inoculation. This "resistance" to a second tumor challenge is inversely proportional to the size of the tumor inoculum and directly proportional to the size of the first tumor. A threshold size is necessary for the inhibitory effect to occur.

At least three hypotheses have been advanced to explain these diverse observations and experiments: (1) "concomitant immunity" -- a primary tumor induces an immunologic response against a secondary tumor or a metastasis in the same host; (2) depletion of nutrients by the primary tumor; or (3) production of antimitotic factors from the primary tumor that directly inhibit the proliferation of the secondary tumor. However, none of these ideas offers a molecular mechanism to explain all of the experiments cited above, and overall they have not been confirmed. Concomitant immunity has been ruled out as a mechanism because tumors can suppress metastasis in mice with severe combined immunodeficiency (SCID).

Once it was realized that a tumor could generate both positive and negative regulators of angiogenesis, then it also became clear that a primary tumor, while stimulating angiogenesis in its own vascular bed, could possibly inhibit angiogenesis in the vascular bed of a distant metastasis. However, at least two conditions would be necessary: (1) the primary tumor (i.e., the first tumor to grow) would need to generate an angiogenic promoter in excess of an inhibitor in its own local vascular bed, and (2) the putative inhibitor would need to have a longer half-life in the circulation than the angiogenic promoter. Research done over the past decade has identified angiostatin, endostatin, and antiangiogenic antithrombin as negative regulators of angiogenesis.

Lewis lung carcinoma generated angiostatin, a 38-kDa cleavage product of plasminogen. Systemic administration of purified angiostatin completely inhibited growth of metastases, producing dormant tumors of microscopic size (<200 μm in diameter) in the lung, and inhibited the growth of primary tumors. Angiostatin is not secreted by tumor cells but is generated through proteolytic cleavage of circulating plasminogen by